Dimension bounds for hyperbolic Coxeter polytopes

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Abstract

We prove that a finite-volume hyperbolic Coxeter polytope has dimension at most fifty-seven. For compact polytopes we obtain the bound seventeen, and fifteen when the reflection group is arithmetic. The arguments combine the local structure of Coxeter diagrams with hyperbolic geometry and exact finite computations.

1 Introduction

A hyperbolic Coxeter polytope is a convex polytope in real hyperbolic space whose dihedral angles are integer submultiples of π. Reflection in its facets generates a discrete group for which the polytope is a fundamental domain. Throughout this paper polytopes have finitely many facets and finite volume, and hyperbolic space has curvature -1. We prove the following dimension bounds.

Theorem 1.1. *Let $P \subset \mathbb{H}^n$ be a finite-volume hyperbolic Coxeter polytope.*

1. *The dimension satisfies $n \leq 57$.*
2. *If P is compact, then $n \leq 17$.*
3. *If P is compact and the group generated by its facet reflections is arithmetic, then $n \leq 15$.*

The classical upper bound for the dimension of a finite-volume hyperbolic Coxeter polytope is 995 [1]. Its proof belongs to the combinatorial approach based on small faces and the incidence estimates of [3]. For compact polytopes the previous upper bound is 29 [8, 9]. This bound also applies to the compact arithmetic case. The arguments below refine the restrictions imposed by the local Coxeter diagrams in these three settings.

Finite-volume arithmetic reflection polytopes exist in dimension 21 [2]. Thus compactness is essential in the third statement of Theorem 1.1. The arithmetic statement here concerns precisely the compact case; the first two statements concern all hyperbolic Coxeter polytopes in their respective classes.

1.1 The local approach

The proofs share the description of a polytope by the Lorentz inner products of its outward facet normals. The facets meeting a finite vertex give a spherical Coxeter diagram. At an ideal vertex, a horospherical section decomposes into a product of Euclidean Coxeter simplices, described by affine diagrams. The classifications of these diagrams organize the possible configurations into classical and exceptional families. Inverse Gram matrices and the Lorentz signature then restrict the ways in which further walls can meet such configurations.

After these common preliminaries, the finite-volume and compact arguments use different global comparisons. For the finite-volume bound we assign nonnegative weights to the corners of two-dimensional faces. Every triangle receives total weight at least two and every quadrilateral at least one. If $s(F)$ denotes the number of sides of a two-face, summing these weights bounds

$$\sum_F (5 - s(F)).$$

Spherical and affine vertex diagrams give upper bounds for the weight at each vertex. Incidence formulas give a conflicting lower bound in dimension 58. A separate argument treats all higher dimensions. Finite and ideal vertices enter this comparison through different local estimates.

For the compact bound, the comparison is geometric. We choose a vertex farthest from a fixed interior point and retain the resulting vector throughout the argument. Projections and inverse Gram matrices express the restrictions from the components of its spherical diagram as quadratic-form inequalities. Contributions from different components are compared using this same vector. The resulting estimates exclude the remaining diagrams in dimensions at least 18.

The arithmetic proof begins with the compact bound already established. Integrality and positivity at the other real embeddings restrict the ground field and the possible root norms. These restrictions exclude the remaining configurations in dimensions 16 and 17. This is the point at which the arithmetic hypothesis enters; the preceding compact argument is independent of it.

The finite computations are stated as mathematical problems with specified domains and exact arithmetic. Their algorithms and the arguments for termination and completeness accompany the required outputs. Uniform geometric estimates handle the unbounded ranks and labels before the remaining finite cases are considered.

1.2 Organization

Part I proves the finite-volume bound in Theorem 2.1. It develops the corner weights, the finite and ideal vertex estimates, and the incidence contradiction. Part II proves the compact bound in Theorem 8.1 by the farthest-vertex comparison. Part III proves the arithmetic bound in Theorem 19.1, using the independently proved result of Part II.

Three corresponding groups of proof appendices supply the detailed arguments. The first gives the local weight rules, forest and cusp calculations, and the higher-dimensional finite-volume exclusions. The second gives the component estimates and the remaining compact diagram cases. The third gives the field and norm restrictions and the remaining arithmetic cases. Each part introduces its method before the case analysis; the appendices complete the proofs of the three statements in Theorem 1.1.

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Part I
Finite-volume polytopes

2 The finite-volume bound

Theorem 2.1. *Every finite-sided, finite-volume real hyperbolic Coxeter polytope has dimension at most 57.*

The theorem belongs to the combinatorial approach to dimension bounds developed by Vinberg, Prokhorov and Khovanskii: small faces force local contributions that a high-dimensional vertex cannot accommodate [1, 3]. The largest presently known dimension of a finite-volume example is 21; Borcherds constructs an arithmetic reflection group in that dimension [2]. Thus the present bound still leaves a substantial existence gap.

This is a computer-assisted argument. The external ingredients are the spherical and affine classifications, the bounds for minimal indefinite diagrams, finite-volume vertex-link geometry, the incidence formulas, and Nikulin's narrow-facet theorem [5, 6]. The local weight inequalities and extension arguments are proved below.

The main argument excludes dimension 58. Appendices A.4 and A.5 treat every higher dimension: first by a finite set of payment families, and then by an all-rank narrow-facet argument. We do not infer this tail from the single excluded dimension, or pass to a facet whose angles have not been proved Coxeter. The marked local rules and their exact arithmetic certificates are internal parts of the argument, not additional published assumptions.

For a two-face F with $s(F)$ sides, write

$$\mathcal{D}_5(P) = \sum_F (5 - s(F)).$$

A simultaneous nonnegative corner rule paying at least 2 on each triangle and 1 on each quadrilateral gives an upper bound for $\mathcal{D}_5$. The final rule has finite-vertex charge at most 377. At an ideal vertex its actual charge C_{act} satisfies $\kappa - C_{\mathrm{act}} \geq 36$, where κ is the link shape curvature. The global lower bound is

$$\mathcal{D}_5 \geq 377 v_f + \sum_{v \text{ ideal}} \kappa(v) + 4205.$$

The contradiction is therefore

$$0 \geq 4205 + 36 v_i > 0.$$

The remainder proves the local and arithmetic interfaces that make this subtraction valid.

2.1 The common counting argument

An angle in this paper is a corner (v, F) of an actual two-face. A payment is a nonnegative weight $w(v, F)$. Write $C(v) = \sum_{F \ni v} w(v, F)$ for its total charge at a vertex. The elementary local-to-global step is worth stating independently of the particular rule and of the certificate files.

Lemma 2.2 (Payment comparison)**.** *Suppose every triangular face receives total payment at least 2 and every quadrilateral receives at least 1. Then*

$$\mathcal{D}_5(P) \leq \sum_v C(v).$$

More generally, suppose an incidence argument gives

$$\mathcal{D}_5(P) \geq Bv_f + \sum_{v \text{ ideal}} \kappa(v) + \Delta, \qquad \Delta > 0,$$

while a simultaneous payment rule has $C(v) \leq B$ at every finite vertex and $C(v) \leq \kappa(v) - \varepsilon$ at every ideal vertex, where $\varepsilon \geq 0$. These hypotheses are inconsistent.

Proof. A face with s sides needs payment at least $5 - s$. This is2 for a triangle and 1 for a quadrilateral. For $s \geq 5$ it follows from nonnegativity. Sum over faces and then over their corners. Combining the resulting upper bound with the proposed lower bound gives $0 \geq \Delta + \varepsilon v_i > 0$. □

The geometry enters in choosing a rule that works at every actual face. The computation enters after that reduction, when the possible vertex diagrams and their contextual costs have been specified. The same corner may not be spent twice merely because two local arguments could have used it: all priorities and compensations must belong to one simultaneous rule. Appendix A.2 gives that rule explicitly, as a function of a marked whole vertex diagram. For the preliminary finite count we use the same tables with the stated B_{13} endpoint restored and without the final three rank-15-guarded changes. We call this preliminary function the preceding rule below.

The proof first separates into finite and ideal vertices. At a finite vertex we maximize the rule over spherical forests of total rank 58. At an ideal vertex we bound each affine factor, multiply by its actual corner-fiber multiplicity, and compare with the curvature of the product link. These branches reunite in the payment comparison. The final changes to the rule use whole components: a visible path inside a larger component does not carry the same contextual bonus. Figure 2 records this division.

3 Corner weights and spherical diagrams

We use outward norm-two roots, with diagonal Gram entries2 and nonpositive off-diagonal entries. At an actual two-face in dimension n, the supporting white roots form a positive-definite rank $n-2$ diagram. White roots together with one black side give a strictly positive-definite edge diagram; adjacent corners are positive semidefinite. A finite corner has an elliptic forest of rank n.

These assertions refer to actual faces of the original chamber. We record two elementary feasibility arguments that will also control additional supporting roots. Let p be a strict interior

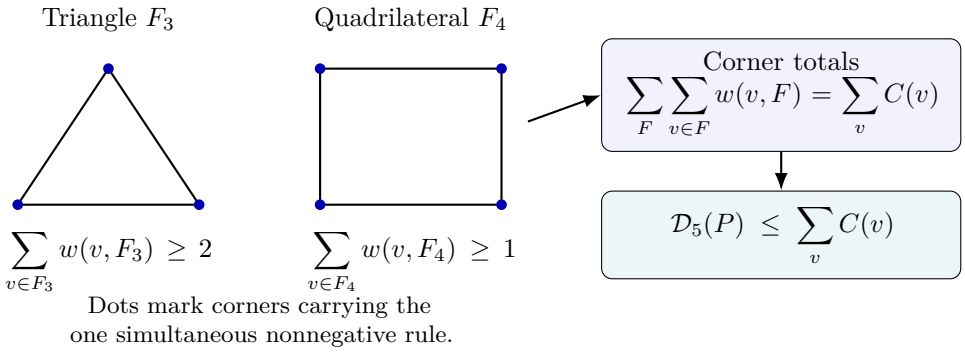


Figure 1: Corner weights in Lemma 2.2. Summing first over faces and then over vertices gives the upper bound for $\mathcal{D}_5(P)$.

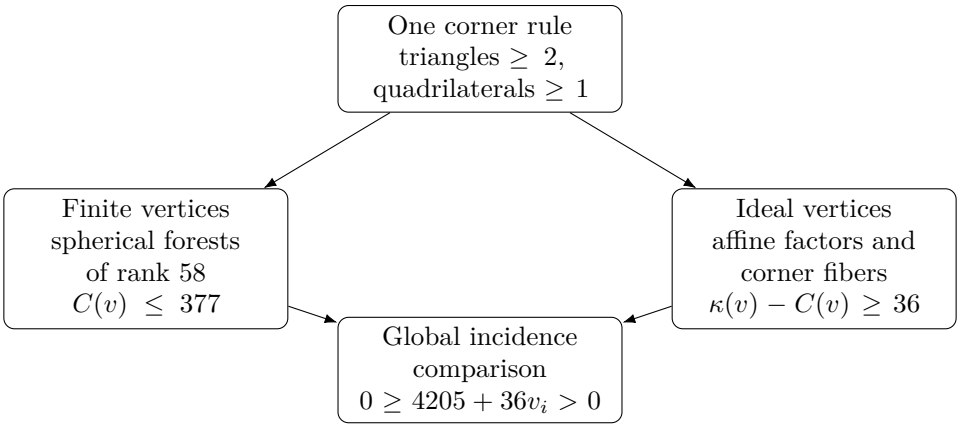


Figure 2: The common geometric rule is established before the finite and ideal calculations separate. The strict correction 4205 comes from the two extrema of a generic linear function on the polytope.

timelike point. A positive-definite principal Gram matrix B has nonpositive off-diagonal entries and an entrywise nonnegative inverse. If $b_i = -(p, e_i) > 0$, then

$$q = p + \sum_i (B^{-1}b)_i e_i$$

is orthogonal to the selected roots and has $q^2 = p^2 - b^T B^{-1} b < 0$. It is future pointing and strict against every other chamber wall. Thus a positive principal set has an actual finite common point, rather than merely a formal Gram model.

An irreducible affine principal set has a positive kernel vector (u_i). Its root combination $\nu = \sum_i u_i e_i$ is null and nonzero, since $(p, \nu) < 0$. It is feasible against every chamber wall. Its projective point is an ideal vertex: an ideal-sphere point is extreme in the closed Klein ball and hence in the chamber. The affine diagram is a whole factor of that cusp, because every proper principal subdiagram of an irreducible affine diagram is positive definite. This is also the containment input of [7, Section 1].

Lemma 3.1 (Opposite sides and minimal cores)**.** *For either opposite pair of sides of an actual quadrilateral, the supporting whites and that pair form an indefinite principal set. An inclusion-minimal indefinite subset contains both marked sides. Its order is at most ten; if all proper principal subsets are elliptic, its order is at most five.*

Proof. A positive set would have an actual common finite point by the projection argument. A positive-semidefinite singular set would have a common ideal point by its affine kernel. Opposite closed sides of a nondegenerate quadrilateral are disjoint, so neither is possible. Deleting either marked side leaves a subset of an actual finite edge diagram. Hence both marks lie in every minimal indefinite subset. The published Lannér/quasi-Lannér order bounds now apply; the classification through order ten is [4]. □

Two such opposite cores cannot be disjoint and orthogonal, since that would give two negative directions. Their overlap or a joining path supplies an adjacent-side route. The local theorem must then restore all the other supporting whites; an argument on the displayed cores alone would not justify a whole-component payment.

A whole component is a connected component of the full actual corner diagram, not merely of a displayed core. Presence conditions in a weight refer only to such whole components. This distinction is essential when additional supporting whites are restored.

Lemma 3.2 (Two ways to seal a component)**.** *Let G be the Gram matrix of an actual indefinite core.*

1. *A Gram-kernel vector z of this core gives the actual zero root relation $\sum_i z_i e_i = 0$. Every further wall therefore satisfies $\sum_i z_i(e, e_i) = 0$.*

2. *Suppose G is nonsingular, an added forest contacts it only at a position p, and $(G^{-1})_{pp} = r > 1$. If every connected component of the added forest contacts p, its total rank is at most $1/(r-1)$.*

Proof. The span of an indefinite core contains a timelike vector. The root combination associated with a Gram-kernel vector is null and orthogonal to that timelike vector. The perpendicular space is positive definite, so the combination is zero. This differs from the nonzero feasible null vector of an affine block.

For the second statement let B be the positive Gram matrix of all added supporting whites, including indirect tails, and let u be their contact column at p. Schur elimination and the one-negative- direction condition give $B - ruu^T \succeq 0$, whence $u^T B^{-1} u \le 1/r$.

For a connected square-two finite Coxeter Gram of rank k, every inverse diagonal is at least $k/(k+1)$. This follows by induction: delete the chosen node, let the remaining component ranks be k_j, and use a contact of magnitude at least one into each component. Inverse positivity gives Schur cost at least $\sum_j k_j/(k_j+1) \ge (k-1)/k$. The resulting inverse diagonal is at least $1/(2-(k-1)/k) = k/(k+1)$; the rank-one base is $1/2$. For an added forest of total rank M contacted in each component, the same bound gives $u^T B^{-1} u \ge M/(M+1)$. Thus $M/(M+1) \le 1/r$, proving $M \le 1/(r-1)$. □

The second argument takes all roots of the growing forest at once. A bound on its immediate neighbors would leave indirect extensions uncontrolled. For a radical seal, nonpositive contact signs and the displayed zero relation restrict every further root; the local short-path alternatives then force either a persistent payment or orthogonality. An indirect extension has a first contacting root, so this is the step that turns a core component into a whole one.

At an ideal vertex with affine factor ranks r_i, the link is $\prod_i \Delta^{r_i}$, with $\sum r_i = n-1$. A marked pair in factor i occurs in $\prod_{j \ne i}(r_j+1)$ actual corner fibers. Hence an affine factor envelope $(r+1)c(r)$ gives

$$C_{\mathrm{act}} \le C_{\mathrm{up}} = V \sum_i c(r_i), \qquad V = \prod_i (r_i + 1). \tag{3.1}$$

The factor-fiber theorem includes nonsimple ideal vertices. Whole affine factors used locally are established by the positive kernel and actual corner semidefiniteness, before applying this multiplicity. Indeed an edge of the product link varies in just one simplex factor. A fixed pair of its facet labels determines an edge in that factor and an arbitrary vertex in every other factor. These are precisely the $V/(r_i+1)$ fibers. The marked pair restores the varying affine factor; all other factors appear with one node deleted. Summing an unweighted copy of each affine table would miss this multiplicity.

3.1 The finite forest algorithm

For the finite calculation multiply charges by64. Its alphabet is

$$\{A_r : 1 \le r \le 58\} \cup \{B_r : 2 \le r \le 58\} \cup \{D_r : 4 \le r \le 58\} \cup \{E_6, E_7, E_8, F_4, H_3, H_4, I_2^*\}.$$

The rank-two symbol I_2^* represents all remaining $I_2(m)$, $m \ge 5$, in the cost calculation. The local all-label theorem gives their common relevant charge and interactions; this notation is not a finite cutoff on m or an identification of their metrics. An admissible formal forest has multiplicities

$m_T \geq 0$ with $\sum_T \operatorname{rank}(T) m_T = 58$. Actual finite vertices are among these forests. The calculation needs no converse realization theorem.

The ordinary cost consists of a component table $b(T)$ and a symmetric interaction table $j(T, U)$; context bonuses are separate. In particular $j(T, T)$ concerns two distinct copies of T. The cost is

$$C_{64}(m) = \sum_T b(T) m_T + \sum_T j(T,T) \binom{m_T}{2} + \sum_{T<U} j(T,U) m_T m_U + \Psi(m). \tag{3.2}$$

The tables and the context function belong to the simultaneous local rule. Their identification with actual corner charges is a geometric obligation, separate from maximizing this expression. Appendix A.1 supplies the actual-face coverage and triangle arguments. Appendix A.3 specifies the complete context function and the finite refinement used in this calculation.

Fix all context-sensitive counts before the residual calculation. After this split the residual alphabet $\mathcal{U}$ has positive integer ranks $r(T)$ and additive integer weights $w(T)$, with no remaining pair interaction. For a residual rank $R \leq 58$ define

$$V(0) = 0, \qquad V(s) = \max_{\substack{T \in \mathcal{U}, r(T) \leq s \\ s - r(T) \text{ reachable}}} \{V(s - r(T)) + w(T)\} \quad (1 \leq s \leq R). \tag{3.3}$$

An empty maximum is an unreachable state, not an ordinary integer. Evaluate in increasing rank, recording a last item for each finite maximum.

Proposition 3.3 (Residual certificate)**.** *The algorithm* (3.3) *terminates and computes the exact maximum over residual multisets. A saved table certifies the same claim if it has base value zero, dominates every allowed last-item transition, and has a tight last-item witness at every positive rank carrying a finite stored value. At positive ranks, a state is unreachable exactly when it has no reachable last-item predecessor; rank zero is the reachable base.*

Proof. There are at most 59 ranks and finitely many transitions at each rank. Every predecessor has smaller rank. By induction, removing one component from any nonempty multiset bounds its weight by one of the candidates in (3.3). Conversely, append the chosen component to an attaining predecessor multiset. This proves both inequalities and reachability. The same induction applies to a saved table using all its transition inequalities and tight witnesses at every stored finite value. Positive item ranks make each tight predecessor smaller, so strong rank induction rules out spurious finite entries. Following witnesses reaches zero after at most 58 steps. Checking only an attaining path would not bound omitted multisets. □

In the actual context split, fixed representative counts have $\sum_{r=9}^{17} r c_r \leq 58$, with a separate A_{12}/D_{12} split, fixed A_{17}, and $0 \leq k_5 \leq \lfloor (58 - \sum r c_r)/5 \rfloor$. The residual ranks are $1, 2, 3, 4, 6, 7, 8$ and 17 through 58. Expand every class that can exceed 377 into all actual-type fibers. Within each fiber fix the complete A_1, A_2 counts. For the only remaining presence guard, split into no A_3 or reserve one A_3 and allow further copies without repeating its D_{14} bonus. The two exceptional increasing families are bounded separately, rather than omitted from the context cover. All count ranges are bounded by rank 58. These facts prove termination of the full context algorithm; the geometric domination theorem establishes that its cover includes every actual forest.

Proposition 3.4 (Computed finite interface)**.** *The preceding simultaneous rule in ambient 58 pays every triangle at least2 and every quadrilateral at least 1. Every actual rank 58 finite forest has*

charge at most 377 except precisely the following eight formal forests: one B_{13} or D_{13} together with any multiset of three A_{15} or D_{15} components. Each of those eight has charge378. All ideal weights admit the envelope of Section 6.

Certificate and completeness argument. The frozen coupled count starts from the complete 3298 fixed-vector/rank 5 cover, retaining fixed A_{17} separately from residual D_{17} and separating the A/B_{12} and D_{12} interaction types. Simultaneous actual base and cross-cost domination covers exceptional finite types and the arbitrary-label rank-two family. Its two potentially increasing context families are bounded separately:759 rows for the A_1/D_5 upgrade, and 539 rows for the A_2/A_4 guarded family, with all 14 costly rows refined through 77 actual assignments. Both bounds are strictly below377.

The complementary complete 39 costly classes have748 fixed actual-type fibers. All actual A_1, A_2 multiplicities are fixed explicitly. The remaining A_3 presence guard is split into no A_3 and one reserved A_3; the latter may contain more A_3 without multiplying the presence bonus. All other context guards are fixed, and residual pair interactions vanish. The 1549 fresh residual tables certify every allowed last-item inequality, unreachable rank and attaining predecessor. Induction on an item's removal therefore proves their upper bounds for arbitrary residual multisets. Reconstructing every winning forest verifies its full contextual cost. All 38 other classes are at most 377; the residual-zero rank 13-plus-three15 class gives exactly the eight listed forests. Their charge is $78 + 3(100)$.

The local inequalities establishing that these costs give valid corner weights are proved in Appendix A.1. □

A separate frozen B_{13} endpoint theorem removes pair $(0, 12)$, with no new coupon. Its complete selected domain has no affected image, but that absence alone is insufficient: two older affine-B_{13} singleton routes really consumed the endpoint unit. Their whole cores instead give distinct already-unit A_1/D_{12} short-leaf contributions. The other endpoint-like pair $(1, 12)$ needed by the current D_{12} theorem is retained. Thus the four B_{13} forests in Proposition 3.4 drop to 377.

4 The dimension-specific final local rule

Theorem 4.1 (Three-rank 15 transfer)**.** *In ambient dimension 58, make these changes to the simultaneous rule:*

1. *In a whole finite D_{13}, give pairs $(0, 11), (0, 12)$ weight $1/2$ instead of 1 only if its actual corner has three distinct whole unmarked A_{15} or D_{15} components.*

2. *Add $1/2$ to whole D_{28} pair $(0, 12)$ only if its corner has two such whole unmarked rank 15 components.*

3. *Add $1/2$ between an endpoint of whole A_{12} and the long endpoint of whole D_{16} under the same two-component presence guard.*

All ideal weights and other finite weights remain. Every triangle still pays at least 2 and every quadrilateral at least 1.

Witnesses are not counted with multiplicity. The negative guard uses $13 + 3 \cdot 15 = 58$ ranks, so the corner has exactly those four components. The two positive guards similarly force exactly $D_{28} + 2 \cdot 15$ or $A_{12} + D_{16} + 2 \cdot 15$. The theorem is not asserted in larger ambient dimensions.

4.1 A general inverse bound for attached forests

Lemma 4.2. *For a connected positive-definite Coxeter Gram matrix B of rank k, with diagonal2 and every nonzero native magnitude at least 1, $(B^{-1})_{ii} \geq k/(k+1)$ for every i. If a finite forest of total rank M has a nonnegative contact vector u with at least one unit contact in every component, then $u^t B^{-1} u \geq M/(M+1)$.*

Proof. Inverse matrices of these positive-definite symmetric Z-matrices are entrywise nonnegative. Delete the chosen root; let the remaining connected components have ranks k_j. Every component has a contact with the root. By induction and inverse positivity its contribution to the root's Schur projection is at least $k_j/(k_j+1)$. Since

$$\sum_j \frac{k_j}{k_j+1} \geq \frac{\sum_j k_j}{1+\sum_j k_j} = \frac{k-1}{k},$$

Schur complementation gives $(B^{-1})_{ii} \geq (2-(k-1)/k)^{-1} = k/(k+1)$. The rank-one base is $1/2$. Larger native magnitudes or multiple contacts only increase the lower bound. Apply it in every contacted component of a forest and use the same sum inequality to prove the second assertion. □

4.2 Complete selected cases and arbitrary additional whites

The fresh complete target domain contains801 patterns,862D_{13} maps, 33 affected contexts and 76 affected pair maps. The physical cores have norm-two roots; full Lorentz, white and all four edge signatures and actual corner classifications are reconstructed exactly. The native-domain completeness is inherited from the universal reduction, rather than claimed from these records alone.

There are five proper completions. Their missing fork leaf is genuinely new: old corner roots outside the displayed A_{12} are orthogonal to it. Its forced unit contact has inverse projection square6,12,30 or102, exceeding2. Additional same-end contacts cannot lower the bound. Mixed-end contacts give an independently paid short path. Thus these completions are impossible.

Four radical cores, numbered1985,2378,2575,2581, seal the entire displayed core. Indeed the whole selected D_{13} forbids contacts on its nodes; every remaining contact position has a nonzero radical coefficient. A new white must pair to zero with the radical. Nonempty contacts therefore meet both signs and give a retained adjacent-black shortcut of length at most8. Otherwise all contacts vanish and the unmarked A_4 is whole. It cannot coexist with the three rank 15 witnesses.

Twelve contexts similarly have a small displayed block with no permitted new contact of projection at most 2. It is whole: any indirect extension would have a first new root contacting it. The rank-capacity guard again excludes it.

In the final twelve contexts the old small factor K has rank $s = 1, 2, 3$. Its unique permitted contact position p has core-inverse square $r = 4/3, 3/2, 2$, respectively. The native gap excludes magnitudes greater than 1; positive-end inverse bounds exclude two same-end contacts. Four cases also permit an opposite contact position q, with $J_{pp} = J_{qq} = 4/3$ and $J_{pq} = -4/3$. Two new roots using p, q have residual squares $2/3$, so their full product is at most $-4/3 + 2/3 = -2/3$. Their forced native bond supplies the retained short path. Thus, without that escape, all new whites in a growing K component contact the old core only at p; some may be orthogonal to it.

Let Y contain all new whites in this whole component, including every indirect tail, and let M be its rank. Every component of Y contacts K. With contact vector u, the full Lorentz condition gives the positive semidefinite residual Schur matrix

$$G_Y - ruu^t \succeq 0, \qquad u^t G_Y^{-1} u \leq 1/r.$$

Lemma 4.2 implies $rM/(M+1) \le 1$, hence $M \le 3, 2, 1$ respectively. In all cases the whole K component has rank at most 4, contradicting the guard. This controls arbitrary supporting whites, not just immediate neighboring roots.

4.3 The universal affine-D12 consumer

The old opposite-fork-leaf affine-D_{12} theorem genuinely used the newly lowered whole finite D_{13} unit. It must be repaired separately. Number its affine factor F of order 13 with spine0 through 8, short leaves 9,10 at 0 and 11,12 at 8. Choose ideal marks $a = 9, b = 11$ and opposite black roots c, d. Let Z be the elliptic spectator forest, orthogonal to F, and x, y the nonnegative contacts of c, d with Z.

An adjacent-black bond or a shared spectator neighbor is an independently paid short path. Otherwise x, y have disjoint supports. The complete all-label end-profile reduction gives each end cost at most 8; a nonextreme end has cost at most 4. The exact Schur identity is

$$\frac{\det S}{h^2k^2} = 1 - \frac{g_c + g_d}{12} + \frac{1}{12}(x/h - y/k)^t G_Z^{-1}(x/h - y/k).$$

Its determinant vanishes at the actual four-black star.

If an end is nonextreme, equality forces costs8 and 4 and $x = y = 0$. A quiet root attached to the selected old leaf gives the retained whole A_1/D_{12} short-leaf unit. If it attaches to the mate, it gives whole D_{13}. Without the new guard its unit is unchanged. With the guard, Z consists of exactly three orthogonal whole rank 15 factors. The three nonextreme equality profiles give, at a distinct corner, respectively a retained A_1/D_{12} unit, an independent whole affine B_{13} priority, or a second guarded D_{13} half. These exhaust24 physical cases. The current B_{13} endpoint repair uses its D_{12} cross unit, not a newly financed D_{13} route; there is no circularity.

When both ends are extreme,$h = k = 1/2$, and the determinant becomes

$$(x - y)^t G_Z^{-1}(x - y) = 1. \tag{4.1}$$

The old two-nonempty-support theorem gives retained short paths or small sealed alternatives. The changed D_{13} output belongs to the arbitrary one-sided branch, say $y = 0$. The quiet corner excludes c and retains Z orthogonally. If its D_{13} guard holds,Z is exactly three whole A_{15}/D_{15} factors.

Lemma 4.3. *Under the strict edge conditions and disjoint-support hypothesis, energy1 on three whole A_{15}/D_{15} factors forces a single unit contact at the long endpoint of one D_{15}, by one black root. The other two factors have zero contact with both black roots.*

Proof. For A_{15} the inverse diagonal entries below2 occur only at nodes 0,1,13,14. They are 15/16, 7/4, 7/4, and 15/16, respectively. Two native contacts give energy at least 2 and violate strict edge positivity. The complete profiles are simple contacts at these four nodes or doubled contacts at 0,14. The next native squared magnitude $(3 + \sqrt{5})/2$ is already excluded. For D_{15} only its long endpoint has inverse diagonal below2, equal to 1, and only its simple contact fits.

Every allowed nonzero component energy is at least 15/16; no A_{15} profile has energy1. The finite exact inverse table also checks every disjoint two-sided profile difference: all are at least 15/16 and none is1. Equal-node contacts are forbidden by disjoint support. Two active components would have total at least $15/8 > 1$. Equation (4.1) therefore forces precisely the stated D_{15} contact. □

The two inactive components are orthogonal to the entire star and the active component. They remain whole, without indirect bridges. If c attaches to selected a, the opposite corner is whole A_{12} at an endpoint and whole D_{16} at its long endpoint. If it attaches to a's mate, that corner is

whole D_{28} at $(0, 12)$. Both physical30-root models have signature $(28, 1, 1)$, positive white rank 26, positive edge rank 27 and semidefinite corners. In either case the new half, guarded by the two inactive rank 15 factors, supplements the quiet D_{13} half at a different actual corner. This proves the required total1.

4.4 Chosen-output and triangle retention

The previous selected and nonselected routes must retain their chosen positive contributions, including every later replacement route. The low66 contexts have order bound at most 12. The 74 protected and 9 adaptive paths have length9, whereas the changed D_{13} role has distance11. The current finite6/7/8 and cycle/tree routes terminate at short paths, protected leaves, small whole factors or unchanged ideal contributions.

The older affine-D_{11} repairs use $D_{13}(1, 11)/(1, 12)$ units, retained here. The current center/leaf theorem uses $D_{13}(0, 10)$, at its fork center. The coupled order 11 unmixed boundary gives D_{11+r} marks $(r, n-2)/(r, n-1)$ of distance9; for $r = 2$ these are disjoint D_{13} roles. The other current finite and contextual outputs are similarly retained. The new producer reconstructs all 216 literal configurations,564 later corners,24D_{14} variants,332 positive D_{12} outputs and the newer sealed families. These finite checks support the named interfaces; universal completeness remains the written routing theorem. The genuine unstored affine-D_{12} exception was repaired above.

For triangles use the complete actual-face theorem, not three pairs in one diagram. Lemma A.1.3 first excludes the three reduced small affine patterns. Low cores give two distinct corner paths of length at most 5. The 1025 high-core triples give paths of length at most 7 or one of four protected E_8 alternatives, with finite marked length6 or7 or whole exceptional affine order at most 9. Current finite6 deletions retain all three exceptional length6 alternatives through their nearby whole leaf; the length7 alternative and other persistent-feature outputs survive. The present distance11 reduction changes none of these chosen payments. Ideal contributions are unchanged. This completes Theorem 4.1.

5 The new complete finite maximum

Proposition 5.1. *Every actual finite rank 58 forest under the final simultaneous rule has charge at most 377. The exact formal maximum is377.*

Proof. The B_{13} theorem lowers four forests in Proposition 3.4 from 378 to 377. The new D_{13} guard lowers the other four by1. Every other forest was already at most 377, except that the two new positive coupon families must be checked.

Rank capacity makes those families exact. They have respectively

$$170 + 2(100) + \tfrac{1}{2} = 370.5, \qquad 66 + 108.5 + 2(100) + 1 = 375.5.$$

The second total counts both A_{12} endpoint coupons. The previous A_{12}/D_{16} cross cost is zero. Each family has three possible multisets of two A_{15}/D_{15} factors, so all six actual forests are covered. No additional context witness can fit. Outside them the change is pointwise nonincreasing. Thus the bound377 holds everywhere, and the eight changed forests attain it formally. A geometric realization of an attainer is neither required nor asserted. □

6 Positive ideal surplus

For affine ranks r_i summing to 57 set

$$H(t)=\prod_i(1+t+\cdots+t^{r_i})=\sum_{j=0}^{57}h_jt^j,\qquad V=\prod_i(r_i+1),$$
$$G=\sum_j h_j\min(j,57-j),\qquad \kappa=\frac{5G-57V}{2}.$$

The geometric all-family envelope is unchanged by every final finite transfer. Its complete numerical table is

$$c(r)=r/2\quad(1\le r\le 8),\qquad c(r)=\frac{5r+56}{r+1}\quad(24\le r\le 57),$$

together with the following entries:

r	9	10	11	12	13	14	15	16
$c(r)$	71/16	53/11	31/6	11/2	40/7	367/60	411/64	227/34
r	17	18	19	20	21	22	23	
$c(r)$	119/18	128/19	137/20	146/21	155/22	163/23	85/12	

The finite arithmetic below assumes the proved factor-charge envelope (3.1); polynomial evaluation does not establish its geometric meaning.

6.1 The terminal-domain algorithm

Let $\mathcal{P}(n,\ell)$ be the list of nondecreasing partitions of n with parts at least ℓ. Generate it by

$$\mathcal{P}(0,\ell)=\{()\},\qquad \mathcal{P}(n,\ell)=\bigcup_{q=\ell}^{n}\{(q,p):p\in\mathcal{P}(n-q,q)\}\quad(n>0).$$

The union is empty if $\ell>n$. The terminal cusp domain is

$$\mathcal{T}_{57}=\bigcup_{\ell=0}^{7}\{(\ell,p):p\in\mathcal{P}(57-\ell,8)\},\tag{6.1}$$

where a zero part is omitted. For each profile compute its polynomial by repeated convolution: start with $h_0^{(0)}=1$, and when adjoining rank r set

$$h_j^{(k+1)}=\sum_{a=0}^{r}h_{j-a}^{(k)},$$

with out-of-range entries zero. Then compute V,G,κ from their displayed definitions and set $m=\kappa/V-\sum_i c(r_i)$.

Proposition 6.1 (Complete terminal calculation)**.** *The partition and convolution algorithms terminate, list every profile in* (6.1) *exactly once, and compute its margin by exact rational arithmetic. Their recorded complete output is2552 profiles, with*

$$\min_{\mathcal{T}_{57}}m=\frac{87}{10240}>0,\qquad \min_{\mathcal{T}_{57}}Vm=36.$$

Algorithm correctness and computed output. Every recursive call lowers n by a positive integer, and every loop is finite. The unique first part of a sorted partition and its unique tail prove completeness and absence of duplicates by induction on n. Its exceptional part below8, if present, determines the outer branch ℓ uniquely. Polynomial convolution is exactly multiplication by $1 + t + \cdots + t^r$, so induction on the factors proves every coefficient. Degrees never exceed57. All remaining operations are finite integer or rational sums and products.

The stated cardinality and minima are the computed proposition: the complete saved certificate records every rank tuple and coefficient list, and the independent endpoint checker reconstructs the domain and every convolution before comparing them. It evaluates all 2552 margins, rather than checking only degree, symmetry or a sample of rows. The two minima are separate assertions and need not have the same attaining profile. The implementation and records are identified in the reproducibility section. □

Lemma 6.2 (Concentrating two simplex ranks)**.** *Let $1 \le a \le b$ be integers, and let U_r be uniform on $\{0, \dots, r\}$. With independent summands, $U_{a-1} + U_{b+1}$ dominates $U_a + U_b$ in convex order. Adding any common independent summand preserves this order.*

Proof. Both sums have mean $(a+b)/2$ and are symmetric about that mean. For an integer threshold t, let Δ_t be the difference of their expectations of $(X - t)_+$. Equal means allow us to use $(t - X)_+$ instead, and symmetry gives $\Delta_t = \Delta_{a+b-t}$. It suffices to take $0 \le t \le (a+b)/2$. Put $D = (a+1)(b+1)$ and $D' = a(b+2)$. For $0 \le t \le a$, direct summation of the increasing side of the two convolution triangles gives

$$\Delta_t = \binom{t+2}{3}\left(\frac{1}{D'} - \frac{1}{D}\right) \ge 0, \qquad D - D' = b - a + 1 > 0.$$

For $a \le t \le (a+b)/2 \le b$, summing over the first uniform instead gives

$$\Delta_t = \frac{t(a+b-t) - (a-1)(a+2b+2)/3}{2(b+1)(b+2)}.$$

The numerator is increasing up to $(a+b)/2$. At $t = a$ it is $(a+2)(b-a+1)/3 > 0$. Outside the common support the difference is zero; between integer thresholds it is linear. Thus all stop-loss differences are nonnegative. On the integer support the values of a convex function are an affine function plus a nonnegative sum of these stop-loss functions, proving convex order. Conditioning on a common independent summand proves the last assertion. □

Lemma 6.3 (Positive low8 reduction)**.** *Every actual ideal vertex has $\kappa - C_{\mathrm{act}} \ge 36$.*

Proof. For $1 \le a \le b \le 7$, replacing (a, b) by $(a-1, b+1)$ preserves normalized envelope cost, since all these ranks stay at most eight. Lemma 6.2, including its independent-summand assertion, gives the required convex order. Since

$$\frac{G}{V} = \frac{57 - \mathbb{E}|2S - 57|}{2},$$

the transfer does not increase normalized curvature or normalized margin. Its product decreases from $(a+1)(b+1)$ to $a(b+2)$. Iteration terminates at ranks at least 8 and at most one positive rank below8. Indeed the integer potential $\sum_i r_i^2$ increases by $2(b-a+1) \ge 2$ at every step and is bounded above by 57^2. Its initial value is at least 57, so at most 1596 steps occur. Rank zero is omitted. No transfer passes through the changed rank 9 envelope. The stopping domain is exactly (6.1).

Proposition 6.1 gives positive normalized terminal margin and absolute envelope surplus at least 36. If the original and terminal quantities are (V, m) and (V', m'), then $V \ge V'$ and $m \ge m' > 0$. Thus $Vm \ge V'm' \ge 36$. This multiplication uses positivity. Finally (3.1) transfers the same lower bound to actual cusp charge. □

7 Shape curvature and the contradiction

Lemma 7.1. *For a proposed ambient 58 polytope,*

$$\mathcal{D}_5 \geq 377v_f + \sum_{v \text{ ideal}} \kappa(v) + 4205.$$

Proof. Choose a generic linear function on the bounded Klein closure of the polytope. Let E_v be the number of edges of its simple vertex figure and E_v^0 the number not cut by the induced level section. If I counts all polygon vertices with multiplicity, the published incidence identities give $I = \sum_v E_v$ and $2f_2 = \sum_v E_v^0$ [3, Theorem9]. Hence $\mathcal{D}_5 = \sum_v (5E_v^0/2 - E_v)$.

Orient each vertex figure by its induced generic linear function. Assign an uncut edge below the section to its upper endpoint and one above the section to its lower endpoint. A vertex of index j receives j or $57 - j$ contributions, at least their minimum. The index distribution is its h-vector [3, Theorems2 and 6], so $E_v^0 \geq G_v$. At the distinct global minimum and maximum the whole vertex figure is uncut, and symmetry gives

$$E_v - G_v = \sum_{j>57/2} h_j(2j - 57) \geq \sum_{j=29}^{57} (2j - 57) = 841.$$

Every coefficient of a simplex or product-of-simplices h-polynomial is at least 1. Adding the two extrema and multiplying by 5/2 gives4205, counted once. For a finite vertex $h_j = 1$, so

$$G_0 = \sum_{j=0}^{57} \min(j, 57 - j) = 812, \qquad \frac{5G_0 - 57 \cdot 58}{2} = 377.$$

The remaining ideal terms are precisely κ. □

For faces with at least five sides,$5 - s(F) \leq 0$. The nonnegative local rule and Proposition 5.1 therefore give

$$\mathcal{D}_5 \leq 377v_f + \sum C_{\text{act}}.$$

Subtract Lemma 7.1 and use Lemma 6.3:

$$0 \geq 4205 + \sum(\kappa - C_{\text{act}}) \geq 4205 + 36v_i > 0.$$

This also excludes the all-finite case. Thus dimension 58 is impossible. Appendices A.4 and A.5 exclude all dimensions at least59. This proves Theorem 2.1.

Part II
Compact polytopes

8 The theorem and the organization of its proof

Our main result is the following dimension bound.

Theorem 8.1 (Dimension bound)**.** *Every compact real hyperbolic Coxeter polytope has dimension at most* 17*.*

Here compact means that every vertex is finite.

The central object is one vector x, obtained from an original farthest vertex p. Energies of orthogonal components of its incident diagram can be added because they are projections of this same vector. Pressures arising from different auxiliary points cannot in general be optimized separately and then added. Instead, we combine them in one reference vector and retain its complete squared norm.

The case split occurs at the finite irreducible components of the diagram at p. Small components pay uniform energy debits. A large A-component is handled in its zero-sum chain coordinates. A large B- or D-component has a long common prefix; its terminal nodes then require separate analysis. Section 12 records a fresh complete cover at dimension 18, and the separate higher-dimensional cover. A *cover* always means necessary possibilities, not realized examples. Section 13 explains the whole-B_{18} terminal partition. Sections 14–17 give the whole-D_{18} closure.

The final composition has two parts. We exclude every dimension at least 19 by its actual-rank forest cover and local consumers. At dimension 18, the complete cover leaves only a whole D_{18} diagram at the same selected farthest vertex.

9 Farthest vertices and projections

Use the hyperboloid model with Lorentz signature $(-+\cdots+)$. Every wall has an outward unit spacelike normal n, and the polytope is given by $\langle r, n\rangle \le 0$. Distinct actual wall normals have nonpositive products. Fix a strict interior point z, choose an actual vertex p farthest from z, and write

$$z = E(p - x), \qquad E = -\langle z, p\rangle > 0, \qquad x \in p^{\perp}, \qquad \|x\|^2 = 1 - E^{-2} < 1. \tag{9.1}$$

The space $p^{\perp}$ is Euclidean. The incident normals at p form a positive definite spherical diagram; compact acute simplicity implies that there are exactly n of them in dimension n. We retain these original normals and this original choice of p, z, x, E throughout an energy argument.

The function $r \mapsto -\langle z, r\rangle$ takes its maximum on a compact hyperbolic polytope at a vertex. Its restriction to a unit speed geodesic has positive second derivative equal to the function itself, so a maximum on a segment occurs at an endpoint; successively passing to proper faces proves the assertion. Consequently

$$-\langle z, r\rangle \le E \quad \text{for every actual point } r \in P. \tag{9.2}$$

At an adjacent vertex q, write $z = E_q(q - y)$. Equation (9.2) gives a comparison factor E/E_q, not a claim that q is farthest.

9.1 Positive inverse matrices and feasible projections

Lemma 9.1 (Inverse positivity)**.** *If a positive definite symmetric matrix B has nonpositive off-diagonal entries, then B^{-1} is entrywise nonnegative.*

Proof. Choose s greater than the largest eigenvalue of B and all its diagonal entries. Then $M = sI - B$ is entrywise nonnegative and has spectral radius $s - \lambda_{\min}(B) < s$. Thus $B^{-1} = s^{-1} \sum_{k \geq 0} (M/s)^k$ is entrywise nonnegative. □

Lemma 9.2 (An actual positive projection)**.** *Let n_j, $j \in J$, be actual wall normals with positive definite Gram matrix B_J, and let z be strict interior. Put*

$$q_0 = z - \sum_{i,j \in J} (B_J^{-1})_{ij} \langle z, n_j \rangle n_i.$$

Then q_0 is future timelike, orthogonal to all selected walls, and has negative product with every unselected actual wall. Its normalization is a point of the original polytope.

Proof. The added coefficients are nonnegative by Lemma 9.1. For an unselected wall every added product is nonpositive and the original product is negative; the selected products vanish. Moreover

$$q_0^2 = z^2 - (\langle z, n_i \rangle)_{i \in J}^T B_J^{-1} (\langle z, n_i \rangle)_{i \in J} < 0.$$

Since $\langle z, q_0 \rangle = q_0^2 < 0$, it has future orientation. All original wall inequalities survive normalization. If fewer than n walls are selected this is a point on a face, not an asserted full-dimensional vertex. □

The same sign check applies to positive combinations of actual normals below: selected products are calculated explicitly; every unknown original wall has nonpositive product with every summand. A nonzero feasible future null vector would give a point of the ideal boundary of P, contradicting compactness. A positive affine Coxeter kernel on actual walls is an obstruction after its future sign and all remaining wall products have been checked.

Lemma 9.3 (Future-point pressure)**.** *Suppose $U \in p^\perp$, $\xi = Ap - U$ is future timelike with $\xi^2 = -k < 0$, and its normalization is feasible against every actual wall. Then the same original vector satisfies*

$$\langle x, U \rangle \geq A - \sqrt{k}.$$

At a different actual vertex q, use the decomposition $\xi = Aq - U$ with $U \in q^\perp$. With all-point comparison factor κ, the corresponding inequality is $\langle y, U \rangle \geq A - \kappa\sqrt{k}$.

Proof. The spatial hypothesis gives $A = -\langle p, \xi \rangle > 0$. Apply (9.2) to $\xi/\sqrt{k}$: $-\langle z, \xi/\sqrt{k} \rangle = E(A - \langle x, U \rangle)/\sqrt{k} \leq E$. The modified comparison is the identical calculation. □

If $U \neq 0$, its support lies in actual orthogonal components and $\|U\|^2 = A^2 - k$; Cauchy therefore also gives

$$\|x_{\operatorname{supp} U}\|^2 \geq \frac{A - \sqrt{k}}{A + \sqrt{k}}. \tag{9.3}$$

Normalize a future vector, not a spacelike correction. For a cap $c = Ap - v$, clear a positive definite old block J:

$$w = \sum_{i,j \in J} (B_J^{-1})_{ij} \beta_j n_i, \quad \beta_j = -\langle c, n_j \rangle \geq 0, \quad \eta = c + w, \quad \delta = \eta^2 = 1 - \beta^T B_J^{-1} \beta.$$

If $\delta < 0$, η is the future vector to normalize. If $\delta > 0$, the actual finite projection instead uses

$$p + \frac{A}{\delta}\eta, \qquad \left(p + \frac{A}{\delta}\eta\right)^2 = -1 - \frac{A^2}{\delta}.$$

When its selected products are nonpositive, its positive coefficients check all unselected walls. The pressures are

$$\langle x, v - w\rangle \geq A - \sqrt{-\delta}, \qquad \langle x, v - w\rangle \geq \frac{A}{1 + \sqrt{\delta/(A^2 + \delta)}}. \tag{9.4}$$

Old normals lie in $p^\perp$, so $v - w$ denotes the spatial part of the corresponding displayed Lorentz vector.

9.2 Opposite walls and common components

Let B be the original incident Gram matrix, $C = B^{-1}$, and

$$\omega_i = \sum_j C_{ij} n_j, \qquad d_i = C_{ii}, \qquad u_i = \omega_i/\sqrt{d_i}.$$

An actual opposite wall has normal

$$c = Ap - v, \quad A > 0, \quad v = \sum_i \beta_i \omega_i, \quad \beta_i = -\langle c, n_i\rangle \geq 0, \quad A^2 = \beta^T C \beta - 1.$$

Strict interiority gives

$$\langle x, v\rangle < A. \tag{9.5}$$

Along the original edge in direction $-u_i$, of length L_i, the original farthest comparison gives

$$\langle x, u_i\rangle \geq \tanh(L_i/2). \tag{9.6}$$

Whole-component projection energies sum to $\|x\|^2$.

Proposition 9.4 (Reciprocal shadows; internal geometric interface)**.** *For an actual cap put $T_i = (C\beta)_i/(A\sqrt{d_i})$. If $T_i > 1$, it is the opposite wall of edge i, and $\tanh L_i = 1/T_i$. If it is not that opposite wall, then $\tanh L_i \geq T_i$. The value $T_i = 1$ is incompatible with a finite endpoint. For distinct actual opposite walls,*

$$\tanh L_i \tanh L_j \geq K_{ij}, \qquad K_{ij} = C_{ij}/\sqrt{d_i d_j}. \tag{9.7}$$

Opposite walls of orthogonal original directions cannot coincide.

Proof. The Schur identity says that $C - C_{\cdot i}C_{i\cdot}/d_i$ vanishes on row and column i; elsewhere it is the inverse of the retained old Gram matrix. It is entrywise nonnegative. Let $d = Dp - \sum_j \gamma_j \omega_j$ be the actual opposite of edge i, distinct from c. Acuteness and the Schur identity give

$$AD \geq \beta^T C \gamma \geq (C\beta)_i (C\gamma)_i / d_i.$$

The actual endpoint equation is $\tanh L_i = D\sqrt{d_i}/(C\gamma)_i$, so $\tanh L_i \geq T_i$. If $T_i > 1$ this alternative is impossible, forcing $c = d$; its endpoint equation gives $\tanh L_i = 1/T_i$. If $T_i = 1$, the distinct alternative is impossible and the same-wall equation would give an ideal endpoint. This is excluded by compactness. For the pair inequality, first use the contacts of the actual opposite of i to get $T_j \geq K_{ij}/\tanh L_i$; the distinct alternative at edge j yields the assertion.

Finally suppose the same wall closes orthogonal directions i, j. Clearing every other old normal adds positive coefficients and gives

$$c_F = Ap - \frac{A}{r_i}u_i - \frac{A}{r_j}u_j, \qquad r_i = \tanh L_i, \quad r_j = \tanh L_j.$$

Its product with the strict interior point remains negative, so $\langle x, u_i\rangle/r_i + \langle x, u_j\rangle/r_j < 1$. But (9.6) makes this sum at least $1 + (\tau_i^2 + \tau_j^2)/2 > 1$, where $\tau_i = \tanh(L_i/2)$. This contradiction proves the orthogonal assertion. □

The displayed pair inequality concerns full tangents, not half tangents. For an opposite wall, its retained foreign component can have at most one contact: two would make a cycle in its actual finite replacement. A positive foreign cost is $q = \beta_j^2 C_{jj} \geq 1/4$. In a nontrivial component $C_{jj} \geq 4/3$, so the only cost below $1/3$ is a simple 3-contact on an A_1, of cost $1/4$. Moreover

$$q < A^2. \tag{9.8}$$

Otherwise the actual reciprocal test would close that foreign direction too, violating the orthogonal assertion, with the equality case giving a null endpoint. Set $F(t) = 2t/(1+t^2)$. For unit directions of correlation k in a component of energy at most e, positive heights satisfy

$$\tau_i + \tau_j \leq \sqrt{2e(1+k)}, \qquad F(\tau_i)F(\tau_j) \leq \frac{4s}{(1+s)^2}, \quad s = \frac{e(1+k)}{2}. \tag{9.9}$$

The second inequality follows from concavity of $\log F$ on $(0,1)$. If its upper bound is smaller than K_{ij}, Proposition 9.4 forces a common actual opposite. Its finite-white extension or positive affine kernel then gives additional actual pressures.

10 Quadratic estimates from reference vectors

The elementary estimate behind every certificate is

$$\|x\|^2 \geq 2\langle x, W\rangle - \|W\|^2. \tag{10.1}$$

Every term in $\langle x, W\rangle$ must be justified. Write $w = \sum_i b_i P_i$, $b_i \geq 0$, with actual pressures $\langle x, P_i\rangle > a_i$, and demand $A_w = \sum_i b_i a_i$. For a new actual pressure $\langle x, Z\rangle \geq h$, choose

$$Q = Z - \sum_i t_i P_i, \qquad W = w + \mu Q.$$

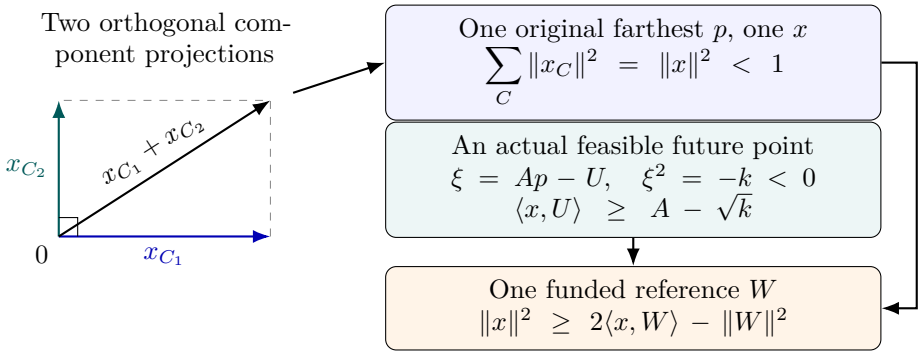


Figure 3: The orthogonal decomposition in $p^\perp$ and the point estimate of Lemma 9.3. Contributions from the components are combined using the same vector.

Before using pressures require $\mu \ge 0$ and check every $b_i - \mu t_i \ge 0$. Then

$$\langle x, W\rangle \ge A_w + \mu(h - \sum_i t_i a_i), \quad \|W\|^2 = \|w\|^2 + 2\mu\langle w, Q\rangle + \mu^2\|Q\|^2. \tag{10.2}$$

The available price b_i is individual to anchor i. The first inequality is strict if some strictly constrained old anchor retains a positive coefficient, or if $\mu > 0$ and the new pressure is strict. For multiple corrections every new weight must be nonnegative; their combined debit at each anchor and all mutual cross terms must be retained.

Put $P_i = e_1 + \cdots + e_i$, $S_i = \langle x, P_i\rangle$. If selected indices satisfy $S_i > f_i$, concave interpolation through $(0,0)$ and the selected (i, f_i) gives a reference whose coordinates are successive slopes. Its P_i coefficients are successive slope decreases, hence nonnegative; its demand is its squared norm. Omitting one unknown internal floor replaces two adjacent slopes by their average, losing half their squared difference. For $\sqrt{t}$ at $t \ge 5$, the loss is less than $1/3200$. Nonadjacent omissions have disjoint open intervals, so losses add. A missing floor has coefficient zero. References are Euclidean vectors, not asserted actual points: actual points prove pressures, then positive coefficients combine them in (10.1).

11 Component estimates

11.1 Ordinary components and exceptional actual defects

All bounds below concern the same original projection, with arbitrary finite spectators.

Component	Energy greater than	Component	Energy greater than
ordinary nonshort A_1	$1/10$	ordinary A_2	$319/2000$
ordinary $I_2(m), m \ge 4$	$1/7$	A_3	$1/4$
B_3, H_3	$2/7$	A_4	$2/7$
B_4, D_4, F_4	$1/3$	H_4	$11/36$
A_5	$3/8$	A_6	$3/7$
B_5, B_6	$3/7$	D_5, D_6	$5/14,\ 3/7$
B_7, D_7	$1/2$	B_8, D_8	$4/7$
E_6, E_7, E_8	$9/20,\ 5/9,\ 2/3$		

An actual isolated dotted short A_1 is denoted I. A second short-wall profile J uses two old A_1 factors whose joint energy exceeds .34. The two typed rank-two exceptions, each of energy $> .09$, are

$$N_7 = (\text{old } A_2,\ \text{new isolated native 7}), \qquad O_7 = (\text{old } I_2(7),\ \text{new isolated native 3}).$$

A bare $I_2(7)$ is not automatically O_7. There is at most one short I/J profile or one typed exception; they do not coexist. Acuteness of their actual isolated walls is essential: two on orthogonal supports would have product $-\sqrt{h_1 h_2}$ strictly between $-1/2$ and 0. Appendix B.1 proves the endpoint estimates, the J case and the ordinary rates.

Useful joint bounds, counting I only once, are

$$I + A_1 > 1/5, \quad I + 2A_1 > 3/10, \quad I + T_2 > 21/100, \quad I + T_3 > 2/7. \tag{11.1}$$

Here T_j is ordinary irreducible of rank j. Both typed exceptions force another T_2 to cost $> 1/5$. Another A_1 costs $> 1/5$ beside N_7 and $> 2/15$ beside O_7. For the latter, isolated height $h = \frac{1}{4}\csc^2(\pi/7) - 1 < 1/3$ gives squared full tangent greater than $3/7$. Squared half tangent exceeds $2/15$, since $4(2/15)/(1 + 2/15)^2 = 120/289 < 3/7$. This is a real geometric argument. The complete contact and defect proofs, including the actual-wall height constraints behind (11.1), are given in Appendix B.1. The uniform actual B/D floor is proved in Appendix B.2.

11.2 The A-chain branch

For A_r, use the zero-sum subspace of $\mathbb{R}^{r+1}$ with roots $(e_i - e_{i+1})/\sqrt{2}$. The inverse Gram matrix is

$$C_{ij} = \frac{2\min(i,j)(r+1-\max(i,j))}{r+1}.$$

A finite-white classification of actual opposite walls gives strong dual floors except for a short list of literal helper caps. Adjacent weak helpers are excluded using their actual mutual product and a positive two-cap projection. Concave interpolation uses only strong floors; helper pressures are paid at surviving anchors. The small end variance yields stronger component bounds:

r	$r \ge 7$	$r \ge 8$	$r \ge 13$	16	17
$e(A_r) >$	1/2	4/7	5/7	199/250	6/7

$e(A_{18}) > 91/100.$

Later local refinements give $e(A_{12}) > 13/20$ and $e(A_{15}) > 413/500$, uniformly with arbitrary spectators. The shared actual feet, common-pair inventories, uniform $1/2, 4/7, 5/7$ bounds, and the $N = 17, 18, 19$ central-window and paid-reference arguments are proved in Appendix B.3. The newer $A_{16} > .86$ proof in particular implies the older .796 bound without requiring its separate historical reference. The fresh smaller-rank proofs are in Sections B.4.5 and B.4.4. Their full native-cosine inventories and individual actual-helper payments are included. Later exact-17 research remains excluded.

11.3 The common B/D prefix branch

For B_r, use long roots $(e_i - e_{i+1})/\sqrt{2}$, $i < r$, and short root e_r. For D_r, use the long chain and the fork root $(e_{r-1} + e_r)/\sqrt{2}$. On a long initial segment, $\omega_i = \sqrt{2}P_i$, $C_{ij} = 2\min(i,j)$. In type D, $x_1 > \cdots > x_{r-1} > |x_r|$; the last coordinate is signed and the forks are distinct actual duals.

Proposition 11.1 (Generic prefix theorem)**.** *Let $6 \le M \le 16$. Suppose an actual component is B_r, $r \ge M+5$, or D_r, $r \ge M+6$. Put*

$$f_i = \sqrt{i} - \frac{61}{100\sqrt{2}}, \quad E_M = \frac{f_5^2}{5} + \sum_{i=5}^{M-1} (\sqrt{i+1} - \sqrt{i})^2, \quad H_M = \left\lfloor \frac{M-5}{2} \right\rfloor.$$

With arbitrary finite spectators, $\sum_{j=1}^{M+1} x_j^2 > E_M - H_M/3200$.

Proof. Appendix B.5 proves that a weak site $5 \le i \le M+1$ has a pure cap

$$v = \gamma\omega_i + \omega_1/2, \quad A^2 = 2i\gamma^2 + 2\gamma - \tfrac{1}{2}, \quad 101/125 < \gamma < 6/5,$$

with own label 5, finite label at least 7, or dotted own contact, and no spectator contact. Actual weak sites are nonadjacent. This is proved at these ranks, not transferred from a whole-component exclusion. If 5 is strong, choose a strong $K = M$ or $M+1$. Interpolate through actual strong knots. Each missing internal floor loses $< 1/3200$; at $K = M+1$ the extra increment $(\sqrt{M+1} - \sqrt{M})^2 > 1/(4M+2) \ge 1/66$ pays the possible extra omission.

If 5 is weak, the first-six-root point table in Appendix B.5 gives $x_1 > .5965$, and 6 is strong, with $S_6 > 2.01815$. Interpolate through $(1, .5965), (6, 2.01815)$ and a later strong endpoint. The slopes decrease: $.5965, .28433$, then $1/(\sqrt{K} + \sqrt{6}) < 1/5$. Its energy exceeds

$$L_M = .7600299945 + \frac{(\sqrt{M} - \sqrt{6})^2}{M-6},$$

the last term being zero at $M = 6$. Now $E_6 < .7$. The $k = M - 6$ increments from 6 onward lie in $(1/8, 1/5)$, so their variance is at most $k(1/5-1/8)^2/4 \le 9/640$. Thus $L_M - E_M > .76-.7-9/640 > 0$, directly also at $M = 6$. This proves the weak-5 branch without first assuming strong 5. □

Appendix B.5 supplies the complete finite-white inventory, zero-own repair, actual two-cap nonadjacency, all four weak-5 pressures and continuous seals, with every spectator retained. Its original rank inequalities are part of the theorem. A short-tail continuation at an excluded rank still requires its own new local proof.

12 The remaining spherical diagrams

12.1 A fresh cover in exact dimension 18

Call A_r, $r \ge 7$, or B_r, D_r, $r \ge 9$, dominant. Each costs $> 1/2$, so at most one is possible. With none the ordinary rank-over-14 baseline, even allowing the unique defect, exceeds 1. Exactly one is therefore present. The fresh total-rank-18 necessary cover is:

Dominant	All retained spectator alternatives
A_{12}	$I + (A_5$ or $D_5)$, or $N_7/O_7 + T_4$
A_{15}	ordinary A_3; $T_2 + A_1$; $I + T_2$; or $O_7 + A_1$
A_{16}	ordinary T_2; two ordinary A_1; $I + A_1$; or N_7/O_7
A_{17}	one ordinary nonshort A_1, or I
A_{18}	none
B_{16}	N_7/O_7
B_{17}	I
B_{18}	none
D_{16}	N_7/O_7
D_{17}	one ordinary nonshort A_1, or I
D_{18}	none

Here ordinary $T_2 = I_2(m)$, $m \ge 3$, and $T_4 \in \{A_4, B_4, D_4, F_4, H_4\}$. H_4 and native 5 are not discarded by field arguments.

For completeness, the accounting steps are as follows. A_7 leaves too many ordinary ranks; $A_8, \dots, A_{11}$ are excluded by 4/7 plus the rank-over-14 baseline. Beside A_{12}, ordinary or J rank-six spectators cost $> 3/7$. With I, (11.1) excludes all rank-five partitions except A_5, D_5. With a typed exception, partition rank 4: two T_2's cost $> .4$, $T_3 + A_1$ costs $> 1/4 + 2/15$, and the other reducible partitions are stronger. Only irreducible T_4 remains. For A_{13}, A_{14}, the 5/7 bound and (11.1) exclude all spectators. At A_{15}, exclude B_3, H_3, three ordinary A_1's, $I + 2A_1$, $J + A_1$, and $N_7 + A_1$. At A_{16}, .796 excludes J but retains precisely the listed rank-two alternatives. At A_{17} retain the remaining A_1.

For B_r, D_r, the 9/14 bound forces spectator rank at most 5, hence $r \ge 13$. Proposition 11.1, within its rank domain, gives

r	13	14	15	16
$e(B_r), e(D_r) >$	.7349975	.7683275	.797425	.823735.

At $13, 14, 15$, every ordinary, I/J, and typed remainder exceeds the residual budget; $I + T_2 > .21$ closes rank 15. At 16, two ordinary A_1's and $I + A_1$ are excluded; ordinary T_2 and both typed defects initially remain. The B_3, D_4, D_3 layer checks in Appendix B.6 extend helper coverage to 13, giving energy $> .8689525$; ordinary $T_2 > 1/7$ closes that case. The fresh B_{17} layers reach a

strong 14/15 endpoint and energy $> .90715$, excluding a nonshort ordinary A_1. This reuses the interpolation proof after new local coverage, not Proposition 11.1 outside its legal ranks.

The first rows close with small shared consumers:

$$A_{12} + I + (A_5/D_5) > 13/20 + 5/14 = 141/140, \quad A_{12} + N_7/O_7 + T_4 > 13/20 + .09 + 2/7 = 359/350.$$

The improved 413/500 bound closes every A_{15} family: ordinary total rank three costs $> 17/70$, $I + T_2 > .21$, and beside $O_7 + A_1$ the total exceeds 787/750. Thus the older separate ordinary-rank-three consumer is unnecessary. The other internal proof homes are:

Row	Excluding project argument
A_{16}	Proposition B.3.5, ordinary and $I + A_1$; Appendix B.10.1, both actual typed defects
A_{17}	Appendix B.10.2, ordinary and actual I
whole A_{18}	Appendix B.7, full two-end native-gap argument
B_{16}	Appendix B.10.3, both actual typed defects
B_{17}	Appendix B.10.5, actual I; local head and helper kernels
D_{16}	Appendix B.10.3, both actual typed defects
D_{17}	Appendices B.10.4.1 and B.10.6, ordinary and actual I
whole B_{18}	Section 13
whole D_{18}	Sections 14–17

Appendix B.6 proves the local layers used in this cover, including the distinct signed D_4/D_3 forks and the actual A_{17} neighbor for B_{17}. The table identifies the complete proof of every retained row, including both short rank-one cases. The independent higher-dimensional exclusions are proved in Appendices B.11–B.14. These arguments use no field, lattice or integral dotted-product hypothesis.

12.2 The separate complete higher-dimensional tail

The exact-18 cover does not exclude higher dimensions. The earlier internal package establishes the separate theorem:

Theorem 12.1 (Internal higher-dimensional exclusion)**.** *There is no compact hyperbolic Coxeter polytope of dimension at least* 19*.*

Appendix B.11 proves the component caps, shared windows and complete actual-rank exclusions from 29 through 20. The published ceiling is used only at dimensions at least 30. At dimension nineteen, Proposition B.11.35 leaves whole A_{19}, B_{19}, D_{19}. Their proofs occupy Appendices B.12–B.14: the A proof splits by the three weak-end patterns; the B proof splits by its two exceptional terminal walls; and the D proof splits by six head types after a common centre analysis. Each case keeps one original vector and every spectator in its cover. Section B.14.22 joins these exclusions.

The higher ranks have their own complete covers in Appendix B.11. The short B_3/B_2 and D_4/D_3 continuations are proved before the prefix estimates that consume them. Section B.14.22 combines these actual-rank arguments with the published compact ceiling. This is a logical conjunction of exclusions, not geometric dimension heredity. No arithmetic theorem or field census enters that conjunction.

13 The whole-B_{18} terminal partition

Classify the actual opposite of old node 17 without assuming a strong head. Section B.8.5 leaves:

(i) a pure far-three helper with finite own 5, finite own at least 7, or dotted own $1 < g < 4/3$;

(ii) near-three with own 6;

(iii) no left contact, with short white label $3, 4, 5, 6, 7, 8$.

All other isolated, left, bridge and affine/null roles are excluded in that inventory. Sections B.8.6–B.8.10 exclude the first branch, and Section B.8.11 excludes the second. In the last branch, Sections B.8.13–B.8.14 exclude distinct terminal caps and then every common-cap role. The last common cap is treated at its actual adjacent whole-A_{18} vertex, with comparison $11/8$; that vertex is never assumed farthest.

Thus an original farthest whole-B_{18} vertex cannot occur. Appendix B.8 gives the complete finite-white inventory, each terminal exclusion, all original-wall checks and the individual coefficient payments. Together with the preceding row consumers, this leaves only whole D_{18} at the selected original vertex.

14 The last case: an actual whole-D_{18} vertex

We treat the final dotted-head case using the complete inventory proved in Appendix B.9. Keep the original p, z, x and use actual old roots

$$n_i = (e_i - e_{i+1})/\sqrt{2} \quad (1 \le i \le 17), \qquad n_{18} = (e_{17} + e_{18})/\sqrt{2}.$$

Write

$$P_i = e_1 + \cdots + e_i, \quad \Omega_i = \sqrt{2}P_i \ (i \le 16), \quad \Omega_{17} = (P_{17} - e_{18})/\sqrt{2}, \quad \Omega_{18} = (P_{17} + e_{18})/\sqrt{2}.$$

In particular x_{18} is signed and $x_1 > \cdots > x_{17} > |x_{18}|$. In this part only, set

$$S_i = \langle x, P_i \rangle, \qquad f_i = \sqrt{i} - \frac{.58}{\sqrt{2}}. \tag{14.1}$$

All terminating decimals in this paper denote exact rational numbers.

The head is the actual opposite wall of the long-end edge 1. Section B.9.1 proves that its normal is

$$H = A_H p - \eta\Omega_1, \qquad A_H^2 = 2\eta^2 - 1 > 0.$$

The same section gives the complete own-contact list. Sections B.9.8 and B.9.9 prove the strong terminal floor $S_{16} > f_{16}$, its funded reference append and the complete fork-role list. The finite-head cases close in Section B.9.22. We treat every dotted head $\eta > 1$ without a parameter cutoff.

Appendix B.9 proves these earlier inputs in geometric dependency order: actual head isolation, the fresh threshold window, strong 16, finite-head closure and the every-head pair exclusions. In particular it proves the changed-threshold strong-site statements afresh rather than inheriting them unconditionally.

14.1 A future point valid for every dotted head

Put $u = \eta^{-2} \in (0, 1)$, and consider

$$\Xi = H + n_1/\eta.$$

This is a positive combination of two actual normals. Its products with H, n_1 are respectively 0 and $-\eta + 1/\eta < 0$, and $\Xi^2 = -(1 - u) < 0$. Its time is $A_H > 0$. Every other actual wall has nonpositive

product with each summand. Hence its normalization is an actual future point of the original polytope, with no unknown wall omitted. Lemma 9.3 gives

$$\langle x, Z_u\rangle \ge h(u), \qquad Z_u = \frac{(2-u)e_1 + ue_2}{2}, \quad h(u) = \frac{\sqrt{2-u} - \sqrt{u(1-u)}}{\sqrt{2}}. \tag{14.2}$$

Indeed the spatial part of Ξ, divided by $\sqrt{2}\eta$, is Z_u. We will use

$$\sum_j (Z_u)_j = 1, \qquad R(u) = \|Z_u\|^2 = 1 - u + u^2/2.$$

Equation (14.2) is a point pressure, not an energy lower bound, and is valid on the entire open interval.

14.2 The four original reference cases

Section B.9.2 gives the following complete initial alternatives: strong 5; a weak pure far-left-three helper at 5; or an isolated dotted weak 5 with a chosen strong $J = 6$ or 7. Thus there are four reference rows. Here is why the initial classification has this precise scope. Left own 4, including own 4/white 3, gives a positive actual affine-B null on the cap and old chain $5, \dots, 18$; the extra left contact preserves the remaining wall signs. The newly possible near-three/own-five at 5 is excluded by its actual right-H_4 point and independent old strong 11/12, before the new references are used. Other finite-white and isolated finite roles are strong. A historical fifth paid row concerned a native-five head and is outside $\eta > 1$.

We emphasize the *bare* references for the three nonhelper rows. They interpolate through the strong knots, are flat on their first J coordinates, and have endpoint sum f_{16}. Sections B.9.4 and B.9.8 give

initial case	B_0	first mean $b <$	unused P_J price $>$	J
strong 5	.9565188	.3652	.151	5
isolated dotted 5, strong 6	.9373243	.3399	.1435	6
isolated dotted 5, strong 7	.9196357	.31938	.1366	7

More explicitly, each reference w has

$$\langle x, w\rangle > A_w, \quad 2A_w - \|w\|^2 > B_0, \quad \operatorname{supp} w \subset \{1, \dots, 16\}, \quad \langle w, P_{16}\rangle = f_{16}, \quad \beta_{16} > .127.$$

No head pressure has been paid. Their raw endpoint-15 lower bounds are $.9403998, .9212053, .9035167$. The legal endpoint-16 append adds at least $.127^2 - .00001 = .016119$; if 15 is weak, its fictitious knot is deleted before pressure. All late repairs preserve the flat first block and its individual prices.

In the isolated-5 rows there is also an unused actual pressure

$$U_5 = P_5 + \frac{v}{2}(e_6 - e_5), \quad .4 < v < 1, \qquad \langle x, U_5\rangle > 1.8185.$$

For

$$Q_H = Z_u - P_J/J, \qquad Q_5 = U_5 - (5/J)P_J$$

the exact identities are

$$\|Q_H\|^2 = R - 1/J, \quad \langle Q_H, Q_5\rangle = 1 - 5/J, \quad \|Q_5\|^2 < \begin{cases} 77/150, & J = 6, \\ 194/175, & J = 7. \end{cases} \tag{14.3}$$

Both residuals have zero product with the bare flat reference. The positive mutual cross in (14.3) will not be discarded.

For the fourth row, the exact *unheaded* weak-helper reference interpolates from $(1, a_0)$, $a_0 = .5455$, to $(6, f_6)$, then through the later actual strong knots to 16, replacing each missing knot by a chord. Its first six coordinates are

$$w_1 = a_0, \quad w_2 = \cdots = w_6 = b_0 = (f_6 - a_0)/5. \tag{14.4}$$

The actual helper proves $x_1 > a_0$ and strong 6. The unheaded reference satisfies

$$\langle x, w\rangle > A_w, \quad L_w := 2A_w - \|w\|^2 > .987759, \quad \langle w, P_{16}\rangle = f_{16}, \quad \beta_{16} > .127.$$

Weak 7, later omissions, and the late 15 repair do not alter (14.4). Again no head pressure has been paid. Appendices B.9.4, B.9.8 and B.9.20 supply the raw-reference increments, omission tables, isolated-5 pressure and individual coefficient prices used here. These are internal geometric lemmas with written proofs, not script-output assumptions.

15 Bounds for the head and fork edges

15.1 The first three rows

In the strong-5 row choose $\mu = .33$ on Q_H. Its debit $.33/5 = .066 < .151$ fits the individual price. Equations (10.2)–(14.3) give

$$L_W > .9565188 + .66(h - .3652) - .1089(R - 1/5) = F(u) := .7372668 + .66h(u) - .1089R(u).$$

For isolated 5 with $J = 6$, use weights $\mu = .33, \nu = .10$ on Q_H, Q_5. The combined debit at J is $.83/6 < .1435$. For $J = 7$ use $\mu = .33, \nu = .11$, with debit $.88/7 < .1366$. In each case all remaining old coefficients are nonnegative *before* the new pressures are invoked, and

$$\begin{aligned} L_W > B_0 &+ .66(h - b) + 2\nu(1.8185 - 5b) - .1089(R - 1/J) \\ &- \nu^2\|Q_5\|^2 - .66\nu(1 - 5/J). \end{aligned} \tag{15.1}$$

The constant terms before $.66h - .1089R$ in the two rows are $22164209/30000000$ and $51729823/70000000$, both greater than $.7372668$. Thus F is a common lower bound. No optimized helper and head energies have been added.

We prove the bound uniformly on $0 < u < 1$. Put $a = \sqrt{2 - u}$, $b = \sqrt{u(1 - u)}$. Direct differentiation gives

$$h''(u) = \frac{1/(4b^3) - 1/(4a^3)}{\sqrt{2}} > \frac{7}{4\sqrt{2}}, \tag{15.2}$$

because $b \le 1/2$, $a > 1$. Hence $F'' > .66 \cdot 7/(4\sqrt{2}) - .1089 > .7$, using $\sqrt{2} < 1.415$. At $u_0 = 2/3$,

$$\begin{gathered} h(u_0) = \sqrt{2/3} - 1/3, \quad R(u_0) = 5/9, \quad F(u_0) > .9956502, \\ F'(u_0) = .2013 - .0825\sqrt{6}, \quad |F'(u_0)| < .0008. \end{gathered}$$

The value bound uses $.81649^2 = .6666559201 < 2/3$, and the derivative bound uses $2.4494 < \sqrt{6} < 2.4495$. Taylor's inequality and completion of the square on the segment between u_0 and u give

$$F(u) > .9956502 - \frac{.0008^2}{1.4} > .99564 \qquad (0 < u < 1). \tag{15.3}$$

No endpoint derivative or sampled minimum is used.

15.2 The weak-helper row: two separate anchor prices

The unheaded reference (14.4) has two available prices

$$\beta_1 = a_0 - b_0 > .24672, \qquad \beta_6 \geq b_0 - (\sqrt{7} - \sqrt{6}) > .1023. \tag{15.4}$$

Indeed $2.0392 < f_6 < 2.0394$, so $.29874 < b_0 < .29878$, while $\sqrt{7} - \sqrt{6} < 2.6458 - 2.4494 = .1964$. An adjacent omitted knot decreases the next slope and increases β_6. The bounds follow from $2.4494 < \sqrt{6} < 2.4495$ and $.4101 < .58/\sqrt{2} < .4102$, by positive squaring.

Define

$$Q = Z_u - (1 - 3u/5)e_1 - (u/10)P_6.$$

Its first coordinate is zero, its second is $.4u$, and its third through sixth coordinates are $-.1u$. Therefore

$$\|Q\|^2 = u^2/5, \qquad \langle w, Q\rangle = 0. \tag{15.5}$$

Choose $\mu = .22$. First pay the *separate* anchor debits

$$\mu(1 - .6u) < .22 < \beta_1, \qquad \mu u/10 < .022 < \beta_6 \quad (0 < u < 1).$$

The remaining original coefficients are nonnegative. Only then add the positive new head pressure μZ_u. The terminal coefficient at 16 remains unused. The full one-vector square gives

$$\begin{aligned} L_{w+\mu Q} &> .987759 + .44[h - a_0 + (3a_0/5 - f_6/10)u] - .00968u^2 \\ &> G(u) := .747739 + .44h(u) + .0542784u - .00968u^2, \end{aligned} \tag{15.6}$$

since $3a_0/5 - f_6/10 > .12336$. This argument rewrites a positive-coefficient pressure decomposition before using it; it does not subtract old lower floors from an unknown signed pressure.

We give the exact small-margin seal. By (15.2), $G'' > .44 \cdot 7/(4\sqrt{2}) - .01936 > .5$. At $u_1 = .642$, positive square tests give

$$\sqrt{1.358} > 1.1653325, \qquad \sqrt{.229836} < .4794122,$$

whose comparison squares are 1.35799983555625 and .22983605750884. The difference exceeds .6859203. Since

$$.4850188 \cdot 1.4142136 = .68592018321568 < .6859203, \qquad 1.4142136^2 > 2,$$

we have $h(.642) > .4850188$, and hence

$$G(.642) > .99200425728.$$

For the derivative use

$$.858 < 1/\sqrt{1.358} < .859, \quad .592 < .284/\sqrt{.229836} < .593, \quad .353 < 1/(2\sqrt{2}) < .354.$$

They follow by multiplying the positive radical brackets $1.1653 < \sqrt{1.358} < 1.1654$ and $.4794121 < \sqrt{.229836} < .4794122$. Thus

$$-.094518 < h'(.642) < -.093545, \qquad .00026136 < G'(.642) < .00068948 < .0007,$$

because $G' = .44h' + .0542784 - .01936u$. Strong convexity on the entire interval now gives

$$G(u) > .99200425728 - .0007^2 = .99200376728 > .992. \tag{15.7}$$

Combining (15.3) and (15.7), every one of the four original initial cases supplies a single legally funded reference W with

$$L_W > .992, \quad \operatorname{supp} W \subset \{1, \ldots, 16\}, \quad \langle W, P_{16}\rangle = f_{16}, \quad \text{unused } \beta_{16} > .127. \tag{15.8}$$

All corrections have sum zero and support at most 7; the late omissions and the original endpoint-16 price survive.

15.3 Both signed fork directions

Let $\Delta = S_{16} - f_{16} \geq 0$. The pressure decomposition of W keeps its original terminal coefficient. Therefore $\langle x, W\rangle > A_W + \beta_{16}\Delta$, and the same norm identity gives

$$\|x - W\|^2 < .008 - .254\Delta, \qquad x_{17}^2 + x_{18}^2 < .008 - .254\Delta. \tag{15.9}$$

The two true unit fork duals are

$$\frac{\Omega_f}{3} = \frac{P_{17} \pm e_{18}}{\sqrt{18}}, \qquad f = 17, 18.$$

For either actual fork height,

$$\begin{aligned} t_f := \langle x, \Omega_f/3\rangle &< \frac{f_{16} + \Delta + \sqrt{.016 - .508\Delta}}{\sqrt{18}} \\ &< \frac{3.59 + .1265}{\sqrt{18}} < .876. \end{aligned} \tag{15.10}$$

The numerator decreases with Δ: its derivative is $1 - .254/\sqrt{.016 - .508\Delta} < 0$ on its nonempty domain. The endpoint is handled by continuity. The seals are $f_{16} < 3.59$, $\sqrt{.016} < .1265$, $\sqrt{18} > 4.2426$, and $.876 \cdot 4.2426 = 3.7165176 > 3.7165$. No absolute value has replaced the signed last coordinate.

For an actual fork edge with distance cosine C, (9.6) gives $t_f \geq \sqrt{(C-1)/(C+1)}$. Hence

$$C < \frac{1 + .876^2}{1 - .876^2} < 7.6, \tag{15.11}$$

since $.876^2 = .767376$ and $7.6 \cdot .232624 = 1.7679424 > 1.767376$. In Section B.9.9, every far-next-three role has $g \geq \cos(\pi/5) > .809$ and exact cosine $C = 9g + 1 > 8.281$. Equation (15.11) removes that entire role, including all its previously classified native/null own alternatives.

16 A local A_{18} lemma at an adjacent actual vertex

Lemma 16.1. *Let q be an actual vertex of a compact Coxeter 18-polytope whose whole incident diagram is A_{18}. Write $z = E_q(q - y)$, $\|y\| < 1$. Suppose*

$$-\langle z, r\rangle \leq \kappa E_q \quad \textit{for every actual point } r \textit{ of the original } P, \qquad \kappa < 11/8.$$

Then $\langle y, u_{18}\rangle > .78$ is impossible for the positive normalized endpoint-dual direction.

The lemma is local at q, which is not assumed farthest. Appendix B.4 supplies its local inventory; no original B_{18} hypothesis is used.

Proof. In the zero-sum subspace of $\mathbb{R}^{19}$, for $i \leq j$,

$$D_i = \|\omega_i\|^2 = \frac{2i(19-i)}{19}, \qquad C_{ij} = \frac{2i(19-j)}{19}.$$

Call $5 \leq i \leq 14$ weak if $\langle y, \omega_i\rangle \leq \sqrt{D_i} - .87$. The complete inventory at comparison $11/8$, established before any endpoint pressure, leaves only:

(a) near-four, meaning own 4 and white 3 at the adjacent old node, in the four literal side-order-5/6 orientations;

(b) a pure far-end-three helper

$$d_i = A_i q - \gamma_i \omega_i - \omega_e/2, \quad e = 1 \text{ or } 18, \quad \varphi/2 \le \gamma_i < 17/16,$$

with own 5, finite own at least 7, or dotted own.

Here $\varphi = (1+\sqrt{5})/2$. All isolated caps are strong. The inventory includes own zero, every finite own, parallel and unbounded dotted owns. Every positive point checks every unknown original wall.

Some calculations locate the numerical boundary. Every ordinary edge ratio at least $3/2$ is strong: with $a = 3/2, k = 11/8$, the square comparison for $q_0 = \sqrt{D_i} \in [2.7, 3.1]$ is the positive polynomial

$$-.21 q_0^3 + 1.1876 q_0^2 - 1.74 q_0 + .7569.$$

It is concave with endpoint values $.583074, .519626$. The near-four ratio of side order ℓ is

$$\frac{1/\sqrt{2} + (\ell-1)/(2\ell)}{\sqrt{(\ell+1)/(2\ell)}}.$$

It increases in ℓ and exceeds $3/2$ at $\ell = 7$, since $(67 + 42\sqrt{2})/56 > 9/4$. Only $i = 5, 6$ with white $i-1$, and $i = 14, 13$ with white $i+1$, can be weak. For pure helpers the identity

$$\frac{(\gamma + 1/(2\ell))^2}{(\ell+1)/(2\ell)} - (4\gamma - 2) = \frac{(\gamma - 1 - 1/(2\ell))^2}{(\ell+1)/(2\ell)} \ge 0$$

gives the stated cutoff at the weakened comparison.

The local input retains genuine helper nonadjacency at this factor. Its proof uses the following exact actual products. For adjacent helper owns $A, B \in [.808, 1.08]$, the mutual product R satisfies

$$\begin{array}{c|l} \text{same end} & R \ge 1 - \frac{18}{19} AB - \frac{|A-B|}{19} > -1/2 \\ \text{outward} & R \ge \frac{1}{2} - AB + \frac{(A-1)(B-1)}{19} > -.7 \\ \text{reversed} & R \ge \frac{1}{2} - AB + \frac{(A+1)(B+1)}{19} > -.7. \end{array}$$

The native alphabet leaves $R = 0$ in the first case and $R = 0, -1/2$ in the others. Outward $R = 0$ contradicts

$$(D_i - 4/5)(D_{i+1} - 4/5) - C_{i,i+1}^2 = \frac{2}{5} C_{i,i+1} - \frac{24}{25} + \frac{8}{95} \ge 884/475 > 0.$$

For reversed $R = 0$, set $l = i, r = 18 - i$. The exact determinant would require

$$4lr A^2 B^2 - (l+1)(r+1)(A^2 + B^2) - 2AB - (l+1)A - (r+1)B + 21/4 = 0.$$

Its own derivatives are positive on that rectangle. If one own is not 5, it exceeds .9; evaluating the other at .808, the endpoints $l = 5, 13$ give $.9601824, 1.6961824 > 0$, and concavity in l excludes all. For two native-five owns the norm-two calculation in basis $1, \varphi$ has nonzero rational determinant coefficient 475 outward and -361 reversed. This uses only native cosine algebra. Thus remaining products are 0 for a common helper and $-1/2$ for opposite helpers.

After clearing old roots, the actual two-cap Schur matrices are $I - t\mathbf{1}\mathbf{1}^T$, $t = (\ell-1)/(2\ell)$, or the matrix with diagonal $\delta_l, \delta_r \in (1/2, 3/5]$, off-diagonal $-1/2$. Both are positive with positive inverse.

Their positive projections are feasible against every original wall, including the reverse wall at q. The cap-time and weight estimates in Appendix B.4 give weighted mean loss less than

$$1/10 + (11/8)(7/16) = .7015625 \quad \text{or} \quad 1/10 + (11/8)(3/10) = .5125.$$

Both are less than .87, excluding adjacent weak helpers. None of these steps uses endpoint height $> .8$.

Now assume the new height $\langle y, u_{18}\rangle > .78$. Each weak near-four has an actual mate, formed by releasing its white neighbor and retaining its cap. Positive finite Gram inversion gives an actual adjacent vertex feasible against every wall. Its edge cosine exceeds 11; for example the left mates have

$$C_j = \sqrt{2(i-1)/19}\left((20-i)/2 + (19-i)/\sqrt{2}\right) > 11, \qquad i = 5, 6.$$

At comparison 11/8, the normalized mate height exceeds 7/8. A left mate has correlation $< .15$ with u_{18}. The sum of these heights exceeds 1.655, whose square exceeds 2.3, while the squared norm of the sum of the directions is < 2.3. This contradicts $\|y\| < 1$. No left near-four remains.

Some $k = 5, 6$ is now strong by helper nonadjacency, with normalized height $> .677$. For a right mate $j = 14, 15$, the correlation formula

$$\rho_{ab} = \sqrt{\frac{a(19-b)}{b(19-a)}} \quad (a \le b)$$

gives $\rho_{18,j} < .46, \rho_{18,k} < .161, \rho_{j,k} < .407$. For one vector $V = .5u_{18} + .6u_j + .4u_k$,

$$\langle y, V\rangle > .5(.78) + .6(7/8) + .4(.677) = 1.1858, \quad \|V\|^2 < .77 + .276 + .0644 + .19536 = 1.30576.$$

Equation (10.1) gives $\|y\|^2 > 1.06584$. All near-four orientations are excluded at .78. Only nonadjacent pure weak helpers remain.

Use rational heights $H_{19-i} = H_i$, linear on 0–5 and 14–19, with

$$H_5 = 1.844,\ H_6 = 1.995,\ H_7 = 2.103,\ H_8 = 2.173,\ H_9 = H_{10} = 2.207.$$

Their positive slopes are $.3688, .151, .108, .070, .034$, then zero. The squares $(H_i + .87)^2$, $5 \le i \le 9$, are

$$7.365796,\ 8.208225,\ 8.838729,\ 9.259849,\ 9.467929,$$

strictly below $140/19, 156/19, 168/19, 176/19, 180/19$. Thus every retained positive dual is paid by a strong floor. If 5 is weak use chord 0–6, reflect at 14, and omit each internal weak knot by its neighbor chord. After chords set $w_i = (H_i - H_{i-1})/\sqrt{2}$. The reference is zero-sum with demand $E_0 = \|w\|^2$.

Before internal holes, the exact energies for zero, one and two weak ends are $.7205882, .70082285, .6810575$. Internal curvatures are $.043, .038, .036, .034$ and reflections; even summing all eight losses gives

$$2(.043^2 + .038^2 + .036^2 + .034^2)/4 = .0028725 < .002873.$$

Nonadjacency makes the losses legitimate. End chords do not change any eligible internal curvature. Hence

$$E_0 > \begin{cases} .6781845, & \text{14 weak: } m = 6,\ K = 13,\ H_K = 1.995, \\ .69794985, & \text{14 strong: } m = 5,\ K = 14,\ H_K = 1.844. \end{cases} \tag{16.1}$$

The surviving individual anchor prices are

$$\beta_{13} \ge (7H_6/6 - H_7)/2 = .11225, \qquad \beta_{14} \ge (6H_5/5 - H_6)/2 = .1089.$$

Adjacent holes increase the anchor price. Omitted knots have price zero; no missing strong 7 or 18 floor is paid.

Define

$$G_0 = -e_{19} + \frac{1}{m} \sum_{\text{last } m} e_i,$$

with 19 included. Then

$$\|G_0\|^2 = (m-1)/m, \quad \langle w, G_0 \rangle = 0, \quad G_0 = \omega_{18}/\sqrt{2} - \omega_K/(m\sqrt{2}).$$

Use the new weight $\mu = .6$. The debit $.6/(m\sqrt{2})$ is $< .071$ for $m = 6$, $< .085$ for $m = 5$, below the separate anchor prices. The positive endpoint coefficient is paid by its actual height. Here

$$-y_{19} > .78\sqrt{18/19} > .7591,$$

since $19(.7591)^2 = 10.94842339 < 18(.78)^2 = 10.9512$. The last-block means are $H_K/(m\sqrt{2}) < .2352$ or $< .261$. The same local-y square, retaining $\langle w, G_0 \rangle = 0$, gives

$$\begin{aligned} m = 6: &\quad \|y\|^2 > .6781845 + 1.2(.7591 - .2352) - .36(5/6) = 1.0068645, \\ m = 5: &\quad \|y\|^2 > .69794985 + 1.2(.7591 - .261) - .36(4/5) = 1.00766985. \end{aligned}$$

Both contradict $\|y\| < 1$, proving the lemma with the local inventory established in Appendix B.4. □

Appendix B.4 supplies the complete local own/white inventory, isolated-cap point bounds, positive mixed weights and native-five determinant expansion used above. It uses only real geometry and comparison 11/8, with no original-B_{18} premise. The original-x energy is never added to this local-y budget.

17 The near/far pair and the final composition

17.1 Every actual near/far fork edge is short

For a near-three or far-three fork role, the complete finite-white classification in Section B.9.9 gives an actual other endpoint q with whole diagram A_{18}. The positive reverse direction at q is exactly its normalized new endpoint dual: p has nonzero product with only that new incident root. Let C be the original edge cosine, $\beta = \sqrt{C^2 - 1}/C$, and $t = t_f < .876$.

Suppose $C \geq 5.6$. Then

$$E_q/E = C - \sqrt{C^2 - 1}\, t > 5.6 - .876(5.51) = .77324.$$

The function increases for $C \geq 5.6$. Here $5.51^2 = 30.3601 > 30.36$ and $(11/8)(.77324) > 1$, so $E/E_q < 11/8$. Also $\beta > .9839$, because $31.36(1 - .9839^2) = 1.0016631744 > 1$. The positive reverse height satisfies

$$\langle y, u_{18} \rangle = \frac{\beta - t}{1 - \beta t} > .78.$$

Its denominator is positive and its limiting gap is

$$.9839 - .876 - .78(1 - .9839 \cdot .876) = .000179192 > 0.$$

The original maximum supplies all-point comparison at q. Lemma 16.1 excludes $C \geq 5.6$. Every actual near/far fork edge therefore has $C < 5.6$.

17.2 The entire near/far family

Suppose the actual fork opposites are near and far:

$$c_N = Np - \gamma\Omega_f - \Omega_o/2, \qquad c_F = Fp - \delta\Omega_o - \Omega_1/2,$$

with f, o the different actual forks. Their times and exact edge cosines are

$$N^2 = 9\gamma^2 + 8\gamma + 5/4, \quad F^2 = 9\delta^2 + \delta - 1/2, \quad C_N = \frac{18\gamma + 8}{\sqrt{19}}, \quad C_F = \frac{18\delta + 1}{\sqrt{19}}.$$

The short-edge bounds and $\sqrt{19} < 4.359$ give $\gamma < .912, \delta < 1.301$. Put $U = 3\gamma + 4/3, V = 3\delta + 1/6, a^2 = 19/36$. Then $N^2 = U^2 - a^2, F^2 = V^2 - a^2$, and

$$(UV - a^2)^2 - N^2F^2 = a^2(U - V)^2 \geq 0, \qquad UV - a^2 > 0.$$

The actual mutual magnitude consequently satisfies

$$\begin{aligned} \lambda = -\langle c_N, c_F \rangle &= NF - 8\gamma\delta - \gamma/2 - (9/2)\delta - 1/4 \\ &\leq \delta(\gamma - 1/2) - 5/9 \\ &< 1.301(.412) - 5/9 = .536012 - 5/9 < 0. \end{aligned}$$

This contradicts all-wall acuteness. If $\gamma \leq 1/2$ the upper bound is already negative; the accepted own list in fact has $\gamma \geq \cos(\pi/4) > 1/2$. Zero, every finite, parallel and unbounded dotted mutual are excluded at once, for all finite or dotted near/far owns.

17.3 Exhausting the actual head and fork covers

For a dotted head, (15.11) removes every far-next-three role from Section B.9.9. What remains is isolated, near-three, or far-three. Their six unordered pairs are exhausted by

Actual pair	Complete excluding argument
isolated/isolated	Sections B.9.14 and B.9.15
isolated/near	Section B.9.16
isolated/far	Sections B.9.17 and B.9.18
near/near	Section B.9.9
near/far	The complete argument above
far/far	Section B.9.18

Both assignments of the genuine signed forks are included. Every pair is impossible, so no dotted head remains.

The head cover also includes own zero, finite and parallel cases. Section B.9.1 removes own $0, 3, 4$ by time constraints and own 6 by a positive affine $[6, 3]$ kernel. At own 6 an actual null is $\sqrt{3}H + 2n_1 + n_2$; at parallel own it is $H + n_1$. The remaining finite labels $5, 7, 8, 9$ and every integer label at least 10 are excluded by the complete conjunction in Section B.9.22. Every dotted own has just been excluded. Thus the whole-D_{18} diagram is impossible.

Theorem 12.1 excludes every dimension at least 19. At exact dimension 18, the fresh complete cover and its row consumers, including Section 13, leave only whole D_{18} at the same original farthest vertex. The head/fork argument excludes it. Hence dimension 18 is impossible, proving Theorem 8.1. Each step uses its own actual-rank cover and the same original vector within a case; no dimension heredity is used.

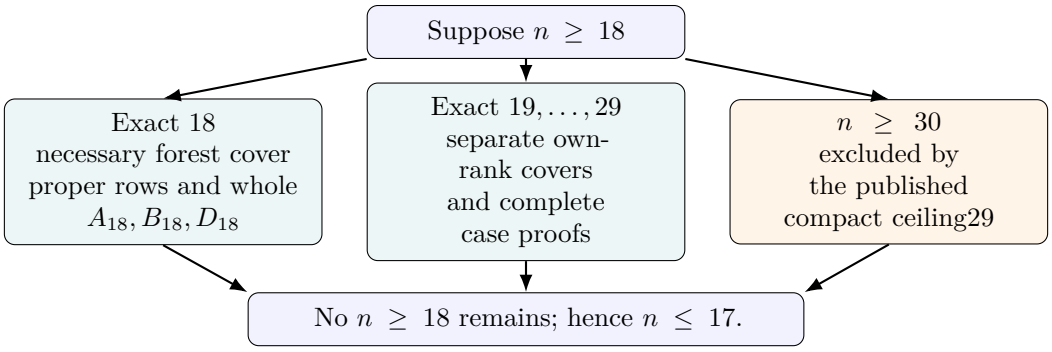


Figure 4: The compact dimension bound. The exclusion of dimension 18 is combined with the higher-dimensional exclusions proved in Section B.14.22.

18 Finite algorithms and conditional verification

18.1 The mathematical finite-cover calculation

The rank covers are finite combinatorial arguments supported by real inequalities, not field enumerations. For fixed total rank n, the forest calculation is:

1. List partitions of n into ranks of spherical irreducibles. Retain all exceptional finite types and classical types at their actual ranks. Keep $I_2(m)$ as a symbolic ordinary rank-two family for unbounded m, distinguishing its exceptional actual typed profiles.

2. Mark at most one allowed $I/J/N_7/O_7$ profile by the actual-wall compatibility lemma. Apply joint bounds only to disjoint sets of components, counting each defect once.

3. Sum proved same-vector energy bounds. Discard a forest only when a strict lower bound is at least 1. If a dominant rank needs a fresh local layer, retain it until that actual-rank lemma is established.

4. Return the retained grouped necessary cover and the inequality discarding every other partition.

Termination follows from finitely many rank partitions and finite type choices at each rank. The unbounded native parameter is represented by inequalities proved over its entire domain. Completeness uses spherical classification, compact acute simplicity, complete actual-defect compatibility, and spectator-uniform component bounds. The output is a necessary cover; no realization follows. A program testing only finitely many native labels would not establish it.

Local own/white calculations have a different domain. After deleting a specified original node, classify every finite extension of the retained spherical forest by its actual opposite wall, including own zero, all finite owns, parallel and dotted owns. Finite white extensions are bounded by spherical classification. Infinite own families are closed by analytic inequalities over complete intervals. Termination of a finite classification does not justify discarding a continuous branch. Completeness must include spectator attachments, affine kernels and every original wall for every auxiliary point.

Part III
Compact arithmetic polytopes

19 The result and the common argument

All polytopes are convex, finite-sided and full-dimensional. Arithmeticity refers to the lattice generated by their facet reflections.

Theorem 19.1. *Every compact arithmetic hyperbolic Coxeter polytope has dimension at most fifteen.*

The argument separates geometry from arithmetic at a definite point. Geometry provides one vector whose squared norm is less than one. Its projections onto whole finite diagram components are constrained by actual neighboring walls. Arithmeticity then limits the fields and integral parameters of those walls. When pressures overlap, their coefficients are paid together before one completed square is used.

Proposition 19.2 (Compact dimension bound)**.** *Every compact real hyperbolic Coxeter polytope has dimension at most seventeen, without an arithmetic hypothesis.*

The general compact bound is proved independently in Part II. It excludes dimension eighteen and every higher dimension, so only dimensions seventeen and sixteen remain here. We first split by the whole finite forest; the final D_{16} argument splits by the inert, ramified or split behavior of two. No argument assumes arithmeticity or Coxeter angles for an arbitrary face.

Section 20 gives the rank-seventeen reduction, and Section 21 states the finite algorithms and their numerical output premises. Section 22 gives the rank-sixteen reduction; Sections 23–26 prove the final dyadic obstruction. The appendices develop the component estimates, actual-domain inclusions and complete forest dispatches used in these reductions.

19.1 One original vector

Use the Lorentz form with negative time coordinate and outward unit normals. Distinct actual wall normals have nonpositive products. Compact acute simplicity gives a rank-n finite spherical forest at each vertex. Choose a strict interior point z and an actual vertex p farthest from it. Write

$$z = E(p - x), \quad E > 0, \quad x \in p^{\perp}, \qquad \|x\|^2 = \sum_C e_C < 1, \quad e_C = \|x_C\|^2.$$

The sum is over whole orthogonal finite components. Overlapping subdiagrams do not supply independent summands. If $q = \cosh(L)p - \sinh(L)u$ is an actual adjacent vertex, then

$$x \cdot u \geq (\cosh L - 1)/\sinh L = \tanh(L/2). \tag{19.1}$$

More generally a future unit point $q = ap - v$, with $v \in p^{\perp}$, satisfying every wall of the original polytope gives $x \cdot v \geq a - 1$. Distance from z is convex on geodesic segments, so its maximum over the compact polytope is attained at a vertex. For any reference vector w,

$$\|x\|^2 \geq 2x \cdot w - \|w\|^2, \tag{19.2}$$

by $\|x - w\|^2 \geq 0$. Each coefficient removed from an original pressure must fit its individual available coefficient. Positive combinations of actual normals satisfy every unselected wall by acuteness; future orientation and the finite or negative square still need proof. The companion and the precisely cited supplements make these checks before using (19.2).

19.2 What arithmeticity adds

An arithmetic reflection lattice has an admissible quadratic-form model over its actual totally real field K; the structural reflection theorem is recorded as Lemma 5.2 of [10]. It supplies K-valued root lines and positive definite nonphysical forms. For roots r_i on those lines put

$$a_{ij} = 2(r_i, r_j)/(r_i, r_i).$$

Invariant cyclic products are algebraic integers and generate the actual ground field. Individual physical unit roots need not be K-valued. Disconnected roots need not have equal raw norm squareclasses.

If an actual principal Gram matrix, after its specified permitted K-rescaling, is defined over $k \subset K$ and indefinite physically, then $K = k$: an additional real embedding fixing k would preserve indefiniteness and contradict positive definiteness. A nonzero algebraic integer also has nonzero integral norm, so bounds at *every* nonphysical embedding can force a degree bound. Neither argument assumes a convenient overfield or an integer power basis.

The rational field is excluded in the dimensions used here: compactness requires anisotropy, while Meyer's theorem makes an indefinite rational form in at least five variables isotropic. These arithmetic inputs are reviewed in [11]. Finite/affine classification and acute wall geometry are external foundations; component estimates, forest reductions and computed propositions below are internal results.

20 The complete rank-seventeen reduction

Write $T_2 = I_2(m)$, $m \geq 3$, for an ordinary irreducible dihedral; $T_3 = A_3, B_3, H_3$ and $T_4 = A_4, B_4, D_4, F_4, H_4$ unless a stated field restriction removes a type. Put $\varphi = (1 + \sqrt{5})/2$. Ordinary nonshort A_1 costs $> 1/10$, ordinary $T_2 > 1/7$, $T_3 > 1/4$, $T_4 > 2/7$. A total-rank-three forest need not cost $> 1/4$; its ordinary bound is $> 17/70$.

The exceptions retain actual-wall meanings. Type I is an isolated short rank-one opposite

$$c_I = \sqrt{h}\, p - \sqrt{1+h}\, e, \qquad 0 < h < 1/2.$$

Type J couples two rank-one directions with joint energy $> .34$. The typed rank-two defects are N_7, old A_2/new isolated-seven, and O_7, old $I_2(7)$/new isolated-three. Each costs $> .09$; they cannot

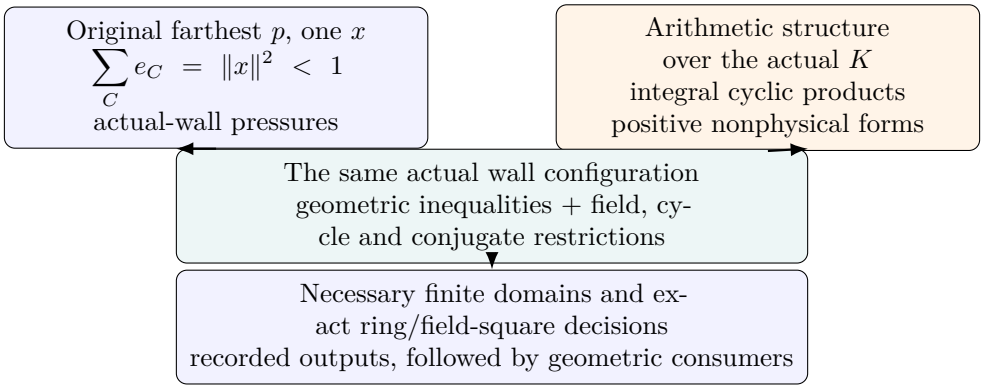


Figure 5: Arithmetic restrictions on the wall parameters. The finite algebraic domains and their outputs are given in Proposition 21.1 and Section 21.

coexist with one another or the short I/J alternatives. A bare old seven is not a typed defect. These are the complete small-component alternatives of Proposition 20.5. Their geometric compatibility proof uses the native gap between distinct actual walls, not merely the abstract old forest.

20.1 Shared costs and arithmetic prefix seals

We specify the component inputs before carrying out either rank count. The ordinary geometric rates are the component theorems of the general compact companion, applied to whole orthogonal components at the same original vertex:

A_1	$1/10$	T_2	$1/7$
A_2	$319/2000$	A_3	$1/4$
B_3, H_3	$2/7$	A_4	$2/7$
H_4	$11/36$	B_4, D_4, F_4	$1/3$
A_5	$3/8$	D_5	$5/14$
B_5	$3/7$	A_6, B_6, D_6	$3/7$
B_7, D_7	$1/2$	B_8, D_8	$4/7$
E_6	$9/20$	E_7	$5/9$
E_8	$2/3$		

Every entry is a strict lower bound. In particular every ordinary component outside A_r, $r \geq 7$, and B_r, D_r, $r \geq 9$, costs more than its rank divided by fourteen. The universal large-component rates used here are $A_r > 1/2$ for $r \geq 7$, $A_r > 4/7$ for $r \geq 8$, $A_r > 5/7$ for $r \geq 13$, and $B_r, D_r > 9/14$ for $r \geq 9$. These are geometric companion inputs, not assertions that a higher-rank whole-diagram exclusion holds at a smaller rank. The stronger B_3, H_3 rate is needed only for the rank sixteen count; the rank seventeen count uses $1/4$.

The actual short-I wall gives parameters u, t with

$$e_I \geq u^2, \quad 0 < u < 1/3, \quad (1-u^2)(1-t^2) = 8ut, \quad t + 3u/2 > .767. \tag{20.1}$$

Every normalized old dual direction outside I has projection at least t. Its squared-time parameter h also gives the following geometric alternative: at $h \leq 1/5$ all these squared projections are at least $1/5$; at $h > 1/5$, $e_I > 3/70$. These are the companion's actual opposite-wall barrier, with no arithmetic hypothesis. We record the needed joint consequences here to make their accounting explicit.

Lemma 20.1. *For ordinary irreducible T_j the following bounds hold, each counting I once:*

$$e_{I+T_2} > .21, \qquad e_{I+T_3} > 2/7, \qquad e_{I+T_4} > 23/70, \qquad e_{I+A_1} > .20204, \qquad e_{I+2A_1} > .29.$$

Proof. For T_2 use $e_{T_2} > 1/7$ and $e_{T_2} \geq t^2$. If $u \geq 519/2000$, then $u^2 + 1/7 > .21$. Below that point $u^2 + (.767 - 1.5u)^2$ is decreasing and has endpoint value $.2100353125 > .21$.

For T_3, its two endpoint correlations are $1/3$, $1/\sqrt{3}$ and ρ with $\rho^2 = (3\varphi+2)/(3\varphi+6) < 16/25$, respectively. Cauchy on the sum of the two unit directions gives $e_{T_3} > (10/9)t^2$, as well as $e_{T_3} > 1/4$. At $u \geq .19$ use $u^2 + 1/4 > 2/7$; below it the decreasing quadratic $u^2 + (10/9)(.767 - 1.5u)^2$ is at least $132407/450000 > 2/7$.

For A_4, the low-h direction cover costs more than $1/3$; the high-h case gives $3/70 + 2/7 = 23/70$, strictly exceeded. The B_4, D_4, F_4 rates already exceed $1/3$. For H_4 we need $t+2u > 6/7$. Substituting $6/7 - 2u$ for t in the left-minus-right side of the identity in (20.1), and multiplying by 49, gives

$$13 - 168u + 575u^2 - 168u^3 + 196u^4.$$

With $v = u - .15$ this is $.269725 - 4.194v + 525.86v^2 - 50.4v^3 + 196v^4$. For $-3/20 < v < 11/60$ it is at least $.269725 - 4.194v + 516.62v^2 > 0$, the last quadratic having negative discriminant. The original identity decreases in t, proving the barrier. At $u \geq 19/125$, $u^2 + 11/36 > 23/70$; below it the decreasing quadratic $u^2 + (6/7 - 2u)^2$ is at least $50389/153125 > 23/70$.

For $I + A_1$, put $f(s) = (\sqrt{1+s} - 1)/(\sqrt{1+s} + 1)$. The actual edge barrier gives half-tangent squares $q = f(h)$ and at least $r = f(1/(4h))$ on the other component. Writing $P = \sqrt{qr}$, inversion gives $q + r = 1 + P^2 - 8P$, while AM–GM gives $q + r \geq 2P$. As $0 < P < 1$, this implies $P \leq q_0 = 5 - 2\sqrt{6}$ and

$$q + r - 2q_0 = (q_0 - P)(8 - P - q_0) \geq 0.$$

Thus $e_I + e_{A_1} \geq 2q_0 > .20204$; the rational seal is $2.44949^2 > 6$. Finally, for two ordinary A_1 components, at $u \geq .3$ use $u^2 + .2 \geq .29$, strictly exceeded. Below it use $q(u) = u^2 + 2(.767 - 1.5u)^2$; then

$$q(u) - q(.3) = (.3 - u)(2.952 - 5.5u) \geq 0, \qquad q(.3) = .290978 > .29.$$

Each calculation estimates one joint sum of orthogonal whole-component energies; no endpoint has been charged a second copy of I. □

Lemma 20.2 (Propagation from the actual old-seven defect)**.** *Suppose a whole $I_2(7)$ component has an actual opposite supported only there, with a native-three contact at one node and zero contact at the other. Every other whole rank-one component has energy $> 2/15$, and every other whole irreducible rank-two component has energy $> 1/5$. This lemma is geometric and allows arbitrary finite spectators.*

Proof. Put $d_7 = \csc^2(\pi/7)$. The typed wall is $c_D = \sqrt{h_D}p - v_D$, with $h_D = d_7/4 - 1 \in (0, 1/3)$. Let $c = \sqrt{h}\,p - v$ close an actual edge of another component. Its finite endpoint retains the whole old $I_2(7)$. A positive contact there contributes at least $d_7/4 > 1$ to the white Schur cost, impossible. Thus v has zero projection to that whole component, and c is distinct from c_D. The native actual-wall gap gives

$$(c_D, c) = -\sqrt{h_D h} \leq -1/2, \qquad h > 3/4. \tag{20.2}$$

This proves the height bound for every opposite outside the defect, without assuming an isolated profile for those other walls. In a rank-one direction its spatial derivative is at most $\sqrt{1+h}$, so $\sinh^2 L \geq h > 3/4$. Hence its half-tangent square is greater than $(11 - 4\sqrt{7})/3 > 2/15$.

Suppose a whole $T = I_2(m)$, $m \geq 3$, instead has energy $e \leq 1/5$. Its normalized duals have correlation $k = \cos(\pi/m)$. Write $s_i = x_T \cdot u_i$ and $F(s) = 2s/(1 + s^2)$. For $m \geq 4$, distinct actual opposites would require

$$k \leq F(s_1)F(s_2) \leq F(\sqrt{1/5})^2 = 5/9 < 1/\sqrt{2} \leq k,$$

impossible. For $m = 3$, Cauchy gives $(s_1 + s_2)^2 \leq 3/5$, and concavity of $\log F$ gives the incompatible bound $1/2 \leq F(s_1)F(s_2) \leq 240/529 < 1/2$. Therefore one actual wall closes both directions. Both lengths have

$$\cosh L_i \leq (1 + 1/5)/(1 - 1/5) = 3/2. \tag{20.3}$$

Let $b, g \geq 0$ be its contacts in T and Q its entire foreign Schur cost. With $d = (1 - k^2)^{-1}$, the finite endpoints give

$$Q + b^2 < 1, \quad Q + g^2 < 1, \qquad h = d(b^2 + 2kbg + g^2) + Q - 1 > 3/4,$$
$$C_b^2 = \frac{d(b + kg)^2}{1 - Q - g^2}, \qquad C_g^2 = \frac{d(g + kb)^2}{1 - Q - b^2}.$$

Every positive b, g is consequently a finite native cosine. At least one is positive. A positive whole foreign contribution costs at least $1/4$, so $Q = 0$ or $Q \geq 1/4$.

For a single contact, say $b = \beta > 0, g = 0$, $C_g^2 = 1 + h/(1 - Q - \beta^2)$. Equations (20.2)–(20.3) imply $Q + \beta^2 < 2/5$; hence $Q = 0$ and the native gap forces $\beta = 1/2$. Then $h = d/4 - 1 > 3/4$ gives $d > 7$ and $m \geq 9$, since $d_8 = 4 + 2\sqrt{2} < 7$. But $\sin t < t$ and $\pi^2 < 10$ give $d > 81/10 > 31/4$, so $C_g^2 = (d-1)/3 > 9/4$, a contradiction.

If both contacts are positive and $Q > 0$, let g be the larger. When $g \geq 1/\sqrt{2}$, $C_b^2 \geq (4/3)(1 + 1/\sqrt{2})^2 > 9/4$. Otherwise $b = g = 1/2$. At old label three, $h = Q < 3/4$; at old label at least four, $C_b^2 \geq (1+k)/(2(1-k)) > 5/2$. Both are impossible.

It remains $Q = 0$. If the larger contact has label at least five, then $g > 4/5$, $b \geq 1/2$, $k \geq 1/2$, $d \geq 4/3$ give $C_b^2 > 3$. If both contacts have label four, $C_b^2 = (1+k)/(1-k) \geq 3$. The remaining finite contact pairs are $(3,4)$ and $(3,3)$. For $(3,4)$, old label at least four gives $C_b \geq 2$; old label three gives $h = \sqrt{2}/3 < 3/4$. For $(3,3)$, old label at least five gives $C_b^2 = (1+k)/(3(1-k)) > 3$; old labels three and four give $h = 0, 1/\sqrt{2} < 3/4$. Every contact and foreign case contradicts (20.2) or (20.3). Thus $e_T > 1/5$, completing both conclusions. □

Both typed defects force the actual field K_7. Their three actual roots form the indefinite $(7, 3, 2)$ principal. Odd bonds permit a square-two normalization over K_7; an extra embedding fixing K_7 would preserve that indefinite principal. Hence actual-field descent gives $K = K_7$. In particular H_3, H_4 cannot occur beside a typed defect, since their five-bond would put $\sqrt{5}$ in K_7. For N_7 every other normalized dual has squared projection $> 1/5$. For O_7, Lemma 20.2 gives every other A_1 cost $> 2/15$ and every other ordinary irreducible T_2 cost $> 1/5$. The hypothesis is the actual isolated-three wall, not the old label seven alone.

Lemma 20.3 (Literal arithmetic helpers)**.** *At a classified long-prefix index $i \geq 6$ of B_r or D_r, an actual far helper $v = \gamma\omega_i + \omega_1/2$ with $202/125 < 2\gamma < 12/5$ is impossible. At a D fork-helper site with $r \geq 9$, the corresponding profile $v = \gamma\omega_i + \omega_f/2$ can only have $2\gamma = \varphi$.*

Proof. Normalize the connected long roots to square two over K. The simple-three helper contact permits the same normalization of the actual outside root; its path cycle makes $t = 2\gamma$ integral. The short B root has square one, so the entire old Gram is rational. All embeddings of $\mathbb{Q}(t)$ extend to K, and every nonphysical one must give a strictly positive Schur complement.

For the far helper the raw inverse entries $i, 1, 1$ give $it_\sigma^2 + 2t_\sigma - 1 < 0$. At $i \geq 6$ this forces $|t_\sigma| < .61$ and $t_\sigma < .28$. Degree at least three would give a nonzero integral norm of magnitude less than $(12/5)(.61)^2 < 1$. In degree two the integral trace is strictly between $503/500$ and $67/25$, hence two; its norm $t(2-t)$ lies strictly between -1 and one, again impossible. Degree one leaves $t = 2$, an actual parallel pair with a future feasible null sum, contrary to compactness.

For the fork helper the inverse entries $i, i/2, r/4$ give

$$4it_\sigma(t_\sigma + 1) + r < 8, \qquad -1 < t_\sigma < 0.$$

The integral nonzero element $t(t+1)$ has physical value $< 204/25$ and other conjugates of magnitude at most $1/4$. Its norm excludes degree at least three. In degree two the trace of t is one or two. Trace one gives $t < 2$ and norm in $(-2, -.995456)$, hence norm -1 and $t = \varphi$. Trace two gives norm in $(-24/25, 0)$, impossible. The rational $t = 2$ is again parallel. This proves both assertions without identifying $\mathbb{Q}(t)$ with the full field in advance. □

Lemma 20.4 (The mixed B_2 tail)**.** *At a B_r site $i = r - 2 \geq 12$, the actual own-three/short-three profile $v = (\omega_i + \omega_r)/2$ is impossible.*

Proof. Normalize the old long roots to square two and the short root to square one over K. The simple-three short contact normalizes the outside root to square one; its own-three product is $-1/\sqrt{2}$. Thus $\sqrt{2} \in K$. At an embedding reversing $\sqrt{2}$, while fixing the rational old Gram, its Schur cost is

$$i/2 - i/\sqrt{2} + r/4 = i(3 - 2\sqrt{2})/4 + 1/2 > 1.$$

At $i = 12$ the last inequality follows from $\sqrt{2} < 17/12$, and it increases with i. This contradicts strict positivity. □

For later prefix counts put

$$f_j = \sqrt{j} - .61/\sqrt{2}, \qquad E_J = f_5^2/5 + \sum_{j=5}^{J-1} (\sqrt{j+1} - \sqrt{j})^2.$$

The generic geometric inventory in the companion applies to B_r with $M = r - 5 \geq 6$ and to D_r with $M = r - 6 \geq 6$, through $M + 1$. Every weak index is the literal far helper just excluded. Hence $6, \ldots, J$ are strong, with $J = r - 4$ in B and $J = r - 5$ in D. If five is strong, positive concave interpolation gives E_J. If it is weak, its actual point gives $x_1 > .5965$, $S_6 > 2.01815$; interpolation through $1, 6, J$ gives

$$L_J = .7600299945 + \frac{(\sqrt{J} - \sqrt{6})^2}{J - 6} > E_J,$$

with the quotient zero at $J = 6$. This follows from the variance comparison: $E_6 < .7$, and at most ten increments in $[1/8, 1/5]$ have variance sum at most $9/640$, less than the initial gain $> .06$. The rational lower seals used below are

J	6	7	8	9	10
$E_J >$	.69685	.73531	.76864	.79805	.82436
J	11	12	13	14	15
$E_J >$	.84816	.86989	.88989	.9084	.92564.

(20.4)

These statements retain the precise rank hypotheses of the companion inventory; the smaller terminal ranks are handled below, rather than being silently inserted into its generic range.

Proposition 20.5 (Necessary cover in rank seventeen)**.** *The whole forest at the original farthest vertex of a compact arithmetic* 17*-polytope is one of Table 1.*

The proof below starts at rank seventeen. The arithmetic baseline component estimates used in its partition stage are $e_{A_{15}} > .79$, $e_{A_{16}} > .83$, $e_{A_{17}} > 6/7$, and, under either actual typed defect, $e_{A_{12}} > .703101$, $e_{A_{14}} > .7897732$. Section C.14.1 derives all these inputs from the stronger printed whole-component proofs, with arbitrary spectators. They are not higher-rank exclusions.

Proof of Proposition 20.5. Call A_r, $r \geq 7$, or B_r, D_r, $r \geq 9$, dominant. There is exactly one dominant. With none, discarding only the one possible I leaves cost $> 16/14$; a typed defect gives $> .09 + 15/14$. Ordinary or J accounting is stronger. Two dominants each cost more than $1/2$. The complete ordinary table includes E, F, H, so this step loses no exceptional irreducible component.

An A_7 leaves ten spectator ranks and is impossible by the same budget. At A_8, A_9, A_{10}, even discarding I leaves at least six ordinary ranks, giving $4/7 + 6/14 = 1$, strictly exceeded. The typed and J alternatives are stronger. Thus $11 \leq r \leq 17$.

For A_{11}, ordinary spectator rank six already costs $> 3/7$. With I, the complete ordinary rank-five partitions are 5, $4+1$, $3+2$, $3+1+1$, $2+2+1$, $2+1+1+1$, 1^5. Only $A_5, D_5, (A_4 \text{ or } H_4) + A_1, T_3 +$

Group	Component	Complete spectators
1	A_{11}	$I + (A_5$ or $D_5)$
2	A_{11}	N_7/O_7 plus A_4, B_4, D_4, F_4
3	A_{12}	$A_5, D_5, (A_4$ or $H_4) + A_1, T_3 + T_2, 2T_2 + A_1$
4	A_{12}	$I + T_4,\ I + T_3 + A_1,\ I + 2T_2$
5	A_{14}	$T_3,\ T_2 + A_1,\ I + T_2$
6	A_{15}	$T_2,\ 2A_1,\ I + A_1,\ N_7/O_7$
7	A_{16}	ordinary A_1 or actual I
8	A_{17}	none
9	B_{15}	N_7/O_7
10	B_{16}	actual I
11	B_{17}	none
12	D_{15}	N_7/O_7
13	D_{16}	ordinary A_1 or actual I
14	D_{17}	none

Table 1: The remaining spherical forests. Unmarked components are ordinary.

$T_2, 2T_2 + A_1$ can initially fit below $3/7$. Lemma 20.1 removes the last three by $23/70 + .1 = 3/7$, $2/7 + 1/7 = 3/7$, and $.21 + 1/7 + .1 > 3/7$. Hence group1 is complete. With a typed defect, ordinary rank four initially leaves T_4 or $2T_2$; the other partitions cost at least .35, $12/35$ or .4 before the defect is added. Propagation makes each $T_2 > .2$, removing $2T_2$; actual-field descent removes H_4. This gives group2.

For A_{12}, the complete cheap ordinary rank-five list is exactly that just displayed. Indeed $B_5 > 3/7$, $B_4, D_4, F_4 + A_1 > 13/30$, $T_3 + 2A_1 > .45$, $T_2 + 3A_1 > 31/70$, and $5A_1 > .5$. A J pair plus three remaining ranks costs $> .34 + 3/14 > 3/7$. These give group3. With I, the rank-four partitions are $T_4, T_3 + A_1, 2T_2, T_2 + 2A_1, 4A_1$. The last two cost $> .29 + 1/7 > 3/7$ and $> .29 + .2 > 3/7$, respectively, using $I + 2A_1$ only once. Thus group4 is complete. A typed defect would leave ordinary rank three and give $.703101 + .09 + 17/70 > 1$, so none remains here.

For $A_{13} > 5/7$, ordinary or J rank four already costs $> 2/7$. With I, the remaining rank-three partitions are $T_3, T_2 + A_1, 3A_1$; their joint costs exceed $2/7$ by Lemma 20.1 or by the ordinary three-A_1 cost. A typed defect plus T_2 costs $> .09 + .2 > 2/7$; two ordinary A_1 cost at least $4/15$ and give an even stronger contradiction. No A_{13} row remains.

At $A_{14} > 5/7$, ordinary spectator rank three leaves T_3 or $T_2 + A_1$; three A_1 and $J + A_1$ are too expensive. With I, only $I + T_2$ remains since $I + 2A_1 > .29 > 2/7$. A typed defect gives $.7897732 + .09 + 2/15 > 1$. These are exactly group5. At $A_{15} > .79$, rank two leaves $T_2, 2A_1, I + A_1$, or either typed defect. A J pair closes; the other bounds do not delete the retained alternatives. At $A_{16} > .83$, its single rank can be ordinary A_1 or actual I. At A_{17} there are no spectators. This proves groups6–8 without an endpoint or field-degree inference.

For B, D, the $9/14$ rate and spectator accounting give $r \geq 12$: at six spectator ranks even the I debit leaves $5/14$; a typed defect gives $4/14 + .09$. The generic arithmetic prefix has last strong knot $J = r - 4$ in B, $r - 5$ in D, within the rank domains specified above. The weakest floors at $r = 12, 13, 14$ are therefore E_7, E_8, E_9. At twelve their possible rank-five remainders cost more than $23/70$, $.09 + 17/70$, or $5/14$, respectively. At thirteen all rank-four remainders cost $> 2/7$, or at least $.09 + .2$ in a typed case. At fourteen the cheapest rank-three possibilities are ordinary $> 17/70$, $I + T_2 > .21$, $I + 2A_1 > .29$, and typed $> .09 + 2/15$. Each closes with its stated prefix floor. Thus no B, D component of ranks twelve through fourteen remains.

We must extend the short tails at their actual ranks before counting the proper higher rows.

For $14 \le r \le 17$, the B_3 finite list has only the extra short-three F_4 role; it is sealed against foreign contacts and is affine or directly strong. At the B_2 site $i = r-2$, the extra own-three/short-three profile is excluded by Lemma 20.4. Own zero, four, six and parallel retain their affine or null principals; native own at least five is directly strong. A bridge through the short endpoint is a full B_r with internal new root and admits no finite foreign arm. Its residual is $1/(2i)$ and its ratio at least $1/\sqrt{2i} + \sqrt{i}/2 > 100/61$ for $i \ge 11$. All the other classified weak roles are excluded by Lemma 20.3. Hence $6, \dots, r-2$ are strong, so $e_{B_r} > E_{r-2}$ in either initial-five branch. At B_{15}, $E_{13} > .88989$ excludes ordinary $T_2, 2A_1, J, I + A_1$, leaving either typed defect. At B_{16}, $E_{14} > .9084$ excludes ordinary A_1 and leaves actual I. Whole B_{17} is retained.

For D_{15} the spectator rank is exactly two, and every allowed remainder costs $> .09$. Its bulk already gives strong $6, \dots, 10$. The local D_4, D_3 lists at indices eleven and twelve retain only an additional own-five/fork-three case. More explicitly, own zero, three, four, six and parallel have the same affine or null principals; middle-three, end-four and the sealed H_4 end-five are directly strong. The full bridge is D_{15}; at D_3 the end-three bridge is A_{15} with both arms at least three, so no finite foreign arm is possible. Its residual $1/8 + 1/(2i) \le 1/6$ and numerator $> .75$ make it strong; its zero-own cycle is affine. The remaining fork helper is golden by Lemma 20.3.

For that golden role the actual positive left H_4 projection has

$$\alpha^2 = \varphi + 5/8 > 56/25, \quad \delta = (5 - 3\varphi)/8, \quad \sqrt{\delta}/\alpha < .091, \quad A^2 = i\alpha^2 - \kappa,$$

where $\kappa = 1/2$ at D_4, $5/8$ at D_3. The positive correction is $\gamma(3n_i/2 + n_{i-1} + n_{i-2}/2)$. The finite projection is future and feasible against all original walls; the correction alone is not used as a hyperbolic point. Its displacement divided by α is flat one on the first $i-3$ coordinates, has square $< i$, and gives pressure $G > \sqrt{i - \kappa/\alpha^2} - .091$. Pay weight $1/6$ at the previous strong knot $K = i-3$. Both initial-five references have endpoint coefficient $> 1/6$, so a single pure-D square gives

$$e_D > E_K + (G - f_K)/3 - (i - K)/36.$$

The complete two-row computation is

	i	K	$G >$	$f_K <$	$e_D >$
D_4	11	8	3.19	2.398	.9493
D_3	12	9	3.33	2.569	.9683.

The radical seals are $3.281^2 < 11 - 25/112$ and $3.421^2 < 12 - 125/448$. The pure energies each exceed .91; adding only the disjoint actual spectator energy $> .09$ excludes both roles. Thus $6, \dots, 12$ are strong and $e_{D_{15}} > E_{12} > .86989$. Ordinary $T_2, 2A_1, J, I + A_1$ close; either typed defect is retained. At D_{16} its ordinary A_1 and actual I are both retained, and whole D_{17} is retained. These are groups12–14.

Every partition, actual defect and permitted dominant rank has now been considered. No root has been deleted from another ambient list, and every sum concerns the same original vector. □

Proposition 20.6 (The other rows in rank seventeen)**.** *Every row of Table 1 except group8 is impossible.*

Proof. The complete dispatch in Section C.14.2 excludes groups1–7 and9–14. Each entry refers to its printed actual-wall, field and individual-payment proof. The shared algebraic census for whole B17 and D17 has separate actual-domain inclusions and separate original-vector consumers. Group8 is treated next. □

Proposition 20.7 (Actual chain endpoints)**.** *If group8 occurs, both actual endpoint opposites are isolated within A_{17}. Their integral parameters U, V generate the same actual field and satisfy*

$$36/17 < U, V < 3969/425, \qquad 0 < U^\sigma, V^\sigma < 36/17 \quad (\sigma \neq \mathrm{id}). \tag{20.5}$$

The field is nongolden, has degree at most five, and at least one endpoint is finite. In fact

$$K \in \{\mathbb{Q}(\sqrt{2}), \mathbb{Q}(\sqrt{3}), K_7, K_9\};$$

in the cubic fields a finite endpoint has label seven or eighteen, respectively.

Appendix C.3 proves the complete endpoint, field and domain-inclusion sequence of Sections C.3.1 and C.3.2, Section C.3.3, Section C.3.4. It retains every finite-white and own case, including the golden time-compatible exception. One decisive step explains the simultaneous accounting. The common body has squared norm .9037838. Both dotted cores are paid at weight .3; the outer coefficients change from .1382 to .0782 and still fund every central foot. On the entire $0 < u < 1$ interval the extra contribution is

$$.6\{\sqrt{17/9 - u} - \sqrt{u(1-u)} - .4174\} - .09(8/5 - 2u + u^2) > .04969.$$

The complete circle bounds in Appendix C.3.4 give $.9037838 + 2(.04969) > 1$ if both ends are dotted. No unavailable strong-five floor is used. The complete finite alphabet, including label30 before its conjugate test, then gives the stated fields, rather than a prescribed field list.

In raw old-root normalization of square two, a cap has square $2U$ and time square $F(U)/18$, where $F(X) = X(17X - 36)$. The inverse endpoint entry is $1/18$. Every actual pair consequently satisfies

$$R = F(U)F(V) \in K^2, \qquad g = (UV - \sqrt{R})/18 \in \mathcal{O}_K, \qquad L = g^2/(UV) \in \mathcal{O}_K. \tag{20.6}$$

The root is physically positive. Acuteness gives $g \leq 0$. If $g \neq 0$, nonphysical positive definiteness gives $0 < L^\sigma < 4$. The branch $g = L = 0$ is retained. Physical owns and mutual parameters must be native or dotted; four would be a compactness-forbidden parallel pair. The endpoints need not be equal.

21 Finite propositions and their complete algorithms

The computations check necessary algebraic domains. An empty domain excludes an actual configuration after geometric inclusion has been proved. A nonempty domain supplies necessary pairs, followed by a separate geometric contradiction, not examples.

21.1 Full quadratic integer rings

For squarefree $d > 1$ use $\omega = (1 + \sqrt{d})/2$ if $d \equiv 1 \pmod 4$, otherwise $\omega = \sqrt{d}$. Let Δ be the fundamental discriminant. If $Y = a + b\omega$ generates K, $Y > Y' > 0$, $Y < B$, then

$$0 < b\sqrt{\Delta} = Y - Y' < B.$$

Thus $\Delta < B^2$, and b and the strip-permitted integers a have explicit finite bounds. Squarefree trial division and these integer intervals cover the full maximal order.

Use rational pairs $a + b\sqrt{d}$. Integrality in the half basis means $2b, a - b \in \mathbb{Z}$; otherwise both coefficients are integral. Signs use exact rational square comparisons. For $b \neq 0$, a field square requires norm n^2 with rational $n \geq 0$ and one of

$$x^2 = (a + n)/2, \quad x^2 = (a - n)/2$$

a nonzero rational square. Set $y = b/(2x)$ and verify the full square identity. Both signed norm branches are necessary and sufficient. For $b = 0$ test rational squares and d times rational squares, including zero. A square norm alone is insufficient.

The complete positive quadratic finite native alphabet is

$$1, 2, 3, (3 + \sqrt{5})/2, (5 + \sqrt{5})/2, 2 + \sqrt{2}, 2 + \sqrt{3}.$$

Indeed $\phi(m) \leq 4$ leaves $m = 3, 4, 5, 6, 8, 10, 12$. Zero is orthogonality, values above four are dotted, and each radical value is retained only in its actual field.

Proposition 21.1 (Recorded finite outputs)**.** *At the indicated complete domains the exact outputs are as follows.*

1. *The quadratic A_{17} domain* (20.5)–(20.6) *has no survivor. There are 25 discriminants $\Delta < 88$, 53 models and 159 ordered pairs: 106 fail field-square membership, 52 long-cycle integrality and one two-cycle integrality.*

2. *The two cubic A_{17} finite-anchor domains have the nine strip models of Table 2, all failing a necessary test.*

3. *The quadratic domain*

$$4 < U < 11,\ 0 < U' < 2, \quad 2 < W < 676/17,\ 0 < W' < 4/17,$$

$K = \mathbb{Q}(U) = \mathbb{Q}(W)$,

$$g = UW - \sqrt{U(U-2)W(17W-4)}, \quad g,\ L = g^2/(2UW) \in \mathcal{O}_K, \quad 0 < L' < 4,$$

with physical native/dotted mutual parameter, has 36 fields $\Delta < 121$ and 120 pairs. Its only survivors are

K	U	W
$\mathbb{Q}(\sqrt{13})$	$(5 + \sqrt{13})/2$	$11 + 3\sqrt{13}$
$\mathbb{Q}(\sqrt{17})$	$(7 + \sqrt{17})/2$	$(21 + 5\sqrt{17})/2$.

4. *The quadratic whole-B_{16} domain*

$$2 < U < 12,\ 0 < U' < 2, \quad 4 < V < 64,\ 0 < V' < 1/4, \quad UV/2 \in \mathcal{O}_K,$$

$$k^2 = \frac{(U-2)V}{4U(4V-1)}, \quad k > 0, \quad G = 4Uk(4V-1) - UV, \quad L = G^2/(2UV),$$

with $k \in K$, integral G, L, actual acuteness and the following exact predicates, has 43 fields $\Delta < 144$: $G \geq 0$; if $G = 0$ then $L = L' = 0$, whereas otherwise $0 < L' < 4$; and U, L must belong to the native finite alphabet of their actual field or be greater than four. In particular $L = 4$ is excluded, and no continuous nonnative value below four is allowed. The identity

$$L + G = ((15U - 32)/2)V - 2U + 4$$

is checked for the physical positive square root k. The complete domain has 100 head models, 85 short models and 273 pairs. Exactly 168 fail half-product integrality and 104 field-square membership. The sole survivor is $U = 5 + 2\sqrt{6}$, $V = 10 + 4\sqrt{6}$.

The quadratic algorithm generates all fields below the proved discriminant bound, lists all strip-permitted integer coefficients, and visits every ordered pair. It applies full field-square, cycle-integrality, conjugate-positivity and native tests, recording every rejection or survivor and complete totals. For (i), $b \le 4$ and the conjugate width 36/17 requires *three* candidates $a = \lfloor -b\omega' \rfloor + 1, +2, +3$. There are at most 3600 pair tests before the strict strips. For (iii), $b_U \le 4$, $b_W \le 17$, with two a_U values and at most one a_W value: at most 4896 pairs. For (iv), safe full boxes are $1 \le a_U \le 7$, $1 \le b_U \le 5$ and $1 \le a_V \le 32$, $1 \le b_V \le 28$. These bounds prove termination. The full-ring strip proof gives coverage and the exact tests justify every rejection. The floors themselves are exact integer operations:

$$\lfloor -b\omega' \rfloor = \begin{cases} \operatorname{isqrt}(b^2 d), & \omega = \sqrt{d}, \\ (\operatorname{isqrt}(b^2 d) - b) \operatorname{div} 2, & \omega = (1+\sqrt{d})/2. \end{cases}$$

Irrationality places $b\sqrt{d}$ strictly between consecutive integers, which proves both formulas, including even b. Trial square division through $\operatorname{isqrt}(d)$ detects squarefree radicands exhaustively. Fundamental discriminants below the proved bound require no larger radicand loop.

21.2 The two cubic finite anchors

Let U be the largest root of

$$P_7 = X^3 - 5X^2 + 6X - 1, \qquad P_{18} = X^3 - 6X^2 + 9X - 3.$$

The full integer rings are $\mathbb{Z}[U]$. Polynomial discriminants are 49 and 81. Translating the first by four gives Eisenstein at seven; the second is Eisenstein at three. At the only possible index prime, an element $(A + B\pi + C\pi^2)/p$ with a coefficient not divisible by p has a unique least numerator valuation among the distinct fractional parts $0, 1/3, 2/3$; division by p makes it nonintegral. Thus no denominator is missing, including at the wild prime three. In the P_7 field the complete physical finite alphabet for the other own and the mutual parameter is

$$0, 1, 2, 3, U, U^2 - 3U + 3;$$

in the P_{18} field it is

$$0, 1, 2, 3, U, U^2 - 4U + 4.$$

The last entries are native fourteen and nine, respectively. Indeed a finite label requires $\phi(m)/2$ to divide three, leaving the rational labels and $7, 9, 14, 18$. The two cubic fields have different discriminants, and contain no quadratic subfield. The double-angle identity gives the displayed polynomial expressions. Values greater than four are dotted, four is forbidden parallel, and zero is retained.

Write $V = a + bU + cU^2$. Rational root brackets and Lagrange interpolation of the three embedding values give the complete boxes

	a	b	c	trials
P_7	$[-4, 4]$	$[-9, 9]$	$[-3, 3]$	1197
P_{18}	$[-6, 6]$	$[-9, 9]$	$[-2, 2]$	1235.

Here is the coefficient proof, before any enumeration. The three roots, in increasing order, have disjoint rational brackets

P_7	$(.19, .20)$	$(1.55, 1.56)$	$(3.24, 3.25)$
P_{18}	$(.46, .47)$	$(1.65, 1.66)$	$(3.87, 3.88)$.

Multiplying the six endpoint values by 10^6 gives respectively

$$\begin{aligned} P_7: &\quad -33641,8000;\ 11375,-11584;\ -35776,15625,\\ P_{18}: &\quad -32264,8423;\ 7125,-19304;\ -70797,4672. \end{aligned}$$

The three sign changes exhaust the roots of each cubic. If r_i is one of them, the denominator in its Lagrange basis polynomial is $P'(r_i)=\prod_{j\neq i}(r_i-r_j)$. The bracket gaps give absolute lower bounds, in physical, middle, small order,

$$(5.1072, 2.268, 4.104), \qquad (7.514, 2.6078, 4.012).$$

Use $0<V_{\text{phys}}<10$ and $0<V_i<9/4$ at the other roots. The coefficient of X^2 in the interpolant is $\sum_i V_i/P'(r_i)$, that of X is $-\sum_i V_i\sum_{j\neq i} r_j/P'(r_i)$, and the constant is $\sum_i V_i\prod_{j\neq i} r_j/P'(r_i)$. Triangle inequalities give the following strict upper bounds on the three absolute terms:

	physical	middle	small	sum below
P_7, c	2	1	.55	4
P_7, b	3.45	3.43	2.64	10
P_7, a	.62	.65	2.78	5
P_{18}, c	1.34	.87	.57	3
P_{18}, b	2.84	3.76	3.11	10
P_{18}, a	1.04	1.58	3.62	7.

Each entry follows by positive rational multiplication with the denominator bound. For example the last constant terms are below $(9/4)3.25\cdot 1.56/4.104<2.78$ and $(9/4)3.88\cdot 1.66/4.012<3.62$. Since a,b,c are integers, these strict estimates give exactly the safe boxes above. Negative coefficients have not been discarded, and the boxes were obtained without looking at the successful strip models.

All field operations reduce modulo the cubic. If $s^2=R$ with R integral, s is integral. Rational R is handled separately: an odd-degree field contains its square root only if that root is rational. Otherwise put

$$A=\operatorname{Tr} R,\ B=((\operatorname{Tr} R)^2-\operatorname{Tr}(R^2))/2, \quad t=\operatorname{Tr} s,\ n=\operatorname{N} s.$$

Try both signed integers n with $n^2=\operatorname{N} R$ and all $-454\le t\le 454$; require

$$z=(t^2-A)/2\in\mathbb{Z}, \quad z^2-2nt=B, \qquad s=\frac{tR+n}{R+z}.$$

The identity follows from the characteristic polynomial of s; its denominator can vanish only for rational R. Verify $s^2=R$ and its physical positive sign. The trace range is complete: the finite anchor has $U<4$ and $V<10$, so $F(U)<128$, $F(V)<1340$ and the physical root has magnitude below 415. Each other root magnitude is below $324/17<20$. Hence $|\operatorname{Tr} s|<455$. Exact interval refinement terminates since no nonzero polynomial of degree at most two vanishes at a cubic generator. The zero element is first recognized by its three zero coefficients. For every other element, evaluate its degree-at-most-two representative on each separate isolating interval and bisect that interval until its sign is fixed. The interval width tends to zero, proving that every sign decision halts at the intended embedding.

All 2432 triples are visited. The recorded first failures are 1287 below the physical lower bound, 919 above its upper bound, 217 nonphysical-strip failures, seven field-square failures and two nonintegral long cycles. They sum to 2432. The separate combination replays every trial with the same implementation.

Field	(a, b, c)	First failure
K_7	$(0, 1, 0)$	long cycle not integral
	$(1, -2, 1), (1, -1, 1), (2, -2, 1), (3, -5, 2)$	not a field square
K_9	$(0, 1, 0)$	long cycle not integral
	$(1, -2, 1), (2, -2, 1), (3, -3, 1)$	not a field square

Table 2: All nine cubic strip models.

Corollary 21.2. *There is no compact arithmetic Coxeter polytope of dimension seventeen.*

Proof. The complete cover and row exclusions leave whole A_{17}. Its actual endpoint reduction gives the quadratic or cubic fields above. Proposition 21.1(i) excludes the former. In a cubic field designate a finite endpoint first, reversing chain labels if necessary; this assumes no chamber symmetry. Every other endpoint lies in the complete box of (ii), which has no survivor. No farthest forest remains. □

22 The complete rank-sixteen reduction

The rank-sixteen classification in Proposition 22.1 starts with all finite components and actual defect compatibility. Two dominant components each cost $> 1/2$. With none, the largest allowed rank-one debit still leaves ordinary cost $> 15/14$; the typed case costs $> .09 + 14/14$. The unique A dominant has rank at least ten and the B/D dominant rank at least eleven. Their small classical tails are checked at their actual ranks, including the direct D_{11} boundary. The D_{14} weak-tail payments give pure energies $> .92531$ or $> .94697$ before the disjoint spectator cost $> .09$ is added. Only then are the stronger tail floors used.

The conditioned $A_{11} > .652761$ component input used below is proved independently of either forest cover in Corollary C.14.1. Its proof reconstructs the no-helper reference directly; the weaker uniform 13/20 theorem would not imply that value.

Proposition 22.1 (Necessary cover in rank sixteen)**.** *The whole forest at the selected original farthest vertex is one of Table 3.*

Table 3: Necessary groups in rank sixteen.

Group	Component	Complete spectators
1	A_{10}	$I + (A_5$ or $D_5)$
2	A_{10}	N_7/O_7 plus A_4, B_4, D_4, F_4
3	A_{11}	$A_5, D_5, (A_4$ or $H_4) + A_1, A_3 + T_2, 2T_2 + A_1$
4	A_{11}	$I + T_4$
5	A_{11}	$O_7 + A_3$
6	A_{12}	A_4, H_4 or $2T_2$
7	A_{12}	$I + T_3$
8	A_{12}	N_7/O_7 plus T_2
9	A_{13}	$A_3,\ T_2 + A_1,\ I + T_2,\ O_7 + A_1$
10	A_{14}	$T_2,\ 2A_1,\ I + A_1,\ N_7/O_7$
11	A_{15}	ordinary A_1 or actual I
12	A_{16}	none
13	B_{14}	N_7/O_7

Group	Component	Complete spectators
14	B_{15}	ordinary A_1 or actual I
15	B_{16}	none
16	D_{13}	$I + T_2$ or $O_7 + A_1$
17	D_{14}	ordinary $I_2(m)$, $m \geq 4$, or N_7/O_7
18	D_{15}	ordinary A_1 or actual I
19	D_{16}	none

Proof of Proposition 22.1. We repeat the rank count from its beginning. If no dominant occurs, the possible I debit leaves fifteen ordinary ranks, costing $> 15/14$; a typed defect gives $> .09 + 14/14$. A J pair is more expensive than the two rank-over-fourteen baseline costs it replaces. Two dominants again each cost $> 1/2$. Thus there is exactly one. For A_7 the nine remaining ranks are too expensive. For A_8, A_9, even I leaves at least six ordinary ranks, giving $4/7 + 6/14 = 1$, strictly exceeded; the typed cost is at least $4/7 + 5/14 + .09 > 1$. Therefore $10 \leq r \leq 16$. For B, D, the $9/14$ rate leaves at most five spectator ranks, so $11 \leq r \leq 16$. All exceptional finite types were already included in the ordinary table.

At A_{10}, ordinary or J rank six closes with $4/7$. With I, enumerate the seven rank-five partitions displayed in the rank seventeen proof. In $4 + 1$ use $I + A_1 > .20204$ plus $T_4 > 2/7$; in $3 + 2$ use $I + T_3 > 2/7$ plus $T_2 > 1/7$; in $2 + 2 + 1$ use $I + A_1 > .20204$ plus $2/7$. The more reducible partitions are stronger. Only $I + A_5, I + D_5$ remain. With a typed defect, the rank-four partitions are $T_4, T_3 + A_1, 2T_2, T_2 + 2A_1, 4A_1$. The propagated costs exclude all but T_4, and actual K_7 excludes H_4. These give groups1 and2.

At $A_{11} > 4/7$, the complete ordinary rank-five list below the remaining $3/7$ budget is

$$A_5, D_5, (A_4 \text{ or } H_4) + A_1, A_3 + T_2, 2T_2 + A_1.$$

The omitted B_5 costs $> 3/7$; $B_4, D_4, F_4 + A_1 > 13/30$; $B_3, H_3 + T_2 > 3/7$; and the other three reducible partitions have costs $> .45$, $31/70$ and $.5$. This is group3. The J cost $.34 + 17/70$ also closes. For actual I use the spectator-safe conditioned component estimate $e_{A_{11}} > .652761$. Of the five rank-four partitions only T_4 remains: $I + T_3 + A_1$ costs $> 2/7 + .1$, $I + 2T_2 > .21 + 1/7$, and the other reducible cases are stronger. Both resulting sums with .652761 exceed one. Thus group4 retains all five irreducible T_4 types.

Under a typed defect the same component floor .652761 holds. At remaining ordinary rank three, $T_2 + A_1 > .2 + 2/15$ and $3A_1 > 2/5$ close. So do $B_3, H_3 > 2/7$; H_3 is also incompatible with K_7. Only A_3 is possible. For N_7 its two normalized endpoint duals have correlation $1/3$ and heights $> 1/\sqrt{5}$. Cauchy on their sum gives

$$e_{A_3} > \frac{4/5}{2 + 2/3} = 3/10,$$

which closes with $.652761 + .09$. Thus only $O_7 + A_3$ remains, giving group5.

At A_{12} use its spectator-safe floor $269/400 = .6725$. Ordinary rank four leaves only $A_4, H_4, 2T_2$: $B_4, D_4, F_4 > 1/3$, $T_3 + A_1 > .35$, $T_2 + 2A_1 > 12/35$, and $4A_1 > .4$. The J alternative is stronger. With I, rank three leaves only $I + T_3$, since $.6725 + .20204 + 1/7 > 1$ and $3A_1$ is stronger. Under a typed defect the component floor is .703101; two ordinary A_1 cost $> 4/15$ and close, whereas a single $T_2 > .2$ gives only .993101 and is honestly retained. These are groups6–8.

At $A_{13} > 5/7$, ordinary B_3, H_3, three A_1, and $J + A_1$ close. The retained ordinary alternatives are $A_3, T_2 + A_1$. With I, only $I + T_2$ remains since $I + 2A_1 > .30204$. With N_7, an $A_1 > .2$ closes;

with O_7, the weaker 2/15 does not. This is group9. At $A_{14} > .76$, the rank-two list is ordinary $T_2, 2A_1, I + A_1$, or either typed defect; J closes. At A_{15} its single spectator is ordinary A_1 or actual I; whole A_{16} has none. This gives groups10–12, with no endpoint or field-degree conclusion smuggled into the cover.

We now check the smaller classical ranks. For B_r and D_r with $r \geq 12$, the legal generic prefix and Lemma 20.3 give strong $6, \ldots, r-4$ and $6, \ldots, r-5$, respectively. At D_{11} the generic parameter $M = 5$ is not allowed. Instead inspect its actual deletion sites five and six: the retained pieces are $A_4 + D_6$ and $A_5 + D_5$. The connected white rank eleven exceeds eight, so the ordinary bulk bridge list still applies. Its D_5, D_6 fork-to-E_6, E_7 contacts remain included and strong; the foreign-far bounds use only indices five and six, within their geometric range $5 \leq i \leq 29$. The left finite-five projection uses the first six long roots, all present. Thus the only weak-five/six profile is again the literal far-three helper, and Lemma 20.3 excludes weak six. This proves strong six without applying the generic theorem outside its rank hypotheses.

Both initial-five branches therefore give the floors (20.4), beginning with E_6 at D_{11}. At classical rank eleven the cheapest rank-five remainders cost $> 23/70$, $.09 + 17/70$ or $5/14$, all closing with .69685. At twelve all rank-four remainders cost $> 2/7$; a typed defect plus T_2 costs $> .29$. These close with $E_7 > .73531$. At thirteen ordinary or J rank three closes with $E_8 > .76864$. Only $I + T_2$ or $O_7 + A_1$ can remain; $I + 2A_1$ and $N_7 + A_1$ are more expensive. In B_{13} the better actual floor $E_9 > .79805$ closes both retained cases. In D_{13} they are group16.

For B_{14} the actual B_3 site is eleven and the B_2 site twelve. The former's extra short-three extension is sealed F_4; own zero is affine and positive own has residual $1/4$ and ratio at least $1 + 1/\sqrt{2}$, hence is strong. At the latter the only extra weak role is excluded by Lemma 20.4. Full bridges have residual $1/(2i)$ and ratio $1/\sqrt{2i} + \sqrt{i}/2 > 100/61$; they admit no foreign arm at their internal new root. The zero/four/six/null cases and the directly strong native owns are those already listed in the rank seventeen proof. Thus $6, \ldots, 12$ are strong, giving $e_{B_{14}} > E_{12} > .86989$. Ordinary T_2, two A_1, $I + A_1$ and J close; either typed defect is retained as group13. The same local sites at B_{15}, B_{16} give strong through $r - 2$, but leave their stated groups14 and15.

For D_{14} the spectator rank is two and always costs $> .09$. Its bulk already gives strong $6, \ldots, 9$ independently of the new tail sites. At site ten the right D_4 has long-leaf-three and fork-three extensions. Their scalars differ: the former has $\sigma = \gamma + 1/2$; the latter has $\delta = 1/2 - Q$, $\sigma = \gamma + 1/4$. At site eleven, $D_3 = A_3$ has middle-three, end-three, end-four and end-five extensions. Middle-three and end-four are affine or directly strong; the sealed H_4 end-five is strong even at own zero. The end-three full bridge is A_{14} with both arms at least three, so admits no finite foreign arm. Its residual is at most $15/88$, and its positive-own numerator $> .75$ makes it strong; own zero is the affine cycle. Full fork bridges are D_{14} and are similarly sealed. All other positive foreign, isolated and finite-own roles are the generic companion list at these actual local ranks.

The only extra weak cases are therefore the literal fork helpers; Lemma 20.3 leaves their golden owns. Use their actual positive H_4 projections with the same α, δ, κ as in the rank seventeen proof. For $K = i - 3$ the previous strong knot has available coefficient $> 1/6$ in both initial-five branches. One pure-D reference gives

$$e_D > E_K + (G - f_K)/3 - (i - K)/36.$$

Here the complete new rows are

	i	K	$G >$	$f_K <$	$e_D >$
D_4	10	7	3.035	2.215	.92531
D_3	11	8	3.183	2.398	.94697.

The seals are $3.126^2 < 10 - 25/112$ and $3.274^2 < 11 - 125/448$. Each pure bound plus the disjoint actual spectator cost $> .09$ exceeds one. Hence both weak roles are impossible, and $6, \dots, 11$ are strong. This gives $e_{D_{14}} > E_{11} > .84816$. Two A_1, $I + A_1$, and J close. Ordinary A_2 also closes, since $.84816 + 319/2000 = 1.00766$. Precisely an ordinary $I_2(m)$, $m \geq 4$, or either typed defect remains as group17. At D_{15} the single ordinary A_1 or actual I is retained, and whole D_{16} has none.

All small types, ranks, integer partitions and actual defect choices were included before this final table. In particular no rank seventeen forest was decremented and no farthest property was transferred to a face. □

22.1 A small component calculation

Lemma 22.2. *At the original farthest vertex of any compact Coxeter polytope, a whole A_3 component has energy $> 13/50$, with arbitrary finite spectators. Arithmeticity is not needed.*

Proof. The inverse unit Gram and normalized old duals are

$$C = \begin{pmatrix} 3/2 & 1 & 1/2 \\ 1 & 2 & 1 \\ 1/2 & 1 & 3/2 \end{pmatrix}, \qquad d_i = C_{ii}, \quad u_i = \omega_i/\sqrt{d_i}.$$

Suppose $e = \|x_{A_3}\|^2 \leq 13/50$. Put $\tau_i = \tanh(L_i/2) \leq x \cdot u_i$ and $H(t) = 2t/(1+t^2)$. The logarithm of H is concave on $(0,1)$: its second derivative is $-1/t^2 - 2(1-t^2)/(1+t^2)^2 < 0$. For a pair of normalized duals with correlation k, Cauchy gives $\tau_i + \tau_j \leq \sqrt{2e(1+k)}$; hence

$$\tanh L_i \tanh L_j \leq H\big(\sqrt{e(1+k)/2}\big)^2.$$

At an adjacent A_3 pair $k = 1/\sqrt{3}$, the right side is at most $5200(1+k)/(113+13k)^2 < k$. The cleared positive difference is $(22876k - 12662)/3 > 0$, using $k > 4/7$. The companion's actual distinct-opposite tangent-product inequality requires the opposite inequality if the two walls are distinct. Thus adjacent directions share their actual opposite, and connectivity gives one wall c closing all three edges.

The individual bound is

$$\tanh^2 L_i \leq \frac{4e}{(1+e)^2} \leq r := 2600/3969.$$

Write $c = Ap - v_A - v_S$, $h = A^2 > 0$, $v_A = \sum \beta_i \omega_i$, and $\beta_i = -(c, n_i) \geq 0$. Put $q = \beta^T C \beta$, $Q = \|v_S\|^2$, so $h = q + Q - 1$. For $v_i = (C\beta)_i/\sqrt{d_i} > 0$ we have

$$q - 1 \leq h \leq r \min_i v_i^2. \tag{22.1}$$

The companion's reciprocal-shadow and orthogonal-opposite lemmas say that every positive whole spectator contribution is at least $1/4$ and strictly below h. In particular $h < 1/4$ forces $Q = 0$.

Two adjacent positive β_i would give, at the finite endpoint of the third edge, a white triangle with all three contacts at least $1/2$. Its Gram is nonpositive, affine at equality, impossible. Thus there are at most two contacts, and two can only be at endpoints. For endpoint values $a, b > 0$,

$$q = \tfrac{3}{2}(a^2 + b^2) + ab, \qquad v_2^2 = (a+b)^2/2.$$

If both are $1/2$, the original principal is affine with a nonzero future feasible null sum. Otherwise the native gap gives $a, b \geq 1/2$ and one at least $1/\sqrt{2}$. Consequently

$$q - r v_2^2 \geq \frac{3(3-r)}{8} + \frac{(1-r)\sqrt{2}}{4} = \frac{27921 + 2738\sqrt{2}}{31752} > 1,$$

using $\sqrt{2} > 7/5$. This contradicts (22.1). There is at least one contact, since each $v_i > 0$.

For one endpoint contact β, $q = 3\beta^2/2$ and $\min v_i^2 = q/9$. Equation (22.1) gives

$$q \le 35721/33121, \quad \beta^2 \le 23814/33121 < 3/4, \quad h \le 2600/33121 < 1/4.$$

Only native three, four or five remains, and each has $q < 1$. For five use $\cos^2(\pi/5) < 2/3$, following from $\sqrt{5} < 7/3$. Then $Q = 0$ and $h = q - 1 < 0$, impossible.

For one middle contact, $q = 2\beta^2$, $\min v_i^2 = q/3$. Thus $q \le 11907/9307$ and $\beta^2 \le 11907/18614 < 16/25 < \cos^2(\pi/5)$. Only native three or four remains, so first conclude $q \le 1$, and only then strengthen the height bound to $h \le r/3 = 2600/11907 < 1/4$. Again $Q = 0$ and $h = q - 1 \le 0$ is impossible. All contact and foreign-spectator alternatives have been exhausted. □

Proposition 22.3 (All rows except whole D_{16})**.** *Groups1–18 are impossible at that same selected original vertex.*

Proof. Groups1–18 are excluded by the complete printed dispatch in Section C.14.3. Its separate actual rank-one branches exhaust the same opposite wall, including the boundary $h = 1/2$. Every overlapping endpoint/head/foot pressure is paid within its local proof before disjoint whole-component costs are added. The universal $A_{12} > 5/7$ estimate supplies both ordinary and typed consumers. Only group19, whole D_{16}, remains for the next section. □

23 The last actual diagram

Proposition 23.1 (Whole-D_{16} reduction)**.** *If group19 occurs, the actual field is quadratic. The actual long head and both genuine signed forks are isolated and dotted. There are integral generators $U, F, G \in \mathcal{O}_K$ satisfying*

$$4 < U < 98/9, \quad 0 < U' < 2, \qquad 4 < F, G < 121/2, \quad 0 < F', G' < 1/2, \tag{23.1}$$

$$UF/4,\ UG/4,\ FG/4 \in \mathcal{O}_K. \tag{23.2}$$

Their raw time squares are

$$T_H = U(U-2), \qquad T_F = 2F(2F-1), \qquad T_G = 2G(2G-1), \tag{23.3}$$

and every pair of time products is a square in K.

We give its field and endpoint reduction below, after specifying the initial geometric inventory and two scalar estimates. The inventory must be proved at actual rank sixteen; it is not supplied by a higher-rank head theorem or by the dyadic calculation.

The original coordinates satisfy

$$x_1 > \cdots > x_{15} > |x_{16}|, \quad P_j = e_1 + \cdots + e_j, \quad S_j = x \cdot P_j, \quad f_j = \sqrt{j} - .61/\sqrt{2}.$$

The same reduction proves strong5 through 14 and one positive prefix reference with

$$E_{14} > .9084, \quad b < .361, \quad \beta_5 > .14, \quad \beta_{14} > .13, \quad f_{14} < 3.311, \qquad S_6 > 2.01815. \tag{23.4}$$

Here b is its first-five mean. These coefficients are spent individually below. The head raw root has square $2U$, spatial part Ue_1; the fork raw roots have squares $2F, 2G$ and spatial parts $(F/2)(P_{15}+e_{16})$, $(G/2)(P_{15} - e_{16})$. They remain two distinct directions even if $F = G$.

The remainder of this exclusion is given in full. No D16 census is used. Every original cycle remains mandatory even when only necessary consequences of it are needed.

Lemma 23.2 (Initial whole-D_{16} inventory)**.** *At the same original whole-D_{16} vertex, knots five through fourteen are .61-strong, and the positive reference has* (23.4)*. The actual long head is isolated, with finite own label at least seven or dotted own. Every actual fork has edge cosh less than twelve and is either isolated dotted or has exactly one simple-three contact at an endpoint of its retained A_{15} path, with positive own contact. Own zero and parallel fork roles are excluded. If both forks are isolated, the actual field has degree two or three.*

The proof below first establishes the actual-rank coarse bounds and head isolation, then closes the late spine and fork inventories. This order keeps every pressure available before it is spent.

Lemma 23.3 (Two exact terminal estimates)**.** *For $0 < u < 1$, put*

$$H(u) = \frac{\sqrt{2-u} - \sqrt{u(1-u)}}{\sqrt{2}}, \qquad R(u) = 1 - u + u^2/2.$$

Then $.8H(u) - .16R(u) > .295$. Also a degree-three algebraic integer with conjugates $A, a, b > 0$, $a, b < 1/2$ and norm eight has $A > 42$.

Proof. For the first assertion let $c = .681$, $t = 2 - c^2 = 1.536239$. The radical difference is positive, and $t - u^2 > 0$. It suffices to prove

$$P(u) = (t - u^2)^2 - 4u(1-u)(2-u) > 0.$$

With $r = u - .7$ the exact expansion is

$$P = r^2(r - .6)^2 + 3.107522r^2 - .0094692r + .002616045121.$$

The last three terms exceed $3r^2 - .01|r| + .0026$. As $.01|r| \le r^2 + .000025$, this is positive. Thus $H(u) > c/\sqrt{2} > .4815351$ on the entire interval.

Moreover R decreases, and H decreases on $(0, .65]$. Below $1/2$ the derivative sign is immediate; between $1/2$ and .65 it follows from

$$\frac{2u-1}{\sqrt{u(1-u)}} \le \frac{.3}{\sqrt{.2275}} < \frac{1}{\sqrt{1.5}} \le \frac{1}{\sqrt{2-u}},$$

where $.09 \cdot 1.5 < .2275$ proves the strict inequality. The following rational radical seals give lower bounds for $H(v)$:

v	$\sqrt{2-v} >$	$\sqrt{v(1-v)} \le$	$H(v) >$
.25	1.3228	.4331	.62
.4	1.2649	.4899	.54
.5	1.2247	.5	.51
.55	1.2041	.4975	.499
.6	1.1832	.4899	.4902
.625	1.1726	.48413	.4867
.65	1.16189	.47697	.4842.

Square the first two columns and use $\sqrt{2} < 1.4143$ to obtain each last entry. Taking H at the right endpoint and R at the left endpoint gives the complete cover

$(0, .25]$	.336	$[.25, .4]$	.307
$[.4, .5]$	.2992	$[.5, .55]$	.2992
$[.55, .6]$	.29596	$[.6, .625]$	.29656
$[.625, .65]$	.29611	$[.65, 1)$	.29542808.

These are strict lower bounds for $.8H - .16R$; the last uses the global radical bound and $R(.65) = .56125$. Every endpoint is covered.

For the second assertion suppose $A \le 42$. Norm eight and $a, b < 1/2$ give $A > 32$ and integer trace $T \in \{33, \dots, 42\}$. Its minimal polynomial is $X^3 - TX^2 + BX - 8$, with integer B. Since $A > T - 1$, AM–GM gives

$$B \ge 2\sqrt{8A} + 8/A > 2\sqrt{8(T-1)} + 8/T.$$

Evaluating the polynomial at $1/2$, between both small roots and the large root, gives $B < T/2 + 63/4$. For even traces the largest permitted B, and the positive seal $32(T-1) - B^2$, are

T	34	36	38	40	42
$B_{\max}$	32	33	34	35	36
$32(T-1) - B_{\max}^2$	32	31	28	23	16.

They contradict the lower bound. At $T = 33$, $B_{\max} = 32$ and the same lower bound is already strictly larger than 32. At odd traces $35, 37, 39$, the permitted maxima are $33, 34, 35$; the exact integers

$$32(T-1)T^2 - (TB_{\max} - 8)^2 = 17191, 14588, 8087$$

are positive and give the same contradiction after squaring. Only $T = 41$ remains. The two bounds force $B = 36$, but $X^3 - 41X^2 + 36X - 8$ has discriminant -2704, incompatible with three real conjugates. Therefore $A > 42$. □

Proof of Lemma 23.2. All coordinates, caps and comparisons in this proof use the original whole-D_{16} vertex. Define its unit-root dual vectors by

$$\Omega_i = \sqrt{2}P_i \quad (1 \le i \le 14), \qquad \Omega_{15} = (P_{15} - e_{16})/\sqrt{2}, \quad \Omega_{16} = (P_{15} + e_{16})/\sqrt{2}, \qquad h_i = x \cdot \Omega_i.$$

Thus both genuine signed fork directions are retained.

The coordinate bound before any terminal classification. The legal rank sixteen bulk and Lemma 20.3 already give strong knots six through eleven. We first use only those knots. The quantities E_J are the explicit scalar sums defined above; in particular the identity

$$\Delta := E_{11} - E_6 = \sum_{j=6}^{10} (\sqrt{j+1} - \sqrt{j})^2 > .15106$$

is an analytic inequality, not an assertion that the actual vector already pays the strong-five reference. Here $E_{11} > .84816$ and

$$E_6 = 12.03721 - .122\sqrt{10} - 2\sqrt{30} < .6971;$$

the rational seals $\sqrt{10} > 3.16227$ and $\sqrt{30} > 5.47722$ prove the latter bound.

If $x_1 \ge .714$, use the positive reference through $(1, .714), (6, 2.01815)$ and then $f_7, \dots, f_{11}$. Its initial slopes are $.714, .26083$, followed by a smaller positive slope and the ordinary decreasing increments. The first increment after six is larger than $\sqrt{7} - \sqrt{6}$, because $f_6 > 2.01815$. Its square is therefore greater than

$$.714^2 + 1.30415^2/5 + \Delta > 1.0010174445,$$

contrary to $\|x\|^2 < 1$. Thus $x_1 < .714$. Actual head comparison then gives $C_H < 77/25$, since $26/51 > .714^2$. Neither the coordinate nor the head bound has used strong five, a weak-five coordinate floor, or a head type.

The literal weak-five seal. At index five the generic finite-white list is the pure far helper $c = Ap - \gamma\Omega_5 - \Omega_1/2$ if the knot is weak. The first-six roots are genuine long roots, and the raw inverse entries are $5, 1, 1$. Thus $q = 2\gamma$ is integral, $202/125 < q < 12/5$, and every other conjugate satisfies

$$5q_\sigma^2 + 2q_\sigma - 1 < 0, \qquad -.69 < q_\sigma < .29.$$

Degree at least four has nonzero norm $< 2.4(.69)^3 < 1$. In degree three the norm is a unit. A positive small conjugate would give magnitude $< 2.4(.29)(.69) < 1$; both are therefore negative, say $-a, -b$, with $ab = 1/q$ and $a, b < .69$. Then $q > 1/(.69)^2$ and the integer trace lies strictly between zero and two, hence is one. Thus $a + b = q - 1 < 1.38$, $q < 2.38$, and the second integer coefficient $1/q - q(q-1)$ lies in $(-3, 0)$. The two cubics have discriminants $-44, -31$, impossible. In degree two the trace is one or two. Trace two gives norm in $(-.96, .621)$; trace one also forces $q < 1.69$, leaving only norm -1 and $q = \varphi$. The rational $q = 2$ is parallel. The actual connected principal is indefinite over $\mathbb{Q}(\varphi)$, so a weak five forces the actual field to be exactly that field.

It also contradicts the coordinate bound already proved. Here the selected old roots are n_5, n_1, n_2. Their correction $w = \gamma n_5 + (2n_1 + n_2)/3$ has positive coefficients and $w^2 = \gamma^2 + 1/3$; the full selected diagram, including c, is H_4. Thus $1 - w^2 = (5 - 3\varphi)/12$. The positive actual left-H_4 projection gives

$$x \cdot \{\gamma(\Omega_4 + \Omega_6)/2 + \Omega_3/6\} \geq \frac{A}{1 + \sqrt{\delta/(A^2 + \delta)}}, \quad A^2 = (4 + 7\varphi)/2, \quad \delta = (5 - 3\varphi)/12.$$

All coefficients in the finite projection are positive, so every unselected original wall remains feasible. Long-arm concavity gives $h_4 + h_6 \leq 2h_5$ and $h_3 \leq 3h_1$. Weakness of five therefore forces $x_1 > \sqrt{2}\{A - \sqrt{\delta} - \gamma(\sqrt{10} - .61)\} > .8$. For a rational seal use $A > 2.768$, $\sqrt{\delta} < .115$, $\gamma < .81$, $\sqrt{10} < 3.163$, and $\sqrt{2} > 1.4$. This contradicts $x_1 < .714$. Consequently five is strong before any head case is excluded.

The coarse fork bound after strong five. Knots five through eleven are now strong. Take their positive concave reference through $f_5, \ldots, f_{11}$, with squared norm $E_{11} > .84816$, endpoint $f_{11} < 2.886$ and available P_{11} coefficient $\sqrt{11} - \sqrt{10} > .15$. For either actual fork put $T_s = x \cdot (P_{15} + se_{16})$. The single tail graft $.15(P_{15} + se_{16} - P_{11})$ has residual square five and zero cross with that reference. Hence

$$\|x\|^2 > .84816 + .3(T_s - 2.886) - .1125.$$

If its edge cosh were at least seventeen, then $T_s \geq 4\sqrt{8/9} > 3.77$, giving $1.00086 > 1$. Thus both actual fork coshes are below seventeen. This uses the strong-five reference only after its hypothesis has been proved, with no head or fork classification assumed.

The exact head and fork white lists. Releasing the long root n_1 retains whole D_{15}. Finite classification allows only an isolated new head or a simple-three contact at its long leaf n_2. A fork attachment would require a large exceptional diagram, an internal contact a second fork, and two contacts a cycle; weighted alternatives are not finite. For an isolated head, $c_H = A_H p - \gamma_H \Omega_1$ and $A_H^2 = 2\gamma_H^2 - 1$. Own zero, three and four have nonpositive time; own six has the actual affine $6, 3$ chain, and parallel has the positive null $c_H + n_1$. The remaining owns are five, finite at least seven, or dotted. With $U_H = 4\gamma_H^2$,

$$K = \mathbb{Q}(U_H), \quad 2 < U_H < 11858/625, \qquad 0 < (U_H)_\sigma < 2.$$

The reflection-defined raw cap has square $2U_H$, own product $-U_H$, spatial part $U_H e_1$ and nonzero time. Its indefinite principal with the rational old system proves actual-field descent and all the stated nonphysical inequalities.

For a nonisolated head write

$$c_H = A_H p - \gamma_H \Omega_1 - \Omega_2/2, \qquad t_H = 2\gamma_H, \quad A_H^2 = t_H(t_H+2)/2, \quad C_H = t_H + 1.$$

The helper permits square-two normalization; its old path cycle makes t_H integral and gives $K = \mathbb{Q}(t_H)$. Strict nonphysical positivity gives $-2 < (t_H)_\sigma < 0$. Own zero has zero time; own three is an actual affine triangle; parallel is a feasible null.

For either fork f, with the other called k, deleting f retains the actual A_{15} path $n_1, \dots, n_{14}, n_k$. The complete finite one-node extensions are isolation, endpoint three, next-to-end three, or endpoint four. Two contacts make a cycle, and no exceptional finite diagram contains this path. There are no foreign components. For positive own put $t = 2\gamma$ at a white-three contact, $a = 2\sqrt{2}\gamma$ at white four. The following table gives UNIT time squares. For a white-three cap the helper permits a raw K-root of norm two and time square $2A^2$. For an isolated fork the reflection-defined raw norm is $2U_f$ and its time square is $2U_f(2U_f - 1)$. For a white-four cap use a raw K-root of norm four, with white product -2, own product $-a$ and time square $4A^2$. Indeed, if a K-root r makes angle $\pi/4$ with an old raw root α of norm two, then $(r,\alpha)^2 = (r,r)$; rescaling by $2/|(r,\alpha)|$ gives this norm-four representative without assuming $\sqrt{2} \in K$.

white contact	A^2	actual edge cosh
none	$2U_f - 1,\ U_f = 4\gamma^2$	$\sqrt{2U_f}$
3 at n_1	$(4t^2 + t - 1)/2$	$(8t+1)/\sqrt{17}$
3 at n_2, n_{14}	$(4t^2 + jt + j - 2)/2$	$4t + j/2,\ j = 2, 14$
3 at n_k	$(4t^2 + 7t + 2)/2$	$(8t+7)/\sqrt{17}$
4 at n_1	$(2a^2 + a)/2$	$4a + 1$
4 at n_k	$(2a^2 + 7a + 6)/2$	$4a + 7$.

The white-three normalization gives integral t and $K = \mathbb{Q}(t)$. White four gives integral a, integral $a^2/2$, and $K = \mathbb{Q}(a)$. These are original cycle statements. An isolated fork has $K = \mathbb{Q}(U_f)$, all other $0 < (U_f)_\sigma < 1/2$, and raw time square $2U_f(2U_f - 1)$. These follow from its full indefinite old-system/cap principal.

Own zero has negative or zero time in the isolated, far-three, far-next-three and far-four cases. Its remaining cases have the following positive affine nulls:

3 at n_{14}	$c + n_{13} + n_f + n_k + 2n_{14}$
3 at n_k	$2c + 4n_k + 6n_{14} + 3n_f + 5n_{13} + 4n_{12} + 3n_{11} + 2n_{10} + n_9$
4 at n_k	$\sqrt{2}c + 2n_k + 2n_{14} + n_f + n_{13}$.

Their products on the selected roots vanish. Their time is positive, and every other original wall has nonpositive product term by term. Parallel always gives $c + n_f$. This excludes own zero and parallel without a head assumption.

Every nonisolated dotted head is impossible. Set $s = t_H + 1 = C_H$. Physically $3 < s < 77/25$; all its other conjugates lie in $(-1, 1)$. The nonzero integral element $s^2(s^2 - 1)$ has physical value < 81 and other conjugate magnitudes at most $1/4$, excluding degree at least five by its norm. Degree one is excluded by Meyer. At degree two the integer trace is three or four, giving norms $s(3-s) \in (-.2464, 0)$ or $s(4-s) \in (2.8336, 3)$, both impossible.

At degree four its absolute norm is one, so $s, s-1, s+1$ are units. For the minimal polynomial $P = X^4 - TX^3 + BX^2 - CX + D$, $D = \pm 1$, the real strips force $P(1) = -1$, $P(-1) = 1$. Hence $B = -1 - D$, $C = 1 - T$. For $D = 1$, $T = s - (s^2 + s - 1)/(s^3 - s)$ lies strictly between two and three:

the fraction is below one and above $1/s > 25/77$. For $D = -1$, $T = s + (s^2 - s - 1)/(s^3 - s)$ lies strictly between three and four, since its added fraction is in $(0, 1/2)$. Both contradict integrality.

At degree three write $P = X^3 - TX^2 + BX - C$ and put $a = -P(1) > 0$, $b = -P(-1) > 0$. The integer norm satisfies $C^2ab < 81/16 < 6$. For $|C| = 1$ the complete equal-parity pairs are $(1,1), (1,3), (3,1), (1,5), (5,1), (2,2)$; for $|C| = 2$ only $(1,1)$ remains. Here $T = (a+b)/2 - C$, $B = (b-a)/2 - 1$. The complete seals are

C	(a,b)	(T,B)	seal
1	$(1,1)$	$(0,-1)$	$P(3) = 23$
1	$(1,3)$	$(1,0)$	$P(3) = 17$
1	$(3,1)$	$(1,-2)$	$P(3) = 11$
1	$(1,5)$	$(2,1)$	$P(3) = 11$
1	$(5,1)$	$(2,-3)$	$\operatorname{disc} P = -23$
1	$(2,2)$	$(1,-1)$	$P(3) = 14$
-1	$(1,1)$	$(2,-1)$	$P(3) = 7$
-1	$(1,3)$	$(3,0)$	$P(3) = 1$
-1	$(3,1)$	$(3,-2)$	$P(77/25) = -4.401088$
-1	$(1,5)$	$(4,1)$	$P(77/25) = -4.647488$
-1	$(5,1)$	$(4,-3)$	$P(77/25) = -16.967488$
-1	$(2,2)$	$(3,-1)$	$P(77/25) = -1.321088$
2	$(1,1)$	$(-1,-1)$	$P(3) = 31$
-2	$(1,1)$	$(3,-1)$	$P(77/25) = -.321088.$

Every displayed polynomial has derivative at least $3X^2 - 8X - 3 = (3X+1)(X-3) > 0$ on the interval. The sign seals or the negative discriminant exclude all fourteen. This exhausts every dotted nonisolated head.

For finite own, the same norm bounds its actual degree by four. The complete $\phi(2m)/2 \le 4$ list, after own three, is $m = 4, 5, 6, 7, 8, 9, 10, 12, 15$. At $7, 8, 10, 12, 15$ the embeddings with cosine multipliers $3, 3, 3, 5, 7$ give a positive nonphysical t_H, forbidden. Only the finite heads $4, 5, 6, 9$ remain.

The four finite nonisolated heads. Their parameters are $t_H = \sqrt{2}, \varphi, \sqrt{3}, 2\cos(\pi/9)$ and their actual fields are generated by these parameters. The first-five block of the strong reference has mean $b < .361$. An actual floor $x_1 > X$ permits rebuilding this block at fixed sum, gaining $\frac{5}{4}(X - .361)^2$, with all its new slopes positive. For $X = .643, .681, .695$ the remaining first-five slope exceeds .27, larger than the next ordinary slope $< .214$. The original individual P_{11} price still funds the signed tail. The resulting complete fork bounds are

head	X	forbidden fork $C \ge$	single norm lower
4	.643	7	1.001265
6	.681	6	1.01186
9	.695	6	1.023305.

The head squares satisfy respectively $x_1^2 \ge \sqrt{2}-1$, $x_1^2 \ge 2\sqrt{3}-3$, and $x_1^2 \ge t_H/(t_H+2)$ with $t_H > 1.87$. The signed tail demands used are > 3.44 at $C = 7$ and > 3.38 at $C = 6$. For head five use $x_1 > .668$ and the reference through $1, 6, 7, \ldots, 11$, whose energy is $> .668^2 + 1.35015^2/5 + .84816 - .6971 = .9618650045$. At $C = 49/8$, $T_s > 3.39$, giving norm > 1.0005650045. Thus both fork coshes are below seven, $49/8$, six, six, in the four respective cases.

No fork can be isolated in these cases. Its original cycle with the nonisolated head gives, for integral g, L,

$$L + g(t_H + 2) = (15t_H^2 + 28t_H - 4)U_f/4 - 2t_H(t_H + 2).$$

At heads $5, 6, 9$, t_H is a dyadic unit, forcing $U_f/4$ integral. The positive norm bounds are $2401/4096$, $9/16$ and $(18/4)(1/8)^2$, each below one. At head four normalize $v(\sqrt{2}) = 1$. The displayed coefficient has valuation two, so only $U_f/2$ is integral. For $W = U_f/2 \in \mathbb{Z}[\sqrt{2}]$, its strips are $W < 49/4$, $0 < W' < 1/4$. The radical coefficient is $1, \ldots, 4$; only $W = 3 + 2\sqrt{2}$ fits, giving $U_f = 6 + 4\sqrt{2}$. The original raw time product has square $1256 + 888\sqrt{2}$, whose norm 448 is nonsquare.

All remaining forks must be endpoint-three. Indeed near-next-three and near-four have cosh at least eleven and greater than twelve. Far-next-three has $t < 3/2$, leaving only own three or own four. Own three gives the actual fifteen-node all-three cycle $c, n_2, n_3, \ldots, n_{14}, n_f, c$, where f is the released fork. The sum $c + n_2 + \cdots + n_{14} + n_f$ is null: its positive all-ones kernel annihilates every selected unit-Gram row. Its time is the positive time of c, and acuteness makes its product with every other original wall nonpositive. It is therefore a feasible ideal ray of the whole chamber, contradicting compactness. Own four has a conjugate violating its negative time strip. Far-four has $a < 3/2$, leaving only $a = \sqrt{2}$; its time numerator $2a^2 + a$ is positive at $-\sqrt{2}$, also forbidden. Own zero and parallel were already excluded.

For an endpoint-three cap its raw time square is $M_f(t) = 4t^2 + t - 1$ far or $M_n(t) = 4t^2 + 7t + 2$ near, with $K = \mathbb{Q}(t)$ and integral $t \geq 1$. Writing $z = t + 1$ near, every nonphysical conjugate lies in $(-13/20, 2/5)$ far or $(-2/5, 13/20)$ near. The full quadratic integer strips give

head	far t	near z	time-product norms
4	$1 + \sqrt{2}$	$2 + \sqrt{2}$	72, 36
5	φ, φ^2	φ^2	5, 5, 95
6	none	none	

At head four the physical bounds are $t < 7/2$, $z < 15/4$; the positive integral radical coefficient is one, giving exactly the displayed values. The square-norm near product $102 + 72\sqrt{2}$ would require rational square part 54 or 48. At head five the physical parameters are below $10/3$; the positive φ coefficient is one, giving far integer coefficients zero or one and near coefficient one. At head six the only strip value is $2 + \sqrt{3}$, above its respective bounds three and $13/4$. No quadratic endpoint survives.

At head nine the actual degree is three. A far parameter has $1 \leq t < 3$ and norm magnitude one; a positive small conjugate would make the norm $< 3(.4)(.65) < 1$. Thus both are negative, $-a, -b$, with $ab = 1/t$, $a, b < .65$, so $t > 400/169$. Its integer trace is strictly greater than $400/169 - 1.3 > 1$ and less than $3 - 2/\sqrt{3} < 2$, impossible. Near, $2 \leq z < 13/4$ has unit norm and both small conjugates positive, since a negative one would give norm magnitude $< (13/4)(.4)(.65) < 1$. The integer trace is four. Then $z > 2.7$ from $a + b < 1.3$, and $z < 3$ because $z + 2/\sqrt{z} > 4$ at $z \geq 3$. The second integer coefficient $1/z + 4z - z^2$ decreases on $(2.7, 3)$ and lies strictly between $10/3$ and $10477/2700 < 4$. This final contradiction excludes head nine. Both actual fork opposites must exist, so the long head is isolated in every case.

White-four, next-three and finite isolated forks. The head is now isolated; its weak upper strip $U_H < 11858/625$, $0 < (U_H)_\sigma < 2$ is retained. A far white-four fork has integral a, integral $a^2/2$, $a < 4$, and every other $-.5 < a_\sigma < 0$. Degree at least three has nonzero norm magnitude < 1. In degree two the magnitude is below two, but integrality of $a^2/2$ forces its norm even. Degree one is impossible. Near, $z = a + 2$ has $z < 9/2$, all other $0 < z_\sigma < 1/2$, and integral $z^2/2$. Degree at least four is excluded by norm. At degree three its positive norm is one, incompatible with integrality of

$z^2/2$. At degree two that integrality forces norm two. Then $z > 4$ and its integer trace $z + 2/z$ lies in $(9/2, 5)$. This excludes both white-four roles over every actual field.

For next-three at n_2, $1 \le t < 4$ and all other conjugates lie in $(-1/2, 0)$. Degree at least three is excluded by norm; degree two has norm -1 and trace two or three. Thus $t = 1 + \sqrt{2}$ or $(3 + \sqrt{13})/2$. For next-three at n_{14} put $z = t + 2$. Then $3 \le z < 9/2$ and all other conjugates lie in $(0, 1/2)$. Degree at least four is excluded by norm. Degree three would have norm one, $z > 4$, and integer trace strictly between five and $11/2$. At degree two norm two gives the same forbidden trace interval as white four; norm one gives trace four, $z = 2 + \sqrt{3}$, $t = \sqrt{3}$.

For these three next-three parameters the actual head path cycle and two-cycle give integral $g = -H \cdot C$, $L = g^2/U_H$, and

$$\begin{aligned} L + g(t+2) &= (15t^2 + 4t - 4)U_H/4 - 2M \qquad \text{far}, \\ L + g(t+2) &= (15t^2 + 52t + 44)U_H/4 - 2M \quad \text{near}. \end{aligned}$$

Each listed t is a dyadic unit, so $U_H/4$ is integral. Writing $W = U_H/4$, its full strips are $W < 11858/2500$, $0 < W' < 1/2$. Use the full integral bases $1, \sqrt{2}$ and $1, \sqrt{3}$ in the first and third fields, and $1, (1 + \sqrt{13})/2$ in the second. Writing $W = a + b\eta$ in the appropriate basis, the positive integer coefficient b is one in all three cases: $W - W' < 11858/2500$, whereas $b \ge 2$ would give respectively at least $4\sqrt{2}$, $2\sqrt{13}$ or $4\sqrt{3}$, all larger than this bound. In particular the coefficient of $\sqrt{13}$ itself is $1/2$, not one. The first two conjugate strips require $\sqrt{2} < a < \sqrt{2} + 1/2$ or $(\sqrt{13} - 1)/2 < a < \sqrt{13}/2$, neither admitting an integer. The last gives $a = 2$, hence $W = 2 + \sqrt{3}$ and $U_H = 8 + 4\sqrt{3}$. Then $C_H = 1 + \sqrt{3}$, $x_1 > .681$. The near-next fork has $C = 7 + 4\sqrt{3} > 13$, hence $T_s > 3.7$. The reference through $1, 6, 7, \ldots, 11$ has energy $> .681^2 + 1.33715^2/5 + .84816 - .6971 = .9724150245$; its legal signed tail graft gives $\|x\|^2 > 1.1041150245$. No next-three role survives.

For an isolated finite fork $F = 4\cos^2(\pi/m)$, with actual degree d, its trace is $2d + \mu(m)$. The nonphysical strip $F_\sigma < 1/2$ and physical $F < 4$ give $3d < 7 - 2\mu(m)$, hence $d \le 2$. The complete quadratic native list and the rational-field exclusion leave only $F = \varphi^2$ or $2 + \sqrt{3}$; native eight and ten have conjugates above $1/2$. Both retained parameters are units. The actual isolated head/fork cycle satisfies

$$L + g = (15U_H F - 32F - 8U_H + 16)/4,$$

so $U_H/4$ is integral. The full integer strips give $U_H = 4\varphi^2$ or $8 + 4\sqrt{3}$, respectively. For the first, $x_1 > .625$ and the positive left-H_4 projection onto c_f, n_f, n_{14}, n_{13} gives

$$V = P_{12} + (e_{13} + e_{14} + e_{15} - se_{16})/2, \quad V^2 = 13, \quad x \cdot V > \sqrt{8\varphi} - (2\varphi - 3) > 3.36.$$

Its old correction is $(\varphi/2)(3n_f/2 + n_{14} + n_{13}/2)$, with positive coefficients and residual $(5 - 3\varphi)/8$. Thus the finite projection is future and feasible against every original wall. The reference through $(1, .625), (6, 2.01815)$ and the later knots has energy $> .9298583845$. Its $.15P_{11}$ graft has residual square two and gives norm > 1.0270583845. For native twelve the same original head gives $x_1 > .681$. Projection onto c_f, n_f has divided displacement P_{14} and demand $\sqrt{8\sqrt{3}} - \sqrt{2}(2 - \sqrt{3}) > 3.34$. Its positive correction is $\gamma_f n_f$ and residual $1 - \gamma_f^2 > 0$. The reference energy is $> .9724150245$; the $.15P_{11}$ graft has square three and gives norm > 1.0411150245. Thus every isolated finite fork is excluded. At this stage a fork is isolated dotted or endpoint-three.

Three head feet used by the remaining spine payments. For an isolated head write $c_H = A_H p - \gamma_H \Omega_1$. The actual first-three/four-root constructions give $x \cdot Z \ge H$, nonnegative Z

with sum one, and

own	Z	H	$R = Z^2$
5	$P_4/4$	$A_H/\sqrt{2} > .393$	$1/4$
$m \geq 7$	$(1-u/2)(e_1+e_2)+(u-1)e_3$	$[\sqrt{2-u}-\sqrt{7u-4-3u^2}]/\sqrt{2}$	$3-4u+3u^2/2$
dotted	$(1-u/2)e_1+(u/2)e_2$	$H(u)$	$R(u)$.

Here $u = \gamma_H^{-2}$; the dotted functions are those of Lemma 23.3. The native-five row is the finite positive H_4 projection. The finite row uses $\xi = c_H + n_1/\gamma_H + 2(\gamma_H^{-1} - \gamma_H)n_2$. Its square is $-(7 - 4\gamma_H^2 - 3/\gamma_H^2) < 0$; its products with c_H, n_1 vanish and that with n_2 is $(3 - 4\gamma_H^2)/(2\gamma_H) < 0$. The dotted row uses $\xi = c_H + n_1/\gamma_H$. The time is positive in both cases, and all other original walls have nonpositive products termwise.

For every finite $m \geq 7$ and every dotted own,

$$.8H - .16R > .2893. \tag{23.5}$$

For seven, its cubic gives $.9009 < \gamma_H < .901$. Rational squares give $A_H > .7894$, $\sqrt{7 - 4\gamma_H^2 - 3/\gamma_H^2} < .2395$, $\gamma_H\sqrt{2} < 1.2743$, hence $H > .4315$; $1.2318 < u < 1.233$ gives $R < .349$ and the result $> .28936$. For all finite $m \geq 8$, $1 < u \leq 4 - 2\sqrt{2} < 1.172$. Put $D = (u-1)(4-3u)$ and $a = .8/\sqrt{2}$. The derivative of $.8H - .16R$ is

$$a\{-1/(2\sqrt{2-u}) - (7-6u)/(2\sqrt{D})\} + .64 - .48u.$$

It is negative on $(1, 7/6]$: the first term is $< -.28$, the second nonpositive, and the last at most .16. On $[7/6, 1.172)$, $\sqrt{D} > .284$, the positive second term is $< .032$, and the other terms are $< -.28 + .08$. The minimum is therefore at native eight, where $\sqrt{2-u} > .910$, $\sqrt{D} < .289$, $R < .373$, giving $H > .4388$ and $.8H - .16R > .29136$. For dotted own Lemma 23.3 is stronger. The graft $.4(Z - P_5/5)$ spends $.08 < \beta_5$, is zero-sum on the first-five flat block, and gives the single-reference gain $> .2893 - .8(.361) + .032 = .0325$.

The complete late D_4, D_3 inventories. At twelve the retained pieces are $A_{11} + D_4$, with right long leaf thirteen, centre fourteen and the two genuine forks. At thirteen they are $A_{12} + D_3$, the right A_3 having middle fourteen and endpoints the forks. Finite classification gives: left endpoint-three, next-three or endpoint-four; right D_4 three at any leaf; right D_3 middle-three or endpoint-three, four or five; and bridges joining a left endpoint by three to a right leaf or middle by three. The full bridges are D_{16} or A_{16}. Other bridges have two forks, a weighted internal bond or large unequal exceptional arms. Two contacts in one retained component make a nonpositive cycle. These are all white cases.

With left contact at position j, the unit residual/ratio data are

role	Δ	σ
left end-three	$(i+1)/(2i)$	$\gamma + j/(2i)$
left next-three	$2/i$	$\gamma + j/(2i)$
left end-four	$1/i$	$\gamma + j/(\sqrt{2}i)$
D_4 long-three or D_3 middle-three	$1/2$	$\gamma + 1/2$
D_4 fork-three	$1/2$	$\gamma + 1/4$
D_3 end-three	$5/8$	$\gamma + 1/4$
D_3 end-four	$1/4$	$\gamma + 1/(2\sqrt{2})$
D_3 end-five	$(5-3\varphi)/8$	$\gamma + \varphi/4$.

The actual edge cosh is $\sqrt{2i}\sigma/\sqrt{\Delta}$. The required strong ratio is $T(q) = 100/61-(q-.61)/(q(2q-.61))$. It increases on this range. Substitution gives $T(\sqrt{28}) < 39/25$ and, at twelve/thirteen, $T(q) < 31/20$ using $q < 26/5$; the latter cleared numerator is $25414 > 0$.

For single-helper own zero at a long $j \geq 2$, the actual affine D principal consists of c, n_{j-1}, the trunk $j, \ldots, 14$ and both forks. A right fork helper has the affine E_8 arms $1, 2, 5$. Far-left-three own zero has negative time; far-left-four has zero time. The near-four zero case has its affine $4, 3, 3$ star. Right D_3 end-four has kernel coefficients $(\sqrt{2}, 2, 2, 1, 1)$ on $(c, f, 14, 13, k)$. End-five is directly strong even at own zero. Own three with a single three-helper gives an affine all-three cycle. Own four with a fork helper gives the affine F_4 path $i-2, i-1, i, c, f$; with no right helper the old right D and c give the long-leaf-four affine principal. Own six uses the $6, 3$ chain through a neighbor distinct from its helper; parallel gives $c + n_i$. Bridges at own zero have their all-three cycles. Each null has positive coefficients, future time, and nonpositive products with all omitted original walls.

Positive-own bridges have ratio > 1.9. Right long/middle-three and end-four have ratio at least $1 + 1/\sqrt{2}$. End-five has $\Delta < .02$, $\sigma > .4$. Left near-three after its null cases has $\gamma \geq 1/\sqrt{2}$ and ratio $> 31/20$; next-three and end-four are stronger. The only weak left role is the classified literal far helper, excluded by Lemma 20.3.

Isolated caps at these sites are also strong, without an arithmetic field cutoff. Own three has the affine D formed by the new leaf, the old left leaf, the right trunk and both forks; own four has the long-leaf-four affine principal; own six and parallel have the nulls just described. At native five, the positive left-H_4 projection has displacement $\gamma(3\Omega_{i+1}/4 + \Omega_{i-3}/4)$, whose product with x is at most γh_i by concavity. With $u = \gamma^{-2} < 1.53$, its demand therefore gives $h_i > \sqrt{2i-u} - \sqrt{u-3/2}$, of loss $< .336$ at $i \geq 12$. All roots through $i+1 \leq 14$ exist. For finite own at least seven, the actual core $c + n_i/\gamma + 2(\gamma^{-1} - \gamma)n_{i-1}$ gives $h_i > \sqrt{2i-u} - \sqrt{7u-4-3u^2}$. Here $u < 1.235$, $7u - 4 - 3u^2 \leq 1/12$, and the loss is $< 1.235/9.5 + .289 < .42$. For dotted own the core $c + n_i/\gamma$ has loss at most $u/9.5 + \sqrt{u(1-u)} \leq (2 + \sqrt{365})/38 < .56$. All these are below .61, proving strong isolation.

Only a pure single-fork-three packet can remain weak. Its square-two parameter $t = 2\gamma$ is integral and all other conjugates satisfy $it_\sigma(t_\sigma + 1) + 2 < 0$. The physical weak ratio gives $\varphi \leq t < 1.98$. The norm of $t(t+1)$ excludes degree at least three; trace one, norm minus one gives $t = \varphi$, and trace two is impossible. Thus each late weak role is exactly native five/fork three.

Its actual positive left-H_4 projection has

$$\alpha^2 = \varphi + 5/8 > 56/25, \quad \delta = (5 - 3\varphi)/8, \quad A^2 = i\alpha^2 - \kappa, \quad \sqrt{\delta}/\alpha < .091,$$

where $\kappa = 1/2, 5/8, 3/4$ at $i = 12, 13, 14$. The correction $\gamma(3n_i/2 + n_{i-1} + n_{i-2}/2)$ is positive. Its divided displacement V equals one through $K = i - 3$, has $V^2 < i$, and gives $x \cdot V > G > \sqrt{i - \kappa/\alpha^2} - .091$. Consequently a paid tail reference has

i	K	μ	$G >$	$f_K <$	one norm lower
12	9	1/6	3.34	2.569	.9717166666
13	10	.16	3.475	2.732	.98532.

Its formula is $E_K + 2\mu(G - f_K) - 3\mu^2$. The endpoint coefficients $d_8 > 1/6$, $d_9 > .16$ fund the individual payments. For finite head at least seven or dotted, the zero-sum first-five head graft above has zero cross with each tail and adds $> .0325$ to that same reference; both exceed one. For native-five head the actual field is $\mathbb{Q}(\varphi)$ and the head raw time square is φ. The two tail raw time squares are $14 + 24\varphi$, $15 + 26\varphi$; their products with φ have nonsquare norms $44, 61$. Thus neither weak role exists, and twelve and thirteen are strong.

The entire centre-fourteen list. Deleting fourteen leaves A_{13} and the two actual fork A_1's. The complete finite white list is

left contact	fork contacts
none	none; one finite $m \geq 3$; $(3,3),(3,4),(3,5)$
end-three at $1, 13$	none; one three/four; both three
next-three at $2, 12$	none
end-four at $1, 13$	none.

Finite path and three-node classification proves the list; an extra fork on a next-three branch has large unequal exceptional arms. For left coefficient η at j and fork coefficients a, b, the original inverse gives

$$\Delta = 1 - 2\eta^2 j(14-j)/14 - a^2 - b^2 > 0, \quad \sigma = \gamma + \eta j/14 + (a+b)/2, \quad C_{14} = \sqrt{28}\sigma/\sqrt{\Delta}.$$

The strong target is below $39/25$. Isolation has finite parameter $B = 4\gamma^2$ with actual $K = \mathbb{Q}(B)$ and all other $0 < B_\sigma < 1/7$. Finite $B < 4$ has nonzero norm $< 4/7$ in degree at least two; rational K is excluded. Dotted isolation has loss $< .56$ from the actual core $c + n_{14}/\gamma$, so is strong. Left-only contacts have the same exact nulls, strong ratio bounds, and excluded literal far helper as above. Left end-three plus one fork-three has $\Delta = 2/7$; own zero is an affine cycle, own three an affine triangle, and own at least four has ratio > 1.8. One fork-four or both fork-three gives $\Delta = 1/28$ and ratio > 2 even at the far own-zero boundary.

With no left contact, own zero and one finite fork makes c the actual isolated cap of that original fork edge: the future timelike point $p + Ac$ is zero on c and every old wall except that fork, strictly negative on it and every unknown wall. Its finite isolated role was excluded above. Own zero with fork pair $(3,3)$ is the affine four-cycle; pair $(3,4)$ is reciprocally near white-four and was excluded; pair $(3,5)$ is already strong at own zero. All positive-own pairs are stronger except own three with $(3,3)$, its actual affine triangle.

For one fork at positive own, own four has the affine $4,3,3$ star on $c, 14, 13$ and the uncontacted fork; own six has the $6,3$ path; parallel is null. Own three/fork three is an affine triangle. Own three/fork four is reciprocally near-next-three and was excluded. Each reciprocal claim uses positive projection onto the finite full retained $A_{15} + c$ system. The omitted fork product is strictly negative and all unknown walls stay strict, so the point is the endpoint of an original fork edge. Own three/fork five has actual field $\mathbb{Q}(\varphi)$; the raw root $\varphi\sqrt{2}c$ has time square $34 + 52\varphi$, positive at $\varphi' = 1 - \varphi$, violating admissibility. Own three/fork at least six has ratio at least $1 + \sqrt{3}/2 > 39/25$. For own at least five and fork at least four the minimum is native-five/fork-four, with ratio $(.809 + \sqrt{2}/4)/\sqrt{1/2} > 39/25$.

Only single-fork-three remains. Its complete weak strip is $\varphi \leq t < 2.21$, and $14t_\sigma(t_\sigma + 1) + 2 < 0$. The same integral $t(t+1)$ norm excludes degree at least three. Quadratic trace one leaves only $t = \varphi$; trace two gives norm in $(-.4641, 0)$ and is impossible. For that golden packet use the positive H_4 point above at $i = 14, K = 11, \mu = .15$, obtaining the one-reference lower bound $> .84816 + .3(3.60 - 2.886) - .0675 = .99486$. The isolated finite-at-least-seven/dotted head gain adds $> .0325$ with zero cross, giving > 1.02736. For native-five head the tail raw time square $16 + 28\varphi$ has product-with-head norm 80, nonsquare. Thus fourteen is strong. All knots five through fourteen are now strong, and direct interpolation gives every coefficient in (23.4), including $\beta_{14} = d_{13} > .13$.

Native-five head and the final fork/degree bounds. At a native-five head the point $Z = P_4/4$, $H > .393$ permits the payment $.56(Z - P_5/5)$, with debit $.112 < .14$ and gain $> 1.12(.393 - .361) - .3136(.25 - .2) = .02016$. The separate signed $.13P_{14}$ tail has square penalty .0338 and zero

cross with this early graft. A fork cosh at least fourteen would give $T_s > 3.72$ and $\|x\|^2 > .9084 + .02016 + .26(3.72 - 3.311) - .0338 = 1.00110$. Thus both fork coshes are below fourteen. The actual field is $\mathbb{Q}(\varphi)$; the head raw time square is φ. Only isolated dotted or endpoint-three roles remain.

For an isolated dotted fork the head/fork cycle gives $F/4$ integral. For $W = F/4$ its strips are $W < 49/2$, $0 < W' < 1/8$. In $W = a + b\varphi$, $1 \le b \le 10$, the strict conjugate strip permits only $b = 8, a = 5$, so $F = 4\varphi^6$. Its time-product-with-head norm is $5056 = 64 \cdot 79$, nonsquare. Far endpoint-three has $t < (14\sqrt{17} - 1)/8 < 36/5$ and conjugate in $(-13/20, 2/5)$. The full ring gives $1 \le b \le 3$ and exactly $t = \varphi, \varphi^2, \varphi^3, \varphi^4$. The product-with-head norms are $1, 1, 76, 149$. The first two positive time products are φ^3, φ^4, but the spatial products are $2\varphi + 3/2$, $5\varphi/2 + 2$. Thus $g = -H \cdot C$ is $-1/2$ or $\varphi/2$: one violates acuteness, the other original path-cycle integrality. Near, $z = t + 1 < (14\sqrt{17} + 1)/8 < 15/2$ has conjugate in $(-2/5, 13/20)$. The same full-ring strip gives $z = \varphi^2, \varphi^3, \varphi^4$. Their product norms are $19, 36, 191$. The square-norm middle product is $54 + 24\sqrt{5}$; its rational square parts would be $30, 24$, neither square. No actual fork remains, excluding native-five head.

Every isolated head now has finite own at least seven or dotted own and therefore the zero-cross gain $> .0325$. A fork cosh at least twelve gives $T_s > 3.67$, and

$$\|x\|^2 > .9084 + .0325 + .26(3.67 - 3.311) - .0338 = 1.00044.$$

Thus both coshes are below twelve. Each fork is isolated dotted or endpoint-three with positive own, as asserted.

Finally suppose both forks are isolated. Their original cycles give integral $Y = FG/4$, with $Y < 1296$ and every other $0 < Y_\sigma < 1/16$. Degree at least five has nonzero norm $< 1296/65536 < 1$. At degree four each positive integer $\mathrm{N}\,F, \mathrm{N}\,G < 9$ is at most eight, but Y integral forces $256 \mid \mathrm{N}\,F\,\mathrm{N}\,G$. The two norms have total dyadic valuation at most six, a contradiction. Degree one is excluded by Meyer. Hence the actual degree is two or three under the stated both-isolated hypothesis. This proves the entire initial inventory at its original-rank and actual-wall scope. □

Proof of Proposition 23.1, from Lemma 23.2. Use raw old roots of square two. An isolated head has raw square $2U$, spatial part Ue_1, time square $U(U-2)$, and actual field $K = \mathbb{Q}(U)$; its nonphysical U_σ lie in $(0, 2)$. An isolated fork has the data stated above, actual $K = \mathbb{Q}(F)$, and $0 < F_\sigma < 1/2$ at every nonphysical place. The only nonisolated forks left by the initial inventory have raw square two and one endpoint-three white contact. Their complete data are

role	spatial vector	time square M	edge cosh
far	$(t/2)(P_{15} + se_{16}) + e_1$	$4t^2 + t - 1$	$(8t+1)/\sqrt{17}$
near	$(t/2)(P_{15} + se_{16}) + (P_{15} - se_{16})/2$	$4t^2 + 7t + 2$	$(8t+7)/\sqrt{17}$.

Here $s = \pm 1$ denotes a genuine fork. In the far role put $z = t$; in the near role put $z = t + 1$. Path-cycle integrality gives integral z, and actual-field descent gives $K = \mathbb{Q}(z)$. To justify these normalizations, the connected simple-three old system has one raw norm squareclass, so a totally positive rescaling of the form puts its roots at square two over K. An isolated own contact has integral two-cycle $4\gamma^2$; rescaling its root line by its nonzero old product gives the raw head or fork displayed above. The raw principal consisting of that cap and the full old system is defined over $\mathbb{Q}(U)$ or $\mathbb{Q}(F)$ and is physically indefinite, since the cap has nonzero time. The descent argument in Section 19 proves the asserted field equalities. For an endpoint cap its simple-three helper permits square-two normalization directly; the closed old path gives integral t, and the same

indefinite-principal argument gives $K = \mathbb{Q}(t)$. The cosh bound twelve gives $z < 6.311$; positive own gives $z \geq 1$ far and $z \geq 2$ near. Negative nonphysical time squares give

$$-.65 < z_\sigma < .4 \quad \text{far}, \qquad -.4 < z_\sigma < .65 \quad \text{near}. \tag{23.6}$$

All these formulas concern actual K-roots, so every positive raw time product is K-valued and its square must be a full K-square.

First sharpen the head bound. If $x_1 \geq .632$, rebuild only the first-five flat block of the positive reference, preserving its sum. The new coordinates are $.632, (f_5 - .632)/4$ four times. They exceed the next slope and preserve nonnegative prefix prices. Their single-reference gain gives

$$\|x\|^2 > E_{14} + \tfrac{5}{4}(.632 - .361)^2 = 1.00020125,$$

a contradiction. Thus $x_1 < .632$. The actual head edge gives $C_H < 7/3$, since $.632^2 < 2/5$, and hence always $U < 98/9$; if dotted, $U > 4$.

Endpoint cycles and the actual degree. For an endpoint cap let $g = -H \cdot C$ and $L = g^2/U$. The original path and own two-cycle make both integral. Expanding the common time product gives

$$\begin{aligned} L + g(t+2) &= (15t^2 - 8)U/4 - 2M && \text{far}, \\ L + g(t+1) &= (15z^2 - 4z - 4)U/4 - 2M && \text{near}. \end{aligned}$$

All remaining terms are integral, and 15 is coprime to four; therefore $z^2U/4$ is integral in both roles. If $|\,\mathrm{N}\, z|$ is odd, z is a unit at every dyadic prime, forcing $U/4$ integral. This is impossible in degree at least three. Indeed the sharpened head bound just proved gives

$$0 < \mathrm{N}(U/4) < \frac{49}{18}(1/2)^{d-1} \leq \frac{49}{72} < 1 \qquad (d \geq 3),$$

contradicting its positive integral norm. This uses the original head bound and all nonphysical places, with no integer-basis assumption.

The nonzero endpoint norm is less than $6.311(.65)^{d-1}$. It excludes $d \geq 6$; at degrees four or five it is below two and hence has odd absolute norm one, just excluded. In degree three only $|\,\mathrm{N}\, z| = 2$ remains. Degree one is excluded by Meyer. If neither endpoint role occurs, both forks are isolated and the initial lemma already gives degree two or three. Thus these are the only actual degrees in every case.

Every cubic endpoint is impossible. In degree three norm magnitude two and $6.311 \cdot .4 \cdot .65 < 2$ force both small conjugates negative far, positive near. Write their positive magnitudes $a, b < 13/20$; then $zab = 2$ and $z > 800/169$. In the far role its integer trace $z - a - b$ is four or five, by AM–GM and $z < 6.311$. The second coefficient $B = 2/z - z(a + b)$ satisfies $B \leq -6$ by AM–GM. The polynomial $X^3 - TX^2 + BX - 2$ is negative at $-13/20$, so $-B < (13/20)^2 + T(13/20) + 40/13 < 7$ and $B \geq -6$. Thus $B = -6$. The two resulting cubics, with $T = 4, 5$, have discriminants $-44, -424$, impossible in a totally real field. Near, the function $z + 2\sqrt{2/z}$ is increasing for $z > 4$. The integer trace is therefore seven, since $z + 2\sqrt{2/z} > 800/169 + 13/10 > 6$ and $z + a + b < 8$. Thus $z > 57/10$. Evaluation at $13/20$ gives $B < 7.205$, whereas $B \geq 2\sqrt{2z} + 2/z > 6.75 + .316 > 7$. The incompatible integer bounds exclude this role too.

Every finite isolated head is impossible. For finite own m, $U = 4\cos^2(\pi/m)$ has degree $\phi(m)/2$. The complete totient-at-most-six list and the initial small-own exclusions leave $m = 7, 8, 9, 10, 12, 14, 18$. At $m = 9, 14$ a nonphysical $U_\sigma > 2$ is forbidden. At cubic $m = 7, 18$, $\mathrm{N}\, U = 1, 3$ is odd. The cubic endpoint exclusion makes both forks isolated; their cycle $UF/4$ forces $F/4$ integral, so $64 \mid \mathrm{N}\, F$. But $F < 72$ and its other conjugates below $1/2$ give $\mathrm{N}\, F < 18$, a contradiction.

The remaining heads have $U = 2 + \sqrt{2}, 2 + \varphi, 2 + \sqrt{3}$. The full endpoint strips, in the maximal orders, give

K	far t	near z	$U(U-2)$
$\mathbb{Q}(\sqrt{2})$	$1 + \sqrt{2}, 3 + 2\sqrt{2}$	$2 + \sqrt{2}, 3 + 2\sqrt{2}$	$2 + 2\sqrt{2}$
$\mathbb{Q}(\varphi)$	$\varphi, \varphi^2, \varphi^3$	φ^2, φ^3	$1 + 3\varphi$
$\mathbb{Q}(\sqrt{3})$	$2 + \sqrt{3}$	$2 + \sqrt{3}$	$3 + 2\sqrt{3}$.

For $\sqrt{2}$ the radical coefficient is one or two, for $\sqrt{3}$ one. In the golden integral basis the coefficient is one or two; the next cover already exceeds the exact cosh bound. These finite integer intervals prove the list, including dotted owns. The corresponding time-product norms are $72, 400, 36, 544$; $5, 5, 380, 95, 180$; and $78, 150$. Only $400, 36$ are squares. Their products are $340 + 240\sqrt{2}$ and $102 + 72\sqrt{2}$; the rational square parts would be $180, 160$ or $54, 48$, none square. Thus no endpoint is possible for a finite head.

For an isolated fork in the $m = 10, 12$ fields the head is a dyadic unit, so $F/4$ is integral. At $m = 8$, normalize the unique dyadic valuation by $v(\sqrt{2}) = 1$. The quarter-cycle gives $v(F) \geq 3$; the time-product valuation is $4 + v(F)$ and must be even, hence $v(F) \geq 4$ and again $F/4$ is integral. For $W = F/4$, the full strips are $W < 18$, $0 < W' < 1/8$. In $\mathbb{Z}[\sqrt{2}]$ no coefficient with $1 \leq b \leq 6$ fits; in $\mathbb{Z}[\varphi]$ only $W = 5 + 8\varphi$ fits; in $\mathbb{Z}[\sqrt{3}]$ only $W = 7 + 4\sqrt{3}$ fits. The last two give time-product norms $25280, 9024$, nonsquares. Thus the actual head is dotted.

One dotted-head reference excludes the remaining cubic. Put $u = 4/U \in (0, 1)$. The positive actual core $c_H + n_1/g_H$ has zero product with c_H, negative product with n_1, future time and nonpositive product with every other original wall. Normalizing it gives

$$x \cdot Z \geq H(u), \quad Z = (1 - u/2)e_1 + (u/2)e_2, \quad Z^2 = R(u).$$

The single graft $.4(Z - P_5/5)$ spends $.08$ at P_5, below its available $.14$, and has zero cross with the flat block. Lemma 23.3 therefore gives gain $> .295 - .8(.361) + .032 = .0382$, hence one reference of energy $> .9466$. A signed $.13(P_{15} + se_{16} - P_{14})$ graft has square residual two and zero cross. If its fork cosh were at least eleven, its actual pressure would exceed 3.65, giving

$$\|x\|^2 > .9466 + .26(3.65 - 3.311) - .0338 = 1.00094.$$

Both fork coshes are therefore below eleven.

In a cubic field all endpoint roles were excluded. The isolated forks satisfy $4 < F, G < 121/2$ and $\mathrm{N}\, F, \mathrm{N}\, G \in \{1, \ldots, 15\}$. The quarter-cycle gives $64 \mid \mathrm{N}\, F\, \mathrm{N}\, G$, forcing both norms eight. Lemma 23.3 gives $F, G > 42$. Their coshes exceed $\sqrt{84} > 9.16$ and their signed pressures exceed 3.584, since $3.584^2 < 1632/127$. Averaging the two genuine fork inequalities gives $S_{15} > 3.584$. The average tail graft has residual square one, so the same reference gives

$$\|x\|^2 > .9466 + .26(3.584 - 3.311) - .0169 = 1.00068.$$

Thus the actual field is quadratic.

Every quadratic endpoint is impossible. The improved cosh bound eleven gives $z < 5.796$, so $|\,\mathrm{N}\,z| < 4$. If its norm is odd, $U/4$ is integral. The full head strip then gives $U = 4\varphi^2$, with time square $8\varphi^5$. In this field the complete endpoint strips give far $t = \varphi, \varphi^2, \varphi^3$ and near $z = \varphi^2, \varphi^3$. The first two far time products are $8\varphi^{10}, 8\varphi^{12}$, requiring $\sqrt{2}$ in the field; the third has nonsquare norm 4864. The near products have nonsquare norm 1216, or equal $48\varphi^{10}$, requiring $\sqrt{3}$. None is possible.

Only norm magnitude two remains. Integer trace and the exact role-specific cosh/conjugate strips give the full list

role	$\mathrm{N}\,z$	$\mathrm{Tr}\,z$	z
far	-2	$3, 4, 5$	$(3+\sqrt{17})/2,\ 2+\sqrt{6},\ (5+\sqrt{33})/2$
near	2	$4, 5, 6$	$2+\sqrt{2},\ (5+\sqrt{17})/2,\ 3+\sqrt{7}$
near	-2	5	$(5+\sqrt{33})/2.$

For example far positive norm two would require $z > 5$ and $5 < z + 2/z < 6$, so is impossible. The other rows follow from $z + \mathrm{N}(z)/z$ and the strict conjugate intervals, not a field search.

In the ramified fields $\sqrt{2}, \sqrt{6}, \sqrt{7}$, z has dyadic valuation one. Integrality of $z^2U/4$ forces $U/2$ integral. Writing $W = U/2$, its strips are $2 < W < 49/9$, $0 < W' < 1$. The only radical coefficient is one: the candidates are $2+\sqrt{2},\ 3+\sqrt{6},\ 3+\sqrt{7}$. The last two exceed $49/9$; the first gives $U = 4 + 2\sqrt{2}$. Its near cap has time square $21 + 15\sqrt{2}$ and the head $16 + 12\sqrt{2}$; their product norm 288 is nonsquare.

In the split fields $\sqrt{17}, \sqrt{33}$, taking norms in $z^2U/4$ gives $4 \mid \mathrm{N}\,U$. The full head strips retain precisely

$\mathbb{Q}(\sqrt{17})$	$(7+\sqrt{17})/2,\ 5+\sqrt{17}$
$\mathbb{Q}(\sqrt{33})$	$(7+\sqrt{33})/2,\ (9+\sqrt{33})/2.$

In the first field the full integral-basis coefficient is one or two, and the omitted strip heads have norms two and nineteen; in the second it is one. No maximal-order class is missing. For $s = \sqrt{17}$ the head time squares are $(19+5s)/2$, $32+8s$. The far cap has time square $(53+13s)/2$. Its first time product is $(16+4s)^2$, but the spatial product is $(33+7s)/4$; hence the actual path cycle would be $(31+9s)/4$, not integral. The second product is $1732+420s$, whose possible rational square parts $882, 850$ are nonsquares. The near cap has time square $(77+19s)/2$ and product norms $832, 3328$, also nonsquares. For $s = \sqrt{33}$ both head time squares have norm -24; the far and near caps have norms $-98, -8$, giving nonsquare product norms $2352, 192$. Thus every endpoint is excluded.

Both genuine forks are consequently isolated dotted. Their original cycles, in these raw normalizations, give

$$L_{HF} + g_{HF} = (15UF - 32F - 8U + 16)/4,$$
$$L_{FG} + 7g_{FG} = 15FG/4 - 8(F+G) + 4,$$

with integral g, L; the analogous head/G identity holds. These yield all three quarter-products. The common actual K-time line supplies every field-square condition. The strict physical and nonphysical strips were retained throughout. This proves the claimed necessary quadratic triple from the initial inventory. □

23.1 Norm bounds, integer orders and valuations

Put

$$N_H = \mathrm{N}\,U, \quad T = \mathrm{Tr}\,U, \quad A = -\mathrm{N}(U-2) = 2T - 4 - N_H.$$

The strict strips imply

$$5 \le T \le 12, \quad 1 \le N_H \le 21, \quad 1 \le A \le 17.$$

For a fork put $n = \mathrm{N}\, F$, $S = \mathrm{Tr}\, F$ and $B = 2S - 4n - 1 > 0$ odd. Then

$$1 \le n \le 30, \quad 5 \le S \le 60, \qquad \mathrm{N}\, T_H = -N_H A, \quad \mathrm{N}\, T_F = -4nB. \tag{23.7}$$

A nonsquare rational norm disproves a field square; a square norm still requires the full square test.

Use the full integer order of $K = \mathbb{Q}(\sqrt{d})$. The gap between the embeddings of U is a positive integral multiple of $\sqrt{\Delta}$, so

$$\Delta < (98/9)^2 < 119, \qquad \Delta \le 118. \tag{23.8}$$

For a prime $\mathfrak{p}$ over two, normalize $v_{\mathfrak{p}}(\mathfrak{p}) = 1$ and write

$$e = v_{\mathfrak{p}}(2),\ r = v_{\mathfrak{p}}(U),\ s = v_{\mathfrak{p}}(U-2),\ k = v_{\mathfrak{p}}(F),\ l = v_{\mathfrak{p}}(G).$$

The factors $2F - 1, 2G - 1$ are units there. Quarter integrality and the time squares give

$$r + k, r + l, k + l \ge 2e, \qquad r + s + e + k,\ r + s + e + l \equiv 0 \pmod 2. \tag{23.9}$$

Also $s = r$ for $r < e$, $s = e$ for $r > e$, and $s \ge e$ for $r = e$. Norm valuations are $\sum f_{\mathfrak{p}} v_{\mathfrak{p}}$. The quadratic possibilities are inert $(e, f) = (1, 2)$, ramified $(2, 1)$, or two split primes $(1, 1)$. No PID or unramifiedness assumption enters.

23.2 Three complete integer-strip lemmas

If $F/4$ is integral, $16 \mid n < 30.25$ gives $n = 16$. For $W = F/4$, norm one and $0 < W' < 1/8$ give $8 < W < 121/8$, and $\mathrm{Tr}\, W = W + 1/W \in \{9, \dots, 15\}$. This gives the complete table

$\mathrm{Tr}\, W$	K	Δ	F when $\Delta \le 118$
9	$\mathbb{Q}(\sqrt{77})$	77	$18 + 2\sqrt{77}$
10	$\mathbb{Q}(\sqrt{6})$	24	$20 + 8\sqrt{6}$
11	$\mathbb{Q}(\sqrt{13})$	13	$22 + 6\sqrt{13}$
12	$\mathbb{Q}(\sqrt{35})$	140	excluded
13	$\mathbb{Q}(\sqrt{165})$	165	excluded
14	$\mathbb{Q}(\sqrt{3})$	12	$28 + 16\sqrt{3}$
15	$\mathbb{Q}(\sqrt{221})$	221	excluded.

If $Y = U/2$ is integral with odd norm, then $2 < Y < 49/9$, $0 < Y' < 1$ and $\mathrm{N}\, Y \in \{1, 3, 5\}$. Its integer trace t is in $3, \dots, 6$ and $1 - t + \mathrm{N}\, Y < 0$, so $t > \mathrm{N}\, Y + 1$. The complete list is

$\mathrm{N}\, Y$	t	Y	2 in K
1	3	$(3 + \sqrt{5})/2$	inert
1	4	$2 + \sqrt{3}$	ramified
1	5	$(5 + \sqrt{21})/2$	inert
3	5	$(5 + \sqrt{13})/2$	inert.

Trace six gives $3 + 2\sqrt{2}$ or $3 + \sqrt{6}$, both above $49/9$; norm five requires trace above six. These bounded integer comparisons prove completeness.

Finally, if $U/4$ is integral with $\mathrm{N}\, U = 16$, then $W = U/4$ has norm one, $1 < W < 49/18$, $0 < W' < 1/2$. Thus $W > 2$, $\mathrm{Tr}\, W = 3$, and

$$U = 4\varphi^2, \quad K = \mathbb{Q}(\varphi), \qquad \varphi = (1 + \sqrt{5})/2. \tag{23.10}$$

24 Odd head norm and inert two

24.1 Odd norm

If N_H is odd, at every dyadic prime $r = s = 0$. In an inert or split field, (23.9) makes both fork valuations odd and at least three; their norms are divisible by 64, contrary to $n \le 30$. Thus two ramifies. Now k, l are even and at least four. Both fork norms are sixteen and $F/4, G/4$ integral. The unit-fork table leaves only $\mathbb{Q}(\sqrt{3})$ or $\mathbb{Q}(\sqrt{6})$. The full head strips with odd norm give

K	U	$\mathrm{N}\,T_H$	$\mathrm{N}\,T_F$
$\mathbb{Q}(\sqrt{3})$	$5+2\sqrt{3}$	-39	-3008
$\mathbb{Q}(\sqrt{6})$	$3+\sqrt{6},\ 5+2\sqrt{6}$	$-15,\ -15$	$-960.$

The radical coefficient is one or two; coefficient three exceeds the physical bound. The conjugate strip and norm parity remove the other rational coefficients. In $\mathbb{Q}(\sqrt{6})$, $6+2\sqrt{6} > 98/9$ and $4+\sqrt{6}$ has even norm.

The $\sqrt{3}$ product norm is $64 \cdot 1833$, a nonsquare. For $U = 3+\sqrt{6}$ the exact time product is $58200 + 23760\sqrt{6}$, of norm 120^2. Its possible rational square parts are 29160 and 29040, strictly between 170^2 and 171^2, so it is not a field square. The remaining triple is $U = 5+2\sqrt{6}$, $F = G = 4U$. The actual head and two genuine fork edges have coshes

$$C_H = (2+\sqrt{6})/2 > 11/5, \qquad C_+ = C_- = 2(2+\sqrt{6}) > 44/5,$$

using $\sqrt{6} > 12/5$. Equation (19.1) gives $x_1 \ge \sqrt{3/8} > .61$ and, for both $s = \pm 1$,

$$x \cdot (P_{15} + s e_{16}) > 4\sqrt{39}/7 > 3.56.$$

The last comparison follows from $6.23^2 < 39$. Averaging the two signed inequalities gives $S_{15} > 3.56$; Equation (23.4) gives $S_6 > 2.01815$. One positive reference gives

$$\|x\|^2 > .61^2 + 1.40815^2/5 + 1.54185^2/9 = 1.032821887 > 1.$$

The last pressure averages the two genuine fork edges, not a virtual root. This excludes odd norm.

24.2 Even norm with inert two

Here $e = 1, f = 2$, and $r = 1$ or two since $r \ge 3$ forces $N_H \ge 64$. If $r = 1$, the head strip gives

$$U = 3+\sqrt{5}, \quad 5+\sqrt{21}, \quad 5+\sqrt{13}.$$

In each case $s = 1$, so (23.9) makes k, l odd and positive. The norm bound gives $k = l = 1$. Thus $W = F/2$ is integral with odd norm, $W < 121/4$, $0 < W' < 1/4$. In $W = (a+b\sqrt{d})/2$, $a \equiv b \pmod 2$, the full strip $0 < a - b\sqrt{d} < 1/2$ gives

d	b range	(a,b) before norm filter	F
5	$1,\dots,13$	$(7,3),(18,8),(25,11)$	$2\varphi^4,\ 2\varphi^6,\ 14+22\varphi$
21	$1,\dots,6$	$(5,1),(23,5)$	$5+\sqrt{21},\ 23+5\sqrt{21}$
13	$1,\dots,8$	$(11,3),(22,6)$	$11+3\sqrt{13}.$

The $d = 13$ pair $(22,6)$ has $\mathrm{N}\,W = 4$, not odd. Each b bound follows from $b\sqrt{d} < W < 121/4$; the displayed a is the only possible integer with the required parity. The close pair $(28,6)$ at $d = 21$ fails since $6\sqrt{21} < 55/2$.

At $d = 5$, product norms are 256 times $11, 55, 95$, all nonsquares. The other square-norm exceptions are

d	F	$T_H T_F$	$\sqrt{\mathrm{N}(T_H T_F)}$	possible square parts
21	$5 + \sqrt{21}$	$12648 + 2760\sqrt{21}$	48	6348, 6300
21	$23 + 5\sqrt{21}$	$303000 + 66120\sqrt{21}$	240	151620, 151380
13	$11 + 3\sqrt{13}$	$52872 + 14664\sqrt{13}$	144	26508, 26364.

These lie strictly between $79^2, 80^2$; $389^2, 390^2$; and $162^2, 163^2$, respectively. No fork exists.

If $r = 2$, then $\mathrm{N}\,U = 16$, $U/4$ integral, and (23.10) holds. Now $s = 1$, so k, l are even. The fork/fork quarter-product gives $k + l \geq 2$, so one fork is divisible by four with norm sixteen. Its $W = F/4$ in $\mathbb{Q}(\varphi)$ has norm one, $W < 121/8$, $0 < W' < 1/8$. Writing $W = a + b\varphi$ gives $b = 1, \ldots, 6$; none permits $0 < a + b(1 - \varphi) < 1/8$. The inert case is exhausted.

25 Even head norm with ramified two

Here $e = 2, f = 1$ and $r = v_2(N_H) \in \{1, 2, 3, 4\}$. If $r = 1$, $s = 1$ and both fork valuations are even and at least four. The unit-fork table leaves $\sqrt{6}$ or $\sqrt{3}$. The head strips with $v_2(N_H) = 1$ give only $U = 4 + \sqrt{6}$ or $3 + \sqrt{3}$. Their product norms are $256 \cdot 75$ or $256 \cdot 141$, nonsquares.

If $r = 2$, the odd-norm $U/2$ table forces $U = 4 + 2\sqrt{3}$. Then $s = 3$, so k, l are odd and at least three. Norm bounds permit only valuation three and norm eight or twenty-four. For $F = 2W$, therefore, $\mathrm{N}\,W = 2$ or six, $W < 121/4$, $0 < W' < 1/4$. In $\mathbb{Z}[\sqrt{3}]$, the strip for $W = a + b\sqrt{3}$, $1 \leq b \leq 8$, allows only $(7, 4), (14, 8)$, of norms one and four. Neither required norm occurs.

If $r = 3$, $U/2$ is integral of norm two and trace four, five or six. Trace six gives $3 + \sqrt{7} > 49/9$; trace five gives the split field $\mathbb{Q}(\sqrt{17})$. Thus ramification forces $U = 4 + 2\sqrt{2}$. Here $s = 2$, so k, l are odd, at most three. Since $k + l \geq 4$, at least one is three and its norm is eight or twenty-four.

For that head $\mathrm{N}\,T_H = -32$, so a product norm is $128n(2S - 4n - 1)$ with S even in $\mathbb{Z}[\sqrt{2}]$. At $n = 8$, a square would require $2S - 33$ square, but it is three modulo four. At $n = 24$, the strip gives $49 \leq S \leq 60$; the even values yield

$$3(2S - 97) \in \{9, 21, 33, 45, 57, 69\}.$$

Only nine is square, at $S = 50$. But even norm for $F = a + b\sqrt{2}$ forces a even, hence $S = 2a$ divisible by four. This excludes fifty.

Finally $r = 4$ gives (23.10), whose field has inert two. Larger r forces $N_H \geq 32 > 21$. All ramified cases are excluded.

26 The complete split branch

Here $d \equiv 1 \pmod 8$, squarefree, and (23.8) leaves $d = 17, 33, 41, 57, 65, 73, 89, 97, 105, 113$. Write $U = (a + b\sqrt{d})/2$, $a \equiv b \pmod 2$. The strip gives $0 < a - b\sqrt{d} < 4$ and $b = 1$, except at $d = 17$

where $b = 1$ or two. The complete even-norm heads are

d	U	T	N_H	A
17	$(5+\sqrt{17})/2$	5	2	4
17	$(7+\sqrt{17})/2$	7	8	2
17	$5+\sqrt{17}$	10	8	8
33	$(7+\sqrt{33})/2$	7	4	6
33	$(9+\sqrt{33})/2$	9	12	2
41	$(7+\sqrt{41})/2$	7	2	8
41	$(9+\sqrt{41})/2$	9	10	4
57	$(9+\sqrt{57})/2$	9	6	8
57	$(11+\sqrt{57})/2$	11	16	2
65	$(9+\sqrt{65})/2$	9	4	10
65	$(11+\sqrt{65})/2$	11	14	4
73	$(9+\sqrt{73})/2$	9	2	12
73	$(11+\sqrt{73})/2$	11	12	6
89	$(11+\sqrt{89})/2$	11	8	10
97	$(11+\sqrt{97})/2$	11	6	12
105	$(11+\sqrt{105})/2$	11	4	14
113	$(11+\sqrt{113})/2$	11	2	16.

For $d = 17$, all coefficient pairs are $(5,1),(7,1),(10,2),(12,2)$; the last has odd norm nineteen. For the other fields two odd a values pass the conjugate strip; at $89, 97, 105, 113$ the larger exceeds $98/9$. These explicit finite coefficient comparisons prove the table, without an assumed field-search output.

26.1 Odd trace

Let $\mathfrak{p}$ be the prime where U is nonunit and $\mathfrak{q}$ its conjugate. At $\mathfrak{q}$, $r = s = 0$, so both fork valuations are odd and at least three. Put $c = v_2(N_H) = r_{\mathfrak{p}}$. A fork has total norm valuation at most four.

If $c = 2$ or four, $s_{\mathfrak{p}} = 1$ and both fork valuations at $\mathfrak{p}$ are even. The only possible ordered valuations are $(0,3)$ for each fork, contradicting $FG/4$ at $\mathfrak{p}$.

If $c = 1$, both forks have valuations $(1,3)$ and norm sixteen. Parity requires $v_2(A)$ odd. The table leaves only

$$d = 41,\ U = (7+\sqrt{41})/2; \qquad d = 57,\ U = (9+\sqrt{57})/2.$$

Now $W = F/2$ is integral of norm four, $W < 121/4$, $0 < W' < 1/4$. In $W = (a + b\sqrt{d})/2$, $b = 1, \dots, 4$, the strip $0 < a - b\sqrt{d} < 1/2$ with parity allows only $(26,4)$ at 41, of norm five, or $(23,3)$ at 57, of norm four. Thus the only candidate is $F = 23 + 3\sqrt{57}$ in $K = \mathbb{Q}(\sqrt{57})$.

For that head $U > 8$, so $x_1 > 1/\sqrt{3} > .577$. Rebuilding only the first-five block of (23.4), with its sum fixed, gives

$$E_{14} + \frac{5}{4}(.577 - .361)^2 > .96672.$$

If a fork had $\cosh L \geq 9$, its actual signed pressure would exceed 3.577. The separately available .13 coefficient at P_{14} has residual square two and zero cross with that first-block change. One square would give

$$\|x\|^2 > .96672 + .26(3.577 - 3.311) - .0338 = 1.00208 > 1.$$

Both fork coshes are therefore below nine, so $F < 81/2$, contrary to $23 + 3\sqrt{57} > 44 > 81/2$.

If $c = 3$, $s_{\mathfrak{p}} = 1$ and both forks again have valuations $(1, 3)$, norm sixteen. The remaining heads have $d = 17$ or 89. For $W = F/2$ the full strip/parity pairs are

d	b range	(a, b)
17	$1, \dots, 7$	$(21, 5), (29, 7)$
89	$1, \dots, 3$	none.

The 17-field norms are four and two, leaving only $F = 21 + 5\sqrt{17}$. The close pair $(17, 4)$ fails since $4\sqrt{17} < 33/2$. The surviving product norm is $(-16)(-1216) = 1024 \cdot 19$, a nonsquare. There are no other odd-trace cases.

26.2 The sole even-trace head

The last row is $U = 5 + \sqrt{17}$. Its two prime valuations are one and two. At the valuation-two prime $s = 1$; $\mathrm{N}(U - 2) = -8$ makes $s = 2$ at the other. Equation (23.9) makes both fork valuations even at both primes, and at the valuation-one prime each is at least two. The fork/fork quarter-product forces one fork also to have valuation at least two at the other prime. It is divisible by four globally, with norm sixteen. For $W = F/4$ we have

$$\mathrm{N}\, W = 1, \quad W < 121/8, \quad 0 < W' < 1/8, \qquad W = (a + b\sqrt{17})/2, \quad 1 \le b \le 3.$$

None admits an integer a in $0 < a - b\sqrt{17} < 1/4$. The final split row is impossible.

Proposition 26.1. *Whole D_{16} cannot occur at the original farthest vertex of a compact arithmetic Coxeter 16-polytope.*

Proof. The actual reduction gives the necessary quadratic triple. Its head norm is odd or even. The odd case and every inert even case were excluded in Section 24; Sections 25 and 26 exclude ramified and split even cases. These exhaust quadratic fields. Each contradiction uses a necessary actual cycle, full field-square condition or one paid original-vector norm. No D16 census is used. □

Proof of Theorem 19.1. The complete acyclic implication is given in Section C.14.4: the independent general compact companion bounds n by seventeen, the rank seventeen dispatch excludes that dimension, and the fresh rank sixteen cover and manual whole-D_{16} obstruction exclude dimension sixteen. All steps retain the original farthest vertex and the stated external and finite-output inputs. □

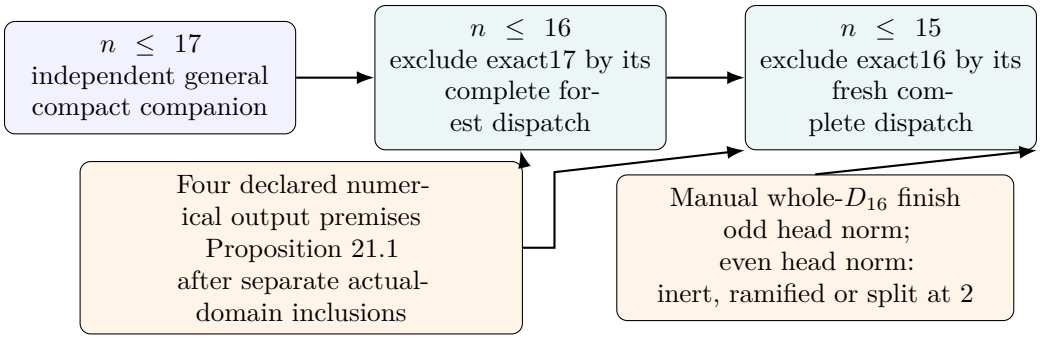


Figure 6: The proof of Theorem 19.1: the compact bound and the separate exclusions in dimensions 17 and 16.

27 Supplementary verification

The finite algorithms, their domains, completeness arguments and outputs are stated in the relevant parts of the paper. Accompanying code, finite certificates and verification records are available in the repository [13].

Selected Lean 4 modules verify internal logical and algebraic deductions. The geometric statements, published results and specified finite-data outputs are supplied as explicit premises.

Part IV
Proof appendices: finite-volume polytopes

A.1 Geometric coverage and exact local certificates

The finite objects in this appendix are marked Gram diagrams and certificates for their extensions. They are not a finite list of all supporting forests. The extension arguments are what allow arbitrarily many additional supporting roots. We first describe the domains and checking predicates, then state the recorded finite propositions. The same organization applies to the preceding rule and to the final rule, with the final guarded consumer treated in the body of the paper.

A.1.1 The complete core domain

Generate the connected affine diagrams of ranks four through eight: the four classical families in each rank, together with $\widetilde{F}_4, \widetilde{E}_6, \widetilde{E}_7, \widetilde{E}_8$. These are 24 sources. To each source on ℓ roots attach a root by every nonzero contact vector in $\{0, 1, \sqrt{2}\}^\ell$. Keep an extension exactly when it is indefinite and every one-root deletion is a finite or affine forest. Identify labelled isomorphic results, and retain every ordered pair of marks whose two deletions are positive definite. This is a finite exact algorithm; isomorphism testing may be done by trying all permutations of at most ten roots.

It is complete. A minimal indefinite core of order at least six cannot be a compact Lannér diagram. It has a proper affine block; the positive kernel and connectivity show that adjoining one adjacent root already makes an indefinite block. Minimality therefore makes this the whole core. A label at least five would occur in a connected proper five-root subdiagram, forbidden by finite/affine classification. A parallel pair with a third contacting root is already indefinite, and a dotted pair is indefinite alone. The contact alphabet is therefore a consequence of classification, rather than a cutoff. Checking all one-root deletions checks every proper principal block.

For orders four and five generate every connected graph with labels respectively 2 through 6 and 2 through 5, and apply the same acceptance and marking tests. These alphabets follow from the finite/affine three- and four-root lists. Order-two dotted cores are analytic cases. For an order-three core retain the full Gram family

$$\begin{pmatrix} 2 & -x & -u \\ -x & 2 & -v \\ -u & -v & 2 \end{pmatrix}, \qquad x = 2\cos(\pi/m),$$

including the parallel endpoint $x = 2$. The black–black label is unbounded; it is not enumerated up to a numerical maximum.

A.1.2 All unions and all additional cross bonds

In a marked core C put $\delta_C(v) = \min(d(v, a), d(v, c))$. The high-core catalogue has $\delta_C \le 7$. In the branch with no adjacent marked path of length at most eight, a shared white pair (v, w) must satisfy $\delta_C(v) + \delta_D(w) \ge 9$. Generate every injective partial matching of white nodes satisfying this condition and preserving all labelled edges and nonedges. No black node is identified. This bounded matching loop includes failed identifications.

In the resulting union test all remaining permitted cross bonds and all permitted multi-position subsets. A cross bond at too small a combined distance gives an earlier short path. The inverse

tests on the positive white blocks and actual black deletions give a finite native alphabet at every other position. For example a bond of magnitude b between two positive white blocks requires

$$b^2(G_{C,W}^{-1})_{vv}(G_{D,W}^{-1})_{ww} < 1.$$

Inverse positivity makes the corresponding necessary one-contact bounds valid even in the presence of further nonnegative contacts.

A pruning certificate must exhibit one of the following: a nonzero nonnegative vector of nonpositive square in a block required positive definite; a vector of strictly negative square in a corner required positive semidefinite; negative index at least two; or a retained adjacent marked short path. A nonnegative-vector witness at minimum native magnitudes remains valid when further contacts are added, since its square can only decrease. Negative index by itself is not used as a monotonicity assertion for further bonds; those branches are enumerated separately.

The unbounded order-three clause has an exact symbolic certificate. Its only surviving high port has inverse diagonal zero. Eliminating the high core gives zero Schur correction to the displayed small Gram for every physical x, u, v. Its negative direction then supplies a second negative direction. This disposes of every finite label and the parallel endpoint at once.

A.1.3 What an extension certificate means

The common geometric lemmas in Section 3 give the following complete meanings to the certificate types. A radical certificate lists the possible contact positions and a nonzero full-core kernel vector. Since its represented vector is zero, every new contact annihilates it. If all allowed coefficients are nonzero, a nonempty nonnegative contact must meet both sign sets. The certificate lists the resulting short paths. Otherwise every extra root is orthogonal to the core, including the first root of any indirect extension.

An inverse certificate gives the physical core inverse J. A new norm-two root with contact column q has residual square $2 - q^t Jq \geq 0$ in a positive-definite complement. At equality its residual vector is zero, so all further products are fixed. For two prospective contacts at i, j, compare J_{ij}^2 with $(2 - J_{ii})(2 - J_{jj})$; Cauchy either excludes them or forces the displayed bond and short path. Multiple-contact exclusions use nonnegative inverse entries in the appropriate end block.

An attached-forest certificate gives a single remaining old contact position p with $J_{pp} = r > 1$. The argument of Lemma 4.2 bounds the entire new rank by $1/(r-1)$, including indirect tails. A protected-role certificate instead forbids a first new contact at specified incident neighbors, thereby preserving a feature or a leaf of the *whole* component. An affine certificate supplies a positive kernel at an actual cusp; all further supporting contacts are orthogonal to that entire factor.

Each certificate is checked in the physical Gram normalization. Positive diagonal congruences may be used only with their norm scales recorded. An unbounded parameter is handled by a proved symbolic identity or a polynomial nonnegativity certificate on its whole stated range, never by sampling.

Lemma A.1.1 (A terminal affine-B tail)**.** *Let K be a nondegenerate Lorentzian subspace of the physical root space. Suppose that a root e of squared norm 2 has orthogonal residual $e_\perp$ with $e_\perp^2 = \rho < 1$. There cannot be a chain $f_1, \ldots, f_k$ of roots in $K^\perp$, each of squared norm 2, with $k \geq 1$ and the following native products. For $k \geq 2$, consecutive products are -1, except for $(f_{k-1}, f_k) = -\sqrt{2}$; also $(e, f_1) = -1$. For $k = 1$, instead $(e, f_1) = -\sqrt{2}$. All other products between distinct roots in the chain, and between e and f_j for $j > 1$, are zero.*

Proof. The complement $K^\perp$ is positive definite. Put

$$v = f_1 + \cdots + f_{k-1} + \frac{f_k}{\sqrt{2}}.$$

For $k \geq 2$, its diagonal terms sum to $2(k-1)+1$ and its consecutive cross terms sum to $-2(k-1)$. Thus $v^2 = 1$. For $k = 1$ the same identity follows from $f_1^2 = 2$. The prescribed products give $(e, v) = -1$ in both cases. Since $v \in K^\perp$, positive-space Cauchy gives the contradiction

$$1 = (e_\perp, v)^2 \leq e_\perp^2 v^2 = \rho < 1.$$

Only the coefficient of the final root has changed: every physical root still has squared norm 2. □

This lemma treats a terminal chain of any length, including cases where the first two new roots alone do not violate Cauchy. Its orthogonality hypothesis must be proved before it is used. After excluding an adjacent white-interior path of length at most eight, the complete contact certificate permits a unique first root, or permits that root together with one exterior root. In the latter case the full no-bridge estimate bounds the whole component in the presence of the exterior root; that bound is strictly smaller than the actual target order. This excludes the exterior root and every indirect bridge. First-contact uniqueness then proves successively that all later roots are genuinely new and orthogonal to the old core. Their native zero products can be used only after this step. If the sole old-core contact of the first root is at i with magnitude 1 and $J_{ii} > 1$, the inverse certificate gives $\rho = 2 - J_{ii} < 1$.

The exact application checks the complete supplied terminal-suffix map, all its edges and nonedges, and the original physical contact-profile selector. A contact theorem computed for another target is used only when its original core, ordered corner and full contact hypotheses coincide; its old target map is never relabelled as the present one. The terminal congruence in the calculation of v^2 is a rational check, with the scale $1/\sqrt{2}$ recorded explicitly.

For every bounded whole component, enumerate each classical target at every allowed rank or order, every exceptional target at its rank, and every induced labelled embedding of its displayed subdiagram. Assign target vertices successively and check each edge and nonedge against the earlier assignments. Finite target size proves termination; deleting the last assigned vertex proves completeness by induction. The marked images select individual entries of Appendix A.2, including zeros and fractions. The displayed whole-component bound, rather than an arbitrary search cap, proves that every actual completion is represented.

Proposition A.1.2 (Recorded core and extension calculations)**.** *The preceding bounded algorithms have the following exact outputs. The high-core generator tests 146 262 attachments and gives 26 isomorphism classes, with order counts* $12, 3, 4, 4, 3$ *in orders 6 through 10 and 768 ordered finite-deletion markings. The low order-four and order-five counts are 32 and 14. The complete high-overlap domain has 82 197 consistent matchings; 75 947 inverse and 3636 edge prunings leave 2614 contexts on 154 physical cores. They have respectively 1329 radical, 823 positive-residual, 259 bounded-component, 120 isolated-inverse, 74 protected and 9 adaptive certificates. All free cross bonds are discharged by 154 502 short-path and 5528 edge witnesses.*

Every disjoint high/high union is excluded, including all 10 990 multi-position subsets. The low/high domain is excluded or gives a whole component of order at most twelve or an independent short path. Bounded affine target searches use all 1153 eligible patterns in orders 14–17, with whole-bound multiplicities $313, 354, 346, 140$*. The exact extension certificates satisfy the predicates just specified. In each retained terminal configuration their chosen* current *corner contributions sum to at least one.*

This is a computed proposition, supported by the separate exact local certificates in the reproducibility package. Its assertion includes the receipt identities and inequalities, not merely the listed counts. The finite orders, alphabets, matchings and target bounds prove termination. The core, union and extension arguments prove coverage. Successful rows alone would prove neither.

A.1.4 Triangles and their four exceptional markings

Lemma A.1.3 (The reduced small affine pairs are absent from triangles)**.** *No actual triangular two-face has an ideal corner whose varying affine factor and marked pair are one of the following:*

1. *a cycle of order ten, marked at antipodal nodes;*
2. *a cycle of order eleven, marked at distance five;*
3. $\widetilde{D}_9$ *of order ten, marked at leaves of opposite forks.*

This assertion allows arbitrary additional supporting roots and other factors of the cusp.

Proof. Let a, b be the two marked roots at the ideal corner, let F be its varying affine factor, and let c be the third side root of the triangle. Normalize every root to square two. All nodes of $F \setminus \{a, b\}$ support the triangle. The other two actual corner diagrams are positive semidefinite, so both principal sets $(F \setminus \{a\}) + c$ and $(F \setminus \{b\}) + c$ are positive semidefinite.

For either positive-definite deleted-factor matrix B, put $p_j = -(c, r_j) \geq 0$. Schur complementation gives

$$p^T B^{-1} p \leq 2.$$

The inverse is entrywise nonnegative, and every nonzero native contact has magnitude at least one. Consequently a position with $(B^{-1})_{jj} > 2$ cannot be contacted. This is a necessary condition on a principal submatrix; extra supporting roots cannot repair its failure.

Number a cycle of order m by $0, \ldots, m-1$, and take $a = 0, b = 5$. Deleting a node gives A_{m-1}, whose inverse diagonal at path position k is $k(m-k)/m$. For $m = 10, 11$, all positions at least three steps from both path ends have diagonal at least $21/10, 24/11$, respectively, both greater than two. The two deleted-factor tests therefore give these possible contact sets:

m	after deleting 0	after deleting 5
10	$\{1, 2, 8, 9\}$	$\{3, 4, 6, 7\}$
11	$\{1, 2, 9, 10\}$	$\{3, 4, 6, 7\}$.

In each row the sets are disjoint. Contact to either removed mark is forbidden by the other deletion: the relevant diagonal is $5/2$ at order ten and $30/11$ at order eleven. Thus c is orthogonal to every root of F.

For the last case number F as the spine $0, \ldots, 5$, with leaves $6, 7$ at 0 and $8, 9$ at 5, and take $a = 6, b = 8$. Each deletion is finite D_9. After deleting 6, its only inverse diagonals at most two occur at the long endpoint 7 and its neighbor 0, with values one and two. After deleting 8, they occur at $9, 5$ with the same values. The short-leaf diagonals are $9/4$, and all other diagonals exceed two. Hence the possible contact sets are $\{0, 7\}$ and $\{5, 9\}$, which are disjoint. The other deletion also forbids contact to each removed mark, again by its short-leaf diagonal $9/4$. Thus c is orthogonal to all of F in this case as well.

Let $\nu = \sum_{j \in F} \delta_j r_j$, with every $\delta_j > 0$, be the positive affine-kernel combination. It is a nonzero future null vector representing the actual ideal corner: nonzeroness follows by pairing with a strict interior point. The third side of a nondegenerate actual triangle does not contain that corner, so $-(c, \nu) > 0$. The orthogonality just proved instead gives $(c, \nu) = 0$, a contradiction. Rotation and reflection cover all cycle pairs, and fork interchange and spine reversal cover all four opposite-fork pairs. The argument has used no restriction on spectator rank or on the other cusp factors. □

We now compare the inherited triangle witnesses with the actual weights of Appendix A.2. The two cycle orbits above are the only current reductions of ideal paths of length at most five. The reduced order-ten D orbit has distance seven and is also absent from triangles by the lemma. Every other printed replacement in affine B, C, D of orders ten through seventeen has distance at least eight. For larger orders the selected length-six or length-seven feature paths retain their units: their branch or nonsimple-bond witness persists within five steps of a mark. Such a witness cannot lie in an affine A cycle. Every affine factor of order at most nine still pays a unit on every pair.

The four alternate witnesses below have an additional protection. Their fixed simple leaf with a degree-two neighbor excludes every classical affine factor, while exceptional affine factors have order at most nine and retain full units. If the marked component is finite, the protected leaf is at distances $2, 1, 0, 0$ from a mark in the four displayed cases. It remains a whole-component endpoint, so the deep finite deletions, which require both marks to be at least six steps from every finite feature, do not remove the unit. The other selected finite length-six or length-seven witnesses already have a persistent branch or nonsimple feature within five steps. The sparse finite replacements in ranks twelve through seventeen are all longer than these selected paths. Thus the two-unit triangle argument below applies to the current rule, rather than assuming that every historical short ideal pair still has weight one.

A minimal indefinite core of an actual triangle contains all three black roots, since every deletion of one black lies in a positive semidefinite corner. Up to order six, a Steiner tree through the marks gives two distinct corner paths avoiding the third black, each of length at most five. In higher orders generate every marked triple in every core: there are $3\binom{7}{3} + 4\binom{8}{3} + 4\binom{9}{3} + 3\binom{10}{3} = 1025$. Delete the third black when evaluating each corner path and feature. The exact elementary test gives two persistent payments except at four triples in the simple ten-node tree

$$0{-}1,\ 0{-}2{-}3,\quad 0{-}4{-}5{-}6{-}7{-}8{-}9.$$

Their protected alternate pairs are

Triple	Alternate pair	Length	Fixed leaf and neighbor
$(0, 1, 9)$	$(0, 9)$	6	$3, 2$
$(0, 2, 9)$	$(0, 9)$	6	$1, 0$
$(1, 2, 8)$	$(1, 8)$	6	$1, 0$
$(1, 2, 9)$	$(1, 9)$	7	$1, 0$

In the first, second and fourth cases, the other corner contains affine E_8 on $0, \dots, 8$ with positive kernel $(6, 3, 4, 2, 5, 4, 3, 2, 1)$. Kernel orthogonality forbids all additional supporting contacts there. In the third case the finite E_8 inverse on $0, \dots, 7$ has diagonal $(30, 8, 14, 4, 20, 12, 6, 2)$ and is entrywise nonnegative. A new native contact at any of $0, \dots, 6$ has projection above two; only position 7 can be contacted. The displayed leaf and its neighbor therefore remain protected in every case.

No classical affine diagram has a simple leaf with a degree-two neighbor. Thus the alternate lies in a finite component or in an exceptional affine factor of order at most nine, and its unit persists. The complete elementary test also verifies the persistent feature for all 54 distance-six cases and the distance-seven cases. The deep finite deletions and the sparse long-pair changes affect none of these selected units. This proves triangle payment two.

A.1.5 The order of the quadrilateral arguments

The local statements are composed by complete face inequalities, not by recursively replacing an obsolete corner unit. Write S_k for the assertion that a suitable adjacent marked path of length at most k makes the *whole face* receive at least one, and F_k for the replacement of a deleted deep

finite length-k payment. The assertion S_k does not give unit weight to that particular pair. Direct literal roles and the independent cycle-ten and cycle-eleven arguments first prove S_5. Their second halves occur at distinct corners, and their whole finite leaves and small-factor roles have the current printed weights. The independent near-terminal affine-C zero proofs likewise pay the entire face; their zero corner is never called a unit.

One more independent priority is essential. An opposite-fork pair in the affine D diagram of order ten contributes $27/32$, rather than its historical unit. Its complete four-profile determinant argument has at least one extreme end. With exactly one extreme, full-metric Cauchy either gives a retained short path or makes its entire spectator column empty or a single unit endpoint contact on a whole A_r, $r \le 3$. With two extremes, the difference energy is $7/4$. A close mixed component gives a distinct short-five half. The complete far inverse spectrum leaves whole A_{15} or D_7; their literal corners give a restored A_{17} endpoint, a protected D_{25} unit, or a whole exceptional affine unit. In the unmixed case choose a side of energy at most $7/8$, giving either an empty column or one unit endpoint contact on a whole A_r, $r \le 7$. In either unmixed branch put $r = 0$ for the empty column. If the extreme root's old unit attachment meets the selected leaf, the paying corner has a whole A_{r+1} endpoint and a whole D_9 short leaf. The printed coupon is $5/32$. If instead it meets the old mate, the corner is whole D_{10+r} with a marked short leaf at distance eight, a retained unit for every $0 \le r \le 7$. These are two distinct attachment cases, not two descriptions of the same coupon. The immediate affine alternatives also supply a distinct literal half or $27/32$. Thus every branch pays at least $27/32 + 5/32 = 1$, without recursively treating this corner as a unit.

The computed inputs are the complete far inverse blocks and their strict-gradient inequalities, all sixteen whole finite coupons, and all twelve mixed whole-corner Gram records. Their finite domain is: for an A component, unit difference energy $d(N-d)/N \le 7/4$ and strict separate inverse diagonals imply

$$4 \le d \le 14, \qquad d+2 \le N \le \left\lfloor \frac{4d^2}{4d-7} \right\rfloor .$$

Enumerate every position in those ranges satisfying the strict diagonal tests. Complete finite inverse formulas leave only D_6, D_7 and the eligible E_6 block outside A; other types have no far allowable pair. Nonunit quadrants have their minimum at $(\sqrt{2}, 1)$ or $(1, \sqrt{2})$, by the recorded positive gradients and positive quadratic remainder. Their minima exceed $7/4$. All loops are finite and the inverse bounds prove coverage; the literal Gram records check every actual corner and its whole marked role. These inputs are not inferred from the counts in Proposition A.1.2.

The remaining dependency order is

$$S_5 \longrightarrow F_6 \longrightarrow S_6 \longrightarrow F_7 \longrightarrow S_7 \longrightarrow F_8 \longrightarrow S_8.$$

Here S_6 uses the cycle cases with shorter arc six, including the order-fourteen case, together with F_6. The protected affine-D order-ten terminal of F_7 invokes the independent priority just proved. Before using S_7, prove the cycle arcs $(7, 8)$ rectangle bound separately: a required neighboring contact gives a Cauchy correction on one fixed side of the entire rectangle. The corrected quadratic remains concave in reciprocal heights. This calculation has no F_8 premise, even though its certificate shares a file with that later composition. For arcs $(7, 7)$ the sealed equality corners currently pay $41/64 + 23/64 = 1$, rather than two historical halves. The affine-tree cases use their persistent branch or nonsimple-bond features and the independent small-factor priorities. Finally the strict F_8 corridor ends at S_7, after which the remaining cycle and tree cases prove S_8. All additional supporting contacts are controlled by the whole-component seals, not by an ambient-rank cap. The large finite-D leaf-nine half has its own distinct small/D_{10} half; it is not used as a unit.

For a small affine factor, its positive kernel first splits off the spectator forest. Adjacent black contact or a shared spectator neighbor gives an earlier short path. Otherwise the two spectator columns have disjoint support. The complete inverse-diagonal profiles and four-root Schur determinant give a strict obstruction or a closed equality system. At equality Gram Cauchy determines the active whole spectator types. These are the bounded profile calculations in Proposition A.1.2, with unbounded native magnitudes retained symbolically. A changed affine-B corner invokes its independent full-face theorem before a later affine-D theorem can spend that corner's old half. An immediate affine alternative within one equality receipt contributes its current fraction at a distinct corner and makes no recursive theorem call.

Finally the finite sparse substitutions are single inequalities on all four actual corners. Their relevant contribution lists are $(1/2, 1/4, 1/4, 0)$, $(0, 1/8, 1/4, 5/8)$ and $(0, 1/4, 1/4, 1/2)$, each summing to one. The extension seals prove the whole context witnesses at the positive corners. If a witness is absent, that branch is excluded or independently paid; an induced small subgraph cannot serve as a witness. The full chosen-role checks cover the 216 literal stars, 564 later corners, 24 D_{14} variants and 332 positive D_{12} outputs. Universal outputs outside those fixed Gram lists use their whole-factor profile proofs. In particular, the affine-B singleton consumers of $B_{13}(0, 12)$ have a distinct sealed A_1/D_{12} unit, and the quiet affine-D_{12} consumer of the guarded D_{13} half is proved in the body of the paper.

This order is well founded: independent face payment, shorter-path or protected terminal payment, whole-factor profile system, then joint finite receipt. A joint receipt is one terminal inequality, not a call back to another decreased role. Every actual quadrilateral either ends at an earlier face theorem or reaches a union and extension certificate in the complete generated domain. Its distinct current contributions total at least one. This proves the local coverage used in Proposition 3.4 and, with Theorem 4.1, the final rule. It uses the recorded distributed certificates; no new flattened certificate run is assumed or claimed.

A.2 The intrinsic corner weights

We specify a single function of the actual marked vertex diagram. At a finite vertex let $\mathcal{F}$ be the full spherical forest, with two marked nodes a, b. In one whole component use the same-component entry below. In distinct components use the ordinary cross entry and the stated contextual additions. At a cusp use only the affine table for its varying factor, with the multiplicity proved in Section 3. An omitted role has weight zero.

Number A_r by $0, \dots, r-1$, and B_r by the same path with its last bond doubled. Number D_r by the spine $0, \dots, r-3$ with leaves $L_r = \{r-2, r-1\}$ at the center $r-3$; its long end is 0. For an A_r factor put $E_r = \{0, r-1\}$ and $P_k(A_r) = \{k-1, r-k\}$, removing repetitions. The long ends of E_6 lie on its two length-two arms. The tables are invariant under diagram automorphisms.

A.2.1 Finite same-component entries

Every pair in a component of rank at most 11 has weight one; this includes $I_2(m)$ for every $m \geq 5$. For classical rank $r \geq 18$, a feature is a vertex of degree other than two, or an endpoint of a nonsimple bond. Put $\delta(v) = \mathrm{dist}(v, \mathrm{features})$. Start with one at distance at most eight and at distance nine containing a whole-component leaf. Delete the unit at distance 6, 7 or 8 when both marks have $\delta \geq 6$. In D_r, $r \geq 20$, halve each distance-nine pair containing a short fork leaf. These

are all ordinary large-rank entries. Their sums, with $x_+ = \max(x, 0)$, are

$$B(A_r) = 8r - 34 - (r-18)_+ - (r-19)_+ - (r-20)_+,$$
$$B(B_r) = 8r - 34 - (r-19)_+ - (r-20)_+ - (r-21)_+,$$
$$B(D_r) = 8r - 32 - [r \geq 20] - (r-20)_+ - (r-21)_+ - (r-22)_+.$$

For ranks 12–17 start with all units and make exactly the following replacements. Z, H, Q, O mean $0, 1/2, 1/4, 1/8$; A13 means A_{13}, and similarly for the other labels. Unlisted entries retain one. The B_{17} convention is stated separately.

```
D12 H: (0,10),(0,11).

A13 Z: (0,12).
B13 Z: (0,12).
D13: no ordinary replacement.

A14 Z: (0,12),(1,13),(0,13).
B14 Z: (0,13),(1,13),(2,13),(3,13).
D14 Z: (0,12),(0,13); H: (0,11).

A15 Z: (0,12),(0,14),(2,14),(0,13),(1,14).
B15 Z: (0,12),(0,13),(0,14),(1,13),(1,14),(2,14),(3,14),(4,14).
D15 Z: (0,11),(1,13),(1,14);
    H: (0,12),(0,13),(0,14),(1,12).

A16 Z: (0,12),(0,14),(0,15),(1,14),(1,15),(3,15),(1,12),(3,14);
    H: (0,13),(1,13),(2,14),(2,15);
    O: (0,11),(4,15).
B16 Z: (0,11),(0,12),(0,13),(0,14),(0,15),
       (1,12),(1,13),(1,14),(1,15),(2,13),(2,14),(2,15),
       (3,14),(3,15),(4,15),(5,15).
D16 Z: (0,10),(0,11),(0,12),(0,13),(1,11),(2,12),(2,14),(2,15);
    H: (0,14),(0,15),(1,12),(1,13),(1,14),(1,15),(2,13).

A17 Z: (0,10),(0,11),(0,12),(0,14),(0,15),
       (1,10),(1,11),(1,13),(1,14),(1,16),
       (2,12),(2,14),(2,15),(2,16),(3,15),(4,14),(4,16),
       (5,15),(5,16),(6,15),(6,16);
    H: (0,16),(2,13),(3,14);
    weight7/8: (1,12),(4,15).
D17 Z: (0,10),(0,11),(0,12),(0,13),
       (1,10),(1,11),(1,12),(1,13),(1,14),(1,15),(1,16),
       (2,12),(2,14),(3,15),(3,16),(4,14),(4,15),(4,16);
    H: (0,14),(2,13),(2,15),(2,16),(3,13),(3,14);
    Q: (0,15),(0,16).
```

For B_{17} keep one exactly at distance at most eight, at distance nine containing a whole-component leaf, and at $(2, 11), (3, 12), (4, 13)$. All other entries are zero. The ordinary same-component sums are

r	12	13	14	15	16	17
A_r	66	77	88	100	433/4	453/4
B_r	66	77	87	97	104	105
D_r	65	78	177/2	100	217/2	227/2

For Proposition 3.4 alone restore $B_{13}(0, 12) = 1$, so that its preliminary sum is 78. The final rule uses the zero entry above. This is the only ordinary-table difference in that preliminary calculation.

A.2.2 Ordinary cross-component roles

The table is symmetric in the two marked components. A product of mark sets means every pair in that product. In the first seven rows S is any irreducible component of rank at most five, and all(S) is its node set. Apply those broad rows first, then replace values where a later row applies. Later rows are replacements, not additions; overlaps never disagree. All unlisted roles are zero. An arbitrary $I_2(m)$ is a small rank-two type, but is not an A_2 where that type is expressly named.

First component and marks	Second component and marks	Weight
S,all(S)	D_{10},L_{10}	1/2
S,all(S)	D_{11},D_{12},D_{13},D_{14}, center and short leaves	1/2
S,all(S),rank(S) ≤ 2	D_{15},center and short leaves	1/2
S,all(S),$3 \leq$ rank(S) ≤ 5	D_{15},L_{15}	1/2
A_1	D_{16},center and short leaves	1/2
S,all(S)	A_{13} central node 6 or A_{15} central node 7	1/2
S,all(S)	A_{17},marks5,11	1/2
A_1 or A_2,all marks	D_{11},L_{11}	1
A_4,E_4	D_{11},L_{11}	1
A_5,P_2	D_{11},L_{11}	1
E_6,long ends	D_{11},L_{11}	1
D_5,long endpoint0	D_{11},L_{11}	1
A_1	D_{12},L_{12}	1
D_5,L_5	D_{12},center and short leaves	1
A_2,all marks	D_{13},L_{13}	1
A_2,all marks	D_{13},center10	23/64
A_1 or A_4,E_4	D_{13},mark9	1/2
A_1 or A_2,all marks	D_{14},mark10	1/2
A_1	D_{14},mark9	1/16
A_1	D_{15},mark11	1/16
A_2,all marks, or A_3,central node 1	D_{10},center7	1
A_1 or A_2,all marks, or A_3,E_3 or A_4,E_4	B_9,doubled terminal8	1/2
A_1	B_{10},doubled terminal9	1
A_2,all marks	B_{10},doubled terminal9	1/2
A_1	B_{11},mark9	1/2
A_1	B_{12},mark9	1/2
A_r, E_r, $1 \leq r \leq 8$	D_9,L_9	5/32
A_r, E_r, $1 \leq r \leq 4$	A_9,central node 4	1/2
A_1	A_{11},central node 5	1
A_2,all marks	A_{11},central node 5	1/2
A_1 or A_2,all marks, or A_4,E_4	A_{11},E_{11}	1/2
E_6,long ends, or D_5,long end0	A_{11},P_2	1
A_1	A_{12},marks5,6	1
A_1	A_{12},E_{12}	1/4
A_4,E_4	A_{12},E_{12} or A_{12},P_3	1
D_5,long end0	A_{12},P_2	1
D_5,long end0	A_{12},E_{12}	3/8
D_5,L_5	A_{12},P_3	1
A_1	A_{13},central node 6	41/64
A_2,all marks	A_{13},central node 6	23/64

First component and marks	Second component and marks	Weight
A_1	A_{13},E_{13}	1/2
A_1	A_{13},P_2	1/4
A_2,all marks	A_{13},E_{13}	1
A_4,E_4	A_{13},P_2	1
A_4,internal marks1,2	A_{13},P_2	1/2
A_2,all marks	A_{14},P_2 or A_{14},P_3	1/2
A_1	A_{14},P_3 or A_{14},P_4	1/2
A_1	A_{15},P_3	1/2
A_1	A_{16},P_4	1/2
A_1	A_{16},E_{16}	1/4

For example, the D_5 long-end/D_{12} center entry is one half; the upgrade to one concerns short D_5 leaves. There is no ordinary A_1/B_8 payment.

A.2.3 Contextual entries

Every witness is a distinct unmarked whole component at the actual corner. Presence is Boolean. The first table replaces ordinary same-component values; the second adds to cross roles whose ordinary values are zero.

Marked component and pair	Presence guard	Value
A_{13},(0,12)	A_1 present	1
A_{14},(0,13)	A_1 present	1/2
A_{15},(0,13),(1,14)	A_1 present	1/4
D_{15},(1,13),(1,14)	A_1 present	1/4
A_{16},(1,12),(3,14)	A_1 present	5/8
D_{16},(0,14),(0,15)	A_1 present	1
D_{14},(0,12),(0,13)	A_1 or A_2 or A_3 present	1/2
D_{13},(0,11),(0,12)	at least three A_{15}/D_{15} components	1/2
D_{28},(0,12)	at least two A_{15}/D_{15} components	ordinary +1/2

Marked pair	Presence guard	Addition
A_4 endpoint/A_{12} second node	A_1 present	1/8
A_2 mark/A_4 endpoint	A_1 and A_{10} present	1/4
A_2 mark/D_5 long endpoint	A_1 and A_9 present	1/2
A_1/D_5 long endpoint	A_{11} present OR both A_9,A_2 present	3/8
A_1/D_5 long endpoint	A_9 and A_2 present	5/8
two distinct A_2 components,any marks	A_{10} and at least two A_1 present	9/32
A_{12} endpoint/D_{16} long endpoint	at least two A_{15}/D_{15} present	1/2

The two A_1/D_5 additions deliberately add to one under the joint A_9, A_2 guard; A_{11} does not repeat the 3/8 term. The A_2/A_2 marks lie in distinct components.

These tables define $b(T)$ and $j(T,U)$ in (3.2): $b(T) = 64B(T)$ sums ordinary same-component entries, and $j(T,U)$ is 64 times the sum of ordinary cross entries on two distinct components. For the preliminary calculation make only the stated B_{13} restoration and omit the three rank-15-guarded

changes. Counting the other contextual pairs gives exactly Ψ in Appendix A.3; for example the A_2/A_2 row costs $64 \cdot 4(9/32)\binom{a_2}{2} = 36a_2(a_2-1)$. The final additional context cost is

$$-64d_{13}[a_{15}+d_{15}\geq 3]+(32d_{28}+64a_{12}d_{16})[a_{15}+d_{15}\geq 2].$$

This identifies one function feeding both geometry and maximization.

A.2.4 Affine factors

Let m be the order, so the affine rank is $m-1$. Number $\widetilde{A}$ as the cycle $0,\ldots,m-1$; $\widetilde{B}$ as spine $0,\ldots,m-3$, double bond $(m-4,m-3)$, and leaves $m-2,m-1$ at 0; $\widetilde{C}$ as the path $0,\ldots,m-1$ with both end bonds doubled; and $\widetilde{D}$ as spine $0,\ldots,m-5$ with leaves $m-4,m-3$ at 0 and $m-2,m-1$ at $m-5$. In the following table A, B, C, D mean these affine families.

Every pair in order at most nine has weight one, including every exceptional affine type. For $\widetilde{A}$ at order ten halve its five antipodal pairs; at order eleven halve its eleven distance-five pairs; at every order at least twelve retain exactly distance at most five. For B, C, D at orders 10–17 start with all units and make these replacements. XxY is a Cartesian product.

```
m10:
 B H:(7,8),(7,9).
 C Z:(0,9).
 D weight27/32: {6,7}x{8,9}.

m11:
 B Z:(8,9),(8,10).
 C Z:(0,9),(0,10),(1,10).
 D H:{7,8}x{9,10}.

m12:
 B Z:(0,9),(9,10),(9,11); H:(8,10),(8,11).
 C Z:(0,10),(0,11),(1,10),(1,11).
 D Z:{8,9}x{10,11}.

m13:
 B Z:(0,10),(9,11),(9,12),(10,11),(10,12);
   H:(1,10),(8,11),(8,12).
 C Z:(0,10),(0,11),(0,12),(1,11),(1,12),(2,10),(2,12).
 D Z:(0,8),{9,10}x{11,12};
   H:(0,11),(0,12),(8,9),(8,10).

m14:
 B Z:(0,10),(0,11),(1,10),(1,11),(2,11),
     (9,12),(9,13),(10,12),(10,13),(11,12),(11,13).
 C Z:(0,10),(0,11),(0,12),(0,13),(1,11),(1,12),(1,13),
     (2,11),(2,12),(2,13),(3,13).
 D Z:(0,9),(0,12),(0,13),(9,10),(9,11),{10,11}x{12,13};
   H:(1,12),(1,13),(8,10),(8,11).

m15:
 B Z:(0,11),(0,12),(1,11),(1,12),(2,12),(3,12),
     {9,10,11,12}x{13,14}.
 C Z:all path pairs of distance>=10 except (2,12).
 D Z:(0,10),(0,13),(0,14),(1,13),(1,14),
```

```
     (9,11),(9,12),(10,11),(10,12),{11,12}x{13,14};
   weight15/16:(2,13),(2,14),(8,11),(8,12).

m16:
 B Z:all pairs of graph distance>=10.
 C Z:all path pairs of distance>=10 except (1,11),(4,14).
 D Z:(0,11),(0,14),(0,15),(2,14),(2,15),(9,12),(9,13),
     (11,12),(11,13),{12,13}x{14,15};
   H:(0,9),(0,10),(1,11),(2,11);
   weight7/16:(1,14),(1,15),(10,12),(10,13).

m17:
 B Z:all pairs of graph distance>=10.
 C Z:all path pairs of distance>=10.
 D Z:(0,10),(0,12),(0,15),(0,16),(1,11),(2,12),
     (3,15),(3,16),(9,13),(9,14),(12,13),(12,14),
     {13,14}x{15,16};
   H:(0,11),(1,12),(1,15),(1,16),(11,13),(11,14);
   weight1/8:(2,15),(2,16),(10,13),(10,14).
```

For classical affine order at least 18, features are branching vertices and endpoints of nonsimple bonds. A simple endpoint alone is not an affine feature. With δ the distance to this set, keep a unit exactly when

$$d(a,b) \le 5 \quad \text{or} \quad 6 \le d(a,b) \le 9 \ \text{ and } \ \min(\delta(a),\delta(b)) \le 5.$$

All other entries are zero. A featureless cycle retains only distance at most five. There are no ideal context bonuses. Summing each table, taking the family maximum and dividing by order gives exactly $c(r)$ in Section 6; the large-rank counts give its formula for $r \ge 24$. Its geometric use is the factor-fiber argument establishing (3.1).

A.3 Contexts in the finite-forest calculation

This appendix specifies the finite algorithm behind Proposition 3.4. It concerns the rule before the last B_{13} and three-rank-15 changes. All charges are multiplied by 64. Put $a_i = m_{A_i}$ and $d_i = m_{D_i}$, and write $[E]$ for the indicator of a condition E. The complete additional context cost in (3.2) is

$$\begin{aligned}\Psi(m) = {} & 36a_2(a_2-1)[a_1 \ge 2,\ a_{10} > 0] + 64a_2d_5[a_1 > 0,\ a_9 > 0] \\ & + 24a_1d_5[a_{11} > 0 \text{ or } (a_9 > 0,\ a_2 > 0)] + 40a_1d_5[a_9 > 0,\ a_2 > 0] \\ & + 32(a_{15}+d_{15})[a_1 > 0] + 64d_{14}[a_1+a_2+a_3 > 0] \\ & + [a_1 > 0](80a_{16} + 32a_{14} + 64d_{16} + 64a_{13} \\ & \qquad + 32a_4a_{12} + 64a_2a_4[a_{10} > 0]).\end{aligned}$$

Witness presence is Boolean. Additional witnesses multiply a bonus only where an actual multiplicity occurs in this formula.

A.3.1 Representatives and actual fibers

Use E_6 as the rank-six representative and B_9 as the rank-nine representative. At rank 12 identify A_{12}, B_{12} with A_{12}, but keep D_{12} separate. Keep A_{17} separate from the other rank-17 types. At all

remaining ranks at most eight use the type A of that rank; at larger ranks use the type D of that rank. The fixed representatives are generated by the integer domain

$$c_9, \dots, c_{17} \geq 0, \quad \sum_{r=9}^{17} r c_r \leq 58, \quad 0 \leq k_{12} \leq c_{12}, \quad 0 \leq k_5 \leq \left\lfloor \frac{58 - \sum_{r=9}^{17} r c_r}{5} \right\rfloor.$$

They consist of c_9 copies of B_9, c_{10} copies of D_{10}, c_{11} copies of D_{11}, k_{12} copies of A_{12}, $c_{12} - k_{12}$ copies of D_{12}, c_r copies of D_r for $13 \leq r \leq 16$, c_{17} copies of A_{17}, and k_5 copies of A_5. The split k_{12} counts representative A_{12} components: before actual-type expansion it combines actual A_{12} and B_{12}, and is distinct from the actual multiplicity a_{12} in Ψ. The residual ranks are

$$\{1, 2, 3, 4, 6, 7, 8\} \ \cup \ \{17, \dots, 58\}.$$

Thus residual rank 17 represents B_{17}, D_{17}, whereas the A_{17} count is already fixed. These conventions matter for the context bonuses.

A.3.2 The numerical coarse screen

The coarse coefficients differ from the actual coefficients b, j: they include allowances for some presence guards. We define them here, so the preliminary screen is a finite numerical problem as well as a description of its output. Barred quantities are unscaled payments; multiplication by 64 gives integer coefficients. For the fixed multiset F and remaining rank R just defined, put

$$\mathcal{R} = \{1, 2, 3, 4, 6, 7, 8\} \cup \{17, \dots, 58\}, \qquad u_r = \begin{cases} E_6 & r = 6, \\ A_r & r \leq 8, \ r \neq 6, \\ D_r & r \geq 17. \end{cases}$$

The fixed base coefficients are

T	A_5	B_9	D_{10}	D_{11}	A_{12}	D_{12}	D_{13}	D_{14}	D_{15}	D_{16}	A_{17}
$\bar{b}(T)$	10	36	45	55	66	65	78	179/2	201/2	219/2	3613/32

For residual types, $\bar{b}(u_r) = \binom{r}{2}$ for $r \in \mathcal{R}$, $r \leq 8$, while

$$\begin{aligned} \bar{b}(D_{17}) &= 919/8, \\ \bar{b}(D_r) &= 8r - 32 - [r \geq 20] - (r-20)_+ - (r-21)_+ - (r-22)_+ \qquad (r \geq 18), \end{aligned}$$

where $x_+ = \max(x, 0)$. The symmetric interaction table is as follows. Omitted entries are zero, including every interaction of two large fixed types and every interaction involving a residual D_r, $r \geq 17$.

	A_5	B_9	D_{10}	D_{11}	A_{12}	D_{12}	D_{13}	D_{14}	D_{15}	D_{16}	A_{17}
A_1	3/8	1/2	1	5/2	5/2	5/2	137/64	33/16	25/16	3/2	1
A_2	0	1	4	5	0	3	151/32	4	3	0	2
A_3	0	1	4	9/2	0	9/2	9/2	9/2	3	0	3
A_4	0	1	4	8	17/2	6	8	6	4	0	4
A_5	0	5/8	5	19/2	27/4	21/2	15/2	15/2	5	0	5
E_6	0	5/8	0	4	0	0	0	0	0	0	0
A_7	0	5/8	0	0	0	0	0	0	0	0	0
A_8	0	5/8	0	0	0	0	0	0	0	0	0

Thus the A_1/A_5 entry is an unconditional upper allowance, not an ordinary payment. Set

$$K(F) = \sum_{T \in F} \bar{b}(T) + \sum_{\{T,U\} \subset F} \bar{j}(T,U), \qquad w_F(r) = \bar{b}(u_r) + \sum_{T \in F} \bar{j}(u_r, T).$$

Multiset sums count distinct occurrences, including equal types. In particular $K(F)$ includes all interactions of the fixed A_5 copies with the large fixed components. Residual representatives have no mutual interaction. Let V_F be the exact-rank Bellman table with items $(r, w_F(r))$, $r \in \mathcal{R}$, and let $V_F^{\neg 2}$ omit rank two. Both are instances of Proposition 3.3, after scaling by 64.

There remain two context terms. For $s \in \{1, 2\}$ set

$$G_0 = [s = 2,\ c_{10} > 0], \qquad G_1 = [c_9 k_5 > 0].$$

An admissible reservation has $s \le R$, $G_0 + G_1 > 0$, and $s = 2$ only if $G_0 = 1$. For every $0 \le j \le \lfloor (R - s)/2 \rfloor$ form

$$L_{s,j} = s w_F(1) + j w_F(2) + V_F^{\neg 2}(R - s - 2j) + \frac{9}{16} j(j-1) G_0 + j k_5 G_1.$$

Here j counts all rank-two representatives: every actual rank-two type, including B_2 and the common dihedral symbol, maps to $u_2 = A_2$. It can exceed the actual A_2 count. Ignoring unreachable states, the coarse value is

$$U(F) = K(F) + \frac{11}{32} c_{17} + \max\{V_F(R),\ L_{s,j} : (s,j) \text{ admissible}\}. \tag{A.3.1}$$

The $11/32 = 22/64$ increment per fixed A_{17} is separate from $\bar{b}(A_{17})$. These formulas define the recorded 3298-way screen and its 39 classes with $U(F) > 377$.

Here is the complete domination test behind this upper problem. Write $B(T) = b(T)/64$, $J(T,U) = j(T,U)/64$ for the preliminary actual rule, and let ρ be the representative map. Put

$$g(A_{13}) = g(D_{14}) = g(D_{16}) = 1, \quad g(A_{14}) = g(A_{15}) = g(D_{15}) = \tfrac{1}{2}, \quad g(A_{16}) = \tfrac{5}{4},$$

and $g = 0$ otherwise. These are the additional same-component allowances from the presence guards, rather than their full replacement values. Let h be symmetric, with $h(A_1, D_5) = 3/8$, $h(A_4, A_{12}) = 1/2$, and all other entries zero. The finite comparison checks, for every actual type and every actual type pair,

$$\begin{aligned} B(T) + g(T) &\le \bar{b}(\rho T) + \tfrac{11}{32}[T = A_{17}], \\ J(T,U) + h(T,U) &\le \bar{j}(\rho T, \rho U). \end{aligned}$$

This states the full domain and inequalities of the coefficient check; it is not a comparison only of the eventual maximizing forests.

Outside the two separately bounded families below, the remaining contexts are exactly the A_2/A_2 and A_2/D_5 terms in $L_{s,j}$. If the first occurs, reserve two A_1 and choose $s = 2$; this also includes the second term whenever it occurs. If only the second occurs, reserve one A_1 and choose $s = 1$. Choose j to be the complete rank-two representative count. Actual A_2 counts are at most j, actual D_5 counts are at most k_5, and actual A_9, A_{10} presence implies the corresponding representative presence. Monotonicity of $j(j-1)$ and jk_5 on nonnegative integers proves domination. If neither context occurs, $V_F(R)$ suffices. Further A_1 copies remain available in the residual table; they do not multiply a Boolean witness bonus. The reservations jointly cover every actual forest in this complement; an individual branch need not bound another formal reconstruction. The additional

5/8 on A_1/D_5 under the joint A_9, A_2 guard and the A_1, A_2, A_4, A_{10} family are covered by their separate complete conditional calculations, including overlap. No actual attainment of the coarse upper bound is asserted.

The ordinary component and interaction inequalities give an upper relaxation for every representative choice. The table check uses every actual type, including the common-cost dihedral symbol; it is not a comparison of only the eventual maximizers. It retains the separate increment of 22 per fixed A_{17}. There are 1295 fixed large vectors, 3298 choices after adjoining the rank-five count, and two residual tables per choice, one allowing A_2 and one excluding it. Every such table is checked by Proposition 3.3.

A.3.3 The actual fibers of the remaining classes

Exactly 39 coarse classes can exceed 377, outside the two context families below. Expand *every* component of each such class into its full actual-type fiber. For repeated representatives choose a nondecreasing list of actual types. This enumerates all 748 fixed actual multisets F without losing mixed assignments. Let $R = 58 - \operatorname{rank}(F)$ and enumerate the complete actual counts

$$0 \le a_1 \le R, \qquad 0 \le a_2 \le \left\lfloor \frac{R - a_1}{2} \right\rfloor.$$

Remove both A_1 and A_2 from the residual alphabet. If $a_1, a_2 > 0$ and F contains A_{10}, also exclude residual A_4: those forests belong to the second family below.

At these fixed counts the only remaining presence condition is the D_{14} bonus triggered by A_3, when $a_1 = a_2 = 0$ and F contains D_{14}. Split this case into no A_3, or at least one A_3. In the second branch reserve one A_3 and charge the bonus $64m_{D_{14}}$ once; further A_3 copies remain ordinary residual items. In the first branch exclude A_3 entirely.

For an allowed residual type T, set

$$w_F(T) = b(T) + \sum_{U \in F} j(T, U).$$

Every ordinary interaction between residual types is zero, as checked on the whole residual alphabet. If $a_1 > 0$, a residual A_4 additionally receives $32m_{A_{12}}$. All other terms of Ψ have become fixed. With $\epsilon \in \{0, 1\}$ denoting the reserved-A_3 branch, evaluate

$$\begin{aligned} C_{64}^{\mathrm{ord}}(F) + a_1 w_F(A_1) + a_2 w_F(A_2) + \Psi(F + A_1^{a_1} + A_2^{a_2}) \\ + \epsilon(w_F(A_3) + 64m_{D_{14}}) + V(R - a_1 - 2a_2 - 3\epsilon). \end{aligned}$$

Here C_{64}^{ord} omits Ψ, and V uses the adjusted residual weights. Ignore unreachable final ranks. The two A_3 branches are used only in the specified case, so the displayed Ψ has not already charged their bonus.

Every forest in this domain determines a branch: remove its fixed components, all its A_1, A_2 components, and one A_3 exactly when required. Conversely a reachable residual multiset reconstructs a forest with the displayed cost. This proves completeness and exactness of the refinement. All counts are bounded by rank 58, and each actual fiber has at most eight residual flags and 59 rank states. This also proves termination.

A.3.4 The two separately bounded context families

The extra A_1/D_5 charge under the joint A_9, A_2 guard is bounded in all 759 conditional rows. The checked upper bound is $2837/8 < 377$, with the A_{17} increment included. This is an upper relaxation; attainment is not claimed.

For the family containing A_1, A_2, A_4, A_{10}, fix the same large representatives and rank-five count. At remaining rank R enumerate

$$1 \le a_1 \le R-6, \quad 1 \le a_4 \le \left\lfloor \frac{R-a_1-2}{4} \right\rfloor, \quad 1 \le a_2 \le \left\lfloor \frac{R-a_1-4a_4}{2} \right\rfloor.$$

Apply the no-A_2 table to the remaining rank and add all explicit costs and context terms, including $64a_2a_4$. Permitting extra A_1 or A_4 in that table only enlarges the domain and therefore gives a valid upper bound. This has 539 coarse classes and 37 985 count choices. Refine all 14 classes whose coarse value exceeds 377 into their 77 actual fixed assignments containing A_{10}, and repeat the bounded loops with the actual tables. Every one of the 23 376 refined choices is at most $2861/8 < 377$. This also bounds any overlap with the first family.

The complementary calculation checks 1549 residual tables, 96 528 contextual choices, 758 838 last-item upper inequalities and 31 545 equality predecessors. Its only class above 377 has residual rank zero; its eight actual fibers are precisely those in Proposition 3.4. The finite loop bounds and the preceding removal argument explain why these are all the exceptions, rather than eight examples found by a search.

A.4 The bounded higher-dimensional tail

The purpose of this appendix is to state the mathematical objects behind the higher-dimensional certificates. The rules change only at a small number of ranks. Their shared comparison and algorithms are more useful than the order in which those changes were discovered.

A.4.1 The common comparison and terminal algorithm

Put $d = n-1$ and

$$\beta_n = \begin{cases} n(n-6)/8 & n \text{ even}, \\ (n-1)(n-5)/8 & n \text{ odd}. \end{cases}$$

The incidence argument of Lemma 7.1 applies at every rank:

$$\mathcal{D}_5 \ge \beta_n v_f + \sum_{v \text{ ideal}} \kappa(v) + \Delta_n, \qquad \Delta_n > 0.$$

The weaker bound $\Delta_n \ge 5(n-1)$ suffices here. For a simultaneous rule with finite bound U_n and ideal charge $C(v)$, it is enough that $U_n \le \beta_n$ and $\kappa(v) - C(v) > 0$ at every cusp. Subtraction as in Lemma 2.2 then gives a contradiction, even if there are no ideal vertices.

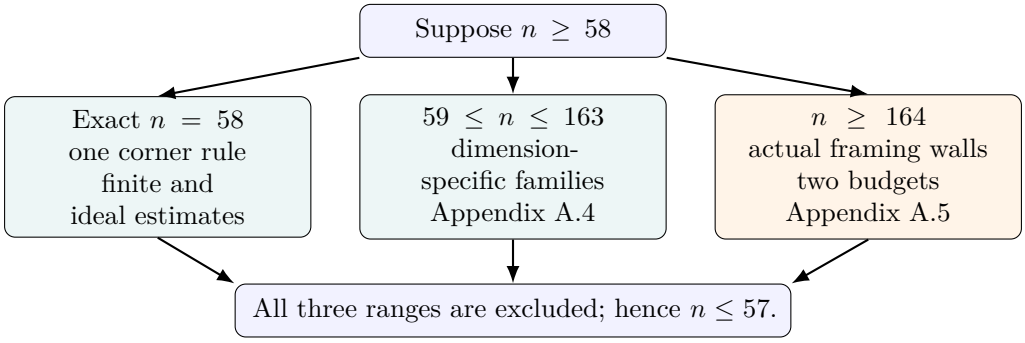


Figure 7: The higher-dimensional exclusions, using the bounded weight families and the narrow-facet argument.

For a cusp profile $\boldsymbol{r} = (r_1, \dots, r_s)$ of total rank d, define

$$H_{\boldsymbol{r}}(t) = \prod_i (1 + \cdots + t^{r_i}) = \sum_j h_j t^j, \quad V_{\boldsymbol{r}} = \prod_i (r_i + 1), \quad G_{\boldsymbol{r}} = \sum_j h_j \min(j, d - j).$$

The normalized surplus associated with an affine-factor envelope c is

$$m_c(\boldsymbol{r}) = \frac{5G_{\boldsymbol{r}} - dV_{\boldsymbol{r}}}{2V_{\boldsymbol{r}}} - \sum_i c(r_i). \tag{A.4.1}$$

The factor-fiber theorem gives $C(v) \le V_{\boldsymbol{r}} \sum_i c(r_i)$. This multiplicity is a geometric statement; a graph pair count alone would not establish it.

Lemma 6.2 permits the move $(a, b) \mapsto (a - 1, b + 1)$, $1 \le a \le b$, whenever $c(a - 1) + c(b + 1) \ge c(a) + c(b)$. The normalized curvature, and hence (A.4.1), does not increase. The product of the two link vertex counts decreases, since $(a + 1)(b + 1) - a(b + 2) = b - a + 1 > 0$. Every move raises the integer sum of squared ranks by $2(b - a + 1)$, while that sum is at most d^2. Thus concentration halts. A positive terminal margin extends to every original cusp. If an absolute surplus is used, its normalized positivity is checked before multiplying by the larger original vertex count.

The terminal enumeration is the sorted-partition recursion in Section 6, with the following additional multiplicity conditions. For a linear/cap envelope $c(r) = \min(r/2, L/2)$, terminal profiles have arbitrary copies of L, at most one part below L, and at most one above L. For a low-only reduction, they have at most one part below L and arbitrary parts at least L; high parts are not concentrated. In a piecewise rule the freely repeated ranks are listed below, and each other named exceptional rank occurs at most once. Adjoin each part only if these conditions survive. The remaining rank decreases at every recursive call. The unique first part of a sorted partition proves completeness and uniqueness.

For each profile compute H by integer convolution and V independently by its product. Check the degree, symmetry and coefficient sum; compute G from its weighted sum and independently from cumulative probabilities of the sum of uniforms. Then evaluate (A.4.1) exactly. A certificate verifies equality of its profile domain with this recursion, every coefficient identity and every positive margin. An advertised minimum also carries an attaining profile. These are finite algorithms with proved domains, not a search for a favorable cusp decomposition.

Finite-vertex costs use the same typed-forest algorithm as Appendix A.3. The fixed counts and residual state retain every multiplicity or guard that affects a later charge. All last-item inequalities are checked, and equality predecessors are supplied whenever an exact maximum is asserted.

A.4.2 The elementary families in dimensions98–163

A feature is a node of degree at least three or an endpoint of a nonsimple bond, including a parallel bond. Distances are graph distances. A condition on a whole component always concerns the whole actual endpoint diagram.

For $98 \le n \le 108$, finite weights are one at distance at most9 or in a whole component of order at most17. Their total is at most $9n$. At a cusp, pay one at distance at most8, in a whole component of order at most17, or at distance9 if a mark is within14 steps of a feature. These are alternative conditions for one unit, not separate charges. For an affine component of order $m \ge 18$, the exact counts are

A	$8m$
C	$\min(9m - 45, 8m - 4)$
B	$\min(9m - 44, 8m - 2)$
D	$\min(9m - 43, 8m)$.

For example, a C spine contributes $8m - 36$ short pairs; the extra distance-nine starts occupy the first16 and last16 available positions, giving $\min(m - 9, 32)$. In B the spine windows have lengths15 and16 and the leaves add two pairs. In D both windows have length15 and the leaves add four pairs. The union of the first a and last b positions in a list of length h has size $\min(h, a + b)$; this also covers overlapping windows. Thus these are all-rank counts. Orders at most17 cost at most $\binom{m}{2} \le 8m$, so $c(r) = \min(8, r/2)$.

For $k = 9, \dots, 13$, use unit weights at distance at most k or in whole components of order at most17, at both finite and ideal vertices. The finite bound is kn and $c(r) = \min(k, r/2)$. For $k = 14$ the distance condition alone suffices. The complete local replacement theorem for these distance families is the core-routing theorem of Appendix A.1, with the corresponding retained short-path threshold. It includes overlapping cores and arbitrary additional supporting whites. The affine graph count does not prove that local theorem.

The terminal algorithm above, with $L = 2k$, gives the following exact positive minima over the indicated dimensions and their entire terminal domains:

n	U_n	$c(r)$	$\min m_c$
98 : 108	$9n$	$\min(8, r/2)$	112101/1419857
109 : 116	$9n$	$\min(9, r/2)$	45873/2476099
117 : 126	$10n$	$\min(10, r/2)$	145234/4084101
127 : 136	$11n$	$\min(11, r/2)$	40603/559682
137 : 146	$12n$	$\min(12, r/2)$	172096/1953125
147 : 156	$13n$	$\min(13, r/2)$	830407/9565938
157 : 163	$14n$	$\min(14, r/2)$	1476616/20511149

In every row $\beta_n > U_n$. This proves the exclusion of 98–163.

A.4.3 Labelled rules for dimensions66–97

We give the marked rules, not just their component sums. The finite supplement tables used here are lists of unordered node pairs. They have a mathematical generation algorithm stated below.

Number an affine component of order m as follows. Type A is the m-cycle. Type C is the path $0, \dots, m - 1$, with label4 on the first and last bonds. Type B has spine $0, \dots, m - 3$, its last bond labelled4, and simple leaves $m - 2, m - 1$ at 0. Type D has spine $0, \dots, m - 5$ and leaves $m - 4, m - 3$ at 0 and $m - 2, m - 1$ at $m - 5$. All other bonds are simple and all unlisted pairs are orthogonal. Finite A_r, B_r use the path $0, \dots, r - 1$, with only the last B bond doubled; finite D_r has spine $0, \dots, r - 3$ and leaves $r - 2, r - 1$ at $r - 3$. Positions are zero-based.

Let q_b pay one at distance at most b, or at distance $b+1, \dots, 9$ with a mark within b of a feature. Let $P_b(m, X)$ be its preferred-image completion in affine type X, with the validated removals described below. At all other marked pairs the weight is zero. The small ideal tables are:

n	all pairs paid	exceptional tables	larger orders
92 : 97	$m \le 16$	$P_7(17, X)$	q_7
87 : 91	$m \le 15$	$P_7(16, X), P_7(17, X)$	q_7
81 : 86	$m \le 14$	$P_6(15, X), P_6(16, X), P_6(17, X)$	q_6
78 : 80	$m \le 13$	$P_5(14, X), \dots, P_5(17, X)$	q_5.

The totals, in the order A, B, C, D, are

$P_7(17)$	119	111	108	127
$P_7(16)$	112	107	103	111
$P_6(15)$	90	100	95	96
$P_6(16)$	96	107	103	111
$P_6(17)$	102	111	108	127
$P_5(14)$	70	87	84	82
$P_5(15)$	75	100	95	96
$P_5(16)$	80	107	103	111
$P_5(17)$	85	111	108	127.

These totals check the tables; they do not define their marked entries.

Write R_9 for unit finite weight at distance at most9 or whole rank at most17. For81–97 add a half between any mark in a whole component of rank at most4 and a mark in $D_{12}, \dots, D_{16}$ at its fork center or short leaf, or in A_{13}, A_{15} at its middle. The full cost is at most $9n$ for $n \le 108$. This slightly enlarges the earliest cross rules, preserving every local payment.

Write R_8 for unit weight at distance at most8, at distance9 with a whole-component leaf mark, or in whole rank at most17. Let F_8 add a half between any mark in whole rank at most5 and a mark of $D_{11}, \dots, D_{16}$ at its fork center or short leaf, of A_{13}, A_{15} at its middle, or of A_{17} at position5 or11. Also add a half between an A_4 endpoint and the unique long-arm neighbor of the D_{13} fork center. Count each marked pair once. The complete finite bound is $8n$ for $n \le 80$.

For78–80 use F_8 enlarged from R_8 to R_9. The added nonleaf distance-nine pairs number $r - 11$ in a rank $r \ge 18$ path and $r - 12$ in D_r, at most r. Ranks at most17 already pay all pairs. Thus its cost is at most $9n$. The D_{11} and A_{17} coupons must be retained: the ideal-five cycle replacements can require them. For76–77 use F_8 itself. For70–75 also upgrade the half for A_1/D_{12} at a short leaf to1, and add a half for either A_2 mark paired with the short terminal of B_{10} incident to its doubled bond. The bound remains $8n$ for $n \le 75$.

For69 additionally delete the finite A_{17} endpoint pair, upgrade A_1/D_{11} short-leaf pairs to1, and pay1 for A_1/A_{11} or 1/2 for A_2/A_{11} at the A_{11} middle. Its finite maximum at 69 is544. For67–68 replace the rank 17 same-component tables by $T_{17}(A), T_{17}(B), T_{17}(D)$ of costs114,105,127 defined below; the maxima become504,512. For66 also change short-leaf distance-nine pairs in whole D_r, $r \ge 20$, from1 to 1/2, add a half between whole rank at most5 and a short leaf of D_{10}, and upgrade A_2/D_{11} short-leaf halves to1. Its maximum is495. All these are actual whole-component conditions. A representative count may allow B_{17} the larger D_{17} cost127, but may not identify the coupon-bearing A_{17} with those types.

For the ideal weights below78 start with P_5 at orders14–17, all pairs at orders at most13, and q_5 at orders at least18. Apply the following setting operations for the listed dimensions:

n	Patch
66–77	D,order 17: set $(1, 11)$ to0.
66–75	B,order 15: set $(0, 11), (11, 13), (11, 14)$ to0.
66–74	Use the order 13 tables below; set H_{17} to 1/2.
66–71	D,order 16: set H_{16} to 1/2.
66–70	B,order 14: set $(0, 10)$ to0; B,order 15: set $(1, 11)$ to0.
66–69	Use the order 12 tables below.
66	B,order 12: set $(9, 10), (9, 11)$ to0; C,order 12: set $(0, 10), (1, 11)$ to0; B,order 14: set $(10, 12), (10, 13)$ to0.

Here

$$\begin{aligned} H_{17} = \{&(0,10),(2,12),(0,11),(1,12),(1,15),(1,16),\\ &(11,13),(11,14),(2,15),(2,16),(10,13),(10,14)\},\\ H_{16} = \{&(0,10),(1,11),(1,14),(1,15),(10,12),(10,13),\\ &(2,14),(2,15),(9,12),(9,13)\}. \end{aligned}$$

At order 13 use cycle distance at most5 in A; all pairs except $(0,10),(10,11),(10,12)$ in B; all except $(0,10),(0,11),(0,12),(1,12),(2,12)$ in C; and all except $\{9,10\}\times\{11,12\}$ in D. The costs are65,75,73,74. At order 12 use cycle distance at most5 in A; all pairs with halves at $(9,10),(9,11)$ in B; all except $(0,11)$ in C; and all except $\{8,9\}\times\{10,11\}$ in D. The costs before the 66 patch are60,65,65,62 and after it 60,64,63,62. Orders at most11 remain wholly paid. Each marked set is invariant under the labelled target's automorphisms; equivalently apply it under every identification with that target.

A.4.4 Finite-table generation and verification

A preferred record consists of two labelled minimal cores, their induced overlap, the two marked black pairs, the selected connected corner component J, its two marks, and its proved whole-completion order bound $h \le 17$. Appendix A.1 defines the admissible core domain and its selected routes. For every record with $h \ge m$ and $|J| \le m$, enumerate all induced labelled injections into the target of order m. Assign an unassigned pattern node and try every unused target node having the correct labels to all nodes already assigned, including zero labels. The recursion has depth at most17 and branching at most17. Each injection has a unique branch, so induction proves termination, completeness and absence of missed orientations. Take the union of marked images, not their multiplicity.

The table $P_b(m,X)$ is this union with q_b, followed by the validated cycle removal above distance b and, in the applicable D targets, removal of

$$\{p : \operatorname{dist}(p) \ge m-4\} \ \cup\ \{\text{the pair of degree-three nodes}\}.$$

The latter has nine pairs. The finite supplement contains the labelled pattern records and paid-set records (b,m,X,P), where P is a list of distinct unordered pairs $0 \le i < j < m$. Missing entries mean zero. The preferred-image algorithm and the setting operations above therefore specify the actual weights without executing an import.

The initial ideal-five domain has949 overlap records,571 with bound at least14. The distance-eight domain has2614 records,1153 with bound at least14; its entire marked image union lies in those same paid or validated-removal sets. It changes no table. Ideal-six uses437 eligible records; ideal-seven uses260 at order 16 and80 at order 17. These numbers are consistency checks, not completeness proofs in place of the stated domains.

For T_{17} take the union of R_8's short/leaf-nine pairs, all bound17 selected marked images, and all marked images whose retained replacement is a unit in the complete literal-interface ledger. Then delete the A_{17} endpoint pair by its separate validated transfer. It occurs in neither the basic nor the selected high-pattern union; all literal totals are recounted after deletion. This finite definition is stronger than merely recording the costs114,105,127. Equivalently its complete literal description is the basic short/leaf-nine set together with

$$\begin{array}{c|l} A_{17} & (2,11),(2,12),(2,13),(2,14),(3,12),(3,13),\\ & (3,14),(4,13),(4,14),(5,14),(0,13),(3,16)\\ B_{17} & (2,11),(3,12),(4,13), \end{array}$$

with unit weights, while D_{17} pays every pair except

$$(0,10),(0,11),(1,10),(1,11),(1,12),$$
$$(1,13),(1,14),(1,15),(1,16).$$

This explicit alternative fixes every marked entry, including old zeros.

The supplement verifier reconstructs all target graphs and all induced injections, compares the resulting pair sets, checks every patch entry and automorphism image, and sums every actual component and cross-role cost. It then checks the Bellman upper inequalities for the complete typed forest domain. Its correctness is the preceding union argument and induction on residual rank. Its halting follows from the explicit order 17 target bound and the finite rank limit of the particular interval. The geometric proof separately shows that every removed preferred or literal image has a retained actual replacement and that every added-white contact is covered; neither the pair-list verifier nor the forest maximum proves that statement.

A.4.5 Envelopes and recorded bounded calculations

For ideal-five factors of order $m \geq 18$, the all-rank counts are

$$\begin{aligned}
p_A(m) &= 5m,\\
p_C(m) &= 5m - 15 + \sum_{k=6}^{9} \min(m-k, 14),\\
p_B(m) &= 5m - 6 + \sum_{k=8}^{11} \min(m-k, 13),\\
p_D(m) &= 5m + 3 + \sum_{k=10}^{13} \min(m-k, 12).
\end{aligned}$$

The same initial/final-window argument used above proves these formulas including overlap. Put $g(r) = \max_X p_X(r+1)/(r+1)$. Their five linear branches have slopes9,8,7,6,5 and give $g(r) \leq 163/23$, with equality at $r = 22$, and $g(r) = (5r+56)/(r+1)$ for $r \geq 24$.

For66–80 use $r/2$ through the stated low rank, the following successive values up to rank 16, and the constant 163/23 at every rank at least17:

n	linear through	successive exceptional values
66	10	16/3, 75/13, 6, 32/5, 107/16, 120/17
67:69	10	65/12, 75/13, 43/7, 32/5, 107/16, 120/17
70	11	75/13, 43/7, 32/5, 107/16, 120/17
71	11	75/13, 87/14, 97/15, 107/16, 120/17
72:74	11	75/13, 87/14, 97/15, 111/16, 120/17
75	12	87/14, 97/15, 111/16, 126/17
76:77	12	87/14, 20/3, 111/16, 126/17
78:80	12	87/14, 20/3, 111/16, 127/17.

For81–86 the envelope is linear through 13, then $20/3, 111/16, 127/17$, then constant 95/13. For87–91 it is linear through 14, then $7, 127/17$, then constant 217/29. For92–97 it is $\min(15/2, r/2)$.

The freely repeated terminal ranks are respectively

n	repeated ranks	at most once
66	$10, 11, 12, 14, 15, 16, 17$	13; one < 10, one > 17
67:69	$10, 11, 13, 14, 15, 16, 17$	12; one < 10, one > 17
70:74	$11, 13, 14, 15, 16, 17$	12; one < 11, one > 17
75:80	$12, 14, 16, 17$	$13, 15$; one < 12, one > 17
81:86	$13, 16, 17$	$14, 15$; one < 13, one > 17
87:91	$14, 16, 17$	15; one < 14, one > 17.

The special moves $2j \mapsto (j-1, j+1)$ have cost gains $11/65$ at $j = 13$ for66; $23/1092$ at $j = 12$ for67–69; $19/182$ at $j = 12$ for70; $16/91$ at $j = 12$ for71–74; and at $j = 13, 15$, respectively, $(4/105, 7/2040)$ for75, $(5/21, 83/408)$ for76–77, and $(5/21, 107/408)$ for78–80. For81–86 the gains at 14,15 are $5/48, 107/408$; for87–91 the gain at 15 is $8/17$. All are positive. The rank 12 duplicate move at 66 is deliberately not used: its gain would be negative.

The recorded exact calculations are:

n	U_n	$\min m_c$
66	495	13367/495924
67:68	504, 512	851/165888
69	544	3641/13608
70	$8n$	1819/51840
71	$8n$	23641/338688
72:74	$8n$	6875/49152
75	$8n$	63/74698
76:77	$8n$	11615/448188
78:80	$9n$	409611/2171546
81:86	$9n$	940675/11294304
87:91	$9n$	146417/911250
92:97	$9n$	10121/4194304.

For an interval the minimum is taken over all its dimensions and all terminal profiles. The finite comparisons satisfy $U_n \le \beta_n$. Consequently the single comparison at the start of the appendix excludes66–97.

A.4.6 The complete ambient59–65 layer table

Write R_N for the rule used at ambient dimension N. Starting from the literal ambient66 rule above, apply the following assignments in the order 65 down to59. An assignment *sets* a weight; it is not a repeated debit. Other entries inherit. In the cross table later rows replace earlier ones, not add to them. There are no context guards until R_{59}. Put $E_r = \{0, r-1\}$ and $P_k(A_r) = \{k-1, r-k\}$.

Finite features in this subsection include endpoints, branch nodes and endpoints of nonsimple bonds in the whole component. Let δ be distance to this set. Outside the small-rank overrides below, the complete changes to the basic distance rule are

N	distance6	distance7	distance8
65	unchanged	unchanged	1/2 if both depths ≥ 17
$64, 63, 62$	unchanged	unchanged	0 if both depths ≥ 11
61	unchanged	unchanged	0 if both depths ≥ 8
60	unchanged	0 if both depths ≥ 6	0 if both depths ≥ 8
59	0 if both depths ≥ 6	0 if both depths ≥ 6	0 if both depths ≥ 8.

In particular the ambient59 distance-eight depth is8, not the depth-six value of the final ambient58 rule.

The finite supplement has complete tables $\mathsf{F}_{N,X,r}$, keyed by ambient layer, finite type and rank. They are lists of records (i, j, w_{ij}) with $0 \le i < j < r$ and exact rational weights; absent pairs mean zero. A retained-pair list instead means its unit indicator. A changed-pair list is only a patch, never a full table. Target graphs and all their zero labels must agree with the canonical labels. These are finite mathematical arrays, with exact selectors specified in the supplement receipt; one does not call a producer to evaluate a corner. The following assignments either print the full patch or name a complete F table.

At65 set A_{16} pairs $(0,14),(0,15),(1,14),(1,15)$ to zero. Set to zero in B_{16}

$$(0,11),(0,12),(0,13),(0,14),(0,15),(1,12),(1,13),(1,14),(1,15),$$
$$(2,13),(2,14),(2,15),(3,14),(3,15),(4,15),(5,15).$$

Halve D_{16} pairs $(1,12),(1,13),(1,14),(1,15)$ and D_{17} pairs $(2,13),(2,14),(2,15),(2,16)$. Their costs are $A_{16}/B_{16}/D_{16} = 116/104/118$ and $A_{17}/B_{17}/D_{17} = 114/105/125$. At64 set $D_{16}(0,11),(0,12)$ to zero and restore $D_{17}(1,12),(1,13)$ to one. At63 set to zero $A_{16}(0,12),(3,15)$, $D_{16}(0,10),(0,13)$ and $D_{17}(1,12),(1,13)$.

At62 halve $A_{15}(0,13),(1,14)$ and $D_{15}(0,13),(0,14)$, delete $B_{15}(0,14)$, halve $A_{16}(0,11),(4,15)$ and $D_{16}(2,13),(2,14),(2,15)$, and set $A_{17}(1,12),(4,15)$ to halves. In D_{17} set $(0,12),(0,13)$ to zero and $(0,14),(0,15),(0,16),(4,14),(4,15),(4,16)$ to halves.

At61 replace the finite tables for $A_{15}, B_{15}, D_{15}, A_{16}, D_{16}, D_{17}$ by their complete $\mathsf{F}_{61,X,r}$ arrays; all earlier zeros and halves remain part of those arrays. At60 replace the three rank 15 arrays by $\mathsf{F}_{60,X,15}$. For A_{16}, A_{17} take the complete full-payment exchange arrays and set respectively $(0,11),(4,15)$ to $1/8$ and $(1,12),(4,15)$ to $7/8$. For D_{16}, D_{17} take the complete eighth-balance arrays and set respectively $(0,14),(0,15)$ to $7/8$ and $(1,15),(1,16)$ to $1/8$. These setting operations, not the arrays' earlier $5/8, 3/8$ versions, define F_{60}.

At59 use complete F_{59} arrays for all three rank 14 types,$A_{15}, D_{15}, D_{16}, A_{17}$. Relative to R_{60}, set $D_{17}(2,12)$ to $5/8$, $(2,14)$ to0, and $(3,15),(3,16)$ to0. The two disjoint D_{17} patches are accumulated; their separate earlier whole arrays must not overwrite each other. The base costs, useful for checking these full marked arrays, are

N	A_{15}	B_{15}	D_{15}	A_{16}	D_{16}	D_{17}
62	104	104	104	113	225/2	120
61	103	103	103	111	111	119
60	101	97	201/2	441/4	443/4	467/4
59	100	97	100	441/4	439/4	919/8.

At59 the rank 14 costs are $89, 87, 181/2$ in A, B, D, and the A_{17} cost is $451/4$. These sums do not define the marked entries.

Here is the entire accumulated ordinary cross-role update table. Every component named is whole; a small rank-five allowance includes all irreducible types where the broad rule allows them. An entry with a product of mark sets applies to every pair in that product. All unlisted entries remain their earlier values.

N	First marked component	Second marked component	Weight
65	A_1 or A_2, all	D_{14},node10	1/2
64	E_6,long ends	D_{11},short leaves	1/2
64	A_1	B_{10},terminal9	1/2

N	First marked component	Second marked component	Weight
63	A_4, E_4	A_{12}, E_{12}	1
63	A_4, E_4	A_{13}, P_2	1
63	D_5,long0	A_{12}, P_2	1
63	A_2,all	A_{13}, E_{13}	1/2
63	A_2,all	A_{14}, P_2	1/2
63	E_6,long ends	A_{11}, P_2	1
63	A_2,all	B_9,terminal8	1/2
63	A_2,all, or A_3,middle1	D_{10},center7	1/2
62	A_2,all, or A_3,middle1	D_{10},center7	1
62	E_6,long ends	D_{11},short leaves	1
61	A_2,all	A_{13}, E_{13}	1
61	A_1	A_{13}, E_{13}	1/2
61	A_2,all	A_{14}, P_3	1/2
61	A_4, E_4	A_{12}, P_3	1/2
61	A_1	B_{10},terminal9	1
60	A_2,all	D_{13},short leaves	1
60	A_4, E_4, or A_5, P_2	D_{11},short leaves	1
60	D_5,short leaves3,4	D_{12},center9 or leaves10,11	1
60	D_5,long0	D_{12},short leaves10,11	7/8
60	D_5,long0	A_{11}, P_2	1
60	A_4, E_4	B_9,terminal8	1/2
60	D_5,long0	D_{11},short leaves9,10	1
60	A_1	B_{11},node9	1/2
60	all marks, any rank 2–5 component	D_{16},center13 or leaves14,15	0
59	A_2,all	A_{13},middle6	7/16
59	A_2,all	D_{13},center10	7/16
59	A_1	A_{13},middle6	9/16
59	A_1	D_{13},node9	1/2
59	D_5,long0	D_{12},short leaves10,11	1/2
59	D_5,long0	A_{12}, E_{12}	3/8
59	A_1	D_5,long0	3/8
59	D_5,short leaves3,4	A_{12}, P_3	1
59	A_1	$A_{14}, P_3 \cup P_4$	1/2
59	A_1	A_{15}, P_3 or A_{16}, P_4	1/2
59	A_1	D_{14},node9 or D_{15},node11	1/16
59	A_1	A_{12},nodes5,6	1
59	A_1	B_{12},node9	1/2
59	A_1	A_{12}, E_{12} or A_{13}, P_2	1/4
59	A_4,internal1,2	A_{13}, P_2	1/2
59	A_4, E_4	A_{11}, E_{11}	1/2

The D_5 long-end/D_{12} center remains its broad half: the 7/8 at 60, and its replacement half at 59, concern only the short D_{12} leaves. The full D_5/D_{12} costs are therefore 45/4 and 21/2. The D_5/D_{11} cost is 17/2 and the D_5/A_{11} cost is2. These distinctions matter to which actual corner can pay. The 59A_1/D_5 entry is unconditional; the later58 guards are not imported. There is no A_1/A_{16} endpoint quarter here, and the A_4/A_{12} third-node value at 59 is still a half.

At59 add exactly two Boolean context roles: a marked pair in two distinct A_2 components receives 1/8 if an unmarked A_{10} and two unmarked A_1 components are present; an A_2 mark paired with the D_5 long end receives 1/2 if unmarked A_1, A_9 components are present. Multiple witnesses

do not multiply a payment. Their forest totals are

$$\tfrac{1}{4}a_2(a_2-1)[a_1 \geq 2,\ a_{10} > 0] + a_2 d_5[a_1 > 0,\ a_9 > 0].$$

There are no other context terms, in particular no rank 15-presence same-component guards.

For ideal vertices introduce complete supplement arrays $\mathsf{I}_{N,m,X}$ with the same pair schema and zero default. At65 use the order 12–17 arrays of the short-seven small-factor interface. This is a replacement, not the assumption that the older66 order 15 tables already agree. Orders at most11 are all unit, orders at least18 use q_5. At64 halve order 11 cycle distance-five pairs, B pairs $(8,9),(8,10)$, the C endpoint pair $(0,10)$, and D pairs $\{7,8\} \times \{9,10\}$; also halve D,order 17,$(3,15),(3,16),(9,13),(9,14)$. At63 set C,order 11,$(0,10)$ to zero; B,order 12,$(0,9)$ to zero; and D,order 13,$(0,8)$ to a half. At62 set that last pair to zero, set B,order 16,$(1,11),(2,12)$ to zero, delete the four just-halved order 17D pairs, and set B,order 14,$(9,12),(9,13)$ and C,order 14,$(1,12)$ to zero. At61 set B,order 11,$(8,9),(8,10)$ and C,order 11, $(0,9),(1,10)$ to zero, and set D,order 16,$(0,9),(2,11)$ to halves.

At60 use complete I_{60} tables for the type/order pairs

$$(B,12),(C,12),(B,13),(C,13),(D,13),(C,14),$$
$$(B,15),(D,15),(B,16),(D,16),(D,17).$$

The order 17D table is the final rebalance table: its two center pairs $(0,10),(2,12)$ are restored to halves, and $(2,15),(2,16),(10,13),(10,14)$ have weight $1/8$. At59 use complete I_{59} replacements for

$$(B,13),(C,13),(B,14),(C,14),(D,14),$$
$$(B,15),(D,15),(C,16),(D,16),(D,17).$$

Equivalently the latter ideal rule is precisely the intrinsic58 table of Appendix A.2, except that every order 10 pair has weight1. This is checked against the complete snapshots, not inferred from family totals. Orders at most10 remain wholly unit throughout59–65 and orders at least18 keep q_5.

The full finite and ideal arrays above are retained mathematical supplement tables, not descriptions by component cost. Their receipt keys specify type, order/rank and canonical labelled graph, with all edges and nonedges checked. Each finite snapshot is a complete pair-domain replacement, or is followed by the explicitly printed patch. Each array is bound by the corresponding immutable composition's source manifest. The supplement verifier selects by these keys, reads exact rational records, checks the pair domain and default, and applies the stated setting operations. All loops have at most $\binom{17}{2}$ entries per small table, and there are seven layers; hence evaluation halts and determines every marked weight. It is distinct from the proof that an actual face reaches those marked roles.

For reference the small-affine family maxima at orders10–17 are

N	10	11	12	13	14	15	16	17
65	45	55	64	74	84	94	107	119
64	45	109/2	64	74	84	94	107	118
63	45	54	63	147/2	84	94	107	118
62	45	54	63	73	83	94	106	116
61	45	53	63	73	83	94	105	116
60	45	53	62	145/2	82	92	103	229/2
59	45	53	62	143/2	80	367/4	411/4	227/2.

Divide an entry by its order to obtain $c(r)$ at rank $r = m-1$. For59–62 use the exact $g(r)$ above rank 16, with low-nine-only concentration. For63–65 use the constant 163/23 above rank 16; the

freely repeated ranks are9–17 at 63,64 and10–17 at 65, with one lower and one higher part. The earlier all-rank formula gives the remaining entries. The exact arithmetic interface is

n	U_n	$\min m_c$	$\min Vm_c$
59	783/2	1/340	91/2
60	404	31/6000	54
61	420	3/250	64
62	434	2741/371712	73
63	449	3505/201344	139/46
64	464	2385/146432	240/23
65	479	61/1936	445/23.

All $U_n \le \beta_n$, so the common comparison excludes59–65. The local proof uses the same core domain, arbitrary-support seals and actual-corner priorities as Appendix A.1, but with each layer's own marked table. Every newly reduced affine corner, including a nonzero fractional reduction, is handled by its whole-face theorem first. If the actual face has no such corner, its ideal weights agree pointwise with the preceding layer. Finite sparse changes then use their distinct retained whole-component alternatives. Deep corridors use the independent shorter-path route and preserve its feature and leaf outputs. Guards are justified by physical whole-component seals before entering the finite-state count. Thus no final58 restoration or extra coupon is spent in an earlier tail rule.

A.5 The unbounded tail: actual framing walls and two budgets

The range $n \ge 164$ requires a proof at all ranks, not a finite list of partitions. We use Nikulin's narrow-facet theorem [5, pp. 654–655]; its unrestricted statement is also given in [6, Section 4, p. 10]. A finite-volume chamber has a facet F whose actual framing hyperplanes have pairwise distance strictly less than $\operatorname{arccosh} 7$. With outward normals of square2 this implies

$$-(x, y) < 14 \tag{A.5.1}$$

for any two framing walls. This is the external narrowness input. We prove the incidence, endpoint and scale statements needed to apply it; they are not additional hypotheses on the cusp.

A.5.1 Outgoing edges really give framing walls

Suppose first that F contains a cusp v. Its horospherical link is a product of affine Coxeter simplices. Write their ranks r_a, with $\sum_a r_a = N = n - 1$, and normalize every old root to square2. In each factor choose two omitted nodes in advance. In the factor containing the normal α_s of F, both choices avoid s. The two choices need not be distinct. If that factor has affine rank one, use its other node twice; its modified target $2(r_a - 1)/9$ is zero. Deleting one node per factor gives a positive-definite support S of N roots and a vertex of the horospherical link. It therefore defines an actual outgoing edge in F. Finite volume gives a finite or ideal second endpoint.

At a finite endpoint choose any new wall x terminating that edge. It and α_s belong to a spherical diagram, so their rank-two Gram is positive definite. At an ideal endpoint let T_b be its affine factors. The positive-definite support S contains at most $\operatorname{rank} T_b$ nodes from each. The total rank N forces equality in every factor: exactly one node of each T_b is missing. If the factor containing α_s has rank at least2, choose its missing node as x; every pair in such a factor has finite angle. If it is affine A_1, choose the missing node of another factor, which is orthogonal to α_s. Another factor exists in the range at issue. Thus again the pair is positive definite.

To see that this is an actual ridge, start with a strict interior timelike point z. Project it off the span of x, α_s. The projection correction is a nonnegative combination of those normals, because their Gram is a positive-definite matrix with nonpositive off-diagonal entries and nonnegative inverse. Every other wall inequality stays strict, and the projected square is even more negative. It gives a point of precisely their common face. Hence x is in the framing set controlled by (A.5.1). Apply the same construction to the second omission tuple and call its terminating wall y.

Neither x nor y is an old deleted cusp wall: that wall is strict along the outgoing edge away from v. Their old contact rows $p_j = -(x, \alpha_j)$ and $q_j = -(y, \alpha_j)$ are nonnegative, and every nonzero entry is at least1. For each affine factor let $c_a = \sum_j \delta_{aj}\alpha_j$, with $\delta_{aj} > 0$ normalized by $\min_j \delta_{aj} = 1$. These are nonzero future null vectors, since their pairing with an interior point is negative. They are all positive multiples $k_a c$ of one cusp vector c. Therefore

$$\sigma_a = \delta_a \cdot p = -(x, c_a), \quad \tau_a = \delta_a \cdot q = -(y, c_a), \quad q_0 = \sigma_a/\tau_a$$

have a common positive ratio. Indeed, if a row height vanished, all its nonnegative contacts with that factor would vanish; proportionality of the c_a would force every factor height and every old contact to vanish. A square-two normal would then lie in the null orthogonal complement of the full cusp span, impossible.

For each endpoint the principal Gram $S{+}x$ or $S{+}y$ is positive semidefinite. Its Schur complement gives the separate budgets

$$\sum_a C_{0a} \le 2, \qquad \sum_a C_{1a} \le 2, \qquad C_{\ell a} = p_{\ell,S_a}^t G(S_a)^{-1} p_{\ell,S_a}. \tag{A.5.2}$$

The inverse is entrywise nonnegative. No bound on a deleted-node contact or on its Coxeter label has been used.

A.5.2 The common-height Lorentz identity

Choose a null partner d_0 with $(c, d_0) = -1$, and write $x = ac + bd_0 + u$, $y = a'c + b'd_0 + u'$ in the orthogonal Euclidean quotient. Then $b = -(x, c) > 0$, $b' = -(y, c) > 0$. Since $(x/b - y/b') \perp c$ and $x^2 = y^2 = 2$,

$$\|u/b - u'/b'\|^2 = 2/b^2 + 2/(b')^2 - 2(x, y)/(bb').$$

The quotient is the orthogonal sum of the affine factor spaces. In factor a put $E_a = \|u_a/\sigma_a - u'_a/\tau_a\|^2$. Since $\sigma_a = k_a b$ and $\tau_a = k_a b'$, the k_a^2 cancels and the preceding identity becomes

$$-(x, y) = \frac{1}{2}\sum_a \sigma_a \tau_a E_a - q_0 - q_0^{-1}. \tag{A.5.3}$$

This proves the scale cancellation for a product cusp; no commensurability or chosen horosphere is required. Let $M = \max(q_0, q_0^{-1})$ and $W_a = \min(\sigma_a, \tau_a)^2 E_a$. Then $\sigma_a \tau_a E_a = M W_a$ and $q_0 + q_0^{-1} \le 2M$.

The prescribed omission choices have targets

$$T_a = 2r_a/9 \quad \text{ordinarily}, \qquad T_a = 2(r_a - 1)/9 \quad \text{in the sole avoid-node factor.}$$

Thus $\sum T_a = (2N - 2)/9$. We next prove

$$\sum_a W_a \ge \begin{cases} (2N-2)/9 - 4 & 1 \le M \le 21/20, \\ (2N-2)/9 - 16/3 & M \ge 21/20. \end{cases} \tag{A.5.4}$$

It follows from(A.5.3) that $-(x, y) \ge M(N - 37)/9$ in the first range and $-(x, y) \ge M(N - 43)/9$ in the second. At $N = 163 + t$ these are at least $14 + t/9$ and $14 + 7t/60$, respectively. They contradict (A.5.1) for every $t \ge 0$.

A.5.3 Point potentials and the complete local domain

For r Euclidean coordinates put $s_k = e_1 + \cdots + e_k$. The classical point potentials and normalized kernel squares are

	v_k	(δ_k^2)
B	$v_0 = 0,\ v_1 = e_1,\ v_k = s_k/2\ (k \ge 2)$	$(1, 1, 4, \dots, 4, 2)$
C	$v_k = s_k,$ metric one half the usual metric	$(1, 2, \dots, 2, 1)$
D	$v_0 = 0,\ v_1 = e_1,\ v_k = s_k/2\ (2 \le k \le r-2)$	
	$v_{r-1} = (s_{r-1} - e_r)/2,\ v_r = s_r/2$	$(1, 1, 4, \dots, 4, 1, 1).$

Write $R(i,j) = \|v_i - v_j\|^2$. For an A cycle of order m use $\delta = 1$ and $R(i,j) = h(m-h)/m$, $h = (i-j) \bmod m$. Ordinary omissions are $(0, \lfloor m/2 \rfloor)$ in A and $(0, r)$ in B, C, D. Modified B, C use $(r-1, 0)$ and $(1, r)$. Rotations and fork automorphisms avoid a prescribed node in A, D without weakening the ordinary geometric estimate.

Exceptional diagrams use the central node0 and consecutive labels along arms $(2,2,2)$, $(1,3,3)$, $(1,2,5)$ for affine E_6, E_7, E_8. Their rational symmetric matrices have diagonal2 and edge -1. For F_4 use diagonal $(2,2,2,4,4)$, successive edges $(-1,-1,-2,-2)$ and scales $(1,1,1,2,2)$; for G_2 use

$$A = \begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -3 \\ 0 & -3 & 6 \end{pmatrix}, \qquad (s_i) = (1, 1, 3).$$

The physical Gram is $A_{ij}/\sqrt{s_i s_j}$. If $k > 0$ spans $\ker A$, put $\delta_i^2 = s_i k_i^2 / \min_j(s_j k_j^2)$. Ground node0 and solve $Au = f$, $u_0 = 0$, for $f = e_i/k_i - e_j/k_j$; then $R(i,j) = \min_l(s_l k_l^2) f^t u$. This defines every resistance by finite rational linear algebra.

Ordinary exceptional omissions are $E_6 : (2,4)$, $E_7 : (3,6)$, $E_8 : (1,6)$, $F_4 : (0,4)$, $G_2 : (0,2)$. To avoid f, take the first two of $(2,4,6) \setminus \{f\}$ in E_6; $(2,5)$ if $f \in \{4,7\}$ and $(4,7)$ otherwise in E_7; $(1,6)$ if $f \in \{3,7\}$ and $(3,7)$ otherwise in E_8; $(0,4)$ if $f \in \{1,3\}$ and $(1,3)$ otherwise in F_4; and the other two nodes in G_2. Modified B_2 has matrix $\left(\begin{smallmatrix} 2 & 0 & -2 \\ 0 & 2 & -2 \\ -2 & -2 & 4 \end{smallmatrix}\right)$, scales $(1,1,2)$ and omissions $(0,1)$, avoiding node2.

The next enumeration is for the *residual* product after a failing factor has been selected below. Its ordered budgets are at most $3/2$ on each side. The original product still has budgets2; the failing-factor classification below is established separately and does not apply this residual cap to the selected factor. For the near range let $B = 21/20$. The finite model list consists of A ranks1–16, B, C ranks2–7, D ranks4–17, all the ordinary and avoid-node exceptional cases, modified B ranks3–7, modified C ranks2–7, and modified B_2. For an omission o a retained contact k is eligible only if $\delta_k^2 R(o,k) \le 3/2$. Enumerate every subset S of these nodes and retain it only if

$$c_o(S) = \sum_{k \in S} \delta_k^2 R(o,k) + \sum_{k < l \in S} \delta_k \delta_l [R(o,k) + R(o,l) - R(k,l)] \le 3/2. \tag{A.5.5}$$

The retained inverse is nonnegative, so this is a lower bound for every row with that support and magnitudes at least1. Include or omit the omitted-node contact independently on each side; included means magnitude at least1. Each full row is nonempty. The two-empty-retained case has the direct bound $W \ge T$.

For full supports U, V, their normalized potentials lie in two finite convex hulls. Form their positive-semidefinite signed point Gram K and the two row-sum constraints $Az = (1,1)$, $z \ge 0$. An exact lower-distance certificate consists of

$$Az = (1,1), \quad Kz + A^t\lambda \ge 0, \quad z_i(Kz + A^t\lambda)_i = 0, \quad e = z^t K z = -\lambda_1 - \lambda_2.$$

Expanding the quadratic at z proves that e is the global minimum on the two hulls. Enumerate active subsets and solve their rational KKT equations; verify a witness by these identities. Exhaustion always halts; failure to find a witness is a failed certificate, never an omitted support.

Singular active systems may be skipped for the following reason. Choose a minimum on the product of the two simplices having the fewest positive coordinates, and restrict to that support I. Both side-sum constraints are nonempty and independent. If its bordered KKT matrix were singular, a nonzero null pair (h, ν) would satisfy

$$A_I h = 0, \qquad K_I h + A_I^t \nu = 0.$$

Multiplying by h gives $h^t K_I h = 0$. Positive semidefiniteness implies $K_I h = 0$, and full row rank then gives $\nu = 0$. Thus $h \neq 0$ and preserves both sums. Moving along h until a positive coordinate first reaches zero preserves feasibility and the minimum, contradicting the choice of support. Some minimizing active system is therefore nonsingular. Its rational solution is found by the exhaustive search. This argument proves search completeness; the displayed slack identities independently prove soundness of each supplied certificate.

Writing $s = \sum_U \delta_k$, $t = \sum_V \delta_k$, the positive row magnitudes and common ratio bound give

$$W \geq \max\{\min(s,t)^2, \max(s,t)^2/B^2\}\, e. \tag{A.5.6}$$

This covers arbitrary unbounded contact magnitudes by convex normalization, not a finite angle list.

The saved exact calculation has91 finite models,1116 retained support pairs,3630 nonempty continuous domains and1395 positive losses. Radicals in support rejection and height bounds are rounded outward to rational bounds with squared inequalities; every surviving support cost is also checked exactly. The counts are checksums. Coverage comes from the complete model list, all subsets and both omitted-node states.

The remaining ranks have no positive loss. For B ordinary $r \geq 8$ the eligible retained sets are $\{1\}, \{r-1\}$; for modified B they are $\{r-2, r\}, \{1\}$. The energy lower bounds are $(r-2)/4$ and $(r-3)/4$, with height-square at least $2/B^2$. For ordinary C they are $\{1\}, \{r-1\}$, and for modified C, $\{0,2\}, \{r-1\}$; energy is at least $(r-2)/2$ and $(r-3)/2$, with height-square at least1. For D, $r \geq 18$, the sets are $\{1\}, \{r-1\}$ and energy is at least $(r-2)/4$ with height-square1. Omitted-node contacts may also occur. Each difference of full convex-hull points has the indicated fixed middle coordinates, of magnitude $1/2$ in B, D or1 in C. Their number proves the energy bounds. Subtracting T gives a nonnegative constant at the initial rank and a positive coefficient of the excess rank.

For an A cycle, $m \geq 18$ even or $m \geq 19$ odd, cost at most $3/2$ permits only one adjacent retained node per side. Singleton full supports have separation at least $\lfloor m/2 \rfloor - 2$; the margin numerator is $m^2 + 8m - 9(4+\varepsilon)^2$ over $36m$. If a full row has two nodes, its height is at least $2/B$ and the margin numerator is $(9-2B^2)m^2 + (2B^2-72)m - 81\varepsilon$ over $9B^2 m$. Here $\varepsilon = m \bmod 2$. Substituting $m = 18 + 2u$ or $19 + 2u$, $u \geq 0$, gives nonnegative coefficients and a positive constant. This proves the tails at every rank.

A.5.4 The classical unit-charge estimate at the original budgets

The residual calculation above starts only after a failing factor has been selected. To justify that selection, we first need a statement at the original endpoint budgets. In this subsection each retained row has cost at most two; neither row is restricted to the residual budget $3/2$.

Lemma A.5.1 (Full-budget classical unit charge)**.** *For the prescribed ordinary or avoid-node omissions of a classical affine factor, let $C_0, C_1 \leq 2$ be its actual retained costs, and let $W =$*

$\min(\sigma,\tau)^2 E$. *If the common height ratio is at most* $B = 21/20$, *then*

$$W + C_0 + C_1 \geq T,$$

where $T = 2r/9$ *ordinarily and* $T = 2(r-1)/9$ *in an avoid-node factor. This includes arbitrary real contact magnitudes with every nonzero contact at least one, and both possibilities for each omitted-node contact.*

Proof. The point potentials and kernel squares are those already displayed above. For omissions i, j, put

$$R = R(i,j), \qquad c_{ik} = \delta_k^2 R(i,k), \qquad \ell_{ik} = R(i,j) + R(i,k) - R(k,j).$$

The number c_{ik} is the unit-contact inverse diagonal, not the actual cost of an arbitrary contact. Normalizing a full contact row by its affine height gives nonnegative convex weights w_k, of sum one. The omitted node itself is allowed in this convex combination; its displacement and its linear loss are zero. Write the two normalized point means as $v_i + a, v_j + b$. Direct polarization gives

$$E = R - \sum_k w_k \ell_{ik} - \sum_l z_l \ell_{jl} + \|a - b\|^2. \tag{A.5.7}$$

Every row height is at least one, so $W \geq E$. The positive retained inverse and the contact gap give

$$C_0 \geq \sum_{k \text{ retained, contacted}} c_{ik}, \qquad C_1 \geq \sum_{l \text{ retained, contacted}} c_{jl}.$$

Here all same-row cross terms are nonnegative. In particular, since $0 \leq w_k, z_l \leq 1$, a bound on each unit diagonal controls the corresponding weighted linear losses without any upper bound on a contact magnitude.

For B, C, D the following table is the complete contact partition. An asterisk denotes the prescribed avoid-node omissions. The column "trivial" marks a contact for which $c_{ik} > T$; if such a contact is present, $C_0 + C_1 > T$ already proves the assertion. In every other row $\ell_{ik} \leq c_{ik}$.

Type	(i,j)	retained k	c_{ik}	ℓ_{ik}	trivial
B	$(0,r)$	1	1	1	
B	$(0,r)$	$2 \leq k \leq r-1$	k	$k/2$	
B	$(0,r)$	r	$r/2$	$r/2$	
B	$(r,0)$	0	$r/4$	$r/2$	yes
B	$(r,0)$	1	$r/4$	$r/2-1$	yes
B	$(r,0)$	$2 \leq k \leq r-1$	$r-k$	$(r-k)/2$	
B^*	$(r-1,0)$	0	$(r-1)/4$	$(r-1)/2$	yes
B^*	$(r-1,0)$	1	$(r-1)/4$	$(r-3)/2$	yes
B^*	$(r-1,0)$	r	$1/2$	0	
B^*	$(r-1,0)$	$2 \leq k \leq r-2$	$r-1-k$	$(r-1-k)/2$	
B^*	$(0,r-1)$	1	1	1	
B^*	$(0,r-1)$	r	$r/2$	$(r-1)/2$	
B^*	$(0,r-1)$	$2 \leq k \leq r-1$	k	$k/2$	
C	$(0,r)$	$1 \leq k \leq r-1$	k	k	
C	$(0,r)$	r	$r/2$	r	yes
C^*	$(1,r)$	0	$1/2$	0	
C^*	$(1,r)$	$2 \leq k \leq r-1$	$k-1$	$k-1$	
C^*	$(1,r)$	r	$(r-1)/2$	$r-1$	yes
C^*	$(r,1)$	0	$r/2$	$r-1$	yes

Type	(i,j)	retained k	c_{ik}	ℓ_{ik}	trivial
C^*	$(r,1)$	$1 \le k \le r-1$	$r-k$	$r-k$	
D	$(0,r)$	1	1	1	
D	$(0,r)$	$2 \le k \le r-2$	k	$k/2$	
D	$(0,r)$	$r-1$	$r/4$	$r/2-1$	yes
D	$(0,r)$	r	$r/4$	$r/2$	yes

The ordinary ranks start at $2,2,4$ for B,C,D; the modified B rows start at rank three. Empty integer ranges in the table contribute no contact. The other ordinary C row follows by reversal, and the other D row by the automorphisms interchanging the two ends and the fork leaves. These same fork automorphisms avoid a prescribed node while preserving the stronger ordinary D estimate.

The omission distances are $r/4$ in ordinary B,D, $(r-1)/4$ in modified B, $r/2$ in ordinary C, and $(r-1)/2$ in modified C. Each is at least its target. If no trivial contact is present, applying the table to both sums in (A.5.7) gives $W + C_0 + C_1 \ge R \ge T$. The exceptional modified B_2 matrix displayed earlier has $R(0,1) = 1$ and every retained inverse diagonal at least $1/2 > T = 2/9$. Thus a nonempty retained support is trivial; if both supports are empty, the omitted-only rows give $W \ge 1$. This proves all B,C,D ranks. Notice that this is a coefficient-one estimate; the later coefficient-$4/3$ estimate alone would not prove the present lemma.

For A write $m = r+1$, $h = \lfloor m/2 \rfloor$ and $\varepsilon = m \bmod 2$. The omissions are $0, h$, and $R(i,j) = d(m-d)/m$, where $d = (i-j) \bmod m$. We first state the finite calculation at small orders with its full domain.

For each $2 \le m \le 17$ and each omission $o \in \{0,h\}$, enumerate all subsets of

$$\{k \ne o : R(o,k) \le 2\}.$$

Keep S exactly when its complete unit-contact cost

$$c_o(S) = \sum_{k \in S} R(o,k) + \sum_{k<l \in S} [R(o,k) + R(o,l) - R(k,l)]$$

is at most two. Pair the two retained sets, keeping pairs with $c_0(S_0) + c_h(S_1) < T$; other pairs are already trivial. Include or omit each omitted node independently to form full supports U, V, and require both full supports nonempty. These are exactly the possible support domains relevant to a failure: actual costs bound their unit costs below, and positive affine height excludes an empty full row. No residual cap, finite angle cutoff or bound on an omitted contact has entered this construction.

Use the Euclidean potentials

$$v_k = e_1 + \cdots + e_k - \frac{k}{m}(e_1 + \cdots + e_m).$$

Their Gram is $K_{ij} = \min(i,j) - ij/m$, and their squared distances are $R(i,j)$. For a pair U, V, form the signed point Gram K, with positive signs on U and negative signs on V. It is positive semidefinite. A rational certificate for its closest convex means consists of $z \ge 0$, the two row-sum conditions $Az = (1,1)$, and multipliers λ satisfying

$$Kz + A^t\lambda \ge 0, \qquad z_i(Kz + A^t\lambda)_i = 0.$$

Then $e = z^tKz = -\lambda_1 - \lambda_2$ is a global lower bound for the squared distance: for any feasible z', expansion gives

$$(z')^tKz' - e = (z'-z)^tK(z'-z) + 2(z')^t(Kz + A^t\lambda) \ge 0.$$

This also covers singular Gram matrices. With $s = |U|$, $t = |V|$, the common ratio bound and the contact gap imply

$$\min(\sigma, \tau)^2 \geq H_B(s,t) := \max\{\min(s,t)^2, \max(s,t)^2/B^2\}.$$

Consequently $W + C_0 + C_1 \geq H_B(s,t)e + c_0(S_0) + c_h(S_1)$.

The recorded full-domain calculation has *580 cases*. For every one, its supplied rational witness satisfies the displayed KKT conditions and

$$H_{21/20}(|U|, |V|)e + c_0(S_0) + c_h(S_1) \geq 2(m-1)/9. \tag{A.5.8}$$

This is the finite computed proposition used here. Its verifier reconstructs all the preceding sets and both omitted-node states, matches the witness keys (m, U, V), and verifies their exact rational identities and inequalities. Within each selected m, the witness map rejects missing, duplicate or out-of-domain support keys. The separate coverage guard also requires one wrapper for each $m = 2, \ldots, 17$ and the exact widened scalar-key set; a wrapper lookup alone would not reject redundant duplicate receipt rows. There are sixteen values of m, finitely many subsets of at most sixteen retained nodes, four omitted-node states per pair, and finitely many rational operations per supplied witness. Thus verification halts. Completeness is the exhaustive support construction, not the number 580. The KKT identity and the height bound prove correctness for arbitrary contact magnitudes.

Crucially, (A.5.8) is the calculation at $B = 21/20$. An earlier height window $31/30$ cannot be substituted: increasing B weakens H_B. The same rational closest-point witnesses are used, but every height and unit-charge inequality is rechecked at the wider window.

It remains to prove the A tail at all orders. For $m \geq 18$, a retained contact at cyclic distance $d \geq 3$ and $d \leq m/2$ has unit cost at least $3(m-3)/m > 2$. Hence each contacted retained node has distance one or two from its omission. Direct substitution in the resistance formula gives

$$\frac{\ell_{ik}}{c_{ik}} \leq \frac{m+\varepsilon}{m-d} \leq L_m := \frac{m+\varepsilon}{m-2}.$$

The same holds in the other row. Therefore polarization gives

$$W + C_0 + C_1 - T \geq R(0,h) - T - (L_m - 1)(C_0 + C_1) \geq R(0,h) - T - \frac{4(2+\varepsilon)}{m-2}.$$

For even $m = 18 + 2u$, $u \geq 0$, the last quantity is

$$\frac{m^2 + 6m - 304}{36(m-2)} = \frac{128 + 84u + 4u^2}{36(m-2)} > 0.$$

For odd $m = 19 + 2u$ it is

$$\frac{m^3 + 6m^2 - 457m + 18}{36m(m-2)} = \frac{360 + 1708u + 252u^2 + 8u^3}{36m(m-2)} > 0.$$

These are all-order identities, not boundary tests or extrapolation. Rotation chooses an opposite omission pair avoiding any prescribed node when $m \geq 3$, so the stronger ordinary estimate covers the modified target too. At $m = 2$ the repeated allowed omission has modified target zero, for which nonnegativity suffices. This completes the proof. □

The lemma closes the classical part of the first-failing-factor classification at the original budgets. The separate exceptional enumeration that follows is consequently sufficient: any factor with $T - W > C_0 + C_1$ must occur among its exceptional supports. Only after selecting one of those factors are the residual 3/2 budgets and the ordered packing table applied.

A.5.5 Ordered packing proves loss at most four

Each positive-loss domain is dominated by a record with no larger ordered cost floors, no smaller loss bound, and the same modified flag. The complete28-record table is:

Type	(C_0, C_1) floors	loss bound
ordinary A_1	$(1/2, 0)$	$2/9$
ordinary A_2	$(2/3, 0)$	$4/9$
ordinary A_4	$(4/5, 4/5)$	$8/9$
ordinary $B_r,\ 2 \le r \le 5$	$(0, r/4)$	$2r/9$
ordinary $D_r,\ r = 4, 5$	$(r/4, 0)$	$2r/9$
ordinary E_6	$(0, 4/3), (4/3, 0)$	$4/3$ each
ordinary E_7	$(1/2, 1), (1, 1/2)$	$14/9$ each
ordinary E_8	$(8/9, 2/3)$	$16/9$
modified $B_r,\ 3 \le r \le 7$	$((r-1)/4, 0)$	$2(r-1)/9$
modified E_6	$(0, 4/3)$	$10/9$
modified E_7	$(0, 3/2), (2/3, 5/6), (5/6, 2/3)$	$4/3$ each
modified E_8	$(1, 1/2), (8/9, 2/3)$	$14/9$ each
modified F_4	$(0, 1), (1/2, 3/4)$	$2/3$ each
modified G_2	$(0, 1/2)$	$2/9$

The verifier supplies and checks a dominating record for each of the 1395 positive domains. It does not assume the table is geometrically realizable by one chamber.

For records with indices at least i, ordered remaining budgets p, q, and at most a modified records, the maximum residual loss is the maximum of 0 and

$$\ell_j + F(j, p - p_j, q - q_j, a - \epsilon_j), \qquad j \ge i,$$

over admissible records. Each has $p_j + q_j \ge 1/2$, so the recursion has length at most $\lfloor 2(p+q) \rfloor$. All costs have a common denominator, hence the state set is finite. Induction on remaining length proves completeness; checking every upper inequality and an equality predecessor proves the recorded maxima

$$F(0, 10/9, 4/3, 1) = 20/9, \qquad F(0, 1, 3/2, 1) = F(0, 3/2, 1, 1) = 22/9.$$

One additional complete exceptional check is needed. If $T - W > C_0 + C_1$, then it is an ordinary E_8 with cost floors $(8/9, 2/3)$ and target $16/9$, or an ordinary E_7/modified E_8 with floors $(1/2, 1)$ or $(1, 1/2)$ and target $14/9$. Its finite algorithm enumerates every grounded exceptional support below the trivial threshold $C_0 + C_1 < T$, including all inverse cross terms, and checks the resulting convex-quadratic bounds. There are15 candidate exceptional supports. Two ordinary E_7 same-row supports at nodes4,7 are excluded by exact convex certificates, in both omitted-node states, giving $W + 3/2 > T$. For the omissions $(3, 6)$ the four ordered full-support pairs are

$$((4, 7), (6)), \quad ((4, 7, 3), (6)), \quad ((3), (4, 7)), \quad ((3), (4, 7, 6)).$$

The coverage guard requires these four distinct keys, not merely four individually valid certificates. The remaining13 have exactly the stated floors. This support check is separate from the residual domination table.

If no factor fails, the two budgets already bound total loss by4. Otherwise choose any failing factor. An E_8 leaves budgets at most $(10/9, 4/3)$, and its own loss at most $16/9$ plus the residual $20/9$ is4. In the other case the residual is at most $22/9$ and the own loss at most $14/9$, again4. All additional failures are included in this residual calculation. Permitting one modified residual record even if the selected factor was modified only enlarges the upper bound. This proves the first line of (A.5.4).

A.5.6 The four-thirds estimate for the other height range

For the same omission choices the all-rank local inequality is

$$W + \frac{4}{3}(C_0 + C_1) \geq T.$$

To prove it, write the two normalized point potentials as $v_i + a, v_j + b$. With convex contact weights w_k, z_k,

$$E = R_{ij} - \sum_k w_k(R_{ij} + R_{ik} - R_{kj}) - \sum_k z_k(R_{ij} + R_{jk} - R_{ki}) + \|a - b\|^2.$$

A nonzero contact has magnitude at least1, every convex weight is at most1, and the nonnegative retained inverse makes its cost dominate the sum of unit diagonal costs. Also $W \geq E$, since both row heights are at least1.

For B, C, D, use the coefficient-one argument in Lemma A.5.1. That part of its proof uses only that both row heights are at least one, and hence is independent of the near-range ratio bound. Its complete contact table also covers modified B_2. Increasing the cost coefficient to $4/3$ preserves the inequality. The A and exceptional factors in this height range require the following separate arguments.

For A put $C = C_0 + C_1$ and $T = 2(m-1)/9$. If $C \geq 3T/4 = (m-1)/6$, nonnegativity of W proves the claim. Only $C < (m-1)/6$ needs further work. For $3 \leq m \leq 6$ the smallest retained unit cost $(m-1)/m$ already reaches this threshold, so no retained contact is possible. The literal A_1 case $m = 2$ is separate, and its modified target is zero. For $7 \leq m \leq 10$ a distance-two contact likewise reaches the threshold; only distance one remains. For $m \geq 10$ the original per-side budget two excludes every cyclic distance at least three, since $3(m-3)/m > 2$ and resistance increases up to the antipode. For a remaining distance $d = 1, 2$, polarization leaves the margin

$$\frac{4}{3}\frac{d(m-d)}{m} - \frac{d(m+\varepsilon)}{m} = \frac{d(m-4d-3\varepsilon)}{3m} \geq 0.$$

The nontrivial domain starts at $m = 7$ for $d = 1$ and $m = 11$ for $d = 2$ in odd parity, where the respective margins are zero; the even cases are nonnegative too. Smaller odd ranks with a negative formal margin were already in the trivial domain. The finite exceptional omission cases enumerate every cheap retained support and both omitted-node states. Single contacts give a convex quadratic whose endpoint and stationary minima are checked exactly; the sole cheap two-contact support has energy $(1 + u^2 + v^2)/2$. These finite certificates use the explicit grounded matrices above, so have no missing high-rank or finite-angle branch.

Summing and using(A.5.2) proves the second line of (A.5.4). The certificate verifies the stated finite quadratic and symbolic polynomial identities. Their domains, the contact-gap argument and the coordinate identities explain why they imply the all-rank inequality.

A.5.7 A narrow facet without a cusp

For completeness we prove the compact-facet alternative rather than import a finer compact-chamber weighting. Let Q be a compact two-face. Its white normals W number $n-2$, and every consecutive black pair with W is positive definite. A positive-definite principal block gives an actual common face by the interior-point projection already proved. A singular positive-semidefinite block has a positive null combination in an affine component. It is nonzero, future null and nonpositive against every other chamber wall, so gives an ideal point of its face.

For a triangle, a minimal indefinite star subdiagram contains all three black nodes: omitting any one puts it in a spherical corner. By the order-ten classification it has at most seven whites. In

a spanning tree joining the three marks, deleting either of two outer marks leaves the other two joined. The resulting two paths avoid the third mark and have at most eight edges, so lie in two actual corner diagrams.

For a quadrilateral split opposite black pairs as U, V. The block $W + U$ is indefinite. If positive definite, its actual face would make opposite sides meet. If positive semidefinite and singular, it would give an ideal point of the compact Q. Both are impossible. Likewise $W + V$ is indefinite. Minimal indefinite subdiagrams $L \subset W + U$, $M \subset W + V$ contain their respective black pairs and at most eight whites. They cannot be orthogonal, since their negative vectors would give negative index at least two. A shortest path in their overlapping or adjacent union between U and V has only white interior nodes and length at most $8 + 1 + 8 = 17$. Its black endpoints are consecutive sides, so the path belongs to an actual corner.

Thus unit weight at ambient spherical distance at most17 pays two on a compact triangle and one on a compact quadrilateral. Its cost at an ambient finite vertex is at most $17n$: spherical components of order at most35 have at most $\binom{r}{2} \leq 17r$ pairs, and larger components are paths or D forks with at most r pairs at each fixed distance. This uses only the finite Coxeter classification, allowing affine subdiagrams elsewhere in the chamber.

A compact facet has dimension $d = n - 1$ and is simple, because its vertices are finite vertices of the Coxeter chamber. Its two-faces are compact faces of the ambient chamber; its angles count a subset of the ambient vertex pairs. Hence their total is at most $17(d + 1)f_0$. Its nonnegative symmetric h-vector has $h_0 = h_d = 1$. Pairing the coefficients $\binom{k}{2}$ and $\binom{d-k}{2}$ gives

$$\frac{f_2}{f_0} > \begin{cases} d(d-2)/8 & d \text{ even}, \\ (d-1)^2/8 & d \text{ odd}. \end{cases}$$

The inequality is strict because the nonzero endpoint coefficients exceed the middle minimum. Consequently the average required payment $5f_2/f_0 - \binom{d}{2}$ exceeds $d(d-6)/8$ or $(d-1)(d-5)/8$. After subtracting $17(d + 1)$, the differences at $d = 143 + 2t$ and $144 + 2t$ are respectively $t^2/2 + 36t + 3/2$ and $t^2/2 + 73t/2 + 19$, positive for $t \geq 0$. Thus a compact facet forces $n \leq 143$.

A finite-volume facet with no ideal vertices is compact. Apply Nikulin's theorem and split by whether the chosen facet has a cusp. The cusp branch above excludes every $n \geq 164$, and the compact-facet branch already excludes $n \geq 144$. This completes the unbounded tail without assuming that the facet itself is a Coxeter polytope.

Part V
Proof appendices: compact polytopes

B.1 Ordinary component proofs and actual defects

Throughout the proof appendices, $F(t) = 2t/(1+t^2)$ denotes the full-/half-tangent conversion. This appendix collects the contact calculations behind the small-component rates. Each is an argument on an actual whole component with arbitrary finite spectators, using Section 9. No field condition is present. We use the spherical/affine classification and the standard Lannér classification: a minimal indefinite Coxeter diagram with all proper subdiagrams positive definite has at most five vertices. A positive affine principal is already impossible by the actual null argument in Section 9.

Proposition B.1.1 (Ordinary costs and the complete small defects)**.** *All the ordinary component bounds in Section 11 hold with arbitrary finite spectators. For the moment use the weaker bound $B_3, H_3 > 1/4$; the improvement to $2/7$ is proved immediately below. The only exceptions to the rank-over-14 small-component baseline are one possible isolated short I, its separately funded J alternative, or one of the two typed rank-two defects.*

Write $m@j$ for actual contact $\cos(\pi/m)$ at old node j. A missing contact is zero. For a wall put $h = A^2$, own cost q and full foreign cost Q, so $h = q + Q - 1 > 0$. The following calculations prove the proposition; their tables are finite mathematical case analyses rather than empirical output.

Lemma B.1.2 (Common-wall forcing and isolation)**.** *Suppose the energy of a component is at most e. A pair for which the right side of* (9.9) *is below K_{ij} has one actual opposite. If one wall closes every direction of the component, then, with $v_i = (C\beta)_i/\sqrt{d_i}$ and $R_e^2 = 4e/(1+e)^2$,*

$$q - 1 \le h \le R_e^2 \min_i v_i^2. \tag{B.1.1}$$

If $h < 1/4$, it has no foreign contact. In a path, a wall closing every direction can have multiple contacts only at the two endpoints; in a D_4 star it has at most one.

Proof. The first assertion is (9.9). For a closed direction, $\tanh L_i = A/v_i$ and $\tau_i^2 \le e$, which gives (B.1.1). Every nonzero foreign cost is at least $1/4$ and is strictly below h by (9.8); hence the isolation assertion. If the old path between two contacts omits a direction closed by the wall, deleting that direction leaves a cycle in an actual finite replacement. This is impossible. The same argument applies to every two contacts in a D_4 star. Finally, a single-contact own cost $q = 1$ gives an affine principal set, impossible even in the presence of further spectators. □

B.1.1 The short directions and their exact debits

For a short A_1 opposite put $b = \beta^2$, $h = A^2$, and let Q be its full foreign cost. Then

$$h = b + Q - 1, \qquad h < b/3, \qquad Q < 1 - 2b/3.$$

Equation (9.8) rules out two foreign components. With one, its cost must be the sole value below $1/3$, namely an A_1 contact 3 of cost $1/4$. Thus the only possibilities are

$$I:\ \text{isolated dotted } A_1, \quad 0 < h < 1/2; \qquad J:\ \text{dotted } A_1 + A_{1,3}, \quad 1/3 < h < 3/8.$$

For completeness, with one foreign component, $Q < h = b + Q - 1$ forces $b > 1$, and $Q < 1/3$ then forces $Q = 1/4$. For J, the heavy edge contributes more than $A/2$ to $x \cdot v$. The light shadow

contributes at least $A(1-\sqrt{1-1/(4h)})$; this is at least $A/2$ if $h \le 1/3$. Strict (9.5) therefore gives the asserted lower bound. Native acuteness excludes two short walls. Indeed their height product is below $1/2$, so their product must be zero. Disjoint supports give a negative product, heavy overlaps give a spatial product above $1/2$, and a sole light/light overlap of two J walls gives $1/4$, below their height product. None is zero. A J heavy direction has $\tanh^2 L > 4/13$, hence energy $> 9/100$; its light direction has $\tanh^2 L > 2/3 > 16/25$, hence energy $> 1/4$. Thus J funds both A_1 factors above their combined ordinary baseline $2/10$. Every nonshort direction pays at least $5-2\sqrt{6} > 1/10$, since $\sqrt{6} < 49/20$. We may conservatively discard the one short direction, but only the isolated type I can create an unfunded debit.

The rank-two exceptions are

old block / new isolated contact	$h = A^2$	energy
$A_2/7$	$(4/3)\cos^2(\pi/7) - 1 \in (0, 1/12)$	$> 9/100$
$I_2(7)/3$	$(1/4)\csc^2(\pi/7) - 1 \in (0, 1/3)$	$> 9/100$.

(B.1.2)

Their spatial supports lie entirely in their own blocks. Two defects in different blocks, or a defect and a short A_1, would have actual wall product $-\sqrt{h_1 h_2} \in (-1/2, 0)$. That interval is forbidden: distinct acute unit walls have product 0 or at most $-1/2$. Therefore

$$\text{at most one short } A_1, \quad \text{or one rank-two defect, but never both.} \tag{B.1.3}$$

This argument is independent of every other component type.

B.1.2 The two rank-two estimates

For A_2, set $e_2 = 319/2000$ and $\sinh^2 L_0 = 31/50$. If both edges have length at least L_0, then their half-tangents exceed $173/500$ and energy is greater than $4\cdot 173^2/(3\cdot 500^2) > e_2$. A shorter edge cannot have two positive own contacts: $3/3$ is affine, while $3/4$ or larger gives $\sinh^2 L \ge 4\sqrt{2}/9 > 31/50$, increasing with either contact and with Q. If only the other node is contacted, apply the same test to that shorter contacted direction. For one contact of square b, the shorter-edge condition is

$$h = 4b/3 + Q - 1 < 124b/243, \qquad Q < 1 - 200b/243. \tag{B.1.4}$$

Every positive foreign cost must be below h. A label 3 gives $h < 1/4$, hence isolation and negative height. For a label 4, $h < 62/243$ forces each foreign cost to be $1/4$; the only positive possibility is $Q = 1/2$, but then $h = 1/6 < 1/4$. A label 5 has $Q < 1/2$, hence at most one foreign component, whose cost exceeds h because its own cost $\varphi^2/3$ is below one. A label 6 is affine. For a finite label $m \ge 7$, $b > .81$ gives $Q < 1/3$; thus $Q = 0$ or $Q = 1/4$. The latter is possible in (B.1.4) only for $b < 729/800$, namely $m = 7, 8, 9, 10$. Indeed $\cos^2(\pi/11) > 45/49 > 729/800$, using $\pi < 22/7$. A dotted contact has $Q < 1/4$ and is isolated.

For an isolated finite contact the same wall closes the mate, since its reciprocal ratio squared is $b/(4b-3) > 1$. For $m \ge 8$ that actual mate has $\tanh^2 L \ge 6\sqrt{2} - 8 > 400/841$, so its energy exceeds $4/25 > e_2$. For an isolated dotted contact, $b < 243/200$ and its distinct mate has shadow $\tanh^2 L \ge b/(4b-3) > 81/124 > 16/25$. For a mixed $m = 7, 8, 9, 10$ contact, the mate's white cost is $b + 1/4 > 1$; the distinct shadow instead gives $\tanh^2 L \ge b/(4b - 9/4) > 11/17 > 16/25$. Both give energy greater than $1/4$. Thus the only exception is isolated new 7 at old A_2. The root $t = \cos(2\pi/7)$ of $8t^3 + 4t^2 - 4t - 1$ lies in $(623/1000, 5/8)$: the polynomial changes sign there and is increasing on $[1/2, 1]$. Hence $b > 1623/2000$ and its actual mate satisfies $\tanh^2 L = 4 - 3/b > 164/541 > 3600/11881$, proving its cost $> 9/100$.

For an old $I_2(m)$, $m \geq 4$, put $k = \cos(\pi/m)$ and $d = (1-k^2)^{-1}$, so $C = d\left(\begin{smallmatrix} 1 & k \\ k & 1 \end{smallmatrix}\right)$. At energy at most $1/7$, (9.9) forces one common opposite. Two contacts give an actual $\cosh L \geq \cot(\pi/(2m))/\sqrt{3} > 11/8$, with individual energy $> 3/19 > 1/7$. For one contact β,

$$q(1 - 7k^2/16) \leq 1, \qquad q = d\beta^2. \tag{B.1.5}$$

Its minimum left side is $(9d+7)/64$. Since $m \geq 8$ gives $d \geq 4 + 2\sqrt{2} > 19/3$, only four old labels remain:

m	consequence of (B.1.5)
4	$\beta^2 \leq 16/25 < \cos^2(\pi/5)$: 3, 4; $h \leq 7/32 < 1/4$, $q \leq 1$;
5	$d > 25/9$: only 3; $h < 7/32$, $q < 1$;
6	only 3; $q = 1$;
7	$d > 5$: only 3; $Q < 3/16$.

Isolation or the affine principal excludes the first three rows. The last is isolated new 3 at old $I_2(7)$, with $0 < h < 1/3$. Its actual mate has $\cosh L = \cot(\pi/7)/\sqrt{3} > 109/91$ by the same cubic bracket; the squared comparison has positive numerator 2652. Thus its energy also exceeds $9/100$, completing (B.1.2) and (B.1.3).

B.1.3 Rank three and four: one common-wall calculation

For energy $1/4$, adjacent correlation at least $1/\sqrt{3}$ forces a common opposite: the lowest cleared comparison is $74/\sqrt{3} > 39$. At energy $1/3$ the threshold $1/\sqrt{2}$ works, since $51/\sqrt{2} > 34$. For A_4 at energy $2/7$, the adjacent correlations are $\sqrt{3/8}, 2/3, \sqrt{3/8}$; the forcing test is $k^3+16k^2+36k-28 > 0$, whose lowest value is positive because $291\sqrt{6} > 704$. Direct inversion verifies the indicated correlation thresholds for A_3, B_3, H_3 and B_4, D_4, F_4. For H_4 use energy $11/36 < 1/3$. Thus a common wall closes all directions in every case.

Number path bonds as $A_3 : (3,3)$, $B_3 : (4,3)$, $H_3 : (5,3)$, $A_4 : (3,3,3)$, $B_4 : (4,3,3)$, $F_4 : (3,4,3)$, and $H_4 : (5,3,3)$. By Lemma B.1.2, multiple contacts can only be at the endpoints. Two endpoint 3 contacts in A_3, A_4 are affine; the next minimum is $3/4$. In the other rows below even $3/3$ is excluded. The displayed values are lower bounds for $q - R_e^2 v_j^2$, contradicting (B.1.1):

T	j	lower bound
A_3	2	$177/200 + 9\sqrt{2}/100 > 1$
A_4	2	$(230 + 15\sqrt{2})/243 > 1$
B_3	2	$(77 + 18\sqrt{2})/100 > 1$
B_4	3	$15/16 + \sqrt{2}/8 > 1$
F_4	2	$15/16 + \sqrt{2}/4 > 1$
H_3	2	$(72\varphi + 95)/100 > 1$.

D_4 already has at most one contact. In H_4, two endpoint contacts retain, after deleting old node 4, a path at least $(3,5,3)$ of determinant $(5-4\varphi)/16 < 0$. Hence all contacts are single.

For a single contact at j let $\sigma_j = \min_i(v_i^2/q)$. Formula (B.1.1) gives $q \leq (1 - R_e^2\sigma_j)^{-1}$. The complete inverse calculation is as follows; entries are in path order, and D_4 means leaf,center.

T	d_j	σ_j	upper q
A_3	$3/2, 2, 3/2$	$1/9, 1/3, 1/9$	$225/209, 75/59, 225/209$
A_4	$8/5, 12/5, 12/5, 8/5$	$1/16, 1/6, 1/6, 1/16$	$162/155, 243/215, 243/215, 162/155$
B_3	$3, 4, 2$	$1/3, 1/2, 1/3$	$75/59, 25/17, 75/59$
D_4	$2, 4$	$1/4, 1/2$	$16/13, 8/5$
B_4	$4, 6, 4, 2$	$1/4, 1/3, 1/2, 1/4$	$16/13, 4/3, 8/5, 16/13$
F_4	$4, 12, 12, 4$	$1/2, 2/3, 2/3, 1/2$	$8/5, 2, 2, 8/5$

Here R_e^2 is respectively 16/25, 56/81 for A_4, and 3/4 for the 1/3 rows. Dividing the upper q by d_j gives the finite native lists: A_3 endpoints 3/4/5, middle 3/4; A_4 endpoints 3/4, inner 3; B_3 nodes $3; 3; 3/4$; D_4 leaf 3/4, center 3; B_4 nodes $3; \varnothing; 3; 3/4$; F_4 endpoints 3, inner $\varnothing$. Each has $q \le 1$. Equality is affine; in every other listed case (B.1.1) gives $h < 1/4$ (the largest needed bounds are $16/75, 7/155, 28/405, 3/16$). Isolation then gives $h = q - 1 < 0$. This proves the six crystallographic and A rows.

For H_3, the minimum contact $\beta = 1/2$ at node 1 or 2 gives tests at node 3 equal to $(199\varphi + 263)/500$ and $(61\varphi + 77)/125$, both above one. At node 3, $\beta \ge 1/\sqrt{2}$ gives the test at node 1 at least $(27\varphi + 118)/150 > 1$. Only 3@3 remains, with $q = (\varphi + 2)/4 < 1$ and $h \le 4\varphi^4/75 < \varphi/4$. No spectator gives negative height; one has cost $Q > h$; two have $Q \ge 1/2$ and hence $h \ge q - 1/2 = \varphi/4$. All are impossible. For H_4, $R_e^2 = 1584/2209$; at $\beta = 1/2$ the four tests (contact node, test direction) are

$$(1,4) : \frac{4938\varphi + 3688}{2209}, \quad (2,4) : \frac{7209\varphi + 5334}{2209}, \quad (3,1) : \frac{2500\varphi + 2667}{2209}, \quad (4,1) : \frac{625\varphi + 1219}{2209}.$$

All exceed one, the last by $\varphi > 198/125$; larger contacts only increase these positive tests. This proves $H_4 > 11/36$.

B.1.4 The two intermediate A components

Use $C_{ij} = 2\min(i,j)(N - \max(i,j))/N$, with $N = 6, 7$. For A_5, distinct 2/3 and 3/4 pairs each demand a sum above 109/100. The vector $u_2 + 2u_3 + u_4$ has square $7 + 4\sqrt{2} < 443/35$, and

$$(109/50)^2 - (3/8)(443/35) = 209/35000 > 0.$$

For A_6, the three pair demands are

$$2249/2000, \qquad 11547/10000, \qquad 2249/2000;$$

the vector $u_2 + 2u_3 + 2u_4 + u_5$ has square $< 33781/1250$, with demand 34037/10000 and Cauchy gap 2181583/700000000 at energy 3/7. These pair bounds follow by substitution in $F(s/2)^2 < K$. Thus a common 2/3 occurs for A_5, and a common 2/3 or 3/4 for A_6, up to reversal.

Each retained path allows one contact. The full two-contact list is

Component/pair	positions	necessary finite-white labels
$A_5, 2/3$	$1, 3$	3/3 or $4@1 + 3@3$
	$1,4; 1,5; 2,4; 2,5$	both 3
	$2, 3$	both in $3, 4, 5$
$A_6, 2/3$	$1, 3$	3/3 or $4@1 + 3@3$
	$1,4; 1,5; 1,6; 2,4; 2,5; 2,6$	both 3
	$2, 3$	$3, 4, 5$ at 2; $3, 4$ at 3
$A_6, 3/4$	$3, 4$	both in $3, 4, 5$
	other eight left/right pairs	both 3

Indeed deletion of either closed node leaves a connected finite path or D/E_6 tree: a 4 can only be terminal, and D/E bonds are all 3. All-3 doubles are affine cycles. The adjacent non-3/3 doubles force an actual edge of cosh at least $(3 + \sqrt{2})/2 > 11/5$ in A_5; for A_6 the minimum formulas are $(5 + 4\sqrt{2})/\sqrt{7}$, $(4\sqrt{2} + 6)/\sqrt{14}$, and $(12 + 9\sqrt{2})/\sqrt{42}$, all above 5/2. The A_5 terminal-4 row has $Q = 0$ and cosh $(3 + \sqrt{2})/2$. The A_6 version has white cost 9/10, hence $Q = 0$, and is treated below. These cosh thresholds imply individual energies above 3/8 and 3/7.

For single 3 contacts the complete own costs and strict white bounds are

component/pair	j	q	Q upper
$A_5, 2/3$	$1,2,3,4,5$	$5/12, 2/3, 3/4, 2/3, 5/12$	$2/3, 2/3, 5/8, 1/2, 5/8$
$A_6, 2/3$	$1,2,3,4,5,6$	$3/7, 5/7, 6/7, 6/7, 5/7, 3/7$	$2/3, 2/3, 3/5, 2/5, 2/5, 3/5$
$A_6, 3/4$	$1/6; 2/5; 3/4$	$3/7; 5/7; 6/7$	$5/8; 1/2; 5/8.$

End and small-Q rows have $h < 1/4$, or only one foreign component with cost $Q > h$. In A_5 at $j = 2, 3$ two foreign components would each exceed $1/3, 1/4$, respectively, forcing $Q \geq 2/3$; the same gap treats $A_6, j = 2$. In the remaining $A_6, j = 3, 4$ rows, two foreign bonds must be 3 since $Q < 5/8$; together with the contacted old node and its two neighbors they form affine two-fork D_5. This excludes all single 3 cases. A single internal 4 gives the affine $(3, 3, 4)$ star; an end 4 has $Q \leq 1/3$ and $h < 1/4$, again impossible.

A single 5 is at a retained path end; its H_3/H_4 white cost exceeds $3/4$, so $Q = 0$. All cases are covered by these actual positive replacements:

component/pair	contact	actual direction	$\cosh L$
$A_5, 2/3$	1 or 5	4 or 2	$\varphi^3/\sqrt{3}$
	2	3	φ^2
	3	2	$\sqrt{3}\varphi^3$
$A_6, 2/3$	1	4	$3\varphi^3/\sqrt{21}$
	2	3	$4\varphi^2/\sqrt{14}$
$A_6, 3/4$	1 or 6	4 or 3	$3\varphi^3/\sqrt{21}$
	3 or 4	4 or 3	$9\varphi^3/\sqrt{21}.$

The first group exceeds $11/5$, the second $5/2$, using $\varphi > 8/5$, $\sqrt{21} < 23/5$, $\sqrt{14} < 15/4$. No new wall is inferred: each replacement contains the already actual c. Finally the A_6 double $4@1 + 3@3$ has the positive actual future point

$$\xi = c + (\sqrt{2} - 1/2)n_1 + (\sqrt{2} - 1)n_2 + (\sqrt{2}/2)n_3, \qquad \xi^2 = 5/4 - \sqrt{2} = -k.$$

Its products with c, n_1, n_3 vanish and its product with n_2 is $(\sqrt{2} - 3)/4 < 0$. Here $A^2 = (5 + 4\sqrt{2})/7$ and $A^2 - 9k = (335 - 236\sqrt{2})/28 > 0$; (9.3) gives energy $> (A - \sqrt{k})/(A + \sqrt{k}) > 1/2$.

B.1.5 The four intermediate B components

Use $u_i = (e_1 + \cdots + e_i)/\sqrt{i}$ and, for $i < r$, $\omega_i = \sqrt{2}(e_1 + \cdots + e_i)$, while $\omega_r = e_1 + \cdots + e_r$. At energies $3/7, 1/2, 4/7$ the relevant forcing signs are $127\sqrt{2/3} - 100$, $(69\sqrt{6/7} - 52)/7$, and $(199\sqrt{6/7} - 176)/7$, all positive. Thus B_5, B_6 have a common $2/3$, and B_7, B_8 a common $6/7$.

For B_5, B_6, deletion 2 retains $A_1 + B_3/B_4$. The right contact is $3@3$, with the sole B_5 exception $3@5$ giving F_4; deletion 3 allows one left $3/4/5$. For $3@3$, own 0 is affine, own 3 is a triangle, and own ≥ 4 gives $\cosh L_3 \geq 2 + 3/\sqrt{2} > 3$. With no right contact the complete choices are $3/4/5$ at 1 or 2: $3@2$ and either 4 are affine; $3@1$ has $h = Q - 1/2 < 1/6$ and closes a forbidden foreign direction. The two 5 cases have $Q = 0$: $5@1$ also closes 4 with $\cosh L_4 = \varphi^3 > 3$, and $5@2$ has $\cosh^2 L_3 = 2\varphi^2/(2 - \varphi) > 9$. For the B_5 short-end $3@5$, finite F_4 forbids left 1 and all spectators; own 0 is affine and the other replacement forces own 3, giving $\cosh L_2 = 2 + \sqrt{2}$. Every branch exceeds the individual threshold $11/5$.

For the $6/7$ pair, a wall without a right contact has a single left 3 or endpoint 4. Apart from far $3@1$ these give an affine fork, double-4, or star. A minimal nonpositive principal for the far

branch must contain $c, 7$, since deletion 7 is finite; their connecting path $c, 1, \dots, 7$ has eight vertices. Spectators attach through c and do not shorten it. This violates the Lannér order bound, excluding the branch with all spectators. In B_7, let β be the contact to the retained right A_1:

β/own 6	left contacts	pressure or rejection
$\beta \geq \cos(\pi/5)$, any	none	$\cosh L_6 \geq \sqrt{6}\cot(\pi/5) > 3$
$\beta_4, 0$	any allowed	affine $(4,4)$
$\beta_4, > 0$	none	$\cosh L_6 \geq \sqrt{24}$
$\beta_3, > 0$	none	$\cosh L_6 \geq 2 + \sqrt{2}$
$\beta_3, 0$	none or 1	affine F_4
$\beta_3, 0$	one $j = 2, \dots, 5$	$\cosh L_6 \geq 1 + 3/\sqrt{2} > 3$.

Here $\beta_m = \cos(\pi/m)$. The last row retains the full foreign cost: its white residual is $\Delta \leq 3/4 - j(6-j)/12 \leq 1/3$, and $v_6 \geq \sqrt{12}(j/12 + 1/(2\sqrt{2}))$. Thus no allowed E_7 extension is silently removed.

In B_8 the retained right B_2 has one simple 3 at either end. At long end 7, own $0/3$ is affine and own 4 gives $\cosh L_7 \geq 6 + 7/\sqrt{2} > 7$. At short end 8, own > 0 gives $\cosh L_6 \geq \sqrt{6} + \sqrt{3} > 11/3$. At own 0, the left-white budget permits only 3@1 or 3@5: none or 1 gives affine F_4, but 3@5 is an allowed bridge, with $\Delta \leq 1/12$ and $v_6 = \sqrt{12}(5/12 + 1/(2\sqrt{2}))$. It has $\cosh L_6 \geq 5 + 3\sqrt{2} > 9$; it is estimated, not removed by an incorrect F_4 sealing claim. These complete lists prove the four asserted B costs.

B.1.6 Extended stars, D_7, and the D_8 closure

Let S be a central D_4 star with an extended arm. If one wall closes its center and three neighbors, every minimal nonpositive principal contains c and all four nodes of S, since deletion of any of them is an actual finite replacement. Compactness and the Lannér order bound make that principal exactly $S + c$. Two contacts give a nonpositive proper triangle or cycle. A sole center 3 is affine, and a larger center contact has a nonpositive proper star. A sole leaf contact is $3/4/5$ from its finite proper path: 3 gives finite D_5, 4 affine, leaving only 5. If that leaf extends, it retains a nonpositive $(5,3,3)$ star; if another arm extends, it retains a nonpositive $(5,3,3,3)$ path. Each omits a supposedly closed S direction, a contradiction. Further negative contacts cannot repair the positive nonpositive quadratic witness. Thus the four opposites cannot all coincide.

For D_5, D_6 the minimum central correlations are $\sqrt{3/5}, \sqrt{2/3}$. At energies $5/14, 3/7$ the forcing signs are $544\sqrt{3/5} - 362$ and $127\sqrt{2/3} - 100$, positive. For E_6, E_7, E_8 the arm lengths are $(1, 2, L)$ with $L = 2, 3, 4$. In raw norm-two coordinates the central Schur residual is $\delta = 2 - [1/2 + 2/3 + L/(L+1)]$; on an arm of length m, squared central correlation is $t/(\delta + t)$, $t = m/(m+1)$. The minima are $\sqrt{3}/2, \sqrt{6/7}, \sqrt{15}/4$. At energies $9/20, 5/9, 2/3$ even $4e/(1+e)^2$ lies below these correlations. All four opposites would coincide, proving the five extended-star costs.

For D_7 write $o = r - 2$, $a = r - 3$, with fork leaves b, f. At energy at most $1/2$, correlation $\sqrt{4/5}$ forces a common o, a. Deletion a retains $A_3 = b - o - f$, so its right contact is none, middle 3, or endpoint $3/4/5$; deletion o allows one left contact. The complete surviving profiles are

$$4@a + 3@o, \qquad \text{isolated } 5@b, \qquad 3@a + 4@b, \tag{B.1.6}$$

up to exchanging the fork leaves. To check the list: middle 3 forces $4@a$ (own $0/3$ is affine); endpoint 5 seals all white extensions by H_4, and a contact at the removed a leaves a nonpositive $(3,3,3,5)$ path. Endpoint 4 forces $3@a$, otherwise an affine terminal-4 fork remains. Endpoint 3 excludes every left contact by cycles or retained F_4; a foreign 3 is affine E_7, a foreign 4 is an affine terminal-4

fork, and larger foreign bonds are nonfinite. Its isolated height is negative for $r = 7$ and it contains affine E_8 for $r \geq 8$. With no right contact, internal-left 3 and endpoint 4 are affine; far-left 3 needs a Lannér path of at least six vertices. This accounts for all spectators as well as the removed own contact. The first and third roles have $\cosh L_o = a + (r-2)/\sqrt{2} > 7$. The isolated 5 has $\tanh^2 L_a = 1 - (9 - 4\sqrt{5})/(r-3)$; at $r = 7$ its half-tangent square is $4/\sqrt{5} - 1 > 3/4$. Each role therefore contradicts energy at most $1/2$, proving the D_7 cost.

For D_8, common $4/5$ has exactly $4@4 + 3@5$. Indeed deletion 4 permits simple 3 at a right D_4 leaf, and deletion 5 allows one left 3 or endpoint 4. Long-leaf 3 forces $4@4$ and forbids foreign contacts by the retained $(3, 4, 3, 3)$ path. A fork 3 with a left 3 gives an affine cycle; without a left contact it is affine E_8; a left 4 is nonpositive. With no right contact only far $3@1$ survives the affine tests, but its path to 5 has six vertices. Common $5/6$ has the preceding center-neighbor list. Both common pairs thus force an actual half-tangent square above $3/4$. At energy at most $4/7$, their opposites are distinct.

Here is the remaining three-direction calculation. Put $q = \sqrt{4/5}$, $k = \sqrt{5/6}$, $\lambda = (1+k)/2$, $\alpha = k\sqrt{1-k}/2$, $e = 4/7$, and $P(t) = t/(1+\sqrt{1-t})^2$. Write $h_i = x \cdot u_i$ and decompose $x = aw + bv + zt_0 + y$, where w, v are the normalized sum and difference of u_5, u_6, $u_4 = qu_5 + t_0/\sqrt{5}$, and $y \perp w, v, t_0$. The full-tangent laws imply

$$h_4 h_5 \geq P(q), \qquad T := (1+\alpha)b^2 + \lambda z^2 \leq \delta := \lambda e - P(k) < 1/400.$$

For the second inequality take the tangent line to $4p/k - (1+p)^2$ at its first zero $p = P(k)$, with $p = h_5 h_6 = \lambda a^2 - (1-\lambda)b^2$. Keeping the negative quadratic terms in $h_4 h_5$ gives

$$h_4 h_5 \leq q\lambda e + A_0|b| + B_0|z| - qkb^2 - q\lambda z^2, \qquad A_0^2 = 8/105, \quad B_0^2 = 4/35.$$

Weighted Cauchy bounds the linear part by $\sqrt{C_0 T}$, where $C_0 = (8/105)/(1+\alpha) + (4/35)/\lambda < 47/250$. Also $qk/(1+\alpha) > .71$, $q > .71$, and $q\lambda e < .489$. Consequently

$$h_4 h_5 < .489 + \sqrt{(47/250)T} - .71T < 101781/200000 < .509 < P(q),$$

a contradiction. The rational seals are $.8944 < q < .8945$, $.9128 < k < .9129$, $.29 < \sqrt{1-k} < .2953$, $.13 < \alpha < .14$, and $.95 < \lambda < .95645$. The tight δ comparison reduces to $91280000 \cdot 70000 - 167780209 \cdot 38083 = 26300653 > 0$; the last square-root comparison uses $\sqrt{47/100000} < 271/12500$. This proves $D_8 > 4/7$ and completes Proposition B.1.1.

B.1.7 The strengthened rank-three rates

We prove the improvement $B_3, H_3 > 2/7$ used in the forest accounting. Suppose the component energy is at most $2/7$. The adjacent correlations are at least $1/\sqrt{2}$, so the forcing argument above gives one actual opposite for the whole component, with

$$h \leq R^2 \min_i v_i^2, \qquad R^2 = 56/81.$$

All foreign costs remain included in $h = q + Q - 1$. Multiple contacts can occur only at the two path endpoints.

For B_3, endpoint contacts u, w give

$$q - R^2 v_2^2 = (131u^2 + 106w^2 + 50\sqrt{2}uw)/81.$$

If either label is at least 4, its minimum is $13/9$ or $443/324$, both greater than 1. Larger contacts increase the expression. At $u = w = 1/2, Q = 0$, the actual half-tangent squares in directions $2, 3$

are $(2\sqrt{2}-1)/7$; increasing Q increases both lengths. Their correlation is $1/\sqrt{2}$, so the positive two-direction cover costs at least $(10\sqrt{2}-12)/7 > 2/7$. For one contact the diagonals are $3, 4, 2$, with minimum ratios $v_i^2/q = 1/3, 1/2, 1/3$. Thus the maximum own costs are $243/187, 81/53, 243/187$. Nodes $1, 2$ admit only native 3, and node 3 only $3, 4$, since $243/374 < \varphi^2/4$. Every $q = 1$ case is affine; every remaining $q < 1$ case has $h < 1/4$, forcing $Q = 0$ and $h = q - 1 < 0$.

For H_3, the endpoint quadratic has positive coefficients; its minimum at $u = w = 1/2$ is $(200\varphi + 287)/324 > 1$. At node 1 or 2, even contact $1/2$, tested in direction 3, gives respectively

$$(599\varphi + 823)/1620 > 1, \qquad (181\varphi + 237)/405 > 1.$$

At node 3, contact at least $1/\sqrt{2}$, tested in direction 1, gives $(75\varphi + 374)/486 > 1$. Only contact $1/2$ remains, with

$$q = (\varphi + 2)/4 < 1, \qquad h \le 14\varphi^4/243.$$

No foreign component gives $h < 0$; exactly one is impossible by $Q < h$. Two or more give

$$h \ge q - 1/2 = \varphi/4 > 14\varphi^4/243,$$

the final difference being $(75\varphi - 112)/972 > 0$. This covers all native/dotted contacts and all spectators.

B.1.8 A rank-free joint payment for an isolated I

Let the actual isolated short wall be

$$c_0 = \sqrt{h}\, p - \sqrt{1+h}\, e, \qquad 0 < h < 1/2,$$

with e the old unit I normal. For an actual opposite outside I, write its time as A and its contact with e as γ. The finite white endpoint retains e, so $\gamma = 0$ or $1/2 \le \gamma < 1$. These opposites are distinct. For the chosen unit dual $u \perp e$, its spatial derivative has square at most $1 + A^2 - \gamma^2$. If $\gamma = 0$, the native gap gives $A^2 \ge 1/(4h)$; if $\gamma > 0$, acuteness gives $A^2 \ge (1+h)\gamma^2/h$. Since $\gamma^2 < 1$, the derivative comparison yields in both cases

$$\tanh^2 L \ge \frac{1}{1+4h}. \tag{B.1.7}$$

For the nonzero case, substitute the lower bound on A^2; the difference from $1/(1+4h)$, after positive denominators are cleared, has numerator $h((5+4h)\gamma^2 - 1) > 0$.

Let q be the actual I-edge half-tangent square. Then $h = 4q/(1-q)^2$, $0 < q < 1/5$. If θ is the half-tangent square of the other edge, the increasing function $4s/(1+s)^2$ and (B.1.7) show $\theta > 1/5 - q$. The exact cross numerator for the strict comparison is

$$(1-q)^2(6-5q)^2 - 20(1-5q)(1+14q+q^2) = (39q-4)^2 + q^2(40 - 10q + 25q^2) > 0.$$

Thus the same original vector satisfies $e_I + \langle x, u\rangle^2 > 1/5$. For another whole A_1 this proves $I + A_1 > 1/5$; adding a second ordinary $A_1 > .1$ proves $I + 2A_1 > .3$.

For the sharper $I + T_2$ bound set $s = \sqrt{q} \in (0, 1/3)$ and $t = \sqrt{1+4h} - 2\sqrt{h}$. Equation (B.1.7) gives normalized height at least t in every other actual edge direction, and

$$e_I \ge s^2, \qquad (1-s^2)(1-t^2) = 8st.$$

We need $t + \frac{3}{2}s > .767$. At $t_0 = .767 - 1.5s > 0$, the left-minus-right of that relation is

$$P(s) = .411711 - 3.835s + 9.338289s^2 - 2.301s^3 + 2.25s^4.$$

Putting $v = s - .2 \in (-.2, 2/15)$, its coefficients are $.00343456, -.3038044, 8.497689, -.501, 2.25$. Its cubic term is at least $-.0668v^2$, so

$$P(s) > .0034 - .304|v| + 8.43v^2 > 0,$$

as the last quadratic has negative discriminant. The original relation decreases strictly in positive t, proving the barrier. This is a scalar real argument; no arithmetic chapter of its source is used.

The ordinary T_2 energy is greater than $1/7$. If $s \geq 519/2000$, $s^2 + 1/7 > .21$. Below that split, $s^2 + (.767 - 1.5s)^2$ decreases to $.2100353125 > .21$. Thus $I + T_2 > .21$, counting I once.

For $I + T_3 > 2/7$, B_3, H_3 already have the intrinsic stronger bound. For A_3, if $h > 1/5$, the actual I-edge gives $e_I > 3/70$, and $3/70 + 1/4 > 2/7$. If $h \leq 1/5$, all three other half-tangent squares are at least $1/5$. The normalized A_3 dual Gram K has $K^{-1}\mathbf{1} > 0$ and $\mathbf{1}^T K^{-1}\mathbf{1} = 5 - 2\sqrt{3}$; positive Cauchy gives energy at least $(5 - 2\sqrt{3})/5 > 3/10 > 2/7$. The same positive-cover principle will be used below: if $K^{-1}\mathbf{1} > 0$ and every height is at least $\sqrt{\theta}$, energy is at least $\theta \mathbf{1}^T K^{-1}\mathbf{1}$.

B.1.9 Propagation from each actual typed rank-two defect

For N_7, put $b = \cos^2(\pi/7) < 5/6$ and $h_N = 4b/3 - 1 < 1/12$. An opposite in another component retains the whole old A_2, so has no contact there or one contact of white cost $q \in [1/3, 1)$. In the contact case its spatial product with the isolated wall has square at least $bq/3$. Acuteness and the bound on the chosen derivative give

$$A^2 \geq \frac{bq}{3h_N}, \qquad \tanh^2 L \geq \frac{b}{b + 6h_N} = \frac{b}{9b - 6} > \frac{5}{9}.$$

The first ratio is minimized at $q = 1/3$; all denominators are positive. Without contact the native gap instead gives $\tanh^2 L \geq 1/(1 + 4h_N) > 5/9$. Every other half-tangent square therefore exceeds $1/5$. In particular another A_1 or irreducible T_2 has energy greater than $1/5$.

For O_7, put $d_7 = \csc^2(\pi/7)$ and $h_O = d_7/4 - 1 \in (0, 1/3)$. No opposite in another component can contact the retained old $I_2(7)$: its minimum white cost $d_7/4 > 1$ violates positive definiteness. Its actual product with the isolated typed wall is thus $-\sqrt{h_O}A < 0$, and the native gap gives

$$A^2 \geq 1/(4h_O) > 3/4 \tag{B.1.8}$$

for every opposite in the other component. This immediately gives another A_1 cost $> 2/15$ by the full-/half-tangent comparison in the main text.

To prove another irreducible $T_2 = I_2(m)$ costs $> 1/5$, suppose the contrary. With $k = \cos(\pi/m)$ and $d = (1 - k^2)^{-1}$, distinct opposites are impossible: for $m \geq 4$, their tangent product is at most $F(\sqrt{1/5})^2 = 5/9 < k$; for $m = 3$, use the pair sum bound to get $F(\sqrt{3/20})^2 = 240/529 < 1/2$. There is one common actual opposite, and both actual edge cosines are at most $3/2$. For own contacts $b, g \geq 0$ and entire foreign cost Q, the finite endpoint equations are

$$Q + b^2 < 1, \quad Q + g^2 < 1, \quad h = d(b^2 + 2kbg + g^2) + Q - 1 > 3/4,$$
$$C_b = \frac{\sqrt{d}(b + kg)}{\sqrt{1 - Q - g^2}}, \qquad C_g = \frac{\sqrt{d}(g + kb)}{\sqrt{1 - Q - b^2}}.$$

Every nonzero own is consequently a finite native cosine. There is at least one, and $Q = 0$ or $Q \geq 1/4$.

If only $b = \beta > 0$, the mate has $C_g^2 = 1 + h/(1 - Q - \beta^2)$. The bounds $h > 3/4, C_g \le 3/2$ force $Q + \beta^2 < 2/5$, hence $Q = 0, \beta = 1/2$. Then $h = d/4 - 1 > 3/4$ implies $d > 7 > d_8$, so $m \ge 9$. But $d > 81/\pi^2 > 81/10 > 31/4$, giving $C_g^2 = (d-1)/3 > 9/4$, a contradiction.

If both contacts are positive and $Q > 0$, either their maximum is at least $1/\sqrt{2}$, giving $C_b^2 \ge 2 + 4\sqrt{2}/3 > 9/4$, or both are $1/2$. For old $m = 3$, the latter gives $h = Q < 3/4$; for $m \ge 4$, it gives $C_b^2 \ge (1+k)/(2(1-k)) > 5/2$. If both contacts are positive and $Q = 0$, a contact of label at least 5 gives $C_b^2 > 3$; two 4's give $C_b^2 = (1+k)/(1-k) \ge 3$. For $3/4$, old $m \ge 4$ gives $C_b \ge 2$, while old $m = 3$ gives $h = \sqrt{2}/3 < 3/4$. For $3/3$, old $m \ge 5$ gives $C_b^2 = (1+k)/(3(1-k)) > 3$; old $m = 3, 4$ give $h = 0, 1/\sqrt{2} < 3/4$. This exhausts all finite labels and all foreign costs, contradicting the common-wall low-energy branch. Thus $O_7 + T_2$ has its other component energy $> 1/5$.

B.2 The uniform actual B/D component floor

The next proof supplies the $9/14$ input used before the long-prefix argument. Its small boundary cases refer only to the complete ordinary contact calculations in Appendix B.1. All displacements remain in the indicated component, and all positive actual-root constructions retain every original spectator wall.

Proposition B.2.1 (Actual B/D floor)**.** *Every whole B_r or D_r component with $r \ge 9$ has actual projection energy greater than $9/14$, with arbitrary finite spectators.*

Proof. Assume energy at most $e = 9/14$. In B it suffices to bound the first seven coordinates; all displacements below lie there. The normalized directions in either type are $u_i = (e_1 + \cdots + e_i)/\sqrt{i}$. We first establish distinctness of $4/5, 6/7$, then use those laws to exclude all common $1/2, 2/3$ profiles. This order is noncircular.

For an internal common $i/(i+1)$ with right retained rank at least four, finite classification permits a right long-leaf 3, or none. The first forces own 4: own $0/3$ is affine fork/triangle, and any foreign contact retains the nonpositive $(3,4,3,3)$ path. Without a right contact, internal-left 3 and endpoint 4 are affine. Only far 3@1 remains, but its path to $i+1$ has at least six vertices, contradicting the Lannér bound. Right D-fork contacts give affine E_8 or a cycle with the left contact. Thus the sole common profile is $4@i + 3@(i+1)$, $Q = 0$, with $\cosh L_{i+1} = i + (i+1)/\sqrt{2} > 7$. The D_8 right-D_4 and center-neighbor boundary calculations are in Section B.1; the center-neighbor calculation also handles D_9 at $6/7$. In B_9, the extra right short-end 3@9 gives F_4, excluding left and foreign white contacts. The removed own 6 remains to check: own 0 is affine F_4, own 4 the affine $(4,3,3)$ star, and own 3 gives $\cosh L_6 = (1 + 1/\sqrt{2})\sqrt{12} > 11/2$ and $\tau_6^2 > 9/13 > e$. Consequently $4/5$ and $6/7$ are distinct.

Write $h_i = x \cdot u_i$. Distinct $6/7$ gives $h_6 h_7 \ge k/(1 + \sqrt{1-k})^2 > 9/16$, where $k = \sqrt{6/7} > 576/625$ with squared margin 21318. Hence $x \cdot w > 3/2$ for $w = u_6 + u_7$. We record four local tests with their exact margins.

Edge 2 *with* $\cosh L_2 \ge 1 + \sqrt{2}$. Then $h_2^2 \ge \sqrt{2} - 1$. Projection of w about u_2 has $\rho = 1/\sqrt{3} + \sqrt{2/7} < 1.114$ and residual square $S = 29/21 + 4\sqrt{2/21} < 2.62$. Its Cauchy envelope $\rho t + \sqrt{S(e - t^2)}$ decreases for $t \ge q = \sqrt{\sqrt{2} - 1}$: its derivative condition follows from $\rho^2 e < 13/16 < 6/5 < \|w\|^2 q^2$. Since $q < .644$, $e - q^2 < .23$, and $\sqrt{2.62 \cdot .23} < .777$, $x \cdot w < 1.114(.644) + .777 = 1.494416 < 3/2$. These follow by squaring, using $1.414 < \sqrt{2} < 1.4143$, $1/\sqrt{3} < .578$, $\sqrt{2/7} < .535$, $\sqrt{2/21} < .309$.

Isolated 5@1. Its height square is $(\varphi - 1)/2$ and its actual direction-4 shadow has $\tau_4^2 = \varphi - 1$. For the same w, projection about u_4 has $\rho = 2(1/\sqrt{6} + 1/\sqrt{7}) < 787/500$ and $S = 16/21 + 4/\sqrt{42} < 69/50$. Use $1/\sqrt{6} < .409$, $1/\sqrt{7} < .378$, $\sqrt{42} > 162/25$; the last residual comparison has cleared

margin 23. Since $\sqrt{e} < 401/500$, $e - (\varphi - 1) < 1/40$ and $(93/500)^2 > 69/2000$, Cauchy gives $x \cdot w < 362087/250000 < 3/2$.

Isolated 5@2. Its height square is φ. The direction-1 shadow gives $x_1 > 31/50$, since $F(31/50)^2 < 161/200 < \varphi/2$ with cleared margin 6541881. Its actual edge 3 has $\cosh L_3 = \sqrt{2}\varphi^2 > 37/10$; hence $S_3 = x_1 + x_2 + x_3 > 131/100$, using $3\tau_3^2 > 81/47 > (131/100)^2$, of margin 3433. Distinct 4/5 gives $x \cdot w_0 > 57/40$, $w_0 = u_4 + u_5$. Put $v = e_1 + e_2 + e_3$, $k_0 = 1/2 + 1/\sqrt{5} < 237/250$, and $x_0 = (31/50, 69/200, 69/200, 0, \ldots)$. Then $x_0 = (11/40)e_1 + (69/200)v$, $x \cdot x_0 > x_0^2 > 31/50$. The residual $w_0 - k_0 v$ has square $13/20 + 1/\sqrt{5} < 11/10$ and excess demand $> 9/50$. All coefficients of $x_0 + (w_0 - k_0 v)/6$ remain positive, and its Cauchy margin is at least

$$(31/50 + (9/50)/6)^2 - e(31/50 + (11/10)/36) = 3/700 > 0.$$

Contacts 4@2 + 3@3. The white A_2 cost is 2/3, so $Q = 0$ or $1/4$; the latter has $\cosh L_3 = 4 + 3\sqrt{2} > 7$. At $Q = 0$, put $s = \sqrt{2}$. The actual floors satisfy $\tau_2^2 = (1 + 2s)/7$, $\tau_3^2 = (2s - 1)/3$, $\tau_2\tau_3 = 1/\sqrt{3}$. Their positive equality vector has $E_0 = (6s - 4)/7 > .6407$. The positive old-A_2 correction of c is an actual future point of square $-s/3$, giving pure displacement

$$U = \frac{2+s}{2}e_1 + \frac{1+s}{3}(e_2 + e_3 + e_4), \qquad x \cdot U \geq A - \sqrt{s/3}, \quad A^2 = (5 + 4s)/2.$$

Its projection coefficients on u_2, u_3 are $\rho_2 = (2s + 1)/6$, $\rho_3 = (1 + s)/\sqrt{3}$; its residual square is $(7 + 4s)/12 < 1.055$, with excess demand $\Delta > .06$. Indeed $1.4142 < s < 1.4143$ gives $A > 2.3083$, $\sqrt{s/3} < .6867$, $\tau_2 \in (.7395, .7396)$, $\tau_3 \in (.7806, .7808)$, $\rho_2 < .6381$, $\rho_3 < 1.394$, and excess $> .06122604$. The old positive coefficients exceed .3, .52; subtracting $(\rho_2, \rho_3)/16$ removes less than .04, .088. The corrected dual therefore has Cauchy margin at least

$$(.6407 + .06/16)^2 - e(.6407 + 1.055/256) = 8825113/11200000000 > 0.$$

This is an actual-point constraint, not an edge-only assertion.

Now exhaust common 1/2. Deletion 1 retains right rank at least eight, so the right contact is none or long-leaf 3@2. The latter forces own $\gamma \geq 1/\sqrt{2}$ (own 0/3 is affine), and $\cosh L_2 = (1 + \gamma)/\sqrt{1 - \gamma^2 - Q} \geq 1 + \sqrt{2}$, already excluded. Without a right contact, $A^2 = 2\gamma^2 + Q - 1 > 0$ and $\gamma^2 + Q < 1$. Own 0 has zero derivative; own 3 has height below 1/4 and forces a foreign closure; own 4/6 is affine. Own 5 with $Q > 0$ has $\cosh L_2 \geq \varphi^2 > 1 + \sqrt{2}$; at $Q = 0$ use the isolated-5 test. Finite $m \geq 7$ forces $Q = 0$; $m \geq 8$ uses the edge-2 test, leaving only isolated 7.

For common 2/3, deletion 2 retains right B/D rank at least seven; deletion 3 allows one left 3/4/5 at 1 or 2. A right long 3@3 forces own 3/4/5: own 0 is affine, own 3 a triangle, own 4 is the tested $4 + 3$ profile, and own 5 has $Q = 0$, $\cosh L_3 = (2 + 5\varphi)/\sqrt{2} > 7$. Without a right contact the full left list is $3@1, 4@1, 5@1, 3@2, 4@2, 5@2$. The first has $A^2 = Q - 1/2 < 1/6$ and forces foreign closure; $4@1, 3@2, 4@2$ are respectively affine terminal-4, two-fork, and $(4, 3, 3)$ star. The 5 cases have $Q = 0$ by white cost $\varphi^2/3 > 3/4$ and were tested above. For D_9 one extra right-fork 3 gives retained $D_7 + c = E_8$. It excludes old-1 and foreign contacts, but the removed own 2 must still be checked: own 0/3/4 gives affine E_8/cycle/star, and own 5 retains nonpositive $(3, 5, 3)$. At higher rank the fork attachment already contains affine E_8. No finite branch is omitted.

For the final isolated 7@1, put $t = \cos(2\pi/7)$, $g = \cos(\pi/7)$, and

$$\xi = c + g^{-1}n_1 + (1 - t)g^{-1}n_2 = \sqrt{t}\,p - \frac{V}{\sqrt{1+t}}, \qquad V = t(e_1 + e_2) + (1 - t)e_3.$$

It has $\xi \cdot c = \xi \cdot n_1 = 0$, $\xi \cdot n_2 = (1/2 - t)/g < 0$ and $\xi^2 = -(1 - t)(2t - 1)/(1 + t) < 0$. All coefficients and its time coefficient are positive, proving actual feasibility and future orientation. Thus

$x \cdot V \geq \sqrt{t(1+t)} - \sqrt{(1-t)(2t-1)}$. The increasing cubic $8t^3+4t^2-4t-1$ gives $.6234 < t < .6235$, with endpoint values $-.000926392768, .000105223$. Hence the first radical is > 1.0059, the second $< .305$, and $.95 < \|V\| < .9588 < .959$, so $x \cdot \widehat{V} > .73$. The projection of w onto $\widehat{V}$ is $\rho = (1/\sqrt{6} + 1/\sqrt{7})(1+t)/\|V\| \in (1.3, 1.35)$, and residual square is $< 11/5$. The Cauchy envelope decreases above .73, since $\rho^2 e < 6561/5600 < 3(.73)^2 < \|w\|^2(.73)^2$. It gives $x \cdot w < 1.35(.73) + .492 = 591/400 < 3/2$; the final squared radical margin is $277/1750000$. Thus $1/2, 2/3, 6/7$ all have distinct actual opposites.

Their exact pair sums are

$$x \cdot (u_1 + u_2) > 1.091, \quad x \cdot (u_2 + u_3) > 1.265, \quad x \cdot (u_6 + u_7) > 1.512.$$

Set $W = .1(u_1 + u_2) + .055(u_2 + u_3) + .305(u_6 + u_7)$. Direct positive Gram estimates give

$$x \cdot W > .639835, \qquad W^2 < .636435825,$$
$$.639835^2 - \frac{9}{14}(.636435825) = \frac{70423123}{280000000000} > 0.$$

To verify the pair sums, use $F(a)F(b) \leq F((a+b)/2)^2$. The correlation lower bounds

$$7071/10000, \qquad 81649/100000, \qquad 9258/10000$$

have squared margins $959, 322397, 26052$ below the exact values. At sums $m/1000$, $m = 1091, 1265$, 1512, the cleared differences

$$p(4000000 + m^2)^2 - 16dm^2\, 1000000$$

are respectively

$$40828209713231, \qquad 358399613480625, \qquad 52443946917888.$$

The six upper Gram entries of the three pair vectors are $3.415, 3.633, 3.852, 3.102, 1.899, 2.474$, obtained by positive squaring of their prefix correlations. Their expansion is the stated bound on W^2. This contradicts Cauchy and proves the proposition. □

B.3 Coupled actual pressures in an A-chain

The crucial feature of this section is that a foot at one end is tested against *actual* opposites at the other end. A collection of separate edge estimates would lose that coupling.

Theorem B.3.1 (Uniform actual A pressure)**.** *With arbitrary finite spectators,*

$$\|x_{A_r}\|^2 > \frac{1}{2} \qquad (r \geq 7).$$

We first prove the uniform step for rank at least eight, then give the rank-seven base in Subsection B.3.3. Put $N = r + 1$, and realize

$$n_i = (e_i - e_{i+1})/\sqrt{2}, \quad C_{ij} = \frac{2\min(i,j)(N - \max(i,j))}{N}, \quad h_i = x_A \cdot u_i.$$

The root space is the sum-zero subspace of $\mathbb{R}^N$. All its inequalities below use the actual projection x_A.

B.3.1 A forcing window and the complete common-pair inventory

Suppose first that $N \geq 9$ and $\|x_A\|^2 \leq 1/2$. Consider the four pairs $1/2, 3/4, (r-3)/(r-2), (r-1)/r$. If their respective opposites are distinct, the outer pair sums exceed 1, and the inner sums exceed $19/16$. These follow from (9.7), concavity of $\log F$, and the minimum correlations $\sqrt{3/7}$, $\sqrt{3/5}$ at $N = 8$. For the inner comparison,

$$F(19/32)^2 = \frac{1478656}{1918225} < \frac{387}{500} < \sqrt{3/5}.$$

For the outer one, $F(1/2)^2 = 16/25 < \sqrt{3/7}$. Set

$$L = 2u_1 + 2u_2 + u_3 + u_4, \qquad R = 2u_r + 2u_{r-1} + u_{r-2} + u_{r-3}.$$

The correlations on the left increase to $\sqrt{i/j}$, whence

$$L^2, R^2 < 12 + 6\sqrt{2} + 4/\sqrt{3} + 4\sqrt{2/3} + \sqrt{3} < 28.$$

The last bound follows, for example, by the positive rational upper bound $2351/84$. For $a = (2, 2, 1, 1)$, direct coordinates give

$$L \cdot R = \left(\sum_{i=1}^{4} a_i \sqrt{\frac{i}{N-i}} \right)^2.$$

This decreases in N; at $N = 9$ its square root is $\sqrt{2} + 2\sqrt{2/7} + 2/\sqrt{5} < 17/5$. Thus $(L + R)^2 < 80$, whereas distinctness of all four pairs gives $x_A \cdot (L + R) > 51/8$. The Cauchy gap is $41/64 > 0$. After reversing the numbering, a common pair is therefore $1/2$ or $3/4$. This is a relabeling, not a symmetry assumption on P.

Here is the finite inventory used for the target in Theorem B.3.1. It is useful to state it before making any energy-dependent discard. Write Q for the total spectator cost. Every retained finite diagram is a forest, so there is at most one contact on each retained own path. The foreign costs satisfy (9.8), are at least $1/4$, and have the gap $(1/4, 1/3)$.

For a common $3/4$ opposite, its own contact positions are in $\{1, 2, 3\}$ and $\{4, \ldots, r\}$. A double profile away from positions $3, 4$ survives one deletion as a connected rank-r finite diagram. Its two new bonds must be 3: a weighted bond would be internal in a B path, and D, E_8 are simply laced. The two 3's instead give an affine cycle in the full diagram. This also covers the rank-eight E_8 boundary. At positions $3, 4$, the labels are

$$p \in \{3, 4, 5\}, \qquad q \in \{3, 4\}.$$

The pair $3, 3$ is affine. The two smallest remaining assignments have actual endpoint cosines

$$\sqrt{\frac{3}{2N}}(N - 3 + \sqrt{2}(N - 4)), \qquad (3\sqrt{2} + 4)\sqrt{\frac{N-4}{2N}}, \tag{B.3.1}$$

both greater than 3 at $N = 9$ and increasing thereafter. Increasing a contact or Q strengthens the appropriate endpoint.

A single 3 at positions $3, \ldots, r-2$ contains an affine E_8 tree with arms $1, 2, 5$, or an affine E_7 tree with arms $1, 3, 3$, obtained by truncation if needed. The other single-3 possibilities satisfy

position	Q bound	A^2
1	$Q < 5/8$	$Q - (N+1)/(2N) < 1/8$
2	$Q < 1/2$	$Q - 2/N$
$r-1$	$Q < 2/(N-3)$	$Q - 2/N$
r	$Q < 1/2 + 1/(2(N-3))$	$Q - (N+1)/(2N) < 3/(2N(N-3))$.

The end heights are below $1/4$; the middle rows allow at most one foreign component and its cost exceeds A^2. All contradict (9.8). A single 4 is an affine $3,3,4$ star internally; at either end its white condition gives $Q < 1/4$, so $Q = 0$ and its height is negative. A single 5 can occur only at 1 or 3, has $Q = 0$ from its H_4 cost $3\varphi^2/8 > 3/4$, and at 3 closes direction 4 with

$$\cosh L_4 = 3\varphi^3\sqrt{(N-4)/N}. \tag{B.3.2}$$

At 1 it gives foot F below. A zero own contact cannot close an own edge. This completes the common $3/4$ list.

For a common $1/2$ opposite there is possibly a contact at 1 and at most one on $2,\dots,r$. For double positions $1,j$, $j \geq 3$, deletion 2 gives a full connected finite path or branch. A branch has only 3's; a path can have a 4 only at its terminal bond $1-c$, with $j = 3$ or r. All 3's give an affine cycle. The profile $4@1 + 3@r$ contains the affine F_4 path $2,1,c,r,r-1$. The remaining $4@1 + 3@3$ has $Q < 1/(2(r-1))$, hence $Q = 0$, and gives foot H. Adjacent contacts $1,2$ have an arbitrary finite native first label and second label 3 or 4; $3,3$ is affine.

For single 3, the same internal affine exclusions apply. At $1, r, r-1$, respectively, the bounds are

$$Q < 3/4, \quad Q < 1/2 + 1/(2r), \quad Q < 2/r.$$

The first two give height below $1/4$, and the last leaves no foreign cost and negative height. At 2, $Q < 1/2 + 1/(2r)$ allows at most two foreign components. One exceeds $A^2 = Q - 2/N$. With two, both contacts are 3; c, n_2, n_1, n_3 and their two foreign neighbors give the exact affine two-fork diagram. A single 4 is affine internally; at 1, $Q < 1/2$ allows at most one cost greater than $A^2 = Q - 1/N$; at r, $Q < 1/r$ leaves negative height. A single label at least 5 is possible only at 1. Label 6 gives the affine $3,6$ path. Labels at least 7 have $Q < 1 - \beta_1^2 < 1/4$, hence $Q = 0$. Label 5 with $Q = 0$ gives F; with $Q > 0$ it remains in L. Thus the remaining end profiles are

$$\text{L}: \text{ adjacent, not } (3,3);\ \text{isolated native } \geq 7;\ \text{isolated 5 with } Q > 0, \qquad \text{H}, \text{F}.$$

No foreign branch has been suppressed by assuming a full star finite.

B.3.2 Three feasible feet prove the half bound

For H and F take the following positive actual combinations:

	ξ	$k = -\xi^2$
H	$c + (\sqrt{2} - \frac{1}{2})n_1 + (\sqrt{2}-1)n_2 + \frac{\sqrt{2}}{2}n_3$	$\sqrt{2} - \frac{5}{4}$
F	$c + \sum_{j=1}^{4}((j+1)\varphi - 2j)n_j$	$34\varphi - 55.$

For H the products with c, n_1, n_3 vanish and the n_2 product is $(\sqrt{2}-3)/4 < 0$. For F the products with c, n_1, n_2, n_3 vanish and the n_4 product is $3\varphi - 5 < 0$. Every other actual inequality follows from positivity. Both have $Q = 0$, future coefficient $A > 0$, and pure A-spatial part, with

$$A_{\text{H}}^2 = \frac{3}{2} + \sqrt{2} - \frac{11/2 + 3\sqrt{2}}{N}, \qquad A_{\text{F}}^2 = \frac{\varphi^2(N-1)}{2N} - 1.$$

At $N = 9$,

$$A_{\text{H}}^2 - 9k_{\text{H}} = \frac{437 - 300\sqrt{2}}{36} > 0, \qquad A_{\text{F}}^2 - 9k_{\text{F}} = \frac{50(89 - 55\varphi)}{9} > 0.$$

The time norms increase with N. Equation (9.3) therefore excludes these profiles under energy at most $1/2$. The central doubles and (B.3.2) already have $\cosh L > 3$, so are excluded by (9.6).

For L, first clear all foreign components positively to obtain $c_F = Ap - v_A$, $c_F^2 = 1 - Q$. Put

$$b = \beta_1,\ d = \beta_2,\ B = b + 2d, \quad h = \frac{4}{3}(b^2 + bd + d^2) + Q - 1, \quad z_0 = h/B^2,$$

$$\lambda_1 = (4b + 2d)/3 - h/B, \qquad \lambda_2 = 2B/3 - 2h/B, \qquad \xi = c_F + \lambda_1 n_1 + \lambda_2 n_2 = Ap - U.$$

The endpoint condition gives

$$B\lambda_1 = 2bd + 1 - Q > 0, \quad B\lambda_2 = 2(1 - b^2 - Q) > 0,$$

and direct multiplication gives

$$\xi \cdot c = \xi \cdot n_1 = \xi \cdot S = 0, \quad \xi \cdot n_2 = -3h/(2B) < 0, \quad k = -\xi^2 = B^2(z_0 - 3z_0^2).$$

Here S denotes all cleared spectator roots. The complete L range is

$$\frac{8}{81} < z_0 < \frac{1}{3}. \tag{B.3.3}$$

Indeed adjacent contacts with one label at least 4 give $z_0 > 1/10$; isolated labels at least 7 have $b > 9/10$, so $z_0 = 4/3 - b^{-2} > 8/81$; isolated 5 with $Q \geq 1/4$ gives $z_0 \geq 3\varphi - 14/3 > 2/15$. The upper bound is precisely $1 - b^2 - Q > 0$. Thus ξ is actual, future timelike and has no foreign spatial part.

For $a_N = 2/3 - 2/N$, its full vector data are

$$A^2 = B^2(a_N + z_0), \quad U^2 = B^2(a_N + 3z_0^2), \quad U_{\text{raw}} = \frac{B}{\sqrt{2}}(2/3 + z_0, 2/3 + z_0, 2/3 - 2z_0, 0, \ldots, 0). \tag{B.3.4}$$

Subtracting $\sqrt{2}B/N$ from every coordinate centers the raw vector. Its normalized actual demand $g = (A - \sqrt{k})/\sqrt{U^2}$ satisfies $g^2 > 9/20$. At $a_N = 4/9$ the equivalent quadratic is $2523z_0^2 - 720z_0 + 484/9 > 0$, of negative discriminant $-72976/3$; increasing N strengthens it.

Let $W = u_{r-1} + u_r$. Direct coordinates give

$$\rho = \frac{U \cdot W}{\sqrt{U^2}} = \frac{2/\sqrt{N(N-2)} + \sqrt{2/(N(N-1))}}{\sqrt{a_N + 3z_0^2}}. \tag{B.3.5}$$

The numerator decreases and denominator increases with N. At $N = 9, z_0 > 8/81$, its squared upper seal is $48114/129500 < 3721/10000$, so $\rho < .61$. If the far opposites are distinct, $x_A \cdot W > 1$. Perpendicular Cauchy instead gives

$$x_A \cdot W < \frac{.61}{\sqrt{2}} + \sqrt{(1/2 - 9/20)4} < .89 < 1.$$

If they are common, their inventory supplies another L foot: H and F have already been excluded. Normal reversal changes the raw sum from $\sqrt{2}B$ to $-\sqrt{2}B'$. The disjoint raw supports therefore give the *positive* centered product

$$U \cdot U' = 2BB'/N, \qquad \operatorname{corr}(U, U') < \frac{2/N}{a_N} = \frac{3}{N - 3} \leq \frac{1}{2}.$$

The two actual demands force energy greater than $4(9/20)/3 = 3/5$. This proves the half bound.

B.3.3 The rank-seven base

Set $N = 8$ and suppose $\|x_A\|^2 \le 1/2$. The correlations of pairs $1/2, 6/7$ are $\sqrt{3/7}$ and those of $3/4, 4/5$ are $\sqrt{3/5}$. The pair-sum calibrations already proved therefore apply. If all four pairs have distinct opposites, put

$$w = 2u_1 + 2u_2 + u_3 + 2u_4 + u_5 + 2u_6 + 2u_7.$$

Then $x_A \cdot w > 51/8$, whereas direct Gram summation gives

$$\begin{aligned} w^2 = {} & \frac{64}{7} + \frac{32}{3} + \frac{16}{5} + 4 \\ & + \frac{64}{\sqrt{21}} + \frac{64}{\sqrt{105}} + \frac{16}{\sqrt{7}} + \frac{64}{3\sqrt{5}} \\ & + \frac{16}{\sqrt{3}} + 8\sqrt{3/5} < \frac{1208}{15}. \end{aligned}$$

The first four terms sum to less than 28, and the last six are respectively less than 14, 32/5, 32/5, 10, 28/3, 32/5; each comparison follows by squaring positive rational bounds. Cauchy is impossible since

$$(51/8)^2 - (1208/15)/2 = 359/960 > 0.$$

Reversing the numbering if necessary, a common pair is $1/2$ or $3/4$.

The central pair. For a common $3/4$ opposite there is at most one contact on each of the paths $1, 2, 3$ and $4, 5, 6, 7$. Native bonds have labels $3, 4, 5$: a larger label with a retained old bond is not finite. For two contacts outside positions $3, 4$, the following finite white tests exhaust the position pairs:

$(1,4), (1,7), (2,4), (2,7), (3,5), (3,7)$	a rank-seven path after deleting 3 or 4
$(1,5), (1,6), (2,5), (2,6)$	D_7 or E_7 after deleting 3
$(3,6)$	D_7 after deleting 4.

Both new bonds must be 3 in these rows; their cycle in the full diagram is affine. At positions $3, 4$, the $3, 3$ pair is affine; every other assignment gives an endpoint cosine greater than 3, already for the minimal $3, 4$ and $4, 3$ assignments in (B.3.1) at $N = 8$. It therefore forces energy greater than $1/2$.

For a single 3-contact, the complete own costs and sufficient white bounds are

j	1	2	3	4	5	6	7
$\beta^t C \beta$	7/16	3/4	15/16	1	15/16	3/4	7/16
$Q <$	5/8	1/2	5/8	3/5	2/5	2/5	3/5.

Here $A^2 = \beta^t C \beta + Q - 1$. One foreign component is impossible by $q < A^2$ in (9.8). Every row except $j = 3, 4$ allows at most one, or has $A^2 < 1/4$. In rows $3, 4$, two foreign components must both start with a 3-contact; the new wall, the contacted old node, its two old neighbors and the two foreign neighbors give the exact affine two-fork D_5 diagram. The case $Q = 0$ has nonpositive height.

A single 4 is an affine $3, 3, 4$ star internally. At an end $Q < 1/4$, so $Q = 0$ and the height is negative. A single 5 is possible only at positions 1 or 3, and its retained H_4 cost $3\varphi^2/8 > 3/4$ forces $Q = 0$. Position 3 gives direction-four cosine greater than 3. Position 1 gives foot F above. At $N = 8$ its time norm satisfies

$$A_F^2 = (7\varphi - 9)/16, \qquad A_F^2 - 9k_F = (7911 - 4889\varphi)/16 > 0.$$

For example $\varphi < 16181/10000$, proved by positive squaring, gives $7911 - 4889\varphi > 1091/10000$. Thus the actual-point inequality (9.3) excludes this last row. The central common-pair case is complete.

The end pair. For common $1/2$, a double profile has positions $1, j$. For $j = 2$ its second label is 3 or 4 and its first label is any finite native label. At $j = 3, 7$ the labels are 3 at j and 3 or 4 at 1. At $j = 4, 6$, deletion 2 gives E_7 and both labels are 3. At $j = 5$, deletion 2 already contains affine E_6. Two 3-labels always give an affine cycle. The $4@1 + 3@7$ row contains the affine F_4 path $2, 1, c, 7, 6$; the $4@1 + 3@3$ row has $Q = 0$ and is foot H. The displayed H formula gives $A_H^2 - 9k_H = (193 - 134\sqrt{2})/16 > 0$ at $N = 8$.

For a single 3, the own costs are the same seven entries above, while the sufficient Q bounds at positions $1, \dots, 7$ are

$$3/4, \quad 4/7, \quad 2/7, \quad 1/7, \quad 1/7, \quad 2/7, \quad 4/7.$$

All except position 2 have $A^2 < 1/4$. At position 2, one foreign component contradicts $q < A^2$; with two, each exceeds $1/4$, so each is at least $1/3$, contrary to $Q < 4/7$. A single 4 is affine internally. At 1, $Q < 1/2$ permits at most one foreign component, whose cost exceeds $A^2 = Q - 1/8$; at 7, $Q < 1/7$ leaves none and gives negative height.

A single label at least 5 is possible only at 1. Label 6 contains the affine $3, 6$ path. A label at least 7 forces $Q < 1 - \beta_1^2 < 1/4$, hence $Q = 0$. Label 5 with $Q = 0$ was excluded by F. Thus the only remaining profiles are exactly L: adjacent contacts other than $3, 3$, isolated label at least 7 with $Q = 0$, or isolated 5 with $Q > 0$.

The far-end comparison. The positive spectator clearing and actual L-foot formulas (B.3.3)–(B.3.4) remain valid at $N = 8$, now with $a_N = 5/12$. The lower bound $z_0 > 8/81$ follows as before: adjacent contacts give $z_0 > 1/10$, an isolated label at least 7 has $\beta_1 > 9/10$, and isolated 5 with $Q > 0$ has $z_0 \geq 3\varphi - 14/3 > 2/15$. For the label-seven bound, $8t^3 - 4t^2 - 4t + 1$ is $-1/125$ at $9/10$ and strictly increasing from $\sqrt{3}/2$ to 1.

The actual demand now satisfies $g^2 > 11/25$: equivalently $A^2/k > 324/49$, whose cleared difference is

$$972z_0^2 - 275z_0 + 245/12 > 0.$$

Its discriminant is -3755. With $W = u_6 + u_7$, direct coordinates give

$$0 < \rho^2 = \frac{5/42 + 1/(2\sqrt{21})}{5/12 + 3z_0^2} < \frac{603612}{1170300} < \frac{324}{625}, \qquad W^2 = 2 + 2\sqrt{3/7} < 10/3.$$

Here the numerator is less than $23/100$ and $z_0^2 > 64/6561$ gives the stated rational seal. If the far opposites are distinct, their pair law gives $x_A \cdot W > 1$. Perpendicular Cauchy instead gives

$$x_A \cdot W < \frac{18}{25\sqrt{2}} + \sqrt{(1/2 - 11/25)10/3} = \frac{18}{25\sqrt{2}} + \frac{1}{\sqrt{5}} < \frac{24}{25}.$$

If they are common, the reversed end inventory supplies another L foot. Normal reversal is $e_j \mapsto -e_{9-j}$, so its raw sum is $-\sqrt{2}B'$ on the disjoint final three coordinates. Consequently $U \cdot U' = BB'/4 > 0$, and their normalized correlation is less than $3/5$. The sum of these two normalized directions has square less than $16/5$ and actual scalar product greater than $2\sqrt{11/25}$. Cauchy forces

$$\|x_A\|^2 > \frac{44/25}{16/5} = \frac{11}{20} > \frac{1}{2}.$$

Both far-end alternatives are impossible. This proves the rank-seven base and completes Theorem B.3.1.

B.3.4 The paired-end Gram and the uniform $4/7$ bound

Proposition B.3.2. *Every actual A_r component, $r \geq 8$, has energy greater than $4/7$, with arbitrary finite spectators.*

Proof. Put $N = r+1 \geq 9$ and suppose energy is at most $4/7$. For distinct outer $1/2$ pairs and inner $3/4$ pairs, the height sums exceed $41/40$ and $1219/1000$, and likewise at the opposite end. At $N = 9$ the correlations are $\sqrt{7}/4$, $\sqrt{5/8}$; the exact tests are

$$F(41/80)^2 = \frac{43033600}{65302561} < 33/50 < \sqrt{7}/4,$$
$$F(1219/2000)^2 = \frac{23775376000000}{30095768093521} < 79/100 < \sqrt{5/8}.$$

The cross-product margins are $3304513, 28079388159$. The correlations increase with N. For $M_i = u_i + u_{N-i}$, $1 \leq i \leq j \leq 4$,

$$\langle M_i, M_j \rangle = 2N\sqrt{\frac{i}{j(N-i)(N-j)}}.$$

Its logarithmic derivative is $1/N - 1/(2(N-i)) - 1/(2(N-j)) < 0$. Thus every paired-end Gram entry decreases with N. For $W = 2M_1 + 2M_2 + M_3 + M_4$,

$$W_9^2 = \frac{1116}{35} + \frac{60}{\sqrt{7}} + \frac{36}{\sqrt{10}} + \frac{72}{\sqrt{70}} < \frac{747}{10}.$$

The last three terms are $< 227/10, 57/5, 87/10$; these imply the sharper $2614/35 < 747/10$. If all four selected pairs were distinct, then

$$\langle x_A, W \rangle > 3269/500, \qquad (3269/500)^2 - (4/7)(747/10) = 104527/1750000 > 0,$$

contradicting Cauchy. Reversing the numbering, a common $1/2$ or $3/4$ therefore occurs.

Use the complete common-pair inventory above before its half-energy discards. At $4/7$, the individual edge cutoff is $C = 11/3$. The two minimal central-double cosines in (B.3.1) exceed $11/3$ already at $N = 9$, and increase thereafter. The single central 5 cosine in (B.3.2) also exceeds $11/3$. The remaining left profiles are precisely L, H, F. The same finite-white, affine and full-spectator checks apply; no earlier small-target exclusion of H/F is used.

For $\xi = Ap - U$, write $g^2 = (A - \sqrt{k})/(A + \sqrt{k})$ and $d = U^2/B^2$, where $\sqrt{2}B > 0$ is its raw coordinate sum. At $N = 9$ the exact shared feet satisfy

type	$g^2 >$	$d >$	$\rho = \langle U/\|U\|, u_{r-1} + u_r \rangle <$
L	$9/20$	$4/9$	.61
H	$53/100$	$1/3$	$3/4$
F	$11/20$	$2/9$	$9/10$.

For L, its formulas and negative-discriminant seal above prove the first two entries. The correlation numerator has square

$$(2/\sqrt{63} + 1/6)^2 = 23/252 + 2/(3\sqrt{63}) < 22/125,$$

giving $\rho^2 < 48114/129500 < 3721/10000$. For H, $A^2 = (8 + 6\sqrt{2})/9 > 9/5$, $k = \sqrt{2} - 5/4 < 1/6$, so $A^2/k > 54/5 > 23409/2209$. Also

$$d = \frac{77 - 12\sqrt{2}}{9(11 + 6\sqrt{2})} > 1/3$$

by $22 > 15\sqrt{2}$. For F, $A^2 = (4\varphi - 5)/9 > 184/1125$, $k = 34\varphi - 55 < 167/12500$; their ratio exceeds $961/81$, with integer margin 5752125. Its density $d > 2/9$ is equivalent to $979 > 605\varphi$. These seals follow from $809/500 < \varphi < 40451/25000 < 16181/10000$, proved by positive squaring. The correlation bounds then follow from $(22/125)/(1/3) < 9/16$ and $(22/125)/(2/9) < 81/100$. At fixed native/foreign parameters, A^2, d, g^2 increase with N, while k, B stay fixed.

Let $W_f = u_{r-1} + u_r$. Its square is $< 2 + \sqrt{2} < 7/2$. For every foot,

$$\rho = \frac{2/\sqrt{N(N-2)} + \sqrt{2/(N(N-1))}}{\sqrt{d}}.$$

For $N \geq 10$, L has the improved $\rho < 8/15$: the numerator square is at most $5/36$ at 10, and $d > 7/15 + 64/2187 > 49/100$; thus $\rho^2 < 125/441 < 64/225$. If the far pair is distinct, its demand is $> 41/40$. Perpendicular Cauchy, with $v = \langle x_A, U/\|U\| \rangle$, gives

$$\langle x_A, W_f \rangle \leq \rho v + \sqrt{(4/7 - v^2)(W_f^2 - \rho^2)}.$$

For L this decreases with v above its demand floor and increases with ρ up to its bound; the sign condition is $v^2 W_f^2 > (4/7)\rho^2$. At $N \geq 10$ it is

$$< (8/15)\sqrt{9/20} + \sqrt{17/40} < 9/25 + 33/50 = 51/50 < 41/40.$$

At $N = 9$, use $\rho < .61, W_f^2 < 10/3$ instead: the two terms are $< .41, .6$, the latter square being $1510229/4200000 < 9/25$. For H and F, without derivative estimates, use $v < \sqrt{4/7} < 19/25$ and their pressure floors. For $N \geq 10$ the bounds are $.57 + \sqrt{29/200} < .97$ and $.7 + \sqrt{3/40} < .98$. At $N = 9$ they improve to $.95, .97$. Every far-distinct case is impossible.

If the far pair is common, it supplies a right L/H/F foot. Normal reversal gives negative raw sum $-\sqrt{2}B'$. Except F/F at $N = 9$, their raw supports are disjoint and

$$\langle U, U' \rangle = 2BB'/N, \qquad \operatorname{corr}(U, U') = \frac{2}{N\sqrt{dd'}}.$$

The following complete positive-sum Cauchy seals hold at $N = 9$, and improve thereafter:

pair	$\rho <$	sum demand $>$	energy $>$
L/L	$1/2$	$2\sqrt{9/20}$	$3/5$
L/H	$3/5$	$139/100$	$19321/32000$
L/F	$5/7$	$141/100$	$139167/240000$
H/H	$2/3$	$36/25$	$3888/6250$
H/F	$5/6$	$73/50$	$15987/27500$.

Every last entry exceeds $4/7$. For F/F, at any $N \geq 9$ the two actual isolated end-five walls have exact product

$$1 - \varphi^2/2 + \varphi^2/N \in (-1/2, 0).$$

The lower inequality uses $\varphi^2 < 3$; the upper follows at $N = 9$ from $\varphi > 8/5$ and strengthens with N. This violates the native gap without an unsupported overlap formula. All cases are excluded. □

B.3.5 A fourth actual foot and the rank-13 threshold

Proposition B.3.3. *Every actual A_r component, $r \geq 13$, has energy greater than $5/7$, with arbitrary finite spectators.*

Proof. Put $N = r + 1 \geq 14$ and suppose the energy is at most $5/7$. The outer and inner selected correlations increase with N. At $N = 14$ the exact pair calibrations are

$$F(263/500)^2 = (263000/319169)^2 < 6791/10000 < \sqrt{6/13},$$

$$F(641/1000)^2 = (1282000/1410881)^2 < 8257/10000 < \sqrt{15/22}.$$

Thus distinct outer and inner pairs have sum demands $> 263/250$ and $> 641/500$. For $M_i = u_i + u_{N-i}$, the paired Gram identity above decreases entrywise with N. At 14, $W = M_1 + M_2 + 2M_3 + 2M_4$ has diagonal contribution $55489/2145 < 25.869$. Its six off-diagonal contributions, in order $12, 13, 14, 23, 24, 34$, are

$$\frac{28}{\sqrt{78}},\ \frac{112}{\sqrt{429}},\ \frac{56}{\sqrt{130}},\ \frac{112}{\sqrt{198}},\ \frac{28}{\sqrt{15}},\ 112\sqrt{3/110},$$

respectively less than $3.171, 5.408, 4.912, 7.960, 7.230, 18.497$. Hence $W^2 < 73.047 < 1461/20$ at every $N \geq 14$. Four distinct pairs would give

$$\langle x_A, W\rangle > 904/125, \qquad (904/125)^2 - (5/7)(1461/20) = 53923/437500 > 0.$$

A common left $1/2$ or $3/4$ follows after relabeling.

Recheck the full common-pair inventory at the new individual cutoff $C = 6$. A central double with second label at least 4 has the first cosine in (B.3.1), greater than $.3(11 + 14) > 6$ at $N = 14$. The $5/3$ central double and isolated central 5 have cosines

$$(3\varphi + 4)\varphi^2\sqrt{(N-4)/N}, \qquad 3\varphi^3\sqrt{(N-4)/N},$$

both > 6 at 14, using $\sqrt{5/7} > 4/5, \varphi > 8/5, \varphi^2 > 13/5, \varphi^3 > 4$. They increase with N. The central $4/3$ case must now be retained. Its foreign cost is zero and its actual positive foot is

$$\mathrm{C}: \quad \xi = c + n_3 + (2 - \sqrt{2})n_4, \qquad k = 3\sqrt{2} - 4.$$

Products on c, n_3 vanish; the n_4 product is $1 - \sqrt{2} < 0$, and all unknown wall products follow from positivity. The time is positive, with

$$A_C^2 = 4 + 3\sqrt{2} - \frac{17 + 12\sqrt{2}}{N}, \quad U_C^2 = 8 - \frac{17 + 12\sqrt{2}}{N}, \quad B_C = 3\sqrt{2}/2 + 2.$$

The complete surviving profiles are L,H,F,C. All L spectators are cleared in their entirety.

For L, at $N = 14$, $a_N = 11/21$ and $8/81 < z_0 < 1/3$. Its bound $g_L^2 > 12/25$ is equivalent to

$$4107z_0^2 - 1200z_0 + 1859/21 > 0,$$

whose discriminant is $-99884/7$. Also $d_L > 11/21 + 64/2187 > 11/20$, with last positive comparison $26880 > 24057$. The other exact data at 14, with $s = \sqrt{2}$, are

	A_{14}^2	k	B^2	d_{14}
H	$(31 + 22s)/28$	$s - 5/4$	$(11 + 6s)/4$	$(66 - 6s)/(7(11 + 6s))$
F	$(13\varphi - 15)/28$	$34\varphi - 55$	$(\varphi + 1)/4$	$(1525 - 939\varphi)/(7(\varphi + 1))$
C	$(39 + 30s)/14$	$3s - 4$	$(17 + 12s)/2$	$(95 - 12s)/(7(17 + 12s))$.

For H, $A^2 > 11/5, k < 33/200$, so $A^2/k > 40/3$. For F, $A^2 > 21/100, k < 167/12500$, so $A^2/k > 2625/167$. For C, $A^2 > 5, k < 1/4$. All exceed $961/81$, proving $g^2 > 11/20$. The density

bounds $d > .3$ reduce to $143 > 62s,\ 15229 > 9411\varphi,\ 593 > 372s$. Use $s < 3/2$ and $\varphi < 40451/25000$; the cancellation-sensitive middle comparison has integer margin 40639. All these demands and densities increase with N.

If either far selected pair is common, its reflected inventory supplies a right L/H/F/C foot. The raw supports, each of length at most 5, are disjoint. Normal reversal gives positive correlation $\rho = 2/(N\sqrt{dd'})$. The complete sum-direction tests are

pair type	$\rho <$	energy $>$
L/L	13/50	16/21
L/non-L	9/25	20449/27200
non-L/non-L	10/21	231/310.

For the middle row, $\rho^2 < 200/1617 < (9/25)^2$ and the demand sum is $> \sqrt{12/25} + \sqrt{11/20} > 1.43$. Every energy exceeds 5/7.

Otherwise both far selected pairs are distinct. Their weighted demand is

$$\langle x_A, R\rangle > 452/125, \quad R = u_{N-1} + u_{N-2} + 2u_{N-3} + 2u_{N-4}.$$

Direct within-end Gram bounds give $10 < J = R^2 < 29$. For a left foot,

$$\rho = \langle U/\|U\|, R\rangle = \frac{K_N}{\sqrt{d}},$$

$$K_N = 2\sqrt{\frac{8}{N(N-4)}} + 2\sqrt{\frac{6}{N(N-3)}} + \sqrt{\frac{4}{N(N-2)}} + \sqrt{\frac{2}{N(N-1)}}.$$

Its terms decrease with N; at 14 they are $< .479, .395, .155, .105$, so $K_N < 1.134 < 8/7$. Thus $\rho < 31/20$ for L and $< 21/10$ otherwise. The square seals are $1280/539 < (31/20)^2,\ 640/147 < (21/10)^2$. Perpendicular Cauchy gives

$$\langle x_A, R\rangle \le \rho v + \sqrt{(5/7 - v^2)(J - \rho^2)}, \qquad v = \langle x_A, U/\|U\|\rangle.$$

On the admissible rectangle it decreases with v above its pressure floor and increases with ρ below its bound, since $v^2 J > (5/7)\rho^2$. This holds already with $J > 10$. Increasing J to 29 is safe. For L the resulting bound is

$$(31/20)\sqrt{12/25} + \sqrt{436199/70000} < 217/200 + 5/2 = 717/200.$$

For every other foot it is

$$(21/10)\sqrt{11/20} + \sqrt{56557/14000} < 63/40 + 201/100 = 717/200.$$

Both are $< 452/125$, contradicting the far demand. This exhausts all actual profiles and proves the nonarithmetic component theorem. □

B.3.6 One central-window lemma for $N = 17, 18, 19$

The next lemma separates the geometric inventory from the reference chosen for a particular rank.

Lemma B.3.4. *Let $N \in \{17, 18, 19\}$, and let an actual A_{N-1} component have the original farthest-vector projection x_A. On $5 \le i \le N-5$, call i weak when $\langle x_A, \omega_i \rangle \le \sqrt{D_i} - .61$. Before any contrary energy bound, every weak opposite is:*

$$\begin{aligned} &\textit{near-four:} && v = \omega_i/\sqrt{2} + \omega_{i-1}/2,\ i = 5, 6,\ \textit{or a reflected role};\\ &\textit{pure helper:} && v = \gamma\omega_i + \omega_e/2,\ e = 1, N-1, \quad Q = 0,\\ &&& \gamma \ge \varphi/2, \quad \gamma < 17/16,\ \textit{own } 5,\ \textit{finite } \ge 7,\ \textit{or dotted.} \end{aligned}$$

Literal weak helpers are nonadjacent, with all spectators retained. If the energy is at most .86, 6/7, or .91, respectively for $N = 17, 18, 19$, no near-four role remains either.

Proof. Both retained old paths have rank at least 4. The complete finite-white list is: none; a simple 3 at an end or an admissible D/E position; an endpoint 4; or two simple end 3's, one on each path. An H/F completion has insufficient rank, and a joined completion has rank $N - 1 > 8$. Finite classification forces the ordinary A-path bridge. An additional branch there cannot be finite. This classification is applied to the actual endpoint, with the entire spectator forest.

For zero own, internal 3/4 and double-white roles give affine stars or cycles. A far 3 has time square $< 1/(2m)$, $m \ge 5$; a next-to-far 3 has $Q < 2/m$ and time square less than its positive foreign cost. Either forces an orthogonal closure. End 4 has $Q < 1/m \le 1/5$, hence no spectator and negative time square. Thus own contact is positive.

For an isolated own, clear all spectators as in the prefix proof, using the sharper A-finite pressure

$$h_i > \sqrt{D_i - 1} - \sqrt{u - 1} \quad (u > 1), \qquad h_i > \sqrt{D_i - u} - \sqrt{u(1-u)} \quad (0 < u < 1).$$

At $u = 1$ there is an actual feasible null. Here $D_i \ge 120/17$, and

$$a_i = \frac{1}{\sqrt{D_i} + \sqrt{D_i - 1}} < .1955.$$

Indeed the denominator exceeds $2.656 + 2.461 = 5.117$ and $5.117 \cdot 391 > 2000$. For $u < 1$ the loss is $< (.1955 + 1.019)/2 = .60725 < .61$. For foreign $u > 1$, $u - 1 \le (2 - \varphi)^2$, so the loss is $< .1955 + 2 - \varphi < .5775$. For pure finite owns at least 8, singleton loss is $< .1955 + \tan(\pi/8) < .1955 + .4145 = .61$, since $(2829/2000)^2 > 2$. Own 5 has its actual H_4 loss $< 25/76 + 5/24 < .61$; own 7 has its actual A_2-core loss $< 2000/8181 + 9/25 < .61$. The latter uses $\sqrt{D_i} > 53/20$, $\sqrt{D_i - 100/81} > 12/5$. Own $3, 4, 6$ and parallel retain their affine/null principals. Thus every isolated cap is strong.

The strength ratio $T(q)$ increases here, and

$$q \le \sqrt{19/2} < 25/8, \qquad T(25/8) = 3/2 - 1441/430050 < 3/2.$$

The residual table of the preceding finite-white classification gives ordinary ratios $> 761/500$: near simple-3/own at least 5 gives at least $(101/125 + 2/5)/\sqrt{3/5}$, D-position gives $(101/125 + 1/5)/\sqrt{2/5}$, and E, end-4 and two-white rows are stronger. Smaller owns have the same affine principals. Foreign far helpers have ratio $> 227/(25\sqrt{35}) > 761/500$; foreign near-four has ratio square $> 121/35$. Consequently every weak cap has $Q = 0$. The near-four ratio increases with side order m, and at $m = 7$ has square $(67 + 42\sqrt{2})/56 > 9/4$. Thus only side orders $5, 6$ remain. For a pure helper the residual identity

$$\frac{(\gamma + 1/(2m))^2}{(m+1)/(2m)} - (4\gamma - 2) = \frac{(\gamma - 1 - 1/(2m))^2}{(m+1)/(2m)}$$

and $T < 3/2$ give $\gamma < 17/16$.

For adjacent literal helpers, the general-N AM–GM product bounds in the common-pair calculation leave mutual 0 for the same helper end, and $0, -1/2$ for opposite ends. Outward mutual 0 is excluded by the positive projected determinant

$$(D_i - 4/5)(D_{i+1} - 4/5) - C_{i,i+1}^2 = \tfrac{2}{5}C_{i,i+1} - 24/25 + 8/(5N) > 0.$$

For reversed mutual 0, put $l = i, s = N - i - 1$. The required determinant is

$$P = 4lsA^2B^2 - (l+1)(s+1)(A^2+B^2) - 2AB - (l+1)A - (s+1)B + (N+2)/4 = 0.$$

Its own derivatives are positive on $[.8, 1.08]$, and it is concave in l, $5 \le l \le N-6$. At $N = 17$, both owns at least 7 give $P(.9, .9) \ge 14.632 > 0$. One 5 and one own at least 8 give, at $(.809, .923)$ or its reverse, minimum $1.53076164678 > 0$. These endpoint values increase for $N = 18, 19$: at $l = 5$ the coefficient of N is $20A^2B^2 - 6(A^2+B^2) - B + 1/4 > 0$ at both comparison pairs and their reversals. If owns are $5, 7$, the determinant is a polynomial of degree at most 2 in $\cos(\pi/7)$ over $\mathbb{Q}(\sqrt{5})$, with nonzero linear coefficient. But $2\cos(\pi/7)$ has irreducible polynomial $t^3 - t^2 - 2t + 1$, and degree 3 over $\mathbb{Q}(\sqrt{5})$. Thus it cannot vanish.

For two native-five helpers, use raw norm-two duals $C_{ij} = \min(i,j) - ij/N$ and $v_i = \varphi\omega_i + \omega_e$. Same-end adjacent difference square is $\varphi^2(N-1)/N < 4$, forcing positive wall product. For opposite ends, with orientation $\epsilon = \pm 1$,

$$\|v_i - v_{i+1}\|^2 = \varphi^2(N-1)/N + 2(N-2)/N - 4\epsilon\varphi/N < 6.$$

The raw native gap forces mutual 0. For $H_i = v_i^2 - 2$, $S = v_i \cdot v_{i+1}$, the rational coefficient of $N^2(H_iH_{i+1} - S^2)$ in basis $1, \varphi$ is $N(N+6)$ outward and $-N^2$ reversed. It cannot vanish. Hence opposite helpers have mutual exactly $-1/2$ whenever they survive.

The retained two-cap matrices are the common-helper $I - t\mathbf{1}\mathbf{1}^T$, or the opposite-helper matrix with diagonal in $(1/2, 3/5]$, cross $-1/2$. They have positive inverses. Positive actual projection retains the entire old spectator block. Put $\lambda = H^{-1}(A_i, A_j)^T > 0$, $\mathcal{D} = A_i\lambda_i + A_j\lambda_j$ and $Z = W_i\omega_i + W_j\omega_j$. For a common left helper, $W_i = \lambda_i\gamma_i + (\lambda_i + \lambda_j)/(2i)$, $W_j = \lambda_j\gamma_j$; reflect for a common right helper. Outward helpers give $W_i = \lambda_i(\gamma_i + 1/(2i))$, $W_j = \lambda_j(\gamma_j + 1/(2(N-j)))$. Reversed helpers give $W_i = \lambda_i\gamma_i + \lambda_j/(2i)$, $W_j = \lambda_j\gamma_j + \lambda_i/(2(N-j))$. Thus $W = W_i + W_j > .8L$, $L = \lambda_i + \lambda_j$. All cap times exceed 2: the smallest time square is $> (16/25)(120/17) + (4/5)(10/17) + 8/17 - 1 = 379/85 > 4$. For a common helper the same explicit inverse as in the prefix proof gives $\sqrt{1+\mathcal{D}}/W < 7/16$. For opposite helpers, $5/6 < \lambda_i/\lambda_j < 6/5$, and

$$\mathcal{D}/L^2 \le 3/55, \quad A_i + A_j \le L/10, \quad L > 40.$$

The first inequality follows by maximizing $(.6(t^2+1) - t)/(t+1)^2$ on $[5/6, 6/5]$. Consequently $\sqrt{1+\mathcal{D}}/W < \sqrt{15/176 + 1/1024} < 3/10$. The spatial rationalization loss is $< W/10$: use $\sqrt{120/17} > 2.65$, $\sqrt{110/17} > 2.54$. The weighted mean losses are thus $< 43/80$ or $< 2/5$, both $< .61$. Adjacent weak helpers are impossible independently of an energy target.

A near-four has an actual positive finite mate with cosine > 7, hence normalized height $> .86$. At the opposite end there is another mate, or a strong anchor of normalized height $> .77$. The correlations are bounded respectively by $< .5, < .44, < .41$ at $N = 17, 18, 19$. They follow by substituting mate index at most 5 and anchor index at least $N - 6$ in $\sqrt{j(N-k)/(k(N-j))}$. At the three contrary energies, Cauchy would give

$$1.63^2 < .86(3), \quad 1.63^2 < (6/7)(2.88), \quad 1.63^2 < .91(2.82),$$

each false. The reflected roles satisfy the same bounds. This removes all near-fours. □

B.3.7 Shared actual endpoint cores and three reference tables

For a weak end $i = 5$ with helper $e = 1, N-1$, let j be the old neighbor of the helper end. All three old selected roles are present in each $N = 17, 18, 19$. Set $v = \gamma\omega_5 + \omega_e/2$. The actual time squares are

$$A^2_{\text{far}} = \frac{10(N-5)\gamma^2 + 10\gamma - (N+1)/2}{N}, \qquad A^2_{\text{near}} = A^2_{\text{far}} + \frac{2(N-10)}{N}\gamma. \tag{B.3.6}$$

These follow directly from the actual dual Gram. Reflected weak ends have the reflected formula.

There are three positive actual constructions:

own	old correction b_c	residual
5	$\gamma n_5 + (2/3)n_e + (1/3)n_j$	$\delta = 2/3 - \gamma^2 > 0$
finite ≥ 7	$\gamma n_5 + 2(1-\gamma^2)n_e$	$-k,\ k = (1-\gamma^2)(4\gamma^2 - 3) > 0$
dotted	n_5/γ	$-k,\ k = 1 - \gamma^{-2} > 0.$

For own 5, $\eta = c + b_c$ is spacelike and the actual future finite point is $p + (A/\delta)\eta$, of square $-1 - A^2/\delta$. It is orthogonal to all four selected roots, while all other wall products are nonpositive. For finite owns at least 7, put $d = 1 - \gamma^2$. The core $\xi = c + b_c$ has products zero with c, n_5 and $-1/2 + 2d < 0$ with n_e. The dotted core has products zero with c and $-\gamma + \gamma^{-1} < 0$ with n_5. All time coefficients are $A > 0$. Positivity checks every other actual wall, including the other helper, every spectator and all unknown walls. Thus, for $U = v - b_c$, the actual point pressures give

$$\langle x_A, U\rangle > A - \sqrt{\delta} \quad \text{(own 5)}, \qquad \langle x_A, U\rangle \geq A - \sqrt{k} \quad \text{(other rows)}. \tag{B.3.7}$$

Dropping the positive old-normal products yields the same lower bounds for $\langle x_A, v\rangle$.

The rational concave references needed below have linear end segments, reflection $H_{N-i} = H_i$, and the following central heights:

N	H_5	H_6	H_7	H_8	H_9	H_{10}
17	2.046	2.176	2.259	2.300	2.300	
18	2.077	2.218	2.314	2.371	2.389	
19	2.104	2.255	2.363	2.433	2.467	2.467.

Only the independent entries before reflection are used. Positive square tests $(H_i + .61)^2 < D_i$ pay every retained strong floor. If 5 is weak use its chord to 6; reflect at the other end; omit every internal weak knot by its neighbor chord. Nonadjacency proves that all losses add. The complete zero/one/two-weak-end tables are

N	number of weak ends	initial energy	maximum internal loss
17	0	.8626932	$3(.047^2/4) = .00165675$
17	1	.8302129333…	.00165675
17	2	.7977326666…	$2(.042^2/4) = .000882$
18	0	.8954558	.002025
18	1	.8640827333…	.00151875
18	2	$H_6^2/6 + .012789$	$3(.039^2/4)$
19	0	.9258842	$4(.043^2/4) = .001849$
19	1	.8955541833…	.001849
19	2	.8652241666…	$3(.038^2/4) = .001083.$

The repeating decimals abbreviate exact rational sums of squared rational slopes. These tables are complete: after zero/one/two end chords, the eligible internal weak intervals and nonadjacent

cardinalities give the indicated maximal counts. Each loss is one quarter of the squared second difference. End chords preserve the neighboring strong anchor; an adjacent omission can only increase its available individual price. Every omitted knot still has price zero.

Proposition B.3.5 (The $N = 17$ paid reference)**.** *An actual A_{16} component has energy $> 43/50$ with arbitrary finite spectators.*

Proof. Assume contrary $e \le .86$. The central-window lemma removes near-fours. Use the $N = 17$ table. Zero weak ends already give $e \ge .86103645$. For two weak ends the reference is flat on both six-coordinate end blocks, has energy $E \ge 1195276/1500000$, and separate anchor capacities

$$\beta_6, \beta_{11} \ge (7H_6/6 - H_7)/2 = 839/6000.$$

For one weak end the block lengths are $6, 5$, $E > .8285$, and $\beta_6 \ge 839/6000, \beta_{12} \ge 349/2500$.

For a left weak helper, project its core displacement U onto vectors constant on each end block. In the six/six case,

$$\Pi U = (5\gamma/6)\omega_6 + \omega_{11}/12 \quad \text{(far)}, \qquad \Pi U = (5\gamma/6 + 1/12)\omega_6 \quad \text{(near)}.$$

All old-root corrections have mean zero in their own block. Thus $Y = U - \Pi U$ sums to zero on each block, $\langle w, Y\rangle = 0$, and its square is

own	$\|Y\|^2$ in the far orientation
5	$(2/3)\gamma^2 + 1/12$
finite ≥ 7	$4\gamma^4 - (16/3)\gamma^2 + 29/12$
dotted	$(5/3)\gamma^2 - 19/12 + \gamma^{-2}$.

The near orientation adds $\gamma/3$. These identities follow by expanding the actual $N = 17$ Gram and

$$b_c^2 - 2\langle v, b_c\rangle = k_c^2 + l_c^2 + m_c^2 - l_c m_c - 2\gamma k_c - l_c,$$

where $b_c = k_c n_5 + l_c n_e + m_c n_j$. The finite polynomial increases for $\gamma^2 > .81$ and is $< 13/12$. The dotted one increases for $1 < \gamma^2 < 289/256$ and is $< 229/768 + 256/289 < 148/125$, with cleared margin 271. Consequently every orientation has $\|Y\|^2 < 148/125 + 17/48 < 8/5$.

The projection demand is $H_6(5\gamma/6+1/12)$. Let Δ be the actual core demand minus this quantity. For native 5, the fresh time bound $A_{\text{far}} > 2.136$ and radical $< .111$ give

$$\Delta > 2.136 - .111 - H_6(91/120) > .327.$$

For finite owns at least 7, use $A_{\text{far}}(.9) > 2.391, \sqrt{k} \le 1/4$; for dotted, use $A_{\text{far}}(1) > 2.667$ and radical $< .338$. They give, respectively,

$$\Delta > 2.391 - .25 - 5H_6/6 = 983/3000 > .327, \quad \Delta > 2.667 - .338 - 11H_6/12 > .327.$$

These bounds subtract fixed radical caps from an increasing time-minus-linear envelope; they do not assert that the actual deficit, with its varying radical, increases. Indeed $A_{\text{far}} - H_6(5\gamma/6 + 1/12)$ increases because

$$(A'_{\text{far}})^2 - 120/17 = 1105/(289A_{\text{far}}^2) > 0, \qquad \sqrt{120/17} > 5H_6/6.$$

Subtract the uniform cap $1/4$ from that envelope starting at $\gamma = .9$, or the uniform cap $.338$ starting at $\gamma = 1$, respectively. The near time is larger with the same cap and reference pairing. This proves the displayed lower bounds on all finite/dotted intervals and both orientations.

For one weak end, the far projection instead uses $(5\gamma/6)\omega_6 + \omega_{12}/10$. Its residual square decreases by $1/60$, and its deficit decreases by exactly $H_5/10 - H_6/12 = 349/15000$. Hence $\|Y\|^2 < 8/5, \Delta > .303$.

Use weight $\mu = .1$, paying each original anchor before adding its actual core pressure. One near foot spends at most $31/320$ at 6; a far foot spends less there and $1/100$ at 12. For two ends the combined debit at each 6/11 is at most

$$.1(5(17/16)/6 + 1/6) = 101/960 < 839/6000.$$

All surviving original coefficients are nonnegative. With $W = w + \mu \sum Y$, one original-vector square gives

$$e \geq E + 2\mu \sum \Delta - \mu^2 \|\sum Y\|^2.$$

For one weak end the gain exceeds $.2(.303) - .01(8/5) = .0446$, giving $e > .8731$. For two ends retain the full cross by $\|Y_L + Y_R\|^2 \leq 2\|Y_L\|^2 + 2\|Y_R\|^2 < 32/5$. The gain exceeds $.2(2 \cdot .327) - .01(32/5) = 167/2500$, and

$$e > 1195276/1500000 + 167/2500 = 1295476/1500000 > .86.$$

All three cases contradict the contrary bound. □

B.3.8 Raw helper-end pressures at $N = 18, 19$

For weak 5, put $z = x_1$ for helper 1 and $z = -x_N$ for helper $N - 1$. Then $\langle x_A, v \rangle = \gamma h_5 + z/\sqrt{2}$. The shared actual cores give deficits $> .313$ at $N = 18$ and $> .306$ at $N = 19$. Here are the complete seals. The weak floors are $h_5 < 2.078$ and $h_5 < 2.105$. For native 5, radical is $< .111$, own $< .81$, and times are $> 2.155, > 2.17$; the deficits exceed $.36082, .35395$. For finite owns at least 7, the core radical has square $k = (1 - \gamma^2)(4\gamma^2 - 3) \leq 1/16$:

own	γ bracket	$\sqrt{k} <$	$A_{18,\text{far}} >$	$A_{19,\text{far}} >$	deficit at 18; 19
7	$(.9009, .901)$	.2158	2.4154	2.434	.327322; .321595
8	$(.9238, .9239)$	.2464	2.4795	2.499	.3132358; .3077905
≥ 9	$\gamma > .93$	1/4	2.497	2.517	.31446; .30935.

The native-seven polynomial $8\gamma^3 - 4\gamma^2 - 4\gamma + 1$ increases above $\sqrt{3}/2$; its signs at $.9009, .901$ have numerators $-569738168, 257608$ and denominators $10^{12}, 10^9$. Native eight is $\sqrt{2 + \sqrt{2}}/2$. The residual polynomial increases on both brackets; substitution of upper endpoints gives the radical bounds. The tight eight comparison is $14640879 \cdot 41436484 < 607129600000000$. Times come from (B.3.6) at the lower endpoints. Every finite label at least 9 has $\gamma > .93$, by monotonicity and $8(.93)^3 - 6(.93) - 1 < 0$. The functions $A_{\text{far}} - 2.078\gamma$ or $A_{\text{far}} - 2.105\gamma$ increase, since $A'_{\text{far}} > \sqrt{D_5}$. Thus the last row covers every finite label.

For dotted $1 < \gamma < 17/16$, retain the exact floor $h_5 \leq \sqrt{D_5} - .61$. At 18,

$$A_{\text{far}} - \sqrt{D_5}\gamma = \frac{(5/9)\gamma - 19/36}{A_{\text{far}} + \sqrt{D_5}\gamma} > 1/216.$$

Its numerator is $> 1/36$, denominator < 6. At 19 the same difference is positive. The function $.61\gamma - \sqrt{1 - \gamma^{-2}}$ decreases throughout the interval: its radical derivative exceeds 1. Using $\sqrt{33}/17 < .338$, its endpoint lower bound is $> .310125$, and adding $1/216$ at 18 gives $> .313$. Consequently

$$z > .442 \quad (N = 18), \qquad z > .432 \quad (N = 19).$$

A near helper at 19 has the same radical and time larger by

$$A_{\text{near}} - A_{\text{far}} > \frac{(18/19)(.8)}{7} > .108,$$

using $A_{\text{far}} < 3, A_{\text{near}} < 4$. Its deficit is $> .414$, so $z > .58$.

Proposition B.3.6 (The last two ordinary A-rates)**.** *With arbitrary finite spectators,* $e(A_{17}) > 6/7$ *and* $e(A_{18}) > 91/100$.

Proof. At $N = 18$, assume $e \le 6/7$. The central-window lemma removes near-fours. The reference table excludes zero and one weak end: the retained energies are .8934308 and $> .8625639833\ldots$. With two weak ends, the reference is flat on both six-coordinate end blocks, of signed mean b with $1/4 < b < .262$, and

$$E \ge H_6^2/6 + .012789 - 3(.039^2/4) = 9978827/12000000.$$

One actual weak helper supplies endpoint $z > .442$ at either end. Change that signed endpoint of the reference to .442 and its other five block coordinates to $(6b - .442)/5$. The block sum is preserved; its last coordinate is $> .2116 > .096$, above the next coordinate. All new positive duals are paid by the old strong floors and this actual endpoint pressure. The single reference's energy gain is

$$(6/5)(.442 - b)^2 > (6/5)(.442 - .262)^2 = .03888.$$

Hence

$$e > 9978827/12000000 + .03888 = 10445387/12000000 > 6/7.$$

Only one helper is used; either orientation works.

At $N = 19$, assume $e \le .91$; no near-four remains. The reference table gives

$$E_0 \ge .9240352, \quad E_1 > .8937, \quad E_2 > .86414.$$

A strong end is a length-five block with $.29 < b_0 < .298$; a weak end is a length-six block with $1/4 < b_1 < .266$. Changing a signed endpoint to $z_0 > b$ raises energy exactly $m(z_0 - b)^2/(m-1)$, preserving the block sum. For $m = 5, z_0 = .432$, the other coordinates exceed .2545, above $.151/\sqrt{2}$; for $m = 6, z_0 \le .58$, they exceed .184, above $.108/\sqrt{2}$. Neighbor omissions only decrease the next coordinate. Opposite end blocks are disjoint.

Zero weak ends already exceed .91. With one weak end and a near helper,

$$e > .8937 + (6/5)(.58 - .266)^2 = 1.0120152.$$

Otherwise its helper is the opposite strong end:

$$e > .8937 + (5/4)(.432 - .298)^2 = .916145.$$

With two weak ends, any near helper gives $.86414 + (6/5)(.58 - .266)^2 = .9824552$. If both helpers are far, they use distinct opposite endpoints. Paying both in the disjoint length-six blocks gives

$$e > .86414 + 2(6/5)(.432 - .266)^2 = .9302744.$$

These exhaust the end patterns and orientations, always on one modified reference for the same x_A. □

B.4 The local A_{18} inventory at comparison $11/8$

This appendix supplies the earlier geometric body used in Lemma 16.1; it does not use that lemma's endpoint pressure or its original B_{18} application. We work at the actual whole-A_{18} vertex q with the same strict interior point and comparison $-\langle z, r\rangle \le \kappa E_q$, $\kappa < 11/8$.

B.4.1 All zero-own and isolated boundaries

On $5 \le i \le 14$, deletion leaves path orders $\ell = i, r = 19 - i$, both at least 5. The complete one-node finite-white list is none; one simple 3 at an end, a D position or an admissible E position; one terminal 4; or two simple end 3's joining the two paths. Two contacts in one path make a nonfinite cycle. A joined completion has rank 18, hence is classical with only new simple end bonds. There are no foreign components at this vertex.

At zero own, a global endpoint 3, next-to-end 3, or endpoint 4 has time square $9/19 - 1, 17/19 - 1, 18/19 - 1 < 0$, respectively. Other simple-3 placements retain affine E_8 arms $1, 2, 5$ or E_7 arms $1, 3, 3$. Internal 4 gives the $4/3/3$ star; two simple white contacts close an all-three cycle. No contact gives zero derivative and time square -1. The affine kernels are positive: for E_7 take center 4 and arms $2; (3, 2, 1); (3, 2, 1)$; for E_8, center 6, arms $3; (4, 2); (5, 4, 3, 2, 1)$. For a $4/3/3$ star, take center $\sqrt{2}$, doubled leaf 1, simple leaves $1/\sqrt{2}$; all-three cycles have coefficients 1. Positive cap time makes every such null nonzero future, and all unknown original wall products are nonpositive. Compactness excludes them.

Isolated own 3 retains the affine E_7 star with the cap as short arm; own 4 has the $4/3/3$ star; own 6 the $6/3$ chain, with kernel $(\gamma, 1, 1/2)$; parallel own has null $c + n_i$. White 3/own 3 closes an all-three cycle. Far white 4/own 3 retains the E_7 star and far white 4/own 4 the doubled star. For pure helpers own $3, 4, 6$ or parallel retains respectively the path cycle, doubled star, $6/3$ chain or $c + n_i$. Additional unselected contacts preserve all wall signs in these actual null arguments.

For isolated positive own put $\tau = \gamma^{-2}$. For $0 < \tau < 1$, the actual positive core $c + \gamma\tau n_i$ gives

$$h_i > \sqrt{D_i - \tau} - \kappa\sqrt{\tau(1 - \tau)}.$$

At $\tau = 1$ it is a feasible null. For $\tau > 1$, positive finite projection on c, n_i gives the exact pressure

$$h_i \ge \sqrt{\frac{D_i - 1}{D_i - \tau}}\,(\sqrt{D_i - 1} - \kappa\sqrt{\tau - 1}).$$

The future point, not its spacelike correction, is normalized. Every unknown wall is checked by the positive selected-root coefficients. Here $D_i \ge 140/19$, $a_i = (\sqrt{D_i} + \sqrt{D_i - 1})^{-1} < 5/26 < .193$. For $0 < \tau < 1$, the loss is

$$< a_i\tau + \kappa\sqrt{\tau(1 - \tau)} < (.193 + 1.389)/2 = .791 < .87.$$

The radical seal is $.193^2 + (11/8)^2 < 1.389^2$. For finite owns at least 8, the positive-bracket finite formula has loss $< 5/26 + (11/8)(.415) < .764$.

Native 5 uses its positive H_4 projection, old coefficients $\gamma(3/2, 1, 1/2)$, residual $\delta_5 = (7 - 3\sqrt{5})/16 < 1/36$, and cap time square > 3.7. The finite-point radical factor $\sqrt{1 + \delta_5/A^2} < 1.01$ gives loss

$$< 25/76 + (11/8)\frac{1.01}{6(.8)} < .619.$$

Native 7 uses the actual negative core $c + (4\gamma/3)n_i + (2\gamma/3)n_{\text{neighbor}}$, whose loss is

$$< 10000/41877 + (11/8)(9/25) < .735.$$

Use $.9 < \cos(\pi/7) < 10/11$, verified by its increasing cubic, and $\sqrt{D_i - 100/81} > 2.47$. Thus every isolated finite or dotted cap is strong at loss .87 and comparison 11/8.

B.4.2 Every nonisolated ratio and the surviving helpers

The ordinary ratio threshold 3/2 at this comparison is proved in Lemma 16.1 by its explicit concave polynomial. For one white 3 at side position t in a path of order ℓ,

$$R = \frac{\gamma + t/(2\ell)}{\sqrt{1 - t(\ell - t)/(2\ell)}}.$$

Near 3, after the own-3 cycle, has ratio $> (.808 + 2/5)/\sqrt{3/5} > 3/2$, except the own-4 near-four role. At D positions $t = 2, \ell - 2$, after the own-4 star, ratio is

$$\geq \frac{\gamma + 1/\ell}{\sqrt{2/\ell}} > \frac{.808 + 1/5}{\sqrt{2/5}} > 3/2.$$

The lower expression increases for $\ell \geq 5$. At the allowed E positions $\ell = 6, 7, 8$, denominator is at most 1/2 and own $> .8$. Near endpoint 4 exceeds 3/2 even at own 3. Far endpoint 4 has own $3, 4$ nulls, and otherwise ratio $\geq \gamma\sqrt{\ell} > 3/2$. Two-white-3 has ratio at least $\gamma\sqrt{D_i} + 1/\sqrt{D_i} > 3/2$ already at $\gamma = 1/2$. All larger finite and dotted owns strengthen these tests. This exhausts the finite-white table.

Only near-four with side order $5, 6$, or pure far helpers, remain weak. The near-four and pure-helper formulas, the order-7 seal, and the entire helper cutoff $\gamma < 17/16$ are derived in the main proof. This completes its own/white inventory without a farthest hypothesis at q.

B.4.3 Native-five determinant and positive mixed weights

For two adjacent native-five helpers use raw norm-two duals $C_{ij} = \min(i, j) - ij/N$. Put $H_i = v_i^2 - 2, S = v_i \cdot v_j, U = H_i - S, V = H_j - S$ and $P = i(N - i - 1) + 1$, with $j = i + 1$. For outward helpers, the exact expansions are

$$NS = P + (P + N - 2)\varphi, \quad NU = (i - N - 2) + (N - i + 1)\varphi, \quad NV = -i - 3 + (i + 2)\varphi.$$

For reversed helpers they are

$$NS = P + (P + N)\varphi, \quad NU = (i - N - 2) + (3i - N - 1)\varphi, \quad NV = -i - 3 + (2N - 3i - 4)\varphi.$$

Since $H_iH_j - S^2 = S(U + V) + UV$, substituting $\varphi^2 = \varphi + 1$ gives rational coefficients $N(N + 6)$ and $-N^2$, respectively. At $N = 19$ these are $475, -361$, both nonzero. The raw difference-square argument in the central-window lemma forces mutual zero first, so neither determinant can satisfy that zero mutual. No arithmetic criterion is used.

For the surviving mixed pairs, use the explicit positive Schur matrices, coefficients $\lambda = H^{-1}(A_i, A_j)^T$, and all four orientation formulas for W_i, W_j in the central-window lemma. They are purely geometric, so remain valid at comparison κ. Cap times still exceed 2; hence $\sqrt{1 + \mathcal{D}}/W < 7/16$ for a common helper and $< 3/10$ for opposite helpers. With $N = 19$, $\sqrt{D_i} > 2.7$ and $\sqrt{C_{i,i+1}} > 2.6$. Rationalizing $E_c - \|Z\|$, as in the prefix proof, bounds the spatial loss by $W/10$. The actual mixed-point pressure now is

$$\langle y, Z \rangle \geq 1 + \mathcal{D} - \kappa\sqrt{1 + \mathcal{D}}.$$

Thus its weighted mean loss is $< .7015625$ or $< .5125$, respectively. Both are less than .87, proving the required nonadjacency against every original wall. Together with the new endpoint/mate/reference argument already written, this completes all earlier local inputs of Lemma 16.1.

B.4.4 The fresh $N = 16$ inventory and $A_{15} > .826$

Proposition B.4.1. *With arbitrary finite spectators, an actual A_{15} component has energy greater than* 413/500.

Proof. Use the exact interval $5 \le i \le 11$, loss .61, and $55/8 \le D_i \le 8$. The finite-white list is the two-path list already proved: each retained path has rank at least 4, and a joined completion has rank $15 > 8$. The zero-own affine and foreign-shadow repairs need only side order at least 5, so give precisely the same exhaustive cases at this rank.

For isolated caps, put $a = (\sqrt{D} + \sqrt{D-1})^{-1} < 1/5$. The cleared $0 < u < 1$ loss is $< (.2 + \sqrt{1.04})/2 < .61$. For foreign $u > 1$ it is $< .2 + 2 - \varphi < .582$. For pure finite labels at least 8, use the exact singleton pressure

$$T = r\sqrt{\frac{r-s}{r+s}}, \quad r = \sqrt{q^2-1}, \quad s = \tan(\pi/m) \le \sqrt{2} - 1 < .415.$$

Its loss $q - T$ decreases with q and increases with s. For the first monotonicity, the positive derivative comparison reduces to

$$(1-s^2)r^4 + 2sr^3 + (2s^4 - s^2)r^2 - 2s^3 r + s^4 \ge r^4/4 > 0$$

on $r \ge 1, 0 \le s \le 1/2$. At $q_0 = \sqrt{55/8} < 2.623$, $2.423 < r_0 = \sqrt{47/8} < 2.425$, and

$$T^2 > (47/8)(2.008/2.84) = 11797/2840 > (2.03)^2.$$

The loss is $< .593 < .61$. Own 5 has its positive H_4 loss $< 25/76 + 5/24$; own 7 has $< 10000/40257 + 9/25 < .61$. Own $3, 4, 6$, parallel and the $u = 1$ case have their actual affine/null exclusions. Thus all isolated caps are strong. The ordinary white ratios exceed $3/2 > T(q)$ using only side order at least 5. The complete weak list is near-four at $5, 6$ or its reflection, and pure helpers $\gamma\omega_i + \omega_e/2$, $e = 1, 15,\ Q = 0,\ \varphi/2 \le \gamma < 17/16$. All positive-foreign roles are strong.

For adjacent helpers, the same actual-product bounds leave mutual 0 for a common helper and $0, -1/2$ otherwise. Outward zero contradicts

$$(D_i - 4/5)(D_{i+1} - 4/5) - C_{i,i+1}^2 = (2/5)C_{i,i+1} - 24/25 + 1/10 \ge 41/25 > 0.$$

For reversed zero, the polynomial of the central-window proof has $l + s = 15$ and constant $9/2$. It increases in each own on $[.8, 1.08]$ and is concave in l. At its endpoint $l = 5$ or 10, both owns at least 7 give $P(.9, .9) = 11.88 > 0$. One 5 and an own at least 8 give the exact minimum $P(.809, .923) = .0907947698 > 0$; the reflected endpoint is larger by $5(.923 - .809)$. Two native-five helpers are impossible by the raw determinant proof, valid for $N \ge 15$. Native 5/7 is impossible by the same nonzero quadratic over $\mathbb{Q}(\sqrt{5})$ versus cubic degree argument. No field restriction is imposed on any other wall. The surviving opposite-helper mutual is therefore $-1/2$. The positive mixed Schur point appends all spectators; its times satisfy $A_i^2 > 699/160 > 4$. The explicit inverse weights give the same mean losses $< 43/80$ or $< 2/5$. The spatial term is $< W/10$, since $\sqrt{55/8} > 2.6, \sqrt{25/4} = 2.5$. Helpers are consequently nonadjacent before any energy target is used.

Assume now $e \le 413/500$. A near-four's actual mate has demand $> .86$. At the other end, a mate or a strong 10/11 has demand $> .76$; their correlation is at most $\sqrt{3/11} < 11/20$. Cauchy would give $1.62^2 < (413/500)(31/10) = 2.5606$, whereas $1.62^2 = 2.6244$. Thus all weak sites are pure helpers.

Use reflected rational heights

$$H_5 = H_{11} = a = 2.012, \quad H_6 = H_{10} = C = 2.128, \quad H_7 = H_9 = 2.196, \quad H_8 = 2.218$$

and linear ends. Flatten an outer weak end to its strong neighbor; omit all internal weak knots by chords. The exact complete bounds are

weak ends	maximum internal loss	$E >$
0	$3(.048^2/4) = .001728$	.8264648
1	$2(.048^2/4) = .001152$	$.7928637333\ldots$
2	$2(.046^2/4) = .001058$	1138171/1500000.

Every positive dual uses an actual strong floor. For one weak end the block lengths are $6, 5$, with separate prices $\beta_6 \geq 43/300, \beta_{11} \geq 179/1250$. For two weak ends they are $6, 6$, and $\beta_6, \beta_{10} \geq 43/300$. Adjacent omissions increase these capacities.

Use the same three actual endpoint-core constructions of Section B.3.7, now with the freshly computed times

$$A^2_{\text{far}} = (220\gamma^2 + 20\gamma - 17)/32, \quad A^2_{\text{near}} = A^2_{\text{far}} + 3\gamma/4.$$

In the six/six reference, the core's block projection is $(5\gamma/6)\omega_6 + \omega_{10}/12$ far, and $(5\gamma/6 + 1/12)\omega_6$ near. Fresh substitution of the $N = 16$ dual Gram gives exactly the three residual polynomials in the $N = 17$ payment proof, with the same near addition $\gamma/3$. This is the fixed-block variance identity, not an imported different-rank energy bound. Thus $Y = U - \Pi U$ is orthogonal to the reference and $\|Y\|^2 < 8/5$.

Let Δ be the core demand minus $C(5\gamma/6+1/12)$. All orientations and own intervals have $\Delta > 1/3$. Indeed $(A'_{\text{far}})^2 - 55/8 = 15/(4A^2_{\text{far}}) > 0$. At native 5, use time > 2.11, radical $< .111$, projection $< C(91/120)$; the deficit is $> 5779/15000$. For finite owns at least 7, use $A_{\text{far}}(.9) = \sqrt{28/5} > 2.366$ and radical $\leq 1/4$, giving $> 257/750$. For dotted owns, use $A_{\text{far}}(1) = \sqrt{223/32} > 2.639$ and radical $< .338$, giving $> 1051/3000$. Monotonicity of A_{far} minus its linear projection makes these full-interval bounds; near times are larger. In the six/five case, a far projection uses $\omega_{11}/10$ instead. Its variance decreases $1/60$, and its deficit decreases $a/10 - C/12 = 179/7500$. Hence $\Delta > 2321/7500 > .309$.

Pay $\mu = .1$ before using each core pressure. One foot costs at most $31/320 < 43/300$ at the weak-side strong anchor and, if far, $1/100 < 179/1250$ at the opposite anchor. For two feet, the combined cost at each $6/10$ is at most $101/960 < 43/300$. Thus $W = (w - \mu \sum \Pi U) + \mu \sum U$ has a nonnegative old strong-floor part and the actual core pressures, giving

$$e \geq E + 2\mu \sum \Delta - \mu^2 \|\sum Y\|^2.$$

One weak end gains $> .0458$, hence $e > .7928 + .0458 = .8386$. For two ends retain the full cross through $\|Y_L + Y_R\|^2 < 32/5$. The gain exceeds $26/375$, so

$$e > 1138171/1500000 + 26/375 = 1242171/1500000 = .828114 > 413/500.$$

Zero weak ends already exceed the target. This proves the component theorem.

Every ordinary rank-three forest costs $> 17/70$, so this new component rate also excludes its ordinary spectators directly. The other retained spectators $I + T_2$ and $O_7 + A_1$ cost $> .21$ and $> .09 + 2/15$; each sum likewise exceeds 1. A separate historical ordinary-rank-three consumer is therefore unnecessary. □

B.4.5 The fresh $N = 13$ component and its native-cosine split

Proposition B.4.2. *With arbitrary finite spectators, an actual A_{12} component has energy greater than* 13/20.

Proof. Use $N = 13$, the exact window $5 \le i \le 8$, loss .65, and $80/13 \le D_i \le 84/13$. The same complete two-path finite-white classification applies at these actual ranks: each path has rank at least 4, a joined extension has rank $12 > 8$, and its only connected finite completion is the ordinary path. The zero-own repairs are unchanged: internal 3/4 and two-white roles have affine principals; endpoint 3 or next-to-end 3 has time square below its positive foreign cost, forcing an orthogonal closure; endpoint 4 has $Q < 1/5$, hence negative time square. Every own contact is positive.

The strength ratio now satisfies, for the entire window,

$$T(q) = 20/13 - \frac{q - .65}{q(2q - .65)} < T(3) = 5809/4173 < 7/5.$$

Near simple 3, D/E positions and endpoint 4 have the same larger ordinary residual ratios as before. Even the near-four exception has minimum $(1/\sqrt{2} + 2/5)/\sqrt{3/5} > 7/5$, so it is strong. Far white 4/own 3 retains affine F_4. Two-white-3 has ratio at least $q/2 + 1/q > 7/5$ at positive own; zero own is affine. Positive foreign cost strengthens these tests; the exceptional foreign far/helper ratios remain $> 761/500$ or have square $> 121/35$.

For isolated owns put $a_0 = (\sqrt{D} + \sqrt{D-1})^{-1} < .212$. The actual cleared $0 < u < 1$ pressure has loss $< (.212 + 1.023)/2 = .6175$; foreign $u > 1$ has loss $< .212 + .383 = .595$. Pure finite owns at least 8 have loss $< .212 + \tan(\pi/8) < .627$. The native-five H_4 point has loss $< 125/352 + 5/24 = 595/1056 < .57$. Native seven's A_2 core has loss $< .264 + \sqrt{13/120} < .601$, using $.9 < \gamma < .902$ and $\gamma^{-2} > 49/40$. Own $3, 4, 6$, parallel and $u = 1$ have their actual affine/null exclusions. All spectators are cleared positively and every unknown wall is checked as in the shared proofs. Thus the complete weak list is pure

$$v = \gamma\omega_i + \omega_e/2, \quad e = 1, 12, \quad Q = 0.$$

The identity $\sigma^2/\delta \ge 4\gamma - 2$ and $T < 7/5$ force $\varphi/2 \le \gamma < .99 < 1$. Only own 5 or finite own at least 7 remains; dotted owns have been excluded by the actual edge inequality.

We prove fresh nonadjacency, without importing a larger-rank determinant. For adjacent owns A, B, the same-helper product satisfies

$$R \ge 1 - AB + (AB - |A - B|)/13 > 1 - .99^2 > 0,$$

contradicting acuteness. Both opposite-helper AM–GM bounds are $> -1/2$, so native gap forces mutual $R = 0$. For outward helpers, the retained parameters obey $\eta < 1/(2\gamma^2) < 25/32 < 4/5$, whereas

$$(D_i - .8)(D_{i+1} - .8) - C_{i,i+1}^2 = (2/5)C_{i,i+1} - 24/25 + 8/65 \ge 428/325 > 0.$$

This makes zero mutual impossible by the actual time/spatial determinant identity.

For reversed helpers put $l = i, s = 12 - i$; $l + s = 12$, $ls = 35$ or 36. The exact time and spatial formulas are

$$A_c^2 = \frac{2l(s+1)A^2 + 2lA - 7}{13}, \quad B_c^2 = \frac{2(l+1)sB^2 + 2sB - 7}{13},$$

$$S = \frac{2lsAB + (s+1)A + (l+1)B + 1/2}{13}, \qquad A_c^2 B_c^2 - S^2 = P/13,$$

$$P = 4lsA^2B^2 - (l+1)(s+1)(A^2 + B^2) - 2AB - (l+1)A - (s+1)B + 15/4.$$

Zero mutual requires $P = 0$. The polynomial increases in either variable on $[.8, 17/16]$: its derivative bracket is at least $1.56ls - 13 \ge 41.6$. If neither own is 5, $A, B > .9$, and the minimum $P(.9, .9)$ at $ls = 35$ is $3.624 > 0$. If both are 5, direct reduction gives $P = ((ls - 56)\varphi - 13)/4 < 0$, also nonzero.

It remains $A = \varphi/2$, $B = \cos(\pi/m)$, $m \geq 7$, including its reversed placement. As a polynomial in B, P has leading coefficient $a = ls\varphi - 13 \neq 0$ and linear coefficient $b = -\varphi - s - 1 \neq 0$. If it vanished, B would have degree at most 2 over $\mathbb{Q}(\varphi)$, hence degree at most 4 over $\mathbb{Q}$. The real cyclotomic degree formula gives $\phi(2m) \leq 8$. Its complete even list with $2m \geq 14$ is $14, 16, 18, 20, 24, 30$. Indeed prime divisors can only be $2, 3, 5, 7$; the totient product bounds leave exactly these even integers. Thus $m = 7, 8, 9, 10, 12, 15$. For $7, 9$, the native cubic degree remains 3 over the quadratic $\mathbb{Q}(\varphi)$. For 8, the quartic real cyclotomic field has unique quadratic subfield $\mathbb{Q}(\sqrt{2})$: its generator $\cos(\pi/8)$ has a cyclic order-four conjugation, whose square fixes $\sqrt{2}$. For 12, the field is $\mathbb{Q}(\sqrt{2}, \sqrt{3})$, with quadratic subfields $\sqrt{2}, \sqrt{3}, \sqrt{6}$. Neither contains $\sqrt{5}$; both native degrees remain 4 after adjoining φ. For 10, $B^2 = (\varphi + 2)/4$ has nonsquare rational norm $5/16$; its minimal quadratic has zero linear coefficient, unlike P. For 15, the minimal quadratic is

$$4B^2 - 2(\varphi - 1)B - (\varphi + 1),$$

whose discriminant squareclass $3(2 + \varphi)$ has nonsquare rational norm 45. Proportionality to P would require

$$2b + a(\varphi - 1) = ls + 11 - 2s - 15\varphi = 0,$$

impossible. All native cases are excluded. This is algebra of literal finite cosines, not an arithmeticity restriction on the polytope. Weak sites are consequently nonadjacent.

Use heights $H_5 = H_8 = a = 1.830$, $H_6 = H_7 = b = 1.891$, linear ends and reflection. Their strong-floor tests are $2.480^2 < 80/13$, $2.541^2 < 84/13$. Flatten a weak outer end and omit internal weak knots by neighbor chords. The complete reference bounds are

weak outer sites $5, 8$	internal omissions	$E \geq$
0	at most one, loss $.061^2/4$	2690283/4000000
1	at most one, same loss	1901431/3000000
2	$6, 7$ both strong	3575881/6000000.

Zero outer weak sites already gives $e > .65$. For one outer weak site, orient it to 5. The block lengths are $6, 5$, with prices $\beta_6 \geq b/12$, $\beta_8 \geq .1525$. For two outer weak sites they are $6, 6$, with $\beta_6 = \beta_7 = b/12$. Neighbor omissions only increase capacities; no weak floor is charged.

Choose one actual weak outer helper. Use the finite-five or finite-at-least-seven actual core in Section B.3.7; their selected roots are still present at $N = 13$, with all-wall signs and future orientations unchanged. The freshly computed times are

$$A^2_{\text{far}} = (80\gamma^2 + 10\gamma - 7)/13, \qquad A^2_{\text{near}} = A^2_{\text{far}} + 6\gamma/13.$$

Project its displacement onto the constant end blocks. In the six/six reference,

$$\Pi U = (5\gamma/6)\omega_6 + \omega_7/12 \quad \text{far}, \qquad \Pi U = (5\gamma/6 + 1/12)\omega_6 \quad \text{near}.$$

For six/five, replace the far $\omega_7/12$ by $\omega_8/10$. The residual $Y = U - \Pi U$ has zero mean in each block and $\langle w, Y \rangle = 0$. Fresh block variance gives

$$\|Y\|^2 = (2/3)\gamma^2 + 1/12 \quad (5), \qquad 4\gamma^4 - (16/3)\gamma^2 + 29/12 \quad (\text{finite } \geq 7),$$

with near addition $\gamma/3$. A six/five far residual decreases $1/60$. Thus every case has $\|Y\|^2 < 17/12 < 8/5$. Both reference pairings obey

$$\langle w, \Pi U \rangle \leq b(5\gamma/6) + a/10.$$

For native five, $A_{\mathrm{far}} > 2.027$, radical $< .111$, pairing $< 1.459425 < 1.46$, so its deficit exceeds .456. For finite owns at least seven, $A_{\mathrm{far}}(.9)^2 = 334/65 > 2.266^2$; the pairing envelope at $\gamma = .9$ is 1.60125. The envelope $A_{\mathrm{far}} - (5b/6)\gamma - a/10$ increases because

$$(A'_{\mathrm{far}})^2 - 80/13 = 585/(169A_{\mathrm{far}}^2) > 0, \qquad \sqrt{80/13} > 5b/6.$$

Subtracting the uniform radical cap 1/4 gives deficit $> 2.266 - .25 - 1.60125 = .41475$ throughout this finite-own interval. Monotonicity of the actual radical-dependent deficit is not required. Near times are larger and pairings smaller. Uniformly $\Delta > .4$. In particular the far-native-five pairing is bounded by 1.46, not the false value 1.438.

Pay $\mu = .1$ for this one helper. A far foot spends $< .0825$ at ω_6 and at most .01 at the opposite strong anchor; a near foot spends $< .090834$ at ω_6. They fit $b/12 > .15758$ and .1525 separately before pressure is used. The single vector $W = w + .1Y$ gives

$$e > E + 2(.1)(.4) - (.1)^2(8/5) = E + .064.$$

Even the worst reference yields

$$e > 3575881/6000000 + 8/125 = 3959881/6000000 = 13/20 + 59881/6000000.$$

All weak sets, orientations and native labels are exhausted, with one original-vector budget. □

B.5 The actual long-prefix inventory and its point pressures

We supply the geometric input of Proposition 11.1. Its exact domain is $6 \le M \le 16$, B_r with $r \ge M + 5$, or D_r with $r \ge M + 6$. For $5 \le i \le M + 1$, deletion of old i leaves a left A_{i-1} of rank at least 4, and a right B_s, $s \ge 4$, or D_s, $s \ge 5$. All roots used at 5 and its neighbors are actual long roots. No short-tail exclusion is imported. Write $D_i = 2i, q_i = \sqrt{2i}$ and $h_i = \langle x, \omega_i \rangle = \sqrt{2}S_i$.

B.5.1 The strong-ratio test

An edge is strong if

$$\frac{\cosh L_i}{q_i} > T(q_i), \qquad T(q) = \frac{1 + (1 - .61/q)^2}{q(1 - (1 - .61/q)^2)} = \frac{100}{61} - \frac{q - .61}{q(2q - .61)}.$$

Its half-tangent then exceeds $1 - .61/q_i$. In particular $T(q) < 100/61$. For the small left layers, clearing positive denominators gives $T(q) < 38/25$ on $[5/2, 10/3]$, and $T(q) < 8/5$ on $[3, 4]$. The resulting quadratics are concave and positive at both endpoints.

B.5.2 Clearing all spectators from an isolated own

For an isolated own γ, retain all contacted spectator blocks, with spatial sum v_S and full cost Q. Put

$$c = Ap - \gamma\omega_i - v_S, \quad u = (1 - Q)/\gamma^2 > 0, \quad A^2 = \gamma^2(D_i - u).$$

If $u < 1$, the positive actual-normal combination

$$\xi = c + v_S + \frac{1 - Q}{\gamma} n_i$$

has products zero on c, S, product $\gamma(u-1) < 0$ on n_i, square $-\gamma^2 u(1-u)$, and time $A > 0$. Inverse positivity expands v_S in actual spectator normals with nonnegative coefficients. Every unknown wall product is nonpositive. Its actual point gives

$$h_i > \sqrt{D_i - u} - \sqrt{u(1-u)}.$$

At $u = 1$ it is a nonzero feasible future null. For $u > 1$, the positive finite projection on $S + n_i + c$, with residual $\gamma^2(u-1)$, gives

$$h_i > \sqrt{D_i - u} - \sqrt{u-1}.$$

Own $3, 4, 6$ retain an affine fork, the $4/3/3$ star, or the $6/3$ path; a parallel own has null $c + n_i$. Thus $\gamma \geq \varphi/2$. If $Q > 0$, $u \leq 3/\varphi^2 = 6 - 3\varphi < 7/6$. For $u > 1$ the loss is $< 1/5 + 2/5 < .61$. For $u < 1$ it is at most

$$u/6 + \sqrt{u(1-u)} \leq (1 + \sqrt{37})/12 < 3/5.$$

Every foreign isolated cap is strong.

At $Q = 0$, isolated own 5 has an actual H_4 correction on three consecutive old nodes, with $\delta_5 = (7 - 3\sqrt{5})/16 < 1/36$. Concavity bounds its corrected spatial product by γh_i; its loss is $< 25/76 + 5/24 < .61$. Own 7 has the negative A_2 core, of loss $< 400/1701 + 9/25 < .61$, using $.9 < \gamma < 10/11$. Finite own at least 8 has singleton pressure $h_i > \sqrt{D_i - 1} - \tan(\pi/m)$, of loss $< 4/21 + 5/12 < .61$. For dotted own use $c + n_i/\gamma$ and $u = \gamma^{-2}$. Each correction is positive in actual roots. No projected mirror is declared native.

B.5.3 The complete finite-white lists

A left A_{i-1} permits one contact: endpoint 3, endpoint 4, next-to-end 3, or an admissible E_6, E_7, E_8 position 3. Their Schur residuals are

$$(i+1)/(2i), \quad 1/i, \quad 2/i, \quad 1/4,\ 1/7,\ 1/16.$$

For a simple contact at position j from the far end, $\sigma = \gamma + j/(2i)$; for endpoint 4, $\sigma = \gamma + j/(\sqrt{2}i)$. The actual edge ratio is $\sigma/\sqrt{\delta - Q}$. The right list is

$B_s,\ s \geq 4$	long-leaf 3
$D_s,\ s \geq 5$	long-leaf 3; fork 3 only for $s = 5, 6, 7$.

Two contacts to a retained component create a nonfinite cycle. A white bridge joining left and right has rank $r > 8$; classification forces finite type B or D, with simple endpoint attachments. A further branch at its internal new node is not finite. Exceptional fork pieces are retained with their full allowed foreign costs.

Zero own needs a separate test. Internal simple/doubled left contacts, right long contacts and full bridges retain affine forks, stars or cycles. Right D-fork contact retains affine E_8. For the separate far-3 case,

$$A^2 = Q - \tfrac{1}{2} < 1/(2i) \leq 1/10.$$

Any foreign contact has derivative square at least $1/4 > A^2$, forcing an orthogonal closure; at $Q = 0$ time square is negative. Zero own has therefore not been omitted.

For positive own, the complete direct-strength check uses the displayed residuals. Near simple-3 has $\delta \leq 3/5, \sigma \geq \gamma + 2/5$; next-to-end has $\delta = 2/i, \sigma \geq \gamma + 1/i$; E-rows have $\delta \leq 1/4$. For $\gamma > 101/125$, near simple-3 exceeds $100/61$ from $i = 8$, and next-to-end from $i = 7$. At $i = 5, 6, 7$, the smaller interval tests for T give the same conclusion. Own $3, 4, 6$ are affine except near-4/white-3,

whose B cosine $(i-1)+i/\sqrt{2}$ is already strong for $i \geq 5$. In D, own 4 without right contact is affine on the actual right fork. White 4 is stronger, except far white 4/own 3, with affine $3,4,3,3$ path.

Right long 3 has $\delta = 1/2 - Q, \sigma = \gamma + 1/2$. Own $0,3$ is affine; the next contact has ratio at least $1 + 1/\sqrt{2} > 100/61$. For right fork 3 in D_s, $s = 5,6,7$,

$$\delta = 1 - s/8 - Q, \qquad \sigma = \gamma + 1/4.$$

Own $0,3,4,6$ retains affine E_8, a cycle, the $3,3,4,3$ path, or the $6,3$ path. The next own gives ratio $(\cos(\pi/5) + 1/4)/\sqrt{3/8} > 1.72 > 100/61$. All finite foreign extensions of the E_6, E_7, E_8 piece remain included through Q. A full bridge has residual at most $1/(2i)$, numerator at least $\gamma + 1/2 + 1/(2i)$; positive own is strong, and zero own gives an affine cycle.

A far simple-3 cap with $Q > 0$ has

$$\cosh L_i = \frac{2i\gamma + 1}{\sqrt{i + 1 - 2iQ}}, \quad \cosh L_i/q_i > R_0(i) := \frac{(809/500)i + 1}{\sqrt{i(i+2)}}.$$

Both $R_0(i)$ and $T(\sqrt{2i})$ increase; the first derivative has sign $(309/500)i - 1 > 0$. The exact intervals cover $5 \leq i \leq 17$:

i	q_i upper	$R_0(i)$ lower	$T(q_i)$ upper
5–8	4	1.536	1.530
9–12	5	1.563	1.550
13–29	7.62	1.577	1.577.

The squared lower margins are $.052740, .321913, .546001$. The upper comparisons follow by multiplying the numerator/ denominator pairs $(27.4921, 18.0316), (44.2721, 28.6395), (107.2045, 68.003166)$. Every foreign far cap is strong.

Consequently every weak cap has exactly

$$v = \gamma\omega_i + \omega_1/2, \quad A^2 = 2i\gamma^2 + 2\gamma - \tfrac{1}{2}, \quad Q = 0, \quad 101/125 < \gamma < 6/5. \tag{B.5.1}$$

Its own is 5, finite at least 7, or dotted. For the upper cutoff, $\sigma^2/\delta \geq 4\gamma - 2$, with difference numerator $[2i(\gamma-1)-1]^2$. At $\gamma = 6/5$ this is $14/5 > (100/61)^2$.

B.5.4 Nonadjacency from one actual mixed point

For adjacent weak owns a, b, direct weighted AM–GM gives

$$\langle c_i, c_{i+1} \rangle \geq \tfrac{1}{2} - ab + \tfrac{1}{4}(a/b + b/a) \geq 1 - ab > -11/25 > -1/2.$$

The actual product must be zero. Clearing all old roots except $i, i+1$ gives the two-cap Schur matrix

$$H = I - t\mathbf{1}\mathbf{1}^T, \qquad t = (i-1)/(2i) > 0.$$

It is positive with positive inverse. Put $\lambda = H^{-1}(A_i, A_{i+1})^T > 0$ and $\mathcal{D} = A_i\lambda_i + A_{i+1}\lambda_{i+1}$. The actual positive projection has displacement $Z = W_i\omega_i + W_{i+1}\omega_{i+1}$, with

$$W_i = \lambda_i a + (\lambda_i + \lambda_{i+1})/(2i), \quad W_{i+1} = \lambda_{i+1} b, \quad Z^2 = \mathcal{D}(1 + \mathcal{D}).$$

Since $Q = 0$, all spectator blocks remain unchanged positive orthogonal blocks; positivity checks every unselected original wall. Write $W = W_i + W_{i+1}$, $A_+ = A_i + A_{i+1} > 4$. Then

$$\lambda_i + \lambda_{i+1} = iA_+, \quad \mathcal{D} = [iA_+^2 + (A_i - A_{i+1})^2]/2, \quad W > (4/5)iA_+.$$

It follows that

$$\frac{1+\mathcal{D}}{W^2} < \frac{(i+1)/2+1/16}{(16/25)i^2} \le 49/256,$$

the last expression decreasing from $i=5$. For $E_c = W_i\sqrt{2i} + W_{i+1}\sqrt{2i+2}$,

$$E_c - \|Z\| = \frac{2W_iW_{i+1}(\sqrt{2i(2i+2)}-2i)}{E_c+\|Z\|} < W/(4\sqrt{10}).$$

The actual point gives $\langle x, Z\rangle \ge 1+\mathcal{D}-\sqrt{1+\mathcal{D}}$, and $\|Z\| < 1+\mathcal{D}$. The weighted mean loss is therefore less than $7/16 + 1/(4\sqrt{10}) < .61$, contradicting weakness at both sites. This proves nonadjacency before using any component-energy contradiction.

B.5.5 The four actual weak-five pressures

At $i=5$, put $A^2 = 10\gamma^2+2\gamma-1/2$ and $H(A,\delta) = A/(1+\sqrt{\delta/(A^2+\delta)})$. The actual point pressures $\langle x, U\rangle \ge T$ are

own	U	T
5	$\gamma(\omega_4+\omega_6)/2+\omega_3/6$	$H(A, 2/3-\gamma^2)$
7, 8	$\gamma(\omega_4+\omega_6)/2+\omega_2/4$	$A-\sqrt{\gamma^2-3/4}$
finite ≥ 9	$\gamma(\omega_4+\omega_6)/2+\omega_1/2$	$H(A, 1-\gamma^2)$
dotted	$(\gamma-\gamma^{-1})\omega_5+(\omega_4+\omega_6)/(2\gamma)+\omega_1/2$	$A-\sqrt{1-\gamma^{-2}}$.

The selected old sets are $\{5,1,2\}, \{5,1\}, \{5\}$, and the dotted pair. In the first three cases clear those roots with their positive inverse coefficients. For the dotted pair use $c+n_5/\gamma$. The selected products are zero or nonpositive and time is positive. All unknown actual-wall products are nonpositive by positivity; $Q=0$ leaves pure component displacement. Finite points are normalized as in (9.4), not as spacelike corrections.

Concavity gives $\langle x, \omega_4+\omega_6\rangle \le 2\langle x,\omega_5\rangle$, $\langle x,\omega_2\rangle \le 2\langle x,\omega_1\rangle$, $\langle x,\omega_3\rangle \le 3\langle x,\omega_1\rangle$. Weak 5 therefore implies

$$x_1 > \sqrt{2}\{T-\gamma(\sqrt{10}-.61)\} > .5965.$$

For own $5,7,8$, the lower bounds exceed $.8, .7, .65$, using $.809 < \cos(\pi/5) < .81$, $.9 < \cos(\pi/7) < .902$, $.923 < \cos(\pi/8) < .9239$. For finite owns at least 9, the deficit increases from $\gamma = .939$: $A' > \sqrt{10}, H_A > 9/10, H_\delta < 0$. At that endpoint $A > 3.1929, \sqrt{A^2+\delta} > 3.211, \sqrt{\delta} < .344$; thus $H > 2.88$, giving the lower bound $> .65$. For $1 < \gamma < 6/5$, the dotted deficit decreases: $A' < 13/4$, the radical derivative exceeds 1, and $\sqrt{10}-.61 > 5/2$. Its endpoint lower bound is

$$\sqrt{2}\{\sqrt{163/10}-\sqrt{11}/6-(6/5)(\sqrt{10}-.61)\} > .5965.$$

Positive square tests $\sqrt{163/10} > 4.03732$, $\sqrt{11}/6 < .552771$, $\sqrt{10} < 3.162278$, $\sqrt{2} > 1.41421$ prove it. Together with nonadjacency this supplies both initial branches of Proposition 11.1 without assuming strong 5.

B.6 Fresh exact-18 classical layers

This appendix supplies the short layers used in the fresh rank-18 cover, at their actual ranks rather than by shifting a larger whole-component theorem. All prices and energies concern the original vector.

B.6.1 A local reference payment

The long-prefix inventory already reaches 12 in B_{16}, and 11 in D_{16}. Any actual strong endpoint K supplies the interpolation proof, in either initial-5 branch. Its endpoint sum is at most $f_K = \sqrt{K} - .61/\sqrt{2}$. Its individual terminal coefficient is $> 1/7$ for $K = 8, \dots, 12$, and $> 7/50$ for $K = 13$. The ordinary coefficient is at least $\sqrt{K} - \sqrt{K-1}$; omitting the preceding knot increases it. The weak-5 reference ends with slope $1/(\sqrt{K} + \sqrt{6}) > 1/7$. The exact conservative data are

K	8	9	10	11	12	13
$q_K >$	.7683275	.797425	.823735	.8472225	.8689525	.88864
$f_K <$	2.398	2.570	2.732	2.886	3.034	3.175.

Here q_K is the certified effective reference energy; the entries follow by adding the positive rational increment floors and permitted omission losses in the prefix proof.

For $K = 8, \dots, 12$, if an actual point/core gives $\langle x, V \rangle \geq G$, V equals 1 on the first K coordinates, and $V^2 \leq R$, pay $\mu = 1/7$ at that individual terminal price. The residual $V - P_K$ has zero product with the reference and square at most $R - K$. One square gives

$$e_{\text{classical}} > q_K + 2(G - f_K)/7 - (R - K)/49. \tag{B.6.1}$$

The weak-5 interpolation has higher energy and no larger endpoint. No extra overlapping first-coordinate gain is added to it. At $K = 13$ use only a nonnegative weight μ within its actual individual capacity β_K; the corresponding formula is

$$e_{\text{classical}} > q_K + 2\mu(G - f_K) - \mu^2(R - K), \qquad 0 \leq \mu \leq \beta_K.$$

The other strong reference anchors retain positive coefficients, which supplies strictness. The $K = 13$ data above are used in the later B_{17} extension with its separately funded new slope $< .39/3 = .13 < 7/50$, not with weight $1/7$.

B.6.2 B_{16} with ordinary T_2

At $i = 13$, deletion retains $A_{12} + B_3$. The right finite extensions are long 3 or short 3, the latter giving F_4. Long 3 has $\sigma = \gamma + 1/2, \delta = 1/2 - Q$; own $0, 3$ is affine and larger owns are directly strong. Short 3 has $\sigma = \gamma + 1/(2\sqrt{2}), \delta = 1/4 - Q$. At zero own, the actual path $c, n_{16}, n_{15}, n_{14}, n_{13}$ has affine labels $3, 4, 3, 3$. Positive own has ratio at least $1 + 1/\sqrt{2}$. The F_4 white block has no finite foreign arm. No middle or doubled extra attachment to B_3 is finite. The full B_{16} bridge has internal c and is foreign-sealed; own zero is the cycle, and positive owns have the direct strong ratio. All left and isolated tests retain their already proved actual matrices and full spectator. Thus pure-helper coverage reaches 13. The interpolation proof now gives $e(B_{16}) > .8689525$, using the actual long neighbor 14. Adding ordinary $T_2 > 1/7$ contradicts the norm. This is a fresh local extension, not an out-of-domain application of Proposition 11.1.

B.6.3 D_{16} with ordinary T_2

At $i = 12$, the right white D_4 has three simple-3 leaf extensions: long leaf 13, and the distinct signed forks $15, 16$. Long 13 has $\sigma = \gamma + 1/2, \delta = 1/2 - Q$; its zero-own two-fork affine tree and own-3 triangle leave only directly strong cases. A fork 3 has $\sigma = \gamma + 1/4, \delta = 1/2 - Q$. Own $0, 3, 4, 6$ gives respectively affine E_8 arms $5, 2, 1$, the five-cycle, the $4/3/3$ star, or the $6/3$ path; parallel gives $c + n_{12}$. All remaining $Q > 0$ cases have ratio > 2.1; pure finite owns at least 7 or dotted have ratio

$> 1.15\sqrt{2} > 25/16$. Only $Q = 0$, own 5/fork-3 remains. Full bridges have residual $1/24$; zero own is a cycle and positive owns are strong. Their internal new node cannot take a finite foreign branch.

For own 5, put $\gamma = \varphi/2$ and $\alpha^2 = \varphi + 5/8 > 56/25$. The actual left H_4 correction is $\gamma(3n_{12}/2 + n_{11} + n_{10}/2)$, with residual $\delta_5 = (5 - 3\varphi)/8$. The finite future point is the positive projection, not this spacelike correction. Its pure scaled displacement is flat through 9, with

$$A^2 = 12\alpha^2 - \tfrac{1}{2}, \quad R < 12, \quad G > \sqrt{12 - 25/112} - .091 > 3.34.$$

The radical bound is $\sqrt{\delta_5}/\alpha < .091$. It is feasible against both actual forks, the entire ordinary T_2 and every unknown wall by positive actual-root coefficients. Choose a strong $K = 8$ or 9. Equation (B.6.1), plus $1/7$, gives totals greater than

$$.7683275 + 2(3.34 - 2.398)/7 - 4/49 + 1/7 > 1.098,$$

or

$$.797425 + 2(3.34 - 2.570)/7 - 3/49 + 1/7 > 1.098.$$

Thus helper coverage reaches 12 before considering the next layer.

At $i = 13$, the right $D_3 = A_3$ list is middle 3, end 3, end 4, end 5. Middle 3 has the affine four-leaf star at zero own, the triangle at own 3, and direct strength otherwise. End 4 at zero own has the actual null

$$\sqrt{2}c + 2n_f + 2n_{14} + n_{13} + n_h,$$

where h is the other actual fork. Positive owns are directly strong. End 5 is the sealed finite H_4 white block, of ratio > 2.99 even at zero own. The only full bridges are middle-3/full D_{16}, and end-3/full A_{16}, with residuals $1/26$ and $1/8 + 1/26$. Zero own gives a cycle; positive owns are strong. At rank $16 > 8$, their internal node cannot take another finite branch.

For no-left end-3, $\sigma = \gamma + 1/4, \delta = 5/8 - Q$. The zero-own, own $3, 4, 6$ and parallel principals are the same E_8, four-cycle, doubled star, 6/3 path and null. Positive foreign cases are strong, and $\gamma \geq 1$ has ratio $\geq \sqrt{5/2} > 25/16$. The actual strength threshold obeys $T(\sqrt{26}) < T(6) < 25/16$. Only $Q = 0$, own 5 or finite own at least 7, remains. Own 5 has its left H_4 point, now flat through 10, with

$$R < 13, \qquad G > \sqrt{13 - 125/448} - .091 > 3.47.$$

For finite $.9 < \gamma < 1$, use both actual negative A_2 cores, corrections $\gamma(4n_{13}/3 + 2n_{\text{side}}/3)$ for sides $12, 14$, and average their spatial pressures. With $\alpha = \sqrt{2}(\gamma + 1/4)$, $d = 4\gamma^2/3 - 1 > 0$, direct products give

$$A^2 = 13\alpha^2 - 5/8, \quad Y^2 = A^2 + 1 - 14\gamma^2/9 < 13\alpha^2, \quad \langle x, Y\rangle \geq A - \sqrt{d}.$$

The average is flat through 11. Both original cores satisfy every wall, including both forks and all spectators. Uniformly $G > \sqrt{13 - 125/529} - 1/3 > 3.23$, $R < 13$. Choose strong $K = 9$ or 10; even this weaker G, with ordinary T_2, gives totals > 1.047 in (B.6.1). All extra D_3 roles are excluded. Coverage reaches 13, so interpolation gives $e(D_{16}) > .8689525$, again contradicting its ordinary T_2. No typed defect is charged as ordinary.

B.6.4 B_{17} with a nonshort A_1

The generic inventory reaches 13. At 14, the preceding complete B_3 layer reaches 14, with the actual old long and short roots retained. Choose strong $K = 12$ or 13. At the right B_2 layer 15, the extra short 3 has its affine F_4 path at zero own; own $4, 6$ gives a doubled star or 6/3 path; parallel

gives a feasible null. Every own $\gamma \geq \cos(\pi/5)$ is strong. Positive foreign cost makes own 3 strong as well. Bridges are internal and foreign-sealed. The sole extra weak case is therefore

$$Q = 0, \qquad c = Ap - (\omega_{15} + \omega_{17})/2.$$

Releasing old 16 and retaining this actual c gives finite A_{17}: old A_{15}, c, and the true short old 17 form its chain. Positive inversion gives its actual future endpoint, feasible against released 16 and every other original wall. Its exact cosine and prefix consequence are

$$C_{16} = (16 + 15\sqrt{2})/\sqrt{18} > 8.7, \quad S_{16}^2 > 16(77/97) > (3.56)^2.$$

Append $S_{16} > 3.56$ to the strong $K = 12/13$ reference, skipping $14, 15$. New slopes $< .53/4, .39/3$ fit the individual capacities $> 1/7, > 7/50$. The weak-five endpoint differs from f_K by less than .00001, so the same strict junction bounds apply. The two totals, including nonshort $A_1 > .1$, exceed

$$.8689525 + .526^2/4 + .1 > 1.038, \qquad .88864 + .385^2/3 + .1 > 1.038.$$

Thus this case is impossible. Pure-helper coverage reaches 15; interpolation gives $e(B_{17}) > .90715$, whose addition to .1 again contradicts 1. Actual I has not been assigned a nonshort debit, so $B_{17} + I$ remains for its separate consumer.

B.7 The whole-A_{18} endpoint argument

Here the entire diagram at the original farthest vertex is A_{18}, with no spectators. Fix that vertex p and the original strict interior point z, and write

$$p^2 = -1, \qquad z = E(p - x), \qquad E > 0, \quad x \perp p, \quad \|x\|^2 < 1.$$

The spatial space is the zero-sum subspace of $\mathbb{R}^{19}$. Its unit incident normals and duals satisfy

$$n_i = (e_i - e_{i+1})/\sqrt{2}, \qquad \langle \omega_i, \omega_j \rangle = \frac{2\min(i,j)(19 - \max(i,j))}{19}, \qquad D_0 = \omega_1^2 = \omega_{18}^2 = 36/19.$$

Interiority gives $x_1 > \cdots > x_{19}$. Let the actual endpoint opposites be the walls terminating the edges obtained by releasing n_1 and n_{18}. Compactness makes these endpoints finite. For each opposite c, its positive time is $A = -\langle p, c \rangle > 0$. We retain every original wall throughout. Reversing the coordinate formulas gives their right-end versions; no chamber symmetry or arithmeticity is assumed. Distinctness of the two opposites, when their mutual native gap is used, will follow from their actual contacts.

B.7.1 The reference under the full norm bound

The $N = 19$ central inventory and helper nonadjacency above precede any contrary energy assumption. Its near-four roles can also be excluded under only $\|x\| < 1$, using their sharper actual mates. For a left near-four at $i = 5, 6$, release $j = i - 1$. The positive replacement has diagram $B_{20-i} + A_{i-2}$ and cosine

$$C_j = \sqrt{2(i-1)/19}\big((20 - i)/2 + (19 - i)/\sqrt{2}\big) > 11.$$

Its normalized height exceeds $\sqrt{5/6} > .91$. At the other end a mate or a strong $13/14$ has demand $> 209/270 > .774$. Their correlation is at most $\sqrt{15/91} < .41$. Cauchy contradicts

$1.684^2 = 2.835856 > 2.82$. Thus all central weak sites are nonadjacent pure helpers already under the full norm bound.

Use the same $N = 19$ rational heights and chord omissions. If k is the number of weak end sites $5, 14$, one reference w has energy

$$E_0 \geq .9240352, \qquad E_1 > .8937, \qquad E_2 > .86414.$$

Its left and right block lengths m are 5 or 6, with signed means

$$b_5 = H_5/(5\sqrt{2}) \in (.29, .298), \qquad b_6 = H_6/(6\sqrt{2}) \in (.25, .266).$$

The separate boundary dual prices satisfy

$$a_5 \geq (.4208 - .151)/2 = .1349, \qquad a_6 \geq (451/1200 - .108)/2 > .1339.$$

Every omission is included and can only increase the adjacent available price.

For an actual endpoint pressure $\langle x, V\rangle \geq K$, subtract $B = \sum b_i\omega_i$, with $b_i \geq 0$ supported on the retained boundary duals, so that $Y = V - B$ has zero mean on each end block. Write $w = \sum a_i\omega_i$, with $a_i \geq 0$. Every retained height satisfies $\langle x, \omega_i\rangle \geq H_i = \langle w, \omega_i\rangle$, and $\langle w, Y\rangle = 0$. For $\mu \geq 0$, the individual conditions $\mu b_i \leq a_i$ give

$$\langle x, w + \mu Y\rangle = \sum_i (a_i - \mu b_i)\langle x, \omega_i\rangle + \mu\langle x, V\rangle \geq w^2 + \mu(K - \langle w, B\rangle).$$

Applying $\|x - (w + \mu Y)\|^2 \geq 0$ therefore gives the reference energy plus the gain

$$2\mu(K - \langle w, B\rangle) - \mu^2 Y^2.$$

Opposite-block residuals have zero cross. For a coordinate pressure $x_{\text{end}} \geq t$, the legally optimized reference gain is $m(t - b_m)^2/(m - 1)$. For all thresholds used below, the debit is less than .061, below each separate price. If either coordinate were $\geq .603$, the three energy bounds plus this gain would exceed 1; the tight case is

$$.86414 + (6/5)(.603 - .266)^2 = 1.0004228.$$

Therefore $x_1, -x_{19} < .603$. The genuine endpoint-edge inequality and $.603^2 < 90/247 = (18/19)(5/13)$ imply

$$C_L, C_R < 9/4. \tag{B.7.1}$$

This bounds actual edge cosines, not cap owns.

B.7.2 Complete local isolated-endpoint gains

An isolated left endpoint has $c = Ap - g\omega_1$, $A^2 = D_0 g^2 - 1 > 0$. Own zero, $3, 4$ has nonpositive time; own 6 and parallel have the positive actual nulls $\sqrt{3}c + 2n_1 + n_2$ and $c + n_1$. The full remaining list is finite 5, finite 7, finite at least 8, or dotted. For a pressure normalized to raw coordinate sum $\sqrt{2}$, subtract ω_m/m; the residual is zero-sum on the relevant block. Every price below is paid before applying its actual pressure. The resulting complete gain table is

own	μ	gain on $m = 5 >$	gain on $m = 6 >$
5	.6	.02904	.059
7	.4	.05152	.07677
finite ≥ 8	1/4	.0471	157/2400
dotted	1/4	.06	4699/60000.

We prove every row, including continuous dotted owns and the unbounded finite-label tail.

For dotted $g > 1$, put $u = g^{-2} \in (0, 1)$. The actual positive core $c + n_1/g$ gives

$$V = \omega_1 - un_1, \quad K(u) = \sqrt{D_0 - u} - \sqrt{u(1-u)}, \quad Y^2 = 2 - 2/m - 2u + u^2.$$

It has square $u - 1 < 0$, positive time, zero cap product and negative old-1 product; positivity checks every unknown wall. At $\mu = 1/4$ the gain is

$$G_m(u) = \tfrac{1}{2}(K(u) - H_m/m) - (2 - 2/m - 2u + u^2)/16.$$

Its shift is exactly $G_6 - G_5 = 1099/60000$. To prove $G_5 > .06$ uniformly, it suffices that

$$K(u) > .7408 - u/4 + u^2/8.$$

The upper bound $\sqrt{u(1-u)} \le 1/2 - (u - 1/2)^2$ follows by positive squaring, with residual $(u - 1/2)^4$. Use lower radical values $1.376, 1.1809, 1.0699, .9459$ at $u = 0, 1/2, 3/4, 1$; their squares are $1.893376, 1.39452481, 1.14468601, .89472681$, strictly below $D_0 - u$. Concavity places $\sqrt{D_0 - u}$ above the successive chords. The three resulting strict quadratic lower bounds are

$[0, 1/2]$	$.3852 - 1.1402u + (7/8)u^2$
$[1/2, 3/4]$	$(7/8)(u - 597/875)^2 + 8357/1750000$
$[3/4, 1]$	$.4511 - 1.246u + (7/8)u^2$.

The first decreases to $677/20000 > 0$; the last increases from $703/80000 > 0$. This proves the entire dotted row.

For finite $g < 1$, put $u = g^{-2}$, $d = D_0 - 1 = 17/19$. The actual finite projection onto c, n_1 gives

$$V = \omega_1 - n_1, \quad K_f = \frac{d - \sqrt{d(u-1)}}{\sqrt{D_0 - u}}, \quad Y^2 = 1 - 2/m.$$

Writing $s = \sqrt{u - 1}$, $K_f^2 = d(\sqrt{d} - s)/(\sqrt{d} + s)$, increasing with g. For every native own at least 8, $s \le \sqrt{2} - 1 < .414214$. Using $.8947 < d < .8948$, $\sqrt{d} > .9459$,

$$(d - .59^2)\sqrt{d} > .51702894, \qquad (d + .59^2)s < .5148265806.$$

Hence $K_f > .59$. Quarter-payment gives $G_{f,5} > .0471$ and $G_{f,6} > .0471 + 1099/60000 = 157/2400$.

For native 5, let $u = 6 - 2\sqrt{5}$ and $b = D_0 - 3/2 = 15/38$. Its actual H_4 correction $\eta = c + g(3n_1/2 + n_2 + n_3/2)$ has square $\delta = 1 - 3g^2/2 > 0$. Use the actual finite point $p + (A/\delta)\eta$, not η. The pressure is

$$V = \omega_4/4, \qquad K_5 = \frac{b - \sqrt{b(u - 3/2)}}{\sqrt{D_0 - u}}, \qquad Y^2 = 1/2 - 2/m.$$

For $s = \sqrt{u - 3/2} < .167$, $.3947 < b < .3948$, $\sqrt{b} > .6282$ give

$$(b - .475^2)\sqrt{b} > .106212915, \qquad (b + .475^2)s < .103610975.$$

Thus $K_5 > .475$. The payment $.6/m \le .12$ fits each boundary coefficient and gives

$$G_{5,5} > 1.2(.475 - .4208) - .36(.1) = .02904, \qquad G_{5,6} > 1.2(.475 - 451/1200) - .36/6 = .059.$$

For native 7, $.900968 < g < .900969$ and $1.231 < u = g^{-2} < 1.232$. The positive actual core $c + g^{-1}n_1 + 2(g^{-1} - g)n_2$, divided by g, gives

$$V = \omega_1 - un_1 - 2(u-1)n_2, \quad K_7 = \sqrt{D_0 - u} - \sqrt{7u - 3u^2 - 4},$$

$$Y^2 = 6 - 8u + 3u^2 - 2/m.$$

Its selected products are zero on c, n_1, negative on n_2, n_3, and nonpositive on every unknown wall; time is positive. On the stated interval, $D_0 - u > .814^2$ and $0 < 7u - 3u^2 - 4 < .070917 < .2664^2$, so $K_7 > .545$. The residual bounds are $Y_5^2 < .299$, $Y_6^2 < .366$. Payment $.4/m \le .08$ gives gains

$$.8(.545 - .4208) - .16(.299) = .05152, \qquad .8(.545 - 451/1200) - .16(.366) > .07677.$$

This proves the table and all its actual-wall and individual-price hypotheses.

B.7.3 The complete two-finite endpoint exclusion

Suppose both isolated endpoint owns are finite. For $D = 36/19, e = 2/19$, their actual mutual magnitude is

$$M(g,h) = \sqrt{Dg^2 - 1}\sqrt{Dh^2 - 1} - egh.$$

It increases strictly in each own for $g, h \ge g_5 = \cos(\pi/5)$. For example the derivative sign follows by squaring positive terms from $g/\sqrt{Dg^2 - 1} > 1/\sqrt{D}$ and $(D^2 - e^2)h^2 > D$, since $D^2 - e^2 = 68/19$ and $g_5^2 > 9/17$.

If either own is 5, monotonicity gives

$$0 < \frac{17\sqrt{5} - 25}{76} = M(g_5, g_5) \le M(g_5, h) < M(g_5, 1) < \sqrt{(9\sqrt{5} - 11)/38} < 1/2.$$

That is forbidden for distinct actual walls. Thus both own labels m, n are at least 7. Relabel so $m \le n$. Now $M > 0$ already at $m = n = 7$, while $M < M(1,1) = 15/19 < \cos(\pi/5)$. The full mutual alphabet therefore leaves exactly native 3 or native 4. Put

$$H(g,h) = 272g^2h^2 - 144(g^2 + h^2) - 8gh + 57,$$

$$F_S(g,h) = 136g^2h^2 - 72(g^2 + h^2) + 19 - 4Sgh.$$

They have the signs of $M - 1/2$ and $M - 1/\sqrt{2}$, respectively, with $S = \sqrt{2}$. This follows by the difference of squares against egh plus the target; the other factor is positive. Both increase in each argument on $[.9, 1]^2$, for $1.4 \le S \le 1.5$, and F_S decreases with S.

For mutual 3, $m \ge 8$ gives

$$M(g_8, g_8) = (17\sqrt{2} - 4)/38 > 1/2.$$

At $m = 7$ the exact rational native enclosures below give

$$M(g_7, g_8) < 1/2 < M(g_7, g_9).$$

At the upper 7/8 pair the bounds $u < .8117452, v < .8535543,\ u+v > 1.6652993, uv < .692869, gh > .832387$ yield $H < -10018272/10^7 < 0$. At the lower 7/9 pair, $u > .8117433, v > .8830210,\ u + v < 1.6947645,\ uv > .716786, gh < .846633$ yield $H > 1146640/10^6 > 0$. Thus no integer $n \ge m$ gives mutual 3.

For mutual 4, if $m \le 10$, increase g to the upper $g_{10}^{+} = .951057$ and h to 1. Since $(g_{10}^{+})^2 < .9045095$ and $4(1.414213)g_{10}^{+} > 5.37997$,

$$F_{\sqrt{2}}(g,h) \le F_{1.414213}(g_{10}^{+},1) < -.491362 < 0.$$

If $m \ge 15$, monotonicity gives

$$M \ge (34/19)\cos^2(\pi/15) - 1 > 1/\sqrt{2}.$$

Use $\cos(\pi/15) > 10783/11025$, from $\cos x > 1 - x^2/2, \pi < 22/7$, and $\sqrt{2} < 99/70$. The rational remaining margin is

$$(34/19)(10783/11025)^2 - 1 - 99/140 = 1498515515/323324662500 > 0.$$

Only $m = 11, 12, 13, 14$ remain. Their complete consecutive-integer gaps are

m	last n below target	first n above target
11	31	32
12	20	21
13	16	17
14	15	16.

Here is full rational arithmetic for these gaps. For $g = g_m, S = \sqrt{2}$, set

$$A = 136g^2 - 72, \quad B = 4Sg, \quad C = 64g^2 - 53 - B, \quad T = 2A - B.$$

At $h = 1 - \delta$, $F_S(g,h) = C - T\delta + A\delta^2$. The native enclosures give

m	g^2 lower	g^2 upper	B lower	B upper
11	.9206248	.9206269	5.42770	5.42773
12	.9330110	.9330131	5.46409	5.46412
13	.9427264	.9427284	5.49246	5.49249
14	.9504826	.9504847	5.51501	5.51504.

Thus the following looser bounds hold:

m	C_-	C_+	T_-	T_+	A_+
11	.49225	.49243	100.981	100.984	53.2053
12	1.24858	1.24875	104.314	104.317	54.8898
13	1.84199	1.84216	106.928	106.931	56.2111
14	2.31584	2.31602	109.016	109.018	57.2660.

For last-below n, use its upper cosine and $S = 1.414213$. The δ numerators over 10^6 are 5130, 12311, 19214, 21852; upper numerators of $A\delta^2$ are 1401, 8400, 20800, 27374, also over 10^6. The exact resulting upper numerators over 10^9 are

$$-24201530, \ -27059654, \ -191554592, \ -38823632.$$

For first-above n, use its lower cosine and $S = 1.414214$. The δ numerators are 4816, 11170, 17027, 19215; dropping the positive quadratic term gives lower numerators over 10^9

$$5911056, \ 83359110, \ 21275863, \ 221059130.$$

All signs are strict. Monotonicity in g, h proves every integer $n \ge m$ lies on one side of the stated gap, so mutual 4 is impossible as well. This proves the two-finite exclusion.

B.7.4 The finite rational angle certificate

For completeness, the only native enclosures used in the preceding finite calculation are:

j	$10^6 g_j^-$	$10^6 g_j^+$	j	$10^6 g_j^-$	$10^6 g_j^+$
7	900968	900969	8	923879	923880
9	939692	939693	10	951056	951057
11	959492	959493	12	965925	965926
13	970941	970942	14	974927	974928
15	978147	978148	16	980785	980786
17	982973	982974	20	987688	987689
21	988830	988831	31	994869	994870
32	995184	995185			

These are exact rational inequalities, not floating-point evaluations. Machin's identity $\pi = 16\arctan(1/5) - 4\arctan(1/239)$ follows by the tangent addition formula and its first-quadrant sign. For

$$A_s = \sum_{r=0}^{s} \frac{(-1)^r}{(2r+1)5^{2r+1}}, \qquad B_s = \sum_{r=0}^{s} \frac{(-1)^r}{(2r+1)239^{2r+1}},$$

alternating remainders and rational cross multiplication give

$$31415926/10^7 < 16A_5 - 4B_2 < \pi < 16A_6 - 4B_1 < 31415927/10^7.$$

For $0 < x < .45$, the alternating cosine Taylor sums give $C_{10}(x) < \cos x < C_8(x)$. Put $a_j = 10^7 j, p_- = 31415926, p_+ = 31415927$, and

$$Q_{10}(a,p) = 3628800a^{10} - 1814400p^2a^8 + 151200p^4a^6 - 5040p^6a^4 + 90p^8a^2 - p^{10}, \qquad Q_8 = Q_{10} + p^{10}.$$

For every displayed row with integers L_j, U_j, the finite exact certificate is

$$10^6 Q_{10}(a_j, p_+) - 3628800 L_j a_j^{10} > 0,$$

$$3628800 U_j a_j^{10} - 10^6 Q_8(a_j, p_-) > 0.$$

There are exactly thirty integer inequalities; their positivity is obtained by direct integer multiplication of the specified finite expressions. The alternating remainder signs and cosine monotonicity then prove all fifteen rows. This is a terminating exact verification recipe with a complete stated output; no execution of a checker is claimed by this manuscript transcription. The historical unused cyclotomic fallback is not an input. Finally $1.414213 < \sqrt{2} < 1.414214$ follows by squaring those two rationals.

B.7.5 The complete both-isolated endpoint conclusion

The local isolated gains, with their separate prices and disjoint residual supports, exclude two dotted endpoints: the three reference-plus-gain lower bounds are

$$.9240352 + 2(.06) > 1, \quad .8937 + .06 + 4699/60000 > 1, \quad .86414 + 2(4699/60000) > 1.$$

For one finite and one dotted endpoint, the full four-pattern totals are

finite own	5/5	6/5	5/6	6/6
≥ 8	1.0311352	> 1.01911666	> 1.01911666	> 1.00787333
5	1.0130752	1.0127	> 1.00105666	> 1.00145666
7	1.0355552	> 1.03047	> 1.02353666	$> 1.01922666.$

In the middle two columns the block order is finite/dotted. The gains add within one square because their residuals occupy opposite zero-sum blocks. Together with the two-finite theorem, these exhaust every isolated endpoint own. Thus both actual endpoint opposites cannot be isolated.

B.7.6 The complete actual endpoint inventory

We now classify an endpoint opposite before using its own contact. Release n_1. The other endpoint is finite, so the actual Gram on $c, n_2, \dots, n_{18}$ is positive definite. The retained diagram is the connected A_{17} path. A one-node finite enlargement is either disconnected or of type A, B, D: exceptional spherical types have rank less than eighteen. Thus the new node is isolated, or has one simple bond to an end or a next-to-end node, or one bond of label four to an end. Two white contacts would create a cycle and cannot occur in a finite diagram. This gives the six nonisolated placements below, without imposing a condition on an arbitrary face.

Write a nonisolated opposite as

$$c = Ap - g\omega_1 - \eta\omega_j, \qquad g = -\langle c, n_1\rangle \geq 0,$$

where $\eta = 1/2$ for a simple white bond and $\eta = 1/\sqrt{2}$ for a white bond of label four. Put $C_{1j} = \langle \omega_1, \omega_j\rangle$ and $D_j = \omega_j^2$. The actual norm and retained projection give

$$A^2 = D_0 g^2 + 2g\eta C_{1j} + \eta^2 D_j - 1, \qquad \delta_{\text{white}} = 1 - \frac{2\eta^2(j-1)(19-j)}{18} > 0.$$

The edge cosine is

$$C = \frac{D_0 g + \eta C_{1j}}{\sqrt{D_0 \delta_{\text{white}}}}.$$

For a simple end bond the retained projection cost is 17/36 and the residual is 19/36; they must not be interchanged. The next-to-end and label-four residuals are 1/9 and 1/18. Substituting in the displayed formula and using (B.7.1) yields

white contact	C	consequence
$j = 2,\ 3$	$(36g + 17)/19$	$g < 103/144:$ own $0, 3, 4$
$j = 3,\ 3$	$(36g + 16)/(2\sqrt{19})$	$g < .101:$ own 0
$j = 17,\ 3$	$(36g + 2)/(2\sqrt{19})$	$g < .49:$ own 0
$j = 18,\ 3$	$(36g + 1)/19$	$g < 167/144 < 7/6$
$j = 2,\ 4$	$(36g + 17\sqrt{2})/\sqrt{38}$	$C \geq 17/\sqrt{19} > 9/4$
$j = 18,\ 4$	$(36g + \sqrt{2})/\sqrt{38}$	$g < .35:$ own $0.$

The numerical cuts follow by positive squaring from $\sqrt{19} < 4.36, \sqrt{38} < 6.165, \sqrt{2} > 1.414$. A positive own is at least 1/2; finite owns are $\cos(\pi/m)$, while parallel and dotted owns are respectively one and greater than one.

For near simple-three with own zero, $A^2 = -2/19$; own three has the actual positive null sum $c + n_1 + n_2$. For the next-to-near simple-three with own zero, the actual affine E_8 principal has center n_3, arms

$$(c), \qquad (n_2, n_1), \qquad (n_4, n_5, n_6, n_7, n_8),$$

and positive kernel coefficients

$$6; \qquad 3; \qquad 4, 2; \qquad 5, 4, 3, 2, 1.$$

Each row of its unit Gram annihilates this vector. For the next-to-far simple-three and far label-four with own zero, $A^2 = -2/19$ and $-1/19$, respectively.

For far simple-three, own zero and own three have $A^2 = -10/19$ and zero. Own four has the actual affine path

$$(n_{17}, n_{18}, c, n_1, n_2), \qquad \text{bonds } (3, 3, 4, 3),$$

with positive kernel $(1, 2, 3, 2\sqrt{2}, \sqrt{2})$. Own six has the null sum $\sqrt{3}c + 2n_1 + n_2$; parallel has $c + n_1$. The same nulls exclude isolated own six and parallel, while isolated owns zero, three and four have negative time square.

Every positive null combination just used contains its actual opposite with a positive coefficient. Its time is consequently positive, so it is a nonzero future null vector. Against any unselected original wall its product is nonpositive by acuteness; against its selected walls the kernel equations give zero. It therefore determines an ideal point of the whole chamber, contradicting compactness. This checks all original walls, including those absent from the selected diagram.

The complete surviving list at either end is therefore:

near:	simple-three at the neighboring node, own four;
far:	simple-three at the opposite end, own five, finite own at least seven, or dotted;
isolated:	own five, own seven, finite own at least eight, or dotted.

In every far case $g < 167/144 < 7/6$. The two copies of this classification concern the actual endpoint walls of the original vertex.

B.7.7 Removing the near endpoint and the two-far possibility

The remaining near role has

$$C = (18\sqrt{2} + 17)/19, \qquad x_{\text{end}}^2 \geq \frac{18}{19}\frac{18\sqrt{2} - 2}{18\sqrt{2} + 36} > .601^2.$$

After clearing positive denominators, the last inequality is $200.469258\sqrt{2} - 283.061484 > 0$; its lower bound at $\sqrt{2} > 1.4142$ is .4421406636. Thus $x_{\text{end}} > .601$. A strong end gives coordinate gain $(5/4)(.601 - .298)^2 = .11476125$; a weak end gives gain greater than $(6/5)(.601 - .266)^2 = .13467$. Either removes the zero-weak or one-weak reference pattern: even the smaller one-weak total is $.8937 + .11476125 > 1$.

Only two weak ends could remain. A far opposite at the other end has $g \geq 1/\sqrt{2}$, a weaker bound than its actual surviving own list. Its edge inequality gives $x_{\text{other}} > .394$. The cleared comparison is

$$270.909288\sqrt{2} - 382.98968 > 0;$$

at 1.4142 the margin is .1302350896. Its weak-block gain is therefore greater than $(6/5)(.394 - .266)^2 = .0196608$. An isolated opposite gives weak gain greater than .059 by the complete local table; another near opposite gives the larger near gain. These opposite residuals have disjoint support, zero cross, and separately paid nonnegative coefficients. The weakest combined total is

$$.86414 + .13467 + .0196608 = 1.0184708 > 1.$$

Thus no near role survives, on either orientation.

Suppose next that both endpoints are far, and write

$$c = Ap - g\omega_1 - \omega_{18}/2, \qquad d = Bp - h\omega_{18} - \omega_1/2.$$

Their time squares are

$$A^2 = (36g^2 + 2g - 10)/19, \qquad B^2 = (36h^2 + 2h - 10)/19.$$

For $0 < g < 7/6$,

$$\frac{8}{5}g^2 - A^2 = \frac{50 - 10g - 28g^2}{95} > \frac{2}{855} > 0,$$

because the numerator decreases up to $7/6$. The same holds for h. Their spatial product satisfies

$$\frac{2gh + 18g + 18h + 1/2}{19} > \frac{18(g+h)}{19} > \frac{216gh}{133} > \frac{8}{5}gh > AB.$$

Here $1/g + 1/h > 12/7$ and $1080 > 1064$. The two caps are distinct: coincidence of their old contact rows would force $g = h = 1/2$, for which both time squares vanish. Their positive mutual product contradicts actual acuteness.

The preceding both-isolated theorem and the endpoint inventory now leave exactly one far endpoint and one isolated endpoint. Reflect the coordinate formulas if necessary so the far cap is on the left. An isolated own five is impossible for every far own. Indeed, with $u_5 = 6 - 2\sqrt{5}$,

$$D_0 - u_5 = 2\sqrt{5} - 78/19 < 178/475 < 3/8, \qquad A^2 < \frac{8}{5}(7/6)^2 = 98/45.$$

Thus $A^2(D_0 - u_5) < 49/60 < (18/19)^2$. For an isolated right cap of own σ, its time is $\sigma\sqrt{D_0 - u_5}$, whereas the spatial product divided by σ is $(18 + 2g)/19 > 18/19$. The actual cap product would be positive. The supports show these caps are distinct, so this too contradicts acuteness.

B.7.8 Uniform removal of the far finite and dotted tail

A far own at least seven or a dotted own has $g > .9$. Its actual length consequently gives

$$C > (167/95), \qquad x_{\text{far}}^2 > 648/2489 > .510^2.$$

The last positive rational margin is $648 - 2489(.2601) = .6111$. The legal coordinate gains on the strong and weak blocks are

$$G_{\text{far},5} > .05618, \qquad G_{\text{far},6} > .0714432.$$

The other endpoint is isolated with own seven, finite own at least eight, or dotted. The minimum over the entire local table is

$$G_{\text{iso},5} > .0471, \qquad G_{\text{iso},6} > 157/2400.$$

In particular the weak minimum must not be replaced by the larger own-seven estimate. The two individually funded residuals belong to opposite blocks, so their gains add in one square. All four possibilities give

$(m_{\text{far}}, m_{\text{iso}})$	strict lower bound for $\|x\|^2$
$(5,5)$	$.9240352 + .05618 + .0471 = 1.0273152$
$(6,5)$	$.8937 + .0714432 + .0471 = 1.0122432$
$(5,6)$	$.8937 + .05618 + 157/2400 > 1.01529666$
$(6,6)$	$.86414 + .0714432 + 157/2400 > 1.00099986.$

This removes all these far owns, including arbitrary finite labels and the continuous dotted interval.

B.7.9 The last mutual gap and an actual orthogonal projection

Only far own five remains. Set

$$\phi = (1+\sqrt{5})/2, \quad g = \phi/2,$$
$$A^2 = (10\phi - 1)/19, \quad L = (18+\phi)/19,$$
$$\delta = 1 - 3g^2/2 = (5 - 3\phi)/8 > 0.$$

Substitution in $t^2 - t - 1$ gives $1.618 < \phi < 1.619$. Write the isolated right cap as $d = A_I p - \sigma\omega_{18}$, with $u = \sigma^{-2}$. If it were finite, $u > 1$ and

$$A^2(D_0 - u) < .8(.9) = .72 < 1 < L^2.$$

Its mutual product with the far cap would be positive. Thus $\sigma > 1$ is dotted and $0 < u < 1$. The actual isolated edge has $C_{\text{iso}} = \sqrt{D_0}\sigma < 9/4$, so $u > 64/171$.

The two caps are distinct because the far cap has a positive old contact at n_1 and the isolated right cap has none. Their actual mutual product is

$$\langle c, d\rangle = \frac{L - A\sqrt{D_0 - u}}{\sqrt{u}} \le 0.$$

Since $A < .9, L > 1, D_0 - u < 260/171 < 25/16$, and $\sqrt{u} > 3/5$, this product is strictly greater than $-5/24$. The native alphabet for distinct actual walls has no value in $(-1/2, 0)$. Therefore

$$\langle c, d\rangle = 0, \qquad AA_I = \sigma L, \qquad u = D_0 - \frac{L^2}{A^2} = \frac{323\phi - 361}{19(10\phi - 1)}. \tag{B.7.2}$$

Orthogonality is proved here before it is used. The rational function has derivative with positive numerator 62453. At $1.618, 1.619$ its values are $80807/144210$ and $161937/288610$. The positive comparisons

$$25(80807) - 14(144210) = 1235, \qquad 281(288610) - 500(161937) = 130910$$

give

$$.560 < u < .562. \tag{B.7.3}$$

These are geometric identities; no integrality or field restriction is being inferred from them.

Consider the positive actual correction

$$\eta = c + g(3n_1/2 + n_2 + n_3/2) = Ap - g\omega_4/4 - \omega_{18}/2.$$

Direct products give

$$\langle \eta, n_1\rangle = \langle \eta, n_2\rangle = \langle \eta, n_3\rangle = 0, \qquad \eta^2 = \langle \eta, c\rangle = \delta > 0.$$

Thus η is spacelike. The actual H_4 block c, n_1, n_2, n_3 is positive definite, its Schur complement being δ. Equation (B.7.2) and the right cap's isolation make d orthogonal to this block.

The required chamber point is the projection

$$q_0 = p + (A/\delta)\eta + A_I d, \qquad H = A^2/\delta + A_I^2, \qquad q_0^2 = \langle p, q_0\rangle = -1 - H.$$

It has zero product with every selected wall. Every correction coefficient on an actual normal is positive; against every unselected original wall, acuteness and $\langle p, a\rangle \le 0$ therefore give $\langle q_0, a\rangle \le 0$.

This includes n_{18} and all unknown facets. Also $\langle p, q_0\rangle < 0$, so q_0 is future timelike. Its normalization is a point of the whole chamber. The spacelike vector η itself was never used as a chamber point.

Put

$$a = \phi^{-1} + \frac{\delta L}{A^2 g u} > 0, \qquad V = \omega_4/4 + a\omega_{18}, \qquad U = A^2 + \delta D_0/u.$$

Using $AA_I = \sigma L$, the point has the full decomposition

$$q_0 = (1+H)p - (Ag/\delta)V, \qquad \delta(1+H) = U.$$

Compare its normalization with the same original farthest vertex:

$$-\langle z, q_0/\sqrt{1+H}\rangle \le -\langle z, p\rangle = E.$$

Substitution of $z = E(p-x)$ gives the actual pressure

$$\langle x, V\rangle \ge K, \qquad K = \frac{\delta(1+H-\sqrt{1+H})}{Ag} = \frac{U-\sqrt{\delta U}}{Ag}. \tag{B.7.4}$$

No auxiliary point has been declared farthest.

All subsequent estimates have positive denominators. The enclosure of ϕ and (B.7.3) give

$$.7989 < A^2 < .7995, \qquad .0178 < \delta < .0183,$$
$$.809 < g < .8095, \quad A < .895, \quad L < 1.033.$$

For the lower estimate retain $D_0 = 36/19$ exactly:

$$\frac{\delta D_0}{u} > \frac{.0178(36/19)}{.562} = \frac{3204}{53390} > .06,$$

where $50(3204) - 3(53390) = 30 > 0$. The upper bound is $\delta D_0/u < .0625$, by positive cross multiplication. Consequently

$$.8589 < U < .862, \quad \sqrt{\delta U} < .126, \quad Ag < .7245025, \quad K > \frac{.7329}{.7245025} > 1.01.$$

For the radical comparison, $.0183(.862) = .0157746 < .126^2$, and $1.01(.7245025) = .731747525 < .7329$. Finally,

$$\delta L < .0189039 < .06(.7989)(.809)(.560) = .02171601936.$$

Since $\phi^{-1} < .619$, this proves $0 < a < .679 < .68$.

B.7.10 One final funded vector for every end pattern

For the pressure (B.7.4), subtract

$$B = \omega_{m_L}/m_L + a\omega_{19-m_R}/m_R, \qquad Y = V - B.$$

The left residual is $\omega_4/4 - \omega_{m_L}/m_L$; the right is a times the reflected $\omega_1 - \omega_{m_R}/m_R$. They are supported on the first m_L and last m_R coordinates, have zero sum on each block, and have zero cross. Their full squared norm is exactly

$$Y^2 = \frac{1}{2} - \frac{2}{m_L} + a^2\left(2 - \frac{2}{m_R}\right).$$

In particular the nonzero cross between the original endpoint duals has disappeared only after the proper block subtractions.

Use the single nonnegative weight $\mu = .4$. It costs $.4/m_L \le .08$ at the left anchor and $.4a/m_R < .0544$ at the right, each below its own available coefficient .1349 or $> .1339$. The all-hole reference and the individual-price hypotheses of the completed-square rule are therefore satisfied simultaneously. Write $\ell_5 = H_5/5 = .4208$ and $\ell_6 = H_6/6 = 451/1200$. The gain is strictly greater than

$$.8(1.01 - \ell_{m_L} - .68\ell_{m_R}) - .16\left[\frac{1}{2} - \frac{2}{m_L} + .68^2\left(2 - \frac{2}{m_R}\right)\right].$$

Replacing a by its upper bound lowers the linear deficit and increases the norm penalty. The four rational budgets are

(m_L, m_R)	gain lower bound	strict lower bound for $\|x\|^2$
$(5,5)$	.1080704	1.0321056
$(6,5)$	$> .13337706$	> 1.02707706
$(5,6)$	.1276	> 1.0213
$(6,6)$	$> .15290666$	$> 1.01704666.$

For example the last bracket equals 703/750 and its linear deficit is .3786; the unrounded gain is $.8(.3786) - .16(703/750)$. Every pattern contradicts $\|x\|^2 < 1$. One original vector, one reference and one set of individually available coefficients were used throughout.

Proposition B.7.1. *An actual original farthest vertex of a compact hyperbolic Coxeter polytope in dimension eighteen cannot have entire incident diagram A_{18}.*

Proof. The finite-white inventory leaves only near, far or isolated endpoint opposites. The near role is impossible on either side. Both isolated endpoints are excluded by their complete local and finite-pair arguments; both far endpoints violate acuteness. Thus precisely one endpoint is far and the other isolated. The uniform coordinate/local-gain argument removes every far own at least seven and every dotted far own. In the remaining far-own-five case, every finite isolated opposite violates acuteness. The dotted isolated opposite has mutual product zero by the actual native gap, and the actual $H_4 + A_1$ point gives the final simultaneous-payment contradiction above. These alternatives exhaust the actual endpoint walls, without arithmeticity, a finite-label cutoff, arbitrary-face heredity, or a rank decrement. □

B.8 Terminal walls in a whole B_{18} diagram

Fix the original strict interior unit point z and an original farthest vertex p, and assume that the entire incident diagram is B_{18}. Normalize the actual old normals by

$$n_i = (e_i - e_{i+1})/\sqrt{2} \quad (1 \le i \le 17), \qquad n_{18} = e_{18}.$$

Write

$$p^2 = -1, \qquad z = E(p - x), \qquad E > 0, \quad x \perp p, \quad x_1 > \cdots > x_{18} > 0, \quad \|x\|^2 < 1.$$

For $P_i = e_1 + \cdots + e_i$, put

$$S_i = \langle x, P_i \rangle, \qquad f_i = \sqrt{i} - .58/\sqrt{2}, \qquad d_K = \sqrt{K} - \sqrt{K-1}.$$

The actual duals are $\omega_i = \sqrt{2}P_i$ for $i \le 17$ and $\omega_{18} = P_{18}$. In particular $\langle \omega_i, \omega_j \rangle = 2\min(i,j)$ for $i,j \le 17$. An index is strong when $S_i > f_i$; equality belongs to the weak case. There are no old spectator directions. All subsequent references in the original frame concern this one vector x.

We shall prove that these hypotheses are impossible. The proof first constructs unconditional terminal sum bounds, then classifies the actual seventeenth opposite. Its far-helper, near-six and short-contact branches are treated by shared actual-point and neighboring-comparison arguments. No bound on an arithmetic field occurs.

We use the actual positive projection and future-point pressure lemmas already proved in the main text. Their hypotheses will be checked for each point below. A positive correction on actual old normals leaves every unknown wall inequality nonpositive by acuteness. A spacelike corrected normal is used only inside its positive finite projection; it is never normalized as a chamber point.

B.8.1 The early window and actual head pressures

Deleting i, $5 \le i \le 14$, retains $A_{i-1} + B_{18-i}$, with both ranks at least four. The finite one-node list consists of isolation; one left end bond of label three or four; a left D or permitted E position; one right long-end simple bond; or an end-to-end bridge. Two contacts on one retained component create a cycle. A joined rank-eighteen diagram is classical; exceptional types add no case. The right B_3 at $i = 15$ additionally allows its short-end simple bond, and the right B_2 at $i = 16$ allows either end. These last roles will be checked separately.

For an actual edge with $C = \cosh L = qR$, $q = \sqrt{2i}$, clearing the positive denominators of its directional pressure shows that

$$R > T(q) := \frac{50}{29} - \frac{q - 29/50}{q(2q - 29/50)} \quad \Longrightarrow \quad S_i > f_i.$$

The function increases for $q \ge \sqrt{10}$, and substitution gives

$$T(q) < 83/50 \ (q < 7), \qquad T(q) < 801/500 \ (q < 15/4), \qquad T(q) < 159/100 \ (q < 16/5).$$

The finite-white Schur calculations have the following complete non-helper bounds; $\phi = (1 + \sqrt{5})/2$.

role	lower bound for R
left near three, own at least five	$[i(\phi + 1) - 1]/\sqrt{2i(i+1)}$
D position	$(i\phi/2 + 1)/\sqrt{2i}$
E position	$2\phi/2 + 3/8$
right long three, positive remaining own	$1 + 1/\sqrt{2}$
right B_3 short three, positive own	$1 + 1/\sqrt{2}$
end-to-end bridge	$(i+1)/\sqrt{2i}$.

The near-three bound increases with i. At $i = 6$ it is $(6\phi + 5)/\sqrt{84} > 801/500$, which covers $6, 7$; at $i = 8$ it exceeds $83/50$, covering larger i. At $i = 5$, every own at least seven gives $R > 5/3$. For the D row, $i = 5$ gives $R > 159/100$, and $i \ge 6$ gives $R \ge 2927/1750 > 83/50$. An E position has $i \le 8$ and retained residual at most $1/4$. The bridge residual is $1/(2i)$; its numerator is at least $(i+1)/(2i)$, proving the displayed lower bound, including own zero. Left label-four roles have the affine small-own alternative or the larger direct edge bounds: near own three has $C = (i-1) + i/\sqrt{2}$. The positive-null and impossible-time boundaries in the common prefix inventory apply to the actual retained blocks at these ranks. For the right long-three own-zero boundary one can also see the kernel directly: use coefficient one on the cap and old i, two on old $i+1, \dots, 17$, and $\sqrt{2}$ on old short 18. Every selected product vanishes.

An isolated finite cap has the actual finite/core pressures used in the earlier component lemmas. At own five its loss is less than $25/76 + 5/24 < .58$; own seven has loss $1000/4779 + 9/25 < .57$; the remaining finite interval has its largest loss at $i = 5$, own eight: $\sqrt{10} + \sqrt{2} - 4 < .578$. For an isolated dotted cap put $u = g^{-2}$. Its positive reciprocal core gives

$$\langle x, \omega_i \rangle > \sqrt{2i - u} - \sqrt{u(1-u)}.$$

For $i \geq 6$ the loss is at most

$$a_i u + \sqrt{u(1-u)}, \qquad a_i = \frac{1}{\sqrt{2i} + \sqrt{2i-1}} < 37/250,$$

whose maximum is less than .5795, since $\sqrt{1 + (37/250)^2} < 1011/1000$. Thus only an isolated dotted cap at 5 may remain weak.

There is one early near-three exception, own five at 5. For $g = \phi/2$, the actual cap has displacement $v = g\omega_5 + \omega_4/2$. On the right actual H_4, correct by

$$w = g(3n_5/2 + n_6 + n_7/2), \qquad \delta = (5 - 3\phi)/8 > 0.$$

The corrected normal is spacelike and its finite projection is actual; the extra old 4 product is negative, and all unknown wall signs follow by positivity. Its pure spatial displacement is

$$Y = (1/2 + 3g/4)\omega_4 + (g/4)\omega_8, \qquad A^2 = (7 + 13\phi)/2, \quad Y^2 = (33 + 49\phi)/8.$$

The finite-point pressure gives

$$\langle x, Y \rangle > 18/5, \qquad Y^2 < 281/20, \qquad \sum Y_j < 43/5.$$

Independently, the earlier rank-valid .61 prefix lemma supplies a strong $K = 11$ or 12. Its old floors exceed 72/25 or 303/100. The vector $Y + (3/4)P_K$ therefore has squared demand exceeding squared norm by .0401 or .78625625, respectively, a contradiction. This use of the earlier lemma concerns $B_{18} \geq B_{16}$ and precedes any new .58 or strong-five conclusion.

The two extra B_2 roles at 16 deserve separate treatment. For short-three/own-five put $g = \phi/2$ and $\alpha = \sqrt{2}g + 1/2$. The left actual H_4, on the cap and old $16, 15, 14$, has residual $(5 - 3\phi)/8$, time square $16\alpha^2 - 1/2$, and normalized spatial displacement constant through 13, of norm less than 4. Since $\alpha > 1.643$, its pressure exceeds $3.976 - .083 = 3.893 > 3.89$. The earlier, independently available strong $K = 11, 12$ references have energies $> .8472225, > .8689525$, sums $< 2.886, < 3.034$, and terminal capacities $> 1/7$. Their roots and actual points lie through 14. Paying $1/7$ at that terminal anchor gives

$$.8472225 + 2(3.89 - 2.886)/7 - 5/49 > 1.0320, \qquad .8689525 + 2(3.89 - 3.034)/7 - 4/49 > 1.0318.$$

For short-three/own-three the same cap has displacement $v = (\omega_{16} + \omega_{18})/2$. Deleting actual short 18 instead gives a positive full D_{18} projection. Its sum and time square are $9 + 8\sqrt{2}$ and $23/2 + 8\sqrt{2}$; the squared sum minus eighteen times the time square is 2. Thus its actual short-edge cosine is $8 + 9\sqrt{2}/2 > 14$, whence $S_{18} > 3.949$. The unspent early $K = 14, 15$ references, of effective energies $> .9231, > .94006$, have terminal capacities $> 1/9$. Paying $1/9$ toward P_{18} gives respectively

$$.9231 + 2(3.949 - 3.332)/9 - 4/81 > 1.0108, \quad .94006 + 2(3.949 - 3.463)/9 - 3/81 > 1.0110.$$

The H_4 construction is a positive finite projection; the second point is the positive projection onto the displayed full finite Gram. All original unselected wall products are therefore nonpositive.

Neither new point is asserted farthest. These arguments use the early references before the new B_2 conclusion, so they introduce no circularity.

Consequently every weak ordinary site is a pure far helper

$$c_i = A_i p - g\sqrt{2}P_i - e_1/\sqrt{2}, \qquad A_i^2 = 2ig^2 + 2g - \tfrac{1}{2}, \qquad \phi/2 \le g < 6/5,$$

apart from the isolated dotted 5. The upper cutoff follows from $R^2 \ge 4g - 2 \ge 14/5 > (83/50)^2$ at $g \ge 6/5$.

For adjacent pure helpers the exact time/dual calculation gives

$$\langle c_i, c_{i+1}\rangle \ge \frac{1}{2} - g_i g_{i+1} + \frac{1}{4}(g_i/g_{i+1} + g_{i+1}/g_i) \ge 1 - g_i g_{i+1} > -11/25.$$

Their actual native gap forces orthogonality. The positive two-cap replacement, with every other old wall retained, has positive inverse Schur matrix; its weighted loss is less than $7/16 + 1/(4\sqrt{10}) < .58$. Thus both sites cannot be weak. This uses their actual new parameter range, rather than inclusion in an older weak class. In particular some $K = 14$ or 15 is strong.

The same actual helper feet give $x_1 > .5455$ at a weak helper 5. For an isolated dotted weak 5, its reciprocal core gives

$$U' = P_5 + (u/2)(e_6 - e_5), \qquad \langle x, U'\rangle > T' > 1.8185, \qquad .4 < u < 1.$$

Indeed the loss is bounded by $au + \sqrt{u(1-u)}$, $a = \sqrt{10} - 3 < 13/80$; for $u \le .4$ it is less than $.565 < .58$, and globally it is less than .588. The pressure bound follows from $\sqrt{10} > 3.162$, $\sqrt{2} < 99/70$. These are actual point pressures, not assigned floors at weak nodes.

B.8.2 Raw references, head isolation and the uniform paid prefix

A reference w has support through a strong K, sum f_K, actual pressure $\langle x, w\rangle > A_w$, and effective energy $L = 2A_w - w^2$. Point insertion need not leave $L = w^2$. For a new actual pressure $\langle x, V\rangle \ge G$, weight $\mu \ge 0$, and $Y = V - P_K$, the completed-square rule is

$$\|x\|^2 > L + 2\mu(G - f_K - \langle w, Y\rangle) - \mu^2 Y^2, \qquad \mu \le \beta_K.$$

The inequality is strict when the old strong reference retains a positive coefficient. Every use below checks its individual prices.

Let $d_i = \sqrt{i+1} - \sqrt{i}$. The first-five energy exceeds .6668157, and rational lower numerators for $10^7 d_i^2$, $i = 5, \dots, 16$, are

$$455488, 385186, 333704, 294372, 263340, 238230, 217494, 200080, 185248, 172465, 161332, 151549.$$

For a direct verification, put $a = 4i + 2$ and use $d_i^2(a - d_i^2) = 1$, whence $d_i^2 > (a^2 - 1)/(a(a^2 - 2))$. Thus the unomitted energies satisfy

$$E_{14} > .9241299, \quad E_{15} > .9413764, \quad E_{16} > .9575096, \quad E_{17} > .9726645.$$

A missing knot $j \ge 6$ loses less than $1/[32(j-1)^3]$; nonadjacency makes these losses disjoint. Their total is less than $(1/32)\int_4^\infty t^{-3}\,dt = 1/1024$. Each terminal capacity is at least d_K.

Before any head payment, the three initial-five cases give the following full contracts:

case at five	$L_{14} >$	$L_{15} >$	$w_1 <$
strong	.9231	.9403998	.3652
pure helper	.95255	.96895	.5455
isolated dotted weak	.9291	.9455	.377.

In the strong case interpolate through all actual strong floors, skipping every weak knot. The first mean is $f_5/5 < .3652$. For a helper, interpolate through $(1, .5455), (6, f_6), (K, f_K)$. Its first-six energy exceeds $.5455^2 + (f_6 - .5455)^2/5 > .74385$; the tails to $14, 15$ exceed $.2087, .2251$. All slopes are positive decreasing.

For isolated dotted weak 5, choose strong $J = 6$ or 7 and a reference constant first J, then straight to K. Insert $\mu[U' - (5/J)P_J]$, with $\mu = .21, .2$. The residual squares are bounded by

$$5 - u + u^2/2 - 25/J < 77/150,\ 194/175$$

and the deficits exceed $.119, .221$. Its individual P_J prices are $.175, 1/7$, below the capacities $> .1783, > .1628$ already at $K = 14$. The gains exceed $.027342, .044057$. The bare first-J energies exceed $.69312, .7139$; the tails to 14 exceed $.2087, .1715$ and to 15 exceed $.2251, .18825$. The full first coordinates are less than

$$.3399 + .21(1 - 5/6) = .3749, \qquad .31938 + .2(1 - 5/7) < .377.$$

This uses the straight reference for these prices. A curved reference with a smaller P_6 capacity is not substituted.

The actual long-head opposite is isolated. Deleting old 1 retains B_{17}; its finite enlargement is isolation or a simple bond to its long endpoint 2. In the latter case

$$v = g\omega_1 + \omega_2/2, \qquad A^2 = 2g^2 + 2g, \qquad C_1 = 2g + 1.$$

Own zero has zero time; own three is an actual affine triangle. Every other own gives $x_1 > \sqrt{\sqrt{2} - 1} > .643$. If 5 is strong, replace the first-five mean $< .3652$ by this pressure, gaining $(5/4)(.643 - .3652)^2 > .0964$ over $L_{14} > .9231$. If 5 is weak, choose strong $J = 6, 7$ and interpolate through $1, J, K$; the first energies exceed $.8034, .8361$ and the tails exceed $.2087, .1715$. Each case exceeds one, including isolated dotted weak 5. Therefore a head has time $A_H^2 = 2g_H^2 - 1$. Owns zero, three, four have impossible time; six and parallel have the positive actual $6/3$ and parallel nulls. Only own five, finite own at least seven, and dotted own remain.

For later reference, every non-five head has a point pressure $\langle x, Z\rangle \geq h$, with $\sum Z_j = 1$ and the following full local data:

own	Z	h	$R = Z^2$
finite ≥ 7	$(1 - u/2)(e_1 + e_2) + (u - 1)e_3$	$[\sqrt{2 - u} - \sqrt{7u - 4 - 3u^2}]/\sqrt{2}$	$3 - 4u + 3u^2/2$
dotted	$(1 - u/2)e_1 + (u/2)e_2$	$[\sqrt{2 - u} - \sqrt{u(1 - u)}]/\sqrt{2}$	$1 - u + u^2/2.$

Here $u = g_H^{-2}$; the first row has $1 < u \leq \cos(\pi/7)^{-2}$, the second $0 < u < 1$. Their actual positive cores are $c_H + g_H^{-1}n_1 + 2(g_H^{-1} - g_H)n_2$ and $c_H + g_H^{-1}n_1$, respectively. Their cap product is zero, the stated old-root products are nonpositive and all unknown signs follow termwise. They are future, giving these pure spatial pressures. Both obey

$$.8h - .16R > .2893. \tag{B.8.1}$$

For native seven, $.9009 < g_H < .901$ gives $h > .4315, R < .349$, and the left side exceeds $.28936$. For finite labels at least eight, $1 < u < 1.172$. Writing $D = (u - 1)(4 - 3u)$, the derivative is

$$\frac{.8}{\sqrt{2}}\left(-\frac{1}{2\sqrt{2 - u}} - \frac{7 - 6u}{2\sqrt{D}}\right) + .64 - .48u.$$

It is negative on $(1, 7/6]$; on $[7/6, 1.172)$, $\sqrt{D} > .284$, its positive term is $< .032$ and its other terms total less than $-.20$. At own eight $h > .4388, R < .373$, giving $> .29136$. For dotted u, the

two chords $\sqrt{2-u} \geq 1.4142 - .379u$ on $[0, 1/2]$ and $\sqrt{2-u} \geq 1.4494 - .4494u$ on $[1/2, 1]$ give, respectively,

$$.8h - .16R > .63987152 - .0944u - .5657\sqrt{u(1-u)} > .30587152,$$

$$.8h - .16R > .69978064 - .2142u - .5657\sqrt{u(1-u)} > .29023064.$$

Use $bu + c\sqrt{u(1-u)} \leq (b + \sqrt{b^2 + c^2})/2$, with radical bounds $.5736, .6049$. This proves the entire finite and unbounded dotted head seal.

B.8.3 One uniform paid prefix, without a terminal premise

If the head has own five, its actual unit normal is

$$c_H = A_H p - (\phi/\sqrt{2})e_1, \qquad A_H^2 = (\phi - 1)/2.$$

A weak pure helper at $i = 5, 6, 7$ would have mutual magnitude

$$M_i(g) = A_H\sqrt{2ig^2 + 2g - \tfrac{1}{2}} - \frac{\phi}{2}(2g + 1).$$

This increases in g, i on the allowed range and $M_7(6/5) < 0$, since the positive squared comparison has difference $(814\phi - 1392)/100 < 0$. An isolated dotted weak 5 has own $h < 2$, by its edge pressure and $f_5^2 < 7/2$. Its increasing mutual magnitude

$$A_H\sqrt{10h^2 - 1} - \phi h$$

lies strictly between zero and $1/4$ for $1 < h < 2$: the endpoint squared comparisons are $(7\phi - 11)/2 > 0$ and $377/16 - (29/2)\phi > 0$. Both cases contradict actual acuteness or the native gap. Thus a native-five head makes $5, 6, 7$ strong independently of any terminal hypothesis.

The head's actual finite H_4 projection gives

$$\langle x, P_4/4\rangle \geq h_0, \qquad h_0^2 = (\phi - 1)/4, \qquad h_0 > .39307.$$

At strong $K = 14, 15$ the curved reference, omitting every nonadjacent internal weak knot, has energies

$$w_{14}^2 > .924016575, \qquad w_{15}^2 > .94125107.$$

Indeed the complete raw energies above lose at most $(.0112^2 + .008^2 + .0061^2)/2 = .000113325$ for $K = 14$, and $< .00012533$ for $K = 15$; these bounds include every permitted hole. The P_5 capacity is $> .151$. Pay weight $.56$ on $P_4/4 - P_5/5$, costing $.112$. Its square is $1/20$, it is orthogonal to the raw reference, and the single-square gain exceeds

$$1.12(.39307 - .3652) - .56^2/20 = .0155344.$$

Hence this initial/head case gives

$$L_{14} > .939550975, \qquad L_{15} > .95678547.$$

All other initial/head cases give larger uniform energies. For a non-five head and strong 5, pay $.4(Z - P_5/5)$. Its price is $.08 < .151$, and (B.8.1) gives gain $.2893 - .8(.3652) + .032 > .02914$. For a weak helper 5, the raw contracts above already give $L_{14} > .95255, L_{15} > .96895$, with no further head graft.

For isolated dotted weak 5, choose strong $J = 6, 7$ and a curved reference to K, constant first J. Its first mean, individual capacity, and the square of $Y_5 = U' - (5/J)P_J$ obey

J	$m <$	$\beta_J >$	$Y_5^2 <$
6	.3399	.1435	77/150
7	.31938	.1366	194/175.

The raw first-J energies exceed $.69317, .71400$; the complete tails to 14 exceed $.2117654, .1732468$ after omission losses $< .000267, .000181$. Thus the raw energies exceed $.9046684, .8870658$. At 15, the additional increment is $> .0172465$. Let $Y_H = Z - P_J/J$. The full cross is $\langle Y_H, Y_5\rangle = 1 - 5/J$, while $\langle w, Y_H\rangle = \langle w, Y_5\rangle = 0$. The one vector $w + .4Y_H + .09Y_5$ spends $.85/J$, below each displayed capacity. Its entire gain is

$$(.8h - .16R) - .8m + .16/J + .18(T' - 5m) - .0081Y_5^2 - .072(1 - 5/J).$$

It exceeds $.04930, .06699$, respectively. The resulting $K = 14$ energies exceed $.9539684, .9540558$, and the $K = 15$ energies exceed $.9712149, .9713023$. The earlier $.21/.2$ insertion is not also charged.

In every case, therefore, there is one paid reference V, supported through a strong $K = 14$ or 15, with sum f_K, unspent terminal capacity $\beta_K \geq d_K$, and

$$2A_V - V^2 > L_K, \qquad L_{14} = .939550975, \quad L_{15} = .95678547. \tag{B.8.2}$$

All corrections have zero sum and support through seven. Their individual nonnegative coefficients were checked before the terminal pressure is used.

We shall also need the following independent implication: if 17 is strong, the head has own five. For a non-five head and strong 5, the curved reference to 17 has energy $> .97133$, and its head gain $> .02914$ already exceeds one. If 5 is a helper, the full curved reference gains $> .74385 - .7126 = .03125$ in its first six coordinates, yielding $E_{17} + .03125 - 1/1024 > 1.0029$. If 5 is isolated dotted, the analogous curved references to 17 have energies $> .9530, .9354$; the joint head/isolated-five gains just proved exceed $.04930, .06699$. Again the totals exceed one. Thus every non-five head is incompatible with strong 17. This argument precedes any classification of its terminal opposite.

B.8.4 Unconditional terminal sums and isolated opposites

For $N = 16, 17, 18$, let $Y = P_N - P_K$. It has support after K, square $N - K$, and zero product with the entire corrected reference in (B.8.2). Paying the nonnegative weight $d_K \leq \beta_K$ shows that $S_N \geq G$ would imply

$$\|x\|^2 > L_K + 2d_K(G - f_K) - (N - K)d_K^2.$$

The rational root bounds

$$3.60555 < \sqrt{13} < 3.60556, \quad 3.74165 < \sqrt{14} < 3.74166,$$
$$3.87298 < \sqrt{15} < 3.87299$$

give $.1361 < d_{14} < .1362$, $.1313 < d_{15} < .1314$. Every bound follows by positive squares and $d_K = 1/(\sqrt{K} + \sqrt{K-1})$.

For $S_{17} \geq 3.760$, the extra terms are bounded below by

$$2(.1361)(.428) - 3(.1362)^2 = .06085028, \qquad 2(.1313)(.297) - 2(.1314)^2 = .04346028.$$

Their totals exceed $1.000401255, 1.00024575$. For $S_{18} \geq 3.830$, the increments are .06135384, .04457632, giving totals $> 1.000904815, > 1.00136179$. Thus

$$S_{17} < 3.760, \qquad S_{18} < 3.830, \qquad C_{17} < 10.879, \quad C_{18} < 9.81. \tag{B.8.3}$$

The edge cosines use the actual directions $P_N/\sqrt{N}$: $C_N < (N + G_N^2)/(N - G_N^2)$. The two fractions are $31.1376/2.8624 < 10.879$ and $32.6689/3.3311 < 9.81$. These are simultaneous consequences for the same x, not energies added from different references.

An entirely isolated 17-opposite has

$$c = Ap - \sqrt{2}gP_{17}, \qquad A^2 = 34g^2 - 1 > 0.$$

Own zero has negative time. Own three, four, six and parallel have positive actual null combinations, respectively,

$$c + n_{16} + 2n_{17} + \sqrt{2}n_{18}, \qquad c + \sqrt{2}n_{17} + n_{18}, \qquad gc + n_{17} + n_{16}/2, \qquad c + n_{17}.$$

Selected products vanish, cap time is positive, and all unselected wall products are nonpositive. Compactness forbids them. For own five, correct on the actual H_4 by $g(3n_{17}/2 + n_{16} + n_{15}/2)$, with positive residual $\delta = (5-3\phi)/8$. The actual finite projection gives pressure on $V = P_{14} + \frac{3}{4}(e_{15} + \cdots + e_{18})$ strictly greater than $(A - \sqrt{\delta})/(\sqrt{2}g)$. Ordering implies $\langle x, V\rangle \leq S_{17}$. Since $A^2 = (17\phi + 15)/2$, the rational bounds $A > 4.61, \sqrt{\delta} < .136, \sqrt{2}g < 1.145$ give $S_{17} > (4.61 - .136)/1.145 > 31/8$. Every finite own at least seven has the positive core $c + g^{-1}n_{17} + 2(1 - g^2)g^{-1}n_{16}$. Its norm is $-(1 - g^2)(4g^2 - 3)/g^2$, selected cap/own products vanish, and the other selected product is negative. All unknown wall signs follow from positive actual coefficients. With $u = g^{-2} \in (1, 4/3)$, divide its displacement by $\sqrt{2}g$. Its coordinates are one through fifteen, $2 - u$ at sixteen, and $u/2$ at seventeen and eighteen. Its product with x is at most S_{17} by ordering. The actual pressure is at least $\sqrt{17 - u/2} - \sqrt{(7u - 4 - 3u^2)/2}$. The first radical exceeds 4.04 and the second is at most $1/\sqrt{24} < .2042$; hence $S_{17} > 34/9$. All these finite cases contradict the same prefix rule: the gains at $G = 34/9$ exceed .065, .047.

For every dotted $g > 1$, put $u = g^{-2}$. The actual future reciprocal core $c + n_{17}/g$ has norm $-(1 - u)$. Its displacement is $U = \sqrt{2}gP_{17} - n_{17}/g$. For $r = 17 - K$, choose

$$Y = U - \sqrt{2}gP_K, \qquad \nu = d_K/(\sqrt{2}g), \qquad Y^2 = 2r/u - 2 + u.$$

The residual is supported after K, so is orthogonal to the entire already paid reference. Its price is $\nu\sqrt{2}g = d_K$, allowing equality at the terminal. The gain is exactly

$$G_K(u) = \sqrt{2}d_K\left[\sqrt{34 - u} - \sqrt{2K} + .58 - \sqrt{u(1 - u)}\right] - d_K^2(r - u + u^2/2).$$

The chord $\sqrt{34 - u} \geq \sqrt{34} - .0864u$ follows from $\sqrt{34} > 5.8309, \sqrt{33} > 5.7445$. The circle square gives $\sqrt{u(1 - u)} \leq 1/4 + u - u^2$. Using $\sqrt{28} < 5.2916, \sqrt{30} < 5.4773$, the bracket exceeds $u^2 - 1.0864u + .8693$ or $u^2 - 1.0864u + .6836$, both positive. Also $\sqrt{2}d_{14} > .19247, \sqrt{2}d_{15} > .18568$. Use the upper d_K only in the negative constant and the lower d_K in $d_K^2(u - u^2/2)$. This gives

$$G_{14}(u) > .183208395u^2 - .190576198u + .111662851,$$
$$G_{15}(u) > .177060155u^2 - .184483062u + .092398928.$$

For $F = au^2 - bu + c$, the identity $4a(F - \tau) = (2au - b)^2 + 4a(c - \tau) - b^2$ has positive last constants

$$\frac{808151227201376}{10^{18}} \quad (\tau = .061), \qquad \frac{244086609159516}{10^{18}} \quad (\tau = .044).$$

Thus the full dotted interval gives gains $> .061, > .044$, and totals $> 1.000550975, > 1.00078547$. No isolated 17-opposite survives.

B.8.5 An exhaustive terminal inventory

Deleting old 17 retains actual A_{16} on $n_1, \dots, n_{16}$ and independent short A_1 on n_{18}. The actual finite opposite extends this whole white Gram positively. A one-node finite enlargement of A_{16} is A_{17}, B_{17} or D_{17}; exceptional types have insufficient rank. Thus the complete white list is

left contact	short contact
none	none or one finite label $m \geq 3$
3 at $j = 1, 16$	none, 3, or 4
4 at $j = 1, 16$	none
3 at $j = 2, 15$	none.

Two left contacts create a cycle. Joining the short root to a left-three endpoint gives A_{18} or B_{18}; joining it to a left-four or a next-node branch is not finite. White parallel/dotted contacts are impossible at this finite endpoint. Own contacts still range over zero, every finite label, parallel and dotted.

Write a cap as

$$c = Ap - g\omega_{17} - \eta\omega_j - \delta\omega_{18}, \quad \sigma = g + \eta j/17 + \delta/\sqrt{2}, \quad \Delta = 1 - \frac{2\eta^2 j(17-j)}{17} - \delta^2 > 0.$$

Absent contacts have coefficient zero. Direct retained-block elimination gives

$$A^2 = 34\sigma^2 - \Delta, \qquad C_{17} = \sqrt{34}\,\sigma/\sqrt{\Delta}.$$

There is no residual cross between the actual left and short blocks. Own parallel is excluded everywhere by the actual positive null $c + n_{17}$. Isolation was excluded above.

For left-four alone, the far and near lengths are $17\sqrt{2}g + 1$ and $17\sqrt{2}g + 16$. Far own zero has zero time; every positive far own gives length greater than thirteen, while the near length is already at least sixteen. For left next-three, the lengths are $17g + 1$ at $j = 2$ and $17g + 15/2$ at $j = 15$. Far own zero has zero time; far own three is the actual simple cycle $c, n_2, \dots, n_{17}$. Every larger positive far own exceeds thirteen. Near positive own gives length at least sixteen; near own zero has the affine positive kernel with coefficients one on c, n_{14}, two on n_{15}, n_{16}, n_{17}, and $\sqrt{2}$ on n_{18}.

Left-three plus short-four has $\Delta = 1/34$ and $C_{17} = 34g + j + 17 \geq 18$. Left-three plus short-three has $\Delta = 19/68$. Own three closes a simple cycle; every larger positive own has $g \geq 1/\sqrt{2}, \sigma > 1$ and length greater than eleven. At own zero, the far bridge contains the actual affine path $(n_{15}, n_{16}, n_{17}, n_{18}, c)$, with positive kernel $(1, 2, 3, 2\sqrt{2}, \sqrt{2})$. For the near bridge, $v = (\omega_{16} + \omega_{18})/2$. Deleting actual short 18 gives positive finite D_{18}. Positive projection checks the released wall and every unknown wall, so this is the actual short-edge endpoint. Its exact data are

$$A^2 = 23/2 + 8\sqrt{2}, \quad \sum v_i = 9 + 8\sqrt{2}, \quad (\sum v_i)^2 - 18A^2 = 2, \quad C_{18} = 8 + 9\sqrt{2}/2 > 14,$$

contradicting (B.8.3).

For pure far-three at $j = 1$, own zero has negative time, own three gives its old simple cycle, own four has null $c + \sqrt{2}n_{17} + n_{18}$, and own six has null $gc + n_{17} + n_{16}/2$. Parallel was already null. The remaining owns are five, finite at least seven, or dotted, with

$$C_{17} = \frac{\sqrt{2}}{3}(17g + 1/2), \qquad g < 4/3.$$

At $g \geq 4/3$ this would be at least $139\sqrt{2}/18 > 10.9$. For pure near-three at $j = 16$, $C_{17} = \sqrt{2}(17g + 8)/3$. Own zero has the positive kernel with coefficients one on c, n_{15}, two on n_{16}, n_{17},

and $\sqrt{2}$ on n_{18}. Own three gives a triangle; four and parallel have the nulls above. Owns at least seven or dotted have $g > .9$ and length > 10.98362. Only owns five and six need remain.

Near own five is impossible independently of any head. For $g = \phi/2$, the positive actual right-B_2 core is

$$\xi = c + 2gn_{17} + \sqrt{2}gn_{18} = Ap - \sqrt{2}(g + 1/2)P_{16}, \quad A^2 = (49\phi + 31)/2, \quad \xi^2 = 1 - 2g^2 < 0.$$

Its cap product is this same negative square and its selected old products vanish. All other original wall products are nonpositive; positive time makes it future. The pressure gives

$$S_{16} > \frac{A - \sqrt{2g^2 - 1}}{\sqrt{2}(g + 1/2)} > 3.7.$$

Indeed $A > 7.42$, $\sqrt{2g^2 - 1} < .556$ and $\sqrt{2}(g + 1/2) < 1.852$, while $7.42 - .556 > 3.7(1.852)$. The uniform prefix rule gives increments $.06306872, .04497024$, contradicting its norm budget. For near own six, $g = \sqrt{3}/2$ gives $C_{17} > 10.5$, hence $S_{17}^2 > 17(19/23) > f_{17}^2$. Thus this surviving role has strong 17 and the isolated native-five head already proved above.

Finally suppose there is no left contact and $\delta = \cos(\pi/m)$ is the short-white magnitude. The length bound implies

$$\frac{g + \delta/\sqrt{2}}{\sqrt{1 - \delta^2}} < 1.866.$$

For $m \geq 9$, $\pi^2 < 10$ and $\cos t > 1 - t^2/2$ give $\delta > 76/81 > 15/16$; even at $g = 0$ the ratio square exceeds $225/62 > 1.866^2$. The entire conservative own list is

m	own g
3	0, every finite positive own, or $1 < g < 1.264$
4	finite own $3, 4, 5$
5	0 or finite own 3
$6, 7, 8$	0.

For short three, $g < 1.866(.8665) - .3533 < 1.264$. Short four gives $g < .82$; its own zero has positive null $c + n_{17} + \sqrt{2}n_{18}$. Short five gives $g < .53$ from $\delta > .809, \sqrt{1 - \delta^2} < .588, \delta/\sqrt{2} > .571$. For short at least six, $g < .933 - .6 < 1/2$. Every null above is a positive actual-root combination with positive cap time, zero selected products and nonpositive product against every remaining original wall. It is therefore a nonzero feasible future ray, forbidden by compactness.

This proves the exhaustive partition into a far-three helper, near-three/own-six, or the displayed no-left short-contact family.

B.8.6 Finite far-helper owns from raw, unspent references

The far cap is

$$c = Ap - \sqrt{2}gP_{17} - e_1/\sqrt{2}, \qquad A^2 = 34g^2 + 2g - \tfrac{1}{2}.$$

Use the three raw contracts of Section B.8.2, before any head graft. Their terminal prices are at least d_K, and their full first coordinates are bounded there. Pay $\mu_{14} = .136, \mu_{15} = .1313$ for a new pressure $\langle x, V\rangle > G$, with $Y = V - P_K$. The full square retains $\langle w, Y\rangle$; no absent P_{14} coefficient is borrowed when the actual strong knot is 15.

For own five, the actual H_4 correction

$$\eta = c + (3g/2)n_{17} + gn_{16} + (g/2)n_{15} = Ap - U$$

has positive square $\delta = (5 - 3\phi)/8$. Use the actual finite point $p + (A/\delta)\eta$. Its selected products vanish, its correction coefficients are positive and every unselected original wall remains nonpositive. Its pressure exceeds $A - \sqrt{\delta}$. For $b = 1/(2g) = 1/\phi < .619$, the vector $V = U/(\sqrt{2}g)$ has coordinates $1 + b$ at one, one through fourteen, and $3/4$ at fifteen through eighteen. Its pressure exceeds 4.10:

$$A^2 = (19\phi + 16)/2, \quad A > 4.834, \quad \sqrt{\delta} < .136, \quad \sqrt{2}g < 1.145.$$

The complete residual data are

K	$Y^2 <$	$\langle w, Y\rangle$
14	2.634	bw_1
15	2.134	$bw_1 - w_{15}/4.$

Use $w_{15} > .1313$ and the actual raw row's L, w_1. The resulting exact strict norm lower bounds are

initial five	$K = 14$	$K = 15$
strong	1.0217895424	1.02014330566
helper	1.020882792	1.01938584884
isolated dotted weak	1.0258028	1.02332542274.

For example the first column is obtained from $L + .272(4.10 - 3.332 - .619w_1) - .018496(2.634)$. Every row is the same original square with its entire first-coordinate cross retained.

For every finite own at least seven, put $u = g^{-2} \in (1, 4/3)$. The positive actual core

$$\xi = c + g^{-1}n_{17} + 2(g^{-1} - g)n_{16} = Ap - U$$

has norm $-(u - 1)(4 - 3u)/u$, zero cap/own products and negative n_{16} product. Its unknown wall signs and future sheet follow from positive actual coefficients and $A > 0$. For $b = 1/(2g) \in (.5, .556)$, $V = U/(\sqrt{2}g)$ has coordinates

$$1 + b, \quad 1\ (2, \ldots, 15), \quad 2 - u, \quad u/2,\ u/2.$$

Its pressure is

$$G = \sqrt{17 + 2b - b^2} - \sqrt{(7u - 4 - 3u^2)/2} > \sqrt{71/4} - 1/\sqrt{24} > 4.0088.$$

The residuals satisfy $Y_{15}^2 < 7/4,\ Y_{14}^2 < 11/4$; both full crosses are bw_1. Indeed $Y_{15}^2 = 4 - 15u/4 + 3u^2/2$, whose difference from $7/4$ is $(3/2)(u - 1)(u - 3/2) < 0$. The three initial-five rows give

	$K = 14$	$K = 15$
strong	1.0010956736	1.00023617738
helper	1.003278544	1.0024615677
isolated dotted weak	1.005311136	1.0036135113.

The tight entry uses precisely

$$.9403998 + .2626(4.0088 - 3.463 - .556(.3652)) - .01723969(7/4) = 1.00023617738.$$

The weaker energy .94006 or a terminal weight .13 would not justify it. All finite far-helper owns are excluded without a head or short-opposite assumption.

B.8.7 A single table for every actual short opposite

Deleting actual 18 retains A_{17}. The finite one-node classification gives precisely isolation, an endpoint bond three or four at $j = 1, 17$, or a next-to-end bond three at $j = 2, 16$. A second contact creates a cycle; the other branching trees and exceptional diagrams have insufficient rank. For the actual short opposite $d = Bp - v$, write $\delta = -\langle d, n_{18}\rangle \geq 0$. At this stage its own contact can be zero, any finite label, parallel, or dotted. The complete data are

white role	v	B^2	C_{18}
isolated	δP_{18}	$18\delta^2 - 1$	$3\sqrt{2}\delta$
end three j	$\delta P_{18} + P_j/\sqrt{2}$	$18\delta^2 + \sqrt{2}j\delta + j/2 - 1$	$(18\sqrt{2}\delta + j)/\sqrt{19}$
end four j	$\delta P_{18} + P_j$	$18\delta^2 + 2j\delta + j - 1$	$18\delta + j$
next three j	$\delta P_{18} + P_j/\sqrt{2}$	$18\delta^2 + \sqrt{2}j\delta + j/2 - 1$	$9\sqrt{2}\delta + j/2.$

For example the old short residual is $1/18$, and clearing a white three or four leaves residual $1 - j(18 - j)/36$ or $1 - j(18 - j)/18$, respectively. The edge formula gives the last column; thus the table includes the actual short normalization.

The unconditional bound $C_{18} < 9.81$ removes near end-four and every positive-own far end-four. The latter has $B = 0$ at own zero. Near next-three has $C_{18} = 9\sqrt{2}\delta + 8$, so a positive own is already too long; at own zero its null is

$$d + n_{15} + 2n_{16} + 2n_{17} + \sqrt{2}n_{18}.$$

Far next-three has $B = 0$ at own zero and $\delta < .7$ by length, so its only possible positive native own is three. Its null is

$$\sqrt{2}d + 2\sqrt{2}n_{18} + 3n_{17} + 2n_{16} + n_{15}.$$

All coefficients are positive, the cap time is positive, selected products vanish and every other original-wall product is nonpositive termwise. These are forbidden future feasible nulls.

We shall apply the remaining three rows twice. In both applications the selected seventeenth cap clears positively to

$$\bar{c} = Ap - (Q/17)P_{17}, \quad a = A^2, \quad Q^2 = 17a + 9, \quad \bar{c}^2 = r = 9/17, \quad a < 63.$$

For near end-three the actual mutual magnitude is $\lambda = AB - Q(\delta + 1/\sqrt{2}) \geq 0$. The length bound gives $\delta < 1.1$, whereas

$$aB^2 - Q^2(\delta + 1/\sqrt{2})^2 = (a - 9)\delta^2 - 9\sqrt{2}\delta - a - \tfrac{9}{2} < .21a - 15.39 < 0.$$

This contradicts acuteness, including own zero and every allowed dotted value. Only far end-three and isolation remain.

B.8.8 Dotted far helpers: the complete mutual split

Let

$$c = Ap - \sqrt{2}gP_{17} - e_1/\sqrt{2}, \quad 1 < g < 4/3, \quad a = 34g^2 + 2g - \tfrac{1}{2}, \quad Q = (34g + 1)/\sqrt{2}.$$

Here $71/2 < a < 1127/18$, and the positive actual correction is $\bar{c} = c + \sum_{i=1}^{16}(17 - i)n_i/17$. Put $G_{17} = 3.760,\ G_{18} = 3.830$, and

$$t = Q/A = \sqrt{17 + 9/a}, \quad K = A\sqrt{18} - Q, \quad H = A - G_{17}Q/17, \quad J_0 = (t - G_{18})/K.$$

Then $4.14 < t < 4.18,\ K, H, J_0 > 0$, and

$$4HJ_0 > 1, \qquad H + J_0 y^2 - y > 0 \quad (y \in \mathbb{R}). \tag{B.8.4}$$

Indeed the first inequality is equivalent to positivity of $F(t) = 4(t - 3.83)(1 - 94t/425) + t - \sqrt{18}$. Replace $\sqrt{18}$ by its strict upper bound 4.243. The new polynomial has value $3333/2125000 > 0$ at 4.14; its difference from that value is

$$(t - 4.14)\{5 - (376/425)(t + 4.14 - 3.83)\} > 0,$$

since the brace exceeds $10919/10625$. The second inequality is the completed square $(2J_0 y - 1)^2 + 4HJ_0 - 1 > 0$.

We also record a useful full-projection test. Suppose an inverse-positive full actual projection has ray $(1 + T)p - LP_{17} - Me_{18}$, with $T > 200$ and $L > M > 0$. Write $k = M/L$. Its norm and the original farthest comparison give

$$S_{17} + kx_{18} \geq \sqrt{17 + k^2}\frac{\sqrt{T}}{\sqrt{1 + T} + 1} > \sqrt{221/15} > 3.83.$$

For the strict bound use $\sqrt{1 + T} > 14$, so the squared last ratio is $> 13/15$. The left side is $(1 - k)S_{17} + kS_{18} < 3.83$, a contradiction. The existence, future sheet and unknown-wall signs of this point follow from its full positive actual projection.

First suppose that the short opposite is far end-three:

$$d = Bp - \delta P_{18} - e_1/\sqrt{2}, \qquad B^2 = 18\delta^2 + \sqrt{2}\delta - \tfrac{1}{2}.$$

Own zero has negative time. Own three has the affine F_4 null above, and owns four, six and parallel have respectively

$$d + \sqrt{2}n_{18} + n_{17}, \qquad d + (\sqrt{3}/2)n_{18} + n_1/2, \qquad d + n_{18}.$$

These positive actual nulls remain feasible against every original wall. For every remaining own put $\lambda = -\langle c, d\rangle \geq 0$ and $v_0 = 1/(17\sqrt{2})$. Positive clearing gives

$$\bar{d} = Bp - (\delta + v_0)P_{17} - \delta e_{18}, \quad \bar{d}^2 = r, \quad \rho = -\langle \bar{c}, \bar{d}\rangle = \lambda + 8/17 = AB - Q(\delta + v_0).$$

The offset here is $8/17$. If $\lambda > 0$, the actual native gap gives $\lambda \geq 1/2$, hence $\rho > r$. The positive actual core $\Xi = \bar{c} + (\rho/r)\bar{d}$ has square $r - \rho^2/r < 0$, positive time, zero d product, negative c product, and zero old A_{16} products. All other products are nonpositive by acuteness. Its pressure and the two original sum bounds would require

$$H + (\rho/r)J - \sqrt{\rho^2/r - r} \leq 0, \qquad J = B - G_{17}v_0 - G_{18}\delta.$$

But $J = \rho/A + (t - G_{18})\delta + (t - G_{17})v_0 > (t - G_{18})\delta$. Positive squaring gives $B < \sqrt{18}\delta + 1/6$, and

$$36(g + 1/34)^2 - a = 2g^2 + 2g/17 + \tfrac{1}{2} + 9/289 > 0.$$

As $Qv_0 = g + 1/34$, this implies $\rho < K\delta$ and $J > J_0\rho$. With $y = \sqrt{\rho^2/r - r}$, the required gap exceeds $H + J_0(y^2 + r) - y > 0$ by (B.8.4).

If $\lambda = 0$, the cleared two-cap Gram has diagonal $9/17$, cross $-8/17$, and inverse $\left(\begin{smallmatrix} 9 & 8 \\ 8 & 9 \end{smallmatrix}\right)$. Its positive projection has

$$\alpha = 9A + 8B, \quad \chi = 8A + 9B,$$
$$T = 9a + 16AB + 9B^2 > 319.5,$$
$$L = \alpha Q/17 + \chi(\delta + v_0) > M = \chi\delta > 0.$$

The full-projection test excludes it. This handles every far-three short own and every unbounded actual mutual range.

Now let $d = Bp - \delta P_{18}$ be isolated. Acuteness $AB - Q\delta \geq 0$ implies $\delta^2 \geq a/(a-9) > 1$, so its own is derived to be dotted. Put $\lambda = AB - Q\delta$. The cleared Gram is $\left(\begin{smallmatrix} r & -\lambda \\ -\lambda & 1 \end{smallmatrix}\right)$. At $\lambda^2 = r$, $\bar{c} + \lambda d$ is a forbidden positive future feasible null. For $\lambda^2 > r$ the same core is negative. Writing $J = B - G_{18}\delta$, acuteness and $B < \sqrt{18}\delta$ give $J > (t - G_{18})\delta$, $\lambda < K\delta$. Its required pressure gap is $H + \lambda J - \sqrt{\lambda^2 - r}$, contradicted by (B.8.4). If $0 < \lambda^2 < r$, native separation gives $\lambda \geq 1/2$. The positive full projection has

$$T = \frac{a + 2\lambda AB + rB^2}{r - \lambda^2} > \frac{68.5}{19/68} = \frac{4658}{19} > 200,$$

using $B^2 > 17$, $AB > 24$. Its coefficients $\alpha = (A + \lambda B)/(r - \lambda^2)$, $\chi = (\lambda A + rB)/(r - \lambda^2)$ are positive, with $L = \alpha Q/17 + \chi\delta > M = \chi\delta > 0$. The full-projection test excludes this entire interval too. Only $\lambda = 0$ remains.

B.8.9 The orthogonal neighbor and its quantitative comparison

For that last mutual value,

$$\delta^2 = \frac{a}{a-9}, \quad B = \frac{Q}{\sqrt{a-9}}, \quad w = 9/a, \quad q = \frac{p - (A/Q)(P_{17} + we_{18})}{\sqrt{D}}, \qquad D = \frac{9(a-9)}{aQ^2}.$$

The positive actual projection has full forest $A_{17}(c, n_1, \ldots, n_{16}) + I(d)$. The other endpoint retaining its whole A_{17} is the positive finite projection on $A_{16} + c + n_{18}$. These endpoints are distinct because $\langle d, n_{18} \rangle = -\delta < -1$. Consequently n_{18} is the actual opposite of d, orthogonal to the entire new A_{17}, with

$$h_I = \delta^2 - 1 = \frac{9}{a-9} < 18/53 < 1/2.$$

All these are full-dimensional points satisfying every original wall; no face heredity is used.

Write the same original $z = E_q(q - y - \eta d)$, where y is in the A_{17} space. We prove the following three bounds, separately for old 17 strong and weak:

$$E/E_q < 1.061, \qquad \eta^2 > .1164, \qquad \text{a signed } A_{17} \text{ endpoint coordinate of } y \text{ exceeds } .51. \quad \text{(B.8.5)}$$

Every actual chamber point, including each auxiliary point used below, is then compared at q with factor < 1.061, because its height relative to the original z is at most E. The point q is never made farthest.

Suppose first that $S_{17} > f_{17} = s$, so the earlier head argument gives the actual isolated native-five head and strong $5, 6, 7$. The original reference through strong 17, with all internal weak knots omitted, has energy $> .97253117$, first-five mean $b < .3652$, fifth capacity $> .151$, and terminal capacity $b_0 = \sqrt{17} - 4 \in (.1231, .1232)$. The actual head foot is $Z_0 = P_4/4$, with pressure $h_0 > .39307$.

Put $Y_0 = Z_0 - P_5/5$, so $Y_0^2 = 1/20$. Paying .56 costs .112 < .151 and gains > .0155344. The paid effective energy is $L > .988$. Retaining the terminal surplus in this same square gives

$$2b_0(S_{17} - s) + x_{18}^2 < .012, \qquad S_{18} < s + .012/(2b_0) + b_0/2 < 3.82335.$$

We next show $g > 1.12$. For $1 < g \le 1.12$ the reciprocal core $c + n_{17}/g$ has pressure $A - \sqrt{1 - g^{-2}}$. Its residual after payment from P_{17} and P_5 has square $.4 + g^{-2}$, and cross $1/(20\sqrt{2})$ with Y_0. The pressure-minus-reference deficit is

$$D_H = A - \sqrt{1 - g^{-2}} - g(\sqrt{34} - .58) - (\sqrt{10}/10 - .058) + b_0/(g\sqrt{2}).$$

It decreases on this interval: $A'^2 = 34 + 18/a < 5.875^2$; the radical derivative exceeds 1.5, and $\sqrt{34} - .58 > 5.25$, so $D'_H < -.875$. At $g = 1.1$ use $A > 6.5452$, $\sqrt{1 - g^{-2}} < .4166$, the third term < 5.7761, and the last term $> .079$; at 1.12 use $6.66255, .45035, 5.88112, .07767$. Also $\sqrt{10}/10 - .058 < .258228$. Thus $D_H > .173$ on $(1, 1.1]$, and $> .150$ on $[1.1, 1.12]$. With residual coefficient $\nu = .075, .077$ respectively, the full gain over L is

$$2\nu D_H - \nu^2(.4 + g^{-2}) - .56\nu/(10\sqrt{2}) > .015075, \quad > .0127780815.$$

Both contradict $L > .988$. The individual terminal price $\nu\sqrt{2}g < .1231$ fits b_0, while $.112 + \nu/(5\sqrt{2}) < .123 < .151$ fits the fifth anchor. Hence $a > 44.3896$, and $k = 9/a < .20276$.

Suppose $E_q/E \le 20/21$. The explicit q formula gives

$$S_{17} + kx_{18} \ge R(k) := \sqrt{17 + k} - (20/21)\sqrt{k(1 - k)} > 3.764.$$

Indeed R decreases on $0 < k < .20276$: its positive derivative term is $< .122$ and the magnitude of the negative term is $> (20/21).594$. At the upper endpoint use $\sqrt{17.20276} > 4.1476$, $\sqrt{.20276(1 - .20276)} < .4021$. The independent strong $K = 11, 12$ references through strong $5, 6, 7$, after every permitted hole, have raw energies $> .86375298, > .88550238$, terminal capacities $> .1543$, $> .1474$, and fifth capacity $> .151$. After the same .56 head payment, pay .14 from that terminal toward $P_{17} + ke_{18}$. Its residual is orthogonal to the head residual and to the reference, of square $17 + k^2 - K$, with $k^2 < .0412$. Since $f_{11} < 2.907$, $f_{12} < 3.055$, the full original squares give

$$.86375298 + .0155344 + .28(3.764 - 2.907) - .0196(6.0412) = 1.00083986,$$
$$.88550238 + .0155344 + .28(3.764 - 3.055) - .0196(5.0412) = 1.00074926.$$

Thus $E_q/E > 20/21$, so $E/E_q < 1.05$. Moreover $a < 64$, $\delta > 8/\sqrt{55} > 1.078$, $Q/A > 4.14$ imply

$$\eta = (E/E_q)\delta(Q/A - S_{18}) > 1.078(4.14 - 3.82335) > .3413, \qquad \eta^2 > .1164.$$

The known actual q-edge releasing c retains $A_{16} + d$; its other endpoint is $A_{16} + n_{17} + d$. Clearing that block leaves residuals $9/17$ and cross $-g - 1/34$, so its cosine is $C' = (34g + 1)/18 > 977/450 > 2.17$. The signed endpoint coordinate is therefore greater than

$$\sqrt{17/18}\frac{2.17 - 1.05}{\sqrt{2.17^2 - 1}} > .9718(1.12)/1.926 > .56.$$

This proves (B.8.5) in the strong case.

If old 17 is weak, $S_{17} \le f_{17} < 3.713$, and

$$E_q/E = \frac{\sqrt{17 + w} - S_{17} - wx_{18}}{\sqrt{w(1 - w)}}.$$

The two original bounds give $S_{17} + wx_{18} < 3.713 + .117w$. If $g \geq 1.1$, then $w \leq 25/119 < .211$. On this range $\sqrt{17+w} > 4.123 + .12w$, since $\sqrt{17+w} + \sqrt{17} < 25/3$, while $\sqrt{w(1-w)} < .408$. The displayed ratio exceeds $(.410 + .003w)/.408 > 1$, contrary to $E_q \leq E$. Thus $1 < g < 1.1$ and $25/119 < w < 18/71$. The same chord holds, $w > .21$, and the denominator is $< \sqrt{954}/71 < .4351$. Consequently

$$E_q/E > .41063/.4351 > .943, \qquad E/E_q < 1000/943 < 1.061.$$

Here $\delta^2 > 119/94 > 81/64$ and $\sqrt{17+w} > 4.1485$, so $\eta > (9/8)(4.1485 - 3.830) = .3583125$, hence $\eta^2 > .128$. Finally $C' > 35/18 > 1.94$, and its endpoint coordinate exceeds

$$.9718(1.94 - 1.061)/1.663 > .51.$$

All three conclusions of (B.8.5) follow.

B.8.10 One local $A_{17} + I$ contradiction for both height branches

We prove the common consumer of (B.8.5). The hypotheses are an actual whole $A_{17} + I$ vertex q, comparison factor $\kappa < 1.061$, a known isolated actual I opposite with parameter $h_I < 1/2$, and

$$\|y\|^2 + \eta^2 < 1, \qquad \eta^2 > .1164, \qquad \text{one signed endpoint coordinate of } y \text{ is } > .51.$$

We use the finite-white geometry proved in Appendices B.3.6 and B.3.7, not their farthest-vertex energy conclusions. Here are the geometry and the changed numerical comparisons explicitly.

Realize A_{17} in the zero-sum subspace of $\mathbb{R}^{18}$. For its unit roots m_i and duals Ω_i,

$$D_i = i(18-i)/9, \quad C_{ij} = i(18-j)/9 \ (i \leq j), \quad h_i = \langle y, \Omega_i \rangle, \quad e_A = \tfrac{1}{2}\sum_{i=0}^{17} (h_{i+1} - h_i)^2.$$

The sequence h is strictly concave. A site $5 \leq i \leq 13$ is weak if $h_i \leq \sqrt{D_i} - .65$. Deleting such i retains two paths of ranks at least four and the entire actual I. The finite one-node list is isolation; one simple bond at an end, D or permitted E position; an end-four; or two end-threes, one on each path. A joined component has rank seventeen and is classical. The retained foreign I cost Q_I is less than one and, if nonzero, at least $1/4$.

Own zero is removed geometrically before any comparison: internal-three roles have affine arms $1, 2, 5$ or $1, 3, 3$; internal-four gives the $4, 3, 3$ star; two whites give an affine cycle. A cap without an A contact cannot close the edge. End-four requires $Q_I < 1/m \leq 1/5$, hence $Q_I = 0$ and negative time square. In the remaining end-three/node-two cases the full old-component norms give $A_0^2 < Q_I$. The known actual I opposite is $\sqrt{h_I}q - \sqrt{1+h_I}d$; acuteness would require $A_0\sqrt{h_I} \geq \sqrt{Q_I}\sqrt{1+h_I}$, impossible. These are the zero-own calculations of Appendix B.3.6, with this actual opposite replacing its farthest-only orthogonal-opposite step.

For a white coefficient vector b, all remaining length data are computed by

$$\Delta = 1 - b^{\mathsf{T}} G_{\text{retained}}^{-1} b - Q_I, \qquad \sigma = \gamma + \langle \Omega_i, \sum_{j \neq i} b_j \Omega_j \rangle / D_i, \qquad C = \sqrt{D_i}\,\sigma/\sqrt{\Delta}.$$

The finite-tree calculation in that appendix gives ratio at least $3/2$ for near-three/own at least five, D/E, end-four and two-white rows, after their actual affine small-own boundaries. Positive foreign cost increases this ratio. Foreign far-three has ratio $> 227/(25\sqrt{35}) > 3/2$; foreign near-four has squared ratio $> 121/35$. A near-four with side order at least eight has ratio $> 761/500$, and side order seven has ratio $> 3/2$. For a pure helper with side order m,

$$\sigma = \gamma + 1/(2m), \quad \Delta = (m+1)/(2m), \quad \sigma^2/\Delta - (4\gamma - 2) = (\gamma - 1 - 1/(2m))^2/\Delta \geq 0.$$

In particular $\gamma \geq 17/16$ also gives ratio at least $3/2$.

These geometric bounds are sufficient under the new comparison. For $u = \sqrt{D_i} \in (\sqrt{65}/3, 3]$, the positive-square test at $C = 3u/2$ is

$$(3u/2 - 1.061)^2 - (1 - .65/u)^2(9u^2/4 - 1) = F(u) = 1.175096 - .258u - 1.3/u + .4225/u^2 > 0.$$

Indeed $F(3) > .0147$, and

$$F(u) - F(3) = (3 - u)\{.258 - 1.3/(3u) + .4225(3 + u)/(9u^2)\} \geq 0,$$

as $u > 8/3$. The actual directional floor $(C - \kappa)/\sqrt{C^2 - 1}$ increases here because $\kappa C > 1$. Every listed ordinary role is therefore strong.

For isolated A contacts, after positive clearing of the whole I, set $U = (1 - Q_I)/\gamma^2$; $A_0^2 = \gamma^2(D_i - U)$. For $0 < U < 1$ the positive reciprocal core gives

$$h_i > \sqrt{D_i - U} - 1.061\sqrt{U(1 - U)}.$$

Its loss is less than $.194U + 1.061\sqrt{U(1 - U)} < (.194 + 1.079)/2 = .6365 < .65$. For $U > 1$ the positive finite projection gives loss $< .194 + 1.061\sqrt{U - 1}$. For foreign $Q_I > 0$, $U - 1 \leq 5 - 3\phi < (5/12)^2$; for pure finite own at least eight, the same bound follows from $\tan(\pi/8) < 5/12$. The loss is $< .637$. The $U = 1$ core is a positive future feasible null. Pure own-five uses the actual H_4 finite point: its old loss is $< 25/76 + 5/24$, and replacing comparison one by 1.061 adds

$$.061\sqrt{\delta_5}\sqrt{1 + \delta_5/A_0^2}/\gamma < .016, \quad \delta_5 = (7 - 3\sqrt{5})/16 < 1/36, \quad \gamma > .8, \quad A_0^2 > 3.6.$$

Its total loss is $< .554$. Pure seven has old loss $< 2500/10368 + .36$ and added loss $< .061(.36) = .02196$, hence total $< .624$. Own three, four, six and parallel have the actual nulls proved in the same appendix. Thus no isolated or foreign role remains weak.

The provisional weak list is exactly pure near-four at $5, 6$ or reflection, or a pure far helper with

$$v = \gamma\Omega_i + \Omega_e/2, \quad e = 1 \text{ or } 17, \quad \phi/2 \leq \gamma < 17/16, \quad Q_I = 0.$$

This cutoff concerns only weak helpers. Their nonadjacency is the original-vector calculation of Appendix B.3.6. To specify its complete domain, for adjacent sites and owns $\alpha, \beta \in (.809, 1.08)$ the common/outward/reversed products have lower bounds

$$1 - 1.08^2, \qquad \tfrac{1}{2} - \alpha\beta + (\alpha - 1)(\beta - 1)/18, \qquad \tfrac{1}{2} - \alpha\beta + (\alpha + 1)(\beta + 1)/18.$$

Native gaps leave common mutual zero or opposite mutual zero/simple. The reversed zero case has numerator

$$18F = 4\ell s\alpha^2\beta^2 - (\ell + 1)(s + 1)(\alpha^2 + \beta^2) - 2\alpha\beta - (\ell + 1)\alpha - (s + 1)\beta + 5, \quad \ell + s = 17, \quad \ell, s \geq 5.$$

It increases in both owns on the permitted range. Unless both are five, one is at least seven and exceeds .9005; concavity in ℓ reduces to endpoints, whose smaller value is $.05526706486 > 0$ at $(.809, .9005)$. The both-five case is the actual distance-one determinant obstruction proved there, on its declared $N \geq 15$ domain. The outward zero case has determinant at least $404/225 > 0$. The resulting mixed projection is positive with positive inverse; adding the entire old I block leaves it positive. Its actual future finite point is feasible against every original wall. The complete inverse calculation bounds the weighted loss by $\kappa(7/16) + 1/10 < .5641875$ for a common helper and

$\kappa(3/10) + 1/10 < .4183$ for opposite helpers. Both are less than .65, proving nonadjacency before near-four has been removed.

A near-four has an actual mate at $j = 4, 5$, obtained by the full positive $A_{j-1} + B_{18-j} + I$ projection. Its geometric cosine exceeds ten, so its new demand exceeds $(10 - 1.061)/\sqrt{99}$, giving $h_4 > 2.24$ or $h_5 > 2.41$. At the other end there is a mate or a strong 12 or 13 by nonadjacency, with demand > 2.178 or > 2.037. Disjoint end-segment Cauchy has worst case

$$e_A > \tfrac{1}{2}(2.41^2/5 + 2.178^2/6) = .976117 > .8836,$$

contrary to $\eta^2 > .1164$. The other placements are stronger. Every remaining weak site is therefore a nonadjacent pure helper.

Here is the entire reference, including every allowed internal hole. Take symmetric heights

$$H_5 = 2.077,\ H_6 = 2.218,\ H_7 = 2.314,\ H_8 = 2.371,\ H_9 = 2.389, \qquad H_{18-i} = H_i,$$

linear from zero to 5, and scale them by $\rho_0 = .98$. Positive squares $(\rho_0 H_i + .65)^2 < D_i$ put them below the strong floors. Their positive successive slopes before scaling are $.4154, .141, .096, .057, .018$. Flatten the first block through 6 if 5 is weak and reflect if 13 is weak; replace each remaining weak knot by its neighbor chord. Deleting j costs $(H_{j-1} - 2H_j + H_{j+1})^2/4$ in reference energy. Nonadjacency makes these edge pairs disjoint. The following bounds cover all hole sets without enumerating subsets:

weak ends	unscaled initial energy	holes	loss per hole	scaled energy
0	.8954558	≤ 4	$\le .00050625$	$> .858$
1	$.8640827333\ldots$	≤ 3	$\le .00050625$	$> .8284$
2	$.8327096666\ldots$	≤ 3	$\le .00038025$	$> .7986.$

The initial values are exactly, with $a = 2.077,\ b = 2.218,\ T = .096^2 + .057^2 + .018^2$,

$$a^2/5 + (b-a)^2 + T, \quad \tfrac{1}{2}\{b^2/6 + a^2/5 + (b-a)^2 + 2T\}, \quad b^2/6 + T.$$

The available individual end-anchor capacities are at least $\rho_0(821/6000)$ on a six-block or $\rho_0(343/2500)$ on a five-block, both $> .134$. The signed block means satisfy $b_5 < .28812,\ b_6 < .98(.2614)$. Every remaining positive coefficient is on an actual strong site; no weak floor or I energy is in these references.

For completeness we now give the feet, deficits and all mutual residual terms needed to use these capacities. At a weak end let m_i, m_e, m_j be its actual own, far-helper and helper-neighbor roots. Write $U = v - k_0 m_i - \ell m_e - m m_j$. The complete own dispatch is

own	(k_0, ℓ, m)	corrected square	pressure with comparison one
5	$(\gamma, 2/3, 1/3)$	$\delta = 2/3 - \gamma^2 > 0$	$> A - \sqrt{\delta}$
finite ≥ 7	$(\gamma, 2(1-\gamma^2), 0)$	$-(1-\gamma^2)(4\gamma^2 - 3)$	$A - \sqrt{(1-\gamma^2)(4\gamma^2-3)}$
dotted	$(1/\gamma, 0, 0)$	$-(1-\gamma^{-2})$	$A - \sqrt{1-\gamma^{-2}}.$

In the first row the point is $q + (A/\delta)(b + k_0 m_i + \ell m_e + m m_j)$; the spacelike corrected normal itself is not a point. The other rows are future negative cores with zero cap product, nonpositive selected old products and all unknown products nonpositive termwise. Time coefficients are positive. All displacements are pure A_{17}.

For a far orientation the time is $A_0(\gamma)^2 = (65/9)\gamma^2 + (5/9)\gamma - 19/36$; near adds $8\gamma/9$. On two six-blocks the projections are

$$\Pi U = (5\gamma/6)\Omega_6 + \Omega_{12}/12 \quad \text{(far)}, \qquad \Pi U = (5\gamma/6 + 1/12)\Omega_6 \quad \text{(near)}.$$

Let $Y = U - \Pi U$. Its sum is zero separately on each end block, so $\langle w, Y\rangle = 0$ after every internal hole. The far residual squares are exactly

$$Q_5 = \tfrac{2}{3}\gamma^2 + \tfrac{1}{12}, \quad Q_f = 4\gamma^4 - \tfrac{16}{3}\gamma^2 + \tfrac{29}{12}, \quad Q_d = \tfrac{5}{3}\gamma^2 - \tfrac{19}{12} + \gamma^{-2};$$

near adds $\gamma/3$. Every finite far square is $< 13/12$, and every dotted far square is $< 148/125$.

We give rational seals for the pressure deficit against the unscaled pairing $2.218(5\gamma/6 + 1/12)$. For owns five, seven, eight and at least nine the respective lower bounds are

own	own bracket	time/radical bounds	deficit lower bound
5	$\gamma < .81$	$A_0 > 2.155,\ \sqrt{\delta} < .111$	$2.155 - .111 - 2.218(91/120) > .362$
7	$.9009 < \gamma < .901$	$A_0 > 2.4154,\ R < .2158$	$2.4154 - .2158 - 2.218(1001/1200) > .3494$
8	$.9238 < \gamma < .9239$	$A_0 > 2.4795,\ R < .2464$	$2.4795 - .2464 - 2.218(.85325) > .34059$
≥ 9	$\gamma > .93$	$A_0(.93) > 2.497,\ R \leq 1/4$	$2.497 - .25 - 2.218(103/120) > .3432.$

All root bounds are positive-square comparisons, using ϕ, the native-seven cubic $8u^3 - 4u^2 - 4u + 1$, and $\gamma_8^2 = (2 + \sqrt{2})/4$. For the last row the derivative applies to the lower envelope $A_0 - \frac{1}{4} - 2.218(5\gamma/6 + 1/12)$: $A_0' > \sqrt{65}/3 > 5(2.218)/6$. It is not a monotonicity claim for the actual varying-radical deficit. Thus the finite deficit is $> .3405$. For dotted owns split the whole interval $(1, 17/16)$ at $1.03, 837/800, 1687/1600$. The increasing function $A_0 - 2.218(5\gamma/6 + 1/12)$ has derivative $> .839$. Time lower bounds at the first three left endpoints are $2.692, 2.775, 2.82$; radical upper bounds for the four intervals are $.24, .295, .317, .338$. The fourth lower function value is the third plus $.839(13/1600)$. The resulting four deficits exceed $.4188, .3913, .3843, .37016$. This covers the entire dotted range, with deficit $> .37$; near times are larger.

Changing comparison from one to 1.061 costs $< .0071$ for five, at most .01525 for finite at least seven, and $< .0207$ for dotted. Indeed the relevant extra terms are $.061\sqrt{\delta}\sqrt{1 + \delta/A^2}$, $.061/4$, and $.061(.338)$; for five $A > 2, \sqrt{\delta} < .115, \delta < .013$. Scaling the reference down gains at least $.02(2.218)(.7575) = .0336027$. Thus every six/six deficit is, before rounding, strictly greater than

$$.3405 - .01525 + .0336027 = .3588527 > .358.$$

For one weak end, on six/five blocks the exact far projection is

$$\Pi U = (5\gamma/6)\Omega_6 + \Omega_{13}/10.$$

Relative to six/six, the pairing increases by $\rho_0(343/15000) = .022409333\ldots$ and the residual square decreases by $1/60$. The new far deficit is therefore strictly greater than

$$.3588527 - .98(343/15000) = .336443366\ldots > .336.$$

Near is unchanged. In particular the opposite price here is $\mu/10$, not $\mu/12$.

For two feet we retain their entire cross. Within each length-six block decompose the own and helper residuals as

$$a(\gamma, k_0) = \frac{\gamma\sqrt{2}}{6}(P_5 - 5e_6) - k_0 m_5, \qquad b(\ell, m) = \frac{e_1 - P_6/6}{\sqrt{2}} - \ell m_1 - m m_2.$$

Direct products give

$$a^2 = 5\gamma^2/3 + k_0^2 - 2\gamma k_0, \quad b^2 = 5/12 + \ell^2 + m^2 - \ell m - \ell \leq 5/12, \quad \langle a_i, b_j\rangle = \gamma_i/6.$$

For equal orientations, $\|Y_L + Y_R\|^2 = Q_{f,L} + Q_{f,R} + (\gamma_L + \gamma_R)/3$. For mixed orientation it is $Q_{f,L} + Q_{f,R} + 2\langle b_L, b_R\rangle + 2\gamma_{\text{near}}/3$, where $\langle b_L, b_R\rangle \le 5/12$. Thus the equal/mixed bounds are $17/6, 11/3$ for two finite owns, and $3.077, 3.91$ when a dotted own is present. These bounds include unequal parameters and both mixed orientations.

If both ends are weak, pay $\mu = .1$ on each foot. The total individual anchor prices are bounded by $31/320, 101/960, 17/192$, all $< .106 < .134$. For one weak end, pay $\mu = .13$. The near price is at most $.13(31/32) < .126$, with residual square < 1.539. For far, the own-anchor price is at most $.13(85/96) < .116$, the separate opposite price is $.13/10 = .013$, and its residual square is $< 1751/1500 < 1.168$. Each fits its own available anchor; a bound on total price alone would not suffice.

In all cases the payment is one completed square. If the reference energy is E_0, the residuals are Y_j, and each projection has been paid from nonnegative individual strong-site coefficients, then

$$\langle y, w + \mu \sum_j Y_j\rangle \ge E_0 + \mu \sum_j \Delta_j, \qquad \|w + \mu \sum_j Y_j\|^2 = E_0 + \mu^2 \|\sum_j Y_j\|^2.$$

For two weak ends, adding the one I coordinate gives

$$e_A + \eta^2 > .7986 + .2(.358 + .358) - .01(3.91) + .1164 = 1.0191.$$

For one weak end the gain is $> .067$: $.26(.358) - .0169(1.539) = .0670709$ near, and $.26(.336) - .0169(1.168) = .0676208$ far. Thus $e_A + \eta^2 > .8284 + .067 + .1164 = 1.0118$.

If neither end is weak, use the known actual signed endpoint coordinate $> .51$ from (B.8.5). On its five-block the mean b_5 satisfies $.28 < b_5 < .28812$. Adding $\lambda(e_{\text{end}} - P_5/5)$, with $\lambda = 5(.51 - b_5)/4$, spends only $(.51 - b_5)/(4\sqrt{2}) < .042 < .134$ at that anchor. It gains $(5/4)(.51 - b_5)^2 > .061538418$. Consequently $e_A + \eta^2 > .858 + .061538418 + .1164 > 1$. All weak-end counts are exhausted. The I energy was counted once and every comparison used the same original farthest vertex. This excludes the final orthogonal branch and hence every dotted far-helper seventeenth cap.

B.8.11 Near-three/own-six: no short opposite survives

The other left-contact terminal role has

$$g = \sqrt{3}/2, \quad c = Ap - \sqrt{2}gP_{17} - P_{16}/\sqrt{2}, \quad a = 65/2 + 16\sqrt{3}, \quad Q = \sqrt{2}(17g + 8), \quad Q^2 = 17a + 9.$$

The inventory made old 17 strong, so there is the actual isolated native-five head $c_H = A_H p - (\phi/\sqrt{2})e_1$, with $A_H^2 = (\phi - 1)/2$. Use the common short table in Appendix B.8.7. It already excludes end-four, next-three and near end-three. Far end-three has the same owns zero, three, four, six and parallel nulls as before.

For any remaining finite far-three short own, $\phi/2 \le \delta < 1$, its actual mutual magnitude with the head is

$$M_H(\delta) = A_H \sqrt{18\delta^2 + \sqrt{2}\delta - \tfrac{1}{2}} - \phi\delta/\sqrt{2} - \phi/2.$$

It increases strictly: the derivative of its first radical exceeds $\sqrt{18}$, and $\sqrt{18}A_H > \phi/\sqrt{2}$. At $\delta = .8$, positive squares give time product > 1.937 and spatial product < 1.725, so $M_H > .212$. At 1 the corresponding bounds are < 2.419 and > 1.9530878, hence $M_H < .4659122 < 1/2$. Every finite remaining own has $\delta > .8$, so $0 < M_H < 1/2$ contradicts the actual native gap. The caps are distinct by their original contact sets. Thus a far-three short opposite must have dotted own $\delta > 1$.

The following constants are deliberately coarse:

$$60.212 < a < 60.2136, \quad 7.759 < A < 7.76,$$
$$32.13 < Q < 32.14, \quad .1494 < w := 9/a < .1495,$$
$$Q/A > 4.141, \quad H := A - G_{17}Q/17 > .65,$$
$$0 < K := A\sqrt{18} - Q < .8.$$

They follow by positive squares from $1.732 < \sqrt{3} < 1.7321$; for the last bound use $(Q+.8)^2 - 18a = 1.6Q + 9.64 - a > 0$. Positive actual clearing now uses the opposite inverse column:

$$\bar{c} = c + \sum_{i=1}^{16}(i/17)n_i = Ap - (Q/17)P_{17}, \qquad \bar{c}^2 = r = 9/17.$$

This leaves the true short normal untouched.

We shall repeatedly use one quadratic seal. For $s > 0, \rho > 0, \rho^2 > rs$, let $y_0 = \sqrt{\rho^2/s - r}$. If $\rho < .8\delta$ and $J > .31\delta$, then

$$H + (\rho/s)J - y_0 > \frac{13}{20} + \frac{31}{80s}\rho^2 - y_0 = \frac{31}{80}(y_0 - 40/31)^2 + \frac{8853}{42160} > 0. \tag{B.8.6}$$

For an isolated short opposite $d = Bp - \delta P_{18}$, put $\lambda = AB - Q\delta \geq 0$. Acuteness gives $\delta^2 \geq a/(a-9) > 1$, so own is dotted. The cleared Gram is $\left(\begin{smallmatrix} r & -\lambda \\ -\lambda & 1 \end{smallmatrix}\right)$. At $\lambda^2 = r$, $\bar{c} + \lambda d$ is a positive future all-wall feasible null. For $\lambda^2 < r$, including $\lambda = 0$, the full positive projection gives an actual finite point whose equations imply

$$k = w + \frac{Q\lambda}{a\delta} \geq w, \qquad k^2 < w, \qquad S_{17} + kx_{18} \geq Q/A - \sqrt{w - k^2}.$$

The two original sum bounds bound the left side by $3.760 + .07k$. For $k \geq w$, rationalization gives

$$\sqrt{w - w^2} - \sqrt{w - k^2} \geq \sqrt{w}\,(k - w) > .07(k - w) \quad (k > w).$$

Hence the gap is minimized at $k = w$. There it exceeds $4.141 - .357 - 3.760 - .07(.15) = .0135$, since $w - w^2 < .1495 - .1495^2 < .357^2$. Every subcritical mutual is impossible. For $\lambda^2 > r$ the positive actual negative core is $\bar{c} + \lambda d$; its square is $r - \lambda^2$, selected products are $r - \lambda^2, 0$, its time is positive, and all unknown signs hold termwise. Its required pressure gap is $H + \lambda(B - G_{18}\delta) - \sqrt{\lambda^2 - r}$. Acuteness gives $B/\delta \geq Q/A > 4.141$, hence $J = B - G_{18}\delta > .31\delta$; also $\lambda < K\delta < .8\delta$. Equation (B.8.6) with $s = 1, \rho = \lambda$ contradicts it on the entire supercritical range.

Finally consider dotted far-three short: $d = Bp - \delta P_{18} - e_1/\sqrt{2}$, $\delta > 1$. Its mutual magnitude is $\lambda = AB - Q\delta - (g + 1/2)$. It is strictly positive on this whole range. Its derivative is positive by the square comparison

$$18(a-9)\delta^2 + \sqrt{2}(a-9)\delta + 9a + \tfrac{9}{2} > 0,$$

and at $\delta = 1$ the time/spatial square gap is $25/4 + 15g - \sqrt{2}(1+g) > 16.50395$. Clear d by the far inverse column to obtain

$$\bar{d} = Bp - (\delta + v_0)P_{17} - \delta e_{18}, \quad \bar{d}^2 = r, \quad \rho = -\langle \bar{c}, \bar{d} \rangle = \lambda + 1/34, \quad v_0 = 1/(17\sqrt{2}).$$

The offset is now $1/34$, not $8/17$. If $\lambda = 1/2$, then $\rho = r$ and $\bar{c} + \bar{d}$ is a forbidden positive future feasible null. The actual native gap then forces $\lambda \geq 1/\sqrt{2}$ and $\rho > r$. Use $\Xi = \bar{c} + (\rho/r)\bar{d}$. It has

square $r - \rho^2/r < 0$, positive time, selected products that same negative square and zero, and zero old A_{16} products. Every unselected original sign holds termwise. The necessary same-p gap is

$$H + (\rho/r)J - \sqrt{\rho^2/r - r}, \qquad J = B - G_{17}v_0 - G_{18}\delta.$$

The actual identity $\rho = AB - Q(\delta + v_0)$ implies $J > .31\delta$. Moreover $B < \sqrt{18}\delta + 1/6$ and $Qv_0 = g + 8/17 > A/6$, so $\rho < K\delta < .8\delta$. Equation (B.8.6) with $s = r$ excludes the entire remaining range, including arbitrarily large dotted mutuals. Thus no short opposite survives near-three/own-six.

B.8.12 A central A_{16} lemma with a rank-two spectator

The no-left terminal roles will lead to three types of actual $A_{16} + I_2(m)$ neighbors. We prove one local lemma with the precise three comparison regimes:

spectator	comparison factor κ	conclusion on the same local A_{16} projection
$I_2(3)$	$\kappa < 67/50$	$e_A > 1/2$
$I_2(m),\ m \geq 4$	$\kappa < 3/2$	$e_A > 7/16$
$I_2(5)$	$\kappa < 157/100$	$e_A > 15/32$.

Let the actual vertex be q, write $z = E_q(q - y)$, and suppose every original chamber point has height at most κE_q. Only the actual opposite b in central direction 8 is needed. In zero-sum $\mathbb{R}^{17}$,

$$\Omega_i = \sqrt{2}(P_i - iP_{17}/17), \quad D_i = 2i(17 - i)/17, \quad C_{ij} = 2i(17 - j)/17\ (i \leq j).$$

Put $D = D_8 = 144/17$, so $2.91 < \sqrt{D} < 3$ and $\sqrt{D - 1} > 2.73$. Write $b = Hq - v_A - v_S$, own $\gamma = -\langle b, m_8\rangle \geq 0$, and whole-spectator cost $Q = \|v_S\|^2 < 1$.

The complete foreign alphabet follows from the retained dihedral inverse. Two contacts make a nonpositive triangle. For one finite contact ϵ the cost is $\epsilon^2/\sin^2(\pi/m)$, and finiteness requires $\cos^2(\pi/m) + \epsilon^2 < 1$. Thus

m	Q
3	$0,\ 1/3,\ 2/3,\ \phi^2/3$
4	$0,\ 1/2$
5	$0,\ (\phi + 2)/5$
≥ 6	0.

In particular the genuine H_3 cost for $m = 5$ is $(\phi + 2)/5 = (5 + \sqrt{5})/10 > 7/10$. The whole block is cleared by its positive inverse on actual roots.

Deleting 8 retains A_7 and A_8. The complete finite-white overlist is one three at $j = 1, 2, 3, 5, 6, 7, 9, 10, 15, 16$; one four at $j = 1, 7, 9, 16$; or two end-threes, one from $1, 7$ and one from $9, 16$; isolation is also allowed. The left $3, 5$ positions are the E_8 additions, $2, 6, 10, 15$ are D additions. Two contacts on one path make a cycle. A joined rank-sixteen tree is classical and has no new internal four. The exceptional finite classification gives no other extension of either retained full path. Every nonempty white cost is at least $7/16$. Thus its retained Schur remainder is at most $9/16$; for $m = 5$ any white contact forces $Q = 0$.

Here is the complete own-zero boundary. For an endpoint-three the time square is $Q - 9/17$, but retention requires $Q < 9/16$. The displayed alphabets allow only $0, 1/3, 1/2$ under that bound, all too small. For next-to-end three the retained remainder is at most $1/4$, hence $Q = 0$ and time square $-2/17$. For endpoint-four it is at most $1/8$, hence $Q = 0$ and time square $-1/17$. Every other internal-three has a positive affine principal: E_8 with arms $1, 2, 5$ at global $3, 14$, or E_7 with

arms $1,3,3$ at $4,\dots,13$. Their kernel coefficients are respectively center 6, arms 3; $4,2$; $5,4,3,2,1$, and center 4, arms 2; $3,2,1$; $3,2,1$. Interior-four gives the $4,3,3$ star with center $\sqrt{2}$, four-leaf 1, simple leaves $1/\sqrt{2}$. Two whites give an all-three cycle with coefficients one. With no white contact the time square is $Q-1<0$. Every null is a positive actual combination, with positive cap time, selected products zero and every other original product nonpositive. Compactness excludes all of them.

For positive own, an own-three together with white-three closes an actual all-three cycle through 8. For a pure far-three helper, own-four gives the $4,3,3$ star at 8, own-six gives the null $\gamma b+m_8+m_7/2$, and parallel gives $b+m_8$. The same own nulls will be used in isolated cases below. The remaining ordinary edge ratios are obtained from the exact formula

$$R=\frac{C_8}{\sqrt{D}}=\frac{\gamma+\beta C_{8j}/D}{\sqrt{1-\beta^2(D_j-C_{8j}^2/D)-Q}}.$$

For two whites add both numerator terms; their retained cross is zero. We list bounds for every ordinary position, leaving only isolation and pure far-three as separate cases:

white position	numerator lower bound	denominator upper bound before foreign cost
near three left/right	$.7+7/16,\ .7+4/9$	$3/4,\ \sqrt{5}/3$
D left/right	$.7+1/8,\ .7+1/9$	$1/2,\ \sqrt{2}/3$
E left	$.7+3/16$	$1/4$
four left/right	$.5+1/(8\sqrt{2}),\ .5+1/(9\sqrt{2})$	$1/\sqrt{8},\ 1/3$
two threes	$.5+17/144$	$\sqrt{17}/12.$

Smaller owns in these rows are the actual cycle/affine boundaries already disposed of. For a far-three with $Q>0$, the $m=3$ alphabet gives $Q\ge 1/3$; the residual is smaller than $.56^2$ and the numerator exceeds .864. For $m\ge 4$, positive $Q\ge 1/2$ makes this bound stronger. Thus in the first two comparison regimes every ordinary row has $R>3/2$, and $C_8>4.36$. The directional floor is then

$$(C_8-\kappa)/\sqrt{C_8^2-1}>\begin{cases}(4.36-1.34)/4.244>.71,\\(4.36-1.5)/4.244>.67.\end{cases}$$

Its square exceeds $1/2$ or $7/16$, respectively.

For the third regime, white contact already forces $Q=0$. One near-three/own-four needs an actual mate, rather than an incorrect null shortcut. If $v_A=\Omega_8/\sqrt{2}+\Omega_7/2$, delete actual 7; the full finite Gram is $A_6+B_{10}+I_2(5)$. For $\Omega_9/2$ instead, delete 9 and obtain $B_9+A_7+I_2(5)$. Their positive actual projections are feasible against every original wall. The exact edge data are

j	σ_j	Δ_j
7	$1/2+9/(10\sqrt{2})$	$1/10$
9	$1/2+8/(9\sqrt{2})$	$1/9.$

As $\sqrt{D_j}>14/5,\ \sigma_j>9/8,\ \sqrt{\Delta_j}\le 1/3$, their cosines exceed $189/20>9$. The new demand is $>(9-1.57)/\sqrt{80}>.82$, more than sufficient. An own-four with no neighboring single-three has the star null; two-white or white-four roles use their ordinary bounds. Every remaining white-three has own at least five, $>.809$. The same ratio formula gives: near-three $>(\phi/2+7/16)/(3/4)>1.662$; $D>(.809+1/9)/(.5)>1.84$; $E>(.809+3/16)/(.25)$; far-four at least $\sqrt{2}+1/4>1.664$ on the

left and $3/2+1/(3\sqrt{2})>1.7$ on the right; two whites $>(.7+17/144)/.344>2$. Thus all these rows have $R>1.66,\ C_8>4.83$, and demand

$$(C_8-1.57)/\sqrt{C_8^2-1}>(4.83-1.57)/4.726>11/16.$$

Its square is $121/256>15/32$.

It remains to cover isolated A contacts, retaining every permitted foreign cost. Put $\tau=(1-Q)/\gamma^2$; then $H^2=\gamma^2(D-\tau)$. Own three, four, six and parallel have the positive actual nulls already listed, with own-three using the E_7 star at 8. For $0<\tau<1$, positive clearing of the whole spectator gives

$$\Xi=b+w_S+\gamma\tau m_8,\quad U=\gamma(\Omega_8-\tau m_8),\quad \Xi^2=-\gamma^2\tau(1-\tau),\quad U^2<H^2.$$

It has zero cap/foreign products, negative own product, positive time and every unknown sign termwise. The comparison gives $\langle y_A,U\rangle\geq H-\kappa\gamma\sqrt{\tau(1-\tau)}$. If $e_A\leq 1/2$, Cauchy instead forces

$$(H-\langle y_A,U\rangle)/\gamma>(1-1/\sqrt{2})\sqrt{D-\tau}>.2928(2.73)=.799344>\kappa/2,$$

a contradiction for every $\kappa\leq 1.57$. Thus this whole continuous interval gives the stronger $e_A>1/2$ in all regimes. The boundary $\tau=1$ is an actual null.

For $\tau>1$ use the actual positive finite point

$$q_0=q+\frac{H}{\gamma^2(\tau-1)}(b+w_S+\gamma m_8).$$

Its selected products vanish, its norm and q product are $-1-H^2/[\gamma^2(\tau-1)]$, and every unknown wall sign holds by positive actual coefficients. Normalization gives exactly

$$h_8\geq\sqrt{\frac{D-1}{D-\tau}}\,\{\sqrt{D-1}-\kappa\sqrt{\tau-1}\}.$$

For $Q>0$, the native gap gives $\tau-1\leq 5-3\phi<(5/12)^2$. For pure finite own at least eight use the sharper $\sqrt{\tau-1}\leq\tan(\pi/8)<.415$. In the first regime the absolute loss in dual height satisfies

$$\sqrt{D}-h_8<.18+1.34(.415)=.7361<.74.$$

Since $\sqrt{D}>2.91$, normalization gives

$$h_8/\sqrt{D}>1-.74/2.91>.74,$$

and hence energy $>.5476>1/2$; the same bound covers the foreign case with its $\sqrt{5-3\phi}<.383<.415$. In the second, $h_8>2.73-1.5(.415)=2.1075$, whose normalized value is $>2/3$, giving energy $>4/9>7/16$. In the third $h_8>2.73-1.57(.415)=2.07845>2.07$, giving normalized value $>.69$ and energy $>.4761>15/32$.

Pure own-five has the actual H_4 correction $\gamma(3m_8/2+m_7+m_6/2)$, with $\delta_5=(7-3\sqrt{5})/16<1/36,\ H^2>4,\gamma>.8$. Its actual finite point yields

$$h_8>\sqrt{D-\gamma^{-2}}-\kappa\sqrt{\delta_5}\sqrt{1+\delta_5/H^2}/\gamma.$$

For $\kappa\leq 1.34$ its loss is $<25/76+1.34(1.004)/(6(.8))<.611<.74$, so energy $>.5476$. For 1.5, the first radical is >2.5 and the last square root <1.01, giving $h_8>699/320$, enough for $4/9$. For 1.57, the first radical is >2.62, the loss $<15857/48000$, and $h_8>2.07$. Pure seven has the actual core

$b + (4\gamma/3)m_8 + (2\gamma/3)m_7$, whose norm and cap product are $1 - 4\gamma^2/3 < 0$, with old products zero. The cubic $8u^3 - 4u^2 - 4u + 1$ is increasing on $[.89, 1]$; its signs at $.9, 10/11$ give $.9 < \gamma < 10/11$. Thus $\gamma^{-2} < 100/81$ and $\sqrt{4/3 - \gamma^{-2}} < 9/25$. The pressure is

$$h_8 > \sqrt{D - \gamma^{-2}} - \kappa\sqrt{4/3 - \gamma^{-2}}.$$

For the three regimes use respectively loss $< .25 + 1.34(.36) < .74$, first radical > 2.6, or first radical > 2.68. The latter gives $h_8 > 2.1148 > 2.07$. All required isolated bounds follow.

Finally consider every pure far-end-three helper, including its unbounded dotted range:

$$v_A = \gamma\Omega_8 + \Omega_e/2, \quad e = 1, 16, \qquad H^2 = D\gamma^2 + C_{8e}\gamma - 9/17 > D\gamma^2, \quad C_{8e} \geq 16/17, \quad \gamma > .8.$$

Own three, four, six and parallel have the actual cycle/star/chain nulls above. For five, the actual correction $\gamma m_8 + (2/3)m_e + (1/3)m_j$, where $j = 2, 15$, has residual $\delta = 2/3 - \gamma^2 \in (0, 1/36)$. The positive finite point gives $\langle y_A, U\rangle > H - \kappa\sqrt{\delta}\sqrt{1 + \delta/H^2}$, with $U^2 = H^2 + \delta$. Even at $\kappa = 1.57$ the loss is $< 15857/60000 < 2/7$; as $H > 2$, $U^2 < (10/9)H^2$, this gives $e_A > 162/245 > 1/2$. For finite owns at least seven let $d_0 = 1 - \gamma^2 \in (0, .19)$. The core $b + \gamma m_8 + 2d_0 m_e$ has square $-d_0(1 - 4d_0)$, zero cap/own products, and helper product $-1/2 + 2d_0 < 0$. Its radical is at most $1/4$, while $H > 2.5$; the demand is $> H - 157/400 > .84H$, giving $e_A > 441/625 > 1/2$. For dotted $\gamma > 1$, the reciprocal core $b + m_8/\gamma$ has square $-(1 - \gamma^{-2})$. Its radical divided by γ is at most $1/2$. As $H > \gamma\sqrt{D}$, the demand/norm ratio is $> 1 - 1.57/(2\sqrt{D}) > .73$, giving $e_A > .5329 > 1/2$. All cores have positive actual coefficients and time; all selected products were checked, and all other original-wall products are nonpositive termwise. This finishes the three-regime local lemma without a new farthest choice, a helper cutoff on arbitrary caps, or an omitted foreign cost.

B.8.13 No-left caps: actual-neighbor recognition and all common cases

The remaining seventeenth opposite has the form

$$c = Ap - \sqrt{2}gP_{17} - \delta P_{18}, \qquad \delta = \cos(\pi/m), \quad 3 \leq m \leq 8,$$

with the own table in Appendix B.8.5, and

$$A^2 = 34g^2 + 34\sqrt{2}g\delta + 18\delta^2 - 1.$$

First distinguish actual common and distinct caps. The full Gram on old A_{17} and c has Schur residual $1 - 17g^2/9$. If $g = 0, 1/2, 1/\sqrt{2}$, this is positive; its positive full projection is the actual old short-edge endpoint. The actual seventeenth opposite is therefore also the actual short opposite, by uniqueness of the old edge endpoint. For $g = 1/\sqrt{2}$ the reciprocal short cosine is $18\delta + 17 > 9.81$, excluding these common-own-four roles. For $g \geq \phi/2$ the residual is negative, so the cap cannot also close the short edge. This is the complete distinction for the own values left by the table; no native product is applied to a wall paired with itself.

At $g = 0, m = 3$, the actual affine F_4 path $n_{15}, n_{16}, n_{17}, n_{18}, c$ has positive kernel $(1, 2, 3, 2\sqrt{2}, \sqrt{2})$, already excluded. For $g = \phi/2, m = 4$, positive clearing of the whole old A_{17} gives the actual negative core

$$\Xi = Ap - sP_{18}, \quad A^2 = (51\phi + 33)/2, \quad s = \sqrt{2}(18 + 17\phi)/36, \quad \Xi^2 = -(17\phi - 19)/36.$$

Its selected old products vanish, its cap product is its negative square, its time is positive, and every other original sign holds termwise. Its pressure implies

$$S_{18} > \{A - \sqrt{(17\phi - 19)/36}\}/s > 3.97,$$

using $A > 7.599$, radical $< .487$, $s < 1.79$. This contradicts the unconditional sum bound. Thus the only distinct-cap family has $m = 3$ and $\phi/2 \le g < 1.264$.

For each of these remaining families let q be the actual seventeenth endpoint. Its complete actual forest is $A_{16}(n_1, \dots, n_{16}) + I_2(m)(c, n_{18})$. The cap and short root have their original finite mutual $-\cos(\pi/m)$; the full Gram is positive. Its full positive actual projection checks every unknown wall. Put

$$C = \cosh d(p, q), \quad s = \sqrt{C^2 - 1}, \quad t = S_{17}/\sqrt{17}, \quad q = Cp - sP_{17}/\sqrt{17}, \quad E_q/E = C - st.$$

The reverse unit tangent belongs wholly to the dihedral component. Its pressure is

$$\frac{\beta - t}{1 - \beta t}, \qquad \beta = s/C. \tag{B.8.7}$$

It increases in β and decreases in t. Thus the reverse-direction energy and the A_{16} energy from Appendix B.8.12 are orthogonal projections of the same local vector.

For the distinct family $m = 3, g \ge \phi/2$,

$$C = \sqrt{136/3}\{g + 1/(2\sqrt{2})\} > 7.82, \qquad t < .912.$$

The function $C - .912\sqrt{C^2 - 1}$ increases here, so

$$E_q/E > 7.82 - .912(7.756) = .746528 > 50/67.$$

Also $\beta > .9917$, and (B.8.7) is $> .83$. The local $A_{16} + I_2(3)$ lemma gives $e_A > 1/2$, while the actual dihedral projection has energy $> .6889$. Their sum exceeds one, excluding every distinct cap.

Of the seven common cases left by the own table, five are $g = 0, m = 6, 7, 8$ or $g = 1/2, m = 4, 5$. Their exact forward cosine is

$$C = \sqrt{34}\,\frac{g + \delta/\sqrt{2}}{\sqrt{1 - \delta^2}}.$$

For $g = 0, m \ge 6$, $C \ge \sqrt{51} > 7.1$; for $g = 1/2, m \ge 4$, $C \ge \sqrt{68}$. Again $t < .912$, hence

$$E_q/E > 7.1 - .912(7.03) = .68864 > 2/3.$$

Now $\beta > .99$, so the reverse pressure is $> (.99 - .912)/(1 - .99(.912)) > .8$. The second local regime gives $e_A > 7/16$; the dihedral projection costs $> .64$. Their sum is $> 431/400 > 1$. This excludes all five common cases.

For the sixth case, $g = 0, m = 5$, we first improve the original comparison without adding a spent head energy. The actual cap is $c = Ap - (\phi/2)P_{18}$, with $A^2 = (9\phi + 7)/2$. If old 17 were strong the actual head would be native five. The distinct head and cap would then have mutual magnitude

$$M = \sqrt{7\phi + 2}/2 - (\phi + 1)/(2\sqrt{2}).$$

Rational squares from $1.618 < \phi < 1.6181$ and $1.4142 < \sqrt{2} < 1.4143$ give $.899 < M < .900$. But this lies strictly between the native-six and native-seven values, a forbidden actual mutual. Therefore $S_{17} \le f_{17}$, and $t < .901$. For its actual $A_{16} + I_2(5)$ neighbor,

$$C^2 = \frac{17(\phi + 1)}{3 - \phi} > 32.2 > 5.67^2, \quad \beta > .984,$$
$$E_q/E > 5.67 - .901(5.582) = .640618 > 100/157.$$

The reverse pressure is $> (.984 - .901)/(1 - .984(.901)) > .73$. The third local regime gives $e_A > 15/32$, while the orthogonal dihedral energy is $> .5329$. The sum exceeds $20033/20000 > 1$. The head gap was used solely to establish old weakness; no head payment was added to this neighboring energy.

The final common case is $g = 1/2, m = 3$. Here $c = Ap - P_{17}/\sqrt{2} - P_{18}/2$, with $A^2 = 12 + (17/2)\sqrt{2}$. The full positive old-$A_{17}$-plus-$c$ projection is the actual short endpoint q, and its whole incident diagram is A_{18}, not the $A_{16} + A_2$ diagram of the other endpoint. Its Schur residual is $19/36$, and

$$C = \frac{17 + 9\sqrt{2}}{\sqrt{19}}, \quad C^2 = (451 + 306\sqrt{2})/19 > 6.82^2, \quad t = S_{18}/\sqrt{18} < .903.$$

Consequently

$$E_q/E > 6.82 - .903(6.747) = .727459 > 8/11.$$

The reverse unit tangent is the positive normalized dual for the new A_{18} endpoint c. With $\beta > .989$, its pressure is

$$\langle y, u_{18} \rangle > \frac{.989 - .903}{1 - .989(.903)} > .8 > .78.$$

Every original chamber point has height at most $E < (11/8)E_q$. The exact hypotheses of Lemma 16.1 therefore hold at this actual whole-A_{18} vertex. That lemma is local and does not require q farthest. It excludes this last common case. Its endpoint threshold concerns the normalized dual direction; it is not a threshold on a raw signed coordinate. All original wall signs here come from the full positive finite projection.

B.8.14 The terminal composition

The exhaustive seventeenth inventory has three disjoint branches. A far-left-three helper has finite own five or at least seven, or dotted own $1 < g < 4/3$. Appendix B.8.6 excludes every finite own, and Appendices B.8.8–B.8.10 exclude every dotted own, every actual short-white profile and every zero/native/parallel/unbounded dotted mutual. The near-left-three branch has exactly own six and is excluded by Appendix B.8.11. The no-left/short-contact branch is exhausted by the distinct/common recognition and the seven common cases above. Every other white role and every isolated seventeenth opposite was removed before this split. It follows that an original farthest whole-B_{18} vertex cannot occur.

All arguments are geometric. Literal finite/affine classification, compact acute simplicity, inverse positivity and the actual unit-wall angle alphabet are the external mathematical inputs. The finite tables here are explicit rational/algebraic inequalities, not new enumeration outputs or runtime receipts. The one original farthest vector is retained throughout each original reference argument. A neighboring vector is introduced only with its derived comparison factor, and all its orthogonal energies are taken at that same actual neighbor. The whole-D_{18} case and the independent higher-dimensional exclusions are proved in the remaining parts of the article.

B.9 The earlier whole-D_{18} geometry

This appendix supplies the head, prefix and fork inputs used before the final dotted-head argument in the main text. We keep its geometric steps in dependency order. In particular an old loss .61 is used only to establish head isolation; the later loss .58 is proved afresh before it is used.

Fix the original interior point and farthest vertex

$$z = E(p - x), \quad p^2 = -1, \quad x \perp p, \quad \|x\| < 1.$$

The entire incident diagram is D_{18}, with actual unit normals

$$n_i = (e_i - e_{i+1})/\sqrt{2}\ (1 \le i \le 17), \qquad n_{18} = (e_{17} + e_{18})/\sqrt{2}.$$

For $P_i = \sum_{j \le i} e_j$, the duals are

$$\Omega_i = \sqrt{2}P_i\ (i \le 16), \quad \Omega_{17} = (P_{17} - e_{18})/\sqrt{2}, \quad \Omega_{18} = (P_{17} + e_{18})/\sqrt{2}.$$

Thus $x_1 > \cdots > x_{17} > |x_{18}|$; the last coordinate is signed. The fork duals have squares nine and mutual product eight, and $\langle \Omega_i, \Omega_f \rangle = i$ for $i \le 16$, $f = 17, 18$. All original reference arguments concern this one x.

We use the main actual-point and payment lemmas. A positive correction on actual normals retains every unknown original wall sign by acuteness. Every core below has positive time; its negative or null norm selects the future sheet. A spacelike corrected root is used only through its positive finite projected point. Every coefficient payment is made at its individual reference anchor before its pressure is used. A zero-sum residual need not be orthogonal to the reference; the general completed square retains that full cross.

B.9.1 An initial window and actual head isolation

Put $f_i^o = \sqrt{i} - .61/\sqrt{2}$. The generic prefix proof in Appendix B.5 applies through 13, with its actual M_{12} rank condition. It gives a complete own/white list there and nonadjacent weak sites; a weak cap is

$$c = Ap - \gamma\Omega_i - \Omega_1/2, \quad A^2 = 2i\gamma^2 + 2\gamma - \tfrac{1}{2}, \quad \phi/2 \le \gamma < 6/5,$$

with own five, finite at least seven or dotted. In particular a strong $K = 10, 11$ exists before either new tail. There are no spectators.

Deleting old 1 retains the full D_{17}. Finite classification leaves an isolated opposite $H = A_H p - \eta\Omega_1$, $A_H^2 = 2\eta^2 - 1$, or one additional simple bond at old 2, with $A_H^2 = 2\eta^2 + 2\eta$. A second contact makes a cycle; any other branch or four gives a nonfinite principal at this rank. In the isolated row owns zero, three and four have nonpositive time; six has its actual $6, 3$ affine chain, and parallel has $H + n_1$. In the nonisolated row own zero has zero time and own three an all-three triangle. Every remaining own is at least $1/\sqrt{2}$, and its actual edge has cosine $2\eta + 1$, so $x_1 > .643$.

We first extend the old-loss window through $14, 15$. Deleting these retains $A_{13} + D_4$ and $A_{14} + D_3$. There is at most one contact on each connected retained piece. A connected full completion has rank eighteen, so is classical; an exceptional finite diagram adds no bridge. The target ratio is

$$T_o(q) = 100/61 - (q - .61)/[q(2q - .61)] < 1.56 \qquad (q = \sqrt{28}, \sqrt{30}).$$

Isolation and left-only contacts use the same Schur formulas as the generic prefix. At isolated own five the left H_4 finite point has loss $< .147 + .174$; every finite own at least seven has loss

$$\frac{1.232}{\sqrt{28} + \sqrt{28 - 1.232}} + 1/\sqrt{12} < .408;$$

dotted loss is $< .096 + .5$. The actual zero/small-own nulls are the double-fork affine D chain, the four/simple star, the $6, 3$ chain and the parallel pair. For a left-only own zero, far-three has negative time, next-far-three and far-four have zero time; near-three/next-near-three give the affine two-fork D principal and near-four gives affine B. Positive own-three with a left-three closes the intervening cycle. All surviving left-only ratios other than the far helper exceed 1.59, 2, or $\sqrt{14}/2$. These leave exactly the displayed pure helper.

The complete new right-hand data are

i, right contact	σ	Δ
14, long leaf three	$\gamma + 1/2$	$1/2$
14, either fork three	$\gamma + 1/4$	$1/2$
15, middle three	$\gamma + 1/2$	$1/2$
15, end three	$\gamma + 1/4$	$5/8$
15, end four	$\gamma + 1/(2\sqrt{2})$	$1/4$.

At 15 an end-five H_4 is also allowed, with ratio > 2.99 already at own zero. The long/middle-three roles have own-zero double-fork affine principals and own-three triangles; all remaining owns give ratio at least $1+1/\sqrt{2}$. The fork/end-three roles have own-zero affine E_8, own-three cycles, own-four stars, own-six chains and the parallel null. Their only extra old-loss cases are own five at either index and finite own at least seven at 15. The end-four own-zero kernel has coefficients $(\sqrt{2}, 2, 2, 1, 1)$ along its actual cap, contacted fork, center, other fork and old 15. Bridges are only left-end-three joined to a D_4 leaf, or to the middle/end of D_3, with residuals $1/28, 1/30, 19/120$. Own zero closes an actual cycle; every positive own gives ratio > 1.96. This exhausts the finite list, including both forks.

We exclude the extra right cases without assuming head isolation. If old 5 is strong, a strong $K = 10, 11$ reference has energies $> .823735, > .8472225$, sums $< 2.732, < 2.886$, terminal capacity $> 1/7$, and fifth capacity $> .146$. Every possible actual head supplies a separately funded pure gain $> .0135$: isolated five by its H_4 point, finite at least seven by $H + n_1/\eta + 2(\eta^{-1} - \eta)n_2$, dotted by $H + n_1/\eta$, and nonisolated by $x_1 > .643$. The exact scalar seals for these rank-free points are proved in Appendix B.3.7 and the head calculation below; their fifth prices are at most .1, with zero sum on the first five. If old 5 is weak, the actual weak-five point and strong 6 give instead bare first-six energies $> .8870, > .9104$, with no additional head gain. For a pure tail vector V, equal to one through K, with $\langle x, V\rangle > G,\ V^2 < R$, a legal $1/7$ payment gives

$$\|x\|^2 > L_K + 2(G - f_K^o)/7 - (R - K)/49.$$

The extra head gain is included only in the already-strong-five row.

At own five set $\alpha = \sqrt{2}(\phi/2 + 1/4)$; $\alpha^2 = \phi + 5/8 > 56/25$. The positive actual left-H_4 correction is $(\phi/2)(3n_i/2 + n_{i-1} + n_{i-2}/2)$; its residual is $(5 - 3\phi)/8$. The projected point has normalized displacement flat through $i - 3$. It gives $G > 3.62, R = 14$ at 14, and $G > 3.74, R = 15$ at 15. The least strong-five totals exceed 1.009, and the bare weak-five totals are larger. For every finite own at least seven at 15, put $u = \gamma^{-2}$. Use both actual positive cores

$$c + \gamma^{-1} n_{15} + 2(\gamma^{-1} - \gamma)n_{14}, \qquad c + \gamma^{-1} n_{15} + 2(\gamma^{-1} - \gamma)n_{16}.$$

Their square is $-d,\ d = 7 - 3u - 4/u$, and their correction cross is $2 - u$. Averaging their spatial displacements gives $Y^2 = A^2 - 3 + u + 2/u < A^2$. Here $.9 < \gamma < 1$, $d \le 7 - 4\sqrt{3} < .0718$, $\alpha^2 > 2.645$, and $A^2 = 15\alpha^2 - 5/8$. Thus Y/α is flat through 13, with $G > 3.677, R = 15$. The two strong-five totals exceed $1.00519, 1.00508$; the bare weak-five totals exceed 1.054. All coefficients are positive and selected products nonpositive, so these are actual all-wall points or cores.

For adjacent hypothetical weak helpers with owns $a, b < 6/5$, the actual product is at least $\frac{1}{2} - ab + (a/b + b/a)/4 \ge 1 - ab > -11/25$. The native gap forces orthogonality. Clearing the retained left path gives $I - t\mathbf{1}\mathbf{1}^{\mathsf{T}}$, $t = (i-1)/(2i)$, positive with positive inverse. The two-helper mixed-point calculation of Appendix B.5 uses only this matrix, times > 2 and owns $> 4/5$; its weighted loss is $< 7/16 + 1/(4\sqrt{10}) < .61$. The retained right block at $14, 15$ is actual D_3. Hence helpers are nonadjacent through 15.

We next prove old 5 strong; no head isolation is yet used. If it is weak, 6 is strong and its actual point gives $x_1 > .5965$. A finite weak-five own gives the stronger $x_1 > .65$; its reference through $1, 6$ and a strong $K = 14, 15$ has energy > 1.00845, so the own must be dotted, $1 < a < 6/5$. With $a_0 = .5965$, the reference through $1, 6$, then every actual strong knot, has initial energy $> .7600299945$, capacities $P_1 > .312168$, $P_6 > .08806$, and terminal $P_{14} > .1361$. All omitted weak knots are nonadjacent. Their complete worst debit is

$$[(.013586)^2 + (.009296)^2 + (.006871)^2 + (.005344)^2]/2 < .000174.$$

The underlying squared increments satisfy $\sum_{j=6}^{13}(\sqrt{j+1} - \sqrt{j})^2 > .2117655$ and the next one $> .0172465$, giving raw energies

$$q_{14} > .97162, \qquad q_{15} > .988867.$$

These bounds follow directly by rational upper square bounds for $\sqrt{42}, \ldots, \sqrt{210}$, or the increment identity $d_j^2 = 2j + 1 - 2\sqrt{j(j+1)}$. At $a_0 = .65$ the initial energy is $> .7968668845$, which proves the finite-own exclusion above.

For the actual full dotted weak-five foot put

$$U_5 = \sqrt{2}aP_5 - a^{-1}n_5 + \Omega_1/2, \qquad T_5 = \sqrt{10a^2 + 2a - \tfrac{1}{2}} - \sqrt{1 - a^{-2}}.$$

Its pressure-minus-THIS-reference deficit is $> .1203$. To check the whole interval, the time derivative is < 3.245, the coefficient of a in $\langle w, U_5\rangle$ is > 2.4519, and the radical derivative is > 1, so the deficit decreases. At $a = 6/5$ use time > 4.03732, radical $< .552771$ and $\langle w, P_5\rangle < 1.733828$, giving $> .120355$. Pay $\mu = .07$, $r = .088$, and $q = \mu(\sqrt{2}a + 1/\sqrt{2}) - r \in (0, .081)$ at the separate P_1, P_6 anchors. For $Y_5 = \mu U_5 - qP_1 - rP_6$,

$$Y_5^2 = .0392a^2 + .0049a^{-2} - .04928\sqrt{2}a + .02892 < .005141.$$

Convexity reduces this bound to the two endpoints. The general completed square, retaining its full reference cross, gains $> .011701$. If 15 is strong this already gives $\|x\|^2 > 1.000568$.

If 15 is weak, 14 is strong and its pure helper has time $A_{15}^2 = 30g^2 + 2g - 1/2$. For finite own at least nine, the actual positive finite pair point has spatial foot $g(\Omega_{14} + \Omega_{16})/2 + \Omega_1/2$. For all $g > \cos(\pi/9) > .939$ its deficit exceeds

$$4.9529 - .939(4.68151) - .42179 = .13517211.$$

Indeed at .939, the exact time and pair normalization give $H > 4.9529$; the deficit increases since $H' > \sqrt{30}/1.066 > 5.13$. Its tail residual has square < 1. Paying .0962 costs $< .136049$ at P_{14} and $< .069$ at P_1, no P_6; its tail is orthogonal to Y_5. Its gain $> .016752673964$, together with the previous one, gives $\|x\|^2 > 1.00007367$. For dotted 15, weakness forces $1 < g < 10/9$; the positive reciprocal core has tail square < 1.28 and deficit $> .16613$. Its .086 payment fits the same two anchors, has no P_6 price and gains $> .0191$. The entire finite and dotted tails are covered by these derivative and endpoint bounds, not by a finite sample.

For owns five, seven and eight do NOT insert the weak-five foot. Their own tail points suffice on raw q_{14}:

own	actual correction	deficit lower bound
5	$gn_{15} + (2/3)n_1 + (1/3)n_2$	.3776769
7	$gn_{15} + n_1/2$	.28605949
8	$gn_{15} + n_1/2$	.232552911.

The first row is the actual H_4 finite point; the other two are negative cores. Their spatial feet are $g(\Omega_{14}+\Omega_{16})/2+\Omega_3/6$ or $g(\Omega_{14}+\Omega_{16})/2+\Omega_2/4$. The native brackets $.809<g_5<.81$, $.9009<g_7<.901$, $.9238<g_8<.9239$, time/radical pairs $4.5554/.111$, $5.0646/.249$, $5.1913/.322$ and $\langle w,\Omega_3/6\rangle<.2747, \langle w,\Omega_2/4\rangle<.3115$ give the table by direct subtraction. The residual squares are <1, including the initial variances $1/15, 1/10$. Payment .09 costs $<.118$ at P_{14} and $<.009$ at P_6, and gains $>.0337$. No weak-five foot is present, so there is no duplicate initial debit. Thus old 5 is strong in every case.

A strong $K=14,15$ now has the ordinary old-loss reference energy $>.90715$, first-five mean $<.361$ and fifth capacity $>.146$. A nonisolated head would give $x_1>.643$, and its legal five-block coordinate graft gains $>(5/4)(.643-.361)^2>.0994$, contradicting the norm budget. The head is therefore ACTUALLY isolated, with own five, finite at least seven, or dotted. This conclusion did not use any later .58 statement or center/fork hypothesis.

B.9.2 The fresh .58 window, with the two changed tail rows

From now on put $f_i=\sqrt{i}-.58/\sqrt{2}$. The head isolation just proved remains valid; old .61-strength is not silently called new strength. At $5\le i\le 13$, deleting i retains $A_{i-1}+D_{18-i}$. The complete white list is isolation; a left endpoint/D/E three or end-four; right long-three or permitted fork-three; or the full endpoint D_{18} bridge. Every left-only own-four has an ACTUAL affine B principal: coefficients $\sqrt{2}$ on its cap, 2 on the old chain $i,\dots,16$, and 1 on both forks. Any extra left contact preserves all outside signs. It is a forbidden positive future null, not a role declared strong by an inadequate edge ratio.

The new ratio threshold is

$$T(q)=50/29-(q-.58)/[q(2q-.58)].$$

It is increasing, with $T<83/50$ for $q<7$, $T<801/500$ for $q<15/4$, and $T<159/100$ for $q<16/5$. For near-three/own-five the ratio $[i(\phi+1)-1]/\sqrt{2i(i+1)}$ increases with i, exceeds $801/500$ at six and $83/50$ at eight. At five every own at least seven is stronger. D-position ratio is at least $(i\phi/2+1)/\sqrt{2i}$, already $>159/100$ at five and $>2927/1750$ from six. E, end-four, long-three and bridge rows are larger. Right-fork-three at retained D_5, D_6, D_7 has residual at most $3/8$ and numerator $>\gamma+1/4$; surviving $\gamma\ge\phi/2$ gives ratio $>83/50$. The actual affine/time small-own boundaries are unchanged.

The new near-three/own-five at old five is absolutely impossible: its actual right-H_4 point has displacement Y, with $\langle x,Y\rangle>18/5$, $Y^2<281/20$, $\sum Y_j<43/5$. The independently rank-valid OLD strong $11,12$ references precede this new window. Cauchy for $Y+(3/4)P_K$ gives strict gaps $.0401, .78625625$. Isolated five/seven/eight-or-larger have losses $<.58$, including the exact singleton maximum $\sqrt{10}+\sqrt{2}-4<.578$. For isolated dotted $i\ge 6$, reciprocal-core loss is $<a_iu+\sqrt{u(1-u)}$, with $a_i<37/250$; its circle bound is $<.5795$. Only isolated dotted five remains in addition to the pure helper, whose weak own range is again $\phi/2\le\gamma<6/5$. The cutoff follows from $R^2\ge 4\gamma-2$; it is not a cutoff on arbitrary fork opposites.

At fourteen/fifteen retain the COMPLETE white table above. Now $T(\sqrt{30})<1.64$. Isolation, ordinary left-only rows, bridges, long/middle-three, end-four and end-five are still strong by their displayed exact ratios. Own-four is removed by the actual affine principal, not that comparison. The previously excluded own-five and finite-end-three cases remain absolute contradictions, independent of the numerical threshold. The two genuinely new weak candidates are:

$$\begin{array}{r|l} 14,\text{fork-three} & \gamma=\cos(\pi/7) \\ 15,\text{end-three} & 1<\gamma<1.05. \end{array}$$

The second cutoff is $1.64\sqrt{5/8} - 1/4 < 1.05$. No formerly strong dotted interval has been discarded without rechecking.

Bulk nonadjacency supplies a strong $K = 11, 12$ independently of these two candidates. In the strong-five branch the raw energies are $> .8626, > .8843$, fifth capacity $> .151$, and terminal capacity $> 1/7$. Every isolated head gives a pure separate gain $> .0155$: five has $P_4/4$ pressure $> .39307$ and .56 payment, with gain $> .0155344$; other heads have actual sum-one foot Z, with $.8h - .16Z^2 > .2893$, and .4 payment at fifth price .08, gaining $> .02914$. In the weak-helper-five branch $x_1 > .5455$ and strong six give initial energy $> .74385$, stronger than the following bare weak-five bounds. For isolated dotted weak five put

$$U' = P_5 + (u/2)(e_6 - e_5), \quad .4 < u < 1, \quad \langle x, U' \rangle > 1.8185.$$

Choose strong $J = 6, 7$. Its zero-sum residual has norm at most $77/150, 194/175$ and deficits $> .119, > .221$. For these EARLY references pay .19 at $J = 6$ and .2 at $J = 7$. The first debit $19/120$ fits $> .15834$; the old .21 debit would not fit at $K = 11$. The two actual gains exceed $.02668, .04405$. First-J energies $> .69312, > .7139$ and exact straight tails $(\sqrt{K} - \sqrt{J})^2/(K - J)$ give, in every weak-five branch,

$$q_{11} > .8697, \qquad q_{12} > .8912.$$

No additional head gain is charged in those rows.

For the native-seven fourteen candidate use both actual side cores, on side 13 or 15, as above. Let $u = \gamma^{-2}$, $\alpha = \sqrt{2}(\gamma + 1/4)$. The complete rational seal is

$$1.2318 < u < 1.2322, \quad 2.64914162 < \alpha^2 < 2.649602, \quad d = 7 - 3u - 4/u < .05736 < .2395^2.$$

Averaging the displacements with the FULL correction cross gives $Y^2 = A^2 - 3 + u + 2/u$. As $A/\alpha > 3.7163$, $\sqrt{d}/\alpha < .1472$, and $3.5 - u - 2/u > .6438 > .24\alpha^2$, the normalized foot is flat through twelve, with $G > 3.569$, $R < 13.76$. The $1/7$ tail payment gives bare weak-five totals $> 1.0025, > 1.0021$; strong-five plus its separate head gain gives $> 1.0109, > 1.0107$.

For the dotted fifteen candidate use the positive actual core $c + \gamma n_{15}$, with norm and cap product $1 - \gamma^2 < 0$, own product zero, positive time and all other original signs termwise. Here $\alpha = \sqrt{2}(\gamma + 1/4)$, $A^2 = 15\alpha^2 - 5/8$. Its normalized displacement is flat through fourteen and satisfies

$$A/\alpha > \sqrt{14.8} > 3.847, \quad \sqrt{\gamma^2 - 1}/\alpha < .182, \quad R < 15.$$

Thus $G > 3.665$, throughout $1 < \gamma < 1.05$. The same legal payments give bare totals $> 1.0046, > 1.0042$ or strong-five-plus-head totals $> 1.0130, > 1.0128$. Both candidates are impossible. The fresh window through fifteen and helper nonadjacency are proved.

Consequently some $K = 14, 15$ is strong. The all-hole references now give effective energies $> .9231, > .94006$, sums f_K, and terminal capacity $> 1/8$ in all initial cases. The weak-five isolated point may now use the later legal .21, .2 payments; they are not added to the earlier versions in the same norm budget. For either genuine fork put $V = P_{17} \pm e_{18}$, $V^2 = 18$. If its actual cosine were at least twelve, its pressure would exceed 3.902; a $1/8$ terminal payment gives

$$.9231 + (3.902 - 3.332)/4 - 4/64 = 1.0031, \qquad .94006 + (3.902 - 3.463)/4 - 3/64 = 1.002935.$$

Thus BOTH actual fork cosines are less than twelve, without assuming strong five, strong center sixteen or a sign for x_{18}.

B.9.3 The exact center-sixteen inventory

Deleting actual 16 retains A_{15} and BOTH orthogonal fork normals. Write an actual center opposite as

$$c = Ap - g\Omega_{16} - \eta\Omega_j - \delta\Omega_{17} - \epsilon\Omega_{18},$$

omitting absent terms. Finite classification at the actual endpoint gives exactly the following white possibilities:

left contact	fork contacts
none	none; one arbitrary finite label; or $(3,3),(3,4),(3,5)$
endpoint three at $1,15$	none; one three/four; or both threes
next-three at $2,14$	none
endpoint four at $1,15$	none.

Two left contacts make a cycle. A fork added to a next-three position has excessive arms $1,2,13$; an internal four is not a long finite classical tree. The full two-path join has rank eighteen, beyond every exceptional type. The three-node fork list explicitly retains H_3, with labels $3,5$. Thus no field restriction has silently removed a white possibility.

The exact retained Gram gives

$$q_L = 2\eta^2 j(16-j)/16, \quad \Delta = 1 - q_L - \delta^2 - \epsilon^2 > 0, \quad \sigma = g + \eta j/16 + (\delta+\epsilon)/2,$$

and $A^2 = 32\sigma^2 - \Delta$, $C_{16} = \sqrt{32}\sigma/\sqrt{\Delta}$. The fresh target satisfies $T(\sqrt{32}) < 1.641$, by positive multiplication using $5.656 < \sqrt{32} < 5.657$.

An isolated center has owns zero/three/four/six/parallel excluded by time, the four-leaf-three star, the four/simple star, the $6,3$ chain and $c + n_{16}$. The left-H_4 point at five gives $\langle x, \Omega_{16}\rangle > \sqrt{32 - g^{-2}} - .2 > 5.27$. Finite owns at least seven have loss $< .12 + 1/\sqrt{12} < .409$; dotted owns have loss $< .09u + \sqrt{u(1-u)} < .5475$. Thus every isolated center is strong. For left-only own zero, the far-three time is negative, far-next-three and far-four have zero time; the near-three/next-three affine D and near-four affine B kernels use both actual forks. Positive own three closes an all-three cycle, four gives the four/simple star at 16, six the $6,3$ chain and parallel a null. All surviving endpoint-four, next-three and near-three ratios exceed 1.641. The only no-fork weak role is

$$c = Ap - \sqrt{2}gP_{16} - e_1/\sqrt{2}, \quad a = A^2 = 32g^2 + 2g - \tfrac{1}{2}, \quad \text{own five, finite } \geq 7 \text{ or dotted}, \quad g < 6/5.$$

Every left/fork bridge is strong or affine. Endpoint-three plus one fork-three has residual $9/32$; own zero closes the long cycle, own three a triangle, and remaining $g \geq 1/\sqrt{2}$ gives ratio > 1.8. Endpoint-three plus one fork-four has residual $1/32$ and even its smallest own-zero ratio exceeds two. Both fork-threes with the left contact have the same residual and smallest numerator $17/32$. With no left contact, the pair $3,3$ has own-zero four-cycle, own-three triangle and all remaining ratios > 1.7. The pair $3,5$ already has own-zero ratio $\phi^3/2 = \phi + 1/2 > 2.1$. The pair $3,4$ has positive-own ratio > 2.2; at own zero its actual reciprocal fork endpoint is the full B_{18} positive projection, with cosine $8 + 9\sqrt{2}/2 > 12$, impossible.

For exactly one fork with finite contact b, own-three with fork-three is affine. With fork-four/five its actual reciprocal fork endpoint is D_{18}, of cosine $8 + 9b > 12$. Higher white labels give a direct ratio > 1.8. Own four, six and parallel have the same actual star/chain/null, using the UNCONTACTED fork in the star. For $g \geq \phi/2$, every fork label at least four has ratio at least $\phi/\sqrt{2} + 1/2 > 23/14 > 1.641$. For fork-three/own-five the actual left-H_4 finite point gives a normalized foot flat through thirteen, with $G > 3.85, R < 16$. The independent strong $K = 11, 12$

references have energies $> .8626, > .8843$ and prices $> 1/7$, so the two legal norm bounds exceed $1.0299, 1.0298$. Thus the only positive-own nonpure weak role is single-fork-three, with finite own at least seven or dotted and $g < 47/40$. The cutoff follows from $1.641\sqrt{3}/2 - 1/4 < 47/40$.

At own zero, single-fork-three/four/six have the actual E_8, four/simple and $6, 3$ nulls. Single-fork-five uses the actual H_4 on $c, f, 16, o$, with positive correction $b(3n_f/2 + n_{16} + n_o/2)$. Its spatial displacement is exactly $(b/\sqrt{2})P_{15}$. The finite point gives

$$S_{15} > \sqrt{2 + 8\phi} - (\sqrt{5} - 2) > 3.628.$$

This makes fifteen strong; upgrading its actual floor in the reference gives

$$\|x\|^2 > .95678 + 2(.1313)(.165) + .165^2/15 > 1.$$

For every remaining single-fork own-zero cap, the fork length below twelve leaves only finite white labels seven through ten. Consequently the COMPLETE weak-center list is:

pure far-three	finite five/at least seven, or dotted
one fork, own zero	white label $7, 8, 9, 10$
one fork-three, positive own	finite ≥ 7 or dotted $1 < g < 47/40$.

Every boundary was dispatched before this necessary list was formed.

B.9.4 Raw references, legal omissions and the center ellipse

We give the reference construction independently of any selected weak-center hypothesis. Whenever fifteen is strong, interpolate through actual strong knots, omitting every weak knot by its neighbor chord. Use the exact increment lower numerators, at denominator 10^7, for $d_i^2 = (\sqrt{i+1} - \sqrt{i})^2$, $i = 5, \ldots, 16$:

$$455488, 385186, 333704, 294372, 263340, 238230, 217494, 200080, 185248, 172465, 161332, 151549.$$

They follow from $d_i^2(a - d_i^2) = 1$, $a = 4i + 2$, and $d_i^2 > (a^2 - 1)/[a(a^2 - 2)]$. Together with the initial energy $> .6668157$, they give $E_{14} > .9241299$, $E_{15} > .9413764$. The sum of all possible nonadjacent omission costs is $< 1/1024$. With a native-five head, its actual first-seven relations force five, six and seven strong; the sharper complete debit is

$$[(.0112)^2 + (.008)^2 + (.0061)^2 + (.0049)^2]/2,$$

giving $> .94125107$. These are squared-slope formulas for every allowed hole set, not a numerical sample of selected weak sets.

The actual native-five head H_4 foot is $P_4/4$, with pressure $> .39307$, square $1/4$. Its .56 zero-sum correction has fifth debit $.112 < .151$ and gain $> .0155344$. For any other isolated head, its rank-free actual sum-one foot has $.8h - .16R > .2893$, and .4 correction gives gain $> .02914$, at fifth debit $.08 < .151$. Their complete scalar proof is the head-foot proof in Appendix B.8.2; it concerns the identical actual first three/four long roots and is independent of the terminal type. When integrated, that shared proof belongs immediately after the generic prefix's four weak-five pressures, rather than being repeated in both classical terminal appendices.

The full endpoint-fifteen contracts are

initial/head case	L_{15} lower	initial price
strong five, native-five head	.95678547	.112 at P_5
strong five, other head	.96953	.08 at P_5
weak helper five	.97164	actual $(1, 6)$ reference
isolated dotted five, strong six	.96525	.4/6 at P_6
isolated dotted five, strong seven	.96016	.4/7 at P_7.

The weak-helper row uses $x_1 > .5455$, initial energy $> .74385$, and strong six; its improvement over $E_6 < .7126$ is $> .03125$. For the isolated dotted rows, the BARE curved energies are $> .9212053, > .9035167$, first means $< .3399, < .31938$, and capacities $> .1435, > .1366$. They use only the head foot, gaining $> .0440466, > .0566531$. The isolated-five point U' is not also paid in these rows. The native-five head is incompatible with weak isolated dotted five by its actual local mutual in $(0, 1/4)$.

Under the additional assumption that sixteen is weak and fifteen strong, put

$$d = 4 - \sqrt{15} \in (.1270, .1271), \quad \Delta = S_{15} - f_{15} \geq 0, \quad U = f_{16} - S_{16} \geq 0.$$

The common reference has effective energy $> .95678$ and terminal capacity $\beta \geq \sqrt{15} - \sqrt{14} > d$. Retaining that unspent surplus in ONE square gives

$$2\beta\Delta + x_{16}^2 + x_{17}^2 + x_{18}^2 < .04322, \quad x_{16} = d - U - \Delta.$$

Hence

$$x_{17}^2 + x_{18}^2 < .04322 - d^2 + 2dU - U^2.$$

For either signed $t = x_{17} \pm x_{18}$, ordering gives $0 < t < 2d < .2541$, while a direct quadratic comparison gives $t < .233 + 1.1U$: the difference of squares has constant $> .000107$, linear coefficient $> .0042$, quadratic coefficient 3.21. Thus, for $P > 0, z_0 \geq 0, z_0 \leq 10P$,

$$PS_{16} + z_0(S_{16} + t) \leq f_{16}P + (f_{16} + .233)z_0. \tag{B.9.1}$$

This is a consequence of one funded original reference, not a new independently optimized terminal energy.

B.9.5 Pure weak centers: one cleared Gram and its full scalar dispatch

Suppose the center is weak and its cap is the pure far-three helper. Own three/four/six/parallel were excluded in the inventory. If fifteen were also weak, its actual cap is another pure helper. The actual product is $> 1 - gh > -11/25$, hence zero by the native gap. Clearing actual A_{14} gives the positive two-cap Gram $\left(\begin{smallmatrix} 8/15 & -7/15 \\ -7/15 & 8/15 \end{smallmatrix}\right)$; both original forks remain orthogonal positive blocks. The actual mixed-point loss is $< 7/16 + 1/(4\sqrt{10}) < .58$, contradicting both weak floors. Thus fifteen is strong and (B.9.1) is available.

Put

$$a = 32g^2 + 2g - \tfrac{1}{2}, \quad k = 16g + \tfrac{1}{2},$$
$$P = k/(8\sqrt{2}), \quad r = 17/32,$$
$$\bar{c} = c + \sum_{i=1}^{15}(16 - i)n_i/16 = Ap - PP_{16}.$$

Then $\bar{c}^2 = r$ and $k^2 = 8a + 17/4$. A finite own has $.8 < g < 1$. A dotted weak own has $1 < g < 7/6$: the exact center cosine is $(32g + 1)/\sqrt{17}$, and at $7/6$ its ratio already exceeds $821/500 > T(\sqrt{32})$. Set $H = A - f_{16}P,\ K = 3A - k$. All intervals below have $H, K > 0$.

The seven actual fork white profiles and their lengths can be derived before center strength is

known. For either assignment $\{f,o\}=\{17,18\}$, $d=Bp-v$, own $\delta=-\langle d,n_f\rangle\ge 0$, they are

	v	B^2	C_f
I	$\delta\Omega_f$	$9\delta^2-1$	3δ
N	$\delta\Omega_f+\Omega_o/2$	$9\delta^2+8\delta+5/4$	$(18\delta+8)/\sqrt{19}$
N_4	$\delta\Omega_f+\Omega_o/\sqrt{2}$	$9\delta^2+8\sqrt{2}\delta+7/2$	$9\sqrt{2}\delta+8$
N_2	$\delta\Omega_f+\Omega_{16}/2$	$9\delta^2+16\delta+7$	$9\delta+8$
F	$\delta\Omega_f+\Omega_1/2$	$9\delta^2+\delta-1/2$	$(18\delta+1)/\sqrt{19}$
F_4	$\delta\Omega_f+\Omega_1/\sqrt{2}$	$9\delta^2+\sqrt{2}\delta$	$9\sqrt{2}\delta+1$
F_2	$\delta\Omega_f+\Omega_2/2$	$9\delta^2+2\delta$	$9\delta+1.$

Indeed deleting f leaves the full actual A_{17} path; its one-node finite list is isolation, either-end three/four, or either next-to-end three. For a white $\beta\Omega_j$, set $L=\langle\Omega_f,\Omega_j\rangle, J=\Omega_j^2$. The edge formula is

$$C_f=(9\delta+\beta L)/(3\sqrt{\Delta_f}),\quad \Delta_f=1-\beta^2(J-L^2/9),$$

and $(L,J)=(8,9),(16,32),(1,2),(2,4)$ gives the table. Both signs are actual, with $C_f<12$. All own labels are initially retained.

Near-four forces $\delta<1/2$, hence $\delta=0$; its acuteness square is $-a/2-17/8<0$. Near-next-three has $\delta<4/9$ and square difference $(\delta+1)[a(\delta-1)-(17/4)(\delta+1)]<0$. Both are impossible. For the other profiles clear only positive actual retained roots:

	v_0	$s=\bar d^2$	$\rho=-\langle\bar c,\bar d\rangle$
I	0	1	λ
N	$1/2$	$3/4$	λ
F	$1/16$	$17/32$	$\lambda+15/32$
F_4	$\sqrt{2}/16$	$1/16$	$\lambda+15/(16\sqrt{2})$
F_2	$1/8$	$1/8$	$\lambda+7/16.$

Here $\lambda=-\langle c,d\rangle\ge 0$ is the ACTUAL mutual magnitude, and $\bar d=Bp-[(\delta+v_0)P_{16}+\delta(e_{17}\pm e_{18})]/\sqrt{2}$. For N, add $n_o/2$. For the far rows add the positive A_{15} inverse column. The other fork is retained when cleared. No projected normal is called an actual facet.

If $\rho^2=rs$, $\bar c+(\rho/s)\bar d$ is a forbidden positive future feasible null. If $\rho^2>rs$, it is a negative core with square $r-\rho^2/s$, time $A+\rho B/s>0$, selected products that negative square and zero, and zero retained old products. Every unknown wall is nonpositive termwise. With $t=x_{17}\pm x_{18}$, its pressure is contradicted when

$$D=H+\frac{\rho}{s}\{B-v_0f_{16}/\sqrt{2}-(f_{16}+.233)\delta/\sqrt{2}\}-\sqrt{\rho^2/s-r}>0, \tag{B.9.2}$$

provided $z_0\le 10P_0$, where $P_0=P+\rho v_0/(s\sqrt{2}),\ z_0=\rho\delta/(s\sqrt{2})$. This funding condition precedes every use of the ellipse.

If $\rho^2<rs$, use the positive actual finite projection

$$q_0=p+\alpha\bar c+\chi\bar d=(1+T)p-V,\quad \alpha=(sA+\rho B)/(rs-\rho^2),\ \chi=(\rho A+rB)/(rs-\rho^2),\quad T=A\alpha+B\chi.$$

All coefficients are positive, its norm is $-(1+T)$, and its normalization satisfies every original wall. Its exact spatial direction and the SAME original maximum give the subcritical estimates below. Both zero mutual and singular boundaries are kept.

Here are the complete finite-own scalar constants. They follow by positive squares, $A'^2 - 32 = 17/a$, and $H' > .58$.

g interval	H lower	K upper
$(.8, .85]$	.4252	.695
$[.85, .9]$	.4542	.752
$[.9, .95]$	.4832	.809
$[.95, 1)$	.5122	.864.

Also $A/K > 20/3$ throughout. For dotted owns the corresponding complete table is

g interval	H lower	K upper	A/K lower
$(1, 1.05]$	.55235	.920	6.5
$[1.05, 1.1]$	.58135	.974	6.5
$[1.1, 9/8]$	.61035	1	6.5
$[9/8, 1.15]$	.62485	1.028	6.46
$[1.15, 7/6)$	.63935	1.05	6.4.

At a right endpoint the asserted K upper is proved by $(k + K_{\text{upper}})^2 - 9a \geq 0$. The five dotted gaps are $.0484, .037476, 0, .045184, 769/3600$; the four finite gaps are $.012025, .005104, .027081, .008496$. K increases, so these endpoint checks cover the intervals. For A/K, square k/A, using $k^2/a = 8 + 17/(4a)$. These are finite rational certificates for continuous ranges, not sampled bounds.

For isolation, acuteness forces $\delta^2 \geq a/(a - 17/4) > 1$. Positive subcritical mutuals are exactly $1/2, 1/\sqrt{2}$. Put $\zeta = 17/(4a)$ and $j = \zeta + k\lambda/(a\delta)$. The finite projection gives

$$S_{16} + jt \geq \sqrt{16 + 2\zeta} - \sqrt{2(\zeta - j^2)}, \quad j^2 < \zeta.$$

For finite g, put $F(\delta, g) = A\sqrt{9\delta^2 - 1} - k\delta$. It increases in both variables for $\delta > 1$. At $g = .8, \delta = 8/5$, its positive-square comparison with $\lambda = 1/2$ has gap $3137/2500 > 0$. Thus the solution δ_0 for $\lambda = 1/2$ satisfies $\delta_0 < 8/5$. For fixed g, the ratio $\lambda/\delta = A\sqrt{9 - \delta^{-2}} - k$ increases with δ, and the solution δ increases with λ. Consequently BOTH subcritical mutuals satisfy $\lambda/\delta > 5/16$, and hence $j - \zeta > (5/16)(k/a)$. The four finite-g strips therefore have

	ζ upper	$j - \zeta$ lower	S_{16} lower
1	.197	.180	3.611265
2	.175	.170	3.612025
3	.157	.161	3.61531
4	.141	.153	3.62223.

To reproduce each certificate, use $t < .255$, $\sqrt{16 + 2\zeta} > 4 + \zeta/5$; the resulting expression increases with j and decreases with ζ on that strip. The four upper radicals are $.332, .335, .335, .331$, whose squared radicands are $.109742, .11195, .111752, .109128$. For dotted g, the same monotonicity gives $\delta < 8/5,\ k/a > .41,\ \zeta < .127,\ j > \zeta + .128$; the same rationalized test gives $S_{16} > 3.608275$. All contradict weak $S_{16} < 3.59$.

For isolated supercritical mutuals use

$$3\delta - .172/\delta < B < 3\delta - 1/(6\delta), \quad \lambda < K\delta - A/(6\delta), \quad (f_{16} + .233)/\sqrt{2} < 2.7033.$$

Then (B.9.2) exceeds

$$H + (.2967/K)\lambda^2 + \{.2967A/(6K) - .172\}\lambda/\delta - \sqrt{\lambda^2 - r}.$$

For finite g, the actual $C_f < 12$ gives $\delta < 4$ and ensures the ellipse price; $\lambda/\delta > .4$ by the increasing positive-square test at $(g,\delta) = (.8,2)$. Completing one square gives lower gaps $> .1292, .0931, .0593, .0295$ on the four strips. For dotted g, first take $\lambda \le 4$. Monotonicity gives $\delta < 5$, so $z_0 < 20/\sqrt{2} < 10P$. The ratio lower bounds $.45, .55, .6, .7$, respectively on

$$(\lambda, g) > (\sqrt{r}, 1), (1,1), (1.2,1), (1.4, 1.15),$$

follow by contradiction from

$$\lambda^2 \le \frac{t^2 a}{9a - (k+t)^2} < \Lambda^2.$$

The last comparison reduces to the increasing quadratic $9a-(k+t)^2-(t^2/\Lambda^2)a > 0$. Its lower-bound signs are $> .7586/r, > .6637, .715, 1.89$. With these ratios and the dotted strips, one square yields positive lower gaps $> .0156, .0046, .0075, .0004, .0058$. The omitted small-$\lambda$ portions are decreasing up to $1, 1.2, 1.4$; at those endpoints their gaps exceed $.26, .28, .153, .059$. For $\lambda \ge 4$ no refined ellipse is needed: $t < 2d < .2541$, $K < 1.081$, and the coarse core gap exceeds

$$.55235 - .172(1.081) + .2599\lambda^2 - \lambda > .52.$$

Thus every positive isolated mutual is excluded, including the entire unbounded dotted alphabet. Only $\lambda = 0$ remains.

For near-three, acuteness is

$$Q(\delta, a) = (a - 17/4)\delta^2 - (17/4)\delta - 3a/4 - 17/16 \ge 0.$$

Finite g gives $\delta > 1$, dotted g gives $\delta > 39/40$; the latter follows from $Q(39/40, 817/18) = -2693/19200 < 0$. The actual edge gives $\delta < 23/9 < 3$. The only positive subcritical mutual is $1/2$. Its full actual finite projection has $T > 300, j < 1/2$ in $V = K_0(P_{16} + j(e_{17} \pm e_{18}))$; hence the original maximum gives $S_{16} + jt > 3.76$. For finite g, use $\delta < 2,\ B/A < 2,\ j < 160/333$, and T's positive numerator $> 300(19/128)$. For dotted g, the sharper $13/10 < \delta < 3/2,\ B/A < 1,\ j < 1/3,\ T > 8664/19$ follows by the exact tests $AB - k(\delta + 1/2) = 1/2$ at $(g,\delta) = (1, 3/2)$ and below $1/2$ at $(7/6, 13/10)$. Both contradict $S_{16} + jt < 3.59 + .255/2 = 3.7175$.

Every supercritical near-three core is funded since $z_0 - 10P_0 = (4\lambda/(3\sqrt{2}))(\delta - 5) - 10P < 0$. With $J = k/2 - 4A/3$,

$$B > 3\delta + 4/3 - 19/(216\delta), \quad \lambda < K\delta - J, \quad J/K = A/(6K) - 1/2.$$

In the finite strips the gap exceeds $.3238 + (989/2160)\lambda^2 + .3271\lambda - \sqrt{4\lambda^2/3 - r}$; in the dotted strips it exceeds $.4292 + (989/2625)\lambda^2 + .3095\lambda - \sqrt{4\lambda^2/3 - r}$. For $\lambda \ge 1$, completion of one square gives respectively positive gaps $> .1052, > .00406$. Below one the displayed lower functions decrease and at one are $> .2127, > .2194$. Thus near-three also forces mutual zero. Its unique positive Q-root lies in $(1, 3/2)$ for finite g, or $(39/40, 33/32)$ for dotted g. Set $\zeta = 17/(4a) < .2$ or $< .13$. An isolated/near pair then has positive actual product $\delta_I[1/2 - \zeta(\delta + 1/2)]$; two near caps have product $\delta - \zeta(\delta + 1/2)^2 > 0$. Two isolated caps have actual mutual $17/(4a - 17) \in (0, 1/4)$. All caps are distinct by their original fork contacts. Hence at least one fork must have a far profile.

For completeness here is the entire far-profile scalar certificate. The exact positive-root sandwiches are

	M	$M^2 - B^2$
F_2	$3\delta + 1/3$	$1/9$
F_4	$3\delta + \sqrt{2}/6$	$1/18$
F	$3\delta + 1/6$	$19/36$.

Always $B < M$ and $B > M - E_1/\delta$, with $E_1 = 1/54, 1/108, 19/216$, respectively. Both F_2, F_4 are automatically supercritical, even at mutual zero. The actual length gives $\delta < 11/9 < 10(1/8)$ for F_2, and $\delta < 11/(9\sqrt{2}) < 10(\sqrt{2}/16)$ for F_4; thus every such ellipse price fits. Substitution in (B.9.2) gives

	$g < 1$	$g > 1$
F_2	$.2972 + (989/360)\rho^2 + .4906\rho - \sqrt{8\rho^2 - r}$	$.3967 + (1978/875)\rho^2 + .4642\rho - \sqrt{8\rho^2 - r}$
F_4	$.2972 + (989/180)\rho^2 + .6941\rho - \sqrt{16\rho^2 - r}$	$.3967 + (3956/875)\rho^2 + .6568\rho - \sqrt{16\rho^2 - r}$.

For positive mutual use $\rho \geq 15/16$ in F_2; for zero use exactly $\rho = 7/16$, whose radical is one. For F_4, $\rho \geq 15/(16\sqrt{2}) > 5/8$. The identity $c\rho^2 - \sqrt{m\rho^2 - r} \geq cr/m - m/(4c)$ gives gaps $> .0376, > .1852$ in the finite case and $> .032475, > .07256$ in the dotted case. It retains every finite, parallel and dotted mutual.

For far-three with positive mutual $\rho \geq 31/32$. The upper root gives $\rho < K\delta$, and the actual $\delta < 3$ funds $z_0 \leq 10P_0$. Positive-square monotonicity forces $\delta > 1$; then the improved upper $B < 3\delta + 1/6 - 1/(12\delta)$ gives

$$D > H + c\rho^2 + L\rho + C_1\rho/\delta - \sqrt{\rho^2/r - r}, \qquad c = .2967/(rK).$$

In finite strips, $L > .0577, C_1 > .1446, \rho/\delta > .48$; in dotted strips $L > .0546, C_1 > .1322, \rho/\delta > .6$. For the last ratio the exact monotone test at $(g, \delta) = (1, 155/96)$ has square gap $37199/12288$. For the finite ratio the corresponding test at $(.8, 2)$ has gap $23399/102400$. One square leaves respectively

$$H + .157621875/K - K/1.1868 + .1252 > .0047$$

and, on the five dotted strips, gaps $> .0805, .0545, .0574, .0440, .0368$ with $.1321$ in place of $.1252$. Thus every positive mutual fails.

At zero far-three mutual the Schur determinant is $1/16$. The full actual $A_{15} + n_o + c + d$ positive projection has

$$\alpha = (17A + 15B)/2, \quad \chi = (15A + 17B)/2, \quad T = [17(a + B^2) + 30AB]/2.$$

Monotonicity of $AB - k\delta - g - 1/2$ gives $\delta < 3/2,\ B/A < 1$. The spatial tail ratio is

$$j = \frac{(\chi/\alpha)\delta}{k/8 + (\chi/\alpha)(\delta + 1/16)} < 1/2.$$

Its $T > 300$ for finite g, $T > 1733/4$ for dotted g. The actual point therefore demands $S_{16} + jt > 3.76$ against the same upper 3.7175. This removes zero mutual too.

Every actual fork profile has now been covered under a weak PURE center: near-four/next-three are impossible; isolated/near profiles must be orthogonal and cannot form a pair; some far profile is necessary, but every far profile fails. All scalar interval checks above use explicitly displayed root sandwiches, monotonicities and completed squares on complete ranges. They do not posit a maximum native label or assume arithmeticity. The pure weak-center branch is excluded.

B.9.6 Nonpure centers and their actual neighbors

Only the two nonpure weak roles from the complete inventory remain. The own-zero/single-fork labels seven through ten will be excluded directly at their actual neighbors; their older head-specific refinements and separate strong-fifteen bootstrap are unnecessary.

First take single-fork-three with finite own $.9 < g < 1$. We do need fifteen strong for this branch. If fifteen were weak, its actual pure helper has parameter $\phi/2 \le h < 6/5$. Put $y = h+1/30,\ t = g+1/4$; the actual times satisfy

$$A_h^2 = 30y^2 - 8/15, \qquad A_c^2 = 32t^2 - 3/4, \qquad M = -\langle c, d\rangle = A_h A_c - 30yt.$$

This retains the full fork contact. It increases in both y, t: positive derivative squaring gives

$$y^2 A_c^2 - t^2 A_h^2 = y^2(2t^2 - 3/4) + (8/15)t^2 > 0,$$

and $1024t^2A_h^2 - 900y^2A_c^2 = t^2(1920y^2 - 8192/15) + 675y^2 > 0$. At the lower corner $h = .8, g = .9$ the time/spatial square gap is $17.3085 > 0$; at $h = 6/5, g = 47/40$ the comparison with $M = 11/10$ has gap $2861/8000 > 0$. Thus the entire actual range has $0 < M < 11/10$. Native values leave $M = 1/2$ or $M \ge 1/\sqrt{2}$.

Positive clearing of A_{14} and the contacted actual fork gives

$$\bar d = A_h p - \sqrt{2} y P_{15}, \quad \bar d^2 = 8/15, \qquad \bar c = A_c p - \sqrt{2} t P_{16}, \quad \bar c^2 = 3/4, \quad \langle \bar c, \bar d\rangle = -M.$$

The other fork remains a positive retained unit block. Both weak floors give deficits $D_h = A_h - \sqrt{2} y f_{15} > .43$, $D_c = A_c - \sqrt{2} t f_{16} > .60$. At $M = 1/2$, the full positive actual projection has coefficients $43 < \alpha < 65$, $37 < \chi < 56$, parameter $T < 959$; its pressure would require $\alpha D_h + \chi D_c \le \sqrt{1+T} - 1 < 31$, whereas the left side exceeds 40.69. For $M \ge 1/\sqrt{2}$, the positive core $\bar d + (4M/3)\bar c$ has negative square $8/15 - 4M^2/3$. Its pressure gap exceeds $.43 + .8M - \sqrt{4M^2/3 - 8/15}$. This decreases on the range and at $11/10$ is $> .27$. All original wall signs hold by actual positive coefficients. Thus fifteen is strong.

Use the one ending-fifteen reference of Appendix B.9.4, with $L > .95678$ and unpaid terminal coefficient $> .1313$. For finite g the actual positive core is $c + n_{16}/g + n_{15}/(2g)$, of norm $-[1 - 3/(4g^2)]$. Its cap/old-fifteen products vanish, its old-sixteen product is negative, both fork products are negative, and time is positive. With $\alpha = \sqrt{2}(g + 1/4)$, its normalized displacement V has

$$V_j = 1 \ (j \le 14), \quad V_{15} = V_{16} = 1 - a,$$
$$V_{17} = 2a + b, \quad V_{18} = \pm b,$$
$$a = \frac{1}{4g(g+1/4)}, \quad b = \frac{1}{4(g+1/4)}.$$

The full residual square is $1 - 2a + 6a^2 + 4ab + 2b^2 < 1.174$. Its reference cross is $-aw_{15}$, and its pressure-minus-reference deficit is $> .24441$: use $A_c/\alpha > 3.964$, $\sqrt{1 - 3/(4g^2)}/\alpha < .28285$, $a > .2$, $w_{15} > .1313$, $f_{15} < 3.463$. A legal .13 terminal payment therefore gives

$$\|x\|^2 > .95678 + .26(.24441) - .0169(1.174) > 1.000486.$$

No absent P_{14} capacity or omitted negative cross is used. This removes every finite own in the nonpure positive-own branch.

For dotted single-fork-three, $g > 1$, the actual center endpoint has full forest $A_{15} + A_2 + A_1$. For own zero and white $\delta = \cos(\pi/m)$, $7 \le m \le 10$, the actual endpoint has $A_{15} + I_2(m) + A_1$. In both cases the positive original incident dual is $P_{16}/4$:

$$q = Cp - \sqrt{C^2 - 1} P_{16}/4, \quad t = S_{16}/4 < 449/500, \quad E_q/E = C - \sqrt{C^2 - 1}\, t.$$

These are actual full finite projections, so every unknown original wall is satisfied. The reverse positive unit dual $u = (Cq - p)/\sqrt{C^2 - 1}$ lies in the dihedral block and satisfies

$$\langle y, u\rangle = \frac{\beta - t}{1 - \beta t}, \qquad \beta = \sqrt{C^2 - 1}/C, \quad z = E_q(q - y).$$

The point q is never asserted farthest; every original chamber point is compared there with factor E/E_q.

For dotted single-fork-three, $C = \sqrt{32}(g + 1/4)/\sqrt{3/4}$, so $C^2 > 200/3 > (204/25)^2$. The increasing function $C - (449/500)\sqrt{C^2 - 1}$ gives

$$E_q/E > 4431/5000 > 7/8, \quad E/E_q < 8/7, \quad \langle y, u\rangle > 5875/6824 > 43/50.$$

For own-zero/single-fork label seven through ten,

$$C = \frac{\sqrt{8}\,\delta}{\sqrt{1-\delta^2}}, \quad C^2 > 648/19 > (29/5)^2,$$

as $\delta > .9$. Hence

$$E_q/E > 117/175 > 2/3, \quad E/E_q < 3/2, \quad \langle y, u\rangle > 2900/3849 > 3/4.$$

The radical upper at 29/5 is 40/7; the bounds $\beta > 197/200$ or $124/125$ give the stated reverse fractions by positive rational multiplication. These two neighbor cases will use the following local facts.

B.9.7 The two required central A_{15} facts

Use zero-sum $\mathbb{R}^{16}$, with

$$\omega_i = \sqrt{2}(P_i - iP_{16}/16), \quad D_i = i(16-i)/8, \quad C_{ij} = i(16-j)/8 \ (i \le j), \quad D_8 = 8.$$

At any actual $A_{15} + A_2 + A_1$ vertex, the actual central edge has $C_8 > 9/4$, without a farthest assumption. At an actual $A_{15} + I_2(m) + A_1$ vertex, $7 \le m \le 10$, with comparison at most $3/2$, the A_{15} energy exceeds $7/16$. We prove both statements from the actual central opposite.

Write it $d = Hq - v_A - v_S$, own $\gamma \ge 0$, whole foreign cost $Q = \|v_S\|^2 < 1$. Deleting eight leaves two A_7 paths. The finite white list is one three at $1, 2, 3, 5, 6, 7$ or reflection; one four at $1, 7$ or reflection; or one three on each path. The third-from-end three is E_8; the middle fourth position is affine E_7. No white five extends a full A_7, and a rank-fifteen join is classical with both new bonds simple. The exact edge data are

$$\sigma = \gamma + \sum_{j<8} \beta_j j/8 + \sum_{j>8} \beta_j (16-j)/8, \quad \Delta = 1 - Q - \sum_{j<8} \beta_j^2 j(8-j)/4 - \sum_{j>8} \beta_j^2 (16-j)(j-8)/4,$$

with $H^2 = 8\sigma^2 - \Delta$, $C_8 = \sqrt{8}\sigma/\sqrt{\Delta}$.

We first remove own zero directly, without a foreign-shadow rule. Two white paths close an actual affine cycle. Single three at global $3, 13$ gives affine E_8 with arms $1, 2, 5$; at $4, \dots, 12$ use E_7 with arms $1, 3, 3$. White four at $7, 9$ has the four/simple star. At endpoint four $\Delta > 0$ requires $Q < 1/8$, so $Q = 0$ and $H^2 = -1/16$; at next-three $Q < 1/4$ forces $Q = 0, H^2 = -1/8$. At endpoint three the retained bound is $Q < 9/16$, but $H^2 = Q - 17/32$. In $A_2 + A_1$, the possible foreign contributions under that bound are $0, 1/3, 1/4, 1/2$; simultaneous contact costs at least $1/3 + 1/4 = 7/12 > 9/16$. Thus $Q \le 1/2$, impossible. For the high dihedral block, ANY positive contact costs

$$[4(1-\delta^2)]^{-1} > 25/19 > 1,$$

so there is no such contact at all. The remaining A_1 again gives only $Q = 0, 1/4, 1/2$ under the retained bound. With no A contact the time is negative and its derivative along that edge is zero.

Every affine kernel has positive actual coefficients and cap time, selected products zero, all unknown products nonpositive; it is a forbidden future feasible null.

For the first local statement, $\gamma \geq \phi/2$ directly gives $C_8 \geq \sqrt{2}\phi > 9/4$. Own four has a four/simple star at eight unless a white contact hits 7 or 9; then $\sigma \geq 1/\sqrt{2} + 7/16, \Delta \leq 9/16$, giving $C_8 > 4$. Own three has the actual E_7 star at eight unless it contacts one of the two three-node arms. The remaining allowed contacts have $(\sigma, \Delta) \geq / \leq (13/16, 1/16), (7/8, 1/4), (15/16, 9/16)$; the weakest cosine is $5\sqrt{2}/2 > 9/4$. This proves the full central-edge statement.

For the high-dihedral second statement, all foreign cost is in the remaining actual A_1. Own-three plus any white-three closes a cycle. Pure far-three own-four/six/parallel have the actual star, $\gamma d + m_8 + m_7/2$, and $d + m_8$ nulls. The remaining ordinary ratios satisfy

role	σ lower	Δ upper before Q
near three	$.7 + 7/16$	$9/16$
D three	$.7 + 1/8$	$1/4$
E three	$.7 + 3/16$	$1/16$
white four	$1/2 + 1/(8\sqrt{2})$	$1/8$
two threes	$1/2 + 1/8$	$1/8$
far three, $Q > 0$	$.8 + 1/16$	$5/16$.

All exceed $3/2$. Their actual edge demand is therefore $> (3\sqrt{2} - 3/2)/\sqrt{17} > 53/80$, giving energy $> 2809/6400 > 7/16$. Extra foreign cost only increases this demand.

For isolated A roles, own three/four/six/parallel have the E_7, star, chain and pair nulls. Put $\tau = (1 - Q)/\gamma^2,\ H^2 = \gamma^2(8 - \tau)$. For $0 < \tau < 1$ the positive actual core $d + w_S + \gamma\tau m_8$ has spatial part $U = \gamma(\omega_8 - \tau m_8)$, norm $-\gamma^2\tau(1 - \tau)$, and $U^2 < H^2$. If $e_A \leq 7/16$, Cauchy forces pressure loss divided by γ greater than $(1 - \sqrt{7}/4)\sqrt{8 - \tau} > 5/6$, against comparison loss at most $3/4$. At $\tau = 1$ it is a forbidden null. For $\tau > 1$, the actual positive finite point gives

$$h_8 \geq \sqrt{\frac{7}{8 - \tau}}\,\{\sqrt{7} - \kappa\sqrt{\tau - 1}\}.$$

Foreign $\sqrt{\tau - 1} \leq 2 - \phi < .382$; pure finite owns at least eight have $\sqrt{\tau - 1} \leq \tan(\pi/8) < .415$. The bracket is positive, and $h_8 > 13/5 - (3/2)(.415) = 791/400$, whose normalized value is $> 2/3$, giving energy $> 4/9 > 7/16$. Pure five uses the actual H_4 finite point, with $\delta_5 = (7 - 3\sqrt{5})/16 < 1/36,\ H^2 > 4, \gamma > .8$:

$$h_8 > \sqrt{8 - \gamma^{-2}} - \kappa\sqrt{\delta_5}\sqrt{1 + \delta_5/H^2}/\gamma > 699/320.$$

Pure seven uses $d + (4\gamma/3)m_8 + (2\gamma/3)m_7$; the native cubic brackets $.9 < \gamma < 10/11$ give

$$h_8 > \sqrt{8 - \gamma^{-2}} - \kappa\sqrt{4/3 - \gamma^{-2}} > 103/50.$$

Both again give normalized height $> 2/3$.

For every pure far-three helper, without an upper own cutoff,

$$H^2 = 8\gamma^2 + \gamma - 17/32 > 8\gamma^2, \qquad \gamma > .8.$$

Five uses the actual correction $\gamma m_8 + (2/3)m_e + (1/3)m_j$, residual $\delta = 2/3 - \gamma^2 \in (0, 1/36)$; its finite point has $U^2 = H^2 + \delta < (10/9)H^2$ and pressure $> H - .2525 > (6/7)H$, giving energy $> 162/245$. Finite owns at least seven use $d + \gamma m_8 + 2(1 - \gamma^2)m_e$, negative square $-(1 - \gamma^2)(4\gamma^2 - 3)$, radical

at most 1/4. Since $H > 2.5$, energy exceeds 289/400. Dotted owns use $d + m_8/\gamma$; its radical divided by γ is at most 1/2, so the pressure ratio is $> 1 - 3/(4\sqrt{8}) > 2/3$, giving energy $> 4/9$. All selected products and positive time are as displayed; unknown wall signs hold termwise. This completes both local facts.

Return to the original actual neighbor. In the dotted single-fork-three case, $C_8 > 9/4$ and factor $< 8/7$ give central normalized height $> 31/(7\sqrt{65})$, hence $e_{A_{15}} > 961/3185$. The orthogonal reverse A_2 energy exceeds 1849/2500, and their sum is $1 + 65813/1592500 > 1$. In the own-zero/high-dihedral case, the second local fact gives $e_{A_{15}} > 7/16$, while the reverse dihedral costs $> 9/16$. The remaining actual A_1 is nonnegative in each SAME neighbor vector and is never charged twice. Both weak-center families are excluded.

B.9.8 Strong sixteen and the legal terminal append

The complete center inventory, the entire pure-center exclusion, and the finite/dotted single-fork-three and own-zero/high-dihedral arguments cover every possible weak center. Therefore $S_{16} > f_{16}$ unconditionally. No statement about strength of either fork was used.

Reassemble each endpoint-fifteen reference through actual strong sixteen, then OMIT weak fifteen before applying any pressure. Appending contributes $(4 - \sqrt{15})^2$; omitting fifteen costs $(2\sqrt{15} - \sqrt{14} - 4)^2/2 < .00001$. Since $4 - \sqrt{15} > .127$, the net gain is $> .016119$. The initial zero-sum corrections are supported through seven, so their individual prices and full crosses are unchanged. The endpoint-fifteen omission debit may conservatively be retained in addition to this new debit; no fictitious floor is ever used. The worst paid ending-sixteen effective energy is

$$L_{16} > .95678547 + .127^2 - .00001 = .97290447 > .9729.$$

The other four rows exceed $.985649, .987759, .981369, .976279$. The unpaid P_{16} coefficient is its true last slope: d_{15} if fifteen is strong or $(d_{14} + d_{15})/2$ if fifteen is weak. It exceeds .127 in every case. All three late slopes are at least d_{15}.

For either actual fork $V = P_{17} \pm e_{18}$, the residual $V - P_{16}$ has square two and zero reference/initial cross. If its actual cosine were at least ten, pressure would exceed 3.837. A legal $.12 < \beta_{16}$ payment gives

$$\|x\|^2 > .9729 + .24(3.837 - 3.59) - .0144(2) = 1.00338 > 1.$$

Thus BOTH genuine fork cosines are less than ten for every head and every original initial-five case. The subsequent main-text uniform dotted-head argument strengthens these data; it is not needed to prove them.

B.9.9 The genuine fork inventory and the all-head near-pair bound

Return to the seven actual fork profiles in Appendix B.9.5, now with $C_f < 10$. For isolated or far-helper own three, an actual affine E_8 kernel has coefficients 2 on its cap, 4 on f, 6 on center 16, 3 on o, and $5, 4, 3, 2, 1$ along $15, \ldots, 11$. Own four has kernel $(\sqrt{2}, 2, 2, 1, 1)$ on its cap, $f, 16, 15, o$; own six has its $6, 3$ chain. An extra far helper lies outside these principals. All have positive cap time and all unknown wall signs termwise. Parallel gives $d + n_f$. For near-three, own zero is isolated own three at the other fork; own three is the actual four-cycle, own six the same chain. Near-four now forces own zero and is the isolated own-four null at the other fork; near-next-three likewise forces own zero and is the four-leaf-three center star. Far-four has no positive time at zero, and its only possible positive own under length ten is three, already excluded. These dispose of ALL formerly allowed near-four/next-three/far-four values, not merely their finite members.

Isolated own five is exactly the actual H_4 point already computed in Appendix B.9.3. Its displacement gives $S_{15} > 3.628$, hence fifteen is strong. With terminal capacity $> .1313$, the one-reference gain $2(.1313)(.165) + .165^2/15$ contradicts $L_{15} > .95678$. That point's computation did not use weak center sixteen. The complete remainder is therefore

I	finite own ≥ 7 or dotted, $g < 10/3$
N	own four/five, finite ≥ 7 or dotted, $g < (10\sqrt{19} - 8)/18$
F	own five, finite ≥ 7 or dotted, $g < (10\sqrt{19} - 1)/18$
F_2	own five or finite ≥ 7, $g < 1$.

These are necessary roles, not examples.

The original ending-sixteen reference retains its terminal surplus. For $\Delta = S_{16} - f_{16} > 0$,

$$\|x - w\|^2 < .0271 - .254\Delta.$$

Since $w_{17} = w_{18} = 0$, each genuine $t = x_{17} \pm x_{18}$ has $t^2 < .0542 - .508\Delta$. The square comparison $(.233 - \Delta)^2 - (.0542 - .508\Delta) = .000089 + .042\Delta + \Delta^2 > 0$ gives

$$S_{17} + x_{18} < 3.823, \qquad S_{17} - x_{18} < 3.823.$$

At cosine $241/25 = 9.64$ the true fork demand has square $1944/133 > 3.823^2$; hence both fork cosines are < 9.64.

For a near-three or far-three fork cap, its actual other endpoint is whole A_{18}. Its original normalized dual height is $t < 3.823/\sqrt{18} < .903$. If its cosine $C \geq 6.82$, then

$$E_q/E > 6.82 - .903(6.747) = .727459 > 8/11, \qquad \langle y, u_{\text{reverse}} \rangle > .8.$$

The reverse positive dual is the new A_{18} endpoint dual. Lemma 16.1 applies at that actual q, with factor $< 11/8$ and endpoint height $> .78$, and gives a contradiction. That lemma is independent of the original head and never assumes q farthest. Therefore

$$C_N, C_F < 6.82, \qquad g_N < 1.21, \qquad g_F < 1.6.$$

These bounds follow from actual all-wall finite endpoints and the same original maximum, not a whole-A_{18} farthest theorem.

Two near-three caps are impossible for EVERY head. Write their owns $g, h < 1.21 < 3/2$ and times $A_g^2 = 9g^2 + 8g + 5/4$. Their spatial product is $K = 8gh + (9/2)(g+h) + 2$. For $P = gh, S = g+h$,

$$A_g^2 A_h^2 - K^2 = 17P^2 - 9(g^2 + h^2) - (17/2)P - 8S - 39/16 \leq (47/4)P - 8S - 39/16 < 0,$$

because $P < 9/4,\ g^2 + h^2 \geq 2P,\ P < 3S/4,\ S < 3$. Their actual product is positive, violating acuteness. They are distinct: coincidence would force $g = h = 1/2$, whose actual cycle was already excluded. This proof retains both fork signs and unequal owns.

B.9.10 Finite heads at least nine, from their exact bare references

For a non-native-five head, the three BARE ending-sixteen references are

	B	b upper	P_J capacity	J
strong five	.9565188	.3652	$> .151$	5
isolated five, strong six	.9373243	.3399	$> .1435$	6
isolated five, strong seven	.9196357	.31938	$> .1366$	7.

They are constructed by the same .016119 append/omission from bare fifteen energies .9403998, .9212053, .9035167. No head or isolated-five pressure has yet been paid in these rows. They remain flat first J, have sum f_{16} and unspent terminal price $> .127$.

For an isolated finite head let $u = \eta^{-2} \in (1, 4/3)$. The positive actual core

$$H + \eta u n_1 + 2\eta(u-1)n_2$$

has square $-\eta^2(7u - 4 - 3u^2)$, zero head/first-root products, negative second-root product, positive time and all unknown signs. Its pressure is $\langle x, Z\rangle \geq h$, with

$$\begin{aligned} Z &= (1 - u/2)(e_1 + e_2) + (u-1)e_3, \\ h &= [\sqrt{2-u} - \sqrt{7u - 4 - 3u^2}]/\sqrt{2}, \\ R &= Z^2 = 3 - 4u + 3u^2/2, \qquad \sum Z_j = 1. \end{aligned}$$

If the native head label is at least ten, then $1 < u \leq 2 - (2/5)\sqrt{5} < 1.11$. On this WHOLE interval $F(u) = .8h - .16R$ decreases: $(7u - 4 - 3u^2)' > 0$ and $F' < -.4/\sqrt{2} + .16 < 0$. At 1.11, positive square bounds give $\sqrt{.89} > .94339, \sqrt{.0737} < .27149$; thus $h > .4748,\ R = .40815$, and $F > .314536$. Pay $.4(Z - P_J/J)$ from the individual P_J anchor. The general original square gives $B + F - .8b + .16/J$, which in the three rows exceeds $1.0108948, 1.00660696, 1.00152484$. No old head lower gain has been subtracted from a paid lower bound.

In the fourth, weak-helper-five case the existing unheaded reference has energy $> .987759$, $w_1 = .5455$, $w_2 = w_3 < .299$. The exact head cross is

$$\begin{aligned} \langle w, Y\rangle < M(u) &= (1.091 - .2465u)/\sqrt{2}, \quad Y = \sqrt{2}Z, \\ Q = Y - (\sqrt{2}/16)P_{16}, &\quad Q^2 = T(u) = 47/8 - 8u + 3u^2. \end{aligned}$$

Pay $\mu\sqrt{2}/16 < .018$ at the genuinely unspent terminal, retaining this full cross. The norm lower bound is $.987759 + 2\mu(f - M) - \mu^2 T$, $f = \sqrt{2-u} - \sqrt{7u - 4 - 3u^2}$. All three functions decrease on the required interval. Bounding f at the RIGHT and M, T at the LEFT endpoint gives

u interval	$f >$	$M <$	$T \leq$	μ	norm lower
$(1, 1.05]$	.7684	.5972	.875	.2	1.021239
$[1.05, 1.08]$	.7125	.5885	.7825	.15	1.00735275
$[1.08, 1.1]$	.6840	.5833	.7342	.14	1.00156468
$[1.1, 1.105]$	.6778	.5798	.705	.14	1.001381
$[1.105, 1.11]$	.6719	.5789	.698075	.13	1.0001415325.

The square brackets for the two radicals at the successive right endpoints are, in order,

$$(.9746, .2062), \quad (.9591, .2466), \quad (.9486, .2646),$$
$$(.9460, .2682), \quad (.94339, .27149).$$

These prove every f entry; M uses $\sqrt{2} > 1.4142$ and T is exact substitution. The table is a complete interval certificate, not a sampled search. All finite labels at least ten are excluded.

For native nine use its actual cubic: $8\eta^3 - 6\eta - 1 = 0$ and $.93969 < \eta < .9397$. Exact products give $1.1324 < u < 1.1325$. Now $\sqrt{2-u} > .93139, \sqrt{7u - 4 - 3u^2} < .28255$, so $h > .4587,\ R < .394$. The strong-five bare row paid by .4 gives $.9565188 + .8(.4587 - .3652) - .16(.394 - .2) = 1.0002788$. For isolated-five rows set

$$Q_H = Z - P_J/J, \quad Q_5 = U' - (5/J)P_J, \quad \langle Q_H, Q_5\rangle = 1 - 5/J.$$

Their squares are $R-1/J$ and at most $77/150, 194/175$. Pay $\mu=.4, \nu=.09, .1$, respectively. The COMBINED anchor debits are $.85/6<.1435,\ .9/7<.1366$. The full one-vector energy bound

$$B+.8(h-b)+2\nu(1.8185-5b)-.16(R-1/J)-\nu^2Q_5^2-.8\nu(1-5/J)$$

exceeds $1.00125296, 1.00128598$. The positive mutual cross is explicitly subtracted.

A weak pure helper five cannot coexist with the native-nine head. Its own is $g=\phi/2$ or $.9<g<1.2$, and its actual head mutual is $\lambda=A_H\sqrt{10g^2+2g-1/2}-\eta(2g+1)$. On the latter entire interval it increases, since the radical derivative exceeds $\sqrt{10}$ and $A_H\sqrt{10}>2\eta$. Native-nine brackets give $\lambda(.9)>.049875>0,\ \lambda(1.2)<.340629<1/2$, a forbidden native gap. At $g=\phi/2$, direct positive squares give $\lambda<.8755(2.769)-.9396(2.618)<-.0356$, contrary to acuteness. All four original initial cases fail, excluding native nine. The remaining finite heads are five, seven and eight.

B.9.11 The all-head face envelope and an isolated-own cutoff

The paid ending-sixteen reference also controls EVERY asymmetric old A_{16}-face direction. For $|v|\le 1$, pay .127 at its unspent terminal toward $Z=P_{17}+ve_{18}$. The residual has square $1+v^2$ and zero reference cross. If $\langle x,Z\rangle\ge 3.761+.0635v^2$, its original square exceeds

$$.9729+.254(.171)-.016129=1.000205$$

because $.254(.0635)=.016129$. Thus

$$\langle x,P_{17}+ve_{18}\rangle<3.761+.0635v^2,\qquad \frac{\langle x,P_{17}+ve_{18}\rangle}{\sqrt{17+v^2}}<.9124. \tag{B.9.3}$$

For the second inequality the quotient of the upper bound decreases with v^2; its derivative numerator is $-1.602+.0635v^2<0$, and $3.761^2<17(.9124)^2$. The two old fork inequalities force the displacement direction of ANY actual point on old A_{16} to have this form. The original all-point maximum therefore gives cosine <10.94 for every such point.

Every isolated actual fork own g is less than $15/8$. Indeed its actual original fork endpoint is whole $A_{17}+A_1$, with cosine $C=3g$, the new A_1 being the cap itself. At $v=\pm1$, (B.9.3) gives original normalized height $<.90145$. If $g\ge 15/8$, the actual endpoint q has

$$E_q/E>45/8-.90145(5.5355)=.635023525>5/8,\qquad E/E_q<8/5.$$

The reverse unit is exactly its whole A_1 normal and has height

$$>\frac{.984-.90145}{1-.984(.90145)}>.73.$$

We prove the required local contradiction, rather than transporting an original-farthest A_{17} theorem.

At an actual $A_{17}+A_1$ vertex with factor $<8/5$, inspect central node eight in zero-sum $\mathbb{R}^{18}$. Here $D=80/9,\ 2.98<\sqrt{D}<3, \sqrt{D-1}>2.8$. Its retained paths are A_7, A_9, with the finite white list of Appendix B.3.6; explicitly single-three positions $1,2,3,5,6,7,9,10,16,17$, terminal fours $1,7,9,17$, two endpoint-threes or isolation. The foreign whole A_1 cost is $Q=0$ or native $Q\ge 1/4$, with $Q<1$. At own zero the endpoint-three time is $Q-19/36$ with $Q<9/16$, allowing only $0, 1/4, 1/2$, all impossible. Next-three and endpoint-four require $Q<1/4, Q<1/8$, hence zero and negative time. Other zero-own positions have the actual E_8/E_7, four/simple or cycle nulls. All are positive future actual combinations, feasible against every unselected original wall.

For positive owns, ordinary white rows except native-five single-white-three have actual cosine > 5. The near-three residual is at most $9/16$, with numerator $> .9 + 7/16$; D/E residual at most $1/4$, numerator $> .9 + 1/20$; foreign far-three residual at most $5/16$ and the same numerator. A near-four with foreign cost satisfies this latter bound. A pure near-four uses its ACTUAL mate at seven or nine, of cosines $(77 + 70\sqrt{2})/(6\sqrt{7}) > 10$ or $9/2 + 4\sqrt{2} > 10$. White-four own-three at a far end has the affine F_4 path; near own-three/four has cosine > 6 by its ratio. Two whites have residual at most $9/80$, giving cosine > 5. Own three/four/six/parallel at the other positions have their actual cycle/star/chain/null boundary, with helper contacts outside the selected principal. Hence every ordinary row has component energy

$$e_A > (5 - 8/5)^2/24 = 289/600 > 12/25.$$

For isolated own γ, clear the whole foreign root and put $\tau = (1 - Q)/\gamma^2$. The same actual reciprocal core and positive finite point as in Appendix B.9.7 give

$$h_8 > \sqrt{D - \tau} - \kappa\sqrt{\tau(1 - \tau)} \quad (0 < \tau < 1),$$

or

$$h_8 \geq \sqrt{\frac{D - 1}{D - \tau}}\{\sqrt{D - 1} - \kappa\sqrt{\tau - 1}\} \quad (\tau > 1).$$

The first case contradicts $e_A \leq 12/25$: Cauchy would require loss $> .3(2.8) = .84$, but the core costs at most $\kappa/2 < .8$. The null boundary is forbidden. In the finite-point case, foreign radical $< .4$ gives $h_8 > 2.16$; pure finite own at least eight has radical $< .415$, giving $h_8 > 2.136$. Both normalized squares exceed $12/25$. Pure five uses the actual H_4 finite projection, $\sqrt{\delta_5} < .1352, \gamma > .8, H/\gamma > 2.7$, and gives $h_8 > 2.4296$. Pure seven uses the actual two-root negative core, native $.9 < \gamma < 10/11$, and gives $h_8 > 2.76 - (8/5)(.36) = 2.184$. Every selected product and future sign is the same direct actual-root calculation already given; no native projection is assumed.

Pure far-three owns at least seven have the UNEQUAL exact times

$$H_1^2 = D\gamma^2 + (10/9)\gamma - 19/36, \quad H_{17}^2 = D\gamma^2 + (8/9)\gamma - 19/36.$$

Both exceed $D\gamma^2$ for $\gamma > .9$. The finite core has radical at most $1/4$ and $H > 8/3$, giving normalized pressure $> .85$. The dotted core has radical divided by γ at most $1/2$, giving pressure ratio $> 1 - \kappa/(2\sqrt{D}) > .7$. Again energy exceeds $12/25$.

The remaining native-five single-white-three row is paid jointly with the known actual A_1 reverse pressure. Choose the H_4 arm on the side AWAY from the helper. Its positive finite point has full displacement U, including the foreign component, and residual $\delta_5 < .01825$. The smaller cross $C_{8j} \geq 8/9$ gives $H > 2.45$. Therefore the normalized point pressure exceeds $1-(8/5)(.1352)/2.45 > .9$. The actual foreign coefficient is $< 3/4$, so its correlation with the unit A_1 is $< 1/3$. For the ONE vector

$$W = (387/800)u + (591/800)U/\|U\|$$

the positive pressures $> .73, > .9$ give $\langle y, W\rangle > 1.0180125$, $W^2 < 1.0180125$; hence $\|y\|^2 > 1$. All other rows give the orthogonal sum $e_A + e_{A_1} > .48 + .73^2 > 1$. This proves the complete local contradiction and the original cutoff $g < 15/8$, including every head and both signed fork choices.

B.9.12 Two standalone local lemmas for a positive two-dual reverse

The distant fork-pair exits below require a reverse direction which can involve TWO positive duals. It cannot be replaced by a large single endpoint coordinate. We prove the two precise local lemmas here.

First let the whole actual forest be $A_{17} + A_1$, comparison $\kappa < 1.262$, and let

$$u = ae + b\omega_{17}, \quad a, b > 0, \quad a^2 + (17/9)b^2 = 1, \qquad \langle y, u\rangle > .8564,$$

where e is the whole A_1 unit. Call only sites eight and nine weak if $h_i \le \sqrt{D_i} - .8$. If either is strong, its normalized dual has height $> .73$ and correlation with u less than $1/4$ (the exact bounds are $\sqrt{8/(17 \cdot 10)}, 1/\sqrt{17}$). The single vector $.75u + .55v_i$ has pressure > 1.039 and square < 1.07125, giving norm > 1.00675. Both sites must be weak.

Their finite-white list is the same full $N = 18$ list, with all four retained side orders at least eight. Thus zero-own endpoint-three has time $Q - 19/36$, with native $Q < 9/16$ allowing only $0, 1/4, 1/2$; next-three has strict $Q < 2/m \le 1/4$; endpoint-four has $Q < 1/8$. They all have negative time, and every other zero-own role is the actual positive affine kernel already displayed. Using site seven instead would lose that strict side-order bound.

For positive ordinary ratios $R \ge 3/2$, the correct comparison at the weaker $\kappa = 1.27$ and loss .8 reduces to the polynomial

$$-.21q^3 + 1.1729q^2 - 1.6q + .64 > 0 \qquad (2.7 \le q \le 3.1).$$

It is concave and its endpoint values are $.737011, .695459$. Near-four side orders at least eight also have ratio $> 3/2$. Foreign exceptional helpers have ratio $> 227/(25\sqrt{35})$, so cannot be weak. Every isolated actual cap is strong by the same positive points; their complete conservative losses at factor 1.27 are

dotted	$< (.193 + 1.285)/2 = .739$
finite ≥ 8	$< 5/26 + 1.27(.415) < .720$
5 finite point	$< 25/76 + 1.27(1.01)/(6(.8)) < .597$
7 negative core	$< 10000/41877 + 1.27(.36) < .697$
$Q > 0, \tau > 1$	$< .193 + 1.27(.4) = .701.$

These follow from the displayed τ formulas, with $.193^2 + 1.27^2 < 1.285^2$; the null boundary is unchanged. Thus both weak sites must be pure far helpers with $\phi/2 \le \gamma < 17/16$. Their complete common/outward/reversed actual-product calculation and positive two-cap Schur matrices were proved in Appendix B.3.6, before its comparison. In particular the reversed numerator has minimum $2763353243/50000000000 > 0$, unless both owns are five, when its separate geometric distance-one obstruction applies. The actual two-cap point has weighted mean loss

$$< .1 + 1.262(7/16) = .652125 \quad \text{or} \quad < .1 + 1.262(3/10) = .4786,$$

both $< .8$. Its entire actual A_1 block is retained and every original wall satisfied. This contradicts the two weak sites and proves the first standalone two-dual lemma.

For the second, let the whole actual forest be A_{18}, comparison $\kappa < 1.27$, and

$$u = a\Omega_{18} + b\Omega_{17}, \quad a, b > 0, \quad \|u\| = 1, \quad L = a + 2b \le \sqrt{19/17} < 1.058, \quad \langle y, u\rangle > .85.$$

The inequality for L is Cauchy in the genuine $N = 19$ inverse end block. The complete own/white/null geometry in Appendix B.4 is independent of its later single-endpoint consumer. At sites five through fourteen, the same finite ratio polynomial and isolated point losses just displayed are $< .8$. Hence possible weak roles are near-four at five/six or reflection, or pure far helpers with $\phi/2 \le \gamma < 17/16$. For two adjacent PURE far helpers, the actual two-cap comparison gives weighted losses $< .655625$, $.481$, both below .8. Thus pure helpers are nonadjacent. This statement does not yet include any near-four role.

A near-four actual mate has cosine > 11, hence normalized height $> 7/8$. Its reverse correlation is $L\sqrt{2j/[19(19-j)]}$. Left mates $j = 4, 5$ have correlation $< .206$ and their two-direction sum contradicts the norm budget. Every left near-four is therefore excluded first. For a right mate $j = 14, 15$, if both left sites five and six were weak, both would now be pure helpers, contradicting their nonadjacency. Hence some left $k = 5, 6$ is strong, with height $> .703$. The three full correlations are $< .666, .234, .407$. The ONE vector $.5u + .6u_j + .4u_k$ has pressure > 1.2312, square < 1.45856, and gives norm > 1.00384. Thus every weak site is a nonadjacent pure helper.

Take symmetric rational heights $H_{19-i} = H_i$, linear on both ends,

$$H_5 = 1.914,\ H_6 = 2.065,\ H_7 = 2.173,\ H_8 = 2.243,\ H_9 = H_{10} = 2.277.$$

Their positive square tests $(H_i + .8)^2 < D_i$ are the earlier .87-window tests, with heights shifted by .07. Flatten a weak first/last block through six, and omit every remaining weak knot by its chord. The complete nonadjacent omission debit is $2(.043^2 + .038^2 + .036^2 + .034^2)/4 = .0028725$. If fourteen is weak, the relevant last block has length $m = 6, K = 13, H_K = 2.065$, energy $E_0 > .72555$ and anchor $> .11808$. If fourteen is strong, $m = 5, K = 14, H_K = 1.914$, energy $E_0 > .74793$, anchor .1159. The initial energy formulas are ordinary squared slope sums; the worst two-weak-end value before omission is $2.065^2/6 + .01772 = .7284241666\ldots$; the one-weak-end value is $.7508121833\ldots$. Stronger opposite-end patterns only increase these bounds.

Retain the exact reverse residual

$$G = u - (L/m)\Omega_K, \qquad \langle w, G\rangle = 0, \qquad G^2 = 1 - 2KL^2/(19m).$$

Payment $\mu = 1/2$ spends only $L/(2m) < .08817, .1058$ at the appropriate actual strong anchor, fitting its own capacity. One completed square gives

$$\|y\|^2 > E_0 + .85 - (H_K/m)L - 1/4 + KL^2/(38m).$$

The gain decreases up to $L = 1.058$, where it is $> .299695$ or $> .277477$. Thus the norm exceeds 1.025245 or 1.025407. The negative term in G^2 and every individual coefficient are retained. This proves the second standalone lemma.

B.9.13 The local comparison with two orthogonal spectators

We shall need a comparison at a weaker factor than the dihedral comparison in Appendix B.8.12. At an actual $A_{16} + 2A_1$ vertex q, suppose that every chamber point is compared with the original maximum at factor $\kappa \le 33/20$. If u is a unit vector in the whole two-A_1 plane, then

$$\langle y, u\rangle > .738 \tag{D2A1}$$

is impossible. Both spectator coordinates are retained in this statement.

Here are the complete local details. In zero-sum $\mathbb{R}^{17}$ put

$$\omega_i = \sqrt{2}(P_i - iP_{17}/17), \quad D_i = 2i(17-i)/17, \quad C_{ij} = 2i(17-j)/17 \quad (i \le j).$$

Thus $D = D_8 = 144/17$, $2.91 < \sqrt{D} < 3$ and $\sqrt{D-1} > 2.73$. Write the actual opposite to eight as $d = Hq - v_A - v_S$, with own γ, and $v_S = \beta_1 a_1 + \beta_2 a_2$. The a_j are the two actual orthogonal unit roots. Their full cost is $Q = \beta_1^2 + \beta_2^2 < 1$. Each nonzero β_j is a finite native cosine, at least $1/2$. Clearing spectators uses the positive actual sum $w_S = \beta_1 a_1 + \beta_2 a_2$.

Deleting eight leaves $A_7 + A_8$. Its complete one-node finite extensions are one 3 at $1, 2, 3, 5, 6, 7, 9, 10, 15, 16$, one 4 at $1, 7, 9, 16$, or one endpoint 3 in each path. A second contact within either path

makes a cycle; the exceptional extensions supply exactly the displayed E_8 positions. For own zero at a global end 3, $H^2 = Q - 9/17$ whereas positivity requires $Q < 9/16$. The entire native two-spectator alphabet below $9/16$ is $Q = 0, 1/4, 1/2$, so the time is negative. The next-3 and end-4 remainders are $1/4, 1/8$; they force $Q = 0$ and again negative time. Isolation gives $H^2 = Q - 1 < 0$. All other own-zero roles have the actual affine E_8/E_7, 4/3/3 star or all-3 cycle. For positive own, isolated $3, 4, 6$, a single-white own 3, the non-near single-white own 4, and own 6 have the corresponding star, cycle or chain null. The near-own-4 roles at seven and nine are retained. Parallel gives $d + n_8$. Every null is a positive sum of actual normals, has positive time through d, and is nonpositive against all omitted walls, including both spectators.

For a single white 3 in a side of order $l = 8$ or 9, at distance t from its remote end, the exact ratio is

$$\frac{C_8}{\sqrt{D}} = \frac{\gamma + t/(2l)}{\sqrt{1 - t(l-t)/(2l) - Q}}.$$

For white 4 replace $t/(2l)$ by $t/(\sqrt{2}l)$ and $t(l-t)/(2l)$ by $t(l-t)/l$. A two-end join adds the two costs. Reserve native-5 single-white-3 for the paired argument below. Every other surviving single-white own is $> .9$, except the retained near-own-4. The following lower squares all exceed 25:

role	C_8^2 lower bound
near 3	11449/425
other non-far D/E 3	51984/1700
left far 3 with $Q > 0$	853776/34000
right far 3 with $Q > 0$	118336/4675
near-own-4 with $Q > 0$	1192464/34000.

In the two far rows $Q \geq 1/4$, with the unequal residuals $5/16, 11/36$ and numerators $77/80, 43/45$. For near-own-4 at $Q = 0$, the same actual opposite is the seven- or nine-edge mate, with finite forests $A_6 + B_{10} + 2A_1$ or $B_9 + A_7 + 2A_1$ and lengths

$$(70 + 63\sqrt{2})/\sqrt{238} > 9, \qquad (18 + 16\sqrt{2})/\sqrt{17} > 9.$$

These are positive projections, so all omitted wall signs are checked termwise. The mate's own normalized height is $> (9 - 1.65)/\sqrt{80} > .8$. To obtain the central height as well, use concavity of the actual A_{16} heights $h_i = \langle y_A, \omega_i \rangle$, with $h_0 = h_{17} = 0$: $h_8 \geq (9/10)h_7$ or $h_8 \geq (8/9)h_9$, respectively. Since $D_7 = 140/17$ and $D_8 = D_9 = 144/17$, this gives

$$\frac{h_8}{\sqrt{D_8}} > .8 \min\{(9/10)\sqrt{140/144},\, 8/9\} > .68.$$

White 4 forces $Q = 0$; its far-own-3 has the affine F_4 path, while the near-own-3 and every own at least 4 have $C_8 > 5$. The two-end-3 join has residual $17/144 = 1/D$ and $C_8 \geq 89/17 > 5$. Consequently all these ordinary roles have

$$\frac{\langle y_A, \omega_8 \rangle}{\sqrt{D}} > \frac{5 - 1.65}{\sqrt{24}} > .68, \qquad e_A > .4624 > .46.$$

For isolated contact put $\tau = (1 - Q)/\gamma^2$. For $0 < \tau < 1$, the actual core

$$\Xi = d + w_S + \gamma\tau n_8, \qquad U = \gamma(\omega_8 - \tau n_8)$$

has square $-\gamma^2\tau(1 - \tau)$, positive time, zero selected products except a negative n_8 product, and all omitted products nonpositive. If $e_A \leq .46$, its pressure would exceed $(1 - .68)\sqrt{D - \tau} > .8736$,

whereas comparison bounds it by $\kappa/2 \le .825$. At $\tau = 1$ it is a forbidden future null. For $\tau > 1$ the positive finite projection

$$q + \frac{H}{\gamma^2(\tau-1)}(d + w_S + \gamma n_8)$$

gives

$$\langle y_A, \omega_8 \rangle \ge \sqrt{(D-1)/(D-\tau)}(\sqrt{D-1} - \kappa\sqrt{\tau-1}).$$

For $Q > 0$, the radical is at most $2 - \phi < .4$; for $Q = 0$ and finite own at least 8 it is at most $\sqrt{2} - 1 < .415$. The resulting heights are > 2.07 and > 2.04525, respectively, and their normalized values exceed .68. Pure isolated 5 uses the positive H4 correction $\gamma(3n_8/2 + n_7 + n_6/2)$, residual $\delta_5 = (5-3\phi)/8 < .01825$; since $H > 2$, its normalized pressure exceeds $1 - 1.65(.1352)/2 > .88$. Pure isolated 7 uses $d + (4\gamma/3)n_8 + (2\gamma/3)n_7$. The cubic $8a^3 - 4a^2 - 4a + 1$ gives $.9 < a < 10/11$, hence $H/\gamma > 13/5$ and the core radical divided by γ is $< 9/25$. Its pressure exceeds $1003/1300 > .68$.

For a pure far-3 helper at either end the exact unequal times are

$$H^2 = D\gamma^2 + C_{8e}\gamma - 9/17 > D\gamma^2, \qquad C_{8e} \in \{18/17, 16/17\}.$$

For finite own at least 7, with $\delta = 1 - \gamma^2 < .19$, the core $d + \gamma n_8 + 2\delta n_e$ has radical at most $1/4$, $H > 2.6$ and pressure $> .84$. For dotted own use $d + n_8/\gamma$; its radical divided by γ is at most $1/2$, giving pressure $> 1 - 1.65/(2\sqrt{D}) > 83/116$. All these cores are positive actual sums and check every wall.

The only exception to $e_A > .46$ is native-5 single-white-3. Choose the H4 correction on the side opposite its helper, and retain the full $U = v_A + v_S - w_5$, rather than calling it pure A. Its actual projected point gives

$$\langle y, U/\|U\| \rangle > 1 - \frac{1.65(.1352)}{H} > .9, \qquad H^2 \ge (44\phi + 27)/17 > 2.4^2.$$

The full white remainder gives $Q < 9/16$, and hence $|\langle u, U/\|U\| \rangle| < .75/2.4 < 1/3$. One vector

$$W = .49275u + .73575\, U/\|U\|$$

has $\langle y, W \rangle > 1.0258245$ and $W^2 < 1.0258245$. Thus $\|y\|^2 \ge 2\langle y, W \rangle - W^2 > 1$. For every other role, $e_A + e_{2A_1} > .46 + .738^2 = 1.004644$. This proves the local statement without adding overlapping energies.

B.9.14 Isolated pairs: finite and orthogonal mutual products

Write the two actual isolated caps as

$$c = Ap - g\Omega_f, \quad d = Bp - h\Omega_o, \qquad A^2 = 9g^2 - 1, \quad B^2 = 9h^2 - 1,$$

and $\lambda = -\langle c, d \rangle = AB - 8gh \ge 0$. The identity

$$(9gh - 1)^2 - A^2B^2 = 9(g - h)^2$$

gives $\lambda \le gh - 1$; all square roots here are positive. If $0 < \lambda < 1$, it is a native finite cosine, at least $1/2$. The positive projection onto c, d gives an actual $A_{16} + I_2(m)$ vertex with

$$C^2 = \frac{17g^2h^2}{1 - \lambda^2} \ge \frac{17(1 + \lambda)}{1 - \lambda} \ge 51.$$

Its displacement is a positive multiple of $P_{17} + ve_{18}$, $|v| < 1$. Every released root, the head and every unknown wall has nonpositive product termwise with the projection. The original envelope therefore gives

$$\kappa < \frac{3}{2}, \qquad \langle y, u \rangle > \frac{.99 - .9124}{1 - .99(.9124)} > .8$$

in the entire new $I_2(m)$ plane; the original maximum is unchanged. The local A_{16} comparison in Appendix B.8.12 gives $e_A > 7/16$ for $m \geq 4$.

For completeness the missing spectator $m = 3$ introduces no uncontrolled contact. Its full inverse matrix is $\left(\begin{smallmatrix} 4/3 & 2/3 \\ 2/3 & 4/3 \end{smallmatrix}\right)$. Two positive contacts cost at least one, and a single contact gives exactly

$$Q = 0, \quad 1/3, \quad 2/3, \quad \phi^2/3.$$

The A-contact list is exactly that of Appendix B.9.13. At own-zero end-3 the only $Q < 9/16$ are zero and $1/3$, both below $9/17$; the next-3 and end-4 rows force zero. All other own-zero and small-own nulls are the same actual principals. Every positive foreign cost is at least $1/3 > 1/4$. Consequently every ordinary ratio, isolated-core estimate, pure-helper estimate and paired H4 estimate in that appendix remains valid with the positive clearing supplied by the A_2 inverse, at the smaller factor $3/2$. In particular the ordinary and isolated energies exceed .46; in the native-5 exception the full foreign correlation with the unit reverse direction remains $< \sqrt{Q}/H < 1/3$, and its same single-vector proof applies since the reverse height exceeds $.8 > .738$. Thus either one square already contradicts the norm bound, or $e_A + e_{I_2} > 7/16 + .64 > 1$. This closes all positive finite mutuals. For $\lambda = 1$, $c + d$ is a nonzero feasible future null.

We next exclude $\lambda = 0$, with no symmetry assumption. The exact relation

$$(17a - 9)b = 9a - 1, \qquad a = g^2, \quad b = h^2$$

shows that exactly one own is finite and one dotted: two finite owns give $\lambda < 0$, and two dotted owns give $\lambda > 0$. Name the finite one $g = \cos(\pi/m)$, $m \geq 7$, so $a < 1 < b$. The actual finite set $n_1, \ldots, n_{15}, n_f, c, d$ has forest $A_{15} + I_2(m) + A_1$. Its positive projection is

$$q_0 = p + \frac{A}{1 - a} c + \frac{gA}{1 - a} n_f + Bd, \qquad C^2 = \frac{8ab}{1 - a}.$$

Its released products are strictly negative, all unknown products are nonpositive, and $q_0^2 = \langle p, q_0 \rangle = -C^2$. Write

$$q_0 = C^2 p - KV, \quad V = P_{16} + \alpha(e_{17} - \sigma e_{18}), \quad \alpha = \frac{8(1 - a)}{9a - 1} \in (0, 1).$$

The full original reference has $\langle w, V \rangle = f_{16} < 3.5899$. Paying its unspent .127 at the residual $V - P_{16}$, whose square is $2\alpha^2$ and cross with w is zero, proves

$$\frac{\langle x, V \rangle}{\sqrt{16 + 2\alpha^2}} < \frac{3.6967 + .127\alpha^2}{\sqrt{16 + 2\alpha^2}} < .9242.$$

Indeed a violation would give $.9729 + .254(3.6967 - 3.5899) > 1$; the α^2 terms cancel exactly. The displayed envelope decreases with α^2.

The native-7 cubic gives $.9009 < g < .901$. The increasing function

$$C^2 = \frac{9a - 1}{1 - a} + 9\frac{9a - 1}{17a - 9}$$

then exceeds $45.2944 > 6.73^2$, and $\alpha > .238$. The envelope improves to .9228. For $m \geq 8$, $a > (.8535)$, the same function gives $C^2 > 56.49 > 7.5^2$. The exact comparisons are respectively

	E_q/E	$\langle y, u \rangle$
$m = 7$	$> .58848916 > 10/17$	$> (.9888 - .9228)/(1 - .9888(.9228)) > .75$
$m \geq 8$	$> .6294972 > 10/17$	$> (.99 - .9242)/(1 - .99(.9242)) > .75.$

Here $u = (Cq - p)/\sqrt{C^2 - 1}$ belongs to the entire $I_2(m) + A_1$ space. Thus $\kappa < 17/10$ and its energy exceeds $9/16$.

We give the local A_{15} argument at this exact factor. In its zero-sum $\mathbb{R}^{16}$ frame, $D_8 = 8$. A finite-white contact to $I_2(m \geq 7)$ costs at least $1/(4\sin^2(\pi/m)) > 25/19 > 1$, so none is possible. The full foreign vector lies in the retained A_1, with cost $Q = \beta^2$, each positive cost at least $1/4$. Deleting eight leaves $A_7 + A_7$, with single-3 positions $1, 2, 3, 5, 6, 7, 9, 10, 11, 13, 14, 15$, endpoint 4, or one endpoint 3 in each path. For own zero, the end-3 time is $Q - 17/32$ with $Q < 9/16$, so $Q = 0, 1/4, 1/2$ gives negative time. Next-3 and end-4 force zero with times $-1/8, -1/16$. The remaining own-zero and small-own roles have the same actual affine stars, chains and cycles as the central inventories above. Keep near-own-4 and native-5 single-white-3.

For positive single-white-3 other than native 5, the own is $> .9$. The residual/numerator bounds are

near 3	$9/16$	$.9 + 7/16$
D position	$1/4$	$.9 + 1/8$
E position	$1/16$	$.9 + 3/16$
far 3, $Q > 0$	$5/16$	$.9 + 1/16.$

The first three give $C_8 > 5$, and the last $C_8^2 > 5929/250 > 4.86^2$. At factor 1.7 the latter normalized height exceeds $3.16/4.757 > .663$, whose square exceeds $7/16$. Near-own-4 with foreign cost is stronger. Without foreign cost its actual seven/nine mate has length $\sqrt{63/8}(3/2 + 4\sqrt{2}/3) > 9$. White 4 forces $Q = 0$; the far-own-3 affine F_4 null is excluded, while the retained near-own-3 and every own at least 4 have $C_8 > 6$. The two-end-3 join has $C_8 \geq 8\gamma + 1 \geq 5$.

For isolated contact, $\tau = (1 - Q)/\gamma^2$ gives the same actual core $d + w_S + \gamma\tau n_8$. If $0 < \tau < 1$, energy at most $7/16$ would force its pressure $> \sqrt{7} - 7/4 > 17/20$, against $\kappa/2 < 17/20$. The boundary is an actual null. For $\tau > 1$, positive projection gives

$$h_8 \geq \sqrt{7/(8 - \tau)}(\sqrt{7} - \kappa\sqrt{\tau - 1}).$$

For $Q > 0$, $\tau < 75/64$; for pure finite own at least 8, $\sqrt{\tau - 1} < .415$. Thus $h_8 > 1.8945 > 17/9$, giving normalized height $> 2/3$. Pure isolated 5 and 7 use the actual H4 point and the $(4\gamma/3, 2\gamma/3)$ core: their heights exceed 2 and $1.988 > 17/9$. Pure far-3 has $H^2 = 8\gamma^2 + \gamma - 17/32 > 8\gamma^2$. For finite own at least 7 the $d + \gamma n_8 + 2(1 - \gamma^2)n_e$ core has radical at most $1/4$, $H > 2.5$, and unit pressure $> .83$. For dotted own $d + n_8/\gamma$ gives pressure $> 1 - 1.7/(2\sqrt{8}) > 39/56$. Both ends and every unbounded interval are included.

Finally native-5 single-white-3 uses the opposite-side H4 point, retaining its full foreign component U. Here $H^2 \geq (5/2)\phi + 47/32 > 11/2$, $\delta_5 < .01825$, so its unit pressure is $> 1 - 1.7(.1352)/2.34 > .9$. Its absolute correlation with the reverse u is $< .75/2.34 < 1/3$. One vector $(81/160)u + (117/160)U/\|U\|$ has pressure and square bounded on opposite sides of 1.0378125; hence $\|y\|^2 > 1$. All other roles have $e_A > 7/16$, and the orthogonal reverse-space energy is $> 9/16$. This excludes the orthogonal pair. Every local point above is a positive actual projection or core; the wall checks include the old head and all unknown walls.

B.9.15 The complete dotted isolated pair

Only $\lambda > 1$ remains for two isolated caps. Put $P = gh$. The inequality $\lambda \le P - 1$ gives $P > 2$; the individual cutoff $g, h < 15/8$ then gives $g, h > 16/15$, so both owns are dotted. There is no need to use the older auxiliary four-thirds gap. Assume $g \ge h$. The actual feet

$$q_c = \frac{d + \lambda c}{\sqrt{\lambda^2 - 1}}, \qquad q_d = \frac{c + \lambda d}{\sqrt{\lambda^2 - 1}}$$

are future and feasible: the selected product is zero, the other is negative, and every omitted wall is checked termwise. Their squared coshes satisfy

$$C_c^2 = 9g^2 + \frac{17P^2}{\lambda^2 - 1}, \qquad C_d^2 = 9h^2 + \frac{17P^2}{\lambda^2 - 1}.$$

Thus

$$C_c^2 \ge 9P + \frac{17P}{P - 2} > \frac{441225}{6208} > 71.$$

The last expression decreases on $2 < P < 225/64$, so the strict individual cutoffs justify its endpoint evaluation.

Retain the actual edge $A_{16} + c$ obtained by releasing the other fork at the $A_{17} + A_1$ neighbour of c. The perpendicular foot of p is $(p + Ac)/(3g)$, with zero product against that released fork. The chosen q_c has negative product against it. In the retained Lorentz plane this signed product is a positive multiple of $-\sinh t$; hence q_c is on the forward ray. Compactness supplies the next actual endpoint R, at least as far from p. The old A_{16} envelope gives $C_R < 10.94 < 11$. For every such endpoint,

$$\sqrt{71} < C_R < 11, \quad \kappa < 1.262, \quad \langle y, u \rangle > .8564, \tag{DIIdot}$$

where $u = (C_R R - p)/\sqrt{C_R^2 - 1}$. For the two latter bounds use $t < .9124$, the increasing function $C - .9124\sqrt{C^2 - 1}$, and $C_R^2 > 441225/6208$, which gives $\sqrt{1 - C_R^{-2}} > .99294$ since $(1 - .99294^2)(441225/6208) > 1$. For example $(.99294 - .9124) - .8564(1 - .99294(.9124)) > .00000282$. All comparisons retain the same original maximum.

The complete finite one-node extensions of the retained A_{16} are $A_{16} + I_2(m)$, including $m = 2$; $A_{17} + A_1, B_{17} + A_1, D_{17} + A_1$; and whole A_{18}, B_{18}, in their allowed end orientations. This is the complete finite classification, not a list of likely endpoint types. The new wall $e = Tp - v$ is distinct from old walls: the released fork vanished at the starting endpoint only, and the other old fork has a dotted product with c. Thus $T > 0$ and every original contact of e belongs to $0, \cos(\pi/m), 1, (1, \infty)$.

If e is orthogonal to A_{16}, the reverse lies in the whole rank-two plane. For $m \ge 4$ the local A_{16} lemma in Appendix B.8.12, and for $m = 3$ its extension above, give energy $> 7/16$; the reverse energy $> .8564^2$ gives a contradiction. For $m = 2$, Appendix B.9.13 applies directly. For whole B_{18}, the inverse diagonal at c is 18, so $C_R^2 \ge 1 + 18A^2 > 121$, impossible.

Here are the remaining B/D germs without a purity assumption. Put $x = -\langle e, n_f \rangle$, $y = -\langle e, n_o \rangle$. For $B_{17} + A_1$, $T^2 < 120/17$. The near white 4 already gives $T^2 \ge 15$. For the far white 4,

$$T^2 = \sqrt{2}(x + y) + 9(x^2 + y^2) + 16xy, \qquad \frac{T^2}{g^2} = -\tfrac{1}{2} + \sqrt{2}y + 17y^2.$$

If both contacts are positive then $T^2 \ge 8.5 + \sqrt{2}$. If $y = 0$ the second expression is negative. If $x = 0$, its denominator exceeds the first expression, forcing $g^2 < 1$, again impossible. For $D_{17} + A_1$, $T^2 < 240/17$. At the near white 3,

$$T^2 = 13/2 + 15(x + y) + 9(x^2 + y^2) + 16xy.$$

One positive contact forces $T^2 \geq 65/4$; if both vanish, the actual affine D_5 vector with coefficients $1,1,2,2,1,1$ on $e, n_{14}, n_{15}, n_{16}, n_f, n_o$ is a forbidden future null. At the far white 3,

$$T^2 = 2(x+y) + 9(x^2+y^2) + 16xy, \qquad \frac{T^2}{g^2} = -1 + 2y + 17y^2.$$

The zero-contact cases fail as above. If a positive contact is at least $1/\sqrt{2}$, then $T^2 \geq 31/4 + 5\sqrt{2} > 240/17$. The only remaining pair is $x = y = 1/2$, whose actual e, n_f, n_{16}, n_o all-3 cycle gives a feasible null.

The final $A_{17} + A_1$ and whole-A_{18} germs have

$$u = \frac{T\Omega'_e + A\Omega'_c}{\sqrt{C_R^2 - 1}},$$

with both coefficients positive. These are exactly the two standalone local statements of Appendix B.9.12, at the factors and pressure in (DIIdot). Thus all actual endpoints are impossible, completing the isolated/isolated exclusion over every head.

B.9.16 The complete isolated/near-three pair

Let

$$c = Ap - g\Omega_f, \quad d = Bp - h\Omega_o - \Omega_f/2, \quad A^2 = 9g^2 - 1, \quad B^2 = 9h^2 + 8h + 5/4.$$

They are distinct actual walls by their different fork contacts. Write $\lambda = -\langle c, d\rangle \geq 0$. For a positive finite mutual,

$$17g^2h^2 - 9(g^2+h^2) - 16\lambda gh - 8h - 9\lambda g - 5/4 - \lambda^2 = 0.$$

Consequently $gh > (18 + 16\lambda)/17 > 3/2$. The positive pair projection onto c, d, retaining old A_{16}, is an actual $A_{16} + I_2(m)$ vertex and has exactly

$$C^2 = \frac{17g^2h^2}{1-\lambda^2} > 51.$$

Its displacement is a positive fork-dual combination. The same envelope gives $\kappa < 3/2$, reverse pressure $> .8$; the complete dihedral local comparison above excludes it. At $\lambda = 1$, $c + d$ is an actual future null.

Suppose next that $\lambda > 1$. The positive core

$$\Xi = d + c/\lambda, \quad k = \sqrt{1 - \lambda^{-2}}$$

has square $-k^2$, zero d product and negative c product; all other wall products are nonpositive. Its time is $B + A/\lambda > 0$. Put

$$D = g/\lambda + h + 1/2, \quad v = \sigma(g/\lambda + 1/2 - h)/D, \quad w = k/D.$$

Comparison at the genuine point Ξ/k gives

$$\langle x, P_{17} + ve_{18}\rangle \geq \sqrt{17 + v^2 + 2w^2} - \sqrt{2}w. \tag{DINcore}$$

We verify the entire continuous bound $w < 1/4$. Put $z = 1/g^2$, $H = h + 1/2$, and $j = \lambda/g = \sqrt{9 - z}B - 8h - 9/2 > 0$. Then

$$w^2 = \frac{j^2 - z}{(1+jH)^2}, \quad \partial_j = \frac{2(j+Hz)}{(1+jH)^3} > 0, \quad \partial_z = -\frac{1}{(1+jH)^2} < 0.$$

Replace $g < 15/8$ by $15/8$ to increase this value. The original $C_{\mathrm{near}} < 10$ gives $h < 2$. Concavity $B'' = -19/(4B^3) < 0$ and its tangent at 2 give

$$B(h) \le (44h + 37/2)/\sqrt{213}, \qquad j < .901h - .757 =: J(h),$$

using $\sqrt{1961/47925} < .20229$. Since the endpoint $j > 8/15$, this forces $7/5 < h < 2$. For $P = 1 + J(h)(h + 1/2)$ define

$$K = P^2 - 16J(h)^2 + 1024/225.$$

Now $K' = 2PP' - 32(.901)J$ and $K''' = 12(.901)P' > 0$, so K' is convex. Its values at $7/5, 2$ are negative: use $P < 2, P' < 2.22, J > .5$ at the first end, and $P < 3.62, P' < 3.3, J > 1.04$ at the second. Thus K decreases, and $K(2) = 185569/1440000 > 0$. This proves $w^2 < 1/16$, with no native-label sampling. The right side of (DINcore) decreases with w, and after subtracting $.0635v^2$ it increases with v^2. Since $\sqrt{137/8} > 4.138$, $\sqrt{2}/4 < .354$, it exceeds

$$3.784 + .0635v^2,$$

contradicting the original envelope $3.761 + .0635v^2$.

It remains to treat $\lambda = 0$. The exact equation is

$$g^2 = \frac{9h^2 + 8h + 5/4}{17h^2 - 9}.$$

Near own 4 gives a negative denominator. For own 5 it gives $(25\phi + 14)/(17\phi - 19) > 225/64$, contrary to $g < 15/8$. Thus $h > .9$. Since c, d are orthogonal and orthogonal to old A_{16}, the positive projection

$$q_0 = p + Ac + Bd, \qquad C^2 = 1 + A^2 + B^2 = 17g^2h^2$$

is an actual $A_{16} + 2A_1$ vertex. Every released and unknown wall is nonpositive. For all $h > .9$, $C^2 > 37$: after clearing the positive denominator, the difference is

$$f(h) = 153h^4 + 136h^3 - (2431/4)h^2 + 333.$$

Here $f'' > 1000$, $f(1.1) = 2.6458$, $f'(1.1) = -28.798$, so $f(h) \ge 2.6458 - 28.798(h - 1.1) + 500(h - 1.1)^2 > 2.23108$. The displacement again has direction $P_{17} + ve_{18}$. Using $\sqrt{36/37} > .9863$ and $\sqrt{37} > 6.0827$ gives

$$E_q/E > .6083 > 20/33, \qquad \langle y, u\rangle > \frac{.9863 - .9124}{1 - .9863(.9124)} > .738.$$

The whole two-A_1 direction is $u = (Ac + Bd)/\sqrt{A^2 + B^2}$. Appendix B.9.13 excludes this exact comparison. Finite, parallel, dotted and orthogonal mutuals have now all been covered for isolated/near-three pairs.

B.9.17 Isolated/far-three: every nondotted mutual

For an isolated cap and a far-three cap write

$$c = Ap - g\Omega_f, \quad d = Fp - h\Omega_o - \Omega_1/2, \quad A^2 = 9g^2 - 1, \quad F^2 = 9h^2 + h - \tfrac{1}{2}.$$

They are distinct actual walls. Here $g > .9$, and h is native 5, finite at least 7, or dotted. Put $\lambda = -\langle c, d\rangle \ge 0$, $s_0 = \sqrt{19}/6$. The positive square identity, with $X = 3g, Y = 3h + 1/6$, is

$$(XY - s_0)^2 - (X^2 - 1)(Y^2 - s_0^2) = (s_0X - Y)^2.$$

It gives the useful full-domain inequality

$$gh \geq \lambda + s_0, \qquad .726 < s_0 < .727. \tag{DIFproduct}$$

At $\lambda = 0$, direct squaring gives

$$g^2 = \frac{9h^2 + h - 1/2}{17h^2 + h - 19/4}.$$

This function decreases: its derivative numerator is $-8h^2 - (137/2)h - 17/4$. For $h > .9$ it gives $g^2 < 769/992 < .81$. For the remaining own $h = \phi/2$ it gives $(11\phi + 7)/(19\phi - 2)$, strictly between $.86, .8649$. Thus g lies strictly between native labels eight and nine, and is < 1, so cannot be dotted. This excludes mutual zero.

We retain an additional simultaneous coordinate bound only as a constraint on the same x:

$$S_2 < 1.04. \tag{DIFS2}$$

Indeed $S_{16} > 3.589$, and if $S_2 \geq 1.04$, Cauchy on the first two and next fourteen coordinates gives $\|x\|^2 > 1.04^2/2 + 2.549^2/14 > 1$. The right side increases for $S_2 \geq 1.04$; also $S_2 \leq \sqrt{2} < S_{16}$. No extra component energy has been added to a paid reference.

Clear d from old A_{16} by its positive endpoint dual:

$$\bar{d} = d + \omega_1^{A_{16}}/2 = Fp - (h + 1/34)\Omega_o - \Omega_f/34, \qquad \bar{d}^2 = r = 9/17, \quad \langle c, \bar{d} \rangle = -\lambda.$$

For $\lambda = 1/\sqrt{2}$, the positive pair inverse, of determinant $r - \lambda^2 = 1/34$, gives an actual whole-B_{18} point, with

$$C = g(34h + 1)/\sqrt{2} > 12.$$

Its direction is a positive combination of the fork duals; the original envelope gives $E_q/E > .0876C > 1.0512$, contrary to the original maximum.

For $\lambda = \phi/2$, clear only n_1: $d_1 = d + n_1/2$ has square $3/4$. The positive projection on c, d, n_1 has coefficients

$$a = \frac{(3/4)A + \lambda F}{\Delta}, \quad b = \frac{F + \lambda A}{\Delta}, \quad \Delta = (2 - \phi)/4,$$

and $C^2 = 1 + [(3/4)A^2 + 2\lambda AF + F^2]/\Delta > 441$. For this last bound, $AF > 13.089$, $\sqrt{3/4} > .866$, $\lambda > .809$, $\Delta < .0955$. Its full displacement, including the head coordinate, is

$$\frac{D}{\sqrt{2}}(P_{17} + ve_{18} + \epsilon P_2), \quad D = ag + bh, \quad |v| < 1, \quad \epsilon = b/(2D) < 5/22.$$

The last inequality follows from $a/b > \lambda$ and $h + \lambda g > 2.2$. Using (DIFS2) conservatively as $S_2 < 1.1$, the normalized pressure is at most

$$\frac{3.761 + .0635s + 1.1\epsilon}{\sqrt{17 + s + 4\epsilon + 2\epsilon^2}}, \qquad s = v^2.$$

On $0 \leq s \leq 1$, $0 \leq \epsilon \leq 5/22$, it decreases in s and increases in ϵ. For the derivative signs use numerator < 4.08, denominator square < 19.1, and $.127(19.1) - 3.761 < 0$, $18.7 - 4.08(2.46) > 0$. Its maximum is $< 4.011/\sqrt{18} < .95$. Thus $E_q/E > .05C > 1.05$, again impossible. All projected points have positive actual coefficients and check the selected products, positive time and every omitted wall.

For $\lambda = \sqrt{3}/2$, the actual sum $d + n_1/2 + \lambda c$ is a nonzero feasible future null. For $\lambda = 1$, use $c + d$. For every remaining finite mutual $.9 < \lambda < 1$, use

$$\Xi = d + n_1/2 + \lambda c, \quad \Xi^2 = 3/4 - \lambda^2 = -k^2 < 0.$$

Its c, n_1 products are zero and its d product is negative; all other products are nonpositive. With $D = h + \lambda g$, $t = 1/D$, $v = \sigma(\lambda g - h)/D$, comparison gives

$$\langle x, P_{17} + ve_{18} + (t/2)P_2 \rangle \geq \sqrt{17 + v^2 + 2t + (2\lambda^2 - 1)t^2} - \sqrt{2\lambda^2 - 3/2}\, t.$$

The upper bound is $3.761 + .0635v^2 + .52t$. By (DIFproduct), $t \leq t_0 = [2\sqrt{Q}]^{-1} < 5/12$, where $Q = \lambda(\lambda + s_0)$. After subtracting $.0635v^2 + .52t$, the lower bound increases with v^2 and decreases with t: its first derivative is $> 1/9 - .0635$, and its radical t-derivative is at most $17/48$, whereas the subtracted coefficient exceeds .86. Set $v^2 = 0, t = t_0$. The radical square

$$B_0 = 17 + 1/\sqrt{Q} + (2\lambda^2 - 1)/(4Q)$$

decreases on $[.9, 1]$, while

$$O = \frac{\sqrt{2\lambda^2 - 3/2} + .52}{2\sqrt{Q}}$$

increases. For the first derivative the numerator is $2s_0\lambda^2 + 2\lambda + s_0 - 2(2\lambda + s_0)\sqrt{Q}$; its positive part is at most 4.181, and the subtracted part exceeds 6.0624. For the second derivative sign the numerator is $2s_0\lambda^2 + 3\lambda + 3s_0/2 - .52\sqrt{2\lambda^2 - 3/2}(2\lambda + s_0)$; its positive part is at least 4.96512, and its subtracted part is below 1.004. At $\lambda = 1$, $t_0 < .3807$, and direct positive squares give

$$\sqrt{B_0} - O > 4.2315 - 1.2272(.3807) = 3.76430496 > 3.761.$$

This excludes the entire finite tail, including all labels at least seven, without extending the inequality to dotted mutuals.

Finally suppose $\lambda = 1/2$. Positive projection on old A_{16}, c, d has determinant $19/68$ and gives an actual whole A_{18} vertex with

$$C = \frac{g(34h + 1)}{\sqrt{19}} > 19/2.$$

The strict bound follows from $gh \geq 1/2 + \sqrt{19}/6$. The displacement has positive fork-dual coefficients; hence

$$E_q/E > 9.5 - .9124(9.448) = .8796448 > 7/8, \quad \kappa < 8/7, \quad \langle y, u \rangle > \frac{.9944 - .9124}{1 - .9944(.9124)} > .884.$$

In the chain $n_{16}, \ldots, n_1, d, c$, the reverse is exactly

$$u = (A\Omega'_{18} + F\Omega'_{17})/\sqrt{C^2 - 1}.$$

Both coefficients are positive. The standalone local A_{18} statement of Appendix B.9.12 applies at the weaker requirements $\kappa < 1.27, \langle y, u \rangle > .85$. Thus every nondotted mutual of an isolated/far-three pair is excluded.

B.9.18 Far-three pairs and the remaining isolated/far-three pairs

This argument is uniform over every head. For far-three/far-three caps put

$$c_f = F(g)p - g\Omega_f - \Omega_1/2, \quad c_o = F(h)p - h\Omega_o - \Omega_1/2, \quad F(t)^2 = 9t^2 + t - \tfrac{1}{2},$$

where each own is native 5, finite at least 7, or dotted, and each is below 1.6. Their mutual magnitude is

$$\lambda = F(g)F(h) - 8gh - (g+h)/2 - 1/2.$$

Clear both off the actual old A_{16}:

$$b_f = F(g)p - (g + 1/34)\Omega_f - \Omega_o/34, \quad b_o = F(h)p - (h + 1/34)\Omega_o - \Omega_f/34.$$

These are positive old-root corrections, with $b_f^2 = b_o^2 = r = 9/17$ and $\langle b_f, b_o\rangle = -(\lambda + 8/17)$. If $\lambda = 0$, their Schur inverse is $\left(\begin{smallmatrix} 9 & 8 \\ 8 & 9 \end{smallmatrix}\right)$. Positive projection gives an actual whole-D_{18} vertex with

$$C^2 = 1 + 9(F(g)^2 + F(h)^2) + 16F(g)F(h) > 205,$$

contrary to the all-point old-A_{16} bound $C < 10.94$. For $\lambda = 1/2$ use the actual future affine-A_2 null $c_f + c_o + n_1$; for $\lambda = 1$, use $c_f + c_o$. Thus any remainder has $\lambda \geq 1/\sqrt{2}$. The elementary positive Lorentz square, applied to $3g + 1/6, 3h + 1/6$ with residual $19/36$, gives $\lambda \leq gh - 1$. If either own were finite, $gh < 1.6$ would give $\lambda < .6$. Both owns are therefore dotted.

Set $\mu = \lambda + 8/17 > r$. The positive actual sums

$$\Xi_f = rb_o + \mu b_f, \qquad \Xi_o = rb_f + \mu b_o$$

have square $-r(\mu^2 - r^2)$, positive time, zero products with old A_{16} and the respective selected cap, negative product with the other cap, and all other products nonpositive termwise. Since $F(t) > 3t$, their normalized feet satisfy

$$\max(C_f^2, C_o^2) > \frac{17(\lambda+1)^2}{\lambda - 1/17} \geq 72. \tag{DFFfoot}$$

For the last inequality, put $a = \lambda - 1/17$ and use $a + 36/17 + (18/17)^2/a \geq 72/17$. The preceding inequality retains unequal owns through $F(g) + F(h) > 3(g+h) \geq 6\sqrt{gh} \geq 6\sqrt{\lambda + 1}$.

For the dotted remainder of an isolated/far-three pair, write $c = Ap - g\Omega_f$, $d = F(h)p - h\Omega_o - \Omega_1/2$. Now $\lambda > 1$, $g < G = 15/8$, and $\lambda \leq gh - s_0$, $s_0 = \sqrt{19}/6$. In particular $h > (1 + s_0)/G > .92$, including possible finite owns. With

$$b = F(h)p - (h + 1/34)\Omega_o - \Omega_f/34, \quad b^2 = r, \quad \langle c, b\rangle = -\lambda,$$

the positive actual foot from $rc + \lambda b$ has

$$C_Q^2 = 1 + \frac{F(h)^2}{r} + \frac{17g^2(h + 1/34)^2}{\lambda^2 - r}.$$

This is feasible against all walls; its selected b product vanishes and its c product is negative. We prove a lower bound on the whole domain $h > .92$, rather than assuming $h > 1$. Replace λ by $gh - s_0$ to decrease the bound. The ratio $g^2/[(gh - s_0)^2 - r]$ decreases with g, because $s_0^2 - r = -1/612$; replace g by G. Put $c_0 = 2s_0/G$, $b_0 = c_0 + 1/17$. Then $.774 < c_0 < .776 < h$, $b_0 > .832$. Polynomial division and the strict negative residual give

$$\begin{aligned} C_Q^2 &> 17\{h^2 + h/9 + 1 + b_0/(h - c_0)\} + 1/18 \\ &> 17\{(25/9)c_0 - 7/9 + 2\sqrt{25b_0/9}\} \\ &> 17(2.15 - 7/9 + 3.04) = 67507/900 > 75. \end{aligned} \tag{DIFfoot}$$

The middle step uses $h^2 \ge (8/3)h - 16/9$ and AM–GM in $h - c_0 > 0$. The endpoint seals are $25(.774)/9 = 2.15$ and $25(.832)/9 = 104/45 > 1.52^2$.

We now justify passage from either foot to an actual endpoint. Choose the cap d owning the larger foot in (DFFfoot), or the far cap in (DIFfoot). Its positive original-edge neighbour is whole A_{18}. Release the other old fork there, retaining the actual white A_{17} path $d, n_1, \dots, n_{16}$. The perpendicular foot of p on this line is the normalization of $p + (F/r)b$. Its product with the released old fork is negative with magnitude less than $1/(34\sqrt{r})$. The selected far/far foot has magnitude

$$\frac{rk + (r+\mu)/34}{\sqrt{r(\mu^2 - r^2)}} > \frac{1}{34\sqrt{r}},$$

where $k > 0$ is the other cap own. The isolated/far foot has magnitude

$$\frac{rg + \lambda/34}{\sqrt{r(\lambda^2 - r)}} > \frac{1}{34\sqrt{r}}.$$

The released root projects to a spacelike normal in the retained Lorentz plane, and its signed product is proportional to $-\sinh t$ on the forward ray. Thus the selected foot is after the perpendicular foot. The next actual endpoint R is at or beyond it, and distance from p increases there. Consequently

$$\sqrt{72} < C_R < 10.94, \qquad \kappa < 1.262, \qquad \langle y, u\rangle > .8564. \tag{DFexit}$$

The latter comparisons use the same original envelope and maximum. No intrinsic Coxeter structure of an arbitrary face is assumed.

The complete finite extensions of the retained white A_{17} are $A_{17}+A_1$, A_{18}, B_{18} at either endpoint, and D_{18} at either next-to-end node: seven oriented germs. Exceptional finite types have rank at most eight, and two contacts to this path make a cycle. Write the new wall $e = Tp - v$, with full original fork contacts $a = -\langle e, n_f\rangle$, $b = -\langle e, n_o\rangle$. The complete germs are

germ	v	$\langle e, d\rangle$
$A_{17} + A_1$	$a\Omega_f + b\Omega_o$	0
A_{18} at d	$a\Omega_f + b\Omega_o$	$-1/2$
B_{18} at d	$a\Omega_f + b\Omega_o$	$-1/\sqrt{2}$
A_{18} at n_{16}	$a\Omega_f + b\Omega_o + \Omega_{16}/2$	0
B_{18} at n_{16}	$a\Omega_f + b\Omega_o + \Omega_{16}/\sqrt{2}$	0
D_{18} at n_1	$a\Omega_f + b\Omega_o + \Omega_1/2$	0
D_{18} at n_{15}	$a\Omega_f + b\Omega_o + \Omega_{15}/2$	$0.$

Every a, b belongs to the full actual alphabet $0, \cos(\pi/m), 1, (1, \infty)$. The new e is distinct from every old p-wall. The released old fork has its only zero at the starting endpoint; the other old fork joins both ends of the retained path, making a nonpositive cycle even when the far own is finite. Nor is e the other original cap: its corrected two-cap determinant is negative. Thus $T > 0$, and the actual original opposite at either fork is distinct from e.

Inverse positivity at R gives

$$B_{18} : T^2 < 7, \qquad D_{18} : T^2 < 14.$$

Acuteness against each distinct original fork opposite gives both original shadow inequalities $(\langle v, \Omega_f\rangle)/(3T) < 1$ and its reflection. For $v = a\Omega_f + b\Omega_o + \beta\Omega_j$, their exact form is

$$(\langle v, \Omega_f\rangle)^2 - 9T^2 = 9 - 17b^2 - 2\beta jb + \beta^2 j(j-18) < 0. \tag{DFshadow}$$

No opposite-only finite-white restriction has been imposed on e.

For B_{18} at n_{16}, $T^2 \geq 15 > 7$. For B_{18} at d, both contacts exceed $3/\sqrt{17} > 1/\sqrt{2}$, so the native jump gives $a, b \geq \phi/2 > .8$, hence $T^2 > 20.76$. For D_{18} at n_1, (DFshadow) gives $a, b > 1/2$, therefore both at least $1/\sqrt{2}$, and $T^2 \geq 16.5 + \sqrt{2} > 14$. For D_{18} at n_{15},

$$T^2 = 9(a^2 + b^2) + 16ab + 15(a + b) + 13/2.$$

A positive contact gives $T^2 \geq 16.25$. With both zero the positive actual sum $e + n_{14} + 2n_{15} + 2n_{16} + n_f + n_o$ is a feasible future affine-D_5 null.

For A_{18} at n_{16}, the shadow is $(1 + b)(1 - 17b) < 0$, and its reflection; both contacts are at least $1/2$. The value $1/2$ gives the actual affine-A_2 null. Both values $1/\sqrt{2}$ give the affine 4–4 null $\sqrt{2}e + n_f + n_o$. Thus $a, b \geq 1/\sqrt{2}$, at least one strictly larger, and

$$T^2 = 9(a^2 + b^2) + 16ab + 16(a + b) + 7.$$

The two nonzero original time coordinates at this actual whole A_{18} give the ONE full inverse identity

$$C_R^2 = 1 + \frac{36}{19}(T^2 + F^2) + \frac{4}{19}TF. \qquad \text{(DFA18cross)}$$

The cross is retained; no separate cap energy is added.

In both families $h > .92$, so $F^2 > 8$. If one contact is at least native 6 and the other native 4, then $T^2 > 53.212437$; if both are at least native 5, then $T^2 > 55.140354$. These use only $\cos(\pi/6) > .866$, $1/\sqrt{2} > .707$. In either case $TF > 20.63$ and (DFA18cross) gives $C_R^2 > 121.3246 > 119.6836$, impossible. The only remaining assignments are $\{a, b\} = \{\phi/2, 1/\sqrt{2}\}$, in both orders. If $h \geq 1$, then $F^2 \geq 9.5$, $T^2 > 50.796378$, $TF > 21.96$, and the same identity gives $C_R^2 > 119.8689 > 119.6836$. This closes every far/far case.

For the remaining finite far own $.92 < h < 1$ in isolated/far, actual incidence $\langle e, d \rangle = 0$ is exactly

$$f(h) = TF(h) - (9a + 8b + 8)h - (a + b + 1)/2 = 0.$$

The shadow gives $3T > 9a + 8b + 8$, and $F' > 3$, so f increases strictly. Put $K = 9a + 8b + 8, L = (a + b + 1)/2$. At $h = 1$,

$$4\{(K + L)^2 - (19/2)T^2\} = 19a^2 - 53b^2 + 38ab + 38a - 30b + 23.$$

For the two remaining assignments, rational enclosures $.809 < \phi/2 < .81$, $.707 < 1/\sqrt{2} < .708$ make this respectively > 40.171733 and > 22.027294. Both squared quantities are positive, so $f(1) < 0$. Hence $f(h) < 0$ throughout $h < 1$, contradicting incidence.

The last germs $A_{17} + A_1$ and A_{18} at d have

$$u = (T\Omega'_e + F\Omega'_d)/\sqrt{C_R^2 - 1},$$

with both coefficients positive and no old-A_{16} component. Appendix B.9.12 excludes both at the exact factor and height in (DFexit). All seven actual endpoint germs have been covered. Thus far-three/far-three is impossible over every head; together with Appendix B.9.17, so is the whole isolated/far-three family. Every null and projected point used only positive actual normals and checked all original walls.

B.9.19 The complete native-five head

Assume the actual isolated head is $H = A_H p - (\phi/2)\Omega_1$, where $A_H^2 = (\phi-1)/2$. Use $1.618 < \phi < 1.61804$ and $.5558 < A_H < .5559$ when the sharper upper bound is needed. All cap comparisons below are with distinct actual walls. The algebraic use of native cosines is elementary: it imposes no arithmeticity, field of definition or integrality on the polytope.

For an isolated fork,

$$M(g) = A_H\sqrt{9g^2-1} - (\phi/2)g$$

increases, because its derivative exceeds $3A_H - \phi/2 > 0$. For every remaining finite own, $.9 < g < 1$, $.66574582 < M(g) < .763924 < \phi/2$; hence the only possible native value is $1/\sqrt{2}$. But $\cos(\pi/9) < .94$, $\cos(\pi/10) > .951$, and

$$M(.94) < .705712 < 1/\sqrt{2}, \qquad M(.951) > .71513525 > 1/\sqrt{2}.$$

No finite own lies between labels nine and ten. The entire finite-own range is excluded. For dotted owns the same function gives $M(1) > .76275$, $M(1.02) < .782772 < \phi/2$, so $g > 1.02$.

For far-three, put $M(g) = A_H\sqrt{9g^2+g-1/2} - (\phi/2)(g+1)$. Its derivative is positive. Four times the difference of its two positive squared terms is

$$(17\phi-19)g^2 - 4g - 2\phi.$$

At native 5 it equals $-(\phi+2)/4 < 0$, contradicting acuteness. For finite owns at least seven it is positive already at .9 and increases, whereas $M(g) < M(1) < .096148 < 1/2$. These owns also fail. Dotted far owns satisfy $g > 1.4$, since $M(1.4) < .452536 < 1/2$. For far-next-three the analogous squared difference is

$$(17\phi-19)g^2 + (2\phi-6)g - \phi - 1.$$

It is positive at $g = \phi/2$ and increasing thereafter, but $M(g) < M(1) < .226252 < 1/2$. Only finite owns were left in this role, so far-next-three is entirely excluded under this head.

Near-three own 4 and 5 require an exact native-cosine check. Write $M = T - b$, where $T = A_H\sqrt{9g^2+8g+5/4}$, $b = \phi(g+1/2)/2$. For $g = 1/\sqrt{2}$, $.89035 < M < .90928$, so it would be a finite cosine. In the totally real biquadratic field $\mathbb{Q}(\phi, \sqrt{2})$,

$$T^2 = \frac{(\phi-1)(23+16\sqrt{2})}{8}, \qquad b = \frac{\phi(\sqrt{2}+1)}{4}.$$

The embedding $\phi \mapsto 1-\phi$ fixing $\sqrt{2}$ makes T^2 negative. If M were a finite cosine, $T = M + b$ would lie in a compositum of totally real cyclotomic fields, impossible. For $g = \phi/2$,

$$T^2 = (11+14\phi)/8, \quad b = (2\phi+1)/4, \quad T^2-b^2 = (17+20\phi)/16 > 0, \quad (b+1)^2-T^2 = (7-4\phi)/16 > 0.$$

Thus $0 < M < 1$. The norm of T^2 from $\mathbb{Q}(\phi)$ is $79/64$, not a rational square. The extension has the embedding $T \mapsto -T$, which sends M to $-T-b < -1$. Every conjugate of a finite cosine lies in $[-1,1]$, a contradiction. The remaining near owns are therefore $> .9$. Their head products are > 1.0721886 and increase; this dotted mutual range is retained, not misclassified as a finite cosine.

It remains to close near/far pairs for this head. Every far own is between 1.4 and 1.6. Its increasing head product satisfies

$$M(1) > .0948756 > 0, \quad M(1.6) < .6282926 < 1/\sqrt{2},$$

so it must equal 1/2. Since $M(1.45) < .4961522$ and $M(1.46) > .5041674$,

$$1.45 < h < 1.46.$$

For a near own $g \in (.9, 1.21)$ define

$$N(g) = \sqrt{9g^2 + 8g + 5/4}, \quad F(h) = \sqrt{9h^2 + h - 1/2}, \quad \lambda = N(g)F(h) - 8gh - g/2 - 9h/2 - 1/4.$$

On the entire rectangle, $N', F' > 3$, $N'' = -19/(4N^3)$, $F'' = -19/(4F^3)$, and $\partial_g \partial_h \lambda = N'F' - 8 > 1$. The minimum derivatives are bounded by

$$\partial_g \lambda > 3.032(4.457) - 12.1 = 1.413624, \quad \partial_h \lambda > 3.967(3.038) - 11.7 = .351746.$$

Thus corner bounds apply to every parameter:

$$.015919 = 3.967(4.457) - 17.665 < \lambda < 4.91(4.489) - 21.5578 = .48319.$$

This is a forbidden native interval. The native-five head now has only the three individual fork roles isolated, near-three, far-three. Their six unordered pair types are covered: isolated/isolated, isolated/near and isolated/far by Appendices B.9.14–B.9.18; near/near by Appendix B.9.9; far/far by Appendix B.9.18; and near/far by the present rectangle. This proves the full native-five-head exclusion, without the unused older native-five isolated/far continuation.

B.9.20 Finite heads seven and eight: one paid reference

For either head put $\eta = \cos(\pi/h)$, $h = 7, 8$, $u = \eta^{-2} \in (1, 4/3)$. The actual positive core

$$\Xi = H + \eta u n_1 + 2\eta(u-1)n_2, \qquad \Xi^2 = -\eta^2 d(u), \quad d(u) = (u-1)(4-3u) > 0$$

has zero H, n_1 products, negative n_2 product, positive time and all other products nonpositive. Its spatial pressure, after division by $\sqrt{2}\eta$, is

$$\langle x, Z \rangle \geq h_0 = \frac{\sqrt{2-u} - \sqrt{d(u)}}{\sqrt{2}}, \quad \sum Z_i = 1, \quad \operatorname{supp} Z \subset \{1,2,3\}, \quad Z^2 = R = 3 - 4u + \tfrac{3}{2}u^2.$$

For head seven the cubic brackets give $1.2318 < u < 1.2322$. The difference of radicals increases on that interval: $(6u-7)^2(2-u) > .116 > d(u)$. At the left end, $\sqrt{.7682} > .87646$, $\sqrt{.07060628} < .26572$. For head eight use $u = 4 - 2\sqrt{2}$, $d = 34\sqrt{2} - 48$, $R = 23 - 16\sqrt{2}$ and $1.4142135 < \sqrt{2} < 1.4142136$. These give the rigorous full-domain bounds

h	h_0 lower	R upper
7	.4318	.3489
8	.43955	.37259.

Use the bare references before any head or isolated-five payment:

initial case	B_0	b upper	P_J capacity	J
5 strong	.9565188	.3652	$> .151$	5
5 isolated weak, 6 strong	.9373243	.3399	$> .1435$	6
5 isolated weak, 7 strong	.9196357	.31938	$> .1366$	7.

These are the raw references already constructed, including the legal strong-sixteen append and every omission before pressure. Their support is at most sixteen, $\langle w, P_{16}\rangle = f_{16}$, and their unspent terminal coefficient exceeds .127.

In the first row pay .4 at $Z - P_5/5$. The debit .08 fits, and the one-square lower bound is

$$B_0 + .8(h_0 - b) - .16(R - 1/5).$$

For isolated weak five retain its actual previously unused pressure $\langle x, U'\rangle > 1.8185$. Set

$$Q_H = Z - P_J/J, \quad Q_5 = U' - (5/J)P_J, \quad \langle Q_H, Q_5\rangle = 1 - 5/J.$$

Their squares are $R - 1/J$ and at most 77/150 or 194/175. Pay $.4Q_H + \nu Q_5$, with $\nu = .09, .1$. The COMBINED anchor costs are $.85/6 < .1435$, $.9/7 < .1366$; both are checked before either pressure is inserted. The full lower bound, including the positive cross, is

$$B_0 + .8(h_0 - b) + 2\nu(1.8185 - 5b) - .16(R - 1/J) - \nu^2 Q_5^2 - .8\nu(1 - 5/J).$$

The resulting exact rational lower bounds are

h	5 strong	$J = 6$	$J = 7$
7	.9859748	.98694896	.98698198
8	.9883844	.98935856	.98939158.

The fourth initial case, weak-helper five, already has $L > .987759$, and receives no extra head payment. Thus every initial case supplies ONE reference W with

$$L_W > .9859, \quad \langle W, P_{16}\rangle = f_{16}, \quad \operatorname{supp} W \subset [1, 16], \quad \beta_{16} > .127.$$

Put $\Delta = S_{16} - f_{16} \geq 0$. All new payments were at indices at most seven; the terminal price remains unspent. Hence

$$\|x - W\|^2 < .0141 - .254\Delta, \qquad \Delta < .0556.$$

The squared projection of x onto the complement of old A_{15} is bounded by

$$\frac{S_{16}^2}{16} + x_{17}^2 + x_{18}^2 < \frac{(3.59 + \Delta)^2}{16} + .0141 - .254\Delta \leq .83062756 < .9114^2.$$

The quadratic increases throughout $0 \leq \Delta \leq .0556$, which justifies its endpoint substitution. For ANY actual point on old A_{15}, original maximality therefore gives

$$C < \frac{1 + .9114^2}{1 - .9114^2} < 10.81. \tag{DFiniteface}$$

This is an all-point distance cap on the original face, with both signed last coordinates retained.

B.9.21 Finite heads seven and eight: the complete fork dispatch

Write $\eta_h = \cos(\pi/h)$, $A_h = \sqrt{2\eta_h^2 - 1}$. The exact brackets, from the native-seven cubic and the eight half-angle formula, are

h	η_h	A_h
7	$(.9009, .901)$	$(.7894, .7897)$
8	$(.9238, .9239)$	$(.8408, .8410)$.

(DFheadbrackets)

For far-next-three the actual time is $B(g)=\sqrt{9g^2+2g}$. The terminal bound gives $9g+1<9.64$, hence $g<.96$. The complete finite list is therefore

$$m=5,7,8,9,10,11.$$

Indeed own zero has zero time; own $3,4,6$ have the actual affine principals already displayed, parallel has $c+n_f$, and dotted violates the length bound. Also $\cos(\pi/12)=(\sqrt{6}+\sqrt{2})/4>.96575$.

The head/far-next mutual is

$$M_h(g)=A_hB(g)-\eta_h(g+1).$$

Since $B'>3$ and $3A_h-\eta_h>0$, it increases. The entire six-label list is covered by the following consecutive native gaps:

h	own labels	M_h lower	M_h upper
7	$5,7,8$	.5325965	.70481042
7	$9,10,11$	.7260396	.77649556
8	5	.63199495	.6340268
8	7	.78086418	.78208658
8	$8,9,10$	.81780615	.8629634
8	11	.87450304	.87783466.

These intervals are respectively inside $(1/2,1/\sqrt{2})$, $(1/\sqrt{2},\phi/2)$, $(1/2,1/\sqrt{2})$, $(1/\sqrt{2},\phi/2)$, $(\phi/2,\sqrt{3}/2)$, and $(\sqrt{3}/2,\cos(\pi/7))$.

Here are explicit inputs for every row, so the table is a finite rational certificate rather than sampled labels:

g	$B(g)$ bound
g_5	$(2.740,2.741)$
g_7	$(3.0176,3.0180)$
g_8	$(3.0867,3.0872)$
g_9	>3.134
g_{10}	<3.1692
g_{11}	>3.1938
$g<.96$	<3.196.

Use $1.618<\phi<1.619$, $.9396<g_9<.94$, $.9510<g_{10}<.9511$, $.9593<g_{11}<.96$. The ninth-own cubic $8t^3-6t-1$ has signs $-.001406950912,.004672$ at $.9396,.94$. The tenth-own identity is $g_{10}^2=(5+\sqrt{5})/8$. For eleven, $\pi/11<2/7$ and the decreasing alternating lower polynomial give

$$g_{11}>1-2/49+2/7203-4/5294205>.9593.$$

The displayed B bounds follow by substituting the appropriate rational endpoints into $9g^2+2g$ and squaring. For example the first head-seven lower bound is $.7894(2.740)-.901(1.8095)=.5325965$; its first upper bound is $.7897(3.0872)-.9009(1.9238)=.70481042$. For the second block use the lower ninth-own and upper .96 endpoint. The four head-eight rows use respectively the fifth, seventh, eighth/tenth and eleventh endpoints. Thus all actual far-next-three caps are excluded for both heads.

For far-three use $F(g)=\sqrt{9g^2+g-1/2}$ and the increasing mutual

$$M_h(g)=A_hF(g)-\eta_h(g+1).$$

Native 5 gives

$$M_7(g_5)\in(.33508862,.33701975),\qquad M_8(g_5)\in(.42162679,.4233563),$$

both forbidden. The bounds $F(g_5) \in (2.4898, 2.4905)$ follow from $F(g_5)^2 = (11\phi + 7)/4$. For head seven,

$$0 < M_7(g_7) < .47984433 < 1/2, \quad M_7(g_8) > .51383002 > 1/2, \quad M_7(1) < .6328451 < 1/\sqrt{2}.$$

Monotonicity rejects label seven and every finite label at least eight. For head eight, $.57755874 < M_8(g) < .745203 < \phi/2$ throughout finite owns at least seven, leaving only $1/\sqrt{2}$. But

$$M_8(g_{14}) < .7027251 < 1/\sqrt{2}, \qquad M_8(g_{15}) > .707213408 > .7072 > 1/\sqrt{2}.$$

The half-angle identity gives $g_{14} = \sqrt{(1+g_7)/2} \in (.9749, .97494)$. The exact formula $g_{15} = (\sqrt{5} - 1 + \sqrt{30 + 6\sqrt{5}})/8$ gives $g_{15} \in (.97814, .97816)$. The time bounds are $F(g_{14}) < 3.00492$, $F(g_{15}) > 3.01479$. Use $2.23606 < \sqrt{5} < 2.23607$ and $6.58910 < \sqrt{30 + 6\sqrt{5}} < 6.58912$; their positive squares give all enclosures. No finite label lies between fourteen and fifteen. Thus every far-three own for either head is dotted,

$$1 < g < 1.6.$$

Only near/far remains: all other unordered pair types are already excluded over every head. Put

$$\lambda = N(g)F(h) - 8gh - g/2 - 9h/2 - 1/4, \quad 1/\sqrt{2} \le g < 1.21, \quad 1 < h < 1.6.$$

It increases in both variables on this WHOLE rectangle. Indeed $N', F' > 3$, both are decreasing, and the mixed derivative is > 1. At the worst extremes,

$$\partial_g \lambda > 3.032(3.082) - 8.5 = .844624, \quad \partial_h \lambda > 3.377(3.032) - 10.1576 = .081464.$$

The top corner gives $\lambda < 4.91(4.914) - 23.543 = .58474 < 1/\sqrt{2}$. Thus the actual mutual is zero or $1/2$.

For $1/2$, $g \le 1$ would imply $\lambda < 4.273(4.914) - 20.75 = .247522$, impossible. Hence $g > 1$, $N > 4.272, F > 3.082$. Positive projection on old A_{16} and the two actual caps has Schur matrix $\left(\begin{smallmatrix} 1 & -1/2 \\ -1/2 & 9/17 \end{smallmatrix}\right)$ of determinant $19/68$. It is an actual whole-A_{18} vertex, with

$$C^2 = 1 + (36N^2 + 68NF + 68F^2)/19 > 10.8^2.$$

The projection's two original fork coefficients are positive. Using $S_{17} \pm x_{18} < 3.823$, its normalized pressure is less than $3.823/\sqrt{17} < .928$. Thus

$$\rho = E_q/E > C - .928\sqrt{C^2 - 1} > .072C + .464/C > .82,$$
$$\kappa < 50/41 < 1.22, \quad \langle y, u \rangle > 1 - 1.22/10.8 > .887.$$

The reverse is the positive combination $(N\Omega'_{18} + F\Omega'_{17})/\sqrt{C^2 - 1}$. Appendix B.9.12 excludes it.

Suppose finally that $\lambda = 0$. Near owns 4 and 5 would give $\lambda < 3.693(4.914) - 18.223 = -.075598$, contrary to acuteness. For finite near own $g = \cos(\pi/m)$, $m \ge 7$, retain old A_{15}, n_f, the near cap and the far cap. Clearing their positive old-root coefficients gives diagonal Schur matrix $\mathrm{diag}(1 - g^2, 17/32)$. The resulting actual vertex is whole $A_{16} + I_2(m)$, with

$$C^2 = 1 + \frac{N^2}{1 - g^2} + \frac{32F^2}{17}.$$

For $m = 7$, monotonicity and $\lambda(.901, 1.35) < 3.9705(4.154) - 16.5063 = -.012843$ force $h > 1.35$. Thus $N^2/(1 - g^2) > 83.65$, $32F^2/17 > 32.47$, and $C^2 > 117.12 > 10.82^2$. For every $m \ge 8$, the

corresponding bounds are $N^2/(1-g^2) > 111.3$, $32F^2/17 > 17.88$, so $C^2 > 130.18 > 11.4^2$. Both contradict (DFiniteface).

For dotted near own, the actual head/far native gaps force $h > 1.03$:

	$M_h(1)$ lower	$M_h(1.03)$ upper
head 7	.6309308	.67823359
head 8	.7435456	.7946087.

They are forbidden intervals between successive native cosines. At the corner $(1.08, 1.03)$, $\lambda > 4.515(3.1746) - 14.3242 = .009119$; therefore orthogonality forces $1 < g < 1.08$. The positive actual core $c_N + gn_f$, square $1 - g^2 < 0$, is a genuine point on all old A_{15}, with

$$C^2 = \frac{N^2}{g^2 - 1} > \frac{20.3876}{.1664} > 122.5 > 10.81^2.$$

The ratio decreases for $g > 1$, since its derivative numerator is $-8g^2 - (41/2)g - 8 < 0$. This contradicts (DFiniteface). Every own of near/far is now excluded, and hence both finite heads seven and eight are impossible.

B.9.22 How these foundations enter the final argument

The head-isolation argument leaves an actual isolated head with finite own 5, finite own at least 7, parallel, or dotted. The parallel case has its actual positive future null. Appendix B.9.10 excludes finite heads at least nine; Appendix B.9.19 excludes five; Appendices B.9.20 and B.9.21 exclude seven and eight. These proofs cover every initial-five reference case with the individual original prices and every omission retained.

For a dotted head, the present appendix supplies precisely the earlier pair exclusions: isolated/isolated, isolated/near-three, isolated/far-three, far-three/far-three, near-three/near-three. The near/far pair and the remaining far-next-three role in that head regime are handled by the separate final uniform dotted-head argument in the main revision. That argument uses the original actual head point and its four paid references, both individual weak-helper anchor prices, the fresh local whole-A_{18} endpoint lemma and its actual length comparison. It is not reproduced here. In particular no conclusion of that final argument was used in the present all-head pair proofs or in the standalone two-dual lemmas. The selected dependency graph is acyclic.

All component energies concern the single x at the original farthest p, or the single explicitly named y at an actual neighbour with factor derived from that same original maximum. Every projected point, future null and edge foot has its selected products and all omitted original walls checked. Finite inventories used here are the ordinary finite Coxeter classification with their displayed rank and contact restrictions; the rational interval tables are explicit finite inequalities, not execution receipts. No field restriction, arbitrary-face arithmeticity, exact-seventeen frontier census or new dimension research enters this appendix.

B.10 The other exact-dimension-eighteen consumers

The necessary forest cover is proved before any row is excluded. This appendix supplies its component-specific consumers, using the same original farthest vertex throughout. A typed exception includes its actual isolated opposite wall; a bare old seven bond is insufficient.

B.10.1 A_{16} with either actual typed-seven defect

Suppose the entire incident forest is $A_{16} + Z$, where $Z = N_7$ means old A_2 with its new isolated seven opposite, and $Z = O_7$ means old $I_2(7)$ with its new isolated three opposite. There is no arithmeticity hypothesis.

Use zero-sum $\mathbb{R}^{17}$, with actual unit normals and duals

$$n_i = (e_i - e_{i+1})/\sqrt{2}, \quad \Omega_i = \sqrt{2}(P_i - iP_{17}/17),$$
$$D_i = 2i(17-i)/17, \quad C_{ij} = 2i(17-j)/17 \quad (i \le j).$$

Write the original interior point as $z = E(p - x - y)$, with x in this A_{16} space and y in the orthogonal rank-two space. The previously proved typed-defect floor gives

$$e_A + e_Z = \|x\|^2 + \|y\|^2 < 1, \qquad e_Z > .09, \qquad e_A < .91. \tag{T1}$$

All auxiliary points below are compared with this original maximum.

Put $\phi = (1+\sqrt{5})/2$, $\theta = 2\cos(2\pi/7)$, $g_7 = \cos(\pi/7)$, and

$$h_N = (\theta - 1)/3, \qquad h_O = (\theta - 1)/(2 - \theta), \qquad k = 1/h_O = \theta^2 + 2\theta - 1.$$

The actual isolated defect wall has time $\sqrt{h_Z}$ and spatial support only in Z. In the N_7 case it is $c_Z = \sqrt{h_N}p - g_7\omega_a^Z$, for an old A_2 dual. The cubic $f(t) = t^3 + t^2 - 2t - 1$, increasing on $[1.24, 1.25]$, has signs

$$f(1.2469) = -410632291/10^{12} < 0, \qquad f(1.247) = 105223/10^9 > 0.$$

Consequently

$$1.2469 < \theta < 1.247, \quad 0 < h_N < 1/12, \qquad 3.0485 < k < 3.0491 < 61/20. \tag{T2}$$

The identity $(\theta - 1)k = 2 - \theta$ follows by the same cubic. These are real native-angle facts, not field restrictions on the polytope.

B.10.1.1 The central window at its actual energy budget

Call $5 \le i \le 12$ weak if $h_i = \langle x, \Omega_i \rangle \le \sqrt{D_i} - .61$. The target-independent inventory and helper nonadjacency in Appendix B.3.6 leave only pure far helpers or near-four roles at $5, 6$ and their reflections. Its older energy-dependent near-four conclusion is not used at the larger budget .91.

A left near-four has $v = \Omega_i/\sqrt{2} + \Omega_{i-1}/2$, $i = 5, 6$. The same actual wall is the opposite in direction $j = i - 1$. Keeping the other original roots and all of Z, positive finite projection gives its actual endpoint, with

$$\Delta_j = 1/(17 - j), \qquad C_j = \sqrt{2j/17}\{(17-j)/2 + (16-j)/\sqrt{2}\}.$$

For $j = 4, 5$, their squares are respectively greater than $177608/1700$ and $187690/1700$, both greater than 100. Thus the normalized mate height exceeds $\sqrt{9/11} > .9$. At the opposite end, either a second mate has height $> .9$, or nonadjacency supplies a strong $11/12$ direction of height $> .77$. Their correlation is at most $\sqrt{5/22} < 1/2$. Cauchy would give

$$1.67^2 < e_A(2 + 2\rho) < 3(.91) = 2.73,$$

whereas $1.67^2 = 2.7889$. Both orientations are impossible. Every remaining weak site is therefore a nonadjacent pure helper.

Use the shared rational reference of Appendix B.3.7:

$$H_5 = 2.046,\ H_6 = 2.176,\ H_7 = 2.259,\ H_8 = H_9 = 2.300, \qquad H_{17-i} = H_i.$$

The ends are linear. A weak end is flattened through six; every other weak knot is removed by its neighboring chord before pressure. The resulting positive-dual reference w has all coefficients at actual strong floors and obeys

number of weak ends among $5, 12$	$E = \|w\|^2$ lower
0	.86103645
1	$> .8285$
2	$E_{\min} = 1195276/1500000.$

(T3)

The last rows include every permitted internal omission. A weak end block has length $L = 6$, mean $b = H_6/(6\sqrt{2}) < .257$; a strong end block has $L = 5$, mean $b = H_5/(5\sqrt{2}) < .29$. In both cases $b > .25$, the next coordinate is $< .1$, and the individual strong-anchor capacities are at least $839/6000$ and $349/2500$, respectively.

If an original signed endpoint coordinate were at least .565, replace its block by $.565, r, \dots, r$, keeping the block sum. Here $r = (Lb - .565)/(L - 1)$ is greater than .17125 or .187, so every surviving coefficient stays nonnegative. The exact variance gain is $L(.565 - b)^2/(L - 1)$. For a weak end, its total with e_Z exceeds

$$E_{\min} + .09 + (6/5)(.565 - .257)^2 > 1.00068746.$$

For a strong end it exceeds $.8285 + .09 + (5/4)(.565 - .29)^2 = 1.01303125$. Hence both signed endpoint coordinates are $< .565$. The actual edge condition in either direction is

$$x_{\text{end}} \geq \sqrt{16/17}\sqrt{(C - 1)/(C + 1)}.$$

At $C = 41/20$ its square is $336/1037 > .565^2$, with positive cleared numerator 198547 over 41480000. Therefore BOTH actual endpoint cosines satisfy

$$C < 41/20. \tag{T4}$$

This is one reference contradiction per endpoint, not addition of two independently optimized lower energies.

B.10.1.2 The complete endpoint and foreign-contact list

Release old 1, retaining the full old A_{15} and Z. Its actual opposite is isolated within A_{16}, or has a single three at $j = 2, 3, 15, 16$, or a single four at $j = 2, 16$. Two contacts make a cycle; exceptional finite types have insufficient rank. An old A_2 admits at most one new contact, with complete costs

$$Q = 1/3, \quad 2/3, \quad \phi^2/3.$$

An old $I_2(7)$ admits none, since even its smallest contact costs $1/[4\sin^2(\pi/7)] > 1$. When there is also an A_{15} contact, only an endpoint-three with old A_2 cost $Q = 1/3$ remains finite: the full chain is A_{18}. A foreign arm at a D position has excessive arms; a four or five becomes an internal multiple bond in the long path.

Write $c = Ap - g\Omega_1 - \beta\Omega_j - v_Z$, where $\beta = 1/2$ or $1/\sqrt{2}$. For $Q = 0$, the exact data are

white three at j	2	3	15	16
C	$(32g + 15)/17$	$(16g + 7)/\sqrt{17}$	$(16g + 1)/\sqrt{17}$	$(32g + 1)/17$

and white fours at $2, 16$ give $(16g + 15/\sqrt{2})/\sqrt{17/2}$ and $(16g + 1/\sqrt{2})/\sqrt{17/2}$. The residuals are $17/32, 1/8, 1/16$, for endpoint three, D-position three and endpoint four. The one mixed endpoint-three multiplies its pure cosine by $\sqrt{51/19} > 8/5$.

At $j = 3$, own zero has the actual affine-E_8 null

$$2n_1 + 4n_2 + 6n_3 + 5n_4 + 4n_5 + 3n_6 + 2n_7 + n_8 + 3c;$$

every positive own has $C \geq 15/\sqrt{17} > 41/20$. At $j = 15$, own zero has time square $-2/17$, own three has the null $c + n_1 + \cdots + n_{15}$, and owns at least four violate (T4). White four at 2 violates (T4) even at own zero; at 16, own zero has time square $-1/17$, and every positive own violates (T4).

For pure white three at 2, own zero has time square $-2/17$, own three has the actual triangle null, and every own at least four violates (T4). The mixed version retains that triangle; its own-zero null is the actual affine E_7 combination

$$2n_1 + 4n_2 + 3n_3 + 2n_4 + n_5 + 3c + 2z_1 + z_2, \qquad \langle c, z_1\rangle = \langle z_1, z_2\rangle = -1/2.$$

Mixed white three at 16 has own-zero time square $-10/51$, the own-three all-three cycle, and for own at least four has $C > (8/5)(32(.7) + 1)/17 = 936/425 > 41/20$. A parallel own always has the null $c + n_1$. Every displayed null has positive cap time and nonpositive product against every omitted wall. A positive finite projection likewise checks all unselected walls termwise.

The only remaining nonisolated endpoint is consequently PURE far three:

$$c = Ap - g\Omega_1 - \Omega_{16}/2, \quad A^2 = (32g^2 + 2g - 9)/17, \quad 0 < g < 677/640 < 53/50, \quad A^2 < 7/4. \text{ (T5)}$$

Own zero has negative time, own three has zero time, and reflection gives the other end. Finite and dotted own values remain present.

B.10.1.3 Using the actual defect wall

For N_7, a pure endpoint wall is distinct from c_Z and has actual product $-A\sqrt{h_N}$ with it. The native gap forces $A^2 h_N \geq 1/4$, so $A^2 > 3$, excluding (T5). An isolated-within-A endpoint instead has

$$A^2 = Dg^2 + Q - 1, \quad C^2 = Dg^2/(1 - Q), \quad D = 32/17, \quad u = (1 - Q)/g^2 = D/C^2.$$

Own zero has negative time, since the full white cost satisfies $Q < 1$. At $Q = 0$, the same gap gives $u < D/4 < 1$. At $Q > 0$, a contact β to the old A_2 gives $Q = 4\beta^2/3$ and actual defect cross $q = 4g_7\beta/3$ or $2g_7\beta/3$. Acuteness, including orthogonality, gives

$$A^2 \geq q^2/h_N \geq Q(2 + \theta)/[4(\theta - 1)] > 13Q/4.$$

Since $Q \geq 1/3$, $C^2 > [1 + (9/4)Q]/(1 - Q) \geq 21/8 > D$, so again $0 < u < 1$. This covers every foreign orientation and finite or dotted own.

For O_7 all endpoints are pure. Set $M = 4A^2/k$. It is four times the square of an ACTUAL negative two-wall product. Its positive values therefore begin $1, 2, \phi^2, 3, 2 + \theta$, followed by larger finite values, parallel 4 or dotted values greater than four. For the far helper, own four gives $M < 1$; finite owns at least five give $1 < M < 2$. In the dotted case, (T5) gives $1 < M < 7/3 < \phi^2$, so the only possibility is $M = 2$. It forces

$$A^2 = k/2, \quad 32g^2 + 2g - 9 = 17k/2, \quad 1.013 < g < 1.014, \quad 1.2346 < A < 1.235. \qquad \text{(T6)}$$

These brackets follow by positive substitution using (T2).

Two far helpers would have this same g. They are distinct because their own $g > 1$ differs from the other-end helper coefficient $1/2$. Their spatial product $(2g^2+32g+1/2)/17 > 69/34$ exceeds their time product $A^2 = k/2 < 61/40$. This violates acuteness. Thus a far helper would face an isolated endpoint. If that isolated endpoint had finite own $\delta \le 1$, then $0 < M_\delta = 4(D\delta^2-1)/k < 20/17 < 2$, so it must equal one. For labels at most twelve, $4(D\delta^2-1) \le (32\sqrt{3}-4)/17 < 3.0485 < k$. For labels at least thirteen, $\delta > .97$ gives $4(D\delta^2-1) > 3.0844 > k$. No finite label works; own zero has negative time and parallel is the actual null pair. The other endpoint is therefore dotted, with $\delta > 1$ and $\sqrt{D}\delta < 41/20$.

Put $B = (2g+16)/17 \in (1.06, 1.0605)$. Their actual mutual magnitude

$$m(\delta) = A\sqrt{D\delta^2-1} - B\delta$$

increases on $\delta \ge 1$, because its derivative exceeds $1.2346(1.37) - 1.0605 > 0$. It is positive at one, and its endpoint upper is $1.235(1.79) - 1.06(1.49) = .63125 < .7$. The native gap thus forces $m = 1/2$. At 1.35 it is $< .4956$, while at 1.358 it is $> .5006322$; hence $1.35 < \delta < 1.358$. But its own ACTUAL defect-wall product now has

$$3 < M_\delta < (1.313)(2.472) = 3.245736 < 2+\theta,$$

the forbidden native-six/native-seven gap. This excludes the last far helper. Both O_7 endpoints are isolated and dotted.

B.10.1.4 One simultaneous endpoint payment

For either isolated N_7 endpoint clear its full foreign vector with its positive old-A_2 inverse. For O_7, put $Q = 0$. Each end has its OWN parameter $u \in (0,1)$, and the actual core

$$\eta = c + v_Z + gun_1 = g\{\sqrt{D-u}\,p - Z(u)\}, \qquad Z(u) = \Omega_1 - un_1$$

has square $-g^2u(1-u)$, positive time, zero selected cap and foreign products, and negative n_1 product. All other walls, including the opposite endpoint and the defect wall, are nonpositive termwise. Its actual normalization gives

$$\langle x, Z(u)\rangle \ge h(u) = \sqrt{D-u} - \sqrt{u(1-u)}.$$

At a length-L block put $Y = Z - \Omega_L/L$, with the reflected signed expression at the right end. Then

$$\langle w, Y\rangle = 0, \quad \langle Y_l, Y_r\rangle = 0, \quad \langle w, Z\rangle = H_L/L \le .4092, \quad Y^2 = 2 - 2/L - 2u + u^2 \le 5/3 - 2u + u^2.$$

The residuals have zero sum and disjoint first/last block supports. Pay $\mu = .32$ separately at BOTH surviving strong anchors: $\mu/L \le .064$ is below both capacities in (T3). All old coefficients remain nonnegative before either pressure is used. For $W = w + .32(Y_l + Y_r)$, one original-vector square yields

$$e_A > E + G(u_l) + G(u_r), \quad G(u) = .64(h(u) - .4092) - .1024(5/3 - 2u + u^2). \tag{T7}$$

The spectator energy has not been added to either gain.

For completeness the following finite scalar certificate covers every independent $0 \le u \le 1$. Write

$$G_0 = .64(h(u) - .4112) - .1024(8/5 - 2u + u^2), \qquad G = G_0 - 208/37500.$$

Positive square tests for $\sqrt{D-u}$ at $0, 1/2, 3/4, 1$ give lower bounds $1.3719, 1.1757, 1.0641, .9393$. Use the root chords and $u^2 \le (l+r)u - lr$ on each interval:

$[l,r]$	root chord	K	a	$\sqrt{a^2+.64^2}$ upper
$[0,1/2]$	$1.3719 - .3924u$	.451008	.097536	.6475
$[1/2,3/4]$	$1.3989 - .4464u$	.506688	.208896	.6733
$[3/4,1]$	$1.4385 - .4992u$	.570432	.293888	.7043

Here $G_0 > K - au - .64\sqrt{u(1-u)}$. The four root squares are

$$1.88210961, \quad 1.38227049,$$
$$1.13230881, \quad .88228449,$$

each below $32/17, 47/34, 77/68, 15/17$, respectively. The three upper-radical square gaps are

$$.000142978704, \quad .000095351184,$$
$$.000068333456,$$

and all are positive. Finally,

$$au + .64\sqrt{u(1-u)} \le \frac{a + \sqrt{a^2+.64^2}}{2},$$

so the three lower bounds for G_0 are $.07849, .06559, .071338$. Thus, uniformly,

$$G > .06559 - 208/37500 = 18013/300000 > .06.$$

Combining (T1), (T3) and (T7) now gives

$$e_A + e_Z > E_{\min} + .12 + .09 > 1.00685066 > 1,$$

a contradiction. The two end parameters were independent, all actual walls were retained, and the rank-two energy was counted once. This excludes BOTH actual typed-seven alternatives beside A_{16}.

Together with the already proved uniform $A_{16} > .86$ and the ordinary/joint spectator bounds, this closes its entire exact-eighteen row: ordinary T_2, two ordinary A_1's and $I + A_1$ are excluded by those earlier bounds, and the two typed cases by the argument here. It does not assert an arbitrary-spectator $A_{16} > .91$ theorem.

B.10.2 The two rank-one spectators beside A_{17}

We now exclude the two genuinely different rank-one alternatives. An ordinary A_1 has energy greater than .1; an actual isolated short I need not. The latter supplies a wall and additional actual edge pressures instead. Throughout this subsection the entire incident forest is $A_{17} + A_1$, at the original farthest vertex p. Write $z = E(p - x - ye)$, where e is its unit rank-one normal, x belongs to zero-sum $\mathbb{R}^{18}$, and $\|x\|^2 + y^2 < 1$. Set

$$\Omega_i = \sqrt{2}(P_i - iP_{18}/18), \quad D_i = i(18-i)/9, \quad C_{ij} = i(18-j)/9 \quad (i \le j).$$

There is no arithmetic hypothesis.

B.10.2.1 Common geometry and the unscaled references

Call $5 \le i \le 13$ weak when $\langle x, \Omega_i \rangle \le \sqrt{D_i} - .61$. The target-independent inventory of Appendix B.3.6 leaves near-four roles at $5, 6$ or their reflections, and pure helpers

$$c = Ap - g\Omega_i - \Omega_a/2, \quad a = 1 \text{ or } 17, \quad \phi/2 \le g < 17/16,$$

with own five, finite at least seven, or dotted. All foreign and own-zero roles have their dispositions in that inventory. Pure helpers are nonadjacent before near-four caps are removed. The cutoff $17/16$ concerns these weak helpers only, never arbitrary endpoint caps.

We remove near-four caps without any contrary component-energy target. For a left cap its same-wall mate is the actual edge endpoint at $j = 4$ or 5. Its full positive finite projection retains $A_{j-1} + B_{18-j} + A_1$, with

$$\delta = 1/(18-j), \quad \sigma = 1/2 + (17-j)/((18-j)\sqrt{2}), \quad C_{\text{mate}} = \sigma\sqrt{(18-j)D_j} > 10.$$

Indeed $\sigma > 1.14$ and the last root is at least $28/3$. Farthest comparison gives height square greater than $9D_j/11$, so the heights at four and five exceed 2.25 and 2.43. At the opposite end there is a mate or a strong twelve/thirteen: otherwise two adjacent weak pure helpers would remain. Their heights exceed 2.218 and 2.077. Disjoint end-segment Cauchy has worst case

$$\|x\|^2 > \tfrac{1}{2}(2.43^2/5 + 2.218^2/6) = 1.000450333\ldots > 1.$$

The other placements and two mates give larger bounds. Reflection excludes the right near-four. Thus every remaining weak site is a nonadjacent pure helper, at the correct original comparison factor one.

Use the unscaled $N = 18$ reference of Appendix B.3.7, with

$$H_5 = H_{13} = 2.077,\ H_6 = H_{12} = C = 2.218,\ H_7 = H_{11} = 2.314,\ H_8 = H_{10} = 2.371,\ H_9 = 2.389.$$

Flatten a weak end through six and delete every other weak knot by its neighboring chord. Its positive strong-site coefficients give $\langle x, w \rangle \ge E = \|w\|^2$. The complete omission table is

weak ends $5, 13$	E lower bound	end blocks
0	$E_0 = .8934308$	5/5
1	$E_1 > .86256$	6/5
2	$E_2 = 9978827/12000000$	6/6.

Each six-block anchor has capacity at least $821/6000$, and each five-block anchor at least $343/2500$, after every internal omission. Their signed coordinate means satisfy $b_6 < .2614$, $b_5 < .294$.

We use only the explicit comparison-one core and block algebra printed in Appendices B.3.7 and B.8.10, not the latter subsection's neighbouring-vertex conclusion or its 1.061 comparison hypothesis. To specify this reuse, a weak-end foot has $U = v - kn_i - \ell n_a - mn_j$, with corrections

$$(k, \ell, m) = (g, 2/3, 1/3), \quad (g, 2(1-g^2), 0), \quad (1/g, 0, 0)$$

for own five, finite at least seven and dotted, respectively. The first uses its actual finite H_4 projected point; the other two are future negative cores. Their positive original-root coefficients check every omitted wall, including e and its opposite. Their pressures $\langle x, U \rangle \ge H$ have far time $A_0^2 = (65/9)g^2 + (5/9)g - 19/36$; near adds $8g/9$. On two six-blocks the projections are

$$\Pi U = (5g/6)\Omega_6 + \Omega_{12}/12 \quad \text{(far)}, \qquad \Pi U = (5g/6 + 1/12)\Omega_6 \quad \text{(near)}.$$

The residual $Y = U - \Pi U$ has zero sum separately on both end blocks and $\langle w, Y\rangle = 0$. Its far squares are

$$Q_5 = 2g^2/3 + 1/12, \quad Q_f = 4g^4 - 16g^2/3 + 29/12, \quad Q_d = 5g^2/3 - 19/12 + g^{-2};$$

near adds $g/3$. The full scalar seals proved there give the six/six deficit $\Delta = H - C(5g/6 + 1/12) > .3405$ for finite owns and $> .37$ for every $1 < g < 17/16$. For a far six/five foot its deficit decreases by $343/15000$ and its square by $1/60$; near is unchanged.

For clarity, the finite-at-least-nine derivative in that seal is a derivative of the lower surrogate

$$\overline{\Delta} = A_0 - 1/4 - C(5g/6 + 1/12) - 343/15000,$$

not of the varying-radical deficit. It satisfies $\overline{\Delta}' > .839$, $Q_f' \le 16/3$. At $g = .93$, $\overline{\Delta} > .32035$ and the adjusted square is .77940804. Thus a far finite foot at weight .145 gains more than .076; five/seven/eight give the respective deficit/square pairs $(.33915, .505), (.32655, .74), (.3177248, .763)$, which give the same strict bound directly. A far dotted foot at weight .13 gains more than

$$.26(5207/15000) - .0169(1751/1500) = 1057901/15000000 > .0705267.$$

The full two-foot crosses are also coordinate identities. In one six-block write

$$a = (g\sqrt{2}/6)(P_5 - 5e_6) - kn_5, \qquad b = (e_1 - P_6/6)/\sqrt{2} - \ell n_1 - mn_2.$$

Then $a^2 = 5g^2/3 + k^2 - 2gk$, $b^2 = 5/12 + \ell^2 + m^2 - \ell m - \ell \le 5/12$, and $\langle a_i, b_j\rangle = g_i/6$. Equal orientations have total square $Q_{f,L} + Q_{f,R} + (g_L + g_R)/3$; mixed orientations have $Q_{f,L} + Q_{f,R} + 2\langle b_L, b_R\rangle + 2g_{\text{near}}/3$. Hence the equal/mixed bounds are $17/6, 11/3$ for two finite owns and $3.077, 3.91$ if a dotted own occurs. Unequal parameters and both mixed orientations are included.

Every use below pays $\mu\Pi U$ from its individual anchors before using the actual pressure. For $W = w + \mu \sum Y$, one completed square therefore gives

$$\|x\|^2 \ge E + 2\mu \sum \Delta - \mu^2 \|\textstyle\sum Y\|^2.$$

No independently optimized component estimates are added.

B.10.2.2 An ordinary rank-one spectator

Suppose first that $\|x\|^2 \le .9$. With one weak end, weight .1 fits both anchor capacities and gives gain $> .2(.317) - .01(8/5) = .0474$; hence $\|x\|^2 > .90996$. With two weak ends, both individual prices are at most $101/960 < 821/6000$. The full mixed square is less than $32/5$, giving gain $> .4(.34) - .01(32/5) = .072$, and total $> .90356$. Only zero weak ends remain.

Both five-block means are $b < .294$. If one signed endpoint coordinate were at least .4, replace that block by $.4, r, r, r, r$, with $r = (5b - .4)/4 > .2625$. The next coordinate is less than .1, so all surviving coefficients remain nonnegative. The block-sum-preserving gain is $(5/4)(.4 - b)^2 > .014045$, contradicting $E_0 + .014045 > .9$. Thus both signed endpoint coordinates are less than .4, and both actual endpoint cosines are less than $3/2$: otherwise their squared coordinate demands are at least $17/90 > .16$. This temporary graft gain is not reused.

For any endpoint wall $c = Ap - v_A - \beta e$, the actual reciprocal shadow and distinct orthogonal-opposite lemma give $A > \beta$, including exclusion of the boundary shadow one. Thus $\|v_A\|^2 > 1$.

Finite classification after deleting old one leaves isolation, a three at $j = 2, 3, 16, 17$, or a four at $j = 2, 17$. For $v_A = g\Omega_1 + \delta\Omega_j$, their zero-foreign cosines, in that order, are

$$(34g + 16)/18, \quad (34g + 15)/(2\sqrt{18}),$$
$$(34g + 2)/(2\sqrt{18}), \quad (34g + 1)/18,$$
$$(34g + 16\sqrt{2})/6, \quad (34g + \sqrt{2})/6.$$

Foreign cost only increases the actual cosine. Own-zero cases violate $\|v_A\|^2 > 1$ or have the actual affine E_8 null at three-at-three. Own three at two has its triangle null. The remaining native substitutions and $C < 3/2$ leave only pure far-three at seventeen, with own four. Indeed its $1/\sqrt{2} \le g < 13/17$ allows only four; a foreign $\beta^2 \ge 1/4$ multiplies its cosine square by at least $36/19$, giving $> 64/19 > 9/4$. Two such opposite pure caps are distinct and have product $8\sqrt{2}/9 - 1/3 > 0$. At least one endpoint is therefore isolated within A_{17}, with foreign contact still allowed.

Put $D = 17/9$, $Q = \beta^2$, $u = (1 - Q)/g^2$. The inequality $Dg^2 > 1$ removes own zero, three and four; six and parallel have actual feasible nulls. Also $C < 3/2$ gives $u > 68/81 > .83$. The following complete point constructions give a pure A displacement $Z = \zeta - P_{18}/18$, $\sum \zeta = 1$, $\langle x, Z\rangle \ge t_0$, $R = \|\zeta\|^2$.

For pure five use the finite H_4 point from $c + (3g/2)n_1 + gn_2 + (g/2)n_3$, with $\delta = 1 - 3g^2/2 > 0$. Here $\zeta = (1/4, 1/4, 1/4, 1/4)$, $R = 1/4$, and $t_0 = AL/(\sqrt{2}g(L + \sqrt{\delta})) > .33$, where $L^2 = 7g^2/18$, $A > .486, L > .504, \sqrt{\delta} < .136$. For pure finite at least seven use $c + g^{-1}n_1 + 2(g^{-1} - g)n_2$. Its selected products are zero, zero, and $(3 - 4g^2)/(2g) < 0$, and

$$\zeta = (1 - u/2, 1 - u/2, u - 1), \quad R = 3 - 4u + 3u^2/2, \quad t_0 = \frac{\sqrt{D - u} - \sqrt{7u - 4 - 3u^2}}{\sqrt{2}}.$$

Seven has $1.23 < u < 1.233$, $R < .35, t_0 > .38$. Finite at least eight has $1 < u < 1.1716$, $R \le .5, t_0 > .39$; the two radicals are respectively $> .8469$ and $< .28868$.

For any Q with $0 < u < 1$, clear the entire foreign contact in $c + \beta e + ((1 - Q)/g)n_1$. Its square is $-g^2u(1 - u)$, its cap/foreign products are zero and its own product is negative. Its positive time and original coefficients check every other wall. It gives

$$\zeta = (1 - u/2, u/2), \quad R = 1 - u + u^2/2, \quad t_0 = \frac{\sqrt{D - u} - \sqrt{u(1 - u)}}{\sqrt{2}} > .4, \quad R < .515.$$

The last bounds use $u > .83$, radicals $> .9428$, $< .376$. The boundary $u = 1$ is an actual future null. For $Q > 0, u > 1$, the native alphabet forces own five, foreign three and $Q = 1/4$. Its positive finite correction $c + gn_1 + e/2$ has square $\delta = 3/4 - g^2 > 0$. Its actual projected point gives $R = 1/2$ and $t_0 > .42$, using $A > .697, L > .762, \sqrt{\delta} < .31$, $L^2 = 8g^2/9$. These are points, not native projected normals.

At the isolated end project Z onto $\Omega_5/(5\sqrt{2})$. Weight $1/2$ costs less than $.071 < 343/2500$, and the residual has square $R - 1/5$, orthogonal to the reference. Its gain is greater than $t_0 - .294 - (R - 1/5)/4$. For the five rows just listed these strict gains exceed $.0235, .0485, .021, .02725, .051$. Each is greater than $.9 - E_0 = .0065692$. This closes the zero-weak-end branch. We have proved $\|x\|^2 > .9$ for the entire rank-one forest. An ordinary spectator has $y^2 > .1$, giving the contradiction.

B.10.2.3 The actual isolated short spectator

Now let its actual wall be $c_I = \sqrt{h}\, p - \sqrt{1 + h}\, e$, $0 < h < 1/2$. Appendix B.1.8 proves, on this same original vector,

$$y \ge s, \quad x_1, -x_{18} \ge a_I = \sqrt{17/18}\, t, \quad t + \tfrac{3}{2}s > .767.$$

Put $k = \sqrt{17/18}$, $D_0 = .7453706$. Since $k > .9718$, $a_I + (3k/2)s > D_0$. Weighted Cauchy therefore gives, for $c > 0, B < D_0$,

$$s^2 + c(a_I - B)_+^2 > \frac{(D_0 - B)^2}{17/8 + 1/c}. \tag{B.10.1}$$

It counts the spectator energy once. An endpoint graft on a block of length L, from its current coordinate B toward a_I, gains $L(a_I - B)_+^2/(L-1)$ and costs $(a_I - B)_+/((L-1)\sqrt{2})$ at that individual anchor. It may be capped at a smaller coordinate when necessary. Its full cross with previously paid helper residuals is included in B.

For two weak ends pay .1 to each foot. Equal orientations spend at most 31/320 at either anchor and have both endpoint coordinates at most .3454. Mixed orientations spend at most 101/960 and 17/192; the unloaded coordinate is at most .2865. If $a_I \le .6$, the respective additional graft prices are at most $.2546/(5\sqrt{2})$ and $.3135/(5\sqrt{2})$. These fit their separate remaining capacities $821/6000 - 31/320$ and $821/6000 - 17/192$. Equation (B.10.1) gives gains $> .0629$ for $c = 12/5, B = .3454$, and $> .0711$ for $c = 6/5, B = .2865$. If $a_I > .6$, cap those grafts at .6; their gains alone exceed .155, .117. For two finite owns the base totals exceed $E_2 + .1362 - .01(17/6) > .9394355$ and $E_2 + .1362 - .01(11/3) > .9311022$. A dotted own raises the deficit sum above .7105 and gives larger totals. The appropriate graft therefore makes each full total greater than one.

For one weak end orient it to five. A near finite helper at weight .145 spends less than $.145(11/12) < 821/6000$, has square $< 17/12$ and gain $> .06895$. A near dotted helper at .14 spends less than $.14(31/32)$, has square < 1.539 and gain $> .0734$. In each case the other five-block is unloaded. Its full graft is legal for $a_I < .972$, since $(.972 - .294)/(4\sqrt{2}) < .12 < 343/2500$. Equation (B.10.1), $c = 5/4, B = .294$, gives graft plus spectator $> .0696$. Thus even the smaller total is $> .86256 + .06895 + .0696 > 1.0011$.

A far finite helper at weight .145 has gain $> .076$. Its own price leaves more than .016, and its opposite price leaves at least .1227, and its endpoint coordinates are at most .2956, .3761. The weak graft is legal up to increment .113; the other graft is legal throughout $a_I < .972$. If $a_I \le .4086$, the weak graft and (B.10.1) give $> .0683$, with $c = 6/5, B = .2956$. Above that split cap the weak gain at $(6/5)(.113)^2 = .0153228$, and use the strong graft with spectator $> .0466$, $c = 5/4, B = .3761$. Both give an extra $> .0619$ and total > 1.00046.

A far dotted helper at weight .13 has gain $> .0705267$, leaves at least 1043/48000 and .1242 at the two anchors, and has endpoint coordinates at most .294, .3676. The weak graft is legal up to increment .1535. Below $a_I = .4475$, its graft plus spectator is $> .0688$, with $c = 6/5, B = .294$. Above this split its capped gain is $(6/5)(.1535)^2 = .0282747$; the opposite graft with spectator is $> .0487$, with $c = 5/4, B = .3676$. Both totals exceed one. Reflection covers the other weak end.

Finally suppose neither end is weak. One full five-block graft with the spectator gives $> .0696$. If the other actual endpoint cosine were at least 8/5, its signed coordinate would exceed 7/15, since $17/78 > (7/15)^2$. The disjoint grafts give

$$\|x\|^2 + y^2 > E_0 + .0696 + (5/4)(7/15 - .294)^2 > 1.000298.$$

Thus both endpoint cosines are less than 8/5. The complete foreign-safe list already proved above leaves only isolation or pure far-three with own four/five: now $g < 139/170$, and a foreign contact would still give $C^2 > 64/19 > (8/5)^2$. Two such far caps are distinct; their spatial product exceeds $17(g_L + g_R)/18 > 1.32$, while each time square $(68g^2 + 4g - 19)/36 < .81$. Their actual product is positive. At least one endpoint is isolated within A_{17}.

For that isolated endpoint pure five is impossible: its time square $(17\phi - 19)/36 < .237$ makes its strictly negative product with c_I smaller in magnitude than 1/2. The foreign five/three row with $u > 1$ is also impossible, since acuteness would require $h(Dg^2 - 1) \ge 1/4$, whereas $h < 1/2, Dg^2 - 1 < .237$.

All other rows use the same actual cores as above, without the ordinary branch's temporary endpoint bound. Pure seven gives gain $> .0485$. For finite at least eight, split $1 < u < 1.1716$ at 1.1: the bounds $(R, t_0) \le (.5,\ \text{lower } .4407)$ and $(.415,\ \text{lower } .3945)$ give gains $> .0717, .04675$.

For every $0 < u < 1$ the cleared-core gain is $> .06$. Here are complete continuous seals, so no sampled parameter is used. Lower chords for $\sqrt{D-u}$ are $1.3743 - .3916u, 1.4012 - .4454u, 1.4402 - .4974u$ on $[0,1/2], [1/2,3/4], [3/4,1]$; positive endpoint squares prove them. Upper chords for R are $1 - .75u, .8125 - .375u, .625 - .125u$. With $.7071 < 1/\sqrt{2} < .7072$, the gain is bounded below by

$$\begin{array}{c|l} [0,1/2] & .47776753 - .0895u - .7072\sqrt{u(1-u)} \\ [1/2,3/4] & .54366352 - .2212u - .7072\sqrt{u(1-u)} \\ [3/4,1] & .61811542 - .3205u - .7072\sqrt{u(1-u)}. \end{array}$$

The inequality $au + b\sqrt{u(1-u)} \le (a + \sqrt{a^2+b^2})/2$, with radical upper bounds $.713, .7411, .7765$, gives respectively $> .0765, .0625, .0696$. Thus every surviving isolated endpoint supplies gain $> .0467$, paid at its own anchor by weight $1/2$. At the opposite end the independent full graft plus the one spectator energy gives $> .0696$. Their residuals have disjoint support. One square yields

$$\|x\|^2 + y^2 > E_0 + .0467 + .0696 = 1.0097308 > 1.$$

This excludes the actual isolated short spectator as well. All original foreign contacts, own parameters and wall signs have been retained; the same original farthest vector and separate anchor prices are used throughout. Both rank-one alternatives beside A_{17} are now closed.

B.10.3 Typed defects beside B_{16} and D_{16}

These cases share the long-prefix and head geometry. They separate at the final short layer: B_{16} has a genuine short normal and a reciprocal endpoint, while D_{16} has two signed forks and a centre calculation. We prove their common input once.

Use the actual unit roots $n_i = (e_i - e_{i+1})/\sqrt{2}$ along the long chain. For B_{16}, $n_{16} = e_{16}$, $\Omega_i = \sqrt{2}P_i$ for $i \le 15$, and $\Omega_{16} = P_{16}$. For D_{16},

$$n_{16} = (e_{15} + e_{16})/\sqrt{2}, \quad \Omega_i = \sqrt{2}P_i\ (i \le 14), \quad \Omega_{15,16} = (P_{15} \mp e_{16})/\sqrt{2}.$$

The last D coordinate is signed. Write the one original vector as $z = E(p - x - y)$, with the classical and rank-two projections x, y. The actual typed defect gives $\|y\|^2 > .09$ and $\|x\|^2 + \|y\|^2 < 1$.

Put $c_7 = \cos(\pi/7)$, $s_7 = c_7^2$. The actual isolated defect wall $d = A_T p - v_T$ has

	v_T	$t_T = A_T^2$
N_7	$c_7 \omega_j^{A_2}$	$4s_7/3 - 1$
O_7	$\omega_j^{I_2(7)}/2$	$1/[4(1-s_7)] - 1.$

The native-seven cubic gives $.9009 < c_7 < .901$, hence $0 < t_N < 1/12$, $.327 < t_O < .342$. Every classical opposite retains the whole old rank-two component. An old $I_2(7)$ accepts no contact: already a single three has determinant $1 - s_7 - 1/4 < 0$. An old A_2 accepts at most one contact, with label three, four or five and whole cost $Q = 1/3, 2/3, \phi^2/3$. All these cases remain; no field restriction removes the last one.

B.10.3.1 Fresh tails, then strong five

Set $f_i = \sqrt{i} - .61/\sqrt{2}$. Appendix B.5 reaches twelve in B_{16} and eleven in D_{16}, leaving only nonadjacent pure far-three helpers. We reuse the actual local matrices and points in Appendix B.6, before its ordinary-spectator consumer. Their proofs retain all spectators. The positive foreign costs here are at least $1/3$, so none weakens a test that required only $Q \ge 1/4$.

In B, the fresh $A_{12} + B_3$ layer at thirteen has exactly the long-three and short-three finite extensions printed there. The latter is the sealed F_4 white block; own zero has its actual affine $3, 4, 3, 3$ path. The full long bridge has residual $1/26$; its minimum numerator, even at own zero, is $1/26 + 1/2$. No short-end bridge gives a finite long path. Thus strong-or-pure-helper coverage reaches thirteen.

In D, the fresh D_4 layer at twelve retains its long leaf, both distinct fork leaves, bridges and all foreign costs. The only extra own-five/fork-three row has its actual finite H_4 point, flat through nine, with $G > 3.34, R < 12$. The local payment formula with $K = 8, 9$ and the typed cost gives

$$.7683275 + .09 + 2(3.34 - 2.398)/7 - 4/49 > 1.0458,$$
$$.797425 + .09 + 2(3.34 - 2.570)/7 - 3/49 > 1.0457.$$

The D_3 layer at thirteen keeps middle-three and end-three, four and five, including the middle/full-D_{16} bridge. The own-five point has $G > 3.47, R < 13$. For every remaining finite own $.9 < g < 1$, average the two actual A_2 cores whose corrections are $g(4n_{13}/3 + 2n_{12}/3)$ and $g(4n_{13}/3 + 2n_{14}/3)$. Both satisfy every original wall. With $\alpha = \sqrt{2}(g + 1/4)$, $a = 4g^2/3 - 1$, they give

$$A^2 = 13\alpha^2 - 5/8, \quad R < 13, \quad G = (A - \sqrt{a})/\alpha > 13/4.$$

Indeed its derivative has numerator $(5/8)/A - (g/3 + 1)/\sqrt{a} < 0$, and at one its value is $8\sqrt{5}/5 - 2\sqrt{6}/15 > 13/4$, using $\sqrt{5} > 559/250, \sqrt{6} < 49/20$. Choosing actual strong $K = 9$ or 10 gives

$$.797425 + .09 + 2(3.25 - 2.570)/7 - 4/49 > 1.000078,$$
$$.823735 + .09 + 2(3.25 - 2.732)/7 - 3/49 > 1.000510.$$

This sharper comparison is needed for the typed spectator; the weaker ordinary-case 3.23 estimate is not substituted. Coverage reaches thirteen in both types. Helper packing extends because its actual mixed-point proof uses the same inverse $2\min(i, j)$ and positive retained right blocks. It does not require those right blocks to have the generic bulk ranks.

Some twelve/thirteen is strong. If five were weak, the actual weak-five point gives $x_1 > .5965, S_6 > 2.01815$. The legal interpolation through one, six and that strong end has energy at least

$$.7600299945 + (\sqrt{12} - \sqrt{6})^2/6 = 3.7600299945 - 2\sqrt{2} > .93.$$

Adding only $\|y\|^2 > .09$ is impossible. Thus five is strong, before choosing a head or final centre. The ordinary references of Appendix B.6 now apply with every internal helper omitted: $q_{12} > .8689525, q_{13} > .88864$. Their first five coordinates are $b = f_5/5 < .361$, with individual P_5 coefficient $> .146$.

B.10.3.2 A common head argument without arithmeticity

Deleting long node one retains B_{15} or D_{15}. Finite classification allows only isolation or a simple three at its long endpoint two. The latter has

$$c = Ap - g\Omega_1 - \Omega_2/2 - v_Z, \quad A^2 = 2g^2 + 2g + Q, \quad C_1 = (2g + 1)/\sqrt{1 - 2Q}.$$

At own zero, zero foreign cost has zero time; positive foreign cost has future null $c + \Omega_2/2 = \sqrt{Q}p - v_Z$. The actual positive expansions of $\Omega_2/2$ are $n_1 + 2(n_2 + \cdots + n_{15}) + \sqrt{2}n_{16}$ in B, and $n_1 + 2(n_2 + \cdots + n_{14}) + n_{15} + n_{16}$ in D. Own three has its triangle. All other positive owns force $C_1 \geq 1 + \sqrt{2}$ and $x_1 > .64$. Graft the first five-block to $.64, (5b - .64)/4, \ldots$. Its other coordinates exceed .29, above the next slope; its P_5 debit is less than $.07 < .146$. Its gain $> (5/4)(.279)^2 > .0973$ makes $q_{12} + .0973 + .09 > 1$. The head must be isolated within the classical component, with foreign contacts still present.

Write $c_H = Ap - g\Omega_1 - v_Z$, $A^2 = 2g^2 + Q - 1$. For a finite pure N_7 head its actual product with d lies strictly between $-1/2$ and zero. For a finite pure O_7 head its magnitude is less than $\sqrt{.342} < 1/\sqrt{2}$, so it must equal $1/2$. This forces

$$g^2 = (1 + 1/(4t_O))/2, \quad 148/171 < g^2 < 577/654, \quad .93 < g < .9394.$$

But $\cos(\pi/8) < .925$ and $\cos(\pi/9) > .9395$, the latter by $8s^3 - 6s - 1$. No finite native own lies there.

For a finite mixed N_7 head the product is $q - A\sqrt{t_N}$, with $q = 4\beta c_7/3$ or $2\beta c_7/3$. At $\beta \geq 1/\sqrt{2}$, $q > .42$ but $A\sqrt{t_N} < \sqrt{2/12} < .409$. At $\beta = 1/2$ the same-root value $q > .6$ is also impossible. The other-root product exceeds $-1/2$ and must vanish. It gives $g^2 = (3s_7 - 2)/(8s_7 - 6) = (1 + 1/(4t_O))/2$, the same gap. Parallel has its actual future null. Every finite head is excluded.

For dotted mixed heads, $Q \geq 2/3$ forces $C_1^2 > 6$ and hence $x_1 > .64$. At $Q = 1/3$, same-root acuteness gives $A^2 > 108/25$ and $C_1^2 > 187/25$. In the other orientation

$$A^2 t_N - q^2 > (4/3)t_N - s_7/9 = (5s_7 - 4)/3 > 0.$$

The actual negative product therefore has magnitude at least $1/2$, so $A\sqrt{t_N} > .8$, $A^2 > 192/25$, $C_1^2 > 313/25$. Again $x_1 > .64$. Only a pure dotted head remains.

For it put $u = g^{-2} \in (0, 1)$. The actual positive core $c_H + g^{-1}n_1$ has square $-(1 - u)$, zero cap product, negative own product, positive time and every omitted wall nonpositive. It gives

$$\langle x, Z \rangle \geq \frac{\sqrt{2 - u} - \sqrt{u(1 - u)}}{\sqrt{2}}, \quad Z = (1 - u/2)e_1 + (u/2)e_2.$$

Pay weight $\sqrt{2}/5$ on $Z - P_5/5$, costing $\sqrt{2}/25 < .057$ at the individual anchor. The residual is zero-sum on the first five coordinates and orthogonal to the reference. Its completed-square gain is

$$\eta = .4(\sqrt{2 - u} - \sqrt{u(1 - u)} - \sqrt{2}b) - .04(1.6 - 2u + u^2) > .03624.$$

For the full interval, $\sqrt{2}b < .5106$. On the two halves use lower root chords $1.4142 - .379u, 1.4494 - .4494u$ and upper chords $u^2 \leq u/2, 1.5u - .5$. The lower gains become $.29744 - .0916u - .4\sqrt{u(1 - u)}$ and $.33152 - .15976u - .4\sqrt{u(1 - u)}$. The circle inequality with radical upper bounds $.4104, .4308$ gives $> .04644, > .03624$. This one paid head can be combined with any later residual vanishing on the first five coordinates.

B.10.3.3 The B_{16} last-long layer

Delete fourteen, retaining the actual B_2 on long fifteen and short sixteen. A finite contact to it is a three at either end. For long fifteen the Schur data are $\sigma = g + 1/2$, $\Delta = 1/2 - Q$. Own zero has null $c + n_{14} + 2n_{15} + \sqrt{2}n_{16}$; own three has its triangle; every other positive own is strong. For short sixteen the data are $\sigma = g + 1/(2\sqrt{2})$, $\Delta = 1/2 - Q$. Own zero has the actual affine F_4 kernel $(\sqrt{2}, 2\sqrt{2}, 3, 2, 1)$ on $c, n_{16}, n_{15}, n_{14}, n_{13}$. Own four has null $\sqrt{2}c + n_{13} + 2n_{14} + n_{15}$,

six has $\sqrt{3}c + 2n_{14} + n_{13}$, and parallel has $c + n_{14}$. Every remaining own at least five has ratio $\phi/\sqrt{2} + 1/2 > 47/30 > 25/16$. Positive foreign $Q \geq 1/3$ makes own three strong too.

All four left/right endpoint-three bridges are finite B_{16}, with residual $1/28$ and $Q = 0$. Their minimum numerator, including own zero and the short-right bridge, gives ratio

$$\sqrt{28}(1/28 + 1/(2\sqrt{2})) > 27/14 > 25/16.$$

Thus these own-zero bridges are directly estimated, not dismissed by an invalid all-three cycle. Left-only and isolated cases use their actual long-neighbour points through fifteen. The sole extra weak nonhelper is consequently $c = Ap - (\Omega_{14} + \Omega_{16})/2$, with no foreign part.

For this wall put $a = (1 + \sqrt{2})/2$. Its spatial part is $aP_{14} + (e_{15} + e_{16})/2$, and $A^2 = 10 + 7\sqrt{2}$. Release old fifteen instead. The retained chain old one through fourteen, then c, then the actual short sixteen is finite A_{16}, all bonds three. Positive finite projection gives an actual future endpoint with the omitted old fifteen strict: its products with the positive old-fourteen and short-sixteen terms are negative. Every other wall is nonpositive termwise, and the full rank-two block is retained. In the old edge direction $P_{15}/\sqrt{15}$ its data are

$$b_{15} = \frac{15 + 14\sqrt{2}}{2\sqrt{15}}, \quad b_{15}^2 - A^2 = 17/60, \quad C_{15} = \frac{15 + 14\sqrt{2}}{\sqrt{17}} > 42/5.$$

Therefore $S_{15} > \sqrt{555/47} > 3.43$ on the original vector. Append this floor to an actual strong twelve/thirteen reference, skipping intermediate unavailable floors. The slopes are $a_K = (3.43 - f_K)/(15 - K)$. For legality use $f_{12} > 3.03, f_{13} > 3.17$: then $a_{12} < .4/3 < 1/7$, $a_{13} < .26/2 < 7/50$. For energy use the opposite bounds $f_{12} < 3.034, f_{13} < 3.175$, gaining $> .396^2/3$ or $> .255^2/2$. The first five block and its head price are unchanged. One square with the head and the one spectator gives totals > 1.0474645 and > 1.0473925, respectively. This disposes of the extra cap and extends helper coverage through fourteen. Its actual mixed-point packing makes thirteen or fourteen strong. Their energies exceed .88864 and $.9084 - 4/3200 = .90715$. The same head gain and spectator then give at least $.88864 + .03624 + .09 > 1$. Thus the entire typed B_{16} row is impossible.

B.10.3.4 The D_{16} centre and the point where cases split

Delete centre fourteen, retaining A_{13}, both actual fork A_1's and the full rank-two component. For own g, left contact β at j, and fork contacts δ, ϵ, the exact data are

$$\Delta = 1 - Q - q_L - \delta^2 - \epsilon^2 > 0, \quad \sigma = g + \beta j/14 + (\delta + \epsilon)/2, \quad q_L = \beta^2 j(14 - j)/7, \quad C_{14} = \sqrt{28}\sigma/\sqrt{\Delta}.$$

The sufficient strong ratio is less than $25/16$. The complete white list is: no left contact; left three at $1, 2, 12, 13$ or left four at $1, 13$; left endpoint-three plus one fork three/four or both forks three. Without a left contact the two-fork pairs are $(3, 3), (3, 4), (3, 5)$, up to exchange; a single fork has any finite label. Full long bridges are foreign-sealed. A permitted extra old-A_2 arm is retained when its full white Gram is positive.

The left-only and isolated roles have their fresh $D_i = 28$ clearings. The isolated $0 < u < 1$ loss is $< (.096 + 1.005)/2 = .5505$. For $Q > 0, u > 1$, $u \leq 8/(3\phi^2) < 1.019$. Hence $\sqrt{28} - \sqrt{28 - u} < .1$ and $\sqrt{u - 1} < .138$, so the full loss is $< .238$. These follow from $\phi > 1.618$, $\sqrt{28} > 5.29$, $\sqrt{28 - 1.019} > 5.19$ and $.138^2 > .019$. At $u = 1$ is the actual null. Pure five/seven/at-least-eight have respective loss bounds $25/160 + 5/24$, $10/81 + \sqrt{13/120}$, and .511. The left four, near-three and D-position tests are stronger than $25/16$. A positive foreign cost makes far-left three strong by its actual full-foreign clearing. A pure far-left helper at fourteen would make thirteen strong and give $q_{13} + .03624 + .09 > 1$. No field argument enters these tests.

Left-three/fork-three bridges have residual $2/7$, numerator $\ge g + 2/7$; own zero/three have their cycles and owns at least four are strong. Left-three/fork-four or two-fork bridges have residual at most $1/28$ and are directly strong including own zero. Actual simple zero-own cycles, where present, are retained. No finite foreign arm fits. For one fork-three, positive owns three/four/six/parallel have the actual cycle, weighted-star, $6,3$, or pair-null tests. With $Q \ge 1/3$, all remaining owns are strong because

$$\Delta \le 5/12, \quad \sigma \ge \phi/2 + 1/4, \quad \sigma^2/\Delta \ge 3(8\phi + 5)/20 > 625/256.$$

Single fork at least four with own at least five is stronger. Own-three/fork-four or five is retained below; foreign cost makes it strong, and larger forks are directly strong. Own-zero/fork-three has the affine E_8 arms $5, 2, 1$, fork-four has null $\sqrt{2}c + 2n_f + 2n_{14} + n_{13} + n_h$, and fork-six its $6, 3$ null. Fork-five permits $Q = 0, 1/3$; fork at least seven forces $Q = 0$. For two forks own three has its triangle, owns at least four are strong, and own-zero/$(3,3)$ is the affine square. This leaves exactly the rows below.

Every remaining row gives a pure D pressure $\langle x, V\rangle \ge G$, flat one on the first eleven coordinates. Choose actual strong $K = 10$, or eleven if ten is weak. The terminal price $1/7$ and head price lie at different anchors; their residuals have zero full cross because the centre residual vanishes on the first K coordinates. The one-vector total is

$$\|x\|^2 + \|y\|^2 > q_K + .03624 + .09 + 2(G - f_K)/7 - (R - K)/49. \tag{B.10.2}$$

For positive own and single fork-three, $Q = 0$, put $\alpha = \sqrt{2}(g + 1/4)$, $A^2 = 14\alpha^2 - 3/4$. Own five has its positive finite H_4 point from $g(3n_{14}/2 + n_{13} + n_{12}/2)$, giving $G > 3.60, R < 14$. For finite at least seven or dotted, weakness forces $g < 9/8$. Use the actual core $c + g(4n_{14}/3 + 2n_{13}/3)$, of square $-a$, $a = 4g^2/3 - 1$. Its old selected products vanish, its cap product is $-a$, both fork products are negative, and every other wall is nonpositive termwise. Its displacement has

$$R = 14 + (1/4 - 4g^2/3)/\alpha^2 < 14, \quad G = (A - \sqrt{a})/\alpha > \frac{2(\sqrt{835} - \sqrt{11})}{11\sqrt{2}} > 3.28.$$

The same derivative as above, now with $3/4$, is negative, so the endpoint $9/8$ covers the entire weak interval.

For own zero and fork-five, $Q = 0$, project the actual H_4 on c, n_f, n_{14}, n_{13}. Its correction is $\delta(3n_f/2 + n_{14} + n_{13}/2)$, $\delta = \phi/2$. Dividing its displacement by $\delta/\sqrt{2}$ gives

$$V = P_{12} + (e_{13} + e_{14} + e_{15} \pm e_{16})/2, \quad R = 13, \quad G > \sqrt{8\phi} - \sqrt{13 - 8\phi} > 3.36.$$

For the genuine N_7 foreign-three row $Q = 1/3$, clear the entire old A_2 by its positive inverse combination v_Z. The finite point from $c + v_Z + \delta n_f$, with $\Delta = 2/3 - \delta^2 > 0$, has displacement $\delta P_{14}/\sqrt{2}$. Since $A > 2.137, \sqrt{\Delta} < .111, \delta < .81$, it gives $\langle x, P_{14}\rangle > 3.5$, so $G > 3.5, R = 14$. No foreign coordinate or second spectator energy is used.

For own zero and single fork at least seven, $Q = 0$, use $c + \delta^{-1} n_f + 2(\delta^{-1} - \delta)n_{14}$. Its cap/fork products vanish, its centre and other-fork products are negative and its time is positive. With $z_0 = \delta^{-2} \in (1, 100/81)$ its data are

$$V = (1^{13}, 3 - 2z_0, z_0 - 1, \ \pm(1 - z_0)), \quad G = \sqrt{16 - 2z_0} - \sqrt{-8 + 14z_0 - 6z_0^2}, \quad R = 24 - 16z_0 + 6z_0^2.$$

The full interval has $14G - R > 32.51$. Indeed, for $z = z_0 - 7/6$, use $\sqrt{16 - 2z_0} > 4.011 - .27z_0$ and $\sqrt{1/6 - 6z^2} \le 1/\sqrt{6} - 3\sqrt{6}z^2$. Endpoint squares and $1.78^2 < .036(96.8748)$ give $14G - R > 32.528 - 1.78z + 96.8748z^2 > 32.51$.

Own-three/fork-four or five has, after releasing that fork, the actual positive finite D_{16} replacement and fork cosine $7+8\delta>12$. Own-zero/$(3,4)$ has the finite B_{16} replacement, releasing the three-contacted fork, with cosine $7+4\sqrt{2}>12$. Positive inversion checks the omitted fork as well as every other wall, so these are actual endpoints. Both give $\langle x, P_{15}\pm e_{16}\rangle > \sqrt{176/13} > 3.67$, $R=16$. Finally own-zero/$(3,5)$ uses the actual finite H_3 projection on the cap and both forks. Its displacement is αP_{14}, $\alpha=(1+\phi)/(2\sqrt{2})$, and

$$A^2=3+11\phi/2,\quad \Delta=(2-\phi)/4,\quad G=\frac{14+21\phi-\phi\sqrt{7}}{4A\alpha}>3.4,\quad R=14.$$

The last strict comparison follows from $1.618<\phi<1.619, \sqrt{7}<2.646, A<3.451, \alpha<.927$: its numerator exceeds 43.696772, whereas $3.4(4\cdot 3.451\cdot .927)=43.5074472$. Every finite projection uses its positive actual-root inverse, has future time and retains all omitted original wall inequalities.

The smallest constant pressure, $G>3.28, R<14$, makes (B.10.2) greater than 1.0249 at ten and 1.0248 at eleven. The rows $(3.36,13),(3.5,14),(3.67,16),(3.4,14)$ give larger margins. The continuous row instead uses $(14G-R-14f_K+K)/49$, giving $>.949975+(32.51-38.248+10)/49>1.0369$, or $>.9734625+(32.51-40.404+11)/49>1.0368$. Thus centre fourteen is strong. The ending-fourteen reference has energy $>.9084-4/3200=.90715$, retaining every permitted internal hole. The same paid head and one typed spectator give $.90715+.03624+.09=1.03339>1$. This excludes the typed D_{16} row as well. The entire argument uses one original farthest vector, true short/fork normals, individual prices and no arithmeticity or face inheritance.

B.10.4 Rank-one spectators beside the rank-seventeen classical factors

The ordinary rank-one case and the actual isolated-short case use different information. We first treat the ordinary D_{17} alternative without imposing a shape on its rank-one opposite.

B.10.4.1 D_{17} with an ordinary nonshort A_1

Use the actual unit D_{17} roots, centre fifteen and forks sixteen/seventeen. The long duals are $\Omega_i=\sqrt{2}P_i$, $i\le 15$; the fork duals are $(P_{16}\mp e_{17})/\sqrt{2}$. The last coordinate is signed. For the original decomposition $z=E(p-x-ye)$, ordinary rank one gives $y^2>.1$, and hence $\|x\|^2<.9$. Every classical cap retains this old e, with full foreign cost $Q=\beta^2<1$, either zero or at least $1/4$.

The valid bulk parameter is $M=11$, so helper coverage first reaches twelve, with an actual strong eleven/twelve. If five were weak, its actual point supplies $x_1>.5965, S_6>2.01815$. The positive reference through one, six and that strong endpoint has energy

$$>.7600299945+(\sqrt{11}-\sqrt{6})^2/5>.9104024765.$$

Its slopes decrease; the endpoint is below its actual strong floor. This contradicts $\|x\|^2<.9$. Five is strong before any new tail or head argument.

Here are the fresh tail matrices, to specify precisely which local arguments of Appendix B.6 are reused. At thirteen/fourteen deletion retains $A_{12}+D_4$ and $A_{13}+D_3$. In the actual D_4, long-fourteen contact three has $(\Delta,\sigma)=(1/2-Q,g+1/2)$; either fork-three has $(1/2-Q,g+1/4)$. In the actual D_3, middle-three, end-three, end-four and end-five have respectively

$$(1/2-Q,g+1/2),\quad (5/8-Q,g+1/4),\quad (1/4-Q,g+1/(2\sqrt{2})),\quad (1-3\phi^2/8,g+\phi/4).$$

The sufficient strong ratio is $<T(6)<25/16$. These are the same finite local matrices, not a whole-component theorem with its rank changed. Left-only and isolated caps use their long-root

clearings at $D_i = 2i$: every isolated foreign case is strong, the pure five/seven/at-least-eight/dotted points have their .61 loss bounds, and a positive foreign far-three is strong. Own-zero and owns three/four/six/parallel have the actual affine/null or reciprocal-shadow dispositions of that inventory, retaining every foreign cost. Both fork choices remain.

For completeness the right lists leave only these nondirect rows. At thirteen, a fork-three with $Q = 0$ and own five survives; all positive foreign cases have ratio > 2.1, and finite at least seven or dotted has ratio $> 1.15\sqrt{2}$. At fourteen, middle-three is direct. End-four at own zero has kernel $(\sqrt{2}, 2, 2, 1, 1)$ on the cap, contacted fork, centre, other fork and old fourteen; all positive owns are direct. The sealed end-five H_4 white block has ratio > 2.99 even at own zero. End-three has its own-zero E_8, own-three cycle, own-four star, own-six path and parallel null. Positive foreign cost is direct; $Q = 0, g \geq 1$ has ratio at least $\sqrt{5/2}$. Thus only pure own five and finite at least seven remain there.

Full bridges use simple-three left endpoints only. The D_4/middle-D_3 bridge has residual $1/(2i)$ and numerator at least $g + 1/4 + 1/(2i)$. The end-three bridge at fourteen has residual $1/8 + 1/28 = 9/56$ and numerator at least $g + 1/4 + 1/28$. Their own-zero principals are actual all-three cycles, and every positive own is directly strong. A foreign arm at the internal bridge cannot be finite. No exceptional finite type has the required rank. The surviving weak rows are therefore pure far-left helpers and the explicit right exceptions just listed.

The right own-five exception at $i = 13, 14$ uses its actual left H_4 projected point with correction $g(3n_i/2 + n_{i-1} + n_{i-2}/2)$, $g = \phi/2$. With $\alpha = \sqrt{2}(g + 1/4)$,

$$A^2 = i\alpha^2 - \kappa, \quad \kappa = 1/2 \text{ (13)}, \ 5/8 \text{ (14)}, \quad G > \sqrt{i - \kappa/\alpha^2} - .091, \quad R < i.$$

The normalized displacement is flat through $i - 3$. The finite point has positive time and all original wall signs, including the two forks and old rank-one component. Positive squares give $G > 3.48$ and $G > 3.61$. For thirteen choose actual strong nine/ten; the local one-square formula and the one ordinary charge give totals $> 1.0757, > 1.0762$. Coverage thus reaches thirteen.

For every remaining finite $.9 < g < 1$ at fourteen, average the two actual A_2 cores with corrections $g(4n_{14}/3 + 2n_{13}/3)$ and $g(4n_{14}/3 + 2n_{15}/3)$. Both satisfy every wall inequality. Their correction cross $8g^2/9$ is retained in the average. With $a = 4g^2/3 - 1$, its data are

$$A^2 = 14\alpha^2 - 5/8, \quad R < 14, \quad G = (A - \sqrt{a})/\alpha > \sqrt{69/5} - 2\sqrt{6}/15 > 3.388.$$

The derivative numerator $(5/8)/A - (g/3 + 1)/\sqrt{a}$ is negative on the full interval, proving the endpoint bound. At actual strong ten/eleven the local payment at its individual $1/7$ coefficient, plus $y^2 > .1$, gives totals $> 1.0295, > 1.0294$. No dotted parameter is missing: it was directly strong. Coverage reaches fourteen with only pure helpers.

Their nonadjacency extends by the actual two-cap calculation: adjacent helper products exceed $1 - ab > -11/25 > -1/2$, so the native gap forces orthogonality. The retained cap Schur matrix is $I - t\mathbf{1}\mathbf{1}^T$, $t = (i - 1)/(2i)$, with positive inverse. The unchanged prefix replacement gives weighted loss $< 7/16 + 1/(4\sqrt{10}) < .61$, contradicting two weak floors. Retained right blocks remain positive; no larger-rank prefix hypothesis is invoked. There is therefore a strong thirteen/fourteen reference with energy $q > .88864$, first-five mean $b < .361$ and individual P_5 coefficient $> .146$. This gives only the safe q_{13} floor; it does not assert strong fourteen or a q_{14} floor.

Delete long one. The full retained D_{16} permits only isolation or a three at its long endpoint two. The latter has $A^2 = 2g^2 + 2g + Q$, $C_1 = (2g + 1)/\sqrt{1 - 2Q}$. Own zero has zero time or its positive full-D null; own three has its triangle. Every other own gives $x_1 > .643$. Its legal first-five graft has gain $(5/4)(.643 - b)^2 > .0994$, already contradictory.

For an isolated-within-D head, $c = Ap - g\Omega_1 - \beta e$, $A^2 = 2g^2 + Q - 1$. The actual foreign shadow and original-farthest orthogonal-opposite obstruction give $A > \beta$, so $2g^2 > 1$. Owns zero/three/four are impossible; six and parallel have actual future nulls. All remaining own and foreign cases have one of the following actual pure-D pressures. A normalized displacement Z, supported in the first five coordinates with sum one, has $\langle x, Z\rangle \geq H$, $R = \|Z\|^2$; weight μ costs $\mu/5$ and gains $2\mu(H - b) - \mu^2(R - 1/5)$ in this one reference.

Pure five uses the finite H_4 correction $g(3n_1/2 + n_2 + n_3/2)$. It gives $Z = P_4/4$, $R = 1/4$, $H^2 = 1/(4\phi) > 2/13 > .392^2$. Weight .4 gives gain $> .0168$. Pure finite at least seven uses $c + g^{-1}n_1 + 2(g^{-1} - g)n_2$, and, for $u = g^{-2} \in (1, 1.232)$,

$$Z = (1 - u/2)(e_1 + e_2) + (u - 1)e_3, \quad H = \frac{\sqrt{2 - u} - \sqrt{7u - 3u^2 - 4}}{\sqrt{2}}, \quad R = 3 - 4u + 3u^2/2.$$

The complete interval seals for weight $1/2$ are

u	H lower	R upper	gain lower
$[1, 1.1]$	$.6708 - .1871$	.5	.0477
$[1.1, 1.2]$	$.63245 - .204125$	.415	.013575
$[1.2, 1.232]$	$.6196 - .2$	.36	.0186.

They follow by positive squaring and the quadratic radical maximum $1/24$. The native-seven cubic gives the upper 1.232 uniformly for all finite labels at least seven.

For dotted or foreign-cleared $0 < u = (1 - Q)/g^2 < 1$, the positive core $c + \beta e + gun_1$ gives exactly the Z, H pressure in the common dotted-head calculation of Appendix B.10.3. Its scalar proof uses only this pressure, $b < .361$ and the individual first-five capacity. It therefore gives the same legal gain $> .03624$ here at weight $\sqrt{2}/5$; no typed-defect hypothesis or conclusion is used. All original walls are retained by the positive core. At $u = 1$ is a nonzero feasible future null. For $Q > 0, u > 1$, the native alphabet forces own five and foreign three only. Its actual finite H_3 projection on c, n_1, e gives $Z = P_2/2$, $R = 1/2$, $H^2 = 1/(2\sqrt{5}) > 2/9 > .47^2$. Weight $1/2$ gives gain $> .034$. Every foreign finite label and the full dotted own interval is thus included.

Each price is at most $.1 < .146$. Every selected finite point is an actual positive projection; every negative core has future time and nonpositive product with all original walls. All head cases give a legal gain greater than .0135. Hence

$$\|x\|^2 + y^2 > .88864 + .0135 + .1 = 1.00214 > 1.$$

This closes precisely the ordinary D_{17} rank-one case. Tail elimination contradictions have not been added to the final reference, and no short-I energy or arithmeticity is assumed.

B.10.5 B_{17} with an actual isolated short spectator

We separate the local payment lemmas from the short-tail classification. This makes the order of the argument explicit: the local lemmas first dispose of the exceptional B_2 cap, and only afterwards do we prove the unconditional strong-five bootstrap. None of the local lemmas assumes that bootstrap or an already strong terminal wall.

Throughout, $P_i = e_1 + \cdots + e_i$, $f_i = \sqrt{i} - .61/\sqrt{2}$, and $\phi = (1 + \sqrt{5})/2$. The original vector is $z = E(p - x - ye)$, with $\|x\|^2 + y^2 < 1$. The actual isolated I-wall is

$$c_I = \sqrt{h}\, p - \sqrt{1 + h}\, e, \qquad 0 < h < 1/2.$$

Put

$$s^2 = \frac{\sqrt{1+h}-1}{\sqrt{1+h}+1}, \qquad t = \sqrt{1+4h} - 2\sqrt{h}, \qquad h(r) = \frac{4r}{(1-r)^2}.$$

Appendix B.1.8 proves $y \geq s$, and height at least t in every other actual unit dual direction. At a classical long end with $\Omega_1 = \sqrt{2}e_1$, this says $x_1 \geq t$. In particular

$$0 < s < 1/3, \qquad (1-s^2)(1-t^2) = 8st, \qquad h = h(s^2).$$

These facts concern actual walls and the same original vector. The local lemmas below require only their stated long-root, point and reference hypotheses; a B short root is not among those hypotheses. Where the initial roots are used, they are the actual long roots $n_j = (e_j - e_{j+1})/\sqrt{2}$, $j = 1, 2, 3$, with actual long duals $\Omega_1 = \sqrt{2}P_1, \Omega_2 = \sqrt{2}P_2$.

Lemma B.10.1 (The additional real I-barrier)**.** *The actual parameters satisfy $t + \frac{5}{2}s > .923$.*

Proof. The test value $t_0 = .923 - 2.5s$ is positive. Substitution in the left-minus-right of the relation gives

$$P(s) = .148071 - 2.769s + 13.601929s^2 - 4.615s^3 + 6.25s^4.$$

For $v = s - .1 \in (-.1, 7/30)$, its coefficients are $.00320029, -.1620642, 12.592429, -2.115, 6.25$. Since $-2.115v^3 \geq -.4935v^2$,

$$P(s) > .0032 - .163|v| + 12v^2 > 0.$$

The last discriminant is $.026569 - .1536 < 0$. The original expression decreases strictly with positive t, proving the claim. This is separate from the earlier .767-barrier. □

B.10.5.1 Local paid squares, with every initial cross retained

Call W_0 a paid classical vector if its demand D_0 has already been obtained from nonnegative actual-pressure coefficients. Its effective lower energy is $\mathcal{E}_0 = 2D_0 - \|W_0\|^2$. At a retained anchor P_m, let the actual demand be a_m; any debit below is made from its remaining individual coefficient. No I-energy is included in $\mathcal{E}_0$.

Lemma B.10.2 (A paid head square at an arbitrary initial anchor)**.** *Suppose an actual pure classical pressure is $\langle x, Z\rangle \geq H$, with Z supported in the first m coordinates and $\langle Z, P_m\rangle = 1$. Write $R = Z^2$ and*

$$B_Z = \langle W_0, Z\rangle + \frac{a_m - \langle W_0, P_m\rangle}{m}.$$

For $\nu \geq 0$, with debit ν/m available at P_m, the same square improves $\mathcal{E}_0$ by

$$2\nu(H - B_Z) - \nu^2(R - 1/m).$$

Proof. Use $W_0 + \nu(Z - P_m/m)$, after removing ν/m from the anchor coefficient. Its residual square is $R - 1/m$; expanding its full cross with W_0 gives the formula. In particular this formula does not assume a weak-five P_1/P_6 reference is flat. □

Lemma B.10.3 (A joint point and the I-height on a paid reference)**.** *Suppose the actual pressures are $ax_1 + vy \geq F$, $y \geq s_0$, with $a > 0, v \geq 0$. Put*

$$B_0 = (W_0)_1 + \frac{a_m - \langle W_0, P_m\rangle}{m}.$$

For $\eta, \tau \ge 0$, write $k = \eta a$, $\ell = \eta v + \tau$, and require the individual anchor debit k/m to be available. The vector $W_0 + k(e_1 - P_m/m) + \ell e$ improves $\mathcal{E}_0$ by

$$2\eta(F - aB_0) + 2\tau s_0 - (1 - 1/m)k^2 - \ell^2.$$

If $\ell = s_0$, put $X = (F - vs_0)/a$. The improvement becomes

$$s_0^2 + 2k(X - B_0) - (1 - 1/m)k^2, \qquad \tau = s_0 - kv/a \ge 0.$$

Proof. Remove k/m at the original anchor before using the two new pressures. Expand the square, retaining $\langle W_0, e_1 - P_m/m\rangle$. The old vector has no e-component. This proves both identities and keeps the nonnegative multiplier and individual-price conditions visible, also when k is capped. □

Lemma B.10.4 (A capped initial graft at a weak-five reference)**.** *Let $a_0 = .5965$. Suppose W_0 is already paid, its first coordinate is a_0, and $\langle W_0, P_6\rangle$ equals its retained P_6 demand. If the remaining P_6 price is at least $1/60$, then*

$$\lambda = \min\{(6/5)(t - a_0)_+, .1\}$$

gives, using $e_1 - P_6/6$ and the single unused I-height, an improvement

$$s^2 + 2\lambda(t - a_0) - (5/6)\lambda^2 > .015.$$

Proof. The full old-reference cross changes the baseline to a_0; it is not zero in general. In the uncapped branch, weighted Cauchy and Lemma B.10.1 give

$$s^2 + (6/5)(t - a_0)_+^2 > \frac{(.923 - .5965)^2}{25/4 + 5/6} > .015.$$

In the capped branch $t > a_0 + 1/12$. The relation makes t decrease with s, and

$$t(.1) = \frac{\sqrt{11401} - 40}{99} < 67/99 < a_0 + 1/12.$$

Thus $s < .1$. The .923-barrier then gives

$$s^2 + .2(t - a_0) - 1/120 > (s - .25)^2 - 83/15000 > .0169666.$$

The debit is at most $1/60$. These are one paid square and one I-height, not two independent initial-coordinate gains. □

A flat-five reference below means a nonnegative actual-anchor reference w, with demand $\|w\|^2 = q$, first five entries $b = f_5/5 < .361$, and remaining P_5 coefficient $> .146$. The anchor demand there is $f_5 = 5b$. Extra terminal anchors are allowed. Every local gain below preserves their nonnegative coefficients. A head means the actual opposite of the old long node one.

Lemma B.10.5 (The foreign-cleared head scalar)**.** *On a flat-five reference, suppose an actual pure pressure has*

$$Z = (1 - u/2)e_1 + (u/2)e_2, \quad H = \frac{\sqrt{2 - u} - \sqrt{u(1 - u)}}{\sqrt{2}}, \quad R = 1 - u + u^2/2, \qquad 0 < u < 1.$$

Weight $\nu = .36$, costing $.072$ at P_5, gives gain $> .04$.

Proof. Use Lemma B.10.2. Put $.5091 < .36\sqrt{2} < .5092$. Lower chords for $\sqrt{2-u}$ and upper chords for R, on the three successive intervals, are

$$\begin{array}{c|c|c} [0,.5] & 1.4142 - .379u & 1 - .75u \\ [.5,.75] & 1.4381 - .4268u & .8125 - .375u \\ [.75,1] & 1.4721 - .4721u & .625 - .125u. \end{array}$$

Their endpoint squares and concavity/convexity prove the bounds. After $b < .361$, the gains exceed

$$\begin{array}{c|c} .35636922 - .0958u - .5092\sqrt{u(1-u)} & .04936922 \\ .39283671 - .16873u - .5092\sqrt{u(1-u)} & .04022171 \\ .43444611 - .2242u - .5092\sqrt{u(1-u)} & .04414611. \end{array}$$

The right column follows from $pu + c\sqrt{u(1-u)} \le (p + \sqrt{p^2+c^2})/2$, using upper radical seals $.5182, .5365, .5564$. □

Lemma B.10.6 (Finite heads on a flat-five reference)**.** *Suppose the retained classical long-end diagram has only an isolated head or a simple-three extension at old node two. On any flat-five reference, a nonisolated head gives gain* $> .0994$*. For a pure finite isolated head the head gain alone is* $> .0186$*, and its gain including the single* I*-height is* $> .0933875$*. For a positive foreign finite head the combined gain is* $> .0957$*. Consequently* $q \ge .90715$ *excludes every nonisolated or finite isolated head beside actual* I*, without a terminal-strength hypothesis.*

Proof. For the nonisolated head,

$$A^2 = 2g^2 + 2g + Q, \qquad C_1 = \frac{2g+1}{\sqrt{1-2Q}}.$$

Own zero has zero time or the positive null $c + \Omega_2/2 = \sqrt{Q}p - \beta e$; the dual has its actual positive old-root expansion. Own three has the triangle and parallel the pair null. Every other own gives $x_1 > .643$. Replacing the first five entries by $.643, (5b - .643)/4, \dots$ is legal: its debit $(.643 - b)/4$ fits the individual P_5 coefficient, and its gain exceeds $(5/4)(.282)^2 > .0994$.

For an isolated finite head $c = Ap - g\sqrt{2}e_1 - \beta e$, $g < 1$, put $Q = \beta^2$. The distinct actual I-wall gives $\lambda = \sqrt{h}A - \sqrt{1+h}\beta \ge 0$. At $\beta = 0$, $0 < \lambda < 1/\sqrt{2}$, so $\lambda = 1/2$. At $\beta > 0$,

$$\lambda < \sqrt{(1+\beta^2)/2} - \beta\sqrt{3/2} \le (\sqrt{10} - \sqrt{6})/4 < 1/2.$$

The middle upper bound decreases with $\beta \ge 1/2$, and its bound before substitution increases with $h < 1/2$; hence $\lambda = 0$. Squaring gives $\beta^2 = h(2g^2 - 1) < 1/2$, so $\beta = 1/2$. Both cases consequently have

$$h(2g^2 - 1) = 1/4, \qquad h > 1/4.$$

Own zero/three/four is impossible, five gives $h = \phi/2 > 1/2$, six gives $h = 1/2$, and parallel has $c + n_1$. Thus a finite isolated own has label at least seven.

If $\beta > 0$, $u = (1 - Q)/g^2 < 1$. The actual positive core

$$c + \beta e + gun_1$$

has products zero on c, e, product $g(u-1) < 0$ on n_1, square $-g^2u(1-u)$, and positive time. It yields exactly the pure pressure of Lemma B.10.5. The I-height costs $> .0557$, by $h > 1/4$, so the gain including I is $> .0957$.

If $\beta = 0$, put $u = g^{-2} \in (1, 1.232)$. The actual core

$$c + g^{-1}n_1 + 2(g^{-1} - g)n_2$$

has zero cap/first-root products, negative second/third-root products and positive time. Its pure data are

$$Z = (1 - u/2)(e_1 + e_2) + (u - 1)e_3, \quad H = \sqrt{1 - u/2} - \sqrt{(7u - 3u^2 - 4)/2}, \quad R = 3 - 4u + 3u^2/2.$$

The native-seven cubic gives $g > .90096$, and $1.232(.90096)^2 > 1$. Weight $1/2$ gives

u	head gain lower	h lower	I energy lower
$[1, 1.1]$	.0477	1/4	.0557
$[1.1, 1.15]$	.033025	11/36	.066
$[1.15, 1.2]$	.0213875	23/68	.072
$[1.2, 1.232]$	.0186	3/8	.079.

The first radicals are bounded below by $.6708, .6519, .63245, .6196$, the second above by $.1871$, $.204125, .204125, .2$, and R above by $.5, .415, .38375, .36$. Positive squaring proves these; the second radicand has maximum $1/24$ and R decreases. Use $h = u/[4(2 - u)]$ and $h(r)$ for the I-column. Every combined gain is $> .0933875 > 1 - .90715$. Every head pressure has positive actual-root coefficients, and all unselected original walls retain nonpositive products. □

Lemma B.10.7 (An actual joint point for every pure dotted mutual)**.** *For an actual pure dotted head $c = Ap - g\sqrt{2}e_1$, put*

$$\begin{aligned} u &= g^{-2} \in (0, 1), \qquad h_0 = \frac{u}{4(2 - u)}, \\ a &= \sqrt{1 - u/2}, \qquad v_0 = \sqrt{u/(8 - 3u)}, \\ d &= \sqrt{1 - a^2 - v_0^2}, \quad F = 1 - d, \qquad s_0 = s(h_0). \end{aligned}$$

For $u \geq 1/5$, every actual $h \geq h_0$ gives a feasible future point $(p - ae_1 - v_0 e)/d$. Consequently

$$ax_1 + v_0 y \geq F, \quad y \geq s_0, \qquad X = \frac{F - v_0 s_0}{a} = \sqrt{\frac{1 - \sqrt{3u/8}}{1 + \sqrt{3u/8}}}.$$

Proof. The actual mutual product is $-A\sqrt{h} < 0$, hence $h \geq 1/(4A^2) = h_0$. Put $v = \sqrt{h/(1 + h)}$; then $v \geq v_0$, $v < 1/\sqrt{3}$. The denominator $D = u/2 - v_0 v$ is positive because $3u(8 - 3u) \geq 111/25 > 4$ on this interval. For

$$T = 1/D, \quad \rho = \frac{Ta}{\sqrt{2}g} > 0, \quad \sigma = \frac{Tv_0}{\sqrt{1 + h}} > 0$$

direct substitution gives

$$p + \rho c + \sigma c_I = T(p - ae_1 - v_0 e).$$

Its c-product is zero and its c_I-product is $T\sqrt{1 + h}(v_0 - v) \leq 0$. Old wall products are nonpositive by duality; all other actual walls are nonpositive termwise. Its time is positive and $d^2 > 0$. Thus the normalized point is actual even if the actual c, c_I Gram is not positive. The identities $v_0 s_0 = 1 - \sqrt{1 - v_0^2}$ and $d/\sqrt{1 - v_0^2} = \sqrt{3u/8}$ give X. It decreases with u, whereas s_0^2 increases. The comparison $X > r$ at $u = U$ is the positive rational test

$$8(1 - r^2)^2 - 3U(1 + r^2)^2 > 0.$$

□

Lemma B.10.8 (The foreign dotted-head parabola)**.** *For a dotted head with $Q = \beta^2 > 0$, set $v = (1-Q)/g^2$. Then $0 < v < 3/4$, and*

$$h \geq h_{\min}(v) := \frac{v}{6-4v}.$$

The actual foreign-cleared core gives the pure pressure

$$Z = (1-v/2)e_1 + (v/2)e_2, \quad H(v) = \sqrt{1-v/2} - \sqrt{v(1-v)/2}, \quad R(v) = 1 - v + v^2/2.$$

For $F(v) = .8H(v) - .16R(v)$, $\zeta = v - 2/3$,

$$F(v) \geq C_0 + A_1\zeta + .448\zeta^2, \quad C_0 > .2976415, \quad 0 < A_1 < .008385, \quad F(v) > .2976.$$

Also $y^2 > .08625v - .014375$ for $v \leq .6$, and $y^2 > .0385$ for $v \geq .6$.

Proof. Acuteness with c_I gives $h(2g^2-1) \geq Q$; hence

$$h \geq \frac{Qv}{2(1-Q)-v} \geq \frac{v}{6-4v}.$$

The second expression increases with $Q \geq 1/4$. The core $c + \beta e + ((1-Q)/g)n_1$ has zero cap/old-I products, negative n_1-product, square $-g^2v(1-v)$ and positive time. It clears the foreign coordinate before producing the displayed pure pressure; every other wall is nonpositive.

Put $r = \sqrt{6}$ and $\ell = r/3 - r\zeta/8 - .09\zeta^2$. On this interval $\ell \geq 7r/24 - .04 > 0$. The exact identity

$$1 - v/2 - \ell^2 = \zeta^2(3r/50 - 3/32 - (9r/400)\zeta - .0081\zeta^2)$$

has nonnegative right side, since the bracket is at least $.045r - .09735 > 0$. Further, $(\sqrt{v(1-v)/2} - 1/3)^2 \geq 0$ gives

$$H(v) \geq (r-1)/3 + (2-r)\zeta/8 + .66\zeta^2.$$

Thus $C_0 = 4(r-1)/15 - 4/45$ and $A_1 = 19/75 - r/10$. Use $2.449489 < r < 2.449490$ and $.008385^2 < 1.792(.2976415 - .2976)$.

For $v \leq .6$, $h_{\min} \leq 1/6$, and $.23(2.081)^2 < 1$ gives $s(h_{\min})^2 > .23h_{\min}$. The exact tangent inequality is

$$\frac{v}{6-4v} - \left(\frac{3v}{8} - \frac{1}{16}\right) = \frac{3(2v-1)^2}{8(6-4v)} \geq 0.$$

For $v \geq .6$, $h_{\min} \geq 1/6$, whose I-energy is $> .0385$, again checked using $h(r)$. □

Lemma B.10.9 (The head on any sufficiently strong flat-five reference)**.** *Assume the actual long-end whitelist of Lemma B.10.6. On any flat-five reference every actual head gives a legal gain, including I once, greater than .07665. In particular $q \geq .9240775$ excludes all heads. No actual strong-fifteen condition is required.*

Proof. Finite and nonisolated heads are covered by Lemma B.10.6. For a pure dotted head with $u \leq .2$, its actual core $c + g^{-1}n_1$ gives the pressure Z, H, R of Lemma B.10.5. Here $H > .6658, R \leq 1$. Weight .3, with price .06, gives gain $.6(.6658 - .361) - .09(.8) = .11088$.

For pure $.2 \leq u \leq .5$, use weight $\eta = .33$ of the joint point alone. In Lemma B.10.7,

$$F > .5839, \quad a < .949, \quad (4/5)a^2 + v_0^2 < .748.$$

The first bound follows at $u = .5$, since d^2 increases; the last follows at $u = .2$, since that expression decreases. The gain is

$$.66(.5839 - .949 \cdot .361) - .1089(.748) = .07780806.$$

Its individual price is $< .066$; no I-energy is added again.

For $.5 \le u \le .8$, use Lemma B.10.3 with $m = 5, B = .361$, $k = 5(X - B)/4$, $\ell = s_0$. The unused I multiplier is nonnegative because $X - B < .269 < .4a$ and $s_0/v_0 \ge 1/2$. Its price is $< .06725$. Positive rational tests give

u	s_0^2 lower	X lower	gain lower
$[.5, .6]$	.0200	.597	.08962
$[.6, .7]$	.0254	.567	.078445
$[.7, .75]$	.0315	.553	.07758
$[.75, .775]$	.0349	.547	.078145
$[.775, .8]$	.0366	.540	.07665125.

Use the right endpoints for X, and left endpoints in $h(r)$ for s_0^2; for the last two these have $h_0 = 3/20, 31/196$. For $.8 \le u < 1$, the actual pure core gives $H > .491, R \le .52$. Indeed H increases there and R decreases. A half-payment gains $> .05$, and $h_0 \ge 1/6$ contributes $> .0385$.

For a positive foreign dotted contact use Lemma B.10.8 at weight .4. The head gain is $> F - .2568$, at price .08. For $v \le .6$, adding the I-height gives

$$> .0839665 + (A_1 + .08625)\zeta + .448\zeta^2 > .0789665,$$

using $(.09464)^2 < 1.792(.005)$. For $v \ge .6$ the gain is $> .2976 - .2568 + .0385 = .0793$. All prices fit the single available P_5 coefficient, and every parameter branch gives more than the stated .07665. □

B.10.5.2 A conditional actual-helper packet at long index fifteen

Lemma B.10.10 (Actual helper pressures and the retained fifteen packet)**.** *Suppose an actual pure far-three opposite at a long index $i \ge 5$ has*

$$c_i = A_i p - \gamma\sqrt{2}P_i - \Omega_1/2, \qquad A_i^2 = 2i\gamma^2 + 2\gamma - 1/2,$$

where $\sqrt{2}P_i$ is the actual old long dual, $n_i = (e_i - e_{i+1})/\sqrt{2}$, and the old initial roots n_1, n_2 are actual. Its own is assumed to be five, finite at least seven, or dotted; the invoking local classification must establish that list. Write $H_k = P_k/(k\sqrt{2})$. The following are actual pure pressures $\langle x, U\rangle \ge T$:

own	U	T
5	$\gamma(\sqrt{2}P_i - n_i) + H_3$	$H(A_i, 2/3 - \gamma^2)$
$7, 8$	$\gamma(\sqrt{2}P_i - n_i) + H_2$	$A_i - \sqrt{\gamma^2 - 3/4}$
finite ≥ 9	$\gamma(\sqrt{2}P_i - n_i) + H_1$	$H(A_i, 1 - \gamma^2)$
dotted	$\gamma\sqrt{2}P_i - \gamma^{-1}n_i + H_1$	$A_i - \sqrt{1 - \gamma^{-2}}$,

where $H(A, d) = A/(1 + \sqrt{d/(A^2 + d)})$.

In particular suppose $i = 15$ is weak, and the legal reference w ends at strong fourteen, is flat-five with $q > .90715$, and has individual P_{14} coefficient $> .1361$. Put $\alpha = \sqrt{2}\gamma$ and

$$\Pi U = \alpha P_{14} + P_5/(5\sqrt{2}), \quad Y = U - \Pi U, \quad \Delta = T - \alpha f_{14} - b/\sqrt{2}.$$

Then $w.Y = 0$, and the following complete table is valid:

own	Δ *lower*	Y^2 *upper*	μ	$2\mu\Delta - \mu^2 Y^2$ *lower*
5	.39558	.724	.09	.06534
7	.340518	.963	.09	.05349294
8	.2873002	1.005	.09	.043573536
finite ≥ 9	.299602	1.4	.09	.04258836
dotted	.3307	1.68	.08	.04216.

Thus $W_0 = w + \mu Y$ has effective energy $> q + .0421$. Its debits are $< .1273$ at P_{14} and $< .01273$ at P_5. Its initial coordinate is

$$(W_0)_1 = b + \frac{\mu(1/k - 1/5)}{\sqrt{2}} < .412.$$

This lemma does not assume a B short normal or that P_{16} is an actual long dual.

Proof. For own five clear old n_i, n_1, n_2, with positive correction $\gamma n_i + 2n_1/3 + n_2/3$; the actual finite block is H_4. Its residual is $2/3 - \gamma^2 > 0$, so use the finite projected point, not the corrected spacelike root. For owns seven/eight clear $\gamma n_i + n_1/2$; its residual is $3/4 - \gamma^2 < 0$. For finite owns at least nine clear γn_i, and for dotted owns use $c_i + n_i/\gamma$. Selected products are zero or negative, time is positive, and all other actual walls are nonpositive termwise. These include the old I, its opposite, any long head and either classical tail. The displayed U's follow by subtraction in the original Euclidean coordinates.

At fifteen the initial residual is $H_k - P_5/(5\sqrt{2})$; the remaining residual has support only in fifteen/sixteen. It is orthogonal to every retained reference anchor. For finite owns,

$$Y^2 = \gamma^2 + 1/(2k) - 1/10;$$

for dotted owns it is $2\gamma^2 - 1.6 + \gamma^{-2}$. Use $\sqrt{28} - .61 < 4.682$, $b/\sqrt{2} < .256$. For finite five/seven/eight, the respective (A_i lower, radical upper, γ upper) are

$$(4.555, .111, .81), \quad (5.064, .249, .901), \quad (5.191, .322, .9239).$$

Their positive squares give the first three deficits. For finite $\gamma \geq .939$, the deficit increases: $H_A > .99$, $A_i' > \sqrt{30}$, $H_d < 0$, and $d = 1 - \gamma^2$ decreases. At .939,

$$A_i > 5.275, \quad \sqrt{A_i^2 + d} > 5.286, \quad \sqrt{d} < .344, \quad H > 4.952,$$

giving $\Delta > .299602$.

For dotted fifteen, weakness implies $\gamma < 10/9$: at $10/9$ its squared edge ratio is $10609/4320 > (39/25)^2$, whereas $T(\sqrt{30}) < 39/25$. The deficit decreases on this interval, since $A_i' < 5.6$, the radical derivative is > 1, and $\sqrt{28} - .61 > 4.681$. At $10/9$, $A_i^2 = 2093/54 > 6.225^2$ and the radical is $< .436$; thus $\Delta > .3307777$. The norm is < 1.68. All entries and prices now follow by rational multiplication. The largest initial coordinate occurs for $k = 1, \mu = .09$, and is below $.361 + .09(.8)/\sqrt{2} < .412$. □

Lemma B.10.11 (A retained helper, a dotted head and actual I)**.** *Assume the actual helper and reference hypotheses of Lemma B.10.10, with $f_{14} < 3.31034$, and an actual isolated-within-classical dotted head. For a pure head with $u \geq .2$, a single paid square gives helper gain $> .0421$ and joint head/I gain $> .0515$. For a positive foreign head the corresponding gains are $> .0421$ and $> .0521$. For a pure head with $u \leq .2$, the standalone gain $> .11088$ on w suffices without paying the helper. In particular each case contradicts the original norm budget.*

Proof. For a pure head with $u \geq .2$, take $B = .412$ above the actual initial coordinate B_0. In Lemma B.10.3, use $m = 5$, $k = 5(X - B)/4$, $\ell = s_0$. Since the helper residual has zero sum on the first five entries, its full cross changes the baseline to precisely B_0; it is not orthogonal to a head. The multipliers are nonnegative: on $[.2, .4]$, $X < .756, a > \sqrt{.8}$, and on $[.4, 1]$, $X < .665, a > 1/\sqrt{2}$. Thus $X - B < .4a$, while $s_0/v_0 \geq 1/2$. The total P_5 price is $< .086 + .01273 < .146$, and the separate P_{14} price fits. The complete continuous table is

u	s_0^2 lower	X lower	$s_0^2 + (5/4)(X - .412)^2$ lower
$[.2, .5]$	0	.629	.05886125
$[.5, .6]$	.0200	.597	.06278125
$[.6, .7]$	.0254	.567	.05543125
$[.7, .8]$	.0315	.540	.05198
$[.8, .9]$	.0385	.514	.051505
$[.9, 1]$	.0464	.490	.054005.

The X-tests are those of Lemma B.10.7; the I-tests use the left endpoints $h_0 = 1/12, 3/28, 7/52, 1/6, 9/44$. For example the first X-test has positive numerator .0002777057265. This proves gain $> .0515$ on all the displayed intervals. Small u uses the standalone actual core already proved in Lemma B.10.9.

For a foreign head put $v = (1 - Q)/g^2$, and use its cleared pressure at weight $\nu = .4$. The initial helper products with $Z - P_5/5$, for $k = 3, 2, 1$, are respectively

$$\frac{2/15}{\sqrt{2}}, \quad \frac{3/10}{\sqrt{2}}, \quad \frac{.8 - v/2}{\sqrt{2}}.$$

The last is largest on $0 < v < .75$. Hence the complete adverse cross is at most $(.072/\sqrt{2})(.8 - v/2)$. Using $1/\sqrt{2} < .7072$, the head gain is

$$> F(v) - .29753472 + .0254592v.$$

For $v \leq .6$, adding the single I-height and the parabola gives

$$> .06020458 + (A_1 + .1117092)\zeta + .448\zeta^2 > .05215458,$$

since $.1201^2 < 1.792(.00805)$. For $v \geq .6$ the gain is $> .0538408$. The combined P_5 price is $< .09273$; the P_{14} price is unchanged. No I-energy was included in the pure cleared pressure or the helper gain. The full totals exceed

$$.90715 + .0421 + .0515 = 1.00075, \qquad .90715 + .0421 + .0521 = 1.00135.$$

□

B.10.5.3 The fresh B_{17} tail and its exceptional actual mate

We now impose the entire $B_{17} + I$ forest, with actual unit roots $n_i = (e_i - e_{i+1})/\sqrt{2}$, $i \leq 16$, and genuine short root $n_{17} = e_{17}$. Thus $\Omega_i = \sqrt{2}P_i$ through sixteen and $\Omega_{17} = P_{17}$. All classical opposites retain the old I. The valid bulk parameter is $M = 12$, giving pure-helper coverage through thirteen and the actual mixed-point nonadjacency.

At fourteen deletion retains $A_{13} + B_3$. Its only right finite extensions are long-three at fifteen and short-three at seventeen, the latter the sealed F_4. Their (Δ, σ) are $(1/2 - Q, \gamma + 1/2)$ and $(1/4 - Q, \gamma + 1/(2\sqrt{2}))$. Own zero has the actual affine star or $3, 4, 3, 3$ path; own three on the long

side is a triangle, and all other positive cases are directly strong. Full bridges attach by three at left endpoints and the long right endpoint, are finite B_{17} and foreign-sealed. A short-end full bridge would put the old four internally in a long path. Left/isolated/foreign cases retain the complete long-root inventory of Appendix B.5. Consequently coverage and nonadjacency reach fourteen, before assuming that five is strong.

At fifteen the retained right block is the actual B_2 on long sixteen and short seventeen. A finite new contact is three at either end only. The data are

$$(\Delta, \sigma) = (1/2 - Q, \gamma + 1/2), \quad (1/2 - Q, \gamma + 1/(2\sqrt{2})).$$

The long case has its own-zero star and own-three triangle. In the short case, own zero has the affine F_4 kernel $(\sqrt{2}, 2\sqrt{2}, 3, 2, 1)$ on $c, n_{17}, n_{16}, n_{15}, n_{14}$. Own four has $\sqrt{2}c + n_{14} + 2n_{15} + n_{16}$, own six has $\sqrt{3}c + 2n_{15} + n_{14}$, and parallel has $c + n_{15}$. For long own zero use $c + n_{15} + 2n_{16} + \sqrt{2}n_{17}$. Direct products give zero selected products, positive cap time and nonpositive products with every remaining original wall. Every other own at least five is strong. Positive foreign $Q \geq 1/4$ makes the short own-three case strong too. All four left/right endpoint bridges are retained, with residual $1/30$; their minimum ratio, including own zero, is

$$\sqrt{30}(1/30 + 1/(2\sqrt{2})) > 100/61.$$

The full bridges are foreign-sealed. The sole additional weak cap is

$$c = Ap - (\Omega_{15} + \Omega_{17})/2, \qquad Q = 0.$$

Release old sixteen and keep this same actual cap. The retained path old one through fifteen, then c, then the actual short seventeen is finite A_{17}, all bonds three. Positive finite projection is future and satisfies every original wall. Omitted sixteen is strict, because it has negative product with the positive old-fifteen and short-seventeen terms. This is the actual other endpoint of old edge sixteen. Its exact data give

$$C_{16} = \frac{16 + 15\sqrt{2}}{\sqrt{18}}, \qquad S_{16} \geq 4\sqrt{\frac{C_{16} - 1}{C_{16} + 1}} = G := \frac{8 + 12\sqrt{2}}{7}, \quad 3.56722 < G < 3.5673 < f_{16}.$$

In particular this floor is not called strong sixteen.

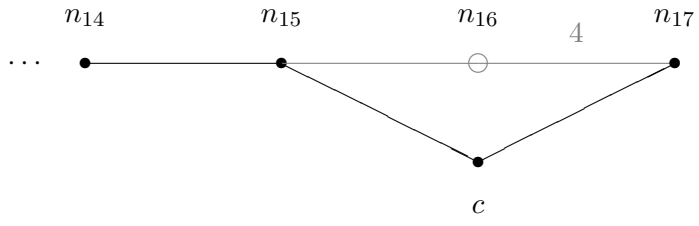


Figure 8: The exceptional cap and the simple-bond path through n_{15}, c, n_{17}. The grey edges belong to the original diagram.

If five is weak, the actual first-six pressure gives $x_1 > a_0 = .5965, S_6 > 2.01815$. The reference through one, six and the actual floor G at sixteen has decreasing slopes and energy

$$> .7600299945 + (3.5672 - 2.01815)^2/10 = .99998558475.$$

Its P_6 coefficient exceeds $.28433 - .154915 > 1/60$. Lemma B.10.4 therefore gives a contradiction. This uses no head reduction or later strong-five conclusion.

If five is strong, take the already available strong thirteen/fourteen reference and append the straight chord to G. Its slopes $< .132, < .129$ fit the individual terminal coefficients. The new energy is respectively

$$> .88864 + .39299^2/3 > .9401, \qquad > .90715 + .25688^2/2 > .9401.$$

For the energy bounds use $f_{13} < 3.17423, f_{14} < 3.31034$. For the slope test instead use $f_{13} > 3.1741, f_{14} > 3.3102$ and $G < 3.5673$, all given by positive squaring. The first five block and its full P_5 coefficient are unchanged. Deleting the long head leaves actual B_{16}, whose finite one-node extension is only isolation or simple-three at long two. Thus Lemma B.10.9 applies to this stronger reference, independently of any strong-fifteen assertion. This contradicts the original norm and excludes the exceptional cap in both initial branches.

B.10.5.4 Unconditional strong five from two retained actual helpers

Coverage through fifteen now consists only of nonadjacent pure helpers. If five is weak with finite own, the actual first-six table gives $x_1 > .65$. With strong six and an actual strong fourteen/fifteen, positive interpolation gives

$$.65^2 + (2.01815 - .65)^2/5 + 1.2921^2/8 = 1.00555718575 > 1.$$

Thus a weak-five own would be dotted, $1 < a < 6/5$. If fifteen is strong, the first-six/strong-fifteen interpolation has energy $> .985$; its P_6 capacity exceeds $1/60$. The capped initial graft adds $> .015$, again impossible. It remains that both five and fifteen are weak, so fourteen is strong.

Let w interpolate through $(1, a_0), (6, f_6)$ and every actual strong knot through fourteen. Put $b_1 = (f_6 - a_0)/5$; then $.28433 < b_1 < .284332$. The rational increment floors give

$$\|w\|^2 > .9703299945, \quad b_{P_1} > .312168, \quad b_{P_6} > .08806, \quad b_{P_{14}} > .1361.$$

For example $b_{P_6} > b_1 - (\sqrt{7} - \sqrt{6})$, and there are at most four nonadjacent omissions among seven through thirteen.

Retain the complete actual dotted-five foot

$$U_5 = \sqrt{2}aP_5 - a^{-1}n_5 + \Omega_1/2, \quad T_5 = \sqrt{10a^2 + 2a - 1/2} - \sqrt{1 - a^{-2}}.$$

Its projection onto the initial reference span is $mP_6 + lP_1$, where

$$m = 4\sqrt{2}a/5 < 1.35766, \qquad l = \sqrt{2}a/5 + 1/\sqrt{2} < 1.04652.$$

The residual Y_5 is orthogonal to w, P_1, P_6, and

$$Y_5^2 = 8a^2/5 - 2 + a^{-2} < 1, \qquad T_5 - \langle w, U_5\rangle > .1203.$$

Indeed $\langle w, P_5\rangle = a_0 + 4b_1 < 1.733828$. The deficit decreases with a: the time derivative is $< 13/4$, the radical derivative is > 1, and $\sqrt{2}\langle w, P_5\rangle > 2.4519$. At $a = 6/5$, use $A > 4.03732, \sqrt{11}/6 < .552771, \sqrt{2} < 1.414214$. Weight $\mu_5 = .05$ consequently gives gain $> .00953$, with debits $< .067883$ at P_6 and $< .052326$ at P_1.

For the complete weak-fifteen packet, project its initial H_k onto P_1, P_6. The projection of U_{15} is

$$\alpha P_{14} + l_k P_1 + m_k P_6,$$

k	l_k	m_k
3	$\sqrt{2}/10$	$\sqrt{2}/15$
2	$\sqrt{2}/5$	$\sqrt{2}/20$
1	$1/\sqrt{2}$	0.

The residual Y_{15} is orthogonal to all original reference anchors. Here $\langle w, H_3\rangle < .275, \langle w, H_2\rangle < .312, \langle w, H_1\rangle < .422$. The proof of Lemma B.10.10, with these initial demands in place of $b/\sqrt{2}$, gives

own fifteen	$T_{15} - \langle w, U_{15}\rangle$ lower	Y_{15}^2 upper	μ_{15}	gain lower
5	.37658	.824	.09	.06111
7, 8	.2313	1.105	.09	.0326835
finite ≥ 9	.1336	1	.09	.015948
dotted	.1647	1.28	.08	.01816.

All points remain actual; the table uses the pure-helper fifteen classification established before this bootstrap.

The full mixed crosses are

$$\langle Y_5, Y_{15}\rangle = \begin{cases} 2a/15 & k = 3, \\ a/10 & k = 2, \\ 0 & k = 1. \end{cases}$$

Their adverse square terms are at most $.00144, .00108, 0$. The first two rows already exceed $1.0395, 1.0114$, without any I-energy. Their combined P_6 debit is $< .067883 + .008486 < .08806$.

For the last two rows, both residuals are orthogonal to $e_1 - P_6/6$. The already paid vector still has first coordinate a_0 and P_6 demand f_6, so apply Lemma B.10.4. The total P_6 debit is $< .067883 + 1/60 < .08455 < .08806$; the P_{14} debit is $< .1273$, and P_1 debits are $< .117 < .312168$. The full original-reference/graft cross is retained in that lemma. One square, with I once, therefore exceeds

$$.9703299945 + .00953 + .015948 + .015 = 1.0108079945,$$

or

$$.9703299945 + .00953 + .01816 + .015 = 1.0130199945.$$

Every weak-five case is impossible. Five is now unconditionally strong.

B.10.5.5 The final composition

Take a strong fourteen/fifteen reference. It has $q > .90715$, first-five mean b, and individual P_5 coefficient $> .146$. The actual B_{16} long-head whitelist, and Lemma B.10.6, exclude all finite or nonisolated heads. The remaining actual head is isolated-within-B and dotted, retaining every finite contact with old I.

If fifteen is strong, its reference has $q > .9240775$, so Lemma B.10.9 excludes every such head. If fifteen is weak, fourteen is strong and its actual cap is a pure helper, so Lemma B.10.11 applies. Its small-parameter branch uses only the standalone actual head point; its other branches retain the full initial helper/head cross and the single I-height. They give the stated strict norm contradictions. This excludes the entire $B_{17}+$ actual-I row.

All local points were positive combinations or positive finite projections of original walls, with every selected sign and future orientation checked before use. The original farthest vector is never replaced. No arbitrary face is declared Coxeter, no disconnected root scale or arithmeticity is

imposed, no actual mate floor is promoted to strong sixteen, and no earlier elimination energy is added again.

B.10.6 D_{17} with an actual short rank-one factor

Keep the actual roots and signed fork coordinates of Appendix B.10.4.1. The original vector is $z = E(p - x - ye)$, with $\|x\|^2 + y^2 < 1$. Its actual isolated rank-one opposite is

$$c_I = \sqrt{h}\,p - \sqrt{1+h}\,e, \quad 0 < h < 1/2, \qquad s^2 = \frac{\sqrt{1+h}-1}{\sqrt{1+h}+1}.$$

We have $y \ge s$, $x_1 \ge t$, where

$$t = \frac{\sqrt{1+14s^2+s^4}-4s}{1-s^2}, \qquad t + 2.5s > .923$$

by Lemma B.10.1. The same y coordinate is counted once; the ordinary debit $y^2 > .1$ is unavailable.

Deleting the head retains actual D_{16}. Its complete extension list is isolation or simple three at long leaf two. The local nulls, points and native-gap arguments from the preceding subsection retain old I and both forks. Thus Lemmas B.10.6 and B.10.9 apply on any legally funded reference constant first five, with mean $< .361$ and individual P_5 coefficient $> .146$. Their head-plus-I gain is $> .07665$, supported in first five plus I, without assuming strong five or any B-terminal conclusion. Finite isolated heads have $h > 1/4$, $y^2 > .0557$; pure owns five/six are impossible.

Fresh tails before strong five. The rank-valid $M = 11$ inventory covers five through twelve, with only nonadjacent pure far helpers weak. There is a strong index in each of $9/10, 10/11, 11/12$. The geometric D_4/D_3 lists from Appendix B.10.4.1 precede its ordinary energy debit. Their only extra weak rows are pure own-five/fork-three at thirteen and pure own-five or finite-own-at-least-seven/end-three at fourteen. Both forks and every foreign contact were retained; dotted end-three owns were directly strong.

The actual left-H_4 finite points are flat through $i-3$, have norm $< i$ and demands $> 3.48, 3.61$ at $i = 13, 14$. If five is strong, their preceding one-square totals before an ordinary debit are $> .9757$; adding $> .07665$ excludes them. If five is weak, the actual pressures $x_1 > .5965, S_6 > 2.01815$ give the stronger bare totals $> 1.039, 1.056$ on the positive P_1, P_6, P_K references. No head gain is needed in that branch.

For every remaining $.9 < g < 1$ end-three own, average the two actual A_2 cores, retaining their full cross $8g^2/9$. The demand is > 3.388, norm < 14, flat through twelve. In the strong-five branch pay $.16/.15$ at actual strong ten/eleven, below the individual terminal prices. With $q_{10} > .823735, q_{11} > .8472225$, the bare totals are

$$.823735 + .32(3.388 - 2.732) - .0256(4) = .931255,$$
$$.8472225 + .30(3.388 - 2.886) - .0225(3) = .9303225.$$

Adding the local head gain closes both. In the weak-five branch the reference norms are $> .8870, .9104$; the capped initial-I graft, Lemma B.10.4, adds $> .015$. Its support first six plus I is disjoint from the tail residual. Its P_6 price is available, and the totals exceed $1.00952, 1.0085$. This removes the tails before any later floor. The pure-helper two-cap calculation extends nonadjacency through fourteen: the actual inverse is $2\min(i,j)$ and both retained fork blocks remain positive. Its unchanged loss is $< 7/16 + 1/(4\sqrt{10}) < .61$. Thus thirteen or fourteen is strong.

Retained weak-five and weak-fourteen feet. Suppose five weak. For a justified $x_1 > a_0$, interpolate through one, six and actual strong knots up to $K = 13, 14$, omitting weak ones. Put $b_1 = (f_6 - a_0)/5$, $C(a_0) = a_0^2 + 5b_1^2$. The increment sums $> .19304, .21155$, minus at most three/four losses each $< 1/3200$, give $q_{13} > C + .1921025, q_{14} > C + .2103$. The needed capacities are

a_0	$C >$	$q_{13} >$	P_1 coefficient $>$	P_6 coefficient $>$
.5965	.7600299945	.9521324945	.312168	.08806
.643	.7916565045	.9837590045	.367968	.07876
.65	.7968668845	.9889693845	.376368	.07736

Terminal thirteen/fourteen coefficients are $> .14144, .1361$. We never assign a terminal coefficient to an internal knot.

If fourteen is weak, its pure far-helper time is $A_{14}^2 = 28g^2 + 2g - 1/2$. The finite/dotted feet use actual long roots thirteen/fourteen/fifteen and the initial local roots. After projection on P_1, P_6, P_{13}, their initial pieces H_3, H_2, H_1 have respective P_1, P_6 coefficients

$$(\sqrt{2}/10, \sqrt{2}/15), \quad (\sqrt{2}/5, \sqrt{2}/20), \quad (1/\sqrt{2}, 0).$$

The residual norm bounds for own five, seven/eight, finite at least nine, and dotted are $.824, 1.105$, $1, 1.28$. Only the last two residuals are confined to fourteen/fifteen; initial crosses in the other rows must be retained. The actual time/radical bounds are

own	$A_{14} >$	radical $<$
5	4.4095	.111
7	4.9017	.249
8	5.0242	.322

For finite own at least nine the deficit increases with $g \geq .939$. At that endpoint $A > 5.1055$, the other time radical > 5.116 and subtracted radical $< .344$ give demand > 4.7838. Dotted weakness implies $g < 10/9$: the squared ratio there is $83521/34020 > (39/25)^2$, above the exact target, which is $< 39/25$. The deficit decreases on this whole interval, since $A' < 5.4$ and the radical derivative exceeds one. At $10/9$, $A^2 = 5879/162 > 6.023^2$, radical $< .436$, so demand > 5.587. We use $\sqrt{26} - .61 < 4.4891$ throughout.

Every finite-own weak-five foot gives $x_1 > .65$. Fourteen strong then gives $q_{14}(.65) > 1$. Otherwise the weak-fourteen gains are $> .06160, .04163, .03203$ for owns five/seven/eight. For finite at least nine the deficit is $> 4.7838 - .939(4.4891) - .460 = .1085351$, and weight .09 gives gain $> .011436318$, total > 1.0004057. The dotted deficit $> .1391$ gives $> .01467$. Their prices fit the table. Hence weak five is dotted, $1 < a < 6/5$. Retain its full point

$$U_5 = \sqrt{2}aP_5 - a^{-1}n_5 + \Omega_1/2, \quad T_5 = \sqrt{10a^2 + 2a - 1/2} - \sqrt{1 - a^{-2}}.$$

Its projection is $mP_6 + lP_1$, with $m = 4\sqrt{2}a/5 < 1.35766, l = \sqrt{2}a/5 + 1/\sqrt{2} < 1.04652$. The residual has

$$Y_5^2 = 8a^2/5 - 2 + a^{-2} < 1, \quad Y_5 \cdot P_1 = Y_5 \cdot P_6 = 0, \quad (Y_5)_2 < .339412.$$

On the .5965 reference, its deficit $> .1203$, at weight .05, gives gain $> .00953$ and P_1/P_6 debits $< .052326/.067883$.

A nonisolated head forces $x_1 > .643$. If fourteen strong, $q_{14}(.643) > 1$. If weak, on $q_{13}(.643)$ the retained five foot gives $> .00466$, while the fourteen finite-at-least-nine and dotted gains are $> .0126789498, .01557$, with zero cross. For owns five/seven/eight the full cross losses $.00144, .00108$

leave gains $> .06034, .04091, .03131$. Combined P_6 price is $< .076369 < .07876$, so all contradict one. A finite isolated head instead has $y^2 > .0557$, already too much beside $q_{13}(.5965)$. Only isolated dotted heads remain under weak five.

On the .5965 reference the full weak-fourteen accounting is

own	deficit lower	weight	net gain $>$
5	.387329	.09	.06160482
7	.2960209	.09	.043253262
8	.24272051	.09	.0336591918
≥ 9 finite	.1465351	.099	.0192
dotted	.1771	.09	.02151

Here $Y_5 \cdot Y_{14} = 2a/15, a/10, 0$ for five, seven/eight, and the last two rows. The first two totals already exceed one. For own eight add an initial-I graft capped at .08. Its unclipped gain is $> .015$ by weighted Cauchy. If clipped, $t > .5965 + 1/15$; since $t(.11) < .65$, we have $s < .11$ and

$$s^2 + .16(t - .5965) - (5/6)(.08)^2 > (s - .2)^2 + .00690666 > .015.$$

Its residual is orthogonal to both feet because P_1, P_6 are retained. The full P_6 price $< .087582 < .08806$ includes the extra .013334. Thus weak fourteen leaves only the last two rows, each with gain $> .0192$, P_1 price $< .070004$, P_{13} price $< .141422$, no P_6 price and a residual supported only on fourteen/fifteen.

Dotted heads on the retained weak-five reference. For a pure dotted head put $u = g^{-2}$, $h_0 = u/[4(2-u)]$, $s_0 = s(h_0)$. The actual native gap gives $h \geq h_0$. The joint point of Lemma B.10.7 supplies

$$ax_1 + v_0 y \geq F = 1 - d, \quad a^2 = 1 - u/2, \quad v_0^2 = \frac{u}{8 - 3u}, \quad d^2 = u/2 - v_0^2.$$

Besides its stated all-h domain $u \geq .2$, it is valid for $.1 \leq u \leq .2, h \leq .2$. Indeed $u(8-3u) \geq .77 > 2/3$ and $h/(1+h) \leq 1/6$, so $D_0 = u/2 - v_0\sqrt{h/(1+h)} > 0$. The actual construction is

$$p + \rho c + \sigma c_I = D_0^{-1}(p - ae_1 - v_0 e), \quad \rho = \frac{a}{D_0\sqrt{2}g} > 0, \quad \sigma = \frac{v_0}{D_0\sqrt{1+h}} > 0.$$

The cap product is zero, the c_I product has sign $v_0 - \sqrt{h/(1+h)} \leq 0$, old n_1, e products are negative and every unknown-wall product is negative. Both actual D forks therefore remain satisfied, for every mutual head/I label.

Put $X = (F - v_0 s_0)/a$. Simplification gives

$$X = \sqrt{\frac{1 - \sqrt{3u/8}}{1 + \sqrt{3u/8}}}, \quad (1 - X^2)(1 + s_0^2) = 4\sqrt{3} s_0 X, \qquad X + 2s_0 > .91.$$

For completeness, $0 < s_0 < \sqrt{5} - 2 < .237$. At $X_0 = .91 - 2s_0 > 0$, the left-minus-right expression is greater than $.1719 - 2.66467 s_0 + 10.0283 s_0^2 + 3.64 s_0^3 - 4 s_0^4$. Writing $w = s_0 - .13$, this is exactly $.00182581 + .092084 w + 11.0423 w^2 + 1.56 w^3 - 4w^4$, bounded below by $.00182581 - .092084|w| + 10.7719 w^2 > 0$. The discriminant is negative. The original expression decreases strictly with positive $X < 1$, proving the full barrier without sampling.

Let W_0 be the reference plus $.05 Y_5$ and any remaining fourteen foot. Its first coordinate is still $a_0 = .5965$, and the feet are orthogonal to P_1, P_6. Set $\delta = \min((X - a_0)_+, .1)$, $\kappa = 6\delta/5$. The joint paid-reference lemma uses nonnegative multipliers $\eta = \kappa/a$, $\tau = s_0 - \eta v_0$ and gives gain

$$s_0^2 + 2\kappa(X - a_0) - (5/6)\kappa^2.$$

Here $s_0/v_0 > 1/2$, $a \geq 1/\sqrt{2}$, $\kappa \leq .12$ ensure $\tau \geq 0$. The P_6 price $\leq .02$ makes the full cost $< .087883 < .08806$; the other prices still fit. If unclipped, the barrier and weighted Cauchy give gain $> (.91 - .5965)^2/(4 + 5/6) > .02033425$. If clipped, $X > .6965 > X(1/3)$ gives $s_0 < .111$; the gain exceeds $(s_0 - .24)^2 + .00564 > .022281$. For $u \leq .1$, the actual head edge gives $x_1 > .7745$, and an initial graft without an I charge gains $> .03072$. For $.1 \leq u \leq .2, h > .2$, I alone costs $> .0455$. Thus all pure dotted parameters are covered. The least totals are

$$.9703299945 + .00953 + .02033425 > 1.0001942,$$
$$.9521324945 + .00953 + .0192 + .02033425 > 1.0011967.$$

For a positive foreign contact put $Q = \beta^2 \geq 1/4$, $v = (1 - Q)/g^2 \in (0, .75)$. Actual acuteness gives $h \geq Qv/[2(1-Q) - v] \geq v/(6 - 4v)$. The positive core $c + \beta e + ((1-Q)/g)n_1$ clears the whole old I, with

$$x \cdot Z \geq H(v) = \sqrt{1 - v/2} - \sqrt{v(1-v)/2}, \quad Z = (1 - v/2)e_1 + (v/2)e_2.$$

Its projection is $(1 - 3v/5)P_1 + (v/10)P_6$, residual norm $v^2/5$. The full retained initial cross is $W_0 \cdot Z \leq .5965 - .14759v$: use $b_1 < .284332, .05(Y_5)_2 < .016971$. The remaining fourteen foot has zero cross. Pay $\nu = .18$ for $v \leq .2$, $.21$ for $.2 \leq v \leq .4$, and nothing thereafter. The gain is $2\nu[H - .5965 + .14759v] - \nu^2 v^2/5$. Adding I once gives

v interval	$H >$	I energy $>$	total gain $>$
$(0, .2]$	.6658	0	.0246888
$[.2, .25]$	.6292	0	.02557275
$[.25, .3]$	.5979	.0121	.02739115
$[.3, .35]$	.5709	.01515	.02191389
$[.35, .375]$	.5590	.0182	.0229054175
$[.375, .4]$	.5480	.0200	.021464225
$[.4, .75)$	unused	.0217	.0217

Positive squares at right endpoints and monotonicity on $(0, .5)$ prove the H bounds. The left endpoints in the lower bound for h give the I bounds. The largest full P_1 price is $< .052326 + .070004 + .1848 = .30713 < .312168$; the P_6 price is $< .067883 + .0084 < .08806$. The terminal price is unchanged. The uniform gain $> .0214$ contradicts one in both fourteen branches. Every weak-five head is excluded. Five is therefore strong.

The pure-helper coverage through fourteen is now available. If centre fifteen were strong, its reference would have $q_{15} > .9240775$, first-five mean $< .361$ and P_5 coefficient $> .146$. The local head-plus-I gain would give > 1.0007275. Thus centre fifteen is weak.

The complete centre-fifteen list. Deleting fifteen retains actual $A_{14} + 2A_1 + I$. Its cap has form

$$c = Ap - g\Omega_{15} - \eta\Omega_j - \delta\Omega_{16} - \epsilon\Omega_{17} - \beta e.$$

There is at most one left contact. Put $Q = \beta^2$; the exact data are

$$q_L = 2\eta^2 j(15-j)/15, \quad \Delta = 1 - q_L - \delta^2 - \epsilon^2 - Q > 0, \quad \sigma = g + \eta j/15 + (\delta + \epsilon)/2, \quad C_{15} = \sqrt{30}\sigma/\sqrt{\Delta}.$$

The sufficient strong ratio is $< T(11/2) < 1.554$. With no left contact, finite classification permits isolation, one arbitrary finite fork label, or pairs $(3,3), (3,4), (3,5)$. Without forks the left possibilities are endpoint three/four at one/fourteen or next-to-end three at two/thirteen. With forks only left

endpoint-three with one fork-three/four or both forks-three is finite. Two left contacts make a cycle; other branches or internal multiple bonds are absent from the full finite classification at this rank. Positive I contacts in a single-fork row are precisely $(m, I) = (3,3), (3,4), (3,5), (4,3), (5,3)$; with two forks only $(3,3,I3)$ is possible. Left endpoint-three may carry $I3/I4$, and endpoint-three/fork-three may carry $I3$; other left bridges are sealed. We retain the whole foreign alphabet.

For isolation, own zero has negative time; three/four/six/parallel have the actual star/path/null principals. On $0 < u < 1$, the cleared core has pressure $\sqrt{30-u} - \sqrt{u(1-u)}$, loss $< .6$. At $u = 1$ it is a forbidden feasible future null. On $u > 1$, the actual finite projection gives $\sqrt{29} - \sqrt{u-1}$. Foreign cost forces $u < 75/64$; pure native at least seven has $u < 100/81$, with losses $< .515, .6$. Pure own five uses the actual left-H_4 projection, whose demand is $> 5.3/1.04 > \sqrt{30} - .61$. Every isolated centre is strong.

For left-only own zero, far-three needs $1/2 < Q < 8/15$, containing no native foreign cost. Far-next-three and far-four have zero time at $Q = 0$; their admissible nonzero costs are too large. The near three/next-three/four rows have actual affine D_5, D_6, B_4 principals. With a simple-three left contact, own three has its actual all-three cycle. At the far endpoint-three, own four/six and parallel have their actual star/path/null dispositions. Every other positive near/next/end-four row is directly strong. In particular, an end-four contact at either end has ratio at least $\sqrt{15}/2 + 1/\sqrt{30} > 2.11$, even at own three. A positive-foreign far-three has ratio $> (.809 + 1/30)/\sqrt{17/60} > 1.57$. For left plus fork-three, $\Delta \le 17/60$: own zero/three has a cycle/triangle; larger owns are strong. With fork-four or both forks-three, $\Delta \le 1/30$ and strength holds even at own zero. Only the pure far helper can remain among left rows.

For own zero with one fork, fork labels three/four/six have the actual affine E_8, B_4 and $6, 3$ principals. For finite fork cosine $b \ge \cos(\pi/7) > .9$, whiteness forces $Q = 0$. Set $u = b^{-2} \in (1, 100/81)$. The actual positive future core $c + b^{-1}n_f + 2(b^{-1} - b)n_{15}$, divided by $b/\sqrt{2}$, has signed displacement $(1^{14}, 3 - 2u, u - 1, \pm(1-u))$, with

$$G = \sqrt{17 - 2u} - \sqrt{-8 + 14u - 6u^2}, \quad R = 25 - 16u + 6u^2.$$

Both fork signs are retained. Complete interval bounds are

u	$G >$	$R \le$
$[1, 1.1]$	3.472	15
$[1.1, 1.15]$	3.4277	14.66
$[1.15, 1.2]$	3.41265	14.535
$[1.2, 100/81]$	3.4119	14.44

Use the maximum second radical $1/\sqrt{6}$, positive squares and decreasing R. Pay .14/.13 at actual terminal thirteen/ fourteen, with residual norm $\le R - K$. Adding $> .07665$ makes all eight totals exceed 1.00135.

For fork five put $b = \phi/2$, $\alpha = b/\sqrt{2}$, $\delta_5 = 1 - 3b^2/2$. The actual finite H_4 consists of cap, contacted fork, centre and other fork. Its positive correction is $b(3n_f/2 + n_{15} + n_{\text{other}}/2)$. The corrected vector is spacelike; the point is the normalized positive finite projection of p. Its pure displacement is αP_{14}. For $Q = 0$, demand > 3.49, norm14, and the local head gain close both references. The genuinely permitted $Q = 1/4$ retains I in this same point:

$$x \cdot P_{14} + dy \ge G > 3.61, \qquad d = 1/(2\alpha) < .875, \quad d^2 < .765, \quad G = \frac{\sqrt{5 + 6\phi}}{1 + \sqrt{(13 - 8\phi)/(18 - 2\phi)}}.$$

At $K = 13, 14$, weights $.14, .13$ give bare mixed gains $> .0874216, .0649831$. Every dotted/foreign head has pure gain $> .04$, and a nonisolated head has $> .0994$, already enough. A pure finite head has

pure gain $> .0186$ and $s > .236$. Append only the shifted I gain $(s-\mu d)^2$, exceeding $.1135^2, .12225^2$. This replaces the existing coefficient μd by s, using a nonnegative additional multiplier; it does not add a second y^2. The least totals exceed $1.00754385, 1.0056781625$. Head and pure tail residuals are orthogonal, and the displayed mixed norm includes the I coordinate.

For positive own with one fork, own-three/fork-three is an affine triangle. Own-three/fork-at-least-six, or fork four/five with foreign contact, is directly strong. With fork four/five and $Q = 0$, the same cap gives the actual opposite of the contacted fork. The positive finite D_{17} replacement is future feasible against every original wall; the omitted fork is strict. Its actual cosh is $15/2 + (17/2)b > 13$. Thus the signed direction $P_{16} \pm e_{17}$, of norm17, has demand $> \sqrt{102/7} > 3.817$. Either funded reference and the local head gain close it. Own four/six/parallel have the actual star/path/ null principals. For larger owns, every fork label at least four is directly strong.

Only fork-three remains. Own five with $Q \geq 1/2$ is strong. At $Q = 0, 1/4$, use the actual left-H_4 point, or respectively the cleared future core

$$c + g(3n_{15}/2 + n_{14} + n_{13}/2) + e/2, \qquad g = \phi/2.$$

It gives a pure displacement flat through twelve, norm < 15, demand > 3.52. Weights $.15/.14$ at actual strong eleven/ twelve, plus $> .07665$, give totals $> 1.0240725, 1.0229116$. For every larger finite/dotted own, foreign contact is directly strong, as is $g \geq 6/5$. On the full remaining $Q = 0, .9 < g < 6/5$ interval use the actual core

$$\xi = c + g^{-1}n_{15} + (2g)^{-1}n_{14}, \quad k = 1 - 3/(4g^2) > 0, \quad \alpha = \sqrt{2}(g + 1/4).$$

Its cap and fourteen products vanish; its fifteen and both fork products are negative. Its square is $-k$, time positive, and every unknown-wall product is nonpositive by positive actual-wall coefficients. With $A^2 = 15\alpha^2 - 3/4$, $A/\alpha > 3.836$, $k/\alpha^2 \leq 575/5046 < .34^2$, the demand is $G > 3.496 > 3.49$, full norm < 15, flat through thirteen. The quotient derivative reduces to $2g^3 - 3g - 3/8 < 0$ throughout. At thirteen pay .14, residual norm < 2. At fourteen the fourteenth coordinate is $1 - a$, $a = [4g(g + 1/4)]^{-1} < 1/4$. Its projection is $(1 - a/14)P_{14}$, reference pairing $\leq f_{14}$, residual norm $< 1 + 2a < 1.5$. Weight .13 is affordable at that actual terminal; the least total is > 1.0051616. No internal terminal coefficient is assumed.

For two forks the finite pairs are $(3,3), (3,4), (3,5)$. Own three has a triangle; owns at least four are directly strong. Own-zero pair-three-three has the actual affine square, including its foreign case. The other pairs are foreign-sealed. Pair-three-four gives an actual reciprocal finite B_{17} endpoint, with cosh $15/2 + (17/4)\sqrt{2} > 13$, hence the same signed demand above. Pair-three-five uses the actual finite H_3 correction $n_{f_1}/2 + (\phi/2)n_{f_2}$. Its pure displacement is αP_{15}, $\alpha = (1+\phi)/(2\sqrt{2})$, and the positive finite residual satisfies $q = \delta_H/\alpha^2 = 2\phi^{-6} < 1/8$. Its pressure $\sqrt{15 - q}/[1 + \sqrt{q/15}] > 3.53$, norm15, closes either funded reference with the local head gain. This exhausts the list.

The final pure helper and the actual-wall interface. The weak centre must therefore be the pure far-three helper at fifteen, $A^2 = 30g^2 + 2g - 1/2$. Nonadjacency with fourteen follows from the same actual prefix Schur calculation, retaining both positive fork blocks. Hence fourteen is strong, $q_{14} > .90715$. Nonisolated and finite isolated heads already contradict this reference by their local gains. For dotted heads use Lemma B.10.11, checking its local geometric interface as follows.

Its finite/dotted helper vectors are $g\sqrt{2}P_{15} - gn_{15} + H_k$ and $g\sqrt{2}P_{15} - g^{-1}n_{15} + H_1$. They use only actual long n_{15}, n_1, n_2; P_{16} in their coordinates is a spatial sum, not a nonexistent long D dual. Both actual D forks have nonpositive product with every positive correction. Their Gram data, fresh times, initial crosses and individual P_5/P_{14} prices are literally those of the local kernel, without requiring a B short normal or a sign for x_{17}.

For pure dotted head $u \le .2$, the kernel bypasses the tail payment and gives gain $> .11088$. For $.2 \le u < 1$, its net tail gain is $> .0421$, including the whole initial residual, and its head-plus-I gain is $> .0515$. Every positive foreign head contact instead gives $> .0521$, with the adverse initial cross and the one I cost included. The least original total is

$$.90715 + .0421 + .0515 = 1.00075 > 1.$$

The P_5 debits $< .09873, .09273$ fit $> .146$, and P_{14} debit $< .1273$ fits $> .1361$. Every actual centre, head, finite foreign label and own range is covered on the same original vector. This excludes $D_{17}+I$ in dimension eighteen, without arithmeticity or an ordinary rank-one debit.

B.11 The independent higher-dimensional tail

The exact-eighteen argument is separate from the exclusion of higher ranks. We start this second argument at the published ceiling twenty-nine. Every remaining dimension is handled by its actual finite forest; no Coxeter structure is inherited by an arbitrary face. The common geometry, ordinary and typed costs, actual A-chain pressures and generic prefix are the lemmas already proved above. Here we supply the additional large-component caps and then join the finite forest cases.

The argument separates at a useful point. An A cap is proved by a central curve and a plateau transfer. A B/D cap is proved by a long prefix and the genuinely different short tails. These two geometric arguments apply to whole components with arbitrary finite spectators. They reunite only in the finite rank and energy accounting. The statements below do not replace an actual component by a convenient abstract completion: every added cap is already an actual wall.

B.11.1 Common large-component caps

Proposition B.11.1 (Large-component caps)**.** *At a farthest vertex, with arbitrary finite spectators, the following components are impossible:*

$$A_r \ (21 \le r \le 29), \qquad B_r \ (23 \le r \le 29), \qquad D_r \ (24 \le r \le 29).$$

We give their finite continuation and energy proofs below. All references to old directions concern the chosen farthest vertex, and all added walls are already actual.

B.11.2 Clearing spectators and the finite continuation list

Put $c_0 = 61/100$. For a long index, write $D_i = \omega_i^2$, $q_i = \sqrt{D_i}$ and $h_i = x \cdot \omega_i$, and call it weak if $h_i \le q_i - c_0$. An actual edge with

$$\frac{\cosh L_i}{q_i} > T(q_i), \qquad T(q) = \frac{1 + (1 - c_0/q)^2}{q[1 - (1 - c_0/q)^2]} = \frac{1}{c_0} - \frac{q - c_0}{q(2q - c_0)},$$

is strong. We use $T(q) < 100/61$ and, on $5/2 \le q \le 31/8$, $T(q) < 761/500$. For the latter, clearing denominators gives the positive concave quadratic $-715800q^2 + 3268319q - 1860500$; check its two endpoints.

First consider an isolated own contact γ with full foreign cost Q. Write $v = \gamma\omega_i + v_S$, $u = (1 - Q)/\gamma^2 > 0$, so $A^2 = \gamma^2(D_i - u)$. If $u < 1$, the positive actual combination

$$\xi = c + v_S + \frac{1 - Q}{\gamma} n_i$$

has products zero with c, S, product $\gamma(u-1) < 0$ with n_i, and square $-\gamma^2 u(1-u)$. It is future and feasible, giving the pure component pressure

$$h_i > \sqrt{D_i - u} - \sqrt{u(1-u)}. \tag{B.11.1}$$

At $u = 1$ the same nonzero feasible vector is null, impossible. For $u > 1$, the actual positive set $S + n_i + c$ has residual $\gamma^2(u-1)$; the finite-projection formula in (9.4) and $x \cdot n_i > 0$ give

$$h_i > \sqrt{D_i - u} - \sqrt{u-1}. \tag{B.11.2}$$

In A the sharper form is $h_i > \sqrt{D_i - 1} - \sqrt{u-1}$: with $B = \sqrt{D_i - 1}$, $t = \sqrt{u-1}$ the exact finite pressure is $\sqrt{B^2 - t^2}/(1 + t/B) > B - t$. No normalized projected mirror is asserted native.

The own labels $3, 4, 6$ and parallel contact are excluded by the local affine fork/star, $(6, 3)$ path, and null sum. Thus $\gamma \geq \varphi/2$. If $Q > 0$, $u \leq 3/\varphi^2 = 6 - 3\varphi < 7/6$. For A, $D_i \geq 85/11$, $q_i > 11/4$, and $\sqrt{D_i - 1} > 5/2$. Formula (B.11.2) has loss $< 4/21 + 5/12 = 17/28 < c_0$, since $u - 1 \leq 5 - 3\varphi < 73/500 < (5/12)^2$. Formula (B.11.1) has loss at most $2/21 + \sqrt{457}/42 < 127/210 < c_0$. For B/D, $D_i \geq 10$; the respective losses are $< 1/5 + 2/5$ and $u/6 + \sqrt{u(1-u)} \leq (1 + \sqrt{37})/12 < 3/5$. Thus every foreign isolated cap is strong.

At $Q = 0$ the isolated estimates are also strong, uniformly for $D_i \geq 85/11$. For own 5, the actual old H_4 segment has $\delta_5 = (7 - 3\sqrt{5})/16 < 1/36$ and, by concavity of the old heights, pressure at most γh_i after its correction. Its loss is $< 25/76 + 5/24 < c_0$; here $q_i + \sqrt{D_i - 25/16} > 19/4$ and $\gamma > 4/5$. For own 7, the adjacent negative A_2 core gives loss $< 400/1701 + 9/25 < c_0$, using $.9 < \gamma < 10/11$. For finite own $m \geq 8$, the old singleton projection gives $h_i > \sqrt{D_i - 1} - \tan(\pi/m)$ and loss $< 4/21 + 5/12 < c_0$. For dotted own contact, $c + n_i/\gamma$ gives (B.11.1) with $u = \gamma^{-2}$, and the same $127/210$ bound. One left H_4 segment suffices: its displacement is $\gamma(3\omega_{i+1}/4 + \omega_{i-3}/4)$ and concavity bounds its scalar product by γh_i.

Here is the complete finite continuation table at a central release. A retained left path A_{i-1} admits endpoint 3, endpoint 4, next-to-end 3, or an admissible E_6, E_7, E_8 position 3. Their Schur residuals are

$$(i+1)/(2i), \quad 1/i, \quad 2/i, \quad 1/4, 1/7, 1/16. \tag{B.11.3}$$

For a left 3 at position j from the far end, $\sigma = \gamma + j/(2i)$; for endpoint 4, $\sigma = \gamma + j/(\sqrt{2}i)$. In each case $\cosh L_i = q_i\sigma/\sqrt{\delta - Q}$. The complete right table is

retained right part	finite attachment
$B_s,\ s \geq 4$	long-leaf 3
B_3	long-leaf 3 or short-leaf 3 (F_4)
B_2	3 at either end
$D_s,\ s \geq 5$	long-leaf 3; fork 3 only for $s = 5, 6, 7$
D_4	3 at any leaf
$D_3 = A_3$	middle 3, end $3, 4, 5$.

Two contacts to a retained component create a cycle. A completion joining both long retained pieces has connected rank at least 21, so finite classification forces the corresponding simple endpoint $A/B/D$ bridge. An additional foreign branch at its internal new node is not finite. The exceptional fork extensions in the table are retained and estimated; an E_7 piece is not declared sealed merely because it is exceptional.

Zero own contact needs a separate check. Internal 3, internal 4, right long 3, and two-piece bridges give the usual affine fork, star or cycle; a right D-fork contact gives affine E_8. A zero-own far 3@1 in B/D satisfies $A^2 = Q - 1/2 < 1/(2i) \leq 1/10$; any foreign component then has derivative

square at least $1/4 > A^2$, contradicting Proposition 9.4. With $Q = 0$ its height is negative. For A, a zero-own far 3 has height $< 1/(2m)$, a next-to-end 3 has $Q < 2/m \le 2/5$ and height $Q - 2/N < Q$, and an endpoint 4 has $Q < 1/m \le 1/5$ and negative height. A positive Q in the first two cases again forces an orthogonal closure. Interior A contacts retain affine E_8 arms $(1, 2, 5)$ or E_7 arms $(1, 3, 3)$. This proves positivity of the own contact without assuming the whole star has no spectators.

For completeness, all other near-position estimates follow directly from (B.11.3). Own 3/4/6 is affine except the near 4/simple-3 role. For own $\gamma > 101/125$, near simple 3 has $\delta \le 3/5$, $\sigma \ge \gamma + 2/5$; the next-to-end row has $\delta = 2/i$, $\sigma \ge \gamma + 1/i$; the E rows have $\delta \le 1/4$. For B/D, the first row exceeds 100/61 from $i = 8$ onward, and the next-to-end row from $i = 7$ onward. At $i = 5$ use $T(q) < 38/25$ on $[5/2, 10/3]$, and at $i = 6, 7$ use $T(q) < 8/5$ on $[3, 4]$; these are positive concave quadratics after clearing denominators. White 4 is stronger, except far 4/own 3, which retains affine $(3, 4, 3, 3)$. The allowed B near 4/own 3 has $\cosh L_i = (i - 1) + i/\sqrt{2}$ and is strong for $i \ge 5$; in D an own 4 with no right contact is already affine on the entire retained right fork. The A near-4 role is kept below.

If a nonisolated far 3 packet has $Q > 0$, the actual data in B/D are

$$\cosh L_i = \frac{2i\gamma + 1}{\sqrt{i + 1 - 2iQ}}, \quad \frac{\cosh L_i}{q_i} > R_0(i) := \frac{(809/500)i + 1}{\sqrt{i(i + 2)}}.$$

R_0 increases for $i \ge 5$, with derivative sign $(309/500)i - 1$; T also increases for $q \ge \sqrt{10}$. The three exact comparison rows are

i	q upper	$R_0(i)$ lower	$T(q)$ upper
5–8	4	1.536	1.530
9–12	5	1.563	1.550
13–29	7.62	1.577	1.577.

The lower squared margins are $.052740, .321913, .546001$; the ($T$ numerator, T denominator) pairs are $(27.4921, 18.0316)$, $(44.2721, 28.6395)$, $(107.2045, 68.003166)$, proving each upper comparison by multiplication. In A, a helper-side parameter $m \ge 5$ instead gives ratio $> (2fm + 1)/\sqrt{m(m + 2)} \ge 227/(25\sqrt{35}) > 761/500$, $f = 101/125$; the derivative sign is $(2f - 1)m - 1 > 0$, with final squared margin 342365. The foreign near-4 role has ratio squared $> 121/35$. Thus every remaining weak packet has $Q = 0$.

B.11.3 The A_{21} certificate and its plateau transfer

Put $N = r + 1 \in [22, 30]$. Then $D_i = 2i(N - i)/N$, $C_{ij} = 2i(N - j)/N$ for $i < j$, and $\|x_A\|^2 = \frac{1}{2}\sum_{i=0}^{N-1}(h_{i+1} - h_i)^2$, with $h_0 = h_N = 0$ strictly concave. For $5 \le i \le N - 5$, $85/11 \le D_i \le 15$. The preceding inventory leaves a near-4/simple-3 packet or

$$v_i = \gamma_i\omega_i + \omega_{e_i}/2, \quad e_i \in \{1, N-1\}, \quad Q = 0, \quad f := 101/125 < \gamma_i < g := 27/25, \quad A_i > 2. \tag{B.11.4}$$

Allowed own contacts are 5, finite $m \ge 7$, or dotted. The dotted upper bound follows because at $\gamma = g$, $\sigma^2/\delta \ge 4g - 2 = 58/25 > (761/500)^2$; the difference numerator is $[2m(g - 1) - 1]^2$. The near-4 edge ratio increases with m and at $m = 8$ is $2\sqrt{2}/3 + 7/12 > 761/500$, so $m \ge 8$ is strong. At $m = 7$, $N = 22, 23$ use $D_7 \le 224/23 < (25/8)^2$ and $T(q) < 3/2$ on $[3, 25/8]$, whose cleared quadratic is $-3400q^2 + 13237q - 7442 > 0$. Every weak near-4 therefore has a mate at index at most 5 for $N = 22, 23$, at most 6 otherwise, or the reversal. That mate is actual by its positive B replacement. Its data are $\delta = 1/j$ and $\cosh L_j = (1/2 + (j - 1)/(j\sqrt{2}))\sqrt{jD_j}$. By reversal take $6 \le j \le N - 4$;

then $jD_j \ge 72(N-6)/N \ge 216/5$ and the coefficient is $> 13/12$. Thus $\cosh^2 L_j > 507/10 > 49$, giving normalized height $> \sqrt{3}/2 > 43/50$.

We next exclude adjacent weak literal packets. Write $j = i+1$, $A = \gamma_i$, $B = \gamma_j$, and $R = c_i \cdot c_j$. Weighted AM–GM of their time coefficients gives

$$\begin{aligned}
\text{same helper}: \quad & R \ge -AB(1-1/N) \ \pm (A-B)/N \\
& \qquad + (N-1)/(2N) + (N+1)(A/B + B/A)/(4N), \\
\text{outward helpers}: \quad & R \ge 1/2 - AB + (A-1)(B-1)/N, \\
\text{reversed helpers}: \quad & R \ge 1/2 - AB + (A+1)(B+1)/N.
\end{aligned}$$

The first is $> 1 - g^2 > -1/2$; the others are $> 1/2 - g^2 - (1-f)(g-1)/22 > -7/10 > -1/\sqrt{2}$. Hence the native gap gives $R = 0$ for one helper and $R \in \{0, -1/2\}$ for opposite helpers. For outward helpers, $R = 0$ is impossible because $\eta_\ell = \delta_\ell/\sigma_\ell^2 < 4/5$ and

$$(D_i - 4/5)(D_j - 4/5) - C_{ij}^2 = (2/5)C_{ij} - 24/25 + 8/(5N) > 0.$$

For reversed helpers, put $l = i$, $s = N - j$, $l + s = N - 1$. The same determinant, multiplied by N, is

$$4lsA^2B^2 - (l+1)(s+1)(A^2+B^2) - 2AB - (l+1)A - (s+1)B + (N+2)/4.$$

It increases in $A, B \ge 4/5$: the derivative uses $4lsB^2 - (l+1)(s+1) \ge (39/25)ls - N > 0$. If either contact is at least .9, concavity in l reduces the minimum to $l = 5$ or $s = 5$; at $(4/5, 9/10)$ it is at least $(509N - 9974)/500 \ge 306/125 > 0$. The last possibility has both contacts $\varphi/2$. In raw norm-two coordinates their difference squared is $\varphi^2(N-1)/N < 4$ for the same helper, so their wall product is positive. For opposite helpers, $R = 0$ would have rational coefficient $N(N+6)$ or $-N^2$ in the expansion of $N^2[(v_i \cdot v_j)^2 - A_i^2 A_j^2]$, according to orientation. Neither vanishes. This proves $R = -1/2$ for opposite helpers.

The retained two-cap Schur matrix is $H = I - t\mathbf{1}\mathbf{1}^T$, $t = (l-1)/(2l)$, for one helper; for opposite helpers it has diagonal residuals $\delta_l, \delta_s \in (1/2, 3/5]$ and off-diagonal $-1/2$. It is positive with positive inverse. Put $\lambda = H^{-1}(A_i, A_j)^T > 0$ and $\mathcal{D} = A_i\lambda_i + A_j\lambda_j$. Its actual mixed point has positive displacement $Z = W_i\omega_i + W_j\omega_j$, and

$$Z^2 = \mathcal{D}(1+\mathcal{D}), \qquad x \cdot Z \ge 1 + \mathcal{D} - \sqrt{1+\mathcal{D}}.$$

For one helper, $W > (4/5)l(A_i + A_j)$ and $\mathcal{D} = [l(A_i+A_j)^2 + (A_i - A_j)^2]/2$; thus $\sqrt{1+\mathcal{D}}/W < 7/16$. For opposite helpers, $5/6 < \lambda_i/\lambda_j < 6/5$, $\mathcal{D}/(\lambda_i + \lambda_j)^2 \le 3/55$, $\lambda_i + \lambda_j > 40$, and $W > 32$; the corresponding bound is $< \sqrt{15/176 + 1/1024} < 3/10$. In either case

$$W_i q_i + W_j q_j - \|Z\| < \frac{W}{2(\sqrt{85/11} + \sqrt{80/11})} < 3W/32.$$

The weighted mean loss of two weak indices is therefore below $17/32$ or $63/160$, both below .61. Adjacency is impossible. Every weak near-4 can now be removed: its mate has normalized height $> 43/50$; an opposite mate or a strong anchor among $N-6, N-5$ has height $> 39/50$ and correlation at most $1/3$. For $N \ge 24$ the latter follows from $6/(N-6) \le 1/3$; for $N = 22, 23$ the mate index is at most five and the opposite anchor at least 16, giving correlation at most $30/\sqrt{8160} < 1/3$. Their sum contradicts $(41/25)^2 > 8/3$. Thus all weak central indices have (B.11.4) and are nonadjacent.

It remains to exclude weak endpoint 5. We give the complete fixed $N = 22$ certificate before its exact lift. Nonadjacency and concavity give strong 6, a middle floor $h_{11} \ge b$, and a strong 16 or 17, where

$$a = \sqrt{96/11} - .61, \quad b = \sqrt{120/11} - .61, \quad d = \sqrt{85/11} - .61,$$
$$2.344 < a < 2.3443, \qquad 2.6928 < b < 2.693, \qquad 2.169 < d < 2.17.$$

If 16 is strong, the minimum reference x_0 through heights a, b, a at $6, 11, 16$ has $E = x_0^2 = a^2/6+(b-a)^2/5 > 47/50$ and positive coefficients $c_6 = c_{16} = (11a-6b)/60 > 4/25$, $c_{11} = (b-a)/5 > 0$. Put $K = 5a/6$, $L = a/12$. For a far helper at 21, each pressure vector in the table has projection $(5\gamma/6)\omega_6 + \omega_{16}/12$, scalar product $\gamma K + L$ with x_0, and height $A^2 = (85/11)\gamma^2 + (5/11)\gamma - 23/44$.

own	pure pressure vector U	target T
5	$\gamma(\omega_4+\omega_6)/2 + \omega_{19}/6$	$H(A, 2/3-\gamma^2)$
7, 8	$\gamma(\omega_4+\omega_6)/2 + \omega_{20}/4$	$A - \sqrt{\gamma^2 - 3/4}$
≥ 9	$\gamma(\omega_4+\omega_6)/2 + \omega_{21}/2$	$H(A, 1-\gamma^2)$
dotted	$(\gamma-\gamma^{-1})\omega_5 + (\omega_4+\omega_6)/(2\gamma) + \omega_{21}/2$	$A - \sqrt{1-\gamma^{-2}}$

Here $x \cdot U \geq T$, $S = \|U - \operatorname{proj} U\|^2$, and $\Delta = T - \gamma K - L$. The residual variances in row order are $2\gamma^2/3 + 1/12$, $2\gamma^2/3 + 1/6$, $2\gamma^2/3 + 5/12$, and $5\gamma^2/3 + \gamma^{-2} - 19/12$. The exact bounds and chosen positive multipliers are

own	5	7	8	≥ 9	dotted
$\Delta >$	3/10	1/4	9/40	13/50	3/10
$S <$	3/4	3/4	3/4	13/12	5/4
μ	1/5	1/5	1/5	24/125	1/6.

The actual sets are respectively $J = \{5, 21, 20\}$, $\{5, 21\}$, $\{5\}$, and the dotted pair; they use the finite-projection formula in (9.4) or the future-core formula in (9.4). The following seals cover the full native ranges, not samples. For 5, $\gamma \in (.809, .81)$ gives $A > 2.21$, $\sqrt{2/3 - \gamma^2} < 1/9$, $\gamma K + L < 1.78$. For 7, $\gamma \in (.9, .902)$ gives $A > 2.47$, radical $< .253$, and projection < 1.96. For 8, $\gamma \in (.923, .9239)$ gives $A > 2.5475$, radical $< .322$, projection < 2.0003, so excess $> .2252$. For finite $m \geq 9$, $\gamma > .939$ by the cubic $8\gamma^3 - 6\gamma - 1$; its deficit increases, since $A' > 11/4$, $H_A > 20/23$, $H_\delta < 0$, and $K < 2$. At .939, $A > 2.5917$, $\sqrt{A^2 + \delta} > 2.6144$, $\sqrt{\delta} < .344$, giving $H > 2.29$ and projection < 2.03. The dotted deficit decreases on $(1, 27/25)$ because $A' < 3$, $K > 1.9$, and the radical derivative exceeds $15/8$. At $27/25$, $A > 2.99$, radical $< .38$, projection < 2.306. Its variance increases to a value below $5/4$.

The vector $x_0 + \mu(U - \operatorname{proj} U)$ has all coefficients nonnegative: the largest old-6 removals are $.154, 4/25, 3/20$, and the old-16 removal is at most $1/60$. Its Cauchy margin is $(E+\mu\Delta)^2 - E - \mu^2 S > 0$, increasing in E. The two tight rows have exact lower margins

$$(47/50 + 9/200)^2 - 47/50 - 3/100 = 9/40000,$$
$$(47/50 + (24/125)(13/50))^2 - 47/50 - (24/125)^2(13/12) = 219/39062500.$$

If 17 is strong, the reference through a, b, d has energy $a^2/12+(b-a)^2/10+(b-d)^2/12+d^2/10 > 24/25$. Its old coefficients exceed $4/25$; replace the helper projection by $\omega_{17}/10$. The deficit decreases by less than $1/40$ and variance decreases by $1/60$. The same multipliers still give positive margins (the tight finite-8 margin is already .01). For a helper at 1, reverse the helper term in every vector. The projection joins the ω_6 coefficient; its value on x_0 is unchanged, while A^2 gains $12\gamma/11$. Since the two time coefficients sum to less than $25/4$, $A_{\text{left}} - A_{\text{right}} > 192/1375$. Targets increase by more than .1, using $H_A > 4/5$. The variance gains $\gamma/3$. All finite rows then have $\Delta > 13/40, S < 3/2$ and work with $\mu = 1/6$; dotted rows have $\Delta > 2/5, S < 5/3$ and work with $\mu = 4/25$. Their old-6 removals are at most $11/72, 59/375 < 4/25$. This treats both helpers and both strong anchors and excludes weak 5.

For $N > 22$, express each reference or dual vector by its potential $t_i = W \cdot \omega_i$ and insert $N - 22$ copies of t_{11} between its two old halves. Its norm $\frac{1}{2}\sum(t_{i+1} - t_i)^2$ is unchanged. The only split coefficient is the positive peak at 11, replaced by nonnegative drops at $11, N - 11$. Residual

potentials are supported in the unchanged first and last six indices, so all projection coefficients and variances above are exactly preserved. Concavity and nonadjacency give the required middle heights at $11, N-11$; the old $6, 16/17$ floors can only increase. For the two helpers the time coefficients are

$$A_N^2 = 10\gamma^2 - 1/2 - (10\gamma-1)^2/(2N), \quad A_N^2 = 10\gamma^2 + 2\gamma - 1/2 - (10\gamma+1)^2/(2N).$$

They increase with N, as do $H(A,\delta)$ and $A - \sqrt{-\delta}$. Every lifted positive certificate therefore keeps its strict margin. Reversal also excludes weak $N-5$.

Finally the $N = 22$ full central reference has

$$E_{\text{full}} = (\sqrt{85/11} - .61)^2/5 + \sum_{i=5}^{10} (\sqrt{D_{i+1}} - \sqrt{D_i})^2 > 1007/1000.$$

Lower/upper pairs for $\sqrt{D_5}, \dots, \sqrt{D_{11}}$ are

$$\begin{aligned} &(2.7797, 2.7799), \quad (2.9541, 2.9543), \quad (3.0895, 3.0897), \\ &(3.1908, 3.1910), \quad (3.2612, 3.2614), \quad (3.3028, 3.3030), \\ &\qquad\qquad (3.3166, 3.3167); \end{aligned}$$

squaring them gives lower energy 1.007192468. For $N \geq 23$ lift this same curve with a constant plateau between $11, N-11$; these two actual heights are at least $\sqrt{11} - .61$ even if weak, since their strong neighbors have dual square at least $D_{10}(23) = 260/23 > 11$. For $N \geq 23$, only $6, \dots, 10$ and their reversals may now be omitted, with at most three weak indices per flank. For $N = 22$, the weak indices may instead be any nonadjacent subset of $6, \dots, 16$, including the middle index 11; this subset has size at most six. Removing one isolated floor, including that middle floor, loses less than $1/1600$, since $q''(t) = -1/q(t)^3$ and $q > 11/4$ on the original central interval. Segment interiors are disjoint; the surviving potential stays concave with nonnegative duals. Consequently $\|x_A\|^2 > E_{\text{full}} - 6/1600 > 4013/4000 > 1$. This proves the A cap, with all spectator pressure already cleared.

B.11.4 Classical packets and the two short tails

For B/D put $D_i = 2i$ and $S_i = h_i/\sqrt{2}$. The continuation table in Section B.11.2 makes every bulk direction strong except

$$v_i = \gamma_i \omega_i + \omega_1/2, \qquad A_i^2 = 2i\gamma_i^2 + 2\gamma_i - 1/2, \qquad Q = 0, \quad 101/125 < \gamma_i < 6/5. \tag{B.11.5}$$

The upper bound follows from $\sigma^2/\delta \geq 4(6/5) - 2 = 14/5 > (100/61)^2$ at $\gamma = 6/5$; the difference numerator is $[2i(\gamma-1)-1]^2$. For a right long 3 the data are $\delta = 1/2 - Q$, $\sigma = \gamma + 1/2$; own $0/3$ is affine and the next contact gives $\sigma/\sqrt{\delta} \geq 1 + 1/\sqrt{2} > 100/61$. For a right fork 3 in D_s, $s = 5, 6, 7$, $\delta = 1 - s/8 - Q$, $\sigma = \gamma + 1/4$. Own $0/3/4/6$ retains affine E_8/cycle/$(3,3,4,3)$ path/$(6,3)$; the next contact gives ratio at least $(\cos(\pi/5) + 1/4)/\sqrt{3/8} > 1.72 > 100/61$. This includes every possible foreign extension of the finite E_6, E_7, E_8 piece. A two-piece white bridge has residual $1/(2i)$ and numerator at least $\gamma + 1/2 + 1/(2i)$, hence is strong; zero own instead gives an affine cycle.

Adjacent packets (B.11.5) are impossible. Weighted AM–GM gives

$$c_i \cdot c_{i+1} \geq 1/2 - ab + (a/b + b/a)/4 \geq 1 - ab > -11/25 > -1/2,$$

so their actual product is zero. Their retained Schur matrix is the same $I - t\mathbf{1}\mathbf{1}^T$, $t = (i-1)/(2i)$, as in Section B.11.3. Its positive actual mixed point has $\sqrt{1+\mathcal{D}}/W < 7/16$ and $W_i\sqrt{2i} + W_{i+1}\sqrt{2i+2} -$

$\|Z\| < W/(4\sqrt{10})$. The derivation uses $A_i > 2$, $\gamma_i > 4/5$ and $i \geq 5$ only: the same explicit inverse and time-coefficient formulas apply. Their sum is below .61, proving nonadjacency.

The index-5 pressure table and its native scalar seals are given in Appendix B.5. Weak 5 forces $x_1 > .5965$, strong 6, and one of $16, 17$ strong. The positive prefix interpolation through $1, 6, 16$ then has energy

$$.5965^2 + (2.01815 - .5965)^2/5 + 1.55051^2/10 = 1.00043812051 > 1. \tag{B.11.6}$$

Endpoint 17 increases the last term. Thus 5 is strong using bulk indices alone, before either short-tail estimate below.

In B, the bulk reaches $r-4$. At $i = r-3$, the extra right B_3 short-end 3 gives F_4, forbidding every white left or foreign extension. Own 0 is affine; otherwise $\delta = 1/4$, $\sigma \geq 1/2 + 1/(2\sqrt{2})$, strong. At $i = r-2$, the extra short-end 3 has $\delta = 1/2 - Q$, $\sigma = \gamma + 1/(2\sqrt{2})$. Own $0/4/6$ is affine, while own ≥ 5 is strong. Own 3 with $Q > 0$ is strong since $\delta \leq 1/4$. The only new weak role is therefore

$$Q = 0, \qquad v = (\omega_{r-2} + \omega_r)/2. \tag{B.11.7}$$

If the short-end contact also joins the left path, the full finite white diagram is B_r and has no foreign extension at c. Its residual $1/(2i)$ and numerator $\geq \gamma + 1/(2i) + 1/(2\sqrt{2})$ are strong even at own 0 for $i \geq 21$. This is a retained zero-own branch, not an assumed affine exclusion. For (B.11.7), deleting $r-1$ and adding the already actual c gives positive A_r, so the same wall actually closes $r-1$. Its exact cosh is

$$C = \frac{(1+\sqrt{2})i + 1}{\sqrt{i+3}}, \qquad i = r-2.$$

For $21 \leq i \leq 23$, $C/\sqrt{2i+2} > 79/50$, $C > 9$, and $\sqrt{2i+2} < 7$. The first comparison reduces to $984i^2 - 19014i - 17498 > 0$, increasing from $i = 21$. Using $\sqrt{C^2-1} > C - 1/C$, its normalized loss is $< 1/(79/50 + 4/63) < .61$. Thus this mate is strong.

In D, the bulk reaches $r-5$. At $i = r-4$ the extra fork 3 has $\delta = 1/2 - Q$, $\sigma = \gamma + 1/4$. Its own $0/3/4/6$ is affine as above; $Q > 0$ is strong. At $Q = 0$, own finite ≥ 7 or dotted has ratio > 1.61, above $T(\sqrt{2i}) < 1.577$ for $19 \leq i \leq 29$. Only own 5 needs the averaged projection below. At $i = r-3$ the full right $D_3 = A_3$ list is middle 3, end $3/4/5$. Middle 3 has $\delta = 1/2 - Q$, $\sigma = \gamma + 1/2$ and is affine or strong. End 4 has $\delta = 1/4 - Q$, $\sigma = \gamma + 1/(2\sqrt{2})$; own 0 is affine, otherwise it is strong. End 5 is finite H_4, forbidding further white contacts, but own 0 is retained: already $\delta = 1 - 3\varphi^2/8$ and $\sigma \geq \varphi/4$ give ratio $> 100/61$. End 3 has $\delta = 5/8 - Q$, $\sigma = \gamma + 1/4$. A left bridge is finite A_r, with no further foreign extension; its residual $1/8 + 1/(2i)$ and numerator $\geq \gamma + 1/4 + 1/(2i)$ are strong for $i \geq 21$ unless own 0, which is an affine cycle. Without that bridge, own $0/3/4/6$ is affine; $Q > 0$ and $\gamma \geq 1$ are strong. Precisely own 5 or finite own ≥ 7, $\gamma < 1$, $Q = 0$, remains.

For own 5 in either short D layer, put $\gamma = \varphi/2$ and $v = \gamma\omega_i + \omega_f/2$ at the contacted fork. The heights are $A^2 = i(\varphi + 5/8) - 1/2$ or $i(\varphi + 5/8) - 5/8$. Select old paths $i, i-1, i-2$ and $i, i+1, i+2$; at the D_3 layer the last node is the other, uncontacted fork. Each actual H_4 correction is $w_\pm = \gamma(3n_i/2 + n_{\text{side}} + n_{\text{second}}/2)$, with $\delta_5 = 1 - 3\gamma^2/2$. The average $Y = v - (w_+ + w_-)/2$ satisfies

$$x \cdot Y > A - \sqrt{\delta_5}, \quad Y^2 = A^2 + 1 - 15\gamma^2/8 < A^2, \quad Y \cdot u_5 = \sqrt{10}(\gamma + 1/4).$$

The cross product is $w_+ \cdot w_- = 3\gamma^2/4$. For $i \geq 19$, $A > 6$, $\|Y\| > 6.4$, $\sqrt{\delta_5} < 1/7$ (use $\varphi > 809/500$), and $\widehat{Y} \cdot u_5 < 3.3602/6.4 < .55$. Thus $x \cdot \widehat{Y} > 41/42 > .97$; strong 5 gives $x \cdot u_5 > .8$. Their sum contradicts $1.77^2 > 3.1$.

For the finite-≥ 7 D_3 role, $i \geq 21$ and $.9 < \gamma < 1$. Now $A^2 = i(2\gamma^2 + \gamma + 1/8) - 5/8$ and the two actual negative A_2 corrections are $w_\pm = \gamma(4n_i/3 + 2n_{\text{side}}/3)$, with norm residual $-d$, $d = 4\gamma^2/3 - 1 > 0$. Their cross product is $8\gamma^2/9$; the average has

$$x \cdot Y \geq A - \sqrt{d}, \quad Y^2 = A^2 + 1 - 14\gamma^2/9 < A^2, \quad Y \cdot u_5 = \sqrt{10}(\gamma + 1/4).$$

At $i = 21$, $Y^2 - 40(\gamma+1/4)^2 = 4\gamma^2/9 + \gamma + 1/2 > 0$; larger i only increases it. Hence correlation is $< 1/2$. The ratio d/A^2 increases in γ and decreases in i; its derivative numerator is $[i(4\gamma^2+13\gamma+3) - 5\gamma]/3 > 0$. Its maximum is $1/195$, giving $x \cdot \widehat{Y} > 1 - 1/\sqrt{195} > .928$. Strong 5 gives $x \cdot u_5 > .807$, so their sum contradicts $1.735^2 = 3.010225 > 3$. Every correction is positive and feasible against all unknown walls. Thus the weak inventory (B.11.5) extends through $r - 3$ in D, with the single B terminal role retained.

We finish the prefix closure explicitly. Let $f_j = \sqrt{j} - .61/\sqrt{2}$ and $E_M = f_5^2/5 + \sum_{j=5}^{M-1}(\sqrt{j+1} - \sqrt{j})^2$. The concave reference has nonnegative duals. The exact seals are $E_{21} > 1.009$, $E_{20} > .99752$, $f_5/5 < .361$: $f_5^2/5 > .6514$ and the increment lower bounds for $j = 5, \ldots, 20$, in units 10^{-5}, are

$$4545, 3846, 3333, 2941, 2631, 2380, 2173, 2000, 1851, 1724, 1612, 1515, 1428, 1351, 1282, 1219.$$

They follow from $(\sqrt{j+1} - \sqrt{j})^2 > 1/(4j+2)$. One isolated missing floor loses $< 1/3200$ by $|(\sqrt{t})''| < 1/40$; two consecutive missing floors lose $< 1/800$, by the variance of their three segment slopes. Disjoint segment interiors make these losses additive.

For a far weak packet at any $5 \leq i \leq 21$, the actual pressure table in Appendix B.5, with $q = \sqrt{2i}$, gives $x_1 > \sqrt{2}[T - \gamma(q - .61)] > .424$. For finite own $5, 7, 8, \geq 9$, use $A - \gamma q > .079$ and the four bracket lower bounds $.61(.809) + .079 - .111$, $.61(.900) + .079 - .253$, $.61(.923) + .079 - .323$, $.61(.939) + .079 - .344$; each times $7/5$ exceeds $.424$. For dotted $\gamma \in (1, 6/5)$ the bracket decreases in both γ, q: $A' - q < (q^2+2)/(4q^3) < .1$, while the radical derivative exceeds 1. Its endpoint bound is

$$\sqrt{2}[\sqrt{3119/50} - (6/5)\sqrt{42} + .732 - \sqrt{11}/6] > .424,$$

sealed by $7.8981 < \sqrt{3119/50}$, $\sqrt{42} < 6.480741$, $\sqrt{11}/6 < .552771$, and $\sqrt{2} > 1.4142$.

In D_r, $r \geq 24$, index 21 is now in range. If strong, $E_{21} - 8/3200 > 1$. Otherwise 20 is strong and $x_1 > .424$; the reference through 20, with this first-coordinate constraint, has energy

$$E_{20} + \frac{5}{4}(.424 - .361)^2 - 7/3200 > .99752 + .00496125 - .0021875 = 1.00029375 > 1. \tag{B.11.8}$$

Its slopes still decrease, so all dual coefficients are nonnegative. For B_r, $r \geq 26$, one of $21, 22$ is strong and the same E_{21} argument applies. At $r = 25$ it still applies through $r - 3$. At $r = 24$, choose strong 21 or 22, or the strong mate 23 of (B.11.7). At $r = 23$, a terminal role at 21 likewise supplies strong 22; a far weak 21 instead uses (B.11.8). In these terminal cases there is at most one two-hole cluster; the excessive loss $9/3200 + 1/800 < .009$ still leaves energy above one. This proves both classical caps and completes Proposition B.11.1.

B.11.5 The same central geometry at ranks nineteen and twenty

Lemma B.11.2 (The actual $N = 20, 21$ central geometry)**.** *Let $N \in \{20, 21\}$, with an actual A_{N-1} component and arbitrary finite spectators. On $5 \leq i \leq N - 5$, define weak by $x_A.\omega_i \leq \sqrt{D_i} - .61$. Before any contrary energy target, its complete weak list is a pure near-four cap $v = \omega_i/\sqrt{2} + \omega_{i-1}/2$, $i = 5, 6$ or its reversal, or a pure far-three helper $v = \gamma\omega_i + \omega_e/2$, $e = 1, N-1$, $101/125 < \gamma < 27/25$, with own five, finite at least seven or dotted. Every weak cap has zero spectator cost. Literal helper weak indices are nonadjacent, for both helper assignments. Each near-four has an actual positive finite mate at four/five or its reversal, with normalized height $> 43/50$.*

Proof. The central interval is still $5 \le i \le N-5$, but its smallest dual square is now $15/2$, rather than $85/11$. We check the changes to the inventory before using the preceding two-cap argument. The exact data are

$$D_i \ge 15/2, \quad q_i > 27/10, \quad q_i < 31/8, \quad C_{i,i+1} \ge 7.$$

Both retained paths have rank at least four. Thus finite classification gives precisely the same no-contact, simple-three, endpoint-four and two-end-three possibilities. A joined finite completion has rank at least nineteen, so it is an ordinary A bridge, with no extra foreign branch. The zero-own repair uses only the helper-side length at least five and the actual affine principals. Its far-three and next-to-far rows have time below their positive foreign cost and force an orthogonal closure; the end-four row has no foreign cost and negative time. Every central own contact is therefore positive.

The isolated-own pressures and selected finite or future points in Subsection B.11.2 are unchanged. Here

$$q_i + \sqrt{D_i - 1} > 26/5.$$

For $u > 1$ with foreign cost, the loss is less than $5/26 + 5/12 = 95/156 < .61$. The same bound holds for pure native labels at least eight, by $\tan(\pi/8) < 5/12$. For $0 < u < 1$, the loss is at most

$$(5/26)u + \sqrt{u(1-u)} \le (5 + \sqrt{701})/52 < 63/104 < .61.$$

The native-seven negative-core loss is less than $20/81 + 9/25 = 1229/2025 < .61$: use $\sqrt{D_i - 100/81} > 5/2$, whose squared margin is $5/324$. The native-five H_4 loss remains less than $25/76 + 5/24 < .61$. The affine owns and the null boundary are unchanged. Thus every isolated cap is strong at these ranks as well.

For the other contact roles the target remains $T(q) < 761/500$ on $[5/2, 31/8]$. All ordinary near-three, next-to-end, exceptional-position and full-bridge estimates above use only helper-side length at least five. Their foreign-cost versions only increase the ratios. The same foreign far-three and near-four comparisons therefore remove every foreign weak role. The complete weak list is a pure near-four/simple-three cap or the literal packet (B.11.4), with $101/125 < \gamma < 27/25$. The dotted upper bound uses the same identity with residual square $[2m(\gamma - 1) - 1]^2$ and only $m \ge 5$. Near-four is strong for helper-side $m \ge 8$. At $m = 7$ the two actual values of q_7 lie in $(3, 25/8)$, where the sharper positive-quadratic seal is $T(q) < 3/2$. Thus a weak near-four has its actual mate at four or five, or a reversal.

We next justify the two-cap argument at these ranks, before removing near-four. Its mutual-product formulas are identities in N. They still force product zero for a common helper and product zero or $-1/2$ for opposite helpers. In the outward case the zero-product determinant is positive because

$$(2/5)C_{i,i+1} - 24/25 + 8/(5N) > 0.$$

In the reversed case the displayed determinant polynomial in Subsection B.11.3 has positive derivatives in the two own parameters. If one own is at least .9, its endpoint minimum is

$$(509N - 9974)/500 \ge 206/500 > 0.$$

Indeed $l, s \ge 5$, $l + s = N - 1$ and $ls \ge 70$ make the positive derivative bracket $(39/25)ls - N$ sufficient even after subtracting $2(27/25) + (l + 1)$. If both owns are five, the raw difference and the nonzero rational coefficients $N(N + 6), -N^2$ in the preceding calculation apply literally. Hence opposite helpers have product exactly $-1/2$.

The positive two-cap Schur matrices and their actual displacements are the same as above. Their time coefficients exceed two. To check this without separating incompatible minima, reverse so $i \le N/2$. The smaller, far-helper time is exactly

$$A_N^2 = 2i\gamma^2 - \tfrac{1}{2} - \frac{(2i\gamma - 1)^2}{2N}.$$

It increases in N and in i on this half interval, and in $\gamma \ge .8$. Its minimum at $N = 20, i = 5, \gamma = .8$ is $187/40 > 4$. The near-helper time is larger by $2\gamma(1 - 2i/N) \ge 0$. We may therefore use the already printed inverse calculation with $A_i, A_j > 2$. It gives $\sqrt{1 + \mathcal{D}}/W < 7/16$ for a common helper and $< 3/10$ for opposite helpers. The spatial rationalization now gives

$$W_i q_i + W_j q_j - \|Z\| < \frac{W}{2(\sqrt{15/2} + \sqrt{7})} < W/10.$$

Thus the weighted loss of two adjacent weak packets is less than $43/80$ or $2/5$, both below .61. Adjacent literal packets are impossible, retaining the entire spectator block and every unknown wall.

The near-four mate is an actual positive finite replacement, with the same cosine formula printed in the cap proof. Reversing its numbering if needed gives $6 \le j \le N - 4$ and

$$jD_j \ge 72(N - 6)/N > 216/5, \qquad \tfrac{1}{2} + \frac{j - 1}{j\sqrt{2}} > 13/12.$$

Its cosine is therefore greater than seven, and its normalized height exceeds $43/50$. None of these steps used an energy target stronger than the original $e_A < 1$. □

Proposition B.11.3. *An actual A_{19} or A_{20} component has projection energy greater than $6/7$, with arbitrary finite spectators.*

Proof. Suppose $e_A \le 6/7$ and apply Lemma B.11.2. At the other end there is another such mate, or, by nonadjacency, a strong anchor in $\{N - 6, N - 5\}$. That anchor has normalized height $> .77$. The correlation is at most

$$\sqrt{\frac{5 \cdot 6}{15 \cdot 14}} = 1/\sqrt{7} < 1/2.$$

The sum of the two heights is > 1.63, while the sum direction has square less than three. This contradicts the assumed energy, since $1.63^2 - 3(6/7) = 5983/70000 > 0$. Reversal removes the other near-four orientation. Only nonadjacent literal helpers remain.

There is now a strong $i \in \{5, 6\}$ and a strong $j \in \{N - 6, N - 5\}$. Their initial and final coordinate intervals are disjoint. The exact height-energy identity and Cauchy give

$$e_A \ge \frac{h_i^2}{2i} + \frac{h_j^2}{2(N - j)}.$$

At $N = 20$, the endpoint floors are already $q_5 - .61 > 2.1$ and $q_6 - .61 > 2.28$, by the positive squares $15/2 > 2.71^2$ and $42/5 > 2.89^2$. Increasing N improves them. Each term is therefore greater than

$$\min\{2.1^2/10, 2.28^2/12\} = 1083/2500.$$

Consequently $e_A > 1083/1250 > 6/7$, with final difference $81/8750 > 0$. This proves the proposition on the same original projection, with no spectator or face-inheritance restriction. □

B.11.6 The one short spectator beside D_{23}

Lemma B.11.4. *The entire forest $D_{23} + I$ cannot occur in dimension twenty-four.*

Proof. The forest is exactly $D_{23} + I$, with no further spectators. Proposition B.11.6 applies because $23 \geq 21$. Its proof uses the rank-valid prefix and local head coupling, independently of the dimension-twenty-four accounting below. □

B.11.7 The finite accounting in dimensions twenty-four through twenty-nine

Proposition B.11.5. *No compact hyperbolic Coxeter polytope has dimension at least twenty-four.*

Proof. The published ceiling twenty-nine leaves only $24 \leq n \leq 29$. Call A_r, $r \geq 7$, or B_r, D_r, $r \geq 9$, dominant. If no component is dominant, each ordinary rank pays its strict rank /14 cost. The common defect theorem leaves at most one discarded short rank, or one typed rank-two defect. The respective total costs exceed $(n-1)/14$ or $.09 + (n-2)/14$, both above one. Thus a dominant block exists. Every dominant block costs more than $1/2$, so there is exactly one. Write its rank as r and its spectator rank as $S = n - r$. Only now may every other block be treated by the small ordinary rates and the single permitted debit. Their complete three conservative costs are

$$S/14, \qquad (S-1)/14, \qquad .09 + (S-2)/14.$$

They represent, respectively, no debit, one short I rank discarded, and one typed rank-two defect. A J profile pays both ranks and does not create a worse case. Empty ordinary sums are zero.

For dominant A, its half-cost first gives $S \leq 7$, hence $r \geq 17$. The shared uniform $A_{\geq 13} > 5/7$ cost then gives $S \leq 4$: at $S = 5$ the weakest typed case already costs $5/7 + .09 + 3/14 = 713/700 > 1$, while the short case exceeds one. Thus $r \geq n - 4$. For $n \geq 25$ this contradicts the cap $r \leq 20$. At $n = 24$ only $r = 20, S = 4$ remains. Proposition B.11.3 and the three spectator costs close it, the weakest short case already giving $6/7 + 3/14 > 1$.

For dominant B/D, the shared $9/14$ floor first gives $S \leq 5$. For $n \geq 25$ we have $r \geq 20$. If $r = 20, 21$, the rank-valid generic prefix gives energy $> .90$, and $S \geq 4$; even the discarded-short cost $3/14$ is too large. If $r = 22, 23$, its energy is $> .94$, with $S \geq 2$. One remaining ordinary rank costs $> 1/14 > .06$; a sole typed rank-two block costs $> .09$. The caps exclude every larger rank. Thus $25 \leq n \leq 29$ fail.

At $n = 24$, $19 \leq r \leq 23$. The exact generic-prefix rows and their full original rank hypotheses give

r	19	20	21
$e_{B/D} >$	.88864	.90715	.9240775

$e_{B/D,\, r=22} > .9401975.$

Here $M = 13, 14, 15, 16$ respectively, so the D hypothesis $r \geq M + 6$ and the B hypothesis $r \geq M + 5$ both hold. The same increment and omission proofs give these rows; they do not apply a prefix below its permitted rank. Together with the three spectator costs at $S = 5, 4, 3, 2$, they exceed one. For $r = 23$ the B cap already applies. The remaining D_{23} has one spectator rank: an ordinary A_1 costs $> .1$, too much beside $> .94$; a J or typed defect needs two ranks. The sole remaining case is the actual isolated I and is exactly Lemma B.11.4. This exhausts every finite forest, without asserting that an abstract forest is realized. Dimensions above twenty-nine use only the named published starting bound. □

B.11.8 One isolated-I argument for all the proper large classical rows

Proposition B.11.6. *An entire farthest-vertex forest $B_r + I$, $r \geq 20$, or $D_r + I$, $r \geq 21$, is impossible. No further spectators are allowed in this statement.*

Proof. Use $M = 15$ in the actual prefix inventory. Its hypotheses are B_r with $r \geq M + 5 = 20$, or D_r with $r \geq M + 6 = 21$; its geometric window is exactly $5 \leq i \leq M + 1 = 16$. Deleting each of those actual long nodes leaves a left path of rank at least four and a right B of rank at least four, or D of rank at least five. The complete white/foreign list and its actual affine or point exclusions leave only the pure far-three helper weak. The actual two-cap argument excludes adjacent weak helpers. These are precisely the stated hypotheses and domain of Appendix B.5; no $M = 17$ or unproved short-tail continuation is being invoked.

First suppose five weak. Then six is strong, the actual weak-five point gives $x_1 > a_0 = .5965$, and one of fifteen/sixteen is strong. Put $b_0 = 2.01815 < f_6$ and interpolate through

$$(1, a_0), \quad (6, b_0), \quad (K, b_0 + \sqrt{K} - \sqrt{6}), \qquad K = 15 \text{ or } 16.$$

The actual endpoint floor is larger than the displayed surrogate. The slopes are a_0, .28433 repeated five times, and $c_K = 1/(\sqrt{K} + \sqrt{6})$ repeated $K - 6$ times, then zero. They decrease strictly. Thus this is a paid nonnegative-anchor reference W_0, with no I coordinate or previously credited I energy. Its first coordinate is a_0 and its retained P_6 demand and pairing both equal b_0. Explicitly

K	$\|W_0\|^2 >$	$c_K <$	remaining individual P_6 coefficient $>$
15	.9851486	.159	.12533
16	1.000435	.156	.12833.

Indeed the common first-six energy is .7600299945. The two tail contributions are greater than $1.4234^2/9$ and $1.5505^2/10$, using $\sqrt{15} > 3.8729$ and $\sqrt{6} < 2.4495$. The coefficient is $.28433 - c_K$, greater than $1/60$ in both rows. Lemma B.10.4 applies to this exact paid reference and adds $> .015$ using the single unused actual I height. Already the weaker fifteen total exceeds 1.0001486; the sixteen row exceeds one even before that graft. Consequently five is strong.

Choose an actual strong $K \in \{15, 16\}$ and now use the concave reference through the actual strong floors, omitting nonadjacent weak knots. The common exact rational increment bounds give

K	E_K lower	energy after all permitted holes $>$
15	.92564	.9240775
16	.94176	.9401975.

At most five isolated internal holes cost $5/3200$. The reference is constant on its first five coordinates, with mean $b = f_5/5 < .361$ and actual P_5 demand f_5. Its individual P_5 coefficient is at least $b - (\sqrt{6} - \sqrt{5}) > .36 - .214 = .146$; omitting six only increases that coefficient. All other anchor coefficients remain nonnegative. No I energy has been used by this reference either.

The head deletion retains B_{r-1} or D_{r-1} of rank at least nineteen or twenty. Finite classification permits exactly isolation or a simple three at the retained long leaf two. A short-end exceptional attachment is too small at these ranks. The entire forest hypothesis leaves just the old I as a possible foreign factor. In the nonisolated row, own zero has zero time or the positive actual null $c + \Omega_2/2$; own three has its triangle. The other own ranges and every isolated foreign/native boundary are exactly the local head inventory of Lemmas B.10.6 and B.10.9.

Their selected long roots n_1, n_2, n_3 and actual old I have the same physical Gram and coordinates here. Their finite projections or future cores have positive actual-wall coefficients, retain all unknown walls, and retain both signed forks in type D. Their joint point covers every actual head/I mutual parameter; it does not presume a finite pair Gram. They do not use a terminal B short normal, a positive final D coordinate or a tail conclusion. The uniform gain is therefore $> .07665$ on this paid flat-five reference, including I exactly once. Its price is at most .1, below the available .146. Hence

$$\|x_{B/D}\|^2 + y^2 > .9240775 + .07665 = 1.0007275 > 1.$$

This proves the proposition with precisely its entire-forest scope. In particular it does not exclude $D_{20}+I$, whose $M = 15$ rank hypothesis fails, or supply a theorem with arbitrary extra spectators. □

B.11.9 One smaller-loss window for whole B_{22}, D_{22}, D_{23}

Lemma B.11.7 (A local whole-diagram .58 window)**.** *Let an entire actual farthest forest be B_r or D_r, $r \geq 19$. On $5 \leq i \leq \min\{20, r-4\}$ in B, or $5 \leq i \leq \min\{20, r-5\}$ in D, use the loss .58. The complete weak list is the pure far-three helper, with own five, finite at least seven or dotted and $\phi/2 \leq g < 6/5$, except that isolated dotted own at five is also permitted. Literal helper weak indices are nonadjacent. Their actual first coordinate bounds are $> .5455$ at five, $> .378$ through twenty, and $> .45$ through nine. No short-tail continuation or whole-rank exclusion is part of this local statement.*

Proof. This time the whole-diagram hypothesis is essential. There is no foreign inverse-Gram cost. Set $c_0 = 29/50$, $F_i = \sqrt{i} - c_0/\sqrt{2}$, and call a long index c_0-strong when $S_i > F_i$. The old .61 floors are still denoted by f_i. We use actual unit roots, the genuine short root in B, and both signed forks in D.

The new inventory is checked locally. The sufficient ratio for c_0-strength is

$$T_0(q) = \frac{50}{29} - \frac{q - 29/50}{q(2q - 29/50)}.$$

It increases for $q \geq \sqrt{10}$. Positive rational multiplication at the right endpoints gives

$$T_0(q) < \begin{cases} 83/50 & q < 7, \\ 801/500 & q < 15/4, \\ 159/100 & q < 16/5, \\ 33/20 & q < 32/5. \end{cases}$$

The local bulk conditions are exactly a right B of rank at least four, or right D of rank at least five. At the stated ranks this gives 5..18 in B_{22}, 5..17 in D_{22}, and 5..18 in D_{23}. Each deletion leaves A_{i-1} and a right B_s, $s \geq 4$, or D_s, $s \geq 5$. The complete one-node finite-white list is the one proved above: left endpoints, next-to-end and exceptional positions, right long leaf, the finite D fork extensions, and full simple endpoint bridges. No generic prefix theorem is being invoked with $M > 16$.

For clarity, at the additional D_{23} index eighteen the retained blocks are exactly $A_{17} + D_5$. A right long-leaf three is the ordinary D extension. A right fork three gives finite E_6, with $(\Delta, \sigma) = (3/8, g + 1/4)$. Own zero has the actual affine E_8 with arms $1, 2, 5$; own three has its all-three cycle; own four has the actual $3, 3, 4, 3$ principal on the cap, the contacted fork and the three old nodes ending at eighteen; own six has the $6, 3$ path; parallel has the pair null. Every other own has ratio $> (.8 + .25)/\sqrt{3/8} > 83/50$. A long-left/fork bridge with those E_6 arms cannot be finite at rank twenty-three. The permitted full bridge instead meets the long leaf and is finite D_{23}, with the same endpoint ratios. This supplies the local endpoint argument at eighteen explicitly.

The complete .58 scalar checks are as follows. Near-three has least ratio $[i(\phi+1)-1]/\sqrt{2i(i+1)}$. It increases with i. At $i = 6$ it exceeds 801/500, handling six/seven; at eight it exceeds 83/50, handling every later index. At five, own at least seven gives ratio $> 5/3$. Thus only own-five/near-three at five is new. Next-to-end positions have ratio at least $(i\phi/2 + 1)/\sqrt{2i}$, above 159/100 at five and 83/50 from six onward. Exceptional positions have residual at most $1/4$, $j \geq 3, i \leq 8$, hence ratio at least $2g + 3/8 > 83/50$. The far-four/own-three principal is affine. Near-four/own-three has

actual cosine $(i-1)+i/\sqrt{2}$; its other owns are stronger. Right long-three and the short-three B_3 exception, when used later as a separate short-layer check, have ratio at least $1+1/\sqrt{2} > 83/50$. A D_s fork, $s \geq 5$, has the same affine own list just checked. The only finite fork extensions have $s = 5, 6, 7$, with residual $1 - s/8 \leq 3/8$. Own zero uses actual affine arms $1, 2, 5$ by extending the retained long arm with at most the old $i, i-1, i-2$; all exist because $i \geq 5$. Own three closes the all-three cycle, own four uses the $3, 3, 4, 3$ path on $i-2, i-1, i, c, f$, and own six/parallel use the local path/null. Every other own has remaining ratio $> (.8+.25)/\sqrt{3/8} > 83/50$. Full bridges have smaller residuals and larger numerators, with their own-zero cycles retained.

The isolated own-five H_4 loss is $< 25/76 + 5/24 < .58$. For seven, the old negative core gives loss $< 1000/4779 + 9/25 < .57$: here $q + \sqrt{D - 100/81} > 3 + 2.9$. For finite owns at least eight, the loss is largest at $D = 10, m = 8$, where it is $\sqrt{10} + \sqrt{2} - 4 < .578$. The monotonic singleton projection proves the whole remaining finite interval. The old owns three/four/six and parallel retain their actual affine/null principals. For isolated dotted own, $u = g^{-2} \in (0, 1)$, the actual core gives loss at most

$$\frac{u}{\sqrt{2i} + \sqrt{2i-1}} + \sqrt{u(1-u)}.$$

At $i \geq 6$ the first coefficient is $< 37/250$, so this is $< (37/250 + 1.011)/2 = .5795$. Only isolated dotted five remains. Thus, apart from that row and near-five, every weak bulk cap is the pure helper $v = g\omega_i + \omega_1/2$, with own five, finite at least seven or dotted.

The new near-five role. For $v = (\phi/2)\omega_5 + \omega_4/2$, put $g = \phi/2$ and use the actual right H_4 correction

$$w = g(3n_5/2 + n_6 + n_7/2), \qquad \delta = (5 - 3\phi)/8, \qquad Y = v - w.$$

Its extra old-four product is negative. Its normalized finite projection retains every original wall. Direct substitution gives

$$A^2 = (7 + 13\phi)/2, \quad Y^2 = (33 + 49\phi)/8, \quad \sum Y_j = \sqrt{2}(2 + 5\phi/2).$$

From $1.618 < \phi < 1.62$, $A > 3.74$ and $\sqrt{\delta} < .136$, we have $x.Y > 3.6, Y^2 < 14.05, \sum Y_j < 8.6$. The independently valid old .61 inventory at $M = 11$ supplies a strong $K = 11$ or twelve, before any new strong-five conclusion. For $Z = Y + (3/4)P_K$ the two demand/square bounds are

K	$x.Z >$	$Z^2 <$	squared-demand excess $>$
11	5.76	33.1375	.0401
12	5.8725	33.7	.78625625.

Here $f_{11} > 2.88, f_{12} > 3.03$. Cauchy excludes both, so the new near-five role is impossible without a .58 strong-five premise.

Helper nonadjacency and three actual consequences. Every weak helper has $\phi/2 \leq g < 6/5$. The upper bound is reproved at this threshold: at $g \geq 6/5$ the squared ratio is at least $4g - 2 \geq 14/5 > (83/50)^2$. Adjacent helper caps are distinct actual walls, and their product is at least $1 - ab > -11/25 > -1/2$. It must be zero. Their positive Schur inverse and actual mixed point give loss $< 7/16 + 1/(4\sqrt{10}) < .58$, using only $i \geq 5, g > .8, A > 2$. Thus new weak helpers are nonadjacent; we have not presumed they were already old .61-weak. This last two-cap calculation uses only the two displayed pure-helper profiles, their actual long duals and the retained positive old blocks. A later short-layer classification may supply those same profiles; after checking that physical interface it may reuse this calculation, without extending the present classification domain.

The actual helper point table gives $x_1 > \sqrt{2}[T - g(\sqrt{2i} - .58)]$. Its complete native ranges imply

$$\begin{array}{c|c} i = 5 \text{ weak helper} & x_1 > .5455 \\ i \le 20 \text{ weak helper} & x_1 > .378 \\ i \le 9 \text{ weak helper} & x_1 > .45. \end{array}$$

For the first subtract at most $.03\sqrt{2}(6/5)$ from the old scalar expression $> .5965$. For the second, the four finite lower brackets are $.58(.809) + .079 - .111$, $.58(.900) + .079 - .253$, $.58(.923) + .079 - .323$ and $.58(.939) + .079 - .344$. Their products with $\sqrt{2}$ exceed .378. The dotted difference decreases in g, q and has enlarged endpoint

$$\sqrt{119/2} - (6/5)\sqrt{40} + .696 - \sqrt{11}/6.$$

The bounds $\sqrt{119/2} > 7.71362, \sqrt{40} < 6.32456$, $\sqrt{11}/6 < .55278, \sqrt{2} > 1.4142$ give $> .3781118$. For $i \le 9$ the finite $A - gq$ term is $> .15$, increasing in g and decreasing in q; the same brackets then exceed .45. For dotted use $\sqrt{1391/50} > 5.274, \sqrt{18} < 4.243$, $\sqrt{11}/6 < .553, \sqrt{2} > 1.4$, giving $> .45556$. All these pressures use actual positive finite projections or future cores. At a last D long node, $\omega_{21} + \omega_{22} = \sqrt{2}P_{21}$ in D_{22} is only a spatial identity; no extra normal is introduced. □

Lemma B.11.8 (A conditional early paid-reference contract)**.** *Keep the whole-diagram and threshold conventions of Lemma B.11.7. Suppose the actual helper-or-strong window, with only its permitted isolated dotted-five exception, has been completed through sixteen. For either actual strong $K \in \{15, 16\}$ there is a paid vector W supported through K, with $\sum W_j = F_K$, remaining individual endpoint-K coefficient $> 1/9$, and effective energy $L_W > Q_K$, where $Q_{15} = .94006, Q_{16} = .95618$. The three initial-five branches, their later conditional extensions to an actual strong nineteen/twenty, and the complete flat-pressure table below have the stated positive-dual and same-vector scope. This lemma supplies no missing short-layer classification or strong endpoint.*

Proof. Use $F_i = \sqrt{i} - .58/\sqrt{2}$ throughout this proof.

Early references in all three five branches. Put $\mathcal{E}_K = F_5^2/5 + \sum_{i=5}^{K-1} d_i^2$. Changing the initial loss from .61 to .58 adds $.03\sqrt{2/5} - .00357 > .0154$. One omitted internal floor j loses $< 1/[32(j-1)^3]$. All holes $j \ge 6$ therefore cost less than $\frac{1}{32}\int_4^\infty t^{-3}dt = 1/1024$.

For either actual strong $K = 15$ or sixteen we need a paid vector W, supported through K, with $\sum W_j = F_K$, endpoint coefficient $> 1/9$ and effective energy

$$L_W = 2D_W - \|W\|^2 > Q_K, \qquad Q_{15} = .94006, \quad Q_{16} = .95618.$$

Here D_W is its pressure demand. Effective energy need not equal the square of the vector. If five strong, the concave reference and the full hole bound give these values; its endpoint coefficient is at least $d_{K-1} > 1/8$. If five a weak helper, interpolate through $(1, .5455), (6, F_6), (K, F_K)$. Its first-six energy exceeds .74385, and its tail energies at fifteen/sixteen/nineteen exceed .225, .2404, .2804. The slopes decrease and the endpoint coefficient $1/(\sqrt{K} + \sqrt{6}) > 1/9$; the first two energies exceed the required Q_K.

If five is isolated dotted, its actual core gives

$$U' = P_5 + (u/2)(e_6 - e_5), \qquad x.U' \ge T' = \frac{\sqrt{10 - u} - \sqrt{u(1-u)}}{\sqrt{2}}.$$

Its loss is $< .588$, since with $a = \sqrt{10} - 3 < 13/80$, $au + \sqrt{u(1-u)} < (a + \sqrt{1 + a^2})/2 < .588$. Weakness forces $u > .4$: at $u \le .4$ the coarser loss $au + .5 < .565$ is too small. In particular

$T' > 1.8185$. Choose actual strong $J = 6$ or seven and $K = 15$ or sixteen. Let W_0 interpolate through $(J, F_J), (K, F_K)$, and put

$$V = U' - (5/J)P_J, \qquad W = W_0 + \mu V, \qquad \mu = .21\ (J = 6), \quad .2\ (J = 7).$$

The residual has sum zero, support through J, and $W_0.V = 0$. Its squared norms are bounded by $77/150, 194/175$, since $U'^2 = 5 - u + u^2/2$ decreases for $.4 < u < 1$. The deficits exceed $.119, .221$. The J payments are $.175, 1/7$, below the available $.1808, .165$; those capacities only increase when K increases. The endpoint-K coefficient stays unchanged. The complete energy data are

J	$F_J^2/J + 2\mu\Delta - \mu^2 V^2 >$	$\text{tail}_{15} >$	$\text{tail}_{16} >$	$\text{tail}_{19} >$
6	.720462	.225	.2404	.2804
7	.757957	.188	.2037	.2445.

Thus both early columns supply $L_W > Q_K$. Every source coefficient is nonnegative, and the construction remains legal at a later actual strong $K = 19$ or twenty. No strong-five or terminal premise has been used in constructing the early references.

For an actual normalized pressure $x.Z > G$, $Z_1 = \cdots = Z_{16} = 1$ and $Z^2 \le R$, pay $1/9$ at the remaining endpoint $K = 15$ or sixteen. The residual $Z - P_K$ has zero product with W because $\sum W_j = F_K$, regardless of whether $L_W = \|W\|^2$. One square gives

$$e > L_W + 2(G - F_K)/9 - (R - K)/81.$$

This uses a positive-dual payment, not the invalid subtraction of two lower pressure bounds. With $F_{15} < 3.463, F_{16} < 3.590$, each row below is sufficient in both early references:

G	R	minimum excess above one $>$
3.956	19	.00023
4	19.7	.0013
4.05	20	.0087
4.1	20	.0198
4.15	21	.0186
4.4	22	.0618.

The remaining task is to verify actual point pressures fitting this table; a diagram count or an abstract Gram completion would not suffice. □

Proposition B.11.9. *None of the entire finite diagrams B_{22}, D_{22}, D_{23} occurs at an actual farthest vertex.*

Proof. The local windows of Lemma B.11.7 reach eighteen/seventeen/eighteen respectively, and hence already contain sixteen. Actual helper nonadjacency supplies a strong fifteen/sixteen, so Lemma B.11.8 applies with all its individual price conditions. We now complete the genuinely different short layers and fund their actual pressures by that table.

The actual B_{22} short tails. At nineteen deletion leaves $A_{18} + B_3$. The new short-three F_4 role has residual $1/4$ and numerator $g + 1/(2\sqrt{2})$. Own zero has its actual affine F_4 kernel, and every positive own has ratio at least $1 + 1/\sqrt{2} > 83/50$. The long-three and all full bridges are strong. At twenty the right B_2 allows three at either end. The long end is the same local row. The short end has residual $1/2$ and numerator $g + 1/(2\sqrt{2})$; own zero/four/six/parallel have their actual affine/null principals, while owns at least seven are directly strong. In particular the own-zero short kernel

on $c, n_{22}, n_{21}, n_{20}, n_{19}$ is $(\sqrt{2}, 2\sqrt{2}, 3, 2, 1)$. Own four/six use respectively $\sqrt{2}c + n_{19} + 2n_{20} + n_{21}$ and $\sqrt{3}c + 2n_{20} + n_{19}$. All four left/right endpoint bridges have residual $1/40$ and minimum ratio $\sqrt{40}(1/40 + 1/(2\sqrt{2})) > 83/50$, including own zero.

Own five uses the actual left H_4 point, with $\alpha = \sqrt{2}g + 1/2$, $A^2 = 20\alpha^2 - 1/2$, $R < 20$. The positive squares $A/\alpha > 4.45$ and $\sqrt{\delta_5}/\alpha < .083$ give $G > 4.3$. Own three is the actual cap $v = (\omega_{20} + \omega_{22})/2$. Releasing old twenty-one gives a positive finite A_{22} endpoint, with its omitted-wall product strict and every unknown wall retained. Its cosine is $(21 + 20\sqrt{2})/\sqrt{23} > 245/24$. It gives $x.P_{21} > 4.15$, $R = 21$: the last comparison is $1511(245/24) > 15289$. Both points fit the paid table. Thus through twenty only helpers, strong indices and possibly isolated dotted five remain in B_{22}.

The actual D_4 and D_3 tails. For D_{22} these are eighteen and nineteen; for D_{23} they are nineteen and twenty. The D_4 finite list allows three at any leaf. The fork-three residual is $1/2$, numerator $g + 1/4$. Own zero/three/four/six/parallel have their exact principals. Own at least eight is strong because $\sqrt{2}(\cos(\pi/8) + 1/4) > 33/20 > T_0(q)$. Retain own five and the newly possible seven. Own five uses its actual left H_4 point, with $A^2 = i\alpha^2 - 1/2$, $\alpha^2 = \phi + 5/8$ and residual ratio $< .01$. At $i = 18$ its first sixteen coordinates are not all flat. Use instead the independently valid old .61 strong-five reference and an old strong eleven/twelve: the first twelve are flat and the old terminal price exceeds $1/7$. The pressure $G > 4.1, R < 20$ gives energy > 1.0102 in both rows. The old strong-five bootstrap here uses only the complete $M = 16$ window through seventeen. Own seven uses

$$\xi = c + g^{-1}n_i + (2g)^{-1}n_{i-1}, \qquad k = 1 - 3/(4g^2) > 0.$$

Its cap and $i - 1$ products vanish; the i product is negative, its square is $-k$ and its time is positive. For $.9 < g < .902$, at $i = 18$ we have $A/\alpha > 4.22$ and $\sqrt{k}/\alpha < .18$, so $G > 4.04, R < 18$. Both quantities only strengthen in the D_{23} layer $i = 19$; its $R < 19$ is still covered by the first paid row. All selected displacements are flat through sixteen in this seven case. Bridges and long leaf have their strong or own-zero cycle dispositions. Both forks remain.

At the right $D_3 = A_3$ layer, middle-three is direct or affine. End-four retains its actual affine own-zero kernel and is otherwise strong. End-five is sealed H_4 and directly strong even at own zero. Full left bridges retain their own-zero cycle and direct positive-own bounds. Thus only no-left end-three remains, with residual $5/8$, numerator $g + 1/4$. Own zero/three/four/six/parallel have the actual E_8/cycle/star/path/null principals. Own five is the actual left H_4 point, $A^2 = i\alpha^2 - 5/8$, giving $G > 4.1, R \le 20$. Every $g \ge 53/50$ is directly strong, since $(131/100)^2(8/5) > (33/20)^2$.

For the whole interval $.9 < g < 53/50$ use both actual A_2 sides, with corrections $g(4n_i/3 + 2n_{i-1}/3)$ and $g(4n_i/3 + 2n_{i+1}/3)$. Their positive future cores have square $-d$, $d = 4g^2/3 - 1 > 0$, and all selected products are zero or negative, including the additional contacted fork. Average their pressures retaining the full correction cross $8g^2/9$. The displacement is flat through sixteen, and

$$x.Y/\alpha \ge \frac{A - \sqrt{d}}{\alpha}, \quad A^2 = i\alpha^2 - 5/8, \quad Y^2 = A^2 + 1 - 14g^2/9 < i\alpha^2.$$

At $i = 19$, splitting at $g = 26/25$ gives respectively $(A/\alpha, \sqrt{d}/\alpha) > (4.33, 0)$ with radical $< .365$, and $A/\alpha > 4.337$ with radical $< .381$. Thus $G > 3.956, R < 19$. At the actual D_{23} index twenty the same positive squares give $A/\alpha > 4.44$ in the first interval and > 4.45 in the second; the radical bounds are unchanged. Hence $G > 4.069 > 4.05, R < 20$, a separate sufficient table row. The increasing ratio d/α^2 justifies both full intervals, not only the endpoints. This proves the complete window through twenty for whole D_{23} without any claim at its later centre twenty-one.

The remaining actual D_{22} centre twenty. Here the retained white diagram is $A_{19}+2A_1$. Its complete finite list is one left endpoint-three/four or next-to-end-three; with one fork only a left endpoint-three and fork-three/four; with both forks the all-three endpoint D_{22} bridge. Their residuals are $1/40$ or $11/40$; the directly strong branches include zero own where appropriate, and the left/fork-three zero/three rows retain the cycle/triangle. Internal branches cannot be finite at this rank. With no fork, the locally rechecked bulk inventory applies. The isolated-five correction uses only actual long roots and the spatial sum $\omega_{21}+\omega_{22}$.

With no left contact, one finite fork label or the pairs $(3,3),(3,4),(3,5)$ exhaust the white possibilities. At own zero, fork three/four/six have their affine principals. Fork five uses the actual left H_4 point: $R=19$, $A^2/\alpha^2=6+8\phi$, $\sqrt{\delta_5}/\alpha<15/56$, giving $G>4.08$. The last bound uses $\phi>79/49$, because $5\cdot 49^2-109^2=124>0$; it is not inferred just from $\phi>8/5$. For fork at least seven put $b=\delta^2\in(.81,1)$. The actual core

$$c+\delta^{-1}n_f+2(\delta^{-1}-\delta)n_{20}$$

has zero cap/fork products, negative centre product, positive time and negative square. Its normalized data are

$$R(b)=30-16/b+6/b^2,\quad G(b)=\sqrt{22-2/b}-\sqrt{-8+14/b-6/b^2}.$$

For $b\le .9$, $R<19.7$ and $G>4.41-.41=4$. For $b\ge .9$, $R\le 20$ and $G>4.44-.39=4.05$. The derivatives have numerators $16b-12$ and $12-14b$; the latter radical square has maximum $1/6$, or $4/27$ on the second interval. These are whole-interval positive-square bounds.

At own three, fork three has its triangle. Fork four/five instead gives the actual finite D_{22} contacted-fork endpoint, with cosine $(20+22\delta)/2>17$, signed direction $\sqrt{2}\omega_f=P_{21}\pm e_{22}$, $R=22, G>4.4$. The released-wall product is strict, so this is an actual endpoint. Fork at least six is direct. Own four has its centre/left/other-fork affine star. For own at least five and fork at least four, direct ratios leave only own-five/fork-four; it has the same left-H_4 data as the B_2 own-five case, $G>4.3, R<20$. It must not be discarded by the old .61 ratio. For fork-three/own-five, the left-H_4 point has $A^2=20\alpha^2-3/4$, $G>4.1, R<20$. For other finite or dotted owns, $g\ge 6/5$ is direct. On $.9<g<6/5$, the actual core above with n_{20},n_{19} has $k=1-3/(4g^2)$, $A^2=20\alpha^2-3/4$ and $R<20$. The full derivative bound $k/\alpha^2\le 575/5046<.34^2$ and $A/\alpha>4.44$ give $G>4.1$.

For two forks, own three has its triangle and owns at least four are direct. At own zero, pair-three-three has its square. Pair-three-four has an actual finite B_{22} contacted-fork endpoint, cosine $(40+22\sqrt{2})/4>17$, again $G>4.4, R=22$. This adjacent vertex is not declared farthest. Pair-three-five uses the actual finite H_3 projection, with displacement αP_{20}, $q=2\phi^{-6}<1/8$ and pressure

$$\sqrt{20-q}/(1+\sqrt{q/20})>4.45/1.08>4.1,\quad R=20.$$

Every listed point fits the paid table. This exhausts the actual centre classification and completes the window through twenty in D_{22} as well. All signs of the final coordinate were retained.

The final reference, without an assumed strong five. Nineteen/twenty are now helper-or-strong, so one is strong. If five a weak helper, its earlier interpolation has energy $>.74385+.2804>1$. If five isolated dotted, its same paid reference has effective energy $>.720462+.2804>1$ or $>.757957+.2445>1$. The endpoint-J payment remains legal because its tail coefficient decreases. Thus five is strong, proved only after the terminal classifications.

If twenty strong, the reference energy exceeds $\mathcal{E}_{20}-1/1024>.99752+.0154-1/1024>1$. Otherwise twenty weak gives strong nineteen, $\mathcal{E}_{19}>.9847+.0154=1.00010$, and $F_5/5<.366$. If a

weak helper occurs at six through nine, its actual $x_1 > .45$ adds $> (5/4)(.45 - .366)^2 = .00882$, more than the full hole loss. If none occurs there, holes start at ten and cost less than $\frac{1}{32}\int_8^\infty t^{-3}dt = 1/4096$. The weak-twenty point gives $x_1 > .378$ and adds $> (5/4)(.378 - .366)^2 = .00018$. The total is

$$e > 1.00010 + .00018 - 1/4096 > 1.0000358 > 1.$$

Each initial adjustment leaves positive decreasing slopes at both junctions. Effective-reference, tail and initial gains above belong to the stated disjoint branches; none is added twice. All three whole diagrams are therefore excluded on the original vector. □

B.11.10 Two small remainders and one extra prefix gain

Lemma B.11.10. *An actual A_{20} component cannot have spectator rank two.*

Proof. The arbitrary-spectator $N = 21$ reference in Lemma B.11.14 gives $e_A > .92274$. An ordinary irreducible rank-two factor costs $> 1/7$; two ordinary A_1's or $I + A_1$ cost $> 1/5$ jointly; a J pair costs $> .34$; and either actual typed defect costs $> .09$. These exhaust rank two and each exceeds .09. Orthogonality on the same original vector gives total energy $> .92274 + .09 = 1.01274 > 1$, a contradiction. □

Lemma B.11.11 (The final packet preserves an extra initial gain)**.** *With exactly the hypotheses of Proposition 11.1, let $G_0 = 3969/800000 = .00496125$. Then its same actual prefix energy is greater than $E_M - H_M/3200 + G_0$, for every $6 \le M \le 16$. This is a replacement bound, not an independent gain to add to a different initial-coordinate coupling.*

Proof. The weak-packet points and full scalar seals in the cap proof give $x_1 > .424$ for a weak packet at every $5 \le i \le 21$. Here we use only $i = M + 1 \le 17$. Its selected actual roots are $i, 1, 2$ for native five, $i, 1$ for seven/eight, i for the remaining finite owns, or the actual dotted pair. Its converted coordinates use old long neighbours $i - 1, i + 1$, which exist because $i + 1 \le r - 3$ in B and $\le r - 4$ in D. All foreign spectators have already been cleared and all point coefficients and wall signs are those of that pressure proof.

If five weak, the generic-prefix proof gives $L_M - E_M > .76 - .7 - 9/640 = 147/3200 > G_0$; at $M = 6$ use its separate direct $L_6 > .76, E_6 < .7$ comparison. No strong-five conclusion is needed in that branch. Now suppose five strong and put $K = M + 1$. If K strong, its reference gains an extra increment $d_M^2 > 1/(4M + 2) \ge 1/66$. At most one extra hole costs $1/3200$, and $G_0 + 1/3200 < 3/500 < 1/66$. This also accounts for the possible hole at six when $M = 6$. If K weak, M strong and the actual packet gives $x_1 > .424$. Modify only the initial five reference slopes to

$$.424, \quad (f_5 - .424)/4, \quad \dots, \quad (f_5 - .424)/4.$$

They are positive and decreasing at both junctions: $1.8 < f_5 < 1.805$, and the next slope is $< 1/4$ even after omitting its neighbour. The exact gain is $(5/4)(.424 - f_5/5)^2 > G_0$. The old hole intervals lie after five and have disjoint interiors, so their losses do not overlap this change. These are legal reference duals on one vector, with no tail or total-norm bootstrap. □

B.11.11 The complete forest accounting in dimensions twenty-three and twenty-two

Proposition B.11.12. *Dimensions twenty-three and twenty-two are impossible.*

Proof. At either dimension there is exactly one dominant component, by the same ordinary/debit and half-cost argument used above. The three spectator costs are still complete. For $S \geq 2$ the typed cost is larger than the short-rank-discard cost by $13/700$. For dominant A, the half-cost gives $S \leq 7$, so $r \geq 16$ at dimension twenty-three and $r \geq 15$ at twenty-two. The caps give $r \leq 20$. The following weaker consequences of already written component theorems suffice:

r	15	16	17	18	19	20
$e_A >$	4/7	.86	6/7	.91	6/7	6/7.

At dimension twenty-three, the weakest short-discard totals for $r = 16, 17, 18, 19, 20$ are respectively $.86 + 6/14$, $6/7 + 5/14$, $.91 + 4/14$, $6/7 + 3/14$ and $6/7 + 2/14$, all at least one and strictly exceeded. Every other debit case is stronger. At twenty-two, ranks fifteen through nineteen similarly use spectator ranks seven through three: $4/7 + 6/14 = 1$, $.86 + 5/14 > 1$, $6/7 + 4/14 > 1$, $.91 + 3/14 > 1$ and $6/7 + 2/14 = 1$, strictly exceeded. The last A_{20} rank-two remainder is exactly Lemma B.11.10. Thus no A remains.

For dominant B/D, the 9/14 floor gives $S \leq 5$. At twenty-three this means $r \geq 18$. The generic $M = r - 6$ floors for ranks eighteen/nineteen/twenty, even for the weaker D requirement, are $> .8689525, .88864, .90715$; their spectator ranks five/four/three give totals above one after either debit. At rank twenty-one the floor $> .9240775$ meets a rank-two remainder. A typed block costs $> .09$. An ordinary irreducible rank-two block costs $> 1/7$. Two A_1's have at least one ordinary cost $> .1$, including the possible short-rank discard; J is stronger. Thus all these totals exceed one. Rank twenty-two with its floor $> .9401975$ permits only isolated I, because an ordinary A_1 costs $> .1$. It is excluded by Proposition B.11.6. At rank twenty-three, B_{23} has its cap and whole D_{23} is excluded by Proposition B.11.9.

At dimension twenty-two, $r \geq 17$. Ranks seventeen/eighteen/nineteen have floors $> .8472225$, $.8689525, .88864$ and spectator ranks five/four/three. Their weakest costs $4/14, 3/14, 2/14$ suffice. At rank twenty the improved standalone prefix floor is

$$.90715 + G_0 = .91211125.$$

Every rank-two remainder costs $> .09$: a typed exception does so directly; an ordinary irreducible costs $> 1/7$; two A_1's include one nonshort cost $> .1$, and J is stronger. All totals exceed one. Rank twenty-one has only one spectator. An ordinary A_1 contradicts even the original $> .9240775$ floor, while actual I is excluded by Proposition B.11.6. At rank twenty-two only whole B_{22}, D_{22} remain, excluded by Proposition B.11.9. Every own-rank finite forest has now been covered. The argument never descends to an arbitrary face or uses a higher-dimensional cover at a different rank. □

B.11.12 One rational curve for the two remaining large-A couplings

We use the same central inventory with a different reference, rather than starting a new component classification. At $N = 20$ the target will be $e_A < .91$; at $N = 21$ no target beyond $e_A < 1$ is needed. The complete pure-helper inventory and its nonadjacency were proved before the 6/7 contradiction. Only the removal of near-four must be revisited. At $N = 20$ its mate and opposite anchor have height sum > 1.63, correlation $< .4$ and hence sum-direction square < 2.8. This contradicts $e_A < .91$, with margin .1089. At $N = 21$ the opposite anchor is fifteen or sixteen. At fifteen its height exceeds .79, since $D_{15} = 60/7 > (61/21)^2$, and its correlation with the near mate is $\leq 1/\sqrt{8} < .354$. Then $1.65^2 > 2.708$. At sixteen its height exceeds .77 and the correlation is $\leq 5/16$, so $1.63^2 > 2.625$. An opposite near-four mate has at least the latter height and smaller correlation. All these comparisons contradict $e_A < 1$. Thus the central weak indices are nonadjacent pure helpers on both stated domains, with arbitrary finite spectators retained.

Use the symmetric rational heights $H_{N-i} = H_i$, linear from zero to five and from $N-5$ to N:

N	H_5	H_6	H_7	H_8	H_9	H_{10}
20	2.128	2.288	2.406	2.488	2.536	2.552
21	2.150	2.317	2.445	2.537	2.597	2.626.

At $N = 21$, $H_{11} = H_{10}$. Every entry is strictly below $\sqrt{D_i} - .61$, by positive squaring, and successive differences decrease. The tight $N = 21, i = 7$ comparison, for example, has margin $28/3 - 3.055^2 = 925/3000000 > 0$. The coordinate reference is $w_j = (H_j - H_{j-1})/\sqrt{2}$. Summation by parts uses nonnegative actual strong-floor coefficients and gives $x_A.w \geq \|w\|^2$ at its available bends.

If end five is weak, replace that entire end by the straight line zero to six; six is strong. Do the reflected replacement at a weak $N-5$. Delete each remaining weak j by averaging its two neighbours. The exact energy loss is $(H_{j-1} - 2H_j + H_{j+1})^2/4$. Nonadjacency gives disjoint affected edge pairs, so these losses add for every permitted weak set. The following finite table records the full bound, not selected weak patterns. In the loss column, count is the maximum number of nonadjacent internal holes and the second factor is the maximum quarter-square on the remaining interval:

N	weak ends	initial energy	internal loss $\leq$	surviving energy $>$
20	0	.9544848	5(.000441)	.9522798
20	1	.9250917333...	4(.000441)	.9233277
20	2	.8956986666...	4(.000324)	.8944026
21	0	.981678	5(.00038025)	.97977675
21	1	.9528575833...	5(.00038025)	.9509563
21	2	.9240371666...	4(.000324)	.9227411.

For an exact rational formula behind every repeating decimal, put $a = H_5, b = H_6, T = \sum_{j=7}^{\lfloor N/2 \rfloor} (H_j - H_{j-1})^2$. The three initial energies are

$$a^2/5 + (b-a)^2 + T, \quad \tfrac{1}{2}[b^2/6 + a^2/5 + (b-a)^2 + 2T], \quad b^2/6 + T.$$

In the two-weak-end rows, six and $N-6$ are strong and the holes lie in $7..N-7$, giving four possible holes in either rank. The zero/one-end interval lengths give the other counts. Direct second differences of the displayed rational heights give the per-hole maxima. This proves completeness of the finite loss table without a combinatorial enumeration or an arithmetic packing premise.

An actual endpoint floor $x_1 \geq z$ improves a constant initial block of length m and coordinate mean b_0 by replacing its entries with $z, (mb_0 - z)/(m-1), \ldots$. If $z > b_0$, the gain is

$$\frac{m}{m-1}(z - b_0)^2.$$

If $z \leq b_0$, leave the block unchanged. All reference sums are preserved. For every $z \leq 1$ the remaining means exceed the next coordinates: at $N = 20$, the first-five and first-six comparisons are $> .126$ versus $< .114$, and $> .123$ versus $< .084$; at $N = 21$, they are $> .130$ versus $< .119$, and $> .127$ versus $< .091$. Deleting an internal neighbour only decreases that next coordinate. Thus the new duals are nonnegative at both junctions. The first and last blocks have disjoint supports, so their changes add without changing any intervening height.

Lemma B.11.13. *An actual A_{19} cannot coexist with either typed rank-two exception.*

Proof. The typed energy $> .09$ gives $e_A < .91$, the stated $N = 20$ domain. Zero or one weak ends already contradict that target by the table. With two weak ends its reference energy exceeds .8944026. Both typed exceptions give every other actual normalized direction height square greater than $(11 - 4\sqrt{7})/3 > 11/80$. For N_7 the stronger $> 1/5$ propagation suffices; for O_7 use the actual no-contact/native-height proof in Appendix B.1.9. The rational seal is $\sqrt{7} < 847/320$, with squared margin $609/102400$. At the two actual A_{19} ends this gives

$$x_1^2, x_{20}^2 > (19/20)(11/80) = 209/1600 > .36^2.$$

The first and last six-block means are $< .27$. Their legally funded disjoint modifications add $> 2(6/5)(.36 - .27)^2 = .01944$. The same-vector energy is therefore $> .8944026 + .01944 + .09 = 1.0038426 > 1$. Both typed roles, rather than a bare dihedral label seven, are covered. □

Lemma B.11.14. *An actual A_{20} has energy $> .92274$ and cannot coexist with an ordinary nonshort A_1 or an actual isolated I.*

Proof. The $N = 21$ window and table apply under $e_A < 1$ with arbitrary finite spectators. Their least energy is $> .9227411 > .92274$. An ordinary nonshort A_1 costs $> .1$, already contradictory.

For actual I, write its universal parameters as s, t: $e_I \geq s^2$, $t + (3/2)s > .767$, $0 < s < 1/3$ and $t < 1$. Both actual normalized end heights are at least t. Put $a = \sqrt{20/21} > .975$; then $x_1, -x_{21} \geq at$. Every endpoint change has $z = at < 1$, so the junction checks just proved apply. In the zero/one-weak-end rows the two absolute block means are $< .305$; in the two-weak-end row they are $< .274$. Each block length is five or six, so their two changes add at least $(12/5)(at - b)_+^2$, using $b = .305$ or $.274$ respectively. Weighted Cauchy, on these two changes and the *single* I cost, gives

$$s^2 + (12/5)(at - b)_+^2 > \frac{(a \cdot .767 - b)^2}{(9/4)a^2 + 5/12}.$$

Indeed $(at - b)_+ + (3/2)as > a \cdot .767 - b$. The denominator is $215/84 < 2.56$. With $b = .305$ the numerator exceeds $.442825^2$, so the gain is $> .076$. With $b = .274$ it exceeds $.473825^2$, so the gain is $> .0875$. The least final total is $> .9227411 + .0875 > 1.01024$; the zero/one-end totals are larger. No separate I charge was added at the second end, and no other component energy was reused. □

B.11.13 The proper D_{19}/D_{20} continuation

Lemma B.11.15. *For an actual D_{19} or D_{20} component, the .61 pure-helper window extends through fifteen or sixteen respectively, with arbitrary finite spectators. This continuation does not assume strong five.*

Proof. The valid $M = 13/14$ bulk reaches fourteen/fifteen. Its complete geometry supplies an actual strong eleven/twelve. In either initial-five branch there is a positive paid reference ending there, with

$$q_{11} > .8472225, \quad q_{12} > .8689525, \quad F_{11} \leq f_{11} < 2.886, \quad F_{12} \leq f_{12} < 3.034,$$

and remaining individual terminal coefficient $> 1/7$. For weak five use the actual $x_1 > .5965, S_6 > 2.01815$ interpolation; it has larger energy and terminal slope $1/(\sqrt{K} + \sqrt{6}) > 1/7$. No ordinary spectator energy has been used in these references.

At $i = r - 4 = 15, 16$, deletion retains $A_{i-1} + D_4$. Its finite extensions are three at any right leaf, or full endpoint D_r bridges. A foreign branch at the internal new node of such a rank-$r > 8$ bridge is not finite. Own zero keeps its actual cycle; positive owns are directly strong. Long-leaf three has its bulk strong/affine disposition. Fork three has residual $1/2 - Q$ and numerator $g + 1/4$;

own zero/three/four/six/parallel have the actual principals, all $Q > 0$ cases and native owns at least seven are directly strong. Both physical forks are retained. The only extra role is pure own-five/fork-three.

Put $g = \phi/2$, $\alpha^2 = \phi + 5/8$ and $\delta_5 = (5 - 3\phi)/8$. The actual left H_4 correction is $g(3n_i/2 + n_{i-1} + n_{i-2}/2)$. Its finite projected point has positive time, all original wall signs, and normalized displacement flat through $i - 3$, with

$$A^2 = i\alpha^2 - 1/2, \quad Y^2 < i\alpha^2, \quad A/\alpha > \sqrt{i - 25/112}, \quad \sqrt{\delta_5}/\alpha < .091.$$

Here $\alpha^2 > 56/25$. At fifteen this gives $G > 3.75, R = 15$; at sixteen it gives $G > 3.88, R = 16$. Both are flat through twelve. Pay $1/7$ at the early reference endpoint before adding the actual point. The two lower bounds $q_K + 2(G - f_K)/7 - (R - K)/49$ exceed 1.0122 or 1.029, respectively, for *both* $K = 11, 12$. Thus the extra role is excluded without a strong-five or later-tail assumption. The actual two-helper Schur calculation then extends nonadjacency to the newly supplied index; all right and spectator blocks stay positive. □

Lemma B.11.16. *The entire forests $D_{19} + N_7$, $D_{19} + O_7$ and $D_{20} + I$ are impossible in dimension twenty-one.*

Proof. For either typed defect beside D_{19}, weak five has actual prefix energy $L_{13} > .7600299945 + 1.156^2/7 > .95$, using $\sqrt{13} > 3.6055, \sqrt{6} < 2.4495$, already too much beside $> .09$. Hence five strong. Lemma B.11.15 supplies coverage through fifteen. If fifteen strong, its reference has energy $> .9240775$. If fifteen weak, fourteen strong and the actual pure packet gives $x_1 > .424$, so the legally modified initial reference gives $> .90715 + .00496125 = .91211125$. In both branches the separate typed energy $> .09$ contradicts one. This repeats the proved prefix-gain calculation only *after* supplying its missing D4 geometry; it does not invoke $M = 14$ at D_{19} outside that theorem's rank hypothesis.

For $D_{20} + I$, the same lemma first supplies coverage through sixteen in both initial-five branches. The positive actual long-root helper points, nonadjacency and initial I parameters are therefore exactly the geometric hypotheses used in the proof of Proposition B.11.6. Its weak-five reference at a strong fifteen/sixteen has energy $> .9851486$ and remaining P_6 price $> .12533$; the capped single-I graft gives $> .015$. Thus five strong. Its strong fifteen/sixteen flat-five reference has $q > .9240775$, actual P_5 demand f_5 and remaining price $> .146$. Deleting long one retains actual D_{19}, with just the same isolated/long-two-three head list and only the old I outside. The selected actual head roots and both genuine forks retain the local point signs, so the q-free head/I gain $> .07665$ closes it. We have replaced the missing $M = 15$ bulk justification by a proved actual D4 continuation, without applying its rank-restricted theorem to D_{20}. No D3 or terminal-seventeen floor is needed. □

B.11.14 One full weak-nineteen payment

Lemma B.11.17. *Suppose an actual pure helper at long node nineteen has own five, finite at least seven or dotted, $g < 6/5$, with actual long dual $\sqrt{2}P_{19}$ and actual selected roots n_{19}, n_1, n_2. Let a paid classical vector W be supported through eighteen, have sum F and retained actual P_{18} demand F, constant first three entries a, no foreign coordinate, effective energy $L > .985613$, and remaining P_{18} coefficient $> .107$. If $\sqrt{2}F \le 38/7$ and $a/\sqrt{2} < .267$, its full actual foot contradicts the norm budget. This is a conditional local pressure lemma, not a classification of node nineteen.*

Proof. The helper time is $A^2 = 38g^2 + 2g - 1/2$. Its actual finite or future cores, using the initial

and own long roots, give

own	U	T
5	$g(\sqrt{2}P_{19} - n_{19}) + H_3$	$H(A, 2/3 - g^2)$
$7, 8$	$g(\sqrt{2}P_{19} - n_{19}) + H_2$	$A - \sqrt{g^2 - 3/4}$
≥ 9 finite	$g(\sqrt{2}P_{19} - n_{19}) + H_1$	$H(A, 1 - g^2)$
dotted	$g\sqrt{2}P_{19} - g^{-1}n_{19} + H_1$	$A - \sqrt{1 - g^{-2}}$,

where $H_k = P_k/(k\sqrt{2})$ and $x.U \geq T$. These are the actual correction/finite-projection recipes already proved, now at their physical index nineteen. They require no B short normal. In type D, $\sqrt{2}P_{20}$ occurring in the coordinate sum is the sum of the two actual fork duals, and both fork products remain nonpositive under each correction.

Put $\alpha = \sqrt{2}g$ and $V = U - \alpha P_{18}$. The four variances are $g^2 + 1/6, g^2 + 1/4, g^2 + 1/2$ and $2g^2 - 3/2 + g^{-2}$; all are < 2.1, the last increasing to $1867/900$ at $6/5$. The helper mass in the first three coordinates is exactly $1/\sqrt{2}$, so the *full* cross is

$$W.U = \alpha F + a/\sqrt{2}, \qquad \Delta = T - W.U > T - g(38/7) - .267.$$

The following complete scalar seals prove $\Delta > .19$. For own five use $A > 5.097, \sqrt{\delta} < .111, g < .81$; for seven use $A > 5.663$, radical $< .253, g < .902$; for eight use $A > 5.8119$, radical $< .323, g < .9239$. Each substituted difference exceeds .19. For finite $g \geq .939$, $H - g(38/7)$ increases because $A' > \sqrt{38}$, $H_A > 47/50$ and $H_\delta < 0$. At .939, $A > 5.906$, the other time radical > 5.916 and $\sqrt{\delta} < .344$ give $H > 5.58$, again sufficient. For dotted owns the deficit decreases on $(1, 6/5)$: $A' - \sqrt{38} < .1$, $\sqrt{38} - 38/7 < .74$ and the subtracted radical derivative exceeds one. At its enlarged endpoint,

$$\Delta > 7.5246 - .55278 - (6/5)(38/7) - .267 > .1905 > .19.$$

These are positive-square and whole-interval derivative bounds, not sampled native labels or a numerical maximum.

Pay $\mu = 1/16$ at the true endpoint-eighteen coefficient. Its price $\mu\alpha < .107$ is available. Adding the actual U pressure and completing the *same* square gives

$$e > L + 2\mu\Delta - \mu^2 V^2 > L + .015546875 > 1.0011598 > 1.$$

The nonzero $W.V = a/\sqrt{2}$ is already included in Δ; no separate early-block or head energy has been added. □

B.11.15 The whole B_{21} terminal

Proposition B.11.18. *The entire farthest forest B_{21} is impossible.*

Proof. Lemma B.11.7 supplies the complete whole .58 bulk through seventeen, with isolated dotted five retained. There is an actual strong fifteen/sixteen, so the full paid-reference contract of Lemma B.11.8 applies. At eighteen the right B_3 permits long-three or sealed short-three F_4. Own zero has its actual affine F_4 principal and every positive own has ratio at least $1 + 1/\sqrt{2}$. Full endpoint bridges are directly strong, including own zero. Thus only pure helpers can be weak there.

At nineteen the right B_2 permits three at either endpoint. The long end is the same local row. At the short end, own zero/four/six/parallel have the actual F_4/star/path/null principals and owns at least seven are strong. Every full bridge has residual $1/38$, with minimum ratio $\sqrt{38}(1/38 + 1/(2\sqrt{2})) > 83/50$, including own zero. Own five has its actual left H_4 foot on

nineteen/eighteen/seventeen, $\alpha = \sqrt{2}g+1/2$, $A^2 = 19\alpha^2-1/2$ and $R < 19$. The squares $A/\alpha > 4.337$ and $\sqrt{\delta_5}/\alpha < .083$ give $G > 4.2$; it is flat through sixteen and is excluded by the early paid table.

Own three instead gives the same actual cap $v = (\omega_{19} + \omega_{21})/2$ at the finite A_{21} replacement of old twenty. Every original wall is retained and the released wall is strict. Its exact cosine $(20 + 19\sqrt{2})/\sqrt{22} > 466/47$ gives $x.P_{20} > 4.04, R = 20$, since $2299(466/47) > 22701$. The early fifteen/sixteen paid formula gives respectively $> 1.00655, > 1.00679$. This is an actual edge from the original vertex, not a farthest assumption at its mate. The complete helper-or-strong window therefore reaches nineteen.

One of eighteen/nineteen is strong. Weak helper five would give the earlier interpolation energy $> .74385 + .2679 > 1$. For isolated dotted five, the same zero-sum residual reference at a strong $K = 18, 19$ has these effective lower bounds:

J	initial base plus gain	$\text{tail}_{18} >$	$\text{tail}_{19} >$
6	.720462	.2679	.2804
7	.757957	.2318	.2445.

Its endpoint-J payment was legal at fifteen and becomes easier. At strong eighteen it has $L > .988362$ or $> .989757$; at nineteen it already exceeds one. Its sum is F_K, endpoint coefficient $> .14$, and its first three coordinates are constant $a = F_J/J + \mu_0(1 - 5/J) < .377$.

If nineteen weak, eighteen strong. For strong five its ordinary concave reference has $L > .97119 + .0154 - 1/1024 > .985613$, sum F_{18}, first-five mean $< .366$ and endpoint coefficient $> 1/9$. The isolated-dotted reference just described is stronger and also has first-three mean $< .377$. Both satisfy $\sqrt{2}F_{18} = 5.42 < 38/7$ and $a/\sqrt{2} < .267$. Every remaining weak-nineteen cap is an actual pure helper with the required physical roots. Lemma B.11.17 therefore excludes it, keeping the full early residual cross.

It remains strong nineteen; weak helper five and isolated dotted five were already excluded in that branch. For strong five, $\mathcal{E}_{19}(.58) > 1.00010$. If an internal helper occurs at six through nine, its actual $x_1 > .45$ gain .00882 pays the full hole loss $< 1/1024$. Otherwise every hole is at least ten; a helper's $x_1 > .378$ gain $> .00018$ pays $< 1/4096$. If there are no holes, no correction is needed. In every branch the legal positive reference has energy greater than one. □

B.11.16 Whole D_{21}: sum the exceptional feet before bounding their errors

Proposition B.11.19. *The entire farthest forest D_{21} is impossible.*

Proof. Use actual long roots through twenty and the other actual fork $n_{21} = (e_{20} + e_{21})/\sqrt{2}$. The genuine long duals stop at nineteen; the fork duals are $(P_{20} \mp e_{21})/\sqrt{2}$. The last coordinate is signed. This proof changes the loss to $c_1 = 4/7$ and writes $G_i = \sqrt{i} - c_1/\sqrt{2}$. We reclassify the weak directions at this threshold; no .58 classification is transported unchanged.

The sufficient ratio is

$$T_1(q) = 7/4 - \frac{q - 4/7}{q(2q - 4/7)}.$$

It increases on $q \geq \sqrt{10}$; positive-denominator comparisons give $T_1(q) < 1.673$ through $q = \sqrt{38} < 37/6$, $T_1(q) < 1.63$ through $q = \sqrt{14} < 15/4$, $T_1(4) < 1.64$ and $T_1(\sqrt{12}) < 1.62$. The actual bulk 5..16 has retained $A_{i-1} + D_{21-i}$, with right rank at least five. Its complete finite-white list, zero-own principals and full rank-twenty-one bridges are already proved local geometric statements. There is no foreign component.

Near-three has the same increasing least ratio $[i(\phi + 1) - 1]/\sqrt{2i(i + 1)}$. At five/six native owns at least seven are strong; at seven the own-five ratio is > 1.63, at eight it is $> 1.66 > T_1(4)$, and

from nine it is > 1.68. Thus only near-three/own-five at five and six need points. Next-to-end white-three has least ratio $(i\phi/2+1)/\sqrt{2i}$, strong from six onward. At five retain its actual floor, rather than call it strong:

$$h_5 \geq \sqrt{10} - c_D, \quad c_D < .57549.$$

Indeed its cosine is at least $1 + 5\phi/2$, giving $h_5^2 \geq 10(5\phi)/(5\phi+4) > 6.6915 > 2.58679^2$; use $\phi > 1.61803$ and $\sqrt{10} < 3.16228$. Exceptional positions have ratio $2g + 3/8 > 1.673$. The far-four/near-four, right long-three, finite D fork and bridge ratios or actual principals are unchanged geometric rows; their lower bounds now exceed the displayed T_1 targets. In particular a fork at retained rank at least five has ratio $> (.8+.25)/\sqrt{3/8} > 1.673$ after its own-zero/three/four/six/null cases. Full endpoint bridges retain their positive-ratio own-zero branch or the actual affine cycle of the other orientation.

Isolated finite five and seven have the preceding losses $< 25/76 + 5/24$ and $< 1000/4779 + 9/25$, both below $4/7$. For finite owns at least eight, use the exact singleton projection. With $r = \sqrt{q^2-1}$ and $s = \tan(\pi/m) \leq \sqrt{2} - 1$, its positive finite correction gives

$$h_i > r\sqrt{(r-s)/(r+s)}.$$

To derive this, select the actual cap and n_i. Their residual is $1 - g^2 > 0$, the displacement after correction is $g(\omega_i - n_i)$, and the finite-point formula divided by g is $\sqrt{r^2-s^2}/(1+s/r)$, the displayed expression. Its loss decreases in q and increases in s. At $q = \sqrt{10}$, $r = 3, s < .415$, it exceeds 2.6 by $9(517/683) > 169/25$, giving loss $< 3.163 - 2.6 = .563 < 4/7$. Owns three/four/six/parallel retain their actual affine/null exclusions.

For isolated dotted own, $u = g^{-2} \in (0,1)$, the positive future core gives the actual pressure

$$x.(\omega_i - un_i) \geq T_i(u) = \sqrt{2i-u} - \sqrt{u(1-u)}.$$

At $i \geq 8$ its loss is at most $a_i u + \sqrt{u(1-u)}$, $a_i = 1/(\sqrt{2i} + \sqrt{2i-1}) < .128$, whose maximum is $< .5681 < 4/7$. Retain isolated dotted five/six/seven. Every other weak bulk row is the pure helper, with $\phi/2 \leq g < 6/5$; the upper cutoff is reproved from $4(6/5) - 2 = 14/5 > (1.673)^2$.

The two near-three points. The own-five near role at five was excluded outright by the actual right-H_4/old-floor argument in Lemma B.11.7. At six use the actual right correction $g(3n_6/2 + n_7 + n_8/2)$, $g = \phi/2$. Its finite point retains old five/nine and every unknown wall. For its displacement Y,

$$A^2 = 9/2 + 8\phi, \quad Y^2 = (41 + 61\phi)/8, \quad \sum Y_j = \sqrt{2}(5/2 + 3\phi),$$

giving $x.Y > 4.04, Y^2 < 17.463, \sum Y_j < 10.401$. Here $A > 4.1766$, $\sqrt{\delta_5} < .136$, $1.618 < \phi < 1.61804$ and $\sqrt{2} < 1.414214$ suffice. The valid old .61 bulk supplies a strong twelve/thirteen, with floors $> 3.03, > 3.17$. For $Y + (3/4)P_K$, the squared demands are $39.84765625, 41.18430625$, and the norm squares are $< 39.8145, < 40.377$. Both contradict Cauchy before a new tail or strong-five conclusion is used.

New weak helpers remain nonadjacent: their actual mutual product is $> 1 - ab > -11/25$, hence zero, and the positive mixed-point loss is $< 7/16 + 1/(4\sqrt{10}) < 4/7$. Isolated dotted five/six/seven need not be nonadjacent. We now retain all their feet together.

The summed-foot identity and its complete error bounds. Let w be a concave formal prefix reference, with nonnegative coefficients β_i in the spatial duals ω_i. At each selected isolated dotted index use its actual pressure, put $Z = \sum \beta_i u_i n_i$ over those indices and $W = w - Z$. Since $w.n_i = \beta_i$ and distinct old root products are nonpositive, $Z^2 \le \sum \beta_i^2 u_i^2$. Summing the actual pressures and completing the single square gives effective energy

$$L_W \ge w^2 + \sum_i \{2\beta_i[T_i - (q_i - c_1)] + (2u_i - u_i^2)\beta_i^2\}.$$

An actual D-position-five floor instead costs $2\beta_5(c_D - c_1)$. These are pressure certificates for the same W, not separate scalar lower bounds to add later. All dotted corrections have support through eight and sum zero; they preserve the late endpoint coefficient and vector sum. Their mutual cross terms have a favourable sign, even for adjacent dots.

For the retained G_i-floor reference when five is not a weak helper, the actual slope differences give $\beta_5 > .1, \beta_6 > .012, \beta_7 > .0095$; the latter follow from $(d_5 - d_6)/\sqrt{2}$ and $(d_6 - d_7)/\sqrt{2}$. Omitting a neighbour increases them. The separate weak-helper-five interpolation below charges only its dotted-six correction, whose actual coefficient exceeds .10 and hence .012. It charges no dotted-seven pressure. These bounds concern the two specified references, rather than an arbitrary concave vector. Write $r(u) = u - u^2/2$. Each signed summand is $2\beta[c_1 - \mathrm{loss}(u) + \beta r(u)]$. For $u \le 3/4$, $r(u) \ge (5/8)u$. At five, $a_5 < .163$ and $\beta \ge .1$ bound the effective loss by $.1005u + \sqrt{u(1-u)} < .5528 < c_1$. At seven, $a_7 < .137, \beta \ge .0095$ similarly give $.1310625u + \sqrt{u(1-u)} < .56983125 < c_1$. For $u \ge 3/4$ the same expressions decrease, so both summands are nonnegative throughout their actual intervals.

At six the summand increases with $\beta \ge \beta_0 = .012$: its derivative is positive because $\mathrm{loss}(u) - 2\beta_0 r(u)$ is bounded by $.133u + \sqrt{u(1-u)} < .57095 < c_1$ on $u \le 3/4$, and decreases thereafter. At β_0 the effective loss is $< .1405u + \sqrt{u(1-u)} < .5752$, again with a decreasing tail. Its entire debit is therefore

$$2(.012)(.5752 - 4/7) = 99/1093750 < .000091.$$

All maxima used here are the elementary identity $\max[au + \sqrt{u(1-u)}] = (a + \sqrt{1+a^2})/2$. No individual dotted-six floor was assumed.

The early reference and the actual terminal prices. First suppose five is not a weak helper. Keep formal five, whether ordinary, D-position or isolated dotted, and keep dotted six/seven. Omit only nonadjacent weak-helper floors. Through a strong fifteen/ sixteen their entire loss is bounded by the five decreasing terms $j = 6, 8, 10, 12, 14$, whose sum is $< .000422$. For D-position five, $\beta_5 < .11425$: the first mean is $< .366402$ and the next retained slope, at six or seven, is $> .20484$. Thus its separate debit is $< 2(.11425)(.57549 - 4/7) < .000929$. The common initial-block change from .61 adds

$$(27/700)\sqrt{2/5} - [(.61)^2 - (4/7)^2]/10 > .0198377.$$

The old exact $E_{15} > .92564, E_{16} > .94176$, minus all three conservative errors, give a paid W supported through K, of sum G_K and effective energy

$$L_W > Q_{15} = .944035, \qquad L_W > Q_{16} = .960155.$$

Subtracting both dotted and D-position errors is conservative even when their five cases cannot coexist. The true individual P_{15} coefficient is at least $d_{14} > 9/70$, and P_{16} at least $d_{15} > 1/8$; the early corrections do not change them.

If five a weak helper, its actual scalar point gives $x_1 > .530$: subtract at most $(.58 - 4/7)\sqrt{2}(6/5)$ from .5455. Six is strong or isolated dotted. Interpolate through $(1, .530), (6, G_6), (K, G_K)$ and use

its actual summed dotted-six foot if necessary. The first-six energy is $> .74018$ because $G_6 > 2.0454$; the fifteen/sixteen tails are $> .225, > .2404$. Even after .000091 these exceed both Q_K. The endpoint coefficient $1/(\sqrt{K} + \sqrt{6})$ also exceeds the same required prices. Thus every initial branch has been covered.

For an actual normalized pressure flat through sixteen, with demand G and full norm square $\leq R$, the positive-dual payment gives

$$e > L_W + 2\tau(G - G_K) - \tau^2(R - K).$$

Use $G_{15} < 3.468923, G_{16} < 3.595940$. The legal price is $\tau = 1/8$ at both endpoints, except native-eight at the centre, where $\tau = 9/70$ is used at fifteen. Effective energy is never identified with W^2.

Actual D4 at seventeen and D3 at eighteen. At seventeen, the complete D_4 leaf-three/bridge classification and the old affine principals leave fork-three owns five/seven/eight. Owns at least nine have ratio $\sqrt{2}(\cos(\pi/9)+1/4) > 1.68 > 1.673$. Own five has its actual left H_4 point, $A^2 = 17\alpha^2 - 1/2$, $G > 4, R < 17$, flat through fourteen. It uses the old .61 early eleven/twelve contract in both initial branches, not the new sixteen-flat formula. Its two totals exceed 1.04. For seven/eight use the actual future core

$$c + g^{-1}n_{17} + (2g)^{-1}n_{16}, \quad k = 1 - 3/(4g^2) > 0.$$

The cap/sixteen products vanish and the seventeen product is negative. For $.9 < g < .924$, $A/\alpha > 4.09$, $\sqrt{k}/\alpha < .21$, so $G > 3.88, R < 17$. It is flat through fifteen, and the old eleven/twelve price $1/7$ gives both totals > 1.0086. No incorrect sixteen-flatness has been used.

At eighteen the right D_3 finite list is middle-three, end-three/ four/five, with every full bridge retained. Middle-three and end-four have their actual affine/direct alternatives; end-five is directly strong even at own zero. A full middle bridge has residual $1/36$ and is directly strong at zero own. An end-three bridge has residual $1/8 + 1/36$, zero-own cycle and positive-own ratio > 1.9. The no-left end-three case has the old own-zero/three/four/six/null principals. Own five has its left H_4 point, $G > 4, R < 18$, flat through fifteen; the old early reference gives both totals > 1.022. Owns $g \geq 43/40$ are directly strong. For $.9 < g < 43/40$ use $c + g^{-1}n_{18} + (2g)^{-1}n_{17}$, with

$$A^2 = 18\alpha^2 - 5/8, \quad A/\alpha > 4.21, \quad k/\alpha^2 < 519200/5193841 < .317^2.$$

This actual core is flat through sixteen, $G > 3.893, R < 18$. The new paid formula exceeds 1.0031 at both early endpoints. Thus eighteen has only helper or strong as well.

The complete actual centre nineteen. Deleting nineteen retains $A_{18} + 2A_1$. With own g, one left contact η at j and fork contacts δ, ϵ, the data are

$$\Delta = 1 - q_L - \delta^2 - \epsilon^2, \quad \sigma = g + \eta j/19 + (\delta + \epsilon)/2, \quad C = \sqrt{38}\sigma/\sqrt{\Delta}.$$

The complete finite-white list is the same actual high-rank tree list already proved: left end-three/four or next-to-end-three; with both forks only the full endpoint D_{21} bridge; with one fork only left endpoint-three and fork-three/four. The full and fork-four bridges have residual $1/38$ and minimum numerators $1/2 + 1/38$, $\sqrt{2}/4 + 1/38$, directly strong. Fork-three bridges have residual $21/76$; own zero/three give the cycle/triangle and all larger owns have ratio > 1.8. With no fork contact, the rechecked left and isolated inventory applies. Its helper feet use $\omega_{20} + \omega_{21} = \sqrt{2}P_{20}$ only as a spatial identity, retaining both actual forks.

With no left contact the white list is one arbitrary finite fork, or pairs $(3,3),(3,4),(3,5)$. At own zero, fork-three/four/six have their actual affine principals. Fork-five uses the actual left H_4 point, $R=18$, $A^2/\alpha^2=5+8\phi$ and $\sqrt{\delta_5}/\alpha<15/56$, giving $G>3.96$. Its corrected rational premise is $\phi>79/49$, as proved above.

For fork at least seven, put $b=\delta^2$. The actual core $c+\delta^{-1}n_f+2(\delta^{-1}-\delta)n_{19}$ has selected products zero/negative, positive time and negative square. Its full signed data are

$$R=29-16/b+6/b^2,\quad G=\sqrt{21-2/b}-\sqrt{-8+14/b-6/b^2}.$$

Monotonicity of R and the quadratic-radical maximum give this complete native split:

fork label	b	$G>$	$R<$ or $\leq$
7	$(.81,.82)$	3.91	18.42
8	$(2+\sqrt{2})/4$	3.91128	18.49034
9	$(.88,.89)$	3.925	18.60
≥ 10	$(.9,.95]$	3.948	18.81
≥ 10	$[.95,1)$	4.04	19.

For seven the two roots are $>4.304,<.388$; for nine, $>4.327,<.402$; for the last two ranges, $>13/3,<.385$ and $>4.346,<.298$. The original native cosine bounds follow from their triple-angle polynomials and positive squares. The critical eight row is not replaced by a coarse continuous bound:

$$R=109-64\sqrt{2},\qquad G=\sqrt{13+4\sqrt{2}}-\sqrt{68\sqrt{2}-96}.$$

The rational enclosure $1.41421356<\sqrt{2}<1.41421357$ gives roots $>4.319358,<.408078$ and the displayed R upper bound. At fifteen the price $9/70$ gives energy >1.00008648; at sixteen the price $1/8$ gives >1.00007843. Every other row gives both totals >1.00085.

At own three, fork-three gives the triangle. Fork-four/five instead gives the actual finite D_{21} contacted-fork endpoint, $C_f=(19+21\delta)/2>16$, signed direction $\sqrt{2}\omega_f$, $R=21, G>\sqrt{315/17}>4.30$. Fork at least six is direct. Own four/six/parallel retain their actual principals. For own at least five and fork at least four, only own-five/ fork-four is not directly strong; its left H_4 point has $\alpha=\sqrt{2}g+1/2$, $A^2=19\alpha^2-1/2$, $G>4.25, R<19$. Own-five/fork-three has $A^2=19\alpha^2-3/4$, $G>4.2, R<19$. Their lowest selected old root is seventeen, so both are genuinely flat through sixteen. For remaining finite/dotted own with fork-three, $g\geq 6/5$ is direct; on $.9<g<6/5$ the actual core $c+g^{-1}n_{19}+(2g)^{-1}n_{18}$ has $R<19$, $A/\alpha>4.326$ and $\sqrt{k}/\alpha<.34$, hence $G>3.98$.

For two forks, own three has its triangle and larger owns are direct. At own zero pair-three-three has its square; pair-three-four has an actual finite B_{21} mate with $C_f=(38+21\sqrt{2})/4>16$, again $G>4.30, R=21$. Pair-three-five has its actual finite H_3 foot with $R=19, q=2\phi^{-6}<1/8$ and $G>\sqrt{19-1/8}/(1+\sqrt{1/152})>4$. Every row is excluded by the stated positive-dual early formula. The actual list is exhausted and nineteen is helper or strong.

The final summed reference. One of eighteen/nineteen is strong. Weak helper five gives effective energy $>.74018+.2679-.000091>1.0079$, so is impossible. Otherwise the same formal reference and summed-dot correction apply through that endpoint. Their finite nonadjacent hole loss is $<.00044$, by appending the sixteen/eighteen terms as needed. The same .000091 and .000929 errors remain valid. Consequently

$$L_{18}>.9895677,\qquad L_{19}>1.0030777.$$

Thus nineteen cannot be strong. In the weak-helper case at nineteen, the paid eighteen reference has sum G_{18}, true retained demand G_{18}, first-three mean $a = G_5/5 < .367$ and unchanged endpoint coefficient $> d_{17} > 1/9$. The summed dots only affect coordinates five through eight. Hence $a/\sqrt{2} < .260 < .267$ and $\sqrt{2}G_{18} = 38/7$. The actual helper uses the physical long n_{19} and both signed fork directions; its P_{20} coordinate sum is not declared a long old dual. All hypotheses of Lemma B.11.17 now hold, with the stronger $L > .9895677$. Its single paid full-foot square contradicts the norm budget. All summed errors and terminal prices have been charged once, with no arithmetic or unproved dotted floor. □

B.11.17 The complete exact-twenty-one cover and its consumers

Proposition B.11.20. *Dimension twenty-one is impossible.*

Proof. The common ordinary/debit costs exclude a forest without a dominant block, and two dominant blocks cost more than one. After establishing uniqueness, write the dominant rank as r and spectator rank as $S = 21 - r$. The complete three spectator budgets remain $S/14, (S - 1)/14, .09 + (S - 2)/14$, with their actual incompatible defect definitions. A dominant A has $S \le 7$ and hence $14 \le r \le 20$. Modern written component bounds make its lower ranks immediate: use $5/7, .826, .86, 6/7, .91$ at $r = 14, 15, 16, 17, 18$ respectively. With the weakest short-discard spectator costs $6/14, 5/14, 4/14, 3/14, 2/14$, every total is strictly greater than one. Typed and fully ordinary cases are stronger. Thus the older detailed rank-fifteen/sixteen spectator partitions need not be reproduced as separate consumer arguments.

At A_{19}, an ordinary irreducible rank-two remainder costs $> 1/7$ beside $> 6/7$. Two ordinary A_1's cost $> .2$; actual $I + A_1 > .2$ is a single joint bound, and J is stronger. Both remaining typed exceptions are exactly Lemma B.11.13. At A_{20}, the sole ordinary or actual-I rank-one remainder is excluded by Lemma B.11.14; J and typed exceptions need two ranks. This exhausts the A side at its own rank.

For B/D, $S \le 5$ and $r \ge 16$. The valid D-admissible $M = r - 6$ floors at sixteen/seventeen/eighteen are $> .823735, .8472225, .8689525$. Their weakest short-discard spectators cost $4/14, 3/14, 2/14$, already too much. At B_{19}, its genuinely valid $M = 14$ improved floor $> .91211125$ closes an ordinary rank-two block, a two-A_1 remainder with its at least one nonshort cost $> .1$, or either typed block $> .09$. At D_{19}, its valid $M = 13$ floor $> .88864$ closes an ordinary irreducible remainder. A two-A_1 remainder costs $> .2$ if ordinary, $> .2$ jointly if it contains actual I, or $> .34$ if it is J. The two typed cases are the separately proved actual D4 continuation in Lemma B.11.16.

At rank twenty even the weaker common floor $> .90715$ contradicts an ordinary nonshort A_1 cost $> .1$. The remaining entire $B_{20} + I$ is covered by Proposition B.11.6; $D_{20} + I$ is covered by Lemma B.11.16, after its own missing D4 layer was supplied. At rank twenty-one there are no spectators. Propositions B.11.18 and B.11.19 exclude those whole diagrams. Every finite farthest forest of rank twenty-one is therefore impossible. There is no appeal to face inheritance or to a necessary cover computed at another dimension. □

B.11.18 The proper classical forests in exact dimension twenty

Lemma B.11.21. *The entire forests $B_{18} + N_7/O_7$, $D_{18} + N_7/O_7$, $B_{19} + I$ and $D_{19} + I$ are impossible in dimension twenty.*

Proof. We use the old loss .61 and true actual-rank continuations. For B_{18}, the valid bulk reaches fourteen. At fifteen the right B_3 permits long-three or the sealed short-three F_4. The latter has residual $1/4$, numerator $g + 1/(2\sqrt{2})$: own zero has its actual affine principal and every positive

own has ratio at least $1 + 1/\sqrt{2}$. All full endpoint bridges are finite B_{18}, foreign-sealed and directly strong, including their zero-own ratio. Thus coverage/nonadjacency reach fifteen.

For D_{18}, the valid bulk reaches thirteen. At fourteen the complete right D_4 list and its affine/direct branches leave only pure own-five/fork-three. Its actual left H_4 point has $\alpha^2 = \phi + 5/8$, $A^2 = 14\alpha^2 - 1/2$, $R < 14$ and $G > 3.62$, using $\alpha^2 > 56/25$ and residual ratio $< .091^2$. It is flat through *eleven*, so choose an actual strong ten/eleven, not twelve. The early positive-dual formula and the separate typed energy $> .09$ give both totals > 1.085. Both initial-five branches admit those same lower reference bounds. The finite white list retains both forks, every foreign cost and every full bridge; the non-direct role has $Q = 0$ by the strong foreign estimate, not by an assumption about the typed block.

At fifteen the right D_3 permits middle-three or end-three/four/five. Middle and end-four have their actual affine/direct branches; sealed end-five is strong even at own zero. Full bridges retain their zero-own cycles and direct positive-own bounds, with no finite extra branch at the internal cap. The end-three no-left role has residual $5/8 - Q$, numerator $g + 1/4$; positive Q and $g \geq 1$ are strong and own zero/three/four/six/parallel have their actual principals. Own five gives its actual left H_4 pressure $G > 3.74, R < 15$, flat through twelve. For finite $.9 < g < 1$, average both actual A_2 cores on sides fourteen/sixteen, retaining their full cross $8g^2/9$. With $d = 4g^2/3 - 1$ their data are

$$A^2 = 15\alpha^2 - 5/8, \quad Y^2 = A^2 + 1 - 14g^2/9 < 15\alpha^2, \quad x.Y \geq A - \sqrt{d}.$$

The bounds $A/\alpha > \sqrt{15 - 125/529} > 3.842$ and $\sqrt{d}/\alpha < 1/3$ give $G > 3.5, R < 15$. At actual strong eleven/twelve, the pure-point formula plus the orthogonal typed cost gives both totals > 1.099 or > 1.030, respectively. Each core is a positive combination of actual walls, future feasible against the entire typed block and every other wall. Thus through fifteen only pure helpers can be weak.

The prefix-gain *calculation*, now with that actual extended coverage, gives $e_{B/D} > .90715 + .00496125 = .91211125$. If five weak its actual reference is stronger; if fifteen strong the extra increment pays the gain; otherwise the actual weak-fifteen packet supplies the legal $x_1 > .424$ modification at strong fourteen. The physical long roots through sixteen exist in both components. We have not applied the old $M = 14$ theorem below its original ranks. Adding either typed cost $> .09$ excludes all four typed forests.

For entire $B_{19} + I$, the valid bulk reaches fifteen. Its actual B_3 layer at sixteen has the same long-three/sealed-F_4/full-bridge list as just proved, now at its physical rank; all remaining weak caps are pure helpers. Their actual nonadjacency extends through sixteen. The scalar argument in Proposition B.11.6 therefore applies after this independently supplied geometric window: weak five has the paid $> .9851486$ reference and unused-I gain $> .015$; strong five has the flat fifteen/sixteen reference $q > .9240775$, true P_5 demand f_5 and individual price $> .146$. Deleting the head retains actual B_{18}, with precisely the isolated/long-two-three whitelist and only the old I outside. The q-free local head/I gain $> .07665$ gives the contradiction. This is a proof at supplied actual geometry, not the generic $B_r \geq 20$ theorem asserted at $r = 19$.

Finally consider entire $D_{19} + I$. Lemma B.11.15 first supplies coverage through fifteen. At sixteen the right D_3 has the same complete finite, foreign, affine and bridge list as above. Only pure own-five or finite $.9 < g < 1$ in the end-three row remain. Own five has its actual left H_4 point, $G > 3.87, R < 16$, flat through thirteen; the bare eleven/twelve formula already exceeds 1.026. For finite owns use both actual A_2 cores on sides fifteen/seventeen, with their full cross, $A^2 = 16\alpha^2 - 5/8$, $R < 16$ and $G > 3.63$, since $A/\alpha > \sqrt{16 - 125/529} > 3.97$ and the radical ratio is $< 1/3$. If five weak, its actual eleven/twelve references have energies $> .9104, > .9314$, so their pure-point totals exceed 1.020. If five strong, the flat-five reference can first use the single initial-I

adjustment: from $t + (3/2)s > 3/4$ its joint gain is

$$s^2 + (5/4)(t - .361)_+^2 > 151321/3050000 > .0496,$$

with a legal clip at .6 when necessary. Its support first five plus I is orthogonal to the actual tail residual. It uses only the available P_5 price, leaving the early terminal price intact. The pure-point formula plus this one joint gain gives both totals > 1.007. Thus coverage/nonadjacency reach sixteen in either branch.

The same independently checked scalar bootstrap and flat-reference argument now prove five strong and give $q > .9240775$. Deleting long one retains actual D_{18}, so the same local head/I kernel, its selected long roots and both signed forks apply and give $q + .07665 > 1$. Earlier tail contradictions were used only to remove their branches; no spent I gain is carried into this new final reference. This whole-forest argument stops before centre seventeen; its older mixed-H_4 catalogue remains preserved but is unnecessary for this composition with the later local head theorem. □

B.11.19 The two rank-one remainders beside A_{19}

Lemma B.11.22. *An actual A_{19} has projection energy $> 563/625 = .9008$, with arbitrary finite spectators. In particular it cannot coexist with an ordinary nonshort A_1.*

Proof. Suppose $e_A \le .9008 < .91$. The complete $N = 20$ window and near-four comparison above apply. Zero or one weak end already has reference energy $> .9522798$ or $> .9233277$. Thus both five/fifteen are weak. The paid rational reference w has energy $E > .8944 = 559/625$, first six coordinates $H_6/(6\sqrt{2})$ and last six their negatives, where $H_6 = 2.288$. Its internal weak holes have already been paid. For the actual weak cap at five, either helper assignment gives

$$A^2 = (15/2)\gamma^2 + C\gamma - 21/40, \quad C \in \{1/2, 3/2\}, \qquad K = w.v = \frac{143}{750}(10\gamma + 1).$$

The positive corrections below use only old n_3, n_4, n_5, so have zero product with this flat reference. For $Y = v - u$ we therefore have $w.Y = K$.

For a finite selected correction, $S^2 = Y^2 = A^2 + \delta$, and its actual finite projected point gives the normalized pressure $x_A.(Y/S) \ge g_0$, with $g_0^2 = (S - \sqrt{\delta})/(S + \sqrt{\delta})$. For a future core of square $-d$, $S^2 = A^2 - d$ and $g_0 = (A - \sqrt{d})/S$. We prove the full overlap gap $g_0 - K/S > .08$ on every own range.

Native five uses the actual left H_4 correction $\gamma(3n_5/2 + n_4 + n_3/2)$, with $\delta = (5 - 3\phi)/8$. The smaller correlation $C = 1/2$ gives $S^2 = 79/40 + 7\phi/4 > 2.19^2$, $\sqrt{\delta} < .14$ and $K < 13013/7500$. Hence $K/S < .8$ and $g_0^2 > 93/107 > .93^2$, a gap $> .13$. Native seven uses $\gamma(4n_5/3 + 2n_4/3)$, $d = 4\gamma^2/3 - 1 > 0$. Its selected products are zero/negative and its time is positive. From $.9 < \gamma < .902$ we have $A > 2.449$, $\sqrt{d} < .292$, $K < 1.911$ and $S^2 = (37/6)\gamma^2 + C\gamma + 19/40 < 977/120 < 9$. Thus $(A - \sqrt{d} - K)/S > .246/3 > .08$.

Every finite own at least eight uses the actual finite pair correction $u = \gamma n_5$, $\delta = 1 - \gamma^2$. Here $S^2 = (13/2)\gamma^2 + C\gamma + 19/40 > 2.5^2$, $\sqrt{\delta} < .4$, so $g_0^2 > 21/29 > .85^2$. The exact lower overlap polynomial is

$$.77^2 S^2 - K^2 \ge \frac{1573}{7200}\gamma^2 - \frac{387563}{900000}\gamma + \frac{4414927}{18000000} > .218(\gamma - 1)^2 + .005\gamma + .027 > 0.$$

Thus $K/S < .77$ and the gap is $> .08$ on the whole finite interval.

For dotted own $1 < \gamma < 27/25$, use the actual future core $c + \gamma^{-1} n_5$. It has zero cap product, negative old-five product and positive time. Now $S^2 = (15/2)\gamma^2 + C\gamma - 61/40 + \gamma^{-2} > 7.325 > 2.7^2$

and $\sqrt{d} < .38$. Since $A > S$, $g_0^2 > 58/77 > .86^2$. Multiplying the overlap difference by γ^2 gives the bound

$$\gamma^2(.77^2S^2 - K^2) > .0319 + .069(\gamma - 1) + 2.632(\gamma - 1)^2 + 2.813(\gamma - 1)^3 + .811(\gamma - 1)^4 > 0.$$

The rounded quartic coefficients come from $146047/180000, -387563/900000, -16929473/18000000$, $.5929$, so every coefficient comparison is rational. This gives a gap $> .09$ for the entire dotted interval.

All selected finite points or future cores are feasible against every actual wall by their positive corrections, and have pure A displacement because $Q = 0$ was proved in the inventory. For their unit $y = Y/S$, pay $\mu = .08$ in the same vector:

$$e_A \geq E + 2\mu(g_0 - w.y) - \mu^2 > 559/625 + 4/625 = 563/625.$$

This contradicts the contrary target. A separate nonshort A_1 costs $> .1$, giving total > 1.0008. □

Lemma B.11.23. *An actual A_{19} cannot coexist with an isolated I, with arbitrary further finite spectators.*

Proof. Write the actual I parameters as s, t, so $y_I \geq s$, $t + (3/2)s > .767$ and both normalized end heights are at least t. Put $a = \sqrt{19/20} > .974$, giving $x_1, -x_{20} \geq at$. We first remove every near-four without assuming $e_A < .91$. Its actual mate has index $j = 4, 5$ and height $> .86$; at the opposite end there is a mate or a strong $k \geq 14$ of height $> .77$. Except $(j, k) = (5, 14)$, the exact dual correlation is at most $1/3$. The positive vector $(543/800)u_j + (435/800)u_k$ has pressure $> 80193/80000$ and norm square at most that same value. One completed square gives energy greater than one.

At the exceptional pair, the actual heights exceed $2.355, 2.231$; their positive squared margins are $.000975, .002999$. The concave reference through those heights at five/fourteen has

$$E = \tfrac{1}{2}[2.355^2/5 + .124^2/9 + 2.231^2/6] > .97.$$

If $s \geq .18$, its orthogonal I energy already gives > 1.0024. If $s < .18$, the barrier gives $t > .497$ and $-x_{20} > .484$. The last-six mean is $< .264$; its legal endpoint modification adds $> (6/5)(.484 - .264)^2 = .05808$. The remaining mean is $> .217$ and the middle slope $< .01$, so the new reference stays concave. This also exceeds one. Reflection treats every orientation. The complete weak window is therefore pure-helper and nonadjacent.

Use the same full rational curve and omission table at $N = 20$. If zero or one ends are weak, its energy is at least the weaker bound $> .9233277$. The two endpoint block means are $< .302$ and each length is five or six. Put $z = (at - .302)_+$. Their disjoint legal changes add at least $(12/5)z^2$. Weighted Cauchy with the single I height gives

$$s^2 + (12/5)z^2 > \frac{(a \cdot .767 - .302)^2}{(9/4)a^2 + 5/12} > \frac{.445058^2}{511/200} > .0775.$$

The exact denominator is $613/240$; the last cleared margin is $.000064123364$. Thus both end indices must be weak.

At five write the actual cap as $v = \gamma\omega_5 + \omega_e/2$, $e = 1$ or nineteen, with $4/5 < \gamma < 27/25$ and $Q = 0$. Its time is $A^2 = (15/2)\gamma^2 + C\gamma - 21/40$, with $C = 3/2$ for the near helper and $1/2$ for the far helper. Native five uses its actual H_4 projection with pressure $H(A, 2/3 - \gamma^2) > A - .111$. For finite owns at least seven, the positive actual core

$$c + \gamma n_5 + 2(1 - \gamma^2)n_e$$

has cap/five products zero, helper product $3/2 - 2\gamma^2 < 0$, square $-(1-\gamma^2)(4\gamma^2-3)$ and positive time. Its radical is at most $1/4$. For dotted own use $c + \gamma^{-1} n_5$ and radical $\sqrt{1-\gamma^{-2}}$. Positive old corrections allow the common left pressure $\gamma h_5 + h_e/2$ to retain each actual demand. Every other wall, including the actual I wall, is still satisfied.

A near helper forces $x_1 > .65$ on all three ranges. For native five use $A > 2.365, \gamma < .81$ and $\sqrt{2}[2.365 - .111 - .81(2.129)] > .65$. For finite owns at least seven, $A - \gamma(q - .61)$ increases from $\gamma = .9$, where $A > 2.6267$; subtracting its radical $\le .25$ still gives $x_1 > .65$. For dotted own the deficit decreases through $1 < \gamma < 1.08$: $A' < 3$, the radical derivative exceeds two, and $q - .61 > 2.12$. At 1.08, $A > 3.1373$, radical $< .37771$, so $\sqrt{2}[3.1373 - .37771 - 1.08(2.129)] > .6508$. The weak end has six-block mean $< .270$; this endpoint gain is $> .17328$, impossible beside $> .8944026$. Neither weak cap can therefore have a near helper.

For either far-helper cap, divide its full actual pressure by $\gamma > .8$. Native five gives $(2.188 - .111)/.81 > 2.39$. Finite owns at least seven have increasing A/γ, initially $\sqrt{200/27} > 68/25$, and radical ratio at most $5/18$. For dotted own, $A/\gamma - \sqrt{\gamma^2-1}/\gamma^2$ decreases; its first derivative term is $< .11$ and its subtracted derivative > 1. At 1.08 it exceeds $(2.9602 - .37771)/1.08 > 2.39$. Since $h_e > 0$ and $1/(2\gamma) < 5/8$, both actual feet give

$$h_5 + (5/8)h_{19} > 2.39, \qquad h_{15} + (5/8)h_1 > 2.39.$$

The full finite and continuous dotted alphabets have been covered.

To spend these pressures together, set $A_0 = .556, B_0 = 2.0425$, $C_0 = 2.288, r = 5/8$; then $B_0 + rA_0 = 2.39$. Build a symmetric reference with heights A_0, B_0, C_0 at one/five/six, then the old central heights and reflection. Its initial slopes are $.556, .371625$ repeated four times, $.2455, .118, \ldots$, decreasing. Delete actual internal weak knots seven through thirteen; their entire loss is $\le .001296$. The positive end coefficients in the spatial duals are

$$\begin{aligned} \beta_1 &= \frac{5A_0 - B_0}{8} = 59/640, \\ \beta_5 &= \frac{5B_0 - A_0 - 4C_0}{8} = 1009/16000, \\ \eta &= \beta_1 - r\beta_5 = 1351/25600 > 0. \end{aligned}$$

All remaining coefficients use actual strong floors. Direct rational calculation of that same central segment gives energy $\ge .021912$, so the reference w_A has

$$E_A \ge A_0^2 + (B_0 - A_0)^2/4 + (C_0 - B_0)^2 + .021912 = .9437388125.$$

Let $\lambda = \sqrt{19/10} > 1.3784$. The single actual-$I$ barrier gives $h_1 + h_{19} + 3\lambda s > 2\lambda(.767)$. Use the one full vector $W = w_A + 3\eta\lambda e_I$. Its two foot coefficients are β_5, and the remaining end coefficient is exactly η times that one barrier. Consequently

$$x.W > E_A + 2\eta[\lambda(.767) - A_0], \qquad W^2 = E_A + (171/10)\eta^2.$$

The reference values A_0, B_0 are used through these positive combined pressures. No separate endpoint floor is required there. The rational bounds $4\eta[\lambda(.767) - A_0] > .105807111375$ and $(171/10)\eta^2 < .047625$ give

$$\|x\|^2 \ge 2x.W - W^2 > .9437388125 + .105807111375 - .047625 = 1.001920923875 > 1.$$

This is one spatial certificate and one I coordinate. Every internal hole and every foot overlap has been charged in that vector. $\square$

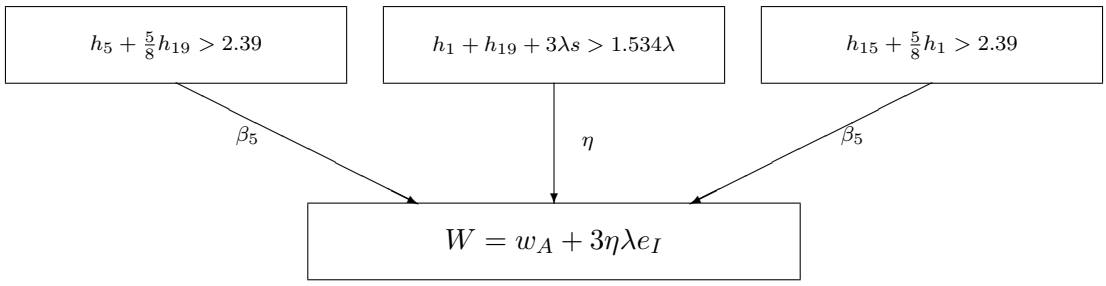


Figure 9: The two far-helper estimates and the I estimate are combined in one vector. The arrows give their positive coefficients.

B.11.20 The whole twenty-dimensional A row

Lemma B.11.24. *The entire vertex forest cannot be A_{20}.*

Proof. Use the $N = 21$ notation of Lemma B.11.2. That lemma and the rational central curve apply before any contrary energy assumption: the weak internal knots are nonadjacent pure helpers, and the near-four roles have already been excluded by norm. Put $C = 2.317$ and $B = 2.150$. According as the end knots five and sixteen are both weak, just one is weak, or both are strong, the concave reference energies, including every actual internal hole, are

$$E_2 > .92274, \qquad E_1 > .9509563, \qquad E_0 > .97977675.$$

These are the $N = 21$ curve rows above. No nonexistent strong end is inserted into the reference.

An actual weak cap at five has own γ and a simple helper at one or twenty, with

$$A^2 = (160\gamma^2 + c\gamma - 11)/21, \qquad c = 32 \text{ (near)}, \quad c = 10 \text{ (far)}.$$

The native-five correction is $\gamma n_5 + (2/3)n_e + (1/3)n_{e'}$, with finite residual $2/3 - \gamma^2$. For finite owns at least seven use the core $c + \gamma n_5 + 2(1 - \gamma^2)n_e$, of negative square $-(1 - \gamma^2)(4\gamma^2 - 3)$; for dotted own use $c + \gamma^{-1}n_5$. All coefficients are positive. The finite row uses the projected point, the other rows a future negative core. Every original wall inequality therefore holds by acuteness. Their pressures on $U = v - w_{\text{old}}$ exceed, respectively,

$$A - .111, \quad A - \sqrt{(1 - \gamma^2)(4\gamma^2 - 3)}, \quad A - \sqrt{1 - \gamma^{-2}}.$$

The finite and continuous dotted domains are those of the window lemma; the latter has $\gamma < 1.08$.

A near helper forces $x_1 > .64$. For five use $A > 2.365, \gamma < .81$. For finite owns at least seven, $A - 2.151\gamma$ increases from $\gamma = .9$, where $A > 2.649$; its radical is at most .25. For dotted own the deficit decreases through 1.08, where $A > 3.1636$ and the radical is $< .37771$. Substituting these bounds into the actual pressure proves the .64 floor throughout. The weak-end six-block mean is $< .274$. Its endpoint graft gains $> (6/5)(.64 - .274)^2 = .1607472$, already impossible beside E_2. Every weak end thus has a far helper.

Suppose both ends are weak. The reference is flat on its first and last six coordinates, with endpoint coefficients $\geq 1549/12000$. The projection of a far foot is $(5\gamma/6)\omega_6 + \omega_{15}/12$. The residual Y has zero sum on both blocks and vanishes in the middle, hence $w.Y = 0$. Its variance and

projected deficit satisfy

own	Y^2	Δ lower
5	$(2/3)\gamma^2 + 1/12 < 1.1$	.30
finite ≥ 7	$4\gamma^4 - (16/3)\gamma^2 + 29/12 < 1.1$	.30
dotted	$(5/3)\gamma^2 - 19/12 + \gamma^{-2} < 1.22$	.32

For five use $A > 2.2$, radical $< .111$, giving $\Delta > .3319$. For the other finite owns split at $.91, .92$: lower times $2.4649, 2.4937, 2.5225$ and upper radicals $.2318, .2434, .25$ give deficits $.302266, .300158, .30305$. The dotted deficit decreases through 1.08, where $A > 2.9794$ and radical $< .37771$ give $> .3233$. These are complete intervals.

Add .13 times both residuals to one reference. At either anchor the combined cost is $< .13(59/60) = 767/6000 < 1549/12000$. Their cross is retained using $\|Y_L + Y_R\|^2 \leq 2(Y_L^2 + Y_R^2)$. With zero, one or two dotted ends the gains are $> .08164, .082784, .083928$. Each added to E_2 exceeds one.

If just five is weak, the six-block and opposite five-block coefficients are at least $1549/12000$ and .1315. The far-foot projection becomes $(5\gamma/6)\omega_6 + \omega_{16}/10$. Its variance decreases by $1/60$ and its deficit by $B/10 - C/12 = 263/12000$. Thus nondotted owns have $\Delta > .278, Y^2 < 1.1$, and dotted owns have $\Delta > .30, Y^2 < 1.22$. Payment .14 costs $< .126$ and .014 at the two anchors and gains $> .05628$ or .060088. Both contradict E_1.

Both ends are now strong. The reference has first and last five-block value $b < .305$, and both anchor coefficients $> .1315$. Releasing the first root retains A_{19}. Finite classification gives isolated, simple-three at $2, 3, 19, 20$, or four at $2, 20$. Two contacts make a cycle; exceptional finite ranks are too small. The complete nonisolated edge-cosh list is

contact	white residual	$\cosh L_1$
3 at 2	21/40	$(40\gamma + 19)/21$
3 at 3	1/10	$(40\gamma + 18)/(2\sqrt{21})$
3 at 19	1/10	$(40\gamma + 2)/(2\sqrt{21})$
3 at 20	21/40	$(40\gamma + 1)/21$
4 at 2	1/20	$(40\gamma + 19\sqrt{2})/\sqrt{42}$
4 at 20	1/20	$(40\gamma + \sqrt{2})/\sqrt{42}$

The minimum allowed owns are $1/2, 0, 1/2, 1/\sqrt{2}, 0, 1/2$; smaller owns have time-zero or actual affine/null obstructions. Every row forces $x_1 > .396$. All but far-three have cosh > 1.5; that last squared floor is $200(\sqrt{2}-1)/[21(10\sqrt{2}+11)] > .396^2$. It reduces to $10441790\sqrt{2} > 14764031$, certified by $\sqrt{2} > 707/500$. If both ends are nonisolated, their disjoint grafts gain $2(5/4)(.396 - .305)^2 = .0207025$, contradicting E_0.

Reflect an isolated end to the left. Its time is $A^2 = (40/21)\gamma^2 - 1$. Own zero, three and four have nonpositive time square; six has the affine six–three path; parallel has the positive feasible null sum. Only five, finite owns at least seven, or dotted remain.

For five take $\gamma(3n_1/2 + n_2 + n_3/2)$. Its finite residual is $1 - 3\gamma^2/2$, and its displacement is $\gamma\omega_4/4$. Its projected-point pressure exceeds .393, since $A > .4966$, the second time root is $> .5146$, and the residual root is $< .1352$. Centering in the first five coordinates leaves deficit $> .0447$ and variance $\gamma^2/10 < .06561$. Payment .6 costs $< .0972$ and gains $> .0300204$.

For finite owns at least seven the actual core is $c + \gamma^{-1}n_1 + 2(\gamma^{-1} - \gamma)n_2$, of square $-[7 - 4\gamma^2 - 3\gamma^{-2}]$. Its centered variance is $(28/5)\gamma^2 - 8 + 3\gamma^{-2} < .6$. For seven, $.9009 < \gamma < .901$ gives $A > .7388$, radical $< .2395$, projected deficit $> .11187$. For larger labels $\gamma > .92$. The radical is strictly concave, because its squared argument has second derivative $-8 - 18\gamma^{-4} < 0$. Its derivative at .92 is $< .66$,

whereas $A' > 1.38$; the deficit derivative is therefore $> 1.38 - .66 - .43 > 0$. Its initial value is $> .1222$. The uniform deficit is $> .111$. Payment .2 costs $< .04$ and gains $> .0204$, contradicting E_0.

For dotted own, $\gamma > 1$ is unbounded. The core $c+\gamma^{-1}n_1$ has centered variance $(8/5)\gamma^2 - 2 + \gamma^{-2}$. Put $u = \gamma^{-2}$, $D = 40/21$, $q = \sqrt{D}$ and $a = q - \sqrt{D-1}$, so $0 < a < .43$. Concavity gives $\sqrt{D-u} \geq q - au$, while

$$au + \sqrt{u(1-u)} \leq (a + \sqrt{a^2+1})/2 < (.43 + 1.089)/2 = .7595.$$

Since $q > 1.38$, the projected deficit divided by γ is $> .1905$, including its .430 block payment. Multiplier $1/(8\gamma)$ costs $1/40$ at the anchor; the variance divided by γ^2 is at most $8/5$. Its gain $> .022625$ contradicts E_0. All corrections have positive coefficients on actual walls; the finite rows use projected points, the negative rows future cores. The full zero-, one- and two-weak-end partition is exhausted. □

B.11.21 The whole twenty-dimensional B row

Lemma B.11.25. *The entire vertex forest cannot be B_{20}.*

Proof. Use the genuine unit B_{20} roots, including the short root $n_{20} = e_{20}$. Lemma B.11.7 covers five through sixteen. At seventeen the retained short piece is B_3: short-three own zero has its affine principal, and positive owns have ratio $\geq 1 + 1/\sqrt{2}$. At eighteen it is B_2: short-three owns zero, four, six and parallel have their actual weighted-star, affine and null exclusions; finite owns at least seven are directly strong. Endpoint bridges have residual $1/36$; their own-zero cycle and positive-own ratios exclude weakness.

For the remaining own-five/short-three role the left H_4 projection is flat through fifteen, has norm < 18 and pressure $G > 4.1$. Old strong knots eleven/twelve have endpoint coefficient $> 1/7$. Old weak five would give actual $x_1 > .5965$, strong six, and a strong endpoint sixteen/seventeen already supplied by the ordinary bulk and B_3 row. Its positive three-knot interpolation then has energy $.7600299945 + 2.2 - (4/5)\sqrt{6} > 1.0004379$. Thus old five is strong. The old eleven/twelve references exclude the $G > 4.1$ row; even the safe norm bound twenty gives > 1.0102.

The exceptional B_2 own-three role has an actual finite D_{20} mate after releasing the *short* root twenty. Positive finite-Gram projection identifies the actual endpoint and gives $\cosh 9 + 5\sqrt{2} > 16$. Its pressure obeys $S_{20}^2 > 300/17 > 4.2^2$. Thus $G > 4.2, R = 20$, flat through both early endpoints, excludes this row by > 1.042. The weaker long-edge mate would not suffice. We have complete .58 coverage through eighteen, with only nonadjacent left helpers and the isolated dotted-five exception. The unused last-long-edge catalogue is unnecessary.

Releasing head one retains B_{19}, so its finite-white list is isolated or simple-three at long leaf two. The latter has $A^2 = 2\gamma^2 + 2\gamma$, $C_1 = 2\gamma + 1$. Own zero has time zero, three the affine triangle; all remaining owns force $x_1 > .643$. The old bulk gives a strong endpoint fourteen/fifteen. If old five is strong, its first-five graft gains $(5/4)(.643 - .361)^2 = .099405$ beside $q > .90715$. If old five is weak, six is strong and $S_6 > 2.01815$. Interpolation through $(1, .643), (6, 2.01815)$ and that endpoint has decreasing positive slopes, initial energy $\geq .7916565045$, and final chord energy $\geq 2.5 - \sqrt{21}/2 > .20871$. This too exceeds one. The head is therefore isolated. Time square, affine six–three and positive null sum exclude owns zero, three, four, six and parallel. Only five, finite at least seven and dotted remain.

Their actual head pressures, normalized to sum one, are

own	Z	h with $x.Z \ge h$
5	$P_4/4$	$> .393$
finite ≥ 7	$(1-u/2)(e_1+e_2)+(u-1)e_3$	$[\sqrt{2-u}-\sqrt{(u-1)(4-3u)}]/\sqrt{2}$
dotted	$(1-u/2)e_1+(u/2)e_2$	$[\sqrt{2-u}-\sqrt{u(1-u)}]/\sqrt{2}$

Here $u=\gamma^{-2}$ is respectively greater than one or in $(0,1)$. The first is an actual H_4 projected point; the others are the positive future cores on actual head roots proved above. Write $R=Z^2$ in the last two rows. The uniform scalar

$$.8h-.16R>.2893 \tag{B.11.9}$$

holds throughout both domains. For seven, $.9009<\gamma<.901$ gives $h>.4315, R<.349$, hence $>.28936$. For finite labels at least eight, $1<u<4-2\sqrt{2}<1.172$. With $D=(u-1)(4-3u)$ the derivative of the left side is

$$\frac{.8}{\sqrt{2}}\left[-\frac{1}{2\sqrt{2-u}}-\frac{7-6u}{2\sqrt{D}}\right]+.64-.48u.$$

It is negative up to $7/6$; above that point its three contributions are $<-.28,.032,.08$. At the eight-endpoint $\sqrt{2-u}>.910, \sqrt{D}<.289, R<.373$, giving $>.29136$. For dotted owns use the chords $1.4142-.379u$ on $u\le.5$ and $1.4494-.4494u$ on $u\ge.5$. Together with $R\le1-.75u$ and $R\le.75-.25u$ they give

$$.63987152-.0944u-.5657\sqrt{u(1-u)}>.30587152,$$
$$.69978064-.2142u-.5657\sqrt{u(1-u)}>.29023064.$$

The circle bounds use $\sqrt{.0944^2+.5657^2}<.5736$ and $\sqrt{.2142^2+.5657^2}<.6049$. No sampled minimum is used.

We next prove fresh .58-strong five. An actual weak helper five supplies the curved reference through $(1,.5455)$, six, and all later strong knots, ending at seventeen/eighteen. Its initial energy is $>.74385$, compared with the ordinary .58 six-energy $<.7126$. Since $E_{17}>.97231$ and all holes cost $<1/1024$, its energy is $>.97231+.74385-.7126-1/1024>1.0025$. The only other weak-five role is an isolated dotted cap. A native-five head forbids helper five/six/seven: their actual wall product would be positive, because

$$A_i^2-(\phi^3/2)(2\gamma+1)^2<(28\gamma^2-32\gamma-13)/5<0.$$

Its product with the isolated weak dotted-five cap lies in $(-.3,0)$, also forbidden. Thus only the other heads need be considered with the latter cap.

Choose actual strong $J=6$ or seven and $K=17$ or eighteen. Interpolate their genuine .58 floors and every strong knot between, omitting and charging all weak knots. The resulting data are

J	initial mean m	w^2	available P_J	Q_5
6	$<.3399$	$>.9530$	$>.1435$	$<77/150$
7	$<.31938$	$>.9354$	$>.1366$	$<194/175$

The squared-increment lower numerators from six through sixteen, over 10^6, are

$$38461, 33333, 29411, 26315, 23809, 21739, 20000, 18518, 17241, 16129, 15151.$$

Their sums from six and seven are $.260107, .221646$. Hole costs are $< .000267, .000181$; endpoint eighteen adds $> 1/70$ and pays any seventeenth hole. The terminal slopes give the displayed available coefficients.

The actual dotted-five foot is $Z_5 = P_5 + (v/2)(e_6 - e_5)$ with demand $T_5 > 1.8185, v > .4$; its centered variance is $Q_5 = 5 - 25/J - v + v^2/2$. Set $Y = Z - P_J/J$, $Y_5 = Z_5 - 5P_J/J$. The head support lies in the first three coordinates, so $Z.Z_5 = 1$ and $Y.Y_5 = 1 - 5/J$. The single vector $W = w + .4Y + .09Y_5$ has total anchor cost $.85/J$, below both available coefficients. Its effective energy is

$$w^2 + .2893 - .8m + .16/J + .18(T_5 - 5m) - .0081Q_5 - .072(1 - 5/J).$$

The two gains are $> .04930, .06699$ and both totals exceed one. Fresh strong five has been proved before any assumed strong eighteen floor.

All new references now start at five, with $b < .3652$ and available P_5 coefficient $> .151$. The head gains are $> .015456$ for five (pay .56 on $P_4/4 - P_5/5$), $> .02914$ for the other finite heads (pay .4 and use (B.11.9)), and $> .03384$ for dotted heads. For the last, the pure dotted scalar $> .03624$ at $\sqrt{2}b < .5106$ loses $< .0024$ when only its linear block term changes to $\sqrt{2}b < .5166$. No typed theorem is imported. If eighteen is strong, its reference has energy $> .985613$ before the head, and even the least gain contradicts the norm.

Otherwise eighteen is an actual helper and seventeen is strong. The reference has $q > .97133$, first-five mean $b < .3652$, and available P_{17} coefficient $> 1/9$. For that actual helper put $V = U - \sqrt{2}\gamma P_{17}$. Its variance is < 2.1, and its pressure deficit, including the nonzero cross $w.V = b/\sqrt{2}$, is $> .21$. For five, seven, eight and larger finite labels the respective deficit seals are $> .3444, .2853, .2306, .2303$. The last interval starts at $\gamma > .939$; its derivative is positive since $A' > 6$, $H_A > .94$ and its radical decreases. The dotted deficit decreases through $6/5$, where $A > 7.3307$ and radical $< .5528$ give $> .2177$. Thus every continuous parameter is covered.

Payment $\nu = 1/16$ costs $\nu\sqrt{2}\gamma < .107 < 1/9$ and gains $> 2\nu(.21) - \nu^2(2.1) = .018046875$. Combine the head and this foot in one vector. If its initial support has length $s = 3, 2, 1$, the residual product is $(\ell_s - .2)/\sqrt{2}$. For head five $\ell_s = 1/4$, giving total cross cost $< .002476$; for the other finite heads $\ell_s \le 1/2$ gives $< .010608$; for dotted heads $\ell_s \le 1$ gives at most .02 exactly. The final totals are

$$\begin{aligned} .97133 + .015456 + .018046875 - .002476 &= 1.002356875, \\ .97133 + .02914 + .018046875 - .010608 &= 1.007908875, \\ .97133 + .03384 + .018046875 - .02 &= 1.003216875. \end{aligned}$$

Every actual head case is impossible, proving the whole B_{20} row. □

B.11.22 The whole twenty-dimensional D row

The three possible isolated long heads have different paid references, but share one finite-white inventory at the last centre. We prove that inventory once, retaining the head-dependent cross products in a small table. This also keeps the endpoint choice honest: one of fourteen/fifteen and one of sixteen/seventeen is strong; no proof below assumes all four floors.

Lemma B.11.26 (The early D_{20} prerequisites)**.** *For the entire forest D_{20} the long-end cap is isolated, of own five, finite at least seven, or dotted. Moreover $S_5 > \sqrt{5} - .61/\sqrt{2}$, without a hypothesis on that cap.*

Proof. Use $n_i = (e_i - e_{i+1})/\sqrt{2}$ for $i < 20$ and $n_{20} = (e_{19} + e_{20})/\sqrt{2}$. Then $\omega_i = \sqrt{2}P_i$ through eighteen and $\omega_{19,20} = (P_{19} \mp e_{20})/\sqrt{2}$. Write $s_i = \sqrt{i} - .61/\sqrt{2}$. The old bulk applies literally

through fifteen: right retained rank is at least five, and all walls are actual. Its only weak rows are nonadjacent pure far helpers. For the head, retained D_{19} admits isolated or simple-three at its long leaf two: other leaves give exceptional finite rank twenty, internal attachments an extra fork, and a four edge is not a large finite D extension. Two contacts give a cycle. The nonisolated row has $A^2 = 2\gamma^2 + 2\gamma$, $C_1 = 2\gamma + 1$, exactly as in the preceding proof. Its $x_1 > .643$ contradicts either the old strong-five graft or the old weak-five three-knot reference there. Those arguments use only the first fifteen long coordinates, so the signed forks do not alter a coefficient. The isolated list follows from $A^2 = 2\gamma^2 - 1$, the actual affine six–three path, and the parallel null sum.

We supply the head-independent strong-five bootstrap before using any head gain. Suppose $S_5 \le s_5$. Its actual far helper gives $x_1 > a = .5965$, $S_6 > B = 2.01815$. Some $K = 11$ or twelve is old-strong. Interpolate through $(1, a), (6, B), (K, B + \sqrt{K} - \sqrt{6})$, with zero slope after K. The slopes $a, .28433, 1/(\sqrt{K} + \sqrt{6})$ strictly decrease. Thus all duals are positive, the endpoint coefficient exceeds $1/7$, and its energy is

$$q_K = .7600299945 + \frac{K - 6}{(\sqrt{K} + \sqrt{6})^2} > .91.$$

The terminal sum g_K is $< s_K < 3.034$. For any actual pressure $x.V \ge G$ with V equal to one on the first K entries and $V^2 \le R$, paying $1/7$ gives

$$\|x\|^2 > q_K + \frac{2(G - g_K)}{7} - \frac{R - K}{49}. \tag{B.11.10}$$

This is one square on this reference; no head is paid.

We need a late actual endpoint. Releasing sixteen gives $A_{15} + D_4$; the right long-leaf-three row has numerator $\gamma + 1/2$, residual $1/2$, with own-zero two-fork affine principal, own-three triangle, and all other owns ratio $\ge 1 + 1/\sqrt{2}$. Only the two short leaves use numerator $\gamma + 1/4$. Releasing seventeen gives $A_{16} + A_3$: middle-three has its affine star/triangle, positive ratios $\ge 1 + 1/\sqrt{2}$; end-four the weighted-star/strong rows; end-five is a sealed actual H_4, strong already at own zero. Full bridges have residual $1/32$ at sixteen, and $1/8 + 1/34$ or $1/34$ at seventeen; own zero is the all-three cycle, positive owns are strong. Left/isolated roles have the complete local list of Lemma B.11.7. Its proof is local at these actual retained pieces: it does not assert a rank condition on a larger-row theorem.

Apart from a pure left far helper, only a right short-leaf/end three remains. Own zero, three, four, six and parallel have respectively the affine E_8, cycle, F_4 path, six–three path and feasible null sum. Own five has the positive left H_4 projection, flat through $i-3$, $R < i$, with $G > \sqrt{i - 125/448} - .091$. At $i = 16, 17$ this is $> 3.745, 3.9$; inserting the safe $q > .91, g_K < 3.034$ in (B.11.10) gives $> 1.0111, 1.0349$.

For finite owns at least seven or dotted, supposed .58 weakness gives $.9 < \gamma < .92$ at sixteen, or $.9 < \gamma < 1.05$ at seventeen. This follows from the exact target $2773138/1687945 < 1.643$ and the residuals $1/2, 5/8$. Use the actual core

$$\xi = c + \gamma^{-1} n_i + (2\gamma)^{-1} n_{i-1}, \quad k = 1 - 3/(4\gamma^2), \quad \alpha = \sqrt{2}(\gamma + 1/4).$$

Its products at c, n_{i-1} vanish, at n_i are negative; its square is $-k < 0$, future time positive. All other wall products are nonpositive term by term. Its displacement is flat through $i - 2$, with

$$G = (A - \sqrt{k})/\alpha, \quad A^2 = i\alpha^2 - (i - 12)/8, \quad R = i - [(i - 12)/8 + k]/\alpha^2.$$

At sixteen, $2.645 < \alpha^2 < 2.738, k < .114$ give $G > 3.768, R < 15.82$. At seventeen, k/α^2 increases through 1.05 to $2350/24843 < .308^2$, while $\sqrt{17 - (5/8)/2.645} > 4.094$. Its derivative sign is

$-4\gamma^4-\gamma^3+6\gamma^2+(9/4)\gamma+3/16>0$ on this interval. Thus $G>3.786, R<16.82$. Formula (B.11.10) gives $>1.0213, 1.0060$. Only pure left helpers can therefore be weak at sixteen/seventeen. Their actual mixed replacements forbid adjacency, so at least one of those two is .58-strong, hence old-strong.

Append that actual endpoint J to the same weak-five reference. Its energy is at least

$$.7600299945+\frac{4-\sqrt{6}}{4+\sqrt{6}}=.7600299945+2.2-(4/5)\sqrt{6}>1.0004379945,$$

using $\sqrt{6}<244949/100000$. This contradiction proves old strong five without any head or centre floor. □

Lemma B.11.27. *The entire vertex forest cannot be D_{20}.*

Proof. Keep the original x and notation of the preceding lemma. There are three head branches, denoted H_5, H_∞, H_f for native five, dotted, and finite label at least seven. We construct their paid references before considering the centre.

For H_5, put $g=\phi/2$, $A_0^2=(\phi-1)/2$. The actual left H_4 projection has displacement $g\omega_4/4$, residual $1-3g^2/2$, and exact pressure gA_0. Hence $S_4\geq 2\sqrt{2}A_0>1.572$. A helper at any of five/six/seven is impossible: its actual head product is positive since $A_i^2-(g^2/A_0^2)(2\gamma+1)^2<(28\gamma^2-32\gamma-13)/5<0$ on $.8<\gamma<1.2$. An isolated weak dotted-five cap has actual product $\phi\gamma-A_0\sqrt{10\gamma^2-1}\in(-.3,0)$, where weakness gives $1<\gamma<2$. Negativity follows from $(7\phi-11)/2>0$; the lower bound from $\phi-\sqrt{5(\phi-1)}>-.15$. Thus fresh .58 five/six/seven are genuinely strong. On any subsequent actual-floor reference pay $.14T$, $T=P_4-(4/5)P_5$. Its anchor cost .112 is below .151, and its effective gain is

$$2(.14)(1.572-.8(1.826))-.8(.14)^2=.015456.$$

For H_∞, $u=g^{-2}\in(0,1)$ and the actual core $c+g^{-1}n_1$ gives $Z=(1-u/2)e_1+(u/2)e_2$ with pressure $h=[\sqrt{2-u}-\sqrt{u(1-u)}]/\sqrt{2}$. On any old reference, flat on the first five with $b<.361$, pay $\lambda(Z-P_5/5)$, $\lambda=\sqrt{2}/5$. Its cost $\lambda/5<.057$ fits the available coefficient $>.146$. The pure scalar proved above gives gain $>.03624$; if $u\leq.5$ it gives $>.04644$. The paid coordinates satisfy

$$w.P_3<1.1962,\quad w.P_2<.8918,\quad w_1<.5166\ (u\geq.5),\quad w_1<.5873\ (\text{all } u).$$

These nonflat entries will be used, not replaced by a flat mean.

For H_f, write $g^{-2}=1+v$. The actual three-wall core is

$$c+g^{-1}n_1+2(g^{-1}-g)n_2;$$

its square is $-[7-4g^2-3g^{-2}]<0$ and its displacement is the same $Z=(1-u/2)(e_1+e_2)+(u-1)e_3$ as the finite head in the preceding lemma. This identity uses no short normal. Here $0<v<.234$; the elementary cosine lower bound $\cos(\pi/7)>.9005$ gives $u<617/500$. Pay $.4(Z-P_5/5)$. The q-free scalar (B.11.9), at identical actual head hypotheses, gives gain $>.2893-.8(.361)+.032=.0325>.031$. For $v\leq.1$, directly

$$\sqrt{1-v}>.9486,\quad \sqrt{v-3v^2}<.2646\quad\Longrightarrow\quad G>.56568(.9486)-.56569(.2646)-.3368>.05.$$

The paid coordinates obey $w.P_3<1.243, w.P_2<.962, w_1<.461$ for $v\geq.1$, and $w_1<.481$ for all v. Only this common local head calculation is reused; the whole B_{20} theorem and its short endpoint are not inputs.

We justify the required late references. The early eleven/twelve old reference, now with old strong five, has $q_{11} > .8472225, q_{12} > .8689525$ and endpoint coefficient $> 1/7$. Pay the appropriate head as just constructed. All own-five right-fork rows at fifteen/sixteen/seventeen are excluded already without that payment: their H_4 pressure is $> \sqrt{i - 125/448} - .091$, flat through $i - 3$, $R < i$, giving $> 1.01101, 1.01087$ at fifteen, and increasing thereafter. The additional finite/dotted rows at sixteen/seventeen have $G > 3.768, R < 15.82$ or $G > 3.786, R < 16.82$ as proved above. Adding even the least head gain .015456 to the old reference excludes them by > 1.01616 and > 1.00089. The remaining weak rows through seventeen are precisely pure left helpers. Their actual nonadjacency gives an actual strong endpoint $K_A \in \{14, 15\}$ and $K_B \in \{16, 17\}$. For H_5 use fresh floors $f_i = \sqrt{i} - .58/\sqrt{2}$; for the other heads use old s_i floors. No individual fifteen/sixteen/seventeen floor has been asserted.

Each missing knot loses $(2\sqrt{j} - \sqrt{j-1} - \sqrt{j+1})^2/2 < 1/[32(j-1)^3]$. In H_5, missing knots start at eight: the total losses at fourteen, fifteen/sixteen and seventeen are $< .000158, .000172, .000182$. In the other heads they start at six; at most $4, 5, 5, 6$ holes cost $1/3200$ each. The old rational increment list gives $E_{14} > .9084, E_{15} > .92564, E_{16} > .94176$, $E_{17} > .94176 + 1/66$. Consequently the same legally paid references have data

	L_{14}	L_{15}	L_{16}	L_{17}	a_{14}	a_{15}	a_{16}	a_{17}
H_5	.939098	.956324	.972444	.9875	3.332	3.463	3.590	3.714
H_∞	.94339	.9603175	.9764375	.991276	3.311	3.442	3.569	3.692
H_f	.93815	.9550775	.9711975	.986036	3.311	3.442	3.569	3.692

Here L_K is a strict lower bound on twice certified pressure minus squared norm, not on that norm itself. The actual coordinate sum is f_K or s_K, strictly below a_K. For small dotted parameter $u \le .5$ add .0102 to its four L entries; for finite $v \le .1$ add .019. Remaining P_{14}, P_{15} coefficients exceed $1/8$; those at sixteen/seventeen exceed .12. The head changes only the first-five coefficients.

If an actual V is flat one through K_A, $x.V \ge G$, $V^2 \le R$, paying $1/8$ gives

$$\|x\|^2 > L_{K_A} + (G - a_{K_A})/4 - (R - K_A)/64. \tag{B.11.11}$$

Both actual fork directions are $P_{19} \pm e_{20}$, of square twenty. Cosh $> 41/5$ gives $G > \sqrt{360/23} > 3.955$. Equation (B.11.11) exceeds one in every head/endpoint row (the least is 1.001098). Thus both actual fork coshes are at most $41/5$, independently of a centre floor. Reciprocal uses below concern actual endpoints: for a positive finite replacement Gram, its inverse is entrywise nonnegative. Projecting p adds positive multiples of actual normals, preserves every unselected strict inequality and makes the released old one strict. The projected point is therefore the actual neighboring endpoint, with no unseen truncating wall.

The centre inventory. Delete eighteen, retaining A_{17} and two orthogonal forks. If the actual centre wall has own γ, left contact β at j, and fork contacts δ, ϵ, then

$$\Delta = 1 - q_{\text{left}} - \delta^2 - \epsilon^2 > 0, \quad \sigma = \gamma + \beta j/18 + (\delta + \epsilon)/2, \quad C_{18} = 6\sigma/\sqrt{\Delta}.$$

For simple three, $q_{\text{left}} = j(18 - j)/36$; endpoint four doubles it. The complete finite-white list is: one left endpoint-three/four at one/seventeen or next-three at two/sixteen; one fork with a left endpoint-three and fork-three/four; both forks with a left endpoint only in the all-three bridge; or, without a left contact, the two-fork pairs $(3, 3), (3, 4), (3, 5)$. More left contacts create a cycle; other branches would require exceptional finite rank twenty. This is classification of the actual white set, not of a face. The sufficient .58 strength target is $< 83/50$.

Mixed left/fork bridges have residual 1/36 (both forks or fork four), with ratios $\geq 19/6$ or $1/6 + 3/\sqrt{2}$. The one-fork-three bridge has residual 5/18: own zero is the all-three cycle, own three the triangle; other owns give ratio at least $(1/\sqrt{2} + 1/36 + 1/4)/\sqrt{5/18} > 83/50$.

With no fork contact the isolated roles have the established .58 feet and affine/null exclusions. This remains the genuine centre: the identity $\omega_{19} + \omega_{20} = \sqrt{2}P_{19}$ is used only in $n_{18} = \omega_{18} - (\omega_{17} + \omega_{19} + \omega_{20})/2$; no extra root is introduced. For far left-four and next-three, own zero has time zero and positive owns are strong. Near left-four is strong already at own zero. Near next-three own zero has the affine two-fork principal; positive owns have ratio $\geq 5/3$. Near endpoint-three owns zero/three/four/six have respectively the two-fork, triangle, weighted-star and six–three affine principals. All other positive owns have ratio $(36\gamma + 17)/(6\sqrt{19}) > 83/50$. Far endpoint-three own zero has $A^2 = -1/2$, three its cycle, four/six the right affine principals, and parallel a null sum. Thus the complete possible weak no-fork list is

$$v = \gamma\omega_{18} + \omega_1/2, \quad A^2 = 36\gamma^2 + 2\gamma - 1/2, \quad \phi/2 \leq \gamma < 6/5,$$

with own five, finite at least seven, or dotted.

For these helpers the positive actual corrections are

own	w_{old}	Y in $V = P_{17} + Y$	G lower, Y^2 upper
5	$\gamma n_{18} + (2/3)n_1 + (1/3)n_2$	$aP_3 + (e_{18} + e_{19})/2$	4.185, .631
7, 8	$\gamma n_{18} + n_1/2$	$aP_2 + (e_{18} + e_{19})/2$	4.08, .655
finite ≥ 9	γn_{18}	$aP_1 + (e_{18} + e_{19})/2$	4.067, .785

Here $V = (v - w_{\text{old}})/(\sqrt{2}\gamma)$ and $a = 1/(6\gamma), 1/(4\gamma), 1/(2\gamma)$ respectively. The finite five and at-least-nine rows use projected points with residuals $2/3 - \gamma^2, 1 - \gamma^2$; the seven/eight row is a negative core of norm $-(\gamma^2 - 3/4)$. Their common time term is $\sqrt{18 + 1/\gamma - 1/(4\gamma^2)} > 4.33$. The normalized residual bounds are $.145, .25, .263$; $Y^2 = ka^2 + 1/2$ gives the displayed variances. All products against every unselected actual wall are nonpositive by the positive corrections, including the actual head.

For dotted helper put $a = 1/(2\gamma) \in (5/12, 1/2)$ and $t = 4a^2$. The future core $c + \gamma^{-1}n_{18}$ gives

$$Y = aP_1 + (1 - t/2)e_{18} + (t/2)e_{19}, \quad Y^2 \leq .75, \quad G = \sqrt{18 + 2a - a^2} - a\sqrt{2 - 8a^2}.$$

For each constant $c \leq .5873$, $(G - ca)' > 7/9 - c > 0$. At $a = 5/12$, positive square bounds $\sqrt{2687} > 51.83$, $\sqrt{22} < 4.691$ give $G - .394a > 3.829$. Hence all head/helper crosses have the bounds

head parameter	$w.Y_5$	$w.Y_{7,8}$	$w.Y_{\geq 9}$	$G - w.Y_{\text{dot}}$
H_5	$< .24625$	$< .219$	$< .211$	> 3.829
$H_\infty, u \geq .5$	$< .2493$	$< .2478$	$< .2754$	> 3.7779
$H_\infty, u \leq .5$	$< .2493$	$< .2478$	$< .3131$	> 3.7484
$H_f, v \geq .1$	$< .259$	$< .2673$	$< .2458$	> 3.8010
$H_f, v \leq .1$	$< .259$	$< .2673$	$< .2564$	> 3.7927

The finite columns follow by multiplying the actual $w.P_k$ bounds above by a; the dotted column uses the increasing $G - ca$ at the common endpoint 5/12. No cross is set to zero.

At $K_B = 17$ pay .12 for Y; at $K_B = 16$ pay for $e_{17} + Y$. The latter has an extra unit square and no extra cross, since the reference stops at sixteen and $e_{17}.Y = 0$. One combined square gives

$$\|x\|^2 > L_{K_B} + .24(G - a_{K_B} - w.Y) - .0144(Y^2 + \varepsilon), \quad \varepsilon = 0 \ (17),\ 1 \ (16). \tag{B.11.12}$$

The minimum over the four helper rows is > 1.0043 in H_5, > 1.001092 in large-u dotted, and > 1.001396 in large-v finite heads. For the small parameter cases, at $K_B = 17$ the reference alone exceeds one; at sixteen the minimum is > 1.0044935 or > 1.0186855 respectively. For example the tight dotted test is $.991276 + .24(3.7779 - 3.692) - .0144(.75) = 1.001092$. Every no-fork weak centre role is excluded.

The fork rows. For one fork and no left contact put $\alpha = \sqrt{2}(\gamma + \delta/2)$, $A^2 = 18\alpha^2 - (1 - \delta^2)$. At own zero, fork labels three/four/six have their actual affine E_8, weighted-star and six–three principals. Fork five has the positive left H_4 projection with $G = \sqrt{4 + 8\phi} - \phi^{-3} > 3.87$, $R = 17 < 18$. For finite fork labels at least seven use the positive core

$$c + \delta^{-1} n_f + 2(\delta^{-1} - \delta) n_{18}.$$

Its square is $-(1 - b)(4b - 3)/b < 0$, $b = \delta^2$, and its products at c, n_f vanish and at n_{18} are negative. The displacement is flat through seventeen, with

$$R = 28 - 16/b + 6/b^2, \qquad G = \sqrt{20 - 2/b} - \sqrt{-8 + 14/b - 6/b^2}.$$

Every finite label is in the following table:

fork label/domain	G lower	R upper
5	3.87	18
$7, .81 < b < .82$	3.798	17.412
8	3.79392	17.49034
$\geq 9, .88 < b \leq .9$	3.808	17.63
$\geq 9, .9 \leq b < 1$	3.831	18

For eight the exact correlated values are $R = 108 - 64\sqrt{2}$ and $G = \sqrt{12 + 4\sqrt{2}} - \sqrt{68\sqrt{2} - 96}$. The rational seal $1.414213562 < \sqrt{2} < 1.414213563$ gives the table. In the other rows R increases for $b > .75$, and the second radicand increases up to $6/7$ and decreases thereafter, proving the interval bounds. Labels seven and at least nine satisfy the indicated disjoint native ranges. Equation (B.11.11) excludes every row with either actual early endpoint and each head: the least test is the native-five/eight/fourteen value $.939098 + (3.79392 - 3.332)/4 - 3.49034/64 = 1.0000414375$. For finite heads its least test is > 1.0041459; for dotted heads > 1.0093859. No head cross occurs because these displacements are flat throughout the head support.

For positive own three, release the contacted fork. The same actual centre wall completes the retained A_{19} to finite D_{20}, so the reciprocal endpoint has $C_f = 9 + 10\delta \geq 14 > 41/5$, impossible. Own four has the affine star on $c, 18, 17$ and the other fork. Own six/fork three has the six–three principal. Own six/fork at least four and own at least seven/fork at least four are directly strong by ratio $\geq \sqrt{2}\gamma + 1/2 > 83/50$; own five/fork at least five is also directly strong. The two exceptions five/three and five/four have the positive left H_4 correction $\gamma(3n_{18}/2 + n_{17} + n_{16}/2)$, finite residual $(5 - 3\phi)/8$ and $\alpha^2 \geq \phi + 5/8$. Their displacement is flat through fifteen, $R < 18$, and $G > \sqrt{18 - 75/224} - .091 > 4.1$, dominating the common row. For single-fork three with finite at least seven or dotted own, $\gamma \geq 6/5$ is directly strong. Throughout $.9 < \gamma < 6/5$ the actual core $c + \gamma^{-1} n_{18} + (2\gamma)^{-1} n_{17}$ has $k = 1 - 3/(4\gamma^2) > 0$, flat displacement through sixteen, $R < 18$, $A/\alpha > 4.208$ and $\sqrt{k}/\alpha < .338$. Thus $G > 3.87$ and the same common row excludes it. Parallel own is independently null; including it in that continuous core interval would also contradict the reference.

For two forks the complete pairs are three/three, three/four, three/five. Own three has the affine triangle, own at least four ratio $\geq 1 + 1/\sqrt{2}$, and parallel the null sum. At own zero, three/three is the affine square. Three/four has, after releasing the three-contacted fork, the actual finite B_{20} reciprocal endpoint $C_f = 9 + 5\sqrt{2} > 41/5$, impossible. Three/five has a positive finite H_3 projection, displacement αP_{18}, $\alpha = \phi^2/(2\sqrt{2})$ and $q = 2\phi^{-6} < 1/8$; its pressure is $\sqrt{18 - q}/(1 + \sqrt{q/18}) > 4.22/(13/12) > 3.87$, $R = 18$. It too is covered by the common row. All original wall inequalities in these finite projections follow from the positive inverse Gram coefficients.

The complete centre domain has been exhausted. Eighteen is therefore genuinely .58-strong in every head branch. Append its fresh floor for H_5, its old floor otherwise, to the reference whose actual endpoint is K_B. At seventeen the added energy is $d_{17}^2 > 1/70$; at sixteen the chord across the possibly weak seventeenth adds $17 - 12\sqrt{2} > .0294$. Both slopes are below the available endpoint coefficient and leave the head payment unchanged. The least resulting totals are the finite-head values $.986036 + 1/70 > 1.0003217$ and $.9711975 + .0294 = 1.0005975$. This contradicts the same original $\|x\| < 1$ in every branch. □

B.11.23 The complete dimension-twenty conjunction

Proposition B.11.28. *There is no compact hyperbolic Coxeter polytope of dimension twenty.*

Proof. We exhaust the forest of an actual farthest vertex in this dimension; no descent from a higher-dimensional face is involved. An irreducible A_r with $r \geq 7$, or B_r, D_r with $r \geq 9$, is called dominant. Each costs more than one half, so at most one occurs. If none occurs, the ordinary rank-fourteen rate, even discarding the single possible isolated I rank, gives $19/14 > 1$. A typed rank-two defect instead gives $.09 + 18/14 > 1$; the funded J pair is stronger than its ordinary rank rate. Thus exactly one dominant occurs.

For an A dominant, rank seven is excluded by its half bound and the ordinary spectator rate. Ranks eight through twelve are excluded by $4/7$ and that same rate: even with I there are seven ordinary ranks at twelve, or with a typed defect six ranks beside its .09 cost. Ranks thirteen/fourteen have energy $> 5/7$ and at least five ordinary ranks after discarding I; a typed defect leaves at least four, so both alternatives exceed one. At ranks fifteen, sixteen and seventeen use respectively $> .826, > .86, > 6/7$. The complete minimum spectator comparisons are

r	ordinary or J	I present	typed present
15	$> 5/14$	$> 2/7$	$> .09 + 17/70$
16	$> 2/7$	$> 2/7$	$> .09 + 1/7$
17	$> 17/70$	$> .21$	$> .09 + .1$

Each sum with its actual dominant floor exceeds one. For the middle I entry the ordinary remainder has rank three; the joint I theorem, rather than discarding that rank, supplies $> 2/7$. The rank-two typed remainders are either an ordinary dihedral or two ordinary A_1 factors, both at least the shown charge. There is at most one defect. Rank eighteen uses $> .91$: an ordinary rank-two remainder is $> 1/7$, $I + A_1 > 1/5$, either typed defect $> .09$, and $J > .34$. The typed sum is strictly greater than one as well. At rank nineteen its remaining A_1 is either ordinary or I; Lemmas B.11.22 and B.11.23 exclude both. Rank twenty is whole and is excluded by Lemma B.11.24.

For a B/D dominant its $9/14$ bound forces rank at least fifteen: at rank fourteen even an I leaves five ordinary ranks, while a typed defect leaves four beside its .09. At fifteen and sixteen the valid weaker D prefix floors are $> .797425, > .823735$. The smallest remainders are $I+$ ordinary rank four $> 2/7$, respectively a typed defect plus an ordinary dihedral $> .09 + 1/7$; all other partitions are stronger. Both rows exceed one. At seventeen the weaker floor is $> .8472225$. Ordinary total

rank three costs $> 17/70$, $I+$ dihedral costs $> .21$, $I+$ two ordinary A_1 costs $> .3$, and a typed defect plus ordinary A_1 costs $> .19$. Each closes the row. At eighteen the weaker floor $> .8689525$ excludes ordinary rank two, $I + A_1$, and J. Both typed remainders are excluded by Lemma B.11.21. At nineteen a nonshort A_1 is excluded by the actual complete coverage through fifteen and its prefix gain $> .91211125$; that same lemma supplies the needed local D_4 continuation before using the prefix proof. Its I alternative is also excluded there. Whole B_{20} and D_{20} are excluded by Lemmas B.11.25 and B.11.27. Finite classification and the complete spectator partitions above exhaust the actual rank-twenty forest. Orthogonal component energies use the same original x throughout. □

B.11.24 The two proper A rows in exact dimension nineteen

Lemma B.11.29. *An A_{17} component cannot coexist with either actual typed-seven defect as its entire rank-two remainder.*

Proof. The actual defect costs $> .09$, so $e_A < .91$. Use the complete $N = 18$ window of Appendix B.3.6, before its weaker contrary-energy conclusion. A near-four actual mate has normalized demand $> .86$; the opposite strong anchor or mate has demand $> .77$ and correlation $< .44$. Cauchy would give $1.63^2 < .91(2.88)$, false. All weak knots five through thirteen are thus nonadjacent pure helpers, with $Q = 0$, own five, finite at least seven or dotted, $\phi/2 \le \gamma < 17/16$. The rational reference of Appendix B.3.7 has end heights $H_5 = 2.077, H_6 = 2.218$ and central heights $2.314, 2.371, 2.389$, reflected. For zero, one or two weak end sites its complete all-hole energies are

$$E_0 \ge .8934308, \quad E_1 > .86256, \quad E_2 \ge 9978827/12000000.$$

Strong ends have five-block mean $.29 < b_0 < .294$; weak ends six-block mean $1/4 < b_1 < .2614$.

Both actual defects force the endpoint floors $x_1 > .392, -x_{18} > .392$. For old A_2/new seven, propagation gives endpoint squared half-tangent $> 1/5$, hence $x_1^2 > 17/90$. For old $I_2(7)$/new three let its isolated opposite have time square $h = (1/4)\csc^2(\pi/7) - 1$. The actual endpoint opposite retains that whole old block. Any finite contact to it would cost at least $1/[4\sin^2(\pi/7)] > 1$, so the endpoint cap has no such contact. Releasing old one leaves A_{16}. Its complete nonisolated contact list is three at $2, 3, 16, 17$, or four at $2, 17$. The corresponding coshes and minimum allowable owns are

$3:2$	$(17\gamma + 8)/9$	$1/2$
$3:3$	$(34\gamma + 15)/(6\sqrt{2})$	0
$3:16$	$(17\gamma + 1)/(3\sqrt{2})$	$1/2$
$3:17$	$(34\gamma + 1)/18$	$1/\sqrt{2}$
$4:2$	$(17\gamma + 8\sqrt{2})/3$	0
$4:17$	$(17\gamma + 1/\sqrt{2})/3$	$1/2$

All but the far-three row have cosh $> 3/2$, giving $x_1^2 > 17/90$. The remaining row gives $289(\sqrt{2} - 1)/[18(17\sqrt{2} + 19)] > (49/125)^2$; clearing denominators leaves $3780919\sqrt{2} > 5336767$, certified by $\sqrt{2} > 707/500$.

For an isolated endpoint $A^2 = (17/9)\gamma^2 - 1$ and its product with the actual defect wall is $-A\sqrt{h}$. For finite own, $hA^2 < 8/27 < 1/2$; the native gap forces $hA^2 = 1/4$. The literal seven algebra gives $7h^3 + 7h^2 - 1 = 0$, $.327 < h < .329$, with degree three over $\mathbb{Q}$. Consequently $.93 < \gamma^2 < .94$. Only label twelve can lie in this interval: for labels at most eleven, $\sin(\pi/11) > 717/2662$ gives squared cosine $< .93$; for labels at least thirteen, $\pi/13 < 22/91$ gives squared cosine $> 7797/8281 > .94$. At twelve equality would imply $h = 9/(17\sqrt{3} - 2)$, impossible for its cubic degree. This is literal-angle algebra, not an arithmetic ground-field restriction. Parallel own is excluded by the same gap and

interval. For dotted own $hA^2 > 1/4$; the next native possibility gives $A^2 \geq 1/(2h) > 3/2$, so its cosh squared exceeds $5/2$ and again supplies the floor. Reflection covers the other endpoint.

Every weak cap gives a stronger floor at its helper end. The complete raw-pressure seals in Appendix B.3.8 are $.36082, .327322, .3132358, .31446$ for five, seven, eight and larger finite owns. For dotted own its exact weak floor gives $> .310125 + 1/216 > .3132358$. Thus the helper endpoint exceeds .4429, since $1.4142(.3132358) = .44297806836$. A near helper has larger time square by $8\gamma/9$. As the two times are $< 3, 4$, its time increases by $> 32/315 > .1$, with the same residual. Its endpoint exceeds .58 for every own parameter.

On a constant block of length m and mean b, grafting a signed endpoint value $z > b$ gains $m(z-b)^2/(m-1)$. All the following grafts preserve positive ordered slopes: the other coordinates stay $> .2517$ in a five-block or $> .184$ in a six-block, above the next reference coordinate. Opposite end blocks are disjoint, so they are changed in one reference, including all previous hole charges. With no weak ends use .392 twice: $E_A > .8934308 + 2(5/4)(.392 - .294)^2 = .9174408$. With one weak end a near helper alone gives $> .98$. Otherwise graft .4429 in the opposite strong block and .392 in its weak block, giving $> .86256 + (5/4)(.4429 - .294)^2 + (6/5)(.392 - .2614)^2 = .9107416445$. With two weak ends a near helper alone gives $> .95$; otherwise their far helper endpoints are distinct, giving $> 9978827/12000000 + 2(6/5)(.4429 - .2614)^2 > .91063$. All cases exceed .91, contradicting the orthogonal defect cost. □

Lemma B.11.30. *An actual A_{18} component cannot coexist with an isolated I; further orthogonal finite spectators are allowed.*

Proof. Use $N = 19$ and the one actual I coordinate $y \geq s$. Its rank-free parameters give both endpoints at least ηt, where $\eta = \sqrt{18/19} > .973$ and $t + (3/2)s > .767$. We first strengthen the near-four removal to the norm-only domain needed here. Its actual finite mate, replacing $j = i - 1$ at $i = 5, 6$, has diagram $B_{20-i} + A_{i-2} + S$ and cosh

$$\sqrt{2(i-1)/19}\,[(20-i)/2 + (19-i)/\sqrt{2}] > 11.$$

Finite positive projection retains every spectator and recognizes this actual endpoint. Its normalized height exceeds .91. The opposite mate or strong knot thirteen/fourteen has height $> .774$ and normalized correlation $< .41$. Thus $1.684^2 = 2.835856 > 2.82$ contradicts $e_A < 1$. The complete shared $N = 19$ window now has only nonadjacent pure helpers, before any other energy assumption.

The shared rational heights are $H_5 = 2.104, H_6 = 2.255, H_7 = 2.363, H_8 = 2.433, H_9 = H_{10} = 2.467$, reflected. With zero, one or two weak end sites the complete energies are $E_0 \geq .9240352, E_1 > .8937, E_2 > .86414$. The strong five-block mean is $b_0 < .298$, weak six-block mean $b_1 < .266$, each $> .29$ or $> 1/4$ respectively. Available dual coefficients are $\beta_6 \geq 1607/12000$ and $\beta_{14} \geq .1349$; deleting the next internal knot only increases them.

A weak-five far helper has $A^2 = (140\gamma^2 + 10\gamma - 10)/19$. Use the full positive actual feet of Appendix B.3.7, not their raw normal sums. Their corrections are $\gamma n_5 + (2/3)n_e + (1/3)n_{e'}$ for five, $\gamma n_5 + 2(1-\gamma^2)n_e$ for finite owns at least seven, and $\gamma^{-1}n_5$ for dotted own. Their actual pressures are bounded below by A minus the respective finite/core radical; the full I block remains an unselected acute block. When both ends are weak, projecting the spatial foot gives

$$PU = (5\gamma/6)\omega_6 + \omega_{13}/12, \quad Y = U - PU, \quad w.Y = 0.$$

Its variance V_2 is $(2/3)\gamma^2 + 1/12$ for five, $4\gamma^4 - (16/3)\gamma^2 + 29/12$ for other finite owns, and $(5/3)\gamma^2 + \gamma^{-2} - 19/12$ for dotted own. The exact cross between two far feet is $Y_L.Y_R = (\gamma_L + \gamma_R)/6$: each head residual pairs with the opposite helper's mass $1/\sqrt{2}$, and all same-side block sums vanish.

Pay $\mu = .138$ for each far foot. At either anchor the total cost is $< .138(31/32) < 1607/12000$. For $\Delta_2 = H - 2.255(5\gamma/6 + 1/12)$ the gain per foot, including half the exact cross, is $F = 2\mu\Delta_2 - \mu^2(V_2 + \gamma/3)$. The full finite/dotted seals are

own	Δ_2 lower	$V_2 + \gamma/3$ upper	F lower
5	67/192	.791	.081248696
7	.3371541666	1.026	.0735154059
8	.32852125	1.087	.069971037
≥ 9	1591/4800	1.107	.070400792
dotted	.36109375	1.54	.070334115

The first three use actual time/radical bounds $(2.17, .111), (2.434, .2158), (2.499, .2464)$ and upper native cosines $.81, .901, .9239$. For larger finite owns evaluate at .93, where $A > 2.517$, then use $A' > 2.714$ and $V_2' \leq 16/3$ to see that the lower gain increases throughout the remaining finite interval. For dotted owns use $1 < \gamma < 17/16$; its deficit decreases to an endpoint $> .36109375$, while $V_2 < 1.185, \gamma/3 < .355$. This proves every row continuously, not by sampling. Both far weak ends therefore give $E_A > .86414 + 2(.0698) > 1.00374$, without spending I.

For exactly one weak end, a near helper already gives $E_A > 1.0120152$ by the shared raw endpoint floor .58. For a far helper the projection becomes $(5\gamma/6)\omega_6 + \omega_{14}/10$. Its deficit and variance are $\Delta_1 = \Delta_2 - 1349/60000$, $V_1 = V_2 - 1/60$. Payment $\mu = .11$ gives a gain $> .058$ in every row: the conservative five/seven/eight/larger/dotted gains are $.065714, .06064868, .05809604, .0585244, .06034939$. The same finite derivative and dotted endpoint bounds prove the complete intervals. Retain the propagated left endpoint at the same time. Let $R = e_1 - P_6/6$, so $R^2 = 5/6$ and $Y.R = \gamma\sqrt{2}/6$. Put $B = b_1 + \mu\gamma\sqrt{2}/6 < .294$, $z = (\eta t - B)_+$ and $\nu = \min((6/5)z, 1/4)$. The full vector is $w + \mu Y + \nu R$. Its coefficient-six cost is $\mu 5\gamma/6 + \nu/(6\sqrt{2}) < .128 < 1607/12000$; the opposite cost is $.011 < .1349$. Its full mixed square therefore gives

$$E_A + y^2 > E_1 + .058 + s^2 + 2\nu z - (5/6)\nu^2.$$

If not clipped, $z + (3\eta/2)s > .452$ and weighted Cauchy gives $s^2 + (6/5)z^2 > .452^2(57/169) > .068$. If clipped the graft alone is $> 5/96$. Both totals exceed $.8937 + .058 + 5/96 > 1$.

With no weak ends graft both propagated endpoint floors in the two five-blocks. Their gain is $(5/2)z^2$, with $z = (\eta t - b_0)_+$ and $z + (3\eta/2)s > .448$. Its sum with s^2 is $> .448^2(190/481) > .0792$, contradicting E_0. The grafts stay ordered even for $\eta t < 1$, because their other block entries exceed .1125 and the next reference entry is $< .108$. Finally if both ends are weak with a near helper, its .58 graft gives $E_A > .9824552$. If the opposite propagated floor exceeds .58 the other .58 graft already contradicts the norm. Otherwise graft its actual propagated floor there; with $z = (\eta t - b_1)_+$ one has $z + (3\eta/2)s > .480$, so its gain with s^2 is $> .480^2(57/169) > .077$. Both block changes are legal and disjoint. All end patterns, internal weak sets and actual own/foreign possibilities are covered, spending the one I energy only once. □

B.11.25 The proper classical rows in exact dimension nineteen

Lemma B.11.31. *The entire forest $B_{18} + I$ is impossible.*

Proof. Use the actual short root $n_{18} = e_{18}$ and long roots before it. The genuine generic bulk covers five through fourteen. At fifteen retained B_3 admits its long-three and short-three F_4 extensions. The latter has residual $1/4$, numerator $\gamma + 1/(2\sqrt{2})$, own-zero affine F_4, and positive-own ratio $\geq 1 + 1/\sqrt{2}$. Its full white F_4 cannot acquire an I arm; full B_{18} bridges have internal new node and

are also foreign-sealed. Left/isolated/foreign estimates use only genuine long roots and retain their full old-I clearing.

At sixteen the right piece is B_2. A second four is not a finite extension, so the only new contacts are a three at either end. Long-three has numerator $\gamma + 1/2$, residual $1/2 - Q$, affine own zero/three and otherwise strong ratio. Short-three has numerator $\gamma + 1/(2\sqrt{2})$, same residual; owns zero/four/six have the affine F_4/star/six–three principals, parallel the actual null sum. Own at least five apart from affine six is strong; at own three a positive $Q \geq 1/4$ is strong. Thus the sole extra row is $Q = 0, v = (\omega_{16} + \omega_{18})/2$. Full endpoint bridges have internal new node, no foreign extension, and direct strong ratio already at own zero.

For the extra row release old seventeen. The same actual wall completes finite A_{18}, with the old I retained. Positive finite projection identifies the actual endpoint and gives $C_{17} = (17 + 16\sqrt{2})/\sqrt{19} > 9.09$, hence $S_{17}^2 > 13753/1009 > 3.6919^2$, stronger than f_{17}. If five is weak the actual interpolation through one, six and seventeen already has energy > 1.0004379945. If five is strong, extend its actual fourteen/fifteen reference by the chord to f_{17}. The energy is $> .9550775$, the first five entries remain $b < .361$ and their coefficient $> .146$. The single actual-I initial graft adds $> .0496$: for a nonclipped five-block, weighted Cauchy with $t + (3/2)s > .767$ gives $s^2 + (5/4)(t-b)_+^2 > (.767 - .361)^2/(61/20) > .0496$; clipping the endpoint at .6 already gives $> .0714$. This rules out the extra row before using sixteen in a bootstrap.

Coverage through sixteen is now precisely nonadjacent pure left helpers. If five is weak, one of fifteen/sixteen is strong. At sixteen its weak-five reference alone exceeds one; at fifteen it has effective energy $> .9851486$, first coordinate .5965 and remaining P_6 coefficient $> 1/60$. Lemma B.10.4 adds $> .015$ using the unused actual I. Thus five is genuinely strong.

Choose the actual strong endpoint fifteen or sixteen. The ordinary old-floor reference has $q > .9240775$, first five entries $b = f_5/5 < .361$, assigned P_5 pressure f_5 at the old .61 loss, and available P_5 coefficient $> .146$. We now check the local head lemma's actual whitelist, rather than importing a B_{17} row theorem. Deleting one retains $B_{17} + I$. Its only nonisolated finite extension is simple three at old two, with optional I contact only three. The isolated-within-B row may meet I by any finite label. These are exactly the two raw equations

$$A^2 = 2\gamma^2 + Q - 1, \qquad A^2 = 2\gamma^2 + 2\gamma + Q,$$

with Q arbitrary finite in the first row and $Q = 0, 1/4$ in the second. All head points use only genuine long normals one/two/three; old short root eighteen remains an unselected acute wall. The whole head, including positive foreign contact, therefore satisfies Lemma B.10.9. Its q-free gain $> .07665$, with I once, gives $q + \text{gain} > 1.0007275$. There is no need to prove sixteen individually strong or to repeat a terminal head/I scalar catalogue. □

Lemma B.11.32. *The entire forest D_{17} with either actual typed-seven defect is impossible.*

Proof. Use the genuine signed forks sixteen/seventeen, centre fifteen, and the one defect projection of energy $e_7 > .09$. For old $I_2(7)$, white positivity prohibits any foreign contact, since its smallest cost is $1/[4\sin^2(\pi/7)] > 1$. For old A_2 both old nodes cannot be contacted (the triangle is affine); a single contact gives precisely $Q \in \{0, 1/3, 2/3, \phi^2/3\}$. In particular every positive foreign cost is at least $1/3$. This is not the I cost $1/4$.

The bulk covers five through twelve, with only nonadjacent pure helpers. Choose actual strong $K = 10$ or eleven, and separately nine or ten. In both initial-five branches a positive reference has endpoint no greater than f_K, energy

$$q_9 > .797425,\ q_{10} > .823735,\ q_{11} > .8472225, \quad f_9 < 2.569, f_{10} < 2.732, f_{11} < 2.886.$$

The endpoint price exceeds .16 at nine/ten or 1/7 at ten/eleven. For every actual pure pressure $x.V > G$ flat one through K, $V^2 \le R$, its one-square payment gives

$$\|x\|^2 + e_7 > q_K + .09 + 2\mu(G - f_K) - \mu^2(R - K). \tag{B.11.13}$$

For $K = 10, 11, \mu = 1/7$, it suffices that $14G - R > 32.6$; the two resulting margins above one exceed $25003/9800000, 47961/19600000$.

At thirteen retained D_4 permits three at a leaf. The long leaf fourteen uses numerator $\gamma+1/2$, not the fork numerator $\gamma+1/4$. Own zero/three is affine, otherwise the long-leaf row is strong. In the fork row owns zero/three/four/six and parallel have their actual E_8/cycle/weighted-star/six–three/null obstructions; positive Q and pure finite at least seven or dotted are strong. Its only extra is pure own five. The left H_4 correction on thirteen/twelve/eleven gives $\alpha^2 = \phi + 5/8$, $A^2 = 13\alpha^2 - 1/2$, flat displacement through ten, $R < 13$, $G > 3.48$. Pay .16 at actual nine/ten; the totals in (B.11.13) exceed $1.076545, 1.076295$. Full bridges are finite D_{17} with internal new node, so cannot acquire a finite foreign arm; residual 1/26 and the own-zero cycle cover them.

At fourteen right A_3 adds middle-three/end-three/four/five. Middle-three/end-four have their affine or direct strong rows, end-five is a sealed actual H_4 strong already at own zero. Full bridges are finite D_{17} or A_{17}, with the new node too far from the ends to add an A_2 arm. The lone end-three has residual $5/8 - Q$, numerator $\gamma + 1/4$. Positive Q is strong; zero/three/four/six and parallel have the actual affine/null obstructions. Pure dotted owns are strong. Own five has the left H_4 foot $G > 3.61, R < 14$, flat through eleven, hence $14G - R > 32.6$. For the remaining $.9 < \gamma < 1$ select both actual A_2 cores with old corrections $\gamma(4n_{14}/3 + 2n_j/3)$, $j = 13, 15$. Their averaged pressure is pure, flat through twelve, with

$$\alpha = \sqrt{2}(\gamma + 1/4), \quad G > \sqrt{14 - 125/529} - .327 > 3.38, \quad R = 14 - \frac{14\gamma^2/9 - 3/8}{2(\gamma + 1/4)^2} < 13.67.$$

The subtracted quotient increases from $177/529 > .33$; the radical quotient increases to $8/75 < .327^2$. Thus $14G - R > 33.65$ on the complete interval. All cores have positive coefficients, negative square, future time and nonpositive products at every original wall, including the two defect walls. Coverage now reaches fourteen.

Delete centre fifteen. Retained whites are A_{14} plus the actual two forks and the defect. With left contact η at j, forks δ, ϵ and total foreign cost Q,

$$\Delta = 1 - q_{\text{left}} - \delta^2 - \epsilon^2 - Q > 0, \quad \sigma = \gamma + \eta j/15 + (\delta + \epsilon)/2.$$

The sufficient .61 ratio target is $< 25/16$. The complete old-piece list is left endpoint-three/four, next-three, neither; left plus one fork only endpoint-three with fork-three/four; left plus both forks only the all-three bridge; without left contact, one fork or the pairs $(3,3), (3,4), (3,5)$. Additional cycles, long arms and interior four bonds are not finite. Full bridges are foreign-sealed; the one-fork-three bridge cannot acquire a foreign A_2 arm of lengths fourteen/one/two. Their residuals $1/30, 17/60$ give the affine or direct strong rows. In the no-fork list zero-own far-three would need $1/2 < Q < 8/15$, absent from the exact cost list; far-four/next-three require $Q < 1/15, 2/15$, so have time zero. The other zero-own rows are actual affine principals. The isolated and foreign-cleared feet use genuine old fifteen/fourteen/thirteen, and only the pure left helper remains weak; the sum of the two fork duals is solely a spatial identity.

The complete hard no-left centre rows have the following pure displacements, all flat through at

least eleven:

row	actual construction	G lower	R upper
own 0, fork 5	left H_4	3.49	14
own 0, fork ≥ 7	three-wall core	3.4	15
own 3, fork $4,5$	actual reciprocal D_{17}	3.81	17
own 5, fork 3	left H_4	3.73	15
own ≥ 7/dot, fork 3	special core	3.49	15
own 0, forks $3,4$	actual reciprocal B_{17}	3.81	17
own 0, forks $3,5$	finite H_3	3.53	15

Every row has $Q = 0$: a positive cost makes the five/own-three rows strong and is prohibited by white positivity in the others. For the fork-five row, its normalized time and finite residual are $1 + 8\phi, 13 - 8\phi$, giving $G > 3.49$. For fork at least seven let $z = 1/\delta^2 \in (1, 100/81)$. The positive core is $c + \delta^{-1} n_f + 2(\delta^{-1} - \delta) n_{15}$, with

$$G = \sqrt{17 - 2z} - \sqrt{-8 + 14z - 6z^2} > 3.81 - .41 = 3.4, \quad R = 25 - 16z + 6z^2 < 15.$$

The second radicand is $1/6 - 6(z - 7/6)^2$. For own three and fork four/five the same actual centre wall completes a finite D_{17} endpoint with $\cosh 15/2 + (17/2)\delta > 13$; positive inverse projection retains the defect and recognizes that endpoint. For the three/four pair the finite mate is B_{17} with $\cosh 15/2 + (17\sqrt{2})/4 > 13$. Both give the stated $G > \sqrt{17(6/7)} > 3.81$. Own-five/fork-three has the left H_4 correction and $G > \sqrt{15 - 75/224} - .091 > 3.73$. For the remaining $.9 < \gamma < 6/5$ single-fork-three interval use $c + \gamma^{-1} n_{15} + (2\gamma)^{-1} n_{14}$, of norm $-[1 - 3/(4\gamma^2)]$. Its normalized radical is $< .34$ and time $> \sqrt{15 - 150/529}$, giving $G > 3.49$. Larger owns are directly strong. The three/five pair uses the positive H_3 projection with $q = 2\phi^{-6} < 1/8$ and pressure $\sqrt{15 - q}/(1 + \sqrt{q/15}) > 3.53$. Own zero fork-three/four/six and the two three forks have their affine principals; positive own four/six and parallel their star/path/null exclusions. Remaining positive-own/fork at least four rows have direct ratio $\geq \phi/\sqrt{2} + 1/2$. Thus every own/finite-white/foreign case is either listed, strong, or affine/null impossible. Each listed row satisfies $14G - R > 32.6$, so (B.11.13) excludes it.

Only nonadjacent pure helpers remain through fifteen. Weak five would give a strong fourteen/fifteen endpoint and energy $> .968$ from its actual first-six reference; adding the one .09 defect contradicts the norm. Thus five is strong. If fifteen is strong its reference $q > .9240775$ suffices. Otherwise fourteen is strong, $q > .90715$. The actual weak-fifteen feet use only n_{15}, n_1, n_2 and positive corrections, with both fork products separately negative; the spatial sum $\omega_{16} + \omega_{17}$ is not an added wall. More explicitly, put $\Omega_{16} = \omega_{16} + \omega_{17}$ solely as a spatial sum. Their displacements are $\gamma(\omega_{14} + \Omega_{16})/2 + \omega_3/6$ for five, the same with $\omega_2/4$ for seven/eight, or $\omega_1/2$ for finite owns at least nine. For dotted own the displacement is $(\gamma - \gamma^{-1})\omega_{15} + (\omega_{14} + \Omega_{16})/(2\gamma) + \omega_1/2$. Their respective pressures are $H(A, 2/3 - \gamma^2)$, $A - \sqrt{\gamma^2 - 3/4}$, $H(A, 1 - \gamma^2)$, $A - \sqrt{1 - \gamma^{-2}}$, with $A^2 = 30\gamma^2 + 2\gamma - 1/2$. Concavity of the original heights bounds the dual sum by $2h_{15}$ and $h_2 \leq 2h_1, h_3 \leq 3h_1$. The full native/dotted scalar in the proof of Lemma B.11.11, valid for $\sqrt{10} \leq q \leq \sqrt{42}$, therefore gives $x_1 > \sqrt{2}[T - \gamma(\sqrt{30} - .61)] > .424$. Grafting only the first five slopes gains $(5/4)(.424 - .361)^2 = 3969/800000$. The same reference and one defect therefore give $\|x\|^2 + e_7 > .90715 + 3969/800000 + .09 = 1 + 1689/800000 > 1$. Both typed defects are excluded. □

Lemma B.11.33. *The entire forest $D_{18} + I$ is impossible.*

Proof. Use the actual centre sixteen and signed forks seventeen/eighteen; $x_1 > \cdots > x_{17} > |x_{18}|$, with no assumed sign of x_{18}. Let s, t be the same rank-free actual-I parameters. We will need the slightly stronger elementary barrier

$$t + (8/5)s > .788. \tag{B.11.14}$$

To prove it, substitute $t_0 = .788 - 1.6s > 0$ into $(1-s^2)(1-t^2) - 8st$. At $z = s - 1/5 \in (-1/5, 2/15)$ its value is

$$.00093696 - .0586944z + 8.962384z^2 - .4736z^3 + 2.56z^4 > .0009 - .06|z| + 8z^2 > 0.$$

The cubic is bounded below by $-.064z^2$ and the last quadratic has negative discriminant. Since the original expression decreases with positive t, the barrier follows. On any strong-five reference with $b < .361$, a legal first-five graft and the one I energy consequently add

$$s^2 + (5/4)(t-b)_+^2 > (.788 - .361)^2/[4/5 + (8/5)^2] > .05426.$$

If $t \geq .6$, clip that graft at .6; it alone gains $> .0714$. All other block entries stay above the next slope, so this is an actual positive-dual payment, not an independent norm estimate.

The genuine bulk is complete through thirteen. Choose an actual strong $K = 10$ or eleven before classifying either tail. In the strong-five branch the sharper ordinary references have $q_{10} > .8244, q_{11} > .8482$, endpoints $f_{10} < 2.731, f_{11} < 2.8854$, and available price $> 1/7$. Indeed their unomitted energies exceed $.824619, .848442$; nonadjacent holes at six/eight/ten lose respectively at most half the squares of $.0172, .01111, .008$. The total losses are $< .000209637, .000241637$. In the weak-five branch the actual first/six references give $L_{10} > .8870, L_{11} > .9104$, without any I charge. For every actual pure pressure $x.V \geq G$, flat one through K, $V^2 \leq R$, the two branches therefore give

$$\|x\|^2 + y^2 > \begin{cases} q_K + .05426, & 5 \text{ strong}, \\ L_K, & 5 \text{ weak}, \end{cases} \quad + \frac{2(G - f_K)}{7} - \frac{R - K}{49}. \tag{B.11.15}$$

The pure residual vanishes on the initial support, so its cross with the first-five/I adjustment is zero.

At fourteen retained D_4 admits three at a leaf. Distinguish its long-leaf numerator $\gamma + 1/2$ from short-leaf $\gamma + 1/4$. The former is affine or strong; in the latter positive foreign cost, finite at least seven and dotted owns are directly strong. Zero/three/four/six and parallel have their actual affine/null obstructions. The remaining pure five has a positive left H_4 foot, flat through eleven, $G > 3.62, R < 14$, excluded in both branches of (B.11.15) by > 1.05. Full bridges have internal new node and are foreign-sealed.

At fifteen the right A_3 admits middle-three, end-three/four/five. Middle-three/end-four are affine or strong; end-five is the actual sealed H_4, strong even at own zero. Their full bridges are finite D_{18} or A_{18}, internal too far from the ends for a foreign I arm. The lone end-three has residual $5/8 - Q$, numerator $\gamma + 1/4$. Positive $Q \geq 1/4$ is strong, as are pure owns $\gamma \geq 1$. Own zero/three/four/six and parallel are affine/null. Pure five has $G > 3.74, R < 15$, flat through twelve, and is excluded by (B.11.15). For $.9 < \gamma < 1$ use both actual A_2 cores with corrections $\gamma(4n_{15}/3 + 2n_j/3)$, $j = 14, 16$. Their average has

$$\alpha = \sqrt{2}(\gamma + 1/4), \quad G > \sqrt{15 - 125/529} - .327 > 3.515, \quad R = 15 - \frac{14\gamma^2/9 - 3/8}{2(\gamma + 1/4)^2} < 14.67.$$

The saving increases from $177/529 > .33$ and the radical quotient increases to $8/75$. Inserting the full saving gives > 1.0065 in the strong-five branch and > 1.0152 in the weak branch. All feet are pure D and feasible against the entire actual I block by positive corrections. Thus coverage through fifteen consists only of nonadjacent pure left helpers, before any centre or strong-five floor.

Delete centre sixteen, leaving A_{15}, both forks and I. The exact Schur data are

$$\Delta = 1 - q_{\text{left}} - \delta^2 - \epsilon^2 - Q > 0, \quad \sigma = \gamma + \eta j/16 + (\delta + \epsilon)/2, \quad C_{16} = \sqrt{32}\,\sigma/\sqrt{\Delta},$$

and the sufficient ratio target is < 1.577. Finite classification gives the same complete structural centre list as in Lemma B.11.32, now with A_{15} and the actual I foreign costs. No-left fork pairs are again $(3,3),(3,4),(3,5)$. A left-plus-both-forks bridge has residual $1/32$ and is foreign-sealed; left-plus-fork-four also has residual $1/32$ and is strong at own zero. Left-plus-fork-three has residual $9/32$ at $Q=0$, and may have the retained I-three extension with $Q=1/4$; its smaller residual $1/32$ only strengthens the affine/direct ratios. It is not falsely declared sealed. With no fork the zero-own far-three would need $1/2 < Q < 17/32$, containing no native I square; far-four/next-three need $Q < 1/16, 1/8$, so have time zero. The other zero-own/isolated/foreign-cleared rows have the actual local affine/foot proofs. Only the pure left helper can remain weak without a fork.

For own zero and one fork, labels three/four/six have the actual affine E_8/weighted-star/six–three principals. Fork five has $Q=0$ or $1/4$. Its positive left H_4 projection, with $\alpha = \phi/(2\sqrt{2})$, has

$$Y_D/\alpha = (1^{14}, 1/2, 1/2, 1/2, \pm 1/2), \quad Y_D^2/\alpha^2 = 15, \quad \delta_5/\alpha^2 = 13 - 8\phi.$$

At $Q=0$ its time ratio is $\sqrt{2+8\phi}$, giving $G > 3.62, R = 15$, enough in (B.11.15). At $Q=1/4$ retain the same point's I coordinate rather than clearing and reusing it. Its ambient normalized data are

$$G = \frac{\sqrt{6+6\phi}}{1+\sqrt{(13-8\phi)/(19-2\phi)}} > 3.74, \quad R = 19 - 2\phi < 15.764, \quad d = \sqrt{2}/\phi < .875,$$

where the pressure is $x.V_D + dy \geq G$, V_D flat through both early endpoints. If five is weak the bare ambient graft already exceeds 1.057. If strong, its bare value exceeds .995 using the sharper q_K, f_K above. For this one mixed row form the ambient vector with classical part $w + (V_D - P_K)/7 + \lambda(e_1 - P_5/5)$ and I coordinate $d/7 + \nu$. Optimizing the unused nonnegative multipliers gives the legal additional gain

$$(5/4)(t-b)_+^2 + (s-d/7)_+^2 > \frac{[.788 - b - (8/5)d/7]^2}{84/25} > \frac{.227^2}{84/25} > .01533.$$

This uses (B.11.14) and includes the full shifted I cross. If $t \geq .6$ its clipped initial gain already suffices. The mixed row is therefore excluded.

For own zero and any fork label at least seven, white positivity forces $Q=0$. The positive actual core $c + \delta^{-1} n_f + 2(\delta^{-1} - \delta) n_{16}$ gives, with $b = \delta^2 \in (.81, 1)$,

$$G = \sqrt{18 - 2/b} - \sqrt{-8 + 14/b - 6/b^2}, \quad R = 26 - 16/b + 6/b^2.$$

We need the correlated continuous bound $14G - R > 34.1904$, not independent coarse bounds for G, R. Set $z = 1/b - 7/6$. Then the second radicand is $1/6 - 6z^2$ and $R = 15.5 - 2z + 6z^2$. Concavity gives $\sqrt{18 - 2/b} \geq 4.252 - .252/b$; the endpoint at $100/81$ has squared margin $1311/5062500$. The tangent bound on the other root gives

$$14G - R \geq 39.912 - 14/\sqrt{6} - 1.528z + (42\sqrt{6} - 6)z^2 > 34.1965 - 1.528z + 96.8748z^2 > 34.1904.$$

The last step completes a square, using $1.528^2 < .0061(4 \cdot 96.8748)$. Equation (B.11.15) then exceeds $1.000219, 1.000313$ for strong five, and 1.008 for weak five. Every native label in this continuous interval is covered.

The remaining fork rows have the following actual pure feet:

row	construction	G lower	R upper
own 3, fork 4, 5	reciprocal D_{18}	3.94	18
own 5, fork 3	left H_4 or cleared core	3.64	16
own ≥ 7/dot, fork 3	special core	3.62	16
own 0, forks 3, 4	reciprocal B_{18}	3.94	18
own 0, forks 3, 5	finite H_3	3.66	16

For the first row, positive Q is directly strong; at zero the same cap has actual reciprocal cosh $8+9\delta > 14$, giving $G > \sqrt{18(13/15)} > 3.94$. For five/three, $Q = 0$ or $1/4$ remains, with $A^2 = 16\alpha^2 - 3/4 + Q$, $\alpha^2 = \phi + 5/8$. At $Q = 1/4$ the actual core is $c+\gamma(3n_{16}/2+n_{15}+n_{14}/2)+e/2$, of square $\delta_5 - 1/4 < 0$; it is pure D after clearing. Its time ratio exceeds 3.971 and radical ratio is $< .322$, so $G > 3.64$; the zero-cost projected point is stronger. For single-fork-three and $.9 < \gamma < 6/5$, $Q = 0$ remains; the actual special core has norm $-[1 - 3/(4\gamma^2)]$, time ratio > 3.964 and radical ratio $< .34$, hence $G > 3.62$. Positive Q and larger owns are strong. The two-fork three/four actual mate has cosh $8 + (9/2)\sqrt{2} > 14$. The three/five H_3 projection has $q = 2\phi^{-6} < 1/8$ and pressure $\sqrt{16-q}/(1+\sqrt{q/16}) > 3.66$. The affine triangles, stars, six–three paths, squares and parallel nulls exclude all remaining small-own rows; positive own with fork at least four has direct ratio $\geq \phi/\sqrt{2}+1/2$. Each displayed pure foot is flat through at least thirteen, has positive actual correction coefficients and is feasible at both actual signed forks and every unknown wall. Its substitution in (B.11.15) exceeds one in both initial-five branches. This completes every finite-white/foreign/own case at centre sixteen.

Coverage through sixteen is now exactly the pure left-helper list, with actual nonadjacency. The prefix inverse entries $2\min(i,j)$ hold through the centre; both actual forks are retained positive old blocks in the mixed projections. No spatial dual sum is used as a new root. The same weak-five bootstrap as in Lemma B.11.31 therefore proves five strong: strong sixteen gives bare energy > 1, while strong fifteen gives $> .9851486$ and the capped initial-I gain $> .015$. Choose the actual strong endpoint fifteen or sixteen; it supplies $q > .9240775$, first-five value $b < .361$ and remaining P_5 coefficient $> .146$. Finally delete head one. Retained $D_{17} + I$ has isolated or simple-three at old two, the latter optionally meeting I only by three. Other leaves give exceptional arms too large, and four or a further fork is not finite. Its raw equations are exactly those printed in the preceding lemma, with every pure head point using actual D long roots one/two/three. The actual signed forks remain unselected acute walls. All conditions of the q-free Lemma B.10.9 hold. Its gain $> .07665$ with the one unused I gives a total > 1.0007275. We do not need the older separate proof that sixteen is individually strong. □

Lemma B.11.34 (The remaining ordinary and B proper rows)**.** *The forests $D_{17}+T_2$, $B_{17}+N_7/O_7$, and $B_{18}/D_{18}+$ ordinary A_1 are impossible. Here T_2 is ordinary; both actual typed defects are distinguished.*

Proof. For $D_{17} + T_2$, the genuine bulk reaches twelve. At thirteen the retained D_4 whitelist is exactly that in Lemma B.11.32, with a general ordinary dihedral retained. Any positive foreign cost is at least $1/4$ and makes the short-leaf row strong. Pure finite owns at least seven or dotted have ratio $> 1.15\sqrt{2} > 25/16$, above the old target at $\sqrt{26}$. The sole extra pure five has the actual left H_4 foot $G > 3.48, R < 13$, flat through ten. Pay $1/7$ at actual nine/ten, in either initial-five branch; the pure D energy exceeds .9757, incompatible with $e_{T_2} > 1/7$. Full bridges have internal new node and are foreign-sealed. Hence only nonadjacent pure helpers remain through thirteen. The *proof* of the prefix-gain lemma now applies with $M = 12$: its actual weak-thirteen foot uses only old one/two/twelve/thirteen/fourteen, all genuine long roots. It gives $e_D > .8689525 + .00496125 = .87391375$; the ordinary dihedral again excludes it. No D3 floor is used.

For B_{17} with either typed defect, the bulk reaches thirteen and the right B_3 layer fourteen is completed by its actual long-three or sealed F_4 list. Thus pure-helper coverage reaches fourteen before any bootstrap. Weak five would give an actual strong thirteen/fourteen endpoint and energy $> .7600299945 + (19 - 2\sqrt{78})/7 > .95$; the defect's .09 excludes it. Five is strong. At fifteen the new B_2 short-three row is affine at owns zero/four/six, strong for the other owns at least five, and

strong at own three with positive foreign cost. Its sole extra pure own-three cap is $v = (\omega_{15} + \omega_{17})/2$. Release old sixteen: this same cap, with the entire defect retained, completes finite A_{17} and is the actual endpoint by positive projection. Its cosh is $(16 + 15\sqrt{2})/\sqrt{18} > 8.7$, giving $S_{16} > 3.56$. Choose actual strong $K = 13$ or fourteen; their reference energies are $> .88864, > .90715$, endpoint upper bounds $3.175, 3.311$ and available coefficients $> .14, > .13$. Append the chord to 3.56 at sixteen, skipping fifteen. The slopes $< .13, 1/8$ fit those individual coefficients. The complete totals with the one defect exceed $.88864 + .385^2/3 + .09 > 1.028$ and $.90715 + .249^2/2 + .09 > 1.028$. The extra cap is impossible; coverage now reaches fifteen. The actual prefix-gain proof at $M = 14$ gives $e_B > .91211125$, again incompatible with .09.

For B_{18} with a nonshort A_1, the B_3 layer fifteen is the same sealed affine/strong list as in Lemma B.11.31; it does not use a short-I parameter. For D_{18} distinguish the D4 long leaf at fourteen, complete its bridges and isolate the pure own-five fork row. Its actual foot has $G > 3.62, R < 14$, flat through eleven. The old ten/eleven references give $e_D > .9957$, so the ordinary A_1 cost $> .1$ excludes this extra. At fifteen the right A_3 middle-three/end-four/five and full bridge lists are those printed above. An ordinary-A_1 contact has cost at least $1/4$, strengthening the direct ratios; full bridges have internal new node with long arms and cannot acquire that foreign arm. The pure end-three own-five has $G > 3.74, R < 15$; other remaining $.9 < \gamma < 1$ owns use the two actual A_2 cores, flat through twelve, $G > 3.5, R < 15$. The old eleven/twelve references give respectively $e_D > 1.0094$ and $e_D > .9408$, so both are excluded by the ordinary .1. Thus both B_{18} and D_{18} have complete pure-helper coverage through fifteen in this spectator setting. The prefix-gain proof at $M = 14$, using actual existing long neighbors rather than a false generic rank invocation, gives $e_{B/D} > .91211125$. Adding the nonshort A_1 is impossible. All pressures in this proof are pure in the classical block; the named orthogonal spectator energy is counted once. □

B.11.26 The complete dimension-nineteen forest cover

Proposition B.11.35. *If an actual farthest vertex in compact dimension nineteen exists, its entire forest is one of A_{19}, B_{19}, D_{19}.*

Proof. We supply the own-rank exhaustion, using current component bounds to avoid an unnecessary succession of older weaker frontier tables. As in dimension twenty a dominant component is A_r with $r \geq 7$ or B_r, D_r with $r \geq 9$. Each costs more than one half, so two cannot occur. Without one, discard at most the single I rank: the ordinary cost is $> 18/14$. A typed defect instead gives $.09 + 17/14 > 1$; the funded J pair is stronger than its ordinary rank baseline. Hence exactly one dominant occurs.

An A_7 dominant is excluded by its half bound and remaining ordinary ranks. At eight through twelve use $4/7$: even discarding I leaves six ordinary ranks at twelve; a typed defect leaves five besides its .09. Both costs strictly exceed the complementary budget. At thirteen/fourteen the actual $5/7$ theorem and four ordinary ranks suffice even after discarding I. A typed defect leaves total ordinary rank at least three; its cost $> 17/70$ together with .09 closes this alternative. For rank fifteen use $> .826$. Its ordinary rank-four remainder is $> 2/7$, an I plus ordinary rank-three is $> 2/7$, and a typed defect plus an ordinary dihedral or two ordinary A_1 costs $> .09 + 1/7$. Every sum exceeds one. Rank sixteen has $> .86$; the complete rank-three remainder possibilities are an ordinary irreducible, ordinary dihedral plus A_1, three A_1, $I+$ dihedral, $I+$ two A_1, $J + A_1$, or a typed defect plus ordinary A_1. Their weakest cost is $> .19$, so every sum again exceeds one. This includes the four older residual A16 cases in one line.

At seventeen the ordinary irreducible dihedral costs $> 1/7$, which with $e_{A_{17}} > 6/7$ suffices strictly. Two ordinary A_1, $I + A_1$ and J are stronger. Either actual typed rank-two defect is excluded by

Lemma B.11.29. At eighteen an ordinary A_1 is excluded by $.91 + .1 > 1$; the actual I alternative by Lemma B.11.30. Rank nineteen is the whole A_{19} row.

For a B/D dominant its 9/14 theorem forces rank at least fourteen: at thirteen the worst I remainder leaves five ordinary ranks, or a typed defect leaves four beside .09. At fourteen/fifteen/sixteen the valid weaker D prefix floors are respectively $> .7683275, > .797425, > .823735$. The smallest sufficient spectator charges are respectively $> 2/7$, $> .09 + 1/7$, and $> .19$. These account for all ordinary partitions, I, J, and both distinct typed defects: the rank-five or rank-four I remainders use the proved joint $I+$ rank-three or ordinary-rank rates, not a fictitious ordinary cost for I. Every row exceeds one.

At seventeen a two-A_1 or $I + A_1$ remainder costs $> 1/5$, already sufficient with the weaker prefix $> .8472225$; J is stronger. An ordinary dihedral is excluded for B_{17} by its genuine $M = 12$ floor $> .8689525$, and for D_{17} by Lemma B.11.34. Both typed defects are excluded for B_{17} by that lemma and for D_{17} by Lemma B.11.32. At eighteen the remaining rank-one factor is ordinary or I. The ordinary rows are excluded by Lemma B.11.34; the actual I rows by Lemmas B.11.31 and B.11.33. Rank nineteen is whole.

Finite irreducible classification supplies no additional exceptional types beyond the small components already charged in the ordinary rate. Defect incompatibility forbids a second defect; every surviving old rank-two seven without its specified isolated opposite is ordinary. The rank partitions above are therefore exhaustive and necessary, not assertions that the three retained whole forests are realized. □

The final exact-nineteen conjunction uses this complete cover and the three separate whole-case propositions B.12.1, B.13.1, and B.14.3. Each concerns an actual whole forest at this same rank, with its own complete head/end classification. No twenty-dimensional theorem is decremented and no Coxeter structure is inherited by a face.

B.12 The whole A_{19} case

The next three appendices treat the whole classical forests at rank nineteen. In each case every point and pressure belongs to one original farthest vertex p, chosen from one strict interior point. We retain $z = E(p - x)$, $x \in p^{\perp}$ and $\|x\| < 1$. No arithmetic restriction is used.

B.12.1 A whole A_{19}: the window and the three reference shapes

Use the zero-sum subspace of $\mathbb{R}^{20}$, with actual old roots and duals

$$n_i = (e_i - e_{i+1})/\sqrt{2}, \quad \Omega_i = \sqrt{2}(P_i - iP_{20}/20), \quad D_i = \Omega_i^2 = i(20 - i)/10.$$

Write $h_i = x \cdot \Omega_i$, $h_0 = h_{20} = 0$. The sequence is strictly concave and

$$\|x\|^2 = \frac{1}{2}\sum_{i=0}^{19}(h_{i+1} - h_i)^2.$$

A central site $5 \le i \le 15$ is weak when $h_i \le \sqrt{D_i} - .61$, equality included.

We use the shared complete .61 window of Lemma B.11.2 at $N = 20$, before any contrary component-energy bound: every weak cap is pure, with no spectator projection, and is either

$$v_i = g\Omega_i + \Omega_e/2, \quad e = 1 \text{ or } 19, \quad \cos(\pi/5) \le g < 27/25,$$

or a near-four cap at $i = 5, 6$ or a reflected site. The first row has own native five, finite native at least seven, or dotted; the smaller own labels and parallel case have their actual affine or feasible-null obstructions. The literal helpers are nonadjacent. These are the actual finite-white, own-zero, foreign-clearing and mixed two-cap conclusions of the shared window proof, with retained paths of rank at least four and full connected rank nineteen; they do not follow by applying a higher-rank whole-forest exclusion here. Its full internal proof precedes the higher-tail energy consumers; it is not a published classification or an additional assumed field condition.

There is a useful stronger removal of near four at this rank. For a left near-four cap at $i = 5, 6$, its positive finite replacement at $j = i - 1$ is an actual endpoint. Its white forest is $B_{21-i} + A_{i-2}$; positive inverse projection proves feasibility against every original wall, including the released j. The exact edge cosine is

$$\cosh L_j = \sqrt{2(i-1)/20}\left\{(21-i)/2 + (20-i)/\sqrt{2}\right\} > 11.$$

For $i = 5$, use $.632(8 + 7.5(1.414)) > 11$; for $i = 6$, use $.707(7.5 + 7(1.414)) > 11$. Its normalized dual demand is $> \sqrt{5/6} > .91$. At the opposite end, either a second near-four mate has that same demand, or helper nonadjacency gives a strong $k = 14$ or fifteen, with demand $> 1 - .61/2.7 > .774$. The correct normalized correlation is

$$\rho = \sqrt{\frac{j(20-k)}{(20-j)k}} \le 1/\sqrt{7} < .38.$$

Cauchy would give $(.91 + .774)^2 < 2 + 2(.38) = 2.76$; instead $1.684^2 = 2.835856$. Thus near four is impossible under $\|x\| < 1$, without a stronger component assumption.

All remaining weak sites are therefore nonadjacent pure helpers. Their edge-ratio target is

$$R(q) = 100/61 - \frac{q - .61}{q(2q - .61)}.$$

It increases for $q \ge 5/2$, while $q \le \sqrt{10} < 16/5$ and $R(16/5) = 3/2 - 251/565104 < 3/2$. For side order $m \ge 5$, $\sigma = g + 1/(2m)$, $\delta = (m+1)/(2m)$, and

$$\frac{\sigma^2}{\delta} - (4g - 2) = \frac{(g - 1 - 1/(2m))^2}{\delta} \ge 0.$$

Weakness consequently sharpens the entire helper range to $g < 17/16$. Native five has $.809 < g < .81$; every other finite own has $.9 < g < 1$.

Take rational symmetric heights

$$H_5 = 2.128,\ H_6 = 2.288,\ H_7 = 2.406,\ H_8 = 2.488,\ H_9 = 2.536,\ H_{10} = 2.552, \qquad H_{20-i} = H_i.$$

They lie strictly below the strong floors by positive rational squaring. Linear interpolation on the tails gives nonnegative dual coefficients. Omit every actual internal weak knot by replacing its height by the mean of its neighbours. The holes are nonadjacent, so their open affected edge pairs are disjoint and their exact quarter-square losses add. If an end site five or fifteen is weak, flatten that entire end through six or fourteen instead. The complete reference data are

weak end count	$E = \|w\|^2$ at least	end block lengths	individual capacities
0	$.9544848 - 5(.000441) = .9522798$	$5, 5$	$\beta_5, \beta_{15} \ge 83/625$
1	$3462479/3750000$	$6, 5$	$\beta_6 \ge 79/600,\ \beta_{15} \ge 83/625$
2	$1341604/1500000$	$6, 6$	$\beta_6, \beta_{14} \ge 79/600.$

(A19refs)

For count one orient the weak end to the left by relabelling the old path. This assumes no symmetry of the polytope. There are at most five internal holes in the first row, four in the other rows. The one-end row loses at most 4(.000441); the two-end row loses at most 4(.000324). Their exact undeleted energies follow from

$$a^2/5 + (b-a)^2 + T, \quad \tfrac{1}{2}[b^2/6 + a^2/5 + (b-a)^2 + 2T], \quad b^2/6 + T,$$

where $a = 2.128$, $b = 2.288$ and $T = .118^2 + .082^2 + .048^2 + .016^2$. The boundary coefficients are

$$(a/5 - (b-a))/2 = 83/625, \qquad ((7/6)b - H_7)/2 = 79/600.$$

An adjacent hole increases these capacities. All positive coefficients are paid by actual strong floors, so $x \cdot w \geq E$ in every allowed hole pattern. The constant signed means obey

$$.3 < b_5 < .302, \qquad 1/4 < b_6 < .27.$$

B.12.2 The actual helper feet and their complete continuous bounds

For a weak site five, write $c = Ap - v$, $v = g\Omega_5 + \Omega_e/2$. The fresh exact times are

$$A^2_{\text{far}} = (300g^2 + 20g - 21)/40, \qquad A^2_{\text{near}} = (300g^2 + 60g - 21)/40. \tag{A19times}$$

Both formulas concern $N = 20$.

The following constructions give actual pure spatial pressures $x \cdot U \geq H$. At native five use the finite H_4 correction

$$k = gn_5 + \tfrac{2}{3}n_e + \tfrac{1}{3}n_{\text{adj}}, \qquad \delta = 2/3 - g^2 > 0, \quad U = v - k.$$

The normalized finite projected point is obtained from $p + (A/\delta)(c + k)$. Its future time is positive, its square is $-1 - A^2/\delta$, and all selected products vanish. Every omitted original-wall product is nonpositive by the positive normal coefficients and acuteness. The pressure is

$$H = A + \delta/A - \sqrt{\delta(A^2 + \delta)}/A > A - \sqrt{\delta} > A - .111.$$

The spacelike correction $c + k$ alone is not a hyperbolic point.

For finite own at least seven, put $d = 1 - g^2$ and use $\xi = c + gn_5 + 2dn_e$. Its selected products are zero, zero and $-1/2 + 2d < 0$; its square is $-d(1 - 4d) < 0$. It is future and satisfies every other original wall termwise. Thus $U = v - gn_5 - 2dn_e$, $H = A - \sqrt{d(1-4d)}$, with the last radical at most $1/4$. For dotted own use $\xi = c + g^{-1}n_5$, of square $-(1 - g^{-2})$, zero cap product and negative own-root product. It gives $U = v - g^{-1}n_5$, $H = A - \sqrt{1 - g^{-2}}$. All vectors U are spatial in the original $p^{\perp}$.

A near helper forces $x_1 > .64$. Indeed weakness gives $h_5 < 2.129$, and dropping the positive old-root corrections from the pressure yields $x_1 \geq \sqrt{2}(H - gh_5)$. At native five, $A > 2.365$, giving

$$x_1 > 1.414(2.365 - .111 - .81(2.129)) > .64.$$

For finite $g \geq .9$, $A'_{\text{near}}(g) > \sqrt{7.5} > 2.7$; thus $A_{\text{near}} - 2.129g$ increases. At .9, $A > 2.626$, giving $1.414(2.626 - .25 - .9(2.129)) = .6502986$. For dotted $1 < g < 17/16$, $A'_{\text{near}} < 3$, whereas

$$\frac{d}{dg}\sqrt{1 - g^{-2}} = \frac{1}{g^3\sqrt{1 - g^{-2}}} > 2.$$

Here $g^3 < 5/4$, the radical $< 2/5$. The difference $A_{\text{near}} - \sqrt{1-g^{-2}} - 2.129g$ decreases. At 17/16, its time is > 3.0879 and radical $< .338$, which again gives $x_1 > .64$. All time/derivative comparisons follow by positive squaring in (A19times).

If site five is weak, its first-six reference block has mean $b_6 < .27$. Grafting .64 into its first coordinate, preserving the sum, raises its energy by $(6/5)(.64 - .27)^2 = .16428$. The other five coordinates remain $> (1.5 - .64)/5 = .172$, above the next old coordinate $< .118$. This is a legal ordered dual reference, and even the weakest row of (A19refs) then exceeds one. Every weak end helper is therefore FAR, in both orientations.

For a far helper on six/six blocks, block averaging gives

$$\Pi U = (5g/6)\Omega_6 + \Omega_{14}/12, \quad Y = U - \Pi U, \quad w \cdot Y = 0, \quad \Delta = H - 2.288(5g/6 + 1/12).$$

The three residual squares are

own	$Q = \|Y\|^2$
5	$(2/3)g^2 + 1/12$
finite ≥ 7	$4g^4 - (16/3)g^2 + 29/12$
dotted	$(5/3)g^2 - 19/12 + g^{-2}$.

(A19Q)

At native five, $A_{\text{far}} > 2.188$, so $\Delta > .34 > .30$, $Q < .521 < 13/12$. For all other finite owns, $A_{\text{far}} - 2.288(5g/6 + 1/12)$ increases, since $A'_{\text{far}} > \sqrt{7.5} > 5(2.288)/6$. The complete radical/time seals are

g interval	A_{far} at left $>$	$\sqrt{(1-g^2)(4g^2-3)} \leq$	$\Delta >$
$(.9, .91]$	2.449	.2318	.3105333…
$[.91, .92]$	2.478	.2434	.3088666…
$[.92, 1)$	2.5065	.25	.3117.

The first two radical bounds use the increasing quartic on those short intervals; its global maximum is 1/16. The tight middle lower bound is 23165/75000. The finite residual polynomial increases for $g^2 > .81$ and tends to 13/12. Thus every finite case has $\Delta > .30$, $Q < 13/12$.

In the dotted interval the same derivative argument makes Δ decrease. At 17/16, $A_{\text{far}} > 2.9108$, the radical is $< .338$, and $2.288(5g/6 + 1/12) = 2.2165$. Thus $\Delta > .3563 > .35$. Its Q increases with $g^2 \geq 1$ and at 17/16 equals $229/768 + 256/289 < 148/125$; the cleared margin is $271 > 0$. These bounds cover the whole continuous dotted interval.

B.12.3 Two weak ends: the exact nonzero cross and both prices

Suppose both five and fifteen are weak. Use the six/six reference in (A19refs). For a left far-helper foot, its scaled first and last residual blocks are

$$\sqrt{2}Y_{\text{own}} = (g/3, g/3, g/3, g/3, g/3 - k, -5g/3 + k),$$
$$\sqrt{2}Y_{\text{helper}} = (1/6, 1/6, 1/6, 1/6 - m, 1/6 - l + m, -5/6 + l).$$

Here (k, l, m) equals $(g, 2/3, 1/3)$ at native five, $(g, 2(1-g^2), 0)$ for other finite owns, and $(1/g, 0, 0)$ for dotted. The blocks have zero sum and direct expansion gives (A19Q). For the right foot of parameter d, reverse and negate the two blocks. In the product of an own block with a reversed helper block only its final two deviations survive; their scaled product is $-g/3$, independently of k, l, m. Consequently for EVERY type pair,

$$Y_L \cdot Y_R = (g + d)/6, \qquad \|Y_L + Y_R\|^2 = Q_L + Q_R + (g + d)/3.$$

We have not assumed orthogonality of the two actual feet.

Choose $\mu = 13/100$ and form one vector $W = w + \mu(Y_L + Y_R)$. At each of the two individual anchors, the combined debit is

$$\mu(5g/6 + 1/12) < 403/3200 < 79/600.$$

The remaining coefficient is nonnegative before using its original floor. Both actual pressures and one completed square give the following complete three-pair table:

types	$\|Y_L + Y_R\|^2 <$	gain over $E >$
finite/finite	17/6	6487/60000
finite/dotted	17729/6000	.11906
dotted/dotted	9229/3000	.13000.

Native five belongs to the finite column. For example the first gain is $.26(.60) - .0169(17/6)$. All three gains follow from the full cross above and the uniform deficit/norm bounds. Thus

$$\|x\|^2 > \frac{1341604}{1500000} + \frac{6487}{60000} = 1 + \frac{3779}{1500000} > 1.$$

This excludes the two-weak-end case with every internal hole retained.

B.12.4 Zero or one weak end: the actual endpoints and their feet

For either of the other reference rows, if $x_1 \geq .55$ graft .55 into its constant first block, preserving the block sum. For length five the gain exceeds $(5/4)(.55 - .302)^2 = .07688$; for length six it exceeds $(6/5)(.55 - .27)^2 = .09408$. Both exceed $1 - E$ in their respective rows. The remaining block coordinates stay $> .2375$ or $> .19$, above the next coordinate $< .16$; a neighbouring hole only improves that junction. Reflection gives $x_1, -x_{20} < .55$. The actual endpoint comparison is

$$x_1^2 \geq (19/20)\frac{\cosh L - 1}{\cosh L + 1}.$$

Since $19/60 > .55^2$, both actual endpoint cosines are less than two.

Deleting one retains A_{18}. Finite classification gives exactly isolation; one simple-three at $2, 3, 18, 19$; or one simple-four at two or nineteen. Two contacts make a nonpositive cycle; exceptional finite types cannot contain this rank-eighteen retained path. For $v = g\Omega_1 + \beta\Omega_j$ the complete nonisolated data are

white contact	$\cosh L$	small-own obstruction
3@2	$(38g + 18)/20$	$g = 0:\ A^2 < 0; \quad g = 1/2:$ affine triangle
3@3	$(38g + 17)/(2\sqrt{20})$	$g = 0:$ affine E_8
3@18	$(38g + 2)/(2\sqrt{20})$	$g = 0:\ A^2 < 0$
3@19	$(38g + 1)/20$	$g = 0:\ A^2 < 0; \quad g = 1/2:\ A^2 = 0$
4@2	$(38g + 18\sqrt{2})/\sqrt{40}$	already > 2 at $g = 0$
4@19	$(38g + \sqrt{2})/\sqrt{40}$	$g = 0:\ A^2 < 0.$

For the affine E_8 at old three, the actual arms have lengths two (old two/one), one (the cap), and five (old four through eight). The positive affine combination has future time through the cap and nonpositive omitted-wall products. Every other positive native contact is at least $1/2$; in the first row the next allowed own is at least $1/\sqrt{2}$. Substitution makes every row except 3@19 exceed two.

A remaining nonisolated left endpoint is therefore the far-three cap

$$v_L = g\Omega_1 + \Omega_{19}/2, \quad A_L^2 = (19/10)g^2 + g/10 - 21/40, \quad 1/\sqrt{2} \le g < 39/38 < 1.1.$$

If both endpoints had this form, their time squares would be $< (8/5)g^2, (8/5)d^2$, since $.3g^2 + .1g - .525 < 0$ for $g < 1.1$. But

$$v_L \cdot v_R = (gd + 1/4)/10 + (19/20)(g + d) > (19/11)gd > (8/5)gd > A_L A_R.$$

The actual walls are distinct: coincidence would force $g = d = 1/2$. Their positive mutual contradicts acuteness. At least one endpoint is isolated.

For an isolated endpoint, $v = \sigma\Omega_1$, $A^2 = (19/10)\sigma^2 - 1 > 0$. Own zero/three/four has nonpositive time, six has its actual affine $6, 3$ principal, and a parallel own has the nonzero feasible future null $c + n_1$. The complete remaining own list is five, finite at least seven, or dotted. For dotted own, the actual edge bound also gives $19/40 < u = \sigma^{-2} < 1$.

Normalize each actual isolated foot by $\sqrt{2}\sigma$. On the left write $Z = \zeta - P_{20}/20$, $\sum \zeta_i = 1$; the right formula is negative reversal. The complete profiles, pressure $x \cdot Z \ge t$ and square $R = \zeta^2$ are

own	ζ	t	R
5	$(1/4, 1/4, 1/4, 1/4)$	$> .34$	$1/4$
finite ≥ 7	$(1 - u/2, 1 - u/2, u - 1)$	$\{\sqrt{1.9 - u} - \sqrt{7u - 4 - 3u^2}\}/\sqrt{2}$	$3 - 4u + 3u^2/2$
dotted	$(1 - u/2, u/2)$	$\{\sqrt{1.9 - u} - \sqrt{u(1 - u)}\}/\sqrt{2}$	$1 - u + u^2/2.$

(A19endfeet)

For native five, use the finite H_4 correction $(3/2)\sigma n_1 + \sigma n_2 + (\sigma/2)n_3$. Its residual $\delta = 1 - (3/2)\sigma^2 > 0$ gives the actual finite projected point $p + (A/\delta)(c + k)$, of square $-1 - A^2/\delta$. It is future and satisfies every original wall by the selected product calculation and positivity. Its raw pressure is $H = AL/(L + \sqrt{\delta})$, $L^2 = A^2 + \delta = .4\sigma^2$. The positive-square bounds $A > .4934, L > .5112, \sqrt{\delta} < .1352$ give $H > .390$ and $t > .34$.

For finite own at least seven, use $c + \sigma^{-1}n_1 + 2(\sigma^{-1} - \sigma)n_2$. Its cap/own products vanish, the next product is $(3 - 4\sigma^2)/(2\sigma) < 0$, and its square is $-(7 - 4\sigma^2 - 3/\sigma^2) < 0$. The dotted core is $c + \sigma^{-1}n_1$, square $-(1 - u)$, zero cap product and negative own product. Both have positive future time, positive coefficients on actual normals, and every omitted-wall product nonpositive. They prove the remaining two profiles, for this same x.

The following scalar seals cover all their parameters. Native seven satisfies $.9009 < \sigma < .901$, hence $1.23 < u < 1.233$,

$$\sqrt{1.9 - u} > .8167, \quad \sqrt{7u - 4 - 3u^2} < .2671, \quad t > .388, \quad R < .34935.$$

The cosine bracket follows by signs of $8\sigma^3 - 4\sigma^2 - 4\sigma + 1$; it asserts no field restriction. For finite own at least eight, $u < 4 - 2\sqrt{2} < 1.1716$, $\sqrt{1.9 - u} > .85346$, and

$$7u - 4 - 3u^2 = 1/12 - 3(u - 7/6)^2 < 1/12.$$

Thus $t > .399$, $R \le .5$. At $u \ge 1.1$, $R \le .415$; at $u < 1.1$, the radicals improve to $> .8944$ and $< .265$, giving $t > .4448$.

For dotted own put $q = \sqrt{1.9}$, $a = q - \sqrt{.9} = 1/(q + \sqrt{.9}) < .43$. The lower chord and circle maximum give

$$\sqrt{1.9 - u} - \sqrt{u(1 - u)} \ge q - au - \sqrt{u(1 - u)} > .6185,$$

using $q > 1.378$, $\sqrt{.9} > .948$, $\sqrt{1 + a^2} < 1.089$. Hence $t > .437$. On $u \ge .5$ the sharper chord $\sqrt{1.9 - u} \ge 1.4178 - .4692u$ and $\sqrt{1 + .4692^2} < 1.105$ give $t > .445$. On $u < .6$, use $1.14 - .5 = .64$

to get $t > .4525$. Also $R < .6378125$ throughout, and $R \le .58$ when $u \ge .6$. All square-root and chord comparisons are strict rational square checks.

On a constant end block of length m, the projection of Z is its boundary dual divided by $m\sqrt{2}$. Thus for $V = Z - \Pi Z$,

$$w \cdot V = 0, \quad \|V\|^2 = R - 1/m, \quad \Delta_h = t - b_m.$$

A paid coefficient $\nu \ge 0$ costs $\nu/(m\sqrt{2})$ at that individual boundary. Combining the actual pressure with the remaining original floors BEFORE its square gives

$$\|x\|^2 \ge E + 2\nu\Delta_h - \nu^2(R - 1/m). \tag{A19endpay}$$

B.12.5 No weak end: one isolated endpoint already suffices except at five

For zero weak ends, take $\nu = 1/2$ in the isolated finite case. Its price is $< 1/14 < 83/625$. At native seven the gain is $> .086 - .15/4 = .0485$. For all finite labels at least eight, the complete subdivision is

u	$t >$	$R - 1/5 <$	$\Delta_h - (R - 1/5)/4 >$
$(1, 1.1)$	.4448	.3	.0678
$[1.1, 1.14)$	.4122	.215	.05645
$[1.14, 1.1716)$	.399	.1894	.04965.

The middle interval uses radicals $> .8717, < .28868$, with $.4122(1.4143) < .8717 - .28868$. The last uses $> .85346, < .28868$; the polynomial R decreases here, with $R(1.1) = .415, R(1.14) = .3894$. Thus every finite isolated own at least seven gives

$$\|x\|^2 > .9522798 + .0485 > 1.$$

For an isolated dotted own, take $\nu = 1/3$, price $< 1/21 < 83/625$. At $u < .6$ the gain is $> (2/3)(.1505) - .438/9 = 31/600$. At $u \ge .6$ it is $> (2/3)(.143) - .38/9 = 239/4500$. Both gains exceed $1 - .9522798$.

Only isolated native five remains. It cannot coexist with a nonisolated far-three opposite: its time is $< .497$, while the latter has time < 1.257. Their spatial product is $\sigma(d/10 + 19/20) > .95(.809) = .76855$, whereas the time product is $< .497(1.257) = .624729$. Their actual mutual would be positive. Thus the other endpoint is isolated as well, and the preceding cases force it too to have native five.

For each normalized native-five foot, $\Delta_h > .038$, $\|V\|^2 = .05$. Use $\nu = 3/4$ at BOTH ends; each price is $< 3/28 < 83/625$. The two residuals have disjoint first/last five-block support and zero block sum, so their cross is exactly zero. The single vector $w + (3/4)(V_L + V_R)$ gives

$$\|x\|^2 > .9522798 + 2\{(3/2)(.038) - (9/16)(.05)\} = 1.0100298 > 1.$$

This exhausts zero weak ends.

B.12.6 One weak end: the helper and endpoint must be paid jointly

Orient the weak site to five. It is far by the helper argument. On the six/five reference its block projection and residual are

$$\Pi U = (5g/6)\Omega_6 + \Omega_{15}/10, \quad Y = U - \Pi U, \quad w \cdot Y = 0, \qquad \Delta = H - 2.288(5g/6) - 2.128/10.$$

Relative to six/six, the square decreases by 1/60 and the deficit decreases by 83/3750. The exact finite middle-row bound 23165/75000, and the stronger native-five/dotted bounds, give

$$\Delta > .2867, \qquad Q = \|Y\|^2 < 1.168.$$

The dotted upper square is $148/125 - 1/60 = 1751/1500 < 1.168$.

We must retain the full cross with the isolated endpoint. Write the far-helper correction at old nineteen/eighteen as $(l, m) = (2/3, 1/3)$, $(2(1-g^2), 0)$, or $(0, 0)$. Its scaled last-five residual is

$$\sqrt{2}Y_{\text{helper}} = (1/5, 1/5, 1/5 - m, 1/5 - l + m, -4/5 + l).$$

For an isolated STRONG-side endpoint, its residual there is $1/5$ minus the reversed ζ. Direct multiplication gives

isolated own	$Y \cdot V$
5	$1/(20\sqrt{2})$
finite ≥ 7	$\{.8 - u/2 + m(1.5u - 2)\}/\sqrt{2} \leq (.8 - u/2)/\sqrt{2}$
dotted	$\{.8 - u/2 - l(1-u) - mu/2\}/\sqrt{2} \leq (.8 - u/2)/\sqrt{2}$.

The finite sign uses $u < 1.233 < 4/3$. For an isolated WEAK-side endpoint, its ζ ends before coordinate five; multiplication with the own six-block row printed earlier gives

$$Y \cdot V = g/(3\sqrt{2}) < 85/336.$$

These formulas retain every correction coefficient, including its sign.

For paid nonnegative μ, ν, one vector $W = w + \mu Y + \nu V$ therefore yields

$$\|x\|^2 \geq E + 2\mu\Delta + 2\nu\Delta_h - \mu^2 Q - 2\mu\nu(Y \cdot V) - \nu^2(R - 1/m). \tag{A19joint}$$

Both individual anchor debits must fit before either old floor is used.

If an isolated endpoint is on the strong side, choose $\mu = .148$. Its own-six debit is

$$\mu(5g/6) < 3145/24000 < 79/600.$$

At fifteen, the helper AND endpoint together cost

$$\mu/10 + \nu/(5\sqrt{2}) < 37/2500 + 4/35 < 83/625$$

for all choices $\nu \leq .8$ below. The common helper gain exceeds .059279328. Using the full cross, the complete table is

isolated strong-side own	ν	$\Delta_h >$	$R - 1/5 <$	$Y \cdot V <$	total gain over $E >$
5	.8	.038	.05	1/28	.07962218
7	.3	.086	.15	.133	.085568928
finite ≥ 8, $u \geq 1.1$	.3	.097	.215	.179	.082234128
finite ≥ 8, $u < 1.1$	.3	.1428	.3	.215	.098867328
dotted, $u \geq .6$	.25	.143	.38	.358	.080537328
dotted, $u < .6$	.25	.1505	.438	.402	.077406328.

For instance the last row is exactly $.059279328 + .5(.1505) - .0625(.438) - .074(.402)$. All cross bounds use $\sqrt{2} > 7/5$ and the stated intervals. Every gain is $> .0774$.

If the strong endpoint is not isolated, the weak endpoint is isolated by the actual endpoint classification. It cannot have native five: the same time/spatial comparison with a far-three opposite, just proved, would force that opposite to be isolated. The remaining finite/dotted weak-side cases use $\mu = .1, \nu = .35$. Now BOTH feet spend the SAME six coefficient; the full debit is

$$\mu(5g/6) + \nu/(6\sqrt{2}) < 17/192 + 1/24 = 25/192 < 79/600.$$

At fifteen the cost is only .01. The common helper gain is $> .04566$ and the full cross penalty is $< .07(85/336) = 119/6720$. The complete table is

isolated weak-side own	$\Delta_h >$	$R - 1/6 <$	total gain over $E >$
7	.118	.183	.08813416
finite ≥ 8	.129	1/3	.07741833
dotted	.167	.472	.08703166.

The tight finite row is $.04566 + .7(.129) - .1225/3 - 119/6720 = .077418333\ldots$. All gains again exceed .0774. Thus in every one-weak-end case,

$$\|x\|^2 > \frac{3462479}{3750000} + .0774 = 1 + \frac{2729}{3750000} > 1.$$

Proposition B.12.1. *A whole A_{19} forest cannot occur at the original farthest vertex of a compact Coxeter nineteen-polytope.*

Proof. The number of weak sites among the two central end sites five/fifteen is exactly zero, one or two. The three preceding arguments exclude these cases, with equality assigned consistently to weak. Their shared local window, endpoint and foot lemmas precede all three consumers and do not use any end-case conclusion. Every remaining internal weak site was retained through its actual omission, and every new pressure was paid inside one original-vector reference. □

B.13 The whole B_{19} case

B.13.1 The short layers before the terminal analysis

Use actual unit roots

$$n_i = (e_i - e_{i+1})/\sqrt{2}\ (i \leq 18), \qquad n_{19} = e_{19}, \quad \omega_i = \sqrt{2}P_i\ (i \leq 18), \quad \omega_{19} = P_{19}.$$

Now $x_1 > \cdots > x_{19} > 0$. Put $S_i = x \cdot P_i$, $f_i = \sqrt{i} - .58/\sqrt{2}$, $d_i = \sqrt{i+1} - \sqrt{i}$, and

$$E_K = f_5^2/5 + \sum_{i=5}^{K-1} d_i^2.$$

Weak includes equality $S_i \leq f_i$. The local whole-diagram lemma B.11.7 applies literally through fifteen: both retained old components have rank at least four. Its complete weak list is

$$v = g\omega_i + \omega_1/2, \quad A^2 = 2ig^2 + 2g - 1/2, \quad \cos(\pi/5) \leq g < 6/5, \tag{B19packet}$$

with own five, finite at least seven, or dotted, together with isolated dotted five. Pure helpers are nonadjacent. We must extend this at the actual B_3, B_2 boundary.

At sixteen the right B_3 has a long-three extension or the short-three F_4 extension. The latter is foreign-sealed; own zero has the actual affine F_4 principal, whereas every positive own has ratio

$\geq 1 + 1/\sqrt{2} > 83/50$. A full bridge is B_{19} and directly strong. Thus the same list holds through sixteen, also at the old .61 threshold. In particular some new strong $K = 15$ or sixteen exists. The full conditional early-reference Lemma B.11.8 therefore gives a paid vector supported through K, of sum f_K, remaining individual P_K coefficient $> 1/9$, and effective energy

$$L_{15} > .94006, \qquad L_{16} > .95618, \quad f_{15} < 3.463, \ f_{16} < 3.590. \tag{B19early}$$

Effective energy means $2D_W - \|W\|^2$, where D_W is its actual pressure demand; it need not equal $\|W\|^2$.

At seventeen the right block is B_2, with simple-three at either end. The long-three row is strong or affine. For short three, own zero/four/six have actual affine principals, parallel has its positive future null, and finite own at least seven has directly strong ratio $\sqrt{2}g + 1/2 > 83/50$. A full bridge has residual $1/34$ and numerator $\geq g + 1/34 + 1/(2\sqrt{2})$, strong even at own zero. Only own three and five remain.

For own five, let $g = \varphi/2$, $\alpha = \sqrt{2}g + 1/2$. The LEFT actual H_4, using seventeen/sixteen/fifteen, has

$$\delta = (5 - 3\varphi)/8, \quad A^2 = 17\alpha^2 - 1/2, \quad V_1 = \cdots = V_{14} = 1, \quad V^2 < 17, \quad x \cdot V > G > 4.$$

The point is the positive finite projection, not its spacelike correction; every omitted-wall product is nonpositive. Use $\alpha > 1.643$, $\sqrt{\delta}/\alpha < .083$ for the last bound. An OLD strong eleven or twelve exists by the original bulk. Their positive references, in both old initial-five branches, have

$$q_{11} > .8472225, \ q_{12} > .8689525, \quad f_{11}^{\text{old}} < 2.886, \ f_{12}^{\text{old}} < 3.034, \quad \text{individual endpoint price} > 1/7.$$

These are the actual prefix construction with its holes retained, not an assertion that old five is strong. Their support lies in the flat first fourteen. One paid square gives

$$.8472225 + 2(4 - 2.886)/7 - 6/49 > 1.043, \qquad .8689525 + 2(4 - 3.034)/7 - 5/49 > 1.042.$$

For own three, the SAME cap has a finite D_{19} replacement after releasing the short old nineteen. Positive inverse projection gives an actual reciprocal short-edge endpoint, with

$$\cosh L_{19} = 17/2 + 19\sqrt{2}/4 > 15, \qquad S_{19}^2 > 133/8 > (4.07)^2.$$

Use (B19early), and the pressure $Z = P_{19}$. The legal $1/9$ endpoint payment gives

$$.94006 + 2(4.07 - 3.463)/9 - 4/81 > 1.025, \qquad .95618 + 2(4.07 - 3.590)/9 - 3/81 > 1.025.$$

Thus coverage reaches seventeen.

Nonadjacency is also actual at this boundary: replacing two adjacent helper caps at sixteen/seventeen retains the long A_{15} and the independent old B_2. Adding their actual far-one leaves gives the positive $D_{17} + B_2$. The later seventeen/eighteen pair instead gives $D_{18} + A_1$. The positive two-cap inverse and all-wall projection therefore apply. No long nineteenth old normal is invented. Every weak site five through seventeen is now (B19packet), except isolated dotted five.

B.13.2 An isolated long-end foot is a local geometric kernel

We record the point formulas once. They apply to any actual isolated long-end cap with initial long roots one/two/three as above, in B or D, whenever the listed own occurs. Write

$$c_L = Ap - \gamma\omega_1, \qquad A^2 = 2\gamma^2 - 1.$$

For each remaining own type the actual normalized pressure is $x \cdot Z \geq h$, $\sum Z_j = 1$:

own	Z	h	$R = Z^2$
5	$P_4/4$	$\sqrt{(\varphi-1)/4}$	$1/4$
finite ≥ 7	$(1-u/2)(e_1+e_2)+(u-1)e_3$	$\{\sqrt{2-u}-\sqrt{7u-4-3u^2}\}/\sqrt{2}$	$3-4u+3u^2/2$
dotted	$(1-u/2)e_1+(u/2)e_2$	$\{\sqrt{2-u}-\sqrt{u(1-u)}\}/\sqrt{2}$	$1-u+u^2/2.$

(B19headfeet)

Here $u = \gamma^{-2}$, in $(1, \cos(\pi/7)^{-2}]$ for finite own, and $(0,1)$ for dotted. All coordinates of Z are nonnegative.

At native five, the actual H_4 correction is $\gamma(3n_1/2+n_2+n_3/2)$, with residual $\delta = 1-3\gamma^2/2 > 0$. The future finite projected point $p+(A/\delta)(c_L+k)$ has square $-1-A^2/\delta$ and every original wall product nonpositive. Its displacement is $(\gamma/4)\omega_4$, and its pressure is $H = AB/(B+\sqrt{\delta})$, $B = \gamma/\sqrt{2}$. The real native-five identity $4\gamma^2 = 2\gamma+1$ gives $\sqrt{\delta} = (1-\gamma)/\sqrt{2}$, hence $H = \gamma A$ and the first row exactly. In particular $h > .393075$, by positive rational squaring.

For finite own at least seven, the actual negative core is

$$c_L + \gamma^{-1}n_1 + 2(\gamma^{-1}-\gamma)n_2.$$

Its cap/own products vanish, the next product is $(3-4\gamma^2)/(2\gamma) < 0$, and its square is $-(7-4\gamma^2-3/\gamma^2) < 0$. For dotted own use $c_L+\gamma^{-1}n_1$, square $-(1-u)$, zero cap product and negative own product. Both have positive future time and positive coefficients on actual normals. Every omitted original-wall product is nonpositive termwise. Their spatial displacements, divided by $\sqrt{2}\gamma$, give the other two rows on the SAME original x. This proves the kernel without an I spectator or an arithmetic hypothesis.

For every finite own at least seven and every dotted own,

$$.8h - .16R > .2893. \tag{longF}$$

At native seven, $.9009 < \gamma < .901$ gives $A > .7894$, $\sqrt{7-4\gamma^2-3/\gamma^2} < .2395$, $\sqrt{2}\gamma < 1.2743$; thus $h > .4315$. Also $1.2318 < u < 1.233$ and $R < .349$, giving $> .28936$.

For finite labels at least eight, $1 < u < 1.172$. Put $a = .8/\sqrt{2}$, $D = (u-1)(4-3u)$. The derivative of the left side is

$$a\left\{-\frac{1}{2\sqrt{2-u}} - \frac{7-6u}{2\sqrt{D}}\right\} + .64 - .48u.$$

On $(1, 7/6]$, the first term is $< -.28$, the second is nonpositive, and the last is at most .16. On $[7/6, 1.172)$, $\sqrt{D} > .284$, the positive second term is $< .032$, and the other terms are $< -.28 + .08$. Thus the scalar decreases with u; its minimum is at native eight. There the radical bounds are $> .910, < .289$, and $R < .373$. Hence $h > .4388$ and the scalar is $> .29136$.

For dotted own, $.5656 < a < .5657$. On $u \leq .5$, use $\sqrt{2-u} \geq 1.4142 - .379u$, $R \leq 1 - .75u$, giving

$$.8h - .16R > .63987152 - .0944u - .5657\sqrt{u(1-u)} > .30587152.$$

The circle bound uses $\sqrt{.0944^2 + .5657^2} < .5736$. On $u \geq .5$, use $\sqrt{2-u} \geq 1.4494 - .4494u$, $R \leq .75 - .25u$, giving

$$.8h - .16R > .69978064 - .2142u - .5657\sqrt{u(1-u)} > .29023064,$$

with $\sqrt{.2142^2 + .5657^2} < .6049$. The chords lie below the concave root by their endpoint square checks. This proves the entire continuous dotted range.

B.13.3 The B_{19} long head and its paid gains

The actual long head is isolated. Indeed deleting one retains B_{18}, whose complete finite one-node list is isolation or simple-three at its long leaf two. A second contact makes a cycle; another four or an additional fork is not finite; exceptional finite ranks are too small. The nonisolated cap would have

$$v = \gamma\omega_1 + \omega_2/2, \quad A^2 = 2\gamma^2 + 2\gamma, \quad \cosh L_1 = 2\gamma + 1.$$

Own zero has zero time and own three is the actual affine triangle, so $\gamma \geq 1/\sqrt{2}$ and $x_1 > \sqrt{\sqrt{2}-1} > .643$.

The OLD actual bulk through fifteen has some strong $K = 14$ or fifteen. If old five is strong, its paid reference has energy $> .90715$, with every actual weak hole omitted. Graft .643 into its first-five block of mean $< .361$. The other four coordinates remain $> .2904$, above the outgoing slope $< .214$, and the gain is $(5/4)(.643 - .361)^2 = .099405$. If old five is weak, six is strong. Begin only with the FORMAL strong-five curve, whose scalar energy is E_K^{old}. Here $f_i^{\mathrm{old}} = \sqrt{i} - .61/\sqrt{2}$ and $E_K^{\mathrm{old}} = (f_5^{\mathrm{old}})^2/5 + \sum_{i=5}^{K-1} d_i^2$. Before assigning any pressure to it, replace its ENTIRE first-six block by .643 and five equal slopes of sum $f_6^{\mathrm{old}} - .643$. Their energy exceeds

$$.643^2 + (2.01815 - .643)^2/5 = .7916565045,$$

whereas the formal first-six energy is $E_6^{\mathrm{old}} < .6971$. The new slopes are ordered: $.643 > (f_6^{\mathrm{old}} - .643)/5 > .275$, above every outgoing slope $\leq d_6 < .197$. There is NO coefficient at five. All later actual weak knots lie in $7, \ldots, K-1$; at most four are nonadjacent, each costing $< 1/3200$. Their affected open intervals are disjoint from the first-six replacement, even for a hole at seven. Only now combine the actual pressures at one, six and the remaining strong knots. One square gives

$$\begin{aligned} \|x\|^2 &> E_K^{\mathrm{old}} - E_6^{\mathrm{old}} + .7916565045 - 4/3200 \\ &> .9084 - .6971 + .7916565045 - .00125 \\ &= 1.0017065045 > 1. \end{aligned}$$

Both references are ordered and paid by the original x. Thus the nonisolated head is impossible.

For the isolated cap, $A^2 = 2\gamma^2 - 1 > 0$ excludes own zero/three/four. Own six has the affine $6, 3$ principal on the actual first two roots; parallel has its feasible positive future null. The remaining owns are exactly native five, finite at least seven, or dotted. Their actual feet are (B19headfeet).

A native-five head forces true new strong five/six/seven. For a helper at $i \leq 7$,

$$c_L \cdot c_i = \gamma_0(2g+1) - A_0 A_i, \qquad \gamma_0 = \varphi/2, \quad A_0^2 = (\varphi - 1)/2.$$

Since $\gamma_0^2/A_0^2 = \varphi^3/2 > 21/10$,

$$A_i^2 - \frac{\gamma_0^2}{A_0^2}(2g+1)^2 < (28g^2 - 32g - 13)/5 < 0 \qquad (.8 < g < 1.2).$$

The convex quadratic is negative at both endpoints. The actual mutual would be positive. For the remaining weak isolated dotted five,

$$c_L \cdot c_5 = \varphi g - A_0\sqrt{10g^2 - 1}.$$

Its sign is strictly negative, since the squared difference in the opposite order is $(4\varphi - 6)g^2 - (\varphi - 1)/2 \geq (7\varphi - 11)/2 > 0$. Weakness gives $g < 1/.58 < 2$; $\varphi - \sqrt{5(\varphi - 1)} > -3/20$ then bounds the mutual

in $(-3/10, 0)$, a forbidden native interval. This proves the stated three strong sites without a later terminal floor.

For any strong-five .58 reference with first-five mean $b = f_5/5 < .3652$ and individual P_5 coefficient $> .151$, a head correction $\mu(Z - P_5/5)$ costs $\mu/5$ before its floor is used. It is orthogonal to that constant block and has square $R - 1/5$. Its effective gain is

$$G_L = 2\mu(h - b) - \mu^2(R - 1/5).$$

The complete paid gains are

head	μ	$G_L >$
5	.56	.015456
finite ≥ 7	.4	.02914
dotted	$\sqrt{2}/5$	.03384.

(B19headgains)

The first uses $h > .393$; the second is (longF) minus $.8b$ plus .032. For the dotted row, expand its gain as

$$.4\{\sqrt{2 - u} - \sqrt{u(1 - u)} - \sqrt{2}b\} - .04(1.6 - 2u + u^2),$$

where $\sqrt{2}b < .5166$. For $u \leq .5$, its lower bound is $.29504 - .0916u - .4\sqrt{u(1 - u)} > .04404$, using $\sqrt{.0916^2 + .4^2} < .4104$. For $u \geq .5$, it is $.32912 - .15976u - .4\sqrt{u(1 - u)} > .03384$, using $\sqrt{.15976^2 + .4^2} < .4308$. The same two concave root chords prove these inequalities. All three individual prices fit.

B.13.4 Curved early references with the whole initial-five debt

Some subsequent pressures need a stronger early reference than (B19early). We give it by the actual initial five alternatives, retaining all holes and full cross terms. Each resulting vector is supported through $K = 15$ or sixteen, has sum f_K, and keeps endpoint coefficient $> 1/9$.

First record a useful exact scalar:

$$E_{17} > .9726645. \tag{B19E17}$$

Its first block is $f_5^2/5 = 1.03364 - .116\sqrt{10} > .6668157$, using $\sqrt{10} < 3.162278$. For $a = 4i + 2$, $u = d_i^2$ satisfies $u(a - u) = 1$. Three positive reciprocal substitutions give $u > (a^2 - 1)/(a(a^2 - 2))$. At denominator 10^7, the full list of lower numerators and cleared positive margins is

i	L_i	margin	i	L_i	margin
5	455488	5248	11	217494	13464
6	385186	536	12	200080	8000
7	333704	14240	13	185248	115712
8	294372	20208	14	172465	14860
9	263340	21360	15	161332	72272
10	238230	27080	16	151549	273164.

The margin is $10^7(a^2 - 1) - L_i a(a^2 - 2)$. The numerators sum to 3058488, proving (B19E17). This is a finite rational hand certificate, not a script-output premise. Also $d_{17}^2 > 1/70$, giving $E_{18} > .98659$.

If five is strong, its curved reference loses $< 1/1024$ over all nonadjacent internal holes. Adding the actual head pressure inside that reference gives

$$L_{15} > .955516, \qquad L_{16} > .971636.$$

These conservative values use the smallest gain in (B19headgains); the appropriate finite/dotted head is stronger.

If five is a weak helper, the actual packet gives $x_1 > .5455$ and six is strong. Use the first-six reference through these actual heights, followed by the ENTIRE curved retained window. Its first-six energy is $> .74385$, rather than the ordinary $< .7126$. Since $d_{15}^2 < .016134$, $d_{16}^2 < .015155$, (B19E17) gives $E_{15} > .9413755, E_{16} > .9575095$. After all holes,

$$L_{15} > .9716, \qquad L_{16} > .9877.$$

No long-head energy was added to this initial-coordinate gain.

If five is isolated dotted, the native-five head has already been excluded by its compatibility. For the other head types keep its actual foot $x \cdot Z \geq h$ and the initial-five foot

$$Z_5 = P_5 + (v/2)(e_6 - e_5), \quad x \cdot Z_5 > T_5 > 1.8185, \quad .4 < v < 1.$$

This is the actual dotted-five negative core; the weak condition forces $v > .4$ by the full continuous loss bound in the shared early-reference lemma. Choose a genuinely strong $J = 6$ or seven. The reference is constant $m = f_J/J$ on the first J coordinates and follows every retained curved knot through K. The complete conservative data are

J	$m <$	$w_{15}^2 >$	$w_{16}^2 >$	individual P_J coefficient $>$	$Q_5 <$
6	.3399	.9217	.9378	.1435	77/150
7	.31938	.9041	.9202	.1366	194/175.

For the energies, $f_6^2/6 > .69317$, $f_7^2/7 > .71400$. The bound $d_i^2 > 1/(4i+2)$ gives at denominator 10^6 the lower numerators

$$38461, 33333, 29411, 26315, 23809, 21739, 20000, 18518, 17241, 16129$$

for $i = 6, \ldots, 15$. Those from six through fourteen sum to 228827, and from seven to fourteen to 190366; $d_{15}^2 > .016129$. The maximal omission losses are $< .000267$ after six and $< .000181$ after seven, by the decreasing sums $1/[32(j-1)^3]$ over their earliest permitted nonadjacent sets. These prove every energy entry. The P_J price is $m - d_J$; the end-K price stays $> 1/9$. Finally

$$Q_5 = \|Z_5 - (5/J)P_J\|^2 = 5 - 25/J - v + v^2/2 < 4.68 - 25/J.$$

Let $Y = Z - P_J/J$, $Y_5 = Z_5 - (5/J)P_J$. Their first-J sums vanish, so they are orthogonal to w. Because Z is supported on the first three coordinates and $Z_5 = 1$ there,

$$Y^2 = R - 1/J, \qquad Y \cdot Y_5 = 1 - 5/J.$$

The one vector $w + .4Y + .09Y_5$ spends the COMBINED P_J price $.85/J$, which fits both rows. Its gain is

$$.8h - .16R - .8m + .16/J + .18(T_5 - 5m) - .0081Q_5 - .072(1 - 5/J).$$

The nonzero last cross is retained. Using (longF), the gain is $> .04930$ at six and $> .06699$ at seven. Thus both initial branches give

$$L_{15} > .9710, \qquad L_{16} > .9871.$$

These are effective energies of the same paid vector, not new independent norm summands.

The identical curved construction at a genuinely strong seventeen has initial reference squares $> .9530, .9354$ in the last case, so its gains give totals $> 1.00230, 1.00239$. For helper five it gives $> .97231 + .03125 - 1/1024 > 1.0025$. At a later actual strong eighteen, the extra endpoint increment dominates the possible additional hole, so the conclusions remain valid. These are conditional uses of an ACTUAL strong endpoint; none is used to assume that endpoint beforehand.

B.13.5 The zero-own short-only terminal walls are impossible

Suppose an actual last-long cap is

$$c_T = Bp - \delta P_{19}, \quad B^2 = 19\delta^2 - 1, \qquad \delta = \cos(\pi/m), \quad m = 5, 6, 7.$$

We do not yet assume that this exhausts the terminal possibilities. If seventeen were weak, its actual helper would have $A_H^2 = 34g^2 + 2g - 1/2$. The two walls are distinct by their old-one contacts. Put

$$M = A_H B - \delta(34g + 1)/\sqrt{2} = -c_H \cdot c_T.$$

For $b = \delta^2 > .65, g > .8$, the difference of the two positive squared terms is

$$(2b - 1)(34g^2 + 2g) + 1/2 - 10b.$$

It increases in both variables and at $(.65, .8)$ equals 1.008. Thus $M > 0$. Positive squaring of its derivatives also proves monotonicity in both: the respective positive differences are

$$(68b - 34)A_H^2 + 18(19b - 1), \qquad (38b + 17)A_H^2 - 171b + 9.$$

For the latter use $A_H^2 > 22.86$, $.65 < b < .812$. At $g = 6/5, \delta = .901$, rational square bounds give $A_H < 7.132, B < 3.798$, and $\delta(34g + 1)/\sqrt{2} > 26.6$. Consequently

$$0 < M < 7.132(3.798) - 26.6 = .487336 < 1/2,$$

a forbidden actual-wall product. Seventeen is therefore genuinely strong before any head split.

If the isolated long head were finite at least seven or dotted, the three initial-five constructions of Section B.13.4 at that actual strong seventeen would contradict the norm. For strong five, the reference square is $> .97133$ and its respective head gain is $> .02914$ or $> .03384$. The helper-five and dotted-five branches already exceed one by their full curved/joint references. Thus only a native-five long head remains.

Write it as

$$c_L = Ap - qe_1, \quad A^2 = (\varphi - 1)/2, \quad q = \varphi/\sqrt{2}, \quad x \cdot (P_4/4) \geq h = \sqrt{(\varphi - 1)/4} > .393075.$$

The true strong five/six/seven compatibility was proved before this terminal assumption. Use the ordinary reference through strong seventeen, omitting all actual internal weak helpers. The largest possible curvature loss is bounded by holes $8, 10, 12, 14, 16$:

$$\begin{aligned} \sum_{j=8,10,12,14,16} \frac{1}{32(j-1)^3} &< .000091108 + .000042867 + .000023479 \\ &\quad + .000014224 + .000009260 \\ &< .000181. \end{aligned}$$

The decreasing weights and nonadjacency make this earliest permitted independent set maximal. Hence the actual reference has

$$w^2 > .9724835, \quad b = f_5/5 < .365190, \quad f_{17} < 3.712985, \qquad \text{prices } P_5 > .151,\ P_{17} > .12. \tag{B19tightref}$$

No unproved strong eighteen was used.

At terminal short five, the actual mutual magnitude is exactly

$$-c_L \cdot c_T = \frac{\sqrt{4+15\varphi}-(\varphi+1)}{2\sqrt{2}} \in (.953, .956).$$

Use $1.618 < \varphi < 1.619$, $1.414 < \sqrt{2} < 1.415$ and positive squaring for this interval. But $\cos(\pi/10) < .952$, while $\cos(\pi/11) > 47/49 > .956$: the latter follows from $\pi < 22/7$ and $\cos t \geq 1 - t^2/2$. There is no finite native magnitude in this gap, and the magnitude is neither zero nor at least one. The two actual walls cannot have this product.

At short six put $\delta = \sqrt{3}/2$, $B = \sqrt{53}/2$ and $k = AB - q\delta$. The complete rational seals

$$.5558 < A < .556, \quad 1.1440 < q < 1.1442,$$
$$.8660 < \delta < .8661, \quad 3.6400 < B < 3.6401,$$
$$1.032 < k < 1.034$$

follow from positive squaring. The ACTUAL positive combination $\xi = c_L + c_T$ is future timelike, square $-2(k-1)$, with both selected cap products $1 - k < 0$. Every old and unknown wall product is nonpositive termwise. It gives the original pressure

$$U = qe_1 + \delta P_{19}, \qquad x \cdot U \geq H = A + B - \sqrt{2(k-1)} > 3.9348.$$

The reference pairing is $w \cdot U = qb + \delta f_{17} < 3.634$; its deficit exceeds .3. After paying BOTH anchors, the actual residual is

$$V = U - \delta P_{17} - (q/5)P_5 = q(e_1 - P_5/5) + \delta(e_{18} + e_{19}), \qquad w \cdot V = 0, \quad V^2 = (4/5)q^2 + 2\delta^2 < 2.548.$$

With $\nu = .1$, the individual prices are $< .023$ at five and $< .087$ at seventeen. A single square gives

$$\|x\|^2 > .9724835 + .06 - .02548 > 1.007.$$

No separate head-energy estimate was added here.

At short seven we need the full head/core cross. Let $t = 27/25$, $k = AB - q\delta$. The exact polynomial $8\delta^3 - 4\delta^2 - 4\delta + 1 = 0$ gives the rational seals

$$.9009688 < \delta < .9009690, \quad .5558929 < A < .5558931, \quad 1.1441227 < q < 1.1441231,$$
$$3.79778 < B < 3.79779, \qquad 1.08033 < k < 1.08035.$$

For the first bracket the cubic increases on $(.9, .901)$; at $.901 - e$ it equals $.000257608 - 8.275224e + 17.624e^2 - 8e^3$, whose signs at $e = .0000312, .0000310$ are negative/positive. The other seals are positive square/product comparisons.

Now $\xi = c_L + tc_T$ is future, with selected cap products $1 - tk < 0, t - k < 0$, and square $-s$, $s = 2tk - 1 - t^2 > 0$. All other original-wall products are nonpositive. The same original vector satisfies

$$U = qe_1 + t\delta P_{19}, \qquad x \cdot U \geq H = A + tB - \sqrt{s} > 4.2486453,$$

because $\sqrt{s} < .40885$. Also $w \cdot U = qb + t\delta f_{17} < 4.03074$, so the actual deficit is $> .2179$.

Use the independent actual native-five H_4 pressure through $Y = P_4/4 - P_5/5$. Its square is $1/20$, its cross with w is zero, and its deficit is $h - b > .027885$. For the two-cap point pay the residual

$$V = U - t\delta P_{17} - (q/5)P_5 = q(e_1 - P_5/5) + t\delta(e_{18} + e_{19}).$$

Its full products are

$$w \cdot V = 0, \quad V^2 = (4/5)q^2 + 2t^2\delta^2 < 2.941, \qquad Y \cdot V = q/20 < .057207.$$

Choose $\mu = .48, \nu = .065$ and use ONE vector $w + \mu Y + \nu V$. Its complete individual debits are

$$P_5 : \ (\mu + \nu q)/5 < .111 < .151, \qquad P_{17} : \ \nu t\delta < .064 < .12.$$

The actual primitive pressures, with these coefficients removed before their floors are used, give the single completed-square gain

$$.96(.027885) - .2304(.05) + .13(.2179) - .004225(2.941) - .0624(.057207) \\ = .0275811582.$$

The last term is the NONZERO head/core cross. Together with (B19tightref), the total is $1.0000646582 > 1$. Thus all three zero-own short-only native terminal walls are impossible.

B.13.6 The complete actual B_{19} last-long classification

Release eighteen. The retained old diagram is A_{17} on one through seventeen and the short A_1 at nineteen. A finite completion has at most one contact to A_{17}: simple-three at $1, 2, 16, 17$, or endpoint four at one/seventeen. Exceptional finite diagrams cannot contain A_{17}. A short-only contact can have ANY finite native label. If both retained components are joined, full rank-nineteen classification permits only a simple-three at left endpoint one/seventeen and short-three/four. An internal left attachment introduces a second inadmissible fork; an internal four is not a large finite B diagram. This exhausts the actual finite white list before any own cutoff.

Write the own product magnitude as $g \geq 0$. With no short contact, a white three at j has

$$\Delta = 1 - j(18 - j)/36, \quad \sigma = g + j/36;$$

a white endpoint four has $\Delta = 1/18$, $\sigma = g + j/(18\sqrt{2})$. The strong ratio target at $q = \sqrt{36} = 6$ is $< 33/20$.

For completeness keep the entire small-own list. Own-zero far three has negative time; far next-to-end three and far four have zero time. Near-three/next-to-end zero-own has the actual affine two-leaf fork ending in four, and near four has the $4, 3, 4$ principal. Own three with a white three has its actual affine cycle; far-four/own-three has the affine $3, 3, 4, 3$ path, while near four/own-three is directly strong. Own four without a short contact has the $4, 4$ principal through old eighteen/nineteen. Own six has the left $6, 3$ principal if its selected neighbour is uncontacted; the other near roles already have a directly strong ratio. Parallel has the positive future null $c + n_{18}$. Every affine null has a positive actual-cap coefficient, future time and nonpositive products with all omitted walls.

For every remaining $g \geq \varphi/2$, near three, next-to-end three and endpoint four are directly strong by the displayed residuals. Only the pure far-three helper (B19packet) can remain weak on the left. Its changed-loss cutoff is $g < 6/5$, reproved by the actual edge ratio, not inferred from an old weak set.

An isolated cap at this final index is also strong or impossible. Use the LEFT old roots eighteen/seventeen/sixteen. Native five has divided finite H_4 loss

$$6 - \sqrt{36 - u} + \sqrt{u - 3/2} < .13 + .168 < .58.$$

Native seven has the negative left-A_2 loss $6-\sqrt{36-u}+\sqrt{4/3-u}<.105+.319$. Finite labels at least eight have singleton loss $<.102+.415$. For dotted own,

$$6-\sqrt{36-u}+\sqrt{u(1-u)}<.084u+\sqrt{u(1-u)}<(.084+1.004)/2<.58.$$

The positive finite points or negative cores are the same actual constructions proved above, now at dual square thirty-six. Their products with short nineteen are nonpositive through the old eighteen term. The spatial $\sqrt{2}P_{19}$ used in a displacement is not an old long normal or its short dual.

If both a left endpoint $j=1,17$ and the short component are contacted, with short magnitude $\delta=1/2$ or $1/\sqrt{2}$, the exact data are

$$\Delta=19/36-\delta^2,\qquad \sigma=g+j/36+\delta/\sqrt{2}.$$

Short four is strong even at own zero. For short three and every positive own,

$$\sigma/\sqrt{\Delta}\geq\frac{1/2+1/36+1/(2\sqrt{2})}{\sqrt{5/18}}>33/20.$$

At own zero with $j=1$, the actual selected sixteen/seventeen/ eighteen/nineteen/cap path is affine F_4. At $j=17$, the cap is exactly $(\omega_{17}+\omega_{19})/2$, already excluded by its ACTUAL short-edge mate in the boundary proof. No mixed-left/short weak cap remains.

With no left contact write $v=g\omega_{18}+\delta\omega_{19}$. Its outgoing ratio is

$$\frac{g+\delta/\sqrt{2}}{\sqrt{1-\delta^2}}.$$

For own zero this is exactly $\cot(\pi/m)/\sqrt{2}$. Short three has the affine F_4 path and short four the double-four principal. At label eight the ratio is $1+1/\sqrt{2}>33/20$, increasing thereafter. The remaining short-five/six/seven rows were proved impossible in Section B.13.5.

For positive own and short at least five, the ratio is $>53/30>33/20$. For short four and own at least four it is $\geq 1+1/\sqrt{2}$; only own-three/short-four remains. For short three, own-three remains. Own four has a finite B_{19} replacement after releasing short nineteen, with actual reciprocal edge cosine $55/2>27$. Positive finite projection makes that endpoint actual, giving $S_{19}^2>247/14>(4.2)^2$. The early references (B19early) with the $1/9$ price exclude this pressure in both K rows. Parallel again has its actual positive feasible null. We next exclude every finite own at least five and dotted own in the remaining short-three row.

Put $\alpha=\sqrt{2}g+1/2$; its actual time is $A^2=18\alpha^2-3/4$. For finite $\varphi/2\leq g<1$, put $d=1-g^2$, $D=g/\sqrt{2}+1/2$, $\eta=d/D$, $\lambda=g+\eta/\sqrt{2}$. The actual positive combination

$$\xi=c+\lambda n_{18}+\eta n_{19}$$

has products zero with the cap and eighteen, and $-D+\eta/2<0$ with nineteen. Its square is $-k$, $k=d-\eta^2/2>0$, because $2D^2-d=\alpha^2-3/4>0$. It is future and satisfies every omitted wall termwise. Its normalized spatial displacement Z is one on the first seventeen coordinates, has square <18, and gives

$$x\cdot Z\geq G=(A-\sqrt{k})/\alpha>3.87.$$

Here is the full interval seal. For $\alpha>1.643$,

$$\frac{k}{\alpha^2}=\frac{d(2-1.5/\alpha^2)}{(\alpha+.5)^2}<.11,\qquad d=.875+\alpha/2-\alpha^2/2.$$

The numerator derivative is at most $-1.143(1.44)+.347(.677)<0$ and its denominator increases. At 1.643 the value is $<.347(1.445)/4.59<.11$. Also $A/\alpha>4.209$, while $\sqrt{.11}<.332$. Thus the stated bound is strict on every finite own in the range.

For ALL dotted $g>1$, use instead $\xi=c+g^{-1}n_{18}$. Its cap product is zero, both old eighteen/nineteen products are negative, its square is $-(1-g^{-2})$, and its time is positive. The normalized displacement again has first seventeen coordinates one and square <18. Since $\alpha>1.9$, $A/\alpha>4.21$. For $t=1/g\in(0,1)$,

$$\frac{\sqrt{1-g^{-2}}}{\alpha}=\frac{t\sqrt{1-t^2}}{\sqrt{2}+t/2}<1/3.$$

For $t\le 1/2$, its square is $\le 3/32$; for $t\ge 1/2$ it is $<100/1089$, both $<1/9$. Hence $G>3.876>3.87$ throughout the unbounded dotted interval.

Use the refined early references of Section B.13.4. Every earlier residual is supported through K, so $Z-P_K$ vanishes on its ENTIRE support. It is exactly orthogonal to all early head/dotted-five corrections, including their mutual nonzero cross already paid there. The remaining $1/9$ endpoint price is available. The new pure tail gain exceeds

$$\begin{cases}2(3.87-3.463)/9-3/81>.0534, & K=15,\\ 2(3.87-3.590)/9-2/81>.0375, & K=16.\end{cases}$$

Each exceeds one when added to EVERY strong/helper/isolated-dotted five effective-energy row above. This closes all short-three finite-own-at-least-five and dotted-own cases.

Thus the full classification leaves either strong eighteen, a pure far helper there, or precisely the two weak walls

$$(b)\quad v_T=(\omega_{18}+\omega_{19})/2,\qquad (c)\quad v_T=\omega_{18}/2+\omega_{19}/\sqrt{2}. \tag{B19survivors}$$

B.13.7 Strong eighteen and a weak far helper are also excluded

If eighteen is strong, the actual window through seventeen justifies the full curved references there. A helper at five gives $E_{18}+.03125-1/1024>1$. An isolated dotted five gives the two-early-foot $J=6,7$ contradictions at that ACTUAL strong endpoint. For strong five,

$$w^2>E_{18}-1/1024>.985613.$$

The smallest paid head gain $>.015456$ already contradicts the norm.

If eighteen is a weak far helper, seventeen is strong. At this boundary, nonadjacency uses the positive finite D_{18} replacement from the retained A_{16} and two simple leaves, together with the independent old short A_1. Both helper caps have zero contact to the short root. This checks the actual mixed two-cap hypotheses; no virtual long nineteenth root is used.

The full curved/early-foot constructions at strong seventeen exclude helper-five and isolated dotted-five. Hence five is strong. Its reference through seventeen has $w^2>.97133$, sum f_{17}, first-five mean $b<.3652$, P_5 price $>.151$ and P_{17} price $>1/9$.

For the actual weak eighteen helper $A^2=36g^2+2g-1/2$. Let $\omega_{19}^*=\sqrt{2}P_{19}$ denote a SPATIAL vector only. With $\alpha=\sqrt{2}g$, the complete actual pressure feet are

own	U	$x\cdot U\ge T$	$V^2,\ V=U-\alpha P_{17}$
5	$g(\omega_{17}+\omega_{19}^*)/2+\omega_3/6$	$H(A,2/3-g^2)$	$g^2+1/6$
7, 8	$g(\omega_{17}+\omega_{19}^*)/2+\omega_2/4$	$A-\sqrt{g^2-3/4}$	$g^2+1/4$
finite ≥ 9	$g(\omega_{17}+\omega_{19}^*)/2+\omega_1/2$	$H(A,1-g^2)$	$g^2+1/2$
dotted	$(g-g^{-1})\omega_{18}+(\omega_{17}+\omega_{19}^*)/(2g)+\omega_1/2$	$A-\sqrt{1-g^{-2}}$	$2g^2-3/2+g^{-2}$.

Here $H(A,\delta)$ is the actual positive finite-projection pressure. The normal corrections are respectively

$$gn_{18}+\tfrac{2}{3}n_1+\tfrac{1}{3}n_2,\quad gn_{18}+n_1/2,\quad gn_{18},\quad g^{-1}n_{18}.$$

They use ONLY actual long roots one/two/eighteen. At native five their finite H_4 residual is $2/3-g^2$; at seven/eight the core has negative square $-(g^2-3/4)$; the next finite pair has residual $1-g^2>0$, and the dotted core has square $-(1-g^{-2})$. Every selected point is future, and every omitted-wall product, including that with actual SHORT nineteen, is nonpositive. The finite pressure comes from a point, never from a spacelike correction by itself.

All four rows satisfy

$$T-w\cdot U>.21,\qquad V^2<2.1,\qquad w\cdot U=\alpha f_{17}+b/\sqrt{2}. \tag{B19helper18}$$

Use $\sqrt{2}f_{17}<5.251$, $b/\sqrt{2}<.259$. Native five has $A>4.9678,\sqrt{\delta}<.111,g<.81$, giving deficit $>.3444$. Native seven has $A>5.5245$, radical $<.249$, $g<.901$, giving $>.2853$. Native eight has $A>5.663$, radical $<.322$, $g<.9239$, giving $>.2306$. For finite labels at least nine, at $g=.939$, $A>5.75,\sqrt{A^2+\delta}>5.765$, $\sqrt{\delta}<.344$, so $H>5.42$ and deficit $>.2303$. The difference $H-5.251g$ increases thereafter: $A'>6,H_A>47/50,H_\delta<0$, and $\delta=1-g^2$ decreases. For dotted $1<g<6/5$, the difference decreases since $A'<6.1$ and the radical derivative >1. At $6/5$, $A^2=2687/50,A>7.3307$, radical $<.5528$, giving deficit $>.2177$. The finite V^2 bounds are immediate; the dotted square increases to $1867/900<2.1$. These positive-square/derivative seals cover every own interval.

Insert BOTH primitive head and helper pressures in one vector

$$W=w+\mu(Z-P_5/5)+\nu V,\qquad \nu=1/16,$$

with the head-specific μ in (B19headgains). The individual prices are $\mu/5$ at five and $\nu\alpha<.107<1/9$ at seventeen. The residual V has an initial piece $P_s/(s\sqrt{2})$, $s=3,2,1$. If ℓ_s is the mean of the first s coordinates of Z, then EXACTLY

$$(Z-P_5/5)\cdot V=(\ell_s-1/5)/\sqrt{2}.$$

The total subtracted cross is $<.002476$ for native-five head, $<.010608$ for other finite head, and $\le .02$ for dotted head. The helper gain from (B19helper18) is $>.018046875$. Thus the single squared-reference totals exceed respectively

$$.97133+.015456+.018046875-.002476>1.0023,$$
$$.97133+.02914+.018046875-.010608>1.0079,$$
$$.97133+.03384+.018046875-.020000>1.0032.$$

No independent head optimum was added to an already optimized helper energy. Strong eighteen and its pure-helper weak case are now both impossible.

B.13.8 Each terminal survivor supplies its own genuine strong seventeen

Only the two actual walls in (B19survivors) remain. Their first eighteen coordinates are a, last coordinate δ, with

$$(b):\ a=(\sqrt{2}+1)/2,\ \delta=1/2;\qquad (c):\ a=\sqrt{2},\ \delta=1/\sqrt{2}.$$

Their time square is $A_T^2=18a^2+\delta^2-1$.

If seventeen were weak, let its actual helper have parameter g, $A_H^2 = 34g^2 + 2g - 1/2$. The mutual magnitude is

$$M = A_H A_T - a(34g + 1)/\sqrt{2}.$$

The actual walls are distinct by their old-one products. With $\kappa = a^2 + \delta^2 - 1$, the positive squared difference is $A_H^2\kappa - 9a^2$. For $g > .8$, it is positive in both rows: in (b), $\kappa = 1/\sqrt{2}$ and it exceeds $22.86(.707) - 9(1.458) > 3$; in (c), it exceeds $22.86(1.5) - 18 > 16$. So $M > 0$. The squared derivative comparison $34\kappa A_H^2 + 18A_T^2 > 0$ shows that M increases with g.

For (b), $g < 1.2$ gives

$$M < 7.132(5.048) - .8533(41.8) = .334396 < 1/2,$$

using $A_T < 5.048, a/\sqrt{2} > .8533$. This violates the native gap. For (c), $A_T = \sqrt{71/2} < 5.959$ and $a/\sqrt{2} = 1$, so

$$0 < M < 7.132(5.959) - 41.8 = .699588 < .7.$$

The only possible native magnitude is $1/2$. But $M(1) = 1/2$ exactly, so strict monotonicity forces $g = 1$. The actual helper then has the nonzero positive future null $c_H + n_{17}$, impossible in a compact polytope. Thus seventeen is strong in BOTH survivors.

With a finite head at least seven or a dotted head, the actual strong-seventeen references now close every initial-five branch: helper five and isolated dotted five by their complete constructions, strong five by $.97133 + .02914 > 1$ or $.97133 + .03384 > 1$. Therefore both survivors force the isolated native-five long head, its H_4 pressure $x \cdot P_4/4 > .393075$, and the genuine strong five/six/seven/seventeen sites. These inputs precede the fresh changed-loss arguments below.

B.13.9 A common complete white list at the newly tested index seventeen

Deleting seventeen retains A_{16} on one through sixteen and B_2 on eighteen/nineteen. The full finite-white list is isolation; left simple-three at $1, 2, 15, 16$; left endpoint-four at one/sixteen; right simple-three at eighteen or nineteen; or a left endpoint-three joined to either right simple-three. Two contacts to one old tree make a cycle. A right-four or larger extension of B_2 is nonpositive. Full bridges have rank nineteen, so exceptional finite types, an internal four and a second fork supply no further alternatives. There are no spectator components.

For a unit actual cap $c = Ap - v$ write $g = -c \cdot n_{17} \geq 0$. The left-only exact data are

white	Δ	σ
$3@j,\ j = 1, 16$	$9/17$	$g + j/34$
$3@j,\ j = 2, 15$	$2/17$	$g + j/34$
$4@j,\ j = 1, 16$	$1/17$	$g + j/(17\sqrt{2})$.

A full bridge has residual $1/34$; even at own zero its numerator is $.5 + 1/34$ for right-long three, or $1/(2\sqrt{2}) + 1/34$ for right-short three. Both resulting ratios exceed two. This complete list will be tested AFRESH under each terminal wall; its old .58-strong labels do not imply the new floors.

The common actual small-own reductions, when not replaced by a direct mutual calculation below, are as follows. Zero-own far-three has negative time, far next-to-end three and far four have zero time. The near next-to-end zero-own has the affine two-leaf fork ending in four; near-four zero-own has the affine chain ending in four at both ends. Own-three with a white-three has the actual all-three cycle. When both right contacts vanish, own-four has the selected $4, 3, 4$ path through $c, 17, 18, 19$. When the contact with n_{18} vanishes, own-six has the $6, 3$ path through $c, 17, 18$. For right-short-three with own-four and zero contact at sixteen, the actual affine star instead

has centre n_{17}, orthogonal leaves c, n_{16}, n_{18} and weights $4, 3, 3$; its positive null is future feasible. Right-long-three with own-four or own-six is tested by its direct ratio below, without using a chorded affine path. Parallel has the positive feasible future null $c + n_{17}$. Each null has positive time through the actual cap and every omitted-wall product nonpositive. The endpoint-four positive-own rows are directly tested below, including own three.

B.13.10 Short-four terminal: a new .56 floor at the one actual index

Assume survivor (c), $v_T = \sqrt{2}P_{18} + e_{19}/\sqrt{2}$, $A_T^2 = 71/2$. We prove

$$S_{17} > F = \sqrt{17} - .56/\sqrt{2}. \tag{B19F56}$$

At $q = \sqrt{34} < 6$, its sufficient ratio target satisfies

$$T_{.56}(q) = 25/14 - \frac{q - .56}{q(2q - .56)} < 25/14 - 34/429 < 1.707.$$

The first function increases on the relevant range.

For isolated caps, the complete small-own/time/affine/null reductions leave five, seven, finite at least eight, or dotted. Use the actual LEFT corrections at seventeen/sixteen/fifteen. For native five, the divided finite-H_4 loss is

$$q - \sqrt{34 - u} + \sqrt{u - 3/2} < .133 + .168 < .56, \quad u = 8 - 4\varphi < 1.528.$$

For native seven the left-A_2 negative core gives loss

$$q - \sqrt{34 - u} + \sqrt{4/3 - u} < .107 + .319 < .56, \quad 1.2318 < u < 1.233.$$

For finite at least eight the actual pair projection gives $< .102 + .415 < .56$. For dotted own the actual core $c + g^{-1}n_{17}$, together with concavity, gives loss

$$q - \sqrt{34 - u} + \sqrt{u(1 - u)} < .087u + \sqrt{u(1 - u)} < (.087 + 1.004)/2 = .5455 < .56.$$

The core/finite-point signs are exactly those proved earlier; all corrections use actual long roots and have nonpositive product with short nineteen. Thus every isolated cap satisfies the new floor, including native eight.

For left-only rows, the small owns in the common list are removed first. For all remaining $g \geq \varphi/2 > .809$, near endpoint-three has ratio

$$(g + 8/17)/\sqrt{9/17} > 1.2795/.728 > 1.757.$$

The next-to-end rows exceed 2.5 and endpoint-four exceeds 3.5. Only far-three at one needs its actual terminal interaction.

For that actual cap, $v_H = g\omega_{17} + \omega_1/2$, $A_H^2 = 34g^2 + 2g - 1/2$, and

$$M(g) = A_H\sqrt{71/2} - (34g + 1) = -c_H \cdot c_T.$$

The two walls are distinct by their old-one products. For $g > .8$, its squared-product difference is $3A_H^2/2 - 18 > 0$, so $M > 0$. Its squared derivative comparison $51A_H^2 + 18(71/2) > 0$ proves strict increase. We have $M(1) = 1/2$. At $g = 243/200 = 1.215$,

$$A_H^2 = 52.12165 < (7.21954)^2, \quad \sqrt{71/2} < 5.9582,$$
$$M < 7.21954(5.9582) - 42.31$$
$$= .705463228 < .707 < \cos(\pi/4).$$

Thus for $g \le 1.215$ the only possible native magnitude is $1/2$; strict increase forces the forbidden parallel helper $g = 1$. Every remaining actual far helper has $g > 1.215$, hence

$$\frac{34g+1}{\sqrt{612}} > 42.31/24.739 > 1.71 > T_{.56}(q).$$

This tests the entire new cap domain; the earlier .58 cutoff $g < 1.2$ has not been imposed.

For right-long three, own zero has its affine star and own three its affine triangle. Every remaining own has $g \ge 1/\sqrt{2}$, ratio $\ge 1 + 1/\sqrt{2} > 1.707$. For right-short three, own zero/four/six/parallel has the actual affine/null obstruction. Own-three and own-five are impossible by the previous ACTUAL short-edge mate and left H_4 boundary arguments, which did not require the new seventeen floor. Every other finite own at least seven has ratio $\sqrt{2}g + 1/2 > 1.77$; dotted owns are stronger. Finally all full bridges have ratio > 2, including own zero. The complete new finite-white/ own domain is exhausted, proving (B19F56).

B.13.11 Short-four terminal: pool the tail before raising its endpoint

Use the old .58 reference through seventeen, but deliberately omit fifteen/sixteen as well as every actual weak knot. Because helpers are nonadjacent, the last retained K before seventeen is fourteen or thirteen. If fourteen is weak, it merges with the forced terminal cluster; all earlier omissions stay isolated.

All earlier holes cost $< .000181$. The terminal cluster averages at most four slopes $d_{13}, \dots, d_{16}$, whose successive gaps are

$$< 1/(4 \cdot 13\sqrt{13}) < .0054.$$

For four numbers with successive gaps at most κ, their variance about the mean is $\le 5\kappa^2$, by $(1/4)\sum_{i<j}(a_i - a_j)^2$ and distances one/two/three. The shorter three-slope cluster costs at most $2\kappa^2$. Thus the terminal loss is $< .000146$, and FULL loss $< .00035$. The affected open intervals are disjoint even when they share a retained endpoint.

Now raise only seventeen by $\Delta = .02/\sqrt{2}$, $.01414 < \Delta < .01415$. Both new terminal slopes are $< .133$:

$$\frac{\sqrt{17} - \sqrt{14} + \Delta}{3} < .133, \qquad \frac{\sqrt{17} - \sqrt{13} + \Delta}{4} < .133.$$

Their incoming slopes are at least $d_{13} > .136$ and $d_{12} > .14$, including any earlier isolated omission. The reference remains concave, with every coefficient nonnegative. Its endpoint coefficient is $> d_{16} > .123$; its first-five block and P_5 coefficient $> .151$ are unchanged.

The old terminal chord has slope $\ge d_{16} > .123$. Lifting its endpoint adds

$$2(\text{old slope})\Delta + \Delta^2/(17 - K) > .246(.01414) = .00347844.$$

Together with the scalar (B19E17),

$$w^2 > .9726645 - .00035 + .00347844 = .97579294, \qquad \sum_j w_j = F.$$

Only seventeen uses the new floor; all other active floors are original strict .58 floors.

The SAME actual terminal cap is also the short-nineteen mate. Deleting old nineteen and adjoining it to the retained A_{18} by its actual simple-three at eighteen gives positive finite A_{19}. Its positive inverse projection is future, strict against every unknown original wall. The omitted old

nineteen starts at zero and becomes strict through the positive cap coefficient and cap-short product $-1/\sqrt{2}$. Thus this is an actual endpoint of that original edge, with

$$\cosh L_{19} = 37/\sqrt{20}, \qquad S_{19} > 3.860.$$

For the last bound, using $\sqrt{20} < 4.47214$, the exact cleared comparison is

$$(19 - 14.8996)37 = 151.7148 > (19 + 14.8996)4.47214 = 151.603757144.$$

We do not assume this neighbouring A_{19} vertex is farthest.

Insert the native-five head pressure by $w_{\mathrm{head}} = w + .56(P_4/4 - P_5/5)$. The individual P_5 price is $.112 < .151$. Its residual has square $1/20$, zero sum on the first-five block and zero cross with w. Its gain is

$$1.12(.027885) - .56^2/20 = .0155512.$$

The modified reference still has support through seventeen, sum F, and P_{17} coefficient $> .123$.

Extend it by the last-two slope $r = (3.860 - F)/2$. The bounds $3.720 < F < 3.72713$ give $0 < r < .07$, so its P_{17} debit fits. The head residual and new tail residual have disjoint support and cross zero. One final square yields gain

$$.0155512 + (3.860 - F)^2/2 > .0155512 + (.13287)^2/2.$$

Consequently

$$\|x\|^2 > .97579294 + .0155512 + .00882721845 = 1.00017135845 > 1.$$

This excludes the entire short-four terminal survivor.

B.13.12 Short-three terminal: retain the orthogonal dotted exception

Assume survivor (b). Put $s = \sqrt{2}$, $a = (1 + s)/2$, $k = as = 1 + s/2$. Then

$$v_T = aP_{18} + \tfrac{1}{2}e_{19}, \qquad A_T^2 = 51/4 + 9s, \qquad 34A_T^2 - 289k^2 = 17s.$$

The actual index-seventeen cap is distinct from c_T: the latter contacts both retained B_2 roots with inverse cost $1 + s/2 > 1$, impossible for that cap's positive finite white block. Write

$$L = v \cdot v_T > 0, \quad M = AA_T - L = -c \cdot c_T, \qquad D = A^2A_T^2 - L^2.$$

Actual acuteness/native products give $M = 0$ or $M \geq 1/2$. Since $A, A_T, L > 0$, D has the sign of M. In particular $D < 0$ means a POSITIVE actual-wall product.

We prove the fresh floor

$$S_{17} > F = \sqrt{17} - .50/\sqrt{2}. \tag{B19F50}$$

Its sufficient ratio target is $T_{.50}(q) = 2 - (q - .50)/(q(2q - .50)) < 2$, $q = \sqrt{34}$.

For isolated caps,

$$v = g\omega_{17}, \quad A^2 = 34g^2 - 1, \quad L = 17kg, \qquad D = 17sg^2 - A_T^2.$$

Own zero has negative time. Every positive finite own and parallel has $g \leq 1$, giving $D \leq 8s - 51/4 < 0$; all violate acuteness. Thus an isolated cap is dotted.

The function $M(g) = A_T\sqrt{34g^2-1} - 17kg$ strictly increases: its squared derivative comparison is $578sg^2 + 289k^2 > 0$. If $M = 0$, the exact parameter is

$$u = g^{-2} = 17s/A_T^2, \qquad .94 < u < .95.$$

Orthogonality is RETAINED. The actual positive core $c + g^{-1}n_{17}$ has future time, square $-(1-u)$, zero cap product, negative own product and every omitted-wall product nonpositive. Its spatial displacement is

$$Y = g\{(1-u)\omega_{17} + (u/2)(\omega_{16} + \omega_{18})\}.$$

Original concavity gives $x \cdot Y \le gh_{17}$. Farthest comparison consequently gives

$$h_{17} \ge \sqrt{34-u} - \sqrt{u(1-u)}.$$

The loss below $\sqrt{34}$ is $< .087u + \sqrt{u(1-u)} < .083 + .238 = .321 < .50$, so the orthogonal case satisfies the new floor.

If $M > 0$, then $M \ge 1/2$. At $g = 7/4$,

$$(119k/4 + 1/2)^2 - A_T^2(825/8) = (684 - 451s)/16 > 0,$$

so $M(7/4) < 1/2$. Strict increase forces $g > 7/4$ and $u < 16/49$. The same actual foot has loss

$$< .087(16/49) + 4\sqrt{33}/49 < .029 + .469 = .498 < .50.$$

This covers every isolated dotted parameter and all mutual labels, including zero.

For left-only roles use the complete residual table of Section B.13.9. Keep its actual zero-own and affine/null reductions, except near endpoint-three at sixteen, which will be tested directly. Every positive-own endpoint-four has ratio $\ge \sqrt{17}/2 + 1/\sqrt{34} > 2$. Every remaining next-to-end three has $g \ge \varphi/2 > .809$ and ratio > 2.5.

For near endpoint-three at sixteen,

$$v = g\omega_{17} + \omega_{16}/2, \quad A^2 = 34g^2 + 32g + 7, \quad L = k(17g+8),$$
$$D = 17sg^2 + 16sg - 27/4 - s.$$

At $g = 0$, $D < 0$. For finite $g \ge 1/2$, $D > 0$, starting with $(45s-27)/4 > 0$. The identity $(34g+16)^2 = 34A^2 + 18$ gives the positive derivative comparison $17sA^2 + 18A_T^2 > 0$ for M. At $g = 1$,

$$(25k + 1/2)^2 - 73A_T^2 = 32 - 39s/2 > 0.$$

Thus $0 < M < 1/2$ for all finite owns and parallel, impossible. Every remaining dotted $g > 1$ has ratio $> 25/\sqrt{153} > 2$, since $625 > 612$.

For far endpoint-three at one, the common small-own/null reductions leave $g \ge \varphi/2 > .8$. The exact data are

$$A^2 = 34g^2 + 2g - 1/2, \quad L = k(17g + 1/2),$$
$$D = 17sg^2 + sg - 27/4 - 19s/4.$$

This polynomial is positive at .8 and increases thereafter, so $M > 0$. The identity $(34g+1)^2 = 34A^2 + 18$ again gives the positive derivative comparison. At $g = 3/2$,

$$(26k + 1/2)^2 - 79A_T^2 = 33 - 22s > 0.$$

Thus every admissible remaining helper must have $g > 3/2$, whose ratio exceeds $52/\sqrt{612} > 2$. This tests the full role, not only its former .58-weak parameters.

For right-long three at eighteen, the exact data give

$$A^2 = 34g^2 + 34g + 8, \quad L = k(17g + 9),$$
$$D = 17sg^2 - (51/2)g - 39/2 - 9s.$$

The convex quadratic is negative at both zero and one; at one it is $8s - 45 < 0$. It is therefore negative on the WHOLE interval $[0, 1]$. Own zero, all finite owns and parallel violate acuteness. A remaining dotted $g > 1$ has ratio $\sqrt{2}g + 1/\sqrt{2} > 2$.

For right-short three at nineteen,

$$A^2 = 34g^2 + 17sg + 15/4, \quad L = 17kg + (19 + 18s)/4,$$
$$D = 17sg^2 - (17/2 + 17s)g - 61/4 - 9s.$$

This convex quadratic is negative at zero and $5/4$, the latter value being $-207/8 - 59s/16$. Thus only $g > 5/4$ can survive actual acuteness, with ratio $\sqrt{2}g + 1/2 > 2$. Finally all full bridges are directly strong by their ratio > 2, including own zero. This exhausts every actual white and own parameter, and proves (B19F50). The .56 floor from the other terminal case was not used.

B.13.13 Short-three terminal: pool every possible final weak cluster

Keep the old .58 floors, omit every actual weak knot, and deliberately omit thirteen through sixteen as well. Since at least one of eleven/ twelve is strong, the last retained K before seventeen is twelve if twelve is strong, otherwise eleven. This includes every old weak set.

Earlier isolated omissions cost $< .000181$. The terminal cluster averages at most six slopes $d_{11}, \ldots, d_{16}$. Their consecutive gaps are

$$< \kappa = 1/(4 \cdot 11\sqrt{11}) < .00686.$$

For n such numbers the exact variance identity gives

$$\sum_i (a_i - \overline{a})^2 = \frac{1}{n} \sum_{i<j} (a_i - a_j)^2 \le n(n^2 - 1)\kappa^2/12.$$

At $n = 6$, this is $(35/2)\kappa^2 < .000824$; the five-slope case is smaller. The clusters have disjoint open edge intervals, so TOTAL loss is $< .00101$.

Lift seventeen by $\Delta = .08/\sqrt{2}$, $.05656 < \Delta < .05657$. Both outgoing terminal slopes are $< .144$:

$$\frac{\sqrt{17} - \sqrt{11} + \Delta}{6} < .144, \qquad \frac{\sqrt{17} - \sqrt{12} + \Delta}{5} < .144.$$

Their incoming slopes are $\ge d_{10} > .154$, respectively $\ge d_{11} > .147$, including any earlier isolated omission. Thus the final interpolation is concave, paid and supported on old strong floors, with only seventeen using the new .50 floor. No assertion is made about the unknown thirteen–sixteen floors.

The old last chord has slope $\ge d_{16} > .123$, so its endpoint lift adds $> .246(.05656) = .01391376$. The new reference satisfies

$$w^2 > .9726645 - .00101 + .01391376 = .98556826.$$

Its first-five mean is still $b < .365190$, with individual P_5 coefficient $> .151$. Insert the actual native-five H_4 pressure in the SAME vector

$$W = w + .56(P_4/4 - P_5/5).$$

Its price .112 fits. Its residual has square $1/20$, zero block sum and zero cross with w. The one original-vector square gives

$$\|x\|^2 > .98556826 + 1.12(.027885) - .56^2/20 = 1.00111946 > 1.$$

No short-edge energy or second farthest vector is added. This excludes survivor (b).

Proposition B.13.1. *A whole B_{19} forest cannot occur at the original farthest vertex of a compact Coxeter nineteen-polytope.*

Proof. The actual window and full last-long finite-white classification give strong eighteen, a weak pure far helper there, or the two walls (B19survivors). Every zero-own short-only exception was excluded using its own actual head/terminal points. The strong and helper end states are impossible. Each survivor supplies its genuine strong seventeen and isolated native-five head before the fresh single-index arguments use them. The separate short-three and short-four proofs then close both remaining walls. Every own, white, finite, parallel and dotted role has been kept, and every point is feasible against the entire original polytope. □

B.14 The whole D_{19} case

B.14.1 The signed frame and the head reduction

Use unit actual roots and duals

$$n_i = (e_i - e_{i+1})/\sqrt{2}\ (i \le 18), \quad n_{19} = (e_{18} + e_{19})/\sqrt{2}, \quad \omega_i = \sqrt{2}P_i\ (i \le 17),$$
$$\omega_{18} = (P_{18} - e_{19})/\sqrt{2}, \qquad \omega_{19} = (P_{18} + e_{19})/\sqrt{2}.$$

The original vector satisfies $x_1 > \cdots > x_{18} > |x_{19}|$, $\|x\| < 1$. The last coordinate remains signed. The centre is seventeen and the two actual forks are eighteen/ nineteen; no artificial long fork normal is introduced.

At loss .58, the complete local whole-diagram lemma B.11.7 applies literally at five through fourteen: the retained right D_{19-i} has rank at least five. It retains every finite fork-to- E_6, E_7, E_8 extension; a surviving positive-own fork ratio is

$$> (.8 + .25)/\sqrt{3/8} > 83/50.$$

Thus every weak cap there is the pure far-three helper

$$v = g\omega_i + \omega_1/2, \quad A^2 = 2ig^2 + 2g - 1/2, \quad \varphi/2 \le g < 6/5,$$

with own five, finite at least seven or dotted, except for isolated dotted five. Actual helper nonadjacency uses the retained positive right block including BOTH fork normals. Hence some $K = 13$ or fourteen and some $J = 6$ or seven is strong, before any head or late-layer conclusion. No fifteen/sixteen/centre floor has yet been used.

Let $f_i = \sqrt{i} - .58/\sqrt{2}$, $E_J = f_5^2/5 + \sum_{i=5}^{J-1} d_i^2$. The old $E_{13} > .88989$, and changing .61 to .58 adds $-.00357 + .006\sqrt{10} > .0154$. Thus $E_{13} > .90529$. The elementary bounds needed below are

$$E_6 < .7126, \quad E_7 < .7510, \quad 2.0393 < f_6 < 2.0394, \quad 2.2356 < f_7 < 2.2357,$$
$$f_5/5 < .3652, \qquad d_6 < .197, \quad d_7 < .183.$$

They follow, for example, from $f_5^2/5 = 1.03364 - .116\sqrt{10} < .6668248$, $d_5^2 < .0456, d_6^2 < .03852$. All remaining internal holes cost

$$\sum \frac{1}{32(j-1)^3} < 1/4000 + 1/10976 + 1/23328 + 1/42592 < .000408 < 1/1024.$$

Only actual weak knots are omitted.

For a true first-coordinate floor $x_1 > X$, use the reference through $(1, X), (J, f_J)$, then every retained strong knot through the actual $K = 13, 14$. Its first slopes are X, then $m = (f_J - X)/(J-1)$. For $X = .643, .668$, they are positive decreasing and $m > d_J$. The single-reference energy is

$$> E_{13} - 1/1024 + C_J(X) - E_J, \qquad C_J(X) = X^2 + (f_J - X)^2/(J-1).$$

At $X = .643$ this gives

J	$C_J(X) >$	$q_J = w^2 >$	individual (P_1, P_J) prices $>$
6	.803379738	.99509	(.3637, .08226)
7	.8361781266	.98949	(.3775, .08243).

(D19headrefs)

Indeed the prices are $X - m$ and at least $m - d_J$; omitting the next knot increases the latter. At $X = .668$, the corresponding $C_6 > .822316738, C_7 > .8557856266$ already give reference energies $> 1.01403, 1.00909$. These are actual original-vector references, not a second energy budget.

Deleting old one retains D_{18}. Its complete finite one-node list is isolation or simple-three at its long leaf two. Two contacts make a cycle; an additional fork/internal double is not a large finite D extension; a short-fork attachment would require an exceptional finite rank nineteen. For a nonisolated head,

$$v_L = g_L\omega_1 + \omega_2/2, \quad A_L^2 = 2g_L^2 + 2g_L, \quad \cosh L_1 = 2g_L + 1.$$

Own zero has zero time, three has the actual affine triangle, and parallel has its feasible positive null. Every other own has $g_L \geq 1/\sqrt{2}$, hence $x_1 > \sqrt{\sqrt{2} - 1} > .643$. For finite own at least five and dotted own,

$$x_1^2 \geq g_L/(g_L + 1) > .668^2.$$

If new five is weak, choose the independently available strong $J = 6, 7$ and use the preceding $X = .668$ reference, contradiction. If new five is strong, even $X = .643$ suffices: graft it into the first-five block of mean $< .3652$, preserving its sum. The remaining coordinates stay $> .2957 > d_5$, and the gain is

$$(5/4)(.643 - .3652)^2 = .09646605.$$

The full reference then gives $.90529 - 1/1024 + .09646605 > 1.00077$. Thus only the nonisolated native-four head could remain:

$$g_L = 1/\sqrt{2}, \quad A_L^2 = 1 + \sqrt{2}, \quad x_1 > .643.$$

If five is a pure weak helper, let its own be r, with $A_5^2 = 10r^2 + 2r - 1/2$. The actual distinct-wall magnitude is

$$M(r) = A_L A_5 - (2+\sqrt{2})r - (1+\sqrt{2})/2.$$

Its squared-product difference is

$$(4+6\sqrt{2})r^2 - (2+\sqrt{2})r - 5/4 - \sqrt{2},$$

positive already at .8 and increasing. Thus $M > 0$. The identity $(10r+1)^2 = 10A_5^2 + 6$ gives a positive squared derivative comparison $(4+6\sqrt{2})A_5^2 + 6(1+\sqrt{2}) > 0$. So M strictly increases.

At native seven,

$$M(r_7) < 1.5538(3.0693) - (3.4142(.9009) + 1.2071) = .48612556 < 1/2.$$

At native eight, positive squaring shows $M(r_8) > 1/2$: the required difference is equivalent to $3 + (5/2)\sqrt{2} > 2(2+\sqrt{2})r_8$, whose square difference is $3/2 + \sqrt{2} > 0$. Finally $M(1) < \sqrt{2}/2$, since the positive square difference is $11/4 - 3\sqrt{2}/2 > 0$. There is no own label between seven and eight. Every finite helper own therefore puts the mutual strictly in $(0, 1/2)$ or $(1/2, \cos(\pi/4))$, impossible. This is a discrete native-angle argument over the real numbers.

For dotted helper own $r > 1$, its exact weak ratio at five is $(10r+1)/\sqrt{60}$. Since

$$T(\sqrt{10}) < T(16/5) = 50/29 - 655/4656 < 317/200$$

and the ratio at $r = 1.13$ has square $15129/6000 > (317/200)^2$, weakness forces $r < 1.13$. Put $t = 1/r$, so $t > .8849$.

The ACTUAL positive future core $c_5 + r^{-1}n_5$ has square $-(1-t^2)$, zero cap product, negative own product and every other actual-wall product nonpositive, including that with the head. Its normalized pressure is

$$Z = P_5 + (t^2/2)(e_6 - e_5) + (t/2)P_1,$$
$$x \cdot Z \geq T(t) = \{\sqrt{10 + 2t - t^2/2} - t\sqrt{1-t^2}\}/\sqrt{2}.$$

Use the $J = 6$ reference of (D19headrefs); six is strong by helper nonadjacency. Projection to its first coordinate and constant next-five block gives

$$\Pi Z = (1/5 + t/2)P_1 + (4/5)P_6, \qquad Q = \|Z - \Pi Z\|^2 = 4/5 - t^2 + t^4/2 < .35.$$

Its true reference pairing is $w \cdot Z = .8f_6 + (.2 + t/2)(.643) < 1.76012 + .3215t$. The complete continuous deficit satisfies $T(t) - w \cdot Z > .0482$. Indeed $T(t) - .3215t$ increases for $t \geq 7/8$: the derivative contains $(2t^2-1)/(\sqrt{2}\sqrt{1-t^2}) > 1/\sqrt{2}$ and a positive further root derivative. At $t_0 = .8849$, the two radicals give $T(t_0) > 2.0929$. Thus the deficit is $> 2.0929 - .3215(.8849) - 1.76012 = .04828465$. The square checks use $.783 < t_0^2 < .784$, $10 + 2t_0 - t_0^2/2 > 3.373^2$ and $.784(.217) < .413^2$.

Insert $w + .1(Z - \Pi Z)$. It spends at most .07 at P_1 and exactly .08 at P_6, both below the separate available prices. The residual is orthogonal to this NONCONSTANT first reference. One square gives

$$\|x\|^2 > .99509 + .2(.0482) - .01(.35) = 1.00123 > 1.$$

If five is instead isolated dotted, its actual first-six core supplies

$$Z_0 = P_5 + (u/2)(e_6 - e_5), \qquad x \cdot Z_0 > T_0 > 1.8185, \quad 2/5 < u < 1.$$

These are the original local roots; no B_{20} conclusion has been transferred. At a strong $J = 6$,

$$\Pi_6 Z_0 = P_1/5 + 4P_6/5, \quad Q_6 < .48, \quad w \cdot Z_0 < 1.76012, \quad T_0 - w \cdot Z_0 > .05838.$$

The coefficient .1 costs .02 at one and .08 at six, giving

$$\|x\|^2 > .99509 + .2(.05838) - .01(.48) > 1.001966.$$

At strong $J = 7$,

$$\Pi_7 Z_0 = P_1/3 + 2P_7/3, \quad Q_7 < 76/75, \quad w \cdot Z_0 < 1.7048, \quad T_0 - w \cdot Z_0 > .1137.$$

Coefficient .12 costs .04 at one and .08 at seven, giving

$$\|x\|^2 > .98949 + .24(.1137) - .0144(76/75) > 1.002186.$$

The P_1 projections and debits are essential; a constant-block projection $Z_0 - (5/J)P_J$ would be wrong for these references.

The nonisolated head is now completely excluded. The head is isolated with time $A_L^2 = 2g_L^2 - 1 > 0$. Own zero/three/four has nonpositive time; six has the actual affine $6, 3$ path; parallel the nonzero positive future null. Its complete remaining own list is native five, finite at least seven, or dotted. No centre, tail or outgoing-fork floor was assumed in this proof.

B.14.2 An independent old strong-five floor in whole D_{19}

We next prove

$$S_5 > \sqrt{5} - .61/\sqrt{2} \tag{D19old5}$$

without any head hypothesis. It is the OLD .61 floor, not the new .58 floor.

Assume contrariwise old five is weak. The rank-valid generic inventory at $M = 11$, $19 \geq M + 6$, gives a pure helper

$$c_5 = Ap - g\omega_5 - \omega_1/2, \quad A^2 = 10g^2 + 2g - 1/2, \quad \varphi/2 \leq g < 6/5,$$

own five, finite at least seven, or dotted. Its actual first-six points give

$$x_1 > a_0 = .5965, \qquad S_6 > B_0 = 2.01815.$$

There is an independent old strong $K = 11$ or twelve. Let $f_i = \sqrt{i} - .61/\sqrt{2}$, $F_K = B_0 + \sqrt{K} - \sqrt{6} < f_K$. The interpolation through $(1, a_0), (6, B_0), (K, F_K)$ has positive decreasing slopes $a_0, .28433, \ldots, .28433, 1/(\sqrt{K} + \sqrt{6})$. Its precise paid data are

K	$q_K >$	$F_K <$	individual endpoint price $>$
11	.9104	2.886	1/7
12	.9300	3.034	1/7.

The first-six energy is .7600299945. The eleven tail exceeds $.8671^2/5$; the twelve tail exceeds .17. A pressure $V = 1$ through K, $x \cdot V \geq G, V^2 < R$, therefore adds legally

$$2(G - F_K)/7 - (R - K)/49. \tag{D19weak5ref}$$

This is payment at the actual anchor, not subtraction of two independent lower bounds.

Under this assumed weak five we can classify the two late layers afresh. Releasing fifteen retains $A_{14} + D_4$; releasing sixteen retains $A_{15} + D_3$. The right D_4 permits simple three at any leaf. The right $D_3 = A_3$ permits middle-three, end-three/four/five. Two contacts to one right piece make a cycle. The long left path admits only the usual endpoint/next-to-end extensions, and the full rank-nineteen bridges are D_{19} or the end-three A_{19} bridge. No exceptional full diagram, internal four or second fork is missing. There is no foreign component.

Here are explicit one-sided seals before the new right rows. For $i = 15, 16$, an isolated own has $q^2 = 2i \geq 30$. The actual left H_4, left A_2, singleton and dotted cores give losses respectively

$$< .15 + .168, \qquad < .12 + .319, \qquad < .11 + .415, \qquad < .093u + \sqrt{u(1-u)} < (.093 + 1.005)/2.$$

Each is below .61. The finite ranges are $u_5 < 1.528$, $1.2318 < u_7 < 1.233$, and $1 < u_{\geq 8} < 1.172$; positive squaring at $q^2 = 30$ proves the first loss terms. The native-five concavity correction uses $i + 1 \leq 17$, an ACTUAL long dual here.

For left-only contact at j, the exact data are $\sigma = g + j/(2i)$ and $\Delta = 1 - j(i-j)/(2i)$ for simple-three, or $\sigma = g + j/(\sqrt{2}i)$, $\Delta = 1/i$ for endpoint-four. After the actual own-zero, three/four/six affine and parallel-null reductions, near endpoint-three has $\sigma > 1.266, \sqrt{\Delta} < .731$; next-to-end three has $\sigma > .8, \sqrt{\Delta} < .366$; endpoint-four has a larger ratio. The positive-own-three near-four row is directly strong, while far-four/own-three retains its affine principal. These ratios exceed $25/16$. Only the pure far LEFT helper is retained. Own parallel has its feasible positive future null.

At fifteen the right LONG-leaf three has numerator $g + 1/2$, residual $1/2$. Own zero has the actual affine double fork on $c, 15, 16, 17, 18, 19$, kernel $(1, 1, 2, 2, 1, 1)$. Own three is an actual triangle; every own at least four has ratio $\geq 1 + 1/\sqrt{2}$. At sixteen the right middle-three has its analogous own-zero four-leaf affine star, own-three triangle and same strong ratio for larger owns. Right end-four at own zero has the weighted affine principal on $c, f, 17, \overline{f}, 16$, kernel $(\sqrt{2}, 2, 2, 1, 1)$; positive own has ratio $\geq 1 + 1/\sqrt{2}$. End-five is the sealed finite H_4, directly strong EVEN at own zero. Full bridges have their own-zero cycles or middle-three star. For positive own, the worst end-three bridge has residual $5/32$, numerator $> 3/4$; the other bridge residuals are smaller. The old target is $< 25/16$ through these indices, because $T(\sqrt{32}) < T(6) < 25/16$. All these rows are strong.

The remaining one-sided right fork-three at fifteen or end-three at sixteen has own-zero affine E_8 with arms five/two/one. Own three gives the actual FIVE- cycle at fifteen, FOUR-cycle at sixteen. Own four has the affine star centred at the released root, with squared leaf weights $1/2, 1/4, 1/4$. Own six has its actual $6, 3$ principal, and parallel its feasible positive null.

For native five use the LEFT actual H_4 projection on c, n_i, n_{i-1}, n_{i-2}. Its positive correction gives

$$\alpha^2 = \varphi + 5/8 > 56/25, \quad \delta_5 = (5 - 3\varphi)/8, \quad V_1 = \cdots = V_{i-3} = 1, \quad V^2 < i,$$
$$x \cdot V > G > \sqrt{i - \kappa/\alpha^2} - .091, \qquad \kappa = 1/2 \ (i = 15), \quad 5/8 \ (i = 16).$$

The actual finite point is future and all omitted-wall products are nonpositive. The displacement is not presumed to be a point by itself. The demands are $> 3.753, 3.873$, from $3.844^2 < 15 - 25/112$, $3.964^2 < 16 - 125/448$. They are flat on BOTH early reference supports. Even the conservative

$$.91 + 2(3.75 - 3.034)/7 - 5/49 > 1.0125$$

contradicts the norm.

At the D_4 row every finite own at least seven or dotted own is directly strong: $\sqrt{2}(g + 1/4) > 1.626 > 25/16$. At the D_3 row its ratio is $(g + 1/4)/\sqrt{5/8}$. Weakness forces $g < 1$, so dotted own is directly strong. For the remaining $.9 < g < 1$, use the actual core

$$\xi = c + g^{-1} n_{16} + (2g)^{-1} n_{15}.$$

Its cap and fifteen products vanish, its sixteen product is $-g+3/(4g)<0$, and its square is $-k$, $k=1-3/(4g^2)>0$. It is future and satisfies BOTH original forks and every other wall termwise. At this actual rank,

$$\alpha=\sqrt{2}(g+1/4),\quad A^2=16\alpha^2-5/8,\quad V_1=\cdots=V_{14}=1,\quad V^2<16.$$

Since $\alpha^2>529/200$, $\sqrt{16-125/529}>3.97$; also $\sqrt{k}/\alpha<.308$. Thus $G>3.662$. The two early-reference bounds are

$$.9104+2(3.662-2.886)/7-5/49>1.0300,\qquad .9300+2(3.662-3.034)/7-4/49>1.0277.$$

Every non-left-helper possibility at fifteen/sixteen is excluded.

Those two remaining left helpers cannot both be weak. Their actual mutual satisfies

$$c_{15}\cdot c_{16}\geq 1/2-ab+(a/b+b/a)/4\geq 1-ab>-11/25>-1/2,$$

so distinct actual walls are orthogonal by the native gap. After retaining the left A_{14} and ENTIRE right D_3, their common helper cost is $14/30$, and their two-cap Schur matrix $I-(14/30)\mathbf{1}\mathbf{1}^T$ is positive with positive inverse. Its acute positive projection is an actual feasible point, with the original two signed forks in a positive retained block. The already proved mixed-packet formula, with $C_{ij}=2\min(i,j)$, times >2 and own parameters $>.8$, gives weighted loss $<7/16+1/(4\sqrt{10})<.61$. This contradicts two weak sites. Thus fifteen or sixteen is genuinely OLD strong, under the assumed weak five.

If sixteen is strong, the reference through one/six/sixteen already has

$$\|x\|^2>.7600299945+(4-\sqrt{6})^2/10>1.0004379945,$$

using $\sqrt{6}<2.44949$. Only strong fifteen remains.

Every FINITE weak-five own gives $x_1>.63$. Here $\overline{h}=\sqrt{10}-.61<2.553$. Native five uses the actual finite H_4, time $A>2.768$ and radical $<.111$, yielding

$$x_1>1.414(2.768-.111-.81(2.553))>.63.$$

Native seven uses the actual negative core $c_5+gn_5+n_1/2$, with $A>3.0688$, radical $<.2486, g<.901$, giving $x_1>1.414(3.0688-.2486-.901(2.553))>.63$. Native eight is RETAINED and uses that same actual core: $A>3.1435$, radical $<.322$, $g<.9239$, giving $x_1>1.414(3.1435-.322-.9239(2.553))>.6543$. For every finite label at least nine, the actual pair projection gives

$$x_1>\sqrt{2}\{A-\sqrt{1-g^2}-g\overline{h}\}.$$

The brace increases with g: $A'>\sqrt{10}>\overline{h}$, and the negative radical has positive derivative. At $g=.939$, $A>3.192$, $\sqrt{1-g^2}<.344$, giving $x_1>1.414(3.192-.344-.939(2.553))>.6373$. The elementary native-nine lower bracket follows by signs of $8g^3-6g-1$. All these feet are future/all-wall points constructed with original roots one through six.

The positive reference through $(1,.63),(6,B_0),(15,B_0+\sqrt{15}-\sqrt{6})$ has energy

$$.63^2+(B_0-.63)^2/5+(\sqrt{15}-\sqrt{6})^2/9>.7822920845+1.4234^2/9>1.0074.$$

Its heights lie below actual strong fifteen, its slopes $.63, .27763, 1/(\sqrt{15}+\sqrt{6})$ decrease. Thus every finite weak-five own is impossible.

For dotted $1 < g < 6/5$, retain the actual foot BEFORE dropping the chamber difference $r = e_5 - e_6$. The future core $c_5 + g^{-1}n_5$ gives

$$T = P_1 + 2gP_5 - g^{-1}r, \qquad x \cdot T \geq K = \sqrt{2}\{A - \sqrt{1 - g^{-2}}\}.$$

The function $A - \sqrt{1 - g^{-2}} - g\overline{h}$ decreases: $A' < 13/4$, the radical derivative exceeds one and $\overline{h} > 5/2$. At $6/5$ it gives the strict bound

$$K - 2gf_5 > \sqrt{2}\{\sqrt{163/10} - \sqrt{11}/6 - (6/5)\overline{h}\} > .5965.$$

A rational certificate uses $\sqrt{163/10} > 4.03732$, $\sqrt{11}/6 < .552771$, $\sqrt{10} < 3.162278$, $\sqrt{2} > 1.41421$, whose lower product is .596535556834.

Set

$$F = 1.80474, \quad B = 3.44164, \quad X = .59647.$$

Positive square bounds give $1.80473 < f_5 < F$, $f_{15} > B$. Since $F - f_5 < .00001$, the actual primitive pressure obeys $x \cdot T \geq K > X + 2gF$. Define exact positive rationals

$$a = (7B - 8F + X)/66 = 10.25003/66, \qquad t = (2F - 3X + B)/22 = 5.26171/22,$$
$$y = 2t - a = 21.32023/66, \qquad \mu = X - y = 18.04679/66.$$

Let

$$W = (X, y, y, y, t, t, \underbrace{a, \ldots, a}_{9}, 0, 0, 0, 0).$$

Then $W \cdot P_5 = F$, $W \cdot P_{15} = B$, $W \cdot r = 0$. For the ACTUAL unknown g, put

$$\lambda = y - a - 2g\mu < 0, \qquad \rho = a + \mu/g - t > 0.$$

Indeed $y - a < .168$, $2g\mu > 2(.2734)$, and $a > .1553, \mu > .2734, t < .2392, 1/g > 5/6$ give $\rho > .1439$. There is the exact original- coordinate identity

$$W = \mu T + \lambda P_5 + aP_{15} + \rho r.$$

Its NEGATIVE coefficient is intentional and legal: the assumed old weakness is the upper bound $S_5 \leq f_5 < F$. The other signs use the actual strong fifteen lower bound and the original strict chamber inequality $x \cdot r > 0$. Consequently

$$x \cdot W > \mu(X + 2gF) + \lambda F + aB = \mu X + (y - a)F + aB = W^2.$$

The last identity follows also by pairing the displayed coordinate decomposition with W, using $W \cdot T = X + 2gF$. No pressure on a residual was invented and no independently optimized energy was added.

Finally $X > .5964, y > .3230, t > .2391, a > .1553$, so

$$W^2 > .5964^2 + 3(.3230)^2 + 2(.2391)^2 + 9(.1553)^2 = 1.00008039 > 1.$$

The one original-vector square $\|x\|^2 \geq 2x \cdot W - W^2 > W^2$ contradicts $\|x\| < 1$. This proves the head-independent OLD floor (D19old5). It asserts neither a .58 five floor nor centre/fork strength.

B.14.3 Native-five D_{19}: the fork bounds actually needed by the centre

Assume the isolated head has native five, and put

$$a = \varphi, \quad g = a/2, \quad c_H = A_H p - g\omega_1, \quad A_H^2 = (a-1)/2.$$

Its actual finite H_4 point, proved in Section B.13.2, gives $S_4 \geq 2\sqrt{2}A_H > 1.572$. We need two fork consequences, rather than the full outgoing- fork taxonomy.

First BOTH actual fork edge cosines are at most ten. Use the independent old strong- five floor (D19old5) and an actual old strong $K = 11$ or twelve. Its curved reference w_0 has first-five mean $b < .361$, individual P_5 price $> .146$, endpoint price $> 1/7$, and $q_{11} > .8472225, q_{12} > .8689525$. Insert the head ONCE through $w_H = w_0 + .64(P_4/4 - P_5/5)$. The price is .128 at five. The orthogonal residual square is $1/20$; its effective gain is $> 1.28(.393 - .361) - .4096/20 = .02048$.

If an actual fork had $C > 10$, its original-edge comparison would give the signed pressure

$$x \cdot V > \sqrt{171/11} > 3.942, \qquad V = P_{18} \pm e_{19}, \quad V^2 = 19.$$

The residual $V - P_K$ vanishes on the entire head-paid support. A legal $1/7$ endpoint payment gives the one-vector totals

$$.8472225 + .02048 + 2(3.942 - 2.886)/7 - 8/49 > 1.00615,$$
$$.8689525 + .02048 + 2(3.942 - 3.034)/7 - 7/49 > 1.00600.$$

Thus both actual fork cosines are at most ten, without a centre floor.

Second no ACTUAL wall whose spatial part is $d\omega_f$, for either fork f, can have finite own $d = \cos(\pi/m)$. This statement does not require the wall to be first on a fork edge. It also applies to a centre wall of own zero with one finite fork contact.

For such a wall, $A^2 = (19/2)d^2 - 1$, and

$$M = A_H\sqrt{(19/2)d^2 - 1} - (a/2)d.$$

The head and wall are distinct by their different old supports. For $1/2 \leq d \leq 1$, M is positive and increasing: at $1/2$ its squared-product difference is $(5a - 6)/8 > 0$; its derivative is positive since $A_H\sqrt{19/2} > a/2$. Furthermore

$$M < M(1) < .556(2.916) - .809 = .812296 < .82 < \cos(\pi/6).$$

The mutual is therefore FINITE, with exactly one of $M = 1/2, \sqrt{2}/2, a/2$. Its strict scalar range excluded orthogonality, parallel and dotted mutuals before algebra.

Only the elementary finite-angle fact is used next: $t = 2d = \zeta_{2m} + \zeta_{2m}^{-1}$ is an algebraic integer for EVERY Coxeter polytope. The fields below are fields of known real angle constants. They are NOT identified with a defining field of the reflection group, and no dotted product or invariant cycle is declared integral.

Put $L = \mathbb{Q}(a)$, $\mathcal{O}_L = \mathbb{Z}[a]$, $B = 19 - 10a$, $\mathrm{N}_L(B) = 71$. Squaring the positive mutual equation gives exactly

M	quadratic for integral t	discriminant
$1/2$	$Bt^2 - 2t + 2a - 6$	$12(45 - 26a)$
$\sqrt{2}/2$	$Bt^2 - 2\sqrt{2}t - 4$	$8(39 - 20a)$
$a/2$	$Bt^2 - 2at + 2a - 8$	$4(173 - 97a)$.

The first and last discriminant norms are $144(179)$, $16(3739)$. They are nonsquares, since $13^2 < 179 < 14^2$, $61^2 < 3739 < 62^2$. Thus the quadratics are irreducible over L. The relative norm of

integral t must be integral in L, yet its rational absolute norm would be $20/71$, respectively $44/71$, impossible.

For the middle row work in the genuine quadratic extension $L_2 = L(\sqrt{2})$. An element $D \in L$ can be square there only if D or $D/2$ is square in L: expand $(r + s\sqrt{2})^2$, whose mixed coefficient forces $rs = 0$. For $D = 8(39-20a)$, the two norms are $64(341), 16(341)$, both nonsquare as $18^2 < 341 < 19^2$. This quadratic is irreducible over L_2. Its integral root would have relative norm $-4/B$, rational absolute norm $256/71^2$, again impossible. Every finite own is excluded.

Own zero has negative time; parallel own has the nonzero feasible future null $c+n_f$. Consequently an isolated fork wall, when encountered, has dotted own. No integrality of that dotted own is used. These are the only outgoing- fork facts needed below.

B.14.4 Native-five D_{19}: one head-paid centre reference

The complete new bulk through fourteen is already available. The actual native-five mutual calculation uses only the first seven long roots, so the proof in Section B.13.3 applies here. Explicitly, for a helper with $r < 1.2$ and $i \le 7$,

$$A_H^2 A_i^2 - [a(r + 1/2)]^2 = [i(a - 1) - a^2]r^2 - 2r - a/2 < .0568r - .809 < 0.$$

It would have positive actual mutual with the head. The remaining isolated dotted five is excluded by the same actual native-gap calculation: weakness forces $1 < r < \sqrt{5/2}$, and its positive increasing head magnitude is $< .556(4.899) - 1.618(1.581) = .165786 < 1/2$. Thus five/six/seven are truly .58-strong.

Choose an actual strong $K = 13$ or fourteen. All internal weak helper holes cost $< .000158$, using the earliest permitted set $8, 10, 12$. The rational certificate (B19E17), restricted to its initial increments, gives

$$E_{13} > .9056051, \quad E_{14} > .9241299, \quad f_{13} < 3.1955, \quad f_{14} < 3.3316, \quad b = f_5/5 < .3652.$$

The reference w_0 has P_5 price $> .151$ and endpoint price $> 1/8$. Insert the head once in $w_H = w_0 + .14(P_4 - .8P_5)$. Its debit is .112 at five, orthogonal residual square .8, and effective gain

$$.28(1.572 - .8(1.826)) - .0196(.8) = .015456.$$

Hence the actual paid references have

$$L_{13} > .9209031, \qquad L_{14} > .9394279.$$

Their first four coordinates are $< .3932$, their sums remain f_K, and the individual end price stays $> 1/8$.

For an actual pressure $x \cdot V \ge G$, $V = 1$ through the chosen K, $V^2 \le R$, the residual $V - P_K$ vanishes on the ENTIRE head-paid support. One paid square gives

$$\|x\|^2 > L_K + \frac{16G - R - 16f_K + K}{64}. \tag{D19fivepay}$$

Therefore the uniform sufficient contradiction is

$$16G - R > 43.19021. \tag{D19fivebudget}$$

The exact thresholds for the two K's are $43.1902016, 43.1822144$. An ACTUAL .50-strong centre would have $S_{17} > 3.7695$; $V = P_{17}, R = 17$ already violates this budget. The following centre classification may thus assume .50 weakness as a CONTRARY hypothesis. No centre floor is being supplied in advance.

B.14.5 Native-five D_{19}: every isolated centre fails

Own zero has negative time. Own three has the actual four- leaf affine star centred at seventeen; own four the weighted star with both forks; six the $6,3$ path through sixteen; parallel the feasible positive null $c+n_{17}$. Each null is future through its positive cap coefficient and nonpositive against all omitted original walls.

At native five use the LEFT finite H_4 on the actual cap/seventeen/sixteen/fifteen. Its positive correction has residual $\delta=(5-3a)/8$; the actual projected point, normalized by $\alpha=\sqrt{2}g$, gives $V=1$ through fourteen, $R<17$, and

$$G>\sqrt{17-1/(2g^2)}-\sqrt{\delta}/\alpha>4.029-.119>3.91.$$

This contradicts (D19fivebudget).

At native seven/eight use $u=r^{-2}$ and the actual core $c+r^{-1}n_{17}+(2r)^{-1}n_{16}$. Its cap/sixteen products vanish, seventeen product is negative, and its square is $-(1-3/(4r^2))$. It is future and every other original-wall product is nonpositive. Its normalized displacement is

$$V=P_{15}+(1-u/4)(e_{16}+e_{17})+(u/2)e_{18},\quad R=17-u+3u^2/8<17,$$
$$G=\sqrt{17-u/2}-\sqrt{u/2-3u^2/8}>4.047-.27=3.777.$$

Indeed $1.168<u<100/81$; the second squared expression decreases and at 1.168 is $.072416<.27^2$. Thus $16G-R>43.432$.

For EVERY finite own at least nine, $r>.939$ and $r^{-2}<1.1352$. The actual finite pair with seventeen has normalized $V=P_{16}+(e_{17}+e_{18})/2$, $R=16.5$. Its TRUE projected-point demand is larger than

$$\sqrt{17-1/(2r^2)}-\sqrt{1/(2r^2)-1/2}>4.053-.26=3.793.$$

The native brackets used here are only ordinary real-angle square/cubic bounds. Every finite case violates the common budget.

For dotted $r>1$, $u=r^{-2}\in(0,1)$, the actual core $c+r^{-1}n_{17}$ gives

$$V=P_{16}+(1-u/2)e_{17}+(u/2)e_{18},\quad R=17-u+u^2/2,$$
$$G=\sqrt{17-u/2}-\sqrt{u(1-u)/2}.$$

The complete continuous estimate is $16G-R>43.19278$. To prove it, use the concave root chord, $\sqrt{17}>4.123105$ and $8/(\sqrt{17}+\sqrt{33/2})<.9775$, obtaining

$$16G-R>48.96968+.0225u-u^2/2-8\sqrt{2}\sqrt{u(1-u)}.$$

For $u\le.45$, $u^2\le.45u$ and monotonicity give the lower bound $48.96968-.091125-\sqrt{31.68}>43.249555$. For $.45\le u\le.55$, use $u^2\le u-.2475$ and the circle maximum

$$Au+B\sqrt{u(1-u)}\le(A+\sqrt{A^2+B^2})/2.$$

With $A=.4775, B=8\sqrt{2}$ and $\sqrt{128.22800625}<11.3238$, the lower bound is $48.96968+.12375-5.90065=43.19278$. For $u\ge.55$, use $u^2\le1.55u-.55$, $A=.7525$ and $\sqrt{128.56625625}<11.33872$, giving 43.19907. All radicals have positive rational square checks. Thus the ENTIRE dotted isolated centre case is excluded.

B.14.6 Native-five D_{19}: the complete finite-white centre list

Deleting seventeen leaves the old A_{16} and both isolated fork A_1's. The complete finite positive white list is:

> isolation;
> left-only $3@1,2,15,16$ or $4@1,16$;
> one fork with any finite native contact;
> two forks $(3,3),(3,4),(3,5)$, either orientation;
> left endpoint-three $j=1,16$ and one fork-three/four, or both fork-threes.

A second left contact makes a cycle; a next-to-end attachment plus another fork has excessive exceptional arms; no exceptional finite diagram has the required full rank. The whole-forest hypothesis leaves no foreign component.

For direct .50 tests use $T_{.50}(\sqrt{34}) < 265/138 < 1.921$. For left-only three,

$$\rho = (\gamma + j/34)/\sqrt{1 - j(17-j)/34}.$$

At $j=2$, own zero has zero time, three an affine cycle; every own at least four is directly strong. At $j=15$, own zero has the actual positive affine kernel $(1,1,2,2,2,1,1)$ on $c,14,15,16,17,18,19$; every positive own is strong. At $j=16$, zero has kernel $(1,1,2,2,1,1)$ on $c,15,16,17,18,19$, three its triangle, four the weighted star using both forks, six its $6,3$ path. Finite owns at least nine and dotted owns are strong; five/seven/eight need the actual point below. At $j=1$, zero has negative time, three the actual long cycle, four the weighted star using sixteen and one fork, six its $6,3$ path. Its remaining far-helper owns are retained for a separate calculation.

For left-only four,

$$\rho = (\gamma + j/(17\sqrt{2}))/\sqrt{1 - j + j^2/17}.$$

Far own zero has zero time; all other far cases and even near own zero are directly strong.

For one fork β,

$$\rho = (\gamma + \beta/2)/\sqrt{1-\beta^2}.$$

Own zero is the actual finite- fork wall already excluded by Section B.14.3, whose proof did not require it to be first on a fork edge. At own three, deleting the contacted fork gives a positive finite D_{19} replacement. Actual positive inverse projection makes it the reciprocal fork endpoint, of cosine

$$C_f = (17 + 19\beta)/2 \geq 53/4 > 10.$$

This contradicts the actual fork bound. The centre and fork opposites are allowed to coincide; no false distinctness was assumed. Own four has the affine weighted star using sixteen and the OTHER fork; six its $6,3$ path; parallel its positive future null. For $\gamma \geq \varphi/2$ and $\beta \geq \cos(\pi/5)$, the ratio is > 2.06. Thus only $\beta = 1/2, 1/\sqrt{2}$ needs further point estimates.

For two forks β, η,

$$\rho = \frac{\gamma + (\beta+\eta)/2}{\sqrt{1-\beta^2-\eta^2}}.$$

Own zero of $(3,3)$ is the actual all-three square, three an affine triangle, six/parallel their actual null obstructions. Own four/five will be treated below; finite at least seven and dotted own are strong. Own zero of $(3,4)$ has an actual finite B_{19} reciprocal endpoint at the three-fork, with $C_f = 17/2 + (19/4)\sqrt{2} > 10$. Positive owns there, and even own zero in $(3,5)$, are directly strong.

For left endpoint-three at $j = 1, 16$ with fork magnitudes β, η, the exact data are

$$\delta = 9/17 - \beta^2 - \eta^2, \qquad \rho = \frac{\gamma + j/34 + (\beta + \eta)/2}{\sqrt{\delta}}.$$

One fork-three at own zero has the actual long/four-cycle. Near positive own is strong. Far own three has its fork triangle, four the weighted star using sixteen and the OTHER fork; larger owns are strong. One fork-four or two fork-threes has residual $1/34$, strong even at own zero. Both signed fork orientations and every mixed own-zero role are now included. Every positive affine/null combination is future through the cap and nonpositive against ALL omitted original walls.

B.14.7 Native-five D_{19}: every remaining centre point exception

For near-left three and own five/seven/eight, $v = \gamma\omega_{17} + \omega_{16}/2$, $A^2 = 34\gamma^2 + 32\gamma + 7$. Use the ACTUAL finite pair c, n_{17}, with residual $\delta = 1 - \gamma^2 > 0$ and projected-point pressure $H = AB/(B + \sqrt{\delta})$, $B^2 = A^2 + \delta$. Its normalized displacement is

$$V = P_{16} + \frac{\gamma}{2\gamma + 1}(e_{17} + e_{18}), \quad \alpha = \sqrt{2}(\gamma + 1/2), \quad R < 16.23.$$

At native five, $A^2 = (49a + 31)/2$, $B^2 = (97a + 65)/4$. Positive-square bounds $A > 7.425, B > 7.448, \sqrt{\delta} < .588, \alpha < 1.852$ give

$$G > 7.425(7.448)/(8.036(1.852)) > 3.715.$$

The sharper $R = 17 - a/2 < 16.2$ gives $16G - R > 43.24$. The weaker $A - \sqrt{\delta}$ pressure would be insufficient. At native seven/eight, $A/\alpha > 4.019$, $\sqrt{\delta}/\alpha < .221$; hence $G > 3.798$, which closes these rows as well.

For one/two fork helpers write

$$v = \gamma\omega_{17} + \beta\omega_f + \eta\omega_h, \quad \alpha = \sqrt{2}(\gamma + (\beta + \eta)/2), \quad A^2 = 17\alpha^2 - (1 - \beta^2 - \eta^2).$$

At own five, the LEFT actual finite H_4 has correction $g(3n_{17}/2 + n_{16} + n_{15}/2)$, residual $(5 - 3a)/8$. For a single helper $\beta = 1/2, 1/\sqrt{2}$, or two helpers $\beta = \eta = 1/2$, its actual projected point has $V = 1$ through fourteen, $R < 17$, and

$$G > 4.082 - .091 > 3.99.$$

Use $\alpha > 1.497$, $A/\alpha > 4.082$, $\sqrt{(5 - 3a)/8}/\alpha < .091$. The contacted fork walls remain outside the selected finite piece, with their actual products nonpositive. Every own-five helper exception is excluded.

For one fork-four and own finite at least seven or dotted, contrary .50 weakness forces $.9 < \gamma < 1.01$. The actual positive core

$$c + \gamma^{-1}n_{17} + (2\gamma)^{-1}n_{16}$$

has zero cap/sixteen products, negative seventeen product, square $-k$, $k = 1 - 3/(4\gamma^2) > 0$. It is future and every omitted wall product is nonpositive. With $\alpha = \sqrt{2}(\gamma + \sqrt{2}/4)$, V is flat through fifteen, $R < 17$, and

$$G = \sqrt{17 - (1/2)/\alpha^2} - \sqrt{k}/\alpha > 4.103 - .291 = 3.812.$$

The complete seals are $\alpha > 1.772, \alpha^2 > 3.141$, $k < .265, \sqrt{k} < .515$.

For one fork-three the entire contrary range is $.9 < \gamma < 1.42$. Use the SAME actual core, with $\alpha = \sqrt{2}(\gamma + 1/4)$, $A^2 = 17\alpha^2 - 3/4$. Its displacement is flat through fifteen and

$$G = \sqrt{17 - (3/4)/\alpha^2} - \sqrt{k}/\alpha > 3.756, \qquad R = 17 - (3/4 + k)/\alpha^2 < 16.8.$$

The continuous proof is explicit. For $\gamma \le 1.1$, the first root is > 4.088, the second $< .324$. The latter squared ratio increases up to 1.1, its derivative having sign $-2\gamma^3 + 3\gamma + 3/8 > 0$. For $\gamma \ge 1.1$, the first root is > 4.098. Uniformly $\sqrt{k}/\alpha < .342$: the inequality $k/\alpha^2 < 7/60$ is equivalent to

$$112\gamma^4 + 56\gamma^3 - 473\gamma^2 + 360 > 0.$$

At $t = \gamma - 5/4$ this becomes $15/4 - 45t + 787t^2 + 616t^3 + 112t^4$. Because $t \ge -.35$, it is at least $15/4 - 45t + (2857/5)t^2 > 0$, with negative discriminant. For the norm, $1.75 - .75/\gamma^2 - .4(\gamma + .25)^2$ is concave and positive at both $.9, 1.42$. Thus $16G - R > 43.296$ throughout, including all dotted parameters.

The two-fork-three/own-four row is a separate actual point. Let $s = \sqrt{2}, \gamma = s/2$ and

$$\xi = c + (4 - 2s)n_{17} + (2 - s)n_{16} + (3 - 2s)(n_{18} + n_{19}).$$

All coefficients are positive. Its cap/seventeen/sixteen products vanish, each fork product is $1/2 - s < 0$, and $\xi^2 = 11 - 8s < 0$. It is future and feasible against every original wall. For $\alpha = 1 + s/2$,

$$V = P_{15} + (5 - 3s)(e_{16} + e_{17} + e_{18}), \quad R = 144 - 90s < 16.721,$$
$$G = \sqrt{14 + 2s} - \sqrt{92s - 130} > 4.102 - .329 = 3.773.$$

Hence $16G - R > 43.647$. All non-far-helper centre cases have now been excluded at their actual changed threshold.

B.14.8 Native-five D_{19}: the only centre left is a dotted far helper

For the far-left-three cap, $v = \gamma\omega_{17} + \omega_1/2$, $A^2 = 34\gamma^2 + 2\gamma - 1/2$. Own zero/three/four/six/parallel already has its actual time, affine or positive-null obstruction. The remaining finite-own pressure table, normalized by $\sqrt{2}\gamma$, is

own	$V = P_{16} + Y$	actual point/core	$G >$
5	$a_3P_3 + (e_{17} + e_{18})/2$	finite H_4, $\delta = 2/3 - \gamma^2$	4.1158
7, 8	$a_2P_2 + (e_{17} + e_{18})/2$	$c + \gamma n_{17} + n_1/2$	3.963
finite ≥ 9	$a_1P_1 + (e_{17} + e_{18})/2$	finite pair c, n_{17}	3.965.

Here $a_3 = 1/(6\gamma) < 5/24$, $a_2 = 1/(4\gamma) < 5/18$, $a_1 = 1/(2\gamma) < .533$. All selected roots are actual. Their positive corrections give future points retaining every unknown wall and both original forks; the finite rows use projected POINTS.

All finite rows satisfy $A/(\sqrt{2}\gamma) \ge \sqrt{17.75} > 4.213$. The first two normalized radical subtractions are $< .097125, .25$. For finite at least nine, $\gamma \ge .95$ gives the crude pair lower bound $> 4.213 - .233 = 3.98$. For $.939 < \gamma \le .95$, the crude lower bound is $> 4.213 - .26$, and the TRUE finite pressure improves $A - \sqrt{\delta}$ by $\delta/(B + \sqrt{\delta})$. Since $\delta \ge .0975$, $\sqrt{2}\gamma < 1.344$, $B < 5.674$, $\sqrt{\delta} < .344$, its normalized improvement is $> .012$. Thus $G > 3.965$ in the whole finite-nine-and-higher range. Native eight was retained geometrically, not removed by a field condition.

The head overlap is NONZERO. For $Y = a_l P_l + (e_{17} + e_{18})/2$,

$$V - P_K = P_{16} - P_K + Y, \qquad w_H \cdot (V - P_K) = a_l(w_H \cdot P_l),$$
$$\|V - P_K\|^2 = 16 - K + la_l^2 + 1/2.$$

With the same legal 1/8 endpoint price, the sufficient barrier is therefore

$$16\{G - a_l(w_H \cdot P_l)\} - (16 + la_l^2 + 1/2) > 43.19021.$$

The true head-paid first coordinates give $w_H \cdot P_l < .3932l$. The three rows have respective lower bounds $45.29059, 43.25856, 43.30270$. All violate the norm budget. No aggregate lower bound was subtracted to manufacture a residual pressure.

For dotted $1 < \gamma \le 10/9$, let $a_0 = 1/(2\gamma) \in [.45, .5)$. The actual core $c + \gamma^{-1}n_{17}$ gives

$$V = P_{16} + a_0 P_1 + (1 - 2a_0^2)e_{17} + 2a_0^2 e_{18},$$
$$G = \sqrt{17 + 2a_0 - a_0^2} - a_0\sqrt{2 - 8a_0^2}.$$

The residual squared norm is $16 - K + 1 - 3a_0^2 + 8a_0^4 \le 16 - K + .75$. The function $G - .3932a_0$ strictly increases, since its derivative is

$$\frac{1 - a_0}{\sqrt{17 + 2a_0 - a_0^2}} + \frac{16a_0^2 - 2}{\sqrt{2 - 8a_0^2}} - .3932 > 0.$$

At $a_0 = .45$, $\sqrt{17.6975} > 4.2068$, $\sqrt{.38} < .6165$, giving $G - .3932a_0 > 3.752435$. The paid barrier is $> 16(3.752435) - 16.75 = 43.28896$, a contradiction on the full small-dotted interval.

For $\gamma \ge 11/8$, its exact edge ratio $(\gamma + 1/34)/\sqrt{9/17} > 1.928$ already gives the contradictory .50-strong centre floor. The ENTIRE remaining actual centre has consequently the profile

$$c = Ap - \gamma\omega_{17} - \omega_1/2, \qquad A^2 = 34\gamma^2 + 2\gamma - 1/2, \qquad 10/9 < \gamma < 11/8, \qquad \text{(D19fivecentre)}$$

with dotted own. No outgoing fork equality, isolation or symmetry was assumed.

B.14.9 Native-five D_{19}: classify short fork edges at the actual adjacent vertex

Replace old seventeen by its ACTUAL opposite (D19fivecentre). The other endpoint q of that original edge has finite forest

$$c \ --3--\ n_1 \ --3--\cdots--3--\ n_{16}, \qquad n_f,\ n_h,$$

an A_{17} and two orthogonal A_1's. This is the actual neighbouring vertex, not a formal completion or a vertex assumed farthest. Set

$$B = 17\gamma + 1/2, \quad C = (34\gamma + 1)/(3\sqrt{2}), \quad s = A\sqrt{17}/3, \quad q = Cp - sP_{17}/\sqrt{17}, \quad C^2 - s^2 = 1.$$

Both forks are unchanged unit normals tangent at q. Put $\sigma = s/\sqrt{34} = A/(3\sqrt{2})$. The centre interval gives

$$A^2 < 2129/32, \qquad \sigma < 77/40 = 1.925, \qquad B > 349/18.$$

We need the products of the new path endpoint duals with the OLD actual wall n_{17}. Let

$$w = (e_1 - P_{17}/17)/\sqrt{2}, \quad \eta = c + w, \quad \eta^2 = 9/17, \quad d_{16} = \sqrt{2}(P_{16} - 16P_{17}/17).$$

These are tangent to q. The new path duals are $\Omega_c = (17/9)\eta$, $\Omega_{16} = d_{16} + \Omega_c/17$: η is orthogonal to old one through sixteen, and $d_{16} \cdot c = -1/17$. Directly,

$$\Omega_c \cdot n_{17} = -B/9, \qquad \Omega_{16} \cdot n_{17} = -(144 + B)/153. \qquad \text{(D19qduals)}$$

These are actual Lorentz tangent identities; no projected mirror has been presumed Coxeter.

Call a q-fork edge short if its actual length satisfies $\cosh L < \sqrt{3/2}$. Let r be its actual opposite, with positive own magnitude β, full finite-white cost $Q < 1$, and time $a_r > 0$. The unit A_1 inverse gives

$$a_r^2 = \beta^2 + Q - 1, \qquad 1 - Q < \beta^2 < (3/2)(1 - Q). \tag{D19qshort}$$

Own zero cannot satisfy this. Parallel own has the actual positive future null $r + n_f$ and is impossible. All finite and dotted owns remain in the displayed strip.

After releasing that A_1, the endpoint retains the ENTIRE new A_{17} and the other A_1. Its complete positive finite white list and costs are

contact outside the released A_1	Q
none	0
only other A_1, finite label $m \geq 3$	$\cos^2(\pi/m)$
A_{17} end-three	17/36
A_{17} next-to-end-three	8/9
A_{17} end-four	17/18
A_{17} end-three plus other-three	13/18
A_{17} end-three plus other-four	35/36.

A second path contact is a cycle. Its rank-eighteen one-node extensions are A, B, D, with no exceptional position. Joining both retained pieces allows only end-three followed by other-three/four, giving A_{19}, B_{19}. A next-to-end attachment plus another edge has excessive exceptional arms, and an end-four followed by another node has a nonterminal double. Thus every finite white type is included.

Applying the exact short strip leaves: isolated dotted own $1 < \beta^2 < 3/2$; path end-three with own five/ six; or other-A_1-only own-five/white-four, own-four/white-five, own-three/white-seven, or white-three with finite own at least seven or dotted, $3/4 < \beta^2 < 9/8$. Here are the complete label checks for the other-A_1 case: white three/four/five/ six/seven/at-least-eight gives respectively the own-square strips

$$(.75, 1.125), \quad (.5, .75), \quad ((3-a)/4, 3(3-a)/8), \quad (.25, .375),$$

a subinterval containing only native own-three at white seven, and upper bound $< .25$ at white at least eight. The other cost rows are empty: $Q \geq 8/9$ has own-square upper bound $< 1/6$, and $Q = 13/18$ has only possible native three below its positive- time lower bound. The mixed 35/36 row is smaller still. This is local finite classification at the ACTUAL q, with no farthest hypothesis there.

Every short opposite r is distinct from OLD n_{17}: that wall has dotted contact $-\gamma$ with new c, which cannot occur in its retained positive finite white endpoint. Actual acuteness/native products therefore apply to r, n_{17}.

For path end-three, $j = c$ or sixteen, write $r = a_r q - \beta n_f - \Omega_j/2$. Own is five/six, so $\beta > .809, a_r < .472$. Using (D19qduals),

$$r \cdot n_{17} = -a_r\sigma + \beta/2 - (\Omega_j \cdot n_{17})/2 > -.472(1.925) + .4045 + .533 = .0289 > 0.$$

For endpoint c the last contribution exceeds one; for endpoint sixteen it exceeds .533. Both are impossible.

For a cap supported on the two q-A_1's,

$$r = a_r q - \beta n_f - \eta n_h, \qquad r \cdot n_{17} = -a_r\sigma + (\beta + \eta)/2.$$

Own-three/white-seven has $a_r < .249$, half spatial sum $> .7$, so the product is $> .7 - .249(1.925) > .22$. Own-five/white-four and its reverse have $a_r^2 = (a-1)/4 < .3931^2$, $(\beta+\eta)/2 > .75805$. The product is $> .75805 - .3931(1.925) = .0013325 > 0$. These are strict actual-wall acuteness contradictions.

Only two short types remain:

$$\begin{array}{c|c} I & r = a_r q - \beta n_f, \quad 1 < \beta^2 < 3/2 \\ J & r = a_r q - \beta n_f - n_h/2, \quad \text{finite own } \geq 7 \text{ or dotted, } .9 < \beta < 1.061, \quad a_r^2 = \beta^2 - 3/4 < 3/8. \end{array}$$

For type J, $\rho = -r \cdot n_{17} = a_r \sigma - (\beta + 1/2)/2 \geq 0$. But $\rho < .613(1.925) - .7 = .480025 < 1/2$. The ACTUAL native gap forces $\rho = 0$. Hence this wall has zero products with ALL old one through seventeen; its exact OLD expansion is

$$r = H_\beta p - \beta \omega_f - \omega_h/2, \qquad H_\beta^2 = (19/2)\beta^2 + (17/2)\beta + 11/8. \tag{D19qJ}$$

It need not be first on an old fork edge; its old dual expansion follows just from the actual products and unit norm.

B.14.10 Native-five D_{19}: one q-fork edge is nonshort

Both q-fork edges cannot be short. The surviving types cover both same-wall and distinct-wall possibilities. An I wall has zero other contact and cannot close the other edge. A J wall has own $> .9$ and other contact $1/2$; in the reverse direction it does not have either surviving short type. Thus two short opposites in the remaining list are distinct.

Two type-I walls have product $-a_r a_t \in (-1/2, 0)$, because both time squares are $< 1/2$. This violates the native gap. For I at one fork and J at the other, the q-product is

$$-a_I a_J + \beta_I/2 > 1/2 - \sqrt{3}/4 > 0,$$

using $a_I^2 < 1/2, a_J^2 < 3/8$ and $\beta_I > 1$.

For two type-J walls, use their OLD expansions (D19qJ). Their negative mutual is

$$M(b,d) = H_b H_d - (17/2)bd - (19/4)(b+d) - 17/8.$$

This function increases in both independent parameters $b, d > .9$. Indeed $H_b' > \sqrt{19/2}$, and

$$(19/2)H_d^2 - [(17/2)d + 19/4]^2 = 18d^2 - 19/2 > 0.$$

Since BOTH actual parameters are < 1.061, monotonicity gives

$$M(b,d) < M(1.061, 1.061) = 1.061^2 - 1.061 - 3/4 = -.685279 < 0.$$

Their actual product would be positive. Equality of the two actual parameters was not assumed; the equal argument is only an upper bound after monotonicity.

At least one actual q-fork edge therefore has

$$D = \cosh L \geq \sqrt{3/2}. \tag{D19qnonshort}$$

No ordinary A_1 energy theorem or farthest-forest classification was applied at q.

B.14.11 Native-five D_{19}: compare that endpoint with the ORIGINAL farthest vertex

For the nonshort q-fork edge, let its ACTUAL other endpoint be $r' = Dq - \sqrt{D^2-1}\, n_f$, in whichever signed orientation has (D19qnonshort). Put

$$\tau = (-z \cdot q)/E = C - (s/\sqrt{17})S_{17}.$$

The ORIGINAL farthest property gives $0 < \tau \le 1$ and

$$\tau D - \sqrt{D^2-1}\, x \cdot n_f \le 1.$$

We retain τ; it is not replaced by one. For fixed $0 < \tau \le 1$, $(D\tau - 1)/\sqrt{D^2-1}$ increases with D: its derivative is $(D-\tau)/(D^2-1)^{3/2} > 0$. The lower endpoint $D_0 = \sqrt{3/2}$ is therefore legal, giving

$$x_{18} \pm x_{19} \ge \sqrt{6}\,\tau - 2.$$

Substitution yields the actual SAME-vector pressure

$$x \cdot V \ge G, \qquad V = P_{17} + t(e_{18} \pm e_{19}), \quad t = \sqrt{17}/(\sqrt{6}\, s),$$
$$G = \sqrt{17 + 6t^2} - 2t, \qquad R = 17 + 2t^2.$$

Every endpoint used is actual and hence satisfies every original and unknown wall. There is no formal replacement or independent energy at q.

Since $s^2 = 17A^2/9$ and $\gamma > 10/9$,

$$t^2 = 3/(2A^2) < 243/7079 < 9/256, \qquad 0 < t < 3/16.$$

The cleared comparison is $243(256) = 62208 < 63711 = 9(7079)$. The function $\sqrt{17+6t^2} - 2t$ decreases on this interval. At its upper endpoint,

$$G > \sqrt{17 + 27/128} - 3/8 > 4.1486 - .375 = 3.7736, \qquad R < 17 + 9/128 = 17.0703125.$$

The positive square check is $4.1486^2 = 17.21088196 < 17 + 27/128$. Thus

$$16G - R > 43.3072875 > 43.19021.$$

The displacement is one through the actual chosen $K = 13, 14$. Its residual is orthogonal to the head-paid reference, and its $1/8$ individual endpoint price is still available. Equation (D19fivepay) gives

$$\|x\|^2 > 1 + (43.3072875 - 43.19021)/64 > 1.0018,$$

contradiction.

Proposition B.14.1. *A whole D_{19} forest at the original farthest vertex cannot have an isolated native-five long head.*

Proof. The complete centre reduction leaves exactly (D19fivecentre). At its actual adjacent vertex the full local short-edge classification and old-wall native gap force one nonshort fork edge. Its actual endpoint violates the single ORIGINAL-vector head-paid norm budget, with the exact distance slack retained. No equality or classification of the outgoing fork opposites was required. □

B.14.12 A reusable centre test for the remaining finite heads

The centre geometry in the preceding argument is useful beyond the native-five head. We separate it from the early payment that changes with the head. Throughout this section the forest is the SAME whole D_{19}, with its signed last coordinate and actual centre seventeen. No field restriction is used.

Suppose an actual strong $K \in \{13, 14\}$ has supplied a legally funded vector w, using only initial coordinates one through six to modify a curved reference. Write L for its effective energy: if its legally available pressure is $x \cdot w \geq d$, then $L = 2d - \|w\|^2$. In particular L need not equal $\|w\|^2$. Assume its INDIVIDUAL P_K coefficient is $> 1/8$, its demand there is $f_K = \sqrt{K} - .58/\sqrt{2}$, and all initial changes have zero coordinate sum. Consequently $w \cdot P_K = f_K$.

For any actual pressure $x \cdot V \geq G$, put

$$B = w \cdot (V - P_K), \qquad R_* = K + \|V - P_K\|^2.$$

Consume $1/8$ of the original P_K coefficient and use the ONE vector $w + (V - P_K)/8$. Its pressure follows from the remaining positive anchor/foot decomposition, rather than subtraction of unrelated lower bounds. Expanding its complete square gives

$$\|x\|^2 \geq L + \frac{16(G - B) - R_* - 16f_K + K}{64}. \tag{D19generalpay}$$

If V is one through K, its residual vanishes there and has no overlap with the early zero-sum changes. The curved reference vanishes after K, so $B = 0$ and $R_* = V^2$. For a nonflat displacement we retain BOTH displayed terms.

Lemma B.14.2. *Suppose the reference above makes* $16(G - B) - R_* > H$ *sufficient to contradict* $\|x\|^2 < 1$. *Either of the following sets of bounds suffices:*

H *at most*	w_1 *at most*	$w \cdot P_2$ *at most*	$w \cdot P_3$ *at most*
42.278	.478	.956	1.291
41.539	.54	.956	1.291.

Then the actual centre opposite is impossible.

Proof. Both ACTUAL fork cosines are less than ten. A fork of cosine at least ten gives $V = P_{18} \pm e_{19}$, $G \geq \sqrt{171/11} > 3.94$, $R = 19$, hence $16G - R > 44.04$. A .50-strong centre gives $V = P_{17}$, $G > 3.7695$, $R = 17$, hence $16G - R > 43.312$. Both clear either threshold. We may classify the centre under its actual .50-weak hypothesis.

The complete positive finite-white list and all direct, affine and reciprocal calculations are explicitly proved in Section B.14.6: isolation; left-only three at one, two, fifteen, sixteen or four at one/sixteen; one arbitrary finite fork; fork pairs $(3,3), (3,4), (3,5)$; and left endpoint-three joined to one fork-three/four or both fork-threes. Each formula depends on its ACTUAL old contacts, not the head. Own-three at one fork has reciprocal cosine $(17 + 19\beta)/2 \geq 53/4$; own-zero at a $(3,4)$ pair has reciprocal cosine $17/2 + (19/4)\sqrt{2} > 10$. The positive inverse proves these are actual endpoints even when a centre and fork opposite coincide. The affine kernels there preserve every original wall term by term. Thus all own-zero/three/four/six/parallel, mixed-left and direct-ratio decisions remain valid EXCEPT own-zero with one finite fork, which was separately removed for native five.

We prove that case without any head-specific obstruction. Let its fork be f, the other fork h, and $c = Ap - \beta\omega_f$, $A^2 = (19/2)\beta^2 - 1$. Fork labels three, four and six have the positive actual nulls

$$2c + 4n_f + 6n_{17} + 5n_{16} + 4n_{15} + 3n_{14} + 2n_{13} + n_{12} + 3n_h,$$
$$\sqrt{2}c + 2n_f + 2n_{17} + n_{16} + n_h,$$
$$\sqrt{3}c + 2n_f + n_{17}.$$

They are future through c, nonpositive against every omitted wall, so compactness excludes them. At label five the ACTUAL H_4 on c, n_f, n_{17}, n_{16} has positive correction $\beta(3n_f/2 + n_{17} + n_{16}/2)$, residual $(5 - 3\varphi)/8$. The finite projected POINT gives

$$V = P_{15} + (e_{16} + e_{17} + e_{18} \pm e_{19})/2, \quad R = 16, \quad G > \sqrt{3 + 8\varphi} - \varphi^{-3} > 3.75.$$

Indeed $1.618 < \varphi < 1.619$ gives first root > 3.992 and $\varphi^{-3} < .238$. Thus $16G - R > 44$. Both signed orientations are included.

For every finite fork label at least seven put $v = \beta^{-2}$. The positive actual core

$$c + \beta^{-1} n_f + 2(\beta^{-1} - \beta) n_{17}$$

has zero products with c, n_f, negative products with n_{17}, n_{16}, n_h, and square $-(1-\beta^2)(4\beta^2-3)/\beta^2 < 0$. All unknown-wall products are nonpositive and its time is positive. Its future normalization gives

$$V = P_{16} + (3 - 2v)e_{17} + (v - 1)e_{18} \pm (v - 1)e_{19},$$
$$G = \sqrt{19 - 2v} - \sqrt{-8 + 14v - 6v^2}, \quad R = 27 - 16v + 6v^2, \quad 1 < v < 100/81.$$

The line $4.364 - .242v$ lies below the first root: its endpoint square checks are $4.122^2 < 17$ and $329.284^2 < 1339(81)$, and concavity puts the chord below the root. With $z = v - 7/6$, the other root is $\sqrt{1/6 - 6z^2} \le 1/\sqrt{6} - 3\sqrt{6}z^2$, and $R = 16.5 - 2z + 6z^2$. Hence

$$16G - R > 42.2746 - 1.872z + 111.57z^2 > 42.2646. \qquad \text{(D19forkcontinuous)}$$

The completion uses $1.872^2 < .04(111.57)$. This bound alone is too weak for the first threshold; we do not round it up. Native seven has $.064 < z < .067$, giving > 42.60; every native label at least nine has $z < -.033$, giving > 42.45. For native eight, $v = 4 - 2\sqrt{2}$ gives

$$G = \sqrt{11 + 4\sqrt{2}} - \sqrt{68\sqrt{2} - 96} > 3.6732094,$$
$$R = 107 - 64\sqrt{2} < 16.490333, \qquad 16G - R > 42.2810174 > 42.278.$$

The positive square seals are $4.0812809^2 < 11 + 4(1.414213562)$, $.4080715^2 > 68(1.414213563) - 96$. These cases exhaust all finite labels without arithmeticity or the native-five finite-fork obstruction.

The remaining point exceptions are already constructed, with all original-wall checks, in Sections B.14.5 and B.14.7. We use their point identities and scalar inequalities, not their head-paid conclusions. Their complete weaker bounds are

actual centre profile	$16G - R$ greater than
isolated five	45.56
isolated seven/eight	43.432
isolated finite at least nine	44.188
isolated dotted	42.328
near-left-three, own five	43.24
near-left-three, own seven/eight	44.538
one/two fork-three/four, own five	46.84
one fork-four, larger finite/dotted own	43.992
one fork-three, larger finite/dotted own	43.296
two fork-threes, own four	43.647.

These vectors are flat through fourteen, hence have no early head cross. For isolated dotted, for example, the actual core gives

$$G=\sqrt{17-u/2}-\sqrt{u(1-u)/2}>4.062-.354=3.708,\quad R=17-u+u^2/2\le 17,\quad 0<u<1.$$

Near-left own five uses the finite pair's projected POINT $AB/(B+\sqrt{1-\gamma^2})$, not the smaller unprojected $A-\sqrt{1-\gamma^2}$. The one-fork continuous intervals $.9<\gamma<1.42$ and $.9<\gamma<1.01$ retain the polynomial/core proofs of Section B.14.7, including every dotted parameter left by weakness. All corrections are positive actual-root combinations or positive finite inverse projections, so neither signed fork nor any unknown wall is discarded.

Only far-left-three remains. The finite point identities of Section B.14.8 are local to $v=\gamma\omega_{17}+\omega_1/2$, not to the head:

own	$V=P_{16}+aP_l+(e_{17}+e_{18})/2$	G greater than
5	$l=3,\ a<5/24$	4.1158
7, 8	$l=2,\ a<5/18$	3.963
finite ≥ 9	$l=1,\ a<.533$	3.953.

They are the actual H_4 point, the negative future core $c+\gamma n_{17}+n_1/2$, and the positive pair projection. Their normalized losses are $<.097125, <.25, <.26$ from a first term $>\sqrt{17.75}>4.213$. For these NONFLAT displacements retain

$$B=a\,w\cdot P_l,\qquad R_*=16+la^2+1/2.$$

Under the first coordinate row the three tests are greater than $44.91, 42.50, 42.387527$; the last is exactly the safe expression $16(3.953-.533(.478))-(16.5+.533^2)$. Under the second coordinate row the first two bounds still apply, and the last becomes

$$16(3.953-.533(.54))-(16.5+.533^2)=41.858791>41.539.$$

Every finite far own is removed with its actual head cross.

All dotted $\gamma\ge 11/8$ are directly .50-strong because $(\gamma+1/34)/\sqrt{9/17}>1.928$. For $1<\gamma\le 10/9$, put $a=(2\gamma)^{-1}\in[.45,.5)$. The actual core $c+\gamma^{-1}n_{17}$ gives

$$V=P_{16}+aP_1+(1-2a^2)e_{17}+2a^2e_{18},$$
$$G=\sqrt{17+2a-a^2}-a\sqrt{2-8a^2},\qquad B=aw_1,\quad R_*\le 16.75.$$

The function $G-aw_1$ increases even for $w_1=.54$: its derivative is

$$\frac{1-a}{\sqrt{17+2a-a^2}}+\frac{16a^2-2}{\sqrt{2-8a^2}}-w_1>0.$$

At $a=.45$, the positive square checks $\sqrt{17.6975}>4.2068, \sqrt{.38}<.6165$ give $G>3.929375$. The two coordinate rows yield left sides >42.6784 and >42.232, respectively. Both suffice.

The ONLY centre left is the literal far-three helper

$$c=Ap-\gamma\omega_{17}-\omega_1/2,\qquad A^2=34\gamma^2+2\gamma-1/2,\qquad 10/9<\gamma<11/8.$$

Sections B.14.9–B.14.11 prove a local geometric statement from precisely this interval. Their derivation uses neither a native-five head nor head energy: the actual adjacent vertex has forest $A_{17}+A_1+A_1$; its two short fork edges are classified with their complete finite-white costs, old n_{17}

acuteness and the actual native gap. At least one is nonshort. Comparing its ACTUAL endpoint with the ORIGINAL farthest vertex retains $0 < \tau \le 1$ and gives a displacement flat through seventeen, with

$$G > 3.7736, \quad R < 17.0703125, \quad 16G - R > 43.3072875.$$

This clears both thresholds in (D19generalpay). The adjacent vertex was never declared farthest. One original x pays every pressure, including all early and nonflat crosses, and the full centre list is exhausted. □

B.14.13 Changing the first knot without losing a weak-six omission

Put $f_i = \sqrt{i} - .58/\sqrt{2}$, $\epsilon = .03/\sqrt{2}$, and

$$F = f_5 - \epsilon = \sqrt{5} - .61/\sqrt{2}, \qquad b = F/5.$$

The independent OLD theorem supplies the INDIVIDUAL floor $S_5 > F$. Useful exact scalar bounds are

$$1.80473 < F < 1.804733, \quad .360946 < b < .3609466, \quad f_5/5 < .36519,$$
$$E_{13} > .9056051, \quad E_{14} > .9241299, \quad f_6 > 2.0393, \quad f_{13} < 3.1955, \quad f_{14} < 3.3316.$$

The E_K are the already proved curved-reference scalar values in (B19E17), not numerical enumeration outputs.

Start with the full .58 reference and lower its knot at five by ϵ, leaving the actual strong later knots. The energy change is $-2\epsilon\beta_5 + (6/5)\epsilon^2$, where

$$\beta_5 = (6/5)\sqrt{5} - \sqrt{6} - .116/\sqrt{2} < .15177.$$

The bounds $\sqrt{5} < 2.236068$, $\sqrt{6} > 2.449489$, $1/\sqrt{2} > .7071$ give this seal, and $\epsilon < .021214$ gives total loss

$$2\epsilon\beta_5 - (6/5)\epsilon^2 < .042428(.15177) - .00054 = .00589929756 < .0059.$$

The new first-five mean is b; the first subsequent slope is at most $\sqrt{6} - \sqrt{5} + \epsilon < .234637$. Thus its original P_5 coefficient is $> .126$. If six is genuinely strong it is $> .126309$, while the INDIVIDUAL P_6 coefficient is $> .0375$, as used below.

Actual .58-weak knots at six through fourteen are pure far-three helpers and nonadjacent. Their ordinary disjoint averaging losses sum to $< .000408$. If six is weak, the interval five–seven changes its loss to $(2\sqrt{6} - \sqrt{5} - \sqrt{7} + \epsilon)^2/2$. Its EXTRA over the ordinary omission is

$$\epsilon(2\sqrt{6} - \sqrt{5} - \sqrt{7}) + \epsilon^2/2 < .021214(.0172) + .000225 = .0005898808.$$

Only this interval touches the changed knot. Hence the COMPLETE omission loss, including weak six, is $< .0010$. The mixed reference therefore has

$$q_{13} > .8987051, \qquad q_{14} > .9172299, \qquad \text{terminal price} > 1/8. \tag{D19mixedenergy}$$

If six is strong all omissions begin at seven; the sharper complete bound is

$$1/6912 + 1/16384 + 1/32000 + 1/55296 < .000256.$$

Omitting later knots only increases the individual capacities just identified. This calculation makes no assertion that five or six is new-strong unless its separate geometric argument proves that fact.

B.14.14 Native seven: exclude a weak six before making the reference

Let $g_0 = \cos(\pi/7)$, $S = g_0^2$, and

$$c_H = A_H p - g_0\omega_1, \qquad A_H^2 = 2S - 1, \qquad u = g_0^{-2}.$$

The polynomial $8g_0^3 - 4g_0^2 - 4g_0 + 1 = 0$, increasing above .9, gives

$$.90096886 < g_0 < .90096888, \quad 1.2319140 < u < 1.2319143, \quad .8116 < S < .8119, \quad A_H < .7897.$$

Expand at .901: value, derivative and quadratic coefficient are $.000257608, 8.275224, 17.624$. Increments $-.00003114, -.00003112$ give opposite signs, and reciprocal-square multiplications give the u interval. These are native-angle scalars, not a field census.

The positive core of Section B.13.2, $c_H + g_0^{-1}n_1 + 2(g_0^{-1} - g_0)n_2$, gives

$$\begin{aligned} Z &= (1 - u/2)(e_1 + e_2) + (u - 1)e_3, \quad \sum Z_i = 1, \\ x \cdot Z \geq H &= \sqrt{1 - u/2} - \sqrt{(7u - 3u^2 - 4)/2} > .4318779, \\ R &= Z^2 = 3 - 4u + 3u^2/2 < .348763. \end{aligned}$$

Its selected products are zero or negative, its square negative, and positive actual-root coefficients check EVERY other wall. The seals use first root $> .6197119$, second $< .187834$, and $R(1.2319140) = .348762155094$.

If six were .58-weak, its actual bulk profile would be

$$c_6 = Ap - \gamma\omega_6 - \omega_1/2, \quad A^2 = 12\gamma^2 + 2\gamma - 1/2, \quad \varphi/2 \leq \gamma < 6/5.$$

It is distinct from c_H by its contact at six. Its mutual magnitude is $M_6 = A_H A - g_0(2\gamma + 1)$. The squared positive-term difference $(20S - 12)\gamma^2 - 2\gamma + 1/2 - 2S$ is positive and increasing: at $\gamma = .809$ it is greater than

$$4.232(.809)^2 - 1.618 - 1.1238 = .027963592.$$

Also $(12\gamma + 1)^2 = 12A^2 + 7$ proves M_6 increases, and

$$M_6 < .7897(4.38) - .9009(3.4) = .395826 < 1/2.$$

The actual native alphabet has no magnitude strictly between zero and one half. This excludes ALL finite and dotted weak-six parameters. Thus six is genuinely strong before any early reference is built; the complete omission loss now starts at seven and is $< .000256$.

A weak-five far helper has $A^2 = 10\gamma^2 + 2\gamma - 1/2$ and $M_5 = A_H A - g_0(2\gamma + 1)$. The derivative is positive, by the squared comparison $(16S - 10)A^2 + 6A_H^2 > 0$; and

$$M_5 < .7897(4.038) - .9009(3.4) = .1257486 < 1/2.$$

Acuteness and the native gap force $M_5 = 0$, hence $(16S - 10)\gamma^2 - 2\gamma + 1/2 - 2S = 0$. This polynomial increases above .809, is negative at one, and positive at 1.04, so $1 < \gamma < 1.04$. The actual core and weak UPPER bound at five, together with $x_5 > x_6$, give

$$x_1 > \sqrt{2}\{A - \sqrt{1 - \gamma^{-2}} - \gamma(\sqrt{10} - .58)\}.$$

The braces decrease: $A' < 13/4$, the radical derivative is greater than one, and $\sqrt{10} - .58 > 2.5$. At 1.04, $A > 3.52$, the radical is $< .275$, and $\sqrt{10} - .58 < 2.58228$, yielding $x_1 > .79114420896 > .79$. The legal reference through $(1, .79), (6, f_6), (K, f_K)$ has energy

$$.79^2 + (2.0393 - .79)^2/5 + 1.156^2/7 > 1.127.$$

Its slopes decrease, and $K = 14$ only raises the final chord energy. This contradicts the SAME norm. Thus weak five can ONLY be the actual isolated dotted cap; it has not been silently discarded.

For that cap put $v = \gamma_5^{-2} \in (0, 1)$. Its actual local first-six-root pressure, proved in the head reduction, is

$$Z_5 = P_5 + (v/2)(e_6 - e_5), \qquad x \cdot Z_5 \geq T_5 > 1.8185.$$

Weakness forces $v > .499$. For $v \leq .499$ the same core gives loss at most

$$\frac{v}{\sqrt{10} + \sqrt{10 - v}} + \sqrt{v(1 - v)} < .16015(.499) + .4999991 = .57991395 < .58.$$

Both upper terms increase there; the denominator seal uses $\sqrt{10} > 3.1622, \sqrt{9.5} > 3.0822$, and $.4999991^2 > .499(.501)$. Such a cap would be strong.

B.14.15 Native seven: pay the head and dotted-five foot jointly

If five is .58-strong, take the ordinary curved reference, with first-five mean $b = f_5/5 < .36519$, omission loss $< .000256$, P_5 price $> .15$ and P_K price $> 1/8$. Use $w = w_0 + .45(Z - P_5/5)$, paying .09 at five. The residual is orthogonal to w_0, and its full effective gain is

$$.9(H - b) - .2025(R - .2) > .0298946025.$$

Thus $L_{13} > .9352437025, L_{14} > .9537685025$.

If five is the retained weak isolated dotted cap, use the OLD floor F, the ACTUAL strong f_6, and every later actual strong knot. Section B.14.13 gives loss $< .0059 + .000256$, individual P_5 price $> .126309$, and P_6 price $> .0375$. Pay .094 at five and set $w_H = w_0 + .47(Z - P_5/5)$. The effective head gain $.94(H - F/5) - .2209(R - .2) > .0338136753$ gives intermediate energies $.9332627753, .9517875753$. The REMAINING P_5 price is $> .0323$, and exactly

$$w_H \cdot P_5 = F, \quad w_H \cdot P_6 = f_6, \quad T_5 - w_H \cdot Z_5 > .013767 + .01615v.$$

Project the second foot onto ONLY the legitimate P_5, P_6 space:

$$\Pi Z_5 = (1 - 3v/5)P_5 + (v/2)P_6, \qquad Q = \|Z_5 - \Pi Z_5\|^2 = v^2/5.$$

With $\nu = .046$, its INDIVIDUAL costs are $\nu(1 - 3v/5) < .0322276 < .0323$ and $\nu v/2 < .023 < .0375$. No head-induced P_3 coefficient is spent as an old anchor. The residual is $v/10$ on coordinates one through four, $-2v/5$ at five, zero at six. Its head cross is $w_H \cdot (Z_5 - \Pi Z_5) = .047v$, not zero. The ONE completed square includes it, giving extra effective gain

$$\begin{aligned} 2\nu(T_5 - w_H \cdot Z_5) - \nu^2 Q &> .001266564 + .0014858v - .0004232v^2 \\ &> .0019026009768 > .0019026. \end{aligned}$$

The polynomial increases on $[.499, 1]$. Final energies are therefore $L_{13} > .9351653753, L_{14} > .9536901753$. The foot is used once, not as an independent norm.

Both final references satisfy

$$w_1 < .453, \quad w \cdot P_2 < .906, \quad w \cdot P_3 < 1.291, \qquad \text{remaining } P_K > 1/8.$$

In the dotted branch the last residual adds at most .0046 to each of the first four coordinates; the preceding bounds were $.4475, .895, 1.2709$. The strong-five branch has $.4481, .8962, 1.2756$. All

changes have zero sum within the first six. The exact sufficient thresholds in (D19generalpay), for $K = 13, 14$, are

strong-five	42.27240304	42.26441584
dotted-five	42.2774159808	42.2694287808.

The common 42.278 and first coordinate row of Lemma B.14.2 apply. This excludes the ENTIRE native-seven head branch in both actual states at five.

B.14.16 Native eight: the strong and weak six cases give different prices

Let $s = \sqrt{2}$, $g_0 = \cos(\pi/8)$, and

$$c_H = A_H p - g_0 \omega_1, \quad A_H^2 = s/2, \quad u = g_0^{-2} = 4 - 2s.$$

The same positive actual head core gives

$$\begin{aligned} Z &= (s-1)(e_1 + e_2) + (3 - 2s)e_3, \quad \sum Z_i = 1, \\ x \cdot Z \geq H &= \sqrt{s-1} - \sqrt{17s - 24} > .43955, \\ R &= 23 - 16s < .372584. \end{aligned}$$

All-wall feasibility and the future negative square follow from the local core proof in Section B.13.2. The numerical seals use $1.414213562 < s < 1.414213563$, with the first root $> .64359$ and second $< .20404$. Use ONLY the independent OLD five floor F, and split six by its ACTUAL .58 status.

If six is strong, the mixed curved reference has omissions beginning at seven, total $< .000256$, and P_5 price $> .126$. Its energies are $> .8994491, .9179739$ for $K = 13, 14$. Pay .09 at five and insert $.45(Z - P_5/5)$. The orthogonal residual gain is

$$.9(.43955 - .3609466) - .2025(.372584 - .2) = .0357948.$$

The final reference therefore satisfies

$$\begin{aligned} L_{13} > .9352439, \quad L_{14} > .9537687, \\ w_1 < .45735, \quad w \cdot P_2 < .9147, \quad w \cdot P_3 < 1.2629. \end{aligned} \tag{D19eightstrong}$$

Its remaining P_K price is $> 1/8$.

If six is weak, its ACTUAL wall is

$$c_6 = Ap - \gamma\omega_6 - \omega_1/2, \quad A^2 = 12\gamma^2 + 2\gamma - 1/2, \quad \varphi/2 \leq \gamma < 6/5.$$

Its distinctness from the head is explicit in its six contact. The mutual magnitude $M = A_H A - g_0(2\gamma+1)$ is positive and increasing: the squared positive-term difference is $(5s-2)\gamma^2 - 2\gamma - (1+s)/2$, positive already at .8; the derivative comparison is $(5s-2)A^2 + 7s/2 > 0$. At $6/5$,

$$M < .8409(4.38) - .9238(3.4) = .542222 < .55.$$

The actual native gap therefore forces $M = 1/2$. Since $M(1) < 1/2$ and $M(1.165) > .502559707$, we have $1 < \gamma < 1.165$.

The actual core and the weak UPPER bound $S_6 \leq f_6$, with $x_6 > x_7$, now give

$$x_1 > \sqrt{2}\{A - \sqrt{1 - \gamma^{-2}} - \gamma(\sqrt{12} - .58)\}.$$

The braces decrease: $A' < 15/4$, the radical derivative is > 1, and $\sqrt{12} - .58 > 2.8$. At 1.165, use $A > 4.2563$, the radical $< .5131$, and $\sqrt{12} - .58 < 2.88411$, obtaining $x_1 > .5419382 > .54$. The actual helper nonadjacency makes seven strong. No strong-six floor was inferred.

Take the legal reference through $(1, .54), (5, F), (7, f_7)$ and the later actual strong knots. Its initial slopes satisfy

$$m = (F - .54)/4 > .3161825, \quad n = (f_7 - F)/2 < .215485, \quad d_7 < .183.$$

Thus its INDIVIDUAL prices at P_1, P_5 are $> .2238, > .1006$. Relative to the mixed reference with six omitted, the first-coordinate gain is $(5/4)(.54 - b)^2 > .04005125$. Keeping the enlarged total omission budget $< .0010$, its energies are

$$q_{13} > .93875635, \quad q_{14} > .95728115.$$

This accounts for weak six before any head pressure is inserted.

The correct projection is onto BOTH available anchors P_1, P_5:

$$\Pi Z = (Z_1 - t)P_1 + tP_5, \quad t = (2 - s)/4, \quad Q = \|Z - \Pi Z\|^2 = 37/2 - 13s < .115224.$$

Use $\mu = .3$. Its separate costs are $\mu(Z_1 - t) < .080331 < .2238$ and $\mu t < .043935 < .1006$. The old reference is constant only on coordinates two through five; we do not project onto a fictitious first-five constant block. Its actual pairing is

$$w_0 \cdot Z = .54(s - 1) + m(2 - s) < .409,$$

so the full head gain is $> .6(.03055) - .09(.115224) = .00795984$. The final weak-six reference satisfies

$$\begin{aligned} L_{13} > .94671619, \quad L_{14} > .96524099, \\ w_1 = .54, \quad w \cdot P_2 < .9366, \quad w \cdot P_3 < 1.2603. \end{aligned} \tag{D19eightweak}$$

It retains the P_K coefficient $> 1/8$.

In the strong-six case the exact sufficient thresholds in (D19generalpay) are 42.2723904, 42.2644032, so $H = 42.273$ suffices. In the weak-six case they are $41.53816384, 41.53017664$, so $H = 41.539$ suffices. The first and second coordinate rows of Lemma B.14.2, respectively, apply. In particular the weak-six case keeps its larger first coordinate in every nonflat centre pressure; it never borrows the strong-six bound. This excludes the ENTIRE native-eight head branch without a new-strong-five assumption or an independent head energy.

B.14.17 Native nine: one mixed reference covers every early omission

Put $g = \cos(\pi/9)$, $c_H = A_H p - g\omega_1$, $A_H^2 = 2g^2 - 1$, and $u = g^{-2}$. The polynomial $8g^3 - 6g - 1 = 0$, increasing above .9, gives

$$.9396 < g < .9398, \qquad 1.1322 < u < 1.1327.$$

The two exact cubic evaluations have the following numerators and denominators:

argument	numerator	denominator
2349/2500	−21983608	15625000000
4699/5000	203952792	125000000000.

The intended root is above .9 by the elementary cosine estimate. The reciprocal checks

$$.9398^2(1.1322) = .999986258088 < 1,$$
$$.9396^2(1.1327) = 1.000002110832 > 1$$

give the second interval.

The positive head core again supplies

$$Z = (1-u/2)(e_1+e_2) + (u-1)e_3, \quad H = \sqrt{1-u/2} - \sqrt{(7u-3u^2-4)/2} > .45865,$$
$$R = 3 - 4u + 3u^2/2 < .3941.$$

Here the first root is $> .6585$, the second $< .19985$; the second radicand is increasing and R decreasing on the interval. Their endpoint values $.039936065, .39401526$ give the stated seals. The normalized core is future and satisfies EVERY original wall by the same actual positive-combination proof as before.

Use the mixed reference of Section B.14.13, including the enlarged weak-six omission. It uses only the OLD five floor, has $q_{13} > .8987051, q_{14} > .9172299$, individual P_5 price $> .126$, and terminal price $> 1/8$. Pay .1 at five and insert $.5(Z - P_5/5)$. The full effective gain is

$$(H-b) - .25(R-.2) > .049125.$$

Hence

$$L_{13} > .9478301, \quad L_{14} > .9663549,$$
$$w_1 = w_2 < .478, \quad w \cdot P_2 < .956, \quad w \cdot P_3 < 1.284.$$

The coordinate formulas are $w_1 = b + .4 - u/4$ and $w \cdot P_3 = 3b + .2$; all other omissions retain the same constant first-five block before this head change. The terminal coefficient is unchanged. The exact sufficient thresholds are $41.4668736, 41.4588864$; the common $H = 41.467$ satisfies the first row of Lemma B.14.2. Lemma B.14.2 therefore applies in its first row and excludes the actual centre opposite. Its proof includes the nonflat far-helper cases and the original-p adjacent-vertex conclusion, using the unchanged terminal coefficient $> 1/8$. This excludes every native-nine head.

B.14.18 Every finite long-head label at least ten

We use OLD loss .61 throughout this branch. Put $f_i = \sqrt{i} - .61/\sqrt{2}$, $b = f_5/5 < .361$. The independent OLD theorem gives $S_5 > f_5$. For an isolated head with $g = \cos(\pi/m)$, $m \geq 10$, put

$$c_H = A_H p - g\omega_1, \quad A_H^2 = 2g^2 - 1, \quad u = g^{-2}, \quad 1 < u < 1.106.$$

The positive core $c_H + g^{-1}n_1 + 2(g^{-1} - g)n_2$ has zero products with the cap and one, negative products with two/three, positive time, and negative square $-[7 - 4g^2 - 3g^{-2}]$. Every other original-wall product is nonpositive term by term. Its normalized pressure is

$$x \cdot Z \geq h(u) = \{\sqrt{2-u} - \sqrt{P(u)}\}/\sqrt{2}, \quad P(u) = (u-1)(4-3u),$$
$$Z = (1-u/2)(e_1+e_2) + (u-1)e_3, \quad \sum Z_i = 1, \quad R = 3 - 4u + 3u^2/2.$$

For any actual OLD reference constant b through five insert $.6Y_H$, where $Y_H = Z - P_5/5$. Its INDIVIDUAL five price is .12, smaller than the available $b - d_5 > .3609 - .214 > .146$. Orthogonality gives effective gain greater than

$$F(u) = 1.2(h(u) - .361) - .36(R - .2). \tag{D19largegain}$$

The function F decreases on $1 < u \le 1.106$. Here $P' \ge .364, P < .073$, so the two negative terms in

$$h' = -\frac{1}{2\sqrt{2}\sqrt{2-u}} - \frac{P'}{2\sqrt{2}\sqrt{P}}$$

have magnitudes $> .35, > .45$. Since $-R' < 1$, $F' < -.96 + .36 < 0$. At $u = 1.106$, positive squares give $\sqrt{2-u} > .9455, \sqrt{P} < .26888$: $P = .072292 < .26888^2$, and $.4784(1.41422) = .676562848 < .9455 - .26888$. Thus $h > .4784$, $R - .2 = .210854$, and

$$F(u) > .06497256. \qquad \text{(D19largecommongain)}$$

For every $m \ge 11$ we have $u < 1.087$. Since $\pi/11 < 2/7$, the alternating cosine lower bound gives

$$\cos(\pi/11) > 1 - 2/49 + 2/7203 - 4/5294205 > .9593.$$

The polynomial decreases on $[0, 2/7]$; $.9593^2(1.087) = 1.00031880463 > 1$, and cosine monotonicity covers all larger labels. At 1.087, $\sqrt{2-u} > .9555, \sqrt{P} < .25357$, because $P = .064293 < .25357^2$ and $.496(1.415) = .70184 < .9555 - .25357$. Hence $h > .496$, $R - .2 = .2243535$, and

$$F(u) > .08123274 \quad (m \ge 11). \qquad \text{(D19largerfinitegain)}$$

Late coverage under this actual head. The OLD rank-valid bulk supplies an independent strong $K = 11$ or twelve, with ordinary curved energies $q_{11} > .8472225, q_{12} > .8689525$, endpoint price $> 1/7$. The ONE head payment gives

$$L_{11} > .91219506, \quad L_{12} > .93392506.$$

We establish late coverage with these references, not with a late conclusion from an earlier assumed-weak-five branch.

The complete rank 19 finite-white classification and all-wall projection identities at fifteen/sixteen are printed in Section B.14.2. Before its numerical comparisons, those local proofs use only the retained $A_{14}+D_4$, respectively $A_{15}+D_3$, and actual released root. Its weak-five assumption supplied the earlier reference, not these identities. We now apply them with the new head-paid reference. For clarity the full coverage is as follows.

Isolated and left-only rows are .61-strong except pure far-left three helpers. The four isolated loss bounds at $q^2 \ge 30$, the near/next-to-end ratios and actual own-zero/three/four/six/parallel principals in that section are unchanged; native-five uses its ACTUAL long $i+1$ dual, at most seventeen. At right D_4, long-leaf-three has residual $1/2$; own-zero is the affine double fork, own-three the triangle, larger owns have ratio at least $1 + 1/\sqrt{2}$. The D_3 middle-three has the same decisions. End-four/own-zero has the printed weighted affine kernel, every positive own is strong, and end-five is finite H_4, directly strong even at own-zero. Both-sided bridges are endpoint-three; own-zero has the long cycle or middle star, while positive own has numerator $> 3/4$, residual at most $5/32$. Two right contacts make a cycle; there is no foreign component or exceptional full-rank bridge.

For right fork-three at fifteen or end-three at sixteen, own-zero has actual affine E_8 arms five/two/one. Own-three gives the FIVE-cycle at fifteen and FOUR-cycle at sixteen; four gives the affine star with squared weights $1/2, 1/4, 1/4$; six/parallel have their actual nulls. At D_4, all remaining finite owns at least seven and dotted owns are directly strong, with ratio $> 1.626 > 25/16$. At D_3, weakness forces a remaining finite $.9 < \gamma < 1$; dotted own is directly strong. There are just two point exceptions:

row	actual construction	$G >$	$V^2 <$
own five/right three at fifteen or sixteen	positive LEFT H_4 point	3.75	16
larger finite own/right end-three at sixteen	$c + \gamma^{-1}n_{16} + (2\gamma)^{-1}n_{15}$	3.662	16.

Section B.14.2 prints their full point/core formulas, finite inverse positivity and EVERY-wall check. They are flat through at least twelve, so have zero head-residual cross. With the ACTUAL separate endpoint data $f_{11} < 2.886$, $f_{12} < 3.034$, the legal $1/7$ payment gives

$$.91219506 + 2(3.662 - 2.886)/7 - 5/49 > 1.0318,$$
$$.93392506 + 2(3.662 - 3.034)/7 - 4/49 > 1.0317.$$

The H_4 demand is larger. Every old-weak late survivor is therefore a literal pure far-left-three helper, with own five, finite at least seven or dotted, parameter $< 6/5$.

Two adjacent such helpers have mutual $> -.44 > -1/2$, hence are orthogonal by the actual native gap. For fifteen/sixteen the retained A_{14} and ENTIRE right D_3 give $I - (14/30)\mathbf{1}\mathbf{1}^T$, positive with positive inverse; fourteen/fifteen retain $A_{13} + D_4$, cost $13/28$. The actual mixed-point calculation printed in Section B.14.2 has prefix inverse $2\min(i, j)$, weighted loss $< 7/16 + 1/(4\sqrt{10}) < .61$, and preserves BOTH forks and all unknown walls. Weak helpers are thus nonadjacent across the bulk/late join; an ACTUAL $J \in \{15, 16\}$ is OLD strong under the finite-head hypothesis.

Labels at least eleven. Use the ordinary OLD reference ending at that actual J. At most five nonadjacent omissions cost $< 5/3200$, giving energy

$$> E_{15} - 5/3200 > .92564 - .0015625 = .9240775.$$

Its five price is still $> .146$. Insert the head foot ONCE. Equation (D19largerfinitegain) yields

$$\|x\|^2 > .9240775 + .08123274 = 1.00531024 > 1.$$

The earlier references established coverage; their energies are not added to this final one.

Native ten. Here $u = 2 - (2/5)\sqrt{5} \in (1.1, 1.106)$, since $447/200 < \sqrt{5} < 9/4$. If sixteen is OLD strong, its ordinary reference has $q_{16} > E_{16} - 5/3200 > .9401975$; the common head gain gives $\|x\|^2 > 1.00517006$. Otherwise sixteen is the actual pure weak helper and fifteen strong. Take the reference ending fifteen and insert the single head correction. It has effective energy $> .98905006$, remaining P_{15} coefficient $> d_{14} > 8/61 > .13$, and ACTUAL coordinate bounds

$$w_1 < .511, \quad w \cdot P_2 < 1.022, \quad w \cdot P_3 < 1.323.$$

They follow from $b + .6(.8 - u/2), 2b + .6(1.6 - u), 3b + .24$.

Write the actual weak helper

$$c = Ap - \gamma\omega_{16} - \omega_1/2, \qquad A^2 = 32\gamma^2 + 2\gamma - 1/2.$$

Its finite feet and negative cores give $x \cdot V \geq G$, $V = P_{15} + Y$, with COMPLETE data

own	Y	$G >$	$Y^2 <$	$w \cdot Y <$
5	$aP_3 + (e_{16} + e_{17})/2,\ a < 5/24$	3.9954	.631	.275625
7, 8	$aP_2 + (e_{16} + e_{17})/2,\ a < 5/18$	3.8426	.655	5.11/18
finite ≥ 9	$aP_1 + (e_{16} + e_{17})/2,\ a < .533$	3.8326	.785	.272363.

These are respectively the actual positive H_4 point, future core $c + \gamma n_{16} + n_1/2$, and finite pair point. Their first normalized term is at least $\sqrt{16.75} > 4.0926$; the losses are respectively $< .097125, .25, .26$. For the last, $\gamma > .939$ and $.939^2(1.1352) = 1.0009296792 > 1$. The displayed norms and overlaps

follow by squaring Y and using the ACTUAL coordinate bounds, retaining every head/helper cross. All finite projections are points with positive coefficients in actual normals, so EVERY original wall is preserved.

For the entire dotted helper range, $a = (2\gamma)^{-1} \in (5/12, 1/2)$, and

$$Y = aP_1 + (1-2a^2)e_{16} + 2a^2 e_{17}, \quad Y^2 \le .75,$$
$$G = \sqrt{16 + 2a - a^2} - a\sqrt{2 - 8a^2}, \quad G - w \cdot Y > 3.5429.$$

Indeed $.75 - Y^2 = (.25 - a^2)(8a^2 - 1) \ge 0$. The function $G - .511a$ increases, since the second term in G' is $(16a^2 - 2)/\sqrt{2 - 8a^2} > 7/9 > .511$, while the first is positive. At $a = 5/12$,

$$G - .511a = \sqrt{2399}/12 - 5\sqrt{22}/72 - .511(5/12) > 4.0816 - .325724 - .212917 = 3.542959.$$

Here $4.0816^2 < 2399/144$ and $\sqrt{22} < 4.69042$. This bounds the ACTUAL overlap, without a separately invented pressure on Y.

Pay $\nu = .13$ from the INDIVIDUAL P_{15} coefficient and use the ONE vector $w + \nu Y$. Its full square, including the already paid head, gives

$$\|x\|^2 > .98905006 + .26(G - f_{15} - w \cdot Y) - .0169Y^2.$$

Since $f_{15} < 3.442$, the finite/dotted deficits are respectively greater than $.2777, .1167, .1182, .1009$; their extra gains exceed $.0615381, .0192725, .0174655, .013559$. Every case satisfies $\|x\|^2 > .98905006 + .013559 = 1.00260906 > 1$. Own-three/four/six and parallel were already excluded by actual affine/null principals, so this helper list is complete. Native ten and every finite $m \ge 10$ are excluded, without borrowing late floors from a different hypothesis.

B.14.19 A dotted head: one exact parabola on its whole parameter interval

Let the isolated actual head be

$$c_H = A_H p - g\omega_1, \quad g > 1, \quad A_H^2 = 2g^2 - 1, \quad u = g^{-2} \in (0,1).$$

The ACTUAL positive core $c_H + g^{-1}n_1$ has zero cap product, negative products with one/two, negative square $u - 1$, positive time. Positivity of its coefficients checks EVERY other original wall. Its future normalization gives

$$x \cdot Z \ge H = \sqrt{1 - u/2} - \sqrt{u(1-u)/2},$$
$$Z = (1 - u/2)e_1 + (u/2)e_2, \quad \sum Z_i = 1, \qquad R = 1 - u + u^2/2.$$

There is no second-neighbour correction of either sign. We treat the WHOLE dotted interval, not a selected compact subinterval.

Put $s = \sqrt{6}$, $z = u - 2/3$, and $q = s/3 - sz/8 - 9z^2/100$. Here $-2/3 < z < 1/3$, and $q \ge s/4 - .04 > 0$. The exact identity

$$1 - u/2 - q^2 = z^2\{3s/50 - 3/32 - (9s/400)z - 81z^2/10000\}$$

is nonnegative: its bracket is at least $.045s - .09735 > 0$, using $|z| \le 2/3$, $s > 2.4494$. Thus the first root is at least q. For $r = \sqrt{u(1-u)/2}$, the elementary square $(r - 1/3)^2 \ge 0$ gives $r \le 1/3 - z/4 - 3z^2/4$. Consequently

$$H \ge (s-1)/3 + (2-s)z/8 + .66z^2, \qquad R = 5/9 - z/3 + z^2/2. \qquad \text{(D19headparabola)}$$

This is a global square comparison, not a sampled minimum.

Return to mixed floors $f_i = \sqrt{i} - .58/\sqrt{2}$, $F = \sqrt{5} - .61/\sqrt{2}$, $b = F/5$. Section B.14.13 includes ALL actual omissions, including enlarged weak six, and supplies

$$q_{13} > .8987051, \quad q_{14} > .9172299, \quad P_5 \text{ price} > .126, \quad P_K \text{ price} > 1/8.$$

Weak isolated dotted five is retained through its OLD individual floor; neither five nor six has been upgraded to .58-strong. Use ONE vector $w = w_0 + .34(Z - P_5/5)$, consuming only .068 of the actual five coefficient. The effective head gain is

$$J = .68(H - b) - .1156(R - .2). \tag{D19dottedgain}$$

Using (D19headparabola), $b < .361$, and $2.4494897 < s < 2.4494898$, the constant/linear/quadratic terms give

$$J > .0419687 + A_J z + .391z^2, \quad 0 < A_J < .0004.$$

The exact linear coefficient is $A_J = .208533333\ldots - .085s$. Completing the square gives

$$J > .04196, \tag{D19dotteduniform}$$

since $.0004^2 < .00000011(4(.391))$. The ACTUAL coordinates retain their correlation with u:

$$\begin{aligned} w_1 = b + .34(.8 - u/2), \quad & .462946 < w_1 < .633, \\ w \cdot P_2 = 2b + .204 < .926, \quad & w \cdot P_3 = 3b + .136 < 1.219. \end{aligned} \tag{D19dottedcoordinates}$$

Every change has zero sum in the first five and preserves the INDIVIDUAL P_K coefficient. No artificial old P_2/P_3 pressure is paid.

B.14.20 The dotted-head centre: retain correlation in the nonflat row

The general square (D19generalpay) applies with the actual $L_K > q_K + J$. The two threshold upper bounds are

$$44.6108736 - 64J, \qquad 44.6028864 - 64J.$$

By (D19dotteduniform), the common sufficient test is

$$16(G - B) - R_* > 41.926. \tag{D19dottedtest}$$

An actual fork of cosine at least ten has test > 44.04; a .50-strong centre has test > 43.312. Thus both fork cosines are less than ten and the centre is .50-weak.

The complete finite-white classification, actual affine/nulls, reciprocal endpoint computations and flat point proofs of Lemma B.14.2 apply before its nonflat coordinate tests. They use the head only through the paid threshold. Every flat point bound is at least 42.328. Own-zero/single-fork with finite label at least seven has the COMPLETE continuous estimate (D19forkcontinuous), > 42.2646, sufficient now without discrete refinements; three/four/six have the actual nulls and five the positive H_4 point. Thus every isolated, near, fork-only, mixed bridge and reciprocal exception is excluded. The actual signed forks and unknown-wall checks are those printed in that shared proof; no native-five-only fork theorem is used.

The remaining far-left-three wall has

$$c = Ap - \gamma\omega_{17} - \omega_1/2, \quad A^2 = 34\gamma^2 + 2\gamma - 1/2.$$

Its complete finite feet are those constructed in the shared proof:

$$V = P_{16} + aP_l + (e_{17} + e_{18})/2, \quad B = a\,w \cdot P_l, \quad R_* = 16 + la^2 + 1/2.$$

Own-five and seven/eight have respectively $(l, a, G) = (3, < 5/24, > 4.1158)$ and $(2, < 5/18, > 3.963)$. The ACTUAL coordinate bounds (D19dottedcoordinates) give tests $> 45.15, 42.638$, sufficient in (D19dottedtest).

For finite own at least nine, $l = 1, a < .533, G > 3.953$. A separately maximized head coordinate would be unnecessarily weak. Retain the ACTUAL correlation: its generalized test is greater than

$$S_9 = 16(3.953 - .533w_1) - (16.5 + .533^2) = 46.463911 - 8.528w_1.$$

Subtract the LARGER unrounded threshold above and substitute the actual J, w_1, using only $b < .361$:

$$\begin{aligned} S_9 - (44.6108736 - 64J) &> D(u), \\ D(u) &= 43.52H - 7.3984R + 1.44976u - 17.7762266. \end{aligned} \tag{D19correlated}$$

This is the complete nonzero head/centre cross calculation. The global parabola (D19headparabola) gives

$$D(u) > .1073 + A_D z + 25.024z^2, \quad A_D = 14.795893333\ldots - 5.44s, \quad 0 < A_D < 1.471.$$

The same two positive square bounds for $\sqrt{6}$ give the constant and linear seals; the quadratic coefficient is exactly 25.024. Therefore

$$D(u) > .1073 - \frac{A_D^2}{4(25.024)} > .1073 - .022 = .0853.$$

The exact last check is $1.471^2 = 2.163841 < .022(100.096) = 2.202112$. This ENTIRE finite-at-least-nine row violates the actual completed square; it need not clear the coarser standalone test41.926.

For a dotted far helper with $1 < \gamma \le 10/9$, put $a = (2\gamma)^{-1} \in [.45, .5)$. Its same actual core gives

$$\begin{aligned} V &= P_{16} + aP_1 + (1 - 2a^2)e_{17} + 2a^2 e_{18}, \\ G &= \sqrt{17 + 2a - a^2} - a\sqrt{2 - 8a^2}, \quad B = aw_1, \quad R_* \le 16.75. \end{aligned}$$

Since $w_1 < .633$, $G - aw_1$ increases on the whole interval: the first derivative term is positive, and $(16a^2 - 2)/\sqrt{2 - 8a^2} - w_1 \ge 1.24/.617 - .633 > 0$. At .45, $G > 3.929375$, as proved in the shared centre lemma. The generalized test is thus greater than

$$S_{\text{dot}} = 46.12 - 7.2w_1.$$

Pointwise it dominates S_9, since

$$S_{\text{dot}} - S_9 = -.343911 + 1.328w_1 > -.343911 + 1.328(.462946) > 0.$$

The correlated contradiction therefore covers every small dotted far helper without substituting unrelated maxima.

Every $\gamma \ge 11/8$ is directly .50-strong. The ONLY centre left is the literal interval $10/9 < \gamma < 11/8$. The head-independent actual-q proof in Sections B.14.9–B.14.11 gives a flat-through-seventeen pressure with $16G - R > 43.3072875 > 41.926$. Its 1/8 terminal payment is still available and its residual has zero cross with the head-paid support. The original slack $0 < \tau \le 1$, both actual fork orientations and actual other endpoint are retained; the adjacent q is never called farthest. Equation (D19generalpay) contradicts the ONE original norm. Every $g > 1$ dotted-head parameter is excluded.

B.14.21 The whole classical alternatives at rank nineteen

Proposition B.14.3. *At the original farthest vertex of a compact hyperbolic Coxeter* 19*-polytope the finite forest cannot be a whole* D_{19}*.*

Proof. Section B.14.1 first excludes every nonisolated head, then isolated own-zero/three/four/six and parallel. Its remaining head list is disjoint and exhaustive: native five, seven, eight, nine, finite label at least ten, or dotted. The independent OLD five floor in Section B.14.2 relies on none of these later exclusions.

Proposition B.14.1 excludes native five. Sections B.14.14–B.14.15 exclude native seven in both actual states at five. Section B.14.16 excludes native eight in both actual states at six. Section B.14.17 excludes native nine while retaining every weak-six omission. Section B.14.18 proves its own fresh late coverage and excludes the entire finite-label tail. Sections B.14.19–B.14.20 exclude the entire continuous dotted interval with correlated payments. Each branch uses actual walls at this SAME rank, one ORIGINAL farthest vector, every original-wall check and legal individual prices. No face is presumed Coxeter, no adjacent vertex presumed farthest, and no arithmeticity or stronger-bound census enters. The complete head partition is exhausted. □

Together with Propositions B.12.1 and B.13.1, this excludes all three whole classical alternatives at nineteen. Proper forests with spectators and the necessary forest cover have separate proofs; this appendix does not replace those obligations. Its conclusions enter that SAME-rank composition, not a rank-decrement principle.

B.14.22 The complete higher-dimensional conjunction

Proof of Theorem 12.1. Suppose a compact Coxeter polytope has dimension at least nineteen. The published compact ceiling puts its dimension at most twenty-nine [8, 9]. Proposition B.11.5 excludes all dimensions from twenty-four through twenty-nine. Proposition B.11.12 excludes twenty-three and twenty-two. Propositions B.11.20 and B.11.28 exclude twenty-one and twenty. At dimension nineteen, Proposition B.11.35 leaves only the whole forests A_{19}, B_{19}, D_{19} at the same original farthest vertex. Propositions B.12.1, B.13.1 and B.14.3 exclude them all. Every dimension in the required interval is therefore impossible. Each cover is proved at its own actual rank, and each whole-component consumer is used only after that cover has supplied its hypothesis. No dimension is inferred by passing to a face, and no arithmetic result enters the conjunction. □

B.15 Proof dependencies and reproducibility

The proof has one common geometric foundation and three genuine branch points. Ordinary components and actual defects are treated in Appendix B.1. The chain and prefix arguments use that foundation but have different coordinate models and terminal classifications. Their component bounds are combined only after an actual-rank forest cover has fixed the allowed spectators.

At exact eighteen the complete cover in Section 12 uses the whole-A_{18} and whole-B_{18} proofs in Appendices B.7 and B.8. Appendix B.9 supplies the remaining head and fork cases; the main text closes the final dotted-head near/far case. The independent higher tail is proved in Appendices B.11–B.14, then joined in Section B.14.22. There is no dependency on the arithmetic companion's dimension bound.

B.16 External mathematical inputs

The external inputs are spherical and affine Coxeter classification, acute-angled compact-polytope simplicity and wall acuteness, the Lannér order bound of five, and the published compact dimension bound 29 for the tail beyond 29 [8, 9]. The actual-wall sign and positive projection conventions are those of [9, p. 42, §2.1]; all project-specific component and terminal statements are proved internally. The elementary real-cyclotomic degree formula $[\mathbb{Q}(\cos(\pi/m)) : \mathbb{Q}] = \phi(2m)/2$ for $m \geq 3$ is a standard algebraic input in the literal native-cosine analysis; it imposes no bound on a polytope ground field. Arithmeticity, field restrictions, integral dotted products, arbitrary Coxeter-face inheritance, and abstract Gram completions being realizable are not premises.

Part VI
Proof appendices: compact arithmetic polytopes

C.1 A strengthened ordinary dihedral comparison

This section supplies the ordinary rank-two comparison needed beside D_{14}. Its component theorem is general real geometry. Arithmeticity enters only when the already established arithmetic D_{14} prefix bound is used in the final corollary. Throughout, p is the ORIGINAL vertex farthest from one strict interior point. The theorem is not asserted at an arbitrary neighbour with a weakened comparison factor.

Proposition C.1.1. *Let a compact hyperbolic Coxeter polytope have an actual farthest vertex p, and write $z = E(p - x)$, $\|x\| < 1$. Let one WHOLE component of the incident finite diagram be $I_2(m)$, $m \geq 4$, and let e be the squared norm of the projection of this SAME x onto that component. If $e \leq 3/19$, then $m = 7$ and one actual opposite wall closes both component directions, has a single old contact of label three there, and has no contact with any other old component. This is precisely the actual O_7 profile. Consequently every ordinary $I_2(m)$, $m \geq 4$, has energy strictly greater than $3/19$, with arbitrary finite spectators.*

Proof. Use outward unit incident roots, finite Gram B, inverse $C = B^{-1} \geq 0$, duals ω_i, and unit inward dual directions $-u_i = -\omega_i/\sqrt{C_{ii}}$. For an actual outgoing edge of length L_i, maximality at the original p gives

$$x \cdot u_i \geq \tau_i = \tanh(L_i/2) > 0, \qquad r_i = \tanh L_i = \frac{2\tau_i}{1 + \tau_i^2}.$$

The old dihedral inverse is

$$C_I = d \begin{pmatrix} 1 & k \\ k & 1 \end{pmatrix}, \qquad k = \cos(\pi/m) \geq 1/\sqrt{2}, \quad d = (1 - k^2)^{-1}.$$

Thus each $\tau_i^2 \leq e$, and the increasing function $4t/(1 + t)^2$ on $0 \leq t < 1$ gives

$$r_i^2 \leq 57/121 =: r. \tag{Dih1}$$

We recall the actual-wall calculation forcing a common opposite. For an outside actual wall $c = Ap - v$, let $\beta_j = -c \cdot n_j \geq 0$ and put

$$T_i(c) = \frac{(C\beta)_i}{A\sqrt{C_{ii}}}.$$

For the actual opposite c_i of edge i, $T_i(c_i) = 1/r_i$. The matrix $C - C_{\cdot j}C_{j\cdot}/C_{jj}$ is entrywise nonnegative: its nonzero block is the inverse of the positive principal Gram obtained by deleting j. Actual acuteness between two distinct opposites therefore gives $T_j(c_i) \leq r_j$, while that same nonnegative matrix gives $T_j(c_i) \geq k/r_i$ within this old component. Hence distinct opposites imply $r_i r_j \geq k$. But (Dih1) gives $r_i r_j \leq 57/121 < 1/\sqrt{2} \leq k$. One actual wall c therefore closes BOTH directions.

Write its full decomposition as

$$c = Ap - v_I - v_S, \quad h = A^2 > 0, \quad b = -c \cdot n_1 \geq 0, \quad g = -c \cdot n_2 \geq 0, \quad Q = \|v_S\|^2 \geq 0.$$

Here S is the entire orthogonal spectator forest. Every positive contact is a native magnitude at least $1/2$; none of its finite, parallel or dotted possibilities is omitted before the actual endpoint constraints are imposed. At least one of b, g is positive.

Suppose both are positive. At the actual endpoint releasing direction one, the retained white Gram has residual $\delta = 1 - Q - g^2 > 0$. Its actual edge formula is

$$\cosh L_1 = \frac{\sqrt{d}(b + kg)}{\sqrt{1 - Q - g^2}} \geq \frac{\sqrt{d}(1 + k)}{\sqrt{3}} = \frac{\cot(\pi/(2m))}{\sqrt{3}} \geq \frac{1 + \sqrt{2}}{\sqrt{3}} > \frac{11}{8}.$$

The last comparison is $128\sqrt{2} > 171$, whose positive squared margin is $32768 - 29241$. It follows that

$$e \geq \tanh^2(L_1/2) > \frac{11/8 - 1}{11/8 + 1} = \frac{3}{19},$$

a contradiction. Hence exactly one old contact $\beta > 0$ remains.

We need one whole-spectator fact at small time. If $h < 1/4$, then c cannot contact any foreign component. Indeed a positive foreign contact gives a normalized derivative

$$\frac{(C_S\beta)_j}{\sqrt{(C_S)_{jj}}} \geq 1/2.$$

This uses the full spectator inverse, with nonnegative entries and diagonal at least one. Its ratio to $\sqrt{h}$ exceeds one. The actual-wall shadow inequality therefore forces this SAME c to be the opposite of that foreign edge as well.

For clarity, the required orthogonal obstruction is specific to the original farthest vertex. Suppose a wall closed orthogonal dual directions u_i, u_j. Clear it positively off all other incident roots W:

$$c_F = c + \sum_{s \in W} \alpha_s n_s, \qquad \alpha = B_W^{-1}(-c \cdot n_W) \geq 0.$$

Strict interiority gives $z \cdot c_F < 0$. In the retained Lorentz three-space its exact form is

$$c_F = Ap - (A/r_i)u_i - (A/r_j)u_j.$$

Therefore

$$\frac{x \cdot u_i}{r_i} + \frac{x \cdot u_j}{r_j} < 1.$$

Original maximality gives the opposite bound $\tau_i/r_i + \tau_j/r_j = 1 + (\tau_i^2 + \tau_j^2)/2 > 1$. This proves the obstruction, including arbitrary contacts in W. Thus $h < 1/4$ really forces $Q = 0$.

Put $q = d\beta^2$. The smaller normalized derivative square is $k^2 q$. The common-wall edge formula and (Dih1) imply

$$q - 1 \leq h = q + Q - 1 \leq rk^2 q.$$

Substituting $k^2 = 1 - 1/d$ and $r = 57/121$ gives the exact strengthened restriction

$$\beta^2(64d + 57) \leq 121. \tag{Dih2}$$

This is derived at the $3/19$ threshold; it is not the older $1/7$ inequality with an unchanged coefficient.

Since $\beta^2 \geq 1/4$, this forces $d \leq 427/64$. For every old $m \geq 8$, $d \geq 4 + 2\sqrt{2} > 34/5 > 427/64$, impossible. The complete remaining old alphabet is $4, 5, 6, 7$. At old $m = 5$, $d = 2 + 2/\sqrt{5} > 185/64$: equivalently $128 > 57\sqrt{5}$, with squared margin139. Since d increases with m, if $\beta \geq 1/\sqrt{2}$ at any of old $5, 6, 7$, (Dih2) would give the opposite $d \leq 185/64$. Thus all three rows have exactly $\beta = 1/2$. Every larger finite contact, parallel and dotted contact has been excluded by the native jump, rather than a sampled label list.

At old four, $d = 2$ and $\beta^2 \leq 121/185 < \cos^2(\pi/5)$. The strict comparison is $413 < 185\sqrt{5}$, with squared margin556. Only contacts three and four remain, giving $q = 1/2$ or 1. Consequently $0 < h \leq 57/242 < 1/4$. The small-time argument just proved gives $Q = 0$, whereas $h = q - 1 \leq 0$, a contradiction.

At old five only $\beta = 1/2$ remains. Since $d < 3$, $q < 3/4$ and $0 < h \leq r(d-1)/4 < 57/242 < 1/4$. Again $Q = 0$, and $h = q - 1 < 0$.

At old six, $\beta = 1/2$, $d = 4$, $q = 1$. If $Q = 0$, then $h = 0$, impossible. For any $Q > 0$, the actual positive sum

$$\nu = c + 2n_1 + \sqrt{3}n_2$$

is a nonzero future null: all three selected products vanish. The coefficient order is for OLD6/NEW3, since $c \cdot n_1 = -1/2$, $n_1 \cdot n_2 = -\sqrt{3}/2$, $c \cdot n_2 = 0$. Every other actual wall pairs nonpositively termwise with ν. It is a feasible ideal ray, contrary to compactness. No foreign contact was discarded in this affine exclusion.

At old seven only $\beta = 1/2$ remains, and

$$0 \leq Q = 1 + h - d/4 \leq \frac{427 - 64d}{484} < \frac{107}{484} < \frac{1}{4},$$

using $d_7 > 5$. Every positive spectator cost is at least $1/4$, by inverse positivity and the native contact lower bound. Thus $Q = 0$. We have exactly the ACTUAL isolated-new-three wall in old $I_2(7)$, closing both directions: the stated O_7. Its time square is $d_7/4 - 1 \in (0, 1/3)$; the bounds $5 < d_7 < 16/3$ follow from $6/5 < 2\cos(2\pi/7) < 5/4$. Its separate inherited energy floor $> 9/100$ is unchanged.

All weak inequalities included $e = 3/19$, so no ordinary equality case is omitted. The proof has retained every finite spectator component, all its contacts and the same original x. No arithmetic hypothesis or weakened-neighbour comparison was used. □

Corollary C.1.2. *At the original farthest vertex in arithmetic dimension sixteen, a whole D_{14} cannot have an ordinary irreducible rank-two spectator.*

Proof. The fresh true-rank-14 argument in the proof of Proposition 22.1 gives

$$e_D > .84816 = \frac{5301}{6250}$$

for its actual whole rank-two spectator. Its bootstrap uses only the independently available spectator floor $> .09$, so it does not depend on the present strengthening. For an ordinary $I_2(m)$, $m \geq 4$, the proposition gives

$$\|x\|^2 = e_D + e_S > \frac{5301}{6250} + \frac{3}{19} = 1 + \frac{719}{118750} > 1.$$

For ordinary A_2, use its geometric companion bound $319/2000$, which exceeds $3/19$ by $61/38000$. Both contradict $\|x\| < 1$. The actual typed N_7/O_7 alternatives retain their separate costs and arguments; they have not been charged as ordinary. □

C.2 Arithmetic chain windows and simultaneous actual feet

The endpoint arguments below share one arithmetic chain calculation. We prove it before specializing to the whole A_{16} and A_{17} cases. The geometric finite-white classification is the one in the general companion; the arithmetic restrictions and the simultaneous treatment of isolated feet are given here. No component dimension is obtained by deleting a node from a previous ambient theorem.

Let $N = 16, 17, 18$, let the original incident forest contain the WHOLE component A_{N-1}, and retain all finite spectators. Use its zero-sum $\mathbb{R}^N$ frame and unit old roots $n_i = (e_i - e_{i+1})/\sqrt{2}$. Then

$$D_i = \omega_i^2 = \frac{2i(N-i)}{N}, \qquad C_{ij} = \omega_i \cdot \omega_j = \frac{2i(N-j)}{N} \quad (i \le j).$$

All heights $h_i = x_A \cdot \omega_i$ concern the SAME original farthest vector. Set

$$\ell_{16} = .601, \qquad \ell_{17} = \ell_{18} = .6.$$

C.2.1 The complete central inventory at the new losses

For $5 \le i \le N-5$, call an index weak when $h_i \le \sqrt{D_i} - \ell_N$. We claim that every central constraint is either strong, an isolated-within-A cap with its actual cleared parameter $0 < u < 1$, or a pure golden end-helper. The last alternative can occur at at most ONE central index.

First establish the full inventory, without assuming a contrary component energy. Deleting i retains two old paths of ranks at least four. Their complete one-node finite extensions are isolation; one admissible end, D- or E-position three; one terminal four; or two endpoint threes, one in each path. Two contacts in a single path make a cycle. A connected two-path completion has rank $N - 1 > 8$, so finite classification forces the ordinary path bridge, with no extra finite foreign arm.

The zero-own repair depends only on side order $m \ge 5$. At a far-end three the time square is less than its full positive foreign cost; the next-to-far three has foreign remainder $< 2/m$ and the same obstruction. If such a positive foreign contact existed, its actual shadow would force the same opposite to close an orthogonal component, contrary to the original-farthest obstruction proved above. Terminal four has foreign remainder $< 1/m \le 1/5$, so no positive spectator cost, and then negative time. The other zero-own D/E, internal-four and two-white roles have their actual positive affine stars or cycles. Their time is positive through the actual cap and every omitted wall product is nonpositive. Own contact is therefore positive.

Write $q = \sqrt{D_i}$. Here $5/2 < q \le 3$. The actual edge ratio needed for a loss ℓ is

$$R_\ell(q) = \frac{1}{\ell} - \frac{q-\ell}{q(2q-\ell)}.$$

It increases with q and decreases with ℓ on this rectangle. Thus

$$R_\ell(q) \le R_{.6}(3) = 41/27 < 761/500.$$

The finite-white residual formulas give the following ordinary lower ratios, with side order at least five: near-three/own at least five has $(101/125+2/5)/\sqrt{3/5}$; a D-position has $(101/125+1/5)/\sqrt{2/5}$; the E, terminal-four and two-end-three rows are stronger. All exceed $761/500$. The smaller owns have the same actual affine principals. For the two exceptional roles WITH positive foreign cost, the complete clearings give ratio $> 227/(25\sqrt{35}) > 761/500$, or ratio square $> 121/35$. These estimates retain the ENTIRE contacted spectator cost. Hence no additional ordinary nonisolated row is weak.

Unlike the older 3/2 test, the present ratio threshold retains near-own-four also at side order seven. Up to reflection all remaining such profiles are

$$v = \omega_i/\sqrt{2} + \omega_{i-1}/2, \qquad i = 5, 6, 7.$$

The simple-three helper normalizes the cap to square two over the actual field, with old product -1; its own product is $-\sqrt{2}$, so $\sqrt{2} \in K$. At the embedding reversing $\sqrt{2}$, the time numerators at $N = 16$ are

$$126 - 88\sqrt{2}, \quad 143 - 100\sqrt{2}, \quad 154 - 108\sqrt{2}.$$

They are positive since $\sqrt{2} < 17/12$. Their increments with N are $(i-1)(3-2\sqrt{2}) > 0$. The conjugate new-root Schur complement is the negative of this numerator divided by N, contradicting positive definiteness. This removes all three side orders and both orientations.

The remaining nonisolated possibility is pure:

$$c = Ap - \gamma\omega_i - \omega_e/2, \qquad e = 1 \text{ or } N-1, \quad Q = 0,$$

with own five, finite own at least seven, or dotted. The exact helper-side identity is

$$\frac{(\gamma + 1/(2m))^2}{(m+1)/(2m)} - (4\gamma - 2) = \frac{(\gamma - 1 - 1/(2m))^2}{(m+1)/(2m)} \geq 0.$$

The ratio threshold and native-five gap therefore give

$$202/125 < t = 2\gamma < 3139/1458 < 54/25. \tag{Astrip}$$

For isolated-within-A contact, retain the whole foreign projection v_S and write

$$c = Ap - \gamma\omega_i - v_S, \quad Q = \|v_S\|^2, \quad u = (1-Q)/\gamma^2, \quad A^2 = \gamma^2(D_i - u).$$

Here $Q < 1$ by finite-white positivity. Before splitting Q, owns three, four, six and parallel are excluded for EVERY foreign cost. Own three has the actual affine E_7 principal with arms $1, 3, 3$; own four has the affine $4, 3, 3$ star; own six has the affine $6, 3$ path. Each positive affine kernel has positive time through the cap, and every omitted original wall, including every foreign wall, has nonpositive product termwise. Parallel similarly has the feasible positive null pair. Compactness forbids all these nulls. Thus $\gamma \geq \varphi/2$ before any foreign-cost estimate. When $Q > 0, u > 1$, the actual positive clearing has loss at most

$$\frac{1}{\sqrt{D_i} + \sqrt{D_i - 1}} + \sqrt{u - 1} < .2 + .383 < \ell_N.$$

This uses $Q \geq 1/4$, $\gamma \geq \varphi/2$, and $u - 1 \leq 5 - 3\varphi < .383^2$. At $u = 1$ the positive cleared vector is a forbidden feasible future null. The interval $0 < u < 1$ is RETAINED, including $Q > 0$; we do not charge a strong floor there.

For the remaining PURE isolated estimates, own five has the actual H_4 projected-point loss $< 25/76 + 5/24 < .6$. For own seven, the increasing cubic $8a^3 - 4a^2 - 4a + 1$ gives $.9 < a < .902$. The actual positive A_2 core gives

$$h_i > \sqrt{D_i - a^{-2}} - \sqrt{4/3 - a^{-2}},$$

and hence loss

$$< \frac{100/81}{2.6 + 2.37} + \sqrt{17/150} < .25 + .337 = .587.$$

Pure own eight is arithmetically impossible: $4\gamma^2 = 2+\sqrt{2}$, and its opposite conjugate would require $D_i(2-\sqrt{2}) < 4$, whereas $D_i \geq 55/8 > 4+2\sqrt{2}$. Every embedding of this minimal field extends to the actual totally real field, so this contradiction applies there. For all finite owns at least nine, the exact singleton finite projection gives $h_i > \sqrt{D_i - 1} - \tan(\pi/m)$. Since $\tan(\pi/9) < 1386/3727 < 2/5$, the loss is $< .6$. This covers the infinite finite-own tail, not a finite label sample. All positive finite and negative-norm projections just used are positive sums of ACTUAL roots; their selected signs, positive time and all omitted walls are retained as in the companion's actual-point lemmas.

C.2.2 Every pure weak helper is golden over the actual field

Normalize old A roots to square two. If r is a K-root on the actual cap line and its simple-three helper is a_e of square two, then $r^2 = 2(r, a_e)^2$. Rescaling by $-(r, a_e)^{-1} \in K$ therefore makes the cap square two with helper product -1. Its own product is $-t$. The path cycle through old simple-three bonds makes t integral. This compares only connected root lines; no disconnected norm squareclasses are equated.

Every nonidentity embedding of $\mathbb{Q}(t)$ extends to K. Strict conjugate positivity of the full old-component Schur complement gives

$$D_i t_\sigma^2 + 2C_{ie} t_\sigma + D_e < 4.$$

After reversing $i \leq N/2$, the cheaper costs for negative and positive t_σ are respectively

$$Q_-(s) = 2is^2 + 4s + 2 - 2(is+1)^2/N, \qquad Q_+(s) = 2is^2 + 2 - 2(is-1)^2/N.$$

At the smallest $N = 16, i = 5$, their values at $-4/5, 12/25$ exceed four by $3/40, 59/1000$, respectively. The costs at those endpoints increase with i and N; their values at zero are below four. Convexity of the quadratics therefore gives the full strict strip

$$-4/5 < t_\sigma < 12/25. \tag{Agoldenstrip}$$

This covers EVERY nonphysical conjugate, not chosen ones.

If the degree is at least five, the nonzero integral norm has magnitude $< (54/25)(4/5)^4 < 1$. At degree four or three its magnitude is below two. Any positive small conjugate would make the magnitude below one, so all small conjugates are negative and their absolute product is $1/t$. At degree four the integral trace lies in $(-1, 0)$: its lower bound is $t - 12/5 > -1$, and its upper bound $t - 3t^{-1/3} < 0$ follows from $(54/25)^4 < 27$. At degree three the trace lies in $(0, 1)$, since $t - 8/5 > 0$ and $t - 2/\sqrt{t} < 1$; the latter follows from $(54/25)(29/25)^2 < 4$. Both degrees are impossible. At degree two the integral trace is one or two. Trace two makes the nonzero norm $t(2-t)$ lie in $(-1, 1)$. Trace one forces $t < 9/5$ and $t(1-t) \in (-2, 0)$; the norm is -1, and $t = \varphi$. Degree one leaves only $t = 2$, the actual parallel pair with a feasible future null. Thus every pure weak helper has $\gamma = \varphi/2$.

The old component and this foreign-free cap have a physically Lorentzian Gram over $\mathbb{Q}(\varphi)$. If K were a proper extension, a nonphysical real embedding fixing that subfield would preserve this indefinite principal Gram, impossible. Consequently the helper forces the ACTUAL field $K = \mathbb{Q}(\varphi)$, rather than a convenient overfield.

C.2.3 The complete golden packing calculation

For a golden cap reversed to $5 \leq i \leq N/2$, its time numerator $M = 2NA^2$ is

$$M_i^{\text{near}} = i(N-i) - (N+1) + (i+2)(N-i)\varphi, \quad M_i^{\text{far}} = i(N-i) - (N+1) + i(N-i+2)\varphi.$$

These formulas retain BOTH helper ends. At $N = 18$, conjugate positivity excludes the midpoint nine and opposite-end indices six through nine. The endpoint costs are $(173 - 84\varphi)/36$ and $(197 - 99\varphi)/36$, both greater than one, and the opposite-end cost is concave in i. At $N = 17$, only far seven is excluded, by $(170 - 84\varphi)/34 > 1$ using $\varphi < 34/21$. At $N = 16$ retain every orientation. The resulting COMPLETE necessary tables are

N	type and M	N(M)
16	near 5 : $38 + 77\varphi$	-1559
	near 6 : $43 + 80\varphi$	-1111
	near 7 : $46 + 81\varphi$	-719
	near 8 : $47 + 80\varphi$	-431
	far 5 : $38 + 65\varphi$	-311
	far 6 : $43 + 72\varphi$	-239
	far 7 : $46 + 77\varphi$	-271
17	near 5 : $42 + 84\varphi$	-1764
	near 6 : $48 + 88\varphi$	-1216
	near 7 : $52 + 90\varphi$	-716
	near 8 : $54 + 90\varphi$	-324
	far 5 : $42 + 70\varphi$	-196
	far 6 : $48 + 78\varphi$	-36
	far 8 : $54 + 88\varphi$	-76
18	near 5 : $46 + 91\varphi$	-1979
	near 6 : $53 + 96\varphi$	-1319
	near 7 : $58 + 99\varphi$	-695
	near 8 : $61 + 100\varphi$	-179
	far 5 : $46 + 75\varphi$	-59.

At $N = 16$, near and far eight are reflections at the same central index, so one row suffices.

Why do these finite tables restrict coexistence? Subtracting a cap's K-valued projection onto the FULL old forest leaves a nonzero vector on the SAME one-dimensional time line over K. The caps have no foreign component. Thus any two time coefficients have ratio in K; their M-ratio must be a square IN K, and its rational norm must be a rational square. This argument retains all spectators.

At $N = 16$ and 18, the positive norm magnitudes in each table are pairwise coprime nonsquares; their ratios are therefore not rational squares. These are direct small integer checks of the displayed complete tables. At $N = 17$, different norm squareclasses $-1, -19, -179$ cannot mix. In class -1, write the four entries as $42\varphi^3, 18\varphi^5, 14\varphi^5, 6\varphi^7$. After squares are removed the rational coefficient classes are $42, 2, 14, 6$. Every pair ratio has an odd valuation at three or seven; none is a rational square or five times a rational square. Those are exactly the positive rational squares in $\mathbb{Q}(\sqrt{5})$, as expansion of $(a + b\sqrt{5})^2$ shows. For the repeated -19 class,

$$\frac{M_{\text{far8}}}{M_{\text{near6}}} = \frac{25 + 33\varphi}{76} = \frac{83 + 33\sqrt{5}}{152}$$

has norm $1/16$. A square would require $a^2 = 121/304$ or $45/304$, both with odd denominator valuation at nineteen. Thus different types cannot coexist, including the repeated norm classes where a norm-only test would be insufficient.

At a fixed index the actual opposite is unique. Reflected copies of the same noncentral type would have

$$-2c_L \cdot c_R = \frac{(N - 2i)(i + (i + 2)\varphi) - (N + 2)}{N}$$

for near helpers, and the same formula with $i-2$ replacing $i+2$ for far helpers. Their simple-three helper bonds and old path make this parameter integral by the invariant cyclic criterion, even for a dotted mutual. But the constant numerators for $i=5,6,7$ at $N=16$ are $12,6,-4$; for $i=5,6,7,8$ at 17 they are $16,11,2,-11$; and at 18 they are $20,16,8,-4$. The far rows have the same respective constants. None is divisible by N, contrary to $\mathcal{O}_{\mathbb{Q}(\varphi)}=\mathbb{Z}[\varphi]$. The central $N=16$ row cannot occupy two different indices. Hence at most ONE central index has a golden helper. This is an all-native-five packing statement, independent of which of those caps are actually weak at the new loss.

C.2.4 Sum every retained isolated foot in one reference

The intervals $0<u<1$ retained earlier cannot be declared strong by the older .61 estimate. For such a cap clear the entire spectator projection by its positive actual-root combination. The actual vector

$$\eta = c + v_S + \gamma u n_i = \gamma\{\sqrt{D_i-u}\,p - (\omega_i - u n_i)\}$$

has square $-\gamma^2 u(1-u)$, positive time, zero contacted spectator products, negative or zero c, n_i products, and all other actual products nonpositive. Its normalization is a genuine point of the whole chamber. Original maximality gives

$$x_A \cdot (\omega_i - u n_i) \geq \sqrt{D_i-u} - \sqrt{u(1-u)}.$$

If $\lambda_i = \sqrt{D_i} - \sqrt{D_i-u} + \sqrt{u(1-u)}$ denotes its loss, then

$$\lambda_i \leq a_i u + \sqrt{u(1-u)}, \qquad a_i = \frac{1}{\sqrt{D_i}+\sqrt{D_i-1}}.$$

The rational constants needed below are

N	$a_i < a$	positive reference coefficient $\beta_i \geq b$	ℓ
16	.1982	.022	.601
17	.1955	.0205	.6
18	.194	.018	.6.

The a bounds are positive-square comparisons at $D_{\min} = 55/8, 120/17, 65/9$. For every $0<u<1$,

$$au + \sqrt{u(1-u)} - \frac{b}{2}(2u-u^2) < \ell. \tag{Aabsorb}$$

For $u \leq 1/2$ use $a/2+1/2$, giving $.5991, .59775, .597$. For $u \geq 1/2$, use $2u-u^2 \geq 3/4$ and $\max(au+\sqrt{u(1-u)}) = (a+\sqrt{1+a^2})/2$. Radical upper bounds $1.0195, 1.019, 1.019$ give respectively $.6006, .5995625, .59975$, all strictly below their ℓ.

Let H be a concave reference with available nonnegative coefficients β_i, positive ones at least b:

$$w_j = (H_j - H_{j-1})/\sqrt{2}, \quad \beta_i = w \cdot n_i, \quad w = \sum_i \beta_i \omega_i, \quad E_H = \|w\|^2 = \sum_i \beta_i H_i.$$

Require $H_i + \ell < \sqrt{D_i}$ at its positive central coefficients. Omit the possible golden index. If S is the set of retained isolated indices with $\beta_i > 0$, define ONE vector

$$Z = w - \sum_{i \in S} \beta_i u_i n_i.$$

Sum all actual point pressures BEFORE estimating its norm:

$$x_A \cdot Z \geq E_H - \sum_{i \in S} \beta_i(\lambda_i - \ell).$$

Because $w \cdot n_i = \beta_i$ and $n_i \cdot n_j \leq 0$ for $i \neq j$,

$$\|Z\|^2 \leq E_H - \sum_{i \in S} \beta_i^2(2u_i - u_i^2).$$

It follows that

$$e_A \geq 2x_A \cdot Z - \|Z\|^2 \geq E_H + \sum_{i \in S} \beta_i\{\beta_i(2u_i - u_i^2) - 2(\lambda_i - \ell)\} > E_H. \qquad \text{(Asummed)}$$

Strictness follows from (Aabsorb) if $S \neq \varnothing$, and from the strict available constraints otherwise. Adjacent isolated indices and positive foreign costs are allowed. No nonadjacency is assumed for this new isolated set, and no independently optimized energies are added.

The references used by the endpoint arguments are

N	$H_5, H_6, \ldots$ to midpoint	E_H	maximum internal golden omission loss
16	$2.021, 2.137, 2.205, 2.227$	.8354522	.000576
17	$2.056, 2.186, 2.269, 2.310$	.8708972	.00055225
18	$2.087, 2.228, 2.324, 2.381, 2.399$	.9037838	.00050625.

Reflect $H_{N-i} = H_i$, use linear end tails, and at $N = 17$ take $H_9 = H_8$. The positive curvatures have exactly the lower bounds b above. All floor inequalities are positive-square comparisons: the new heights are the old .61 heights shifted by $.61 - \ell$, so $H_i + \ell$ is unchanged. A single internal omission sets its coefficient zero by neighbour averaging and increases the adjacent coefficients. An end omission replaces the first tail by the chord 0–6, or its reflection. These are changes in ONE reference. In particular the full and omitted references preserve the summed-foot argument whenever their positive central coefficients have the displayed lower bounds; subsequent endpoint grafts must check that condition again rather than assume it.

C.2.5 A golden end hole pays for its own endpoint

For later proper-component consumers we record how the exceptional end omission is closed. Suppose its central index is five, reversing if necessary, and put $\gamma = \varphi/2 < .81$. For either helper end e, with its adjacent old node j, the actual finite H_4 correction is

$$b_c = \gamma n_5 + \tfrac{2}{3} n_e + \tfrac{1}{3} n_j, \qquad \eta = c + b_c, \quad \eta^2 = \tfrac{2}{3} - \gamma^2 < .111^2.$$

The actual positive finite projected point has pure spatial displacement $v - b_c$. Its normalization and original maximality give $x \cdot (v - b_c) \geq A - .111$. Since $x \cdot b_c > 0$, the helper-end signed coordinate satisfies

$$x_{\text{end}} > \sqrt{2}(A - .111 - \gamma h_5) > .49.$$

Both orientations are covered: the smaller opposite-end time squares, at $N = 16, 17, 18$, are

$$(38 + 65\varphi)/32, \quad (42 + 70\varphi)/34, \quad (46 + 75\varphi)/36.$$

They give $A > 2.115, 2.136, 2.155$. Weakness gives $h_5 < 2.022, 2.057, 2.088$. Using $\sqrt{2} > 1.4$, the corresponding signed-coordinate lower bounds are $.512652, .502362, .493808$. The nearer-end time is

larger. Here the signed coordinate is x_1 at helper one and $-x_N$ at helper $N-1$. No strong floor at the missing index five has been inferred.

Use the chord 0–6 at the omitted end. At the helper end the resulting reference is flat on $L=6$ coordinates in the near case and $L=5$ in the far case, with signed mean $.25 < b_0 < .3$. Replace that block by

$$.49, \quad \frac{Lb_0 - .49}{L-1}, \dots, \frac{Lb_0 - .49}{L-1}.$$

Its sum is unchanged. The latter coordinates lie in $(.19, .262)$; the next old coordinate is below .101. Thus concavity persists, the new central boundary coefficient is $> .062 > b$, and all other positive central coefficients keep their previous lower bound. The new endpoint coefficient is positive and is paid by the actual endpoint pressure just proved. The one-reference energy increases by

$$\frac{L}{L-1}(.49 - b_0)^2 > .04332.$$

The pre-graft chord energy is exactly

$$E_H - \frac{1}{2}\{H_5^2/5 + (H_6 - H_5)^2 - H_6^2/6\}.$$

At $N = 17, 18$ it is respectively 12569249/15000000 and 13079276/15000000; at $N = 16$ it exceeds .8008. These direct rational formulas give, after the graft and simultaneous absorption (Asummed), the conservative component bounds

$$e_{A_{15}} > .8348, \qquad e_{A_{16}} > .8703, \qquad e_{A_{17}} > .9032.$$

If the single golden hole is internal instead, subtract the internal losses in the reference table: the energies are $> .8348762, .87034495, .90327755$. If there is no hole they are stronger. The three conclusions therefore include every omission location, all retained isolated feet, and arbitrary finite spectators. They are the shared component estimates; no historical ambient-dimension consumer has been imported.

C.3 The actual whole-A_{17} endpoint reduction

We now prove the geometric and arithmetic inclusion needed before using the recorded finite A_{17} propositions. The old incident diagram is WHOLE A_{17}, so $N = 18$ and there is no spectator. Keep the original p, z, x throughout. Use the $N = 18$ reference and simultaneous-foot argument of Appendix C.2, with energy $E_0 = .9037838$, central coefficients at least .018, and first/last five-coordinate signed means $.2951 < b < .2952$. The next signed coordinate is below .1. A possible single golden central hole is omitted before its pressure is estimated.

C.3.1 Actual endpoint lengths and the full finite-white list

Even with a golden end hole, the chord reference has energy

$$E_{\min} = \frac{13079276}{15000000} > .8719.$$

Each signed end block has length $L = 5$ or 6 and mean $.25 < b < .3$. If its actual endpoint coordinate were at least .63, replace the block by

$$.63, \quad (Lb - .63)/(L-1), \dots, (Lb - .63)/(L-1).$$

Its sum and every available central height are unchanged. The remaining coordinates exceed .155, and the following coordinate is $< .1$; hence the changed positive central coefficient is $> .055/\sqrt{2} > .018$. All other positive central coefficients still meet .018. Thus the SAME summed-foot proof applies to this ONE new reference, whose energy increases by

$$\frac{L}{L-1}(.63-b)^2 > \frac{6}{5}(.33)^2 = .13068.$$

This exceeds one after adding $E_{\min}$. Therefore both signed actual endpoint coordinates are $< .63$.

For either actual endpoint edge,

$$x_{\text{end}} \geq \sqrt{17/18}\sqrt{(C-1)/(C+1)}.$$

The bound $C \geq 5/2$ would give endpoint square $\geq 17/42 > .63^2$, impossible. If the ACTUAL field is nongolden, no central golden hole exists. Use the full reference and graft instead to $.575 = 23/40$. The remaining block coordinates exceed .225125; all changed central coefficients still exceed .018. The energy gain is $(5/4)(.575-.2952)^2 = .09786005$, giving total > 1.00164385. Since $C \geq 21/10$ would give endpoint square $\geq 187/558 > (23/40)^2$, we obtain

$$\text{always } C < 5/2, \qquad \text{outside } \mathbb{Q}(\varphi) : C < 21/10. \tag{A17C}$$

Release old node one. The retained white path is A_{16} on nodes two through seventeen. Finite one-node classification gives isolation; helper three at $j = 2, 3, 16, 17$; or helper four at $j = 2, 17$. A second contact would make a cycle; no exceptional finite rank-seventeen diagram contains that path. There is no foreign contact in this whole-diagram case. Parallel own always has its actual future-null pair.

For helper three put $t = 2\gamma$. Connected simple-three normalization gives a square-two cap, helper product -1, own product $-t$, and integral t. Its full Lorentz star is defined over $\mathbb{Q}(t)$; all-place definiteness gives ACTUAL $K = \mathbb{Q}(t)$. The unit time square and edge coshes are exactly

$$A^2 = \frac{17t^2 + 2(18-j)t + j(18-j) - 36}{36},$$

j	C
2	$(17\gamma + 8)/9$
3	$(17\gamma + 15/2)/(3\sqrt{2})$
16	$(17\gamma + 1)/(3\sqrt{2})$
17	$(17t + 1)/18.$

These are obtained by clearing ALL retained A_{16} roots. For helper four the two formulas are

$$j = 2 : \ C = (17\gamma + 8\sqrt{2})/3, \qquad j = 17 : \ C = (17\gamma + 1/\sqrt{2})/3.$$

The raw white residuals are $18/17$ at a helper-three endpoint, $4/17$ at a next-to-end three, and $1/17$ at a helper-four endpoint.

At helper $j = 3$, every positive own violates $C < 5/2$. Own zero has the ACTUAL feasible affine-E_8 null

$$2n_1 + 4n_2 + 6n_3 + 5n_4 + 4n_5 + 3n_6 + 2n_7 + n_8 + 3c.$$

Its future time is $3A > 0$, and all omitted products are nonpositive. At $j = 16$, own zero has $A^2 = -1/9$, own three has the actual all-three cycle, and own at least four violates $C < 5/2$. The helper-four rows fail already at own zero for $j = 2$ and own at least three for $j = 17$; the latter own-zero time is $-1/18$. At $j = 2$, own zero has negative time and $C < 5/2$ gives $\gamma < 29/34$, leaving owns three, four or five. Own three is an actual affine triangle. Own four generates $\mathbb{Q}(\sqrt{2})$, but its edge exceeds $21/10$, contrary to the nongolden bound. Only own five in the ACTUAL golden field can remain at this end.

C.3.2 The far-helper strip and the golden exceptions

For the far helper $j = 17$, positive time gives $t > 1$. Outside the golden field, (A17C) gives

$$1 < t < 184/85, \qquad -19/17 < t_\sigma < 1 \quad (\sigma \neq \mathrm{id}).$$

Set $f(s) = s(s^2-1)$. On this nonphysical interval $|f(s)| \leq 2/(3\sqrt{3})$: on $[-1,1]$ use $4/27 - q(1-q)^2 = (1-3q)^2(4-3q)/27$; on $[-19/17,-1]$ the bound $1368/4913$ is smaller. Physically $0 < f(t) < 8$. Degree at least four would give nonzero integer norm of magnitude $< 8[2/(3\sqrt{3})]^3 < 1$. At degree three the magnitude is $< 32/27 < 2$, so $t, t-1, t+1$ would all be units. The signs at one and minus one leave exactly

$$X^3 - X - 1, \quad X^3 - 2X^2 - X + 1, \quad X^3 + X^2 - 2X - 1, \quad X^3 - X^2 - 2X + 1.$$

The first has discriminant -23, so is not totally real. The second is negative at $11/5$, with increasing positive branch there, so its physical root exceeds $11/5 > 184/85$. The latter two have positive values $5347/4913, 2899/4913$ at $-19/17$, contrary to all three roots lying above that endpoint.

In degree two the integral trace is $0, 1, 2, 3$, and the nonzero norm is $\pm 1, \pm 2$. For trace zero the negative conjugate strip forces $1 < t < 19/17$, so $-t^2$ lies strictly between $-2, -1$. For trace one the norm $t(1-t)$ leaves only -1, giving $t = \varphi$, or -2, giving the rational $t = 2$. For trace two the norm $t(2-t)$ is strictly between $-1, 1$, with zero excluded in degree two. For trace three the conjugate strip requires $t > 2$, and $t(3-t)$ lies strictly between one and two. Thus no nongolden quadratic survives. In degree one only $t = 2$ is possible, the forbidden parallel. Hence any far helper forces the actual golden field.

In $K = \mathbb{Q}(\varphi)$, the unconditional bound gives $1 < t < 44/17$ and the same nonphysical strip. Writing $t = a + b\varphi$ in the FULL ring yields $0 < b < 2$; thus $b = 1$, $a = 0$ or one, and $a = 1$ violates the physical bound. Again $t = \varphi$. Every possible nonisolated endpoint is therefore a near- or far-simple-three helper with own five in $K = \mathbb{Q}(\varphi)$.

We now exclude that actual field in full, retaining its isolated alternatives. The unit near/far helper times are $M_N/36, M_F/36$, where

$$M_N = 13 + 49\varphi, \quad \mathrm{N}(M_N) = -1595, \qquad M_F = -2 + 19\varphi, \quad \mathrm{N}(M_F) = -395.$$

Two far helpers have positive actual product $(1 + 4\varphi)/9$. Near/far time-norm ratio $319/79$ is not a rational square. Two near helpers have invariant two-cycle $(500 + 660\varphi)/81 \notin \mathbb{Z}[\varphi]$. Thus both endpoints cannot be nonisolated.

For an isolated endpoint put $U = 4\gamma^2$. Rescale its actual K-root line to raw square $2U$ and sole old product $-U$. The actual two-cycle makes U integral and the full Lorentz star gives $K = \mathbb{Q}(U)$. Its raw time square is

$$H^2 = U(17U - 36)/18, \quad 36/17 < U < 225/17, \quad 0 < U' < 36/17.$$

The FULL ring models $U = a + b\varphi$ are exactly

U	$\mathrm{N}(U)$	$-\mathrm{N}[U(17U-36)]$	time norm squareclass
$1+\varphi$	1	251	251
$2+\varphi$	5	1595	$5\cdot 11\cdot 29$
$2+2\varphi$	4	4880	$5\cdot 61$
$3+2\varphi$	11	4631	$11\cdot 421$
$2+3\varphi$	1	2699	2699
$3+3\varphi$	9	14499	179
$3+4\varphi$	5	16895	$5\cdot 31\cdot 109$
$4+4\varphi$	16	22784	89
$4+5\varphi$	11	38291	11
$5+5\varphi$	25	16475	659.

Indeed $b\sqrt{5} < 225/17$ gives $b = 1,\dots,5$; the strict conjugate strip gives respectively $a = 1,2;\ 2,3;\ 2,3;\ 3,4;\ 4,5$. The entries follow from

$$\mathrm{N}[U(17U-36)] = n(289n - 612T + 1296), \quad n = \mathrm{N}(U), \quad T = \mathrm{Tr}(U).$$

The displayed squareclasses are distinct; their small prime factorizations can be checked by trial divisibility through the respective square roots.

Actual roots on the SAME K-time line require $U(17U-36)/M_N$ or $/M_F$ to be a square IN K. Their rational norms exclude every model except $U = 2+\varphi$ with a near helper; none has the far-helper class $5\cdot 79$. For that exception $U(17U-36) = M_N$ exactly. Its equal positive raw time coefficients and spatial product $U^2/18$ give

$$g = -(4+22\varphi)/9, \qquad g^2/U = (168+164\varphi)/81,$$

whose nonintegral two-cycle excludes it.

Finally two isolated golden endpoints cannot use different rows, since their rational time-norm squareclasses differ. With the same row their equal positive raw times give

$$g = 2U(9-4U)/9, \qquad g^2/U^2 = 4(9-4U)^2/81 \in \mathcal{O}_K.$$

Integral closedness makes $2(9-4U)/9$ integral, hence $8U/9$ integral; Bezout $8\cdot 8 - 7\cdot 9 = 1$ then makes $U/9$ integral. But $0 < \mathrm{N}(U) < 8100/289 < 29$, whereas $\mathrm{N}(U/9) = \mathrm{N}(U)/81$ would be a positive integer. Thus the actual golden field is impossible.

Both endpoints are now isolated in a nongolden actual field, with independent integral generators satisfying

$$K = \mathbb{Q}(U) = \mathbb{Q}(V), \quad 36/17 < U, V < 3969/425, \quad 0 < U^\sigma, V^\sigma < 36/17 \quad (\sigma \neq \mathrm{id}). \quad \text{(A17isolated)}$$

No equality of their parameters, finite-own assumption or common disconnected raw scale was used.

C.3.3 One two-end graft gives actual degree at most five

There is no golden hole after the preceding argument. If BOTH signed endpoint coordinates were at least $1/2$, modify the first and last five-coordinate blocks simultaneously. Replace each signed block mean b by

$$a, r, r, r, r, \quad a = 1/2, \quad r = (5b-a)/4.$$

Both sums and every central height stay unchanged. The supports are disjoint and each residual has sum zero, so the two residuals are orthogonal to one another and to the old flat blocks. Moreover $r > .243875 > .1$, making every changed positive central coefficient $> .018$. The new endpoint coefficients are paid by the two ACTUAL coordinates assumed at least a. The summed-foot proof therefore applies to this ONE new reference, with energy

$$> .9037838 + 2(5/4)(.5 - .2952)^2 = 1.0086414.$$

Thus at least one actual endpoint coordinate is $< 1/2$. For its generator U, the edge formula $C^2 = 17U/36$ and the farthest endpoint comparison give

$$36/17 < U < 66564/10625 < 157/25. \qquad \text{(A17shortU)}$$

The other generator need not satisfy this sharper upper bound; the chosen one still generates the ENTIRE actual field.

Let $F(U) = U(U-1)(U-2)$ and $d = [K : \mathbb{Q}]$. For $d \geq 2$ all factors are nonzero algebraic integers. Physically

$$0 < F(U) < 142.$$

At any other embedding put $t = U^\sigma - 1 \in (-1, 19/17)$. On $[-1, 1]$, $|t^3 - t| \leq 2/(3\sqrt{3}) < 2/5$; on $(1, 19/17)$ it is at most $1368/4913 < 1/3$. If $d \geq 7$, the nonzero integral norm would have magnitude $< 142(2/5)^6 = 9088/15625 < 1$. Hence $d \leq 6$. For $d = 6$, the bound $142(2/5)^5 < 2$ makes its absolute norm one, and all three separate factor norms have absolute value one.

We give the complete degree-six seal. Set $s = U - 1$, and let $r_i = |U^{\sigma_i} - 1|$, $i = 1, \ldots, 5$. Irreducibility excludes $r_i = 0, 1$. The unit norms and (A17shortU) give

$$0 < r_i < 19/17 < 9/8, \quad \prod r_i = 1/s > 3/16, \quad s < 132/25, \quad s^2 - 1 < 16799/625 < 27.$$

The sharper $s < 132/25$, rather than merely $s < 16/3$, is needed in the last inequality.

If m positive numbers $q_i < 1$ have $p = (\prod q_i)^{2/m}$, ordinary AM–GM applied separately to q_i^2 and $1 - q_i^2$ gives

$$p + \{\prod (1 - q_i^2)\}^{1/m} \leq 1.$$

Let k be the number of $r_i > 1$, so $m = 5 - k$. Each large r_i contributes $|1 - r_i^2| < 72/289 < 1/4$. The following six cases exhaust all possibilities:

k	$\prod_{\text{small}} r_i$ lower	p lower	$\prod_{i=1}^5 \lvert 1 - r_i^2 \rvert$ upper
0	3/16	1/2	1/32
1	1/6	2/5	81/2500
2	4/27	1/4	27/1024
3	32/243	1/8	49/4096
4	not needed	not needed	1/256
5	empty product	not needed	1/1024.

For the product bounds use $\prod_{\text{small}} r_i > (3/16)(8/9)^k$. The four root comparisons for p follow respectively from $9/256 > 1/32$, $1/6 > 4/25$, $16/729 > 1/64$, $32/243 > 1/8$; each exponent has been cleared positively. Multiplying the last column by 27 gives

$$27/32, \quad 2187/2500, \quad 729/1024, \quad 1323/4096, \quad 27/256, \quad 27/1024,$$

all below one. But the unit norm of $U(U-2)$ would be

$$1 = (s^2-1)\prod_{i=1}^{5}|1-r_i^2| < 1.$$

This excludes degree six and proves the ACTUAL degree bound $d \le 5$, without a field list or an equality assumption on the two endpoints.

C.3.4 The whole two-dotted interval and the derived finite fields

Suppose both isolated owns are dotted. For the left endpoint put $u = \gamma^{-2} \in (0,1)$. The positive actual core

$$c + \gamma^{-1}n_1 = \gamma\{\sqrt{17/9-u}\,p - (\omega_1 - un_1)\}$$

has negative square $u-1$, zero c product and negative n_1 product. Every other original wall, INCLUDING the other endpoint opposite, has nonpositive product with both summands. After normalization it is a genuine chamber point and gives

$$x \cdot Z_L \ge h(u) = \sqrt{17/9-u} - \sqrt{u(1-u)}, \quad Z_L = \omega_1 - un_1.$$

The right endpoint has its independent parameter $v \in (0,1)$ and reflected vector Z_R.

Let

$$Y_L = Z_L - \omega_5/5, \qquad R(u) = \|Y_L\|^2 = 8/5 - 2u + u^2.$$

This is a zero-sum vector supported in the first five coordinates. Its central-root products are zero except $-1/5$ at five. Its reflected partner is supported in the LAST five coordinates. Consequently $w \cdot Y_L = w \cdot Y_R = Y_L \cdot Y_R = 0$, while $w \cdot Z_L = w \cdot Z_R = .4174$. Pay both actual core pressures at $\mu = .3$:

$$w' = w + \mu Y_L + \mu Y_R, \quad \beta'_5 = \beta'_{13} = .1382 - .06 = .0782, \quad \beta'_i = \beta_i \ (6 \le i \le 12).$$

Every positive central coefficient is still at least .018, and $w' \cdot n_i = \beta'_i$ EXACTLY.

Retain every central isolated foot, with its OWN parameter $v_i \in (0,1)$, using ONE vector

$$V = w' - \sum_{i \in S} \beta'_i v_i n_i.$$

Sum the endpoint and central actual pressures before estimating the norm. The identities just proved and $n_i \cdot n_j \le 0$ give

$$x \cdot V \ge E_0 + \mu\{h(u) - .4174 + h(v) - .4174\} - \sum_S \beta'_i(\lambda_i - .6),$$
$$\|V\|^2 \le E_0 + \mu^2\{R(u) + R(v)\} - \sum_S (\beta'_i)^2(2v_i - v_i^2).$$

Every central summand is nonnegative by the already proved absorption inequality, since $\beta'_i \ge .018$. Thus

$$\|x\|^2 \ge E_0 + G(u) + G(v), \qquad G(u) = .6\{h(u) - .4174\} - .09(8/5 - 2u + u^2).$$

This whole continuous interval has a strict rational seal. Concavity and the endpoint squares give $\sqrt{17/9-u} > 1.3743 - .4315u$ on $[0,1]$. Therefore

$$G(u) > .43014 - .0789u - .09u^2 - .6\sqrt{u(1-u)}.$$

For $u \le 1/2$, use $u^2 \le u/2$ and the circle inequality

$$au + b\sqrt{u(1-u)} \le (a + \sqrt{a^2+b^2})/2.$$

With $a = .1239, b = .6$, $\sqrt{a^2+b^2} < .6127$ gives $G(u) > .06184$. For $u \ge 1/2$, use $u^2 \le 1.5u - .5$; $a = .2139, b = .6$, with radical $< .637$, gives $G(u) > .04969$. The positive square margins are $.00005008, .00001579$, respectively. Hence throughout $(0,1)$,

$$\|x\|^2 > .9037838 + 2(.04969) = 1.0031638,$$

impossible. Parallel owns already have their actual future null. At least one endpoint is a finite native own.

At that endpoint $U = 4\cos^2(\pi/m) = 2 + 2\cos(2\pi/m)$ generates the ACTUAL field. The degree bound makes $\phi(m) \le 10$. The COMPLETE elementary totient list is

$$3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 14, 15, 16, 18, 20, 22, 24, 30.$$

Indeed only prime divisors $2, 3, 5, 7, 11$ are possible; their permissible prime powers and products give that list. Labels three and four have $U \le 2 < 36/17$. Six gives the rational field, excluded by Meyer and compact anisotropy at form rank eighteen. Five and ten give the already excluded golden field. For the following nine labels choose the indicated coprime cyclotomic embedding:

m	k	$\cos(2k\pi/m) = \sin\theta$	$2\theta/\pi$
9	2	$\theta = \pi/18$	1/9
14	3	$\theta = \pi/14$	1/7
15	2	$\theta = 7\pi/30$	7/15
16	3	$\theta = \pi/8$	1/4
20	3	$\theta = \pi/5$	2/5
24	5	$\theta = \pi/12$	1/6
30	7	$\theta = \pi/30$	1/15
11	2	$\theta = 3\pi/22$	3/11
22	3	$\theta = 5\pi/22$	5/11.

Every last entry exceeds $1/17$. Strict concavity on $0 < \theta < \pi/2$ gives $\sin\theta > 2\theta/\pi$, so that conjugate would have $U^\sigma > 36/17$, a contradiction. Label thirty was included before its conjugate test. Only labels $7, 8, 12, 18$ remain. Their actual fields are, respectively, $K_7, \mathbb{Q}(\sqrt{2}), \mathbb{Q}(\sqrt{3}), K_9$; the cubic parameters have polynomials

$$X^3 - 5X^2 + 6X - 1, \qquad X^3 - 6X^2 + 9X - 3,$$

with discriminants $49, 81$ and irreducibility by the rational-root test. This proves the actual field reduction, rather than assuming a list of possible fields.

C.3.5 Actual inclusion in the recorded finite domains

The isolated endpoint normalization in (A17isolated) has raw square $2U$, sole old product $-U$ and time square $F(U)/18$, where now $F(X) = X(17X - 36)$. For the two actual endpoint roots, subtracting their K-valued old projections leaves one common K-time line. Thus

$$R = F(U)F(V) \in K^2, \quad g = \frac{UV - \sqrt{R}}{18}, \quad L = \frac{g^2}{UV}.$$

The square root is PHYSICALLY POSITIVE. The long old-three cycle gives $g \in \mathcal{O}_K$; the actual two-cycle gives $L \in \mathcal{O}_K$. Acuteness gives $g \le 0$. If $g \neq 0$, every nonphysical place satisfies $0 < L^\sigma < 4$; the branch $g = L = 0$ is retained. The physical owns and mutual products retain their native/dotted alphabet, with a parallel pair excluded by its actual future null. These are precisely the stated necessary tests in the quadratic domain; the endpoint parameters need not be equal.

If K is quadratic, this actual pair therefore enters the FULL quadratic domain of Proposition 21.1(i), including its maximal-order classes, square signs and zero mutual. That recorded finite-output proposition says the domain is empty. If $K = K_7$ or K_9, the finite endpoint proved above supplies the actual anchor seven or eighteen, respectively. The same original time-line square, cyclic and native conditions put the other endpoint in the full cubic finite-anchor domain of Proposition 21.1(ii). Its complete ring/box/decision arguments are given in Section 21, and the recorded output is empty.

Thus the stated A_{17} finite-output premises, together with the geometric inclusions proved here, exclude whole A_{17} at the original arithmetic rank-seventeen vertex.

C.4 The separate whole-A_{16} endpoint proof

The whole A_{16} case has $N = 17$ and its own constants. It is not a consequence of reranking the preceding A_{17} theorem. We retain one original farthest vector throughout. The common chain proof gives central reference energy

$$E_0 = .8708972, \quad (H_5, H_6, H_7, H_8) = (2.056, 2.186, 2.269, 2.310), \quad H_9 = H_8,$$

with positive coefficients at least .0205. Its worst single end-hole chord has energy

$$\frac{12569249}{15000000}.$$

All grafts below preserve the common simultaneous-foot proof by checking those coefficients afresh.

C.4.1 Actual lengths and all nonisolated endpoint roles

On either signed end block, of length five or six and mean $.25 < b < .3$, grafting to .67 leaves each remaining coordinate $> .145$; the following coordinate is below .1. The central coefficient stays $> .045/\sqrt{2} > .0205$. The gain $> (6/5)(.37)^2 = .16428$, even after the worst chord omission, exceeds the available norm budget. Both signed endpoint coordinates are therefore $< .67$. The actual edge comparison is now

$$x_{\text{end}} \ge \sqrt{16/17}\sqrt{(C-1)/(C+1)}.$$

At $C = 17/6$ its square $176/391$ exceeds $.67^2$. Hence both endpoints have $C < 17/6$. If both signed coordinates were at least .56, disjoint grafts on the SAME reference would gain $> 2(6/5)(.26)^2 = .16224$, again exceeding one. Since $16/51 > .56^2$, at least one endpoint has $C < 2$.

Outside the actual golden field there is no hole and the full block mean lies in $(.2907, .2908)$. A single graft to .613 gains $> .12976605$, making total > 1.00066325. The retained coordinates are $> .210125$, so the coefficient threshold remains valid. As $32/85 > .613^2$, both endpoints then have $C < 7/3$. Two disjoint grafts to .52 gain $> .1313316$, giving total > 1.0022288. The exact comparison $1296/4777 > (13/25)^2$ therefore proves

$$\text{always both } C < 17/6,\ \min C < 2; \qquad K \neq \mathbb{Q}(\varphi):\ \text{both } C < 7/3,\ \min C < 181/100. \tag{A16lengths}$$

Release node one and retain the whole white A_{15}. Its complete finite one-node extensions are isolation; helper three at $j = 2, 3, 15, 16$; or helper four at $j = 2, 16$. There is no foreign component or exceptional rank-sixteen completion. A second contact makes a cycle. For helper three, the actual connected normalization gives integral $t = 2\gamma$, ACTUAL $K = \mathbb{Q}(t)$, raw time square $M_j/17$, and

$$M_j = 16t^2 + 2(17-j)t + j(17-j) - 34,$$

j	C
2	$(16t+15)/17$
3	$(16\gamma+7)/\sqrt{17}$
15	$(16\gamma+1)/\sqrt{17}$
16	$(16t+1)/17.$

For helper four the lengths are

$$(16\gamma + 15/\sqrt{2})/\sqrt{17/2}, \qquad (16\gamma + 1/\sqrt{2})/\sqrt{17/2}.$$

These follow by eliminating ALL retained roots; the unit white residuals are $17/32, 1/8, 1/16$ in the endpoint-three, next-three and endpoint-four rows.

Helper $j = 3$ with positive own gives $C \geq 15/\sqrt{17} > 17/6$. Its own-zero case has the same actual affine-E_8 vector printed in the preceding endpoint proof. At $j = 15$, own zero has unit time square $-2/17$, own three has the actual all-three cycle, and owns at least four give $C > 17/6$. Helper four at $j = 2$ fails even at own zero. At $j = 16$, its own-zero time is $-1/17$, and every positive own gives $C \geq (8\sqrt{2}+1)/\sqrt{17} > 17/6$. At near $j = 2$, own zero has negative time and own three the actual affine triangle. At far $j = 16$, positive time requires $t > 1$, since $M_{16} = (t-1)(16t+18)$. Parallel has the positive actual future-null pair. All these nulls use UNIT actual normals and are feasible against every omitted wall.

In a nongolden field the near helper has $t < 37/24$, so its only remaining native own is four, $t = \sqrt{2}$. The far helper satisfies

$$1 < t < 29/12, \qquad -9/8 < t_\sigma < 1.$$

As before $f(t) = t(t^2-1)$ has other-place magnitude at most $2/(3\sqrt{3})$; on the additional interval $[-9/8, -1]$, the maximum is $153/512$, smaller. Physically $0 < f(t) < 12$. The integral norm excludes degrees at least four. At degree three $t, t-1, t+1$ would all be units. The same complete four cubics as before occur. The negative-discriminant cubic and the two with a conjugate below $-9/8$ fail; the only survivor is

$$P(X) = X^3 - 2X^2 - X + 1,$$

whose physical root $a > 2$ generates K_7. For degree two, trace $0, 1, 2, 3$ and norm $-2, -1, 1, 2$ exhaust the strip. Trace zero would require $\sqrt{2}$, with conjugate below $-9/8$; trace one leaves only φ or rational two; trace two leaves $1 + \sqrt{2}$, since $1 + \sqrt{3}$ is too large; trace three's irrational positive root exceeds $29/12$. The rational two is parallel. Thus the nongolden helper roles are exactly near silver, far silver, or far K_7. Their actual lengths are respectively

$$(16\sqrt{2}+15)/17, \quad (17+16\sqrt{2})/17, \quad (16a+1)/17,$$

all $> 181/100$. The short endpoint selected in (A16lengths) is consequently ISOLATED.

C.4.2 The full golden endpoint table

For any isolated endpoint, its OWN K-rescaling gives raw norm $2U$, sole old product $-U$, integral $U = 4\gamma^2$ generating ACTUAL K, and

$$F(U) = U(16U - 34), \quad H_U^2 = F(U)/17, \quad C^2 = 8U/17, \quad U > 17/8, \quad 0 < U^\sigma < 17/8. \quad \text{(A16U)}$$

No raw time itself is assumed K-valued; the ratios along the SAME one-dimensional K-time line are K-valued.

Suppose $K = \mathbb{Q}(\varphi)$. The near helper has $t < 199/96$ and conjugate $-2 < t' < 1/8$. The finite native list leaves φ; a dotted full-ring $a + b\varphi$ has $b = 1$ and no integer a in $2 < t < 199/96$. The far helper satisfies $1 < t < 283/96$, $-9/8 < t' < 1$, giving $t = \varphi$ or φ^2. All three types have raw times $M/17$:

helper	M	$\mathrm{N}(M)$
near φ	$12 + 46\varphi$	-1420
far φ	$-2 + 18\varphi$	-356
far φ^2	$16 + 50\varphi$	-1444.

The norm squareclasses $355, 89, 1$ forbid different types. Reflected near copies have nonintegral actual long-cycle $g = -(7 + 41\varphi)/17$. Reflected far-φ copies have positive product $g = (4 + 15\varphi)/17$. Reflected far-φ^2 copies have $g = (19 - 15\varphi)/17 \in (-1, 0)$, forbidden for two distinct square-two actual roots by the native gap. Thus both endpoints cannot be nonisolated.

For the isolated golden end, the universal length cap gives

$$17/8 < U < 4913/288, \qquad 0 < U' < 17/8.$$

Writing $U = a + b\varphi$ gives $b = 1, \ldots, 7$. The complete conjugate intervals give $a = 1, 2;\ 2, 3;\ 2, 3;\ 3, 4;\ 4, 5;\ 4, 5;\ 5, 6$, and the physical upper bound removes only $(6, 7)$. The complete thirteen rows are

U	$\mathrm{N}(U)$	$-\mathrm{N}(F(U))$	squareclass
$1 + \varphi$	1	220	55
$2 + \varphi$	5	1420	355
$2 + 2\varphi$	4	4336	271
$3 + 2\varphi$	11	4180	1045
$2 + 3\varphi$	1	2396	599
$3 + 3\varphi$	9	12924	359
$3 + 4\varphi$	5	15020	3755
$4 + 4\varphi$	16	20416	319
$4 + 5\varphi$	11	34100	341
$5 + 5\varphi$	25	15100	151
$4 + 6\varphi$	4	21744	151
$5 + 6\varphi$	19	50996	12749
$5 + 7\varphi$	11	58036	14509.

For $n = \mathrm{N}(U), T = \mathrm{Tr}(U)$, every entry follows from $\mathrm{N}(F(U)) = n(256n - 544T + 1156)$. The listed classes are squarefree: $1045 = 5 \cdot 11 \cdot 19$, $3755 = 5 \cdot 751$, $319 = 11 \cdot 29$, $341 = 11 \cdot 31$, $12749 = 11 \cdot 19 \cdot 61$, $14509 = 11 \cdot 1319$, $355 = 5 \cdot 71$, $55 = 5 \cdot 11$; the remaining small primality checks are direct trial division. The only repeated class is151. Precisely the FIRST SIX rows have $U < 17/2$, equivalently $C < 2$, and their classes occur nowhere else in this complete table.

If one endpoint is near-φ or far-φ^2, its $C > 2$, so the opposite isolated end is in the first six rows. Class one matches none. The near class355 matches only $U = 2 + \varphi$; there $F(U) = 12 + 46\varphi$, so the actual positive raw times are equal. Their spatial product is $U(\varphi + 2)/17 = U^2/17$, giving the same nonintegral long cycle $g = -(7 + 41\varphi)/17$. For a far-φ helper the opposite end may be ANY of the thirteen rows, but none has class89. All helper/isolated possibilities are excluded.

Both golden endpoints must now be isolated. One belongs to the first six rows by the selected $C < 2$. The SAME time-line square test forces equality of the two U-models; the repeated151 class

cannot occur at the selected end. Equal positive raw times give

$$g = U(34 - 15U)/17, \qquad L = g^2/U^2 = ((34 - 15U)/17)^2 \in \mathcal{O}_K.$$

Integral closedness makes $(34 - 15U)/17$, and then $U/17$, integral, using Bezout for15 and17. But the selected norms $1, 5, 4, 11, 1, 9$ are smaller than 289, so $\mathrm{N}(U/17) = \mathrm{N}(U)/289$ cannot be a nonzero integer. The ACTUAL golden field is impossible.

C.4.3 Both actual endpoints are isolated

The actual field is now nongolden, and the selected endpoint of $C < 181/100$ is isolated. At $K = \mathbb{Q}(\sqrt{2})$, its full possible isolated strip from $C < 7/3$ is

$$17/8 < U < 833/72, \quad 0 < U' < 17/8.$$

Writing $U = A + B\sqrt{2}$, the complete models are

$$2 + s,\ 3 + s,\ 3 + 2s,\ 4 + 2s,\ 5 + 3s,\ 6 + 3s, \qquad s = \sqrt{2}.$$

Indeed $B = 1, \ldots, 4$; the conjugate strip permits respectively $A = 2, 3;\ 3, 4;\ 5, 6;\ 6, 7$, and the physical upper bound removes both $B = 4$ possibilities. Their signed F-norms are

$$-1016, -2212, -1852, -9184, -17444, -13752.$$

The near-silver helper has $M = 28 + 30s$, norm $-1016 = -8 \cdot 127$. Only $U = 2 + s$ has a compatible norm squareclass: none of the other five norms is divisible by127. For that exception $F(U) = M$ and the actual long cycle $g = -(22 + 26s)/17$ is not integral. The far-silver helper has $M = 32 + 34s$, norm $-1288 = -8 \cdot 7 \cdot 23$. None of the six time norms is divisible by23. Both silver helper roles are excluded.

For the remaining far helper in K_7, let $a > 2$ be the root of $P(X) = X^3 - 2X^2 - X + 1$. Its three roots have disjoint rational brackets

$$2.246 < a < 2.247, \quad -.803 < b < -.801, \quad .554 < c < .556.$$

Signs and derivatives of P isolate them. The full ring is $\mathbb{Z}[a]$, as justified below. The selected isolated endpoint has $17/8 < U < 7$, $0 < U_b, U_c < 17/8$. Write $U = A + Ba + Ca^2$. Lagrange interpolation expresses C as the sum of the three values divided by the derivatives at the three roots. The two positive denominators exceed five and four; the magnitude of the negative denominator exceeds $17/8$. Consequently $-1 < C < 7/5 + (17/8)/4 < 2$, so $C = 0, 1$. The small conjugate difference gives

$$|B + C(2 - a)| < (17/8)/(c - b) < 8/5.$$

If $C = 0$, positive physical difference forces $B = 1$, and the conjugate interval forces $A = 1$. If $C = 1$, the physical/conjugate difference < 7 removes $B \geq 1$, leaving $-1, 0$. For $B = -1$, the c-value requires $A \geq 1$, while the b-value requires $A \leq 0$. For $B = 0$, $A = 0, 1$. Thus the COMPLETE selected models are

$$U = 1 + a, \quad a^2, \quad 1 + a^2.$$

Their positive time norms are $5048, 32248, 82264$. The helper time numerator is $(a - 1)(16a + 18)$, of norm $7496 = 8 \cdot 937$. The prime937 divides none of these three norms; therefore no actual time ratio can be a K-square. This excludes the last nongolden helper. BOTH actual endpoints are isolated.

C.4.4 Degree, the whole two-dotted domain and finite anchors

The selected isolated endpoint generates the actual field, with

$$17/8 < U < 556937/80000 < 7, \qquad 0 < U^{\sigma} < 17/8.$$

For $F_3(U) = U(U-1)(U-2)$, physical value $0 < F_3(U) < 210$. At a nonphysical embedding $y = U^{\sigma} - 1 \in (-1, 9/8)$. The same cubic bound gives $|y^3 - y| < 2/5$, including the upper extra interval where the maximum is $153/512$. A degree at least seven would give a nonzero integer norm $< 210(2/5)^6 = .86016 < 1$. Hence ACTUAL degree is at most SIX. No degree-five refinement from the A_{17} case is used.

Suppose both isolated owns were dotted. Their ACTUAL endpoint cores give, for independent $u, v \in (0,1)$,

$$x \cdot Z_L \geq h(u) = \sqrt{32/17 - u} - \sqrt{u(1-u)}, \qquad Z_L = \omega_1 - un_1,$$

and its reflection. The core is future timelike, zero on its cap, negative on the released old root, and nonpositive against EVERY other actual wall, including the opposite endpoint. The residual $Y_L = Z_L - \omega_5/5$ has zero sum, support in the first five coordinates, norm $R(u) = 8/5 - 2u + u^2$, and sole central product $-1/5$ at five. The reflected residual subtracts $\omega_{12}/5$, and is orthogonal to both the first residual and the old flat block. Here $w \cdot Z_L = w \cdot Z_R = .4112$.

Pay each core at weight $\mu = .32$. The individual outer coefficients change from .1406 to $.1406 - .064 = .0766 > .0205$; every other central coefficient is unchanged. In particular the changed reference has its EXACT products equal to these new coefficients. Subtract all retained central isolated-foot terms with their own parameters, and apply the common simultaneous absorption proof to this ONE vector. All endpoint/central and adjacent-central crosses are retained, giving

$$\|x\|^2 \geq E_0 + G(u) + G(v), \quad G(u) = .64\{h(u) - .4112\} - .1024(1.6 - 2u + u^2).$$

The following COMPLETE three-chord certificate covers $[0,1]$:

u interval	$L(u) < \sqrt{32/17 - u}$	K	a	$\sqrt{a^2 + .64^2}$ upper
$[0, 1/2]$	$1.3719 - .3924u$	.451008	.097536	.6475
$[1/2, 3/4]$	$1.3989 - .4464u$	.506688	.208896	.6733
$[3/4, 1]$	$1.4385 - .4992u$	.570432	.293888	.7043.

The strict endpoint values $1.3719, 1.1757, 1.0641, .9393$ have squares below $32/17, 47/34, 77/68, 15/17$. Concavity proves each chord lower bound. On $[l, r]$ use $u^2 \leq (l + r)u - lr$; then $G(u) > K - au - .64\sqrt{u(1-u)}$. The circle inequality used earlier gives the three lower values .07849, .06559, .071338. The squared radical margins are

$$.000142978704, \quad .000095351184,$$
$$.000068333456.$$

Thus $G(u) > .06559$ on the whole dotted interval, and

$$\|x\|^2 > .8708972 + 2(.06559) = 1.0020772.$$

Both owns cannot be dotted. Parallel has its actual future null, so at least one endpoint is a finite native own.

Its integral $U = 2 + 2\cos(2\pi/m)$ generates the actual field, and the degree bound gives $\phi(m) \leq 12$. The full totient list is

$$3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 18, 20, 21, 22, 24, 26, 28, 30, 36, 42.$$

Only prime factors $2,3,5,7,11,13$ and their allowed prime powers occur; checking products with totient at most twelve gives exactly this list. Labels three and four have $U \le 2 < 17/8$. For own six the ACTUAL unit-root null is $\sqrt{3}c + 2n_1 + n_2$, a nonzero future ray feasible against all other walls. Five and ten give the excluded golden field. The remaining forbidden labels have explicit coprime embeddings:

m	k	$\cos(2k\pi/m)$ lower
$9, 11, 13, 15, 21$	2	$\sin(\pi/18) > 1/9$
$14, 16, 20, 22, 26, 28$	3	$\sin(\pi/14) > 1/7$
$24, 36, 42$	5	$\sin(\pi/12) > 1/6$
30	7	$\sin(\pi/30) > 1/15$.

All exceed $1/16$, by strict sine concavity. Their conjugate U would exceed $17/8$. Only finite anchors seven, eight, twelve and eighteen remain, with ACTUAL fields $K_7, \mathbb{Q}(\sqrt{2}), \mathbb{Q}(\sqrt{3}), K_9$.

We use FULL integer rings. For K_7, $P(X+3) = X^3 + 7X^2 + 14X + 7$ is Eisenstein at seven, and the polynomial discriminant is49. For K_9, put $Q(X) = X^3 - 3X - 1$; $Q(X+1) = X^3 + 3X^2 - 3$ is Eisenstein at three and the discriminant is81. The Eisenstein uniformizer criterion proves maximality at the only possible index primes, while the discriminants exclude every other index prime. Equivalently these are the full-ring proofs already given in Section 21, under translation. Thus the bases are the full $\mathbb{Z}[a]$, $\mathbb{Z}[k]$; the two quadratic rings are $\mathbb{Z}[\sqrt{2}]$ and $\mathbb{Z}[\sqrt{3}]$.

C.4.5 All twenty-one actual finite-anchor time norms

The opposite isolated endpoint V, whether finite or dotted, has

$$17/8 < V < 833/72, \qquad 0 < V^{\sigma} < 17/8.$$

For $\mathbb{Q}(\sqrt{2})$, its complete six models were printed above. For $\mathbb{Q}(\sqrt{3})$, the full ring gives $B = 1, 2, 3$, and conjugate intervals $A = 2, 3;\ 4, 5;\ 6, 7$. The physical upper bound removes only $(A, B) = (7, 3)$. The EXACT five models are

$$2 + r,\ 3 + r,\ 4 + 2r,\ 5 + 2r,\ 6 + 3r, \qquad r = \sqrt{3}.$$

For K_7, the leading-coefficient interpolation bound now gives $-1 < C < 3$, so $C = 0, 1, 2$ in $V = A + Ba + Ca^2$. The same small conjugate difference gives $|B + C(2 - a)| < 8/5$. The complete integer possibilities are

C	B	A
0	1	1
1	-1	none
1	0	$0, 1$
1	1	1
2	$-1, 0$	0.

For $C = 2, B \ge 1$, the physical/conjugate difference exceeds $3.047(1 + 2(1.444)) = 11.846736 > 833/72$. This excludes every remaining larger B; the conjugate-difference bound excludes the smaller ones. The resulting EXACT six models are

$$1 + a,\ a^2,\ 1 + a^2,\ 1 + a + a^2,\ 2a^2 - a,\ 2a^2.$$

For K_9, the three roots have brackets

$$1.879 < k < 1.880, \quad -.348 < b < -.347, \quad -1.533 < c < -1.532.$$

The two positive Lagrange denominators exceed $7.5, 4$, and the magnitude of the negative one exceeds $17/8$. Thus again $C = 0, 1, 2$. The full conjugate difference gives $|B - Ck| < 9/5$. The constant-term interval cases are

C	B	A
0	1	2
1	1	1
1	2	$1, 2$
1	3	3 fails the physical upper bound
2	2	none.

For $B = 3, C = 1$, the physical value is $> 3 + 3(1.879) + 1.879^2 = 12.167641 > 833/72$. For $C = 2, B \geq 3$, the physical difference exceeds $3.411(3 + .694) > 833/72$. At $C = 2, B = 2$, the b-strip forces $A \geq 1$ and the c-strip $A \leq 0$. Hence the EXACT four models are

$$2 + k,\ 1 + k + k^2,\ 1 + 2k + k^2,\ 2 + 2k + k^2.$$

Every listed accepted model satisfies all strict strips by the root brackets; every omitted integer coefficient has been excluded. These are complete handwritten domains, not unexecuted worker results.

The original raw times of two isolated roots are $F(U)/17$, $F(V)/17$. The same actual K-time line makes their ratio a K-square. A necessary condition is a square rational norm; BOTH possible signed norms of any putative square root satisfy that condition. Here are ALL twenty-one signed time norms, in the stated model order:

field and finite anchor	$\mathrm{N}(F(V))$
$\sqrt{2},\ 2 + \sqrt{2}$	$-1016, -2212, -1852, -9184, -17444, -13752$
$\sqrt{3},\ 2 + \sqrt{3}$	$-764, -3432, -8688, -12428, -27612$
$K_7,\ 1 + a$	$5048, 32248, 82264, 110552, 1784, 330688$
$K_9,\ 2 + k$	$16872, 66168, 79032, 293624.$

For a quadratic model the reproducible identity is

$$\mathrm{N}(F(V)) = n(256n - 544T + 1156).$$

For a cubic model with coefficient data (T, S, n), it is

$$\mathrm{N}(F(V)) = n(4096n - 8704S + 18496T - 39304).$$

The six K_7 triples are

$$(5, 6, 1), (6, 5, 1), (9, 20, 13), (11, 24, 13), (10, 17, 1), (12, 20, 8),$$

and the four K_9 triples are

$$(6, 9, 3), (9, 18, 9), (9, 6, 1), (12, 27, 17).$$

They follow by multiplication in the displayed full defining bases.

The anchors have odd valuations respectively at the primes $127, 191, 631, 19$:

$$1016 = 8(127), \quad 764 = 4(191), \quad 5048 = 8(631), \quad 16872 = 8 \cdot 3 \cdot 19 \cdot 37.$$

NONE of the other time norms in its field row is divisible by that prime. The exact nonzero residues of the ABSOLUTE time norms are respectively

$$(53, 74, 40, 45, 36), \quad (185, 93, 13, 108), \quad (67, 234, 127, 522, 44), \quad (10, 11, 17).$$

Thus every UNEQUAL anchor/model time ratio has nonsquare rational norm and cannot be a square in K. No square norm is used as a sufficient test, and no native, orthogonal, parallel or dotted mutual case is omitted by this necessary invariant.

Only $V = U$ at the actual finite anchor remains. The endpoint walls are distinct, and their canonical raw norms are $2U$. Equal time squares give equal POSITIVE physical times; their spatial product is $U^2/17$. Therefore the actual two-cycle is

$$g = \frac{U(34 - 15U)}{17}, \qquad L = \frac{g^2}{U^2} = \left(\frac{34 - 15U}{17}\right)^2 \in \mathcal{O}_K.$$

The zero branch would force the irrational anchor $U = 34/15$, impossible, but integral closedness applies in either case. It makes $15U/17$, hence $U/17$, integral: $15 \cdot 8 - 17 \cdot 7 = 1$. The four anchor norms are $2, 1, 1, 3$, in degrees $2, 2, 3, 3$; their quotients by the corresponding powers of seventeen are not integers. This final contradiction excludes every diagonal pair too.

The whole-A_{16} exclusion is complete: actual endpoint lengths and helper/golden cases, BOTH isolated, degree at most six, full two-dotted interval, finite anchors and actual fields, complete full-ring domains, every unequal time ratio, and the actual diagonal two-cycle. This manual proof uses the printed inequalities and norm tables. It adds no census premise and assumes no equality between the raw root scales of disconnected components.

C.5 One actual rank-one opposite

For the full forests $A_{15} + A_1, B_{15} + A_1, D_{15} + A_1$, the correct first split concerns the SAME actual opposite d of the old rank-one direction e. Either it contacts the old large component, or all those products vanish. In the isolated case physical unit normalization gives

$$d = \sqrt{h}\, p - \sqrt{1 + h}\, e, \qquad h > 0.$$

No unrelated wall is chosen in the second case, and this identity does not identify disconnected field-root norm squareclasses. We first close the whole $A_{15} + A_1$ case and record its separate B_{15} nonisolated corollary.

C.5.1 Nonisolated rank-one opposites beside A_{15}

Let the original full forest be $A_{15} + A_1$, with old unit A-roots in zero-sum $\mathbb{R}^{16}$, and write

$$d = Ap - v_A - \gamma e, \quad A > 0, \quad \gamma > 0, \quad Q = \|v_A\|^2, \quad A^2 = \gamma^2 + Q - 1.$$

At the actual endpoint of the rank-one edge, old $A_{15} + d$ is a finite rank-sixteen white diagram. Its COMPLETE nonisolated extensions are

role	v_A	Q
endpoint three	$\omega_j/2,\ j = 1, 15$	$15/32$
endpoint four	$\omega_j/\sqrt{2},\ j = 1, 15$	$15/16$
next-to-end three	$\omega_j/2,\ j = 2, 14$	$7/8$.

These are A_{16}, B_{16}, D_{16}, including both orientations. Two old-path contacts make a nonpositive cycle; no exceptional rank-sixteen completion contains this long path.

We must have $\gamma > 1$. If $\gamma < 1$, the full positive actual replacement

$$r = p + \frac{A}{1-\gamma^2}(d + \gamma e)$$

has negative norm, positive time, zero products with e, d and all retained old A-roots except the contacted node j. Every unknown wall is nonpositive termwise, and its omitted n_j product is strictly negative. It is therefore the ACTUAL endpoint of old edge j. The SAME d would close both that direction and the orthogonal old rank-one direction, contradicting the original-farthest orthogonal obstruction. For $\gamma = 1$, $d + e$ is an actual future null. Thus $Q < A^2$ and the entire surviving own domain is dotted.

The actual rank-one edge length is $C = \gamma/\sqrt{1-Q}$, with energy at least $(C-1)/(C+1)$. At $Q \geq 7/8$, $C > \sqrt{8} > 7/3$ gives energy $> 2/5$. Together with the independently proved same-vector $e_{A_{15}} > .8348$, this excludes endpoint-four and next-three throughout their unbounded own domains. For endpoint-three, if $\gamma \geq 41/40$,

$$C^2 \geq 1681/850 > 49/25$$

gives rank-one energy $> 1/6$, and $.8348 + 1/6 = 1 + 11/7500$. Only $1 < \gamma < 41/40$ remains.

Here the ACTUAL positive core

$$d + \gamma e = Ap - \omega_j/2$$

is future timelike of square $1 - \gamma^2$, zero on e, negative on d, and nonpositive on every other original wall. Original maximality gives $h_j/2 \geq A - \sqrt{\gamma^2 - 1}$. The right side decreases with γ. At $41/40$, $A^2 = 831/1600 > .72^2$ and $\sqrt{\gamma^2 - 1} = 9/40$. Thus the appropriate ACTUAL signed A_{15} endpoint satisfies

$$x_{\text{end}} > \sqrt{2}(.72 - .225) > .69.$$

Rebuild its reference before using the new pressure. Take the common N16 raw heights $2.021, 2.137, 2.205, 2.227$, reflected, with their possible single golden hole. Internal omission loses at most .000576. End omission replaces the first tail by 0–6, giving

$$E_{\text{end}} = 48050651/60000000 > .80084.$$

Thus every RAW reference has energy $> .79$, before any earlier golden-endpoint graft. At the selected endpoint its flat block has length five or six, mean $.25 < b < .3$, and next coordinate $< .101$, including an adjacent internal omission. Graft this block afresh to .69. The other coordinates are $> (L(.25) - .69)/(L-1) \geq .14$; the changed central coefficient is $> (.14 - .101)/\sqrt{2} > .027 > .022$. Every other positive central coefficient retains .022; the golden hole remains coefficient zero. The exact sum-preserving zero-sum change has zero old block cross, so the new ONE reference has energy

$$> .79 + (6/5)(.69 - .3)^2 = .97252.$$

Absorb EVERY retained actual isolated foot again into THIS reference, using the common N16 loss inequality and its NEW coefficients. This gives $e_A > .97252$, including adjacent and foreign feet. No earlier optimized endpoint gain or foot energy was added.

Finally the original rank-one edge has $C > \sqrt{32/17} > 11/9$, so its distinct orthogonal energy is $> 1/10$. Adding that energy ONCE gives $\|x\|^2 > .97252 + .1 > 1$. This excludes the entire nonisolated actual rank-one opposite beside A_{15}, over every actual field.

For later use the corresponding B_{15} argument is separate. A finite rank-sixteen extension of old B_{15} can only be B_{16}, attaching by three at its LONG endpoint. Thus $v_B = \Omega_1/2$, $Q = 1/2$, and $A^2 = \gamma^2 - 1/2$. The same reciprocal actual projection proves $\gamma > 1$. The original rank-one energy is then $> (\sqrt{2}-1)/(\sqrt{2}+1) = 3 - 2\sqrt{2} > .17$. The already written fresh B_{15} bulk in the proof of Proposition 22.1, with arbitrary spectators and both initial-five cases, gives $e_B > E_{13} > .88989$. Their sum exceeds 1.05989. Thus a nonisolated actual rank-one opposite is impossible beside B_{15} as well. No D_{15} conclusion is transported from this corollary.

C.5.2 The isolated A_{15} case and its invariant cycles

Now suppose the SAME actual opposite is isolated: $d = \sqrt{h}\,p - \sqrt{1+h}\,e$. If $h \geq 1$, its actual edge has energy at least $3 - 2\sqrt{2} > .17$; together with $e_A > .8348$ this already contradicts the norm budget. Assume $0 < h < 1$ and put $V = 4h$. The actual two-cycle makes V integral and all nonphysical places satisfy $-4 < V^\sigma < 0$. Physically $0 < V < 4$. Consequently $V/4$ CANNOT be integral, since its nonzero norm would have magnitude below one. The actual field has degree at least two by the stated Meyer/compact-anisotropy input for the ENTIRE rank-seventeen form.

We need pressures only at

$$J = \{5, 6, 7, 9, 10, 11\}, \qquad \ell_i = .49 \ (i \text{ odd}), \quad \ell_6 = \ell_{10} = .496.$$

The midpoint eight will never be assigned a pressure or coefficient.

Let an actual central opposite be

$$c = Ap - v_A - \beta e, \quad Q = \beta^2 < 1, \quad \lambda = -c \cdot d = A\sqrt{h} - \beta\sqrt{1+h} \geq 0.$$

The INVARIANT actual two/three cycles are

$$L = 4\lambda^2, \quad W = 4\beta^2, \quad T = 8\beta\sqrt{1+h}\lambda, \qquad L, W, T \in \mathcal{O}_K.$$

Zero β or λ is included. Direct expansion cancels the mixed radical:

$$L + T = VA^2 - W(1 + V/4). \tag{Rankonecycle}$$

No individual unit coefficient or disconnected raw time is assumed K-valued.

For an isolated-within-A cap, $v_A = \gamma\omega_i$, $B = 4\gamma^2$, this gives the exact full-foreign identity

$$L + T + V + W = \frac{i(16-i)BV}{32} \in \mathcal{O}_K. \tag{A15isocycle}$$

At EVERY nonphysical embedding, positive definiteness on the whole old A_{15} and this cap gives $0 < B^\sigma < 4/D_i$; its additional foreign square is nonnegative and only strengthens the bound. Together with $-4 < V^\sigma < 0$, this gives $|B^\sigma V^\sigma| < 16/D_i$. For one white-three contact, $v_A = \gamma\omega_i + \omega_j/2$, $t = 2\gamma$, and

$$M = i(16-i)t^2 + 2\min(i,j)(16 - \max(i,j))t + j(16-j) - 32,$$

the corresponding identity is

$$L + T + W = MV/32 \in \mathcal{O}_K. \tag{A15whitecycle}$$

Thus the foreign radical is retained through its triangle term. Dropping that term would not prove these congruences.

C.5.3 Finite native parameters and all dyadic places

For $a = 2\cos(\pi/m)$, $m \geq 4$, in any containing field, every dyadic valuation normalized by $v(2) = 1$ satisfies

$$0 \leq v(a), v(a+1) \leq 1/2.$$

Here is the full root-of-unity argument. Write $a = \zeta + \zeta^{-1}$, ζ primitive $2m$-th. Prime-to-two roots keep their order on reduction: a nontrivial odd-order root reducing to one would make its zero geometric sum reduce to a nonzero odd integer. Thus $v(a) > 0$ occurs only when $2m = 2^s$. For $s \geq 3$, $v(a) = v(\zeta^2 + 1) = 1/2^{s-2} \leq 1/2$. The formula follows because conjugate primitive 2^r-root differences have equal valuation and product two. Likewise

$$a + 1 = \zeta^{-1}(\zeta^3 - 1)/(\zeta - 1)$$

has positive valuation only when $2m = 3 \cdot 2^s$; for $m \geq 4$, $s \geq 2$, and $v(a+1) = 1/2^{s-1} \leq 1/2$. For $m = 3$, $a = 1$ and $v(a+1) = 1$; that exception is removed by an actual affine cycle BEFORE this lemma is used. In particular $B = 4\cos^2(\pi/m) = a^2$ satisfies $v(B) \leq 1$ for every finite $m \geq 3$. If $a \notin K$, use the cyclotomic compositum and descend the normalized valuation of $B \in K$. These bounds persist under every actual field extension.

For a finite one-white-three own, the actual three-cycle along the old path excludes own three. For every other finite own, apply the bounds to t. Put $a_i = i(16 - i)$ and $C = \min(i, j)(16 - \max(i, j))$. The following four parity cases prove $v(M) \leq 3$ at EVERY dyadic place.

If i, j are odd, rewrite in $s = t + 1$:

$$M = a_i s^2 + 2(C - a_i)s + [r(16 - r) - 32], \qquad r = |i - j|.$$

The last two coefficients are divisible by four and a_i is odd. The leading term has valuation at most one, while the others have valuation at least two. If i is odd and j even, use t instead: a_i is odd, C even and a_j divisible by four. The same unique-minimum argument gives $v(M) \leq 1$. If i is even and j odd, the constant is odd and the other coefficients are even, giving $v(M) = 0$. If both are even, $i = 6, 10$ makes $k = i/2$ odd. With $l = j/2$,

$$M/4 = k(8 - k)t^2 + 2\min(k, l)(8 - \max(k, l))t + l(8 - l) - 8.$$

For even l use the t-argument; for odd l use $s = t + 1$, whose constant $r(8 - r) - 8$, $r = |k - l|$, is divisible by four. Thus $v(M/4) \leq 1$ and $v(M) \leq 3$. No midpoint-eight assertion has been made.

Equation (A15whitecycle) now forces $v(V) \geq 2$ at every dyadic place. At all other finite places V is already integral. Hence $V/4$ would be integral, impossible. This excludes EVERY finite one-white-three cap at the six required sites, including foreign contacts and both orientations. The same proof applies to a dotted t whenever both $v(t), v(t+1) \leq 1/2$ have been established.

C.5.4 The fresh six-site inventory at losses .49, .496

At each $i \in J$, both retained old paths have ranks at least four. Their COMPLETE finite-white list is no old-A contact; one three at an admissible end or D/E position; one four at a path end; or one endpoint three in each path. The two-path bridge has rank fifteen, so it is A_{15}: a third arm through the new root would have two old arms at least four, incompatible with finite D, and no exceptional finite type has this rank. The old e-contact β is retained throughout. It only restricts the white list further.

Own zero is not silently discarded. Without old-A contact its time $Q - 1$ is negative. For a white three at an internal old node, the full old diagram and cap have actual affine arms $(1, 2, 5)$ or

$(1,3,3)$, unless the contact is at $1,2,14,15$. At global ends one or fifteen, the full finite white path has rank at least five; adding e allows only $\beta = 0, 1/2, 1/\sqrt{2}$. Then $A^2 = 15/32 + Q - 1 < 0$. At two or fourteen, $\beta = 0$ gives time $-1/8$. A positive foreign contact in that finite branched white component forces $\beta = 1/2$, and the full actual roots then have the affine arms $(1,2,5)$. An internal white four has the $4/3/3$ star. At a global end white four, any positive e-contact would put a three BEFORE the four in a path of rank at least six, or make a nonfinite $4,4$ principal; hence $Q = 0$ and time $15/16 - 1 < 0$. Two white threes at own zero give the actual all-three cycle. Every null is a positive sum of actual roots, future through the cap time, and nonpositive against ALL omitted walls.

For every positive own, the parallel value is excluded BEFORE the remaining case dispatch: $c + n_i$ is its actual feasible future null, including positive foreign contacts. For white three on a side of order $m = i$ or $16 - i$, number its node j from that side's far end. The actual residual and numerator are

$$\delta = 1 - j(m-j)/(2m), \qquad \sigma = \gamma + j/(2m).$$

For white four at an end they are $\delta = 1/m$, $\sigma = \gamma + j/(\sqrt{2}m)$. For two white threes, $\delta = 1/D_i$, $\sigma \geq \gamma + 1/D_i$. The ENTIRE old-$e$ cost changes the denominator to $\delta - Q$, and

$$C_i = \frac{\sigma\sqrt{D_i}}{\sqrt{\delta - Q}}.$$

If the desired ℓ_i-pressure fails, this implies

$$\frac{\sigma}{\sqrt{\delta - Q}} \leq R_{\ell_i}(\sqrt{D_i}) < 1.881. \tag{A15ratio}$$

For completeness the ratio target increases with q and decreases with ℓ on $q \leq \sqrt{8} < 2.829$, $\ell \geq .49$. At $q = 2.829, \ell = .49$, positive denominator clearing gives the strict margin119649968. Thus this is a fresh changed-loss inventory, rather than reuse of the old .601 weak list.

Finite one-white-three owns have already been excluded by the invariant parity argument, including own three's affine cycle. Thus only DOTTED $\gamma > 1$ remains there. At a D-position the ratio is $> (1 + 1/m)/\sqrt{2/m} \geq 6/\sqrt{10} > 1.881$. At an E-position $\delta \leq 1/4, \sigma > 1$, stronger still. Foreign cost decreases the denominator and cannot weaken this. For near white four, every positive own gives ratio

$$\geq (1/2 + 4/(5\sqrt{2}))\sqrt{5} > 2.38.$$

For far white four, owns three, four and six have actual affine principals using old roots at most three steps on each side of i, avoiding the helper at one or fifteen. The remaining own is at least $\varphi/2 > .809$, giving $.809\sqrt{5} + 1/\sqrt{10} > 2.12$. For two white threes, own three has an actual cycle. Every other positive own is at least $1/\sqrt{2} > .7$, and its ratio exceeds $.7(2.6) + .35 > 1.881$. Every finite-white role except dotted near/far three has now been included and removed.

C.5.5 Every remaining dotted helper fails over every field

For near white three the ratio is

$$\frac{\gamma + (m-1)/(2m)}{\sqrt{(m+1)/(2m) - Q}}.$$

At $m \geq 7, \gamma > 1$, it exceeds $10/\sqrt{28} > 1.881$. For $m = 5, 6$, weakness gives $2 < t = 2\gamma < 2.11555 < 53/25$. These are $i = 5, j = 4$, $i = 6, j = 5$, and reflections. Lift EVERY nonidentity embedding of $\mathbb{Q}(t)$ to K. Strict full-old-A positivity gives respectively

$$(55/8)t_\sigma^2 + 11t_\sigma + 2 < 0, \qquad (15/2)t_\sigma^2 + (25/2)t_\sigma + 23/8 < 0.$$

The foreign square is nonnegative, strengthening these conditions. Both negative quadratic intervals lie in $(-1.4,-.2)$; check positive endpoint values at $-7/5,-1/5$.

If the minimal degree is at least three and some small conjugate is at least -1, the nonzero integral norm of $t(t+1)$ has magnitude

$$< 6.6144(1/4)(.56)^{d-2} \le .926016 < 1.$$

If all small conjugates are below -1, the norm of $t+1$ has magnitude $< 3.12(.4)^{d-1} < 1$. In degree two the integral trace is one, and the norm $t(1-t)$ lies strictly between $-3,-2$. In degree one no integer lies in $(2,2.12)$. Thus every weak near dotted helper is impossible, including foreign contacts and proper actual fields.

For far white three, if $\gamma \ge 34/25$ its squared ratio is at least its $m=5$ value $5329/1500 > 1.881^2$. To check the minimum, put $r=1/m$. The derivative of $2(\gamma+r/2)^2/(1+r)$ has sign $1-\gamma+r/2<0$, and the ratio increases with γ. Hence a weak far helper has $2<t<68/25$. Conjugate positivity gives the entire strip

$$-4/5 < t_\sigma < 12/25.$$

The N16 endpoint cost checks at $-.8,.48$ have margins $3/40, 59/1000$; they increase over the reflected half-window. Again positive foreign cost only strengthens them.

For $f(t)=t(t^2-1)$, small conjugate magnitude is at most $2/(3\sqrt{3})$. Degree at least four would give norm magnitude

$$\frac{68}{25}\left\{\left(\frac{68}{25}\right)^2-1\right\}\frac{8}{81\sqrt{3}}<1.$$

For an explicit positive seal, eight times the physical factor is139.229184, below $81(1.732)=140.292$. Thus the minimal degree is at most three. At degree three, $|\mathrm{N}(t)| < 2.72(.8)^2 < 2$, so it is a unit. Two positive small conjugates would give norm below one. For opposite signs their absolute product $1/t$ forces

$$t > 125/48, \quad t_{\text{positive}} > 125/272, \quad |t_{\text{negative}}| > 625/816,$$

putting the integral trace strictly between two and three. For two negative conjugates, the trace is one and $t<2.6$. The integral second coefficient $1/t-t(t-1)$ lies in $(-4,-1.5)$, hence is -2 or -3. The two cubics are X^3-X^2-2X-1, of discriminant -31, and $X^3-X^2-3X-1=(X+1)(X^2-2X-1)$. Neither is the minimal polynomial of the totally real cubic. Degree three is excluded.

Degree two has trace two or three. Trace two has norm in $(-1.9584,0)$, hence -1; trace three has positive norm below two, hence one. The exact possibilities are

$$t=1+\sqrt{2}, \qquad t=\varphi^2.$$

Degree one is the forbidden parallel two. For $1+\sqrt{2}$, t is a dyadic unit and $v(t+1)=1/2$. For φ^2, both t and $t+1=\varphi+2$ have odd norm and are dyadic units. These facts persist in every actual field extension. Thus the already proved four-parity argument applies to M for BOTH remaining dotted possibilities. Equation (A15whitecycle) forces $V/4$ integral, a contradiction. All weak dotted helpers are excluded.

C.5.6 The isolated central caps through the full $h<1$ interval

For isolated-within-A contact, use (A15isocycle). At odd sites $i(16-i)$ is coprime to32, so integer Bezout gives $BV/32 \in \mathcal{O}_K$. At six and ten, the gcd is four, giving $BV/8 \in \mathcal{O}_K$. For ANY finite own the dyadic bound $v(B)\le 1$ then forces $v(V)\ge 4$ or ≥ 2, respectively. Thus every finite isolated own

would make $V/4$ integral, impossible; own zero has negative time. Parallel was already excluded globally by its actual null.

For dotted own, $u = (1-Q)/\gamma^2 \in (0,1)$. Its positive ACTUAL core $c + \beta e + \gamma u n_i$ has square $-\gamma^2 u(1-u)$, positive time, zero c, e products, negative n_i product and all other wall products nonpositive. It gives

$$h_i > \sqrt{D_i - u} - \sqrt{u(1-u)}, \qquad \text{loss} \le au + \sqrt{u(1-u)},$$

with $a < .2$ at the odd sites and $a < .19$ at six and ten. If $u \le 1/4$ at an odd site, the loss is $< 1/20 + \sqrt{3}/4 < 39/80 < .49$. Thus an odd weak cap forces $u > 1/4$ and $B < 16$. The integral $BV/32$ has physical magnitude < 2 and EVERY other magnitude $< 1/(2D_i) \le 4/55$. Since the actual degree is at least two, its nonzero norm has magnitude $< 8/55 < 1$, impossible.

At six or ten, for $u \le .27$ the loss is $< .0513 + .444 = .4953 < .496$. An even weak cap therefore has $u > .27$, $B < 400/27$. The integral $BV/8$ has physical magnitude $< 200/27$, and all other magnitudes $< 4/15$. At degree at least three its norm is $< (200/27)(4/15)^2 = 128/243 < 1$. At degree two, instead use the rational integer norms:

$$0 < \mathrm{N}(B) < 640/81 < 8, \qquad 0 < |\mathrm{N}(V)| < 16.$$

The first is positive and the second negative before taking absolute values. Integrality of $BV/8$ requires64 to divide their product. But an integer from one through seven has dyadic valuation at most two, and one through fifteen at most three. Their sum is less than six, a contradiction. This argument permits arbitrary ramification and full maximal order.

Combining the complete own-zero, finite-white, finite, parallel, dotted-helper and isolated cases proves, for EVERY $i \in J$,

$$h_i > \sqrt{D_i} - \ell_i.$$

There is still NO midpoint-eight assertion.

C.5.7 One final reference for every isolated rank-one height

Choose the rational heights

$$H_5 = H_{11} = 2.132, \quad H_6 = H_{10} = 2.2426, \quad H_7 = H_8 = H_9 = 2.3162$$

and linear outer tails. Their six positive constraints lie strictly below the just proved pressures by direct squares at $55/8, 15/2, 63/8$. The midpoint coefficient is zero, the slopes decrease, and

$$E = 2.132^2/5 + .1106^2 + .0736^2 = .92673412.$$

For $1/2 \le h < 1$, the separate rank-one edge energy is at least $5 - 2\sqrt{6} > .1$; hence the original norm exceeds one. Together with the earlier $h \ge 1$ argument this leaves only $0 < h < 1/2$.

We retain BOTH actual A_{15} endpoint pressures in that case. Put

$$T = \sqrt{1+4h} - 2\sqrt{h}, \qquad u = \frac{\sqrt{h}}{\sqrt{1+h}+1}, \qquad \alpha = \sqrt{15/16}.$$

The actual rank-one edge gives energy $\ge u^2$, $0 < u < 1/3$, and direct substitution gives $(1-u^2)(1-T^2) = 8uT$. The rank-free actual-I wall comparison (20.1) applies with every foreign contact retained. Explicitly, for another actual opposite with time A and old-e contact $\beta = 0$ or $1/2 \le \beta < 1$, its chosen spatial derivative square is at most $1 + A^2 - \beta^2$. For $\beta = 0$, the native gap

gives $A^2 \ge 1/(4h)$; for $\beta > 0$, acuteness gives $A^2 \ge (1+h)\beta^2/h$. Both imply full tangent square $\ge 1/(1+4h)$. In the positive-contact case the comparison, after positive denominators are cleared, has numerator $h((5+4h)\beta^2 - 1) > 0$. Thus its normalized half-edge height is at least T. At BOTH signed A_{15} endpoints this gives $x_1, -x_{16} \ge \alpha T$. This is the original wall comparison, not a new field assumption.

The first and last five-coordinate reference means are $b = 2.132/(5\sqrt{2}) < .30152 < .302$. Since $T > \sqrt{3} - \sqrt{2} > .317$ and $\alpha > .968$, $\eta = \alpha T > b$. Graft BOTH disjoint blocks to η. Their other coordinates $(5b - \eta)/4 > 1/8$ exceed the next old coordinate $.1106/\sqrt{2} < .079$. Thus all dual coefficients remain nonnegative, supported only on the six proved pressures and the two ACTUAL endpoint bounds; the midpoint coefficient remains zero. The single rebuilt reference gives

$$e_A > E + (5/2)(\eta - b)^2.$$

For completeness the scalar barrier $T + 3u/2 > .76$ follows directly from the displayed relation. At $T_0 = .76 - 1.5u > 0$, its left-minus-right is bounded below by

$$264/625 - (19/5)u + (21419/2500)u^2 > 0.$$

The quadratic discriminant is $-55964/1562500 < 0$; the cubic term was bounded using $u < 1/3$. The relation decreases strictly with positive T, proving the barrier. With $D = \eta - b > 0$,

$$D + (3/2)\alpha u > .76\alpha - b > .43368.$$

Weighted Cauchy gives

$$(5/2)D^2 + u^2 > \frac{.43368^2}{2/5 + (9/4)\alpha^2} = \frac{.43368^2}{803/320} > .074.$$

The positive comparison is $.1880783424 > .18569375$. Adding the original rank-one energy ONCE to the rebuilt A_{15} reference yields

$$\|x\|^2 > .92673412 + .074 = 1.00073412 > 1.$$

All $h > 0$, including boundaries $1/2, 1$, are excluded for the isolated opposite.

Together with the complete nonisolated argument, this excludes the ENTIRE original $A_{15} + A_1$ family, with either ordinary nonshort rank one or actual short I. The dichotomy concerns one actual wall throughout, and the two case contradictions were never added to one another. No historical frontier-count subtraction is needed. The separate B_{15} nonisolated corollary only supplies its isolation reduction; its isolated ordinary and actual-I branches remain to be proved.

C.5.8 A fresh two-tail comparison for D_{15}

We first supply the missing body estimate for $D_{15} + A_1$. The conclusion is an energy bound, rather than unconditional strength of the last two knots. Use actual unit roots

$$n_j = (e_j - e_{j+1})/\sqrt{2} \quad (1 \le j \le 14), \qquad n_{15} = (e_{14} + e_{15})/\sqrt{2}.$$

For $P_i = e_1 + \cdots + e_i$, the long dual is $\Omega_i = \sqrt{2}P_i$, $i \le 13$. The two fork duals are

$$\Omega_{14} = (P_{14} - e_{15})/\sqrt{2}, \qquad \Omega_{15} = (P_{14} + e_{15})/\sqrt{2}, \qquad \|\Omega_{14}\|^2 = \|\Omega_{15}\|^2 = 15/2.$$

The old rank-one unit root e is orthogonal to this space. Retain the SAME original decomposition

$$z = E(p - x - ye), \quad x_1 > \cdots > x_{14} > |x_{15}|, \quad e_D = \|x\|^2, \quad e_1 = y^2, \quad e_D + e_1 < 1.$$

Set $f_i = \sqrt{i} - .61/\sqrt{2}$ and

$$E_K = f_5^2/5 + \sum_{j=5}^{K-1} (\sqrt{j+1} - \sqrt{j})^2.$$

The shared geometric bulk inventory and Lemma 20.3 give strong $6, \dots, 10$ and $e_D > E_{10} > .82436$, with either initial-five reference. The generic parameter is $M = 15 - 6 = 9$, so its last established index is ten. Eleven and twelve need a fresh check.

For their actual opposite write

$$c = Ap - v_D - \beta e, \quad A > 0, \quad \gamma = -c \cdot n_i \geq 0, \quad Q = \beta^2 < 1.$$

The full retained white Gram is positive definite. Every nonzero foreign cost satisfies $Q \geq 1/4$. If the inverse calculation gives residual $\delta > 0$ and $\cosh L_i = \sqrt{2i}\,\sigma/\sqrt{\delta}$, the original edge comparison makes the index strong if

$$\frac{\sigma}{\sqrt{\delta}} > T(\sqrt{2i}), \qquad T(q) = \frac{100}{61} - \frac{q - .61}{q(2q - .61)}.$$

On this interval T increases and $\sqrt{2i} < 5$, so

$$T(\sqrt{2i}) < T(5) = \frac{442721}{286395} < \frac{31}{20};$$

the final cleared numerator is $23825 > 0$. This actual threshold, rather than $100/61$, retains the full bridge list at this rank.

Deleting eleven leaves $A_{10} + D_4$, with long leaf twelve, centre thirteen and forks fourteen and fifteen. Deleting twelve leaves $A_{11} + D_3$, with middle thirteen and endpoints fourteen and fifteen. Finite classification gives the COMPLETE list: on the left, endpoint-three, endpoint-four or next-to-end-three; on D_4, simple three at any of its three leaves; on D_3, middle-three or endpoint-three, four or five. A two-piece bridge is simple three at a left endpoint and any listed simple-three right contact. Both left orientations and both fork orientations are retained. There is no exceptional finite rank-fifteen diagram. A bridge cannot have an extra old-e contact: that adds two forks to its D_{15} completion or the nonfinite arms $11, 3, 1$ to its A_{15} completion. Thus ONLY for bridges $Q = 0$.

For a left simple three at j, counted from its far end,

$$\sigma = \gamma + \frac{j}{2i}, \qquad \delta = \frac{i+1}{2i} \text{ at endpoints}, \qquad \delta = \frac{2}{i} \text{ at next-to-end sites.}$$

For endpoint four, $\sigma = \gamma + j/(\sqrt{2}i)$, $\delta = 1/i$. Foreign Q subtracts from every displayed residual. The remaining exact inverse-Gram formulas are

right contact	δ before Q	σ
D_4 long three or D_3 middle three	$1/2$	$\gamma + 1/2$
D_4 fork three	$1/2$	$\gamma + 1/4$
D_3 endpoint three	$5/8$	$\gamma + 1/4$
D_3 endpoint four	$1/4$	$\gamma + 1/(2\sqrt{2})$
D_3 endpoint five	$(5 - 3\varphi)/8$	$\gamma + \varphi/4$

and

i, right contact	δ	$\sigma-\gamma$ for near/far left endpoint
11, long three	1/22	21/22, 6/11
11, fork three	1/22	31/44, 13/44
12, middle three	1/24	23/24, 13/24
12, endpoint three	1/6	17/24, 7/24.

The middle-three D_{15} bridge is retained separately; treating D_3 only as an end-contact path would miss it.

Every affine exclusion below applies with arbitrary foreign contacts. Its positive combination of actual roots has positive future time and nonpositive product with every unselected original wall termwise. It would be a feasible null direction, impossible in a compact polytope. For a single three at global long node $j \geq 2$, own zero has the affine D principal on the cap, n_{j-1}, trunk $n_j, \dots, n_{13}$, and BOTH forks; the trunk coefficients are two and the four terminal coefficients one. At a fork, own zero has affine E_8 arms $1, 2, 5$: the other fork is the length-one arm and the contacted fork plus cap is the length-two arm. Far-three at one with own zero has $A^2 = Q - 1/2$ and $Q < (i+1)/(2i) \leq 6/11$; the only native costs are $0, 1/4, 1/2$, so $A^2 \leq 0$. Far-four with own zero forces $Q < 1/i < 1/4$, hence $Q = 0$ and zero time. Near-four with own zero has the affine $4, 3, 3$ star. Right endpoint-four with own zero has affine B_4 kernel $(\sqrt{2}, 2, 2, 1, 1)$ on the cap, contacted fork, centre, old twelve and other fork. Right endpoint-five remains in the direct estimates.

Own three with a single-three helper has the actual all-three affine cycle. Own six has the affine $6, 3$ path using an old neighbour other than the helper. Own four with a right fork helper has affine F_4 path $n_{i-2}, n_{i-1}, n_i, c,$ fork, whose bonds are $3, 3, 4, 3$. With a nonadjacent left helper it has the $4, 3, 3$ star. The ADJACENT near-three/own-four profile is retained; it is not declared an affine star. An isolated-within-D cap has the affine D, star and $6, 3$ principals at owns three, four and six. Parallel own always has the actual feasible null $c + n_i$.

The remaining direct estimates are exhaustive. Right long-three and middle-three have ratio at least $1 + 1/\sqrt{2} > 31/20$. Right fork-three and endpoint-three have $\gamma \geq \varphi/2 > .8$; if $Q > 0$, their largest residual $3/8$ gives $1.05/\sqrt{3/8} > 31/20$. Positive-own right endpoint-four has ratio at least $1 + 1/\sqrt{2}$. Endpoint-five has residual $< .02$ and numerator $> .4$, hence ratio > 2.8, even for own zero. Left near-three with $\gamma \geq \varphi/2$ has ratio

$$\frac{\sigma}{\sqrt{\delta}} > \frac{.8 + 5/11}{\sqrt{6/11}} > 31/20.$$

Its extra own-four case has $\gamma > .7$, $\sqrt{\delta} < .74$, ratio $> 635/407 > 31/20$. Next-to-end-three has $\delta \leq 2/11$ and $\gamma > .8$. Positive-own endpoint-four has ratio at least $\sqrt{11}/2 > 31/20$. Every positive-own bridge has ratio > 1.9: even its weakest numerator is $19/24$ over $\sqrt{1/6}$. Own-zero bridges have the actual affine cycle.

The only unresolved left far-three is the literal packet $v_D = \gamma\Omega_i + \Omega_1/2$. Weakness gives $\gamma < (31/20)\sqrt{6/11} < 1.147$; the own removals give $202/125 < \varphi \leq 2\gamma < 12/5$. Lemma 20.3 excludes it at eleven and twelve, including foreign contacts. That lemma is not applied to the right-fork packet.

We also need all isolated-within-D caps, without assuming $Q = 0$. Their remaining own satisfies $\gamma \geq \varphi/2$ by the ALL-foreign-cost affine removals already given. Put $D_i = 2i$ and $u = (1 - Q)/\gamma^2$. For $0 < u < 1$, the actual positive core

$$\xi = c + \beta e + \frac{1 - Q}{\gamma} n_i$$

has selected products zero except $\xi \cdot n_i = \gamma(u-1) < 0$, square $-\gamma^2 u(1-u)$, and positive time. The original maximum gives, using $x \cdot n_i > 0$,

$$x \cdot \Omega_i > \sqrt{D_i - u} - \sqrt{u(1-u)}.$$

Since $q_i - \sqrt{D_i - u} < .108u$, its loss is at most

$$\frac{.108 + \sqrt{1 + .108^2}}{2} < .557 < .61.$$

At $u = 1$ the same core is a forbidden feasible null. For $Q > 0, u > 1$, positively project onto the ACTUAL finite set c, n_i, e, with residual $\gamma^2(u-1) > 0$. All correction coefficients are positive; clearing the old-e contact leaves a pure-D displacement. Here

$$u \le \frac{3}{\varphi^2} = 6 - 3\varphi < 7/6, \qquad \sqrt{u-1} \le 2 - \varphi < .4, \qquad q_i - \sqrt{D_i - u} < .13,$$

so the full loss is $< .53$.

For $Q = 0$, native five uses the LEFT actual H_4 correction

$$w_5 = \gamma(3n_i/2 + n_{i-1} + n_{i-2}/2), \qquad \delta_5 = 1 - 3\gamma^2/2 > 0.$$

Its actual finite projection gives $x \cdot (\gamma\Omega_i - w_5) > A - \sqrt{\delta_5}$. The divided displacement is $3\Omega_{i+1}/4 + \Omega_{i-3}/4$; long-arm concavity bounds its pressure above by $x \cdot \Omega_i$. With $u = \gamma^{-2} < 25/16$, the loss is $< 25/144 + 1/4 < .61$. Native seven uses the positive LEFT A_2 negative core with correction $\gamma(4n_i/3 + 2n_{i-1}/3)$. Its divided displacement is $2\Omega_{i+1}/3 + \Omega_{i-2}/3$. Using $1.21 < u < 100/81$, its full loss is $< .14 + .36 = .5$. All finite natives at least eight use the actual singleton projection, with loss $< .14 + \tan(\pi/8) < .555$. Dotted own uses the preceding core and loss $< .557$. All corrections use original roots, with $i + 1 \le 13$; neither fork is treated as another long root.

Thus a new weak tail can ONLY be the pure packet

$$v_D = \gamma\Omega_i + \Omega_f/2, \qquad i = 11 \text{ or } 12, \quad f = 14 \text{ or } 15, \quad Q = 0.$$

Weakness gives $\gamma + 1/4 < (31/20)\sqrt{5/8} < 1.24$. For $t = 2\gamma$,

$$202/125 < t < 1.98 < 12/5.$$

Normalize the old D roots to raw square two over the ACTUAL field K. The actual simple-three fork contact permits the same normalization for the cap; its actual path cycle makes t integral. The raw inverse entries $i, i/2, 15/4$ give

$$4i\, t_\sigma(t_\sigma + 1) + 15 < 8, \qquad -1 < t_\sigma < 0$$

at EVERY nonphysical embedding. Every embedding of $k = \mathbb{Q}(t)$ extends to K. The nonzero integer $t(t+1)$ has physical value $< 204/25$ and other conjugates of magnitude at most $1/4$. Degree at least three would give a nonzero integral norm of magnitude $< 51/100$. In degree two the trace is one; trace two would require $t > 2$, contrary to the sharper weak strip. Then $-2 < t(1-t) < -.995456 < 0$ forces norm -1, and $t = \varphi$. There is no physical rational integer in the strip. The new weak tail is therefore native five. Arithmetic positivity by itself does not remove this packet.

For that packet put

$$\alpha = \sqrt{2}(\gamma + 1/4), \quad \alpha^2 = \varphi + 5/8 > 56/25, \quad \delta = (5 - 3\varphi)/8, \quad \frac{\sqrt{\delta}}{\alpha} < .091.$$

The last comparison follows, for example, from $\delta/\alpha^2 < 73/8972 < .091^2$. The full time square is

$$A^2 = i\alpha^2 - \kappa, \qquad \kappa = 1/2 \ (i = 11), \quad \kappa = 5/8 \ (i = 12).$$

The LEFT H_4 correction gives the actual finite point $p + (A/\delta)(c + w_5)$. It has positive future time, negative square and zero selected products. EVERY omitted original wall, including the OTHER fork and old e, pairs nonpositively termwise. For $Y = v_D - w_5$, $V = Y/\alpha$, $K = i - 3$, the SAME original maximum yields

$$x \cdot V > G = \sqrt{i - \kappa/\alpha^2} - .091, \qquad V_j = 1 \quad (1 \le j \le K),$$
$$V^2 = (A^2 + \delta)/\alpha^2 < i.$$

All fifteen coordinates are retained in this norm.

Choose the shared positive prefix reference w on its first $K = 8$ or 9 coordinates. If five is strong it uses knots $5, \dots, K$. Otherwise use the actual weak-five endpoint, strong six and terminal K. Both have $w^2 \ge E_K$, $w \cdot P_K \le f_K$, and terminal INDIVIDUAL coefficient respectively

$$\sqrt{K} - \sqrt{K-1} \quad \text{or} \quad \frac{1}{\sqrt{K} + \sqrt{6}},$$

both greater than $1/6$. Pay that coefficient BEFORE using the terminal floor: $W = w + (V - P_K)/6$. The FULL cross with w is zero since $V - P_K$ vanishes on its first K coordinates. Its residual norm is $< i - K = 3$. One square gives

$$e_D > E_K + \frac{G - f_K}{3} - \frac{1}{12}.$$

At eleven, $3.282^2 < 11 - 25/112$ gives $G > 3.191$; $f_8 < 2.398$, $E_8 > .76864$ yield $e_D > .94964$. At twelve, $3.423^2 < 12 - 125/448$ gives $G > 3.332$; $f_9 < 2.569$, $E_9 > .79805$ yield $e_D > .96905$. If either tail is weak its body bound is stronger than $E_{12} > .86989$. If neither is weak, the ordinary or weak-five reference through twelve gives this last bound directly. We have proved the UNCONDITIONAL conclusion

$$e_D > .86989. \tag{R1D}$$

We have not proved unconditional strength of eleven or twelve.

Let d now be the actual opposite of the old rank-one edge. If it contacts D_{15}, classification of the FULL white diagram $D_{15} + d$ permits only the D_{16} completion: simple three at the long endpoint. Thus

$$d = Ap - \Omega_1/2 - \gamma e, \qquad Q_D = 1/2, \qquad A^2 = \gamma^2 - 1/2.$$

For $\gamma < 1$, the positive actual projection

$$p + \frac{A}{1 - \gamma^2}(d + \gamma e)$$

is future and feasible against ALL original walls. All retained products vanish, and its omitted n_1 product is negative. It is a finite endpoint on old long edge one, so the SAME d closes both that edge and the orthogonal old rank-one direction. This contradicts the original-farthest orthogonal common-opposite obstruction proved above. For $\gamma = 1$, $d + e$ is the forbidden future feasible null. Hence $\gamma > 1$ and $e_1 > 3 - 2\sqrt{2} > .17$. Together with (R1D) this exceeds one. The ACTUAL rank-one opposite is isolated from the ENTIRE D_{15}.

C.5.9 One ordinary rank-one argument for B_{15} and D_{15}

For B_{15} use actual roots $n_j = (e_j - e_{j+1})/\sqrt{2}$, $j < 15$, $n_{15} = e_{15}$; for D_{15} retain the preceding roots. The long duals through thirteen agree. Their actual old rank-one opposites have been isolated by separate proofs and therefore have the form

$$d = \sqrt{h}\, p - \sqrt{1+h}\, e, \qquad h > 0.$$

Ordinary rank one is the range $h \geq 1/2$, INCLUDING its boundary, so its original edge gives

$$e_1 \geq q(h) = \frac{\sqrt{1+h}-1}{\sqrt{1+h}+1} > .1, \qquad q(1/2) = 5 - 2\sqrt{6} > .1.$$

This disjoint original energy is counted once.

The B_{15} short-layer inventory gives strong $6, \ldots, 13$, with $e_B > E_{13} > .88989$ in either initial-five case. If $h \geq .57$, then $\sqrt{1+h} > 5/4$ and $e_1 > 1/9 > .111$, a contradiction. Thus $h < .57$. For D_{15}, (R1D) instead gives $h < .7$: otherwise $e_1 > 3/23$, whereas

$$.86989 + 3/23 - 1 = \frac{747}{2300000} > 0.$$

In this ORDINARY branch only, either weak tail is now impossible: its pure-body bound $> .94964$ or $> .96905$ already exceeds the remaining $1 - .1$ budget. Thus D_{15} has strong $6, \ldots, 12$.

Before selecting the head, eliminate weak five in either component. Its actual pure-component point gives $x_1 > .5965$; strong six gives $S_6 > f_6 > 2.01815$. Interpolate through these pressures and EVERY later strong knot up to $J = 13$ in B, $J = 12$ in D. The first-six energy is

$$.5965^2 + \frac{(2.01815 - .5965)^2}{5} = .7600299945 > .7600.$$

Its middle slope is $> .2843$, above the next ordinary increment $< .197$, so all individual duals are positive. As $E_6 < .6971$, this SINGLE rebuilt reference gives $e_T > E_J + .0629$, $T = B, D$. Even for D this is $> .93279$; adding $e_1 > .1$ is impossible. Five is therefore strong.

Fix the positive ordinary reference w through all knots $5, \ldots, J$:

T	J	E_J lower bound	h upper bound
B	13	.88989	.57
D	12	.86989	.7

and

$$w_1 = \cdots = w_5 = b = f_5/5 < .361, \qquad \text{individual } P_5 \text{ coefficient} = b - (\sqrt{6} - \sqrt{5}) > .14.$$

No previous tail or weak-five gain is added to this w. For $.49 \leq X \leq .643$, replace its first five coordinates by $X, (f_5 - X)/4, \ldots, (f_5 - X)/4$. The other four stay $> .2904$, above the next coordinate $< .214$. Every individual dual remains nonnegative and later knots are unchanged. Whenever ACTUAL $x_1 > X$, the single reference square gives

$$e_T > E_J + \frac{5}{4}(X - b)^2 > E_J + \frac{5}{4}(X - .361)^2. \tag{R1G}$$

Let c be the ACTUAL opposite of old long edge one. It is distinct from d, which has no B/D component. Deleting one retains B_{14} or D_{14} and old e. The COMPLETE finite extension in

that connected white tree is no contact or simple three at its LONG endpoint two. For D_{14}, fork attachment gives nonfinite arms $11, 2, 1$, and an internal attachment adds a second fork. For B_{14}, another weighted bond or fork cannot give a finite classical extension. Two contacts to either tree form a nonpositive cycle; exceptional finite ranks are too small. Foreign contacts to e are still retained.

In the nonisolated branch, with $g = -c \cdot n_1$ and $Q = \beta^2 = (-c \cdot e)^2$, the unit inverse Gram gives

$$c = Ap - g\Omega_1 - \Omega_2/2 - \beta e, \quad A^2 = 2g^2 + 2g + Q, \quad C_1 = \frac{2g+1}{\sqrt{1-2Q}}, \quad Q < 1/2.$$

Own zero and $Q = 0$ has zero time. Own zero and $Q > 0$ has the ACTUAL positive null

$$c + \Omega_2/2 = \sqrt{Q}\, p - \beta e.$$

Its clearing expansions must be kept separate:

$$\frac{\Omega_2}{2} = \begin{cases} n_1 + 2(n_2 + \cdots + n_{14}) + \sqrt{2} n_{15}, & B, \\ n_1 + 2(n_2 + \cdots + n_{13}) + n_{14} + n_{15}, & D. \end{cases}$$

Thus BOTH D forks and the ACTUAL B short root have their correct coefficients. Selected products vanish; the old-e product and every unknown original-wall product are nonpositive. The null is future feasible, contrary to compactness. Own three has the affine triangle c, n_1, n_2. Every other positive own has $g \geq 1/\sqrt{2}$, so $C_1 \geq 1 + \sqrt{2}$ and $x_1 > \sqrt{\sqrt{2} - 1} > .643$. The legal graft (R1G) adds $> .099405$; even the D total is $.86989 + .099405 + .1 = 1.069295 > 1$. The head is therefore isolated WITHIN its entire classical component, but it may still contact old e.

Its remaining form is

$$c = Ap - g\Omega_1 - \beta e, \quad A^2 = 2g^2 + Q - 1, \quad C_1^2 = \frac{2g^2}{1-Q} = 1 + \frac{A^2}{1-Q}.$$

Because its actual finite endpoint retains e, the COMPLETE foreign alphabet splits into

$$Q = 0, \qquad Q = 1/4, \qquad 1/2 \leq Q < 1.$$

The last interval includes EVERY finite foreign label at least four; no typed-seven alphabet is imported. Actual acuteness with d gives

$$\lambda = -c \cdot d = A\sqrt{h} - \beta\sqrt{1+h} \geq 0, \qquad h(2g^2 - 1) \geq Q. \tag{R1N}$$

If $\lambda > 0$, the ACTUAL native gap gives $\lambda \geq 1/2$, including parallel and dotted mutuals. The squared inequality excludes $g < 1/\sqrt{2}$; equality requires $Q = A = 0$. Owns zero, three and four are therefore impossible. Own six has the affine $6, 3$ kernel $(\sqrt{3}, 2, 1)$ on c, n_1, n_2, and parallel own has the future feasible null $c + n_1$. Both apply with every foreign contact present. For $Q = 0$, native five has $A^2 = (\varphi - 1)/2 < .31$; even the weaker D cut $h < .7$ gives

$$0 < \lambda^2 = A^2 h < .217 < 1/4,$$

forbidden by the native gap. The remaining PURE owns are finite labels at least seven and the ENTIRE dotted interval.

With $u = g^{-2}$, their actual pressure table is

own	Z	H	$R = Z^2$
finite ≥ 7	$(1 - u/2)(e_1 + e_2) + (u-1)e_3$	$[\sqrt{2-u} - \sqrt{7u - 4 - 3u^2}]/\sqrt{2}$	$3 - 4u + 3u^2/2$
dotted	$(1 - u/2)e_1 + (u/2)e_2$	$[\sqrt{2-u} - \sqrt{u(1-u)}]/\sqrt{2}$	$1 - u + u^2/2.$

Both Z are nonnegative, sum to one, and use genuine first-three long-root coordinates. For the finite row use the positive combination

$$\xi = c + g^{-1}n_1 + 2(g^{-1} - g)n_2, \qquad \xi^2 = -(7 - 4g^2 - 3g^{-2}) < 0.$$

Its selected products are $0, 0, (3 - 4g^2)/(2g) < 0$. For the dotted row use $\xi = c + g^{-1}n_1$, of square $-(1 - u) < 0$. Both have positive time A. All unselected original walls, including e, d and BOTH D forks, pair nonpositively termwise. Their future normalizations and the SAME original maximum give $x \cdot Z \geq H$.

We prove the scalar estimate on the FULL domain:

$$.8H - .16R > .2893. \tag{R1S}$$

For native seven the elementary cubic gives $.9009 < g < .901$. Positive rational squaring gives $A > .7894$, $\sqrt{7 - 4g^2 - 3/g^2} < .2395$, $g\sqrt{2} < 1.2743$, hence $H > .4315$. Also $1.2318 < u < 1.233$, where R decreases, so $R < .349$, and the expression is $> .28936$.

For every finite native at least eight, $1 < u \leq 4 - 2\sqrt{2} < 1.172$. Put $D = (u - 1)(4 - 3u)$ and $a = 2\sqrt{2}/5$. For $F = .8H - .16R$,

$$F'(u) = a\left(-\frac{1}{2\sqrt{2 - u}} - \frac{7 - 6u}{2\sqrt{D}}\right) + .64 - .48u.$$

On $(1, 7/6]$ the first contribution is $< -.28$, the second is nonpositive and the last is at most $.16$. On $[7/6, 1.172)$, $D > .08066$, so $\sqrt{D} > .284$; the now positive second contribution is $< .032$, while the others total $< -.28 + .08$. Thus F decreases. At the native-eight endpoint, $\sqrt{2 - u} > .910$, $\sqrt{D} < .289$, $R < .373$, giving $H > .4388$ and $F > .29136$.

For dotted $0 < u < 1$, put $a = .8/\sqrt{2}$, so $.5656 < a < .5657$. For $u \leq 1/2$, concavity gives $\sqrt{2 - u} \geq 1.4142 - .379u$, $R \leq 1 - .75u$. Consequently

$$F > .63987152 - .0944u - .5657\sqrt{u(1 - u)} > .30587152.$$

Indeed $\sqrt{.0944^2 + .5657^2} < .5736$, and

$$bu + c\sqrt{u(1 - u)} \leq \frac{b + \sqrt{b^2 + c^2}}{2}.$$

For $u \geq 1/2$, use the chord $\sqrt{2 - u} \geq 1.4494 - .4494u$, $R \leq .75 - .25u$; then

$$F > .69978064 - .2142u - .5657\sqrt{u(1 - u)} > .29023064,$$

using $\sqrt{.2142^2 + .5657^2} < .6049$. The two chord endpoint inequalities are positive rational squarings. This covers the unbounded dotted-own interval and proves (R1S).

Use ONE vector

$$W = w + \frac{2}{5}(Z - P_5/5).$$

Its debit $2/25 = .08$ is below the INDIVIDUAL P_5 coefficient $> .14$, paid BEFORE the floor. The residual has zero sum on the first five coordinates, hence FULL cross zero with w, and norm $R - 1/5$. The actual pressure, remaining individual floors and one square give

$$\begin{aligned} e_T &\geq 2x \cdot W - W^2 \\ &> E_J + .8H - .16R - .8b + .032 \\ &> E_J + .2893 - .8(.361) + .032. \end{aligned}$$

Even for D this is $> .90239$. Adding original $e_1 > .1$ is impossible. This PURE displacement is never used for a mixed head.

For EVERY $Q \geq 1/2$, (R1N) gives

$$C_1^2 > \frac{1 + Q/h_*}{1 - Q}, \qquad h_* = \begin{cases} .57, & B, \\ .7, & D. \end{cases}$$

The expression increases on $[1/2, 1)$. For B it is at least $214/57 > 3$, so $C_1 > 41/25$, $x_1^2 > 8/33 > .49^2$. For D it is at least $24/7 > (37/20)^2$, so $x_1^2 > 17/57 > .54^2$. Their legal graft gains are

$$\frac{5}{4}(.49 - .361)^2 = .02080125 \quad (B), \qquad \frac{5}{4}(.54 - .361)^2 = .04005125 \quad (D).$$

Their original totals exceed 1.01069125 and 1.00994125, respectively. This covers ALL finite foreign labels at least four, finite/dotted head owns and zero or positive mutuals.

For $Q = 1/4, \lambda > 0$, the native gap gives

$$A\sqrt{h} \geq 1/2 + (1/2)\sqrt{1 + h} > 1.1.$$

For B, $1.45^2(.57) < 1.21$ gives $A > 1.45$, hence $C_1 > 1.95$, $x_1^2 > 19/59 > .56^2$. Its legal gain .04950125 is contradictory. For D, $1.3^2(.7) < 1.21$ gives $A > 1.3$, $C_1^2 > 244/75 > (9/5)^2$, $x_1^2 > 2/7 > .53^2$. Its legal gain .03570125 makes the total > 1.00559125.

Finally $Q = 1/4, \lambda = 0$ gives

$$g^2 = \frac{1 + 1/(4h)}{2}.$$

For B, $41/57 < g^2 \leq 3/4$; for D, $19/28 < g^2 \leq 3/4$. Thus $\cos(\pi/5) < .81 < g \leq \cos(\pi/6)$; more precisely $g > .84$ in B, $g > .82$ in D. The full native alphabet leaves only own six at the INCLUDED $h = 1/2$ boundary. Its actual affine null was already excluded. Parallel and dotted $g \geq 1$ are also impossible.

This covers every original wall, head own, foreign contact and mutual parameter for $h \geq 1/2$. Neither B_{15} nor D_{15} can occur with an ORDINARY nonshort rank-one component. The actual short-I cases $0 < h < 1/2$ still need their separate arguments; this ordinary bootstrap does not discard them.

C.5.10 The short-I cycle identity and the B_{15} long window

We now enter $0 < h < 1/2$. Keep the ACTUAL isolated opposite $d = \sqrt{h}\, p - \sqrt{1 + h}\, e$, and put $V = 4h$. Its actual two-cycle and strict nonphysical positivity give

$$V \in \mathcal{O}_K, \qquad 0 < V < 2, \qquad -4 < V_\sigma < 0 \quad \text{at EVERY nonphysical place.}$$

In particular $V/4$ cannot be integral: every conjugate of that nonzero number has magnitude < 1.

For ANY actual classical-edge opposite $c = Ap - v - \beta e$, put $Q = \beta^2 < 1$ and $A_0^2 = A^2 - Q = \|v\|^2 - 1$. Actual acuteness gives $\lambda = -c \cdot d = A\sqrt{h} - \beta\sqrt{1 + h} \geq 0$. The actual cycles

$$L_c = 4\lambda^2, \qquad W_I = 4\beta^2, \qquad T_c = 8\beta\sqrt{1 + h}\lambda$$

are integral. The full radical cancellation is

$$L_c + T_c + W_I = V(A^2 - \beta^2) = V A_0^2 \in \mathcal{O}_K. \tag{IC}$$

This retains $\beta = 0$, $\lambda = 0$ and every positive foreign contact. It does not assume integrality of an individual Cartan entry or identify disconnected root scales.

For B_{15}, strong $6, \dots, 13$ are already available. If five is weak, its COMPLETE geometric profile is pure:

$$v = \gamma\Omega_5 + \Omega_1/2, \qquad t = 2\gamma \in \mathcal{O}_K, \quad 202/125 < t < 12/5, \quad A^2 = (5/2)t^2 + t - 1/2.$$

Every other conjugate of t obeys $5t_\sigma^2 + 2t_\sigma - 1 < 0$, hence $-.69 < t_\sigma < .29$. Degree at least four contradicts $2.4(.69)^3 < 1$. In degree three the norm is a unit; a positive small conjugate would make its magnitude $< 2.4(.29)(.69) < 1$. Writing the other conjugates $-a, -b$, the trace is one: it is $> 1.616 - 1.38 > 0$ and at most $t - 2/\sqrt{t} < 2$. Then $a + b = t - 1 < 1.38$, $t < 2.38$, and the integer second coefficient $1/t - t(t-1)$ is -1 or -2. The corresponding polynomials

$$X^3 - X^2 - X - 1, \qquad X^3 - X^2 - 2X - 1$$

have discriminants $-44, -31$, so are not totally real. In degree two the norm is ± 1 and the trace one or two. Trace two gives $t(2-t) \in (-.96, .621)$, impossible; trace one gives $t = \varphi$. The only rational value, two, is an actual parallel null. Thus weak five is golden.

Its connected raw old-B/cap Gram is defined over $\mathbb{Q}(\varphi)$, with old long and cap squares two and old short square one. It is physically Lorentzian. A proper actual-field extension would give a nonphysical embedding fixing that indefinite principal. Hence ACTUAL $K = \mathbb{Q}(\varphi)$. Here $A^2 = (4 + 7\varphi)/2$, and $\mathrm{N}(4 + 7\varphi) = -5$ is odd. Equation (IC) makes $V/2$ integral at every dyadic place, and therefore everywhere. The strict strips force $\mathrm{N}(V) = -4$. For $T = V/2$, $\mathrm{N}(T) = -1$, $0 < T < 1$, $-2 < T' < 0$. Thus $T' = -1/T$, $T > 1/2$, and the integral trace $T - 1/T \in (-1.5, 0)$ is -1. It follows that

$$V = 2\varphi - 2, \quad h = (\varphi - 1)/2 > .3, \quad q(h) > .065.$$

But the same weak-five reference gives $e_B > E_{13} + .7600 - .6971 > .95279$. Adding $e_I > .065$ is impossible. Thus five is strong, without a long-head hypothesis.

We next classify the ACTUAL long-fourteen opposite. Its retained white pieces are A_{13} on one through thirteen, the genuine short root n_{15}, and old e. The COMPLETE finite possibilities within B are: isolation; left-only endpoint-three/four or next-to-end-three; short-only any finite native contact; or a left endpoint-three bridge to short-three/four. Two left contacts form a cycle. A next-to-end branch joined to short has nonfinite arms $1, 2, 11$; a left-four bridge puts its double bond internally in rank fifteen; short-five or larger cannot complete that long finite path. Every old-e contact remains, including an old-e three arm on a short-three bridge.

Write $\gamma = -c \cdot n_{14}$, $Q = \beta^2$. Weakness at fourteen implies

$$\frac{\sigma}{\sqrt{\delta - Q}} \le T(\sqrt{28}) < \frac{25}{16}.$$

Indeed $\sqrt{28} < 5.292$, T increases, and the rational comparison at 5.292 has positive cleared numerator 76368925.

For isolation, own zero has $A^2 = Q - 1 < 0$. Own three has the affine star on $c, n_{14}, n_{13}, n_{15}$ with kernel $(1, 2, 1, \sqrt{2})$; own four has the $4, 4$ null, own six the $6, 3$ null, and parallel has $c + n_{14}$. Every kernel is future feasible against ALL original walls, regardless of foreign contacts. For the rest put $u = (1 - Q)/\gamma^2$. For $0 < u < 1$, the actual core $c + \beta e + \gamma u n_{14}$ gives

$$x \cdot \Omega_{14} > \sqrt{28 - u} - \sqrt{u(1-u)}.$$

As $(\sqrt{28}+\sqrt{27})^{-1}<.1$, its loss is $<(.1+1.005)/2<.61$. The $u=1$ core is null. For $Q>0, u>1$, $Q\ge 1/4$, $\gamma\ge\varphi/2>.8$, so $u<75/64$. The actual finite projection onto c,e,n_{14} gives pressure $>\sqrt{27}-\sqrt{u-1}$, with loss $<.1+5/12<.61$. For pure finite owns at least seven the singleton projection has loss $<.1+\tan(\pi/7)<.6$; one may use $\tan(\pi/7)<1078/2159<1/2$. Native five uses the actual LEFT H_4 correction on n_{14},n_{13},n_{12}, with residual $(5-3\varphi)/8<.136^2$. Since $\gamma>.8$, $\gamma^{-2}<25/16$, its loss is $<5/32+.17<.61$. Thus all isolated fourteen profiles are strong.

For left-three at $j=1,13,2,12$,

$$\delta=1-\frac{j(14-j)}{28},\qquad \sigma=\gamma+j/28.$$

For left-four at $j=1,13$, $\delta=1/14$, $\sigma=\gamma+j/(14\sqrt{2})$. Subtract Q in each denominator. At left-three $j\ge 2$, own zero has the actual affine B kernel

$$c+n_{j-1}+2(n_j+\cdots+n_{14})+\sqrt{2}n_{15}.$$

At $j=1$ finite classification permits only $\beta=0,1/2,1/\sqrt{2}$, hence $A^2=Q-1/2\le 0$. Own-zero left-four uses the positive affine C kernel $c+\sqrt{2}(n_1+\cdots+n_{14})+n_{15}$ at one, and the local squared-cosine star $1/2,1/4,1/4$ at thirteen. For a left-three, own three has its actual affine cycle and own four the $4,4$ null. At far one, own six has the $6,3$ null; parallel always has its null pair. The remaining far weakness has precisely $\gamma\Omega_{14}+\Omega_1/2$ and $202/125<2\gamma<12/5$; Lemma 20.3 excludes it, including $Q>0$. At near thirteen every remaining own is at least $1/\sqrt{2}$, with ratio $>(.707+13/28)/.732>25/16$. At next-to-end two/twelve the ratio is $>(.7+1/14)/\sqrt{1/7}>25/16$. All positive-own left-four rows have ratio at least $(.5+1/(14\sqrt{2}))\sqrt{14}>25/16$. Thus every left-only role is strong.

For short-only contact $\eta=-c\cdot n_{15}>0$, introduce the actual integral cycles

$$B=4\gamma^2,\quad W=4\eta^2,\quad T=4\sqrt{2}\gamma\eta,\qquad A_0^2=7B+7T+15W/4-1.$$

Equation (IC) implies $WV/4$ integral, because fifteen is odd. For short labels three, five, six and seven, W is respectively $1,\varphi^2,3$, or the unit root of X^3-5X^2+6X-1. Each is a unit at every dyadic place. Thus $V/4$ is integral, impossible. This argument includes EVERY own, including zero, and every old-e contact. For short four, only $V/2$ follows. Own zero has the actual $4,4$ null. For own three, normalize the cap to raw square two; both products with old fourteen and fifteen are -1. Its rational old-B/cap principal is physically Lorentzian, with Schur residual -55. Hence actual $K=\mathbb{Q}$, making $0<V/2<1$ an integer, impossible. Every other positive own has

$$\frac{\sigma}{\sqrt{\delta-Q}}\ge\frac{\gamma+1/2}{\sqrt{1/2}}\ge 1+1/\sqrt{2}>25/16.$$

For every short label at least eight, even own zero has ratio at least $\cot(\pi/8)/\sqrt{2}=1+1/\sqrt{2}>25/16$. All higher finite and unbounded dotted owns are included.

A short-four bridge has residual $1/28$ and $\sigma=\gamma+j/28+1/2$, $j=1,13$; even own zero has ratio >2.8. For a short-three bridge,

$$v=\gamma\Omega_{14}+\Omega_j/2+\Omega_{15}/2.$$

Its original cycles make $a=2\gamma$ integral and $\sqrt{2}$ integral and K-valued. Directly

$$4A_0^2=M=28a^2+(4j+28\sqrt{2})a+2j+11+2j\sqrt{2}\in 1+2\mathcal{O}_K.$$

Equation (IC) gives $MV/4$ integral. Writing $M = 1 + 2Z$ and multiplying by $1 - 2Z$ makes $V/4$ integral. This is valid with arbitrary dyadic ramification; no quotient ring is presumed a field. It includes $a = 0$, both bridge orientations, and the retained old-e three arm.

We have proved strong EVERY $5, \dots, 14$ and

$$e_B > E_{14} > .9084, \quad b = f_5/5 < .361, \quad f_{14} < 3.311. \tag{BW}$$

The genuine reference has individual P_5 price $> .14$, terminal P_{14} price $\sqrt{14} - \sqrt{13} > .13$, and $w_{15} = 0$. Its first-coordinate graft is legal for $.49 \le X \le .67$: the other first-five coordinates are $> .2836 > .214$. Therefore

$$x_1 > X \quad \Longrightarrow \quad e_B > .9084 + \frac{5}{4}(X - .361)^2. \tag{BG}$$

C.5.11 The actual B_{15} head in the short-I branch

The finite B_{14} head list and the own-zero/triangle clearings in Section C.5.9 apply without any ordinary-h premise. A nonisolated head would have $x_1 > .643$. Now (BG) alone gives $e_B > .9084 + .099405 > 1$, without charging I. Thus the ACTUAL head is isolated within B_{15}:

$$c_H = A_H p - g_H \Omega_1 - \beta_H e, \quad A_H^2 = 2g_H^2 + Q_H - 1, \quad Q_H = \beta_H^2.$$

As before,

$$\lambda_H = A_H \sqrt{h} - \beta_H \sqrt{1+h} \ge 0, \qquad (2g_H^2 - 1)h \ge Q_H.$$

This removes owns zero, three and four. The actual $6, 3$ and parallel nulls remove six and one. Pure native five would give $0 < \lambda_H^2 < .31(.5) < 1/4$, impossible.

For pure finite own at least seven, use the actual finite head point and the SINGLE paid reference of Section C.5.9:

$$e_B > E_{14} + F(u) - .8b + .032, \qquad F(u) = .8H - .16R > .2893.$$

This is not a separately optimized head energy. Since $g_H < 1$, its positive mutual satisfies $\lambda_H < 1/\sqrt{2}$; the native alphabet forces $\lambda_H = 1/2$. Thus $h = [4(2g_H^2 - 1)]^{-1} > 1/4$. At native seven or eight, $h \ge \sqrt{2}/4$, so $q(h) > .0755$: use $\sqrt{2} > 1.4142$, $\sqrt{1+h} > 1.1634$ and $.1634/2.1634 > .0755$. The original total is then > 1.0164. For all finite natives at least nine, the largest-root cubic gives $g_H > .939$, hence $u < 1.135$. The already proved monotonicity of the finite scalar gives

$$F(u) > F(1.135) > .302416;$$

at that rational endpoint, $\sqrt{2-u} > .93$, $\sqrt{(u-1)(4-3u)} < .284$, $H > .4565$, $R < .3924$. Also $q(h) > .0557$, since $\sqrt{1.25} > 1.118$. The single original total is > 1.009716. Thus the only pure head left is DOTTED.

For $Q_H \ge 1/2$, acuteness gives $C_H^2 \ge 2 + 1/h$. If $h < .2$, then $C_H^2 > 7$, $x_1 > .67$, and (BG) gives body > 1.02775125. If $.2 \le h < .5$, then $C_H > 2$, $x_1 > .57$, and $q(h) > .045$, using $\sqrt{1.2} > 1.095$. The original total is > 1.00800125. For $Q_H = 1/4, \lambda_H > 0$, native gap gives $A_H\sqrt{h} > 1$. For $h < .2$ this again gives $C_H^2 > 7$. For $h \ge .2$, use $\sqrt{1+h} > 1.08$ to get $A_H\sqrt{h} > 1.04$, so

$$C_H^2 = 1 + (4/3)A_H^2 > 1 + (4/3)(1.04^2/.5) > 1.95^2.$$

Then $x_1 > .56$; its graft plus $q(h) > .045$ makes the total > 1.00290125. These intervals cover all positive foreign labels and every positive finite/parallel/dotted mutual.

The only mixed possibility left is $\beta_H = 1/2, \lambda_H = 0$. For $U = 4g_H^2$ its exact relation is

$$V(U-2) = 2, \qquad U > 3. \tag{BQ}$$

Initially both finite and dotted own are retained here.

We shall need a point valid for BOTH the pure dotted and quarter-foreign heads. Put $\delta_H = 1 - Q_H$, $u = \delta_H / g_H^2$. For the pure dotted head $0 < u < 1$. For the quarter head, (BQ) and $0 < h < 1/2$ give $g_H^2 > 3/4$, again $0 < u < 1$. The positive ACTUAL combination

$$\xi = c_H + \beta_H e + \frac{\delta_H}{g_H} n_1$$

has square $-\delta_H(1-u) < 0$, products zero with c_H, e, and product $-g_H + \delta_H/g_H < 0$ with n_1. Its time is $A_H > 0$. All omitted original walls, including d, have nonpositive product termwise. Its old-e displacement cancels EXACTLY. Its normalized pressure is the SAME dotted-row Z, H, R used in (R1S), now genuinely derived for the mixed head as well.

If this quarter head had finite own, then $u > 3/4$, $h > 1/4$. For the dotted-row scalar on $[3/4, 1)$, the derivative comparison reduces to $-4u^3 + 13u^2 - 10u + 2 > 0$. This polynomial is $1/8$ at $3/4$ and increases; the remaining derivative term $.16(1-u)$ is positive. Hence

$$F(u) > F(3/4) = \frac{2\sqrt{10} - \sqrt{6}}{10} - .085 > .3024.$$

Together with $q(h) > .0557$ and the same paid head reference, the total exceeds 1.0097. Thus BOTH remaining head families are dotted.

For any actual short-fifteen pressure $S_{15} > T$, use the SINGLE vector

$$W = w + .4(Z - P_5/5) + .13 e_{15}.$$

The individual debits are .08 at P_5 and .13 at P_{14}, paid before using either floor. The head residual is zero-sum on the first five coordinates and hence orthogonal to w. The terminal e_{15} is orthogonal to both. The FULL square gives

$$e_B > E_{14} + F(u) - .8b + .032 + .26(T - f_{14}) - .0169. \tag{BT}$$

If the actual short cosh were at least fifteen, its original edge would give $S_{15} \geq \sqrt{105/8} > 3.62$, and the body alone would exceed $.9084 + .0325 + .06344 = 1.00434$. Thus $C_S < 15$. Also the legal $X = .632$ head graft would give $.9084 + (5/4)(.632 - .361)^2 = 1.00020125$; hence $x_1 < .632$, and the actual head comparison gives $C_H < 7/3$. These are necessary ACTUAL edge bounds, not cutoffs imposed on formal roots.

C.5.12 Every actual short-fifteen white profile

The finite endpoint of the actual short-fifteen edge retains A_{14} on one through fourteen and old e. Finite classification gives exactly SEVEN profiles, with both endpoint orientations kept:

contact in A_{14}	v_B	allowed β
none	γP_{15}	any finite native or zero
endpoint three $j = 1, 14$	$\gamma P_{15} + P_j/\sqrt{2}$	$0, 1/2, 1/\sqrt{2}$
next-to-end three $j = 2, 13$	$\gamma P_{15} + P_j/\sqrt{2}$	0
endpoint four $j = 1, 14$	$\gamma P_{15} + P_j$	0.

An extra old-e arm at a next-to-end branch gives nonfinite long E arms. At a white four it puts the double bond internally in rank sixteen. An endpoint-three old-e arm permits only three or four, giving A_{16} or B_{16}. Thus the displayed foreign list is complete. The own $\gamma = -c \cdot n_{15}$ is initially zero, finite native, parallel or dotted.

Own zero is impossible in every row. Isolation has $A^2 = Q - 1 < 0$. Endpoint-three at one has $A^2 = Q - 1/2 \le 0$. For every other white-three $j \ge 2$, use the actual affine B null

$$c + n_{j-1} + 2(n_j + \cdots + n_{14}) + \sqrt{2}n_{15}.$$

For white-four at one use $c + \sqrt{2}(n_1 + \cdots + n_{14}) + n_{15}$; at fourteen use $c + \sqrt{2}n_{14} + n_{15}$. Every selected product vanishes; its positive cap coefficient gives future time, and all unselected original products are nonpositive termwise. Parallel own has the universal null $c + n_{15}$.

For positive own the time squares, before adding Q, are

profile	A_0^2
isolated	$15\gamma^2 - 1$
white three, $a = 2\sqrt{2}\gamma$	$(15a^2 + 4ja + 4j - 8)/8$
white four, $t = 2\gamma$	$(15t^2 + 4jt + 4j - 4)/4.$

These are positive for every $\gamma \ge 1/2$. Normalize the old B long roots to square two and short root to square one. A white-three cap has square two, white product -1 and short product $-a/2$; a white-four cap has square one, white product -1 and short product $-t/2$. These scalings follow from the actual native bonds over K; no disconnected scale is guessed. The old-B/cap principal is physically Lorentzian and defined over $\mathbb{Q}(a)$ or $\mathbb{Q}(t)$. A proper actual-field extension would fix that indefinite principal at a nonphysical embedding. Hence actual K equals that generating field. For isolation normalize c/γ to get actual $K = \mathbb{Q}(B)$, $B = 4\gamma^2$, in the same way.

For an isolated finite own, $0 < B_\sigma < 4/15$ at every nonphysical place. For $B = 2 + 2\cos(2\pi/m)$, degree $d = \phi(m)/2$ and trace $2d + \mu(m)$ give $26d < 56 - 15\mu(m)$, hence $d \le 2$. The complete finite list is $m = 3, 4, 5, 6, 8, 10, 12$. Rational cases three/four/six force actual $K = \mathbb{Q}$, excluded by the compact WHOLE-form Meyer obstruction. For five/eight/ten the other B values

$$(3 - \sqrt{5})/2, \quad 2 - \sqrt{2}, \quad (5 - \sqrt{5})/2$$

all exceed $4/15$. For twelve the other value $2 - \sqrt{3} > 4/15$, using $\sqrt{3} < 26/15$. Thus no isolated finite own survives, regardless of β.

For white three, the actual path makes a integral; the own two-cycle makes $a^2/2$ integral. Write $M_j = 15a^2 + 4ja + 4j - 8$. The cycle identity gives $M_j V/8$ integral. Let $t = 2\cos(\pi/m)$. The root-of-unity dyadic lemma proved in Section C.5.3 says t is a unit at every dyadic place unless m is a power of two. For a non-power-two label, $a = \sqrt{2}t$ has valuation $1/2$, normalized by $v(2) = 1$. The leading term of M_j has valuation one; the other terms have valuation at least two. Thus $v(M_j) = 1$, forcing $V/4$ integral, impossible. For $m = 4$, $a = 2$ gives a rational Lorentzian principal, so actual $K = \mathbb{Q}$. For $m = 2^k \ge 8$, put $B = t^2$. Its even degree d and norm two imply $\mathrm{N}(2B) = 2^{d+1}$ is nonsquare. Hence $\mathbb{Q}(a)/\mathbb{Q}(B)$ has the sign automorphism $a \mapsto -a$. At this nonphysical embedding,

$$M_j(-a) \ge 15a^2 - 56a + 48 > 0$$

because $a > 12/5$, the right polynomial vanishes at $12/5$ and increases thereafter. This contradicts positive definiteness.

For white four the actual native-four path makes $2t$ K-valued, and the own two-cycle makes t^2 integral, so $t \in \mathcal{O}_K$. For $M_j = 15t^2 + 4jt + 4j - 4$, the cycle identity gives $M_j V/4$ integral. A

non-power-two own makes t, and hence M_j, a dyadic unit, again impossible. For the remaining power-two owns, the near $j = 14$ short edge already has $C \geq 15/2 + 14 > 15$. At far one the sign automorphism $t \mapsto -t$ exists for every power-two $m \geq 4$, and its time numerator $15t^2 - 4t > 0$ contradicts admissibility. Thus EVERY finite own has been removed.

For completeness, the actual cosh with a white contact of unit magnitude η is

$$C_S = \frac{\sqrt{15}\gamma + \sqrt{2}\eta j/\sqrt{15}}{\sqrt{\delta - Q}}, \qquad \delta = 1 - \frac{2\eta^2 j(15 - j)}{15}.$$

White-four has pure $C_S = 15\gamma + j$; its dotted range violates $C_S < 15$ at both ends. White-three at near next-to-end thirteen has pure $C_S = 15a/4 + j/2 > 17$. A foreign cost only increases these coshes.

For the remaining far next-to-end three at $j = 2$,

$$2\sqrt{2} < a < 56/15, \qquad -8/15 < a_\sigma < 0, \qquad K = \mathbb{Q}(a),\ a \in \mathcal{O}_K.$$

Degree at least four contradicts $(56/15)(8/15)^3 < 1$. In degree three the norm has magnitude < 2, hence is one. For other roots $-b, -c$, $bc = 1/a$ and $b, c < 8/15$ force $a > 225/64$. The trace is then three, so $b + c = a - 3 < 11/15$, whereas $(b + c)^2 \geq 4/a > 15/14$, impossible. Degree one is excluded by the whole-form rational obstruction. Degree two has norm -1, trace three and $a = (3 + \sqrt{13})/2$. Its time numerator $15a^2 + 8a = 53a + 15$ has odd norm -199. It is a unit at EVERY dyadic place; $M_j V/8$ would make $V/8$ integral.

Only end-three dotted roles remain. Far one has

$$2\sqrt{2} < a < 118/15, \qquad -2/3 < a_\sigma < 2/5.$$

For near fourteen put $S = a + 2$, giving time numerator $15S^2 - 4S - 4$ and

$$2 + 2\sqrt{2} < S < 122/15, \qquad -2/5 < S_\sigma < 2/3.$$

Both a, S are integral and have square divided by two integral. If their actual degree $d \geq 3$, their integer norms are divisible by four, while their magnitudes are

$$< (122/15)(2/3)^{d-1} \leq 488/135 < 4,$$

impossible. In degree two both trace and norm are EVEN: $\mathrm{N}(z)^2/4$ integral forces even norm, and $\mathrm{Tr}(z^2/2) = \mathrm{Tr}(z)^2/2 - \mathrm{N}(z)$ forces even trace.

The far strips allow trace $4, 6, 8$, with $-6 < \mathrm{N}(a) < 4$, nonzero. The near strips allow trace $6, 8$, with $-4 < \mathrm{N}(S) < 6$. The COMPLETE finite hand tables are

far trace	$N = -4$	$N = -2$	$N = 2$
4	$2 + 2\sqrt{2}$ (wrong conjugate)	$2 + \sqrt{6}$	$2 + \sqrt{2}$ (wrong conjugate)
6	$3 + \sqrt{13}$	$3 + \sqrt{11}$	$3 + \sqrt{7}$
8	$4 + 2\sqrt{5}$ (too large)	$4 + 3\sqrt{2}$ (too large)	$4 + \sqrt{14}$

and

near S trace	$N = -2$	$N = 2$	$N = 4$
6	$3 + \sqrt{11}$	$3 + \sqrt{7}$	$3 + \sqrt{5}$ (wrong conjugate)
8	$4 + 3\sqrt{2}$ (too large)	$4 + \sqrt{14}$	$4 + 2\sqrt{3}$.

For their nine retained rows, the integral time-numerator factor $M_j/2^k$ and its odd norm are

role and a	k	$M_j/2^k$	$\mathrm{N}(M_j/2^k)$
far $2+\sqrt{6}$	1	$77+32\sqrt{6}$	-215
far $3+\sqrt{13}$	2	$(169+47\sqrt{13})/2$	-39
far $3+\sqrt{11}$	1	$154+47\sqrt{11}$	-583
far $3+\sqrt{7}$	1	$124+47\sqrt{7}$	-87
far $4+\sqrt{14}$	1	$231+62\sqrt{14}$	-455
near $1+\sqrt{11}$	1	$142+43\sqrt{11}$	-175
near $1+\sqrt{7}$	1	$112+43\sqrt{7}$	-399
near $2+\sqrt{14}$	1	$215+58\sqrt{14}$	-871
near $2+2\sqrt{3}$	3	$50+29\sqrt{3}$	$-23.$

Every factor is a unit at ALL dyadic places, including ramified places in the actual field. The seven $k=1$ rows make $V/4$ integral. The $k=2$ row makes $V/2$ integral. In a quadratic field such a $W=V/2$ has $0<W<1$, $-2<W'<0$, norm -1 and trace -1 or zero. Trace zero is reducible or a boundary; trace -1 gives $\mathbb{Q}(\sqrt{5})$, not the actual $\mathbb{Q}(\sqrt{13})$.

Only the final near row remains. Here actual $K=\mathbb{Q}(\sqrt{3})$ and $\mathcal{O}_K=\mathbb{Z}[\sqrt{3}]$. The full strict V strips give $V\in\{\sqrt{3}-1,\sqrt{3}\}$: writing $V=a+b\sqrt{3}$ gives $0<b<\sqrt{3}<2$, so $b=1$, then $a=-1,0$. Hence $h>.18$, $q(h)>.04$. Its pure short cosh is $15(2+2\sqrt{3})/8+7/2>13$; foreign Q only increases it. Thus $S_{15}>\sqrt{90/7}>3.58$. The SAME joint vector (BT), with I counted once, gives

$$e_B+e_I>.9084+.0325+.05304+.04=1.03394>1.$$

Every nonisolated short profile is excluded. The actual short-fifteen opposite is ISOLATED within B_{15} and DOTTED.

C.5.13 The quarter-foreign head cannot meet an isolated short cap

Suppose $\beta_H=1/2,\lambda_H=0$. Let $U=4g_H^2>4$, so $V(U-2)=2$, and write the isolated dotted short cap

$$c_S=A_Sp-g_SP_{15}-\beta_Se,\qquad B=4g_S^2>4,\quad Q=\beta_S^2.$$

The actual short edge and the nonphysical principal give

$$B<60(1-Q),\qquad 0<B_\sigma<\frac{4(1-Q_\sigma)}{15}.$$

All finite native β_S, including zero, remain.

The native-three head/old-e link fixes the actual K-root squares:

$$r=\frac{c_H}{\sqrt{2}g_H},\quad f=\frac{e}{\sqrt{2}g_H},\quad r^2=f^2=\frac{2}{U},\quad r\cdot f=-1/U,\quad s=\frac{c_S}{g_S},\quad s^2=4/B.$$

The old short product of s is -1. Its old-e coefficient $q_0\in K$ is nonnegative, with $s\cdot f=-2q_0/U$, and

$$q_0^2=\frac{2UQ}{B}\text{ is a FULL }K\text{-square},\qquad r\cdot s=1+q_0/U-J=-D,\quad J>0,\ D\geq 0.$$

These are actual-root normalizations, not assigned relative lengths for disconnected old components.

The ENTIRE long old-B path cycle, the two-cycle, the head/old-e/short triangle and the long old-e/head/old-B/short cycle give, respectively,

$$G=UBD,\quad L_c=UBD^2/2,\quad T_c=q_0BD,\quad P_c=2q_0B\quad\text{integral.}$$

All formulas retain mutual zero. The two time squares are $(2U-3)/(2U)$ and $15+4(Q-1)/B$. Expanding $J = D + 1 + q_0/U$ gives

$$4(Bq_0 + G + T_c + L_c) = 28UB - 45B + 8UQ - 16Q - 8U + 12. \tag{BQ2}$$

Since $4Bq_0 = 2P_c$, $4Q$ is integral and forty-five is odd, this equality forces $B/2$ integral at every dyadic place, with arbitrary ramification. For $Q = 0$ its stronger divisibility forces $B/4$ integral. Degree at least three is impossible because

$$0 < \mathrm{N}(B/2) < 30(2/15)^{d-1} \le 8/15 < 1.$$

Degree one is excluded by compact rational anisotropy. In degree two, $Q = 0$ likewise gives $0 < \mathrm{N}(B/4) < 1$. Only positive Q remains.

The actual head principal is Lorentzian over $\mathbb{Q}(U)$, so actual $K = \mathbb{Q}(U)$. The earlier actual head bound $C_H < 7/3$ gives

$$4 < U < 49/6, \qquad 0 < U' < 3/2.$$

From $V = 2/(U-2)$ integral, $\mathrm{N}(U-2)$ is a negative divisor of four, of magnitude greater than one. It is -2 or -4, and the trace of U is $5, \ldots, 9$. The COMPLETE surviving irreducible rows are

$\mathrm{N}(U-2)$	U	$\mathrm{N}(U)$
-2	$3+\sqrt{3}$	6
-2	$(7+\sqrt{17})/2$	8
-4	$(5+\sqrt{17})/2$	2
-4	$3+\sqrt{5}$	4
-4	$4+2\sqrt{2}$	8
-4	$(9+\sqrt{41})/2$	10.

For verification, writing $T = \mathrm{Tr}(U)$ gives $\mathrm{N}(U) = 2T - 6$ or $2T - 8$. The other $T = 5, \ldots, 9$ rows are rational, boundaries, or violate $U' < 3/2$. Thus this is a derived list of actual quadratic fields, not a prior field restriction.

Put $Z = B/2$, $M_0 = \mathrm{N}(Z) \in \mathbb{Z}_{>0}$. The full short bounds give $M_0 < 4\mathrm{N}(1-Q)$. A finite β_S in a quadratic field has label $3, 4, 5, 6, 8, 10, 12$. For labels four, six, eight, ten and twelve, the right side is respectively at most $1, 1/4, 1/2, 1/4, 1/4$, and the inequality is strict. Only

$$Q = 1/4,\ M_0 = 1 \text{ or } 2; \qquad Q = \varphi^2/4,\ M_0 = 1,\ K = \mathbb{Q}(\sqrt{5})$$

remain; native five has $4\mathrm{N}(1-Q) = 5/4$. In both branches $\mathrm{N}(Q) = 1/16$. The FULL square $q_0^2 = UQ/Z$ requires $\mathrm{N}(U)/M_0$ to be a rational square. In the entire six-row table this leaves only $\mathbb{Q}(\sqrt{2})$ or $\mathbb{Q}(\sqrt{17})$ with $M_0 = 2$, or $\mathbb{Q}(\sqrt{5})$, $U = 3+\sqrt{5}$, $M_0 = 1$.

For $Q = 1/4, M_0 = 2$, the full bounds give $20 < Z < 45/2$, $0 < Z' < 1/10$. Its integer trace is twenty-one or twenty-two, with discriminant 433 or 476. Neither has squareclass two or seventeen. For $\mathbb{Q}(\sqrt{5}), M_0 = 1, Q = 1/4$,

$$10 < Z < 45/2, \qquad 0 < Z' < 1/10.$$

The trace is $11, \ldots, 22$. For $T^2 - 4$ to be five times a square, only $T = 12, 13, 17, 18, 22$ need inspection; the quotients are $28, 33, 57, 64, 96$. Only eighteen survives, giving

$$Z = 9 + 4\sqrt{5} = \varphi^6, \quad U = 3 + \sqrt{5} = 2\varphi^2, \quad q_0^2 = \frac{1}{2\varphi^4}.$$

This is not a square in $\mathbb{Q}(\sqrt{5})$, because two is not a square there. Finally $Q = \varphi^2/4, M_0 = 1$ gives $8.29 < Z < 10.365$, $0 < Z' < 1/8$: use $Z < 30(3-\varphi)/4$, $Z' < (2+\varphi)/30$, and $1.618 < \varphi < 1.6181$. Its only integer traces are nine and ten; discriminants $77, 96$ are not five times squares. Every mixed quarter-head branch is excluded.

C.5.14 The remaining pure dotted pair: a complete norm argument

The actual remaining head is PURE dotted and the short cap is isolated-within-B dotted:

$$c_H = A_H p - \sqrt{2} g_H e_1, \qquad c_S = A_S p - g_S P_{15} - \beta e, \qquad g_H, g_S > 1.$$

Put $U = 4g_H^2$, $B = 4g_S^2$, $Q = \beta^2$, $\delta = 1 - Q$. The original Lorentzian connected principals give $K = \mathbb{Q}(U) = \mathbb{Q}(B)$. At all places the complete necessary strips are

$$4 < U < 98/9, \quad 4 < B < 60\delta, \quad 0 < V < 2,$$
$$0 < U_\sigma < 2, \quad 0 < B_\sigma < 4\delta_\sigma/15, \quad -4 < V_\sigma < 0, \quad 0 < \delta_\sigma \le 1.$$

All U, B, V are integral. Equation (IC) gives

$$UV/2,\ BV/4 \in \mathcal{O}_K.$$

For example subtracting the integral V term from $V(U/2 - 1)$ gives the first statement; for $V(15B/4 - 1)$, the odd coefficient fifteen is invertible at dyadic places and all other places already allow division by four. For the nonzero integer $Y = BV/4$,

$$0 < Y < 30, \qquad |Y_\sigma| < 4/15.$$

Degree at least four contradicts $30(4/15)^3 < 1$. The whole-form rational obstruction removes degree one. Thus actual degree is two or three. Also $K = \mathbb{Q}(V)$: any proper extension would have a nonphysical embedding fixing positive physical V, contrary to the strict $V_\sigma < 0$ strip.

The one paid head reference already proves

$$e_B > .9409.$$

The same head-plus-short vector gives

$$e_B > .9084 + .0325 + .26(T - 3.311) - .0169 \quad \text{if } S_{15} > T. \tag{BP}$$

In particular $S_{15} > 3.61$ forces $e_B > 1.00174$. There are two uniform dispatches. If $V > 6/5$, then $h > .3$, $q(h) > .065$, and $e_B + e_I > 1.0059$. If $B > 225/4$, the ACTUAL isolated short edge has

$$C_S^2 = \frac{15B}{4\delta} \ge 15B/4 > 3375/16 > (29/2)^2.$$

Its original comparison gives $S_{15} > \sqrt{405/31} > 3.61$, contradicting (BP) without charging I. These dispatches use one reference and its two distinct individual anchor prices.

In degree three every conjugate of B is positive, so

$$\mathrm{N}(B) \in \{1, 2, 3, 4\}, \qquad \mathrm{N}(V) \in \{1, \dots, 31\}.$$

Indeed $\mathrm{N}(B) < 60(4/15)^2 = 64/15$, and V has two negative conjugates with norm < 32. Integrality of $BV/4$ requires $64 \mid \mathrm{N}(B)\mathrm{N}(V)$. The entire finite list forces $\mathrm{N}(B) = 4, \mathrm{N}(V) = 16$. As both nonphysical B values are $< 4/15$, the physical $B > 4/(4/15)^2 = 225/4$. The high-short-edge dispatch excludes the entire cubic branch, with every finite β retained.

In degree two write $K = \mathbb{Q}(\sqrt{D})$, $D > 1$ squarefree. In the full quadratic order, the gap between V, V' is < 6. For $D \equiv 1 \pmod 4$, write $V = (a + b\sqrt{D})/2$, same-parity integers, $b > 0$; then $b\sqrt{D} < 6$. The complete possible D are $5, 13, 17, 21, 29, 33$. Otherwise $V = a + b\sqrt{D}$, $b > 0$, and

$b\sqrt{D} < 3$, so $D = 2, 3, 6, 7$. Only $D = 5$, respectively $D = 2$, admits $b = 2$; all other possible b equal one. Imposing BOTH strict strips gives exactly

D	V	$-\mathrm{N}(V)$
2	$\sqrt{2}-1,\ \sqrt{2},\ 2\sqrt{2}-1$	$1,\ 2,\ 7$
3	$\sqrt{3}-1,\ \sqrt{3}$	$2,\ 3$
5	$(\sqrt{5}-1)/2,\ (\sqrt{5}+1)/2,\ \sqrt{5}-1$	$1,\ 1,\ 4$
6	$\sqrt{6}-1$	5
7	$\sqrt{7}-1$	6
13	$(\sqrt{13}-3)/2,\ (\sqrt{13}-1)/2$	$1,\ 3$
17	$(\sqrt{17}-3)/2,\ (\sqrt{17}-1)/2$	$2,\ 4$
21	$(\sqrt{21}-3)/2,\ (\sqrt{21}-1)/2$	$3,\ 5.$

For full coverage: at $D = 5, b = 1$, $a = -1, 1$; at $b = 2$, $a = -2$. At $D = 13, 17, 21$, $a = -3, -1$. At $D = 29, 33$ the strips would require both odd $a \le -3$ and odd $a \ge -1$, impossible. At $D = 2, b = 1$, $a = -1, 0$, and at $b = 2$, $a = -1$. At $D = 3$, $a = -1, 0$, and at $D = 6, 7$, $a = -1$. Thus no coefficient, actual field or strict boundary has been silently omitted.

Now $0 < \mathrm{N}(B) < 16$. Writing $n = -\mathrm{N}(V)$, integrality of $BV/4$ requires $16 \mid \mathrm{N}(B)n$. Odd n is impossible. For $n = 2, 6$, $\mathrm{N}(B) = 8$; for $n = 4$, it is $4, 8, 12$. Only SIX V rows remain:

$$\sqrt{2}, \quad \sqrt{3}-1, \quad \sqrt{5}-1, \quad \sqrt{7}-1, \quad (\sqrt{17}-3)/2, \quad (\sqrt{17}-1)/2.$$

Four exceed $6/5$, and are excluded by the head-plus-I dispatch. The entire quadratic branch is reduced to

$$K = \mathbb{Q}(\sqrt{3}),\ V = \sqrt{3}-1,\ \mathrm{N}(B) = 8; \qquad K = \mathbb{Q}(\sqrt{17}),\ V = (\sqrt{17}-3)/2,\ \mathrm{N}(B) = 8.$$

In $\mathbb{Z}[\sqrt{3}]$, norm eight would require $a^2 - 3b^2 = 8$, impossible modulo three. In the full $\sqrt{17}$ order put $B = (T + s\sqrt{17})/2$, same-parity integers, $s > 0$. Its norm and strict strip give

$$30 < B < 60, \quad 31 \le T \le 60, \quad T^2 - 17s^2 = 32.$$

Modulo seventeen $T \equiv 7, 10$. In that COMPLETE integer interval the only candidates are $T = 41, 44, 58$, giving $s^2 = 97, 112, 196$. Only the last is a square. Thus $B = 29+7\sqrt{17} > 57 > 225/4$, and the high-short-edge dispatch is contradictory. This excludes BOTH quadratic remnants and the entire pure dotted pair.

Together with the ordinary branch, the full actual $B_{15}+A_1$ family is now excluded by the written arguments, whether the isolated rank-one wall has ordinary $h \ge 1/2$ or actual short $0 < h < 1/2$. No numerical-output premise, field census or realization is used in this closure.

C.5.15 The odd-D fork cycle and the actual D_{15} centre

Retain the original D_{15} roots, both signed forks and the actual short I opposite. We use (IC), with every foreign β and its radical-cancelling triangle retained.

First, in WHOLE $D_r + A_1$ with $r \ge 5$ ODD, an actual long-edge opposite cannot have exactly one extra D contact, at a fork with ANY finite native magnitude η. Indeed for $v_D = \gamma\Omega_i + \eta\Omega_f$ put

$$U = 4\gamma^2, \qquad B = 4\eta^2, \qquad Z = 4\gamma\eta.$$

Actual two-cycles and the entire old-three path cycle make these integral, also at $\gamma = 0$. The exact inverse products give

$$8A_0^2 = 4iU + 4iZ + rB - 8 =: N.$$

At every dyadic place, normalized by $v(2) = 1$, a finite native B has $0 \le v(B) \le 1$. To see this in its minimal field, write $B = (1+\zeta)(1+\zeta^{-1})$, $\mathrm{N}(B) = \Phi_m(-1)$. This norm is odd unless m is a power of two, when it equals two; the elementary factorization $n = \prod_{d|n,d>1} \Phi_d(1)$ proves the assertion. The valuation bound persists in EVERY actual extension. As r is odd, rB is the unique lowest term in N. Thus $v(N) = v(B) \le 1$. Integrality of VA_0^2 forces $v(V) \ge 2$ at every dyadic place, hence $V/4$ integral, impossible. This covers own zero, every finite/parallel/dotted own and every old-I contact. Native four is included: its own two-cycle is essential and must not be dropped.

At D_{15}, this removes the ONLY weak right-fork packets at eleven and twelve already classified in Section C.5.8. Together with the shared bulk, strong $6, \dots, 12$ are available without ordinary rank-one energy. It remains to prove thirteen directly.

Deleting thirteen leaves A_{12}, BOTH forks, and old e. Finite classification gives the COMPLETE table:

left contact	fork contacts
none	none; one arbitrary finite label; $(3,3), (3,4), (3,5)$
endpoint three at $1, 12$	none; one fork three/four; both forks three
next-to-end three at $2, 11$	none
endpoint four at $1, 12$	none.

Two left contacts form a cycle. Next-to-end with a new fork has nonfinite unequal long arms. With no left contact, two finite fork contacts must form A_3, B_3, H_3; with an endpoint contact, finite long $A/B/D$ completions give precisely the displayed three/four choices. Adding old e can restrict but cannot enlarge this list. For left coefficient η_L at j, fork coefficients a, b, and foreign $Q = \beta^2$, put

$$q_L = \frac{2\eta_L^2 j(13-j)}{13}, \quad \Delta = 1 - q_L - a^2 - b^2 - Q > 0, \quad \sigma = \gamma + \frac{\eta_L j}{13} + \frac{a+b}{2}.$$

The actual inverse formula is $A^2 = 26\sigma^2 - \Delta$, $C_{13} = \sqrt{26}\sigma/\sqrt{\Delta}$. The exact threshold satisfies

$$T(\sqrt{26}) < \frac{100}{61} - \frac{4490}{48909} < \frac{31}{20},$$

with final positive cleared numerator 146719. Foreign Q only increases every direct ratio.

For isolation, own zero has negative time. Own three has the affine four-leaf-three star on $c, n_{13}, n_{12}, n_{14}, n_{15}$; own four has the $4,3,3$ star; own six the $6,3$ path; parallel its actual null pair. All apply with arbitrary foreign cost. Every remaining finite own is excluded arithmetically. For $B_0 = 4\gamma^2$, actual reflection normalization gives old raw squares two, cap square $2B_0$ and marked product $-B_0$. The original connected principal has Schur residual $B_0(2-13B_0)$. Thus every nonphysical B_0 obeys $0 < (B_0)_\sigma < 2/13$. Cyclotomic trace $2d+\mu(m)$ and physical $B_0 < 4$ force $d \le 2$. The complete labels are $3, 4, 5, 6, 8, 10, 12$. The nonrational other values at five/eight/ten/twelve,

$$(3-\sqrt{5})/2, \quad 2-\sqrt{2}, \quad (5-\sqrt{5})/2, \quad 2-\sqrt{3},$$

all exceed $1/4 > 2/13$. The rational cases were already null. For every dotted isolated own, the actual core $c + \beta e + \gamma u n_{13}$, $u = (1-Q)/\gamma^2 \in (0,1)$, gives

$$x \cdot \Omega_{13} > \sqrt{26-u} - \sqrt{u(1-u)}.$$

Its full loss is $< (.1 + \sqrt{1+.1^2})/2 < .553$. Every isolated row is therefore strong or impossible.

With a left contact and no fork, far-three own zero has $A^2 = Q - 1/2$, $Q < 7/13$, so its native costs $0, 1/4, 1/2$ give nonpositive time. Far-four own zero forces $Q = 0$. Near/next-three own zero

has affine D trunk coefficients two and cap/previous-node/both-fork coefficients one. Near-four own zero has the affine $4,3,3$ star. Positive own three has its all-three cycle, own four the fork star, own six the fork $6,3$ path, and parallel its null pair. The remaining near/next-three owns have $\gamma \ge \varphi/2 > .8$, with ratios at least

$$\frac{.8 + 6/13}{\sqrt{7/13}} > 1.7, \qquad \frac{.8}{\sqrt{2/13}} > 2.$$

Positive endpoint-four has ratio at least $\sqrt{13}/2 > 1.8$. The only left-only possible weak role is the literal far helper at thirteen, with

$$202/125 < 2\gamma < 2(31/20)\sqrt{7/13} < 12/5.$$

Lemma 20.3 excludes it including foreign contacts.

One fork and no left contact is excluded by the odd-D cycle just proved. For fork pair $(3,3)$, own zero is the actual four-cycle and own three the local triangle; all other positive owns have ratio $\ge 1 + 1/\sqrt{2} > 31/20$. For pair $(3,5)$, even own zero has ratio $\varphi^3/2 > 2.1$. For $(3,4)$, every positive own has ratio > 2.2. At own zero finite whiteness forces $Q = 0$. The original four-cycle puts $\sqrt{2}$ in K, and

$$A_0^2 = (37 + 26\sqrt{2})/8.$$

The numerator has odd norm seventeen, so is a unit at EVERY dyadic place, also in proper extensions. The cycle identity forces $V/4$ integral.

For a left endpoint-three and one fork-three, $\Delta \le 15/52$, $\sigma \ge \gamma + 15/52$. Own zero has its actual cycle, own three its triangle; all other owns have ratio > 1.8. For one fork-four, $\Delta \le 1/26$, and even own zero has ratio $(1/26 + 1/(2\sqrt{2}))\sqrt{26} > 1.9$. For both forks-three, $\Delta \le 1/26$, $\sigma \ge 7/13$, ratio > 2.7. Both left and fork orientations are included. Each affine kernel has positive cap coefficient and time, with all omitted ORIGINAL walls nonpositive termwise. No foreign wall was discarded. This exhausts the table and proves strong thirteen.

Hence $6, \dots, 13$ are strong. Both initial-five references give $e_D > E_{13} > .88989$. If five is weak, its curved reference gives the stronger $e_D > E_{13} + .7600 - .6971 > .95279$. Therefore weak five implies $h < .21$, since $q(.21) = 1/21$ already exceeds the remaining budget. Its literal far-helper inverse entries are $5,1,1$, the SAME entries used in Section C.5.10. The complete minimal-degree and actual-field golden argument there therefore applies over old D_{15}, without using a B rank conclusion. It forces $V = 2\varphi - 2$, $h > .3$, contrary to $h < .21$. Five is strong. Fix the genuine reference through $5, \dots, 13$:

$$w^2 = E_{13} > .88989, \quad b < .361, \quad f_{13} < 3.175, \quad w_{14} = w_{15} = 0.$$

Its INDIVIDUAL P_5 and P_{13} prices exceed .14. The first-five graft is legal on $.49 \le X \le .67$, as in the B calculation.

C.5.16 The actual D_{15} head with short I

The COMPLETE actual D_{14} white head list and both-fork positive clearing were proved in Section C.5.9. For a nonisolated head,

$$C_1^2 = \frac{(2g+1)^2}{1-2Q} = 1 + \frac{2A^2}{1-2Q}.$$

If $Q = 0$, native gap with d gives $A^2h \geq 1/4$. If $Q > 0$, acuteness gives $A^2h \geq Q(1+h) \geq 1/4$. For $h < .05$, $C_1^2 > 11 > 7$, $x_1 > .67$; the legal graft gives $e_D > 1.00924125$. For $h \geq .05$, the complete own-zero and own-three nulls leave $g \geq 1/\sqrt{2}$, hence $x_1 > .643$. Now $q(h) > .012$, using $\sqrt{1.05} > 1.0245$. The original total is $.88989 + .099405 + .012 = 1.001295 > 1$. Thus the head is isolated WITHIN D_{15}.

Its form and full foreign split are

$$c_H = A_H p - g_H \Omega_1 - \beta_H e, \quad A_H^2 = 2g_H^2 + Q_H - 1, \quad Q_H = 0,\ 1/4,\ \text{or } [1/2, 1),$$

with $(2g_H^2 - 1)h \geq Q_H$. The original nulls and native gap remove owns zero, three, four, six, parallel, and pure five. For a pure finite head use the SAME proved actual finite point and the ONE paid head reference:

$$e_D > .88989 + F(u) - .2888 + .032.$$

For finite own, native gap forces $h = [4(2g_H^2 - 1)]^{-1} > 1/4$. The complete four native cases are

native	F lower bound	$q(h)$ lower bound	original total lower bound
7	.2893	.083	1.00539
8	.291984	.0755	1.000574
9	.302416	.069	1.004506
≥ 10	.31690336	.0557	1.00569336.

Here are the tight fresh scalar details. For seven, $g_H < .901$ and $4(2(.901)^2 - 1)(.4) < 1$, so $h > .4$. For eight, put $s = \sqrt{2}$,

$$H_8 = \frac{\sqrt{2s-2} - \sqrt{34s-48}}{s}, \quad R_8 = 23 - 16s, \quad h_8 = s/4.$$

The brackets $1.4142135 < s < 1.4142136$ give $\sqrt{2s-2} > .91017$, $\sqrt{34s-48} < .28856$, $H_8 > .4395$, $R_8 < .3726$. Thus $F_8 > .291984$, and $\sqrt{1+h_8} > 1.1634$ gives $q(h_8) > .0755$. For nine, the cubic gives $.939 < g_H < .94$, $u < 1.135$, $h > .32$. The earlier finite scalar at 1.135 gives $F > .302416$; $\sqrt{1.32} > 1.1488$ gives $q(h) > .069$. For EVERY native at least ten, $u \leq 2 - (2/5)\sqrt{5} < 1.106$. At that rational bound, $\sqrt{.894} > .9455$, $\sqrt{.072292} < .269$, $H > .4783$, $R = .410854$, giving $F > .31690336$. The finite scalar's already proved monotonicity supplies the entire infinite native tail. Each table row applies to ONE reference and charges I once.

For $Q_H \geq 1/2$, $C_1^2 \geq 2 + 1/h$. The complete h split is

h range	x_1 bound	$q(h)$ bound	total lower bound
$(0, .2)$	.67	unused	1.00924125
$[.2, .3)$	.62	.045	1.01874125
$[.3, .5)$	.57	.065	1.00949125.

The edge squares are respectively > 7, $> 16/3 > (9/4)^2$, and > 4. For $Q_H = 1/4, \lambda_H > 0$, native gap gives $A_H\sqrt{h} \geq .5 + .5\sqrt{1+h}$. The same three ranges give, respectively, $x_1 > .67, .63, .57$, and totals $> 1.00924125, 1.02534125, 1.00949125$. For the middle row use $A_H\sqrt{h} > 1.04$, $C_1^2 > 49/9$; for the last use $A_H\sqrt{h} > 1.07$, $C_1^2 > 4$. The I bounds follow from $\sqrt{1.2} > 1.095, \sqrt{1.3} > 1.14$. All grafts lie in the PROVED $.49, .67$ interval; these are alternative contradictions, never stacked.

Thus the only remaining heads are pure dotted or $\beta_H = 1/2, \lambda_H = 0, V(U_H - 2) = 2$. The second still includes finite own. For BOTH families $u = (1 - Q_H)/g_H^2 \in (0, 1)$, and the actual cleared point

$$c_H + \beta_H e + \frac{1 - Q_H}{g_H} n_1$$

is future negative, selected-wall nonpositive and feasible against EVERY original wall, including both forks and d. The detailed computation in Section C.5.11 uses only these first long roots and applies here verbatim. It gives the pure first-two-coordinate pressure $x \cdot Z \geq H$, $F = .8H - .16R > .2893$. It does not declare the finite quarter-head dotted.

C.5.17 Both signed forks and the complete isolated-cap argument

The genuine fork duals are $\Omega_\pm = (P_{14} \pm e_{15})/\sqrt{2}$; their squared norm is $15/2$. The ORIGINAL edge comparison gives, independently of the opposite cap's contact pattern,

$$T_\pm = x \cdot (P_{14} \pm e_{15}) \geq \sqrt{15}\sqrt{(C_\pm - 1)/(C_\pm + 1)}.$$

For either ACTUAL signed pressure $T_\pm > T$, use

$$W_\pm = w + .4(Z - P_5/5) + .14(P_{14} \pm e_{15} - P_{13}).$$

Its individual debits are .08 from P_5 and .14 from P_{13}, both affordable BEFORE using their floors. The head residual is zero-sum on the first five coordinates. The signed residual $e_{14} \pm e_{15}$ has norm two and is disjoint from the head residual. Both are orthogonal to w. The FULL one-vector square therefore gives

$$e_D > .88989 + .0325 + .28(T - 3.175) - .0392. \tag{DF}$$

If either $C_\pm \geq 14$, then $T_\pm \geq \sqrt{13} > 3.605$, making $e_D > 1.00359$. Thus BOTH actual fork coshes are < 14, without assuming a fork-cap whitelist.

Fix either genuine fork f, and suppose its actual opposite is isolated within D_{15}:

$$c = Ap - g\Omega_f - \beta e, \quad Q = \beta^2 < 1.$$

Own zero has negative time; parallel own has the actual future feasible null $c + n_f$. For every remaining positive own put $U = 4g^2$; it is integral and $A_0^2 = 15U/8 - 1 > 0$. Here is an explicit actual K-normalization, valid for finite as well as dotted own. Choose old raw roots $\alpha_i = \sqrt{2}n_i$. The actual reflection difference

$$r = s_c(\alpha_f) - \alpha_f = 2\sqrt{2}g\,c$$

is a K-vector on the actual cap line. Its Gram has $r^2 = 2U$, $r \cdot \alpha_f = -U$, other old products zero. The connected original principal is defined over $\mathbb{Q}(U)$, with physical Schur residual

$$2U - (15/4)U^2 = -2UA_0^2 < 0.$$

Any proper actual extension would have a nonphysical embedding fixing that indefinite principal. Hence ACTUAL $K = \mathbb{Q}(U)$, and every nonphysical place has $0 < U_\sigma < 8/15$. The actual finite fork endpoint gives

$$C_f^2 = \frac{15U}{8(1-Q)} < 196, \qquad 0 < U < 1568/15.$$

The cycle identity makes $15UV/8$ integral; as $15 = 16 - 1$, $Y = UV/8$ is integral too. Its full strips are

$$0 < Y < 392/15, \qquad |Y_\sigma| < 4/15.$$

Degree at least four contradicts $(392/15)(4/15)^3 < 1$. Degree one is excluded by the compact WHOLE-form rational-anisotropy obstruction.

In degree three, $0 < \mathrm{N}(U) < 100352/3375 < 32$ and $0 < \mathrm{N}(V) < 32$. Integrality of $UV/8$ requires $512 \mid \mathrm{N}(U)\mathrm{N}(V)$. Each positive integer below thirty-two has ordinary two-adic valuation at most

four, so their product cannot have valuation nine. This excludes EVERY cubic field and ramification pattern.

In degree two $0 < \mathrm{N}(U) < 12544/225 < 56$. The actual V generates K, and the COMPLETE quadratic strict-I list is precisely the sixteen values already derived in Section C.5.14; that derivation used only $0 < V < 2, -4 < V' < 0$, not a B geometric hypothesis. Write $n = -\mathrm{N}(V)$. Now $64 \mid \mathrm{N}(U)n$. Odd n is impossible. For $n = 2, 6$, $\mathrm{N}(U) = 32$; for $n = 4$, it is $16, 32, 48$. The same SIX surviving V values are

$$\sqrt{2}, \quad \sqrt{3}-1, \quad \sqrt{5}-1, \quad \sqrt{7}-1, \quad (\sqrt{17}-3)/2, \quad (\sqrt{17}-1)/2.$$

If $\mathrm{N}(U) \geq 32$, the strict conjugate strip gives physical $U > 60$. Then $C_f^2 > 112.5 > 10.6^2$, and $T_f > \sqrt{360/29} > 3.52$. Equation (DF) gives $e_D > .97979$. Every displayed V is $> 14/25$, hence $h > .14$, $\sqrt{1+h} > 16/15$, $q(h) > 1/31 > .032$. The original total is > 1.01179. If $\mathrm{N}(U) = 16$, necessarily $n = 4$, so $V = \sqrt{5}-1$ or $(\sqrt{17}-1)/2$, both $> 6/5$. Thus $q(h) > .065$. Also $U > 30$, $C_f > 15/2$, $T_f > \sqrt{195/17} > 3.38$. The SAME signed reference gives $e_D > .94059$ and total > 1.00559. No head or fork energy optimized separately has been added to this reference.

Thus neither ACTUAL fork opposite can be isolated within D_{15}, with its entire finite old-I contact and every own range retained. The remaining nonisolated cases are now classified at their own actual finite endpoints.

C.5.18 Every nonisolated fork cap

Fix fork f and denote the other fork by k. Deleting f retains the ACTUAL A_{14} path $n_1, \ldots, n_{13}, n_k$, plus old e. The COMPLETE finite contact and foreign table is

retained D contact	allowed β
simple three at n_1 or n_k	$0, 1/2, 1/\sqrt{2}$
simple three at n_2 or n_{13}	0
native four at n_1 or n_k	$0.$

Two path contacts form a cycle. Other branches would need finite exceptional rank at most eight, while this completion has rank fifteen. A next-to-end old-I arm creates nonfinite long E arms. An old-I arm at white four makes its double bond internal. At endpoint three only old-I labels three or four give finite A_{16}/B_{16}. This retains BOTH path ends and next-to-end sites.

Write $v_D = g\Omega_f + \eta\Omega_j$. Own zero is completely dispatched as follows. Far-three at n_1 has $A^2 = Q - 1/2 \leq 0$; next-three at n_2 has $A^2 = 0$. Near-next-three at n_{13} has the actual affine D_4 null

$$c + n_{12} + n_f + n_k + 2n_{13}.$$

Near-end-three at n_k has affine E_8 kernel

$$2c + 4n_k + 6n_{13} + 3n_f + 5n_{12} + 4n_{11} + 3n_{10} + 2n_9 + n_8.$$

Its arms have lengths $1, 2, 5$. Far-four at one and near-four at k have respectively

$$\sqrt{2}c + 2(n_1 + \cdots + n_{13}) + n_f + n_k, \qquad \sqrt{2}c + 2n_k + 2n_{13} + n_f + n_{12}.$$

Every selected product vanishes; its cap coefficient gives positive future time. Every unselected ORIGINAL wall, including the entire old-I arm, has nonpositive product termwise. Thus these are forbidden feasible nulls. Parallel own has $c + n_f$.

For white three put $t = 2g$. The actual closed three-path cycle makes t integral and K-valued. Actual reflection differences normalize cap and old D roots to square two, with own product $-t$ and white product -1. The physically Lorentzian old-D/cap principal then proves ACTUAL $K = \mathbb{Q}(t)$, with time numerators

$$\begin{array}{c|c} j = 1, 2, 13 & 8A_0^2 = 15t^2 + 4jt + 4j - 8 \\ j = k & 8A_0^2 = 15t^2 + 26t + 7. \end{array}$$

For white four put $a = 2\sqrt{2}g$. The original reflection difference at its white-four neighbour equals $2c$, fixing the cap's K-line. Use raw cap square one, white product -1 and own product $-a/2$. The actual cycles give $a \in \mathcal{O}_K$, $a^2/2 \in \mathcal{O}_K$, and actual $K = \mathbb{Q}(a)$. Now

$$\begin{array}{c|c} j = 1 & 16A_0^2 = 15a^2 + 8a \\ j = k & 16A_0^2 = 15a^2 + 52a + 44. \end{array}$$

All these A_0^2 are positive for positive native or dotted own, so the actual-field descent uses an original Lorentzian principal in its precise domain.

For white three with own three, the original old-three path closes an all-three affine cycle; its all-ones kernel remains future feasible with foreign contacts. For every other FINITE own, the dyadic lemma of Section C.5.3 gives $v(t), v(t+1) \leq 1/2$ at EVERY dyadic place. In the long-j numerator the term $15t^2$ has unique smallest valuation ≤ 1. At $j = k$, putting $s = t + 1$ gives $15s^2 - 4s - 4$, with the same unique-lowest-term argument. Integrality of VA_0^2 forces $v(V) \geq 2$ at all dyadic places, contrary to $V/4$ nonintegral.

For white four and a non-power-two finite own, $t = 2\cos(\pi/m)$ is a dyadic unit, so $v(a) = 1/2$. The leading term in either numerator has unique valuation one; the others have valuation at least two. The cycle identity makes $V/8$ integral, impossible. For $m = 4$, $a = 2$ gives a rational Lorentzian principal, so actual $K = \mathbb{Q}$, impossible. For higher powers of two, the SAME norm-two half-square argument used in Section C.5.12 gives an automorphism $a \mapsto -a$. The far conjugate numerator $15a^2 - 8a$ is positive. The near numerator $15a^2 - 52a + 44$ has roots $22/15, 2$ and is positive since $a^2 \geq 4 + 2\sqrt{2} > 6 > (12/5)^2$. Both contradict strict nonphysical positivity. Every finite own is now excluded.

Only $g > 1$ remains. The complete pure fork coshes, which only increase on adding foreign Q, are

white contact	C_f at $Q = 0$
n_1 three	$(15t + 2)/8$
n_k three	$(15t + 13)/8$
n_2 three	$(15t + 4)/4$
n_{13} three	$(15t + 26)/4$
n_1 four	$15a/4 + 1$
n_k four	$15a/4 + 13/2$.

They follow from $\Omega_f^2 = 15/2$, $\Omega_f \cdot \Omega_k = 13/2$, $\Omega_f \cdot \Omega_j = j$, and retained-path inverse costs $7/15, 13/15, 14/15$. Since $C_f < 14$, $t > 2$, $a > 2\sqrt{2}$, near-next-three and near-four are immediately impossible.

Far-four has

$$2\sqrt{2} < a < 52/15, \qquad -8/15 < a_\sigma < 0.$$

Degree at least three contradicts $(52/15)(8/15)^2 = 3328/3375 < 1$. In degree two the norm magnitude is < 2 but must be EVEN since $a^2/2$ is integral. In degree one an even integer in the physical interval does not exist. Far-next-three has $t = 2g$ in the FULL dotted strip $2 < t < 52/15$, with $-8/15 < t_\sigma < 0$ at every nonphysical place. No half-square condition or $2\sqrt{2}$ lower bound is

imposed on t. Degree at least three is again impossible; degree one is rational, and degree two has norm -1, trace two or three. Thus $t = 1 + \sqrt{2}$ or $(3 + \sqrt{13})/2$. Their time numerators $15t^2 + 8t$ have odd norms $-79, -199$, so are units at all dyadic places. The cycle identity makes $V/8$ integral. Both next-three and both white-four dotted roles are now exhausted.

The only roles left are endpoint-three dotted. For the far one put $z = t$, and for the near one put $z = t + 1$. The COMPLETE actual strips and time numerators are

role	physical z	every other z_σ	$M = 8A_0^2$
far	$2 < z < 22/3$	$-2/3 < z_\sigma < 2/5$	$15z^2 + 4z - 4$
near	$3 < z < 38/5$	$-2/5 < z_\sigma < 2/3$	$15z^2 - 4z - 4.$

Here actual $K = \mathbb{Q}(z)$, z is integral, and $MV/8$ is integral. If $\mathrm{N}(z)$ were odd, z and M would be units at every dyadic place, forcing $V/8$ integral. Thus its nonzero integer norm is EVEN.

We need a small ideal argument to exclude absolute norm two WITHOUT assuming a half-square or unramified dyadic primes. If $|\mathrm{N}(z)| = 2$, then $(z) = \mathfrak{p}$ for one prime of norm two, with exponent one. Let its ramification index be e. For $e \geq 2$, $v_\mathfrak{p}(z) = 1/e$; the leading $15z^2$ in M has unique smallest valuation $2/e \leq 1$. Thus $v_\mathfrak{p}(V) \geq 3 - 2/e \geq 2$. At all other dyadic primes z, hence M, is a unit, giving $v(V) \geq 3$. So $V/4$ is integral. For $e = 1$, the other dyadic primes have total ramification-residue degree $d - 1$. At each, M is a unit and $\mathrm{ord}_\mathfrak{q}(V) \geq 3e_\mathfrak{q}$. Hence

$$2^{3(d-1)} \mid |\mathrm{N}(V)|, \qquad 0 < |\mathrm{N}(V)| < 2\,4^{d-1} = 2^{2d-1}.$$

For $d \geq 2$, $3(d-1) \geq 2d - 1$, a contradiction. This proves the norm-two exclusion in EVERY ramification pattern using the Dedekind ideal norm and $\sum e_\mathfrak{p} f_\mathfrak{p} = d$.

For $d \geq 5$, the full strips give $|\mathrm{N}(z)| < (38/5)(2/3)^4 < 2$, contrary to nonzero even norm. For $d = 3, 4$ its magnitude is $< 152/45 < 4$; only two could remain, already excluded. Degree one is rational and impossible. Thus actual degree is two, with $|\mathrm{N}(z)| = 4$. Its trace is EVEN as well. If odd, z cannot be nonunit at every dyadic prime: conjugation permutes those primes, making both z, z' nonunit there and their integer trace even. Norm four also ensures a nonunit prime exists. A quadratic field would therefore need TWO dyadic primes, both with $e = f = 1$, one unit and one nonunit for z. At the unit prime M is a unit, so eight divides $|\mathrm{N}(V)|$, contrary to its strict quadratic bound < 8.

For far z, positive norm four is impossible since $zz' < (22/3)(2/5) < 4$. Thus norm is -4; $z' > -2/3$ gives $z > 6$. Its even trace is six, so

$$z = 3 + \sqrt{13}, \qquad M = 4(169 + 47\sqrt{13})/2.$$

The displayed factor in the FULL quadratic order has odd norm -39. Thus $MV/8$ integral makes $V/2$ integral. As proved in the short-B analysis, an integral $W = V/2$ with $0 < W < 1, -2 < W' < 0$ must generate $\mathbb{Q}(\sqrt{5})$, not the actual $\mathbb{Q}(\sqrt{13})$. No far endpoint-three dotted cap exists.

For near z, negative norm -4 is impossible since $|zz'| < (38/5)(2/5) < 4$. Norm is four, $z' > 0$, $z > 6$; its even trace is eight. Thus

$$z = 4 + 2\sqrt{3}, \quad t = 3 + 2\sqrt{3}, \quad M = 8(50 + 29\sqrt{3}), \quad K = \mathbb{Q}(\sqrt{3}).$$

The full V strips in $\mathbb{Z}[\sqrt{3}]$ give $V = \sqrt{3} - 1$ or $\sqrt{3}$, so $h > .18$, $q(h) > .04$. Its actual pure fork cosh is

$$(15t + 13)/8 = (29 + 15\sqrt{3})/4 > 13.$$

Any foreign Q increases it. The original signed edge gives $T_f > \sqrt{90/7} > 3.58$. The SINGLE reference (DF) gives $e_D > .99659$ and $e_D + e_I > 1.03659$. This excludes the LAST endpoint-three dotted role.

Every actual fork opposite is either isolated or in the complete nonisolated table. Both classes have now been contradicted for either signed fork, both remaining head families and every finite foreign old-I contact. This proves the short-I D_{15} branch. With its separately written ordinary argument and the unconditional actual rank-one isolation, the whole $D_{15} + A_1$ family is covered for every $h > 0$. These arguments introduce no field-census output premise.

C.6 Actual typed-seven spectators beside B_{14} and D_{14}

Here the ENTIRE old forest is $B_{14} + Z$ or $D_{14} + Z$, where Z is an ACTUAL N_7 or O_7 profile. We do not apply these conclusions to a bare old seven bond. The actual defect field descent in Section 20.1 gives $K = K_7 = \mathbb{Q}(\theta)$, where

$$\theta = 2\cos(2\pi/7), \quad \theta^3 + \theta^2 - 2\theta - 1 = 0, \quad 4g_7^2 = 2 + \theta, \quad g_7 = \cos(\pi/7).$$

Indeed the ACTUAL old-plus-defect-wall $[7, 3]$ principal is Lorentzian over this cubic field; any nonphysical embedding fixing it would preserve its indefinite Gram. The normalized defect wall c_T is orthogonal to the WHOLE classical component, with time squares

$$h_N = (\theta - 1)/3, \qquad h_O = \frac{\theta - 1}{2 - \theta}.$$

Keep the one original decomposition $z = E(p - x - y)$, where x is classical and y is the entire old rank-two projection. Then

$$e_T + e_Z = \|x\|^2 + \|y\|^2 < 1, \qquad e_Z > .09.$$

The already proved actual-defect geometry and propagation are reused at their precise whole-component domain; no duplicate propagation theorem is introduced.

The three cubic roots lie in

$$(31/25, 5/4), \quad (-1/2, 0), \quad (-2, -9/5).$$

The polynomial signs at those six endpoints are $-559/15625, 1/64, 1/8, -1, -1, 1/125$. Thus $0 < h_N < 1/12$, $0 < h_O < 1/3$. The order discriminant is forty-nine, so $[\mathcal{O}_K : \mathbb{Z}[\theta]]$ is odd and K_7 is unramified at three.

For an opposite retaining the whole old spectator, let Q be its FULL inverse unit-Gram cost. The COMPLETE possible list is

$$O_7 : \ Q = 0; \qquad N_7 : \ Q = 0, 1/3, 2/3. \tag{TQ}$$

Even one old-$I_2(7)$ contact costs $1/(4\sin^2(\pi/7)) > 1$. For old A_2, two contacts contain the affine all-three triangle or a stronger nonpositive principal. A single contact permits only labels three/four/five, with costs $1/3, 2/3, \varphi^2/3$. Five is incompatible with the cubic actual field. These are whole-spectator costs, not isolated bond estimates.

C.6.1 A common actual head argument

The fresh rank-fourteen prefix gives strong $6, \ldots, 12$ for B, and $6, \ldots, 11$ for D. The B conclusion uses its actual B3/F4 and B2 sign-conjugate rows, not a larger-rank tail. For D, the actual D4/D3 sites are ten and eleven; their only extra weak fork packets have the actual left-H_4 ONE-reference body bounds $> .92531$ and $> .94697$. Adding the separate spectator $> .09$ excludes them. The

exact local inventory and reference proof are in Proposition 22.1; their terminal indices remain ten and eleven here. No floor at the unclassified D centre twelve is assumed.

If five is weak, its actual pure point gives $x_1 > .5965$. Interpolate through that pressure, strong six and every later genuine strong knot. The first-six energy is $> .7600$, versus $E_6 < .6971$, with middle slope $> .2843$, above the next $< .197$. Even the weaker D total is $.84816 + .0629 + .09 = 1.00106 > 1$. Five is therefore strong BEFORE selecting the head. The positive reference w, zero after J, has

$$\begin{array}{c|c|l} T & J & w^2 = E_J \\ \hline B & 12 & > .86989 \\ D & 11 & > .84816 \end{array}, \qquad b = f_5/5 < .361, \qquad \text{individual } P_5 \text{ coefficient} > .14.$$

For $X = .64, .643$, the first-five graft is legal, preserves every later knot and gives gain $> (5/4)(X - .361)^2$.

Deleting the actual long-head direction retains B_{13} or D_{13} and the ENTIRE spectator. The COMPLETE classical white list is no contact, or simple three at its long endpoint two. In the latter branch

$$c = Ap - g\omega_1 - \omega_2/2 - v_Z, \quad A^2 = 2g^2 + 2g + Q, \quad C_1 = \frac{2g+1}{\sqrt{1-2Q}}, \quad Q < 1/2.$$

Own zero at $Q = 0$ has zero time; at $Q = 1/3$ it has the actual future feasible null $c + \omega_2/2 = \sqrt{Q}\, p - v_Z$. Its original clearing is

$$\frac{\omega_2}{2} = \begin{cases} n_1 + 2(n_2 + \cdots + n_{13}) + \sqrt{2} n_{14}, & B, \\ n_1 + 2(n_2 + \cdots + n_{12}) + n_{13} + n_{14}, & D. \end{cases}$$

The foreign products are nonpositive because its spatial foreign part is the negative of a positive inverse-Gram combination. All omitted ORIGINAL wall products are nonpositive termwise. Own three has its actual affine triangle. Every other positive own gives $C_1 \geq 1 + \sqrt{2}$, $x_1 > .643$. Its legal graft adds $> .099405$, already exceeding one with $e_Z > .09$, even at E_{11}. The head is isolated WITHIN its entire classical component, while foreign contacts are retained.

Write the remaining head

$$c = Ap - g\omega_1 - v_Z, \quad A^2 = 2g^2 + Q - 1, \quad C_1^2 = \frac{2g^2}{1-Q}, \quad U = 4g^2 \in \mathcal{O}_K.$$

Own zero has negative time. Own three has negative time at $Q = 0, 1/3$; at $Q = 2/3$, the ACTUAL chain n_2, n_1, c, z_1, z_2, labels $3, 3, 4, 3$, has affine kernel $(1, 2, 3, 2\sqrt{2}, \sqrt{2})$. Own four at $Q = 0$ has zero time; at $Q = 1/3$, the $3, 4, 3, 3$ chain has kernel $(\sqrt{2}, 2\sqrt{2}, 3, 2, 1)$; at $Q = 2/3$, n_1, c, z_1 is the $4, 4$ null. Own five is absent in K_7; own six has the $6, 3$ kernel $(\sqrt{3}, 2, 1)$; parallel has $c + n_1$. Every affine combination has positive cap coefficient, positive future time and nonpositive product with ALL omitted original walls. Thus only finite owns at least seven or dotted remain.

The COMPLETE finite native invariant alphabet in K_7 is

$$4\cos^2(\pi/m) \in \{1, 2, 3, 2 + \theta, 1 + \theta + \theta^2\}, \qquad m = 3, 4, 6, 7, 14. \tag{TA}$$

Indeed $\phi(m)/2$ divides three, giving the complete totient list $3, 4, 6, 7, 9, 14, 18$. The nine/eighteen field is ramified at three: its polynomial $X^3 - 3X + 1$, translated by minus one, becomes Eisenstein $X^3 - 3X^2 + 3$. It cannot equal K_7, whose discriminant divides forty-nine. The last invariant follows from $2\cos(\pi/7) = \theta^2 + \theta - 1$. Native four is retained; an unsquared $\sqrt{2}$ need not belong to K when root-line squareclasses differ.

For $Q > 0$, a full nonphysical Schur complement requires $U_\sigma/2 + Q < 1$. Normalize the head to raw square $2U$, with own product $-U$ against an old square-two long root; the full Schur complement is $2U(1 - U/2 - Q)$. This compares no disconnected root scales. The remaining finite own seven has a conjugate $U_\sigma > 3/2$, contradicting $Q \geq 1/3$. Fourteen has one $U_\sigma > 61/25 > 2$. Every finite mixed head is excluded.

For a dotted mixed head with $Q = 2/3$, $C_1^2 > 6$, so $C_1 > 12/5$, $x_1 > .64$. For $Q = 1/3$ write $v_Z = \omega_b/2$. The actual N_7 wall gives

$$c \cdot c_T = q - A\sqrt{h_N}, \qquad q = 2g_7/3 \text{ in the same old } A_2 \text{ direction}, \quad q = g_7/3 \text{ in the other.}$$

In the first orientation acuteness yields $A^2 h_N \geq q^2 > 16/45$, so $A^2 > 64/15$. The CORRECT identity is

$$C_1^2 = \frac{3}{2}(A^2 + 2/3) > 37/5.$$

In the other orientation $g > 1$ gives

$$A^2 h_N - q^2 > \frac{4}{3} h_N - \frac{2+\theta}{36} = \frac{5\theta - 6}{12} > 0.$$

The product is strictly negative. Its actual native gap forces $A\sqrt{h_N} \geq q + 1/2 > .8$, hence $A^2 > 192/25$, $C_1^2 > 313/25$. Both orientations give $x_1 > .64$. The one legal graft and $e_Z > .09$ contradict even the D total:

$$.84816 + .09 + \frac{5}{4}(.64 - .361)^2 > 1.03546.$$

Every remaining head is PURE.

For a pure head its product with the actual defect wall is $-A\sqrt{h_T} < 0$. Their actual two-cycle gives

$$M = 4A^2 h_T = 2(U - 2)h_T \in \mathcal{O}_K, \qquad M \geq 1.$$

For N_7, $h_N < 1/12$ forces $U > 8$. For O_7, $M \geq 4$ likewise forces $U > 8$. For its finite mutual alphabet, use (TA). Put $k = \theta^2 + 2\theta - 1$. The cubic gives $1/h_O = k$, so $U = 2 + Mk/2$. The ODD order index means integrality of $Mk/2$ requires its three polynomial coefficients to be even. The COMPLETE table is

M	$[1]Mk$	$[\theta]Mk$	$[\theta^2]Mk$
1	-1	2	1
2	-2	4	2
3	-3	6	3
$2+\theta$	-1	5	3
$1+\theta+\theta^2$	1	6	2.

Only $M = 2$ remains. Then $U = (1 + \theta)^2$ and

$$.79 < 64/81 < u = 4/U < 625/784 < .8.$$

All pure heads, including every original finite own, have thus reduced to DOTTED owns with complete domains

$$0 < u < 1/2 \quad \text{or} \quad .79 < u < .8.$$

The actual positive point $c + g^{-1}n_1$ has square $-(1 - u)$, positive time, product zero with c, negative product with n_1, and nonpositive product with EVERY other original wall, including c_T.

It gives the same dotted-row Z, H, R pressure as before. On these TWO full intervals its scalar improves to

$$F = .8H - .16R > .3055.$$

For $u \le 1/2$, use $a = 2\sqrt{2}/5$, $.56568 < a < .5657$, the chord $\sqrt{2-u} > 1.4142 - .379u$ and $R \le 1 - .75u$. The circle inequality gives

$$F > .639984656 - \frac{.0944 + \sqrt{.0944^2 + .5657^2}}{2} > .305984656,$$

since $.0944^2 + .5657^2 < .5736^2$. For $.79 < u < .8$, direct positive squares give $\sqrt{2-u} > 1.0954$, $\sqrt{u(1-u)} < .4074$, $H > .4864$, $R < .52205$. Thus $F > .305592$. Both continuous domains, including unbounded dotted own, are covered.

Pay ONLY .08 from the individual P_5 price: $Y_H = Z - P_5/5$, $W = w + .4Y_H$. The residual is zero-sum on the first five coordinates, has FULL cross zero with w, and norm $R - 1/5$. The effective gain is

$$F - .8b + .032 > .0487. \tag{TG}$$

For B_{14}, adding spectator ONCE already gives $.86989 + .0487 + .09 = 1.00859 > 1$. This completes BOTH typed B_{14} cases. For D_{14}, the weaker E_{11} reference still needs its own centre-twelve proof.

C.6.2 The full D_{14} centre at its actual rank

Use actual roots $n_i = (e_i - e_{i+1})/\sqrt{2}$, $i \le 13$, $n_{14} = (e_{13} + e_{14})/\sqrt{2}$. The centre is twelve, forks thirteen and fourteen. The signed coordinate x_{14} is retained. At this point ONLY $5, \dots, 11$ are strong; E_{12} is a scalar value, not yet an available floor. The reference through eleven has individual terminal price $\sqrt{11} - \sqrt{10} > 1/7$, $f_{11} < 2.886$, and the head residual Y_H is supported on the first five coordinates.

Deleting twelve retains A_{11}, both genuine fork A_1 roots and the entire old spectator. The COMPLETE finite-white list is: without forks, isolation, simple three at $j = 1, 2, 10, 11$, or endpoint four at $j = 1, 11$; with left contact and forks, only endpoint-three at one/eleven with one fork-three/four or both forks-three; without left contact, one arbitrary finite fork label, or fork pairs $(3,3), (3,4), (3,5)$. Five is absent in K_7. Two left contacts or two contacts within old A_2 form a nonpositive cycle. Next-to-end plus a fork has nonfinite arms $1, 2, 9$; a four internal to a long path is nonfinite. Every left/fork bridge is foreign-sealed: an extra old-A_2 arm makes arms $11, 1, 2$ or adds another branch to a full D_{14} tree. All other foreign contacts retain (TQ).

For the actual opposite write

$$c = Ap - g\omega_{12} - \eta\omega_j - \delta\omega_{13} - \epsilon\omega_{14} - v_Z.$$

A missing contact has coefficient zero. The exact actual Schur and outgoing data are

$$q_L = \frac{2\eta^2 j(12-j)}{12}, \quad \Delta = 1 - q_L - \delta^2 - \epsilon^2 - Q > 0,$$
$$\sigma = g + \eta j/12 + (\delta + \epsilon)/2, \qquad C_{12} = \sqrt{24}\sigma/\sqrt{\Delta}.$$

The .61 sufficient ratio is $< 25/16$. For $a = \sqrt{24}$, $4.8 < a < 5$, its exact expression is

$$\frac{2 - 2(.61)/a + (.61/a)^2}{1.22 - .3721/a} < \frac{1.773}{1.142} < \frac{25}{16}.$$

Thus any ratio above 25/16 proves a true strong-twelve floor.

For no fork and a left contact, own-zero far-three would require $1/2 < Q < 13/24$, next-to-end-three would require $0 < Q < 1/6$, and far-four would require $0 < Q < 1/12$. None occurs in the COMPLETE foreign list; the latter two have zero time at $Q = 0$. Near/next-three own zero has the affine double-fork principal: preceding old node, path to twelve, both forks and cap. Near-four has the actual kernel $(\sqrt{2}, 2, 2, 1, 1)$ on $c, n_{11}, n_{12}, n_{13}, n_{14}$. Own three with white three has the actual cycle; own four the star at twelve with leaves cap and both forks; own six the $6, 3$ path to a fork; parallel the null pair. Own five is absent. Positive endpoint-four ratios are at least $\sqrt{12}/2 + 1/\sqrt{24} > 25/16$; remaining next-to-end-three ratios exceed 2.4, and near-three ratios exceed 1.84. A possible weak far-three is precisely $g\omega_{12} + \omega_1/2$, with $.9 < g < 6/5$. It is excluded by Lemma 20.3 at twelve, which includes arbitrary foreign contacts.

For no fork and isolation, own zero has negative time. Own three has the actual four-leaf-three star; four/six/parallel have the preceding star/path/null, and five is absent. For $Q > 0$ after these removals, or for dotted $Q = 0$, put $u = (1 - Q)/g^2 < 1$. The original positive clearing $c + v_Z + gun_{12}$ is future negative and gives

$$x \cdot (\omega_{12} - un_{12}) \geq \sqrt{24 - u} - \sqrt{u(1 - u)}.$$

Here v_Z is a positive inverse-Gram combination of ACTUAL old spectator normals. At $u = 1$ this is a forbidden feasible null. For $0 < u < 1$, the loss below $\sqrt{24}$ is $< .104u + \sqrt{u(1 - u)} < .555 < .61$; use $\sqrt{24} + \sqrt{23} > 9.69$ and $\sqrt{1 + .104^2} < 1.006$. Because $x \cdot n_{12} > 0$, this proves strength. For every pure finite own at least seven, $1 < u = g^{-2} < 4/3$, use instead

$$\xi = c + g^{-1}n_{12} + 2(g^{-1} - g)n_{11}.$$

It has square $-g^2(7u - 4 - 3u^2) < 0$, products zero with cap and twelve, negative with eleven and both forks, positive future time, and all omitted original-wall products nonpositive termwise. Its displacement is $\omega_{12} - un_{12} - 2(u - 1)n_{11}$; the pressure is $\sqrt{24 - u} - \sqrt{7u - 4 - 3u^2}$. The first radical loses $< .139$, while

$$7u - 4 - 3u^2 = 1/12 - 3(u - 7/6)^2 < .289^2.$$

The total loss is $< .428 < .61$. This covers the ENTIRE finite native tail.

For left endpoint-three plus forks, $Q = 0$. With both forks-three, $\Delta = 1/24$, $\sigma \geq 1/2 + 1/24$; with one fork-four, $\Delta = 1/24$, $\sigma \geq 1/(2\sqrt{2}) + 1/24$. Both are directly strong even at own zero. With one fork-three, $\Delta = 7/24$; own zero has its actual cycle, own three its contacted-fork triangle, and every own at least four is directly strong. The far orientation is used in each lower bound, so near orientations are also covered.

With no left contact and positive own, a single fork-three/own-three has its affine triangle. Single fork at least six with own three is directly strong. Single fork-four/own-three with $Q > 0$ has $\Delta \leq 1/6$ and is strong; its $Q = 0$ case is retained below. Own four has the star at twelve with leaves cap, eleven and the UNCONTACTED fork. Own six has the $6, 3$ path through eleven; parallel has its null pair. For every remaining own at least seven or dotted, fork at least four is directly strong. Fork-three with $Q > 0$ has $\Delta \leq 5/12$ and is strong. At $Q = 0$, a possible weak fork-three has $.9 < g < 6/5$. Its ACTUAL triangle makes $t = 2g$ integral. Normalize the connected simple-three cap to raw square two; all nonphysical K_7 embeddings obey

$$48t_\sigma(t_\sigma + 1) + 14 < 8, \qquad -1 < t_\sigma < 0.$$

Its nonzero integer $t(t + 1)$ has norm $< (204/25)(1/4)^2 = .51 < 1$, impossible. For two forks-three, own three has its triangle and owns at least four are directly strong. Pair three/four is directly strong for every positive own.

With no left contact and own zero, single fork-three has the actual affine E_8 kernel: centre twelve coefficient six, uncontacted fork three, contacted fork four, cap two, and old eleven through seven coefficients five through one. Single fork-four has kernel $(\sqrt{2}, 2, 2, 1, 1)$ on cap, contacted fork, twelve, eleven and other fork. Single fork-six has the $6, 3$ principal. Every null has positive cap time and correct signs against ALL omitted ORIGINAL walls. Single fork-five is absent. For any finite fork label at least seven, $\delta > .9$; finite whiteness $\delta^2 + Q < 1$ forces $Q = 0$. Two forks-three have the actual four-cycle; pair three/four has $Q = 0$ and remains below. Thus the ONLY nonstrong possibilities are: own-three/single-fork-four $Q = 0$; own-zero/pair-three-four; own-zero/single-finite-fork-at-least-seven. All other own, foreign and unbounded dotted cases have been covered.

C.6.3 Reciprocal endpoints and the remaining continuous fork family

For own-three/single-fork-four, the SAME actual cap is also the contacted fork's opposite. Deleting that fork retains A_{13}; adding the cap at its next-to-end node by three gives finite D_{14}. For own-zero/pair-three-four, release the fork-three direction instead; the retained A_{13} and cap give finite B_{14}. In both cases keep the ENTIRE orthogonal spectator. Positive inverse finite-Gram projection from p gives an actual future vertex on that original edge. Every unknown wall remains strict: its product with p is strict and all correction coefficients are nonnegative. The omitted old fork is strict because the cap coefficient and that fork contact are positive. This proves an ACTUAL endpoint, not a formal completion or a new farthest vertex.

Both reciprocal fork coshes are

$$C_f = 6 + 7\sqrt{2}/2 > 10.$$

In the first case $A^2 = (17 + 12\sqrt{2})/2$, $v \cdot \omega_f = 6 + 7/\sqrt{2}$, $(v \cdot \omega_f)^2 - 7A^2 = 1$. In the second, $A^2 = (17 + 12\sqrt{2})/4$, $v \cdot \omega_f = 7/2 + 3\sqrt{2}$, and the difference is $1/2$. These give the displayed cosh by the original edge formula. For either sign, the genuine spatial direction $V = P_{13} \pm e_{14}$, $V^2 = 14$, gives

$$x \cdot V > \sqrt{126/11} > 3.38.$$

Pay $\mu = 1/7$ at the individual P_{11} coefficient $> 1/7$. The residual $V - P_{11}$ vanishes on all first eleven coordinates, has square three and full cross zero with w. One original square WITHOUT any head credit gives

$$e_D + e_Z > .84816 + .09 + \frac{2(3.38 - 2.886)}{7} - \frac{3}{49} > 1.018.$$

Both exceptional reciprocal rows are excluded.

Finally let own be zero and the single finite fork contact have square $b_f = \delta^2 \in (.81, 1)$. Put $u = 1/b_f \in (1, 100/81)$; $Q = 0$ was proved, rather than assumed. The actual positive combination

$$\xi = c + \delta^{-1} n_f + 2(\delta^{-1} - \delta) n_{12}$$

has zero cap and contacted-fork products, negative centre and other-fork products, and square

$$-k, \qquad k = \frac{(1 - b_f)(4b_f - 3)}{b_f} > 0.$$

Its time is positive and every unknown original-wall product is nonpositive termwise. It is therefore a feasible future point of the ENTIRE polytope. For $\alpha = \delta/\sqrt{2}$, its divided spatial displacement

and pressure are

$$V = (1^{11}, 3-2u, u-1,\ \pm(1-u)),$$
$$x \cdot V \geq G = \sqrt{14-2u} - \sqrt{-8+14u-6u^2},$$
$$R = V^2 = 22 - 16u + 6u^2.$$

This retains the ACTUAL fork sign; it never assumes $x_{14} > 0$. The residual $V - P_{11}$ vanishes on the entire reference support and is orthogonal to Y_H.

On the ENTIRE interval,

$$14G - R > 30.58.$$

Here is a continuous rational seal. The chord $\sqrt{14-2u} \geq 3.756 - .292u$ has positive endpoint-square margins 11/15625, 574919/410062500. Put $z = u - 7/6$. Then

$$-8 + 14u - 6u^2 = 1/6 - 6z^2, \qquad R = 23/2 - 2z + 6z^2,$$

and $\sqrt{1/6 - 6z^2} \leq 1/\sqrt{6} - 3\sqrt{6}z^2$. Consequently

$$\begin{aligned} 14G - R &\geq 13618/375 - 14/\sqrt{6} - 2.088z + (42\sqrt{6} - 6)z^2 \\ &> 13618/375 - 5.716 - 2.088z + 96.8748z^2 \\ &> 13618/375 - 5.716 - .012 = 2294/75 > 30.58. \end{aligned}$$

Use $\sqrt{6} > 2.4494$, $5.716(2.4494) > 14$ and $2.088^2 < .048(96.8748)$ for the final completed square.

Now combine the TWO primitive pressures BEFORE taking the norm:

$$W = w + .4Y_H + (V - P_{11})/7.$$

Its INDIVIDUAL prices are .08 at P_5 and 1/7 at P_{11}. Both remaining coefficients are positive. Both residuals are orthogonal to w AND to each other. Using the head gain (TG) and the same spectator ONCE gives

$$\begin{aligned} e_D + e_Z &> E_{11} + .09 + .0487 + \frac{14G - R - 14f_{11} + 11}{49} \\ &> .84816 + .09 + .0487 + \frac{30.58 - 40.404 + 11}{49} = 1.01086 > 1. \end{aligned}$$

This is one original reference, not the addition of independent head and centre optima.

Every ACTUAL centre-twelve cap is now strong or impossible. The positive curve through twelve is available with $E_{12} > .86989$. Its first-five mean and individual P_5 price are unchanged. Apply the SAME actual head point and its primitive .08 payment once to this NEW reference:

$$e_D + e_Z > .86989 + .0487 + .09 = 1.00859 > 1.$$

This excludes BOTH D_{14} actual typed-seven branches. The order was fresh $6, \ldots, 11$, then five, then the actual head, THEN centre twelve and its reciprocal/continuous exceptions. Neither fork was assigned an unsupported prefix floor, and no actual short-I barrier or arithmeticity of a face was used.

C.7 A whole D_{13} component with any rank-three remainder

This argument covers EVERY finite rank-three spectator forest beside a whole D_{13} at the original farthest vertex in dimension sixteen. It does not depend on subtracting rows of a previous frontier. Retain actual unit D_{13} roots, long spine one through eleven, forks twelve and thirteen, and signed coordinates $x_1 > \cdots > x_{12} > |x_{13}|$. Write $e_D + e_S < 1$ for the SAME original vector and its entire orthogonal rank-three spectator.

Every such spectator has

$$e_S > .21. \tag{S3}$$

Here is the COMPLETE partition and exception argument. With ordinary components only, partitions three, two-plus-one and one-plus-one-plus-one cost $> 1/4$, $> 1/7 + .1$ and $> .3$. For actual I, the remaining rank two is one ordinary irreducible T_2, or two ordinary A_1's. The existing rank-free joint costs are $I + T_2 > .21$ and $I + A_1 > .20204$; the latter plus one further A_1 is $> .30204$. The actual I is counted once. A J pair plus the last ordinary A_1 costs $> .34 + .1$. An actual typed rank-two defect costs $> .09$. For N_7, existing propagation gives the last $A_1 > .2$; for O_7, it gives $A_1 > 2/15$. The compatibility theorem permits no additional I/J beside a typed defect and at most the single stated short-rank exception. These are every rank-three partition and exception. The propagation/joint theorems are the SAME whole component statements already used in the shared seals; no new field refinement is needed for this cost.

For D_{13}, choose generic parameter $M = 7$. Its exact rank hypothesis $13 = M + 6$ holds. Deleting $i = 5, \ldots, 8$ retains $A_{i-1} + D_{13-i}$, with both necessary side ranks at least four/five. The full finite connected replacement has rank thirteen, so exceptional high-rank alternatives do not arise; the existing small E_6, E_7, E_8 fork attachments are retained in the local list. The complete generic weak row is the pure far helper, excluded at $i \geq 6$ by Lemma 20.3. Thus six, seven and eight are strong before either short layer or strength at five.

We need only two earlier anchors, $K = 6, 7$. If five is strong use the positive ordinary reference with norm E_K. If five is weak, its actual pure point gives $x_1 > .5965$, $S_6 > 2.01815$. At $K = 6$ use coordinates $(.5965, .28433, \ldots, .28433)$. Their norm is $.7600299945 > E_6$, their sum $2.01815 < f_6$, and terminal coefficient $.28433 > 1/6$. At seven append $\sqrt{7} - \sqrt{6}$; the endpoint is $< f_7$, energy $> E_7$ and terminal coefficient $1/(\sqrt{7} + \sqrt{6}) > 1/6$. All individual duals are nonnegative. The strong-five terminal coefficients at six/seven are also $> 1/6$. In EITHER initial branch we therefore have an actual supported reference w on the first K coordinates satisfying

$$w^2 \geq E_K, \quad w \cdot P_K \leq f_K, \quad \text{individual } P_K \text{ coefficient} > 1/6.$$

The exact scalar seals are

$$E_6 > .69685,\ E_7 > .73531,\ E_{10} > .82436, \qquad f_6 < 2.019,\ f_7 < 2.215.$$

The fresh short layers are $i = 9$, with white $A_8 + D_4$, and $i = 10$, with white $A_9 + D_3$. Their COMPLETE left-only finite extensions are endpoint-three/four and next-to-end-three. Ranks nine/ten preclude a new exceptional type. Two contacts to either retained component form a forbidden cycle. The shared local inverse formulas and original spectator clearing apply at these

ACTUAL side ranks. We give the right and bridge cases explicitly:

right role	δ before Q	σ
D_4 long three	$1/2$	$\gamma + 1/2$
D_4 fork three	$1/2$	$\gamma + 1/4$
D_3 middle three	$1/2$	$\gamma + 1/2$
D_3 end three	$5/8$	$\gamma + 1/4$
D_3 end four	$1/4$	$\gamma + 1/(2\sqrt{2})$
D_3 end five	$(5 - 3\varphi)/8$	$\gamma + \varphi/4$.

Every foreign Q is the ENTIRE spectator cost, not an individual bond, and subtracts from the residual.

The long/middle-three own-zero rows have the actual two-fork/four-leaf affine kernel, own three their triangle; every remaining own at least four has ratio $\geq 1 + 1/\sqrt{2} > 100/61$. End-four own zero has the actual star of squared costs $1/4, 1/4, 1/2$; every positive own has the same direct lower ratio. End-five is the sealed H_4, strong even at own zero, since its residual is $< 1/40$ and numerator $> 2/5$. A left endpoint-three/right-simple-three bridge completes D_{13}, or, for D_3 end-three, A_{13}. The former has residual $1/(2i)$ and positive-own numerator at least $\gamma + 1/4 + 1/(2i)$; the latter at ten has residual $7/40$ and positive-own numerator $> .75$. Both ratios exceed $100/61$. Their own-zero cycles are affine. The bridge cap is internal in a rank-thirteen classical tree, so has no finite spectator arm; the A_{13} bridge's old arms have lengths nine and three. An end-four/five cannot join the long left path: its non-three bond would be internal in a nonfinite rank-thirteen tree.

For the left-only small owns, internal-three and endpoint-four own zero retain their actual affine fork principals. Own three with a white-three contact has its cycle. Own four with no right contact has the affine principal obtained from the retained right D_5/D_4 INCLUDING n_i: its unit long-leaf inverse diagonal is two, so a four attachment has cost one. Own six has its local $6, 3$ path. Parallel always has its future feasible null pair. Far-left-three at own zero with foreign contacts has $A^2 = Q - 1/2 < 1/(2i) < 1/4$. The original small-time shadow argument proved in the dihedral section would make the SAME cap close an orthogonal spectator direction, contrary to the original-farthest common-opposite obstruction. For $Q = 0$ its time square is negative. Thus that own-zero row also is impossible. Every null here is a positive combination of ACTUAL walls with positive cap time and nonpositive product at EVERY omitted original wall. The nonzero left-only ratios and foreign clearings are the legal shared prefix formulas at $i = 9, 10$. The only potentially weak left row is the literal far helper and is removed by Lemma 20.3.

For isolation, the original singleton and LEFT H_4/A_2 projected points and negative cores remain available through $i = r - 3$, with $D_i = 2i \geq 18$. They cover arbitrary foreign cost and all finite/parallel/dotted own, without a fictitious long root. For clarity their scalar margins still suffice here: the $0 < u < 1$ core has full loss

$$< \frac{.12 + \sqrt{1 + .12^2}}{2} < .57 < .61,$$

using $(\sqrt{18} + \sqrt{17})^{-1} < .12$. For foreign $u > 1$, $u < 7/6$, the radical loss is $< .14 + .4 < .61$. Pure native five's LEFT H_4 correction has loss $< .2 + .25 < .61$; native seven's LEFT A_2 correction has loss $< .15 + .36 < .61$; all finite natives at least eight use the singleton loss $< .15 + .415 < .61$. The underlying divided displacements and concavity are exactly those printed in Section C.5.8; only these freshly stated $D_i \geq 18$ scalar bounds are used, not its $i = 11, 12$ estimates.

For the retained single-fork-three/end-three roles, own zero, three, four and six have respectively the actual E_8, cycle, F_4 and $6, 3$ principals; all required old arms exist at nine/ten. Parallel has its

actual null pair. Thus $\gamma \geq \varphi/2$. If $Q > 0$, then $\delta \leq 3/8$, $\sigma > 21/20$, giving ratio $> 42/25 > 100/61$. For $Q = 0, \gamma \geq 6/5$, the ratio again exceeds $100/61$. ALL finite and dotted own values in between are retained. The ONLY possible weak short caps are therefore

$$v = \gamma\omega_i + \omega_f/2, \quad Q = 0, \quad i = 9, 10, \ f = 12, 13, \quad 202/125 < 2\gamma < 12/5.$$

Their actual simple-three contact normalizes the cap to raw square two over ACTUAL K. The original path cycle makes $t = 2\gamma$ integral. At EVERY nonphysical embedding,

$$4it_\sigma(t_\sigma + 1) + 13 < 8.$$

Hence $-1 < t_\sigma < 0$. All embeddings of the minimal field lift to K. The integer $t(t+1)$ has physical value $< 204/25$ and other absolute conjugates at most $1/4$. Degree at least three contradicts its nonzero integral norm bound $< .51$. In degree two the trace is one or two. Trace one gives norm -1 and $t = \varphi$; trace two gives norm in $(-24/25, 0)$, impossible. The rational value two is an actual parallel null. Every weak fork helper is therefore golden, without a field census or a scale comparison with spectators.

For each golden packet use the ACTUAL LEFT finite H_4 correction

$$w_{\text{old}} = \gamma(3n_i/2 + n_{i-1} + n_{i-2}/2), \quad \alpha^2 = \varphi + 5/8 > 56/25, \quad \delta = (5 - 3\varphi)/8, \quad \sqrt{\delta}/\alpha < .091.$$

The actual finite point is the normalization of $p + (A/\delta)(c + w_{\text{old}})$. It has square $-1 - A^2/\delta$, positive future time and zero selected products. Every omitted original-wall product is nonpositive by actual acuteness and the positive coefficients. The correction alone is not a hyperbolic point. For $Y = v - w_{\text{old}}$, $V = Y/\alpha$, $K = i - 3$, the fresh inverse calculations give

$$A^2 = i\alpha^2 - \kappa, \quad \kappa = 1/2 \ (i = 9), \quad \kappa = 5/8 \ (i = 10),$$
$$V_j = 1 \ (j \leq K), \quad V^2 < i, \quad x_D \cdot V > G = \sqrt{i - \kappa/\alpha^2} - .091.$$

Use the already established $K = 6/7$ reference in EITHER initial-five branch. Pay its INDIVIDUAL terminal coefficient $1/6$ BEFORE using its floor. For $W = w + (V - P_K)/6$, the residual vanishes on the first K coordinates, so its FULL cross with w is zero, and its norm is < 3. One square gives

$$e_D > E_K + \frac{G - f_K}{3} - \frac{1}{12}.$$

The COMPLETE fresh table is

layer	i	K	$G >$	$f_K <$	$e_D >$
D_4	9	6	2.871	2.019	.8975
D_3	10	7	3.026	2.215	.92231.

The two radical seals are $2.962^2 < 9 - 25/112$, $3.117^2 < 10 - 125/448$. Adding ONLY the actual disjoint spectator (S3) contradicts the original norm in BOTH cases. Thus all $6, \ldots, 10$ are now genuinely strong.

The final ordinary or legitimate weak-five reference through ten gives $e_D > E_{10} > .82436$. Consequently

$$e_D + e_S > .82436 + .21 = 1.03436 > 1.$$

This excludes WHOLE D_{13} with EVERY rank-three spectator forest at the actual farthest vertex in dimension sixteen. Every pressure belonged to one ORIGINAL vector; the spectator was counted once. No actual-I barrier, typed field lock, field census, arithmeticity of a face or unexecuted computation was used to enlarge this theorem's scope.

C.8 Whole B_{16}: geometry before the finite pair domain

Now the ENTIRE incident forest is B_{16}; there is no spectator to delete or to charge. Use actual unit roots $n_i = (e_i - e_{i+1})/\sqrt{2}$, $i \le 15$, $n_{16} = e_{16}$, and the original vector $z = E(p-x)$, $x_1 > \cdots > x_{16} > 0$, $\|x\| < 1$. The legal fresh bulk and B3/B2 terminal checks give strong $6, \dots, 14$ independently of five and either endpoint. We first prove all physical reductions needed to enter the exact quadratic algorithm.

C.8.1 The actual head and a tight short-edge length bound

For $a = .643$ or $.648$, the reference through $(1, a), (6, f_6), \dots, (14, f_{14})$ has positive decreasing coordinates

$$a, \quad (f_6 - a)/5 \text{ five times}, \quad \sqrt{j} - \sqrt{j-1}\ (7 \le j \le 14), \quad 0, 0.$$

Its first middle slope exceeds .274, hence the next square-root slope $< .2$. If actual $x_1 > a$, ONE primitive square gives

$$\|x\|^2 > a^2 + (f_6 - a)^2/5 + \sum_{i=6}^{13} (\sqrt{i+1} - \sqrt{i})^2.$$

The last sum is $> .2117$. A direct rational proof avoids a numerical receipt: for $A = 4i + 2$, put $L = (A^2 - 1)/(A(A^2 - 2))$. The small root $d = (\sqrt{i+1} - \sqrt{i})^2$ satisfies $d(A - d) = 1$, whereas

$$L(A - L) - 1 = -\frac{1}{A^2(A^2 - 2)^2} < 0.$$

Thus $d > L$. For $i = 6, \dots, 13$, strict lower numerators of L at denominator 10^7 are

$$385186, 333704, 294372, 263340, 238230, 217494, 200080, 185248.$$

Their sum is 2117654. With $f_6 > 2.01815$, the two reference norms are > 1.0033565045, > 1.0070662045.

The ACTUAL head's retained white diagram is B_{15}. Finite classification permits isolation or simple three at its LONG endpoint two, with no symmetric short-end alternative. For the latter, $A_H^2 = 2g^2 + 2g$, $C_H = 2g + 1$. Own zero has zero time, own three has its affine triangle, and parallel has the actual null pair. Every remaining own has $g \ge 1/\sqrt{2}$, so $x_1 > \sqrt{\sqrt{2} - 1} > .643$, impossible by the first reference. The head is isolated:

$$c_H = A_H p - \sqrt{2} g_H e_1, \quad U = 4g_H^2 \in \mathcal{O}_K, \quad A_H^2 = U/2 - 1, \quad U > 2.$$

If $U \ge 12$, its ACTUAL edge has $C_H = \sqrt{U/2} \ge \sqrt{6}$, giving $x_1 > .648$, impossible by the second reference. The strict rational seal uses $\sqrt{6} > 2.449$, $.648^2(3.449) < 1.449$. Thus $2 < U < 12$.

Let C_S be the ACTUAL short-sixteen edge cosh. Its actual finite endpoint and original comparison give $S_{16} \ge 4\sqrt{(C_S - 1)/(C_S + 1)}$. If $C_S \ge 16$, this is > 3.7573. For strong five, interpolate through true knots five through eight and this terminal pressure. The last slope is $< .172 < \sqrt{8} - \sqrt{7}$. Using

$$E_8 > .69685 + .0385186 + .0333704 = .768739, \qquad f_8 < 2.3971,$$

its norm is

$$> .768739 + 1.3602^2/8 = 1.000007005.$$

The upper seal for f_8 follows from $\sqrt{8} < 2.828428$, $\sqrt{2} < 1.414214$, $.431328(1.414214) < .61$; also $f_8 > 2.39$ gives the claimed last-slope bound. For weak five use the actual pressures $x_1 > .59$, $S_6 > 2$ and $S_{16} > 3.7573$. The decreasing slopes $.59, .282, .17573$ give one reference of norm

$$.59^2 + 1.41^2/5 + 1.7573^2/10 = 1.054530329 > 1.$$

Both alternatives contradict the SAME original norm without adding a head gain. Therefore $C_S < 16$.

C.8.2 All seven actual short-wall profiles

The retained ACTUAL A_{15} finite-white list is isolation; helper three at $j = 1, 2, 14, 15$; or helper four at $j = 1, 15$. Two contacts give a cycle. Other simple branches or non-three bonds cannot give a finite rank-sixteen tree. There are no foreign components.

Normalize old long roots to raw square two and the genuine short root to square one: $\alpha_i = e_i - e_{i+1}$, $s_{16} = e_{16}$. All e_i are then K-valued. A helper-three cap normalizes to square two, helper product -1, and short product $-T/2$, where $T = 2\sqrt{2}\gamma$. The ACTUAL cycles give $T, T^2/2 \in \mathcal{O}_K$, not $T/2$ integral. Its exact formulas are

$$c = Ap - \gamma P_{16} - P_j/\sqrt{2}, \quad 2A^2 = F_j(T) = 4T^2 + jT + j - 2,$$
$$C_S = \frac{8T + j}{\sqrt{32 - j(16 - j)}}.$$

The original old-plus-cap Gram is Lorentzian over $\mathbb{Q}(T)$, so actual $K = \mathbb{Q}(T)$, and every other $F_j(T_\sigma) < 0$. A helper-four cap has raw square one and helper product -1, with

$$A^2 = 16\gamma^2 + 2j\gamma + j - 1, \quad C_S = 16\gamma + j, \quad K = \mathbb{Q}(\gamma).$$

These field equalities follow from nonphysical embeddings fixing an original indefinite principal, not from arithmeticity of a face.

Own zero is retained before dispatch. Helper-three at one/two has $A^2 = -1/2, 0$. At fifteen its actual null is $c + n_{14} + 2n_{15} + \sqrt{2}n_{16}$; at fourteen it is $c + n_{13} + 2n_{14} + 2n_{15} + \sqrt{2}n_{16}$. Helper-four at one has zero time; at fifteen use $c + \sqrt{2}n_{15} + n_{16}$. Isolation has $A^2 = -1$. Every null has positive cap time and selected products zero; ALL omitted original-wall signs are nonpositive termwise. Parallel own always has $c + n_{16}$.

Near helper-four positive own has $C_S \geq 23$. Far helper-four has $\gamma < 15/16$, so its complete positive native alphabet is three through eight. The native-nine cubic at $15/16$ is $-17/512$, so all larger finite labels, parallel and dotted are beyond this cutoff. Own three is rational and excluded by WHOLE-form compact anisotropy/Meyer. For owns four through eight use respectively the nonphysical conjugates

$$-\sqrt{2}/2, \quad (1 - \varphi)/2, \quad -\sqrt{3}/2, \quad \cos(3\pi/7), \quad -\cos(\pi/8).$$

The resulting time squares are $8 - \sqrt{2}$, $9 - 5\varphi$, $12 - \sqrt{3}$, $16\gamma_\sigma^2 + 2\gamma_\sigma$, and $16\gamma_\sigma^2 - 2\gamma_\sigma$, all positive. They contradict admissibility in the ACTUAL generating field.

Far next-end-three has $T < 15/4$, every other $-1/2 < T_\sigma < 0$. In degree $d \geq 2$, its nonzero integral norm has magnitude $< 15/8 < 2$, hence one. But $T^2/2$ integral would make $\mathrm{N}(T)^2/2^d$ an integer, impossible. Degree one is rational. Near next-end-three has $T < 9/4$, leaving only own three $T = \sqrt{2}$ or own four $T = 2$. Own five already has $T^2 = 2\varphi^2 > 81/16$. The rational case is

impossible; the other conjugate time numerator is $20 - 14\sqrt{2} > 0$. Both next-end orientations and both helper-four roles are exhausted.

For the remaining far-end-three put $S = T$; for near-end-three put $S = T + 2$. Their time polynomials are $4S^2 + S - 1$ and $4S^2 - S - 1$, with respective nonphysical intervals

$$\frac{-1-\sqrt{17}}{8} < S_\sigma < \frac{-1+\sqrt{17}}{8}, \qquad \frac{1-\sqrt{17}}{8} < S_\sigma < \frac{1+\sqrt{17}}{8}.$$

Both have $0 < S < 42/5$, $|S_\sigma| < .641$ and $S^2/2$ integral. For degree $d \geq 3$, the nonzero integer norm is divisible by four but has magnitude $< (42/5)(.641)^2 = 3.4514004 < 4$. Degree one is rational; actual degree is two. Trace and norm are EVEN by the same half-square argument already proved above.

The hand table uses the EXACT physical short-edge cutoff $C_S < 16$, rather than only the coarser $S < 42/5$ used to bound the degree:

$$\begin{array}{r|l} \text{far} & \sqrt{2} \leq S < (16\sqrt{17}-1)/8 < 65/8 \\ \text{near} & 2+\sqrt{2} \leq S < (16\sqrt{17}+1)/8 < 67/8. \end{array}$$

Indeed the actual edge formulas are $C_S = (8S+1)/\sqrt{17}$ and $C_S = (8S-1)/\sqrt{17}$, respectively; the lower bounds retain the native own gap. For example the far integer $4 + 3\sqrt{2}$ passes the coarse bound but violates this exact physical cutoff and is not a table row.

The far strips give traces $2, 4, 6, 8$, $-6 < \mathrm{N}(S) < 4$. The near strips give traces $4, 6, 8$, $-4 < \mathrm{N}(S) < 6$. Solving ALL even trace/norm pairs and imposing the two strict strips gives the COMPLETE eleven rows:

role	T	$\mathrm{N}(F_j(T))$
far	$2+\sqrt{6}$	-53
	$3+\sqrt{13}$	-25
	$3+\sqrt{11}$	-151
	$3+\sqrt{7}$	-19
	$4+\sqrt{14}$	-117
near	$\sqrt{2}$	-9
	$1+\sqrt{11}$	-43
	$1+\sqrt{7}$	-103
	$2+3\sqrt{2}$	-137
	$2+\sqrt{14}$	-229
	$2+2\sqrt{3}$	$-83.$

For explicit coverage, the far norm columns are $-4, -2, 2$. Trace two gives only inadmissible conjugates or nonreal roots; trace four retains $2+\sqrt{6}$; trace six retains all three $3+\sqrt{13}$, $3+\sqrt{11}$, $3+\sqrt{7}$; trace eight retains only $4+\sqrt{14}$. The near S norm columns are $-2, 2, 4$. Trace four retains only $2+\sqrt{2}$; trace six retains $3+\sqrt{11}, 3+\sqrt{7}$; trace eight retains $4+3\sqrt{2}, 4+\sqrt{14}, 4+2\sqrt{3}$. Subtract two to recover the printed near T. The omitted rows violate a strict conjugate/physical bound or are repeated rational roots. All displayed norms are negative and their physical times positive, so the other time signs are exactly the required negative ones. These are necessary actual models, never examples.

C.8.3 The original connected-basis squares remove every helper

Normalize the actual head to the K-root $h = c_H/(\sqrt{2}g_H)$, with $h \cdot \alpha_1 = -1$, $h^2 = 2/U$. Every nonphysical head Schur complement gives $0 < U_\sigma < 2$. For any of the eleven helper-three rows, the

sixteen old raw roots and $r = \sqrt{2}c$ are an ORIGINAL K-basis. The coefficient of r in h, computed from its positive time coefficients, therefore gives

$$R_0 = \frac{U-2}{UF_j(T)} \in K^2.$$

This derives a common time-line square in one actual connected basis. It does not put separate unit roots or square roots in K.

For quadratic K, the full head strip gives

$$0 < \mathrm{N}(U) < 24, \qquad 0 < -\mathrm{N}(U-2) < 20.$$

For all five NEAR DOTTED rows and the far $\sqrt{6}, \sqrt{11}$ rows, the absolute time norm is a prime at least forty-three. Its valuation in $\mathrm{N}(R_0)$ is odd because neither of the two bounded integer norms can contain it. For far $\sqrt{7}$, the FULL ring $\mathbb{Z}[\sqrt{7}]$ gives the complete head list

$$U = 3+\sqrt{7},\ 4+\sqrt{7},\ 6+2\sqrt{7}, \quad \mathrm{N}(U) = 2, 9, 8, \quad \mathrm{N}(U-2) = -6, -3, -12.$$

Writing $U = a + b\sqrt{7}$ shows $b = 1, 2$; the strict head strips give precisely these a's. None contains prime nineteen. For far $\sqrt{14}$, its full strip similarly gives

$$U = 4+\sqrt{14},\ 5+\sqrt{14}, \quad \mathrm{N}(U) = 2, 11, \quad \mathrm{N}(U-2) = -10, -5.$$

The odd prime thirteen in time norm -117 therefore cannot cancel. These are nine eliminated rows.

The near $T = \sqrt{2}$ row needs its FULL square test rather than a norm-only rejection. Here actual $K = \mathbb{Q}(\sqrt{2})$, because its connected old-plus-cap Gram is physically Lorentzian over that field. Put $s = \sqrt{2}$, $\epsilon = 1 + s$. The ACTUAL cap has

$$A_F^2 = (21+15s)/2 = 3\epsilon^3/2, \qquad c_F = A_F p - P_{16}/2 - P_{15}/s.$$

The complete head strip in $\mathbb{Z}[s]$ is

$$U = 2+s,\ 3+s,\ 3+2s,\ 4+2s,\ 5+3s,\ 6+3s,\ 6+4s.$$

Indeed $U = a + bs$ has $1 \le b \le 4$; the allowed pairs are $b = 1 : a = 2, 3$; $b = 2 : a = 3, 4$; $b = 3 : a = 5, 6$; $b = 4 : a = 6, 7$, with the last $a = 7$ beyond twelve. For $R_0 = (U-2)/(3\epsilon^3 U)$, the norms are

$$1/9,\ 1/63,\ 7/9,\ 1/18,\ 1/7,\ 1/81,\ 4/9.$$

The four nonsquares are impossible. At $U = 2+s$, $R_0 = 1/(3\epsilon^4)$; at $U = 6+4s$, $R_0 = 2/(3\epsilon^4)$. Neither $1/3$ nor $2/3$ is a square in $\mathbb{Q}(s)$: a rational square in that field requires the rational number itself or half of it to be a rational square. The last row $U = 6+3s$ REALLY gives a square, with its positive physical root $k = 1/(3\epsilon)$. For the ACTUAL mutual magnitude $M = -c_H \cdot c_F$,

$$A_H A_F / g_H = \epsilon^2, \qquad M/g_H = (4+3s)/2, \qquad 4M^2 = (174+123s)/2.$$

Its rational norm is $9/2$, so it is not integral. This contradicts the actual two-cycle. The positive time sign is fixed by actual outward roots, not an optional square-root sign. Thus the near own-three model also is excluded.

The far $T = 3 + \sqrt{13}$ model is last. Put $r = \sqrt{13}$, $\eta = (3 + r)/2$. Its time polynomial is $F_1(T) = 5\eta^3$. Use the FULL ring $\mathbb{Z}[(1 + r)/2]$. The complete head strip gives the SIX rows

U	$\mathrm{N}(U)$	$\mathrm{N}(U-2)$	$\mathrm{N}(R_0)$
$(5+r)/2$	3	-3	$1/25$
$(7+r)/2$	9	-1	$1/225$
$4+r$	3	-9	$3/25$
$5+r$	12	-4	$1/75$
$(11+3r)/2$	1	-17	$17/25$
$(13+3r)/2$	13	-9	$9/325$.

Writing $U = (a + br)/2$, same-parity integers, the strip gives $b = 1, 2, 3$ with respectively $a = 5, 7;\ 8, 10;\ 11, 13$. The last four norm entries are nonsquares. For the first, $R_0 = (101 - 28r)/15$. A full field square would have rational coefficient square $52/15$ or $49/15$, neither rational-square. For the second, multiplying by the square η^2 gives

$$\eta^2 R_0 = (7 - r)/90.$$

Its rational coefficient square would be $13/180$ or $1/180$, again impossible. No norm-square is treated as sufficient. All seven actual white profiles are now exhausted except isolation.

C.8.4 The actual isolated pair and the entire cubic branch

The actual short cap is $c_S = A_S p - g_S P_{16}$. Own zero and parallel are impossible as above. Put $V = 4g_S^2$. Rescale the ACTUAL K-root to raw norm V, old-short product $-V/2$: physically $F_S = \sqrt{V}\, c_S$, spatial $(V/2)P_{16}$, time square $V(4V - 1)$. Its original Lorentzian Gram over $\mathbb{Q}(V)$ gives ACTUAL $K = \mathbb{Q}(V)$, and EVERY nonphysical place has $0 < V_\sigma < 1/4$. The short edge bound gives $V < 64$. Rational K is excluded by WHOLE-form Meyer. Finite own would give $V < 4$ and $0 < \mathrm{N}(V) < 4(1/4)^{d-1} \le 1$, impossible. Thus $V > 4$, the short own is dotted. Degree at least four gives $\mathrm{N}(V) < 64(1/4)^3 = 1$. Hence actual degree is two or three.

Normalize $h = c_H/(\sqrt{2}g_H)$, $r_F = c_S/g_S$, so $h^2 = 2/U$, $r_F^2 = 4/V$, spatial parts e_1, P_{16}. The original old sixteen roots plus r_F form a K-basis. The positive coefficient of r_F in h gives

$$k^2 = \frac{(U-2)V}{4U(4V-1)} \in K^2, \qquad k > 0. \tag{B16K}$$

Let $H = Uh$, $F = (V/2)r_F$. The ACTUAL full long-path cycle and two-cycle give

$$\begin{gathered} G = -2H \cdot F = 4Uk(4V-1) - UV \ge 0, \qquad L = \frac{G^2}{2UV}, \qquad G, L \in \mathcal{O}_K, \\ (G + UV)^2 = 4UV(U-2)(4V-1), \\ L + G = ((15U - 32)/2)V - 2U + 4. \end{gathered} \tag{B16C}$$

The old-path factors are -1, followed by $-V, -G/V$, whose complete product is G. Orthogonal $G = L = 0$ is retained. The last identity, subtracting seven integral copies of UV, implies $UV/2$ integral. This is rank sixteen's own half-product, not the odd-rank-seventeen census invariant. Thus every actual pair has the COMPLETE domain

$$\begin{gathered} K = \mathbb{Q}(U) = \mathbb{Q}(V), \quad d = 2 \text{ or } 3, \quad U, V, UV/2 \in \mathcal{O}_K, \\ 2 < U < 12, \quad 4 < V < 64, \qquad 0 < U_\sigma < 2, \quad 0 < V_\sigma < 1/4. \end{gathered}$$

Suppose $d = 3$. The positive norm $\mathrm{N}(V)$ is one, two or three. For odd norm, V is a unit at every dyadic prime, so $UV/2$ integral makes $U/2$ integral. For norm two, (V) is a SINGLE prime $\mathfrak{p}$ of norm two, exponent one. Use integer prime valuations and let $e = \mathrm{ord}_{\mathfrak{p}}(2)$. If $U/2$ were nonintegral, its only deficient prime could be $\mathfrak{p}$, with $\mathrm{ord}_{\mathfrak{p}}(U) = e-1$. Then $\mathrm{ord}_{\mathfrak{p}}(U-2) = e-1$, $\mathrm{ord}_{\mathfrak{p}}(4V-1) = 0$. The nonzero field square (B16K) would have valuation

$$(e-1) + 1 - 2e - (e-1) = 1 - 2e,$$

ODD, impossible for a square. Thus $W = U/2$ is integral in every case, with physical $1 < W < 6$ and both other conjugates strictly between zero and one.

We prove the small cubic-integer fact needed here. Any totally real cubic integer with physical root $w > 1$ and other roots $a, b \in (0, 1)$ has $w > 5$. Assume $w \leq 5$. Put integer trace $S = w+a+b \leq 6$, positive integer norms $A = wab$, $B = (w-1)(1-a)(1-b)$. Its monic polynomial is

$$P(X) = X^3 - SX^2 + (A + S - B - 1)X - A.$$

It is positive between the two distinct small roots. For $0 < X < 1$, it increases with S and decreases with A, B. If $A \geq 2$, then

$$P(X) \leq X^3 - 6X^2 + 6X - 2 < -5(X - 3/5)^2 - 1/5 < 0.$$

If $B \geq 2$, similarly $P(X) < -5(X - 2/5)^2 - 1/5 < 0$. Thus $A = B = 1$. For $S \leq 5$, $P(X) \leq X^3 - 5X^2 + 4X - 1 < -(2X-1)^2 \leq 0$. Only $S = 6$ remains, giving $P(X) = X^3 - 6X^2 + 5X - 1$. But $P(5) = -1 = (5-w)(5-a)(5-b)$ forces $w > 5$, a contradiction. Therefore $U > 10$.

The ACTUAL head edge now gives $x_1 > (\sqrt{5}-1)/2 > .618$. The strict conjugate short bounds give $V > 16\,\mathrm{N}(V) \geq 16$, so its ACTUAL short edge gives $S_{16} > 4\sqrt{7}/3 > 3.5276$. Together with unchanged $S_6 > 2.01815$, use ONE positive reference

$$w_0 = .33797P_1 + .129085P_6 + .150945P_{16}.$$

Its coordinates are .618, then five .28003's, then ten .150945's. The primitive square on the SAME original x gives

$$\|x\|^2 > .618^2 + 1.40015^2/5 + 1.50945^2/10 = 1.00185193475 > 1.$$

This excludes EVERY actual cubic field, with all dyadic ramification allowed and no polynomial field census. Actual K is consequently quadratic.

C.8.5 Exact inclusion in the distinct whole-B_{16} census

Every hypothetical whole B_{16} now satisfies EXACTLY the actual-root strips and cycles of Proposition 21.1(iv), with the original scales $U = 4g_H^2, V = 4g_S^2$. The square in (B16K) is a FULL K-square, with its PHYSICAL positive root. The exact G, L are (B16C). Acuteness gives $G \geq 0$. For $G = 0$, $L = L' = 0$; otherwise strict nonphysical two-root positivity gives $0 < L' < 4$. The own U and mutual L lie in the actual quadratic finite native alphabet or are > 4. Parallel four has its actual feasible null and is excluded; zero mutual is retained. These are necessary conditions of ACTUAL walls, not conditions assigned to a formal projection.

For completeness, this algorithm has a finite full-order domain justified by the physical strip. For the maximal basis $\omega = (1 + \sqrt{D})/2$ or $\sqrt{D}$, $U - U' = b\sqrt{\Delta} < 12$, hence fundamental discriminant

$\Delta < 144$. Trial $2 \le D \le 143$, exact squarefreeness and this discriminant test exhaust all actual fields. The FULL safe integer boxes are

$$U = a + b\omega: \quad 1 \le a \le 7,\ 1 \le b \le 5; \qquad V = a + b\omega: \quad 1 \le a \le 32,\ 1 \le b \le 28.$$

For U, the gap gives $b\sqrt{D} < 12$ in the half basis and $b\sqrt{D} < 6$ otherwise; $0 < U' < 2$ bounds a. For V, replace twelve/six by sixty-four/ thirty-two and use $0 < V' < 1/4$. These give the printed boxes, followed by EXACT strict two-place strips. At most ten head and twenty-eight short models occur per field. Visit EVERY ordered head/short pair. Exact rational-pair arithmetic, both signed-norm square-root branches, verified full square identities, integrality, conjugate and native predicates are the algorithm of Section 21. All loops are finite; there is no unbounded coefficient search or numerical root cutoff. The three disjoint discriminant intervals $[0, 48), [48, 96), [96, 144)$ cover the domain. A separate complete combine recomputes field and pair records; it is a check with the SAME implementation, not an independent algorithm.

At this precisely proved domain we use the recorded finite-output proposition 21.1(iv) as an EXPLICIT computed premise. Its 43 fields,100 head models,85 short models and273 pairs leave one necessary pair,

$$U = 5 + 2\sqrt{6}, \qquad V = 10 + 4\sqrt{6}.$$

It is not an empty census or a realized polytope. For this pair the ACTUAL head/short edges have

$$C_H = (2 + \sqrt{6})/2 > 11/5, \qquad C_S = 2(2 + \sqrt{6}) > 44/5.$$

They give $x_1 > .61$, $S_{16} > 3.56$. With unchanged $S_6 > 2.01815$, ONE reference

$$w = .32837P_1 + .127445P_6 + .154185P_{16}$$

has positive coefficients and norm

$$.61^2 + 1.40815^2/5 + 1.54185^2/10 = 1.00640742675 > 1.$$

The original completed square is contradictory. Every actual short profile, all ground fields and the complete nonempty quadratic pair domain have now been covered for whole B_{16}. The finite output remains the declared numerical premise in Proposition 21.1(iv).

C.9 Whole classical rank seventeen: actual fields and census inputs

The two rank-seventeen classical forests require DISTINCT geometric reductions. Their final quadratic parameter domain happens to agree, but this will be proved from each actual original-root configuration. We first record the common isolated-five lattice interface used in their head reductions.

C.9.1 The actual golden long-end star and all invariant lattices

For the lattice

$$L = \mathbb{Z}[\varphi]^{n+1}, \qquad f = -\varphi t^2 + x_1^2 + \cdots + x_n^2,$$

the subgroup generated by all integral hyperplane reflections has an infinite-volume fundamental chamber for $n \ge 8$ [12]. We show below that the relevant facet reflections preserve this lattice.

Put $u = \varphi - 1$ and use the unit-equivalent form $f = -ut^2 + \sum x_i^2$. A whole D_n long-end-five star has old raw roots $-e_i + e_{i+1}$, $-e_{n-1} - e_n$, and cap $r = e_0 + \varphi e_1$, all square two. Its full Lorentzian Gram forces actual $K = \mathbb{Q}(\varphi)$: any larger field has a nonphysical embedding fixing this indefinite

principal. The old group gives all even spatial sign changes and permutations. Conjugating the cap reflection gives maps R_i whose only nonidentity block, on zero and i, is

$$\begin{pmatrix} \varphi & -\varphi \\ 1 & -\varphi \end{pmatrix}.$$

A whole B_n head-five star has the SAME configuration and contains those even signed permutations, so the argument also applies there. Arithmeticity supplies a full invariant $\mathcal{O}_K$-lattice Λ, for example by summing finitely many translates of a lattice stable under an arithmetic finite-index subgroup.

Let $\mathcal{A}$ be the algebra generated by these lattice-preserving maps. For distinct spatial i, j, k, even sign changes d_{ij} give $(1 - d_{ij})(1 - d_{ik}) = 4E_{ii}$. Permutations give all spatial $4E_{ij}$. The displayed R_i and the unit φ then give $4E_{i0}, 4E_{0i}, 4E_{00}$; explicitly

$$\begin{aligned} 4E_{ii}R_i &= 4E_{i0} - 4\varphi E_{ii}, \\ R_i 4E_{ii} &= -4\varphi E_{0i} - 4\varphi E_{ii}, \\ R_i 4E_{i0} &= -4\varphi E_{00} - 4\varphi E_{i0}. \end{aligned}$$

The golden ring is a PID, for example by its Minkowski bound $\sqrt{5}/2 < 2$. Scale the coordinate-content ideal of Λ to the unit ideal. The matrix units imply $4L \subset \Lambda \subset L$. The FOURfold sandwich is essential for odd n.

Modulo two, $\mathcal{O}_K/2 = \mathbb{F}_4$. Put

$$\ell(t, x) = \varphi t + \sum x_i, \qquad H = \ker \ell, \qquad v = (\varphi^2, 1, \ldots, 1).$$

The only invariant subspaces are $0, \langle v \rangle, H, L/2L$. Indeed any unequal spatial coordinates give $e_i + e_j$ by a transposition difference; their orbit spans the spatial augmentation hyperplane, and R_i supplies a time direction, giving H. If every spatial coordinate is equal, R_i-invariance forces the line $\langle v \rangle$. Here $v \in H$ exactly for odd n.

For the two residue flags

$$W_1 = (\Lambda + 2L)/2L, \qquad W_2 = ((\Lambda \cap 2L)/2)/2L, \qquad W_1 \subset W_2,$$

the content normalization makes $W_1 \neq 0$. If W_1 is full, $\Lambda = L$. If $W_1 = H$, differences under even sign changes of a lift of $e_i + e_k$, divided by two, put a coordinate vector in W_2, forcing it full. If $W_1 = \langle v \rangle$, the same differences force $H \subset W_2$; for even n, W_2 is full, while for odd n it is H or full. Thus, writing E for the inverse image of H, the COMPLETE lattice list is

$$L, \quad E, \quad \mathcal{O}_K v + 2L, \quad \mathcal{O}_K(v + 2z) + 2E \quad (n \text{ odd}, \ z \in L).$$

This includes every lift, not merely every residue subspace.

Intrinsically,

$$E = \{x \in L : f(x) \in 2\mathcal{O}_K\}, \qquad E^* = L + \mathcal{O}_K(v/2), \qquad f(v) = n - 1 - 2\varphi.$$

The latter norm has dyadic valuation zero for even n, exactly one for odd n. Writing $x = y + av/2$ shows

$$\{x \in E^* : f(x) \in \mathcal{O}_K\} = L :$$

if a is not divisible by two, the final norm term has negative valuation and the other terms are integral. Hence every isometry of E, and by duality of $\mathcal{O}_K v + 2L = 2E^*$, preserves L. For a deep odd lattice put $\lambda = v + 2z$; its norm still has valuation one. Writing $x = a\lambda/2 + e$, $e \in E$, gives

$$\{x \in \Lambda/2 : f(x) \in \mathcal{O}_K\} = E.$$

Its isometries therefore preserve E and then L. Every invariant lattice in the COMPLETE list has $O(f,\Lambda) \subset O(f,L)$. Thus the actual reflection group lies inside the literal integral group. Its arrangement chamber refines the assumed compact chamber, contradicting the external literal no-compact-chamber cutoff (or the stronger infinite-covolume statement). This excludes the actual whole-B/D isolated long-end-five configuration, with every compatible invariant lattice retained.

C.9.2 Whole B_{17}: the fresh window and actual short wall

Use actual B_{17} roots and $x_1 > \cdots > x_{17} > 0$ from one original farthest vector. The generic parameter $M = 12$ proves the bulk through thirteen; literal far helpers at indices at least six are excluded arithmetically. At fourteen the right B_3 extra short-three extension is the sealed F_4: own zero has its affine principal, every positive own has ratio $\geq 1 + 1/\sqrt{2}$. At fifteen the right B_2 extra own-three/ short-three profile has the opposite-$\sqrt{2}$ Schur cost $15(3 - 2\sqrt{2})/4 + 1/2 > 1$. Its zero/four/six and parallel owns have actual nulls; larger owns and the full B_{17} bridge are directly strong. The bridge residual is $1/(2i)$, and even own zero has ratio $1/\sqrt{2i} + \sqrt{i}/2 > 100/61$. All required left corrections use genuine long node sixteen. Thus six through fifteen are strong.

If five is weak with finite own, its ACTUAL finite helper point gives $x_1 > .65$. The reference through one, six and every later strong knot gives

$$e_B > E_{15} + .65^2 + (2.01815 - .65)^2/5 - E_6 > .92564 + .7968 - .7 > 1.$$

For dotted weak five put $t = 2g \in (2, 12/5)$. Its actual path cycle makes t integral, with all other conjugates in $(-.69, .29)$. Degree at least four contradicts $2.4(.69)^3 < 1$; degree one has no integer; degree two has trace two and norm $t(2-t) \in (-24/25, 0)$. In degree three a positive small conjugate gives norm magnitude < 1. Two negative small conjugates force norm one; trace two violates $(t-2)^2 \geq 4/t$. Trace one forces $t > 9/4$, where the integer second coefficient $1/t + t - t^2$ lies strictly between $-3, -2$. Thus five is strong too.

The ACTUAL retained B_{16} head list is isolation or long-endpoint-three. The nonisolated case gives $x_1 > .643$, whose legal first-five graft gains $> .099405$ over $E_{15} > .92564$, impossible. For an isolated head, small own/nulls remove zero, three, four, six and parallel; the golden long-star lemma removes five. For finite own at least seven or dotted, the already proved actual first-three-coordinate point gives $x \cdot Z \geq H$ and $F = .8H - .16R > .2893$. Paying .08 at the INDIVIDUAL P_5 coefficient gives gain $> .0325$.

Independently, a legal first-five graft at $x_1 \geq .605$ would give $E_{15} + (5/4)(.605 - .361)^2 > 1.00006$. Hence $x_1 < .605$. For the actual head parameter $U = 4g_H^2$,

$$K = \mathbb{Q}(U), \quad U \in \mathcal{O}_K, \quad 2 < U < 28/3, \quad 0 < U_\sigma < 2.$$

The raw head has square $2U$, spatial Ue_1, time square $U(U-2)$. This original full-rank Lorentzian principal proves ACTUAL field equality. FINITE HEADS ARE RETAINED until excluded below.

We first derive a preliminary short bound without using that head. If actual $C_S \geq 15$, then $S_{17} > 3.856$. For strong five, the positive reference at $(5, 1.804), (17, 3.856)$ has norm 1.0017752. For weak five, use $(1, .59), (6, 2), (17, 3.856)$; its norm is $.74572 + 3.444736/11 > 1.05$. Therefore $C_S < 15$. The actual retained A_{16} short-white list is exactly isolation, helper three at $j = 1, 2, 15, 16$, or helper four at $j = 1, 16$.

For helper three put $a = 2\sqrt{2}\gamma$, raw cap square two and helper product -1. Actual cycles make $a, a^2/2$ integral. Its complete formulas are

$$A_j^2 = (17a^2 + 4ja + 4j - 8)/4, \qquad C_S = \frac{17a + 2j}{2\sqrt{34 - j(17-j)}}.$$

Here A_j is the RAW cap time coefficient. For helper four use raw cap square one, time square $17\gamma^2 + 2j\gamma + j - 1$, $C_S = 17\gamma + j$. The original connected Lorentzian principal gives ACTUAL $K = \mathbb{Q}(a)$ or $\mathbb{Q}(\gamma)$.

Own zero at helper-three one/two has negative/ zero time; near fifteen/sixteen has the actual positive null

$$c + n_{14} + 2n_{15} + 2n_{16} + \sqrt{2}n_{17}, \qquad c + n_{15} + 2n_{16} + \sqrt{2}n_{17},$$

respectively. Helper-four own zero at one has zero time; near sixteen already violates the length bound. Parallel always has the actual future feasible null pair. Every null is feasible against ALL original walls termwise.

At helper-three two, $a < 56/17$, every other $-8/17 < a_\sigma < 0$. Its norm is a unit, incompatible with $a^2/2$ integral. Near next-end fifteen leaves only own three $a = \sqrt{2}$, with inadmissible conjugate $(86 - 60\sqrt{2})/4 > 0$. Far helper-four has $\gamma < 14/17$, leaving only owns three/four/five. The rational case is impossible; the other time conjugates are $17/2 - \sqrt{2} > 0$, $(38 - 21\varphi)/4 > 0$. Thus all four non-end roles are excluded.

For far-end-three put $v = a$; for near-end-three put $v = a + 2$. Their nonphysical strips have absolute bound .617, with smaller one-sided bound .382; physical $v < 7.7$, and $v^2/2$ is integral. Degree at least three would give nonzero integer norm of magnitude $< 2.932 < 3$, but its half-square norm requires divisibility incompatible with norms one or two. Hence actual degree is two. Even trace/norm and the exact negative-time intervals give the COMPLETE tables

role	$(\mathrm{Tr}\, v, \mathrm{N}(v))$	a
far	$(4,-2),(6,2),(6,-2),(6,-4)$	$2+\sqrt{6},\ 3+\sqrt{7},\ 3+\sqrt{11},\ 3+\sqrt{13}$
near	$(4,2),(6,2),(6,-2),(8,4)$	$\sqrt{2},\ 1+\sqrt{7},\ 1+\sqrt{11},\ 2+2\sqrt{3}.$

For coverage: far traces $2, 4, 6$ and norms $-4, -2, 2$ exhaust the quadratic formula; near traces $4, 6, 8$ and norms $-2, 2, 4$ do likewise. All other rows have wrong physical/conjugate signs or are rational/nonreal. The exact time signs use

$$174 - 72\sqrt{6},\ 382 - 106\sqrt{13},\ 348 - 106\sqrt{11},\ 280 - 106\sqrt{7} < 0$$

for far roots, and

$$90 - 64\sqrt{2},\ 324 - 98\sqrt{11},\ 256 - 98\sqrt{7},\ 456 - 264\sqrt{3} < 0$$

for near v's. Thus the full exact intervals, not just safe decimal overstrips, are imposed.

An isolated short cap has invariant $V = 4\gamma_S^2$, raw square V, spatial $(V/2)P_{17}$, time square $V(17V - 4)/4$. Its original Lorentzian principal gives $K = \mathbb{Q}(V)$, $0 < V_\sigma < 4/17$. Finite own would have nonzero integral norm $< 16/17$; parallel is its null pair. Thus it is dotted. The preliminary $C_S < 15$ gives $V < 900/17$; degree at least four gives norm $< 57600/83521 < 1$. So the ACTUAL field, whether the short cap is isolated or one of the helper models, has degree two or three. The rational field was separately excluded by the WHOLE rank-eighteen Meyer obstruction.

C.9.3 Whole B_{17}: exact head/short compatibility

Before excluding any finite head, its actual degree bound and $0 < U_\sigma < 2$ give the COMPLETE remaining finite list

native	U
7	$U^3 - 5U^2 + 6U - 1 = 0$
8	$2 + \sqrt{2}$
10	$(5 + \sqrt{5})/2$
12	$2 + \sqrt{3}$
18	$U^3 - 6U^2 + 9U - 3 = 0.$

Indeed $\phi(m) \le 6$ gives $3,4,5,6,7,8,9,10,12,14,18$; the earlier null/golden exclusions remove three/four/five/six. Nine and fourteen have a nonphysical $U > 2$. In particular finite native eight in $\mathbb{Q}(\sqrt{2})$ MUST still be retained.

The individual P_5, P_{15} prices exceed $.14, .13$. For any actual short pressure $S_{17} > T$, ONE vector

$$w_{15} + .4(Z - P_5/5) + .13(P_{17} - P_{15})$$

spends $.08, .13$ before using either floor. The head residual is zero-sum on the first five; the terminal residual is disjoint with square two. All FULL crosses with the reference and each other vanish. Thus

$$\|x\|^2 > E_{15} + .0325 + .26(T - f_{15}) - .0169(2).$$

If $C_S \ge 11$, the actual short edge gives $S_{17} \ge \sqrt{85/6} > 3.763$. With $E_{15} > .92564, f_{15} < 3.442$, the right side is > 1.0078. Therefore ACTUAL $C_S < 11$.

Of the eight short-helper models, the actual coshes $(17a+2)/(6\sqrt{2})$ and $(17a+32)/(6\sqrt{2})$ leave only far $a = 2+\sqrt{6}$ or near $a = \sqrt{2}$. For both remaining rows the raw cap time norm has prime twenty-three to odd valuation:

role	H_c^2	$-16\mathrm{N}(H_c^2)$
far	$(87+36\sqrt{6})/2$	828
near	$(45+32\sqrt{2})/2$	92.

The ACTUAL field equals that of a. The COMPLETE compatible dotted head strips $4 < U < 11, 0 < U' < 2$, with their $h_0 = -\mathrm{N}(U(U-2))$, are

K	U	h_0
$\mathbb{Q}(\sqrt{6})$	$3+\sqrt{6},\ 4+\sqrt{6},\ 5+2\sqrt{6},\ 6+2\sqrt{6}$	$15, 20, 15, 96$
$\mathbb{Q}(\sqrt{2})$	$3+\sqrt{2},\ 3+2\sqrt{2},\ 4+2\sqrt{2},\ 5+3\sqrt{2},\ 6+3\sqrt{2}$	$7, 7, 32, 63, 36$
$\mathbb{Q}(\sqrt{2})$ finite	$2+\sqrt{2}$	4.

The full rings are the displayed square-root orders. Writing $U = A + B\sqrt{d}$ gives $B < 11/(2\sqrt{d})$, and $A = \lfloor B\sqrt{d} \rfloor + 1, +2$; the physical strip gives exactly the rows. There is no finite head in $\mathbb{Q}(\sqrt{6})$; the finite $\sqrt{2}$ head is explicitly included. Both raw roots occupy the SAME actual K-time line, so the rational time-norm ratio $p/(16h_0)$ must be a square. Its twenty-three valuation is odd in EVERY row. No nonisolated short cap survives. The actual short cap is now isolated dotted:

$$K = \mathbb{Q}(V), \quad 4 < V < 484/17, \quad 0 < V_\sigma < 4/17.$$

Derive the B17 cycles in THEIR OWN actual scales: head raw square $2U$, time square $U(U-2)$; short raw square V, time square $V(17V-4)/4$. The actual long path makes $a = 2(r_H, r_S)$ integral and its actual two-cycle makes $L = a^2/(2UV)$ integral. The original common time line gives

$$\begin{gathered} R_0 = U(U-2)V(17V-4) \in K^2, \qquad a = UV - \sqrt{R_0}, \quad \sqrt{R_0} > 0 \text{ physically}, \\ L - a = 8UV - 17V - 2U + 4 =: d_0, \qquad a^2 = 2UV(a + d_0). \end{gathered} \tag{B17C}$$

No D17 fork normalization was used.

First exclude ALL finite heads. In degree three, $0 < \mathrm{N}(V) < (484/17)(4/17)^2 < 2$, so V is a unit. At every dyadic prime, $d_0 \equiv -V$ is a unit; the cycle equation forces a into the prime and then

$$2\,\mathrm{ord}(a) = \mathrm{ord}(2) + \mathrm{ord}(U).$$

For finite cubic heads seven/eighteen, the actual field has odd discriminant forty-nine/eighty-one, and $\mathrm{N}(U) = 1, 3$. Thus two is unramified and U a dyadic unit; the equality reads an even number equals one. Both finite cubic heads are excluded without assuming $U > 4$.

For finite quadratic heads eight/ten/twelve, the FULL short strips have exactly these seven models and necessary rational time-norm ratios:

head	V	$\mathrm{N}(H_S^2/H_H^2)$
8	$3 + 2\sqrt{2}$	103/64
8	$10 + 7\sqrt{2}$	383/16
12	$7 + 4\sqrt{3}$	647/48
12	$14 + 8\sqrt{3}$	61
10	$2 + 3\varphi$	171/80
10	$5 + 8\varphi$	919/80
10	$7 + 11\varphi$	239/16.

Coverage follows from the full-order coefficient ranges $B = 1, \ldots, 10$ in $\sqrt{2}$, $1, \ldots, 8$ in $\sqrt{3}$, $1, \ldots, 12$ in φ. The corresponding complete ceil arrays for A are

$$2, 3, 5, 6, 8, 9, 10, 12, 13, 15;$$
$$2, 4, 6, 7, 9, 11, 13, 14;$$
$$1, 2, 2, 3, 4, 4, 5, 5, 6, 7, 7, 8.$$

Exact $0 < V' < 4/17$ retains only $B = 2, 7;\ 4, 8;\ 3, 8, 11$, respectively. Rational brackets for the square roots and φ decide every ceil and endpoint. If $n = \mathrm{N}(V)$, $T = \mathrm{Tr}(V)$, and $h_0 = -\mathrm{N}(U(U-2))$, each ratio is

$$\frac{n(68T - 289n - 16)}{16h_0}.$$

None in the COMPLETE table is a rational square: the reduced denominator has odd three/five valuation where indicated, and the remaining integer numerators $103, 383, 61, 239$ are nonsquares. Thus every finite quadratic head is excluded. Only NOW conclude $4 < U < 28/3$.

In degree three the unit V and cycle valuation remain valid. If two is unramified they force $U/2$ integral. Its physical value is in $(2, 14/3)$, both other conjugates in $(0, 1)$, contradicting the complete cubic-integer lemma of Section C.8.4, which requires physical value > 5. If two ramifies, V's monic unit polynomial $X^3 - AX^2 + BX - 1$ has a repeated factor modulo two, forcing A, B ODD. Its two small positive roots in $(0, 4/17)$ and physical $V < 484/17$ give

$$A = 19, 21, 23, 25, 27, \qquad B^2 > 4(A - 8/17), \qquad B < \frac{272A + 4849}{1156}.$$

The least and largest possible odd B are, respectively,

$$(9, 7),\ (11, 9),\ (11, 9),\ (11, 9),\ (11, 9).$$

Every row is empty. This excludes the ENTIRE cubic branch in the actual B17 domain without a field census. The actual field is quadratic.

C.9.4 Whole B_{17}'s actual inclusion in the shared quadratic domain

Rename ONLY the algebraic variable $W = V$, $g = a$ in (B17C). We have proved from the ACTUAL B17 geometry

$$K = \mathbb{Q}(U) = \mathbb{Q}(W), \quad U, W \in \mathcal{O}_K,$$
$$4 < U < 11, \quad 0 < U' < 2, \qquad 2 < W < 676/17, \quad 0 < W' < 4/17,$$
$$g = UW - \sqrt{U(U-2)W(17W-4)}, \quad g, L = g^2/(2UW) \in \mathcal{O}_K.$$

The physical g is strictly negative: $d_0 > (8U-17)2 - 2U + 4 = 14U - 30 > 0$, so $R_0 - (UW)^2 = 2UWd_0 > 0$. Thus $L > 0$. At the other embedding actual positive definiteness gives $0 < L' < 4$. The physical L is a true finite native invariant or > 4; parallel four has its actual null pair. Every predicate of Proposition 21.1(iii) is now satisfied. This domain inclusion uses B17's short raw square V, not D17's fork square $4W$. The common algebraic formula is a proved coincidence of necessary invariants.

The complete full-order algorithm and its explicit output premise are already stated in Section 21. Its discriminant bound $\Delta < 121$, two conjugate-strip candidates for each U coefficient, one for each W, and $b_U \le 4, b_W \le 17$ cover every field and ordered pair and make every loop finite. Both signed-norm square branches are tested; a rational-square norm alone is not sufficient. The old 36-field/120-pair output leaves TWO necessary pairs, in $\sqrt{13}$ and $\sqrt{17}$. For the two actual surviving heads, $u = 4/U$ lies in $(.92, .94)$ and $(.719, .72)$. Their actual dotted head points give, respectively,

$$H > .535,\ R < .51; \qquad H > .482,\ R < .54.$$

Positive rational squaring proves the radical bounds $\sqrt{2-u} > 1.0295, 1.1313$ and $\sqrt{u(1-u)} < .272, .4495$, respectively. Paying .36 at the head costs ONLY .072 from P_5, and gives gains $> .085$, $> .043$. Both actual W's exceed $104/5$; the B17 ACTUAL short edge has $C_S^2 = 17W/4 > 442/5$, hence $S_{17} > \sqrt{357/26} > 3.7$.

Use ONE reference through fourteen:

$$w_{14} + .36(Z - P_5/5) + .13(P_{17} - P_{14}).$$

The individual prices $.072, .13$ are legal. The two residuals are disjoint, orthogonal to the reference and each other; terminal norm is three. Thus the SAME original vector would satisfy

$$\|x\|^2 > .9084 + .043 + .26(3.7 - 3.311) - .0169(3) = 1.00184 > 1.$$

Both complete census survivors are excluded. This closes whole B_{17}, conditional only on the explicitly identified external facts and the recorded complete quadratic output premise.

C.9.5 Whole D_{17}: fresh geometry before any field census

Use ACTUAL unit D_{17} roots, long spine one through fifteen, forks sixteen/seventeen, and

$$x_1 > \cdots > x_{16} > |x_{17}|, \qquad \|x\| < 1.$$

The signed coordinate is retained. The legal generic $M = 11$ window gives strong six through twelve. At thirteen/fourteen the COMPLETE right D4/D3 lists leave only the pure own-five/fork-three or own-five/end-three packets. Here completeness uses the SAME actual finite role list printed in Section C.7, with the FRESH indices thirteen/fourteen. The D4 long-three and D3 middle-three

roles have residual 1/2, shift 1/2; fork/end-three has residual 1/2 or 5/8, shift 1/4. D3 end-four has residual 1/4, shift $1/(2\sqrt{2})$; sealed end-five has residual $(5-3\varphi)/8$, shift $\varphi/4$. The D_{17} bridge residual is $1/(2i)$; the A_{17} end-three bridge at fourteen has residual $1/8 + 1/28 = 9/56$. Their positive-own numerators make them strong, and their own-zero cycles are affine. For single-fork/end-three, own zero, three, four and six retain the actual affine E_8, cycle, F_4 and $6, 3$ principals. The own-three cycle has FIVE actual nodes at D4 and FOUR at D3 end-three; it is not replaced by a triangle. The far-helper arithmetic lemma removes the other nonisolated possibility. The original LEFT isolated-core/H4/A2/singleton proof applies with $2i \geq 26$, so its previously proved $2i \geq 18$ loss margins suffice. The remaining fork-helper raw Schur constant is ACTUAL seventeen; the complete norm/trace argument leaves only its golden own. There is no spectator or extra long root here. Their actual LEFT H_4 points have divided displacement V, flat one through $K = i - 3$, $V^2 < i$, and pressure

$$x \cdot V > G > \sqrt{i - \kappa/(\varphi + 5/8)} - .091, \quad \kappa = 1/2,\ 5/8.$$

Every point uses positive actual walls, positive future time and all original-wall signs. The relevant rows are

i	K	μ	$G >$	$f_K <$
13	10	.16	3.48	2.732
14	11	.15	3.61	2.886.

The radical seals are $13 - 25/112 > 3.571^2$, $14 - 125/448 > 3.701^2$. The INDIVIDUAL terminal prices exceed $.16, .15$, in both initial-five references. The residual $V - P_K$ vanishes on the whole reference support and has norm < 3.

If five is weak, its actual endpoint reference has energies $L_{10} > .8870, L_{11} > .9100$. The two paid squares are

$$.8870 + .32(3.48 - 2.732) - .0256(3) > 1.0495, \qquad .9100 + .30(3.61 - 2.886) - .0225(3) > 1.0597.$$

If five is strong, first classify the ACTUAL head: the retained D_{16} admits only isolation or simple three at its long endpoint two. The nonisolated row gives $x_1 > .643$, hence legal first-five gain $> .0994$. An isolated head has small owns/nulls excluded as before, including five by Section C.9.1. For finite own at least seven, its actual finite head point gives gain $> .0135$ at cost .1 from P_5. Explicitly for $1 < u < 1.232$,

$$\eta = H - .361 - (R - .2)/4 > .0135.$$

On $u \in [1, 1.1], [1.1, 1.2], [1.2, 1.232]$, the respective rational seals are

$$H > .6708 - .1871, \quad .63245 - .204125, \quad .6196 - .2, \qquad R \leq .5, .415, .36.$$

The middle second-radical square has maximum 1/24; all bounds are positive squarings. They give gains $> .0477, .013575, .0186$.

For a dotted head use its ACTUAL positive core, $Z = (1 - u/2)e_1 + (u/2)e_2$, $0 < u < 1$. Paying .36 at P_5, cost .072, gives gain

$$.72(H - b) - .1296(R - 1/5) > .04. \tag{D17H}$$

Here is the full continuous seal rather than a borrowed actual-I theorem. On $[0, 1/2]$, $[1/2, 3/4]$ and $[3/4, 1]$, use lower chords for $\sqrt{2 - u}$

$$1.4142 - .379u, \quad 1.4381 - .4268u, \quad 1.4721 - .4721u$$

and upper chords for R

$$1 - .75u, \quad .8125 - .375u, \quad .625 - .125u.$$

With $.5091 < 9\sqrt{2}/25 < .5092$, the three gain bounds reduce to

$$\begin{aligned} &.35636922 - .0958u - .5092\sqrt{u(1-u)}, \\ &.39283671 - .16873u - .5092\sqrt{u(1-u)}, \\ &.43444611 - .2242u - .5092\sqrt{u(1-u)}. \end{aligned}$$

The circle inequality and squared-root upper bounds $.5182, .5365, .5564$ give minima $> .04936922, .04022171, .04414611$. No I wall or energy enters.

Every head residual is zero-sum on the first five and orthogonal to the H4 residual beyond $K = 10/11$. The bare strong-five H4 bounds are $.98692, .99786$. Adding even the smallest LEGALLY funded head gain $> .0135$ excludes both. All six through fourteen are consequently strong BEFORE deciding five or the final head.

Weak five with finite own gives actual $x_1 > .65$; its rebuilt six-through-fourteen reference exceeds one. Its dotted case is precisely the complete minimal-degree norm/trace argument already proved for the SAME inverse entries in Section C.9.2. Thus five is strong. A nonisolated head now gives $E_{14} + .0994 > 1$, so it is isolated. The legal graft at $x_1 \geq .632$ gives $E_{14} + (5/4)(.632 - .361)^2 > 1.00020125$. Therefore $x_1 < .632$. The original head raw normalization gives

$$r_H^2 = 2U, \quad \text{spatial } r_H = Ue_1, \quad H_0^2 = U(U-2), \quad K = \mathbb{Q}(U), \quad 2 < U < 11, \quad 0 < U_\sigma < 2.$$

Finite heads remain possible at this stage.

For completeness the conjugate interval itself bounds their COMPLETE native alphabet. For $U = 2 + 2\cos(2\pi/m)$, a coprime exponent $1 < a < m/4$ gives a forbidden conjugate > 2. For odd $m \geq 9$ use two. For even $m \geq 14$ not divisible by three use three. For $m = 6k$, $k = 4, 5$, use five/seven; for larger even k use $k-1$, or $k-3$ if $k \equiv 1 \pmod 3$; for larger odd k use $k-2$, or $k-4$ if $k \equiv 2 \pmod 3$. Each is coprime and below $3k/2$. The remaining labels are $3, 4, 5, 6, 7, 8, 10, 12, 18$. The prior null/golden exclusions leave exactly seven/eight/ten/twelve/eighteen, with the parameters already printed for B_{17}.

For either actual signed fork direction $V_f = P_{16} \pm e_{17}$, if $C_f \geq 13$, the ORIGINAL edge gives $x \cdot V_f \geq \sqrt{102/7} > 3.817$. Use the SAME head gain and a separately funded $.13(V_f - P_{14})$, whose residual norm is three and cross with the head/reference is zero. Then

$$\|x\|^2 > .9084 + .0135 + .26(3.817 - 3.311) - .0169(3) = 1.00276 > 1.$$

Thus BOTH actual fork coshes are < 13, without equating their parameters.

C.9.6 The complete actual D_{17} fork table

Fix either genuine fork f. Deleting it retains ACTUAL A_{16}. Finite classification gives isolation, helper three at $j = 1, 2, 15, 16$, or helper four at $j = 1, 16$. For three put $t = 2\gamma$, raw cap square two, white product -1, own product $-t$. For four put $a = 2\sqrt{2}\gamma$, raw square one, white product -1, own product $-a/2$. Actual cycles make the parameter integral, and for four also $a^2/2$ integral. The original Lorentzian principal gives ACTUAL field equality with that parameter. The COMPLETE

raw time and actual edge table is

helper	H_c^2	C_f
3@1	$(17t^2+4t-4)/4$	$(17t+2)/(6\sqrt{2})$
3@2	$(17t^2+8t)/4$	$(17t+4)/4$
3@15	$(17t^2+60t+52)/4$	$(17t+30)/4$
3@16	$(17t^2+30t+9)/4$	$(17t+15)/(6\sqrt{2})$
4@1	$(17a^2+8a)/16$	$(17a+4)/4$
4@16	$(17a^2+60a+52)/16$	$(17a+30)/4$.

The near endpoint THREE formula differs from the B17 short formula and is not substituted for it. Every other conjugate has negative displayed time square. Every null below uses UNIT ACTUAL cap normals; its positive time and all original-wall signs follow termwise.

The 3@15 own-zero kernel is $n_{14}+2n_{15}+n_{16}+n_{17}+c$; own three has its triangle. All other owns have $t \geq \sqrt{2}$, $C_f > 13$. The 4@16 own-zero kernel is $n_{14}+2n_{15}+2n_{16}+n_{17}+\sqrt{2}c$; every positive $a \geq \sqrt{2}$ gives $C_f > 13$. At 3@2, own zero has zero time; positive owns satisfy $1 \leq t < 48/17$, $-8/17 < t_\sigma < 0$. Degree at least three has nonzero integer norm magnitude < 1. Degree two leaves norm -1, trace one or two; the golden root violates the conjugate strip, leaving $1+\sqrt{2}$. Its raw time norm is $-47/16$. For a finite head the only compatible head is eight, of time norm -4, and the ratio $47/64$ is nonsquare. For a DOTTED head, the complete actual $\sqrt{2}$ head strip gives time norms $-7, -7, -32, -63, -36$. Prime forty-seven again has odd ratio valuation. Thus this row is excluded under EVERY head. At 4@1 the same norm strips leave only $a = 1+\sqrt{2}$, but $a^2/2 = 3/2+\sqrt{2}$ is not integral. Its own zero has zero time. Every four non-end role is exhausted, with finite, parallel and dotted own retained; parallel everywhere has $c+n_f$. Far-end own zero has negative time, while near-end own zero gives the forbidden rational actual field.

For the two remaining end-three roles put $v = t$ far, $v = t+1$ near. Their raw times are

$$H_c^2 = (17v^2 \pm 4v - 4)/4,$$

with upper sign far. The COMPLETE strips are

role	physical v	every other v_σ
far	$1 \leq v < 6.372$	$-.617 < v_\sigma < .382$
near	$2 \leq v < 6.607$	$-.382 < v_\sigma < .617$.

The exact nonphysical endpoints are $(\mp 2 \pm 6\sqrt{2})/17$, and the physical upper endpoints are $(78\sqrt{2} \mp 2)/17$. These exact signs are retained whenever a safe decimal overstrip admits an extra row.

Let g_0 be the ACTUAL raw head/helper product. Its actual triangle or complete old-chain cycle makes it integral. The common original time-line identities are

$$4g_0^2 - 4g_0U(t+2) + U^2(t+2)^2 = U(U-2)(17t^2+4t-4),$$
$$4g_0^2 - 4g_0U(t+1) + U^2(t+1)^2 = U(U-2)(17t^2+30t+9).$$

Reducing modulo $4\mathcal{O}_K$ gives the necessary parity statement

$$Uv^2 \in 2\mathcal{O}_K. \tag{D17P}$$

It does NOT justify $Uv/2$ integral at ramified dyadic primes. The physical and conjugate bounds belong to actual $K = \mathbb{Q}(v)$, not merely an unexamined smaller parameter field.

C.9.7 Whole D_{17}: every finite head is excluded

Assume one of the FIVE finite heads retained in Section C.9.5. Its actual field is the corresponding quadratic or cubic field, so the complete fork table applies with that field equality.

For heads seven/ten/eighteen, U is a dyadic unit and two is unramified, by discriminants forty-nine/five/eighty-one. Parity (D17P) makes $v/2$ integral. In the far role its nonzero norm has magnitude $< 3.186(.3085)^{d-1} < 1$ for $d = 2, 3$. For the near role the cubic norm is $< 3.304(.309)^2 < 1$. In the remaining quadratic golden near case, $v/2 = a + b\varphi$ has physical value < 3.304, other value in $(-.191, .309)$. The gap gives $b \le 1$, $b > 0$. For $b = 1$ the interval requires $.427 < a < .927$, no integer. Both end-helper roles are impossible.

For head eight use $K = \mathbb{Q}(s)$, $s = \sqrt{2}$, $U = 2 + s$. At its dyadic prime $\mathrm{ord}(U) = 1$, $\mathrm{ord}(2) = 2$. Parity requires $s \mid v$. The COMPLETE far strip has $t = 1 + s, 3 + 2s$, neither divisible by s. The complete near strip has $v = 2 + s, 3 + 2s$; only $2 + s$ survives, giving $t = 1 + s$. Its raw time square has norm $-23/4$, with nonsquare ratio $23/16$ to the actual head time norm -4. For head twelve, $U = 2 + \sqrt{3}$ is a dyadic unit although two ramifies. Parity requires v in its dyadic prime, where $\sqrt{3} = 1$ in the residue field. The COMPLETE far/near strips leave only $t = 2 + \sqrt{3}$, respectively $v = 2 + \sqrt{3}$; both have nonzero residue one. Thus BOTH actual forks are isolated under EVERY finite head. The assertion is presently conditional on a finite head, not carried into the dotted case.

For an isolated positive fork put $V = 4\gamma_f^2$. Its ACTUAL raw root has square $2V$, sole old product $-V$, time square

$$H_f^2 = V(17V - 8)/4, \quad K = \mathbb{Q}(V), \quad 0 < V_\sigma < 8/17.$$

Own zero has no spatial wall part. Own three has the actual affine E_8 arms one/two/five; own four has positive affine vector $2n_{15} + 2n_f + n_{14} + n_k + \sqrt{2}c$, where k is the other fork. Own six has its $6, 3$ path; parallel has the actual null pair. All are future feasible against EVERY original wall. For finite labels at least seven, degree at least three contradicts $\mathrm{N}(V) < 4(1/2)^2 \le 1$. In degree two the strict strip forces norm one, trace four, namely native twelve. Native five is the other finite candidate. The cubic/silver head fields cannot equal those of these candidates. At golden head ten the fork-five time norm is $-55/16$, giving nonsquare ratio $11/16$ to head norm -5. At head twelve the fork-twelve time norm is $-191/16$, ratio $191/48$ to head norm -3, also nonsquare. Both forks are DOTTED under every finite head.

The actual head/fork LONG cycle gives $g = (r_H, r_f)$ integral; the actual two-cycle gives $L = g^2/(UV)$ integral. Their spatial product is $UV/2$. The SAME original time line gives

$$(g - UV/2)^2 = U(U - 2)V(17V - 8)/4, \qquad L - g = 4UV - 17V/2 - 2U + 4.$$

Hence $W = V/2$ is integral, by subtracting eight copies of V from the integral $17V/2$. This uses BOTH cycles without a dyadic-unit assumption. The actual fork bound gives

$$2 < W < 676/17, \quad 0 < W_\sigma < 4/17, \quad H_f^2 = W(17W - 4). \tag{D17W}$$

For the three quadratic finite head fields, the COMPLETE full-order strips and time-norm ratios are

head	W	$\mathrm{N}(H_f^2/H_0^2)$
8	$3 + 2\sqrt{2},\ 10 + 7\sqrt{2},\ 17 + 12\sqrt{2}$	$103/4,\ 383,\ 2007/4$
10	$2 + 3\varphi,\ 5 + 8\varphi,\ 7 + 11\varphi,\ 10 + 16\varphi$	$171/5,\ 919/5,\ 239,\ 5104/5$
12	$7 + 4\sqrt{3},\ 14 + 8\sqrt{3}$	$647/3,\ 976.$

For coverage, the positive coefficient bounds are fourteen/seventeen/eleven, and for each coefficient the small-conjugate width $4/17 < 1$ permits at most one integer constant term. The physical bound

removes the otherwise small-conjugate $20+14\sqrt{2}$. For $n=\mathrm{N}(W), T=\mathrm{Tr}(W)$, $\mathrm{N}(17W-4)=289n-68T+16$ gives every printed ratio. Each is a nonsquare by an odd denominator three/five valuation or the nonsquare integer numerator; $976=16(61)$. The original common K-time line therefore excludes every row.

For finite heads seven and eighteen the FULL rings are $\mathbb{Z}[U]$. Their polynomials have discriminants $49, 81$; the former translated by four is seven-Eisenstein, the latter is three-Eisenstein. The distinct-valuation uniformizer argument already proved for the cubic bases in Section C.4.4 excludes the only possible index prime. Both are irreducible modulo two, so two is inert. The positive norm of W is $< 10816/4913 < 3$. An even norm would be divisible by eight; hence $\mathrm{N}(W)=1$, $W>289/16$.

Let α, β, χ be physical and two small conjugates of the actual finite-head generator. For head seven the exact root and derivative bounds are

$$3.246<\alpha<3.248, \quad 1.554<\beta<1.556, \quad .198<\chi<.199,$$
$$5.149<f'(\alpha)<5.167, \quad |f'(\beta)|>2.289, \quad f'(\chi)>4.128,$$
$$1.752<\beta+\chi<1.754, \qquad \beta-\chi>1.355.$$

They follow by polynomial signs and products of the disjoint root gaps. Writing $W=A+BU+CU^2$ in the FULL ring, Lagrange interpolation gives

$$C=\sum \frac{W_i}{f'(U_i)}, \qquad B+C(\beta+\chi)=\frac{W_\beta-W_\chi}{\beta-\chi}. \tag{D17L}$$

The complete strip implies $3<C<8$ and $|B+C(\beta+\chi)|<.174$. At $C=4,5,6,7$, only $C=4, B=-7$ survives. Then $7\chi-4\chi^2\in(1.229,1.235)$, so the small strip requires $1.229<A<1.471$, no integer. For head eighteen the corresponding bounds are

$$3.879<\alpha<3.880, \quad 1.652<\beta<1.653, \quad .467<\chi<.469,$$
$$7.590<f'(\alpha)<7.605, \quad |f'(\beta)|>2.63, \quad f'(\chi)>4, \quad 2.120<\beta+\chi<2.121, \quad \beta-\chi>1.183.$$

Interpolation gives $2<C<6$, $|B+C(\beta+\chi)|<.199$. For $C=3,4,5$, the products lie in $(6.360,6.363), (8.480,8.484), (10.600,10.605)$; none is within .199 of an integer. No coefficient B exists. Both cubic finite heads are excluded.

All FIVE finite heads have now been eliminated at their ACTUAL fields. Only NOW conclude that the head is DOTTED, with $4<U<11$, $0<U_\sigma<2$. Fork isolation and the dotted fork assumptions proved conditionally in this subsection are NOT exported to that remaining head. We next prove them anew in the dotted case.

C.9.8 Whole D_{17}: degree reduction under a dotted head

The complete fork table remains valid. For an end helper, the integral v has norm magnitude $<6.607(.617)^{d-1}$. Degree at least five gives norm <1. At degree four it is a unit; parity (D17P) then makes $Y=U/2$ integral. With physical $2<Y<11/2$ and three other conjugates in $(0,1)$,

$$0<|\mathrm{N}(Y(1-Y))|<99/256<1,$$

impossible. For an isolated fork the actual LONG+TWO cycles above remain valid under a dotted head BEFORE assigning its own type. They again give integral $W=V/2$, with $0<W<676/17$ and $0<W_\sigma<4/17$ at every other embedding. The lower bound $2<W$ is not used before the finite-own exclusion below. Degree at least four contradicts $(676/17)(4/17)^3<1$. Thus actual degree is two or three. Rational degree was excluded by the named WHOLE-form Meyer input.

We first remove cubic nonisolated forks. The end-helper norm is $\pm1, \pm2$. Opposite signs of the two small roots give magnitude $< 6.607(.617)(.382) < 2$. Two positive small roots in the far case, or two negative in the near case, give magnitude < 1. Norm magnitude two can therefore occur only as norm $+2$ in the following two patterns. For far roots $t, -a, -b$, $tab = 2$, $0 < a, b < .617$, the trace is five, $6 < t < 6.234$, and the integer second coefficient is -6 or -7. The polynomials $X^3 - 5X^2 - 6X - 2$, $X^3 - 5X^2 - 7X - 2$ have discriminants $-424, 229$. The first is not totally real. At $-.617$, the second has value $.180669887 > 0$, inconsistent with both negative roots above that lower endpoint. For near roots v, a, b with $vab = 2$, the trace is seven, $5.766 < v < 6$, and the second coefficient is seven. Its polynomial $X^3 - 7X^2 + 7X - 2$ has discriminant -59, impossible. Thus a remaining cubic end-helper is a unit, so $U/2$ integral.

Here is the full EVEN-cubic-head conclusion, also reused below. For integral $Y = U/2$, physical $2 < Y < 11/2$ and other roots in $(0, 1)$ give

$$0 < |\mathrm{N}(Y(1 - Y))| < 99/64 < 2.$$

Thus $\mathrm{N}(Y) = 1, \mathrm{N}(1 - Y) = -1$. Its polynomial is $X^3 - TX^2 + (T - 1)X - 1$, $T = 3, \ldots, 7$. The COMPLETE discriminants are $-23, -31, -23, 49, 257$. The first three are not totally real; at $T = 7$, evaluation at six is -1, forcing physical root > 6. Only $T = 6$ remains. Its reciprocal generates the cyclic field K_7, with full basis $1, u, u^2$ for $u^3 - 5u^2 + 6u - 1 = 0$. The square discriminant gives cyclicity, so choose its PHYSICAL largest root as generator. The full-order proof and root/derivative bounds are those just proved above and are algebraic facts of this field, not a finite-head hypothesis.

For a cubic end-helper $v = A + Bu + Cu^2$, the full far/near strips and (D17L) imply $-1 < C < 2$, hence $C = 0, 1$, and $|B + C(\beta + \chi)| < .738$. At $C = 0$, $B = 0$, and the small strip forces $A = 0$, contrary to physical $v \geq 1$. At $C = 1$, only $B = -2$ is possible. Since $2\chi - \chi^2 \in (.356, .359)$, the small strip gives $A = 0$ in either role. But $v_\beta = \beta(\beta - 2) < -.689$, outside BOTH strips. This exhausts every coefficient triple. Cubic end helpers are impossible, so BOTH actual cubic forks are isolated dotted. Finite own in degree three would violate $\mathrm{N}(V) < 4(8/17)^2 < 1$. Each integral W satisfies (D17W), with norm one or two; no equality of the two fork parameters has been assumed.

C.9.9 Whole D_{17}: the entire isolated cubic field domain

Choose ONE actual isolated cubic fork; the other may have a different parameter. Write its monic polynomial $P(X) = X^3 - AX^2 + BX - n$, $n = \mathrm{N}(W) = 1, 2$. Its two other positive roots lie in $(0, q)$, $q = 4/17$, physical $W < 676/17$, hence integer trace $A \leq 40$. Since $P(q) < 0$,

$$64 - 272A + 1156B - 4913n < 0.$$

For $n = 2$, physical $W > 289/8$, and the small-root arithmetic-geometric inequality gives $B \geq 2\sqrt{2W} + 2/W > 17$. The polynomial inequality gives $B < 20642/1156 < 18$, impossible. Thus W is a UNIT.

For the actual head/fork products in the normalization $r_H^2 = 2U, r_f^2 = 4W$, the LONG and TWO cycles give integral g, L, with

$$L - g = (8U - 17)W - 2U + 4 =: d_0, \qquad g^2 = 2UW(g + d_0).$$

At every dyadic prime, $d_0 \equiv -W$ is a unit. The equation forces g into the prime and then $2\,\mathrm{ord}(g) = \mathrm{ord}(2) + \mathrm{ord}(U)$. If two is unramified this makes $U/2$ integral. The even-cubic-head conclusion above gives actual field K_7. Apply the SAME full-order interpolation as for finite head seven, now merely as algebra of the field and the CURRENT unit-W strip. Its complete $C = 4, \ldots, 7$

tests leave only $C = 4, B = -7$, then no integer A. Thus the unramified case is impossible, without assuming the head itself is finite.

If two ramifies, the unit polynomial has a repeated factor modulo two. Its derivative $X^2 + B$ shows this requires BOTH A, B odd. The physical root and two small-root bounds give

$$A = 19, 21, \ldots, 39, \qquad B^2 > 4(A - 8/17), \qquad B < \frac{272A + 4849}{1156}.$$

The COMPLETE lower/upper odd-integer tests are

A	19	21	23	25	27	29	31	33	35	37	39
$B_{\min}$	9	11	11	11	11	11	13	13	13	13	13
$B_{\max}$	7	9	9	9	9	11	11	11	11	11	13.

Only

$$P_{29} = X^3 - 29X^2 + 11X - 1, \qquad P_{39} = X^3 - 39X^2 + 13X - 1$$

remain. Their discriminants are $4(3)(383)$, $4(5021)$, with squarefree odd parts. For an elementary full-order certificate, trial primes below $\sqrt{383}$ are $2, 3, 5, 7, 11, 13, 17, 19$, none dividing383. For5021, all primes below its square root are

$$2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67;$$

the respective remainders are

$$1, 2, 1, 2, 5, 3, 6, 5, 7, 4, 30, 26, 19, 33, 39, 39, 6, 19, 63.$$

The shifted polynomials are two-Eisenstein:

$$P_{29}(X + 1) = X^3 - 26X^2 - 44X - 18, \qquad P_{39}(X + 1) = X^3 - 36X^2 - 62X - 26.$$

Only two can divide the power-order index. At its uniformizer $W - 1$, valuations of $a_0 + a_1\pi + a_2\pi^2$ are distinct modulo three; a numerator not all divisible by two has unique least valuation at most two. Dividing by two cannot be integral. Thus $\mathcal{O}_K = \mathbb{Z}[W]$ in BOTH cases.

The exact small-root brackets for P_{29} are $.14 < u < .15$, $.23 < v < .235$; its physical root satisfies $28.615 < w < 28.63$. They follow from values $-.025656, .000875, .008067, -.003547125$ at those four endpoints. The physical derivative is > 800, both small derivative magnitudes > 2.27. For P_{39}, values $-.040569, .000128, .019361, -.016952$ give $.11 < u < .12$, $.21 < v < .22$, $38.66 < w < 38.68$, physical derivative > 1480, small derivative magnitudes > 3.45.

Write the ACTUAL head $U = a + bW + cW^2$ in these proved FULL rings. Its current strips are physical $(4, 11)$, other values $(0, 2)$. Lagrange interpolation bounds c respectively by

$$-2/2.27 < c < 11/800 + 2/2.27, \qquad -2/3.45 < c < 11/1480 + 2/3.45.$$

Both intervals are strictly inside $(-1, 1)$, so $c = 0$. Then $b = (U(w) - U(u))/(w - u)$ satisfies $0 < b < 11/(w - u) < 1$, impossible for an integer. Every ramified cubic case has been excluded. The ACTUAL field is QUADRATIC.

C.9.10 Whole D_{17}: complete quadratic end-helper exclusion

The dotted-head fork table already excludes all four non-end white roles. For the remaining far/near end-three roles write v's polynomial $X^2 - TX + n$. The COMPLETE far ranges are $T = 1, \ldots, 6$, $n = -3, -2, -1, 1, 2$; near ranges $T = 2, \ldots, 7$, $n = -2, -1, 1, 2, 3, 4$. The quadratic formula and EXACT time signs leave the following fourteen and fifteen rows:

(T, n, d) far	(T, n, d) near
$(2, -1, 2)$	$(3, -1, 13)$
$(3, -1, 13)$	$(3, 1, 5)$
$(3, -2, 17)$	$(4, -1, 5)$
$(4, -1, 5)$	$(4, 1, 3)$
$(4, -2, 6)$	$(4, 2, 2)$
$(4, 1, 3)$	$(5, -1, 29)$
$(5, -1, 29)$	$(5, -2, 33)$
$(5, -2, 33)$	$(5, 1, 21)$
$(5, -3, 37)$	$(5, 2, 17)$
$(5, 1, 21)$	$(6, -1, 10)$
$(6, -1, 10)$	$(6, -2, 11)$
$(6, -2, 11)$	$(6, 1, 2)$
$(6, 1, 2)$	$(6, 2, 7)$
$(6, 2, 7)$	$(6, 3, 6)$
	$(7, 3, 37)$.

Here actual $K = \mathbb{Q}(\sqrt{d})$. At each bounded pair T, n, substitute the larger root and its conjugate into the FULL strips. Every missing pair is rational, nonreal or violates an endpoint. The safe far decimal overstrip admits one extra $(3, 1)$, $v = \varphi^2$. Its exact conjugate time numerator is $(123 - 55\sqrt{5})/2 > 0$, excluded by $123^2 - 5(55)^2 = 4$. Thus the exact, rather than decimal, domain is fully enforced.

For odd n, v is a unit at EVERY dyadic prime, so parity (D17P) makes $U/2$ integral even with ramification. The FULL quadratic head strip for $Y = U/2$ is $2 < Y < 11/2$, $0 < Y' < 1$. Its integer trace is three through six and norm $1 \le k \le T - 2$; at trace six the physical bound removes $k = 1, 2$. The COMPLETE eight actual heads and $h = -\mathrm{N}(U(U - 2))$ are

d	U	h
5	$2\varphi^2,\ 4\varphi^2$	$16, 64$
2	$4 + 2\sqrt{2}$	32
3	$4 + 2\sqrt{3}$	32
13	$5 + \sqrt{13}$	48
17	$5 + \sqrt{17}$	64
21	$5 + \sqrt{21}$	48
6	$6 + 2\sqrt{6}$	96.

Odd-norm helper rows in fields29/10/37 are immediately incompatible. For all others, put $p =$

$-16\mathrm{N}(H_c^2) > 0$. The complete time-norm tests are

role	(T, n, d)	p	h
far	$(2, -1, 2)$	287	32
	$(3, -1, 13)$	711	48
	$(4, -1, 5)$	1271	$16, 64$
	$(4, 1, 3)$	423	32
	$(5, 1, 21)$	983	48
	$(6, 1, 2)$	1679	32
near	$(3, -1, 13)$	207	48
	$(3, 1, 5)$	311	$16, 64$
	$(4, -1, 5)$	599	$16, 64$
	$(4, 1, 3)$	839	32
	$(5, 1, 21)$	1503	48
	$(6, 1, 2)$	2303	32
	$(6, 3, 6)$	503	96.

All $p/(16h)$ are nonsquares. At $h = 32, 96$ the odd numerator gives odd two-adic denominator valuation. The four $h = 48$ reduced cases are $237/256, 983/768, 69/256, 501/256$; the middle denominator has odd three valuation, the other numerators are nonsquares. For $h = 16, 64$, the numerators $1271, 311, 599$ are nonsquares. These are NECESSARY actual common-time-line square tests, not sufficiency of a norm-square.

For even-norm rows DO NOT assume $U/2$ integral. Their complete actual fields, full head strips and h's are

d	U	h
2	$3+\sqrt{2},\ 3+2\sqrt{2},\ 4+2\sqrt{2},\ 5+3\sqrt{2},\ 6+3\sqrt{2}$	$7, 7, 32, 63, 36$
6	$3+\sqrt{6},\ 4+\sqrt{6},\ 5+2\sqrt{6},\ 6+2\sqrt{6}$	$15, 20, 15, 96$
7	$3+\sqrt{7},\ 4+\sqrt{7}$	$12, 27$
11	$4+\sqrt{11},\ 5+\sqrt{11}$	$35, 28$
17	$2+\eta,\ 3+\eta,\ 4+2\eta,\ 5+2\eta$	$8, 16, 64, 19$
33	$3+\xi,\ 4+\xi$	$24, 24$.

Here $\eta = (1+\sqrt{17})/2$, $\xi = (1+\sqrt{33})/2$. The FULL integral bases and $0 < U' < 2, 4 < U < 11$ give coefficient bounds respectively $3, 2, 2, 1, 2, 1$. At fixed positive coefficient, the width-two conjugate interval gives two consecutive constants; the physical strip removes the excess. For example at $\sqrt{7}$, coefficient two already has least physical $6 + 2\sqrt{7} > 11$. Thus no full-order head was omitted.

The COMPLETE remaining even-norm time rows are

role	(T, n, d)	p	all h
far	$(3, -2, 17)$	200	$8, 16, 64, 19$
	$(4, -2, 6)$	828	$15, 20, 15, 96$
	$(5, -2, 33)$	1592	24
	$(6, -2, 11)$	2492	$35, 28$
	$(6, 2, 7)$	252	$12, 27$
near	$(4, 2, 2)$	92	$7, 7, 32, 63, 36$
	$(5, -2, 33)$	72	24
	$(5, 2, 17)$	824	$8, 16, 64, 19$
	$(6, -2, 11)$	668	$35, 28$
	$(6, 2, 7)$	1692	$12, 27$.

Every entry follows by expanding

$$16\mathrm{N}(H_c^2) = (17T \pm 4)^2 n + (17T \pm 4)(-17n - 4)T + (-17n - 4)^2.$$

Every ratio $p/(16h)$ is nonsquare EXCEPT far $(T, n, d) = (3, -2, 17)$, $h = 8$. The far nonexceptional rows retain odd valuations at $23, 199, 89, 7$. In the exceptional field the other ratios are $25/32, 25/128, 25/38$, nonsquares. The near rows retain odd valuations at $23, 3, 103, 167, 47$. Thus a genuine square-norm exception is kept.

For that exception

$$t = (3 + \sqrt{17})/2, \quad U = (5 + \sqrt{17})/2,$$

the ACTUAL time squares are

$$H_c^2 = (225 + 55\sqrt{17})/8, \qquad H_0^2 = (11 + 3\sqrt{17})/2.$$

Their ratio $R = (165 + 35\sqrt{17})/64$ has norm $25/16$. If $R = z^2$ in K, then $\mathrm{N}(z) = \pm 5/4$, and

$$\mathrm{Tr}(z)^2 = 245/32 \quad \text{or} \quad 85/32.$$

Neither is a rational square, by its odd two-adic denominator valuation. But the trace is rational. The last actual common-time square fails. All quadratic end-helper rows are now excluded, in both signed orientations.

Both ACTUAL forks are therefore isolated under the dotted quadratic head. Their original LONG+TWO cycles, applied BEFORE assuming own dotted, give integral $W = V/2$ with $0 < W < 676/17$ and $0 < W' < 4/17$. The lower bound $2 < W$ is deferred until finite owns have been excluded. Finite own would have $0 < W < 2$, other $0 < W' < 4/17$, hence nonzero integer norm $0 < \mathrm{N}(W) < 8/17$, impossible. Parallel has its actual null pair. Both forks are isolated DOTTED, without equating their two W parameters.

C.9.11 Whole D_{17}'s distinct actual inclusion and final budget

Choose ONE of the two actual isolated dotted forks. Its raw square is $4W$, sole old product $-2W$, spatial $W(P_{16} \pm e_{17})$, time square $W(17W - 4)$. The actual head has square $2U$, spatial Ue_1, time square $U(U - 2)$. The original K-time line gives

$$R_0 = U(U - 2)W(17W - 4) \in K^2, \qquad g = UW - \sqrt{R_0},$$

with PHYSICAL positive square root. The original LONG path and TWO cycles give

$$g,\ L = \frac{g^2}{2UW} \in \mathcal{O}_K.$$

All physical/conjugate strips are now

$$K = \mathbb{Q}(U) = \mathbb{Q}(W), \quad 4 < U < 11, \quad 0 < U' < 2, \quad 2 < W < 676/17, \quad 0 < W' < 4/17.$$

The actual independent conjugate root lines give $0 < L' < 4$ when $g \neq 0$. Here the current physical strips force $d_0 = (8U - 17)W - 2U + 4 > 0$, hence $g < 0$, so no orthogonal exception is silently lost. The true physical mutual L is finite native or dotted, while parallel four has its forbidden actual null pair. Thus EVERY predicate of Proposition 21.1(iii) holds for this ORIGINAL D17 geometry.

This inclusion is separate from Section C.9.4: D17 raw fork norm is $4W$; B17 raw short norm was V. No equation of different root scales or arithmeticity of a face supplied either inclusion. The

existing full-order algorithm covers every $\Delta < 121$, every strict-strip model and ordered pair with both field-square norm signs. Its 36-field/120-pair output, explicitly ASSUMED as the recorded computational proposition, has the same TWO necessary survivors. The field/counter/survivor combine does not independently recompute the 118 rejected tests.

For either actual survivor, use its actual head point and ONE chosen signed fork. The head gain at cost .072 from P_5 is $> .043$, by the full rational seals already printed in Section C.9.4. The actual fork has $C_f^2 = 17W/4 > 442/5$, giving

$$x \cdot (P_{16} \pm e_{17}) > 3.7.$$

Pay .13 from the DISTINCT P_{14} price:

$$y = w_{14} + .36(Z - P_5/5) + .13(P_{16} \pm e_{17} - P_{14}).$$

The signed residual has norm three and vanishes on all reference coordinates; it is orthogonal to the head residual. Both individual payments are legal BEFORE using their original floors. ONE square on the SAME original x gives

$$\|x\|^2 > .9084 + .043 + .26(3.7 - 3.311) - .0169(3) = 1.00184 > 1.$$

Every actual field and actual fork type has now been covered. This closes whole D_{17} at the stated external and explicitly computed premises. The argument needed one chosen fork for the last contradiction, while preserving both signed walls and all original constraints.

C.10 Chain component bounds at ranks ten through fourteen

We now finish the smaller whole-A consumers. The shared mechanism is unchanged: classify an ACTUAL central opposite, constrain a pure helper arithmetically, and absorb EVERY retained isolated foot in ONE reference. The index ranges, finite side exceptions, helper packing and coefficient capacities are proved at their own N; no numerical theorem is reranked.

C.10.1 A spectator-safe A_{14} bound greater than $4/5$

Use the zero-sum space in $\mathbb{R}^{15}$,

$$n_i = (e_i - e_{i+1})/\sqrt{2}, \quad \Omega_i = \sqrt{2}(P_i - iP_{15}/15), \quad D_i = 2i(15-i)/15, \quad h_i = x_A \cdot \Omega_i.$$

All spectators are kept; the SAME original farthest vector has $\|x\| < 1$. For $i = 5, \dots, 10$ take loss $\ell = .597$. Here $D_5 = D_{10} = 20/3$, $D_6 = D_9 = 36/5$, $D_7 = D_8 = 112/15$. The exact ordinary sufficient ratio

$$R_\ell(q) = 1/\ell - (q-\ell)/(q(2q-\ell))$$

increases on this interval. Since $q < 11/4$,

$$R_\ell(q) < 1000/597 - 8612/53933 < 761/500;$$

the last positive cleared numerator is 106860761. The older coarse $q < 3$ bound is not used.

Deleting a central node retains two paths of rank at least four. The FULL side-order-$m \geq 5$ finite-white list is no contact, permitted endpoint/D/E simple three, endpoint four, or two end-threes joining the paths. The connected replacement rank is fourteen, so no high-rank exceptional type is missing. The complete own-zero repairs retain internal affine arms, stars and two-contact cycles;

far/next-three use the ENTIRE foreign cost and reciprocal original-wall law; endpoint-four has no positive-time foreign completion. All isolated own-three/four/six/parallel nulls apply for EVERY foreign cost BEFORE splitting it, as proved in the common chain section.

The ordinary nonisolated ratios exceed 761/500, as do the complete foreign far/near bounds $227/(25\sqrt{35})$ and $\sqrt{121/35}$. The only extra near-four orientations are

$$v_A = \Omega_i/\sqrt{2} + \Omega_{i-1}/2, \qquad i = 5, 6, 7$$

and reflections, with zero foreign cost. Their conjugate time numerators at $N = 15$ are $114 - 80\sqrt{2}, 128 - 90\sqrt{2}, 136 - 96\sqrt{2} > 0$. The actual simple-three link puts $\sqrt{2}$ in K; its sign-reversing embedding makes the original Schur complement negative. Near-four of side order at least eight is already strong, so both helper orientations are covered.

For an isolated-within-A cap write $c = Ap - \gamma\Omega_i - v_S$, $Q = \|v_S\|^2 < 1$, $u = (1 - Q)/\gamma^2$. For $Q > 0, u > 1$, the actual positive clearing has loss $< .202 + .382 = .584 < \ell$. At $u = 1$ it is a feasible null. For pure native five the actual H_4 loss is $< 25/76 + 5/24 < \ell$. For EVERY pure finite native at least seven the following original negative core gives a fresh stronger bound, including native eight:

$$\xi = c + \gamma^{-1} n_i + \frac{2(1 - \gamma^2)}{\gamma} n_j, \qquad j \text{ either old neighbour.}$$

Put $d = 1 - \gamma^2 < .19$. The cap and n_i products vanish; $\xi \cdot n_j = (2d - 1/2)/\gamma < 0$; its square is $-d(1 - 4d)/\gamma^2 < 0$. Positive cap time and actual acuteness check EVERY omitted original wall. Its normalized comparison gives loss

$$< \frac{10000}{39771} + \frac{25}{81} = \frac{22275}{39771} < .561 < \ell.$$

Here $\sqrt{d(1 - 4d)} \le 1/4$, $\gamma > .9$, $q > 2.58$, and $\sqrt{D_i - 100/81} > 2.33$. Every remaining isolated foot is retained in the FULL interval $0 < u < 1$, pure or foreign; none is assigned an unsupported strong floor.

Every remaining weak nonisolated cap is a PURE endpoint helper with

$$202/125 < t = 2\gamma < 3139/1458 < 54/25.$$

The fresh cheapest negative/positive conjugate Schur costs are

$$Q_-(t) = 2it^2 + 4t + 2 - 2(it + 1)^2/15, \quad Q_+(t) = 2it^2 + 2 - 2(it - 1)^2/15.$$

At $i = 5$, $Q_-(-4/5) = 4$ EXACTLY, $Q_+(12/25) = 4 + 16/375$. They increase at the relevant signs over the half-window, so strict positivity gives every other conjugate in $(-4/5, 12/25)$. The COMPLETE all-degree norm/trace argument of Section C.2.2 uses exactly these physical/conjugate bounds and proves $t = \varphi$, actual $K = \mathbb{Q}(\varphi)$. No restriction is imposed in a branch without a helper.

The COMPLETE six-type $30A^2 = M$ table is

type	M	N(M)
near 5	$34 + 70\varphi$	-1364
near 6	$38 + 72\varphi$	-1004
near 7	$40 + 72\varphi$	-704
far 5	$34 + 60\varphi$	-404
far 6	$38 + 66\varphi$	-404
far 7	$40 + 70\varphi$	-500.

Different norm-squareclasses cannot coexist on the SAME original connected-block K-time line. For the repeated far-five/six class, the ratio is

$$\frac{97+9\varphi}{101}=\frac{203+9\sqrt{5}}{202}.$$

Its norm is one. A full K-square would have rational coefficient square $405/404$ or $1/404$, both impossible at prime101. For reflected copies, the actual cyclic parameter has constant numerators $8,1,-10$ modulo fifteen, so is nonintegral in $\mathbb{Z}[\varphi]$. Explicitly the near/far formulas are

$$\frac{(15-2i)(i+(i\pm 2)\varphi)-17}{15}.$$

Every different and repeated type is covered; at a fixed index the ACTUAL opposite is unique. Thus at most ONE central index has a weak helper. The number and adjacency of isolated feet are UNRESTRICTED.

For each retained $0<u<1$ foot the positive original combination $c+v_S+\gamma u n_i$ gives

$$x_A\cdot(\Omega_i-un_i)\geq\sqrt{D_i-u}-\sqrt{u(1-u)}.$$

The raw loss satisfies $L_i\leq a_iu+\sqrt{u(1-u)}$, where

$$a_{5,10}<.202,\quad a_{6,9}<.194,\quad a_{7,8}<.190.$$

Each follows from the ACTUAL $a_i=(\sqrt{D_i}+\sqrt{D_i-1})^{-1}$. The full adjusted-loss inequality

$$L_i-\frac{\beta_i}{2}(2u-u^2)<.597$$

holds for capacities

$$\beta_{5,10}\geq.06,\qquad\beta_{6,9}\geq.026,\qquad\beta_{7,8}\geq.0246.$$

For outer sites, $u\leq.4$ has loss $<.5708$; above .4 the circle bound gives $.6111-.03(.64)=.5919$. For sites six/nine, $u\leq.5$ gives $<.597$; above it use $(.194+1.01865)/2-3(.026)/8=.596575$. For seven/eight, the corresponding bounds are .595 and $(.190+1.018)/2-3(.0246)/8=.594775$. Thus EVERY pure/foreign foot parameter is covered.

Use symmetric rational heights

$$H_5=H_{10}=A=1.98498,\quad H_6=H_9=B=2.08628,\quad H_7=H_8=C=2.13551,$$

linear on the outer tails. Their exact floor squares are below the true ℓ-strong heights. The central dual coefficients are

$$.147848,.026035,.024615,.024615,.026035,.147848.$$

A single internal helper is omitted by its neighbour chord, making its coefficient zero and increasing neighbouring ones. The largest energy loss is $(2B-A-C)^2/4=.000677821225$.

For $w=\sum\beta_i\Omega_i$, and EVERY retained foot with positive coefficient, put $Z=w-\sum\beta_iu_in_i$. Primitive actual pressures give $x_A\cdot Z\geq E_H-\sum\beta_i(L_i-\ell)$. ALL adjacent-foot crosses have the correct sign $n_i\cdot n_j\leq0$, so

$$Z^2\leq E_H-\sum\beta_i^2(2u_i-u_i^2).$$

ONE square and the adjusted losses give $e_A > E_H$. Strong constraints supply strictness when no feet remain. The full energy is $A^2/5 + (B-A)^2 + (C-B)^2 = .80071440298$; after the largest internal omission it is

$$> .800036581755 > 4/5.$$

It remains to cover an OUTER helper at five/ten. Reverse labels if necessary, without assuming a chamber symmetry. The actual H_4 positive correction is

$$b_c = \gamma n_5 + (2/3)n_e + (1/3)n_{\mathrm{adj}}, \quad \delta = (c + b_c)^2 = (5 - 3\varphi)/12 < .111^2.$$

The positive FINITE projected point is used; the spacelike $c + b_c$ alone is not a point. For helper end $e = 1$ or fourteen, the signed endpoint $y = x_1$ or $-x_{15}$ obeys

$$y > \sqrt{2}(A_c - .111 - \gamma h_5).$$

The smaller actual time has $A_c^2 = (34 + 60\varphi)/30 > 2.09^2$; weakness gives $h_5 < 1.985$, $\gamma < .81$. Thus BOTH orientations have $y > 1.4(2.09 - .111 - .81(1.985)) = .51961 > .49$.

Omit knot five by the chord zero-to-six. Its reference energy is

$$E_{\mathrm{end}} = \frac{1}{2}[B^2/6 + A^2/5 + (B-A)^2 + 2(C-B)^2] > .764.$$

At the actual helper end there is a flat signed block of length six or five, mean $b_0 = B/(6\sqrt{2}) \in (.24, .25)$ or $A/(5\sqrt{2}) \in (.25, .3)$. Graft the ACTUAL endpoint to .49, keeping the block sum fixed. The other entries $(Lb_0 - .49)/(L-1)$ lie in $(.19, .2525)$, above the next old coordinate $< .073$. All duals remain positive. The changed central boundary coefficient is $> .117/1.415 > .06$, meeting its own capacity; all others retain theirs. The missing-five coefficient remains zero. The gain in this SAME reference is

$$\frac{L}{L-1}(.49 - b_0)^2 > (6/5)(.19)^2 = .04332.$$

Absorb ALL other isolated feet with their individual parameters as above. Then $e_A > .764 + .04332 = .80732 > 4/5$. No second helper exists by packing. The proof covers every actual field and arbitrary finite spectators:

$$e_{A_{14}} > 4/5. \tag{A14U}$$

In particular its ordinary-two-A_1 and actual-I-plus-A_1 rank-sixteen consumers close by adding respectively $> .2$ or the rank-free JOINT cost $> .20204$, with I counted once. In rank seventeen, every rank-three remainder has cost $> .21$ by the rank-free partition proof of (S3), so the ENTIRE older A_{14} row closes at once.

C.10.2 A fresh spectator-safe A_{13} bound

Now $N = 14$, central sites five through nine, with

$$D_5 = D_9 = 45/7, \quad D_6 = D_8 = 48/7, \quad D_7 = 7, \qquad \ell = 1199/2000 = .5995.$$

Deleting a central site again retains two paths of rank at least four. The complete side-order-$m \geq 5$ white list is isolation, permitted simple-three/D/E, endpoint-four, or one endpoint-three on EACH retained path. Two contacts within one path form a cycle; the connected rank-thirteen completion is classical, with no missing exceptional type. The fresh sufficient ratio satisfies

$$R_\ell(q) < 2000/1199 - 4801/32403 < 761/500, \qquad 5/2 < q < 3.$$

This follows from the same positive derivative and a direct positive rational cross product.

Own zero is dispatched BEFORE positive-own ratios. Isolation has $A^2 = Q - 1 < 0$. Internal-three has the actual affine arms $1, 2, 5$ or $1, 3, 3$; internal-four has its weighted star, two threes their cycle. The only residual endpoint/next-three/four times are

A^2	white bound
$Q - 15/28$	$Q < 3/5$
$Q - 1/7$	$Q < 2/5$
$Q - 1/14$	$Q < 1/5.$

The first positive time is $< 9/140 < 1/4$, below any foreign contribution. The second has at most one foreign contribution, larger than A^2. Both violate the actual orthogonal reciprocal opposite law; the last admits no positive foreign cost. At $Q = 0$ all three times are negative. Every affine/null vector has positive cap time and correct signs against EVERY original wall.

The complete ordinary ratios for positive own exceed $761/500$, including

$$\frac{.808 + .4}{\sqrt{.6}}, \quad \frac{.808 + .2}{\sqrt{.4}}, \quad \frac{.808}{\sqrt{.25}}, \quad q/2 + 1/q.$$

The complete foreign far/near bounds are still $227/(25\sqrt{35})$, $\sqrt{121/35}$. Thus each weak nonisolated cap is PURE: an endpoint helper or own-four/simple-three near helper of side order five/six/seven. Near-four of side order at least eight is already strong. We retain the smaller near-four roles rather than use an invalid N15 conjugate exclusion.

For a pure endpoint helper, the exact square ratio identity gives $t = 2\gamma < 54/25 < 9/4$. If its own is not five, $t > 9/5$. Actual simple-three normalization makes t integral. At every lifted nonphysical embedding, the cheapest signed old-A Schur costs are

$$2it^2 + 4t + 2 - 2(it + 1)^2/14, \qquad 2it^2 + 2 - 2(it - 1)^2/14.$$

At the relevant negative/positive endpoints they increase with i and N. Already at $N = 13, i = 5$, costs at $-.83, .5$ exceed four by $69/1625, 2/13$. Hence every other conjugate belongs to $(-.83, .5)$. Degree at least six contradicts $(9/4)(.83)^5 < 1$. In degrees three/four/five the norm magnitude is < 2; a positive small conjugate would make it < 1, so all are negative, of product magnitude $1/t$. AM–GM bounds the integer trace respectively in

$$(.14, 1), \quad (-.69, 0), \quad (-1.52, -1),$$

none containing an integer. The exact underlying comparisons are $t(t - 1)^2 < (9/4)(5/4)^2 < 4$, $t^4 < (9/4)^4 < 27$, $t(t + 1)^4 < (9/4)(13/4)^4 < 256$. In degree two trace two gives a nonzero norm strictly between $-1, 1$; trace one leaves only norm -2, whose polynomial $X^2 - X - 2$ is reducible. Degree one leaves parallel two, an actual null. Therefore every weak endpoint helper is golden, with ACTUAL $K = \mathbb{Q}(\varphi)$ by its original Lorentzian principal.

The COMPLETE $28A^2 = M$ golden table is

type	M	$\mathrm{N}(M)$
near 5	$30 + 63\varphi$	-1179
near 6	$33 + 64\varphi$	-895
middle 7	$34 + 63\varphi$	-671
far 5	$30 + 55\varphi$	-475
far 6	$33 + 60\varphi$	$-531.$

All five norm-squareclasses differ, excluding unlike types on the common actual K-time line. Reflected same-type cyclic parameters are

$$\frac{(14-2i)(i+(i\pm 2)\varphi)-16}{14}.$$

At $i=5,6$ their constant coefficients are $2/7,-2/7$, nonintegral. At the middle there is only one actual index. Thus at most ONE weak golden helper exists.

If such a helper occurs, EVERY other central site is .65-strong. Indeed $R_{.65}(3)<1.4$, below all ordinary and near-four roles; the isolated raw loss bound below is $<.6135$. Any .65-weak endpoint helper is also .5995-weak, so packing excludes a second one. For the actual finite H_4 point use

$$U=v_A-\gamma n_i-(2/3)n_e-(1/3)n_{\mathrm{adj}},\qquad \delta=(5-3\varphi)/12<.111^2,\qquad x_A\cdot U>A-\sqrt{\delta}.$$

Positive finite projection, not the spacelike corrected normal alone, checks all original walls.

For helper index five, set $a=1.885, b=1.968$. Use heights $H_6=H_7=H_8=b, H_9=a$, linear on its two outer tails. The energy is $E_5=(b^2/6+(b-a)^2+a^2/5)/2=.681519$. Project U onto its flat blocks. For far/near helper the projections are

$$(5\gamma/6)\Omega_6+\Omega_9/10,\qquad (5\gamma/6+1/12)\Omega_6.$$

Their reference products are <1.5169. The smaller time is $A>2.061$; the deficit is $>.4331$. The FULL orthogonal residual norms are $(2/3)\gamma^2+1/15$ and $(2/3)\gamma^2+1/12+\gamma/3$, both $<.8$. Pay $\mu=.2$ from the INDIVIDUAL strong coefficients: the largest debit is $.2(5(.81)/6+1/12)<.151667<b/12=.164$, and the far other debit .02 is below $(a/5-(b-a))/2=.147$. ONE square gives $e_A>E_5+.4(.4331)-.04(.8)=.822759$.

For helper at six/seven, use only $H_5=H_9=a$, with flat middle block of four coordinates. Energy $E_0=a^2/5=.710645$, both individual coefficients .1885. Writing $l=i-5$, the flat-block projection is

$$\gamma[(4-l)\Omega_5+l\Omega_9]/4+\Omega_{\text{helper end}}/10.$$

Here the added term is $\Omega_5/10$ for helper endpoint one, and $\Omega_9/10$ for helper endpoint thirteen; these are the flat-block boundaries, not the literal helper-node indices. Its reference product is $a(\gamma+.1)<1.71535$. The smaller time is $A>2.155$, so the deficit is $>.328$. The FULL residual norm is $\gamma^2[2l(4-l)/4-1]+1/15<.724$. At $\mu=.1$, EACH actual debit is $<.091<.1885$. ONE square gives

$$e_A>.710645+.2(.328)-.01(.724)=.769005.$$

Every golden case therefore already exceeds the target energy below.

A weak near-four also costs too much, without a false arithmetic conjugate ban. Its actual simple-three mate $j=4,5,6$ has the reciprocal finite endpoint data

$$\delta_j=\frac{1}{14-j},\quad \sigma_j=\frac{1}{2}+\frac{13-j}{(14-j)\sqrt{2}},\quad C_j=\sigma_j\sqrt{(14-j)D_j}>31/4.$$

Positive finite inverse projection proves an ACTUAL original endpoint, with every omitted wall strict/nonpositive as required. Thus its normalized dual height is $>\sqrt{27/35}>.878$. At site nine every nongolden nonweak role has raw loss $<.618$, hence normalized height $>.75$. If nine itself is weak near-four, use its ACTUAL mate ten, whose height is $>.878$. The other orientation, with mate eight, has side order nine and was already strong. The two chosen unit duals have $j\le 6, k\ge 9$, correlation

$$\rho\le\sqrt{5/12}<2/3.$$

Cauchy on their sum gives $e_A > (.878 + .75)^2/[2(1 + 2/3)] = .7951152$. Reflection only reverses index names.

It remains to absorb ALL isolated feet. Own-three/four/six/parallel have their ALL-foreign actual nulls before the Q split. Pure five's actual H_4 loss is $< 595/1056 < .57$. Every pure finite native at least seven uses the positive adjacent core already proved in Section C.10.1; at this rank its loss is

$$< \frac{4/3}{q + \sqrt{D - 4/3}} + 1/\sqrt{12} < .279 + .289 = .568 < \ell.$$

For $Q > 0, u > 1$, only own five remains, and positive finite clearing gives loss $< .206 + .383 = .589 < \ell$. The $u = 1$ vector is a forbidden feasible null. Retain EVERY $0 < u < 1$, pure or foreign, with its actual pressure

$$x_A \cdot (\Omega_i - u n_i) \geq \sqrt{D_i - u} - \sqrt{u(1 - u)}.$$

Its raw loss is $< a_i u + \sqrt{u(1 - u)}$, with $a_{5,9} < .206$, $a_{6,7,8} < .199$. The raw circle maximum is $< .6135$, justifying its earlier use.

Choose symmetric heights, linear on the ends,

$$H_5 = 1.9359, \quad H_6 = 2.0191, \quad H_7 = 2.0462.$$

The exact floor squares after adding ℓ are $2.5354^2 < 45/7$, $2.6186^2 < 48/7$, $2.6457^2 < 7$. Its central coefficients are

$$\beta_5 = \beta_9 = .15199, \quad \beta_6 = \beta_8 = .02805, \quad \beta_7 = .0271.$$

The energy is

$$E = 1.9359^2/5 + .0832^2 + .0271^2 = .757198412, \qquad E - 53/70 = \frac{97221}{1750000000} > 0.$$

Assume $e_A \leq E$. The golden and near-four contradictions leave only genuine strong constraints or isolated ACTUAL feet at every paid index. Their FULL adjusted losses satisfy

$$a_i u + \sqrt{u(1 - u)} - (\beta_i/2)(2u - u^2) < .5995.$$

For outer sites the proof works already at $\beta = .08$: below .4 raw loss $< .5824$, above it $.6135 - .04(.64) = .5879$. For inner sites below .5, raw loss $< .5995$; above it use

$$(.199 + 1.01961)/2 - .0271(3/8) = .5991425 < .5995.$$

All root comparisons are positive rational squares, covering the ENTIRE intervals.

Form ONE vector $Z = w - \sum \beta_i u_i n_i$ over ALL retained weak feet. The actual pressures and every adjacent-foot cross give exactly the simultaneous-foot inequalities proved above. The single square implies $e_A > E$, contradicting the assumed $e_A \leq E$. Strictness comes from strong floors if no foot is present, otherwise from each own-coefficient adjusted loss. Thus

$$e_{A_{13}} > .757198412 > 53/70 \tag{A13U}$$

with every actual field and arbitrary finite spectators. Only now add the disjoint original ordinary costs: $A_3 > 1/4$, or $T_2 > 1/7, A_1 > .1$. They close BOTH ordinary rank-three alternatives. The actual-I and actual-O_7 alternatives retain their separate hypotheses and proof.

C.10.3 The actual seven-field A_{13} consumer

For the actual $O_7 + A_1$ alternative, the ACTUAL field is K_7. We use a direct full-order integer strip, rather than any computed field census. Put $\vartheta = 2\cos(\pi/7)$, $f(T) = T^3 - T^2 - 2T + 1$. Its full ring is $\mathbb{Z}[\vartheta]$. This follows already from the proved full ring $\mathbb{Z}[u]$, $u^3 - 5u^2 + 6u - 1 = 0$:

$$\vartheta = u^2 - 3u + 1, \qquad u = \vartheta^2.$$

Thus no new external integral-basis assumption is required here.

There is NO algebraic integer $w \in K_7$ with physical $0 < w < 256/25$ and both other values in $(0, 28/45)$. To prove this, exact polynomial signs give roots

$$1.8019 < \vartheta < 1.802, \quad .445 < \alpha < .446, \quad -1.248 < \beta < -1.246.$$

Write $w = a + b\vartheta + c\vartheta^2$ in the FULL ring. For $d = b - (\vartheta - 1)c$,

$$w_\alpha - w_\beta = (\alpha - \beta)d, \qquad |d| < .369,$$

using $\alpha - \beta > 1.691$. Put $\Delta = w - (w_\alpha + w_\beta)/2$. The exact trace identities give

$$\Delta = \frac{(3\vartheta - 1)}{2}d + (3\vartheta^2 - 2\vartheta - 2)c.$$

The first coefficient is < 2.203, its product with d has magnitude $< .812907$; the second coefficient is > 4.136. Since $-28/45 < \Delta < 10.24$, this forces $c = 0, 1, 2$. For $c = 0$, $b = 0$, leaving a positive integer $a < 28/45$. For $c = 2$, the d interval gives $1.2348 < b < 1.973$, no integer. For $c = 1$, it gives $b = 1$. Now $.643025 < \alpha + \alpha^2 < .644916$; positivity of $w_\alpha = a + \alpha + \alpha^2$ forces $a \geq 0$, but then $w_\alpha > .643025 > 28/45$. This exhausts every full-order coefficient.

Reclassify the SAME N14 central sites at the NEW loss $\ell = .574$, not the previous .5995 list. The exact ratio is

$$R_\ell(q) < 500/287 - 1213/8139 < 8/5.$$

The finite-white list and own-zero reciprocal repairs are unchanged, with their full foreign costs. Native five is absent in K_7. Every own-four/simple-three near role, at any position and with any foreign contacts, has the actual $\sqrt{2}$ cycle and is absent in the cubic field. The remaining COMPLETE nonisolated comparisons against the larger threshold $8/5$ are: near-three $> (1.3)/\sqrt{.6}$; D-position-three $> (1.1)/\sqrt{.4}$; E-position > 1.8; far-four $> .9\sqrt{5}$; near-four white contact has $\sigma > 53/50, \delta \leq 1/5$; two-white-three has $q/2 + 1/q > 8/5$. The full FOREIGN far-three bound is

$$\frac{1.8m + 1}{\sqrt{m(m+2)}} \geq 10/\sqrt{35} > 8/5 \quad (m \geq 5).$$

It increases with m, and positive foreign cost only strengthens every other ratio. All own-three/four/six/parallel affine cases retain the original nulls in their proper roles. Thus any weak nonisolated cap is pure endpoint helper with $\gamma > .9$, $9/5 < t = 2\gamma < 57/25$. The signed Schur costs proved above still give every other conjugate in $(-.83, .5)$. In actual K_7, t has degree one or three. Degree three gives unit norm, both small conjugates negative, and integer trace

$$.14 < \mathrm{Tr}(t) \leq t - 2/\sqrt{t} < 1,$$

using $t(t-1)^2 < 58368/15625 < 4$. Degree one leaves only parallel two. No weak nonisolated cap remains.

For an isolated cap keep its ENTIRE spectator projection. Its actual own cycle $w = 4\gamma^2$ is integral in K_7. Original K-rescaling gives new diagonal $2/w$, marked old product -1. At both other embeddings, the old-A Schur complement requires

$$0 < w_\sigma < 4/D_i \le 28/45.$$

Weakness would give $\gamma/\sqrt{1-Q} \le R_\ell(q) < 8/5$, hence physical $0 < w < 256/25$. This is precisely the impossible FULL-order strip just proved. It covers EVERY finite/dotted/pure/foreign isolated cap, not only generating w's. No weak feet remain.

All sites five through nine are now .574-strong. The symmetric positive reference with

$$H_5 = H_9 = 1.9614, \quad H_6 = H_8 = 2.0446, \quad H_7 = 2.0717$$

has genuine floor squares $2.5354^2 < 45/7$, $2.6186^2 < 48/7$, $2.6457^2 < 7$. Its duals are $.15454, .02805, .0271, .02805, .15454$, and ONE square gives

$$e_A > 1.9614^2/5 + .0832^2 + .0271^2 = .777074642 > 233/300.$$

The ACTUAL O_7 spectator costs $> .09$ and propagates the last ordinary $A_1 > 2/15$. Adding these disjoint original projections ONCE gives $233/300 + .09 + 2/15 = 1$, strictly exceeded. A bare old seven bond is never substituted for the actual O_7 profile.

C.10.4 The $A_{13} + I + T_2$ consumer with every remaining foot

Here I is the ACTUAL isolated short wall, and T_2 an ORDINARY irreducible dihedral. The strengthened ordinary theorem proved in Section C.1, together with the separate ordinary A_2 bound, gives $e_T > 3/19$. Use the actual rank-free I parameters from (20.1):

$$e_I \ge u^2, \quad 0 < u < 1/3, \quad t + 3u/2 > .767.$$

Both signed A_{13} endpoint coordinates satisfy

$$x_1, -x_{14} \ge \alpha t, \qquad \alpha = \sqrt{13/14} > .9636.$$

All these pressures concern the SAME original farthest vector. The I energy is counted once.

For endpoint means $b < .2738$, or $b < .2666$, the positive linear lower bounds are

$$\alpha t - b > .4652 - 1.446u, \qquad \alpha t - b > .4724 - 1.446u,$$

respectively. They follow from $.9636(.767 - 1.5u) = .7390812 - 1.4454u$. The following exact completion applies on the FULL continuous domain:

$$(1 + cB^2)[u^2 + c(A - Bu)^2] - cA^2 = [(1 + cB^2)u - cAB]^2.$$

For $B = 1.446$, the positive rational margins

$$(5/2)(.4652)^2 - .0868[1 + (5/2)B^2] = .000498828,$$

$$(5/4)(.4724)^2 - .077[1 + (5/4)B^2] = .000701535$$

therefore give the uniform bounds $u^2 + (5/2)(.4652 - Bu)^2 > .0868$ and $u^2 + (5/4)(.4724 - Bu)^2 > .077$.

The COMPLETE N14 helper/near-four classifications of Section C.10.2 do not depend on its contrary energy assumption. A golden helper at five gives $e_A > .822759$, which closes with the joint $I + T_2 > .21$. For a helper at six/seven, retain its SAME one-reference H_4 square of energy $L > .769005$, with price $< .091$ at each outer coefficient .1885. Choose the five-coordinate block OPPOSITE its helper endpoint. The actual H_4 displacement and its flat block projection are constant there; their residual vanishes POINTWISE on that block. Thus an endpoint residual G has

$$G^2 = 4/5, \quad w \cdot G = Y \cdot G = 0, \quad b = 1.885/(5\sqrt{2}) < .2666.$$

Its individual price is $\kappa/(5\sqrt{2})$, with remaining capacity $> .0975$. One JOINT pressure square gives

$$e_A > L + 2\kappa(\alpha t - b) - (4/5)\kappa^2.$$

For $u \geq .3$, $L + 3/19 + .09 > 1$. For $u \leq .2$, choose $\kappa = .2$; its price is $< .0283$. The total is $> L + 3/19 + u^2 + .4(.4724 - Bu) - .032$, decreasing to an endpoint additive value .08128, already contradictory. For $.2 \leq u \leq .3$, put $d = .4724 - Bu > 0$, $\kappa = 5d/4$; its price is $< .033 < .0975$. The total exceeds $L + 3/19 + .077 > 1$ by the completed-square certificate. Both helper ends and reflected indices are covered with ACTUAL orthogonal residuals.

Every weak near-four gives $e_A > .7951152$; adding joint $I + T_2 > .21$ also exceeds one. We are left with ONLY true strong sites or the actual isolated feet of Section C.10.2.

Use that section's full rational reference, energy $E = .757198412$, outer coefficients .15199, inner $.02805, .0271, .02805$. Its two endpoint residuals are

$$G_L = e_1 - P_5/5, \qquad G_R = -e_{14} + (e_{10} + \cdots + e_{14})/5.$$

They are disjoint, have norm $4/5$, and are orthogonal to w. For $W = w + \kappa(G_L + G_R)$, EACH outer central coefficient becomes

$$\beta'_5 = \beta'_9 = .15199 - \kappa/(5\sqrt{2}).$$

The others are unchanged. For EVERY remaining foot parameter $v_i \in (0, 1)$, form ONE vector $T = W - \sum \beta'_i v_i n_i$. The primitive endpoint/strong/foot pressures, paid BEFORE taking its square, give

$$x_A \cdot T \geq E + 2\kappa(\alpha t - b) - \sum \beta'_i (L_i - \ell),$$
$$T^2 \leq E + (8/5)\kappa^2 - \sum (\beta'_i)^2 (2v_i - v_i^2).$$

Here $b = 1.9359/(5\sqrt{2}) < .2738$. Every endpoint/foot cross is incorporated through the REMAINING coefficient β'_i, and every adjacent-foot cross has the actual nonpositive old-root sign. The proved continuous adjusted-loss inequalities hold at outer coefficients at least .08 and the unchanged inner coefficients. Consequently

$$e_A > E + 4\kappa(\alpha t - b) - (8/5)\kappa^2.$$

For $u \geq .3$, $E + 3/19 + .09 > 1$. For $u \leq .2$, choose $\kappa = .2$; both outer coefficients remain $> .1236 > .08$. The total is $> E + 3/19 + u^2 + .8(.4652 - Bu) - .064$, decreasing to endpoint additive value .1168. For $.2 \leq u \leq .3$, use $\kappa = 5(.4652 - Bu)/4 \leq .22$; both outer coefficients remain $> .1199 > .08$. The exact completion gives total $> E + 3/19 + .0868 > 1$. All actual I parameters, ALL foreign/adjacent feet and both endpoints have been covered. This closes the last A_{13} rank-sixteen alternative with one original vector and one I energy.

C.10.5 A universal A_{12} bound from one paid endpoint

Set $N = 13$, central sites five through eight,

$$D_5 = D_8 = 80/13, \quad D_6 = D_7 = 84/13, \qquad \ell = .6005.$$

Keep arbitrary finite spectators and EVERY actual field. Assume $e_A \le 5/7$. The fresh exact sufficient ratio is

$$R_\ell(q) < 2000/1201 - 77980/458949 < 3/2, \qquad q < 51/20,$$

with positive margin $5105207/(2402 \cdot 458949)$. Deleting a central site leaves two paths of rank at least four. The COMPLETE white list is isolation, permitted endpoint/D/E simple three, endpoint four, or one endpoint-three on EACH path. Connected rank twelve precludes exceptional completions; its two long arms preclude a fork completion. Every spectator contribution remains in the actual retained finite Gram.

The complete own-zero residual repairs are

A^2	Q bound
$Q - 7/13$	$Q < 3/5$
$Q - 2/13$	$Q < 2/5$
$Q - 1/13$	$Q < 1/5.$

The first positive time is $< 4/65 < 1/4$, below any foreign contribution. The second permits at most one foreign component, with contribution $Q > A^2$. Both violate the actual reciprocal orthogonal opposite law. The last has no possible foreign contact. At $Q = 0$ their times are negative. Internal-three/four/two-contact cases have the actual affine arms/star/cycle. ALL isolated owns three/four/six and parallel retain their positive future nulls before splitting the foreign cost.

The ordinary nonisolated ratios exceed $38/25 > 3/2$, including the same first three side-order-five comparisons printed above, and two-path $q/2 + 1/q > 97/60$. Foreign far/near bounds are still $227/(25\sqrt{35})$, $\sqrt{121/35}$. Thus every weak nonisolated cap is PURE: an endpoint helper, or OWN-four/SIMPLE-three near helper, with BOTH orientations retained. The roles are not interchanged.

For an endpoint helper the exact ratio-square identity gives $t = 2\gamma < 17/8$. Non-five own has $t > 9/5$. At $N = 13, i = 5$, the signed conjugate Schur costs at $-.83, .5$ exceed four by $69/1625, 2/13$, and they increase across the half-window. Hence every other conjugate lies in $(-.83, .5)$. The COMPLETE all-degree trace/norm proof of Section C.10.2 uses only these bounds, $9/5 < t < 9/4$, actual integrality and all lifted embeddings. It excludes every non-five helper. A weak endpoint helper is therefore golden, actual $K = \mathbb{Q}(\varphi)$. A near-four simple-three normalization instead forces actual $K = \mathbb{Q}(\sqrt{2})$, so it cannot coexist with a golden helper.

The COMPLETE $26A^2$ golden time rows are

type	$26A^2$	$\mathrm{N}(26A^2)$
near 5	$26 + 56\varphi$	-1004
near 6	$28 + 56\varphi$	-784
far 5	$26 + 50\varphi$	-524
far 6	$28 + 54\varphi$	$-620.$

The four norm-squareclasses differ. Reflected cyclic parameters are $21\varphi/13, 9\varphi/13$ at five, and have constant $-9/13$, coefficient $8/13$ or $4/13$ at six. All are nonintegral in the full golden ring. Thus at

most ONE golden helper occurs. Every other central site is .65-strong, by the complete geometric inventory at that loss; another weak helper would also be .6005-weak.

Use its ACTUAL H_4 correction and positive finite projected point as before, with $\delta = (5 - 3\varphi)/12 < .111^2$. For helper at five, take $a = 1.830, b = 1.891$, heights $H_6 = H_7 = b, H_8 = a$, linear tails. Its energy

$$E = (b^2/6 + a^2/5 + .061^2)/2 = .6347405833\ldots$$

has individual coefficients $b/12 > .15758, .0305, .1525$. The far/near flat-block projections are $(5\gamma/6)\Omega_6 + \Omega_8/10$ and $(5\gamma/6 + 1/12)\Omega_6$. The reference product is < 1.459425, smaller actual time > 2.027, deficit $> .45$. FULL residual norm is $< .8$. Pay $\mu = .2$: largest own price $< .151667$, far other price .02, both individually affordable. ONE square gives $e_A > E + .148 > 5/7$.

For helper at six, take only $H_5 = H_8 = a$. Energy $a^2/5 = .66978$, both outer prices .183. The flat middle block has three coordinates. The projections are $(2\gamma/3)\Omega_5 + (\gamma/3 + .1)\Omega_8$ or the reverse placement of the .1. The reference product is < 1.6653; smaller actual time > 2.106, deficit $> .329$. FULL residual norm is $\gamma^2/3 + 1/15 < .286$. At $\mu = .1$, each price is $< .064 < .183$. ONE square gives $.66978 + .2(.329) - .01(.286) = .73272 > 5/7$. Reflections merely reverse labels. No golden helper occurs under the contrary energy assumption.

For near-four side order m, its exact ratio

$$R_m = \frac{1/\sqrt{2} + (m-1)/(2m)}{\sqrt{(m+1)/(2m)}}$$

increases in m. At seven, $R_7^2 = 67/56 + 3\sqrt{2}/4 > 9/4$, by $42\sqrt{2} > 59$. Weak near-four therefore has only own/helper positions $5/4, 6/5$ or their reflections. Deleting the ACTUAL helper $j = 4, 5$ gives a finite B completion and reciprocal endpoint:

$$\delta_j = \frac{1}{13 - j}, \quad \sigma_j = \frac{1}{2} + \frac{12 - j}{(13 - j)\sqrt{2}}, \quad C_j = \sigma_j \sqrt{(13 - j)D_j} > 31/4.$$

Positive finite projection checks ALL original walls and the reciprocal law identifies the SAME actual opposite. Its normalized height is $> .878$. On the opposite side, index eight has height $> .75$ from loss $< .618$, or use its ACTUAL near-four mate nine with height $> .878$. The two normalized duals have correlation $\le 5/8$, so

$$e_A > \frac{(.878 + .75)^2}{2(1 + 5/8)} > 4/5.$$

No weak near-four remains.

For isolated caps keep the entire Q. Pure five has the actual LEFT H_4 correction with $u = \gamma^{-2} < 25/16$, $\sqrt{\delta_5}/\gamma < 5/24$. The divided displacement is the old concave combination at barycentre i, so it is bounded above by h_i. At this ACTUAL $D_i \ge 80/13$,

$$q + \sqrt{D_i - 25/16} > 12/5 + 2 = 22/5,$$

and its loss is

$$< 125/352 + 5/24 = 595/1056 < .57.$$

This proves the exact local bound used in the preceding N14 theorem; no larger-rank denominator is assumed.

EVERY pure finite native at least seven uses the positive adjacent core $c+\gamma^{-1}n_i+2(\gamma^{-1}-\gamma)n_j$. It is zero on cap and n_i, negative on n_j, future timelike and nonpositive at ALL omitted original walls. Its pressure loses

$$< \frac{4/3}{q+\sqrt{D_i-4/3}} + 1/\sqrt{12} < .286 + .289 = .575 < \ell.$$

Here $q > 2.48$, $\sqrt{D_i - 4/3} > 2.195$. For $Q > 0, u > 1$, only own five is possible, since every other remaining finite $\gamma > .9$ has $u \le .75/.81 < 1$. Positive finite clearing gives loss $< .212 + .383 = .595$. At $u = 1$ its actual vector is a forbidden future feasible null. Retain ALL $0 < u < 1$, pure/foreign, with pressure

$$x_A \cdot (\Omega_i - un_i) \ge \sqrt{D_i - u} - \sqrt{u(1-u)}.$$

Its loss is $< a_i u + \sqrt{u(1-u)}$, where $a_{5,8} < .212$, $a_{6,7} < .205$; the raw circle maximum $< .6175$ supplies the mate argument above.

Fix the symmetric reference

$$H_5 = H_8 = 1.8795, \quad H_6 = H_7 = 1.9405, \qquad \beta_{5,8} = .15745, \quad \beta_{6,7} = .0305, \quad E = .71022505.$$

All heights are strictly below the true strong floors. The FULL continuous adjusted-loss estimate

$$a_i u + \sqrt{u(1-u)} - (\beta_i/2)(2u - u^2) < .6005$$

holds even for ANY outer $\beta_i \ge .08$. For outer sites below .4, raw loss $\le .5848$; above it, $.6175 - .04(.64) = .5919$. For inner sites the expression is at most $.1745u + .01525u^2 + \sqrt{u(1-u)}$. On $(0, .5]$ it is $\le .5910625$. On $[.5, .7]$, $u^2 \le 1.2u - .35$ and the circle bound with coefficient .1928 give $< .6003125$. On $[.7, 1)$, $\sqrt{u(1-u)} < (3-u)/5$, by positive square difference $(2u-1)(13u-9)/25$, giving $< .6$. Thus EVERY endpoint of the continuous split and every foot parameter are covered.

We record the full JOINT endpoint/foot rule. Let $\bar{Z}$ sum to one, supported on the first at most FOUR coordinates, and let its ACTUAL pure pressure be $x_A \cdot Z \ge h$, where $Z = \bar{Z} - P_{13}/13$. Its flat-end projection is $\Pi Z = \Omega_5/(5\sqrt{2})$. For $Y = Z - \Pi Z$,

$$w \cdot Y = 0, \quad S = Y^2 = \|\bar{Z}\|^2 - 1/5, \quad b = w \cdot \Pi Z < .266.$$

For $0 \le \mu \le .4$, spend $\mu/(5\sqrt{2})$ from INDIVIDUAL β_5. Its remainder is $> .1008 > .08$; the opposite outer and both inner coefficients are unchanged. For EVERY weak foot use this REMAINING $\beta_i' = (w + \mu Y) \cdot n_i$, and form

$$T = w + \mu Y - \sum \beta_i' u_i n_i.$$

Primitive pressures and ALL old-root crosses give

$$x_A \cdot T \ge E + \mu(h - b) - \sum \beta_i'(L_i - \ell),$$
$$T^2 \le E + \mu^2 S - \sum (\beta_i')^2 (2u_i - u_i^2).$$

ONE square and the just-proved adjusted losses yield

$$e_A > E + 2\mu(h - b) - \mu^2 S. \tag{A12P}$$

Endpoint/foot crosses have not been discarded: they are paid through the REMAINING individual coefficients. Reflection gives the same rule at the other end.

If actual $x_1 \geq .34$, take $\bar{Z} = e_1$, $S = 4/5, \mu = .1$. Then $e_A > .71702505 > 5/7$, impossible. Likewise $-x_{13} < .34$. The two ACTUAL endpoint edges therefore have half-tangent square

$$< (13/12)(.34)^2 = 3757/30000 < 3/23,$$

hence cosh $< 13/10$. The strict rational margin is 3589/690000.

Deleting endpoint one retains A_{11} and ALL spectators. Its COMPLETE finite list is isolation, simple three at global $j = 2, 3, 11, 12$, or four at $j = 2, 12$. The six pure coshes for positive own γ are respectively

$$\frac{24\gamma + 11}{13}, \frac{24\gamma + 10}{2\sqrt{13}}, \frac{24\gamma + 2}{2\sqrt{13}}, \frac{24\gamma + 1}{13}, \frac{24\gamma + 11\sqrt{2}}{\sqrt{26}}, \frac{24\gamma + \sqrt{2}}{\sqrt{26}}.$$

Foreign Q only increases them. At $\gamma \geq 1/2$, all but far-three exceed 1.3; far-three/own-three has its actual affine cycle, and every remaining own gives $(12\sqrt{2} + 1)/13 > 1.3$. For own zero with white three, the spatial directional square is $(13 - j)^2/312 \leq 121/312$. The actual short length bound gives time square $< 69(121)/(169 \cdot 312) < 1/4$. Any positive foreign contribution violates the reciprocal rule. At $Q = 0$, the remaining times are negative or the actual affine $1, 2, 5$ principal. White-four at two has its affine star of squared weights $1/2, 1/4, 1/4$; far twelve has $Q < 1/12$, hence $Q = 0$ and negative time. Every own-zero and nonisolated role is excluded.

Each ACTUAL endpoint is therefore isolated within A_{12}, with foreign contacts retained:

$$c = Ap - \gamma\Omega_1 - v_S, \quad u = (1 - Q)/\gamma^2, \quad D_1 = 24/13, \quad C^2 = D_1/u < 169/100.$$

Thus $u > 2400/2197 > 1$, excluding the dotted/parallel/$u \leq 1$ domain by an ACTUAL edge bound. Own three has $Q < 1 - 600/2197 < 3/4$, at most two contacted spectator components. Their largest contribution is at least $Q/2$, and $Q/2 > A^2 = Q - 7/13$, forbidden by reciprocity; at $Q = 0$ time is negative. Own four has $Q < 1 - 1200/2197 < 1/2$, so at most one foreign contribution $Q > A^2 = Q - 1/13$. Own six has the actual $6, 3$ null. Only finite owns five or at least seven remain.

For pure five use the ACTUAL positive finite H_4 projection on c, n_1, n_2, n_3, with $\delta = (5 - 3\varphi)/8 > 0$. Its sum-one displacement is $\bar{Z} = P_4/4$, $S = 1/20$, and pressure

$$h = \frac{A^2 + \delta - \sqrt{\delta(A^2 + \delta)}}{A\gamma\sqrt{2}}, \quad A^2 = (24/13)\gamma^2 - 1, \quad A^2 + \delta = 9\gamma^2/26.$$

The exact brackets $.6545 < \gamma^2 < .65475$, $.2083 < A^2 < .2088$, $A < .457$, $.017875 < \delta < .01825$, $A^2 + \delta > .2265$ give $\sqrt{\delta(A^2 + \delta)} < .0644$ and $A\gamma\sqrt{2} < .525$. Hence $h > .308$. Payment (A12P) at $\mu = .4$ gains $> .0256$.

For EVERY pure finite own at least seven use the ACTUAL adjacent negative core at one/two. Its sum-one displacement and pressure are

$$\bar{Z} = (1 - u/2)(e_1 + e_2) + (u - 1)e_3, \quad h = \frac{\sqrt{D_1 - u} - \sqrt{7u - 4 - 3u^2}}{\sqrt{2}}, \quad S \leq .3.$$

Since $u < 100/81$, $D_1 - u > 644/1053 > .782^2$; the other radical is $< .289$. Thus $h > .348$, and $\mu = .4$ gains $> .8(.348 - .266) - .16(.3) = .0176$. This covers native eight and the whole finite tail without a label cutoff.

For EVERY positive-foreign endpoint, $u > 1$ forces native five. Clear ALL contacted spectator normals positively with n_1. The actual finite projection gives the simpler pure pressure

$$x_A \cdot (\Omega_1 - n_1) > \sqrt{D_1 - 1} - \sqrt{u - 1}.$$

Its stronger exact pressure has additional positive factor $\sqrt{D_1-1}/\sqrt{D_1-u}>1$, so the displayed lower bound is valid. Here $\bar{Z}=(e_1+e_2)/2$, $S=.3$, $\sqrt{11/13}>.919$, $\sqrt{u-1}<.383$, so $h>.378$. At $\mu=.3$, gain $>.0402$.

All these points have positive ACTUAL wall coefficients, correct selected products, positive future time and nonpositive products at EVERY omitted original wall. Every $\bar{Z}$ has the support required by the JOINT payment rule. Even the weakest endpoint gain gives the CONTRARY-branch bound $.71022505+.0176=.72782505>5/7$. Thus the assumed $e_A\le 5/7$ is impossible:

$$e_{A_{12}}>5/7 \tag{A12U}$$

with every actual field and arbitrary spectators. The number .72782505 is not asserted as an unconditional component rate.

This one stronger theorem closes the rank-sixteen ordinary-rank-four and actual-$I+T_3$ rows, whose disjoint or JOINT remainder costs $>2/7$. It also closes BOTH typed-seven rank-two rows: the actual defect costs $>.09$ and propagates the remaining ordinary $T_2>.2$, giving $5/7+.09+.2>1$. The weaker earlier typed-only component estimate therefore needs no separate chapter. In rank seventeen it closes every older A_{12} row by the actual rank-five spectator partitions and rank-free joint costs.

C.10.6 A rank-eleven chain: the complete local alternatives

We next prove a uniform estimate at the smaller path size, and then a joint estimate when an ACTUAL short isolated I is present. Both arguments use the original vector x; the second will rebuild its reference rather than add a new gain to an optimized first one. Write $N=12$, $A=A_{11}$, and use

$$n_i=(e_i-e_{i+1})/\sqrt{2},\qquad D_i=\frac{2i(12-i)}{12},\qquad C_{ij}=\frac{2i(12-j)}{12}\quad(i\le j).$$

Only sites $5,6,7$ will be paid. Put $\ell=.61$, $q_i=\sqrt{D_i}$. Then

$$D_5=D_7=35/6,\quad D_6=6,\qquad a_i=\frac{1}{q_i+\sqrt{D_i-1}}<.217. \tag{A11a}$$

Indeed $q_i>2.415$, $\sqrt{D_i-1}>2.198$, and $(2.415+2.198)(.217)=1.001021>1$. A site is strong if $h_i>q_i-\ell$.

Deleting one of these sites retains two paths of ranks at least four. The complete one-node finite extensions are: no white contact; one admissible endpoint, D- or E-position three; one terminal four; or two endpoint threes joining the paths. Two contacts to one path make a forbidden cycle. A connected completion has rank eleven, so exceptional finite diagrams cannot add a missing joined-path role. We retain the ENTIRE contacted spectator forest, writing its squared projection as Q; every positive whole-component contribution is at least $1/4$.

The own-zero cases are excluded before any positive-own ratio is used. An internal three contains the actual affine arms $1,2,5$ or $1,3,3$; an internal four contains the affine $3,3,4$ star; two white threes contain the affine cycle. The remaining far-three role, with side order $m\ge 5$, has time square $<1/(2m)$, below any necessary positive foreign contribution. The next-to-far three has $Q<2/m$, hence at most one contacted whole spectator, and time square $Q-1/6<Q$. Either would make this same actual wall close an edge in that orthogonal spectator, contrary to the actual reciprocal-opposite law. With $Q=0$ its time is negative. Far-four has $Q<1/m\le 1/5$, so $Q=0$ and negative time. These are all residual own-zero roles. The affine vectors have positive time through the actual cap and nonpositive product with every omitted original wall. They are forbidden feasible future nulls, even in the presence of foreign contacts.

For $q \ge 12/5$, the sufficient strong-edge ratio is

$$R_\ell(q) = \frac{100}{61} - \frac{q - .61}{q(2q - .61)}.$$

It increases with q, and

$$R_\ell(q) \le R_\ell(3) < 3/2, \qquad \frac{3}{2} - R_\ell(3) = \frac{1669}{197274} > 0.$$

The complete near-three, D/E, terminal-four and two-white bridge formulas have, respectively, the lower comparisons

$$\frac{.808 + .4}{\sqrt{.6}}, \quad \frac{.808 + .2}{\sqrt{.4}}, \quad \frac{.808}{\sqrt{.25}}, \quad q/2 + 1/q > 97/60.$$

All exceed $38/25 > 3/2$. The smaller positive owns have their actual affine cycle, star or 6–3 obstruction. For the two exceptional profiles WITH $Q > 0$, clearing the entire spectator gives ratio

$$> \frac{227}{25\sqrt{35}} > 38/25 \quad \text{or ratio square } > 121/35.$$

Consequently a weak NONISOLATED cap is PURE, and is either a simple-three END-helper

$$c = A_c p - \gamma\Omega_i - \Omega_e/2, \qquad e = 1 \text{ or } 11, \tag{A11helper}$$

or own-four/simple-three near helper

$$c = A_c p - \Omega_i/\sqrt{2} - \Omega_j/2, \qquad |i - j| = 1. \tag{A11near4}$$

Both orientations are retained. The near-four ratio increases with side order; from order eight it already exceeds $3/2$. Thus its complete list, up to reflection, is $(i, j) = (5, 4), (6, 5), (5, 6)$. It would be incorrect to exclude these three roles by the larger-path conjugate test.

For an isolated-within-A cap retain

$$c = A_c p - \gamma\Omega_i - v_S, \quad Q = \|v_S\|^2, \quad u = (1 - Q)/\gamma^2.$$

Before splitting Q, own-three, four, six and parallel have their actual affine E_7 arms $1, 3, 3$, weighted star, 6–3 path or positive null pair. These nulls are feasible against ALL foreign walls by acuteness. Hence every remaining positive own has $\gamma \ge \varphi/2$, $\varphi = (1 + \sqrt{5})/2$. For $Q > 0, u > 1$, positive clearing gives loss

$$< a_i + \sqrt{u - 1} < .217 + (2 - \varphi) < .599 < \ell.$$

The actual $u = 1$ vector is a forbidden future feasible null.

The PURE finite cases with $u > 1$ are handled by actual points at this rank. For own five the same positive H_4 projection used above has loss

$$< \frac{25/16}{q_i + \sqrt{D_i - 25/16}} + 5/24 < \frac{625}{1792} + \frac{5}{24} = \frac{2995}{5376} < .558.$$

Here the denominator is $> 2.415 + 2.066 > 4.48$; $2.066^2 < 205/48 = D_{\min} - 25/16$. For EVERY finite native at least seven put $d = 1 - \gamma^2$ and choose an old adjacent root n_j. The positive actual combination

$$\xi = c + \gamma^{-1} n_i + (2d/\gamma) n_j$$

is zero on c, n_i, negative on n_j, and has square $-d(1-4d)/\gamma^2 < 0$. It is future timelike and all omitted ORIGINAL wall products are nonpositive. Its comparison gives loss

$$< \frac{100/81}{q_i + \sqrt{D_i - 100/81}} + \frac{25}{81} < \frac{213975}{369279} < .58 < \ell.$$

We used $\gamma > .9$, $\sqrt{D_i - 100/81} > 2.144$ and $\sqrt{d(1-4d)} \le 1/4$. This includes native eight and the entire unbounded finite-native tail.

Retain EVERY remaining $0 < u < 1$, pure or foreign. Positive inverse-Gram clearing expresses v_S as a positive combination of all contacted old spectator roots. The actual vector

$$\eta = c + v_S + \gamma u n_i = \gamma\{\sqrt{D_i - u}\, p - (\Omega_i - u n_i)\}$$

has positive time, square $-\gamma^2 u(1-u)$, and zero products on the cap and cleared spectators. Its product on n_i is $-\gamma(1-u) < 0$; all other original walls have nonpositive product termwise. It therefore supplies the SAME-x pressure

$$x_A \cdot (\Omega_i - u n_i) \ge \sqrt{D_i - u} - \sqrt{u(1-u)} = q_i - L_i, \qquad L_i < .217u + \sqrt{u(1-u)}. \qquad \text{(A11foot)}$$

No restriction on the number or adjacency of these isolated feet is made.

C.10.7 The all-degree end-helper reduction at $N = 12$

For (A11helper), the exact side-order identity is

$$\frac{(\gamma + 1/(2m))^2}{(m+1)/(2m)} - (4\gamma - 2) = \frac{(\gamma - 1 - 1/(2m))^2}{(m+1)/(2m)} \ge 0.$$

Weakness gives $\varphi \le t = 2\gamma < 17/8$; if the own is not five then $t > 9/5$. The actual simple-three normalization makes t integral in K. For a lifted nonphysical embedding of $\mathbb{Q}(t)$, strict old-A Schur positivity gives the cheapest signed costs

$$Q_-(s) = 2is^2 + 4s + 2 - 2(is+1)^2/12, \qquad Q_+(s) = 2is^2 + 2 - 2(is-1)^2/12.$$

They increase with i on the relevant half-window. At $i = 5$,

$$Q_-(-17/20) - 4 = 31/480 > 0, \qquad Q_+(1/2) - 4 = 1/8 > 0.$$

Their values at zero are below four, and the outer quadratic branches increase away from their allowed interval. Thus EVERY other conjugate obeys

$$-17/20 < t_\sigma < 1/2. \qquad \text{(A11strip)}$$

We give the full degree argument, including degree five. In degree at least six a nonzero integral norm has magnitude $< (17/8)(17/20)^5 < 1$, impossible. In degrees three through five its magnitude is less than two. One positive small conjugate would instead make it $< (17/8)(1/2)(17/20)^{a-2} < 1$. All small conjugates are therefore negative and their absolute product is $1/t$. AM–GM bounds their trace above by $t - (a-1)t^{-1/(a-1)}$. The resulting strict trace intervals are

a	trace interval	upper comparison
3	$(0, 1)$	$t(t-1)^2 < (17/8)(9/8)^2 < 4$
4	$(-1, 0)$	$t^4 < (17/8)^4 < 27$
5	$(-2, -1)$	$t(t+1)^4 < (17/8)(25/8)^4 < 256.$

The lower bounds follow from $t > 9/5$ and each small conjugate $> -.85$. None contains an integer. In degree two the norm is 1 or -1. A negative small conjugate gives trace $t - 1/t \in (1, 2)$. A positive one forces $t > 2$ and trace $t + 1/t \in (2, 3)$. Degree one leaves only $t = 2$, the forbidden actual parallel pair. Thus every weak end-helper is native five.

Its actual connected old-path-plus-cap Gram, normalized by its simple three, is Lorentzian over $\mathbb{Q}(\varphi)$. A proper totally real extension would have a nonphysical embedding fixing this negative direction. Admissibility forbids it: the ACTUAL field is $\mathbb{Q}(\varphi)$. A near-four simple-three cap similarly forces ACTUAL $K = \mathbb{Q}(\sqrt{2})$. These restrictions apply only in the stated exceptional branches; no field list is assumed for the other caps.

C.10.8 Uniform A_{11} energy, paying a helper and all feet

We prove $e_{A_{11}} > 13/20$ with arbitrary finite spectators and every actual field. Assume contrariwise $e_A \le .65$. First remove the retained near-four cases by their OWN actual mate, not a field ban. For (A11near4) the three complete time squares, up to reflection, are

$$\frac{39 + 28\sqrt{2}}{12}, \qquad \frac{83 + 60\sqrt{2}}{24}, \qquad \frac{41 + 30\sqrt{2}}{12}. \tag{A11silver}$$

Each exceeds six. With $v = \Omega_i/\sqrt{2} + \Omega_j/2$ and $T_j = (v \cdot \Omega_j)/(A_c\sqrt{D_j})$, direct inverse-Gram multiplication gives

$$T_j^2 - 1 = \begin{cases} 1/((12 - j)A_c^2), & j = i - 1, \\ 1/(jA_c^2), & j = i + 1. \end{cases}$$

In all three roles and their reflections the integer factor is at least six. Consequently $1 < T_j^2 < 37/36 < (51/50)^2$. The full old-root replacement at j is positive definite, its final Schur complement being $(v \cdot \Omega_j/\sqrt{D_j})^2 - A_c^2 > 0$. Its actual off-diagonal entries are nonpositive. Positive finite inverse projection of the strict interior point gives a common point strict against every other ORIGINAL wall. Hence this cap is the ACTUAL opposite at j, rather than a formal cutter behind another facet. Its half-tangent pressure gives

$$x_A \cdot \frac{\Omega_j}{\sqrt{D_j}} \ge T_j - \sqrt{T_j^2 - 1} > \frac{51 - \sqrt{101}}{50} > .81.$$

Thus $e_A > .6561 > .65$, a contradiction. Every weak nonisolated exception is now golden.

The complete golden time table is

type	$24A_c^2$	$\mathrm{N}(24A_c^2)$
5, near	$22 + 49\varphi$	-839
6, either end	$23 + 48\varphi$	$-671 = -11 \cdot 61$
5, far	$22 + 45\varphi$	$-551 = -19 \cdot 29.$

(A11gold)

Both actual norm-two helper caps lie on the same K-time line; their time-square ratio must be a K-square. The three norm squareclasses are distinct, so unlike rows cannot coexist. Two center-six caps cannot be the distinct actual opposites of the same edge. For the reflected near-five and far-five pairs the actual closed helper paths give raw cyclic products $(2 - 7\varphi)/6$ and $(2 - 3\varphi)/6$. Neither is integral in the full ring $\mathbb{Z}[\varphi]$. Thus at most ONE central index has a weak golden helper. This does not limit the isolated feet of (A11foot).

Set $a = 1.805, b = 1.839$. With heights linear on the unmentioned outer tails, there are three reference cases:

helper	positive $\beta_i = w \cdot n_i$	$E = w^2$
none	$\beta_5 = \beta_7 = .1635,\ \beta_6 = .034$	.652761
6	$\beta_5 = \beta_7 = .1805$	.651605
5	$\beta_6 = .17025,\ \beta_7 = .1635$	.60820725.

(A11refs)

In the first case $H_5 = H_7 = a, H_6 = b$. For the second replace H_6 by a, so its helper coefficient is zero. For the third flatten heights on $0, \dots, 6$, with $H_6 = b, H_7 = a$; reflect a helper at seven. Its energy is exactly $(b^2/6 + (b-a)^2 + a^2/5)/2$. All paid heights are strictly below the strong floors, since $2.415^2 < 35/6$ and $2.449^2 < 6$.

Only the end-helper case needs a helper payment. For the actual helper at five put $g = \varphi/2$, let $e = 1$ or 11 and let $j = 2$ or 10 be its adjacent old root. The ACTUAL finite H_4 projection uses

$$w_{\text{old}} = gn_5 + (2/3)n_e + (1/3)n_j, \qquad \delta = 2/3 - g^2 > 0, \qquad U = v - w_{\text{old}}.$$

The positive vector

$$\xi = p + (A_c/\delta)(c + w_{\text{old}})$$

has selected products zero, square $-1 - A_c^2/\delta$, and positive time. Every unselected original wall has nonpositive product, by positivity and acuteness. Comparison with the ORIGINAL p, x gives

$$x_A \cdot U > A_c - \sqrt{\delta}, \qquad U \cdot n_k \geq 0 \quad \text{for ALL old } k. \tag{A11U}$$

The second conclusion follows directly from the selected zero and unselected nonpositive products of the actual projected point. The correction U alone is not claimed to be a hyperbolic point.

Let Π average the first six and last five coordinates, leaving coordinate seven fixed. Then w of the last row of (A11refs) lies in its range. The correction roots have zero block averages. For $Y = U - \Pi U$, direct block variance gives

	ΠU	Y^2
near	$(5g/6 + 1/12)\Omega_6$	$(2/3)g^2 + g/3 + 1/12$
far	$(5g/6)\Omega_6 + \Omega_7/10$	$(2/3)g^2 + 1/15.$

In particular $w \cdot Y = 0$, $Y^2 < .8$. For example the far variances of Ω_5, Ω_{11} are $5/3, 8/5$, their cross variance is zero, and the old correction contributes $-g^2 - 1/3$; the near cross variance contributes $g/3$. These are FULL residual norms. The smaller actual time square is $(35g^2 + 5g)/6 - 13/24$. Using $.809 < g < .81$ gives

$$A_c > 1.987, \quad \sqrt{\delta} < .111, \qquad w \cdot \Pi U \leq .675b + .1a = 1.421825$$

in the far case; in the near case the product is $< (91/120)b = 1.394575$. Both actual helper deficits therefore satisfy

$$\Delta = A_c - \sqrt{\delta} - w \cdot \Pi U > .454175 > .45.$$

Pay $\mu = .1$. Writing $\Pi U = \sum a_i \Omega_i$, define

$$W = w + \mu Y, \qquad b_i = \beta_i - \mu a_i.$$

The near debit at six is $< .075834$. The far debits are $< .0675$ at six and .01 at seven. Each remaining positive b_i exceeds .034, and no helper index is paid. Crucially (A11U) gives

$$W \cdot n_i = b_i + \mu U \cdot n_i \geq b_i. \tag{A11price}$$

This sign retains every helper/foot cross. It is not replaced by an unsupported orthogonality assertion. In the no-helper and center-helper cases set $W = w, b_i = \beta_i, \mu = 0$.

Let S contain ALL retained isolated feet at indices with positive b_i, and set

$$Z = W - \sum_{i \in S} b_i u_i n_i.$$

Before assigning any floor the helper debit has been removed from each individual coefficient. Combining its pressure with all strong and actual foot pressures gives

$$x_A \cdot Z \geq E + \mu\Delta - \sum_{i \in S} b_i (L_i - \ell).$$

Since distinct old normals have nonpositive products, (A11price) gives, INCLUDING adjacent-foot crosses,

$$Z^2 \leq E + \mu^2 Y^2 - \sum_{i \in S} b_i^2 (2u_i - u_i^2).$$

For $0 < u \leq 1/2$, raw loss is $< .217/2 + 1/2 = .6085$. For $1/2 \leq u < 1$, the circle bound and $\sqrt{1 + .217^2} < 1.024$ give

$$L_i - \frac{b_i}{2}(2u - u^2) < \frac{.217 + 1.024}{2} - \frac{3(.034)}{8} = .60775 < \ell.$$

One completed square therefore yields

$$e_A \geq 2x_A \cdot Z - Z^2 > E + 2\mu\Delta - \mu^2 Y^2.$$

If no foot occurs, strict paid strong/helper pressures give the same strictness. The three cases in (A11refs) give respectively $.652761, .651605$, and $.60820725 + .2(.45) - .01(.8) = .69020725$. Each exceeds .65, contradicting the contrary assumption. We have proved

$$e_{A_{11}} > 13/20. \tag{A11Urate}$$

The ordinary-rank-five remainder costs $> 5/14$, so $13/20 + 5/14 = 141/140 > 1$. The typed-seven plus ordinary A_3 alternative also closes directly: its distinct whole blocks cost $> .09$ and $> .26$. The actual-$I + T_4$ case needs the joint argument that follows.

C.10.9 Actual I: six time-denominator obstructions

Suppose an ACTUAL isolated short I spectator has opposite

$$c_I = \sqrt{h}\, p - \sqrt{1 + h}\, e_I, \qquad 0 < h < 1/2.$$

The common $N = 12$ local inventory above was proved before assuming this wall exists. Its only nonisolated weak exceptions are exactly the three golden times in (A11gold) and the three silver times in (A11silver). We now exclude ALL six by actual cycles.

For completeness write $A_c^2 = M/L$. Their fields, denominators and numerator norms are

type	K	M/L	$\mathrm{N}(M)$
5, near	$\mathbb{Q}(\varphi)$	$(22 + 49\varphi)/24$	-839
6	$\mathbb{Q}(\varphi)$	$(23 + 48\varphi)/24$	-671
5, far	$\mathbb{Q}(\varphi)$	$(22 + 45\varphi)/24$	-551
$(5,4)$	$\mathbb{Q}(\sqrt{2})$	$(39 + 28\sqrt{2})/12$	-47
$(6,5)$	$\mathbb{Q}(\sqrt{2})$	$(83 + 60\sqrt{2})/24$	-311
$(5,6)$	$\mathbb{Q}(\sqrt{2})$	$(41 + 30\sqrt{2})/12$	-119.

The golden numerators follow from

$$M_N = i(12-i) - 13 + (i+2)(12-i)\varphi, \quad M_F = i(12-i) - 13 + i(14-i)\varphi;$$

the silver formula is $A_c^2 = D_i/2 + C_{ij}/\sqrt{2} + D_j/4 - 1$. The actual fields equal the displayed quadratic fields by the connected Lorentzian-principal argument already proved. The rings of integers are respectively $\mathbb{Z}[\varphi]$, $\mathbb{Z}[\sqrt{2}]$. In particular this is not a field assumption in the remaining no-exception branch.

The own two-cycle of c_I makes $v = 4h$ integral. Strict positivity of its two-root Gram at the other embedding gives $-4 < v' < 0$; physically $0 < v < 2$. Thus

$$0 < |\mathrm{N}(v)| < 8.$$

Every listed cap is PURE A_{11}, whereas c_I has spatial part only in I. Their product is $-A_c\sqrt{h}$. The scale-invariant actual two-cycle therefore makes $A_c^2 v = Mv/L$ integral. Each displayed $\mathrm{N}(M)$ is coprime to 6, while L is 12 or 24. No principal-ideal hypothesis is needed: multiply by the conjugate of M and use an integer Bezout relation for $\mathrm{N}(M)$ and L. It follows that $v \in L\mathcal{O}_K$. Its nonzero norm is then divisible by $L^2 \geq 144$, contradicting the bound eight. All six types and their reflected orientations are excluded. Consequently every paid central site is strong or has its actual directional foot (A11foot), with no restriction on the actual field in this remaining branch.

C.10.10 One actual endpoint and a joint $A_{11} + I$ square

The rank-free actual-I edge parameters are

$$u^2 = \frac{\sqrt{1+h}-1}{\sqrt{1+h}+1}, \qquad t = \sqrt{1+4h} - 2\sqrt{h}, \qquad 0 < u < 1/3.$$

The original I edge gives $E_I \geq u^2$. Let $d = A_d p - v_A - \beta e_I - v_{\mathrm{other}}$ be the ACTUAL opposite on the first A_{11} edge. It differs from c_I, which stays strictly negative on the entire outgoing A_{11} ray. Acuteness gives

$$-d \cdot c_I = A_d\sqrt{h} - \beta\sqrt{1+h} \geq 0.$$

If $\beta = 0$, this positive actual-wall mutual magnitude is at least $1/2$ by the native gap. If $\beta > 0$, its finite contact with the retained old I root has $\beta \geq 1/2$, so $A_d\sqrt{h} > 1/2$. Thus $A_d\sqrt{h} \geq 1/2$ in every white, own and foreign role. For the actual inward unit first-edge direction $U_1 = \Omega_1/\sqrt{11/6}$, the ENTIRE spatial vector of d has norm $\sqrt{1 + A_d^2}$. Therefore

$$\tanh L = \frac{A_d}{v_{\mathrm{total}} \cdot U_1} \geq \frac{A_d}{\sqrt{1+A_d^2}} \geq \frac{1}{\sqrt{1+4h}}.$$

Taking the increasing half-tangent yields

$$x_A \cdot U_1 \geq t, \qquad x_1 \geq \alpha t, \quad \alpha = \sqrt{11/12}. \tag{A11Iend}$$

Only physical unit-normal geometry is used; no disconnected raw-root scales are identified.

The exact relation between u, t is

$$(1-u^2)(1-t^2) = 8ut.$$

It implies $t + (3/2)u > 3/4$. Indeed substituting $t_0 = 3/4 - 3u/2 > 0$ into left-minus-right and multiplying by sixteen gives

$$7 - 60u + 149u^2 - 36u^3 + 36u^4 > 7 - 60u + 137u^2 > 0.$$

The last quadratic has discriminant -236, and the relation decreases strictly in positive t. Also $t < 1$ and $t > \sqrt{3} - \sqrt{2} > .3$. These estimates hold throughout the full interval $0 < h < 1/2$.

Keep the EXACT central heights $H_5 = H_7 = 1.805, H_6 = 1.839$ and $E = .652761$. Put

$$b = \frac{361}{1000\sqrt{2}}, \quad c = \frac{.034}{\sqrt{2}}, \quad \eta = \alpha t, \quad r = (5b - \eta)/4,$$

and choose the NEW zero-sum reference

$$w = (\eta, r, r, r, r, c, -c, -b, -b, -b, -b, -b).$$

Its first-five sum is still $5b$, so its central dual heights are the three fixed H_i, while its first height $\sqrt{2}\eta$ is paid by the ACTUAL endpoint (A11Iend). The safe rational bounds are

$$.95 < \alpha < .958, \quad .255 < b < .256, \quad \eta > b, \quad \eta < .958, \quad r > .07925, \quad c < .025.$$

All follow by positive squares and $t > .3$. Hence $\eta > r > c > 0$, so all dual coefficients are nonnegative. The ONLY positive ones are

$$\beta_1 = (\eta - r)/\sqrt{2}, \quad \beta_5 = (r - c)/\sqrt{2} > 217/6000 > .034, \quad \beta_6 = .034, \quad \beta_7 = .1635.$$

In particular β_5 is the coefficient of this NEW reference, including its full endpoint/foot cross.

Let S be ANY set of retained feet among $5, 6, 7$ and form $Z = w - \sum_{i \in S} \beta_i s_i n_i$. Every new central coefficient is at least .034. The continuous adjusted-loss bound proved above therefore applies afresh:

$$L_i - \frac{\beta_i}{2}(2s_i - s_i^2) < .61.$$

Combining all foot pressures, strict strong central heights and the actual endpoint gives

$$x_A \cdot Z \geq w^2 - \sum_{i \in S} \beta_i (L_i - .61),$$
$$Z^2 \leq w^2 - \sum_{i \in S} \beta_i^2 (2s_i - s_i^2).$$

Every adjacent-foot cross is nonpositive. The single completed square proves $e_A > w^2$, including the empty-foot case by strict central floors. At the REFERENCE level, the change from the constant first-five block is $(\eta - b, -(\eta - b)/4, \ldots, -(\eta - b)/4)$; it has zero sum and zero cross with that old constant block. Thus exactly

$$w^2 = E + (5/4)(\eta - b)^2.$$

This identity is not added to an independently optimized old-foot norm.

Write $D = \eta - b > 0$. The exact I barrier and $\alpha < 1$ give

$$D + (3/2)u > (3/4)\alpha - b > 913/2000.$$

Weighted Cauchy gives

$$(D + 3u/2)^2 \leq \{(5/4)D^2 + u^2\}(4/5 + 9/4) = \frac{61}{20}\{(5/4)D^2 + u^2\}.$$

Since the ACTUAL I energy is counted ONCE, on the original vector,

$$e_{A_{11}} + E_I > \frac{652761}{1000000} + \frac{833569}{12200000} = \frac{43986266}{61000000} > \frac{18}{25}. \tag{A11Irate}$$

The positive final margin is 66266/61000000. Every ordinary $T_4 = A_4, B_4, D_4, F_4, H_4$ costs $> 2/7$, so $18/25 + 2/7 = 176/175 > 1$. This closes the entire actual-$I + T_4$ alternative with arbitrary other spectators and every actual field.

C.10.11 The fresh $N = 11$ window and its small-side repairs

The remaining rank-ten chain cases require a different finite-white list. At A_{10}, use zero-sum $\mathbb{R}^{11}$,

$$D_i = \frac{2i(11-i)}{11}, \qquad C_{ij} = \frac{2i(11-j)}{11} \quad (i \le j),$$

and ONLY sites $4, 5, 6, 7$. Here $D_i = 56/11$ or $60/11$, so $9/4 < q_i = \sqrt{D_i} < 12/5$. Write an actual cap as $c = A_c p - v_A - v_S$, with full spectator cost $Q = \|v_S\|^2 < 1$. Each positive whole-component contribution is at least $1/4$. The actual reciprocal-opposite law gives the stronger individual restriction

$$Q_C < A_c^2 \tag{A10foreign}$$

for each contacted whole spectator C. If $Q_C > A_c^2$, the original finite replacement on its contacted node has positive Schur residual $Q_C - A_c^2$. Positive finite projection checks EVERY original wall and would make c close two orthogonal components at the original farthest vertex. Equality instead gives the positive-cleared future null. Neither is possible. The cleared projection is never treated as an additional native facet normal.

Take loss $\ell = 13/20 = .65$. The sufficient ratio

$$T(q) = \frac{20}{13} - \frac{q - 13/20}{q(2q - 13/20)}$$

increases on the stated q-interval. Indeed its derivative has positive numerator $2q^2 - (13/5)q + 169/400$. The exact upper comparison is

$$T(q) < \frac{17645}{12948} < \frac{1363}{1000},$$

with cleared positive gap 3124 in the last cross product. Thus ratio at least 1.363 is strong.

Deleting a selected site leaves path ORDERS in $\{4, 5, 6, 7\}$. The COMPLETE finite one-node extensions are: isolation; one admissible endpoint/D/E white-three; terminal white-four; terminal white-five on an A_3, giving H_4; one endpoint-three in each path, giving A_{10}; or, at site four or its reflection seven, a middle-A_3/end-three join, giving D_{10}. This last role is a genuine new small-side case. Two contacts to one retained path make a cycle. Rank ten precludes exceptional joined completions; a weighted B_{10} join would need a rank-one end side, which is absent. An A_3 has no missing one-node F_4 extension with these unweighted old bonds.

At own zero, isolation has $A_c^2 = Q - 1 < 0$. Internal three has the affine E_8 arms $1, 2, 5$, or E_7 arms $1, 3, 3$; internal four has the weighted $4, 3, 3$ star; either two-three join has its actual all-three cycle. The positive E_7 kernel has central coefficient four, short arm two and the two other arms $3, 2, 1$. The E_8 kernel has center six and arms 3, $4, 2$, $5, 4, 3, 2, 1$. The weighted star has center $\sqrt{2}$, four-leaf one and three-leaves $1/\sqrt{2}$; cycle coefficients are all one. Every kernel is future through the positive cap time and nonpositive against all unselected ORIGINAL walls.

The residual single-contact rows are

A_c^2	strict finite-white bound
$Q-6/11$	$Q<5/8$
$Q-2/11$	$Q<1/2$
$Q-1/11$	$Q<1/4.$

In the first, positive time is $<7/88<1/4$, contradicting (A10foreign). The second permits at most one whole foreign contribution, larger than its time. The third has $Q=0$ and negative time. The white-five/own-zero case will instead be directly strong below.

Isolated owns three, four and six have the actual E_7, weighted-star or 6–3 null, BEFORE splitting Q. A parallel actual pair has its positive future null in every profile. For SINGLE white-three, own-three closes the actual all-three cycle; own-six uses an old neighbor different from the helper. Own-four has its star unless that helper is adjacent to the own site. The adjacent near-four role is retained in the next calculation. All other positive owns have $\gamma\ge\varphi/2>4/5$.

For a white-three at position t in a side of order l, discarding foreign cost only decreases the ratio, giving

$$\frac{\gamma+t/(2l)}{\sqrt{1-t(l-t)/(2l)}}.$$

Near-three exceeds $(4/5+3/8)/\sqrt{5/8}>1.363$. The D-position exceeds $(4/5+1/4)/\sqrt{1/2}>1.363$; its formula increases with $l\ge4$. Allowed E-positions have denominator at most $1/2$, hence ratio >1.6. Near own-four/simple-three has minimum at $l=4$, with

$$R_4^2=41/40+3\sqrt{2}/5>1.8734>1.363^2.$$

Its numerator increases and denominator decreases with l, so ALL these roles are strong without using a quadratic-field exclusion.

White-four has ratio $\ge\gamma\sqrt{l}$. For own at least four this is $\ge\sqrt{2}$. Near white-four/own-three already has ratio >2 at $l=4$. Far white-four/own-three has $\sqrt{l}/2+1/\sqrt{2l}$, increasing for $l\ge4$. At $l=5$ its square is $5/4+1/\sqrt{2}+1/10>1.363^2$. Only the far role at $i=4$, or its reflection seven, remains. It has $Q=0$ and

$$v_A=\Omega_4/2+\Omega_1/\sqrt{2},\qquad A_c^2=(13+7\sqrt{2})/11.$$

The actual own-three normalization and closed cycle put $\sqrt{2}$ in the field. At any embedding extending its sign reversal, the connected principal still has positive time square $(13-7\sqrt{2})/11>0$. It therefore retains a negative direction, contradicting strict nonphysical positivity. This argument needs only the contained quadratic field, not equality with it. It is the required small-side repair; the unavailable longer E_7 arm is not used to discard this role.

For white-five on A_3, put $\beta=\varphi/2>4/5$. Its white residual is $\delta_H=(7-3\sqrt{5})/16<1/50$. Even own zero has ratio $\beta/(4\sqrt{\delta_H})$, whose square exceeds two. All such profiles are strong. The A_{10} two-end-three join has ratio $\ge\gamma q_i+1/q_i>113/72$ for positive own. The D_{10} join has white cost $1/2+3/7=13/14$, so $Q=0$. Its two offsets are $19/28, 9/28$; the positive-own ratio is at least $23\sqrt{14}/28>3$. All their own-zero cycles were handled above.

Finally a far-three helper WITH foreign cost has residual at most $3/8$ and numerator $>4/5+1/14=61/70$. Its ratio square exceeds $29768/14700>1.363^2$. Every possible weak NONISOLATED cap is therefore a PURE end-helper

$$v_A=\gamma\Omega_i+\Omega_e/2,\qquad e=1\text{ or }10. \tag{A10helper}$$

This conclusion is independent of both the actual-I and typed-seven hypotheses.

C.10.12 A small conjugate polynomial replaces the degree census

For (A10helper), the exact side-order square identity gives $4\gamma - 2 \le R^2$. Weakness and $T(q) < 1.363 < 1.37$ imply

$$\varphi \le t = 2\gamma < 38769/20000 < 39/20.$$

Actual simple-three normalization makes t integral. The old roots and cap have a common totally positive scale times the norm-two matrix with helper product -1 and own product $-t$. This full-rank connected principal is physically Lorentzian. Hence a nonphysical embedding fixing $\mathbb{Q}(t)$ would preserve a negative direction: ACTUAL $K = \mathbb{Q}(t)$.

At every other embedding the exact Schur inequality is

$$D_i t_\sigma^2 + 2C_{ie} t_\sigma + 20/11 < 4.$$

At $i = 4$, its cheapest signed costs are $(56s^2 + 28s + 20)/11$ for negative s, and $(56s^2 + 16s + 20)/11$ for positive s. At five the increases are $4(s^2 - s)/11$, $4(s^2 + s)/11$, respectively. They are positive in the stated half-windows. Boundary excesses $456/6875$ at $-24/25$ and $263/13750$ at $53/100$ imply

$$-24/25 < t_\sigma < 53/100.$$

Let $f(T) = T(T^2 - 1)$. The physical $f(t)$ is nonzero integral, with $0 < f(t) < 43719/8000$. On the ENTIRE small-conjugate interval,

$$|f(s)| \le \frac{2}{3\sqrt{3}} < \frac{77}{200};$$

the last comparison squares to $160000 < 160083$. Degree at least three would give a nonzero integral norm of magnitude

$$< \frac{43719}{8000}\left(\frac{77}{200}\right)^2 = \frac{259209951}{320000000} < 1,$$

and every further factor is smaller than one. The degree is at most two. Degree one has no integer in $[\varphi, 39/20)$. In degree two the nonzero norm has magnitude $< 1.95(.96) < 2$, hence is 1 or -1. A positive small conjugate gives integral trace $t + 1/t \in (2, 3)$, impossible. A negative one gives trace $t - 1/t \in [1, 2)$, hence exactly one. Thus

$$t = \varphi, \qquad K = \mathbb{Q}(\varphi). \tag{A10gold}$$

This is a complete all-degree argument. No field census or assumed cubic bound is used.

C.10.13 Every isolated A_{10} cap is strong

This part also needs neither spectator hypothesis. Keep $u = (1 - Q)/\gamma^2 > 0$. The small-own/null cases were excluded for all Q, so $\gamma \ge \varphi/2$. For $0 < u < 1$, the positive ACTUAL vector

$$\xi = c + v_S + \frac{1 - Q}{\gamma} n_i$$

has square $-(1 - Q)(1 - u) < 0$, positive time, zero selected products and negative product on n_i. The entire spectator correction is positive, so every original omitted wall is nonpositive. The original farthest comparison gives

$$h_i > \sqrt{D_i - u} - \sqrt{u(1 - u)}.$$

Here

$$a_i = \frac{1}{q_i + \sqrt{D_i - 1}} < \frac{100}{427} < \frac{47}{200},$$

since $q_i > 9/4$ and $\sqrt{D_i - 1} > 101/50$. The full continuous loss is

$$< \frac{.235 + \sqrt{1 + .235^2}}{2} < \frac{.235 + 1.028}{2} = .6315 < .65.$$

Equality $u = 1$ is the feasible future null. For $Q > 0, u > 1$, positive finite clearing has loss $< a_i + \sqrt{u - 1}$. Since $Q \geq 1/4$ and $\gamma \geq \varphi/2$, $u \leq 3/\varphi^2$, hence $\sqrt{u - 1} \leq 2 - \varphi < .4$. The loss is $< .635 < .65$.

For PURE native five the actual finite H_4 point has loss

$$< \frac{25/16}{q_i + \sqrt{D_i - 25/16}} + 5/24 < 625/1648 + 5/24 < .588.$$

The denominator is $> 2.25 + 1.87 = 4.12$. It is the positive finite projected point, not its spacelike correction, that checks all original walls. For EVERY finite native at least eight the finite two-root projected point gives, with $B = \sqrt{D_i - 1}$, $v = \sqrt{u - 1}$,

$$h_i > \frac{\sqrt{B^2 - v^2}}{1 + v/B} > B - v.$$

Its bracket is positive, and $v \leq \tan(\pi/8) = \sqrt{2} - 1 < .415$. Thus the full loss is $< .235 + .415 = .65$, covering the entire infinite native tail.

For native seven, the elementary cubic $8\gamma^3 - 4\gamma^2 - 4\gamma + 1$ and $\pi < 22/7$ give $.9 < \gamma < 10/11$. Consequently $121/100 < u < 100/81 < 4/3$. For an adjacent old root n_j, the positive actual core

$$\xi = c + \gamma u n_i + 2\gamma(u - 1)n_j$$

is zero on c, n_i, negative on n_j, and has square $-\gamma^2(7u - 4 - 3u^2) < 0$. It is future and nonpositive against all other ORIGINAL walls. Its pressure gives

$$h_i > \sqrt{D_i - u} - \sqrt{7u - 4 - 3u^2}.$$

The first loss is $< .294$, using $(100/81)/4.21 < .294$, $q_i > 2.25$, and $\sqrt{D_i - 100/81} > 1.96$. The second radicand decreases for $u > 7/6$; at $121/100$ it is $.0777 < .28^2$. Thus the total loss is $< .574 < .65$. All isolated finite, foreign and dotted profiles are strong; no foot is left unpaid.

C.10.14 The actual-I and typed-seven consumers of the same window

In the actual-I case, only the golden helpers (A10gold) still need to be excluded. The fresh COMPLETE time table, with reflected indices covering six and seven, is

i	e	$11A_c^2$	$\mathrm{N}(11A_c^2)$
4	1	$8 + 21\varphi$	-209
4	10	$8 + 18\varphi$	-116
5	1	$9 + 21\varphi$	-171
5	10	$9 + 20\varphi$	$-139.$

These are direct substitutions in

$$A_c^2 = (\varphi^2/4)D_i + (\varphi/2)C_{ie} + D_e/4 - 1, \qquad D_e = 20/11.$$

The first numerator norm has exactly one factor eleven; none of the others has one. For the ACTUAL isolated I, its own two-cycle gives $V = 4h \in \mathcal{O}_K$, $0 < V < 2$, and $-4 < V' < 0$. We are now in the actual quadratic field $K = \mathbb{Q}(\varphi)$, so $0 < |\mathrm{N}(V)| < 8$. The helper and I cap have disjoint spatial supports. Their actual two-cycle makes $A_c^2 V$ integral, hence $\mathrm{N}(A_c^2)\mathrm{N}(V)$ integral. Every displayed row forces eleven to divide $\mathrm{N}(V)$: its denominators are eleven or eleven squared, with numerators coprime to eleven after cancelling the one factor in the first row. This contradicts the nonzero norm bound eight. No relative primitive scale, PID assertion or extra I-energy charge is involved.

For EITHER actual typed-seven defect, the same generic local geometry and all-degree helper proof remain valid. The previously proved ACTUAL $[7, 3]$ principal instead gives $K = K_7$, a cubic field with no quadratic subfield. Thus (A10gold) is impossible in this branch as well. This reuses the complete generic window; it does not transport the $N = 12$ inventory or infer arithmeticity of an arbitrary face. It also avoids a duplicated cubic-only trace argument for the same already excluded helper domain.

In both cases every site $4, \dots, 7$ is genuinely .65-strong. Take symmetric heights, linear on the tails,

$$H_4 = H_7 = 8/5, \qquad H_5 = H_6 = 42/25.$$

They lie strictly below the proved floors, because $(9/4)^2 < 56/11$ and $(233/100)^2 < 60/11$, with positive margins $5/176, 2821/110000$. The only positive dual coefficients are $\beta_4 = \beta_7 = 4/25$, $\beta_5 = \beta_6 = 1/25$. Every one is paid by an actual strong height. One square on the original vector yields

$$e_{A_{10}} > (8/5)^2/4 + (42/25 - 8/5)^2 = \frac{404}{625}. \tag{A10rate}$$

This rate has the stated actual-I OR actual typed-seven hypothesis; it is not asserted for a bare arbitrary-field A_{10}.

For the actual-$I + (A_5$ or $D_5)$ alternative, add only the ordinary whole rank-five component: $404/625 + 5/14 = 8781/8750 > 1$. The positive I energy is not needed. For actual typed-seven plus ordinary T_4, count the defect once and use

$$404/625 + 9/100 + 2/7 = \frac{17887}{17500} > 1.$$

These are sums of disjoint whole-component energies of the SAME original x. They exclude both entire rank-sixteen A_{10} alternatives, without a field census or a newly executed finite calculation.

C.11 Chain consumers with typed or ordinary spectators

C.11.1 The two actual typed-seven A_{14} forests

Here the WHOLE incident forest is A_{14} plus one actual rank-two defect, with no further spectators. Put $c = \cos(\pi/7)$, $\theta = 2c$. The actual $[7, 3]$ principal proves $K = K_7 = \mathbb{Q}(\theta)$. Its full integer ring and root brackets are those proved in Section C.10.3. The defect wall is isolated from the ENTIRE old A_{14}, and has the form

	$d = \sqrt{h_D}\, p - v_D$	h_D
N_7	$v_D = c\,\Omega_e^{A_2}$	$(4/3)c^2 - 1$
O_7	$v_D = \Omega_e^{I_2(7)}/2$	$1/(4(1-c^2)) - 1.$

Both defects have disjoint energy $> .09$. We first strengthen the central floors, then use this ACTUAL wall to pay both ends.

At $N = 15$, use sites $5, \dots, 10$ and fresh loss $\ell = .55$. Their dual squares are $20/3, 36/5, 112/15$, in reflected order, with $5/2 < q < 11/4$. The increasing sufficient ratio obeys

$$R_\ell(q) < R_\ell(11/4) = 164/99 < 5/3.$$

The full finite-white list remains isolation, admissible endpoint/D/E three, endpoint four, or two endpoint threes joining paths. The retained paths have ranks at least four. The target-independent own-zero residuals are $Q - 8/15, Q - 2/15, Q - 1/15$, with bounds $Q < 3/5, 2/5, 1/5$. The first positive time is below any whole foreign contribution; the second permits at most one contribution greater than its time; the last permits none. The actual reciprocal law excludes those positive cases; at $Q = 0$ times are negative. Internal-three/four and two-three roles retain their all-wall positive affine kernels.

The actual field K_7 has no quadratic subfield. Native five is absent by its two-cycle; own-four/simple-three is absent by its UNSQUARED connected $\sqrt{2}$ cycle. All appropriate smaller owns and parallel retain their all-foreign actual nulls. For the other ordinary roles, the FRESH lower ratios are

role	lower ratio
near-three	$13/\sqrt{60} > 5/3$
D-position	$11/\sqrt{40} > 5/3$
E-position	1.8
far-four	$.9\sqrt{5}$
near white-four	$53\sqrt{5}/50$
two end-threes	$q/2 + 1/q \geq 13/\sqrt{60} > 5/3.$

These use $\gamma > .9$, side order $m \geq 5$, white-three residual $1 - j(m-j)/(2m)$ and offset $j/(2m)$; white-four has residual $1/m$, offset $j/(\sqrt{2}m)$. The last comparison uses the ACTUAL minimum $q^2 = 20/3$, not merely $q > 5/2$. Positive foreign cost strengthens all these ratios. For a foreign far-three helper complete clearing gives

$$\frac{1.8m + 1}{\sqrt{m(m+2)}} \geq 10/\sqrt{35} > 5/3,$$

increasing for $m \geq 5$. Thus a weak nonisolated cap is PURE end-helper with $\gamma > .9$.

Put $t = 2\gamma$. Its exact square-ratio identity gives

$$9/5 < t < 23249/9801 < 19/8.$$

It is integral in the ACTUAL cubic field. The signed Schur costs are

$$2it^2 + 4t + 2 - 2(it+1)^2/15, \qquad 2it^2 + 2 - 2(it-1)^2/15.$$

At $i = 5$, their values at $-4/5$, $12/25$ are four and $4 + 16/375$. They increase across the half-window, so both other conjugates lie in $(-.8, .48)$. In degree three the nonzero norm has magnitude $< (19/8)(.8)^2 < 2$. A positive small conjugate would make it $< (19/8)(.48)(.8) = .912 < 1$. Both are negative with absolute product $1/t$; the integral trace is one, being $> t - 1.6 > .2$ and $\leq t - 2/\sqrt{t} < 2$. Its second coefficient is

$$B = t - t^2 + 1/t, \qquad -3459/1216 < B < -199/225.$$

The function decreases, so $B = -2$ or -1. The candidate cubics $X^3 - X^2 - 2X - 1$, $X^3 - X^2 - X - 1$ have discriminants $-31, -44$, contradicting total reality. Degree one leaves only $t = 2$, its actual parallel null. No weak helper remains.

We now prove directly that no $w \in \mathcal{O}_{K_7}$ satisfies

$$0 < w < 11, \qquad 0 < w_\alpha, w_\beta < 3/5. \tag{A14strip}$$

Use the proved FULL basis $w = a + b\theta + c_0\theta^2$, with roots

$$1.8019 < \theta < 1.802, \quad .445 < \alpha < .446, \quad -1.248 < \beta < -1.246.$$

Set $d_0 = b - (\theta - 1)c_0$ and $\Delta = w - (w_\alpha + w_\beta)/2$. The exact identity is

$$\Delta = A_0 d_0 + B_0 c_0, \quad A_0 = (3\theta - 1)/2 < 2.203, \quad B_0 = 3\theta^2 - 2\theta - 2 > 4.136.$$

The small-value difference gives $|d_0| < .6/(\alpha - \beta) < .355$, hence $|A_0 d_0| < .782065$, while $-.6 < \Delta < 11$. Values $c_0 \le -1$ force $\Delta < -3.353935$; $c_0 \ge 3$ forces $\Delta > 11.625935$. Only $c_0 = 0, 1, 2$ remain. At zero, $b = 0$, then a would be an integer in $(0, .6)$. At two, $1.2488 < b < 1.959$, impossible. At one, $.4469 < b < 1.157$, so $b = 1$. But $.643025 < \alpha + \alpha^2 < .644916$; positivity forces $a \ge 0$, contradicting $w_\alpha < .6$. This proves the entire strip.

For ANY isolated-within-A cap retain its entire foreign cost and put $w = 4\gamma^2$. The actual own two-cycle makes w nonzero integral. Normalize its original root to own raw product -1, square $2/w$. The old inverse entry $D_i/2$ gives

$$0 < w_\sigma < 4/D_i \le 3/5$$

at BOTH other embeddings, even with foreign contacts. Weakness would imply

$$0 < w < 4(164/99)^2 = 107584/9801 < 11,$$

contradicting (A14strip). Every site $5, \ldots, 10$ is therefore genuinely .55-strong, with no helper or weak foot left to omit.

C.11.2 The actual typed wall forces both A_{14} ends

The increasing cubic $8c^3 - 4c^2 - 4c + 1$ has values $-.000569738168, .000257608$ at $.9009, .901$. Thus

$$.9009 < c < .901, \quad .08 < h_N < .083, \quad .32 < h_O < .3284 = 821/2500.$$

Substitution into the displayed height formulas gives these rational seals. The shared exact gap is

$$\psi = \frac{15(3c^2 - 2)}{28(4c^2 - 3)}, \qquad c^2 < \psi < (1 + c)/2 < 1. \tag{A14gap}$$

For the lower bound, $\psi = (1 + 1/(4h_O))/(28/15) > 21690/22988 > .9 > c^2$. For the upper, clearing positive denominators and using the cubic reduces to $39c^2 - 14c - 19 > 0$, increasing for $c > .9$ and equal to $.04061159 > 0$ at $.9009$. In an N_7 branch, h_O in this identity is only its scalar function of c.

The COMPLETE finite actual own-square alphabet in K_7 is

$$\gamma^2 \in \{0, 1/4, 1/2, 3/4, c^2, (1 + c)/2\}.$$

Cyclotomic degree first permits $2, 3, 4, 6, 7, 9, 14, 18$. The 9/18 cubic field differs: $2\cos(\pi/9)$ has polynomial $X^3 - 3X - 1$, whose shift by one is Eisenstein at three, whereas K_7 has discriminant 49. If γ itself is K_7-valued, its finite list further reduces to $0, 1/2, c$; $\cos(\pi/14)$ has degree six.

For EITHER actual endpoint cap $c_A = A_c p - v_A - v'_D$, let $\delta > 0$ be its Schur residual against ALL retained old roots. The actual edge gives

$$C_{\text{end}}^2 = 1 + A_c^2/\delta, \quad 0 < \delta \le 1, \qquad y \ge \sqrt{14/15}\sqrt{\frac{C_{\text{end}} - 1}{C_{\text{end}} + 1}}. \tag{A14end}$$

The two signed coordinates are $x_{A,1}$, $-x_{A,15}$. This cap differs from the actual defect wall d, which cannot close its edge. The two endpoint caps need not differ from each other.

For O_7, positive retained $I_2(7)$ contact would cost at least $\csc^2(\pi/7)/4 > 1$, impossible. Thus $v'_D = 0$, and the positive ACTUAL mutual magnitude $\lambda = -c_A \cdot d = A_c\sqrt{h_O}$ is at least $1/2$. We exclude equality in every endpoint role. An isolated cap has $A_c^2 = (28/15)\gamma^2 - 1$; equality forces $\gamma^2 = \psi$. The finite alphabet, parallel one and dotted $\gamma > 1$ all avoid (A14gap).

For a nonisolated endpoint, retained A_{13} has the COMPLETE finite list: helper-three at $j = 2, 3, 13, 14$, or helper-four at $j = 2, 14$. The exact fresh time polynomials are

$(j, \text{helper label})$	$15A_c^2$
$(2, 3)$	$28\gamma^2 + 26\gamma - 2$
$(3, 3)$	$28\gamma^2 + 24\gamma + 3$
$(13, 3)$	$28\gamma^2 + 4\gamma - 2$
$(14, 3)$	$28\gamma^2 + 2\gamma - 8$
$(2, 4)$	$28\gamma^2 + 26\sqrt{2}\gamma + 11$
$(14, 4)$	$28\gamma^2 + 2\sqrt{2}\gamma - 1.$

(A14endtimes)

Simple-three normalization makes γ itself K_7-valued. At own zero only $j = 3$ is future, with time $1/5$, giving $\lambda^2 < 1/4$. At own $1/2$ the four times are $6/5, 22/15, 7/15, 0$. The first two have mutual square $> 1/4$, the third $< 1/4$, and the last is not future. For every $\gamma \ge c > .9$ all four polynomials increase and their minimum exceeds $16.48/15 > 1$. Thus their mutual squares exceed $1/4$. This covers every remaining finite and every dotted own; parallel has its actual null.

For helper-four, connected normalization gives $2\sqrt{2}\gamma \in K_7$. At finite own labels $3, 6, 7, 14$, the norms of $2w$, $w = 4\gamma^2$, are $8, 216, 8, 56$, all nonsquares. Only own four remains, besides zero, parallel or dotted. At zero, near time $11/15$ has mutual square $< 1/4$, and far time is negative. At own four, times $17/5, 1$ have mutual squares $> 1/4$. At $\gamma \ge 1$, both times exceed one. Hence equality $\lambda = 1/2$ is impossible in EVERY role. The next positive native magnitude is at least $1/\sqrt{2}$, so

$$C_{\text{end}}^2 \ge 1 + 1/(2h_O) > 2071/821 > (397/250)^2.$$

Equation (A14end) gives

$$y^2 > \frac{2058}{9705} > (23/50)^2, \qquad y > .46.$$

The two cleared rational margins are $40511, 11055$. Only connected products within A_{14} were used to normalize helpers; no equal scale of its disconnected defect is assumed.

For N_7, retain the ENTIRE foreign old A_2 block and write

$$\lambda = A_c\sqrt{h_N} - \rho \ge 0, \qquad \rho = v'_D \cdot v_D.$$

At $Q = 0$, the native gap gives $A_c^2 > 250/83$, hence $C_{\text{end}} > 2$. Positive foreign contact has the COMPLETE finite list

label	Q	ρ
3	$1/3$	$c/3$ or $2c/3$
4	$2/3$	$\sqrt{2}c/3$ or $2\sqrt{2}c/3$.

Two contacts make a cycle; six is affine; larger labels are nonfinite, and five is forbidden in K_7. If an A_{13} contact is also present, its full inverse cost is at least $13/28$. Thus $\delta \le 17/84$. Acuteness gives $A_c\sqrt{h_N} \ge \rho > .3$, so $A_c^2 > 90/83$ and $A_c^2/\delta > 7560/1411 > 3$. Without an A_{13} contact, $\delta = 1 - Q$. At $Q = 2/3$, $\rho > .3\sqrt{2}$ gives $A_c^2/\delta > 540/83 > 3$. At $Q = 1/3, \rho = 2c/3$, the same bound follows from $\rho > .6$. These arguments retain EVERY own parameter.

Only $Q = 1/3, \rho = c/3$, with no A_{13} contact, remains. Positive λ gives $A_c\sqrt{h_N} \ge c/3 + 1/2 > .8$ and $A_c^2/\delta > 960/83 > 3$. At $\lambda = 0$, the exact equation $A_c^2 = (28/15)\gamma^2 - 2/3$ instead gives $\gamma^2 = \psi$. Its ACTUAL own two-cycle puts $4\gamma^2$ in K_7; the finite alphabet, parallel and dotted ranges again avoid the gap. Thus every N_7 endpoint has $C_{\text{end}} > 2$, giving $y > \sqrt{14/45} > .55 > .46$. The two strict .46 signed demands therefore hold for BOTH actual defect types.

C.11.3 Both typed endpoints in one legal reference

All central sites are .55-strong. Choose symmetric heights, linear on the tails,

$$H_5 = H_{10} = a = 2.03198, \quad H_6 = H_9 = b = 2.13328, \quad H_7 = H_8 = c_0 = 2.18251.$$

The floor tests are $2.58198^2 < 20/3$, $2.68328^2 < 36/5$, $2.73251^2 < 112/15$. The six central coefficients are $.152548, .026035, .024615, .024615, .026035, .152548$. The base norm is

$$E = a^2/5 + (b-a)^2 + (c_0 - b)^2 = .83847382698.$$

The first and last five coordinates are flat with signed mean $b_0 = a/(5\sqrt{2})$, $.28 < b_0 < .2875$. In EACH block replace the signed endpoint by .46, and redistribute its remaining sum constantly over the other four coordinates. Their value $r = (5b_0 - .46)/4$ lies in $(.235, .245)$, below the endpoint and above the next old coordinate $< .073$. The NEW reference stays concave, with positive endpoint and changed boundary coefficients. Every central dual height is unchanged and genuinely paid. There are no omitted weak feet or helpers.

The two residuals have disjoint block supports, zero block sums and zero cross with the original constant blocks. Each reference gain is exactly $(5/4)(.46 - b_0)^2 > .0371953125$. All actual paid pressures are strict. One square therefore gives

$$e_{A_{14}} > .83847382698 + 2(.0371953125) = .91286445198.$$

Only NOW add the disjoint defect energy $> .09$ of the SAME original vector, counted ONCE. The full norm exceeds 1.00286445198, a contradiction. Both actual typed-seven A_{14} forests are excluded. No K_7 condition is imposed on the separate ordinary rank-two branch.

C.11.4 The ordinary $A_{14} + T_2$ branch has one actual three-wall core

Now the WHOLE forest is $A_{14} + S$, where $S = I_2(m)$, $m \ge 3$, is ORDINARY: neither actual typed defect is present. The universal bound (A14U) gives

$$0 < e_S < 1/5.$$

Let $k = \cos(\pi/m) \ge 1/2$, $d_0 = (1 - k^2)^{-1}$. The old unit Gram inverse is $d_0 \left(\begin{smallmatrix} 1 & k \\ k & 1 \end{smallmatrix}\right)$. The positive unit incident duals u_1, u_2 have correlation k. For their ACTUAL edges put $\tau_i = \tanh(L_i/2)$, $r_i = \tanh L_i$. Original farthest comparison gives $x \cdot u_i \ge \tau_i > 0$; hence $\tau_i^2 < 1/5$ and each actual edge cosine is $< 3/2$.

The two edges have ONE actual opposite. Indeed Cauchy on u_1+u_2 gives $\tau_1+\tau_2 \le \sqrt{2e_S(1+k)}$, so $\tau_1\tau_2 < (1+k)/10$. For $0 < a, b < 1$,

$$\frac{2a}{1+a^2}\frac{2b}{1+b^2} \le \frac{4ab}{(1+ab)^2}.$$

The right side increases in $ab < 1$; therefore

$$r_1 r_2 < \frac{40(1+k)}{(11+k)^2} < k.$$

The last cleared polynomial is $k^3 + 22k^2 + 81k - 40$, increasing for $k \ge 1/2$ and equal to $49/8 > 0$ at $1/2$. The actual two-wall law for distinct opposites requires $r_1 r_2 \ge k$. Thus one wall d closes BOTH directions, including the old A_2 case.

Write

$$d = A_d p - v_S - v_A, \quad h = A_d^2 > 0, \quad b = -d \cdot n_1, \quad g = -d \cdot n_2, \quad Q = \|v_A\|^2,$$

so

$$h = d_0(b^2 + g^2 + 2kbg) + Q - 1.$$

Every positive b, g is finite native, being in a retained positive two-root Gram at the other actual endpoint. The whole A_{14} is retained there, so $Q < 1$; a positive Q is at least $7/15$. Both $b = g = 0$ would give negative time. If both are positive and $Q > 0$, the actual edge formula gives

$$C_1^2 = \frac{d_0(b+kg)^2}{1-g^2-Q} \ge \frac{(4/3)(3/4)^2}{3/4 - 7/15} = \frac{45}{17} > \frac{9}{4},$$

contrary to $C_1 < 3/2$.

If precisely one contact is positive, write $b = \beta, g = 0$. At the other endpoint the connected finite rank-sixteen diagram is classical. Its only possible foreign attachment is a SINGLE old A_{14} END-three, with $\beta = 1/2$ or $1/\sqrt{2}$. An internal attachment would give arms of lengths two and two others totaling thirteen, which are neither D_{16} arms nor exceptional finite arms. An end-four would become an internal weighted bond; a five bond cannot occur. Thus $Q = 7/15$ and $h = d_0\beta^2 - 8/15$. The smaller normalized derivative square is $k^2 d_0 \beta^2$. Since $r_i^2 < 5/9$, we get

$$\beta^2(4d_0 + 5) \le 24/5.$$

At $\beta = 1/\sqrt{2}$, this forces $d_0 \le 23/20 < 4/3$, impossible. At $\beta = 1/2$, $d_0 \le 71/20 < 4$, so $m = 3, 4, 5$. The first two times are $-1/5, -1/30$. At five, $d_0 = 2 + 2/\sqrt{5} < 3$, giving $0 < h < 13/60 < 1/4$. The actual small-height foreign-isolation lemma then forbids this remaining contact: the same wall would close an A_{14} direction in addition to its S directions. Consequently

$$Q = 0. \tag{A14coreiso}$$

This is isolation of an ACTUAL wall from the entire old component, not face heredity.

We classify every native S-contact. With one contact the same derivative law now gives $\beta^2(4d_0+5) \le 9$. Thus $d_0 \le 31/4$. For $m \ge 9$, $\sin(\pi/m) < \pi/m$ and $\pi^2 < 10$ give $d_0 > 81/10 > 31/4$. Hence $m = 3, \ldots, 8$ is complete. If $\beta \ge 1/\sqrt{2}$, then $d_0 \le 13/4$, so $m \le 5$. At four, positive time and the inequality leave only new five. At five, new three has negative time, new four remains, and new five or larger violates $\beta^2(4d_0 + 5) \le 9$. At six only new three remains, with zero time. At seven/eight only new three remains; old seven/new three is ACTUAL O_7 and excluded by ordinary

meaning. At old three, positivity and the inequality give $3/4 < \beta^2 \le 27/31$. Only new seven or eight remains: for all native nine or larger, $\cos(\pi/9) > 76/81$ and $(76/81)^2 > 27/31$. Old three/new seven is ACTUAL N_7 and is excluded. Thus precisely four one-contact rows survive.

With two positive contacts, every old $m \ge 5$ has

$$C_i \ge \frac{\cot(\pi/(2m))}{\sqrt{3}} \ge \frac{\cot(\pi/10)}{\sqrt{3}} > \sqrt{3} > 3/2.$$

At old four, either contact of at least four gives $C_i \ge \sqrt{3}$; both must be three. At old three, both contacts at least four give the same contradiction. One is therefore three; if the other $\beta \ge \cos(\pi/5) > 4/5$, the opposite edge has $C^2 = (1+\beta)/(3(1-\beta)) > 3$. The remaining possibilities are three/three, with zero time, or three/four in either direction orientation. This completes the classification.

Put $\varphi = (1+\sqrt{5})/2$, $s = \sqrt{2}$. The SIX necessary old-role models are

S	d contacts in S	h	K
$I_2(4)$	one 5	$(\varphi-1)/2$	$\mathbb{Q}(\varphi)$
$I_2(5)$	one 4	$1/\sqrt{5}$	$\mathbb{Q}(\varphi)$
A_2	one 8	$(s-1)/3$	$\mathbb{Q}(s)$
$I_2(8)$	one 3	$s/2$	$\mathbb{Q}(s)$
A_2	3, 4 in either orientation	$s/3$	$\mathbb{Q}(s)$
$I_2(4)$	3, 3	$s/2$	$\mathbb{Q}(s)$.

(A14six)

They are necessary actual Gram types, never realized chambers.

We justify the ACTUAL field column. The native two-cycles put the quadratic field inside K in the tree types $(2,4,5)$, $(2,3,8)$. For $(3,3,4)$, the essential actual THREE-cycle is $8kbg = \sqrt{2}$. Choose the degree-two root as base in a tree, and any root with both products nonzero in the triangle. Let its original K-root square be the totally positive q_0. Rescale the two other original roots by ratios of K-valued products so their products with the base are $-q_0$. Their squares become $q_0/k^2, q_0/b^2$; the remaining product is zero or $-q_0 g/(kb)$, whose coefficient is in the displayed field F because $g/(kb) = kbg/(k^2b^2)$. Thus the three ACTUAL root lines have Gram $q_0 M$, M over F. It is Lorentzian by $Q = 0, h > 0$. A proper totally real extension K/F would have a nonphysical embedding fixing this negative direction while preserving the positive scalar. Strict nonphysical positivity forbids it. Hence $K = F$, without identifying raw scales across the old components.

C.11.5 Every ordinary-core A_{14} endpoint exceeds .44

Name the five distinct heights in (A14six)

$$h_a = (\varphi-1)/2, \quad h_b = 1/\sqrt{5}, \quad h_c = (s-1)/3, \quad h_d = s/2, \quad h_e = s/3.$$

Every one is positive and at most $s/2$. For an actual endpoint cap $c = A_c p - v_A - v_F$, let $\gamma = -c \cdot n_1 \ge 0$, $Q = \|v_F\|^2$, and $\delta > 0$ be the full retained-white Schur residual. Its actual edge obeys (A14end). It suffices to prove $C_{\text{end}} > 31/20$, because then its signed coordinate satisfies

$$y^2 > \frac{154}{765} > .44^2.$$

This actual wall differs from the common spectator wall d, which is orthogonal to all A_{14} directions. Their mutual magnitude is

$$\lambda = -c \cdot d = A_c\sqrt{h} - \rho \ge 0, \qquad \rho = v_F \cdot v_S \ge 0. \tag{A14mut}$$

A positive λ is at least $1/2$. No individual unit product is declared K-valued: its invariant square is. The finite positive mutual labels are

K	finite positive labels
$\mathbb{Q}(\varphi)$	$3,4,5,6,10$
$\mathbb{Q}(s)$	$3,4,6,8.$

Indeed cyclotomic degree at most two first permits $3,4,5,6,8,10,12$; the $5/10, 8, 12$ fields are respectively $\mathbb{Q}(\sqrt{5}), \mathbb{Q}(\sqrt{2}), \mathbb{Q}(\sqrt{3})$. The full orders are $\mathbb{Z}[\varphi]$, $\mathbb{Z}[s]$.

First suppose c is PURE isolated: no retained A_{13} or S contact. Then $\delta = 1$, $A_c^2 = (28/15)\gamma^2 - 1$, and $\lambda = A_c\sqrt{h} > 0$. Its own two-cycle requires

$$4\gamma^2 = \frac{15}{7}(1 + \lambda^2/h) \in \mathcal{O}_K.$$

Here are ALL values $z = 1 + \lambda^2/h$ for the finite mutual alphabet:

	3	4	5	6	10
h_a	$(2+\varphi)/2$	$1+\varphi$	$(3+2\varphi)/2$	$(2+3\varphi)/2$	$(3+3\varphi)/2$
h_b	$(3+2\varphi)/4$	$(1+2\varphi)/2$	$(5+3\varphi)/4$	$(1+6\varphi)/4$	$(4+5\varphi)/4$

and

	3	4	6	8
h_c	$(7+3s)/4$	$(5+3s)/2$	$(13+9s)/4$	$(16+9s)/4$
h_d	$(4+s)/4$	$(2+s)/2$	$(4+3s)/4$	$(3+s)/2$
h_e	$(8+3s)/8$	$(4+3s)/4$	$(8+9s)/8$	$(7+3s)/4.$

Each dyadic numerator has a coefficient not divisible by seven. Multiplication by $15/7$ cannot make both full-order coefficients integral. Thus EVERY finite mutual is impossible. Retain parallel/dotted mutuals by the weaker conclusion $\lambda \geq 1$. Then $A_c^2 \geq 1/h \geq s$, so

$$C_{\text{end}}^2 \geq 1 + s > (31/20)^2.$$

For PURE nonisolated caps, the complete endpoint roles and times are (A14endtimes). Their residuals, in row order, are

$$15/28, \quad 1/7, \quad 1/7, \quad 15/28, \quad 1/14, \quad 1/14.$$

At own zero only row $j = 3$, helper-three, and near helper-four have positive time. The first has $\lambda^2 = h/5 < 1/4$, impossible. For the second $\lambda^2 = 11h/15$: it lies in $(0, 1/4)$ at h_a, h_c, in $(1/4, 1/2)$ at h_b, h_e, and in $(1/2, \cos^2(\pi/5))$ at h_d. These are all native gaps. The coarse exact brackets $1.6 < \varphi < 5/3$, $2 < \sqrt{5} < 5/2$, $1.4 < s < 1.5$ seal every strict interval. Every other own-zero time is nonpositive.

For ANY positive own $\gamma \geq 1/2$, the first three helper-three rows already give $C_{\text{end}} \geq 9/5$, $C_{\text{end}}^2 \geq 169/15$, $C_{\text{end}}^2 \geq 64/15$. Both helper-four rows give $C_{\text{end}}^2 > 113/15$. Only far helper-three at $j = 14$ remains. Its actual simple-three normalization gives $2\gamma \in K$. Thus finite own is $0, 1/2, \varphi/2$ in $\mathbb{Q}(\varphi)$, or $0, 1/2, s/2$ in $\mathbb{Q}(s)$. Zero is negative time and $1/2$ has zero time. At $\varphi/2$, time $(8\varphi - 1)/15$ gives mutual squares

$$(9 - \varphi)/30 \in (0, 1/4), \qquad (8\varphi - 1)/(15\sqrt{5}) \in (1/4, 1/2).$$

At $s/2$, time $(6 + s)/15$ gives, for h_c, h_d, h_e,

$$\frac{5s - 4}{45} \in (0, 1/4), \quad \frac{3s + 1}{15} \in (1/4, 1/2), \quad \frac{6s + 2}{45} \in (0, 1/4).$$

All are forbidden native gaps. Parallel own has its positive actual null. For EVERY dotted $\gamma > 1$ all time polynomials increase and the minimum far time exceeds $22/15$. With residual at most $15/28$ this gives $C_{\text{end}} > 29/15$. The entire unbounded own range is covered.

We now keep the ENTIRE foreign S block. At most one S contact is possible in the retained finite Gram; two would make a nonpositive cycle through its old bond. For old $I_2(4)$, $I_2(5)$, only native three is finite, of costs $1/2, d_5/4$, where $d_5 = 2 + 2/\sqrt{5}$. For old $I_2(8)$, any contact costs > 1. For old A_2, the finite labels are three, four, five; its actual $\mathbb{Q}(s)$ excludes five, leaving costs $1/3, 2/3$. White parallel/dotted products cannot occur.

If there is also an A_{13} contact, the full connected finite rank-sixteen diagram is classical. An old five/eight bond is impossible there. For old $S = A_2, I_2(4)$, the ONLY join is an A_{13} END-three and a native-three S contact; an internal or weighted attachment gives excessive arms, an internal double bond, or a cycle. The exact residuals are

$$S = I_2(4) : \ \delta = 1/28, \qquad S = A_2 : \ \delta = 17/84.$$

At old four, $\rho > 1/2, h < 3/4$, so acuteness alone gives $A_c^2 > 1/3$ and $C_{\text{end}}^2 > 31/3$. At old A_2, the complete inverse products give

$$\frac{\rho_{\min}^2}{h} = \frac{1}{3} + \frac{s}{4} > \frac{41}{60}.$$

Thus $C_{\text{end}}^2 > 1 + (41/60)(84/17) > 4$. These estimates retain all own parameters and require no finite mutual assumption.

It remains to consider foreign caps isolated WITHIN A_{14}, with

$$A_c^2 = (28/15)\gamma^2 + Q - 1, \qquad \delta = 1 - Q.$$

The full inverse products yield

S, h	foreign label	$\rho_{\min}^2/h$	$1 - Q$
$I_2(4), h_a$	3	$\varphi^3/4 > 1$	$1/2$
$I_2(4), h_d$	3	$(4 + 3s)/8 > 1$	$1/2$
$I_2(5), h_b$	3	$d_5(d_5 - 1)/(4(d_5 - 2)) > 3/2$	$1 - d_5/4 < 3/10$
A_2, h_c or h_e	4	$2(1/3 + s/4) > 4/3$	$1/3$.

The d_5 inequality reduces to $(d_5 - 3)(d_5 - 4) > 0$, with $2 < d_5 < 3$. Acuteness gives respectively $C_{\text{end}}^2 > 3, 3, 6, 5$, sufficient even when the mutual is zero.

Only old A_2, foreign-three, remains. At h_c, the two inverse products are $a_8/3, 2a_8/3$, $a_8 = \cos(\pi/8)$; at h_e, they are $(s + 2)/6, (2s + 1)/6$. These include BOTH old-node orientations. If $\lambda = 0$, the exact time equation forces the own two-cycle $4\gamma^2$ to be one of

$$\frac{15(4 + s)}{28}, \qquad \frac{15(2 + s)}{7}, \qquad \frac{15(8 + 3s)}{56}.$$

The first occurs for the smaller product in both models; the others are their respective larger products. None is integral in $\mathbb{Z}[s]$, by its nonzero coefficient residue at seven. Thus every zero-mutual exception is impossible for ANY finite or dotted own. If $\lambda > 0$, then $\lambda \geq 1/2$, while $h < 1/2, \rho > .3$. Hence $A_c^2 > (.5 + .3)^2/.5 = 32/25$ and

$$C_{\text{end}}^2 > 1 + \frac{32/25}{2/3} = 73/25 > (31/20)^2.$$

We have covered every white, own, foreign and zero-mutual role. Both actual endpoints therefore satisfy

$$x_{A,1} > .44, \qquad -x_{A,15} > .44. \tag{A14ordends}$$

Reversing index names proves the second demand without requiring a chamber symmetry or distinct opposite walls at the two ends.

C.11.6 Ordinary A_{14}: pay both ends and all feet afresh

We return to the COMPLETE .597 local inventory and directional pressures in Section C.10.1. There is at most ONE weak golden helper. All other weak central sites are the retained pure/foreign $0 < u_i < 1$ feet; no adjacency or packing restriction is imposed on them. Their full continuous adjusted-loss bounds hold whenever the ACTUAL remaining prices satisfy

$$\beta_{5,10} \geq .06, \quad \beta_{6,9} \geq .026, \quad \beta_{7,8} \geq .0246.$$

These are individual coefficient thresholds, not a total debit budget.

The reference must be rebuilt BEFORE any new energy is used. Let $v = \sum \beta_j \Omega_j$ be a concave reference whose endpoint prices are paid by actual endpoint demands, and whose central heights retain their strong floors. For ALL paid isolated feet put

$$Z = v - \sum \beta_i u_i n_i.$$

Its actual pressure and full norm obey

$$x_A \cdot Z \geq v^2 - \sum \beta_i (L_i - .597),$$
$$Z^2 \leq v^2 - \sum \beta_i^2 (2u_i - u_i^2).$$

Here $v \cdot n_i = \beta_i$ is the NEW coefficient, including every endpoint/foot cross. Every adjacent-foot cross is nonpositive. The previously proved full-interval loss bounds give $e_A > v^2$ by ONE square. Strict endpoint pressures give strictness also when no foot remains. We use orthogonality of the block changes only to compute the NEW reference norm, never to ignore their crosses with feet.

Use the base heights

$$H_5 = H_{10} = a = 1.98498, \quad H_6 = H_9 = b = 2.08628, \quad H_7 = H_8 = c_0 = 2.13551.$$

The full base norm is .80071440298. If one internal helper lies at six through nine, omit ONLY its knot by the neighbor chord, losing at most .000677821225. The neighboring coefficients increase. Thus the base norm is at least .800036581755, with its first and last five coordinate blocks still flat at signed mean

$$b_0 = \frac{a}{5\sqrt{2}}, \qquad .28 < b_0 < .3.$$

In each block replace the signed endpoint by its actual paid value .44, keeping the block sum fixed. The other four become $r = (5b_0 - .44)/4$, with $.24 < r < .265 < .44$. They exceed the next old signed coordinate, which is $< .073$. The modified reference is concave, its new endpoint coefficients are positive, and EACH changed outer central price is

$$> .167/1.415 > .118 > .06.$$

Every other paid central coefficient stays unchanged or increases; the helper price remains zero. All central heights are preserved. The two disjoint block gains give

$$v^2 > .800036581755 + 2(5/4)(.44 - .3)^2 = .849036581755. \tag{A14ord0}$$

Apply the simultaneous-foot square to THIS new reference.

For ONE outer helper, relabel it as site five. Its ACTUAL H_4 point already proved in Section C.10.1 gives the signed helper-end demand $> .51961$, in BOTH helper orientations. Pay that endpoint at .51, and the OTHER endpoint at .44 by (A14ordends). Omit only knot five by the chord zero-to-six. Its base norm is $> .764$. The left six-block mean lies in $(.24, .25)$; the right five-block mean lies in $(.25, .3)$. Distribute each remaining block sum constantly after replacing its endpoint.

If the helper end is near, the five other left coordinates exceed $(1.44 - .51)/5 = .186$, while the other block's four exceed $(1.25 - .44)/4 = .2025$. If it is far, the right four exceed $(1.25 - .51)/4 = .185$, and the left five exceed $(1.44 - .44)/5 = .2$. Each is positive, below its endpoint target and above the next old coordinate $< .073$. Thus both new boundary prices exceed $.112/1.415 > .079$. Site six needs only .026; site ten needs .06. All other prices retain their own thresholds; the missing-five price remains zero. No strong-five floor has been used.

For near helper the exact block-gain formula gives gains $> .08112, .0245$. For far helper they are $> .055125, .04332$. Consequently

$$v_{\text{near}}^2 > .86962, \qquad v_{\text{far}}^2 > .862445.$$

The SAME simultaneous-foot square applies, retaining every foot and all boundary crosses with these end changes.

The minimum of the three complete branches is (A14ord0), which exceeds 16/19: it exceeds .849, and $19(.849) > 16$. Only NOW add the disjoint ORDINARY T_2 energy $> 3/19$ proved in Proposition C.1.1. The original norm exceeds one. Thus the entire ordinary $A_{14} + T_2$ branch is excluded. Together with the actual typed-seven proof and the already stated two-A_1 and joint-$I + A_1$ consumers, this covers every rank-sixteen A_{14} alternative. The proof never increases the ordinary dihedral floor or transfers a K_7 assumption to this quadratic-field branch.

C.11.7 The rank-seventeen A_{15} typed alternatives

Here the WHOLE forest is A_{15} plus one actual rank-two typed defect. Thus $K = K_7$, its disjoint defect energy is $> .09$, and the ENTIRE foreign cost is

$$Q \in \{0, 1/3, 2/3\}.$$

Positive values occur only for old A_2: finite foreign labels are three, four, five, and K_7 excludes five. For old $I_2(7)$, even a three costs $\csc^2(\pi/7)/4 > 1$. There are no further spectators.

Use $\vartheta = 2\cos(2\pi/7)$, satisfying $p(X) = X^3 + X^2 - 2X - 1$. The earlier generator $\theta = 2\cos(\pi/7)$ satisfies

$$\vartheta = \theta^2 - 2, \qquad \theta = \vartheta^2 + \vartheta - 1.$$

Hence the already proved FULL integer ring is also $\mathbb{Z}[\vartheta]$. Its roots and derivatives satisfy

$$1.246 < \vartheta_1 < 1.247, \quad -.446 < \vartheta_2 < -.445, \quad -1.802 < \vartheta_3 < -1.801,$$
$$5 < p'(\vartheta_1) < 6, \quad -3 < p'(\vartheta_2) < -2, \quad 4 < p'(\vartheta_3) < 5, \quad \vartheta_2 - \vartheta_3 > 1.355.$$

The three paired endpoint signs are $(-.005049064, .000105223)$, $(.002199464, -.000096125)$, $(-.000257608, .003875599)$; derivative bounds then follow on the brackets. For ANY integer $y = a + b\vartheta + c_0\vartheta^2$, Lagrange interpolation gives

$$c_0 = \sum_j \frac{y_j}{p'(\vartheta_j)}, \qquad \frac{y_2 - y_3}{\vartheta_2 - \vartheta_3} = b - (1 + \vartheta_1)c_0.$$

We prove three COMPLETE integer-strip exclusions.

There is no integer with

$$4 < y_1 < 16, \qquad 0 < y_2, y_3 < 32/55. \tag{A15stripA}$$

Interpolation gives $0 < c_0 < 4$ and $|b - (1 + \vartheta_1)c_0| < .43$. At $c_0 = 1, 2, 3$, respectively, $b = 2$, no integer, or $b = 7$. For the first case

$$-.358399 < \vartheta_3^2 + 2\vartheta_3 < -.356796$$

forces $.356796 < a < 32/55 + .358399 < 1$. For the last case

$$-2.876197 < 3\vartheta_3^2 + 7\vartheta_3 < -2.872388$$

forces $a = 3$, but the physical value then exceeds $3 + 7(1.246) + 3(1.246)^2 = 16.379548$.

There is no integer with

$$9/5 < y_1 < 68/25, \qquad -.8 < y_2, y_3 < .48. \tag{A15stripB}$$

Interpolation gives $-.14 < c_0 < 1.064$; hence $c_0 = 0, 1$. The difference bound is $< .945$. At zero $b = 0$, giving incompatible physical and small rational-integer strips. At one $b = 2, 3$; their values at ϑ_2, before adding a, lie in $(-.693084, -.691975)$, $(-1.139084, -1.136975)$. The small-value strip permits $a = 0, 1$ in the first case, $a = 1$ in the second. Their physical values exceed 4.044516, 6.290516, respectively.

Finally no integer has

$$2 < y_1 < 21/10, \qquad -7/5 < y_2, y_3 < -1/5. \tag{A15stripC}$$

Now $-1 < c_0 < 2$: the interpolation lower and upper bounds are $-1/60$, 1.12. The difference bound is $< .886$. At zero the rational case fails; at one $b = 2, 3$. The same ϑ_2 intervals force $a = 0$, with physical values $> 4.044516, 5.290516$. All three full-ring strips are proved.

At $N = 16$ use sites $5, \dots, 11$ and loss $1/2$. Here $D_i = i(16 - i)/8 \geq 55/8$, $2.6 < q_i < 3$. The full white list is the two paths of ranks at least four, with the usual endpoint/D/E three, endpoint four, or two-end-three join. Rank fifteen excludes exceptional completions. The increasing sufficient ratio is

$$T(q) = 2 - \frac{q - 1/2}{q(2q - 1/2)} < 61/33.$$

Own-zero global end-three has time $Q - 17/32$, $Q < 3/5$, negative for the actual foreign-cost set. Next-end-three has time $Q - 1/8$, $Q = 0, 1/3$; its positive time is less than its foreign cost. End-four has $Q = 0$ and negative time. The actual reciprocal law, affine internal three/four and two-three kernels, and parallel nulls dispose of all those roles.

For a simple-three helper $t = 2\gamma$ is integral in K_7; the only positive finite own cosines themselves in this field are labels three and seven. Three closes the actual affine cycle. For native seven at a NONEND helper node j, $D_j \geq 7/2$, and its positive conjugate $t_\sigma = 2\cos(3\pi/7) > 2/5$ violates Schur positivity already through

$$D_i t_\sigma^2 + D_j > (55/8)(4/25) + 7/2 = 23/5 > 4.$$

The omitted cross and foreign costs are nonnegative. For dotted owns the remaining lower ratios are $6/\sqrt{10} > 61/33$ at D-positions, $2\gamma + 3/8 > 19/8$ at E-positions, and

$$\frac{3m - 1}{\sqrt{2m(m + 1)}} \geq 17/\sqrt{84} > 61/33$$

for near-three of order $m \geq 6$. The last function increases; its cleared square margin at six is 2157. Endpoint-four/own-three would put $\sqrt{2}$ in K_7; other owns have ratio at least $6/\sqrt{10}$. The two-path bridge has $\gamma q + 1/q > .9q + 1/q > 61/33$. Foreign far-three has $Q \geq 1/3$ and ratio square

$$\geq \frac{3(2m\gamma+1)^2}{2m(m+3)} > 15/4 > (61/33)^2.$$

It increases with $m \geq 5$ at $\gamma > .9$. Only pure far helper or near dotted side-order-five helper remains.

For the pure far role the square identity gives $9/5 < t < 5899/2178 < 68/25$. The fresh $N = 16, i = 5$ signed Schur costs exceed four at $-.8, .48$ by $3/40, 59/1000$. Their monotonicity puts both other values in $(-.8, .48)$. Strip (A15stripB) excludes every such cap. For near dotted order five, $2 < t < 1363/660 < 21/10$. Its necessary conjugate inequality is $55t_\sigma^2 + 88t_\sigma + 16 < 0$. That quadratic equals $3/5 > 0$ at $-7/5, -1/5$, so both values lie between these endpoints. Strip (A15stripC) excludes the last nonisolated weak cap.

For isolation within A, the actual own two-cycle $B = 4\gamma^2$ is integral, and full conjugate Schur positivity gives

$$0 < B_\sigma < 4/D_i \leq 32/55.$$

Use the COMPLETE finite-square alphabet already proved in Section C.11.2. Small owns three/four/six have all-foreign actual nulls. The other finite labels seven/fourteen each have a conjugate > 1, using the stated brackets for θ^2, $2 + \theta$. Thus no finite isolated own survives. For a dotted own, $0 < u = (1 - Q)/\gamma^2 < 1$. Its actual all-wall foot has loss

$$< u/5 + \sqrt{u(1-u)},$$

since $(\sqrt{D} + \sqrt{D-1})^{-1} < 1/5$. At $u \leq 1/4$ this is $< 1/20 + \sqrt{3}/4 < .484 < 1/2$. Weakness therefore implies $u > 1/4$ and $4 < B = 4(1 - Q)/u < 16$. Strip (A15stripA) excludes every such dotted foot, including both positive foreign costs.

Every site $5, \ldots, 11$ is now actually half-loss strong. Choose symmetric heights, linear on the tails,

$$H_5 = 2.122, \quad H_6 = 2.238, \quad H_7 = 2.306, \quad H_8 = 2.328.$$

The strict floor tests are $2.622^2 < 55/8$, $2.738^2 < 15/2$, $2.806^2 < 63/8$, $2.828^2 < 8$. The successive slopes $.4244, .116, .068, .022$ and their reflection decrease at all paid knots. One positive reference square gives

$$e_{A_{15}} > 2.122^2/5 + .116^2 + .068^2 + .022^2 = .9191408.$$

Only NOW add the disjoint actual defect energy $> .09$, obtaining > 1.0091408. Both actual typed-seven A_{15} alternatives are excluded, with no remaining weak foot, field census or independently added optimum.

C.11.8 The ordinary $A_{15} + T_2$ rank-seventeen consumer

Here the WHOLE forest is $A_{15} + T_2$, with $T_2 = I_2(m)$ ordinary. If $e_T > 1/6$, the already proved $e_{A_{15}} > .8348$ gives total $> .8348 + 1/6 > 1$. Suppose $e_T \leq 1/6$. Put $k = \cos(\pi/m)$, $d = (1 - k^2)^{-1}$. The same two-direction argument as above gives

$$\tanh L_1 \tanh L_2 \leq \frac{48(1+k)}{(13+k)^2} < k.$$

The last cleared polynomial is $k^3 + 26k^2 + 121k - 48$, increasing on $k \geq 1/2$ and equal to $153/8$ there. Thus the two ACTUAL opposites are one wall.

Write its two T_2 contacts $b, g \geq 0$, its full A_{15} cost Q, and its time square $h > 0$. Any positive foreign cost is at least $15/32$. Two positive T_2 contacts are impossible. At old three, contacts three/three have the actual affine null. Otherwise one is at least native four; the edge coshes are at least $(1+2\sqrt{2})/3 > 5/4$, $(1+\sqrt{2})/\sqrt{3} > 29/21$. Their half-tangents exceed $1/3, 2/5$. The positive dual certificate at correlation $1/2$ gives energy $> 124/675 > 1/6$. At old four both coshes are at least $(1+\sqrt{2})/\sqrt{3}$, giving $e_T > 32/175 > 1/6$, using $k < 3/4$. For old $m \geq 5$, an edge cosine is at least $\cot(\pi/(2m))/\sqrt{3} > 7/5$; its individual energy already exceeds $1/6$. Positive foreign cost only increases these actual coshes.

There is exactly one T_2 contact $\beta > 0$. Put $q = d\beta^2$. The actual own/mate derivative law gives

$$h = q + Q - 1, \quad h \leq (24/49)k^2 q, \qquad \beta^2(25d + 24) \leq 49.$$

Necessarily $q > 1$: if $q \leq 1, Q > 0$, its foreign reciprocal shadow makes the same actual wall close an orthogonal component, or gives a feasible null; $Q = 0$ instead has nonpositive time. For $m \leq 5$, these inequalities give $Q < (24/49)k^2 \leq 18/49$. For $m \geq 6$, $\beta^2 \geq 1/4, d \geq 4$ give $Q \leq (172 - 25d)/196 \leq 18/49$. Both contradict $Q \geq 15/32$ unless $Q = 0$. Also $d \leq 172/25$, so $m \leq 8$; for $m \geq 9$, $d > (63/22)^2 > 172/25$. The complete native table is

m	β^2 upper bound	$q > 1$ own labels
3	$147/172$	$7, 8$
4	$49/74$	5
5	$< 441/841$	4
6	$49/124$	none
7	$< 49/149 < 1/2$	3
8	$< 1/2$	3.

At old three, every native nine or larger has $\beta > 15/16$, from $\pi/9 < 22/63$, $\cos t > 1 - t^2/2$. The other rows use the consecutive native squares and $q > 1$. Parallel or dotted $\beta \geq 1$ violates the same table bounds. The two seven rows are exactly actual typed defects and absent by ordinary meaning. Only old/new pairs

$$(3, 8), (8, 3), (4, 5), (5, 4)$$

remain. Their ACTUAL fields are respectively $\mathbb{Q}(\sqrt{2})$, $\mathbb{Q}(\varphi)$, by the original connected Lorentzian-principal descent proved above for the three-wall core. No relative scale between old components has been asserted.

In both silver rows the mate edge has

$$C_s^2 = (3 + 2\sqrt{2})/3, \qquad e_T > 411/2500 = .1644.$$

For an exact seal, $\sqrt{2} > 707/500$ gives $C_s^2 > 1457/750 > (13937/10000)^2$; the half-tangent comparison with .1644 has positive cleared margin 4393. The golden rows have $C_g = \cot(\pi/5) > 11/8$, hence $e_T > 3/19$.

For silver $K = \mathbb{Q}(\sqrt{2})$, recalibrate the common $N = 16$ loss to $\ell = .6007$. Its proof is unchanged on the full rectangle $.6 \leq \ell \leq .601$: $R_\ell(q) \leq R_{.6}(3) = 41/27 < 761/500$; all finite and foreign $u > 1$ seals printed in the common inventory are below .6, and near-four has the SAME three positive conjugate numerators. Every remaining weak pure helper would be golden and force the other actual field, so none exists. Retain ALL isolated feet, with $a_i < .1982$. Take symmetric heights, linear on the tails,

$$H_5 = 2.0213, \quad H_6 = 2.1373, \quad H_7 = 2.2053, \quad H_8 = 2.2273.$$

Every positive central price is at least .022, and the floor squares are unchanged from the complete N16 reference. Below $u = 1/2$, raw loss is $< .5991$. Above it, adjusted loss is

$$< (.1982 + 1.0195)/2 - 3(.022)/8 = .6006 < .6007.$$

The FULL simultaneous-foot square, including adjacent feet, therefore gives

$$e_A > 2.0213^2/5 + .116^2 + .068^2 + .022^2 = .835694738.$$

The total with the silver spectator exceeds 1.000094738.

For golden $K = \mathbb{Q}(\varphi)$, the exact old spectator is $I_2(4)$ or $I_2(5)$. Its ENTIRE possible foreign cost is

$$Q = 0, 1/2 \quad \text{or} \quad Q = 0, (5 + \sqrt{5})/10,$$

respectively. A retained finite cap has only one native-three contact; a four already costs at least one, and two contacts make a cycle. Use loss $\ell = .59$, sites $5, \ldots, 11$. The fresh sufficient ratio is

$$R_{.59}(q) \le 148081/95757 < 31/20.$$

The complete finite-white list is unchanged, but we check its ordinary roles against THIS stronger target. Own-zero internal-three/four and two-three retain their affine kernels. The residual global end-three, next-three and end-four own costs are $15/32, 7/8, 15/16$, less than one. At $Q = 0$ their times are negative; at positive Q they are smaller than that foreign cost, violating the actual reciprocal law. Parallel is its actual positive null.

A simple-three helper makes 2γ integral in the actual golden field, so finite own is three or five. Three has its cycle; hence $\gamma \ge \varphi/2$. Endpoint-four/own-three would put $\sqrt{2}$ in K, so its remaining owns have $\gamma \ge 1/\sqrt{2}$. The complete nonfar lower comparisons are

D-position	$(\varphi/2 + 1/5)/\sqrt{2/5} > 31/20$
E-position	$2\gamma + 3/8 > 31/20$
near-three	$(\gamma + (m-1)/(2m))/\sqrt{(m+1)/(2m)} > 31/20$
endpoint-four	$6/\sqrt{10} > 31/20$
two-path bridge	$\gamma q + 1/q > 2.$

The near-three expression increases for $m \ge 5$; at five use $\varphi/2 > .809$, $\sqrt{3/5} < 31/40$, giving $1209/775 > 31/20$. Own-four/simple-three is also impossible by its unsquared connected cycle. For foreign far-three, $Q \ge 1/2$ leaves residual at most $1/(2m)$, and ratio $> (\gamma + 1/(2m))\sqrt{2m} > 31/20$. A weak nonisolated cap is therefore a PURE far helper.

Its square identity gives $\varphi \le t = 2\gamma < 1761/800 < 9/4$; the unchanged N16 signed Schur strip is $-.8 < t' < .48$. For $t = a + b\varphi$, these force $b = 0, 1$. At zero only parallel two fits physically. At one, the small strip gives $a = 0, 1$; one gives $\varphi^2 > 9/4$. Thus $t = \varphi$. The COMPLETE N16 golden table and cyclic packing in Section C.2.3 apply independently of weakness: at most ONE weak helper exists.

For isolated-within-A cap, $B = 4\gamma^2 \in \mathbb{Z}[\varphi]$, $0 < B' < 4/D_i \le 32/55$. Its complete finite native alphabet is labels $3, 4, 5, 6, 10$. The conjugate of label ten exceeds $32/55$; small three/four/six have their actual affine principals. Only finite own five remains. At $Q = 0$, its actual H_4 loss is $< 25/76 + 5/24 < .59$. At the two positive foreign costs the actual foot parameter is respectively

Q	u	loss upper bound
$1/2$	$4 - 2\varphi < .764$	.577
$(5 + \sqrt{5})/10$	$4 - 8\sqrt{5}/5 < .423$	.584.

For the first, $\sqrt{u(1-u)} = \sqrt{10\varphi - 16} < .425$; for the second use its full circle loss with $a_i < .1982$. The rational seals $809/500 < \varphi < 40451/25000$, $\sqrt{5} > 2.236$ suffice. Every finite isolated cap is strong.

A dotted foot with $u \le 1/4$ has loss $< 1/20 + \sqrt{3}/4 < .484$. A weak dotted foot therefore has $4 < B < 16$, $0 < B' < 32/55$. Writing $B = a + b\varphi$ gives $b = 2, \dots, 7$; the interval $b/\varphi < a < b/\varphi + 32/55$ leaves, in order, no integer, two, three, no integer, four, no integer. The COMPLETE list is

$$B = \varphi^4, \quad 3 + 4\varphi, \quad 2\varphi^4.$$

The last two have $u < 4/9$. For the first with $Q > 0$, $u < 1/3$. Their loss is $< (4/5 + \sqrt{20})/9 < 44/75 < .59$. Only PURE $B = \varphi^4$ can be weak, with $u_0 = 20 - 12\varphi \in (.58, .59)$.

In the actual connected old-A normalization, the raw dotted time square is

$$H_i^2 = \frac{B\{i(16-i)B - 32\}}{16}.$$

For $i = 5, 6, 7, 8$ its norms are

$$-8271/256, \quad -8816/256, \quad -9119/256, \quad -9216/256.$$

These four squareclasses differ. Here $8816 = 16 \cdot 19 \cdot 29$, while the others are nonzero modulo nineteen; $8684^2 < 8271 \cdot 9119 < 8685^2$; and $9216 = 96^2$, whereas $8271, 9119$ are not integer squares. Reflected copies have the actual K-valued two-cycle parameters

$$-7/4 - 45\varphi/8, \quad -1 - 9\varphi/2, \quad 1/4 - 21\varphi/8.$$

None is integral. Their squares must be integral by the actual two-cycle criterion, which for a K-element forces integrality of the element itself. Thus at most ONE weak dotted index exists.

A weak helper and weak dot cannot coexist. Compare only their ACTUAL old-A-linked root lines on the same K-time line. The following odd prime has odd valuation in each helper time norm, and does not divide ANY dotted numerator:

helper	$-\mathrm{N}(M)$	p	8271 mod p	8816 mod p	9119 mod p	9216 mod p
near 5	1559	1559	476	1021	1324	1421
near 6	1111	101	90	29	29	25
near 7	719	719	362	188	491	588
near 8	431	431	82	196	68	165
far 5	311	311	185	108	100	197
far 6	239	239	145	212	37	134
far 7	271	271	141	144	176	2.

All listed primes have the elementary trial-division checks; $1111 = 11 \cdot 101$. The norm ratio is not a rational square, so the time ratio cannot be a K-square. No freely normalized spectator time is used in this argument.

Take symmetric heights, linear on the tails,

$$H_5 = 2.032, \quad H_6 = 2.148, \quad H_7 = 2.216, \quad H_8 = 2.238.$$

Their exact floor sums are $2.622, 2.738, 2.806, 2.828$, and

$$E = .8443688.$$

There are precisely three complete weak-set cases. With no helper and no weak dot, one square gives $e_A > E$. With one weak dot, use its actual foot in the SAME reference. The loss is $< .61$. At outer five/eleven, $\beta = .1452$ and $2u_0 - u_0^2 > .8$, so its correction gain is positive. At an internal knot $\beta \le .024$, the energy debit is $< 2(.024)(.02) = .00096$. Thus $e_A > .8434088 > .8434$.

With one internal helper there is no weak dot and no second helper. Omitting just its knot loses at most $.048^2/4 = .000576$, giving $e_A > .8437928 > .8434$.

With an outer helper, relabel it as five. Its actual finite H_4 point gives, at its signed helper end,

$$y > \sqrt{2}(A_c - .111 - \gamma h_5) > .49.$$

Indeed the smaller actual time is $(38+65\varphi)/32 > 2.115^2$, $h_5 < 2.033, \gamma < .81$, and $1.4(2.115 - .111 - .81(2.033)) = .500178 > .49$. This recomputes the endpoint demand at the new loss; it does not borrow the older stronger height cutoff. Omit five by the chord zero-to-six. Its base reference norm is $> .805$, by the exact chord formula already printed for the SAME heights. At the actual helper end its flat block has length five or six, signed mean $.25 < b_0 < .3$. Graft that signed coordinate to .49, preserving the block sum. The other coordinates lie in $(.19, .262)$, above the next old coordinate $< .101$. The reference stays concave and paid; its gain exceeds .04332. No weak dot or other helper remains. Hence $e_A > .84832 > .8434$.

Every golden branch gives $e_A > .8434$. Only NOW add its disjoint ordinary spectator energy $> 3/19$, obtaining total $> 1 + 123/95000 > 1$. Together with the silver and high-energy branches this excludes the entire ordinary $A_{15} + T_2$ rank-seventeen forest. In particular no blanket $T_2 > 1/6$ claim or disconnected root-scale comparison has entered the proof.

C.11.9 A whole A_{16} with any rank-one spectator

Suppose the WHOLE forest is $A_{16} + A_1$. The A_1 is initially ordinary or actual short I; we do not identify the two hypotheses. The common component proof gives

$$e_A > .8703, \qquad e_1 < .1297 < .13.$$

Write the ACTUAL rank-one opposite as $c = A_c p - \gamma e - v_A$, with $h = A_c^2 = \gamma^2 + Q - 1 > 0$, $Q = \|v_A\|^2 < 1$. A finite one-node completion permits at most one A_{16} contact; a second makes a cycle, and rank seventeen is too large for exceptional finite types. Thus $Q > 0$ implies $v_A = b\Omega_j$, $b \ge 1/2$, and $Q \ge 8/17$.

In fact $Q < h$. If $\gamma < 1$, put $\delta = 1 - \gamma^2 = Q - h > 0$, $\eta_c = c + \gamma e$. The actual vector

$$y = \delta p + A_c \eta_c$$

has square $-\delta Q < 0$, positive time, zero products on c, e and every old A-root except n_j. It is a positive combination of p and ACTUAL roots, so every omitted original wall product is nonpositive. Its normalization is therefore an actual finite endpoint on the outgoing A edge j, with the SAME opposite as the A_1 edge. The original orthogonal-component theorem forbids this. At $\gamma = 1$, $c + e$ is a feasible future null. Consequently $\gamma > 1$, giving $Q < h$.

Let $\tau = \tanh(L/2)$ on the actual rank-one edge. Its exact equation gives

$$\tanh^2 L = \frac{h}{1 + h - Q} > Q \ge 8/17, \qquad \tanh^2 L = \frac{4\tau^2}{(1+\tau^2)^2} < \frac{5200}{12769} < 8/17,$$

using $\tau^2 \le e_1 < .13$. The last cross-product margin is 13752. Contradiction. Thus $Q = 0$, and the actual opposite is

$$c_1 = \sqrt{h}\, p - \sqrt{1+h}\, e, \qquad 0 < h < 5200/7569 < 7/10. \tag{A16r1}$$

The height bound inverts the same half-tangent equation. This isolated dotted wall has not been called a short I.

Put

$$s^2 = \frac{\sqrt{1+h}-1}{\sqrt{1+h}+1}, \quad t = \sqrt{1+4h} - 2\sqrt{h}, \quad \eta = \sqrt{16/17}.$$

Then $e_1 \geq s^2$. For ANY other actual edge opposite $d = A_d p - v - \beta e$, distinct from c_1, native gap at $\beta = 0$ or acuteness with $\beta \geq 1/2$ otherwise gives $A_d\sqrt{h} \geq 1/2$. Bounding its ENTIRE spatial derivative by $\sqrt{1+A_d^2}$ gives the same physical half-tangent pressure as in the earlier actual-I argument:

$$x_1, -x_{17} \geq \eta t.$$

It uses only $h > 0$, not $h < 1/2$.

The exact scalar barrier $t + 3s/2 > .767$ for $0 < h < 1/2$ is the already proved rank-free barrier (20.1), with its proof in Section B.1.8. For $1/2 \leq h \leq .6$, monotonicity gives $t \geq t(.6) > .294$, $s \geq s(.5) > .317$, and hence sum $> .7695$. The rational seals are $\sqrt{3.4} < 1.844, \sqrt{.6} < .775$, $.294(3.394) < 1$, and $s(.5)^2 = 5 - 2\sqrt{6} > .101 > .317^2$. For $.6 \leq h < .7$, use $t > t(.7) > .275$, $s \geq s(.6) > .34$, giving sum $> .785$. Here $\sqrt{3.8} < 1.95, \sqrt{.7} < .837$, $.275(3.624) < 1$, and $\sqrt{1.6} > 1.264$ gives $s(.6)^2 > 33/283 > .34^2$. Thus on the ENTIRE widened interval

$$t + 3s/2 > .767.$$

Its actual own two-cycle gives

$$v = 4h \in \mathcal{O}_K, \quad 0 < v < 14/5, \qquad -4 < v_\sigma < 0$$

at EVERY nonphysical embedding. No field degree is assumed.

The complete $N = 17, \ell = .6$ inventory has strong sites, all retained isolated feet, or one pure golden helper. For the latter, ACTUAL $K = \mathbb{Q}(\varphi)$; its complete $34A^2_{\text{helper}} = M$ table is already printed in Section C.2.3. Its seven norm magnitudes $1764, 1216, 716, 324, 196, 36, 76$ are all coprime to seventeen. The actual helper/c_1 two-cycle makes $Mv/34$ integral. Conjugating M and using an integer Bezout relation gives $v \in 17\mathcal{O}_K$. But its nonzero quadratic norm has magnitude $< 56/5 < 17^2$, impossible. This uses the widened physical bound explicitly, rather than the short-I norm bound eight. All weak golden helpers disappear. After this branch the actual field is again UNRESTRICTED.

We prove that BOTH actual A endpoint edges have cosine $C < 2$. Use the same central heights $H_5 = 2.056, H_6 = 2.186, H_7 = 2.269, H_8 = H_9 = 2.310$, reflected with linear tails. Their base energy is .8708972; every positive central coefficient is at least .0205. Each five-coordinate end block has signed mean $.2907 < b < .291$; its next coordinate is $< .092$. Suppose one actual signed endpoint is at least .55. Pay that end at .55. Pay the disjoint other end at

$$y = \min(.55, \max(b, \eta t)).$$

Both block sums are preserved by distributing their other four coordinates constantly. All remaining coordinates exceed $.225875 > .092$. Thus the reference is concave and EACH changed central boundary price still exceeds .0205. All other prices retain their values; all central heights remain fixed. The FULL simultaneous-foot square therefore applies to this NEW reference, including every foot/endpoint and adjacent-foot cross.

The first block gain exceeds $(5/4)(.55 - .291)^2 = .08385125$. The disjoint A_1 energy plus the other block gain exceeds .07. If $\eta t \geq .55$, the gain alone is larger. Otherwise put $z_0 = (\eta t - .291)_+$.

Then $z_0 + (3/2)\eta s > .4529$. Weighted Cauchy gives

$$e_1 + \text{other gain} \geq s^2 + (5/4)z_0^2 > \frac{.4529^2}{36/17 + 4/5} > .07.$$

The final rational comparison is $.4529^2(85) = 17.43506485 > 17.36$. The SAME reference and A_1 energy would total > 1.02474845. Thus each signed actual endpoint is $< .55$, by two separate reflected contradictions, not an added pair of optimized bounds. If its actual $C \geq 2$, its signed coordinate square is $\geq 16/51 > .55^2$. Hence BOTH actual endpoint edges have

$$C < 2. \tag{A16r1C}$$

Suppose an endpoint cap is isolated WITHIN A:

$$c = A_c p - \gamma\Omega_1 - \beta e, \qquad 0 \leq \beta < 1.$$

Own zero has negative time. For $U = 4\gamma^2$, rescaling the original K-root against an old norm-two root gives raw square $2U$, own product $-U$. Its actual own two-cycle is integral; nonphysical old-A Schur positivity gives

$$U \in \mathcal{O}_K, \qquad 0 < U_\sigma < 17/8.$$

The actual edge equation and cutoff give $C^2 = 8U/(17(1-\beta^2))$, hence $0 < U < 17/2$. The three DISTINCT actual walls c, e, c_1 have $\lambda = A_c\sqrt{h} - \beta\sqrt{1+h} \geq 0$. Their two/three cycles make

$$L = 4\lambda^2, \quad W = 8\beta\sqrt{1+h}\lambda, \quad V = 4\beta^2$$

integral, including possible zero cycles. Expanding ALL β-terms gives

$$L + W + V = \frac{(8U-17)v}{17} =: \rho \in \mathcal{O}_K.$$

Therefore

$$z = \frac{Uv}{17} = 15\rho - (7U-15)v \in \mathcal{O}_K.$$

Physically $0 < z < 7/5$; EVERY other conjugate lies in $(-1/2, 0)$. If $[K:\mathbb{Q}] \geq 2$, its nonzero integer norm has magnitude $< (7/5)(1/2)^{[K:\mathbb{Q}]-1} < 1$. At degree one, integral U, v have $U \leq 8, v \leq 2$, so $0 < z \leq 16/17 < 1$. Both cases are impossible. Neither endpoint cap can be isolated within A. This includes every finite old-A_1 contact and the rational field directly; no Meyer or field-degree bound is used.

For completeness we exhaust the remaining ACTUAL endpoint roles. Deleting n_1 retains A_{15} on nodes two through sixteen and old A_1. The complete nonisolated list is simple-three at $j = 2, 3, 15, 16$, or four at $j = 2, 16$. Only simple-three at $j = 2, 16$ permits positive old-A_1 contact, with $\beta = 1/2, 1/\sqrt{2}$. At $j = 3, 15$, appending that component would give nonfinite arms $1, 2, 13$; for helper-four it would make the four internal in a long path. Thus the other rows have $\beta = 0$.

For simple-three the exact retained residual and ACTUAL edge cosine are

$$\delta = 1 - \beta^2 - \frac{(j-1)(17-j)}{32}, \qquad C = \frac{32\gamma + 17 - j}{4\sqrt{34}\sqrt{\delta}}.$$

For helper-four replace $17-j$ in the numerator by $\sqrt{2}(17-j)$, and 32 in the subtracted denominator term by 16. The four non-end roles have

(j, helper)	C
$(3,3)$	$(32\gamma+14)/(2\sqrt{17})$
$(15,3)$	$(32\gamma+2)/(2\sqrt{17})$
$(2,4)$	$(32\gamma+15\sqrt{2})/\sqrt{34}$
$(16,4)$	$(32\gamma+\sqrt{2})/\sqrt{34}$.

At positive own $\gamma \geq 1/2$ all exceed two. At zero, far-three and far-four have negative time, and near-four still has $C > 2$. The near next-three zero-own role has the actual affine E_8 ray

$$2n_1 + 4n_2 + 6n_3 + 5n_4 + 4n_5 + 3n_6 + 2n_7 + n_8 + 3c.$$

Every selected product and norm is zero; its time $3A_c > 0$ is future, and all omitted original walls are nonpositive. Compactness forbids it.

Near simple-three $j = 2$ has

$$C = \frac{32\gamma + 15}{17}, \quad \frac{32\gamma + 15}{3\sqrt{17}}, \quad \frac{32\gamma + 15}{\sqrt{17}}$$

at $\beta = 0, 1/2, 1/\sqrt{2}$. At zero foreign cost, own zero has negative time; under $C < 2$ only positive own three remains, with its actual affine triangle. At $\beta = 1/2$, every positive own gives $C \geq 31/(3\sqrt{17}) > 2$. Its own-zero role has the actual affine ray

$$3n_1 + 6n_2 + 5n_3 + 4n_4 + 3n_5 + 2n_6 + n_7 + 4c + 2e.$$

It has the same all-original-wall future null properties. At $\beta = 1/\sqrt{2}$, even own zero has $C \geq 15/\sqrt{17} > 2$.

Only far simple-three $j = 16$ remains. Put $t_0 = 2\gamma$. Its exact data are

$$A_c^2 = \frac{8t_0^2 + t_0 - 9}{17} + \beta^2, \qquad C = \frac{16t_0 + 1}{17}, \ \frac{16t_0 + 1}{3\sqrt{17}}, \ \frac{16t_0 + 1}{\sqrt{17}}$$

at the three foreign costs. Own zero has negative time in all three. The last row contradicts $C < 2$. In the middle row $t_0 < 3/2 < \varphi$; the only positive native values are one and $\sqrt{2}$. One gives its actual all-three cycle. At $\sqrt{2}$, simple-three normalization puts $\sqrt{2}$ in the ACTUAL field. The lifted sign reversal has positive time numerator $7 - \sqrt{2}$, hence a negative old-A Schur complement in a nonphysical positive form. This excludes that row. Thus $\beta = 0$, and

$$1 < t_0 < 33/16, \qquad 17A_c^2 = (t_0 - 1)(8t_0 + 9).$$

BOTH actual endpoint walls now have the form

$$c_L = A_c p - (u\Omega_1 + \Omega_{16})/2, \qquad c_R = B_c p - (v\Omega_{16} + \Omega_1)/2, \quad 1 < u, v < 33/16.$$

They are distinct: equality would force $u = v = 1$, contrary to positive time. Their spatial product is $[uv + 1 + 16(u + v)]/34$. Their time product is $< (8/17)uv$, because $8t_0^2 + t_0 - 9 < 8t_0^2$. Also $u + v > (32/33)uv$. Therefore

$$c_L \cdot c_R > uv/66 > 0,$$

contradicting acuteness of distinct ACTUAL walls. Every original wall, signed endpoint, foreign contact and field degree has been retained. The entire $A_{16} + A_1$ rank-seventeen alternative, ordinary or actual I, is excluded.

C.12 Classical rank-fifteen and rank-sixteen spectator cases

C.12.1 Rank-fifteen classical typed consumers share the local head

The WHOLE forest is B_{15} or D_{15} plus actual N_7 or O_7. The actual field, defect wall and complete foreign costs are exactly (TQ), with disjoint defect energy $> .09$. We reuse the PROVED local head

argument of Section C.6.1, checking all its rank-specific premises. No rank-fourteen whole-polytope conclusion is transported.

For B_{15}, the generic $M = 10$ proof applies through eleven: retained left A and right B ranks are at least four, and joined rank fifteen precludes exceptional completions. Its weak role is PURE far helper

$$v = g\omega_i + \omega_1/2, \quad Q = 0, \quad A_c^2 = 2ig^2 + 2g - 1/2, \quad 101/125 < g < 6/5.$$

Lemma 20.3 excludes these classified indices from six onward. At twelve the complete right-B_3 list has its actual short-three F_4 seal, with all other rows affine or strong. At thirteen the only extra weak B_2 role is PURE own-three/short-three. Its actual long-square-two/short-square-one normalization puts $\sqrt{2}$ in K; sign reversal gives old Schur cost

$$13(3 - 2\sqrt{2})/4 + 1/2 > 1.$$

The bound already holds at twelve by $\sqrt{2} < 17/12$. The full bridge has residual $1/(2i)$ and ratio $1/\sqrt{2i} + \sqrt{i}/2 > 100/61$; it admits no foreign arm because the new node is internal to the large B path. The F_4 piece likewise has no finite enlargement at its cap. All foreign/isolated seals retain the entire spectator. Thus six through thirteen are ACTUALLY strong.

For D_{15}, the generic $M = 9$ proof gives six through ten. The ACTUAL right-D_4, D_3 layers are eleven, twelve. Their rank-local complete lists in Proposition 22.1 retain all affine zero/three/four/six/parallel rows, direct end-four and sealed H_4 rows. Full D_{15} and long-arm A_{15} bridges have no finite foreign arm. The D_3 one-fork bridge has residual $1/8 + 1/(2i) \le 1/6$, positive-own numerator $> .75$, hence is strong; own zero has its actual all-three cycle. The only extra weak packet has $Q = 0$, own-five/right-three: the local arithmetic helper Schur calculation at these layers has $2\gamma < 12/5$ and leaves only the golden own. It is absent in ACTUAL K_7. Thus six through twelve are ACTUALLY strong, without a centre-thirteen or either fork floor.

In either component, weak five has its PURE actual foot $x_1 > .5965$, while strong six gives $S_6 > 2.01815$. Positive interpolation through one, six, eleven has energy

$$.7600299945 + \frac{(\sqrt{11} - \sqrt{6})^2}{5} > .9104.$$

Its slopes are positive and decreasing; the last term exceeds .150372482 from $\sqrt{11} > 3.3166, \sqrt{6} < 2.4495$. Adding the actual defect $> .09$ contradicts the original norm. Five is strong BEFORE the head. The ordinary positive prefix references through thirteen for B, twelve for D, therefore have energies $E_{13} > .88989, E_{12} > .86989$, first-five mean $b < .361$, and individual P_5 price $> .14$.

Deleting the actual head retains B_{14} or D_{14}. Their COMPLETE finite one-node extensions are isolation or simple-three at long leaf two: a second contact makes a cycle; another fork/weighted attachment gives nonfinite arms or an internal multiple bond. Consequently the exact local head formulas are those already proved:

	A_c^2	C_1
helper-three	$2g^2 + 2g + Q$	$(2g+1)/\sqrt{1-2Q}$
isolated within classical	$2g^2 + Q - 1$	$\sqrt{2g^2/(1-Q)}$.

The own-zero positive-Q null uses the FULL rank-fifteen old-root clearing

$$\frac{\omega_2}{2} = \begin{cases} n_1 + 2(n_2 + \cdots + n_{14}) + \sqrt{2}n_{15}, & B, \\ n_1 + 2(n_2 + \cdots + n_{13}) + n_{14} + n_{15}, & D. \end{cases}$$

All coefficients are on ACTUAL roots. Every omitted wall, including both signed D forks, has nonpositive product. All other small-own affine chains in that head proof use only c, n_1, n_2 and the two old spectator roots, present unchanged here.

The finite mixed-head Schur test uses only K_7 and the SAME rational full costs $0, 1/3, 2/3$. The actual N_7-wall mutual test uses only $g_7/3, 2g_7/3$, its two exact old-A_2 inverse products. The pure-head actual two-cycle, full mutual native alphabet and order parity are independent of classical rank. Thus its complete reduction to pure dotted $0 < u < 1/2$ or $.79 < u < .8$ applies to the PRESENT actual walls. Its actual pressure $x \cdot Z \geq H$ has support in the first three coordinates, sum one and full-interval scalar

$$.8H - .16Z^2 > .3055.$$

The chord/circle proof of this inequality is fully written in the cited SAME-part head argument; it is not a new external or computed premise.

Its primitive correction $.4(Z - P_5/5)$ pays ONLY .08 from the current available P_5 coefficient. Its full reference cross is zero. Every earlier graft excluding a mixed or nonisolated head preserves the first-five sum and all later knots. Thus that entire local proof supplies the SAME effective single-square gain

$$> .3055 - .8(.361) + .032 = .0487.$$

This is a completed-square gain, not the squared norm of an independently optimized head vector. It requires neither the outgoing short B wall nor the outgoing D centre or fork walls.

The two original-vector totals are

$$\begin{array}{l|l} B_{15} + \text{actual typed7} & .88989 + .0487 + .09 = 1.02859 > 1 \\ D_{15} + \text{actual typed7} & .86989 + .0487 + .09 = 1.00859 > 1. \end{array}$$

Each actual defect is counted ONCE. No actual-I barrier, arbitrary-face theorem or additional finite output has been assumed.

C.12.2 The actual-I B_{16} row: a bound before classifying the short cap

Here the WHOLE forest is $B_{16} + I$, where I is the ACTUAL isolated short rank-one opposite, not an ordinary A_1. Keep unit long roots $n_i = (e_i - e_{i+1})/\sqrt{2}$, $i < 16$, actual short $n_{16} = e_{16}$, and orthogonal old I-root e. The SAME original vector has $x_1 > \cdots > x_{16} > 0$ and $e_B + e_I < 1$.

The fresh rank-sixteen bulk gives strong six through twelve. At thirteen the right-B_3 short-three F_4 seal applies. At fourteen the only extra B_2 role is own-three/short-three, whose actual sign-conjugate cost $14(3 - 2\sqrt{2})/4 + 1/2 > 1$ is impossible. The complete bridge is internally attached to full B_{16}, with no foreign finite arm; its zero-own ratio is already strong. Thus $S_i > f_i = \sqrt{i} - .61/\sqrt{2}$ for six through fourteen, independently of FIVE or the outgoing short wall.

The actual I parameters satisfy

$$e_I \geq u^2, \quad x_1 \geq t, \quad t = \sqrt{1 + 4h} - 2\sqrt{h}, \quad 0 < h < 1/2, \qquad t + (8/5)u > 197/250.$$

This is the previously proved rank-free actual-I barrier, with its original same-vector point domain. Take the reference linear on its first six coordinates and with actual knots six through fourteen. It has

$$E_0 > .89026, \quad b = f_6/6 < .3364, \quad f_6 > 2.01815, \quad f_{14} < 3.311, \quad \text{individual } P_{14} \text{ price} > .13.$$

Indeed it is the ordinary reference of energy $E_{14} > .9084$ less

$$\frac{(6\sqrt{5} - 5\sqrt{6} - .61/\sqrt{2})^2}{30} < .01814.$$

The positive radical bracket $.43133 < .61/\sqrt{2} < .43134$ bounds the numerator's positive root by .73769. Lift its first coordinate to $\max(t, b)$, preserving the six-block sum. The other five remain $\geq (f_6 - 1)/5 > .2036$, above the next slope $< .197$, so every residual coefficient stays nonnegative. The I energy plus this gain is

$$e_I + \frac{6}{5}(t - b)_+^2 > \frac{.4516^2}{64/25 + 5/6} > .0601.$$

Thus the combined base already exceeds .95036, counting I once.

For the ACTUAL short edge of cosine C, original farthest comparison gives $S_{16} \geq 4\sqrt{(C-1)/(C+1)}$. If $C > 21/2$, then $S_{16} > 3.635$. Pay .13 at P_{14}, adding $.13(P_{16} - P_{14})$. It is orthogonal to the complete first-fourteen reference and its first-six adjustment, with square $2(.13)^2$. ONE square yields

$$\|x\|^2 > .95036 + .26(3.635 - 3.311) - .0338 = 1.0008 > 1.$$

Consequently

$$C \leq 21/2 \tag{B16Icut}$$

for EVERY actual short opposite, before its own or foreign type is assigned.

Deleting the short root retains A_{15} and old I. The COMPLETE nonisolated white list is helper-three at $j = 1, 2, 14, 15$, or helper-four at $j = 1, 15$. For simple-three at two/fourteen, the retained D_{16} extension admits no finite old-I arm, so $\beta = 0$. At one/fifteen it is A_{16}, allowing $\beta = 0, 1/2, 1/\sqrt{2}$. For helper-four, an added old-I arm makes that four internal in a large path, so $\beta = 0$.

For simple-three write

$$c = A_c p - \gamma P_{16} - P_j/\sqrt{2} - \beta e, \qquad a = 2\sqrt{2}\gamma.$$

Positive own has $a \geq \sqrt{2}$, and

$$A_c^2 = \frac{16a^2 + 4ja + 4j - 8}{8} + \beta^2, \quad b_j = \frac{16a + 2j}{8\sqrt{2}}, \quad \delta_j = 1 - \frac{j(16 - j)}{32} - \beta^2, \quad C = b_j/\sqrt{\delta_j}.$$

Actual simple-three normalization keeps long roots and cap square two, short root square one, OVER K. The closed cycle and own two-cycle give

$$a \in \mathcal{O}_K, \qquad a^2/2 \in \mathcal{O}_K.$$

The connected old-B/cap Schur complement is $-(16a^2+4ja+4j-8)/4 < 0$. The Lorentzian-principal descent therefore gives ACTUAL $K = \mathbb{Q}(a)$, and

$$16a_\sigma^2 + 4ja_\sigma + 4j - 8 < 0.$$

Only AFTER this field identification is the rational branch excluded by the named whole-form Meyer/compact-anisotropy input. No field list or square-root extension was assumed.

Own zero at one/two has nonpositive time. At fourteen/fifteen its actual future null kernels are

$$n_{13} + 2n_{14} + 2n_{15} + \sqrt{2}n_{16} + c, \qquad n_{14} + 2n_{15} + \sqrt{2}n_{16} + c.$$

Their selected products vanish, time is positive through the cap, and every omitted original wall is nonpositive. Parallel own has its positive null pair. We may now treat positive own.

At $j = 1$, cutoff (B16Icut) gives $a < 53/10$, and the conjugate quadratic gives

$$-13/20 < a_\sigma < 25/64 < 2/5.$$

The nonzero integer $a^2/2$ is physically < 15, other values in $(0, 2/9)$. Degree at least three would give norm $< 15(2/9)^2 < 1$; degree is two. Its even nonzero norm $N(a)$ has magnitude < 4, hence is 2 or -2. For norm two, the small conjugate bound forces $a > 128/25$, and integral trace lies in $(5, 6)$. For norm minus two the integral trace is between two and six, and $\mathrm{Tr}(a^2/2) = (T^2 + 4)/2$ makes it even. Thus $T = 4$, $a = 2 + \sqrt{6}$. Positive β gives $\delta_j \le 9/32$, hence $C > 11 > 21/2$. At $\beta = 0$, $A_c^2 = (41 + 17\sqrt{6})/2$. The actual I invariant $J = 4(1 + h)$ has $4 < J < 6, 0 < J' < 4$. The full order $\mathbb{Z}[\sqrt{6}]$ leaves only $J = 3 + \sqrt{6}$: its positive radical coefficient is one, the physical strip leaves constants two/three, and conjugate positivity excludes two. The actual two-cycle

$$A_c^2(J - 4) = (61 + 24\sqrt{6})/2$$

is nonintegral. This excludes $j = 1$.

At $j = 2$, $\beta = 0, C = 4a + 1$, so $a \le 19/8$, while $-1/2 < a_\sigma < 0$. Degree at least three has norm < 1; degree two has nonzero even norm of magnitude $< 19/16 < 2$, impossible. At fourteen, positive own gives $C = 4a + 7 \ge 4\sqrt{2} + 7 > 21/2$. Its own-zero null was already printed.

At $j = 15$, $a < 18/5$ and $-12/5 < a_\sigma < -4/3$. For $z = a + 2$, $z^2/2$ is integral, physically < 18, other values $< 2/9$, hence degree is two. The trace of a is $0, 1, 2$; even norm and half-square trace parity leave zero or two. At trace two, $10/3 < a < 18/5$ gives $-6 < N(a) < -4$, whose sole integer is odd. At trace zero, a^2 is positive even integral, between $(4/3)^2$ and $(12/5)^2$, hence two or four. Four would be rational. Thus $a = \sqrt{2}, \gamma = 1/2$, $A_c^2 = (21 + 15\sqrt{2})/2 + \beta^2$. Positive β gives positive conjugate time since $43/2 - 15\sqrt{2} > 0$, contradicting the FULL old-B/I/cap Schur condition. At zero, the complete actual-I strip leaves $J = 3 + \sqrt{2}, 3 + 2\sqrt{2}, 4 + \sqrt{2}$. The actual two-cycle values are

$$(9 + 6\sqrt{2})/2, \quad (39 + 27\sqrt{2})/2, \quad (30 + 21\sqrt{2})/2,$$

all nonintegral in the full order.

For helper-four $\beta = 0$ and

$$A_c^2 = 16\gamma^2 + 2j\gamma + j - 1, \qquad C = 16\gamma + j.$$

At fifteen even own zero violates the cutoff. At one own zero has zero time; positive own has $\gamma \le 19/32$, leaving only native three. Its ACTUAL helper-four normalization keeps cap square one and old-short product $-1/2$; the whole connected old-B/cap Gram is rational Lorentzian. Thus ACTUAL $K = \mathbb{Q}$, excluded by the same whole-form Meyer input. All six nonisolated roles are exhausted.

The actual short cap is consequently

$$c_S = A_S p - \gamma P_{16} - \beta e, \quad B = 4\gamma^2, \quad A_S^2 = 4B + \beta^2 - 1.$$

Its actual finite endpoint has $\beta = 0$ or finite native < 1. Own zero has negative time; parallel has its positive null. The own two-cycle makes B integral. Its OVER-K normalization has raw square B, old-short product $-B/2$, and connected old-B/cap Schur complement $B - 4B^2 < 0$. Thus ACTUAL $K = \mathbb{Q}(B)$ and $0 < B_\sigma < 1/4$. A finite own has $B < 4$, giving nonzero integer norm < 1 in degree at least two; rational degree is already excluded after field descent. The own is dotted. The actual edge $C = 2\sqrt{B}/\sqrt{1 - \beta^2}$ now gives

$$B \in \mathcal{O}_K, \quad 4 < B \le \frac{441}{16}(1 - \beta^2), \quad 0 < B_\sigma < 1/4, \quad [K : \mathbb{Q}] = 2 \text{ or } 3. \tag{B16Istrip}$$

Degree at least four would have norm $< (441/16)(1/4)^3 < 1$. The cubic alternative is retained until its complete argument below.

C.12.3 The entire cubic $B_{16} + I$ short-cap branch

Assume degree three in (B16Istrip). Write its roots a, u, v, with a physical. Its nonzero positive norm satisfies $0 < auv < (441/16)(1/4)^2 < 2$, hence $auv = 1$ and $a > 16$. The minimal polynomial is

$$P(X) = X^3 - SX^2 + TX - 1, \qquad 17 \le S \le 28.$$

Because $u, v < 1/4 < a$, $P(1/4) < 0$; multiplying by sixty-four gives $4T - S \le 15$. At the root u,

$$0 = P(u) < -(S - 1/4)u^2 + Tu - 1.$$

The quadratic is positive there and has negative leading coefficient, so its discriminant is positive, giving $T^2 \ge 4S$. For $17 \le S \le 20$, the first bound gives $T \le 8$, contradicting $T^2 \ge 68$. For $21 \le S \le 24$, $T \le 9$ contradicts $T^2 \ge 84$. For $25 \le S \le 28$, $T \le 10$ forces $S = 25, T = 10$. Thus the COMPLETE cubic domain is

$$F(X) = X^3 - 25X^2 + 10X - 1.$$

Its signs at $0, 1/5, 1/4, 24, 25$ are $-1, 1/125, -3/64, -337, 249$. Its physical root lies in $(24, 25)$. Positive old-I contact would have $\beta \ge 1/2$, but then $a \le 1323/64 < 24$. Hence $\beta = 0$. The rational-root test proves irreducibility, and the polynomial discriminant is the squarefree $473 = 11 \cdot 43$. Therefore $\mathcal{O}_K = \mathbb{Z}[B]$, $\mathrm{N}(B) = 1$, $\mathrm{N}(4B - 1) = 3$. These are necessary algebraic data, not an example of an actual cap or chamber.

The ACTUAL I invariant $J = 4(1 + h)$ lies in this SAME field, with

$$4 < J_a < 6, \qquad 0 < J_u, J_v < 4.$$

Exact signs at $23/125, 369/2000$ and $441/2000, 111/500$ give

$$.184 < u < .1845, \quad .2205 < v < .222, \quad a > 24.5935, \quad v - u > .036.$$

The corresponding exact evaluations are

argument	numerator	denominator
23/125	−333	1953125
369/2000	2193409	8000000000
441/2000	1716121	8000000000
111/500	−144869	125000000.

Put $\eta = 1/B = B^2 - 25B + 10$. The integral bases $1, B, B^2$ and $1, B, \eta$ differ by determinant one. Thus write uniquely $J = l + mB + n\eta$. Since $auv = 1$,

$$\frac{J_a - J_u}{a - u} = m - nv, \qquad \frac{J_a - J_v}{a - v} = m - nu.$$

Both values lie in $(0, .247)$, because their numerators are in $(0, 6)$ and their denominators exceed 24.368. Hence $|n|(v - u) < .247$, so $|n| \le 6$. The COMPLETE integer check is as follows. For $n = 1, \ldots, 4$, the first inequality gives $m \ge 1$, the second $m < .985$. For $n = 5, 6$, it gives $m \ge 2$, while the second gives $m < 1.354$. At zero, $m \in (0, .247)$, impossible. For $n = -k$, $1 \le k \le 4$, the second inequality gives $m \ge 0$; $k \ge 2$ violates the first because $kv > .441 > .247$. At $k = 1$ only $m = 0$ remains. For $k = 5, 6$, both inequalities force $m = -1$; at five the second is negative, at six

the first exceeds $.323 > .247$. Thus only $m = 0, n = -1$ remains. Now $J = l - \eta$, with $l = 5, 6$ from the physical strip. Five violates $J_u > 0$, since $u < 1/5$. Consequently

$$J = 6 - 1/B.$$

The strip is solved completely; it was not falsely declared empty.

It forces $h = (2 - 1/a)/4 > 47/96$. The increasing relation between actual I energy floor r and height is $h = 4r/(1-r)^2$. At $r = .099$, $396000/811801 < 47/96$, with cleared positive margin 138647. Thus $e_I > .099$. The isolated short cap has $\beta = 0$, so its actual edge has $C = 2\sqrt{a} > 9$, giving $S_{16} > 7/2$. Use the UNLIFTED reference of energy $> .89026$, with individual P_{14} price $> .13$, and insert $.13(P_{16} - P_{14})$. This residual has square $2(.13)^2$ and full reference cross zero. The SAME original-vector square gives

$$e_B > .89026 + .26(3.5 - 3.311) - .0338 = .90560.$$

Add I ONCE: total > 1.00460. The entire cubic branch is impossible. Only the quadratic short-cap branch remains.

C.12.4 The complete quadratic $B_{16} + I$ parameter domain

Assume the quadratic branch of (B16Istrip). The ACTUAL I invariant $J = 4(1 + h)$ is integral, $4 < J < 6$, $0 < J' < 4$, and cannot be rational. Write its polynomial $X^2 - sX + m$. The full trace/norm conditions are

$$5 \le s \le 9, \qquad 6s - 36 < m < 4s - 16, \quad m \ge 1.$$

The COMPLETE nineteen pairs are $s = 5 : m = 1, 2, 3$; $s = 6 : m = 1, \dots, 7$; $s = 7 : m = 7, \dots, 11$; $s = 8 : m = 13, 14, 15$; $s = 9 : m = 19$. The square discriminants at $(6, 5), (7, 10), (8, 15)$ are not quadratic. The remaining sixteen actual-I models are as follows; $M = -\mathrm{N}(J(J-4)) > 0$:

K	J	M
$\mathbb{Q}(\sqrt{2})$	$3 + \sqrt{2},\ 4 + \sqrt{2},\ 3 + 2\sqrt{2}$	$7, 28, 7$
$\mathbb{Q}(\sqrt{3})$	$3 + \sqrt{3},\ 4 + \sqrt{3}$	$12, 39$
$\mathbb{Q}(\sqrt{5})$	$(7 + \sqrt{5})/2,\ (9 + \sqrt{5})/2,\ 3 + \sqrt{5}$	$11, 19, 16$
$\mathbb{Q}(\sqrt{6})$	$3 + \sqrt{6}$	15
$\mathbb{Q}(\sqrt{7})$	$3 + \sqrt{7}$	12
$\mathbb{Q}(\sqrt{13})$	$(5 + \sqrt{13})/2,\ (7 + \sqrt{13})/2$	$3, 27$
$\mathbb{Q}(\sqrt{17})$	$(5 + \sqrt{17})/2,\ (7 + \sqrt{17})/2$	$4, 32$
$\mathbb{Q}(\sqrt{21})$	$(5 + \sqrt{21})/2,\ (7 + \sqrt{21})/2$	$3, 35.$

This field list follows from an ACTUAL wall strip; it is not a global field census.

For the short-cap integer B, put $T = \mathrm{Tr}(B)$, $N = \mathrm{N}(B)$. Its physical/conjugate strip gives $N = 1, \dots, 6$, $T \le 27$, $T \ge 4N + 1$. For field discriminant d, write $T^2 - 4N = dk^2$. The full-order parity and small conjugate are exactly

$$T \equiv k \pmod 2 \quad (d \text{ odd}), \qquad T \equiv 0 \pmod 2 \quad (4 \mid d),$$
$$(2T - 1)^2 < 4dk^2 < 4T^2.$$

Comparing these bounded integer squares gives the COMPLETE table

d	k full range	(T,k)
8	$1,\dots,9$	$(6,2),(20,7)$
12	$1,\dots,7$	$(14,4)$
5	$1,\dots,12$	$(7,3),(18,8),(25,11)$
24	$1,\dots,5$	$(10,2),(20,4)$
28	$1,\dots,4$	$(16,3)$
13	$1,\dots,7$	$(11,3),(22,6)$
17	$1,\dots,6$	$(21,5)$
21	$1,\dots,5$	$(5,1),(23,5)$.

For $4 \mid d$, the even trace is at most twenty-six. All loop bounds are explicit integer inequalities; there is no Pell cutoff or unbounded coefficient search.

Here are the resulting fourteen models. Write $H = -\mathrm{N}(B(4B-1)) = N(4T-16N-1)$. The last column keeps ALL possible positive old-I costs $Q = \beta^2$:

$\sqrt{D}$	B	(T,N)	H	$Q>0$
$\sqrt{2}$	$3+2\sqrt{2}$	$(6,1)$	7	$1/4$
	$10+7\sqrt{2}$	$(20,2)$	94	$1/4$
$\sqrt{3}$	$7+4\sqrt{3}$	$(14,1)$	39	$1/4$
$\sqrt{5}$	$(7+3\sqrt{5})/2$	$(7,1)$	11	$1/4,(3+\sqrt{5})/8$
	$9+4\sqrt{5}$	$(18,1)$	55	$1/4$
	$(25+11\sqrt{5})/2$	$(25,5)$	95	none
$\sqrt{6}$	$5+2\sqrt{6}$	$(10,1)$	23	$1/4,1/2$
	$10+4\sqrt{6}$	$(20,4)$	60	none
$\sqrt{7}$	$8+3\sqrt{7}$	$(16,1)$	47	$1/4$
$\sqrt{13}$	$(11+3\sqrt{13})/2$	$(11,1)$	27	$1/4,1/2$
	$11+3\sqrt{13}$	$(22,4)$	92	none
$\sqrt{17}$	$(21+5\sqrt{17})/2$	$(21,4)$	76	none
$\sqrt{21}$	$(5+\sqrt{21})/2$	$(5,1)$	3	none
	$(23+5\sqrt{21})/2$	$(23,1)$	75	none.

For completeness, positive Q initially has $Q < 377/441$, so every native nine or larger is excluded. Native seven has cubic degree. Only labels $3,4,5,6,8$ remain; their own two-cycles must lie in K. Substituting their squares into the physical bound and FULL nonphysical Schur condition $4B'+Q'<1$ gives exactly the last column; in particular all six/eight contacts fail. The old I root has not been assigned a freely chosen raw length.

We eliminate EVERY positive Q. Let that old root's raw square be λ, while the old-B short root has square one. The short cap can be rescaled OVER K to square B, own product $-B/2$. If $Q>0$, its actual old-I product has square $B\lambda Q$, a K-square. The actual I opposite has raw square λJ and time square $\lambda J(J-4)/4$. Both actual cap times lie on the SAME K-vertex line. Taking their ratio and the known square $B\lambda Q$ proves the necessary FULL field-square condition

$$\frac{Q(4B+Q-1)}{J(J-4)} \in K^2.$$

No assumption that λ is a square is made.

For $Q = 1/4$, the rational norm requires $(48T - 256N - 9)/M$ to be a rational square. In field order $\sqrt{2}, \sqrt{3}, \sqrt{5}, \sqrt{6}, \sqrt{7}, \sqrt{13}$, its complete numerator list is

$$(23, 439), \quad 407, \quad (71, 599), \quad 215, \quad 503, \quad 263.$$

Against the complete M-table above, every ratio fails by odd prime valuation: seven for $\sqrt{2}$, three for $\sqrt{3}, \sqrt{6}, \sqrt{7}, \sqrt{13}$ (after cancelling the common five in 215/15); $71, 599$ match none of the squareclasses $11, 19, 1$. For $Q = 1/2$, the numerator is $8T - 64N - 1$. At $\sqrt{13}$ it is twenty-three, incompatible with $M = 3, 27$. At $\sqrt{6}$ the norm passes, but the FULL expression is $B/4$, $B = 5+2\sqrt{6}$. A square root of B would require rational square part three or two; neither is a rational square. The remaining golden $Q = (3 + \sqrt{5})/8$ has norm $1/16$, while $\mathrm{N}(4B + Q - 1) = -139/16$. It would require $139/M$ square for $M = 11, 19, 16$, impossible. Thus the short cap is PURE:

$$\beta = 0. \tag{B16Ibeta}$$

Its ACTUAL negative core $c_S + \gamma^{-1} n_{16}$ is future, zero on the cap, negative on the old short wall and nonpositive against every other original wall. With $u = 4/B \in (0, 1)$, it gives

$$V = P_{16} - u e_{16}, \quad x \cdot V \geq G = \sqrt{16 - u} - \sqrt{u(1-u)}, \qquad R_{\mathrm{tail}} = (V - P_{14})^2 = 1 + (1-u)^2.$$

Use the UNLIFTED reference w of energy $> .89026$. A signed first-coordinate demand $x_1 > X > b$ can be grafted in the first six block, preserving its sum. For EVERY $X \leq 1$ the remaining five coordinates exceed $.2036 > .197$; all coefficients stay nonnegative. The gain is $(6/5)(X - b)^2$. Pay .13 at P_{14} before inserting the actual short-foot pressure. Its residual is orthogonal to the full reference and first-six change. ONE square gives

$$\|x\|^2 > .89026 + r_0 + \frac{6}{5}(X - .3364)^2 + g_S, \qquad g_S = .26(G - 3.311) - .0169 R_{\mathrm{tail}}, \tag{B16Ifinal}$$

where $e_I > r_0$; omit the coordinate term if none is required.

The following COMPLETE elementary interval table seals the short term for the fourteen B-models, in their field order:

$\sqrt{D}$	u intervals	$G >$	$R_{\mathrm{tail}} <$	$g_S >$
$\sqrt{2}$	$(.6856, .6864); (.1998, .2012)$	$3.448; 3.572$	$1.099; 1.641$	.017
$\sqrt{3}$	$(.28704, .2872)$	3.511	1.509	.0264
$\sqrt{5}$	$(.5834, .584); (.2224, .224); (.16116, .1616)$	$3.432; 3.554; 3.610$	$1.174; 1.605; 1.705$	.0116
$\sqrt{6}$	$(.404, .4048); (.202, .2024)$	$3.458; 3.571$	$1.356; 1.637$	.0153
$\sqrt{7}$	$(.2504, .2516)$	3.53	1.563	.0305
$\sqrt{13}$	$(.3664, .367); (.1832, .1835)$	$3.470; 3.588$	$1.402; 1.668$	.0176
$\sqrt{17}$	$(.192, .19225)$	3.581	1.653	.0422
$\sqrt{21}$	$(.8348, .835); (.174, .175)$	$3.521; 3.598$	$1.028; 1.683$	.0372.

All entries follow by positive squares from

$$\sqrt{2} \in (1.4142, 1.4143), \quad \sqrt{3} \in (1.73205, 1.73206),$$
$$\sqrt{5} \in (2.2360, 2.2361), \quad \sqrt{6} \in (2.4494, 2.4495),$$
$$\sqrt{7} \in (2.6457, 2.6458), \quad \sqrt{13} \in (3.6055, 3.6056),$$
$$\sqrt{17} \in (4.1231, 4.1232), \quad \sqrt{21} \in (4.5825, 4.5826).$$

Bound $u(1-u)$ at the interval end nearest $1/2$. For example $\sqrt{17}$ uses $\sqrt{16-u} > 3.9758$, $\sqrt{u(1-u)} < .3941$, giving $g_S > .0422643$. The smaller $\sqrt{2}$ model gives $g_S > .0170469$. These are entire interval seals, not sampled optimizer outputs.

C.12.5 The quadratic $B_{16} + I$ head and final same-vector table

Deleting the actual long head retains B_{15} and old I. Its ONLY nonisolated finite extension is simple-three at long endpoint two. A positive old-I contact there is only three. Thus

$$c_H = A_H p - g_H \omega_1 - \omega_2/2 - \beta_H e, \quad A_H^2 = 2g_H^2 + 2g_H + Q_H, \quad Q_H = 0, 1/4.$$

Own zero with $Q_H = 0$ has zero time; with $Q_H = 1/4$ its COMPLETE positive old-B clearing is the nonzero future null $(p - e)/2$. It is feasible against all original walls. Own three has its affine triangle. Every other positive own gives $C_H \geq 1 + \sqrt{2}$, $x_1 > .643$. The legal first-six graft alone yields

$$e_B > .89026 + (6/5)(.643 - .3364)^2 > 1.$$

The actual head is isolated WITHIN B.

Write

$$U = 4g_H^2 \in \mathcal{O}_K, \quad A_H^2 = U/2 + Q_H - 1, \quad C_H^2 = \frac{U}{2(1 - Q_H)}.$$

Acuteness with the DISTINCT actual I opposite gives the useful complete foreign constraint

$$h(U/2 - 1) \geq Q_H.$$

Own zero/three/four is impossible by time and this constraint. Own six with $Q_H > 0$ contradicts $h < 1/2$; at zero it has its actual affine 6–3 path. Parallel has the positive null pair. The complete finite quadratic-own list therefore leaves only five/ten in $\mathbb{Q}(\sqrt{5})$, eight in $\mathbb{Q}(\sqrt{2})$, twelve in $\mathbb{Q}(\sqrt{3})$. Their necessary costs are $Q_H = 0$ for five, and $0, 1/4$ for the others. In the low-J $\sqrt{3}$ branch, twelve also has $Q_H = 0$, since $(\sqrt{3} - 1)\sqrt{3}/8 < 1/4$.

For a dotted head the actual old-long normalization has raw square $2U$, own product $-U$. Its connected Schur complement is $-U(U - 2) < 0$. Thus ACTUAL $K = \mathbb{Q}(U)$, $U > 4$, $0 < U' < 2$, and U is not rational. The short and head raw time squares, anchored in the SAME old B component, are

$$H_S^2 = B(4B - 1), \qquad H_H^2 = U(U - 2 + 2Q_H).$$

Subtracting their full K-valued old-forest projections leaves the SAME one-dimensional K-vertex line. Hence H_S^2/H_H^2 must be a FULL K-square. We do not compare the freely normalized I time after eliminating the short cap's old-I contact. For $z = a + b\sqrt{D}$ of norm q^2, a square requires one of $(a + q)/2, (a - q)/2$ to be a rational square; both signs will be checked. The actual short and head walls are distinct because only the latter has a positive old-head product.

For $\mathbb{Q}(\sqrt{2})$, finite-eight head time norms are $-4, -7/2$ at $Q_H = 0, 1/4$. Neither matches short norms $-7, -94$. The COMPLETE dotted U values below $5 + 3\sqrt{2}$ are

$$3 + \sqrt{2}, \quad 3 + 2\sqrt{2}, \quad 4 + 2\sqrt{2}.$$

Their pure time norms are $-7, -7, -32$. All fail against short norm -94 by odd forty-seven valuation. Against $B = 3 + 2\sqrt{2}$, the first ratio is $(43 + 30\sqrt{2})/7$, norm one, whose rational square parts $25/7, 18/7$ both fail. The last fails the norm test. At $U = B$ the FULL time-square test PASSES, so a different actual condition is required. For $s = \sqrt{2}$, raw norms are $2B, B$, future time product $27 + 19s$, spatial product $B^2/2$. Their raw mutual product $g = -(37 + 26s)/2$ has $g/B = -(7 + 4s)/2$; the ACTUAL two-cycle is

$$\frac{4g^2}{2B^2} = \frac{81 + 56s}{2},$$

nonintegral. For low $J = 3 + \sqrt{2}$, the foreign constraint excludes positive Q_H at the first two low U, with upper bounds $1/8, (3-\sqrt{2})/8$. At the next U, positive cost gives $C_H > 17/8$, $x_1 > .6$. The pure low values were all excluded, and the next pure U also gives that floor. For $J = 4 + \sqrt{2}$, only the first low U could prevent $x_1 > .49$; its pure case failed, and positive cost gives $C_H > 5/3, x_1 > 1/2$. Every larger U has the same stronger floor. At the high $J = 3 + 2\sqrt{2}$, no head floor will be required.

For $\mathbb{Q}(\sqrt{3})$, low $J = 3 + \sqrt{3}$ forces finite-twelve cost zero. Its time norm -3 mismatches short norm -39. The only dotted $U < 4 + 2\sqrt{3}$ is $3 + \sqrt{3}$. Its pure norm -12 also fails. Positive cost gives $C_H > 71/40$, hence $x_1 > .527$. All larger U give a stronger floor. The high-J branch needs no head floor.

In $\mathbb{Q}(\sqrt{5})$, finite-five time norm is -1; finite-ten norms are $-5, -5/4$. None matches short norms $-11, -55, -95$. The COMPLETE dotted $U < (9 + 3\sqrt{5})/2$ are

$$3 + \sqrt{5}, \quad 4 + \sqrt{5}, \quad (7 + 3\sqrt{5})/2.$$

Their pure time norms are $-16, -11, -9$. Only $U = 4 + \sqrt{5}$, smallest short $B = (7 + 3\sqrt{5})/2$, passes a norm test. Its time ratio is even a field square, but the actual raw products are

$$\text{time} = (69 + 31\sqrt{5})/2, \quad \text{spatial} = (43 + 19\sqrt{5})/4, \quad g = -(95 + 43\sqrt{5})/4.$$

With raw norms $2U, B$, $\mathrm{N}(U) = 11, \mathrm{N}(B) = 1, \mathrm{N}(g) = -55/4$. The ACTUAL two-cycle has norm $275/4$, not integral. At the lowest $U = 3 + \sqrt{5}$, the foreign constraint permits only $Q_H = 0, 1/4, 1/2, \varphi^2/4$. Their time norms are $-16, -11, -4, -19$. Only $Q_H = 1/4$, smallest short B, passes the norm test. Its field ratio $(103 + 45\sqrt{5})/22$ has norm one and rational square parts $125/44, 81/44$, both nonsquares. Thus the lowest U is impossible with EVERY foreign cost. At each of the next two low U, positive cost gives $x_1 > .57$; their pure cases failed above. Every larger U has that floor even at zero cost. Thus $x_1 > .57$ throughout the golden field.

The other fields have no finite head among five/eight/ten/twelve. Their complete low dotted cases are

$\sqrt{D}$	U requiring a test	$\mathrm{N}(H_H^2)$, $Q_H = 0$	$-\mathrm{N}(H_S^2)$	consequence
$\sqrt{6}$	none; $U \geq 3 + \sqrt{6}$	$--$	$23, 60$	$x_1 > .49$
$\sqrt{7}$	none needed	$--$	47	no head floor
$\sqrt{13}$	$(5 + \sqrt{13})/2$	-9	$27, 92$	high J : $x_1 > .48$
$\sqrt{17}$	$(5 + \sqrt{17})/2$, $(7 + \sqrt{17})/2$	$-8, -16$	76	$x_1 > .51$
$\sqrt{21}$	$(5 + \sqrt{21})/2$	-5	$3, 75$	$x_1 > .49$.

The displayed norm ratios are nonsquares $1/3, 9/92, 2/19, 4/19, 5/3, 1/15$. Positive Q_H at a listed low U already gives its required coordinate floor. The next U in the last three fields are respectively $(7 + \sqrt{13})/2, 5 + \sqrt{17}, (7 + \sqrt{21})/2$; these suffice at zero cost and strengthen with cost. All these finite low-U lists are complete by writing $U = (T + k\sqrt{d})/2$ with full-order parity, $U > 4, 0 < U' < 2$: the stated physical threshold bounds $k > 0$, and the conjugate width gives the listed constants. For $\sqrt{6}$, $C_H > 80/49$ gives $x_1^2 > 31/129 > .49^2$. At $\sqrt{21}$, positive cost gives $C_H^2 > 8/3 > (80/49)^2$. The low-J $\sqrt{13}$ case instead uses the direct actual-I pressure $t > .5905 > .59$, from $h < .076$. No unbounded native or head-coefficient domain remains outside these monotone upper ranges.

Insert these ACTUAL floors into the single reference (B16Ifinal). Each r_0 follows by the increasing identity $h = 4r/(1-r)^2$ and the positive rational radical brackets already printed. The

COMPLETE scalar dispatch is

$\sqrt{D},\ J$ branch	r_0	X	$g_S >$	total $>$
$\sqrt{2},\ 3+\sqrt{2}$	.024	.6	.017	1.014641952
$\sqrt{2},\ 4+\sqrt{2}$	.075	.49	.017	1.010571552
$\sqrt{2},\ 3+2\sqrt{2}$	.093	$--$	.017	1.00026
$\sqrt{3},\ 3+\sqrt{3}$	.041	.527	.0264	1.001254032
$\sqrt{3},\ 4+\sqrt{3}$	.089	$--$	.0264	1.00566
$\sqrt{5},$ all	.035	.57	.0116	1.002342752
$\sqrt{6}$	.076	.49	.0153	1.009871552
$\sqrt{7}$	.085	$--$	.0305	1.00576
$\sqrt{13},$ low	.018	.59	.0176	1.003035552
$\sqrt{13},$ high	.070	.48	.0176	1.002605152
$\sqrt{17},$ all	.032	.51	.0422	1.000624352
$\sqrt{21},$ all	.045	.49	.0372	1.000771552.

A dash means that the coordinate gain is OMITTED, not charged again. Every row strictly exceeds one. All old-I contacts, actual head roles, short models, field-square exceptions and positive time signs were retained. The quadratic branch is impossible. Together with the complete cubic and six-role reductions, this excludes the entire $B_{16}+$ ACTUAL-I rank-seventeen forest on ONE original vector.

C.12.6 The ordinary $D_{16}+A_1$ branch and its shared coefficient

The WHOLE forest is $D_{16}+A_1$, with the actual rank-one edge NONSHORT: $L_e \geq \operatorname{arsinh}(1/\sqrt{2})$. This is the ordinary hypothesis, not a property of a bare A_1 diagram. Use genuine unit D-roots, centre fourteen, forks fifteen/sixteen, signed last coordinate, and orthogonal old unit root e. Write the original vector as (x,y), with $y>0$ and $\|x\|^2+y^2<1$. The actual nonshort edge gives

$$y^2 \geq 5-2\sqrt{6} > .1, \qquad y > s_0 = \sqrt{.1}, \quad .316 < s_0 < .317.$$

We will keep this ONE external coordinate inside any mixed head/centre square. For an opposite retaining e, its entire foreign contact is β, $Q=\beta^2<1$. Positive Q is native and at least $1/4$, with no value strictly between $1/4$ and $1/2$.

The generic $M=10$ bulk applies through eleven at the ACTUAL D_{16} rank. Its only weak role is pure far helper, removed from six onward by the arithmetic literal-helper lemma. If five is weak, its actual pure point and strong six/eleven give the positive interpolation energy $>.9104$ already proved above. Adding the nonshort energy $>.1$ is impossible, so five is strong.

The FRESH D_4, D_3 layers are twelve, thirteen. Their full finite lists retain all one-sided foreign contacts. Full D_{16} or long-arm A_{16} bridges admit no extra old-A_1 arm; the latter would create inadmissible branch arms, the former another branch. Own zero/three/four/six/parallel have their actual affine/null or direct strong alternatives; end-four and sealed H_4 are strong. Positive $Q \geq 1/4$ strengthens the weak fork-three/end-three ratios enough to make own-five strong. At $Q=0$, the full all-field helper norm/trace seal leaves only golden own.

For those two golden packets, the ACTUAL LEFT H_4 correction is $g(3n_i/2+n_{i-1}+n_{i-2}/2)$, $g=\varphi/2$, with $\alpha^2=\varphi+5/8>56/25$, $\delta_5=(5-3\varphi)/8$. Its finite PROJECTED POINT, positive and all-original-wall feasible, gives pure displacement V, flat through $i-3$, $V^2<i$, pressure

$G > \sqrt{i - \kappa/\alpha^2} - .091$. The two complete payments are

	i	K	μ	$G >$	total $>$
D_4	12	9	1/6	3.34	$.79805 + .1 + (3.34 - 2.569)/3 - 3/36 > 1.0717$
D_3	13	10	.16	3.47	$.82436 + .1 + .32(3.47 - 2.732) - .0256(3) > 1.0837.$

Here $\kappa = 1/2, 5/8$, respectively. The available terminal slopes are $d_8 > 1/6, d_9 > .16$; each is paid BEFORE the actual pressure. The root seals are $3.431^2 < 12 - 25/112$, $3.561^2 < 13 - 125/448$. Every such packet is impossible. Thus ALL five through thirteen are actually strong, with

$$E_{11} > .84816, \quad E_{13} > .88989, \quad b = f_5/5 < .361, \quad P_5 \text{ price} > .14.$$

At the reference ending eleven, P_{11} price $> 1/7$, $f_{11} < 2.886$. The scalar $E_{14} > .9084$ is not yet an available centre floor.

Deleting the actual head retains D_{15} and old e. Its only nonisolated finite role is simple-three at long leaf two. The exact formula is

$$A_H^2 = 2g^2 + 2g + Q, \qquad C_H = (2g + 1)/\sqrt{1 - 2Q}.$$

Own zero at positive Q has the ACTUAL future null

$$c + \omega_2/2 = \sqrt{Q}\, p - \beta e, \qquad \omega_2/2 = n_1 + 2(n_2 + \cdots + n_{14}) + n_{15} + n_{16}.$$

All omitted wall signs are nonpositive. Own-three has its affine triangle. Other positive owns give $x_1 > .643$; the legal first-five graft gain $> .099405$, together with $E_{13} + .1$, exceeds one. The head is isolated WITHIN D, so

$$A_H^2 = 2g^2 + \beta_H^2 - 1, \qquad C_H^2 = 2g^2/(1 - \beta_H^2).$$

Own zero has negative time. For own three at positive time, its foreign shadow has $\beta_H/A_H > 1$, making it also close the orthogonal rank-one edge, impossible. Own four is zero time if pure, forbidden shadow equality if mixed. Own six/parallel has its actual affine/null principal with every foreign contact present. Only own five, finite at least seven, or dotted remains.

We record a compatible legal head gain

$$\eta_H = .0059. \tag{D16OH}$$

Every head residual is zero-sum on the first five coordinates. For a PURE finite own at least seven or dotted, the actual first-three-root point and FULL scalar (R1S) give gain $> .2893 - .8(.361) + .032 > .0325$ after paying .08 at P_5. No short or fork normal is used. For pure native five, the ACTUAL finite H_4 point has

$$x \cdot (P_4/4) \geq H_H = \frac{1}{2\sqrt{\varphi}} > .393.$$

Indeed its corrected square is $\delta_H = (5 - 3\varphi)/8$, time square $a = (\varphi - 1)/2$; the finite-point formula $H(A, \delta) = A/(1 + \sqrt{\delta/(A^2 + \delta)})$ divided by $\sqrt{2}g$ simplifies to the displayed $1/(2\sqrt{\varphi})$. Its strict rational seal follows from $\varphi > 1.618$. Pay .02 at P_5 for $.1(P_4/4 - P_5/5)$, of square .01/20. The gain is $> .2(.393 - .361) - .01/20 = .0059$.

For mixed finite own at least seven or dotted, $\beta_H \geq 1/2, g > .9$ gives $C_H^2 > (8/3)(.9)^2 > 1.46^2$, hence $x_1 > .43$. Its legal fixed-sum first-five graft gains $> (5/4)(.43 - .361)^2 = .00595125$. It has no external coordinate payment.

For mixed native five keep that coordinate. The ACTUAL H_4 point gives

$$x \cdot (P_4/4) + d_H y \geq H_H, \quad d_H = \frac{\beta_H}{\sqrt{2}g} < .875, \quad H_H = \frac{H(A_H, \delta_H)}{\sqrt{2}g}.$$

Here $A_H^2 = a + \beta_H^2$, $a = (\varphi - 1)/2$. The function $\sqrt{a + \beta^2} - \beta s_0$ increases for $\beta \geq 1/2$. Using $1.618 < \varphi < 1.619$ gives

$$A_H(1/2) > .7476, \quad \sqrt{\delta_H} < .136, \quad \sqrt{2}g < 1.145, \qquad H_H - d_H s_0 > \frac{.7476 - .136 - .5(.317)}{1.145} > .393.$$

Pay .02 at P_5 AND $.1 d_H$ from the positive external coefficient s_0, retaining final coefficient s_0. The ONE ambient completed square gives the SAME gain .0059. A later centre pressure with coefficient $1/7$ and external factor $d_C < .875$ spends $< .125$ more. The combined external payment is $< .0875 + .125 = .2125 < s_0$. Thus this head gain is LEGALLY compatible with that later mixed centre, without spending the rank-one energy twice.

C.12.7 The complete ordinary D_{16} centre

Deleting centre fourteen retains A_{13}, both ACTUAL fork roots and old e. Its COMPLETE finite-white list is: without fork, isolation or admissible endpoint/D-position three or endpoint four on the left; left endpoint-three with one fork-three/four; the full all-three endpoint/two-fork bridge; or no left with one arbitrary finite fork contact, or fork pairs $(3,3), (3,4), (3,5)$. An internal attachment plus fork gives inadmissible arms; a four internal to a long path is nonfinite. Rank excludes exceptional completions. Write the actual residual and ratio as

$$\Delta = 1 - q_L - \delta^2 - \epsilon^2 - Q > 0, \quad \sigma = g + \eta_L j/14 + (\delta + \epsilon)/2, \quad C_{14} = \sqrt{28}\sigma/\sqrt{\Delta}.$$

The sufficient .61 ratio target is $< 25/16$.

With no fork and a left contact, own-zero far-three requires $1/2 < Q < 15/28$, containing no native foreign square. Far D-position and far-four require $Q < 1/7, 1/14$, hence zero and zero time. Other own-zero roles have the actual affine fork/terminal-four principals. Own-three/white-three has the actual cycle; own-four the star at fourteen with cap and both forks; own-six its 6–3 principal; parallel its null. Positive endpoint-four, D-position and near-three are directly strong. Only PURE far-three can be weak, since the complete foreign-far clearing is strong at this index. Lemma 20.3 excludes that pure role at fourteen.

For isolation within D, $A_c^2 = 28g^2 + Q - 1$. Own zero is negative time. Small owns and parallel have the actual four-leaf/star/chain/null witnesses. For $Q > 0$ or dotted own, keep $u = (1 - Q)/g^2$. At $0 < u < 1$, the actual positive clearing $c + \beta e + g u n_{14}$ gives PURE-D pressure

$$h_{14} > \sqrt{28 - u} - \sqrt{u(1 - u)}.$$

Its full loss is $< .1u + \sqrt{u(1-u)} < .5525 < .61$. At one the actual clearing is the forbidden future null. The finite mixed interval $u > 1$ MUST also be retained: it occurs only for own-five and $Q = 1/4$. The ACTUAL finite projection onto c, n_{14}, e, with positive corrections on all three roots, gives instead

$$h_{14} > \sqrt{28 - u} - \sqrt{u - 1}.$$

Here $u \leq 3/\varphi^2 = 6 - 3\varphi < 7/6$, $\sqrt{u-1} \leq 2 - \varphi < .382$. The first loss is $< (7/6)/10 < .117$, so total loss is $< .499 < .61$. The projected POINT, not its spacelike correction, provides the actual-wall comparison. Pure native five uses the left H_4 with loss $< .15 + .168$; pure native seven the left A_2

with loss $< .124 + .319$; every finite native at least eight the finite pair with loss $< .118 + .415$. All omitted original walls remain nonpositive by the same positive actual-root constructions. Thus every isolated own/foreign parameter is strong or impossible.

For left/fork bridges, full two-fork D_{16} is foreign-sealed and strong at own zero. Left/fork-four is sealed, residual $1/28$, and directly strong. Left/fork-three has pure residual $2/7$, own-zero cycle, own-three triangle and own at least four directly strong. Its POSSIBLE old-A_1-three arm is retained: $Q = 1/4$ leaves residual $1/28$ and strengthens the bound. No larger finite foreign label fits that long branch. This row is not incorrectly called sealed.

With no left and positive own, own-three/ fork-three has its affine triangle; own-three/fork at least six is strong. Own-three/fork-four or five is strong with $Q > 0$; its pure case is recognized below at the actual reciprocal fork. Own-four has the star on cap, old thirteen and the UNCONTACTED fork; own-six uses the preceding left node, and parallel its null. For own at least five and fork at least four, the direct ratio is $\geq \varphi/\sqrt{2} + 1/2 > 25/16$. Fork-three/own-five with $Q \geq 1/2$ is directly strong. For $Q = 0, 1/4$ take its ACTUAL left-H_4 correction, with $\alpha^2 = \varphi + 5/8$. At zero, its finite POINT is pure D, flat through eleven, pressure $G > 3.38$, squared norm < 14. At $Q = 1/4$, the positive FUTURE core

$$c + g(3n_{14}/2 + n_{13} + n_{12}/2) + \beta e$$

has square and cap product $\delta_5 - 1/4 < 0$, zero selected old and e-products, and all omitted wall products nonpositive. It clears e to the SAME pure D-displacement. Its pressure satisfies

$$G > \sqrt{14 - 1/(2\alpha^2)} - .322 > 3.38.$$

Use $\sqrt{14 - 25/112} > 3.711$ and $(1/4 - \delta_5)/\alpha^2 = 3(\varphi - 1)/(8\varphi + 5) < .322^2$. Paying $1/7$ at P_{11} gives

$$.84816 + .1 + \frac{2(3.38 - 2.886)}{7} - \frac{3}{49} > 1.028.$$

No head gain is needed for these rows. For every other own finite at least seven or dotted, positive Q is strong; $\gamma \geq 6/5$ is strong. The remaining PURE fork-three role has $202/125 < 2\gamma < 12/5$; the full all-field fork-helper proof in Lemma 20.3 leaves only the already excluded golden own-five.

With no left and own zero, single fork-three has the actual E_8 kernel: centre fourteen six, uncontacted fork three, contacted fork four, cap two, and old thirteen through nine coefficients five through one. Single fork-four has kernel $(\sqrt{2}, 2, 2, 1, 1)$ on cap, contacted fork, fourteen, thirteen, other fork. Single fork-six has the 6–3 principal. Every kernel has positive cap time and correct signs at ALL omitted walls. Fork-five allows exactly $Q = 0, 1/4$ and remains below. Every finite fork label at least seven has $\delta^2 > .81$; retained-white positivity forces $Q = 0$. Two forks/own-three has its actual triangle, own at least four is strong. Own-zero pair-three/three has the affine square; pairs-three/four and three/five have $Q = 0$. These are all remaining finite, native, parallel, dotted and foreign roles.

C.12.8 Ordinary D_{16}: the mixed finite point is paid jointly

For own-zero/single-fork-five, put $\delta = \varphi/2, \alpha = \delta/\sqrt{2}$. Select the ACTUAL H_4 on cap, contacted fork, fourteen, thirteen, with positive correction $\delta(3n_f/2 + n_{14} + n_{13}/2)$. Its finite projected POINT has divided D-displacement

$$V_D = (1^{12}, 1/2, 1/2, 1/2, \ \pm 1/2), \qquad V_D^2 = 13.$$

The sign is the genuine fork sign. At $Q = 0$, its demand is $G > \sqrt{8\varphi} - \sqrt{13 - 8\varphi} > 3.36$, from $\varphi > 1.618$. Pay $1/7$ at P_{11}; the residual norm is $13 - 11 = 2$. With $E_{11} + .1$ alone the total exceeds 1.0427.

At $Q = 1/4$, KEEP the actual external coordinate in the SAME point:

$$x \cdot V_D + d_C y \geq G, \qquad d_C = \sqrt{2}/\varphi < .875, \qquad G = \frac{\sqrt{4+6\varphi}}{1+\sqrt{(13-8\varphi)/(17-2\varphi)}} > 3.48.$$

The numerator exceeds 3.7024, the inner square root is $< .0638$, and $3.48(1.0638) = 3.702024 < 3.7024$. Its TOTAL spatial norm is $17 - 2\varphi = 13 + d_C^2$. There is no artificial pure-D demand for this mixed point.

Let w be the ordinary reference ending eleven. Take the applicable head insertion/change above and centre insertion $(V_D - P_{11})/7$. The FINAL external coefficient is s_0. BEFORE using either mixed pressure, pay $d_C/7 + \mu_H d_H$ from that coefficient, and pay their P_{11}, P_5 prices from w. At mixed-head-five $\mu_H = .1$; the total external debit is $< .2125 < s_0$. Pure or fixed-sum head changes require no external debit. The P_{11} price remains positive, and its P_5 payment is independent. The head residual has zero sum on the first five coordinates; V_D is flat through twelve, so their FULL D-cross is zero. The single ambient completed square gives

$$\begin{aligned}\|x\|^2 + y^2 &> E_{11} + .1 + .0059 + \frac{2(G - d_C s_0 - f_{11})}{7} - \frac{2}{49} \\ &> .84816 + .1 + .0059 + \frac{2(3.48 - .875(.317) - 2.886)}{7} - \frac{2}{49} > 1.0037.\end{aligned}$$

The external component appears ONCE as the positive $s_0 e$ reference with its paid residual coefficient. No separate $y^2 > .1$ is added to a mixed ambient square that already spent it.

C.12.9 Ordinary D_{16}: every remaining finite fork and the final floor

For own zero and ANY finite fork label at least seven, put $u = \delta^{-2} \in (1, 100/81)$; $Q = 0$ was proved. The ACTUAL positive core

$$c + \delta^{-1} n_f + 2(\delta^{-1} - \delta) n_{14}$$

has cap/fork products zero, centre product $(3 - 4\delta^2)/(2\delta) < 0$, negative square and positive time. Its product with the OTHER fork is negative in each signed orientation; every other original wall is nonpositive. It gives pure pressure

$$\begin{gathered}V = (1^{13}, 3 - 2u, u - 1, \ \pm(1 - u)), \\ x \cdot V \geq G = \sqrt{16 - 2u} - \sqrt{-8 + 14u - 6u^2}, \\ R = V^2 = 24 - 16u + 6u^2.\end{gathered}$$

On the ENTIRE interval $14G - R > 32.51$. For a direct continuous proof, the chord $\sqrt{16 - 2u} \geq 4.011 - .27u$ has positive endpoint-square margins: at one $3.741^2 < 14$, and at $100/81$ its margin is $456199/81000000$. Set $z = u - 7/6$. Then the second radicand is $1/6 - 6z^2$, $R = 27/2 - 2z + 6z^2$, and $\sqrt{1/6 - 6z^2} \leq 1/\sqrt{6} - 3\sqrt{6} z^2$. Consequently

$$14G - R > 32.528 - 1.78z + 96.8748 z^2 > 32.528 - .009 > 32.51.$$

Use $\sqrt{6} > 2.4494$, $14/\sqrt{6} < 5.716$, and $1.78^2 < .036(96.8748)$. Paying $1/7$ at P_{11} gives, WITHOUT head credit,

$$.84816 + .1 + \frac{32.51 - 14(2.886) + 11}{49} > 1.0115.$$

This excludes the entire finite-label range, without sampling native cosines.

For pure own-three/single-fork-four or five, the SAME actual cap closes that fork: retained A_{15} plus cap is finite D_{16}. For own-zero/pair-three/four, releasing the fork-three direction gives finite B_{16}. Keep old e throughout. Positive inverse finite projection gives an actual future endpoint, strict against all other original walls; the omitted fork is strict through its positive cap contact. The exact fork coshes are $7+8\delta>12$ or $7+4\sqrt{2}>12$. The genuine signed direction $V=P_{15}\pm e_{16}$, $V^2=16$, has demand $G>\sqrt{176/13}>3.67$. Payment $1/7$ at P_{11} gives

$$.84816+.1+\frac{2(3.67-2.886)}{7}-\frac{5}{49}>1.0701.$$

Own-zero/pair-three/five uses its ACTUAL finite H_3 projection, with positive correction $n_{f_3}/2+(\varphi/2)n_{f_5}$. Its pure normalized displacement is EXACTLY P_{14}. For $q=2\varphi^{-6}<1/8$, its demand is

$$x\cdot P_{14}\geq\frac{\sqrt{14-q}}{1+\sqrt{q/14}}>3.4.$$

Use $\sqrt{111/8}>3.7249$, $\sqrt{1/112}<.095$, and $3.4(1.095)=3.723$. The reference ending eleven with payment $1/7$ gives

$$.84816+.1+\frac{2(3.4-2.886)}{7}-\frac{3}{49}>1.0337.$$

All remaining centre caps are therefore strong or impossible. The full actual curve through fourteen is now available, giving

$$\|x\|^2+y^2>E_{14}+.1>.9084+.1>1.$$

The ENTIRE ordinary/nonshort $D_{16}+A_1$ alternative is excluded. The short actual-I alternative has not received the ordinary coefficient s_0e or its nonshort bootstrap. It requires its own proof next.

C.13 The full actual-I D_{16} case

C.13.1 The actual-I D_{16} foundation

The final classical consumer has ENTIRE forest $D_{16}+I$, with the ACTUAL isolated short opposite $c_I=\sqrt{h}\,p-\sqrt{1+h}\,e$, $0<h<1/2$. Its original parameters are $y^2\geq q(h)$, $x_1\geq t$, and $t+(8/5)\sqrt{q(h)}>.788$. No ordinary rank-one cost is assigned.

At the actual rank sixteen the generic bulk gives strong six through eleven. The D_4, D_3 sites twelve/thirteen have the SAME complete local inventories and actual left-H_4 data proved in the preceding ordinary section. Those geometric lists and points do not use its nonshort energy. All foreign old-I costs are retained; the only further weak rows are PURE golden own-five. Their actual pressures $G>3.34, 3.47$ are flat through nine/ten, with full norms below twelve/thirteen.

If five is strong, its ordinary positive reference and the actual-I coordinate have one joint initial-block gain $>.05426$. Here is its precise payment. The first-five mean is $b<.361$; graft it to $\min(.6,\max(b,t))$. The other coordinates remain above the next slope. At $t\geq.6$, the block gain alone is $>(5/4)(.6-.361)^2>.05426$. Otherwise set $z=(t-b)_+$. Then $z+(8/5)\sqrt{q(h)}>.427$, so weighted Cauchy gives joint gain

$$>\frac{.427^2}{4/5+64/25}>.05426.$$

The head is not used. The separately paid actual left-H_4 packets give totals

$$.79805 + .05426 + (3.34 - 2.569)/3 - 3/36 > 1.0259,$$
$$.82436 + .05426 + .32(3.47 - 2.732) - .0256(3) > 1.0379.$$

Their residuals vanish on the initial block, so neither overlaps its change. If five is weak, its PURE actual point gives $x_1 > .5965, S_6 > 2.01815$. Positive interpolation gives $L_9 > .8610, L_{10} > .8870$; their terminal prices remain $> 1/6, .16$. The SAME actual packets then have pure D-energies $> 1.0346, 1.0463$, without adding any I energy. Thus six through thirteen are strong in BOTH branches before five, head or centre is concluded. At weak five specifically,

$$L_{13} > .7600299945 + \frac{(\sqrt{13} - \sqrt{6})^2}{7} > .95095.$$

We must exclude weak five arithmetically at its actual local rank. It is the literal PURE far helper, with integral $t_5 = 2g$, $202/125 < t_5 < 12/5$, and every other conjugate satisfies

$$5(t_5')^2 + 2t_5' - 1 < 0, \qquad -.69 < t_5' < .29.$$

Degree at least four has norm magnitude $< (12/5)(.69)^3 < 1$. In degree three a positive small conjugate would make it $< (12/5)(.69)(.29) < 1$. Both are negative, with product $1/t_5$, and integral trace one. Indeed the trace is positive and by AM–GM is < 2. Reality of the two small roots requires $t_5(t_5 - 1)^2 \geq 4$, forcing $t_5 > 2.3$. The integer second coefficient $1/t_5 - t_5^2 + t_5$ then lies in $(-883/300, -5877/2300) \subset (-3, -2)$, impossible. In degree two the norm is -1; positive norm would have magnitude $< (12/5)(.29) < 1$. Its trace $t_5 - 1/t_5 \in (0, 2)$ is one, hence $t_5 = \varphi$. Rational two is the actual parallel null. The PURE connected Lorentzian principal therefore forces ACTUAL $K = \mathbb{Q}(\varphi)$.

The weak-five time is $A_5^2 = (4 + 7\varphi)/2$. The actual I own invariant has $J = 4(1 + h) \in \mathcal{O}_K$, $4 < J < 6, 0 < J' < 4$. The actual two-cycle makes $A_5^2(J - 4)$ integral. Since $\mathrm{N}(4 + 7\varphi) = -5$, this factor is invertible modulo two, so $z = (J - 4)/2 \in \mathbb{Z}[\varphi]$. Its full strip is $0 < z < 1, -2 < z' < 0$. Its norm is -1, trace -1 or zero; zero would give the forbidden positive root one. Thus $z = \varphi - 1$, $h = (\varphi - 1)/2 > .3$. The actual I energy then exceeds .06, from $(53/47)^2 < 13/10$. Adding it ONCE to $L_{13} > .95095$ contradicts the original norm. Five is therefore strong. Every individual floor five through thirteen is available before the head.

The head's COMPLETE finite-white list is isolation or simple-three at long leaf two, with all old-I costs kept. For the nonisolated row, $A_H^2 = 2g^2 + 2g + Q$, $C_H = (2g + 1)/\sqrt{1 - 2Q}$. Own zero at positive Q has $c_H \cdot c_I = \beta(\sqrt{1 + h} - \sqrt{h}) > 0$, contrary to acuteness; at zero cost time is zero. Own three has its affine triangle. The first-six-linear reference through thirteen has energy $> .87175$, mean $< .3364$, and any first-coordinate lift $a < 1$ is legal. Positive Q gives $x_1 > .73$; pure own at least five gives $x_1 > 2/3$. Either lift alone makes its energy > 1. The remaining nonisolated own-four, $Q = 0$, gives $x_1 > .643$ and $A_H^2 = 1 + \sqrt{2} < 5/2$. Its actual mutual with c_I forces $A_H^2 h \geq 1/4$, hence $h > .1$ and $y^2 > .023$. The single graft and this disjoint energy total exceed 1.0075. Thus the head is isolated within D.

Write it as

$$c_H = A_H p - g\omega_1 - \beta e, \quad a = 2g^2 - 1, \quad A_H^2 = a + Q, \quad ah \geq Q.$$

Own zero/three/four has nonpositive time or violates the actual acuteness condition. Own six/parallel has its all-wall affine/null principal. Native five is impossible: at positive cost $ah < Q$, while pure time gives the forbidden positive actual mutual $\sqrt{ah} < 1/2$. A remaining finite head has

$a < 1$; at positive cost $h > Q \geq 1/4$, and at zero cost the actual native gap gives $ah \geq 1/4$. Thus EVERY finite head forces $h > 1/4$.

For a mixed head, or pure dotted head, $u = (1-Q)/g^2 \in (0,1)$. The ACTUAL positive clearing $c_H + \beta e + gun_1$ is future, negative norm, and feasible against EVERY original wall including c_I. It gives PURE-D pressure

$$Z = (1-u/2)e_1 + (u/2)e_2, \quad x \cdot Z \geq H = \frac{\sqrt{2-u} - \sqrt{u(1-u)}}{\sqrt{2}}, \quad R = 1 - u + u^2/2.$$

At $\mu = .36$, its individual P_5 debit is $.072 < .14$. Its full-reference gain

$$G_H = .72(H-b) - .1296(R - 1/5) > .04 \tag{D16IH}$$

holds on the ENTIRE interval. For clarity the three chord seals are

u interval	$\sqrt{2-u}$ lower chord	R upper chord
$[0, 1/2]$	$1.4142 - .379u$	$1 - .75u$
$[1/2, 3/4]$	$1.4381 - .4268u$	$.8125 - .375u$
$[3/4, 1]$	$1.4721 - .4721u$	$.625 - .125u$.

The endpoint-square comparisons are positive. Using $.5091 < 9\sqrt{2}/25 < .5092$, substitution gives lower expressions

$$.35636922 - .0958u - .5092\sqrt{u(1-u)},$$
$$.39283671 - .16873u - .5092\sqrt{u(1-u)},$$
$$.43444611 - .2242u - .5092\sqrt{u(1-u)}.$$

The circle inequality gives the following bounds:

square-root upper bound	gain lower bound
.5182	.04936922
.5365	.04022171
.5564	.04414611.

This proves the full scalar without sampling own parameters or spending I.

For a PURE finite head use the actual first-three-root finite pressure already proved with (R1S). Its native-seven/eight seals give gains $> .0325, .0345$. For every finite native at least nine, $\gamma > .939$, $1 < u < 1.135$; the finite scalar decreases with u on this range. At its upper endpoint $\sqrt{2-u} > .93$, the other radical is $< .284$, $R < .393$, so gain is $> .0456$. The actual mutual with c_I gives the compatible complete floors

finite own	$h >$	$y^2 > q_0$	$\eta_H >$
7	2/5	.083	.0325
8	7/20	.074	.0345
all ≥ 9	1/4	.055	.0456.

These heads and the cleared head each use only a zero-sum first-five residual, paid at P_5, and have no I component. They are orthogonal to every centre residual flat through eleven.

C.13.2 Actual $D_{16}+I$: the height reduction and unconditional centre

First suppose $h \geq .23$. Then $y^2 > .0516$, with $\sqrt{.0516} < .228$. Use either that fixed external reference and the cleared-head gain $> .04$, or a pure-finite-head row of the last table. In every case its total $q_0 + \eta_H$ is at least .0916, and its external coefficient $u_0 = \sqrt{q_0} > .22$. The entire SAME-rank centre inventory, actual points and all original-wall proofs are already supplied in Sections C.12.7–C.12.9. Those geometric proofs did not use nonshortness until their coefficient accounting. They apply here with every actual old-I contact retained, including the finite mixed isolated $u > 1$ and the unsealed left/fork-three arm.

Their complete nondirect point data are

centre role	actual pressure	D norm
own-five/fork-three, $Q = 0, 1/4$	$xV > 3.38$	< 14
own-zero/fork-five, $Q = 0$	$xV > 3.36$	13
own-zero/fork-five, $Q = 1/4$	$xV + dy > 3.48,\ d < .875$	13
own-zero/finite fork $\geq 7, Q = 0$	$14G - R > 32.51$	R
reciprocal own-three/fork-four or five; pair $(3,4)$	$xV > 3.67$	16
pair $(3,5)$, own zero	$xP_{14} > 3.4$	14.

Every V is flat through eleven. The mixed point has TOTAL spatial norm $13 + d^2$; it has not become a fictitious pure-D point. Use w_{11}, the applicable pure-D head foot and the centre residual $(V - P_{11})/7$. The individual P_5, P_{11} prices are paid first; the residual cross is zero. In the mixed row spend $d/7 < .125$ from $u_0 > .22$, keeping FINAL external coefficient u_0. The single ambient square is

$$E_{11} + q_0 + \eta_H + \frac{2(G - du_0 - f_{11})}{7} - \frac{R - 11}{49}.$$

Pure rows use $q_0 + \eta_H \geq .0916$; their totals exceed $1.0196, 1.0343, 1.0031, 1.0617, 1.0253$, respectively, with the continuous row using its proved $14G - R$ estimate. For the mixed row the cleared, native-seven, native-eight and native-at-least-nine head choices give totals $> 1.0116, 1.0203, 1.0173, 1.0189$. They follow by substituting the COMPLETE last table and $G > 3.48, d < .875$. Thus all centre exceptions are impossible under this contrary height. Only THEN is the actual E_{14} reference available, giving

$$E_{14} + q_0 + \eta_H > .9084 + .0916 = 1,$$

a contradiction. Therefore $h < .23$. Every finite head would require $h > 1/4$; it is excluded. The head is isolated within D, DOTTED, possibly contacting old I. Its cleared pure-D gain (D16IH) is now uniformly available.

We still must establish centre fourteen WITHOUT the contrary height. The only point rows left by its complete inventory are precisely those in the table above. For own-zero/single-fork finite label $m \geq 7$, $Q = 0$ is already proved. Set $t_f = 4\cos^2(\pi/m)$. The actual fork-product normalization makes the restricted old-D/cap Schur complement $4/t_f - 8 < 0$. Strict nonphysical positivity gives

$$3 < t_f < 4, \qquad 0 < (t_f)_\sigma < 1/2.$$

Degree at least three has nonzero integer norm $< 4(1/2)^2 = 1$. Degree one has no integer in $(3,4)$. In degree two its norm is one and trace four, so $t_f = 2 + \sqrt{3}$: only native twelve remains. The connected Lorentzian principal forces ACTUAL $K = \mathbb{Q}(\sqrt{3})$.

The golden point rows instead force ACTUAL $K = \mathbb{Q}(\varphi)$. For own-five/fork-three, normalize by that ACTUAL three; the old-D spatial norm is $14\varphi + 9 > 1$, independent of its optional old-I contact. For own-zero/single-fork-five, normalization to fork product minus one has Schur

complement $4/(\varphi+1)-8<0$. For own-zero/pair-three/five, normalize by the three; its D-spatial norm is $4+(11/2)\varphi>1$. Each restricted Gram is over the golden field and physically Lorentzian. The already proved original-root descent therefore identifies the ACTUAL field, even in the mixed old-I branch.

The actual invariant now has the sharper COMPLETE strip

$$J=4(1+h)\in\mathcal{O}_K,\qquad 4<J<123/25,\quad 0<J_\sigma<4. \tag{D16IJ}$$

In the golden full ring it leaves only $J=3+\varphi$. Indeed $J=a+b\varphi$ has $b=1,2$; at one $a=3$, at two the physical interval gives $a=1$ but its conjugate is negative. Thus $h=(\varphi-1)/4$, $y^2>.0359$. The exact rational seal is $309(9641)^2-2000(14360000)=1204229>0$. In $\mathbb{Z}[\sqrt{3}]$, the same strip gives only $J=3+\sqrt{3}$, hence $h=(\sqrt{3}-1)/4$, $y^2>.04$. No rational J exists in the strip.

Use again ONE w_{11}, pure-D head gain $>.04$, actual centre point and FINAL external coefficient $u_0=\sqrt{q_0}$. The head debit is $.072<.14$; the centre debit is $1/7$. Their D-cross is zero. For golden rows choose $q_0=.0359$, $.189<u_0<.19$; the mixed debit $d/7<.125$ leaves a positive coefficient. For reciprocal rows use $q_0=u_0=0$ and no I energy at all. The COMPLETE resulting totals are

own-five/fork-three	$.84816+.0359+.04+.988/7-3/49>1.0039$
own-zero/fork-five, $Q=0$	$.84816+.0359+.04+.948/7-2/49>1.0186$
own-zero/fork-five, $Q=1/4$	$.84816+.0359+.04+.8555/7-2/49>1.0054$
reciprocal/pair $(3,4)$	$.84816+.04+1.568/7-5/49>1.0101$
pair $(3,5)$	$.84816+.0359+.04+1.028/7-3/49>1.0096.$

The mixed numerator is $2(3.48-.875(.19)-2.886)=.8555$; its actual total point norm was retained before the coefficient debit.

For native twelve put $z=8-4\sqrt{3}$. The already proved actual finite-fork point has

$$G=\sqrt{16-2z}-\sqrt{-8+14z-6z^2},\quad R=24-16z+6z^2.$$

The positive squares give $1.0716<z<1.072$, first radical >3.722, second $<.337$, so $G>3.385>3.38$, and $R<13.75$. Choose $q_0=.04, u_0=.2$. The total exceeds $.84816+.04+.04+.988/7-(11/4)/49>1.0131$. Every nondirect centre role is impossible. Hence ALL individual five-through-fourteen floors now hold UNCONDITIONALLY.

C.13.3 Joint head height and both genuine D_{16} fork edges

For the actual dotted head let $Q=\beta_H^2$, $u=(1-Q)/g_H^2\in(0,1)$. Acuteness with c_I gives $h(2g_H^2-1)\geq Q$ at $Q>0$. At zero the positive actual mutual has native magnitude at least $1/2$, giving the same stronger inequality $h(2g_H^2-1)\geq 1/4$. Thus in ALL cases

$$h\geq k(u)=\frac{u}{4(2-u)}.$$

Use the actual w_{14} reference, head insertion $.36(Z-P_5/5)$, and the disjoint I energy ONCE. The total exceeds

$$.9084+\eta_0(u)+q(h),\quad \eta_0(u)=.72(H(u)-.361)-.1296(R(u)-1/5).$$

No initial-five/I gain is added to this overlapping head gain.

For $0<u\leq 3/8$, $H'(u)\leq -1/4-1/(4\sqrt{2})<-.425$, so $\eta_0'<-.1764$. Also $q'(h)\leq 1/4$, $k'(u)\leq 32/169$. Hence $\eta_0(u)+q(k(u))$ decreases on the ENTIRE interval. At $u=3/8$,

$$H=(4\sqrt{13}-\sqrt{30})/16>.55906,\quad R=.6953125,\quad k=3/52,\quad q(k)>.014.$$

Use $\sqrt{13} > 3.60555$, $\sqrt{30} < 5.47723$, and $55(493)^2 - 52(507)^2 = 1147 > 0$. The total would exceed $.9084 + .0784107 + .014 = 1.0008107$. Thus

$$u > 3/8, \qquad h > 3/52, \qquad q(h) > .014, \qquad g_H^2 < (8/3)(1-Q). \tag{D16Iu}$$

From $h < .23$ and actual acuteness, $Q < 299/668 < 1/2$. The COMPLETE finite foreign alphabet therefore leaves only $\beta_H = 0, 1/2$. At positive contact, $g_H^2 < 2$ and $h > 1/12$.

For EITHER genuine fork direction $V = P_{15} \pm e_{16}$, $V^2 = 16$, an ACTUAL edge cosine $C \geq 37/4$ would give $xV \geq \sqrt{528/41} > 3.588$. Pay $1/7$ at P_{11}; the residual has square five and is orthogonal to the reference and the head residual. The latter still pays only its separate P_5 coefficient. The ONE square, with I counted once, gives

$$.84816 + .04 + .014 + \frac{2(3.588 - 2.886)}{7} - \frac{5}{49} > 1.00069.$$

Apply the contradictions separately to the two actual edges. Thus BOTH fork cosines satisfy

$$C < 37/4. \tag{D16IC}$$

No fork-classification or field-degree result was used to prove this cut. We will use it first, eliminating a redundant weaker length stage.

C.13.4 The complete actual $D_{16} + I$ fork roles

Release fork sixteen and retain the path $n_1, \ldots, n_{15} = A_{15}$ and old e. Its COMPLETE finite one-node list is isolation, helper-three at $j = 1, 2, 14, 15$, or helper-four at $j = 1, 15$. A second path contact makes a cycle; rank sixteen excludes exceptional completions, and a multiple bond must be terminal. Only endpoint-three at one/fifteen permits an old-I arm, with $\beta = 0, 1/2, 1/\sqrt{2}$. At next-to-end three, appending that arm gives nonfinite arms $1, 2, 13$. At helper-four it makes the four internal in a long path. Thus those four special roles have $\beta = 0$. Reversing the genuine fork sign covers the other outgoing edge separately.

For next-to-end-three at two put $t = 2\gamma$. Its exact data are

$$A_c^2 = 2t^2 + t, \qquad \Delta = 1/8, \qquad C = 4t + 1.$$

Own zero has zero time. At positive own, actual helper-three normalization makes t integral, ACTUAL $K = \mathbb{Q}(t)$, and $-1/2 < t_\sigma < 0$. The connected Schur complement is $-2t(2t+1) < 0$. The stronger INDEPENDENT cut (D16IC) gives $t < 33/16 < 5/2$. Degree at least three has norm < 1. In degree two its norm is -1, and the exact integral trace condition leaves $t = 1 + \sqrt{2}$, whose actual $C = 5 + 4\sqrt{2} > 37/4$, impossible. Degree one gives the empty actual-J strip (D16IJ). For long-end helper-four at one set $t = 2\sqrt{2}\gamma$. Its data are $A_c^2 = t^2 + t/2, \Delta = 1/16, C = 4t + 1$. Here the actual cap normalizes to square one, not two; its cycle through the four makes t integral. Schur complement $-t(2t+1)/2 < 0$ gives the SAME field/conjugate/degree argument, hence the same contradiction. This avoids an unnecessary weaker silver-role energy estimate.

The far next-three at fourteen has $A_c^2 = 2t^2 + 7t + 6, C = 4t + 7$, $t = 2\gamma$. Positive own gives $C \geq 11 > 37/4$; at own zero the actual future null is $c + n_{13} + 2n_{14} + n_{15} + n_{16}$. The far helper-four at fifteen has $A_c^2 = t^2 + (7/2)t + 3, C = 4t + 7$, $t = 2\sqrt{2}\gamma$. Positive own again violates the cut; at zero use $\sqrt{2}c + n_{13} + 2n_{14} + 2n_{15} + n_{16}$. Both kernels have zero selected products and norm, positive cap time, and all omitted ORIGINAL wall signs by positivity/acuteness. The full fork list is reduced to isolation or endpoint-three at one or the other fork fifteen.

For those two nonisolated roles write

$$c = A_c p - (t/2)\omega_{16} - \omega_j/2 - \beta e, \qquad j = 1, 15.$$

The exact unit geometry is

j	A_c^2	C
1	$2t^2 + t/2 - 1/2 + \beta^2$	$(8t+1)/\sqrt{17 - 32\beta^2}$
15	$2t^2 + 7t/2 + 1 + \beta^2$	$(8t+7)/\sqrt{17 - 32\beta^2}$.

Own zero at one has nonpositive time. At fifteen its restricted connected principal is rational Lorentzian even at zero, giving actual rational field and the impossible integral-J strip. Parallel has its positive future null.

At positive own the ACTUAL helper-three normalization gives raw square two, own product $-t$, integral t. The restricted Schur complements are $1 - 4t^2 - t$, $-(4t^2 + 7t + 2)$; both are physically negative. Thus ACTUAL $K = \mathbb{Q}(t)$, and

j	EVERY other t_σ
1	$(-1 - \sqrt{17})/8 < t_\sigma < (-1 + \sqrt{17})/8$
15	$(-7 - \sqrt{17})/8 < t_\sigma < (-7 + \sqrt{17})/8$.

(D16Itstrip)

A helper-four OLD-I contact $\beta = 1/\sqrt{2}$ is impossible: at fifteen positive own has $C \geq 15$; at one the cut leaves only own-three, with its actual all-three cycle. Hence $\beta = 0, 1/2$.

The same cycles exclude own-three in both roles. For every even finite own label $m \geq 4$, the coprime exponent $m - 1$ gives conjugate $-2\cos(\pi/m) \leq -\sqrt{2} < -1.4$, outside both intervals. For odd $m \geq 7$, exponent $m - 2$ gives $-2\cos(2\pi/m)$. It is below minus one for role one; for role fifteen it is below -1.4 when $m \geq 9$. At its remaining seven, exponent three gives a positive conjugate, outside the entirely negative near-fork interval. These inequalities follow from $\pi < 22/7, \cos z > 1 - z^2/2$. Only finite own-five remains: $t = \varphi, K = \mathbb{Q}(\varphi)$.

Its actual-J strip leaves $J = 3 + \varphi, h = (\varphi - 1)/4$. At $\beta = 0$, the two exact A_0^2 values are $(3 + 5\varphi)/2, (6 + 11\varphi)/2$. The actual c, c_I two-cycles are

$$(2 + 3\varphi)/2, \qquad (5 + 6\varphi)/2,$$

nonintegral. At $\beta = 1/2$, their actual mutual magnitude is

$$F = \sqrt{h(A_0^2 + 1/4)} - \sqrt{1 + h}/2.$$

The rational brackets $1.618 < \varphi < 1.619$ give $.4085 < F < .411$ at one, and $.8325 < F < .835$ at fifteen. The first lies in $(0, 1/2)$; the second between $\varphi/2$ and $\sqrt{3}/2$. Both are forbidden ACTUAL native gaps. Thus every nonisolated fork has DOTTED own $t > 2$.

C.13.5 An individually isolated finite fork also fails

An isolated finite-own fork has

$$c_F = A_F p - \gamma\omega_f - \beta_F e, \qquad s = \gamma^2, \quad A_F^2 = 8s + \beta_F^2 - 1.$$

Actual old-D root normalization to square one and fork product minus one gives Schur complement $1/s - 8 < 0$. The Lorentzian-principal descent gives ACTUAL $K = \mathbb{Q}(s)$, $0 < s_\sigma < 1/8$. For small finite three/four/six this is rational, contradicting the integral actual-J strip. Five gives

$s = (\varphi + 1)/4$. For every finite label at least seven, $t_f = 4s$ has physical value in $(3, 4)$, others in $(0, 1/2)$. The norm/trace proof above gives only $t_f = 2 + \sqrt{3}$, native twelve. The ACTUAL I strips then fix

K	s	h
$\mathbb{Q}(\varphi)$	$(\varphi + 1)/4$	$(\varphi - 1)/4$
$\mathbb{Q}(\sqrt{3})$	$(2 + \sqrt{3})/4$	$(\sqrt{3} - 1)/4$.

Both have $q(h) > .0359$.

We also need the COMPLETE old-I contact restriction for this finite fork. Put $B_0^2 = 8s - 1$. Its actual mutual with c_I is

$$F(\beta) = \sqrt{h(B_0^2 + \beta^2)} - \beta\sqrt{1 + h},$$

strictly decreasing. Its sign is the sign of $hB_0^2 - \beta^2$. For own-five $hB_0^2 = s$; acuteness requires $\beta \le \gamma$. At $\beta = 1/2$, $F < 1/2$, because the positive-square difference is $1/2 + \sqrt{1 + h}/2 - s > 0$. Thus every $0 < \beta < \gamma$ has forbidden $0 < F < 1/2$. Only $\beta_F = 0, \gamma$ remains. For own-twelve $hB_0^2 = 1 + h$; at $\beta = 1/2$, $1/2 < F < 1/\sqrt{2}$. At $\beta = 1/\sqrt{2}$, $0 < F < 1/2$; monotonicity covers every larger finite contact. These comparisons use $h < .23$: $\sqrt{1 + 5h/4} < 1.135$ and $\sqrt{(1 + h)/2} - 1/4 - h > .7 - .48 > 0$. Therefore $\beta_F = 0$ for twelve. All foreign finite labels were included, without a separate field-degree census.

Now use the actual dotted head. Its cleared-head parameter u has the pressure $H(u), R(u)$ already proved. For a proposed finite fork, $q(h) > .0359$. At $u \le .48$, the derivative of its effective head gain is negative: $H' \le -1/4$ and $\eta' \le -.18 + .1296(1 - u) < 0$. At .48, $H > .5184, R = .6352$, giving gain $> .05692608$. The single actual reference would total $.9084 + .0359 + .05692608 > 1$. Hence $u > .48$, and the head integer $T = 4g_H^2$ satisfies

$$4 < T < 25/3, \qquad 0 < T' < 2.$$

The COMPLETE golden full-order list is

$$T = 2 + 2\varphi,\ 3 + 2\varphi,\ 2 + 3\varphi,\ 3 + 3\varphi.$$

Writing $T = a + b\varphi$ gives $b \le 3$; $b = 1$ has physical value below four, and $b = 2, 3$ each has precisely the two displayed constants. For $H_0 = h(T/2 - 1)$ their exact values are $1/4, 1/4 + h/2, 3/8, 3/8 + h/2$. All are below $1/2$, so old-I head contact is zero or $1/2$. At zero contact, the last three actual mutuals lie in $(1/2, 1/\sqrt{2})$; at $1/2$, they lie in $(0, 1/2)$. Only $T = 2 + 2\varphi$ remains, with $g_H^2 = (\varphi + 1)/2 = 2s$ and $\beta_H = 0, 1/2$.

For $\mathbb{Q}(r)$, $r = \sqrt{3}$, the full strip leaves $T = 3 + r, 4 + 2r$. Their H_0 values are $1/4, 1/2$. The complete contacts are zero or $1/2$ at the first; zero, $1/2$, $1/\sqrt{2}$ at the second. The middle contact gives its forbidden $(0, 1/2)$ mutual, and the last gives $u = 2 - r < .48$. Thus the remaining head rows are $g_H^2 = (3 + r)/4, \beta_H = 0, 1/2$, or $g_H^2 = (2 + r)/2, \beta_H = 0$.

Both original actual root lines may be normalized OVER K to their nonzero own product minus one. Their ACTUAL mutual product is K-valued, so

$$U = \frac{A_H A_F - \beta_H \beta_F}{g_H \gamma} \in K.$$

This retains the WHOLE combined expression; neither separate unit time nor foreign product is assumed K-valued. For the golden table, equal zero/nonzero contact patterns give $U = 2\sqrt{2}$; the two mixed patterns give $U = \sqrt{6 + 2\varphi}$. Neither belongs to $\mathbb{Q}(\varphi)$; the second has nonsquare rational norm 44. For the three silver head rows, $\beta_F = 0$ and the exact U^2 values are

$$8, \qquad 6 + 2r, \qquad 24 - 8r.$$

The first requires $\sqrt{2} \notin \mathbb{Q}(r)$; the others have nonsquare norms $24,384$. Every remaining actual raw Gram row contradicts $U \in K$. Thus NO isolated fork has finite own. Own zero has negative time; parallel has its actual positive null. Both fork opposites now have DOTTED own, whether isolated or nonisolated.

C.13.6 Every actual $D_{16}+I$ field has degree two, three or four

For a NONISOLATED dotted fork set $v = t$ at helper one, $v = t+1$ at helper fifteen. The actual cut and (D16Itstrip) give integral v, ACTUAL $K = \mathbb{Q}(v)$, $0 < v < 4.9$, and every other $|v_\sigma| < .641$. Degree at least five would have nonzero norm $< 4.9(.641)^4 < 1$. Thus its actual degree is at most four.

For an ISOLATED dotted fork set $T = 4\gamma^2$. Its actual edge is $C^2 = 2T/(1-\beta^2)$, so

$$4 < T < 1369/32 < 121/2.$$

Its actual own two-cycle is integral. Normalize its original root to own product minus one against old raw D-roots of square two. Its square is $2/T$, old dual projection square four, and restricted Schur complement $2/T - 4 < 0$. The connected Lorentzian principal gives ACTUAL $K = \mathbb{Q}(T)$. Strict nonphysical diagonal and Schur positivity give $0 < T_\sigma < 1/2$.

We prove the degree bound on the ENTIRE strip, including degrees five and six. At degree at least seven, the norm is $< (121/2)(1/2)^6 = 121/128 < 1$. For degree five let P be the monic integer minimal polynomial. The four small roots have product $\mathrm{N}(T)/T > 1/61$. Since $61(7/20)^4 < 1$, their average exceeds $7/20$. The four positive deficits $1/2 - T_\sigma$ then have average less than $3/20$. AM–GM gives

$$0 < |P(1/2)| < 61(3/20)^4 = 4941/160000 < 1/32.$$

But a nonzero monic degree-five integer polynomial at $1/2$ has magnitude at least $1/32$. It is nonzero because $T > 4$ and the minimal polynomial is irreducible. At degree six, $61(2/5)^5 < 1$ makes the five small-root average $> 2/5$; the deficit average is $< 1/10$. Thus $0 < |P(1/2)| < 61/100000 < 1/64$, again impossible. This proves degree at most four for an isolated fork, without a coefficient census.

These two roles exhaust either ACTUAL fork, and in each branch the generating field is the ACTUAL defining field. Degree one would make the integral J a rational integer in the empty interval $(4, 123/25)$. Consequently the whole forest has

$$[K : \mathbb{Q}] \in \{2, 3, 4\}.$$

No old-I contact has been set to zero in this isolated case. The nonisolated roles still require their complete exclusions below.

C.13.7 Nonisolated $D_{16}+I$ forks: quartic and old-I contact

Keep $v = t$ in the long-end role, $v = t+1$ in the other-fork role. The actual field is generated by this integral parameter. Its physical bounds are $v < 4.644$, $v < 4.894 < 4.9$, respectively. The other windows are $(-.641, .391)$, $(-.391, .641)$. In degree four, any small conjugate of the wrong sign would give nonzero norm magnitude $< 4.9(.391)(.641)^2 < 1$. Thus the three small roots are all negative for the long-end role, positive for the other-fork role. Their absolute product is $1/v$, since $4.9(.641)^3 < 2$. Write their sum $s < 1.923$. Then $v > 3.79$, and AM–GM gives $s > 30/17$.

In the other-fork role, the integer trace $v + s$ is six. Hence $v > 4.077$, and $v + 3/v^{1/3} \le 6$ gives $v < 4.2$. For the pairwise small-root sum p, the integer third coefficient is $E_3 = vp + 1/v$. AM–GM gives $E_3 > 4.77 + 5/21 > 5$. But $p < 3(.641)^2$, so $E_3 < 4.2(1.232643) + 1/4 < 6$. Impossible. In the

long-end role, integer trace $v-s$ is two. Thus $v<3.923$; AM–GM requires $v-2\geq 3/v^{1/3}$, which forces $v>3.9$ since $3.9(1.9)^3<27$. Its integer second coefficient $E_2=p-vs$ is $<-6.177357<-6$. Since $v<4$, AM–GM gives $p>1$, so $E_2>1-3.923(1.923)>-7$. Again impossible. Every nonisolated fork therefore has ACTUAL degree at most three, with BOTH permitted old-I contacts retained.

We next exclude its remaining positive old-I contact $\beta=1/2$. Then

$$j=1:\ t<107/32,\qquad j=15:\ t<83/32.$$

Equivalently $v<107/32,115/32$. Degree one is already excluded by the integral actual-J strip.

In degree three, a wrong-sign small root would give norm $<(115/32)(.391)(.641)<1$. Thus the two small roots are negative at one, positive at fifteen, with absolute product $1/v$. Their magnitudes a,b satisfy $a+b<1.282$, $v>2.43$, $a+b>1$. At fifteen $v>3$, so integer trace $v+a+b\in(4,5)$, impossible. At one the trace $v-a-b$ is two. Thus $v<3.282$, and $v(v-2)^2\geq 4$ forces $v>3.1$. The integer second coefficient $1/v-v(v-2)\in(-5,-3)$ is minus four. Its polynomial $X^3-2X^2-4X-1=(X+1)(X^2-3X-1)$ is reducible. Positive old-I contact is therefore quadratic if it exists.

The complete quadratic trace/norm windows give EXACTLY

j	t
15	$(1+\sqrt{13})/2,\quad 1+\sqrt{2}$
1	$1+\sqrt{2},\quad 1+\varphi,\quad (3+\sqrt{13})/2.$

For coverage, at fifteen the trace of $v=t+1$ is three/four. Trace three has norm minus one, trace four norm two. At one trace two has norm minus one; trace three has norm one or minus one. These follow from the actual strict root windows and the physical upper bounds above; all absolute norms are strictly below two or three as needed. The ACTUAL full quadratic-J strips leave

$$J=3+\sqrt{2},\qquad 3+\varphi,\qquad (5+\sqrt{13})/2$$

in their respective fields. Write $c\cdot c_I=-F$. For positive old contact,

$$F=A_c\sqrt{h}-\sqrt{1+h}/2,\qquad F^2=hA_c^2+(1+h)/4-\sqrt{h(1+h)A_c^2}.$$

The ACTUAL two-cycle puts F^2 in K, so $h(1+h)A_c^2$ must be a FULL K-square. On the COMPLETE five rows it equals $Z/64$, with

j,t	Z	$\mathrm{N}(Z)$
$15,(1+\sqrt{13})/2$	$29t+37$	-81
$15,1+\sqrt{2}$	$77+56\sqrt{2}$	-343
$1,1+\sqrt{2}$	$47+32\sqrt{2}$	161
$1,1+\varphi$	$77t-33$	-605
$1,(3+\sqrt{13})/2$	$33t+12$	$243.$

Every norm is negative or a positive nonsquare rational integer. Thus no row is a field square. Every nonisolated fork is orthogonal to OLD I:

$$\beta=0.$$

Its contact with the actual I OPPOSITE has been retained and used, rather than declared zero.

C.13.8 Nonisolated $D_{16} + I$ forks: the complete cubic domain

Put $y_0 = J - 4$. In actual degree two or three, $y_0/2$ is NOT an algebraic integer. In degree two its norm would have magnitude $< (23/50)(2) < 1$. In degree three suppose $x = y_0/2$ were integral. Its physical value lies in $(0, 23/50)$, other roots $-b, -c$ in $(-2, 0)$; its norm is one. AM–GM gives $b + c > 2.94$, so integer trace $x - b - c = -3$. Then $b + c = x + 3$, and reality forces $x(x + 3)^2 \geq 4$, so $x > 1/3$. The integer second coefficient $1/x - x^2 - 3x$ lies in $(0, 2)$, hence is one. Its polynomial $X^3 + 3X^2 + X - 1 = (X + 1)(X^2 + 2X - 1)$ is reducible, contradicting actual degree three. This proves the half-integrality obstruction in both stated degrees.

At $\beta = 0$, the two nonisolated time formulas are $A_c^2 = q(v)/2$,

$$q(v) = 4v^2 + v - 1 \quad \text{or} \quad 4v^2 - v - 1.$$

The ACTUAL fork/c_I two-cycle makes $q(v)y_0/2$ integral. If $\mathrm{N}(q(v))$ were odd, its integral cofactor and integer Bezout would force $y_0/2$ integral. Thus $\mathrm{N}(q(v))$ is even. Both $q(v) \equiv v - 1 \pmod{2\mathcal{O}_K}$. For the cubic minimal polynomial P, the determinant norm therefore gives

$$P(1) \equiv 0 \pmod 2. \tag{D16Iparity}$$

This works in the FULL integer ring, without a monogenic-order assumption.

We exhaust every cubic sign pattern. For the long-end role $2 < v < 4.65$. Two positive small roots give norm $< 4.65(.391)^2 < 1$. With two negative roots $-a, -b$, their product is $1/v$, integer trace is two or three. At trace two, $3.1 < v < 3.282$; integer second coefficient is minus four, giving the reducible cubic already printed. At trace three, $4 \leq v < 4.282$; the coefficient is minus five or minus four. Minus five gives $(X + 1)(X^2 - 4X - 1)$. Minus four gives $P(1) = -7$, violating (D16Iparity). For mixed signs $a, -b$, with $a < .391, b < .641$, norm is minus one; $v > 3.98, b > .55 > a$. The integral trace is four, and $v > 4$. Its second coefficient $-1/v + v(4 - v) \in (-4, 0)$ is minus three, minus two or minus one. Parity forces minus two. But

$$P(-b) = 1 + 2b - 4b^2 - b^3$$

for $P = X^3 - 4X^2 - 2X + 1$ is strictly concave on $[0, .641]$ and positive at BOTH endpoints, the second value .375101279. It has no required negative root. All long-end cubic cases fail.

For the other-fork role $3 < v < 4.9$, two negative small roots give norm $< 4.9(.391)^2 < 1$. With two positive roots, norm two forces $v > 4.86$, small-root sum > 1.27, and integer trace strictly between six and seven. For norm one, the trace is five or six. At six, $a + b > 1.1, v > 4$, while $(.641 - a)(.641 - b) > 0$ gives $ab > .294219 > 1/v$, impossible. At five, $3.718 < v \leq 4$; the integer second coefficient lies in $(4, 6)$, hence is five. Its polynomial $X^3 - 5X^2 + 5X - 1 = (X - 1)(X^2 - 4X + 1)$ is reducible. For mixed signs $a, -b$, $a < .641, b < .391$, norm is minus one, $v > 3.98, a > .52$. The trace is five, hence $v > 4.359$. Its positive integer second coefficient $v(5 - v) - 1/v$ is one or two. Parity forces one. Only

$$P(X) = X^3 - 5X^2 + X + 1$$

remains; no polynomial enumeration was assumed.

Its discriminant is $404 = 4 \cdot 101$. The index of $\mathbb{Z}[v]$ in the FULL order is one or two. Index two would supply an integral $(A + Bv + Cv^2)/2$ with binary coefficients not all zero. Since $\mathrm{Tr}(v) = 5$, $\mathrm{Tr}(v^2) = 23$, integral trace leaves the three nonzero even-weight numerators. Their norms are $3/4, 9/2, -3/4$, not integers. Thus $\mathcal{O}_K = \mathbb{Z}[v]$. Exact endpoint signs isolate roots

$$4.74 < v < 4.75, \quad .60 < a < .61, \quad -.35 < b < -.34.$$

Suppose the required integral $J = A + Bv + Cv^2$ exists. Lagrange interpolation and its full strip $4 < J_v < 4.92, 0 < J_a, J_b < 4$ give $-2 < C < 2$, hence $C = -1, 0, 1$. The small-value difference gives $|B + C(a+b)| < 4/.94 < 4.26$; since $.25 < a + b < .26$, $-4 \le B \le 4$. At $C = -1$, $J_v - J_b = (v-b)(B-v-b) < 0$, impossible. At zero, positivity forces $B \ge 1$, so $J_v - J_b > 5.08 > 4.92$. At one, $J_v - J_a > (4.13)(-4 + 4.74 + .60) > 4.92$. Every full-order coefficient case fails. The actual-I strip is empty in the last cubic field. No nonisolated cubic fork exists.

C.13.9 Nonisolated $D_{16} + I$ forks: the quadratic K-basis obstruction

It remains to exclude actual degree two. Normalize the connected old D roots OVER K to square two. Their sixteen spatial coordinate vectors $e_1, \dots, e_{16}$ are then K-linear combinations of these roots. The actual fork root normalizes to $r_F = \sqrt{2}c_F$, square two and helper product minus one. Together with ANY nonzero original K-root r_I on old I, the eighteen vectors

$$e_1, \dots, e_{16}, r_F, r_I$$

form an ACTUAL K-basis: r_F has nonzero time, r_I has its nonzero I-component. Normalize the head to own old-long product minus one, $r_H = c_H/(\sqrt{2}g_H)$, square $2/T$, $T = 4g_H^2$. Its coefficient on r_F is $A_H/(2g_H A_F) \in K$, by comparing the unique time coordinate. Consequently

$$R_{HF} = \frac{A_H^2}{TA_F^2} \in K^2. \tag{D16IHF}$$

Neither the unit e, p, nor separate unit time coordinates have been asserted K-vectors.

The actual head and old-I contact give the complete strips

Q_H	T	other T'
0	$4 < T < 32/3$	$0 < T' < 2$
1/4	$4 < T < 8$	$0 < T' < 3/2$.

The nonphysical Schur complement is $2(1 - Q_H)/T' - 1 > 0$, with the FULL old-I projection retained. At positive cost, actual acuteness also gives $T \ge 2 + 2/(J-4)$. At zero cost the actual head/c_I two-cycle gives $(T-2)(J-4)/2 \in \mathcal{O}_K$.

The complete quadratic fork list comes from the strict physical/conjugate windows, not a field census. At long-end $j = 1$, trace is two/three/four/five. Trace five would have noninteger norm strictly between one and two. Trace two has norm minus one; trace three and four each permit norm one, minus one or minus two. At the other-fork role, trace is one/two/three. Their negative norms are respectively three; one/two/three/four; two/three. Trace two/norm minus three is rational and

excluded. Solving each exact integer pair and the full actual-J strip gives

j	t	J	$A_F^2(J-4)$
1	$1+\sqrt{2}$	$3+\sqrt{2}$	$3+(3/2)\sqrt{2}$, nonintegral
1	$1+\varphi$	$3+\varphi$	$4t-3/2$, nonintegral
1	$(3+\sqrt{13})/2$	$t+1$	$(3t+4)/2$, nonintegral
1	$(3+\sqrt{17})/2$	$t+1$	$(7t+5)/2$, nonintegral
1	$2+\sqrt{3}$	$3+\sqrt{3}$	$6t-1$
1	$2+\sqrt{5}$	$3+\varphi$	$5t+2$
1	$2+\sqrt{6}$	none	empty actual-J strip
15	$(1+\sqrt{13})/2$	$t+2$	$(3t+5)/2$, nonintegral
15	$1+\sqrt{2}$	$3+\sqrt{2}$	$3t+3/2$, nonintegral
15	$1+\sqrt{3}$	$3+\sqrt{3}$	$5t+5$
15	$1+\sqrt{5}$	$3+\varphi$	$9\varphi+6$
15	$(3+\sqrt{17})/2$	$t+1$	$5t+4$
15	$(3+\sqrt{21})/2$	$t+1$	$(14t+15)/2$, nonintegral.

The last field has a NONEMPTY actual J strip; it fails by integrality, not by an invented empty field. Every entry is an ACTUAL fork/c_I two-cycle. The full orders are $\mathbb{Z}[t]$ in the nonintegral rows. Only fields $\sqrt{3}, \sqrt{5}, \sqrt{17}$ and the five unmarked fork rows remain.

Their COMPLETE head lists are

K	Q_H	T
$\mathbb{Q}(s),\ s=\sqrt{3}$	0	$3+s,\ 4+2s$
	1/4	$3+s,\ 4+2s$
$\mathbb{Q}(\varphi)$	0	$2+2\varphi,\ 4+4\varphi$
	1/4	$2+2\varphi,\ 2+3\varphi,\ 3+3\varphi$
$\mathbb{Q}(\sqrt{17}),\ t=(3+\sqrt{17})/2$	0	$2+t,\ 2+2t$
	1/4	$2+t$.

For full coverage, in $\sqrt{3}$ the raw zero-cost list additionally has $5+2s$, excluded because its two-cycle is $(3+s)/2$. The positive-cost lower bound is $T \geq 3+s$. In the golden field the raw zero-cost list is $2+2\varphi, 3+2\varphi, 2+3\varphi, 3+3\varphi, 3+4\varphi, 4+4\varphi$. Its two-cycle is $\{b-a+2+(a-2)\varphi\}/2$; both coefficients require a, b even. At positive cost the strict strips and $T \geq 2+2\varphi$ leave exactly the three displayed rows. In $\sqrt{17}$, raw zero-cost $1+t, 2+t, 2+2t, 3+2t$ has two-cycle $\{(a-2)t+2b-3a+6\}/2$; a is even. At positive cost the raw $1+t, 2+t$ list reduces to $2+t$ by acuteness. Writing $T=a+bw$ in each FULL order, the physical upper and conjugate width bound positive b, so these finite lists exhaust every element, including all dotted own and old-I contacts.

Compute the rational norms of the NECESSARY field square (D16IHF). For $\sqrt{3}$, the far/near fork time norms are $-13/2, -25/2$:

Q_H	T	$\mathrm{N}(R_{HF}),\ j=1$	$j=15$
0	$3+s$	1/78	1/150
0	$4+2s$	1/13	$1/25^*$
1/4	$3+s$	1/208	$1/400^*$
1/4	$4+2s$	23/416	23/800.

For the golden field the time norms are $-19, -9$:

Q_H	T	$j=1$	$j=15$
0	$2+2\varphi$	$1/76$	$1/36^*$
0	$4+4\varphi$	$1/304$	$1/144^*$
$1/4$	$2+2\varphi$	$11/1216$	$11/576$
$1/4$	$2+3\varphi$	$29/304$	$29/144$
$1/4$	$3+3\varphi$	$1/304$	$1/144^*$.

For $\sqrt{17}$, only the near fork remains, time norm minus thirteen: the three displayed head rows have norms $1/208, 1/52, 1/1664$. All unstarred entries are positive rational nonsquares. They are direct applications of

$$\mathrm{N}(R_{HF}) = \frac{\mathrm{N}(T/2 - 1 + Q_H)}{\mathrm{N}(T)\mathrm{N}(A_F^2)}.$$

The starred square-norm cases REQUIRE a FULL field-square test. Their exact radicands and both possible rational square parts are

K, head row, near fork	R_{HF}	r^2 candidates
$\sqrt{3},\ Q_H = 0, T = 4+2\sqrt{3}$	$(7-4\sqrt{3})/5$	$4/5,\ 3/5$
$\sqrt{3},\ Q_H = 1/4, T = 3+\sqrt{3}$	$(2-\sqrt{3})/20$	$3/40,\ 1/40$
$\varphi,\ Q_H = 0, T = 2+2\varphi$	$(13-8\varphi)/6$	$5/6,\ 2/3$
$\varphi,\ Q_H = 0, T = 4+4\varphi;\ Q_H = 1/4, T = 3+3\varphi$	$(5-3\varphi)/12$	$3/16,\ 5/48.$

Each candidate fails rational squareness by an odd prime valuation. Thus no actual head/fork K-basis condition survives, including every norm-square exception and both head old-I contacts. All quadratic nonisolated forks are impossible.

Together with the complete quartic, cubic and old-I arguments, BOTH actual fork opposites are now isolated within D_{16} and DOTTED. Their independently varying own integers $T_i = 4\gamma_i^2$ generate the SAME ACTUAL field of degree two, three or four. No equality of parameters or vanishing contact with c_I has been imposed.

C.13.10 Isolated forks: full foreign principals and the shared actual cycles

Both ACTUAL fork caps are now isolated within D, dotted, with

$$c_i = A_i p - \gamma_i \omega_i - \beta_i e, \quad T_i = 4\gamma_i^2, \quad \omega_i^2 = 8, \quad \omega_1 \cdot \omega_2 = 7.$$

They are DISTINCT, since their positive own products are at different old forks. Each T_i generates the ACTUAL field. The old-I contact is finite or zero, and has not been equated to contact with c_I.

Normalize old D roots over K to square two and each cap to own product minus one. It has raw square $2/T$, old-D projection square four. Keep old I at ANY totally positive raw square $\ell \in K$. The additional contact square is $2\beta^2\ell/T$, so its Schur cost is $2\beta^2/T$, independently of ℓ. Strict positivity of the FULL principal at every nonphysical place gives

$$0 < T_\sigma < (1-\beta_\sigma^2)/2, \qquad 4 < T < \frac{1369}{32}(1-\beta^2). \tag{D16Ifull}$$

All factors are positive; β^2 means its actual invariant embedding, not an assumed K-valued cosine. For degree d, the integral positive norm consequently satisfies

$$1 \le \mathrm{N}(T) < \frac{1369}{32\,2^{d-1}}\mathrm{N}(1-\beta^2).$$

For a positive finite label m, $E=\mathbb{Q}(\beta^2)$ has degree $\phi(m)/2$ dividing d. Pairing inverse cyclotomic roots gives $\mathrm{N}_E(4\sin^2(\pi/m))=\Phi_m(1)$. These formulas retain all disconnected root scales until they cancel.

When BOTH $\beta_i=0$, a second degree-INDEPENDENT lemma will be useful. Let r_i be the above ACTUAL raw caps, $P=r_1\cdot r_2\in K$, $Q=T_1T_2$. Subtracting their old-D projections leaves the SAME one-dimensional K-time line, so

$$(P-7/2)^2=\frac{4(2T_1-1)(2T_2-1)}{Q}.$$

The actual FIVE-cycle $r_1,n_{15},n_{14},n_{16},r_2,r_1$ has Cartan factors $-T_1,-1,-1,-1,T_2P$; hence $X=QP$ is integral. The actual TWO-cycle gives $L=QP^2$ integral. At $P=0$ both assertions are immediate. Expanding the ACTUAL residual determinant gives

$$L-7X=\frac{15Q}{4}-8T_1-8T_2+4,\qquad \frac{Q}{4}=4Q-\frac{15Q}{4}\in\mathcal{O}_K. \tag{D16Iquarter}$$

No degree, ramification, equal-parameter, or individual Cartan-integrality premise entered this identity. It also implies $q(T_1)q(T_2)\in K^2$, $q(T)=T(2T-1)$, by clearing square denominators in the determinant.

C.13.11 The quartic isolated branch ends with the quarter-product

At $d=4$, the complete positive old-I alphabet is

$$m=3,4,5,6,8,10,12,15,16,20,24,30.$$

Indeed $\phi(m)$ is two, four or eight. Any prime divisor has $p-1$ dividing one of those values, so $p=2,3,5$; the allowed prime powers and products give exactly the list. The COMPLETE values of $256\mathrm{N}_K(1-\beta^2)$ are

m	3	4	5	6	8	10	12	15	16	20	24	30
$256\mathrm{N}(1-\beta^2)$	81	16	25	1	4	1	1	1	2	1	1	1.

They follow from the prime-power cyclotomic rule $\Phi_{p^a}(1)=p$, all other $\Phi_m(1)=1$. For EVERY $m\geq 4$, the full norm bound gives $\mathrm{N}(T)<34225/65536<1$. Only $\beta=1/2$ or zero remains.

At $\beta=1/2$, $4<T<4107/128<65/2$, other $T_\sigma<3/8$. The positive norm is less than two, hence one. Its three reciprocal small conjugates $Y_i>8/3$ have product T. For three positive numbers $>a$, their sum is $<2a+\text{product}/a^2$; merge two at a time, since $(y-a)(z-a)/a>0$. Thus

$$Y_1+Y_2+Y_3<16/3+9T/64.$$

The integral unit T^{-1} has trace strictly between eight and $16/3+585/128+27/512<10$, hence nine. Each $Y_i<11/3$. The integral nonzero $T^{-1}-3$ has physical absolute value <3, others in $(-1/3,2/3)$. Its nonzero integer norm has magnitude $<3(2/3)^3<1$, impossible. Thus BOTH quartic forks have $\beta_i=0$.

Their positive integer norms satisfy

$$1\leq\mathrm{N}(T_i)<1369/256<6.$$

The shared actual TWO/FIVE-cycle identity (D16Iquarter) gives

$$256\mid\mathrm{N}(T_1)\mathrm{N}(T_2).$$

But each norm is at most five, so their product is at most twenty-five. This contradiction excludes the ENTIRE quartic branch directly. The longer norm-two/unit/coefficient route is unnecessary once the shared quarter-product is available; its source is preserved rather than duplicated. This argument applies the already proved degree-independent actual-cycle lemma at degree four, and adds no geometry or finite numerical-output premise.

C.13.12 Cubic isolated forks: two reusable coefficient lemmas

We will use two COMPLETE cubic strip arguments several times. First, no cubic integer Z generating a totally real field can satisfy

$$0 < Z < 1369/64, \qquad 0 < Z_\sigma < 1/4.$$

Its positive norm is $< 1369/1024 < 2$, hence one, and $Z > 16$. The reciprocal small conjugates $r, s > 4$ have product Z, giving $r + s < 4 + Z/4$. The integral trace of the unit Z^{-1} lies in $(8, 2409/256) \subset (8, 10)$, so is nine. Writing $a = \mathrm{Tr}(Z)$, its polynomial is $X^3 - aX^2 + 9X - 1$. Its negative value at $1/4$ gives $a > 81/4$, while $a < Z + 1/2 < 1401/64 < 22$. Thus $a = 21$, but that cubic has discriminant -864. This contradicts total reality. In particular NO cubic fork parameter in the full isolated strip belongs to $2\mathcal{O}_K$.

Second suppose a cubic fork parameter V has norm eight, with physical $4 < V < 1369/32$, others in $(0, 1/2)$. Then $V > 32$, and its polynomial $X^3 - aX^2 + bX - 8$ obeys

$$34 \le a \le 43, \qquad b^2 > 32(a - 1), \qquad 2b \le a + 31.$$

The first follows from $a \ge V + 2\sqrt{8/V} > 33$. The last follows by evaluating at $1/2$. The COMPLETE integer comparison is

a	34	35	36	37	38	39	40	41	42	43
b least	33	33	34	34	35	35	36	36	37	37
b greatest	32	33	33	34	34	35	35	36	36	37.

Only $(35, 33), (37, 34), (39, 35), (41, 36), (43, 37)$ remain. Their exact discriminants are

$$-17131, \quad -16124, \quad -11651,$$
$$-2704, \quad 11821.$$

Thus every such actual cubic parameter has the SAME polynomial

$$P_8(X) = X^3 - 43X^2 + 37X - 8. \tag{D16IP8}$$

No field or polynomial enumeration is being assumed.

C.13.13 Cubic isolated forks: every old-I contact vanishes

At degree three the full foreign alphabet is $m = 3, 4, 6, 7, 9, 14, 18$, since $\phi(m)/2$ divides three. The complete values of $64\mathrm{N}(1 - \beta^2)$ are $27, 8, 1, 7, 3, 1, 1$. The shared full-principal norm bound immediately excludes six/nine/fourteen/ eighteen. At native four, T is a norm-one integer with physical $T < 1369/64$ and others $< 1/4$, impossible by the first cubic lemma. For native seven, put $s_0 = \sin^2(\pi/7)$. Its three conjugates satisfy $64s^3 - 112s^2 + 56s - 7 = 0$; the exact endpoint values

$f(3/16) = -1/64$, $f(.189) = .015329216 > 0$, and $f' \geq 12$ on $[0, 1/4]$ give $3/16 < s_0 < .189$. The full principal then forces $\mathrm{N}(T) = 1$,

$$48/7 < T < 81/10, \quad x + y < 25/32, \quad Txy = 1.$$

AM–GM gives $x+y > 2/3$, so integer trace is eight. Its polynomial is X^3-8X^2+bX-1. At $1/2$, negativity gives $1 \leq b \leq 5$. The COMPLETE five discriminants are $-1871, -1563, -1175, -731, -255$. All are negative. Only $\beta = 0, 1/2$ remains at EACH fork.

For $\beta = 1/2$, let $n = \mathrm{N}(T)$. The full strip gives $n = 1, 2, 3, 4$. With cubic polynomial $X^3 - aX^2 + bX - n$, the exact root inequalities give

$$T > 64n/9, \quad a > T + 2\sqrt{n/T}, \quad a < 4203/128, \quad b^2 > n(4a - 3), \quad 192b \leq 72a + 512n - 28.$$

The last uses its negative value at $3/8$. For $n = 4$, traces $30, 31, 32$ have respective least $b = 22, 23, 23$, greatest $b = 21, 22, 22$. No coefficient fits. Thus every positive-old-contact fork has norm one, two or three.

If BOTH forks had $\beta = 1/2$, let their actual own-normalized raw products with old I of square ℓ be $b_1, b_2 < 0$. Then

$$b_i^2 = \frac{\ell}{2T_i}, \qquad W = \frac{\ell}{2b_1 b_2} \in K, \quad W^2 = T_1 T_2.$$

Integral closure makes W integral. The ACTUAL triangle through old I makes $Z = WP$ integral, where $P = r_1 r_2$. The full old-D/old-I residual determinant gives

$$(2Z - 7W - 1)^2 = (8T_1 - 3)(8T_2 - 3).$$

Modulo $2\mathcal{O}_K$, this implies $W^2 = T_1 T_2 \in 2\mathcal{O}_K$. Taking cubic norms forces eight to divide a product of two members of $\{1, 2, 3\}$, impossible. Thus any positive-contact fork has a ZERO-contact partner.

Let its parameter be T, the zero-contact partner V, and $P = r_T r_V$. Their old-I cross projection is zero. The ACTUAL five-cycle gives $X = TVP \in \mathcal{O}_K$, and

$$(P - 7/2)^2 = \frac{(8T - 3)(2V - 1)}{TV}, \qquad (2X - 7TV)^2 = 4TV(8T - 3)(2V - 1).$$

Integral closure gives $TV \in 2\mathcal{O}_K$. At every prime above two, the two last factors are units, so the valuation of TV is EVEN. If $\mathrm{N}(T) = 1$ or three, T is a dyadic unit and $V \in 2\mathcal{O}_K$, contradicting the first cubic lemma. Hence $\mathrm{N}(T) = 2$.

Its principal ideal is a UNIQUE prime $\mathfrak{p}$ of norm two, valuation of T one. Let $e = v_{\mathfrak{p}}(2) \leq 3$. At every other dyadic prime, the product integrality already makes V divisible by two. At $\mathfrak{p}$, the even product valuation requires $v_{\mathfrak{p}}(V)$ odd. If $e = 1$ or three this again gives $V \in 2\mathcal{O}_K$. Therefore $e = 2$. Cubic degree leaves exactly one other dyadic prime of norm two. Avoiding even V forces valuations one at $\mathfrak{p}$, at least two at the other. Since $1 \leq \mathrm{N}(V) \leq 10$, its norm must be eight.

For the norm-two T, write $P_T = X^3 - aX^2 + bX - 2$. The identity $2 = T(T^2 - aT + b)$ proves, WITHOUT a monogenic-order assumption: odd b gives $e = 1$; even b, odd a gives $e = 2$; both even gives $e = 3$. Thus a is odd, b even. The complete bounded coefficient comparison at $15 \leq a \leq 32$ is

a	15	17	19	21	23	25	27	29	31
b least even	12	12	14	14	14	14	16	16	16
b greatest even	10	10	12	12	12	14	14	16	16.

Only $(25, 14), (29, 16), (31, 16)$ remain, with discriminants $-984, 4(5099), 4(2263)$. The norm-eight partner has polynomial (D16IP8) and discriminant 11821. Generators of the SAME actual field must have polynomial discriminants differing by a rational square. But $5099, 2263, 11821$ are respectively three, seven, five modulo eight. Neither relevant odd rational ratio is a square. Every mixed-contact pair is impossible. Therefore BOTH cubic forks have $\beta_i = 0$.

C.13.14 The cubic zero-contact pair contradicts the same quarter-product

The positive cubic norms satisfy $1 \le n_i < 1369/128 < 11$. The degree-independent actual identity (D16Iquarter) forces $64 \mid n_1 n_2$. Their product is at most one hundred; the only factor pair with each at most ten is $n_1 = n_2 = 8$. The complete cubic norm-eight lemma then gives the SAME polynomial P_8 for both parameters. That polynomial has exactly ONE root above four and two below $1/2$, because either actual parameter has those conjugate bounds. Both physical T_i must be its unique large root. The physical embedding is injective, so $T_1 = T_2 = T$ is DERIVED.

Now $T^2/4$ is integral by the shared actual-cycle identity. Integral closure makes $T/2$ integral. It satisfies the first impossible cubic strip of Section C.13.12. This contradiction excludes the ENTIRE cubic branch, including all unequal parameters until equality was forced. No old contact with c_I was removed.

C.13.15 Quadratic isolated forks: the actual field list and square test

It remains to exclude actual degree two. We retain both independently varying fork parameters and the actual long head. The common full principal is (D16Ifull); no contact with the opposite c_I is removed. The old-I invariant already proved above satisfies

$$J = 4(1+h) \in \mathcal{O}_K, \qquad 4 < J < 123/25, \quad 0 < J' < 4.$$

It generates the actual quadratic field, since a rational J could not have these two different strips.

Let D be the positive fundamental discriminant. In the full integer basis, $J - J' = b\sqrt{D}$, b a positive integer. Thus $D < (123/25)^2 < 25$. The complete fundamental list is $5, 8, 12, 13, 17, 21, 24$. At $D = 24$, the physical strip would give $J = 2 + \sqrt{6}$, whose conjugate is negative. At $D = 5$, coefficient two would give $J = 1 + 2\varphi$, also with negative conjugate. The other fields force coefficient one. The resulting complete table is

K	full integer generator w	J	$J(J-4)$
$\mathbb{Q}(\sqrt{2})$	$s = \sqrt{2}$	$3 + s$	$-1 + 2s$
$\mathbb{Q}(\sqrt{3})$	$r = \sqrt{3}$	$3 + r$	$2r$
$\mathbb{Q}(\sqrt{5})$	$\varphi = (1+\sqrt{5})/2$	$3 + \varphi$	$-2 + 3\varphi$
$\mathbb{Q}(\sqrt{13})$	$\eta^2 = \eta + 3$	$2 + \eta$	$\eta - 1$
$\mathbb{Q}(\sqrt{17})$	$\eta^2 = \eta + 4$	$2 + \eta$	η
$\mathbb{Q}(\sqrt{21})$	$\eta^2 = \eta + 5$	$2 + \eta$	$\eta + 1$.

Here every $\eta = (1+\sqrt{d})/2$ has the indicated radicand. This is a derived list for the actual field, not a field-list hypothesis.

We will repeatedly need a full field-square test. If $R \in \mathcal{O}_K$ is a K-square, its square root is integral. Its norm must be a nonnegative rational square d^2. Moreover

$$\left(\operatorname{Tr}\sqrt{R}\right)^2 = \operatorname{Tr} R + 2d \quad \text{or} \quad \operatorname{Tr} R - 2d \tag{D16IQsquare}$$

must be an integer square. Both signs must be tested. A square rational norm alone will never count as a field-square proof. When a square really survives, we display its root.

All interval tables below use the full bases in the first table. For a strip $L < T = a + bw < U$, $0 < T' = a + bw' < V$, the embedding difference bounds the positive integer b. For each such b, the short conjugate interval contains at most one integer a, except in the explicitly wider head

strips. One then checks the physical strip. Thus the coefficient rules printed with each table prove completeness. For locating those integers, it is enough to use the strict bounds

$$1.41421 < \sqrt{2} < 1.41422, \quad 1.73205 < \sqrt{3} < 1.73206, \quad 2.23606 < \sqrt{5} < 2.23607,$$
$$3.60555 < \sqrt{13} < 3.60556, \quad 4.12310 < \sqrt{17} < 4.12311, \quad 4.58257 < \sqrt{21} < 4.58258.$$

Each follows by squaring positive rational endpoints. No polynomial or field enumeration is an input to these tables.

C.13.16 Quadratic isolated forks: every positive old-I contact is excluded

For a positive finite contact $\beta = \cos(\pi/m)$, the degree of $\mathbb{Q}(\beta^2)$ divides two. The complete alphabet is $m = 3, 4, 5, 6, 8, 10, 12$. For an irrational β^2, this subfield is the whole actual K. The full norm bound is

$$1 \le \mathrm{N}(T) < \frac{1369}{64}\mathrm{N}(1 - \beta^2).$$

Native twelve gives the physical upper bound $(1369/128)(2 - \sqrt{3}) < 4$. Native six has norm one, $8 < T < 1369/128$, $0 < T' < 1/8$; its integral trace is nine or ten, giving respectively $\mathbb{Q}(\sqrt{77})$ or $\mathbb{Q}(\sqrt{6})$, neither in our actual list. At native ten the field is golden, $T < 4.086$, $T' < .328$. Writing $T = a + b\varphi$ gives $1 < b = (T - T')/\sqrt{5} < 2$, impossible.

Further information comes from the actual two-cycle with c_I. Its unit mutual has magnitude

$$M = \sqrt{hA^2} - \beta\sqrt{1 + h},$$

and $4M^2$ is an integral invariant. All the squared terms belong to K, so expansion makes $h(1+h)\beta^2A^2$ a K-square. We impose no prescribed label on the actual mutual with c_I.

At native four, clearing rational square denominators requires

$$R = J(J - 4)(4T - 1) \in K^2.$$

The full strips $4 < T < 1369/64$, $0 < T' < 1/4$ give exactly

K	T	R	$\mathrm{N}(R)$ or both trace tests
$\sqrt{2}$	$3 + 2s$	$21 + 14s$	$49;\ 56, 28$
$\sqrt{2}$	$10 + 7s$	$73 + 50s$	329
$\sqrt{3}$	$7 + 4r$	$96 + 54r$	468
φ	$2 + 3\varphi$	$22 + 33\varphi$	$121;\ 99, 55$
φ	$5 + 8\varphi$	$58 + 89\varphi$	605
$\sqrt{13}$	$4 + 3\eta$	$21 + 15\eta$	$81;\ 75, 39$
$\sqrt{17}$	$8 + 5\eta$	$80 + 51\eta$	76
$\sqrt{21}$	$2 + \eta$	$27 + 15\eta$	$9;\ 75, 63.$

The coefficient bounds $b \le 7, 6, 9, 5, 5, 4$ in field order, with the unique integer $0 < a + bw' < 1/4$, give every row after the physical bound. Every isolated norm is nonsquare. In the square-norm rows both integer trace tests fail. Thus no native-four contact survives.

At native five the golden strips give $T = 2 + 3\varphi, 4 + 6\varphi$. Here $b \le 6$ and $0 < a + b\varphi' < (\varphi + 2)/8$. The necessary square

$$(1 + 4\varphi)(8T + \varphi - 3)$$

has values $113+177\varphi$ and $225+361\varphi$, of nonsquare norms $1441, 1529$. At native eight, $K=\mathbb{Q}(\sqrt{2})$, $b\le 2$, and only $T=3+2s$ lies in the full strips. The necessary square is

$$(2+3s)(8T+s-2)=146+100s, \qquad \mathrm{N}=1316,$$

again nonsquare. This removes every positive label at least four, including the genuine native-eight strip survivor.

For native three, $\beta=1/2$, the necessary domain becomes

$$4<T<4107/128, \quad 0<T'<3/8, \qquad R=J(J-4)(8T-3)\in K^2.$$

The coefficient bounds are $b\le 11, 9, 14, 8, 7, 7$; the conjugate strip again supplies the unique a. The complete twenty rows and norms are

K	all T	corresponding $\mathrm{N}(R)$
$\sqrt{2}$	$3+2s;\ 6+4s;\ 10+7s;\ 13+9s$	497; 161; 2401; 1169
$\sqrt{3}$	$7+4r;\ 9+5r;\ 14+8r$	3156; 468; 4884
φ	$2+3\varphi;\ 4+6\varphi;\ 5+8\varphi;\ 7+11\varphi;\ 9+14\varphi$	1045; 781; 3949; 2981; 605
$\sqrt{13}$	$4+3\eta;\ 8+6\eta$	573; 789
$\sqrt{17}$	$5+3\eta;\ 8+5\eta;\ 11+7\eta$	188; 956; 2236
$\sqrt{21}$	$2+\eta;\ 9+5\eta;\ 11+6\eta$	141; 1437; 645.

All norms except 2401 are nonsquare. The remaining case really passes:

$$K=\mathbb{Q}(\sqrt{2}), \quad T=10+7s, \qquad R=147+98s=[7(1+s)]^2.$$

It must be retained. If it occurred at both forks, the two physical parameters would be equal and the actual unit mutual would be $1-T/4$. Its two-cycle would be

$$(T-4)^2/4=67/2+21s,$$

which is not integral in the full ring $\mathbb{Z}[s]$. Thus a positive-contact fork must have a zero-contact partner V.

The complete zero-contact $\mathbb{Q}(\sqrt{2})$ strip gives

$$V=3+2s,\ 6+4s,\ 10+7s,\ 13+9s,\ 16+11s,\ 17+12s,\ 20+14s.$$

For these actual caps let P be their raw product. The old-I cross projection is zero, so the actual residual identity and five-cycle give

$$(P-7/2)^2=\frac{(8T-3)(2V-1)}{TV}, \quad X=TVP\in\mathcal{O}_K, \quad (2X-7TV)^2=4TV(8T-3)(2V-1).$$

Integral closure makes $X-7TV/2$ integral, hence $TV/2$ integral. At the unique prime over two, normalized by $v(2)=2$, the last two factors are units. Thus $v(TV)$ is even and at least two. Since $\mathrm{N}(T)=2$, $v(T)=1$; the partner valuation must be odd and positive. The seven partner norms are $1, 4, 2, 7, 14, 1, 8$. Only

$$V=10+7s, \quad 16+11s, \quad 20+14s$$

remains. But the determinant requires $TV(8T-3)(2V-1)$ to be a field-square. Its respective rational norms are

$$4\cdot 7^3\cdot 31, \quad 4\cdot 7^5, \quad 16\cdot 7^3\cdot 47.$$

All are nonsquare. This excludes the last mixed pair. Both actual quadratic forks therefore have ZERO old-I contact. Their contacts with c_I have never been set to zero.

C.13.17 Quadratic isolated forks: the complete actual head list

We may now use the common quarter-product (D16Iquarter). It is not enough by itself in degree two, so we also retain the actual long head already proved isolated and dotted:

$$c_H = A_H p - g\omega_H - \epsilon e, \quad \omega_H^2 = 2, \quad g > 1, \quad \epsilon = 0 \text{ or } 1/2, \quad U = 4g^2.$$

The preceding original-vector argument gave $(1-\epsilon^2)/g^2 > 3/8$. Actual raw own normalization gives head square $2/U$, raw old-D dual square one. The physical principal is Lorentzian over $\mathbb{Q}(U)$; actual-field descent gives $K = \mathbb{Q}(U)$. The FULL old-I Schur complement at the other embedding is $2(1-\epsilon^2)/U' - 1 > 0$. Consequently

$$U \in \mathcal{O}_K, \qquad 4 < U < \frac{32}{3}(1-\epsilon^2), \qquad 0 < U' < 2(1-\epsilon^2). \tag{D16IQhead}$$

No raw scale of the disconnected old I has been prescribed.

At $\epsilon = 0$, the actual c_H, c_I two-cycle requires

$$(J-4)(U-2)/2 \in \mathcal{O}_K.$$

At $\epsilon = 1/2$, the same expansion as before requires $J(J-4)(2U-3) \in K^2$. Here is the COMPLETE pre-test list in (D16IQhead):

K	$\epsilon = 0$	$\epsilon = 1/2$
$\sqrt{2}$	$3+s;\ 3+2s;\ 4+2s;\ 5+3s;\ 6+3s$	$3+2s;\ 4+2s$
$\sqrt{3}$	$3+r;\ 4+2r;\ 5+2r$	$3+r;\ 4+2r$
φ	$2+2\varphi;\ 3+2\varphi;\ 2+3\varphi;\ 3+3\varphi;\ 3+4\varphi;\ 4+4\varphi$	$2+2\varphi;\ 2+3\varphi;\ 3+3\varphi$
$\sqrt{13}$	$2+\eta;\ 3+\eta;\ 3+2\eta;\ 4+2\eta$	$2+\eta;\ 3+2\eta$
$\sqrt{17}$	$2+\eta;\ 3+\eta;\ 4+2\eta;\ 5+2\eta$	$2+\eta;\ 3+\eta$
$\sqrt{21}$	$2+\eta;\ 3+\eta;\ 4+2\eta;\ 5+2\eta$	$2+\eta;\ 3+\eta.$

The coefficient bounds are $b \le 3, 3, 4, 2, 2, 2$. For each b, one or two integers satisfy $0 < a + bw' < 2$, respectively $3/2$, before the physical bound.

To show every deletion explicitly, the $\epsilon = 0$ cycle values, in the displayed order, are

$\sqrt{2}$	$1/2,\ (3-s)/2,\ 1,\ 3/2,\ (2+s)/2$
$\sqrt{3}$	$1,\ 2,\ (3+r)/2$
φ	$1,\ (1+\varphi)/2,\ 3/2,\ (2+\varphi)/2,\ (3+\varphi)/2,\ 1+\varphi$
$\sqrt{13}$	$(3-\eta)/2,\ 1/2,\ (4-\eta)/2,\ 1$
$\sqrt{17}$	$(4-\eta)/2,\ 1,\ 2,\ (2+\eta)/2$
$\sqrt{21}$	$(5-\eta)/2,\ 3/2,\ 3,\ (4+\eta)/2.$

Every nonretained value is nonintegral in the FULL written integer basis. For $\epsilon = 1/2$, the corresponding square-test norms are

$\sqrt{2}$	$161,\ 49$
$\sqrt{3}$	$36,\ 276$
φ	$121,\ 319,\ 99$
$\sqrt{13}$	$27,\ 81$
$\sqrt{17}$	$52,\ 4$
$\sqrt{21}$	$51,\ 15.$

The square norm 81 in $\sqrt{13}$ is not sufficient: the radicand $9+3\eta$ has trace twenty-one and fails both tests $39, 3$ in (D16IQsquare). The retained square roots in $\sqrt{2}, \sqrt{3}, \varphi, \sqrt{17}$ are $3+s, 3+r, 3+\varphi, 2+\eta$. The full head survivors are therefore

K	(U, ϵ)	$\mathrm{N}(H_U)$ in that order
$\sqrt{2}$	$(4+2s, 0),\ (4+2s, 1/2)$	$-128,\ -56$
$\sqrt{3}$	$(3+r, 0),\ (3+r, 1/2),\ (4+2r, 0)$	$-48,\ -18,\ -128$
φ	$(2+2\varphi, 0),\ (2+2\varphi, 1/2),\ (4+4\varphi, 0)$	$-64,\ -44,\ -256$
$\sqrt{13}$	$(4+2\eta, 0)$	-192
$\sqrt{17}$	$(3+\eta, 0),\ (3+\eta, 1/2),\ (4+2\eta, 0)$	$-64,\ -8,\ -256$
$\sqrt{21}$	$(4+2\eta, 0)$	$-192.$

The constant used in the last column is

$$H_U = \begin{cases} 2U(U-2), & \epsilon = 0, \\ U(2U-3), & \epsilon = 1/2. \end{cases}$$

For each zero-old-contact fork, the SAME actual one-dimensional K-time line requires

$$H_U q(T) \in K^2, \qquad q(T) = T(2T-1). \tag{D16IQheadfork}$$

Indeed the squared raw residual-time norms are $(U-2)/U$, respectively $(2U-3)/(2U)$, for the head, and $2(2T-1)/T$ for a fork. Their product is a K-square: subtract the actual K-valued old-D and old-I projections from the actual raw mutual. Clearing SQUARE denominators gives exactly the displayed condition. It neither identifies disconnected raw root scales nor drops the head's old-I projection.

C.13.18 Quadratic isolated forks: every field and unequal pair closes

The full zero-old-contact strips are $4 < T < 1369/32$, $0 < T' < 1/2$. The coefficient bounds $b \le 15, 12, 19, 11, 10, 9$, with the unique a in the conjugate strip, give the COMPLETE lists

K	all T
$\sqrt{2}$	$3+2s;\ 6+4s;\ 10+7s;\ 13+9s;\ 16+11s;\ 17+12s;\ 20+14s$
$\sqrt{3}$	$7+4r;\ 9+5r;\ 14+8r;\ 16+9r;\ 21+12r$
φ	$2+3\varphi;\ 4+6\varphi;\ 5+8\varphi;\ 6+9\varphi;\ 7+11\varphi;\ 9+14\varphi;\ 10+16\varphi;\ 11+17\varphi;\ 12+19\varphi$
$\sqrt{13}$	$3+2\eta;\ 4+3\eta;\ 7+5\eta;\ 8+6\eta;\ 12+9\eta$
$\sqrt{17}$	$2+\eta;\ 5+3\eta;\ 8+5\eta;\ 11+7\eta;\ 16+10\eta$
$\sqrt{21}$	$2+\eta;\ 4+2\eta;\ 9+5\eta;\ 11+6\eta;\ 13+7\eta.$

We use the quarter-product at every prime over two, and the actual head condition (D16IQheadfork). All valuations give a uniformizer value one. The elementary quadratic splitting of two follows by factoring the full basis polynomial modulo two; the ramified cases can equivalently be checked by the norms and division in their full bases.

For $K = \mathbb{Q}(\sqrt{2})$, the unique dyadic prime has $v(2) = 2$. The seven valuations in table order are $0, 2, 1, 0, 1, 0, 3$. The quarter-product requires a sum at least four, so either fork's valuation-zero rows are impossible. The remaining rows and q-norms are

$$(6+4s, -28), \quad (10+7s, -62), \quad (16+11s, -98), \quad (20+14s, -376).$$

Multiplication by the head norm -56 gives no rational square. For head norm -128, only $T = 16+11s$ passes this necessary norm test. It would occur at both forks, whose valuation sum would be two, contradicting the quarter-product.

For $K = \mathbb{Q}(\sqrt{3})$, the five valuations at the ramified prime, again $v(2) = 2$, are $0, 1, 2, 0, 0$. The quarter-product forces both $T = 14 + 8r$. Its q-norm is -156. None of its products with the three head norms $-48, -18, -128$ is a rational square. Thus the head condition fails.

For $K = \mathbb{Q}(\varphi)$, two is inert. Only $T = 4 + 6\varphi, 10 + 16\varphi$ has positive valuation, in each case one; every other row is a dyadic unit. Both forks must lie in this pair. Their q-norms are $-44, -220$. The head norms rule out $\epsilon = 0$; at $\epsilon = 1/2$ they force both $T = 4 + 6\varphi$. For its remaining head $U = 2 + 2\varphi$,

$$H_U q(T) = 3916 + 6336\varphi, \quad \mathrm{N} = 1936, \quad \mathrm{Tr} = 14168.$$

The two trace-square tests are $14256, 14080$, neither square. This contradicts the full field-square condition.

For $K = \mathbb{Q}(\sqrt{13})$, two is again inert. Only $T = 8 + 6\eta$ has positive dyadic valuation, equal to one. Both forks would have that parameter. The unique head gives

$$H_U q(T) = 38208 + 29328\eta, \quad \mathrm{N} = 20736, \quad \mathrm{Tr} = 105744.$$

Its two tests $106032, 105456$ are nonsquare, so this field also fails.

For $K = \mathbb{Q}(\sqrt{17})$, two splits. Label its primes by $\eta = 0, 1$ modulo two. The five valuation pairs are

$$(1, 0),\ (0, 2),\ (2, 0),\ (0, 1),\ (3, 1),$$

and the corresponding q-norms are $-2, -36, -100, -98, -304$. The quarter-product requires the two valuation pairs to sum componentwise to at least $(2, 2)$. At $\epsilon = 1/2$, head norm -8 permits only the first and fourth rows. No pair from $\{(1, 0), (0, 1)\}$ can meet that requirement. At $\epsilon = 0$, head norm -64 or -256 permits only the second and third rows. Their two primes force one of EACH:

$$T_1 = 5 + 3\eta, \qquad T_2 = 8 + 5\eta.$$

This genuinely unequal pair has not been discarded. But the shared actual fork time line requires $q(T_1)q(T_2)$ square, where

$$q(T_1)q(T_2) = 98940 + 63360\eta, \quad \mathrm{N} = 3600, \quad \mathrm{Tr} = 261240.$$

Its two trace tests are $261360, 261120$. The latter is $511^2 - 1$; the former lies strictly between 511^2 and 512^2. Neither is square.

Finally, in $K = \mathbb{Q}(\sqrt{21})$, two is inert. Only $T = 4 + 2\eta$ has positive valuation, equal to one, so both forks have that parameter. The unique actual head gives

$$H_U q(T) = 9888 + 5520\eta, \quad \mathrm{N} = 2304, \quad \mathrm{Tr} = 25296.$$

Its tests $25392, 25200$ are both nonsquare. For clarity the remaining larger square comparisons used above are

$$\begin{aligned} 119^2 < 14256 < 120^2, \qquad & 118^2 < 14080 < 119^2, \\ 325^2 < 106032 < 326^2, \qquad & 324^2 < 105456 < 325^2, \\ 159^2 < 25392 < 160^2, \qquad & 158^2 < 25200 < 159^2. \end{aligned}$$

This exhausts all six fields derived from the actual old-I strip.

The isolated fork case is therefore impossible in degree two as well as degrees three and four. Together with the already proved degree bound, full actual fork inventory and nonisolated exclusions, this excludes the ENTIRE $D_{16} + I$ alternative at the original farthest vertex. Both forks, the long head and the actual I-opposite belong to that same original configuration. Every geometric height and price used earlier concerns its one original vector; the algebraic branches add no energy from a different point. No intrinsic face was assumed Coxeter or arithmetic, no prescribed lattice was introduced, and no two fork parameters were equated before equality was forced. The finite tables in this quadratic proof are complete hand arguments in explicitly bounded full integer bases, rather than new assumed script outputs.

C.14 Complete forest dispatch and the final dimension deduction

The local arguments now meet the two necessary whole-forest covers. We make this last interface explicit: a list of possible forests is a necessary classification, while each row below is excluded by a theorem at its own actual rank and with its whole spectator forest retained.

C.14.1 The component inputs used before the partition count

Most older component floors in the partition proofs follow at once from stronger results already proved here:

whole component and hypothesis	available strict floor
A_{12}, arbitrary spectators	5/7
A_{14}, arbitrary spectators	4/5
A_{15}, arbitrary spectators	.8348
A_{16}, arbitrary spectators	.8703
A_{17}, arbitrary spectators	.9032.

Thus the older $A_{12} > .6725$, typed $A_{12} > .703101$, $A_{14} > .76$, typed $A_{14} > .7897732$, $A_{15} > .79$, $A_{16} > .83$ and $A_{17} > 6/7$ floors used in Propositions 20.5 and 22.1 require no separate field census or historical numerical chapter. The first two rows are (A12U) and Section C.10.1; the last three were proved by the one-reference simultaneous-foot calculation in Sections C.2.4 and C.2.5. These are whole-component estimates with arbitrary spectators, not higher-dimensional whole-forest exclusions.

The conditioned A_{11} floor needs its own short corollary; it cannot be deduced from the weaker uniform conclusion 13/20.

Corollary C.14.1. *If an actual I, N_7 or O_7 occurs beside a whole A_{11}, then $e_{A_{11}} > .652761$, with arbitrary other finite spectators.*

Proof. The complete $N = 12$ inventory in Section C.10.6 is local and has no contrary component-energy hypothesis. It leaves the isolated feet and exactly the six pure nonisolated golden/silver time types printed in Section C.10.9. With an actual I, that section's two-cycle and denominator proof excludes all six independently of an assumed A_{11} energy.

With either actual typed-seven defect, the indefinite actual three-root principal gives $K = K_7$. Every one of the six nonisolated caps instead has an actual Lorentzian principal over $\mathbb{Q}(\varphi)$ or $\mathbb{Q}(\sqrt{2})$. The permitted original-root normalization and actual-field descent therefore force the whole actual field to be that quadratic field. This contradicts its cubic degree. The argument concerns the actual cap principal, not a unit-root normalizer adjoined to K. Thus all nonisolated exceptions are excluded in every stated branch.

Use the no-helper reference

$$H_5 = H_7 = 1.805, \quad H_6 = 1.839, \qquad E = .652761, \quad (\beta_5, \beta_6, \beta_7) = (.1635, .034, .1635).$$

Every retained site is strong or has its actual isolated foot. For any set of feet, use the SAME vector $Z = w - \sum \beta_i u_i n_i$. The continuous inequalities already proved in Section C.10.8 have no contrary-energy assumption: for $u \le 1/2$, the raw loss is $< .6085 < .61$; for $u \ge 1/2$, its loss after the smallest available coefficient .034 is $< .60775 < .61$. They apply to every pure or foreign foot. Every adjacent-foot cross has the nonpositive old-root sign. The single completed square therefore gives $e_A > E$. With no foot, the strict strong pressures give the same strictness. Neither necessary forest cover has been used in this proof. □

All ordinary geometric rates, defect compatibility and the rank-valid original-point/prefix interfaces still refer to their complete proofs in the general compact companion. The precise interfaces used are its inverse/projection/point-pressure lemmas, reciprocal-shadow proposition, ordinary-component appendix, actual typed-propagation appendix, uniform A-chain and B/D component floors, and actual long-prefix inventory. Our arithmetic helper seals and the explicit local-rank boundary checks then give the stronger prefix floors used in the two partition proofs. This distinguishes the shared geometric foundation from the arithmetic branch.

C.14.2 Every rank-seventeen row has a complete consumer

Apply Proposition 20.5 at the original farthest vertex. For groups one through fourteen of Table 1, the following is a complete dispatch. The cited sections contain the actual-wall, field and individual coefficient arguments; a final scalar is only their last step.

Group	Complete exclusion on the one original vector
1	The uniform $A_{11} > 13/20$ theorem in Section C.10.8 plus the ordinary rank-five cost $> 5/14$ gives $13/20 + 5/14 = 141/140 > 1$. The I energy is not needed.
2	The same uniform theorem, the actual defect counted once, and ordinary $T_4 > 2/7$ give $13/20 + .09 + 2/7 > 1$.
3	Section C.10.5 gives $5/7$. Every retained ordinary rank-five partition costs $> 5/14$. Their disjoint projections exceed one.
4	Use that same $A_{12} > 5/7$ component theorem. For $I + T_4$ the joint cost is $> 23/70 > 2/7$; for $I + T_3 + A_1$ it is $> 2/7 + .1$; for $I + 2T_2$ it is $> .21 + 1/7 > 2/7$. Each counts I exactly once.
5	Section C.10.1 gives $4/5$. Ordinary T_3 or $T_2 + A_1$ costs $> 17/70$; the actual $I + T_2$ costs $> .21$. Each total exceeds one.
6	Section C.11.8 excludes the entire ordinary T_2 branch. Two ordinary A_1 or $I + A_1$ close with $.8348 + .2 > 1$ or $.8348 + .20204 > 1$, respectively. Section C.11.7 excludes both distinct actual typed branches, at their cubic field and with their complete foreign costs.
7	Section C.11.9 treats the SAME actual rank-one opposite, first its nonisolated alternative, then every isolated height. This includes ordinary A_1 and actual short I.

Group	Complete exclusion on the one original vector
8	Sections C.3 and C.3.5 prove actual endpoint isolation, the degree bound, the full two-dotted contradiction and finite-anchor field dispatch. Both quadratic and cubic domains of Proposition 21.1(i),(ii) are empty at the declared numerical output boundaries.
9	Section C.12.1 gives the actual typed head and complete rank-fifteen domain before its use: $.88989 + .0487 + .09 > 1$. Both N_7, O_7 are included.
10	Sections C.12.2 through C.12.5 give the complete original-rank short-cap inventory, actual degree two/three dispatch and full head/foreign/contact cases. Every remaining parameter model is excluded by its actual original head and I compatibility.
11	Sections C.9.2, C.9.3 and C.9.4 prove the B_{17}-specific reduction and actual inclusion in Proposition 21.1(iii). Its two surviving necessary pairs each fail the fully paid original-vector consumer.
12	Section C.12.1 gives $.86989 + .0487 + .09 > 1$ in the distinct D_{15} typed domain, with both actual defects.
13	Sections C.12.6 through C.12.9 exclude the ordinary rank-one branch. Sections C.13.1 through C.13.18 exclude the actual-I branch, including every actual degree two, three and four. The ordinary nonshort energy is never transferred to the short-I proof.
14	Sections C.9.5 through C.9.11 give the separate genuine-fork, head and field reduction for D_{17}. They independently enter the same algebraic domain (iii) and exclude its two survivors by their own original-wall parameter meanings.

All ordinary, J, actual-I and both actual typed alternatives were considered in the preceding cover, before this dispatch. There is no omitted A_{13} row. A bare seven bond has not been identified with a typed defect. Hence no rank-seventeen forest remains, proving Corollary 21.2 by the complete local arguments.

C.14.3 Every rank-sixteen row closes at its own rank

Use the fresh Proposition 22.1; no node is deleted from a rank-seventeen forest. For the complete nineteen groups of Table 3, the dispatch is

Group	Complete exclusion on the original vector
1,2	Section C.10.14 proves $404/625$ under the actual I or either actual typed defect. The totals are $404/625 + 5/14 > 1$ and $404/625 + .09 + 2/7 > 1$.
3	Section C.10.8 gives $13/20 + 5/14 > 1$ for every retained ordinary rank-five forest.
4	Section C.10.10 proves the separate JOINT $A_{11} + I > 18/25$ with all remaining feet. Its disjoint $T_4 > 2/7$ gives $18/25 + 2/7 = 176/175 > 1$.

Group	Complete exclusion on the original vector
5	The uniform $A_{11} > 13/20$, the actual $O_7 > .09$, and Lemma 22.2 give $13/20 + .09 + 13/50 = 1$, strictly exceeded.
6,7,8	The universal Section C.10.5 gives $5/7$. For group6 ordinary rank four costs $> 2/7$; for group7 the JOINT $I + T_3 > 2/7$. For group8 the actual defect propagates the last ordinary $T_2 > .2$, and $5/7 + .09 + .2 > 1$. The older weaker typed component rate needs no duplicate proof.
9	Section C.10.2 excludes ordinary A_3 and $T_2 + A_1$ with its $53/70$ component rate. Section C.10.3 excludes actual $O_7 + A_1$ by the complete cubic full-order strip. Section C.10.4 excludes $I + T_2$ using both endpoints, each remaining foot and the ordinary $3/19$ dihedral floor, with I counted once.
10	The universal $A_{14} > 4/5$ plus $2A_1 > .2$ or the JOINT $I + A_1 > .20204$ excludes those two reducible cases. Sections C.11.1 through C.11.3 exclude both actual typed defects. Sections C.11.4 through C.11.6 exclude the ordinary irreducible T_2 in its distinct field domain.
11	Section C.5 takes the same actual rank-one opposite and exhausts its nonisolated/isolated dichotomy. The isolated proof covers every $h > 0$, including $1/2$ and one.
12	Section C.4 gives the full actual $N = 17$ endpoint, field and finite-anchor proof, including every one of the twenty-one time norms.
13	Sections C.6 and C.6.1 give $.86989 + .0487 + .09 > 1$ with the genuine mixed factor and both actual typed defects.
14,18	Section C.5.8 and the common ordinary Section C.5.9 exclude nonisolated opposites and all isolated ordinary $h \geq 1/2$. Sections C.5.10 through C.5.14 and C.5.15 through C.5.18 give the separate short-I proofs for $0 < h < 1/2$. Equality $h = 1/2$ belongs to the ordinary theorem, so no boundary or actual rank-one opposite is lost.
15	Sections C.8.1 through C.8.5 give the actual whole-B_{16} reduction and cubic exclusion. The complete quadratic output Proposition 21.1(iv) leaves one pair; its actual edge consumer gives $1.00640742675 > 1$.
16	Section C.7 proves the fresh short-layer exclusions before using $e_D > .82436$. Every rank-three remainder, including $I + T_2$ and $O_7 + A_1$, has the proved joint cost $> .21$; the total exceeds one.
17	The ordinary dihedral theorem and its actual D_{14} consumer in Section C.1 exclude the $m \geq 4$ ordinary case. Section C.6 separately proves the full actual D_{14} centre/head argument, then $.86989 + .0487 + .09 > 1$, for both typed branches.

Group	Complete exclusion on the original vector
19	The actual whole-D_{16} reduction Proposition 23.1 and manual dyadic proof Proposition 26.1 exclude the last forest. There is no whole-D_{16} census premise.

For the rank-one rows, taking the same actual opposite is essential. If it has a nonzero old-large-component contact, use the full nonisolated theorem. Otherwise its normalization has the isolated form and lies in exactly one of the stated height intervals. This dichotomy does not replace an actual wall by a formal Gram extension. All sums concern disjoint whole components of the SAME original x, after any overlapping head/end/foot pressures were jointly paid inside the indicated proof.

Thus groups one through eighteen prove Proposition 22.3; group nineteen proves that no rank-sixteen forest occurs.

C.14.4 The acyclic global implication and the verification boundary

The general compact bound is Theorem 8.1. Its proof includes the exact dimension-eighteen exclusion and an independently proved tail for every dimension at least nineteen. That tail uses only the geometric parts of its mixed-source supplements. It uses no arithmetic exact-rank theorem, field census or conclusion of this paper. Hence Proposition 19.2 is a one-way input here.

If a compact arithmetic Coxeter polytope had dimension greater than fifteen, that companion would leave exact dimension seventeen or sixteen. The two complete covers and the preceding local dispatches exclude both. This proves Theorem 19.1. No induction down dimensions, face arithmeticity, hereditary Coxeter claim or transfer of a farthest vertex to a neighbour occurs.

The proof graph is therefore

shared original-vector geometry
$\longrightarrow$ general compact 17,

shared geometry + actual arithmetic structure
$\longrightarrow$ component and endpoint lemmas
$\longrightarrow$ fresh exact-rank covers and actual domain inclusions,

these inclusions + four declared finite outputs
$\longrightarrow$ exact 17 and exact 16 exclusions,

general compact 17 + both exact-rank exclusions
$\longrightarrow$ 15.

The actual-domain inclusions for B_{17} and D_{17} are distinct, although their quadratic algebraic domain is shared. The whole-D_{16} finish is manual. Neither should be conflated with the $D_{16} + I$ degree-two/three/four proof at ambient rank seventeen.

The algorithmic propositions retain their declared four numerical output premises: the quadratic A_{17} empty domain; the two cubic A_{17} empty finite-anchor domains; the two survivors in the shared quadratic B/D_{17} domain; and the one whole-B_{16} quadratic survivor. Section 21 proves domain finiteness, coverage, exact decision rules and termination. The actual geometric inclusion proofs are separate, and every nonempty output is followed by its explicit original-vector contradiction. The producer runs and same-code replays retain the execution scopes stated there.

The admissible quadratic-form model, actual K-root lines, cyclic integrality and positive definiteness at every nonphysical place are the named external arithmetic structure. The exact literal golden-lattice cutoff used in Section C.9.1 is a separate external premise, attributed to [12]; that section proves the passage to every compatible invariant lattice rather than infer it from similarity or commensurability.

C.15 Guide to the arithmetic proof appendices

The following table locates the component estimates and final reductions.

Interface	Printed mathematical proof
A11/A10	Sections C.10.6–C.10.14; uniform 13/20, joint 18/25 and the distinct conditioned Corollary C.14.1.
A12–A17	Sections C.10.5, C.10.2, C.10.1, C.11.1, C.11.4, C.11.7, C.11.8, C.11.9, C.4, C.3.
Classical typed and rank-one	Sections C.6, C.12.1, C.7, C.5.8, C.5.10, C.5.15.
B16/D16 plus I	Sections C.12.2–C.12.5 and C.13.1–C.13.18; ordinary D16 is separate in Section C.12.6.
Whole classical rows	Sections C.8.1, C.9.2, C.9.5 and Section 23; all four computed premises are in Section 21.
Covers and global implication	Propositions 20.5, 22.1 and Sections C.14.2–C.14.4; the general compact bound uses the independent companion interface Theorem 8.1.

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